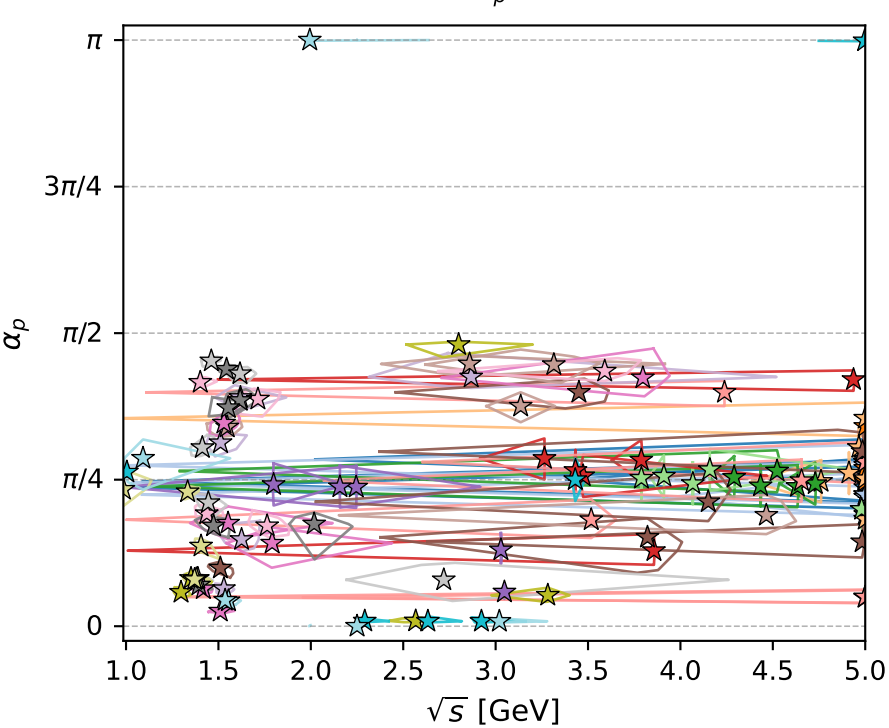
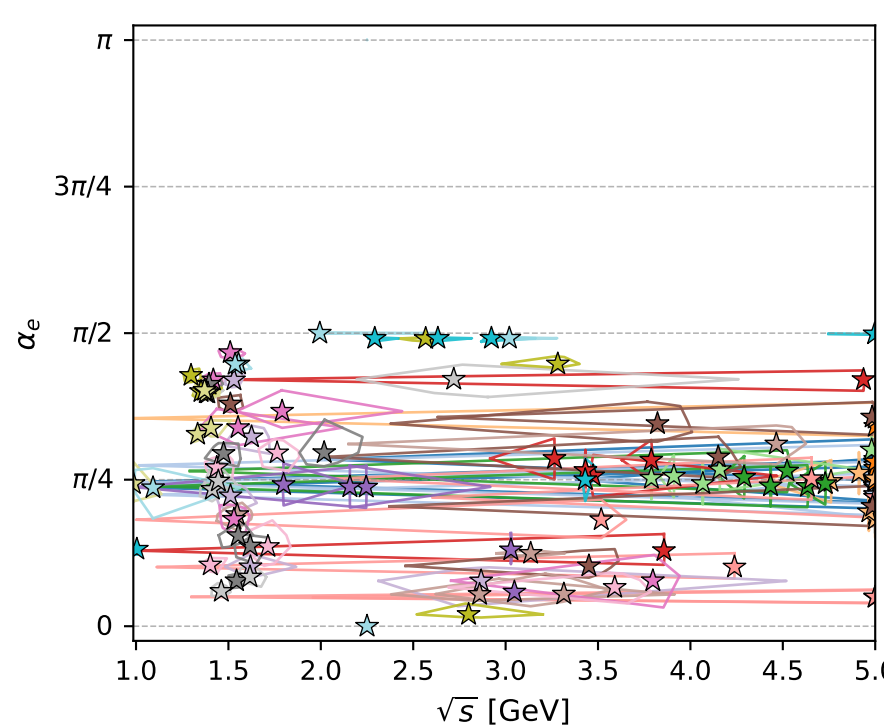
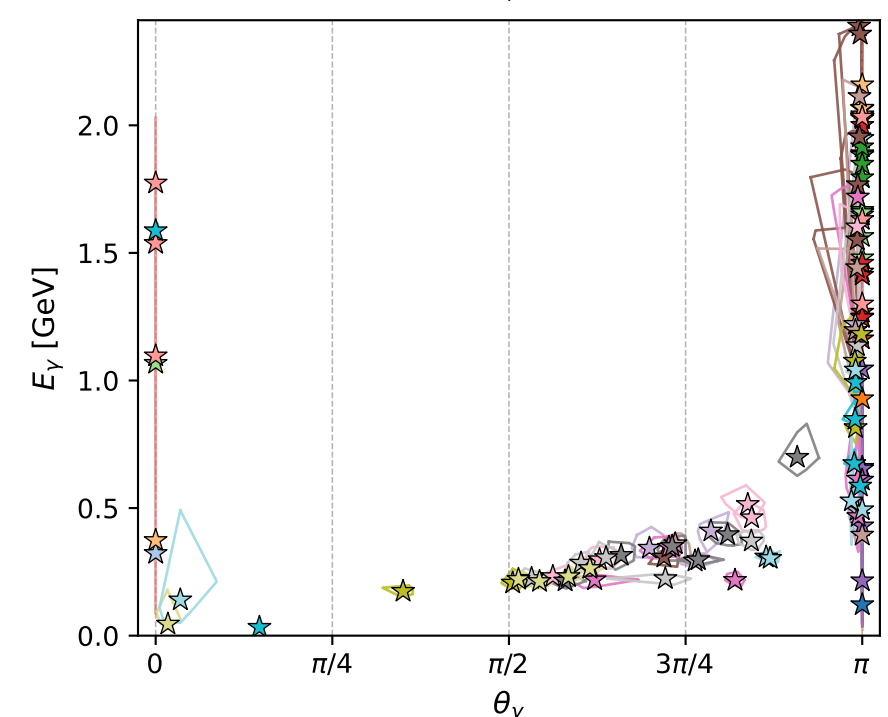
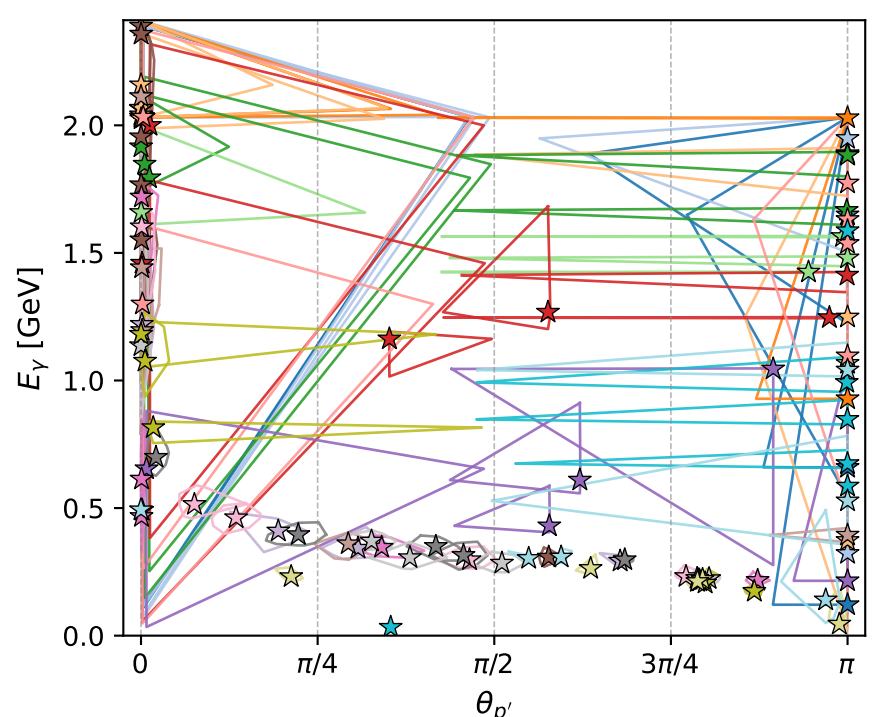
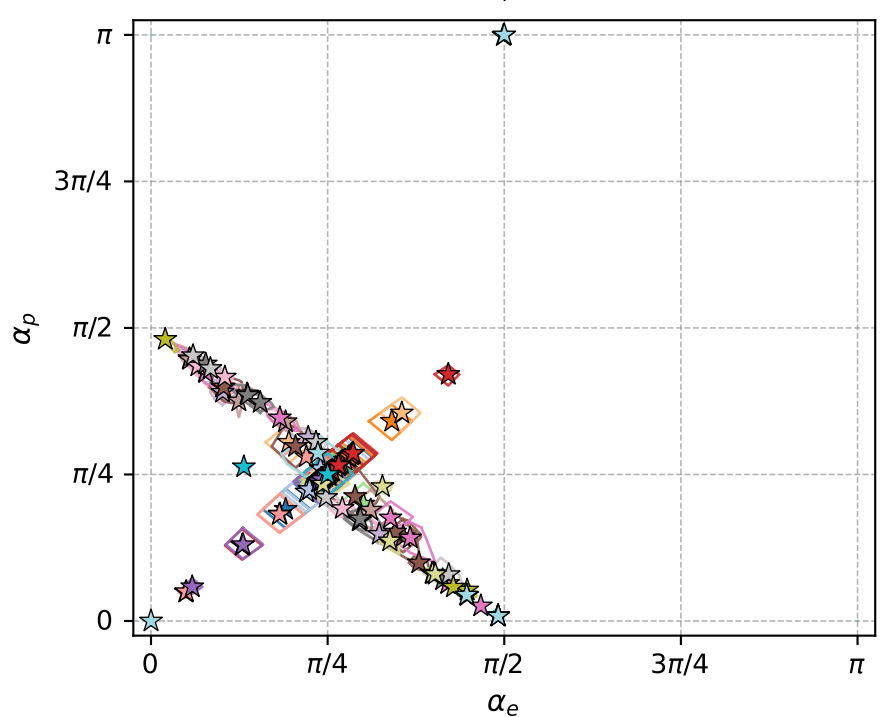
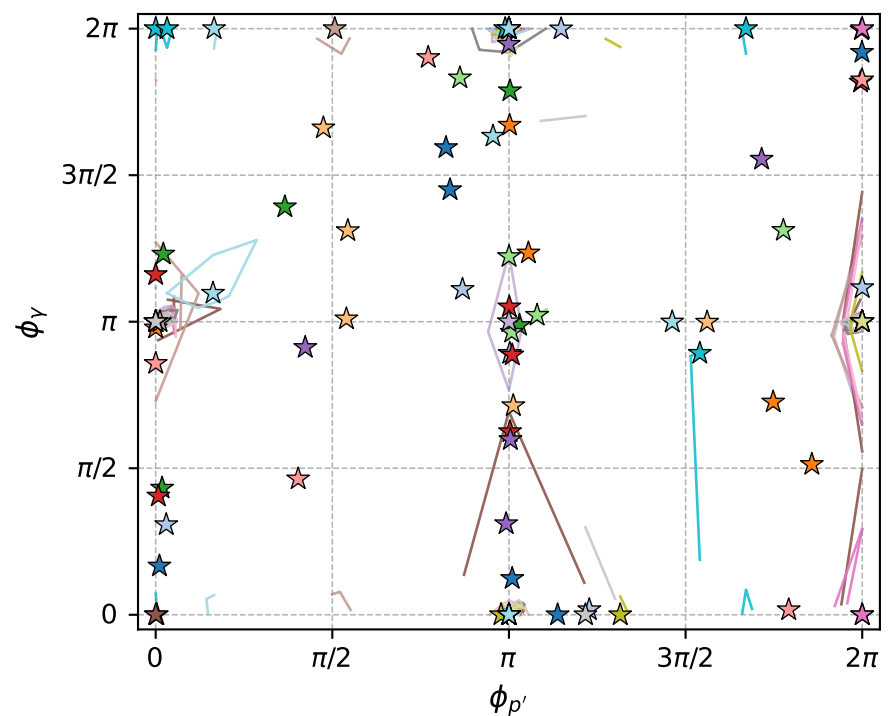
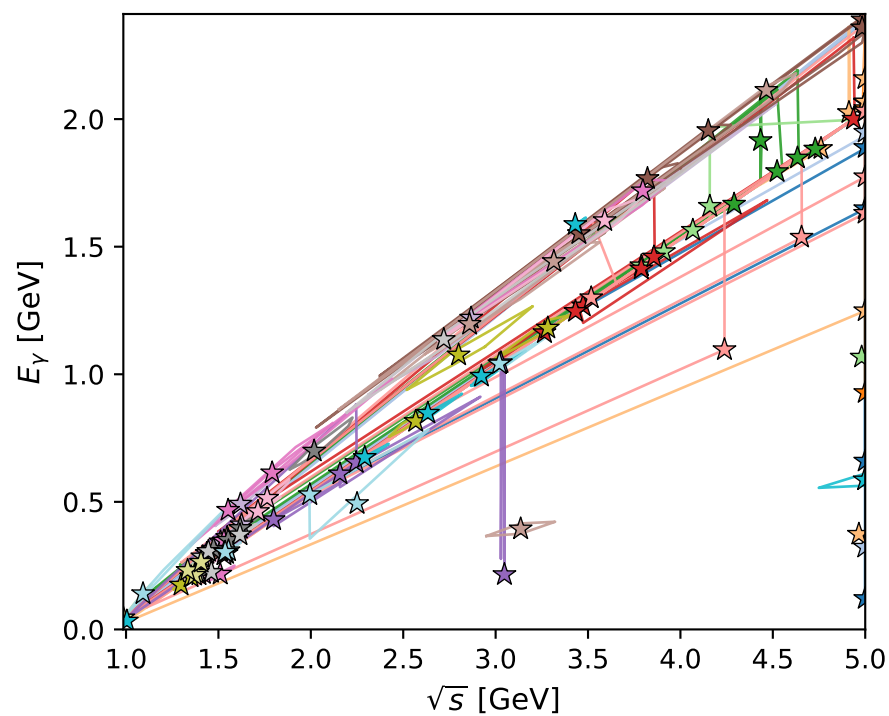
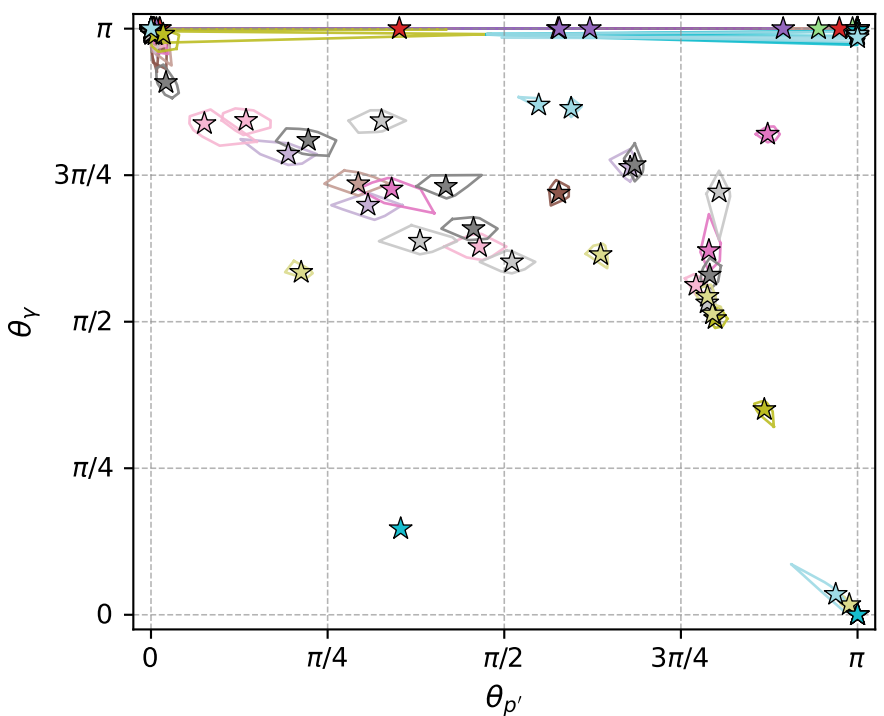


electron: polarization cluster P1; all local-minimum configurations and pairwise projections of their 8D contours



polarization cluster P1: α_e in $[0, \pi/2)$, $\alpha_p = \pi/4$
 111 local minima in this polarization cluster
 contour delta=0.05
 configured 8D samples/minimum=256
 dGHZ range=2.29125e-08 to 0.03006092
 global-minimum offsets=6.290374e-10 to 0.0300609

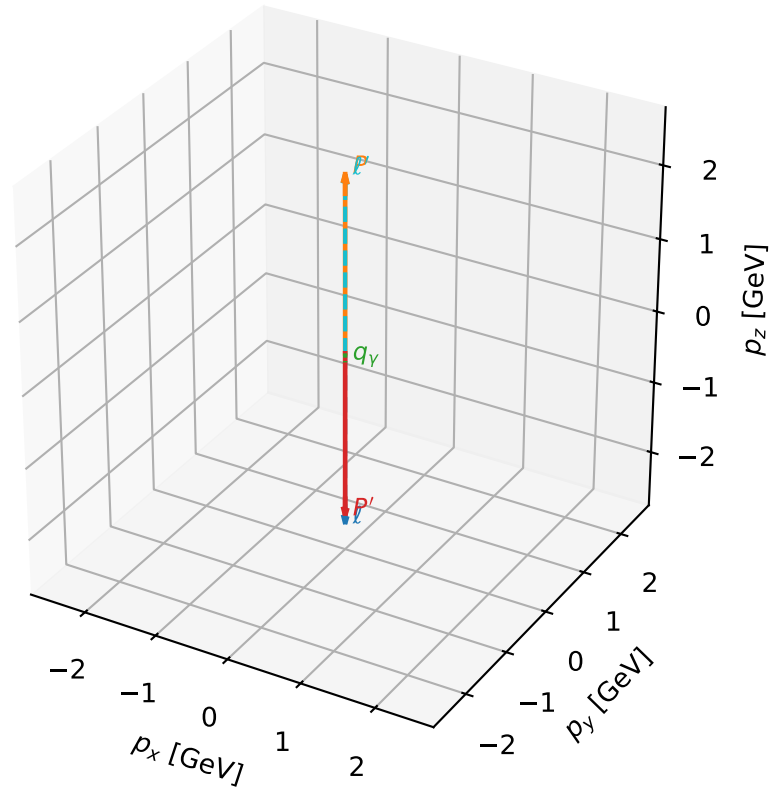
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_2 $d_{\{\text{rm GHZ}\}}=2.29125\text{e-}08$
region: polarization_cluster_01_local_minimum_2
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.87696633$ rad
incoming lepton: $0.639486|+> +0.768802|->$
proton mixing angle: $\alpha_p=0.87697731$ rad
incoming proton: $0.639478|+> +0.768809|->$

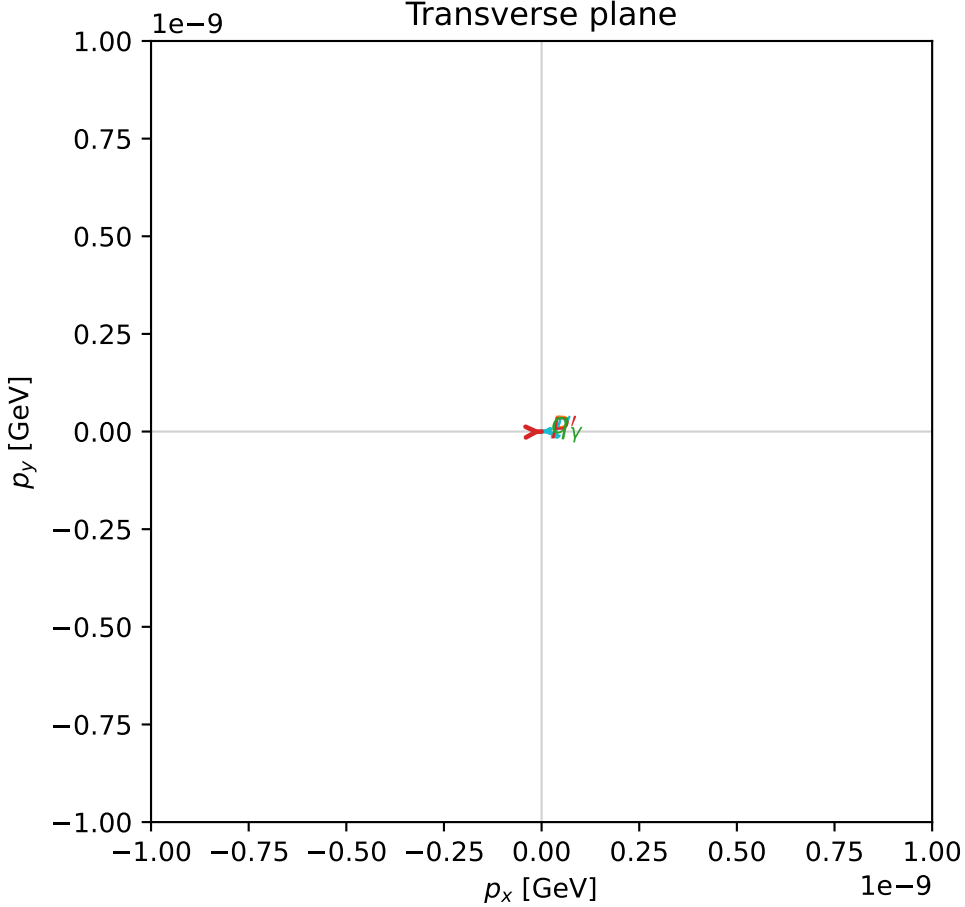
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28627$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=0.0335096$, $\phi_Y=0.522884$
 $E_Y=0.121256$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.19891$
 $Q^2=23.228$, $x_B=0.998073$, $t=-22.0603$, $W^2=0.924694$, $y=0.964872$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.4075$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.4075$
 P' $E= 2.4712$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2863$
 q_Y $E= 0.1213$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1213$

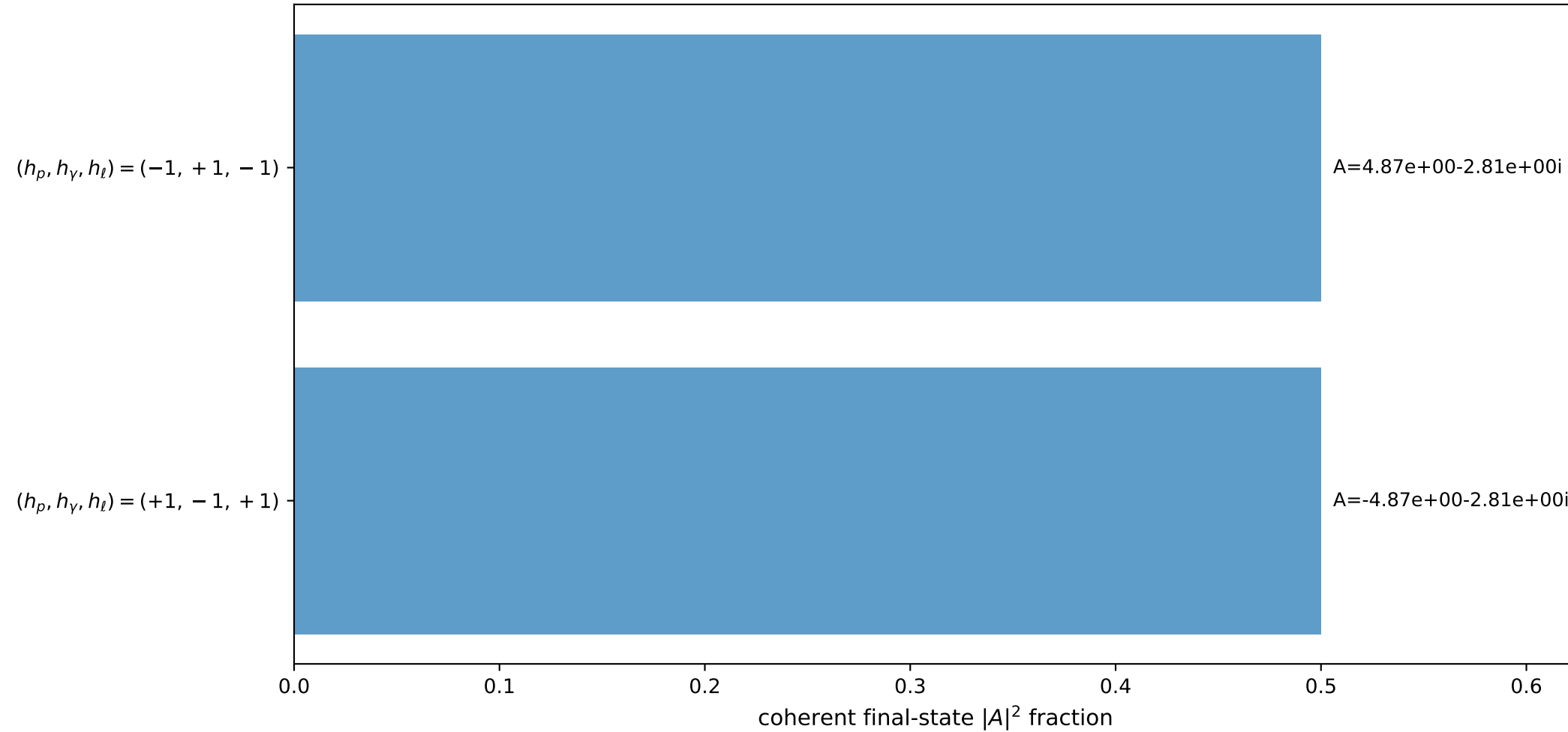
3D momenta



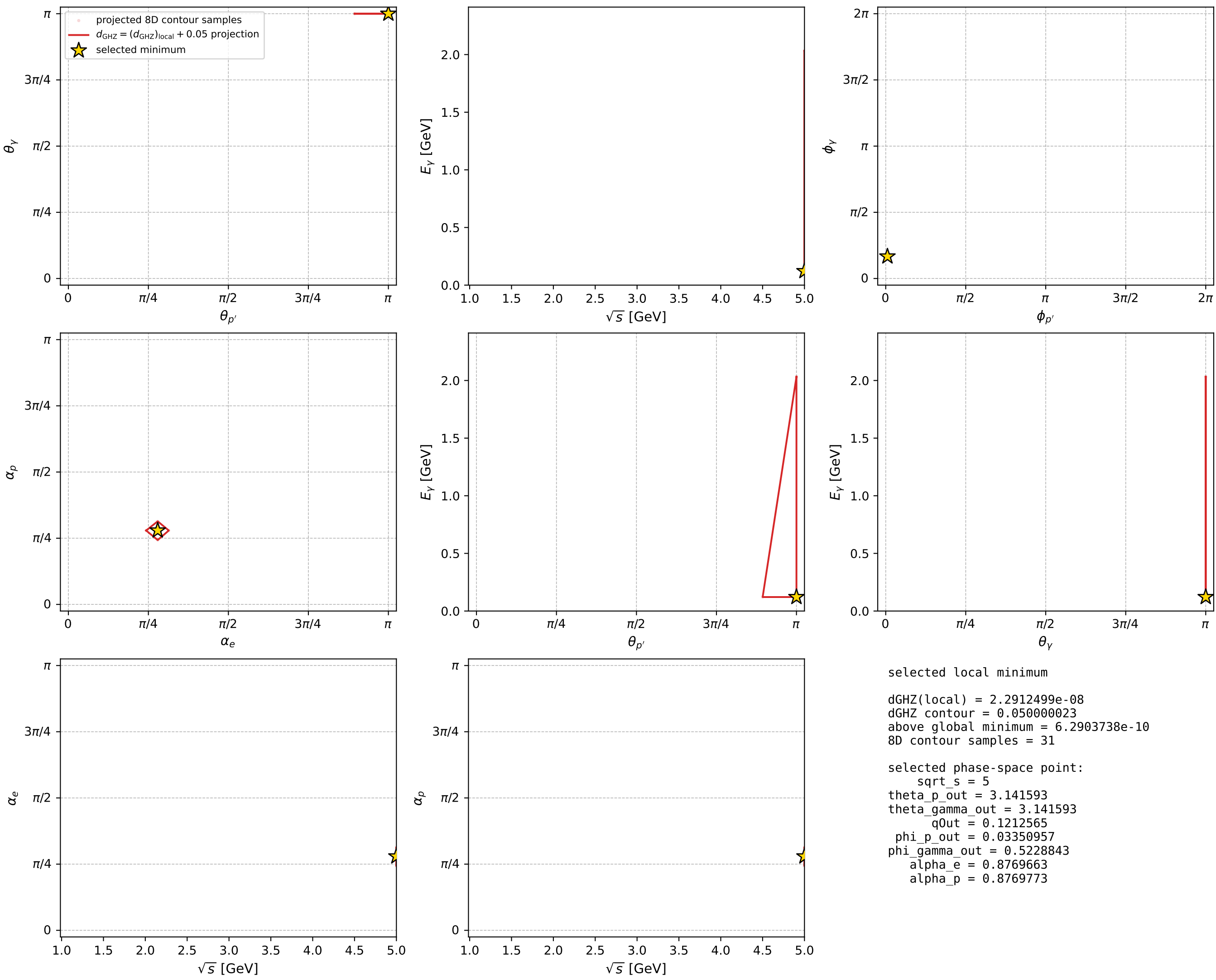
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 2; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

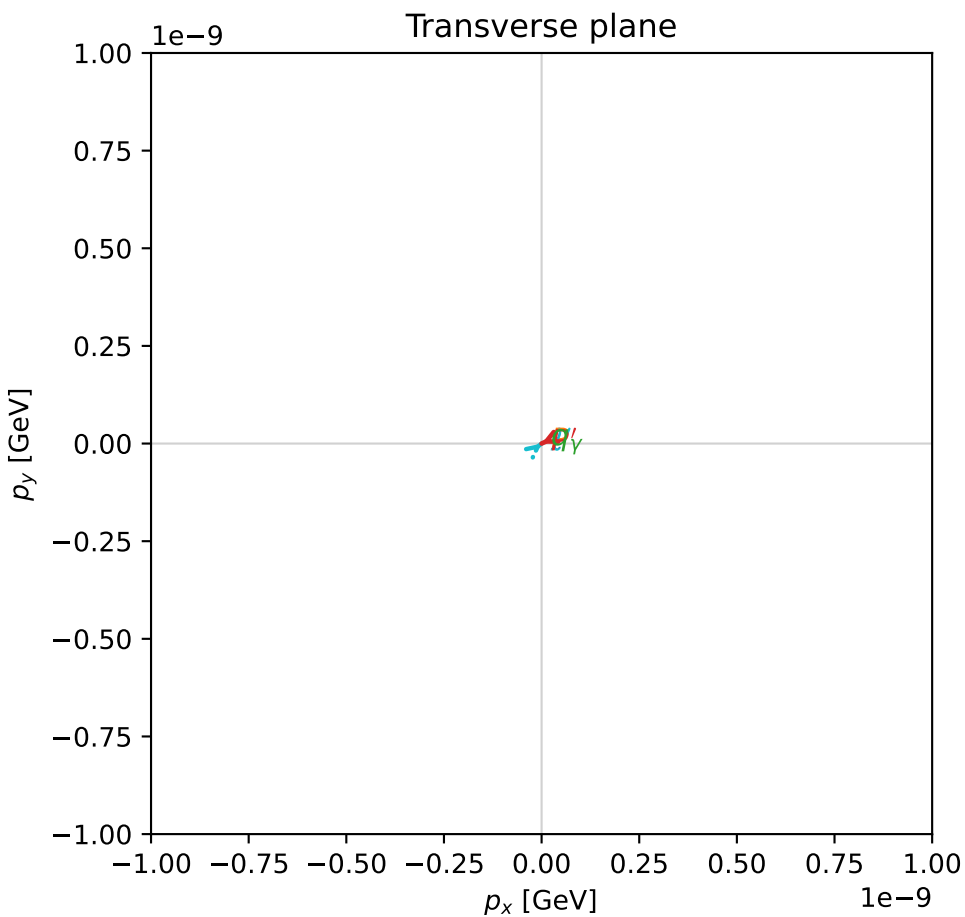
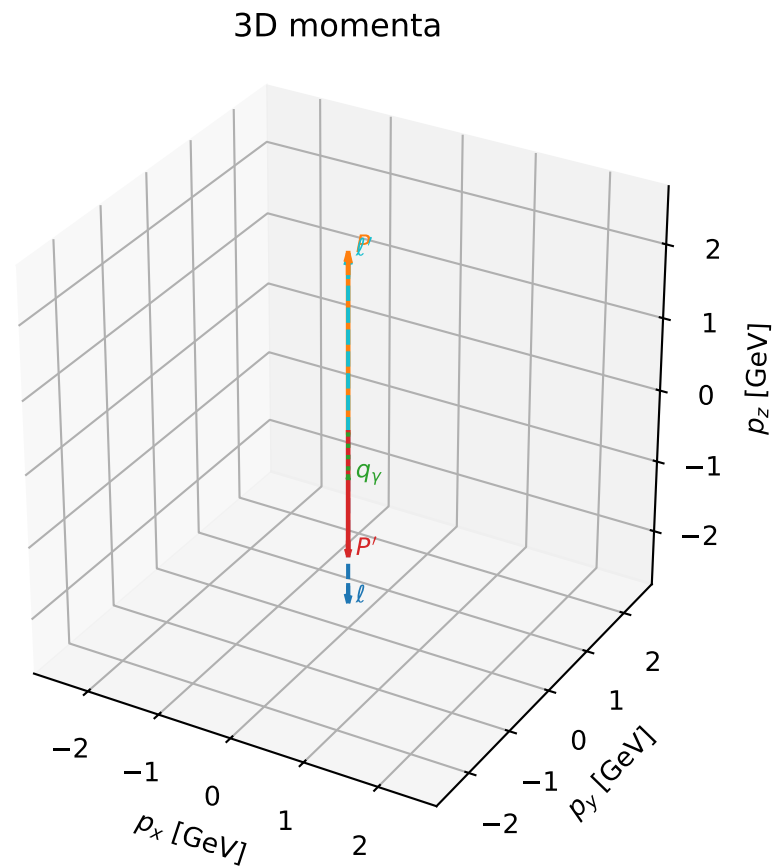


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_11 d_{\rm GHZ}=2.44872e-08
region: polarization_cluster_01_local_minimum_11
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.59862965$ rad
incoming lepton: $0.826109|+\rangle +0.563511|-\rangle$
proton mixing angle: $\alpha_p=0.59861028$ rad
incoming proton: $0.82612|+\rangle +0.563495|-\rangle$

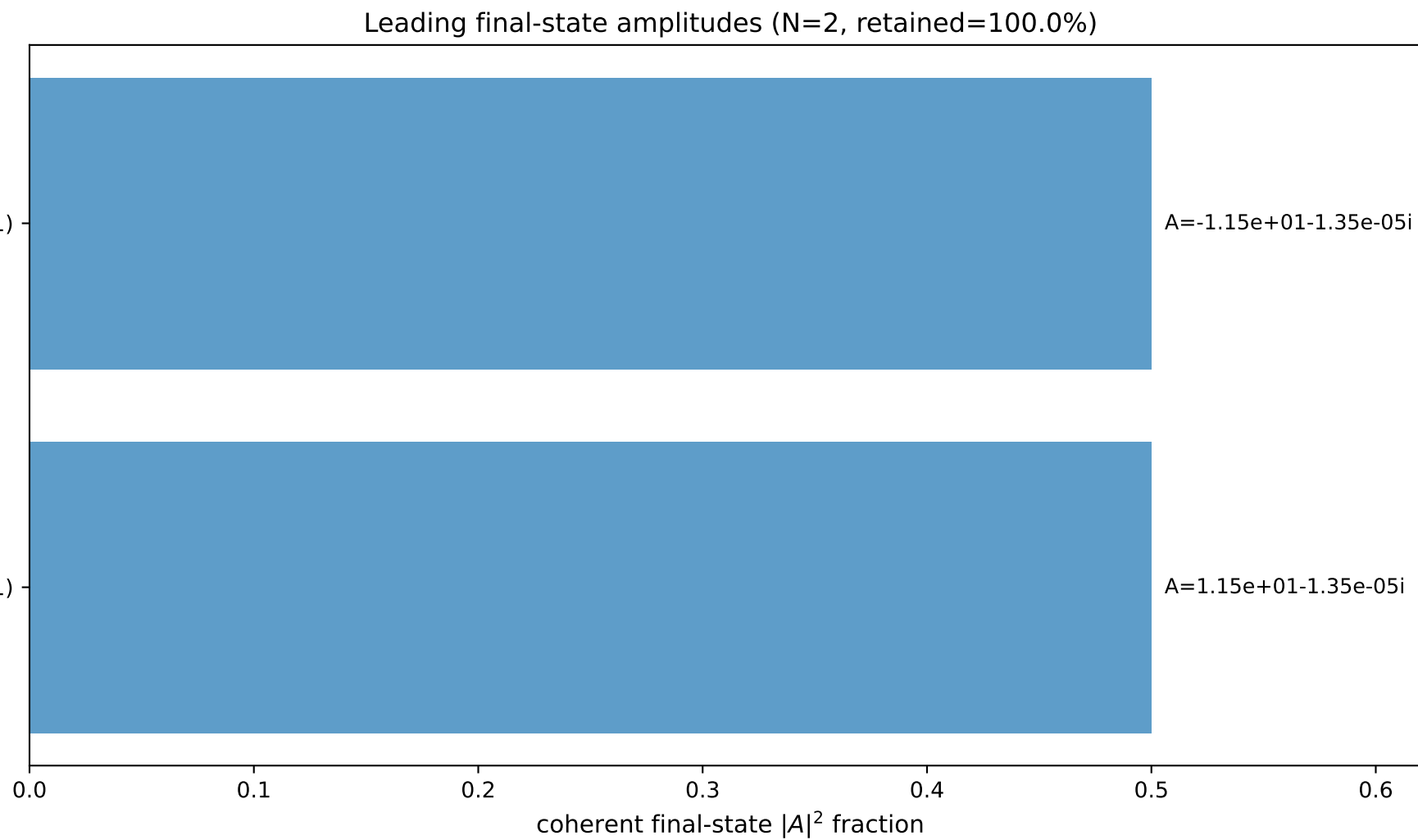
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.72071$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=3.57447$, $\phi_Y=1.17554\text{e-}06$
 $E_Y=0.659762$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.674751$
 $Q^2=22.9669$, $x_B=0.986451$, $t=-16.6848$, $W^2=1.19529$, $y=0.965266$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3805$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3805$
 P' $E= 1.9598$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.7207$
 q_Y $E= 0.6598$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6598$

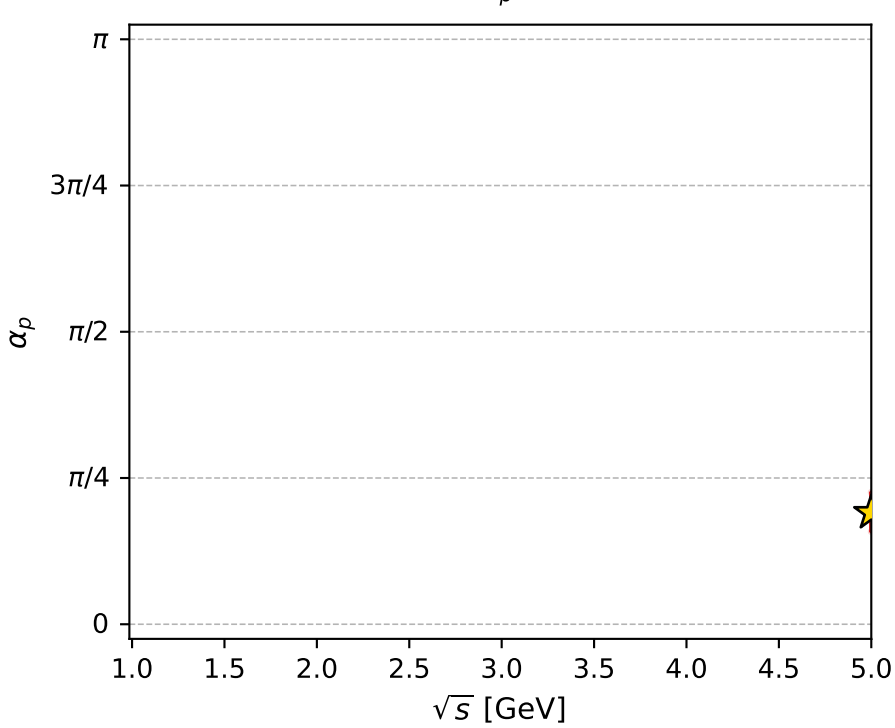
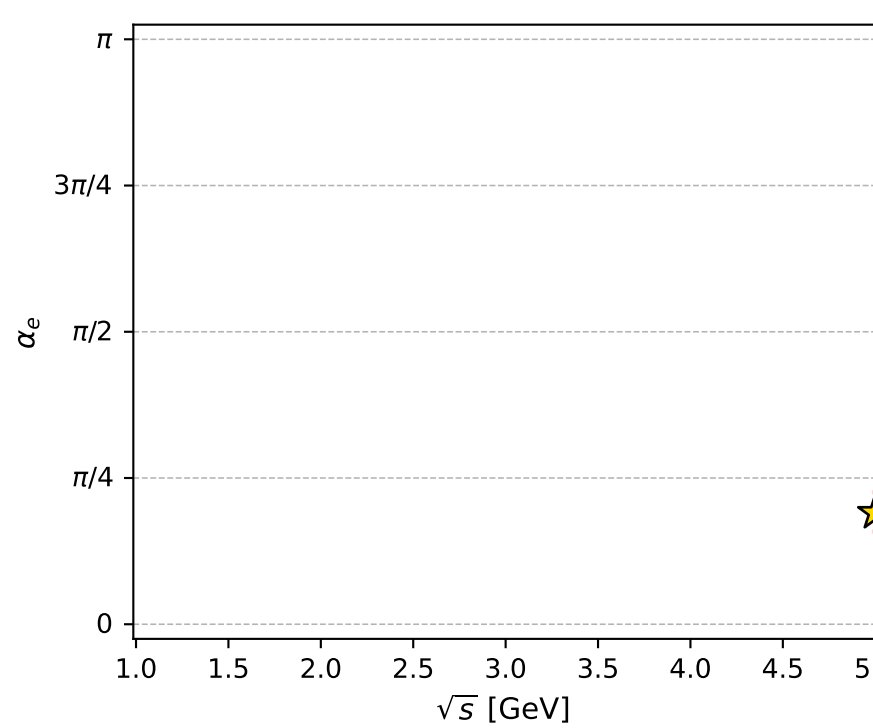
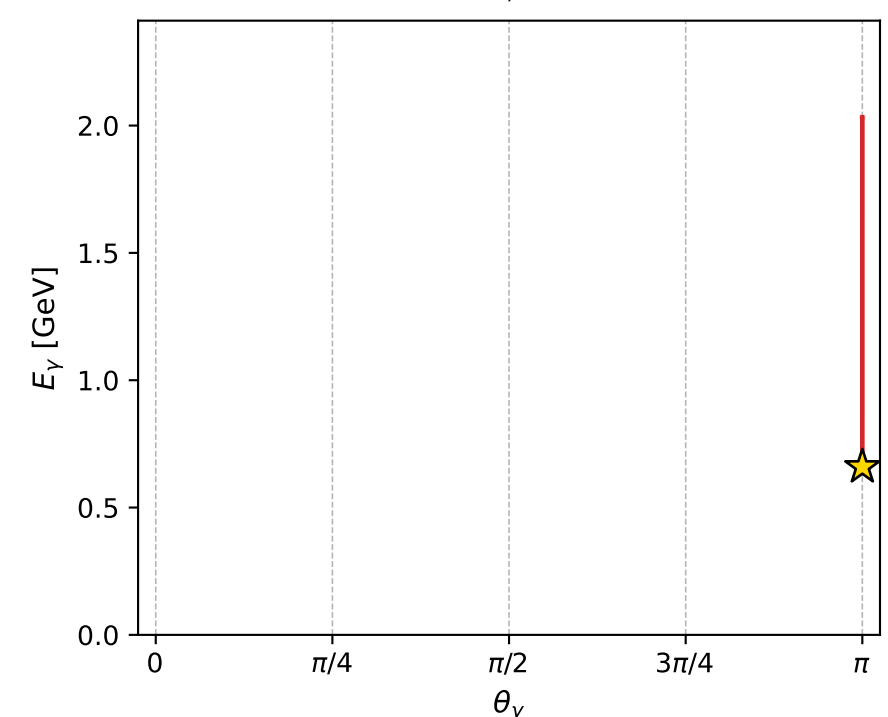
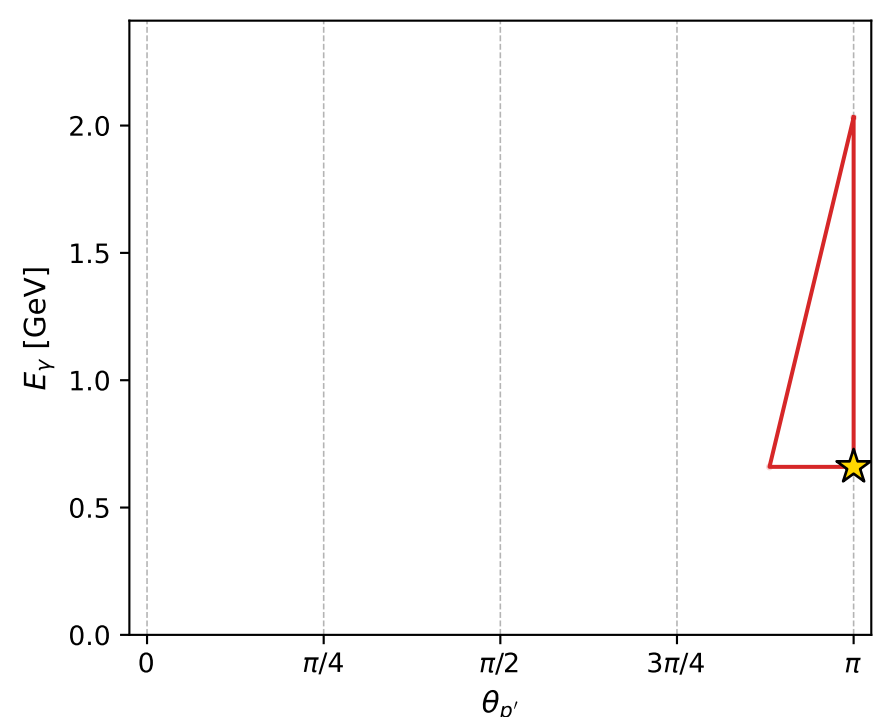
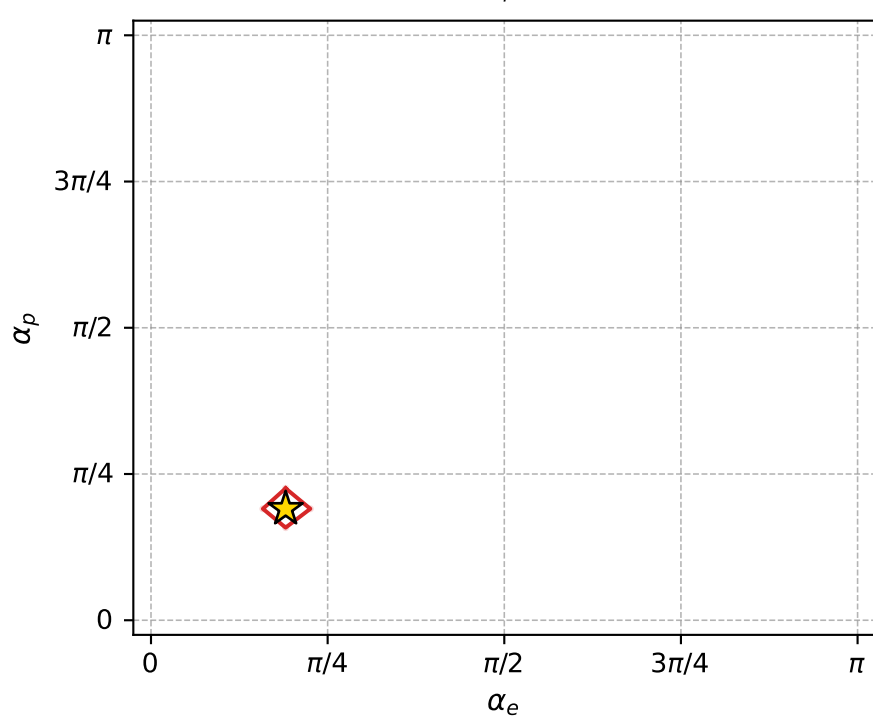
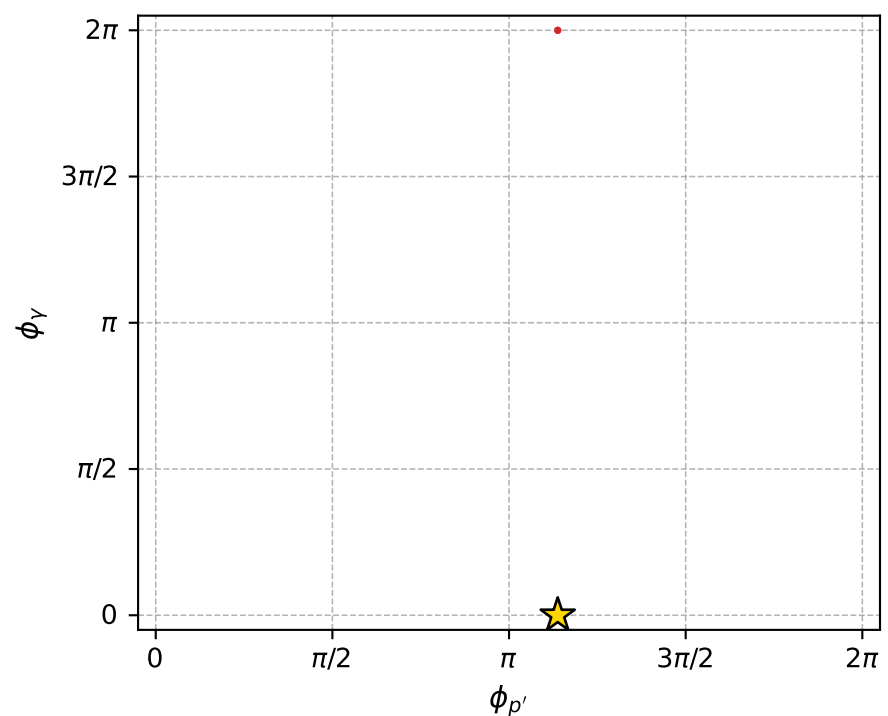
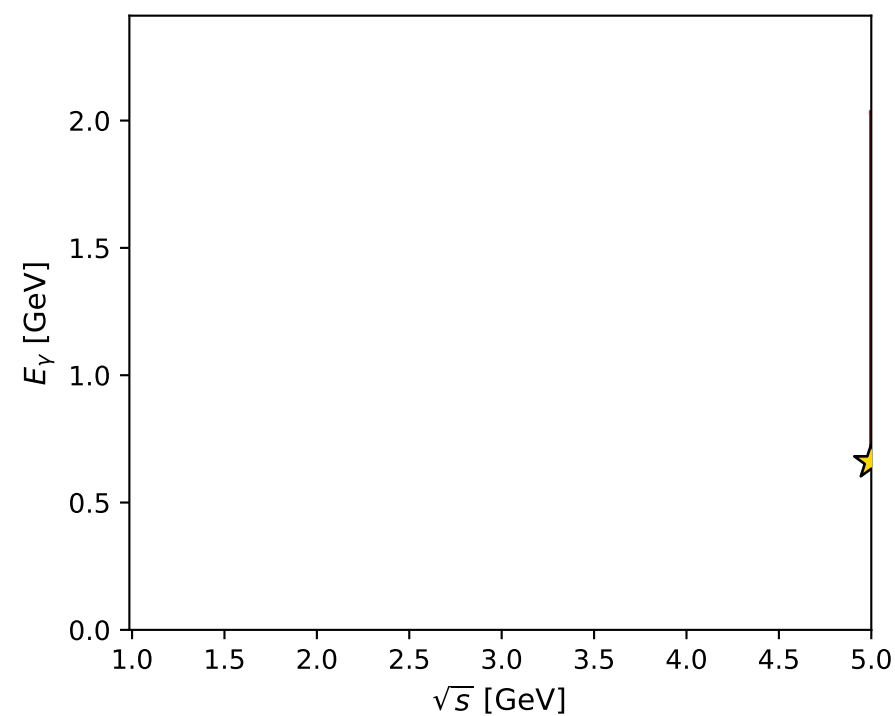
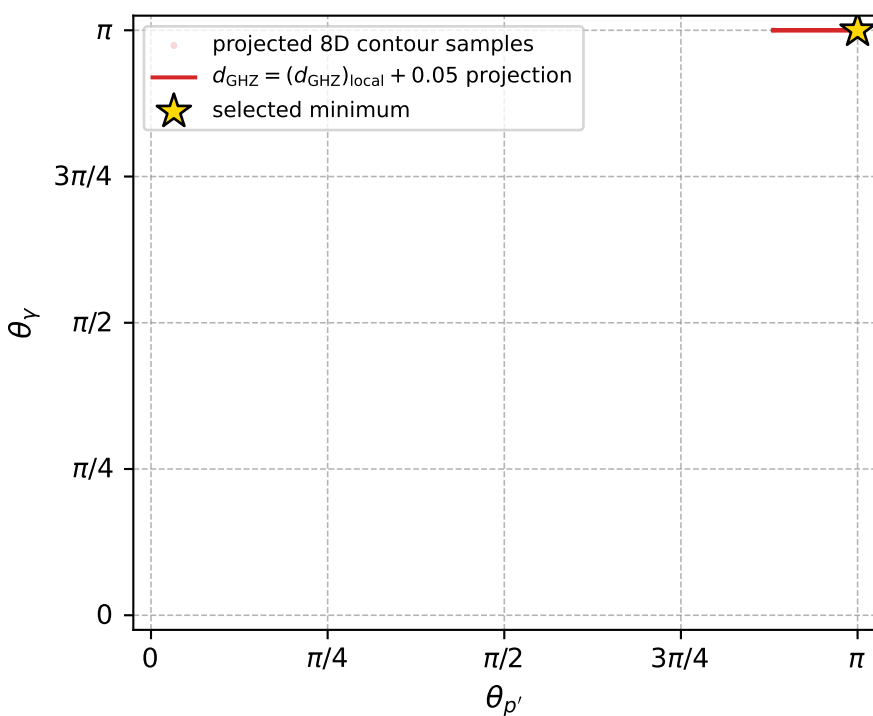


$(h_p, h_Y, h_\ell) = (+1, -1, +1)$

$(h_p, h_Y, h_\ell) = (-1, +1, -1)$



electron: polarization cluster P1, local minimum 11; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.4487161\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000024$
 above global minimum = $2.2036992\text{e-}09$
 8D contour samples = 41

selected phase-space point:

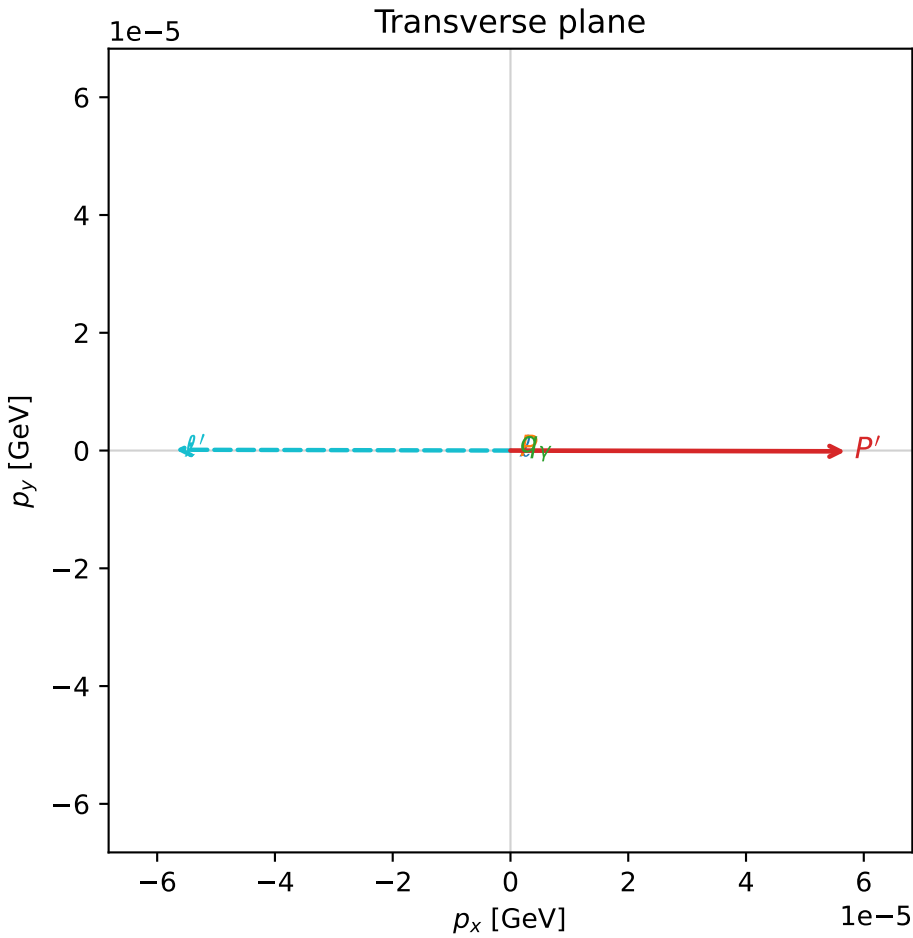
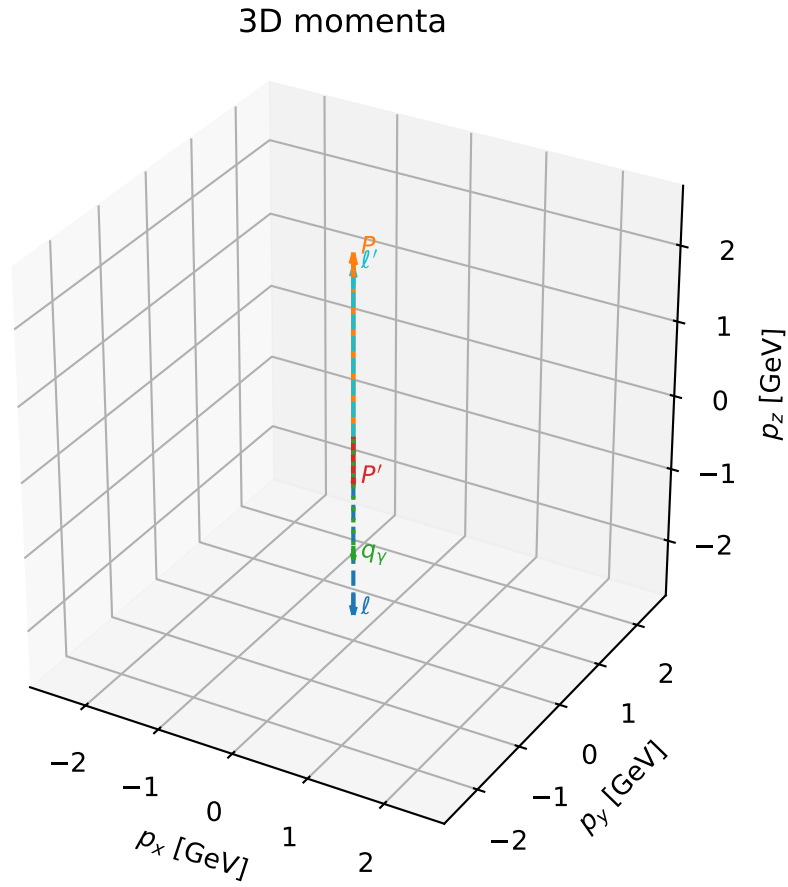
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.6597622$
 $\text{phi_p_out} = 3.574466$
 $\text{phi_gamma_out} = 1.175542\text{e-}06$
 $\text{alpha_e} = 0.5986296$
 $\text{alpha_p} = 0.5986103$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_16 $d_{\{\text{rm GHZ}\}}=2.47608\text{e-}08$
 region: polarization_cluster_01_local_minimum_16
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.89433114$ rad
 incoming lepton: $0.626041|+\rangle +0.779791|-\rangle$
 proton mixing angle: $\alpha_p=0.89433384$ rad
 incoming proton: $0.626038|+\rangle +0.779792|-\rangle$

$s=24.9994$, $\sqrt{s}=4.99994$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.59265$
 $\theta_{P'}=3.1415$, $\theta_Y=3.14159$, $\phi_{P'}=6.28079$, $\phi_Y=6.02949$
 $E_Y=1.64887$, $\theta_{\ell'}=2.53716\text{e-}05$, $\phi_{\ell'}=3.1392$
 $Q^2=21.6261$, $x_B=0.926939$, $t=-6.84211$, $W^2=2.58442$, $y=0.967293$

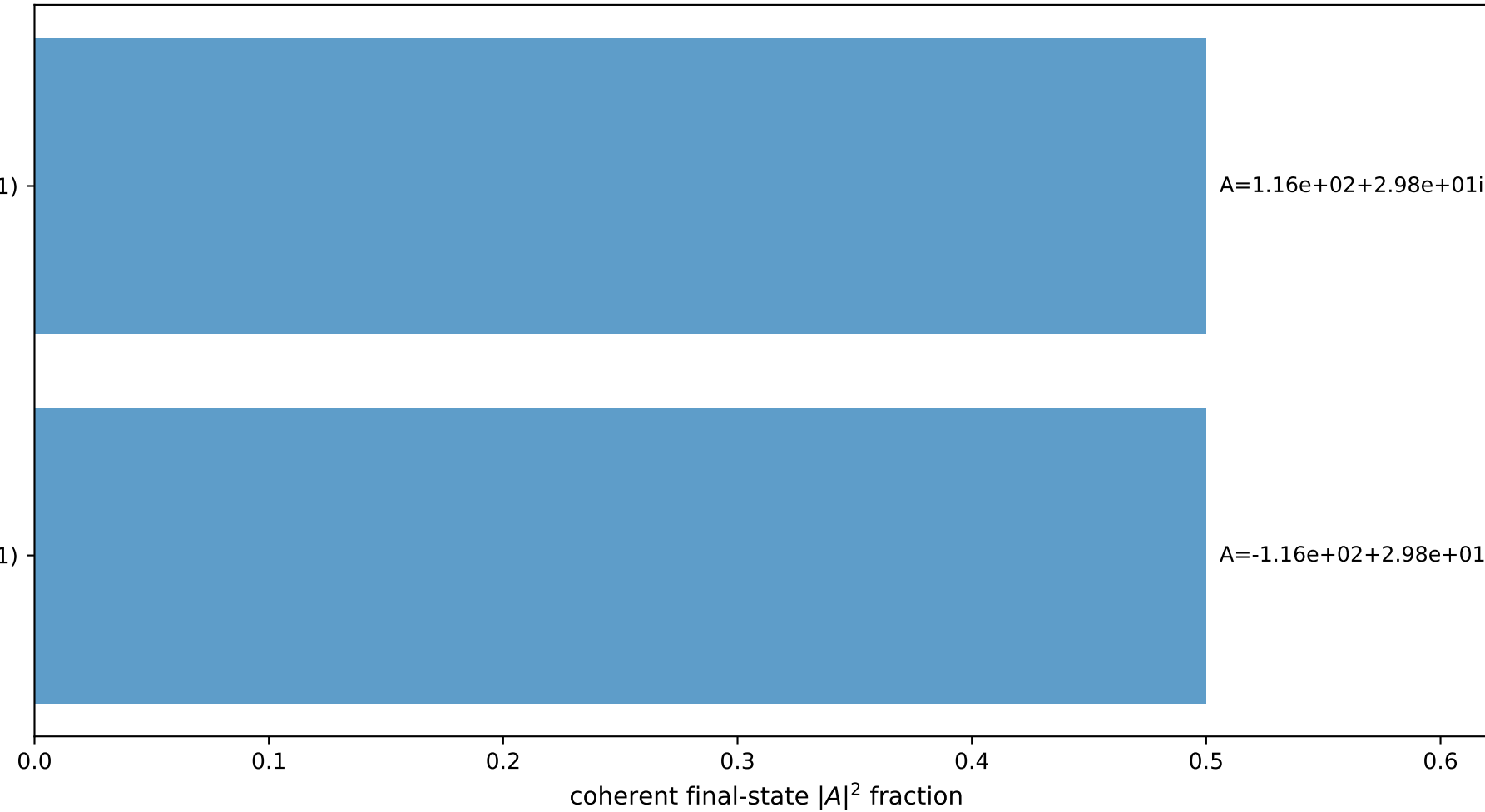
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2415$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.2415$
 P' $E= 1.1095$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.5927$
 q_Y $E= 1.6489$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.6489$



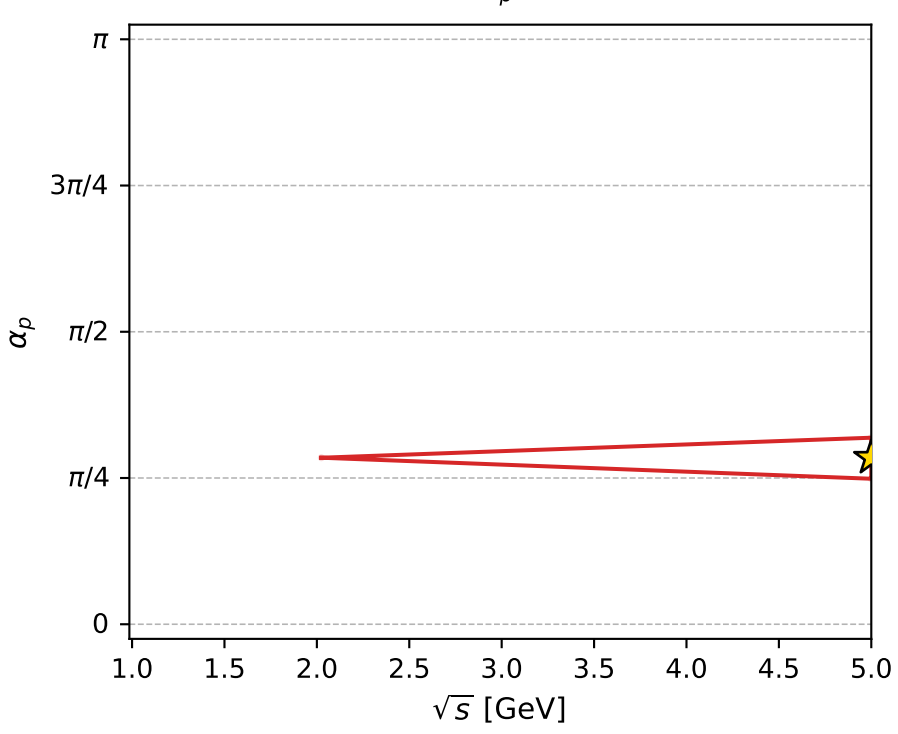
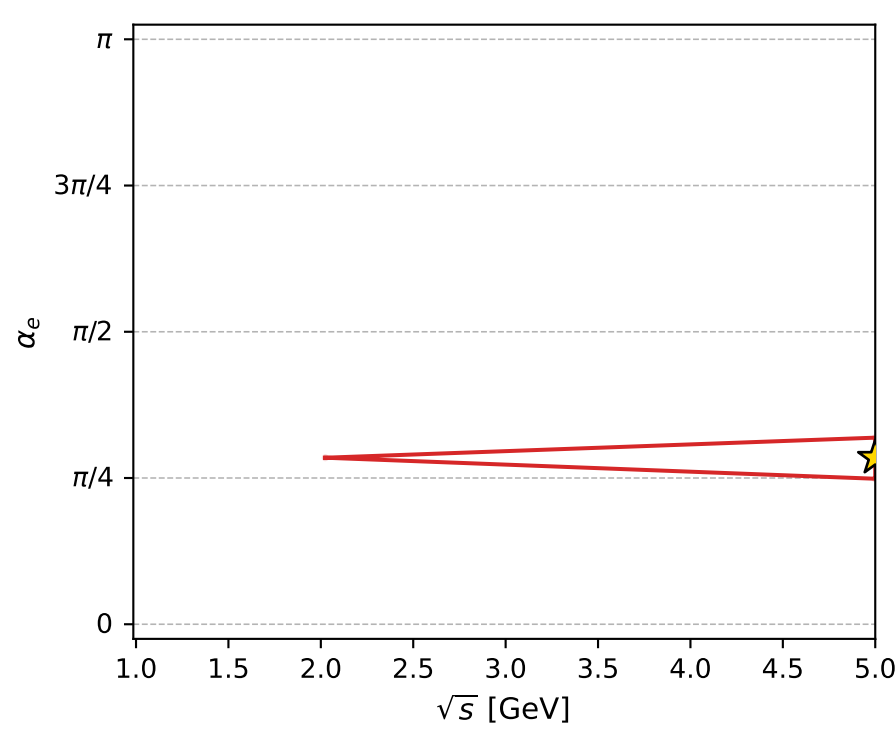
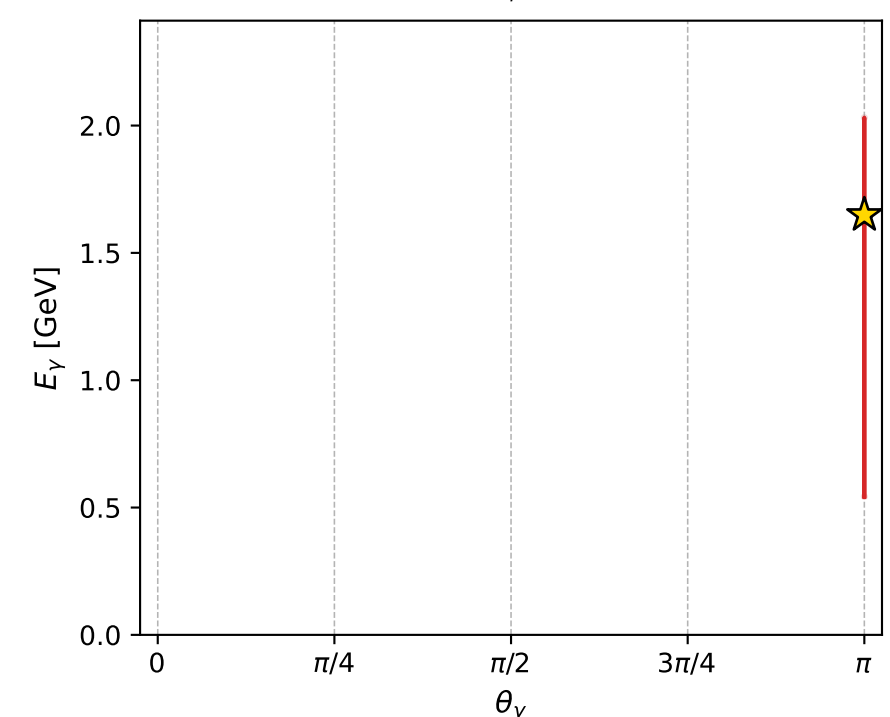
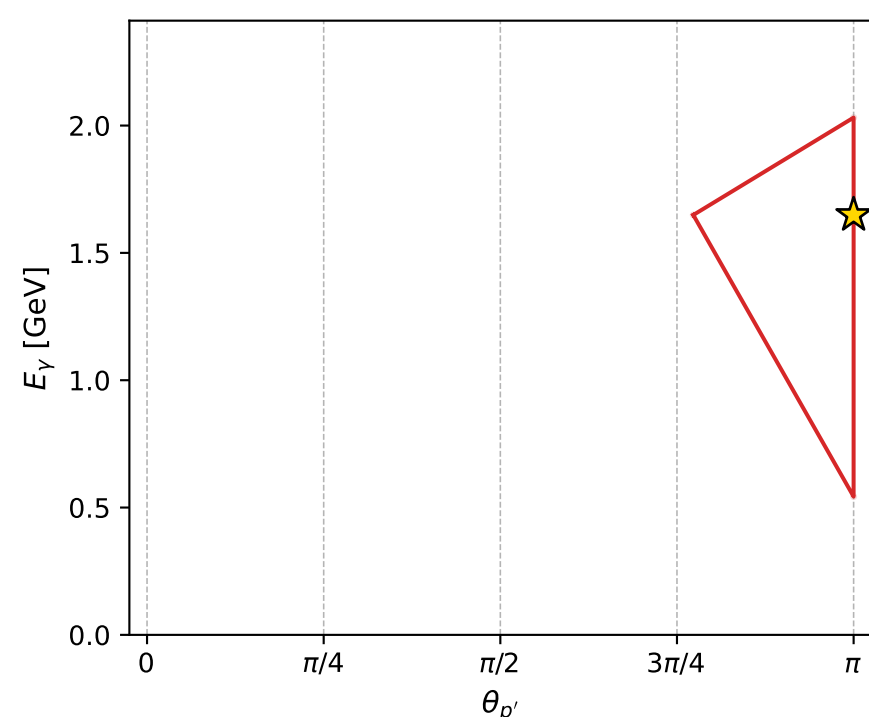
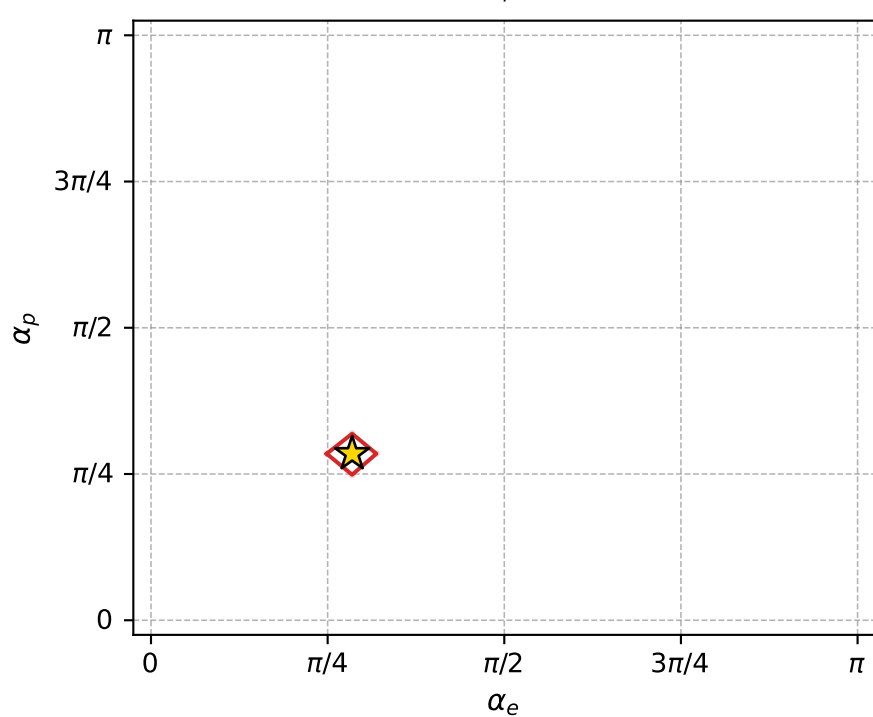
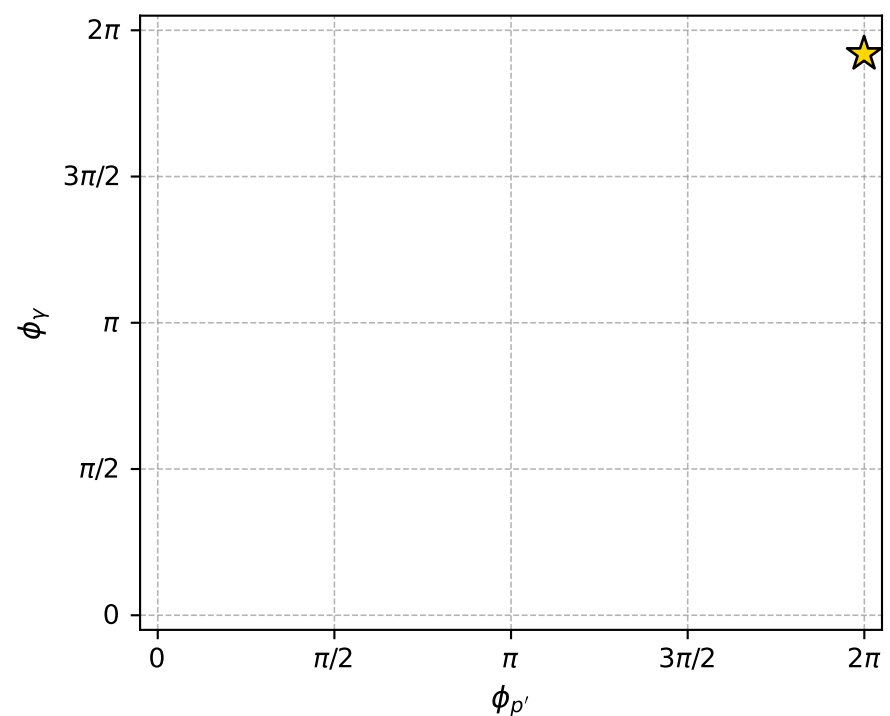
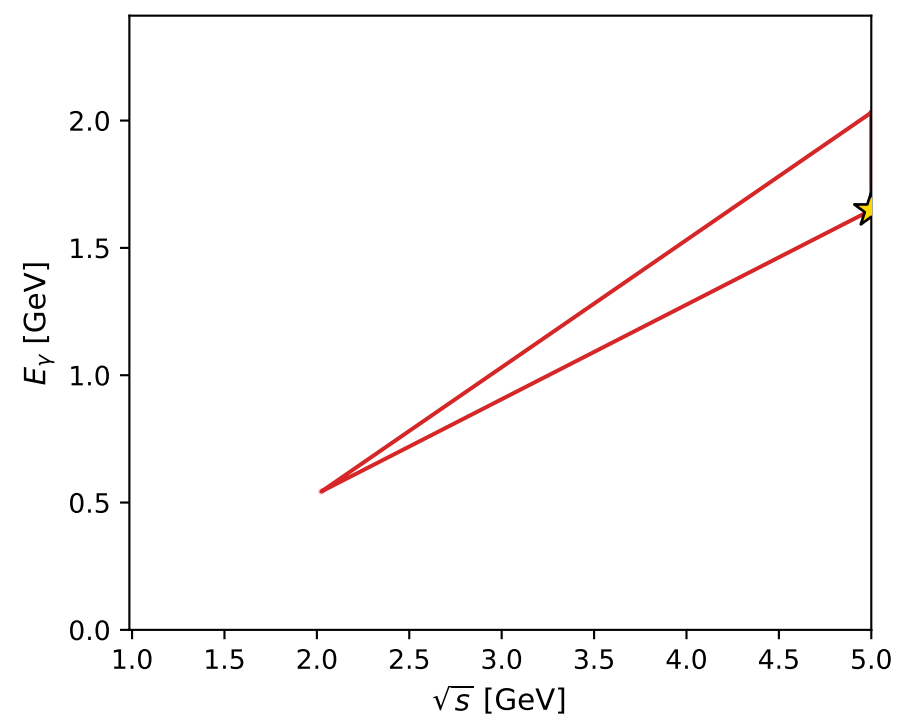
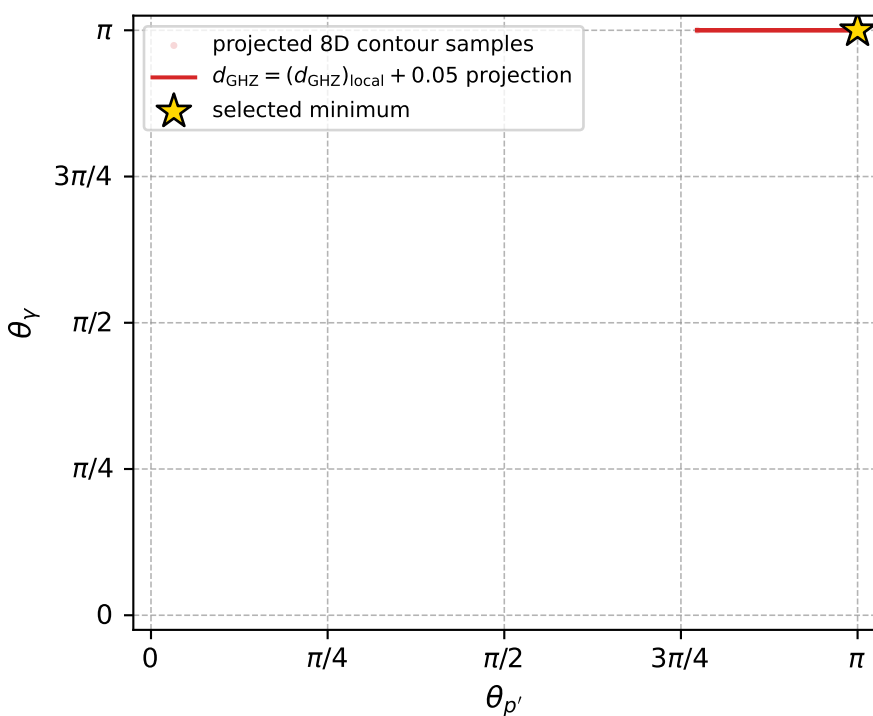
$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 16; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.47608\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000025$
 above global minimum = $2.4773383\text{e-}09$
 8D contour samples = 125

selected phase-space point:

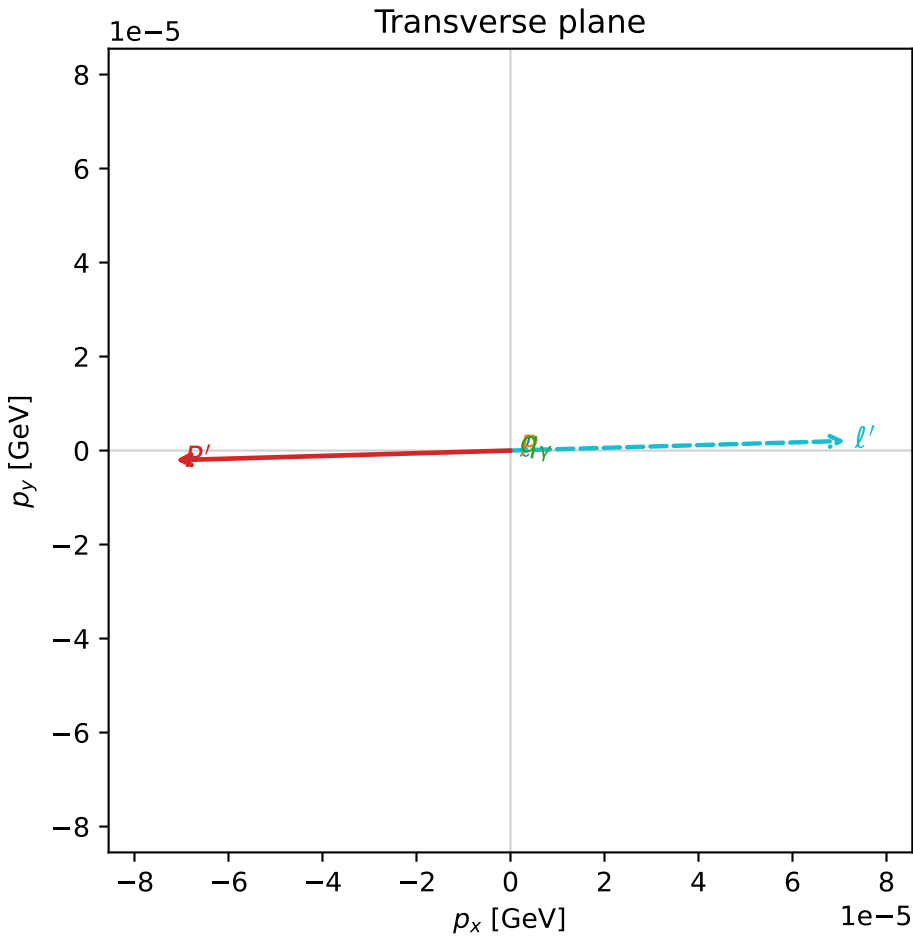
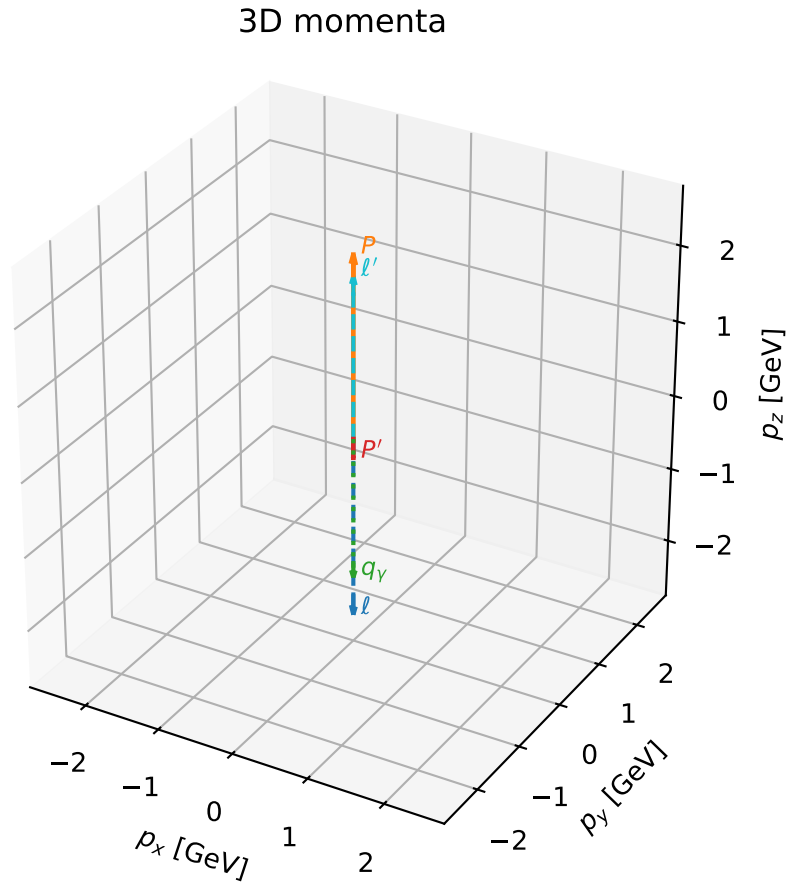
$\text{sqrt_s} = 4.999938$
 $\text{theta_p_out} = 3.141497$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.648874$
 $\text{phi_p_out} = 6.28079$
 $\text{phi_gamma_out} = 6.029488$
 $\alpha_e = 0.8943311$
 $\alpha_p = 0.8943338$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_19 $d_{\{\text{rm GHZ}\}}=2.53539\text{e-}08$
 region: polarization_cluster_01_local_minimum_19
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78519375$ rad
 incoming lepton: $0.707251|+\rangle +0.706962|-\rangle$
 proton mixing angle: $\alpha_p=0.7851987$ rad
 incoming proton: $0.707248|+\rangle +0.706966|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.251318$
 $\theta_{P'}=3.14131$, $\theta_Y=3.14159$, $\phi_{P'}=3.17031$, $\phi_Y=0.388985$
 $E_Y=1.8888$, $\theta_{\ell'}=3.33062\text{e-}05$, $\phi_{\ell'}=0.0287211$
 $Q^2=20.648$, $x_B=0.88364$, $t=-4.47898$, $W^2=3.59883$, $y=0.968774$

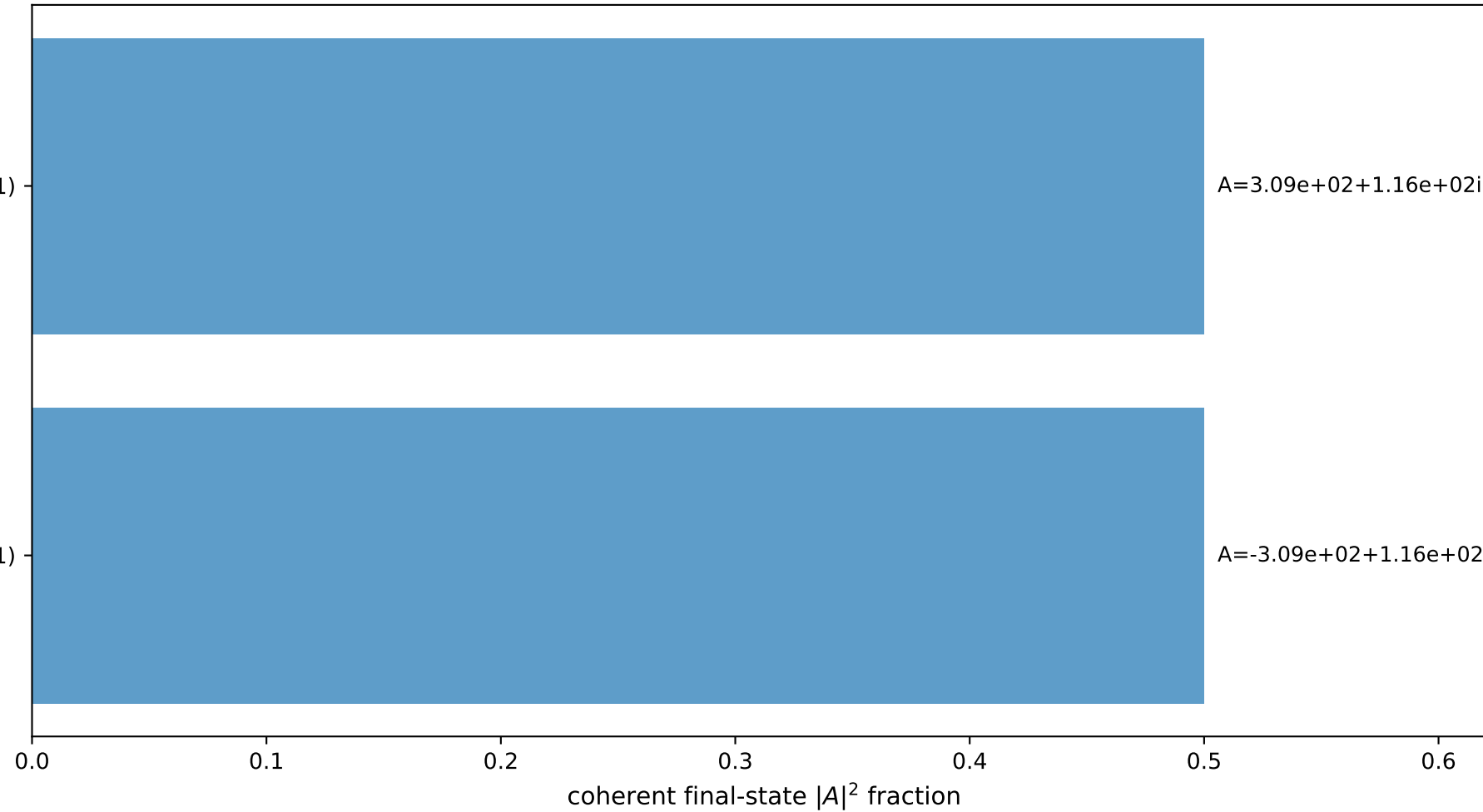
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1401$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.1401$
 P' $E= 0.9711$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.2513$
 q_Y $E= 1.8888$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8888$



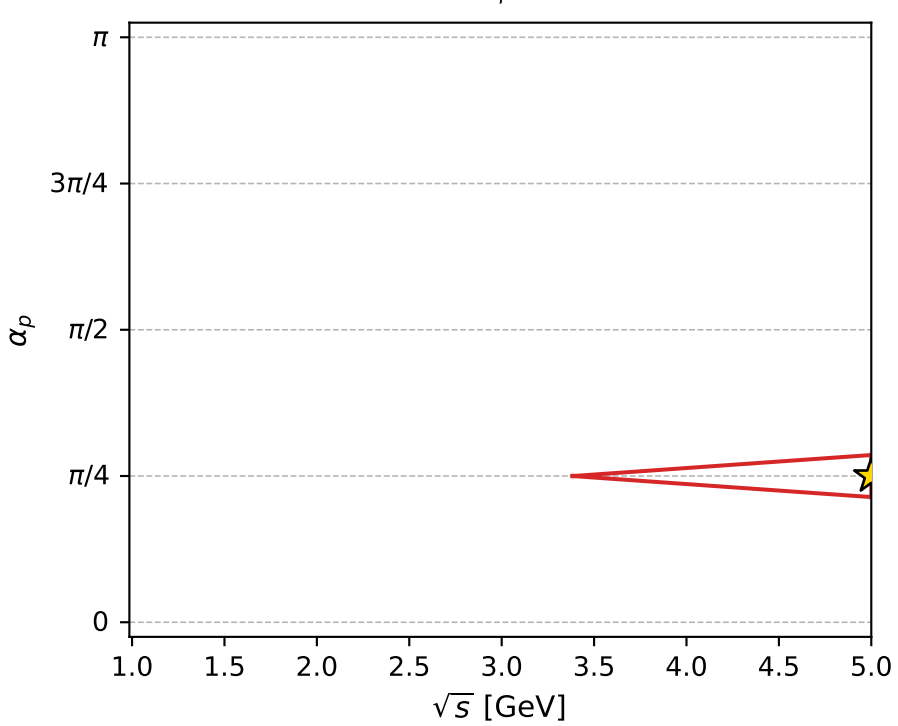
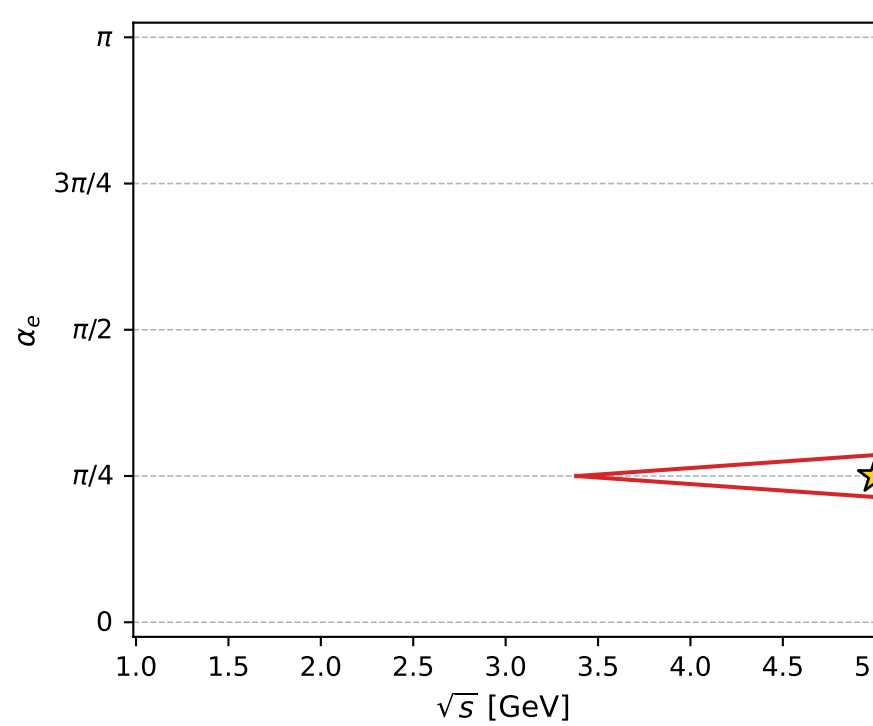
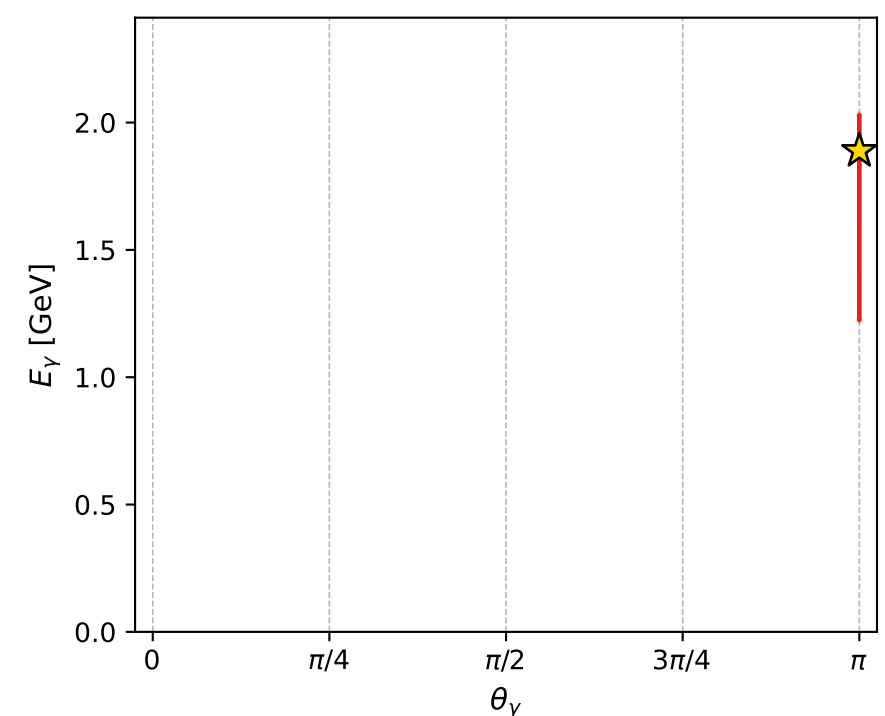
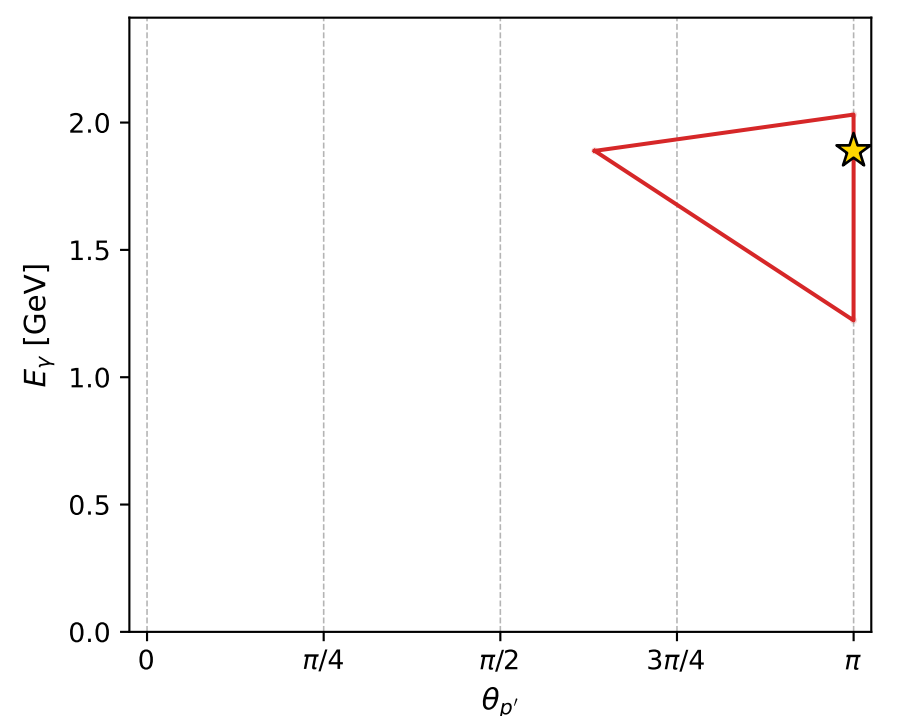
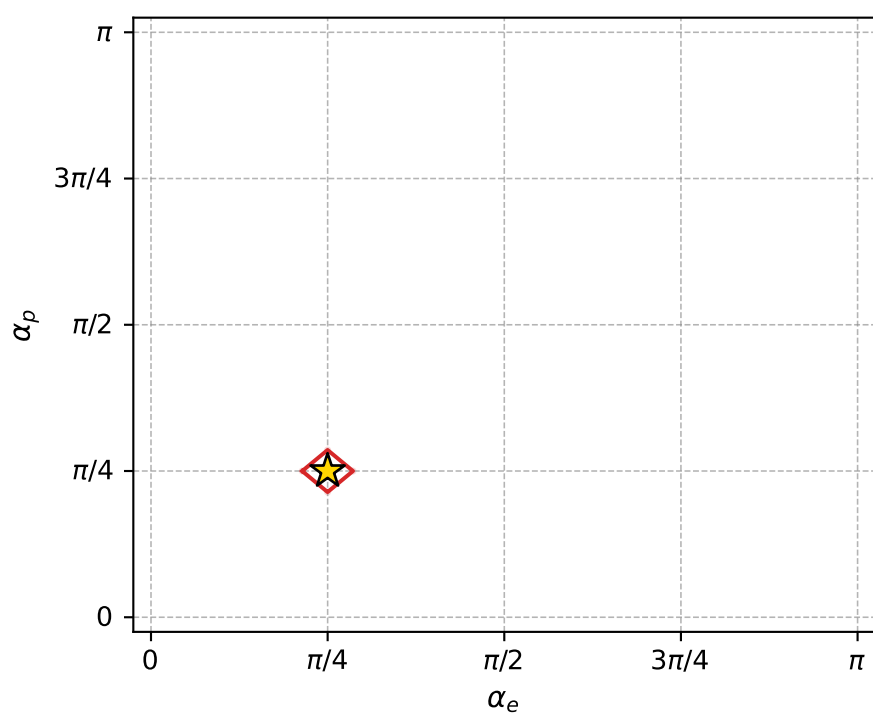
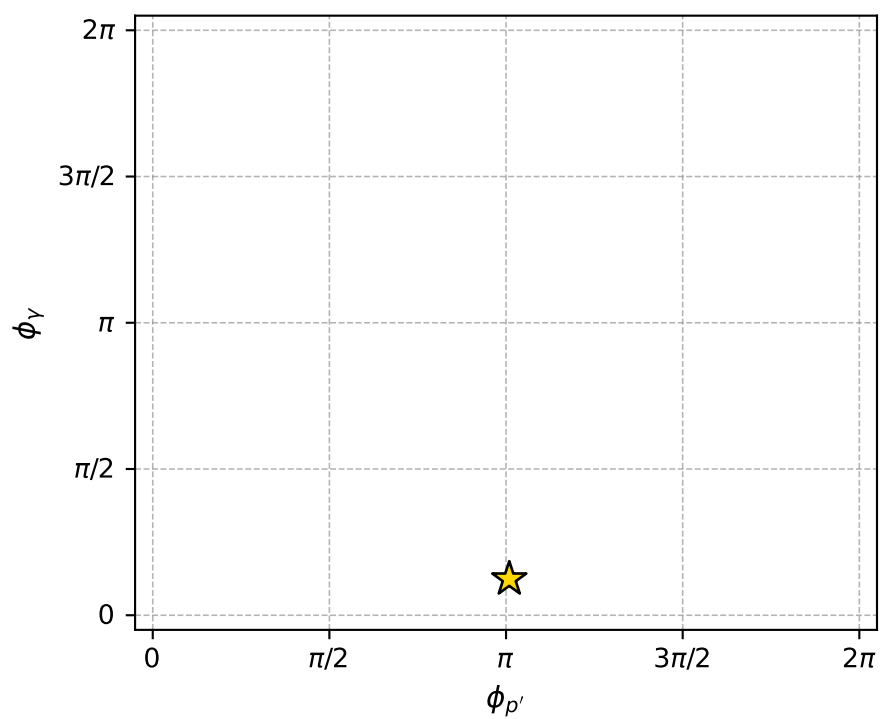
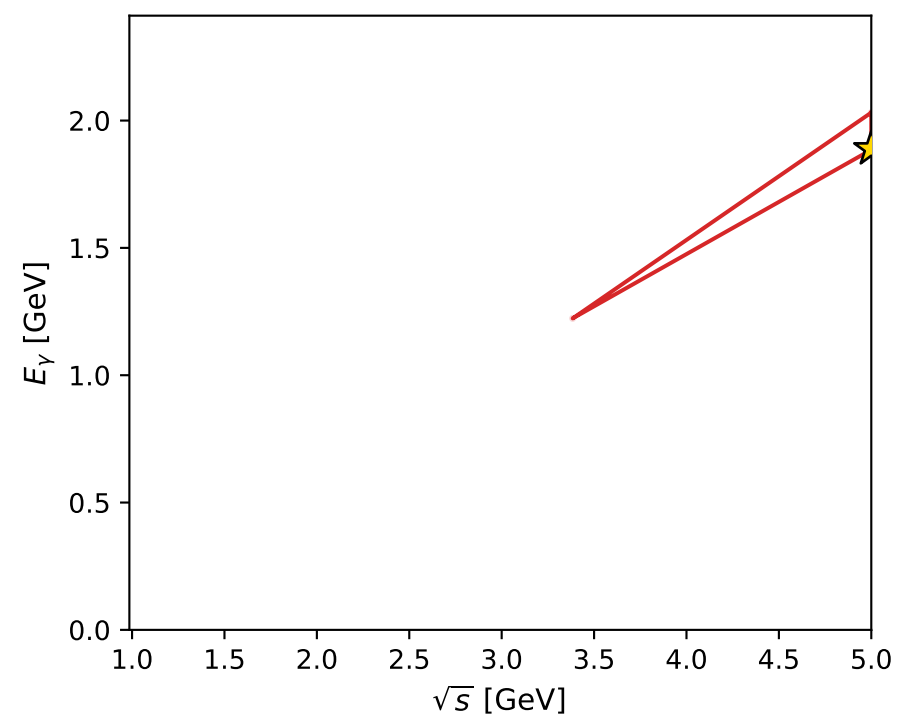
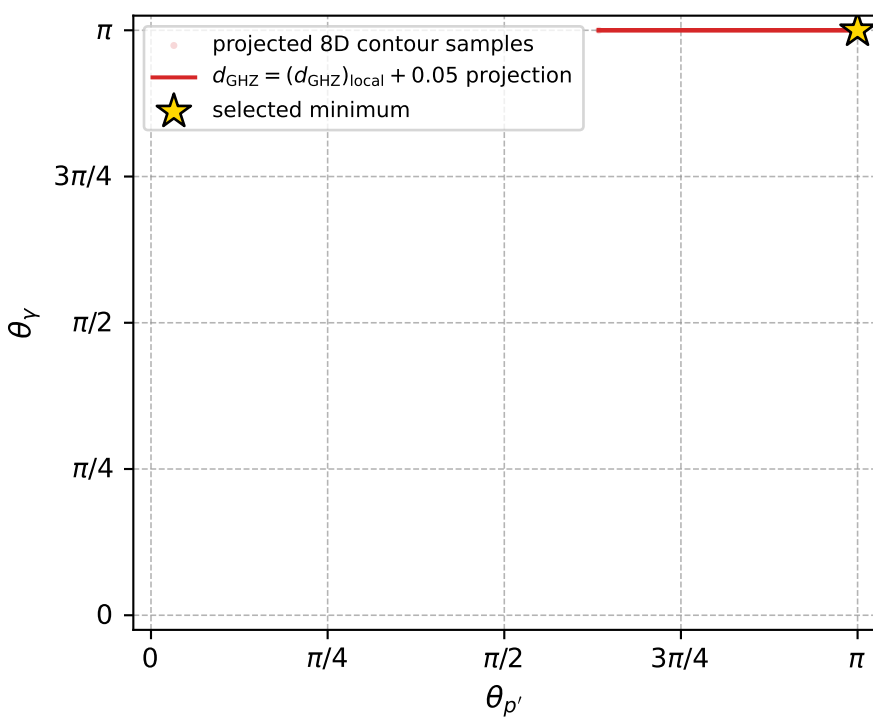
$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 19; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.5353888\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000025$
 above global minimum = $3.0704264\text{e-}09$
 8D contour samples = 73

selected phase-space point:

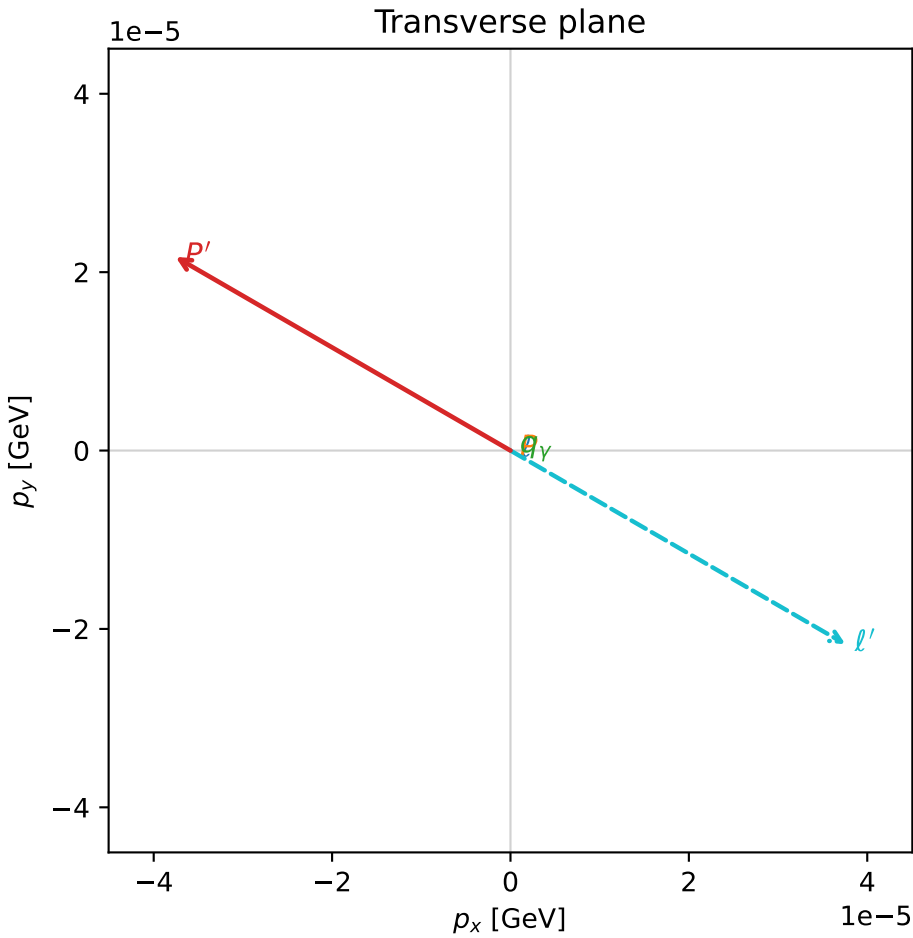
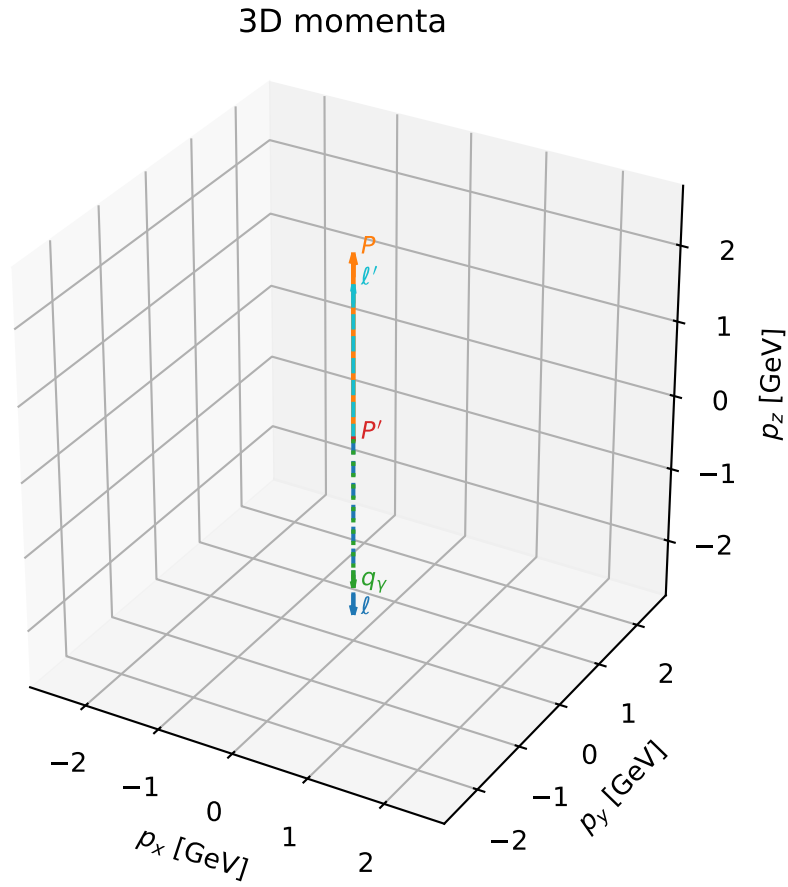
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141309$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.888799$
 $\text{phi_p_out} = 3.170314$
 $\text{phi_gamma_out} = 0.3889854$
 $\alpha_e = 0.7851938$
 $\alpha_p = 0.7851987$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_23 $d_{\{\text{rm GHZ}\}}=2.57818\text{e-}08$
region: polarization_cluster_01_local_minimum_23
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.77571838$ rad
incoming lepton: $0.713918|+\rangle +0.700229|-\rangle$
proton mixing angle: $\alpha_p=0.77571056$ rad
incoming proton: $0.713924|+\rangle +0.700224|-\rangle$

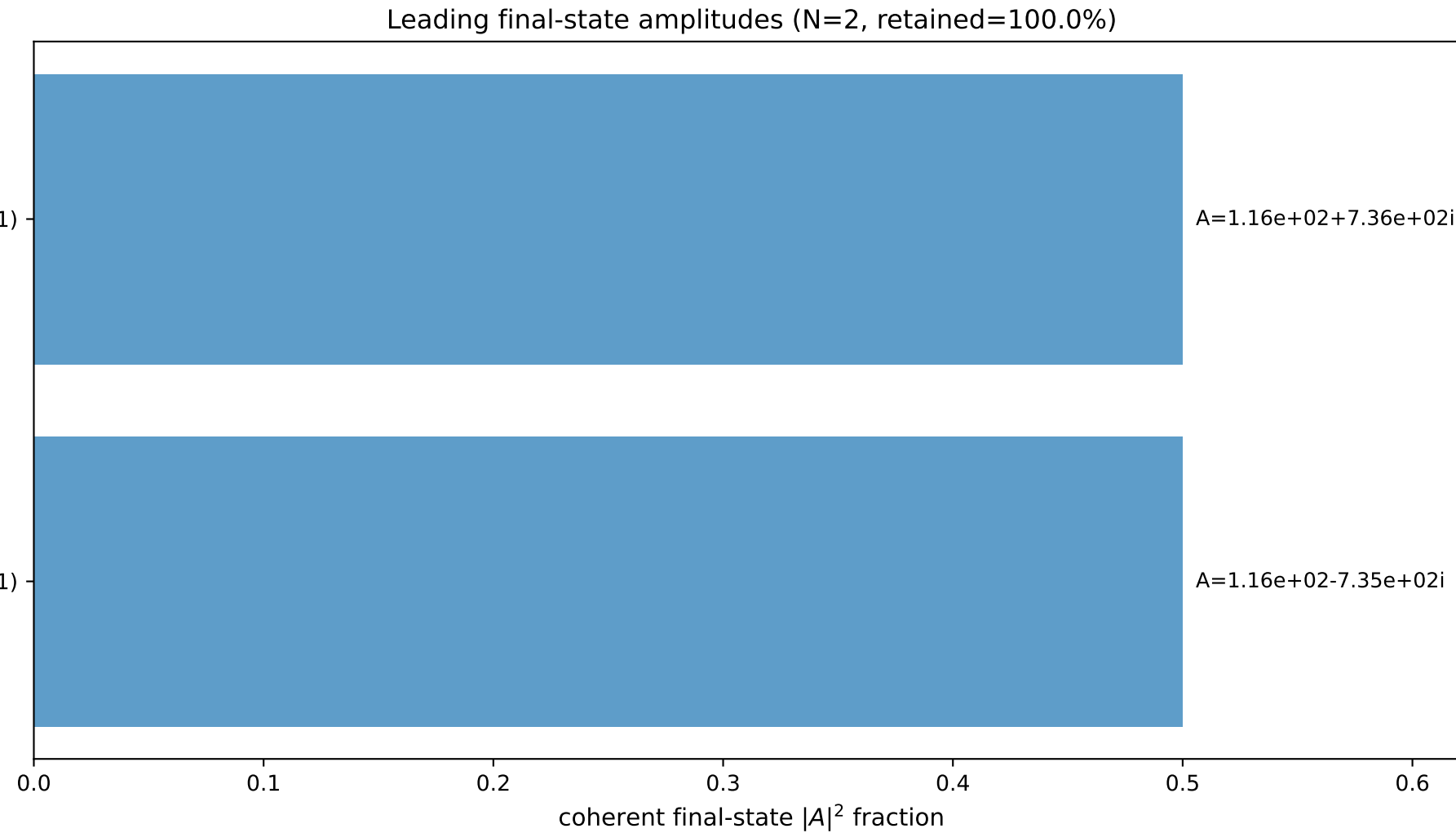
$s=24.9977$, $\sqrt{s}=4.99977$, $|\vec{P}|=2.41189$, $|\vec{P}'|=0.00408939$
 $\theta_{P'}=0.0106052$, $\theta_Y=3.14159$, $\phi_{P'}=2.61744$, $\phi_Y=4.55652$
 $E_Y=2.03292$, $\theta_{\ell'}=2.13758\text{e-}05$, $\phi_{\ell'}=5.75903$
 $Q^2=19.5733$, $x_B=0.836333$, $t=-3.07548$, $W^2=4.71027$, $y=0.970393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0288$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0288$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0041$
 q_Y $E= 2.0329$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0329$

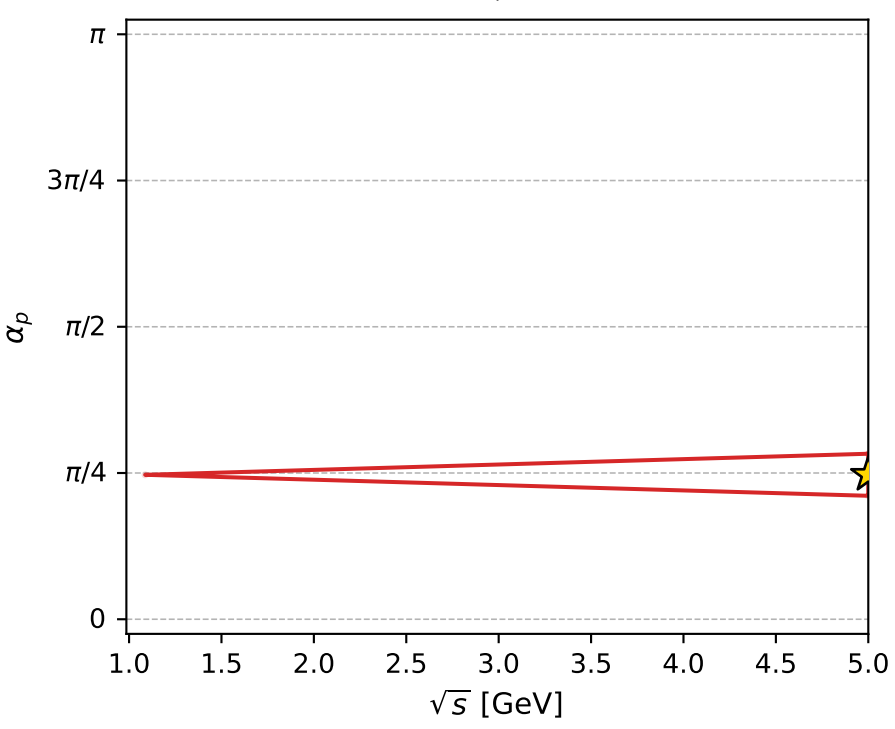
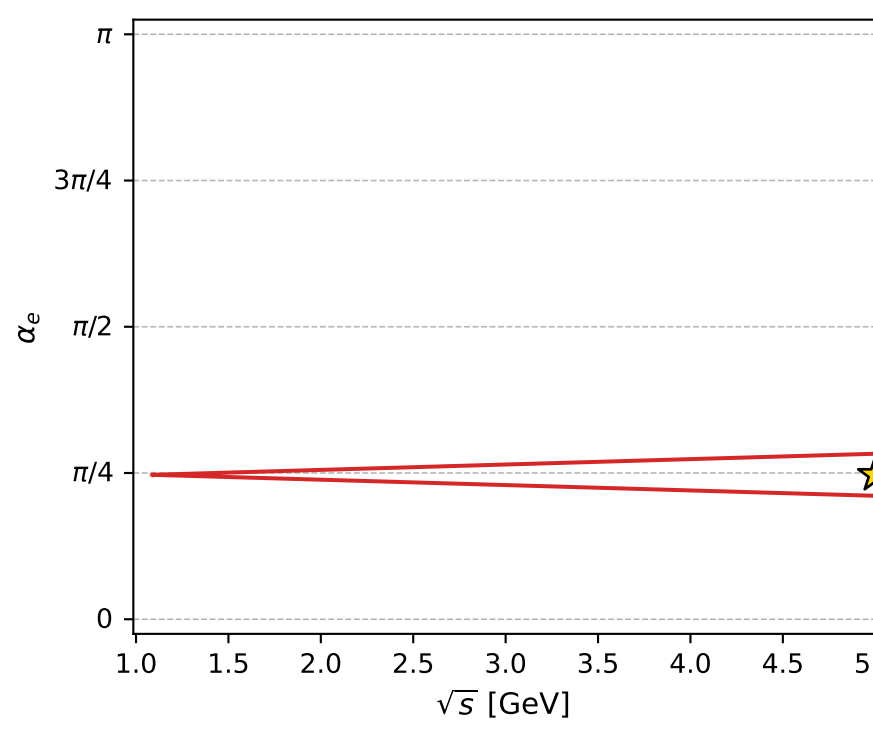
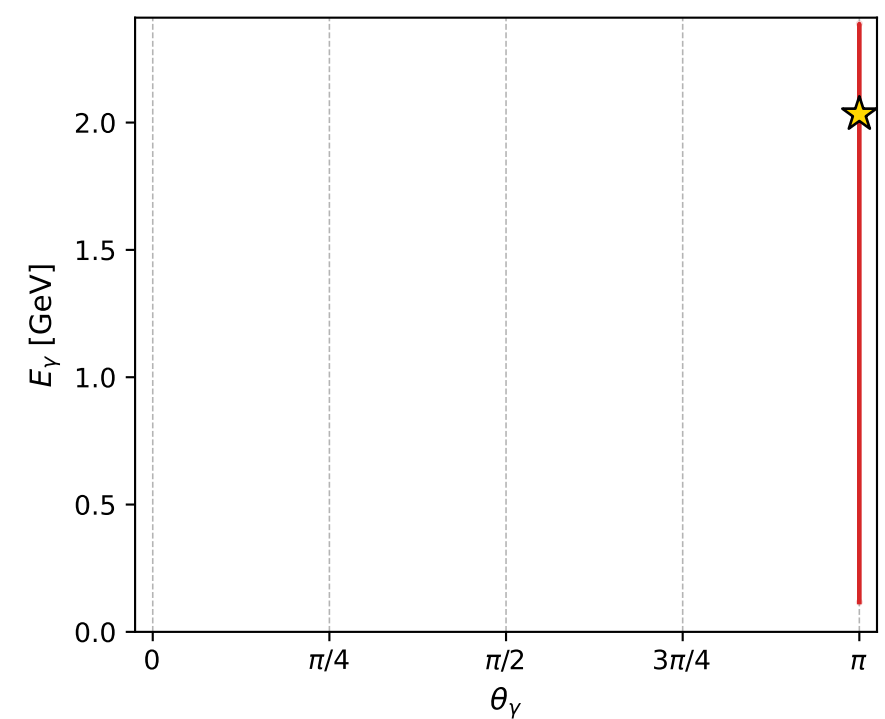
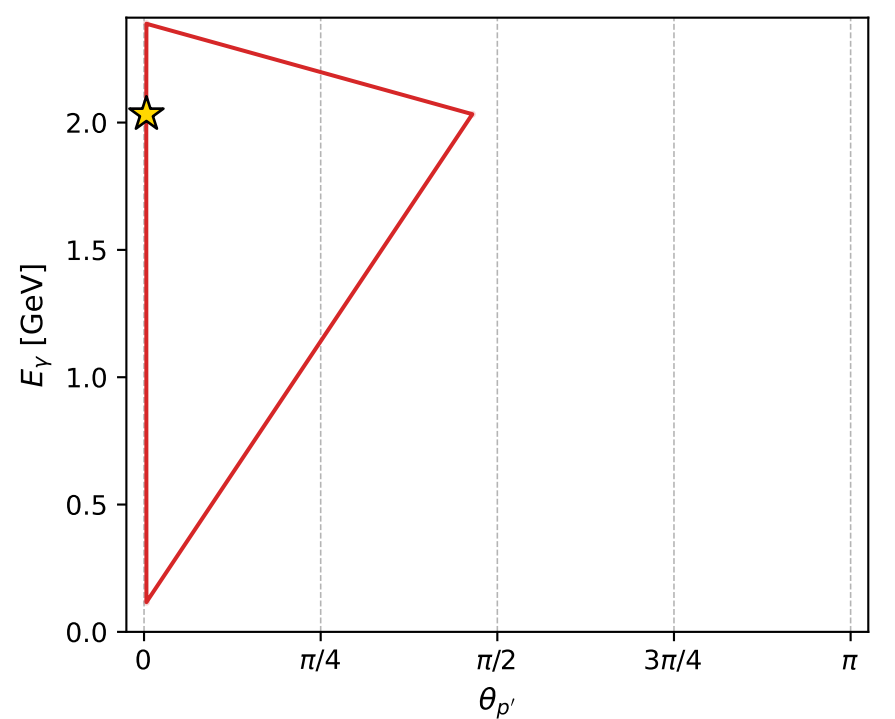
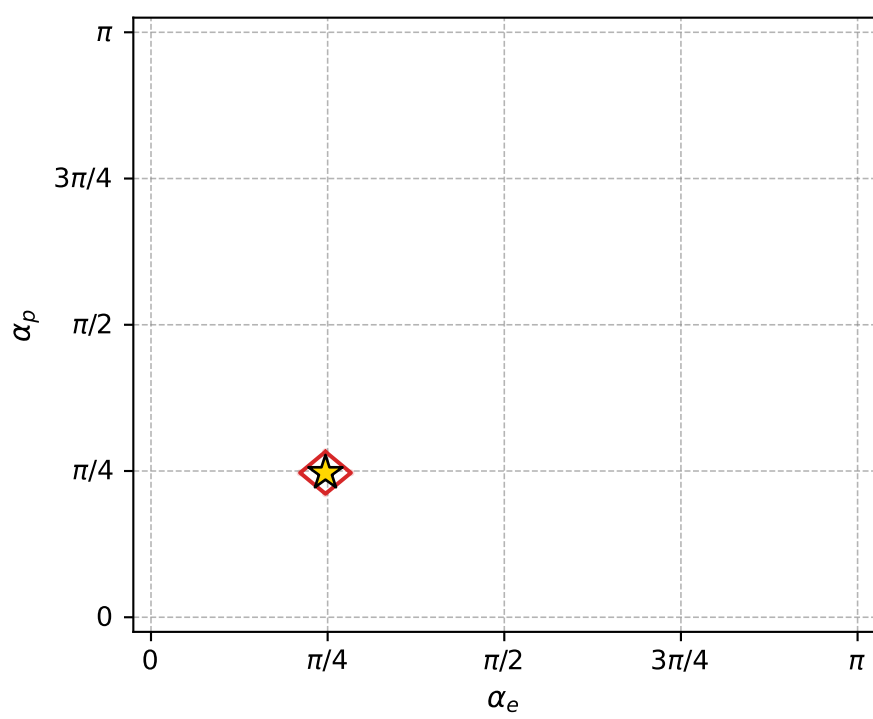
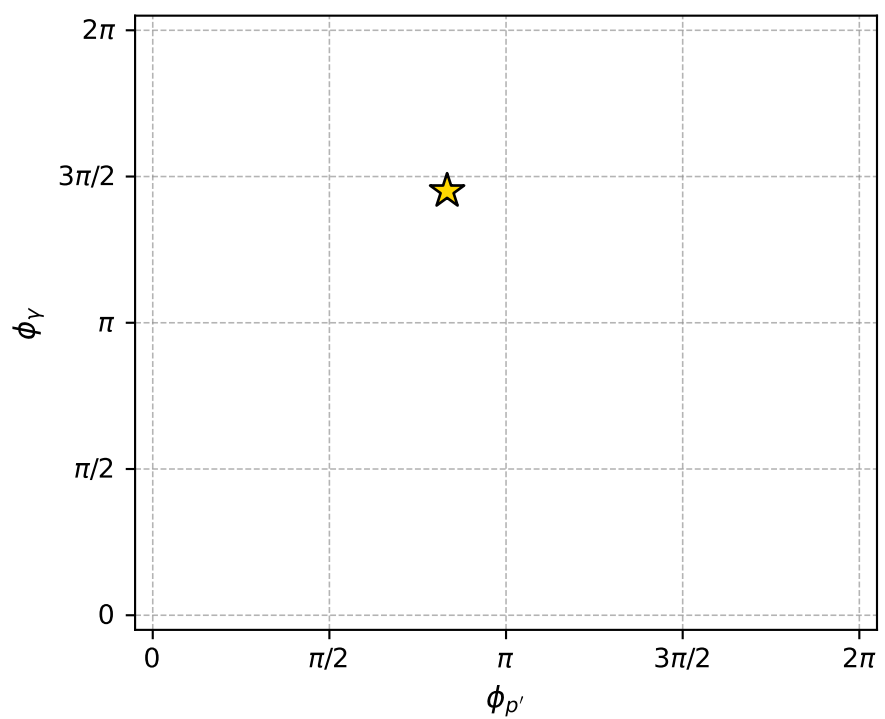
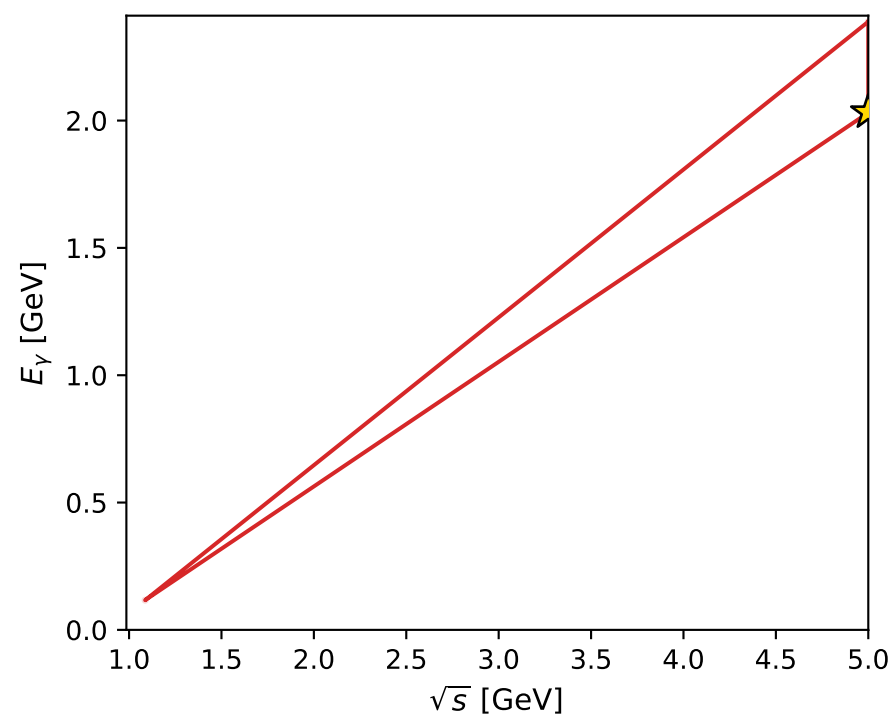
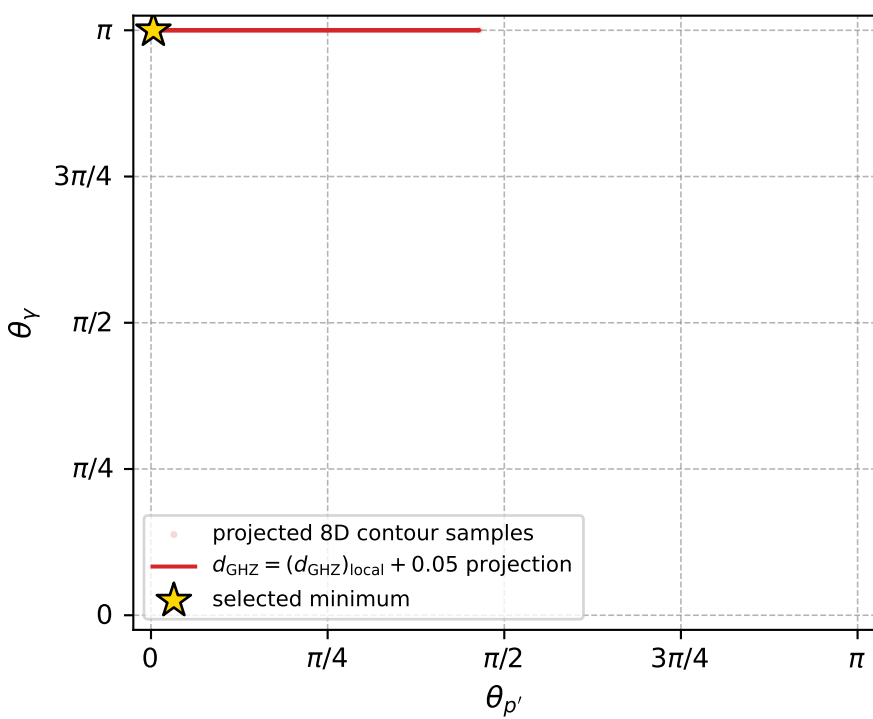


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P1, local minimum 23; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.578178\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $3.4983177\text{e-}09$
 8D contour samples = 128

selected phase-space point:

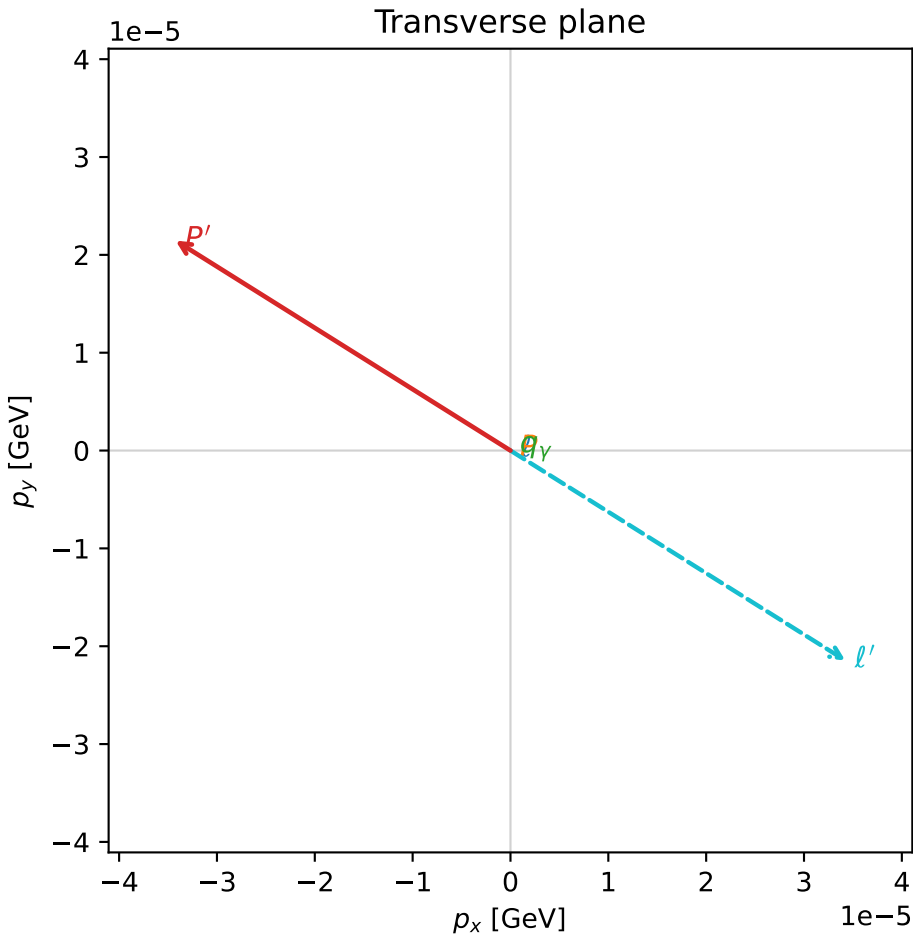
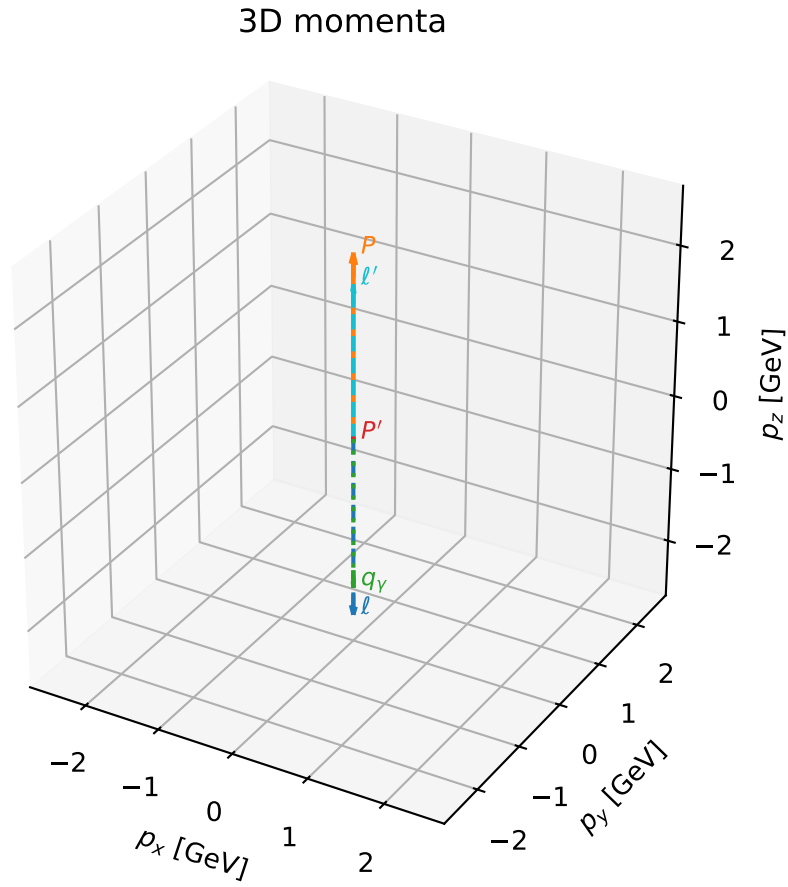
$\text{sqrt_s} = 4.999765$
 $\text{theta_p_out} = 0.0106052$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.032923$
 $\text{phi_p_out} = 2.617436$
 $\text{phi_gamma_out} = 4.556516$
 $\alpha_e = 0.7757184$
 $\alpha_p = 0.7757106$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_24 $d_{\{\text{rm GHZ}\}}=2.58712\text{e-}08$
 region: polarization_cluster_01_local_minimum_24
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.742527$ rad
 incoming lepton: $0.736762|+\rangle +0.676152|-\rangle$
 proton mixing angle: $\alpha_p=0.74253335$ rad
 incoming proton: $0.736758|+\rangle +0.676157|-\rangle$

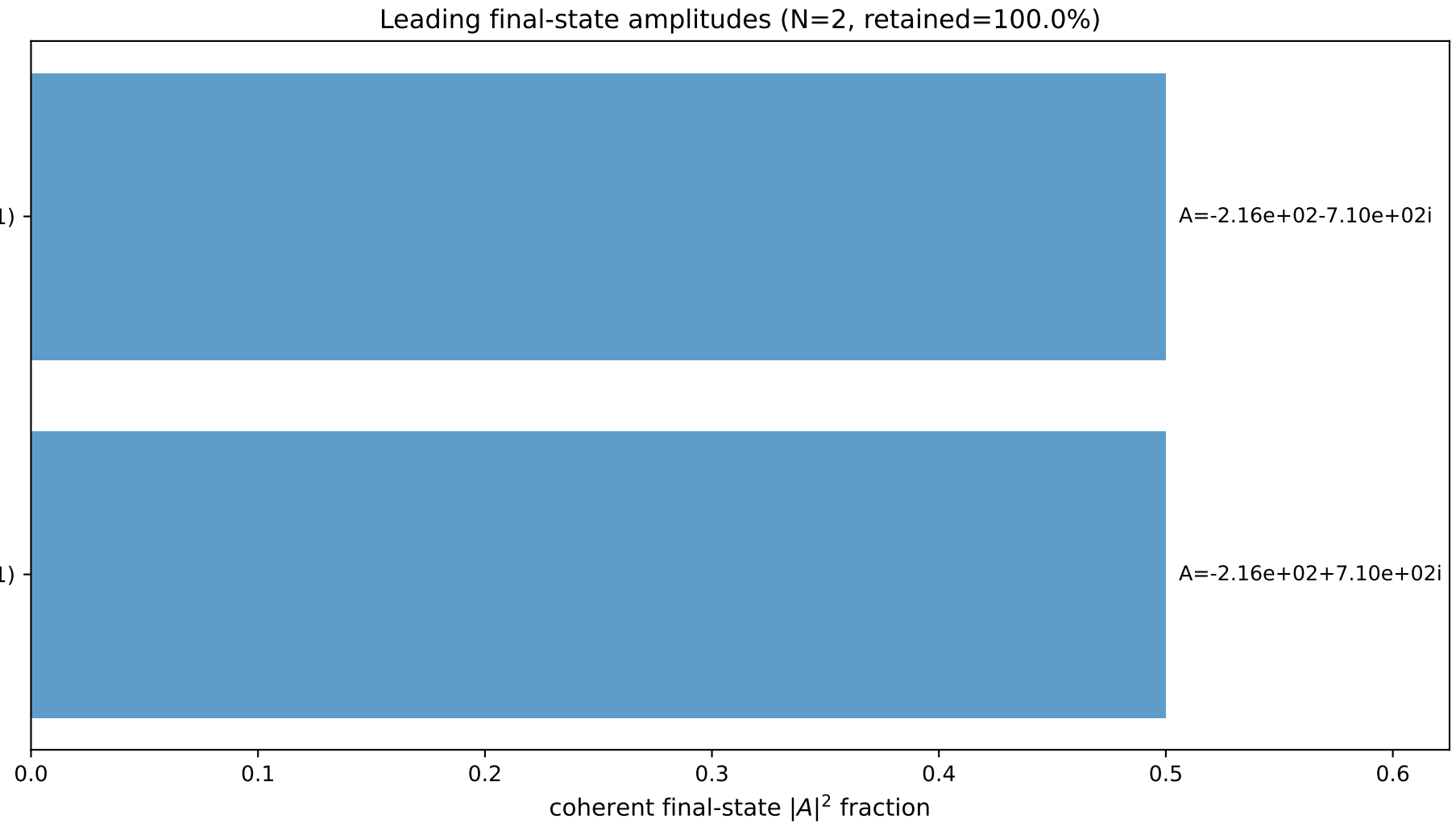
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0041485$
 $\theta_{P'}=0.00973758$, $\theta_Y=3.14159$, $\phi_{P'}=2.58165$, $\phi_Y=5.00822$
 $E_Y=2.03307$, $\theta_{\ell'}=1.99099\text{e-}05$, $\phi_{\ell'}=5.72325$
 $Q^2=19.5752$, $x_B=0.836327$, $t=-3.07541$, $W^2=4.71079$, $y=0.970396$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0289$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0289$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0041$
 q_Y $E= 2.0331$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0331$

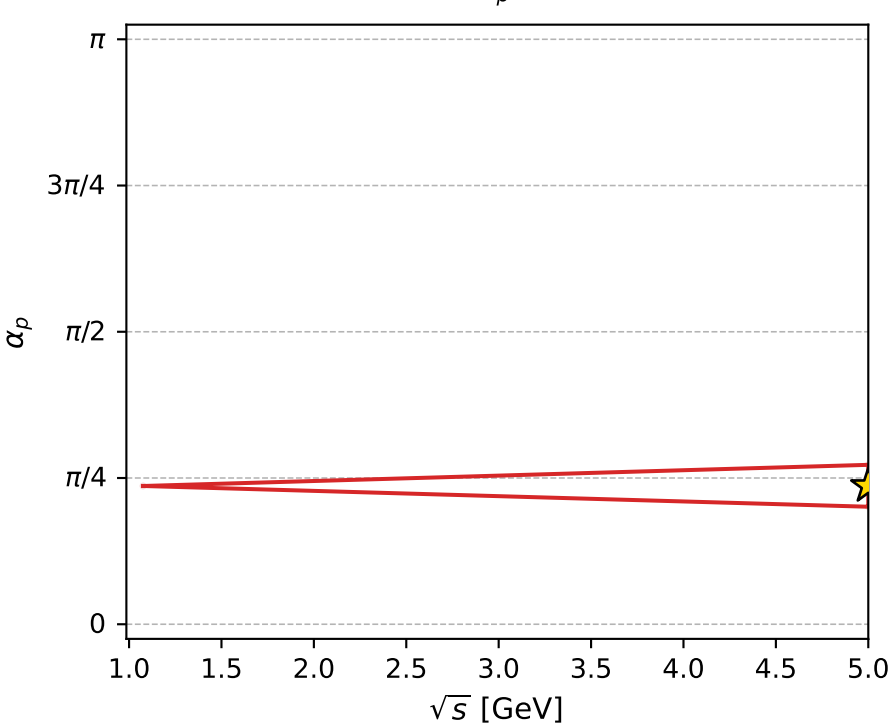
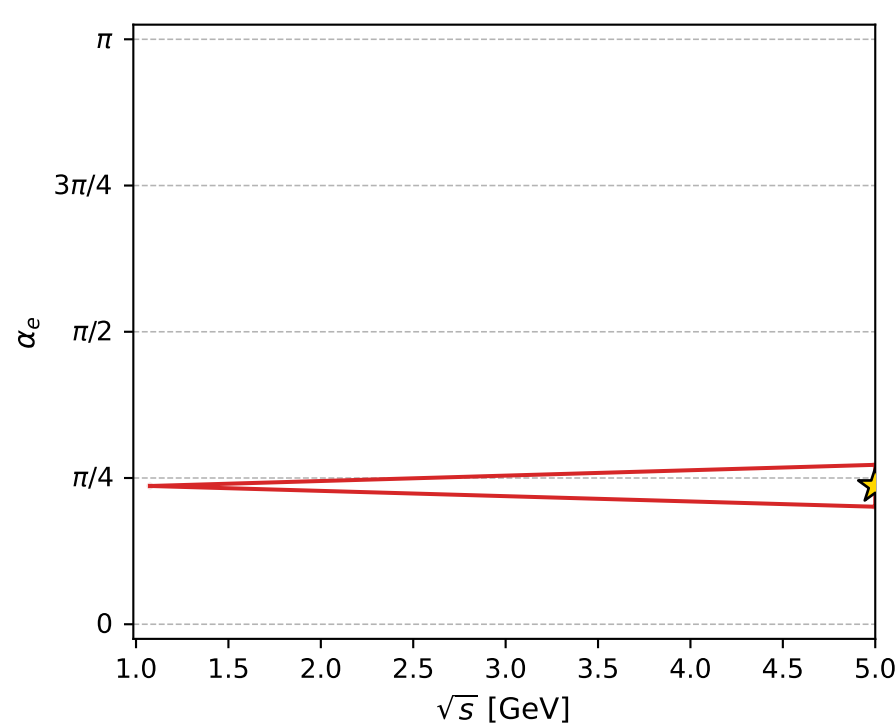
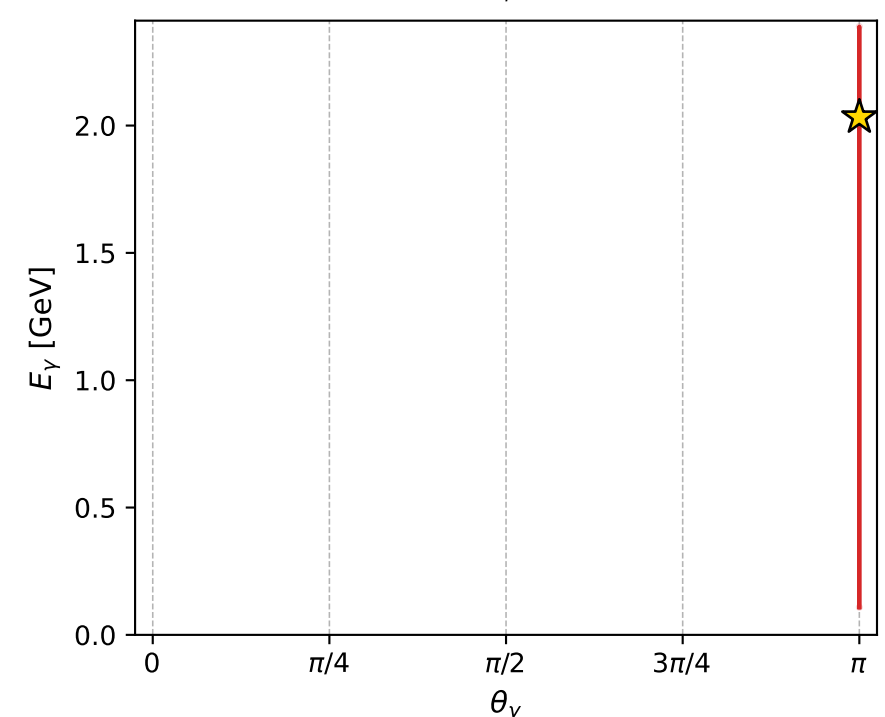
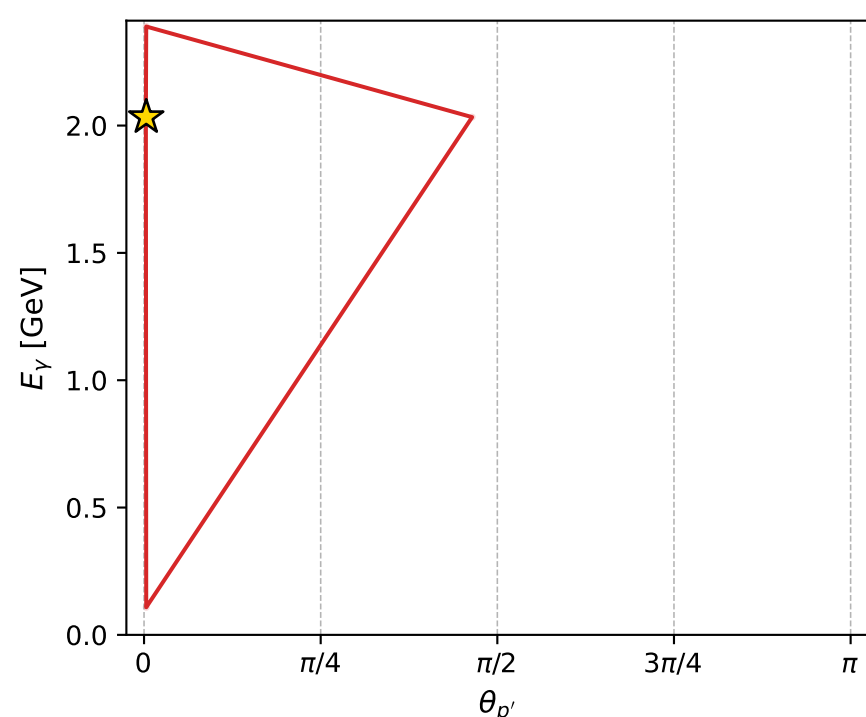
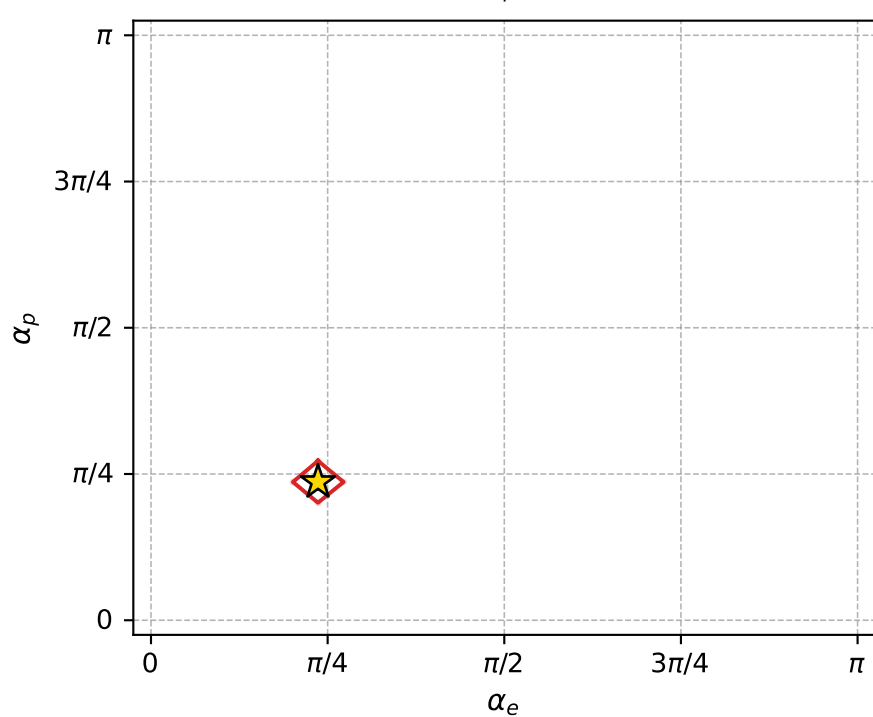
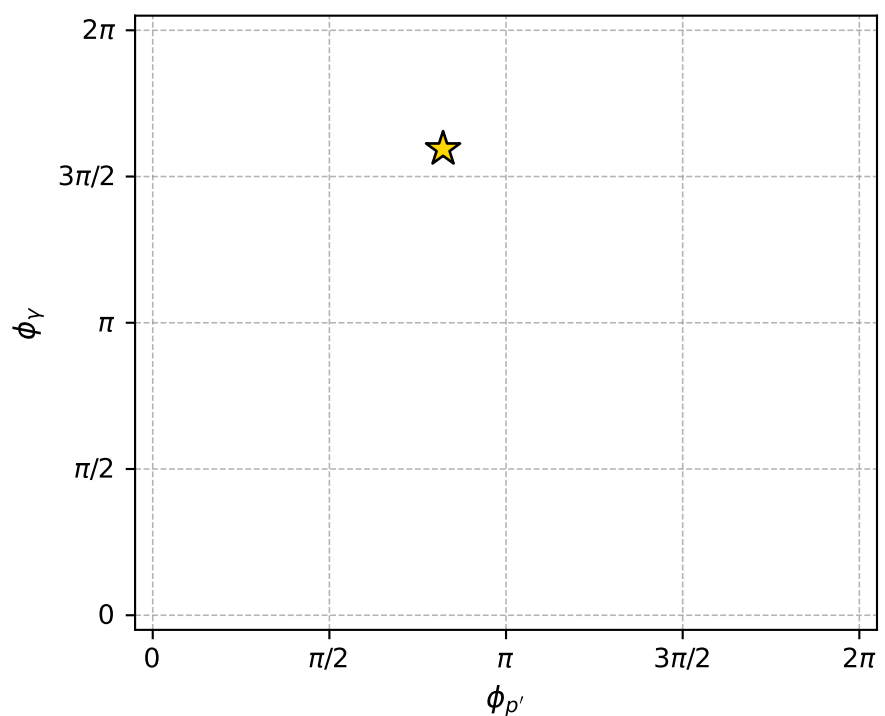
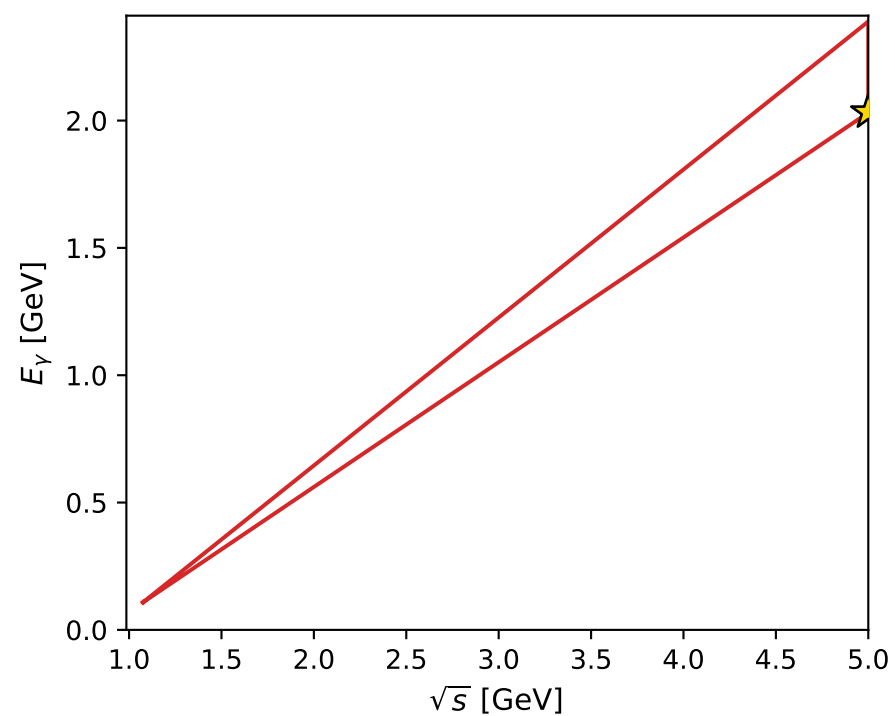
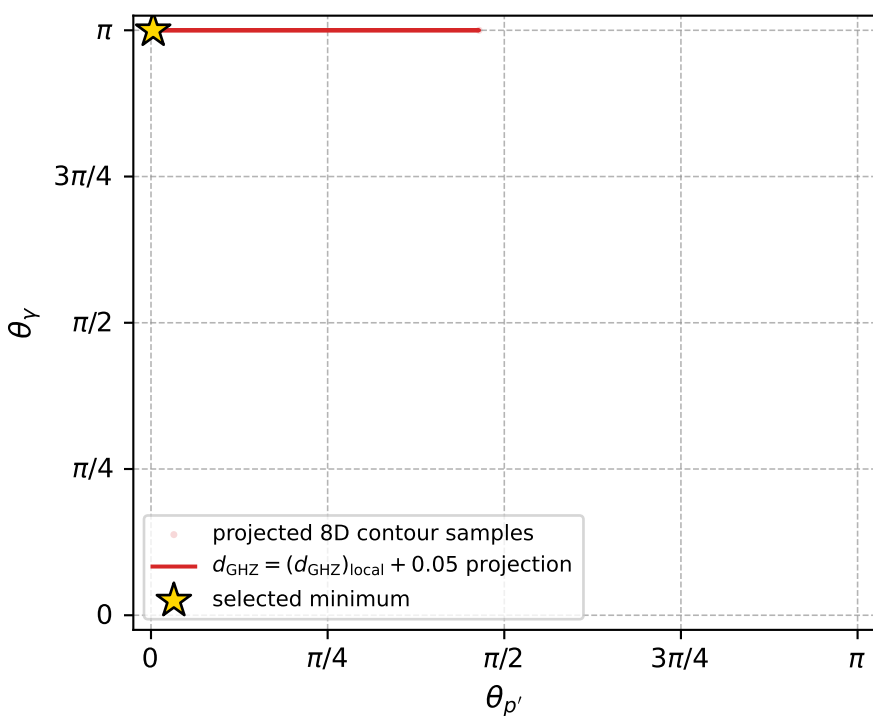


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P1, local minimum 24; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.5871198\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $3.5877363\text{e-}09$
 8D contour samples = 76

selected phase-space point:

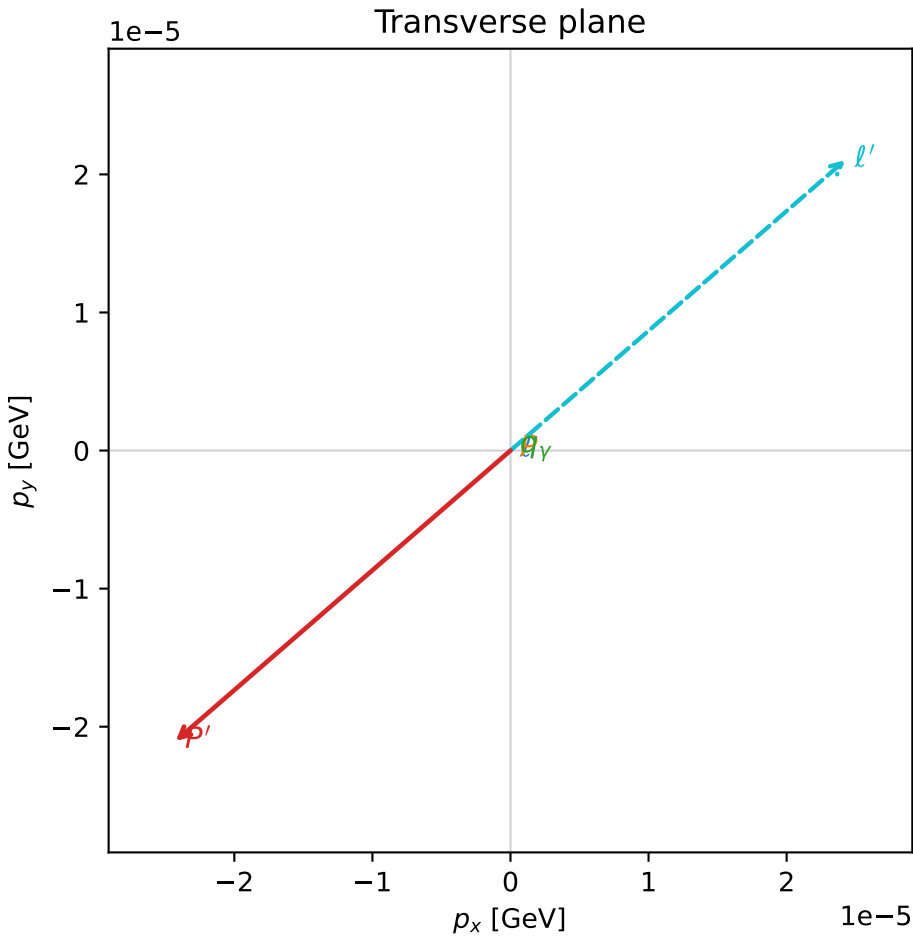
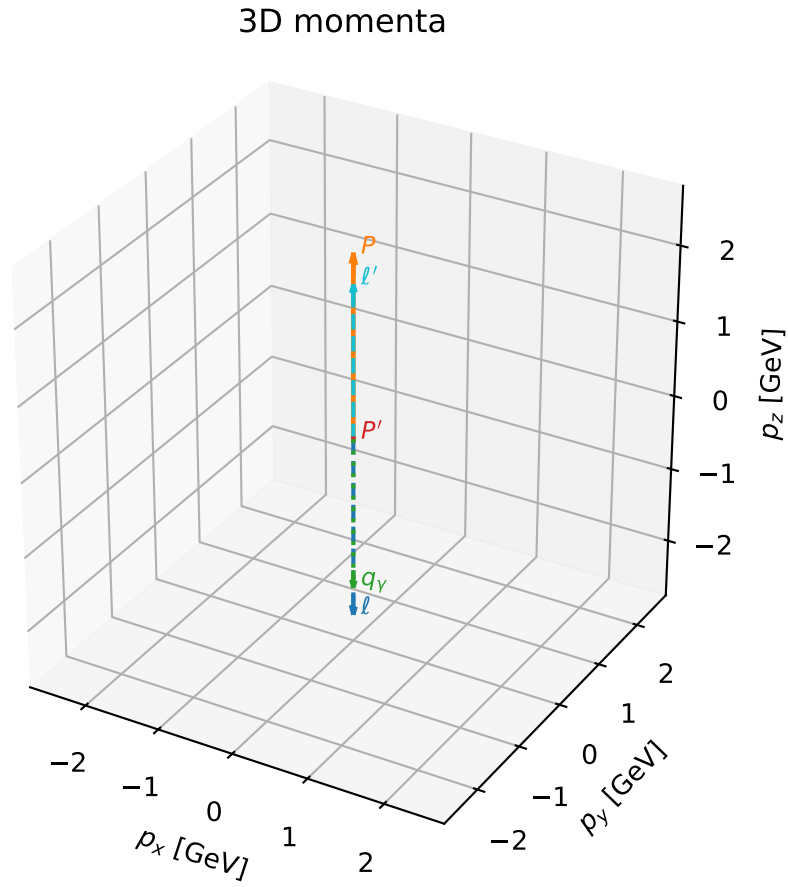
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.009737583$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.03307$
 $\text{phi_p_out} = 2.581654$
 $\text{phi_gamma_out} = 5.008222$
 $\alpha_e = 0.742527$
 $\alpha_p = 0.7425333$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_25 $d_{\{\text{rm GHZ}\}}=2.59286\text{e-}08$
region: polarization_cluster_01_local_minimum_25
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.768686$ rad
incoming lepton: $0.718825|+\rangle +0.695191|-\rangle$
proton mixing angle: $\alpha_p=0.76868388$ rad
incoming proton: $0.718826|+\rangle +0.69519|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00436041$
 $\theta_{P'}=0.00736311$, $\theta_Y=3.14159$, $\phi_{P'}=3.8561$, $\phi_Y=0.0545742$
 $E_Y=2.03318$, $\theta_{\ell'}=1.5825\text{e-}05$, $\phi_{\ell'}=0.714507$
 $Q^2=19.5741$, $x_B=0.836282$, $t=-3.07439$, $W^2=4.71185$, $y=0.970398$

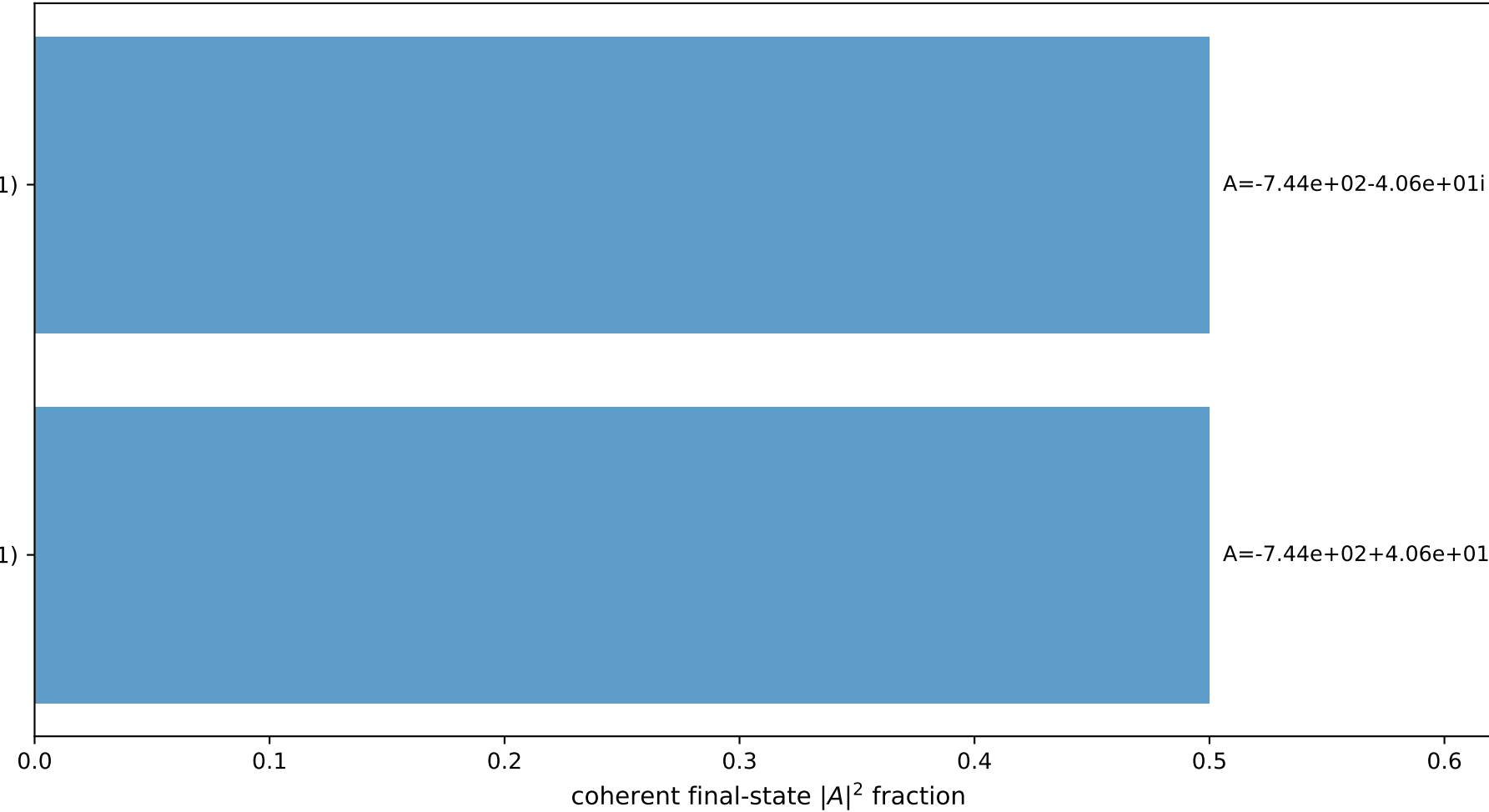
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0288$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0288$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0044$
 q_Y $E= 2.0332$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0332$



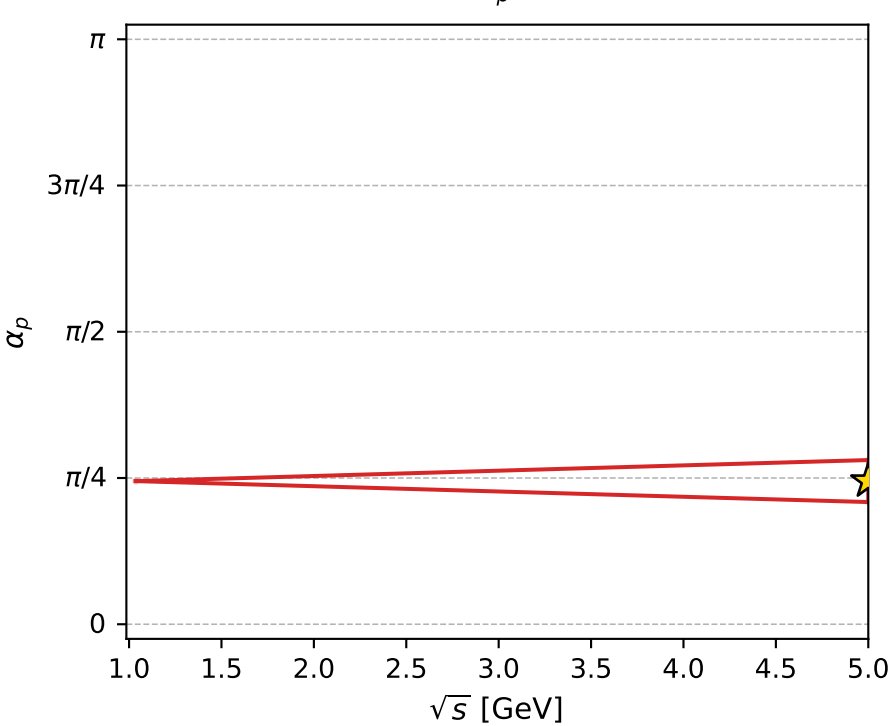
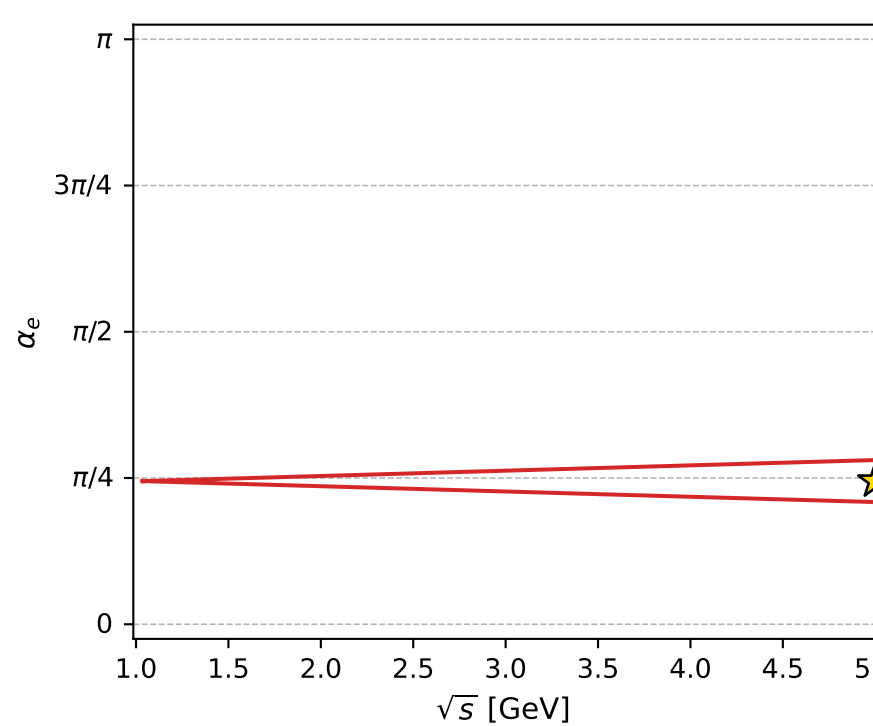
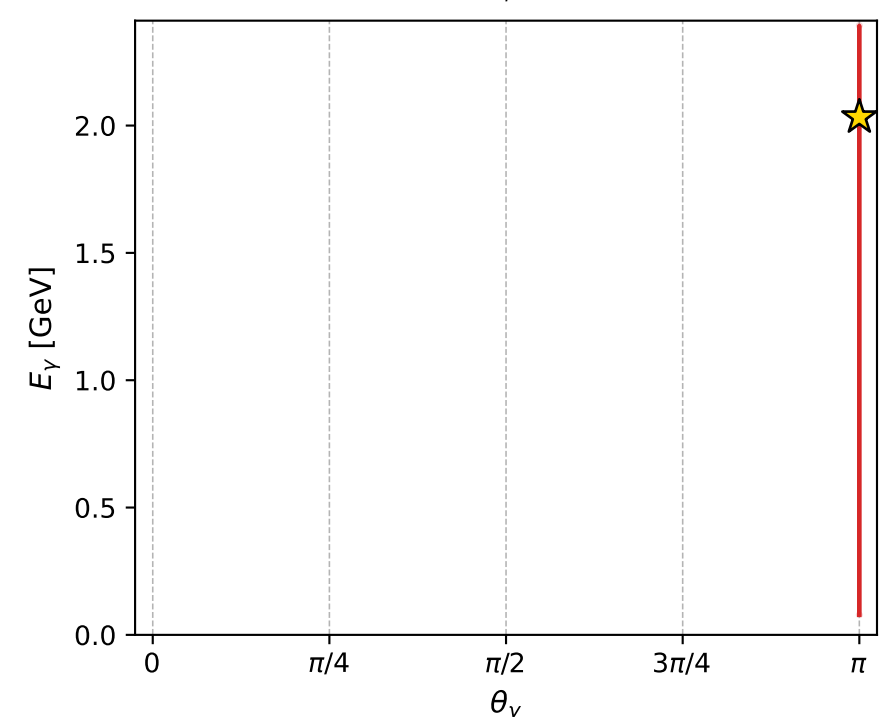
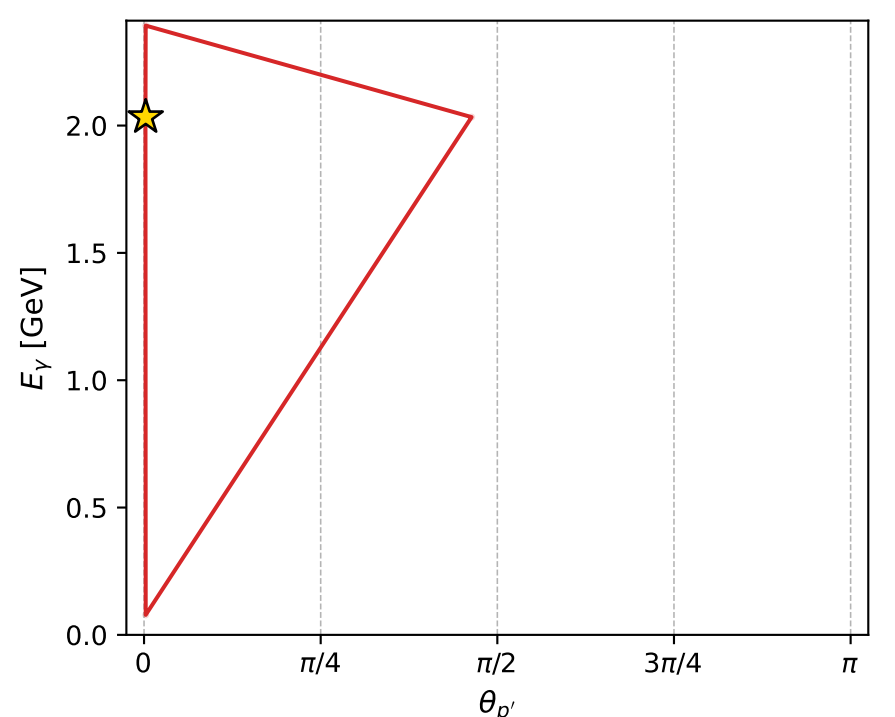
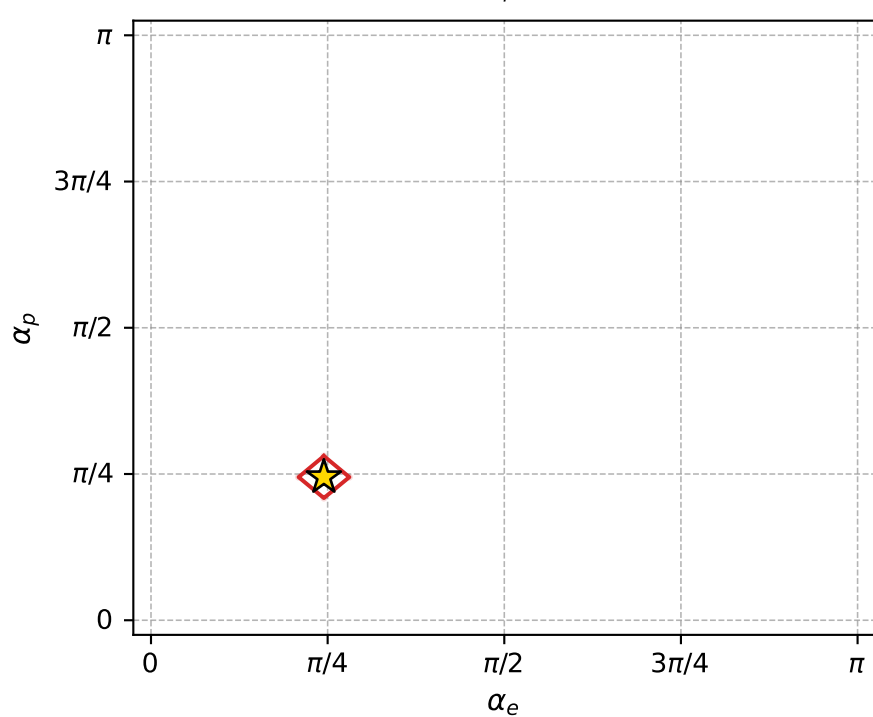
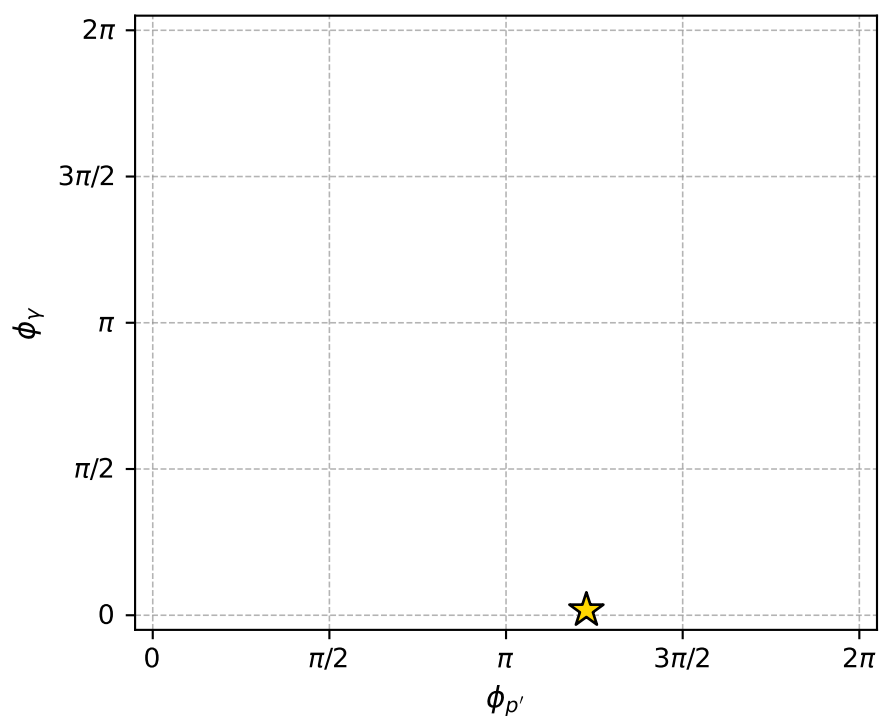
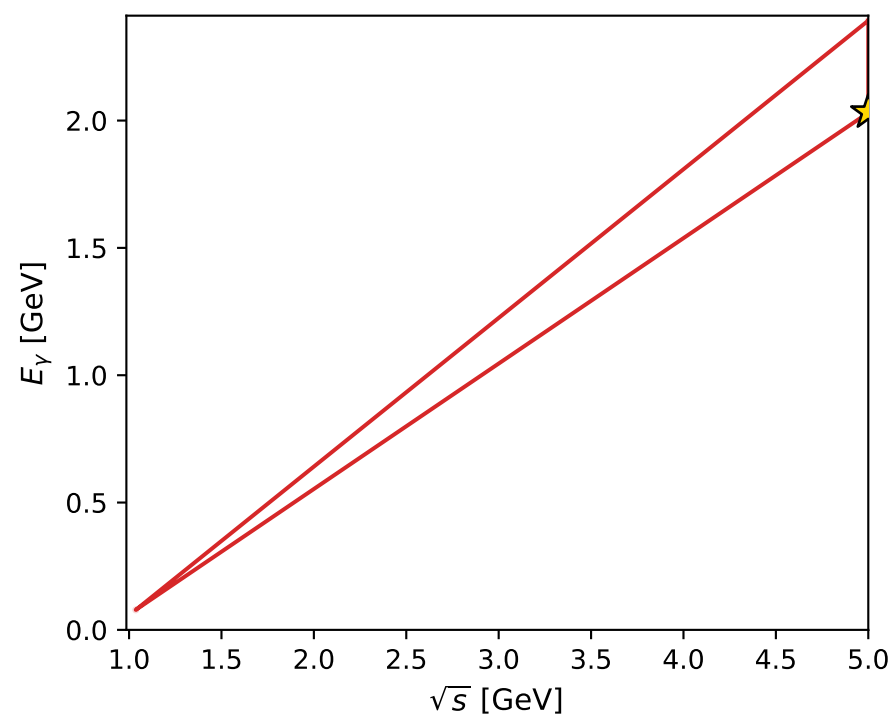
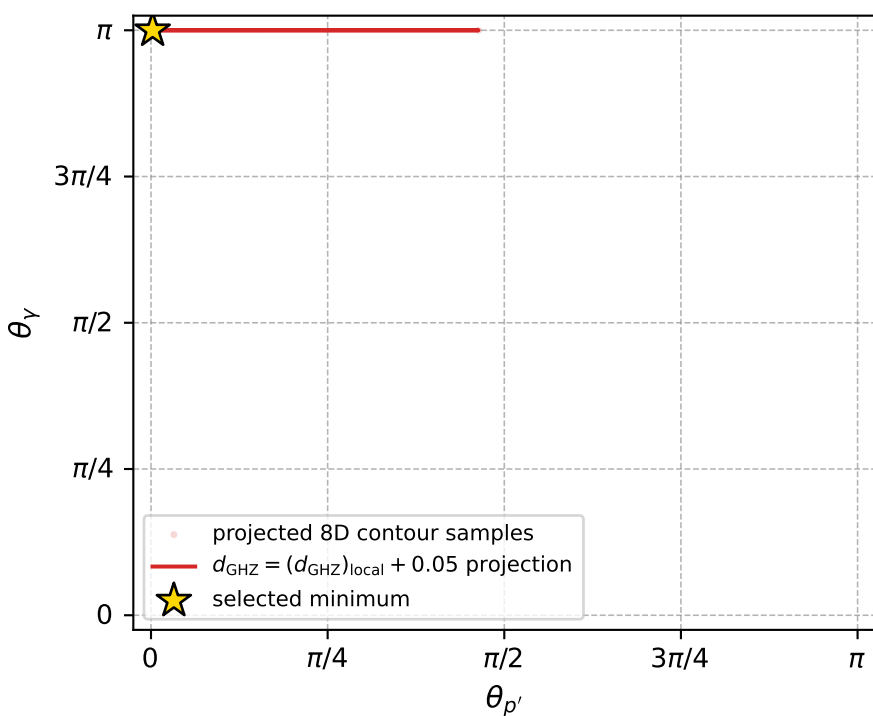
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 25; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.5928622\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $3.6451599\text{e-}09$
 8D contour samples = 76

selected phase-space point:

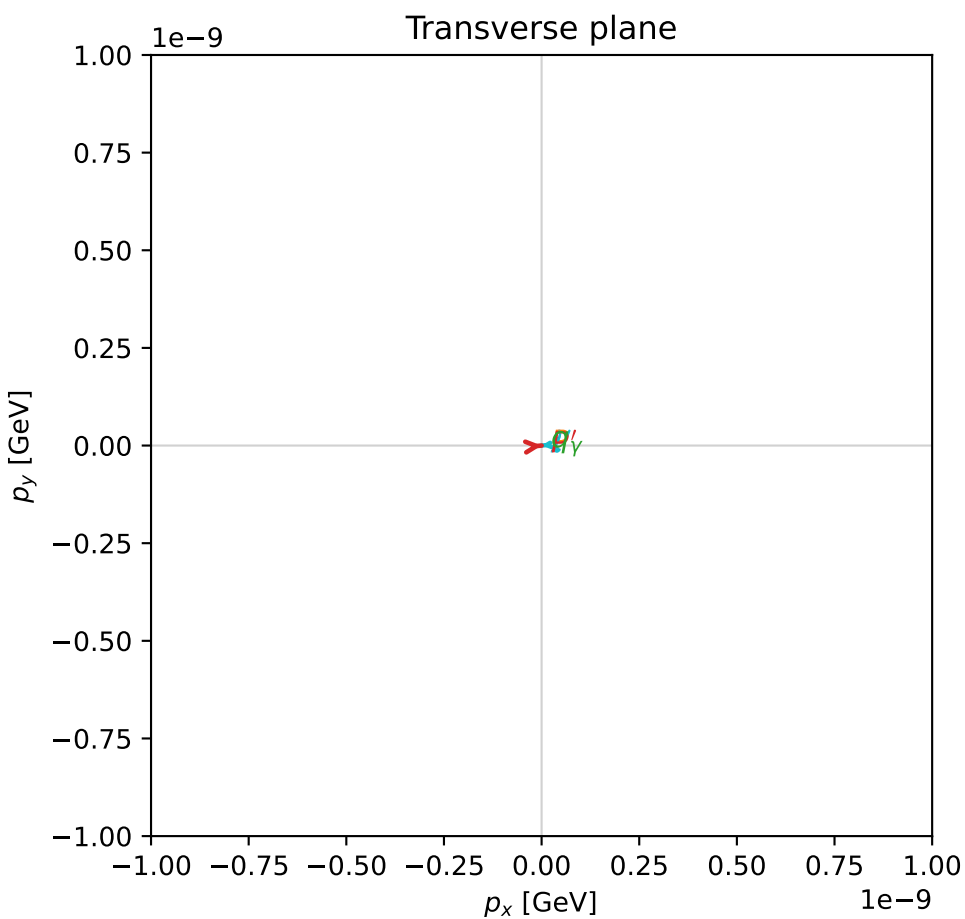
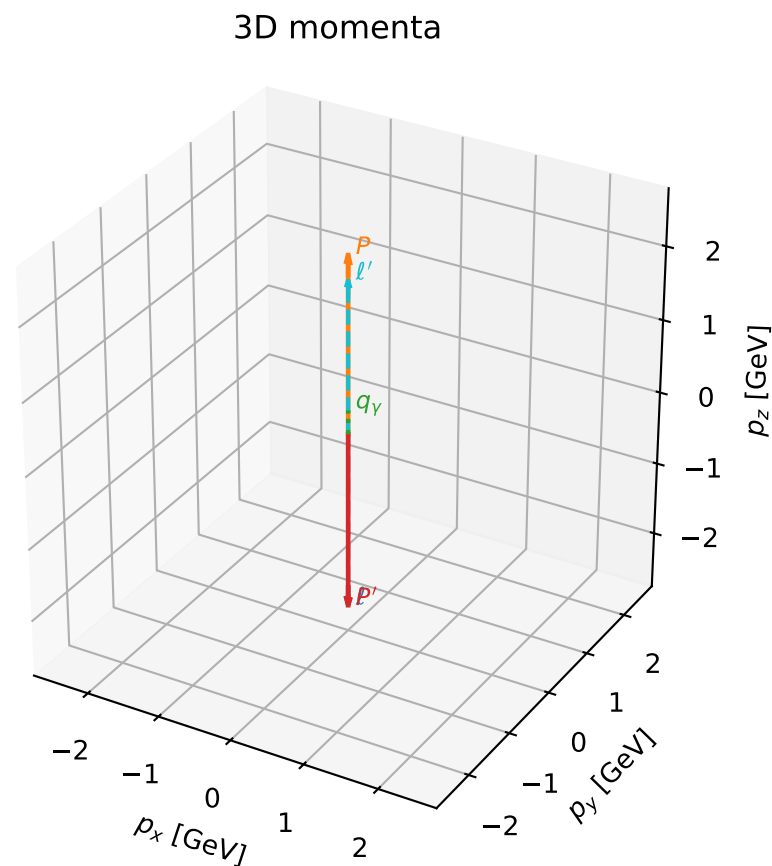
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.007363108$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.033175$
 $\text{phi_p_out} = 3.856099$
 $\text{phi_gamma_out} = 0.05457418$
 $\alpha_e = 0.768686$
 $\alpha_p = 0.7686839$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_27 d_{\rm GHZ}=2.59874e-08
region: polarization_cluster_01_local_minimum_27
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.75349531$ rad
incoming lepton: $0.729302|+\rangle +0.684192|-\rangle$
proton mixing angle: $\alpha_p=0.81730854$ rad
incoming proton: $0.684187|+\rangle +0.729307|-\rangle$

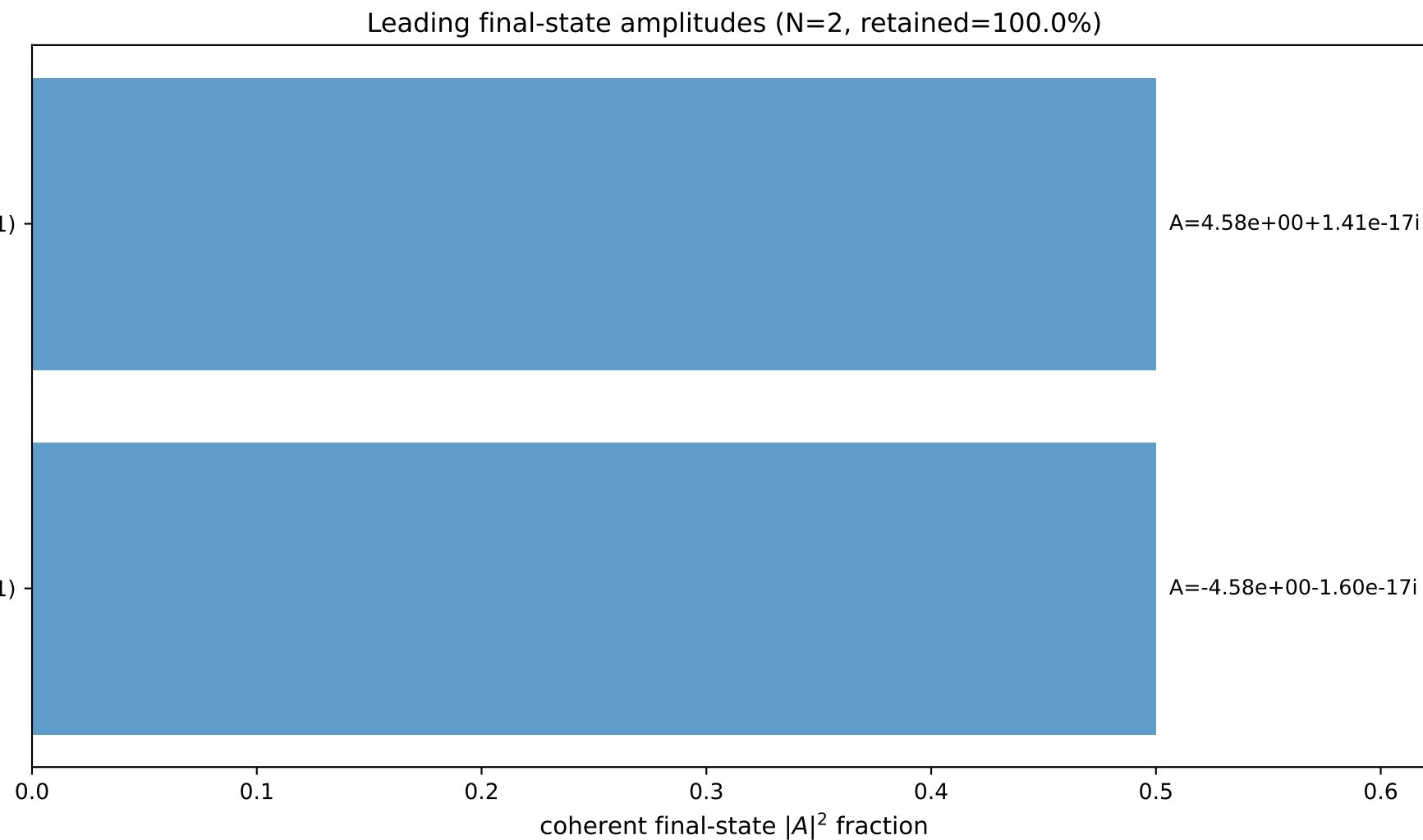
$s=24.9779$, $\sqrt{s}=4.99779$, $|\vec{P}|=2.41087$, $|\vec{P}'|=2.41087$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=0.0939447$, $\phi_Y=0.966166$
 $E_Y=0.322487$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.23554$
 $Q^2=20.1393$, $x_B=0.862026$, $t=-23.2492$, $W^2=4.10329$, $y=0.969487$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4109$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4109$
 P $E= 2.5869$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4109$
 ℓ' $E= 2.0884$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0884$
 P' $E= 2.5869$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4109$
 q_Y $E= 0.3225$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3225$

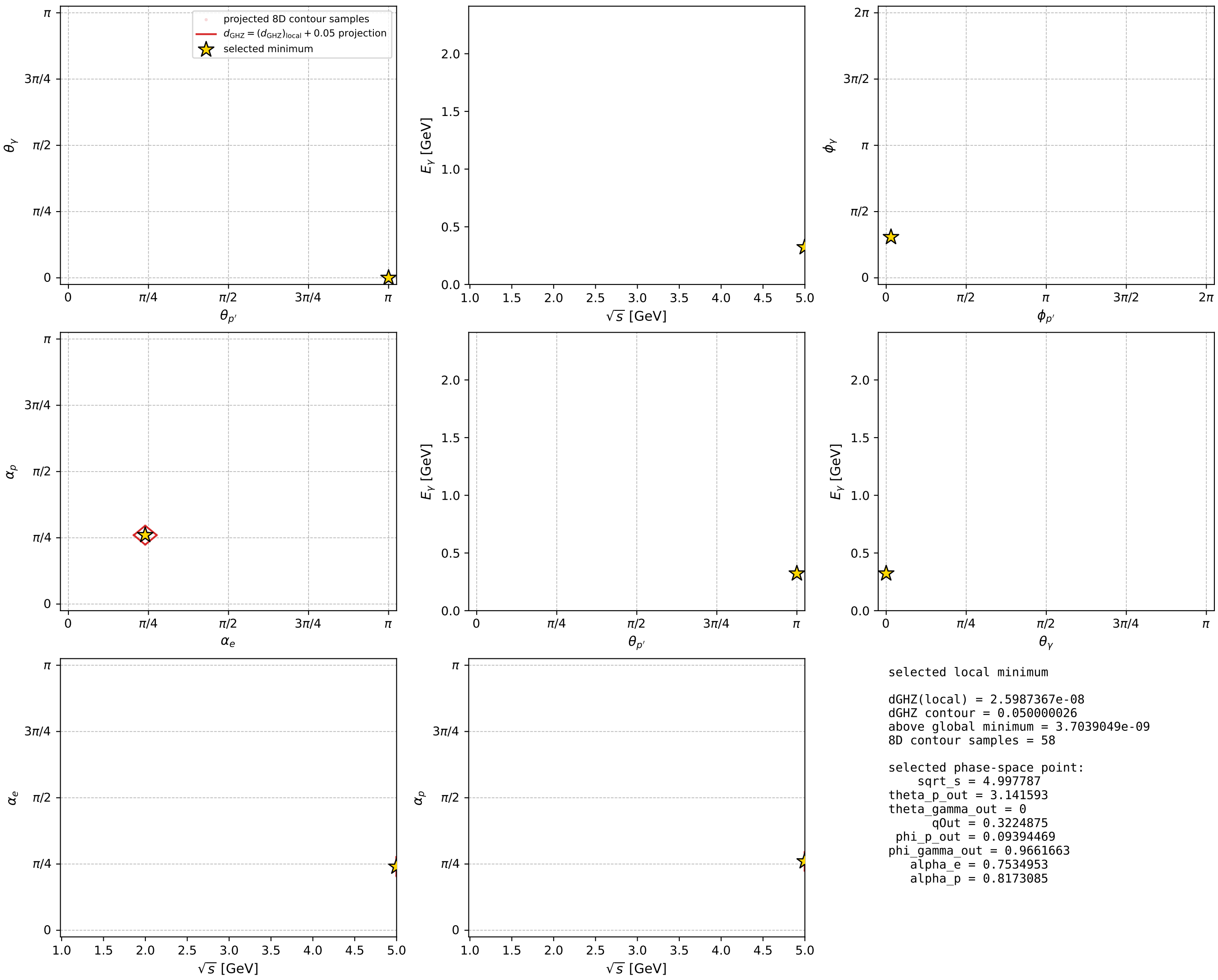


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P1, local minimum 27; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

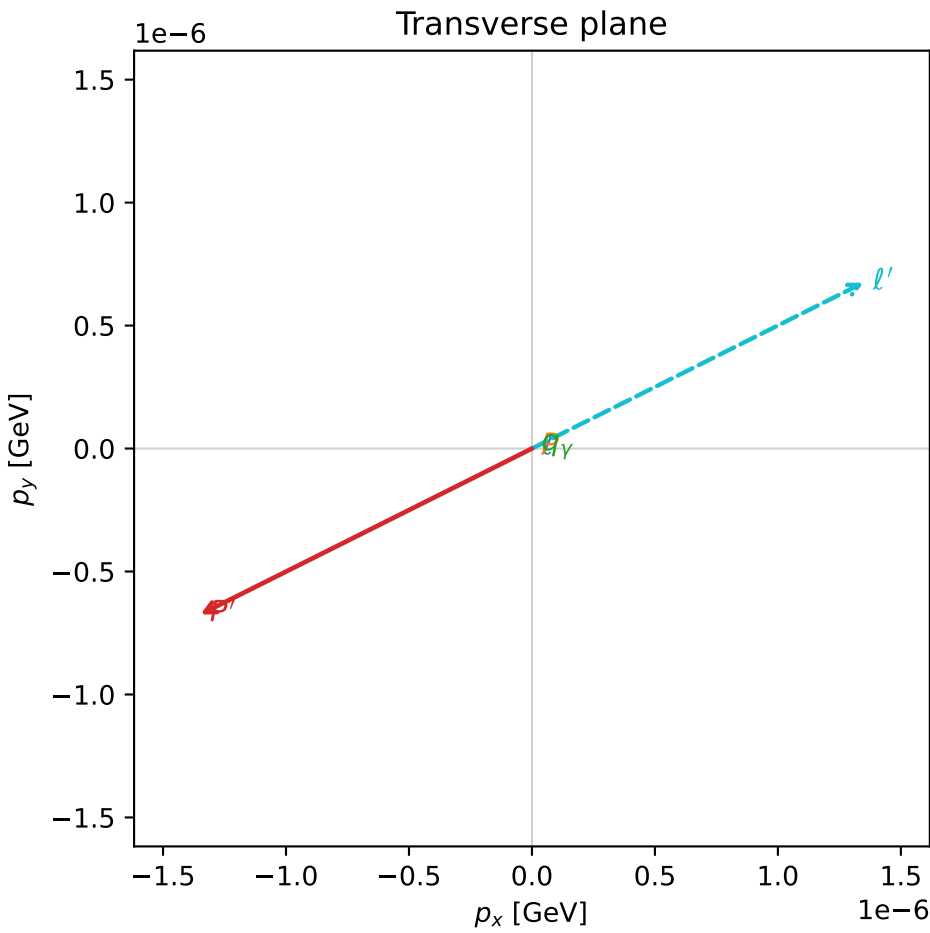
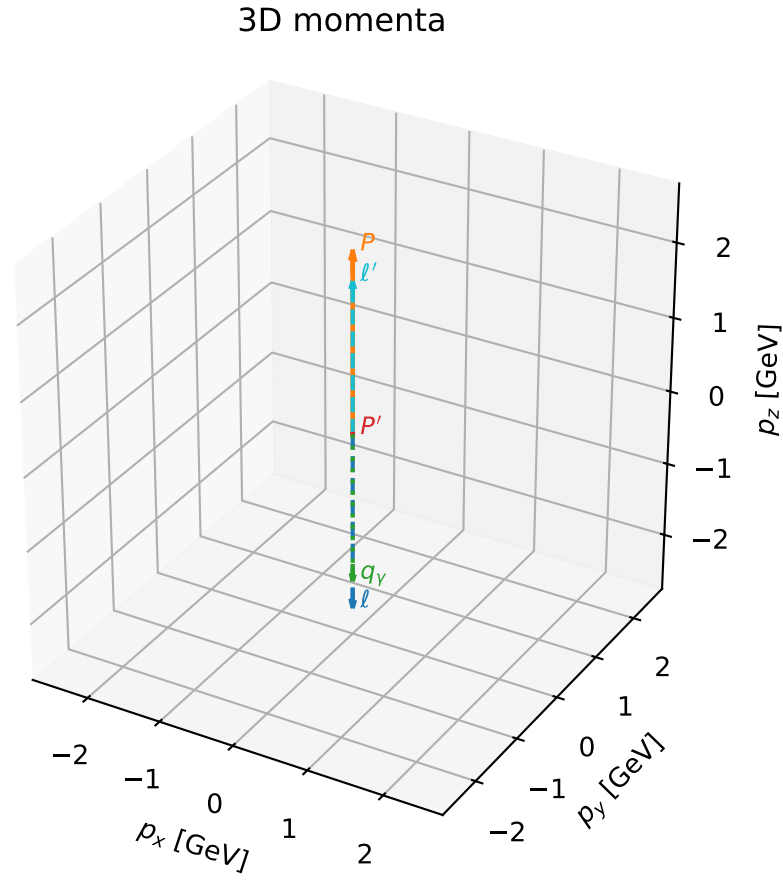


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_28 $d_{\{\text{rm GHZ}\}}=2.60293\text{e-}08$
 region: polarization_cluster_01_local_minimum_28
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.7050247$ rad
 incoming lepton: $0.761596|+\rangle +0.648053|-\rangle$
 proton mixing angle: $\alpha_p=0.70501577$ rad
 incoming proton: $0.761601|+\rangle +0.648046|-\rangle$

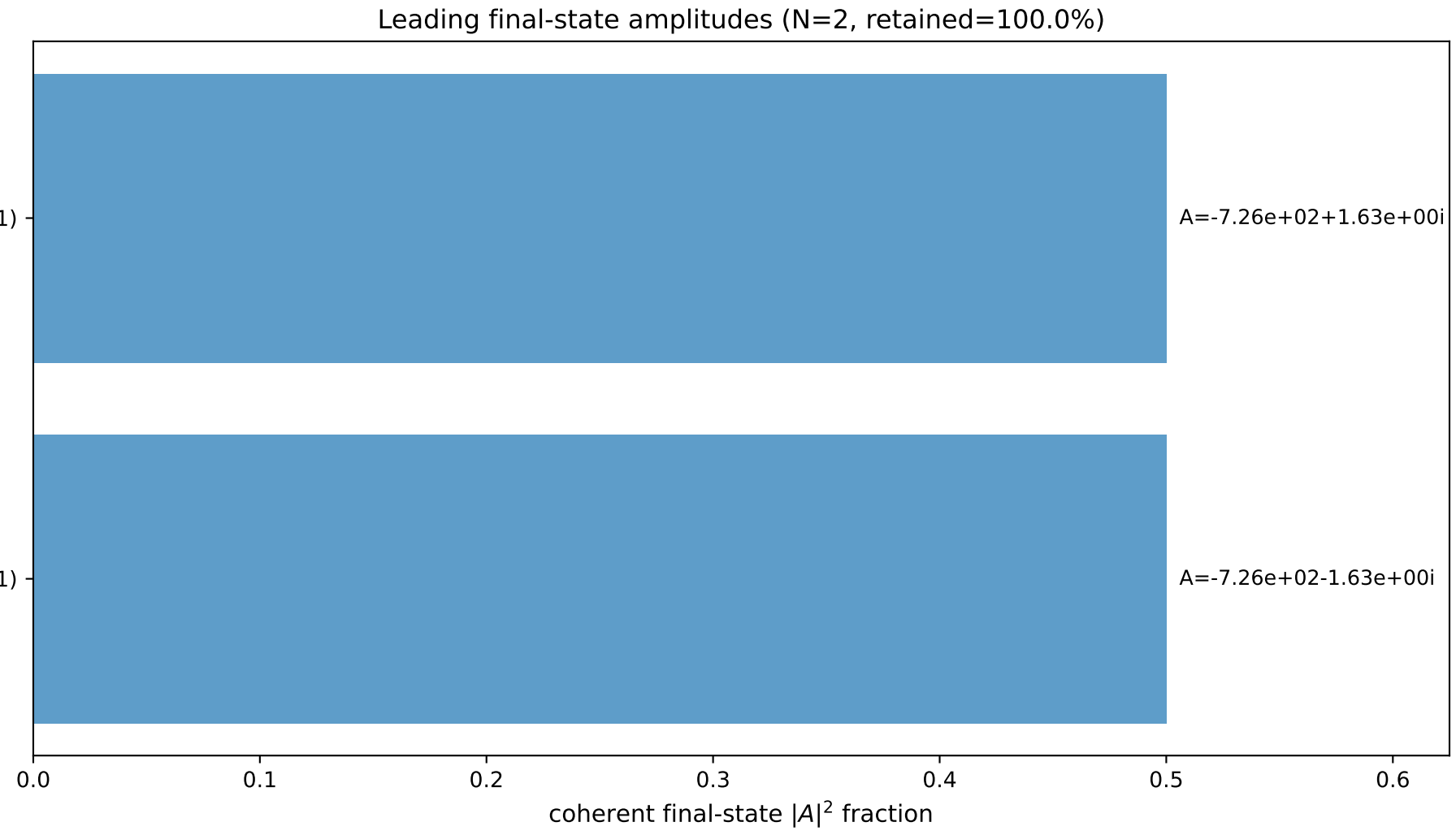
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.000139487$
 $\theta_{P'}=0.0108069$, $\theta_Y=3.14159$, $\phi_{P'}=3.60549$, $\phi_Y=6.28094$
 $E_Y=2.03107$, $\theta_{\ell'}=7.42072\text{e-}07$, $\phi_{\ell'}=0.463893$
 $Q^2=19.5945$, $x_B=0.837181$, $t=-3.0947$, $W^2=4.6907$, $y=0.970367$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0309$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0001$
 q_Y $E= 2.0311$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0311$



$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

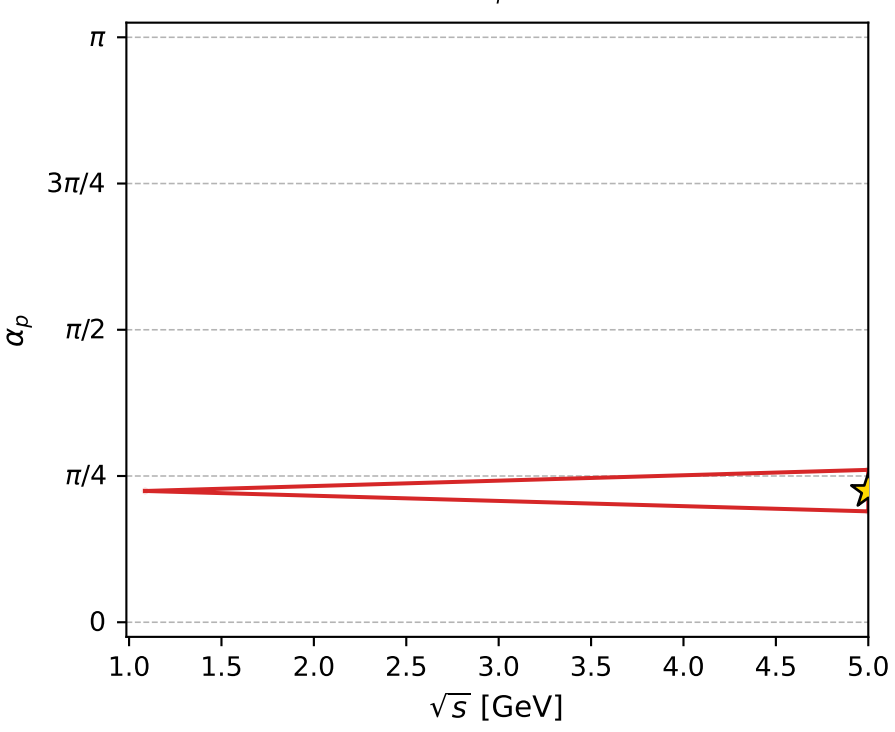
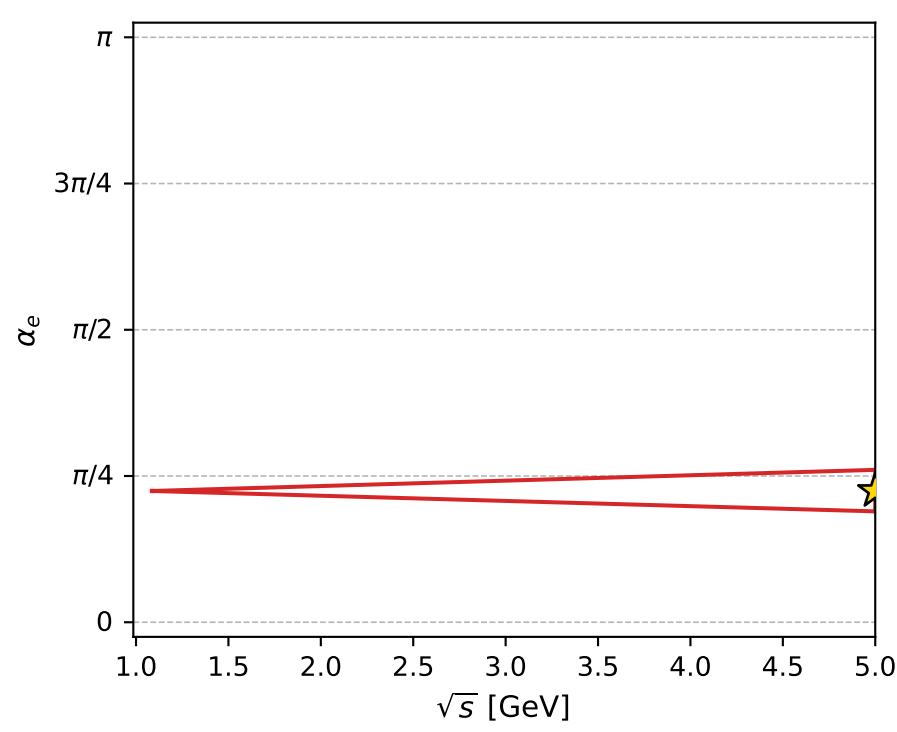
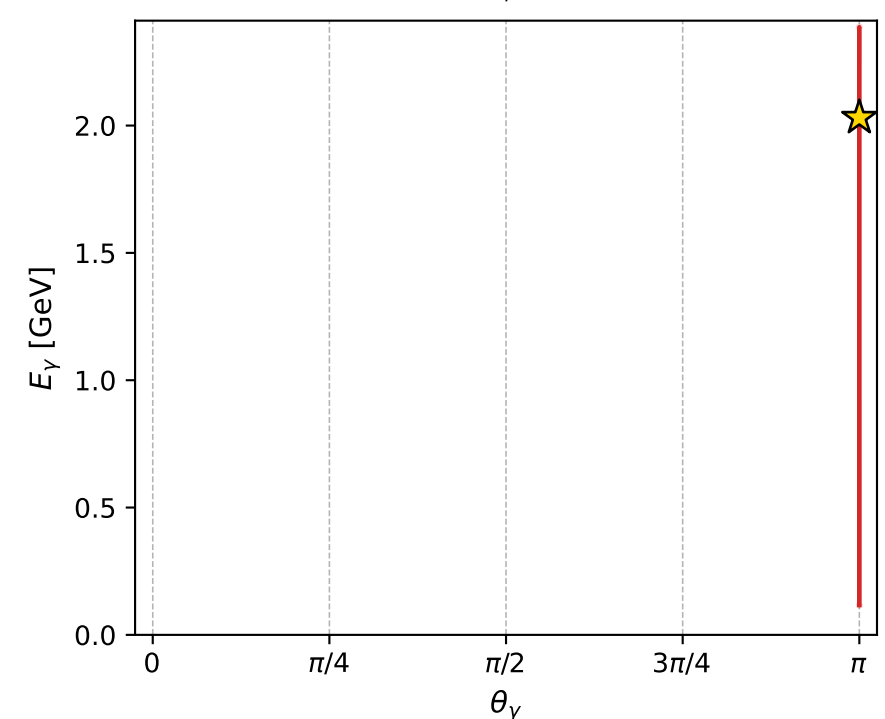
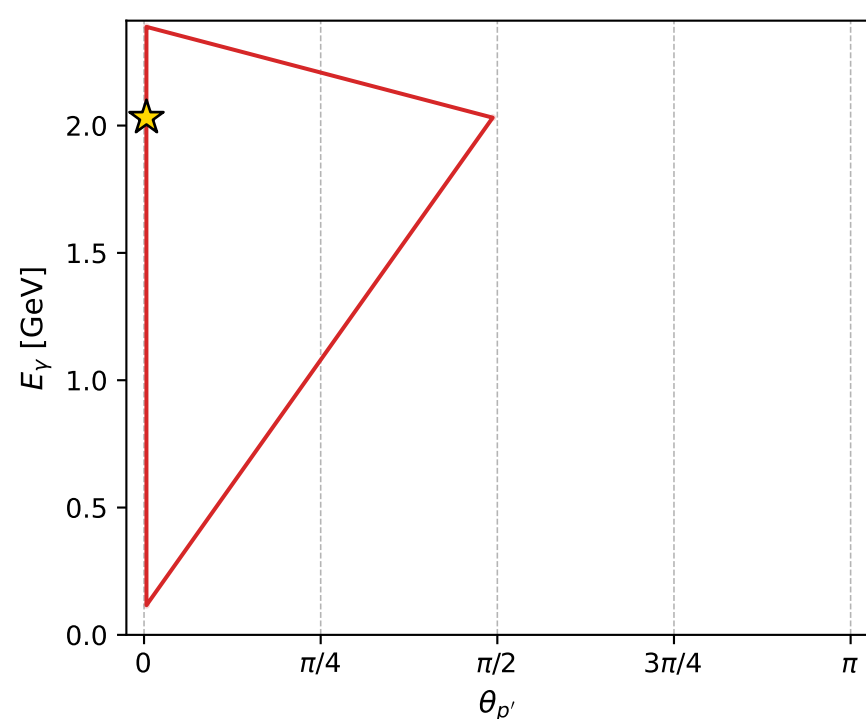
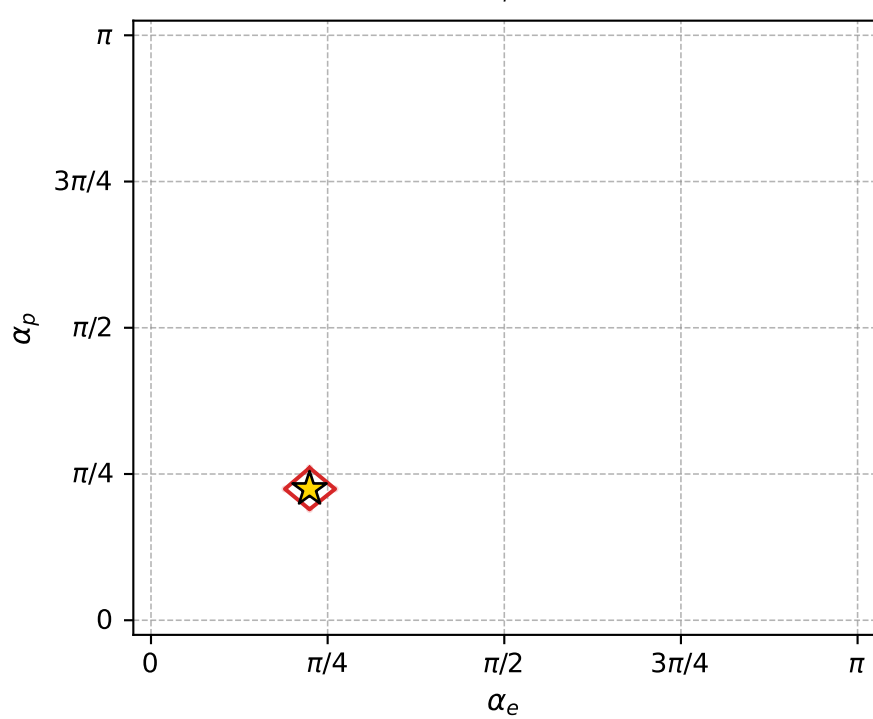
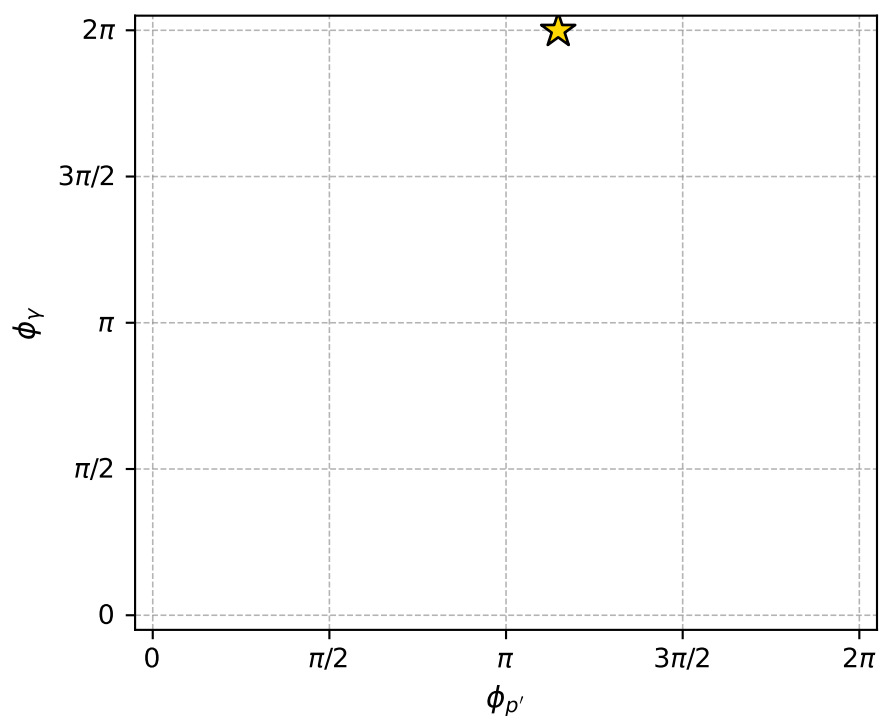
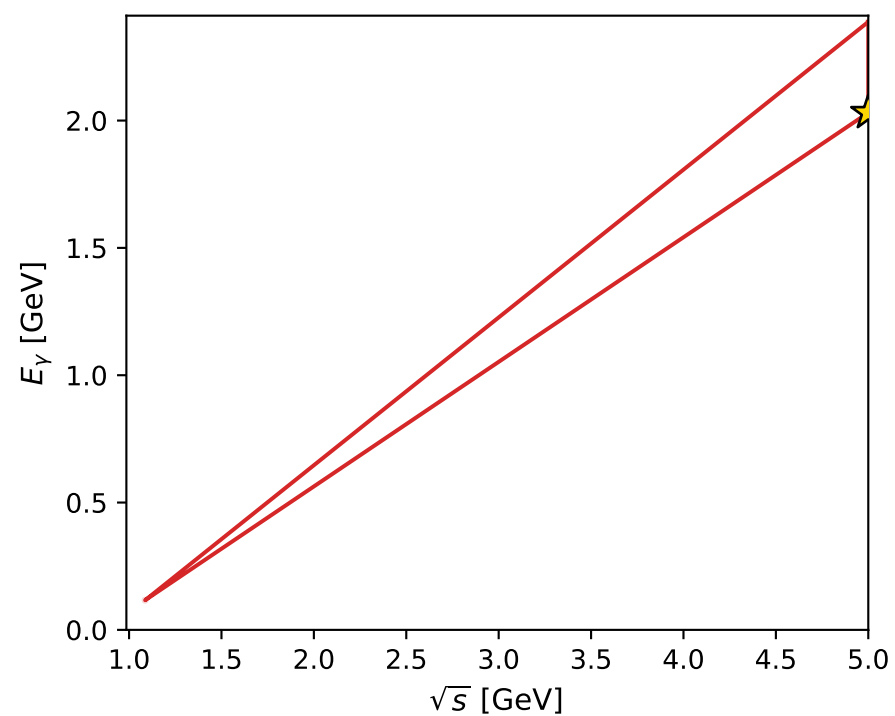
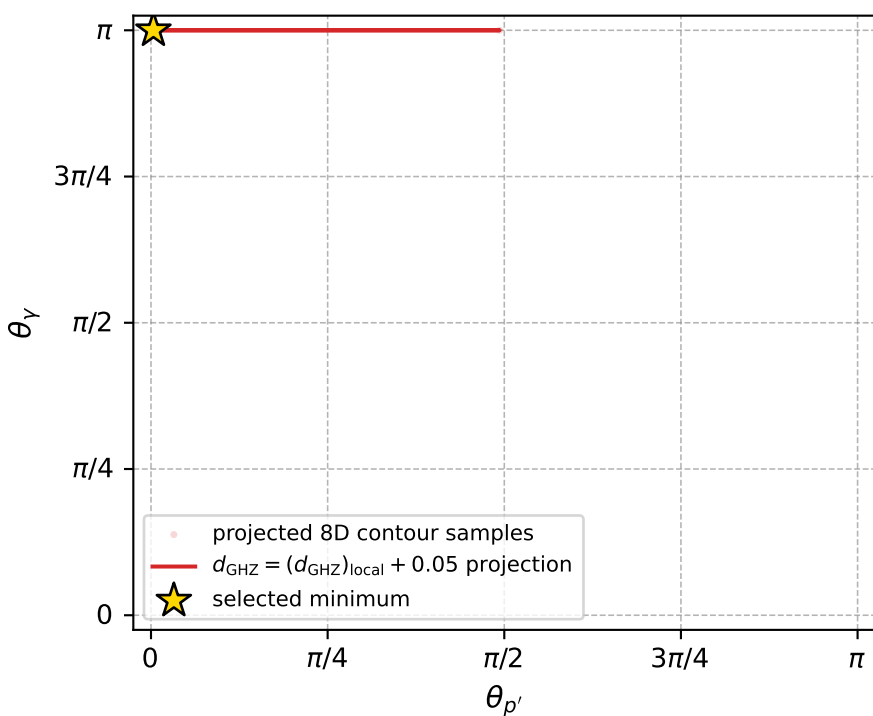
$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$



$A=-7.26\text{e+}02+1.63\text{e+}00i$

$A=-7.26\text{e+}02-1.63\text{e+}00i$

electron: polarization cluster P1, local minimum 28; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.60293\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $3.7458378\text{e-}09$
 8D contour samples = 71

selected phase-space point:

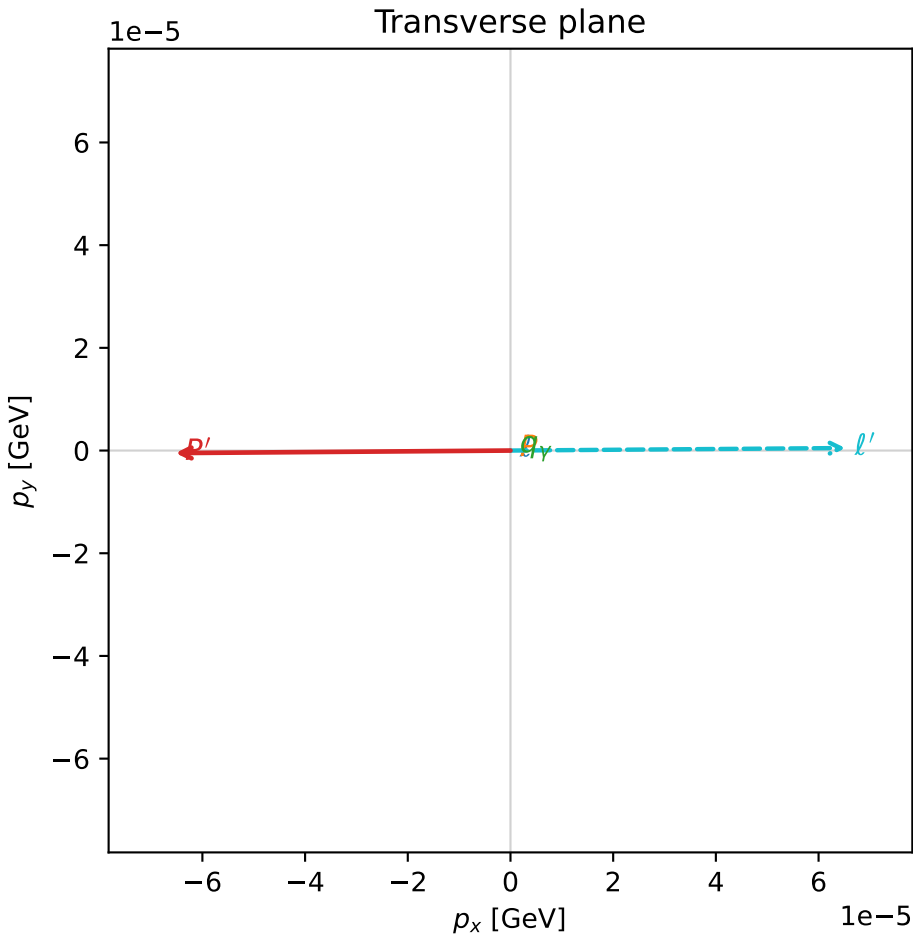
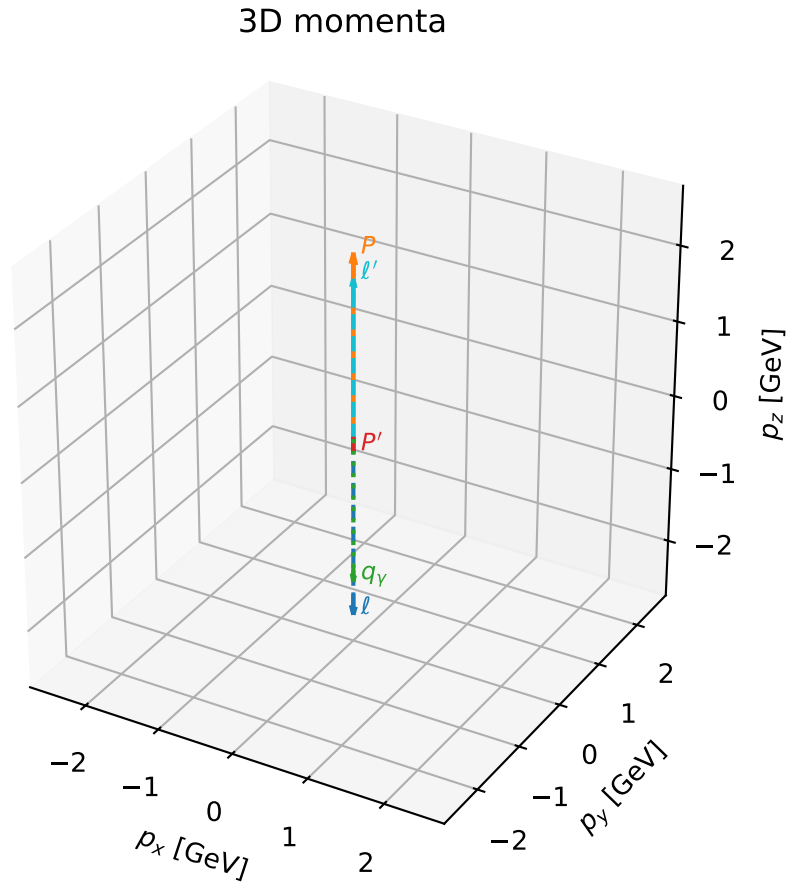
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.01080693$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.03107$
 $\text{phi_p_out} = 3.605486$
 $\text{phi_gamma_out} = 6.28094$
 $\alpha_e = 0.7050247$
 $\alpha_p = 0.7050158$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_32 $d_{\{\text{rm GHZ}\}}=2.62922\text{e-}08$
region: polarization_cluster_01_local_minimum_32
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.6915138$ rad
incoming lepton: $0.770282|+\rangle +0.637704|-\rangle$
proton mixing angle: $\alpha_p=0.69151265$ rad
incoming proton: $0.770282|+\rangle +0.637703|-\rangle$

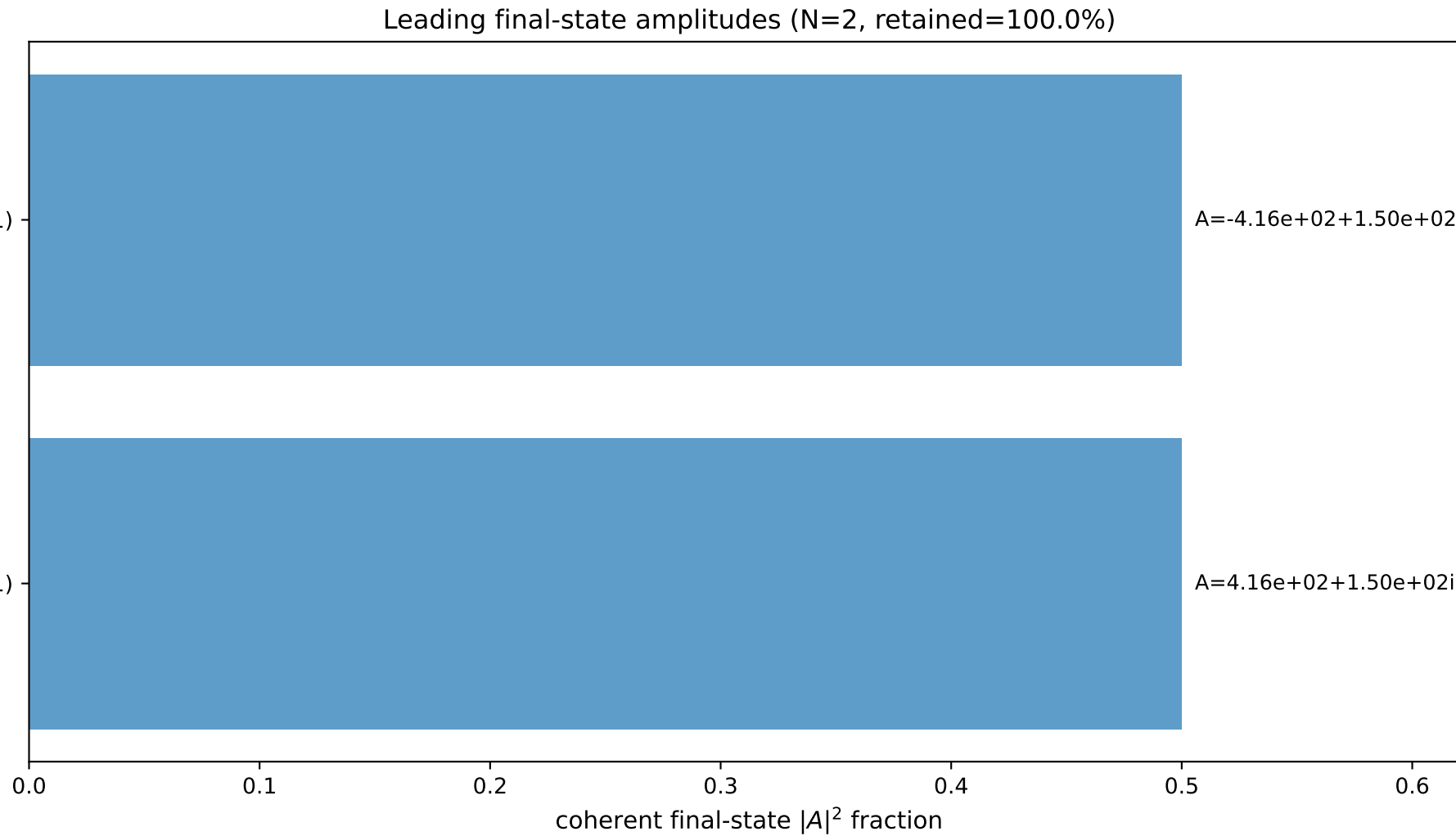
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.152285$
 $\theta_{p'}=3.14116$, $\theta_Y=3.14159$, $\phi_{p'}=3.14913$, $\phi_Y=2.80196$
 $E_Y=1.94872$, $\theta_{\ell'}=3.10422\text{e-}05$, $\phi_{\ell'}=0.00753968$
 $Q^2=20.2706$, $x_B=0.866979$, $t=-3.89357$, $W^2=3.98998$, $y=0.969344$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1010$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.1010$
 P' $E= 0.9503$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.1523$
 q_Y $E= 1.9487$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9487$

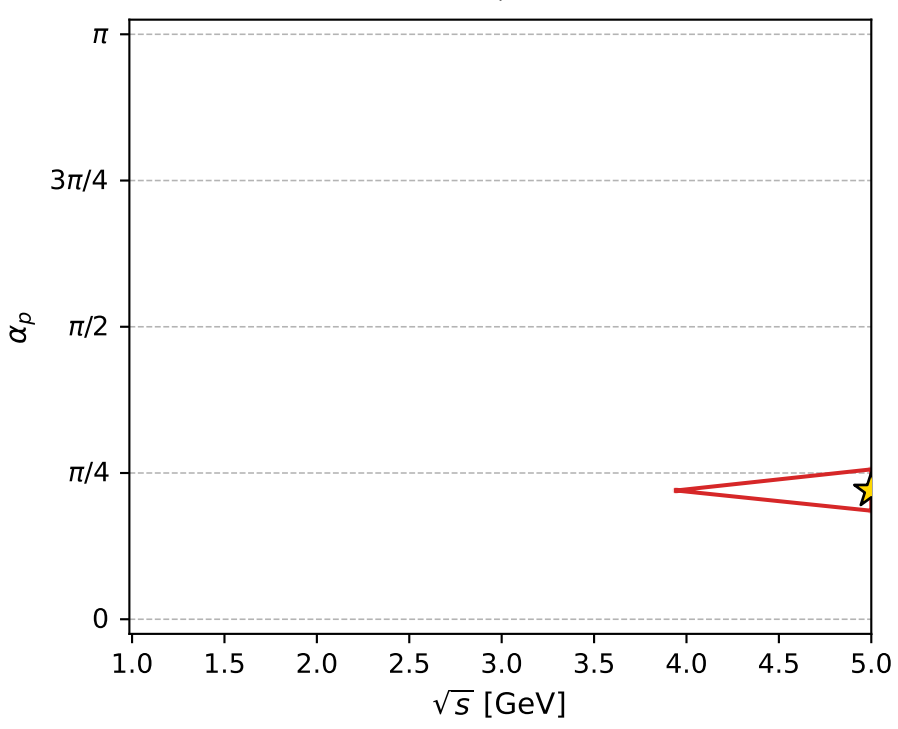
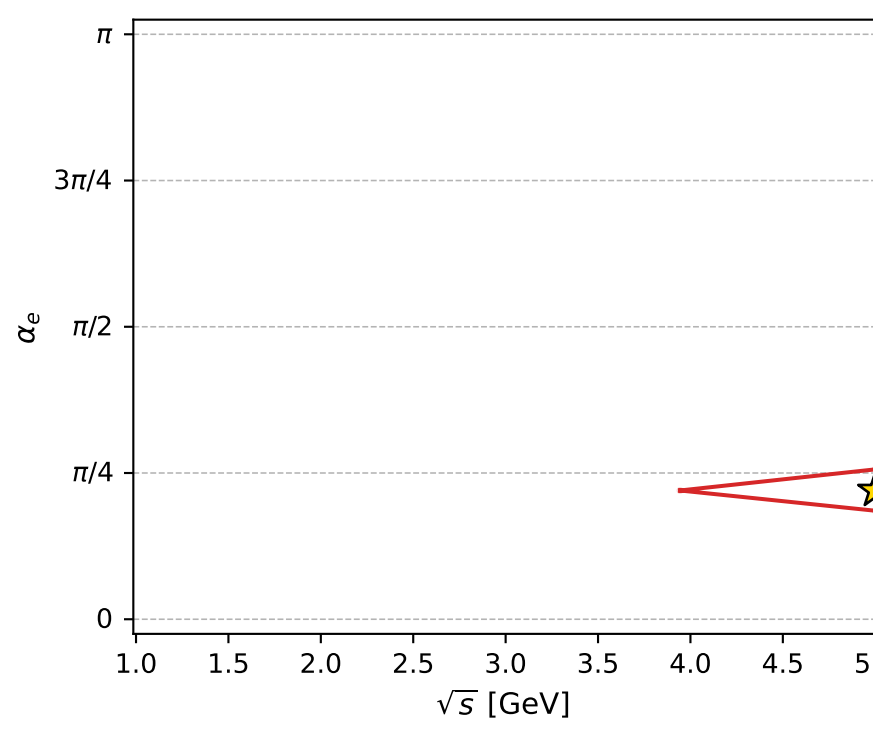
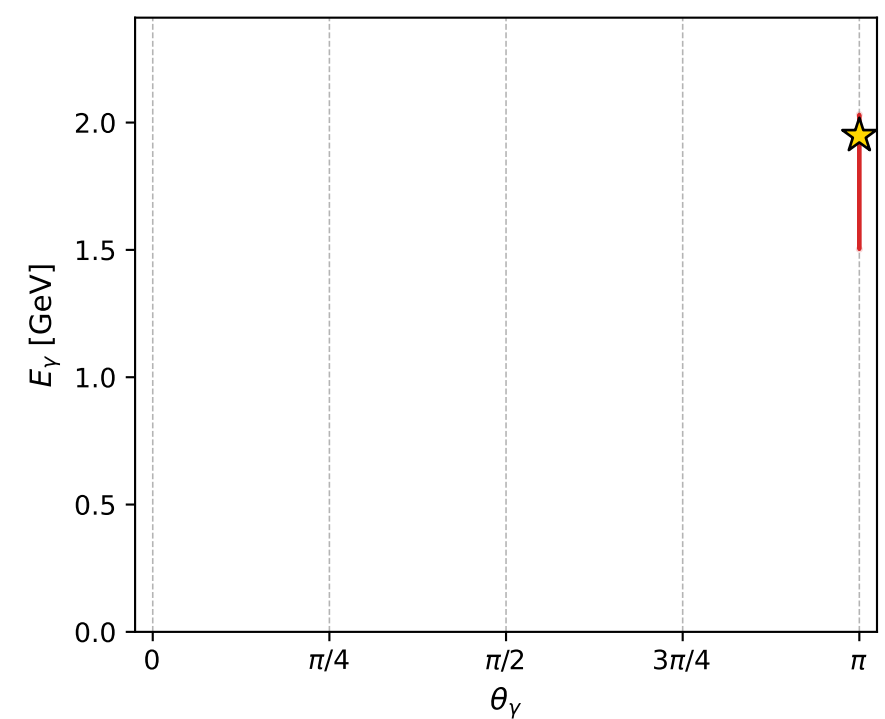
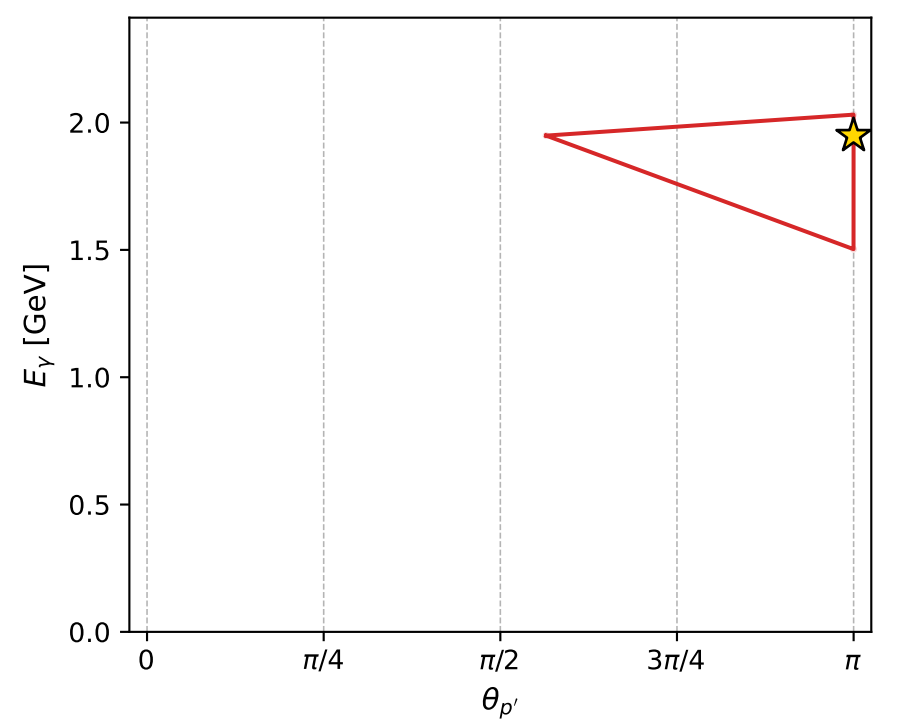
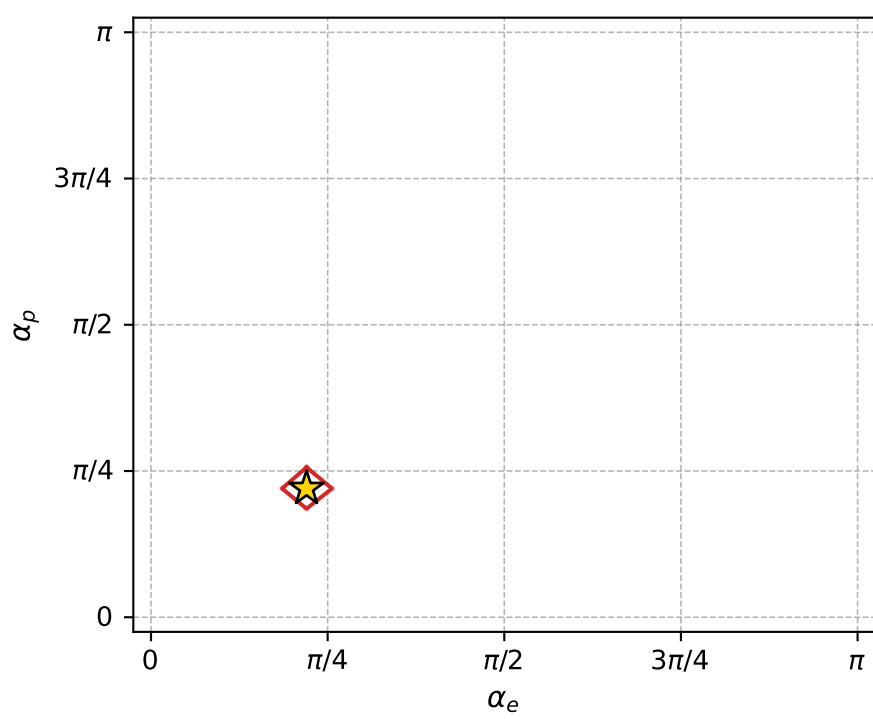
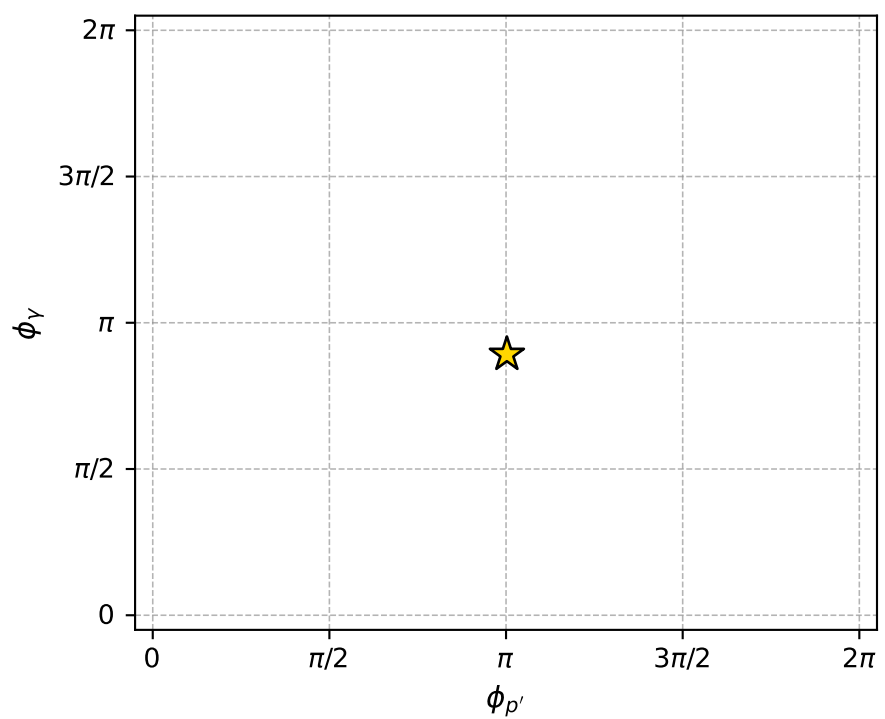
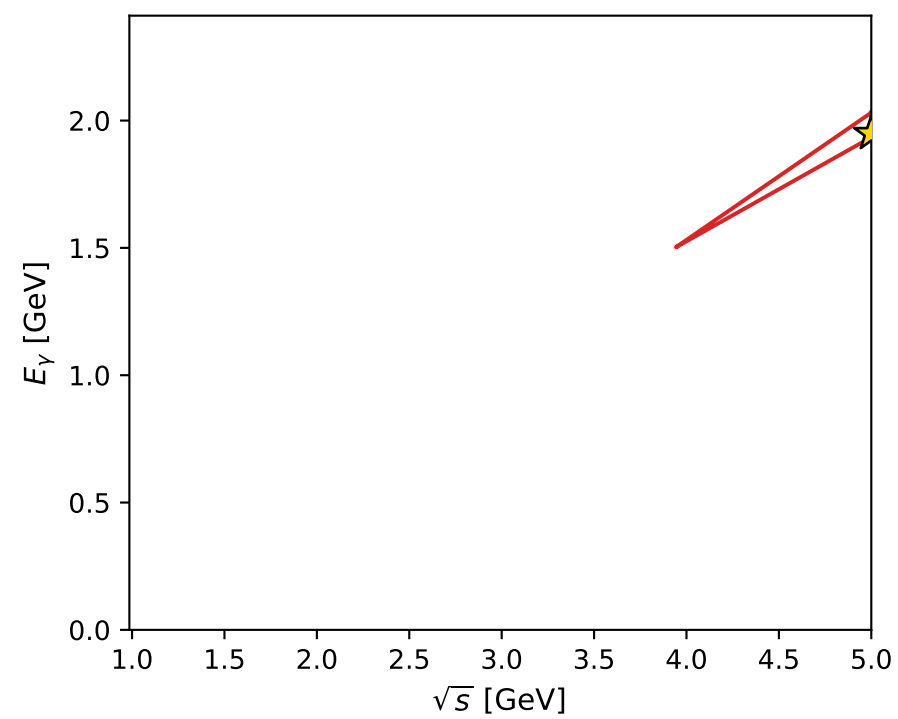
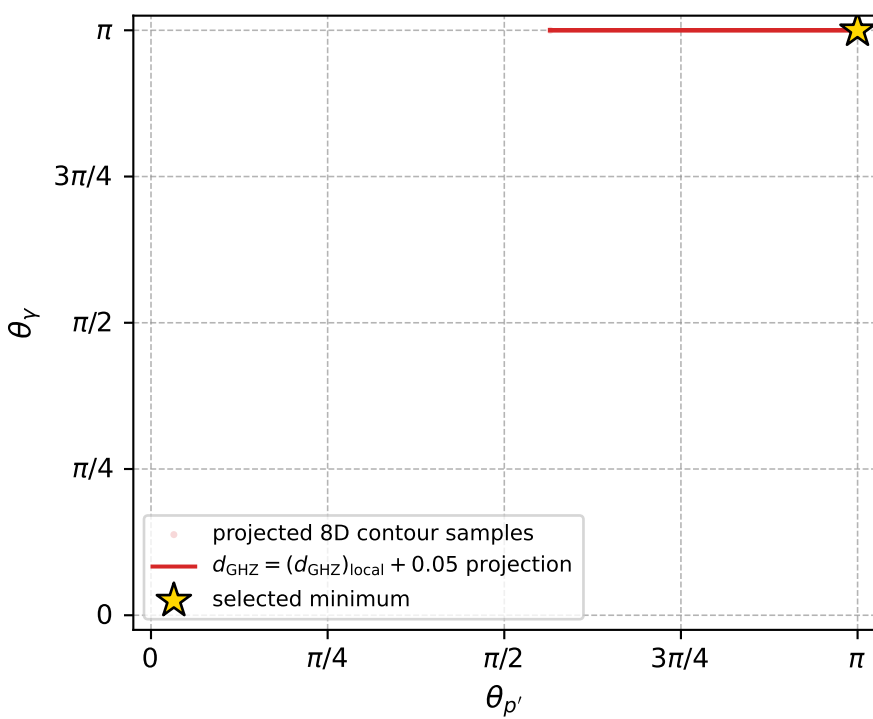


$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$



electron: polarization cluster P1, local minimum 32; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6292177\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $4.0087148\text{e-}09$
 8D contour samples = 67

selected phase-space point:

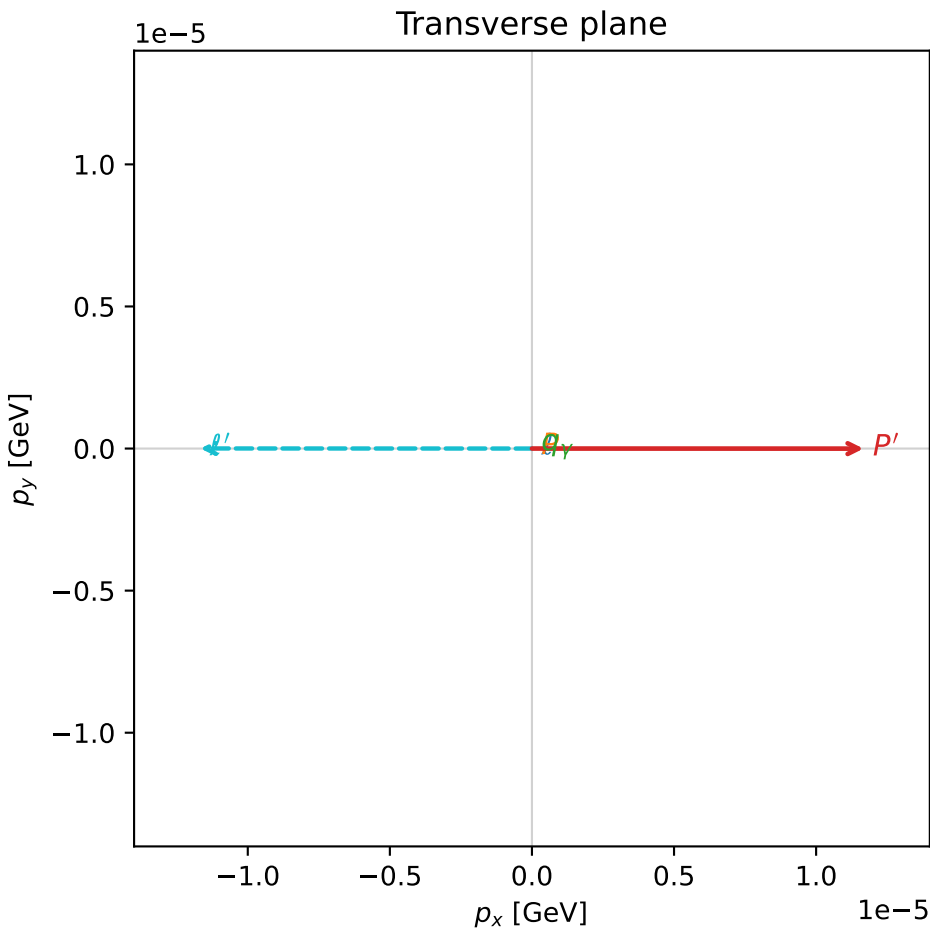
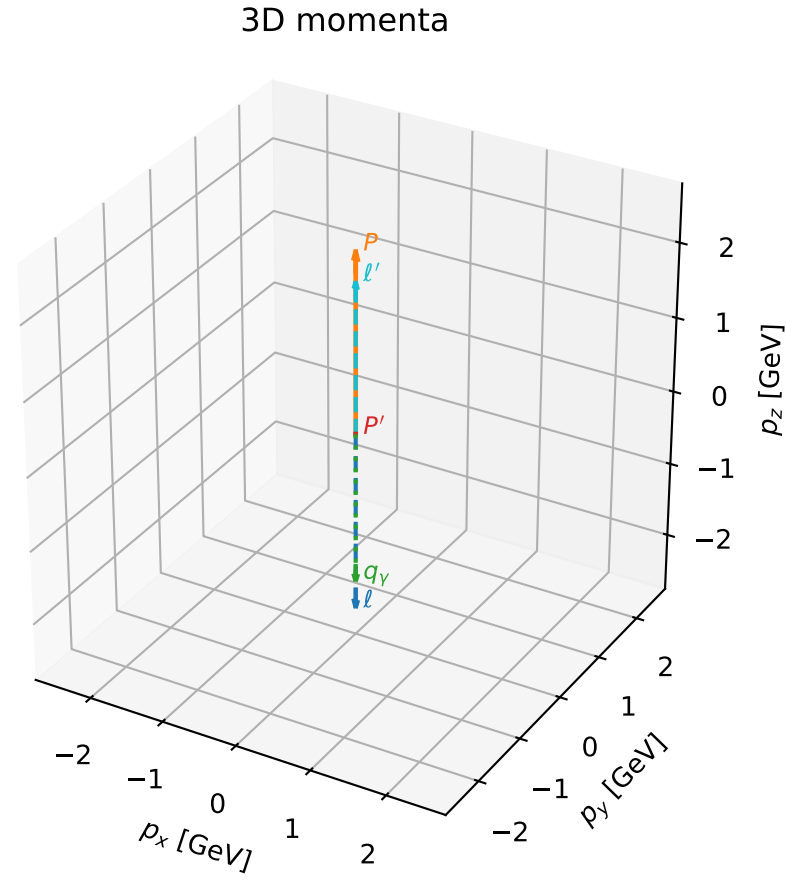
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141164$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.948717$
 $\text{phi_p_out} = 3.149132$
 $\text{phi_gamma_out} = 2.801962$
 $\alpha_e = 0.6915138$
 $\alpha_p = 0.6915126$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_35 $d_{\{\text{rm GHZ}\}}=2.63636\text{e-}08$
 region: polarization_cluster_01_local_minimum_35
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.86247052$ rad
 incoming lepton: $0.650563|+\rangle +0.759452|-\rangle$
 proton mixing angle: $\alpha_p=0.86246408$ rad
 incoming proton: $0.650568|+\rangle +0.759448|-\rangle$

$s=24.999$, $\sqrt{s}=4.9999$, $|\vec{P}|=2.41196$, $|\vec{P}'|=0.00239249$
 $\theta_{P'}=0.00487673$, $\theta_Y=3.14159$, $\phi_{P'}=6.28293$, $\phi_Y=3.50677$
 $E_Y=2.03215$, $\theta_{\ell'}=5.74822\text{e-}06$, $\phi_{\ell'}=3.14134$
 $Q^2=19.5828$, $x_B=0.836699$, $t=-3.08376$, $W^2=4.70189$, $y=0.970382$

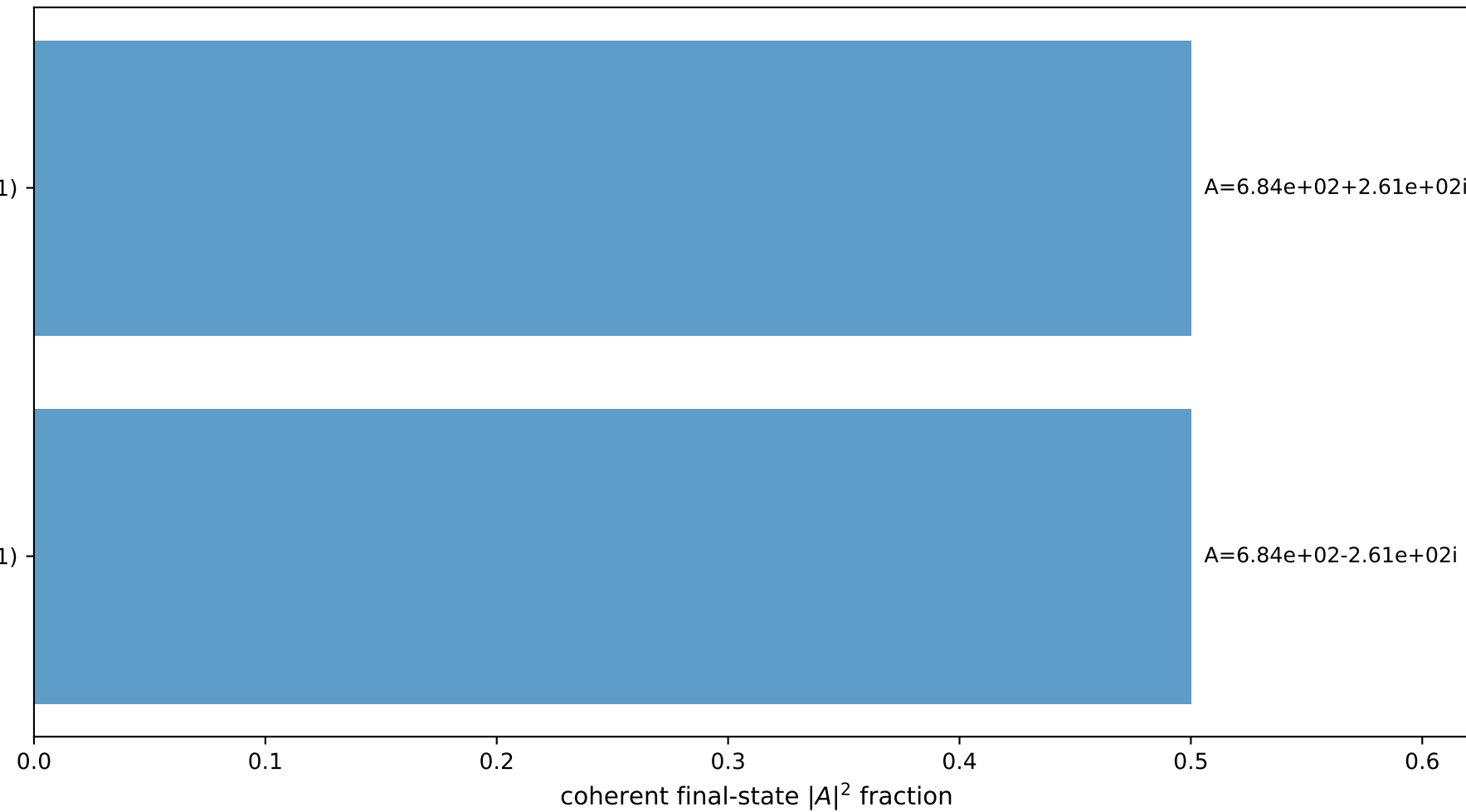
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0298$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0298$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0024$
 q_Y $E= 2.0321$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0321$



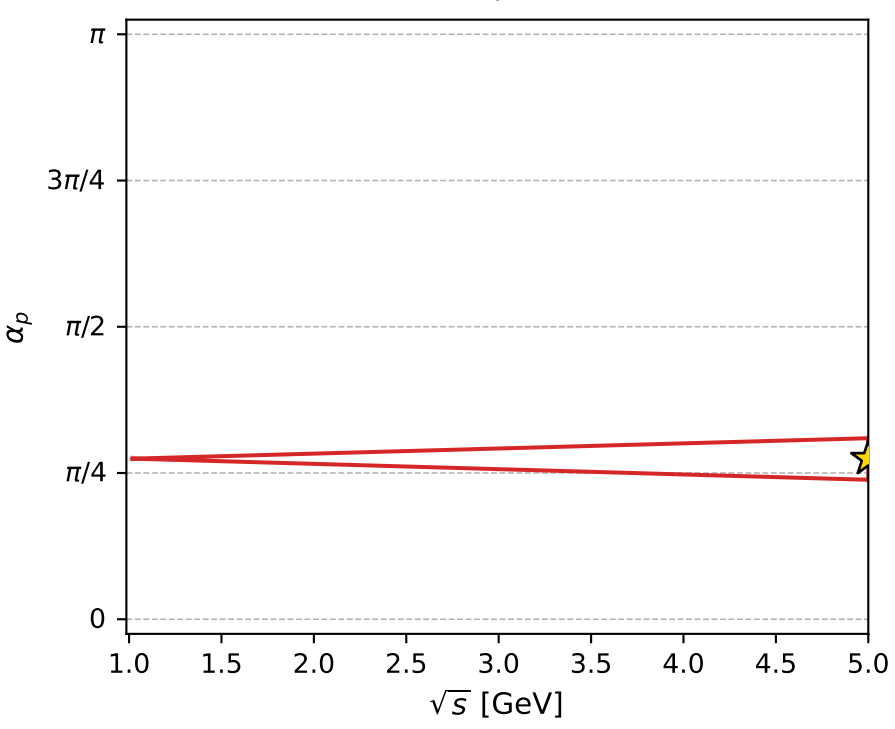
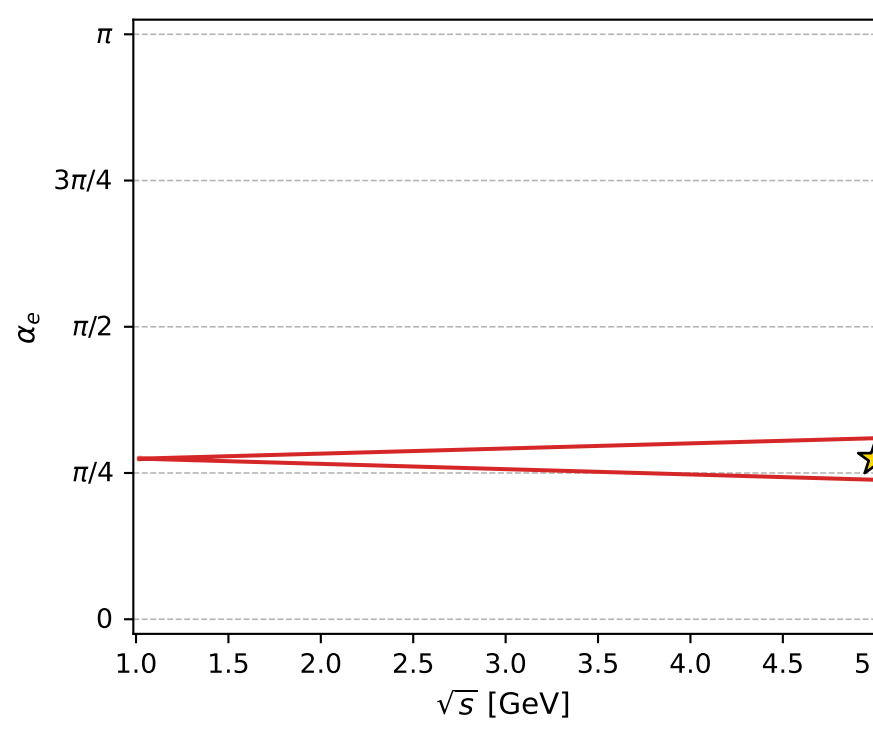
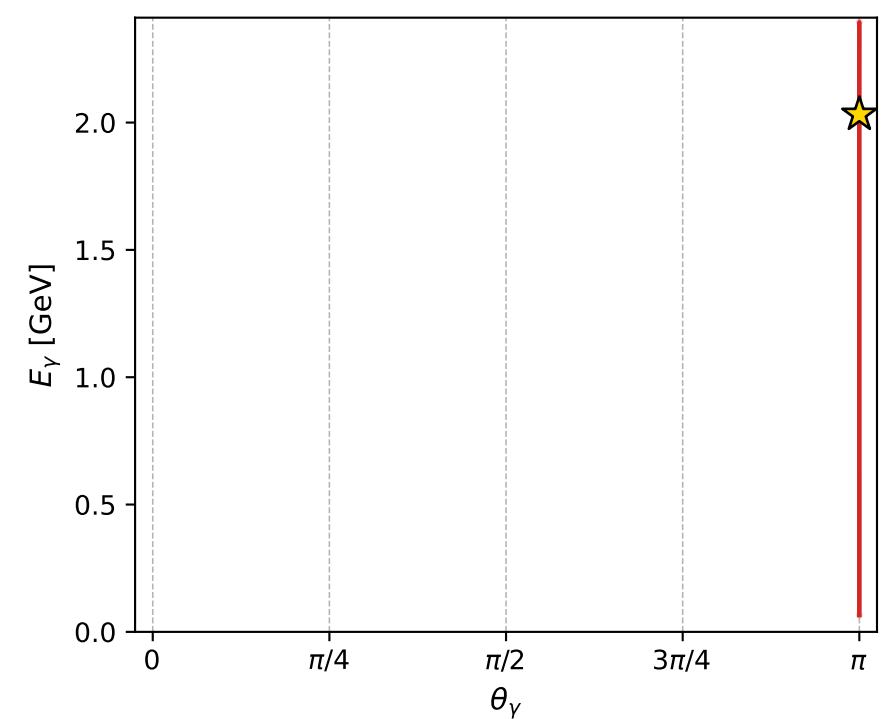
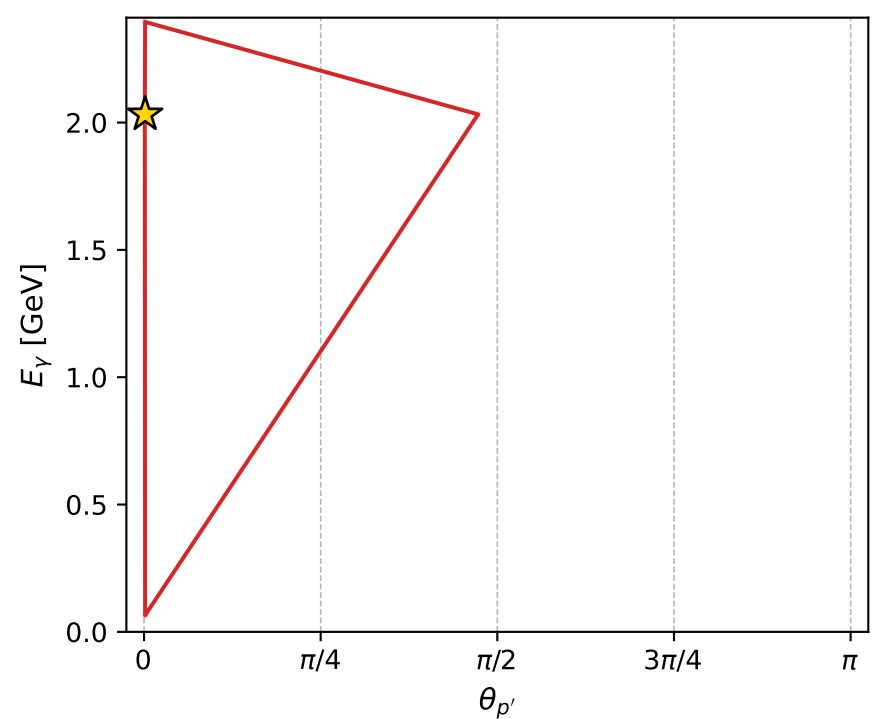
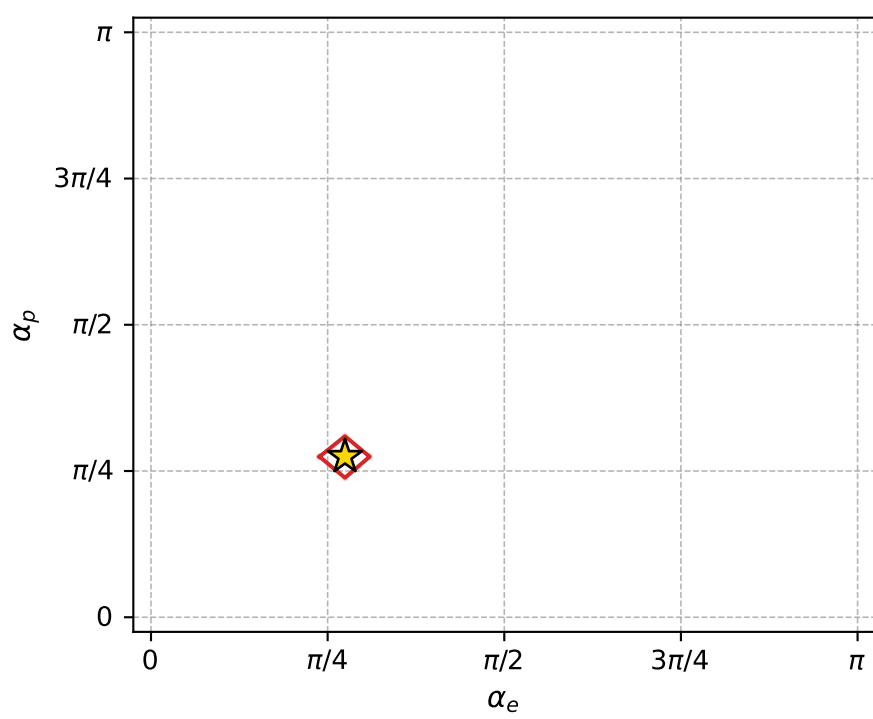
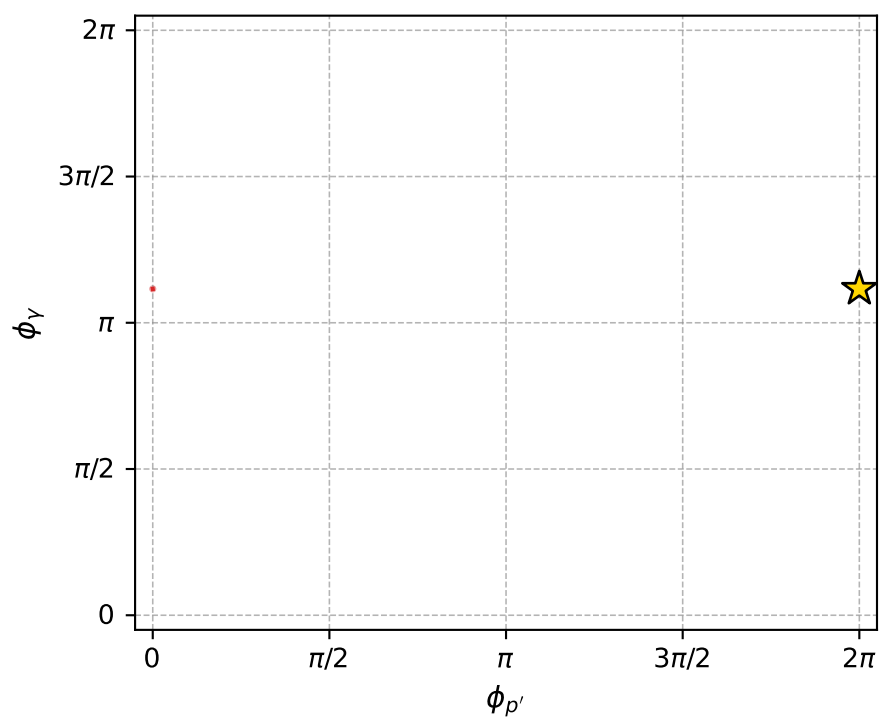
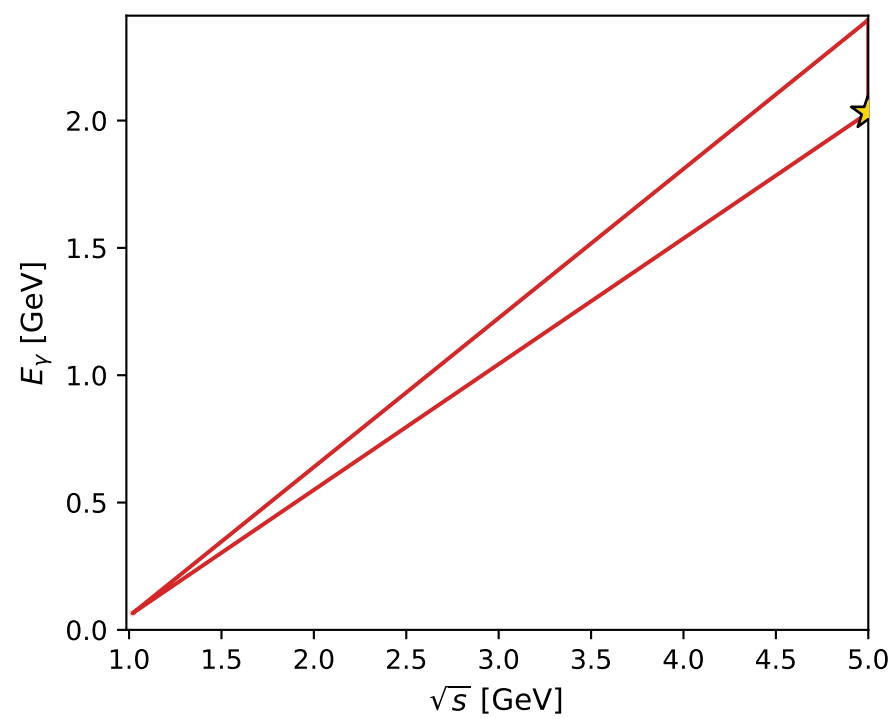
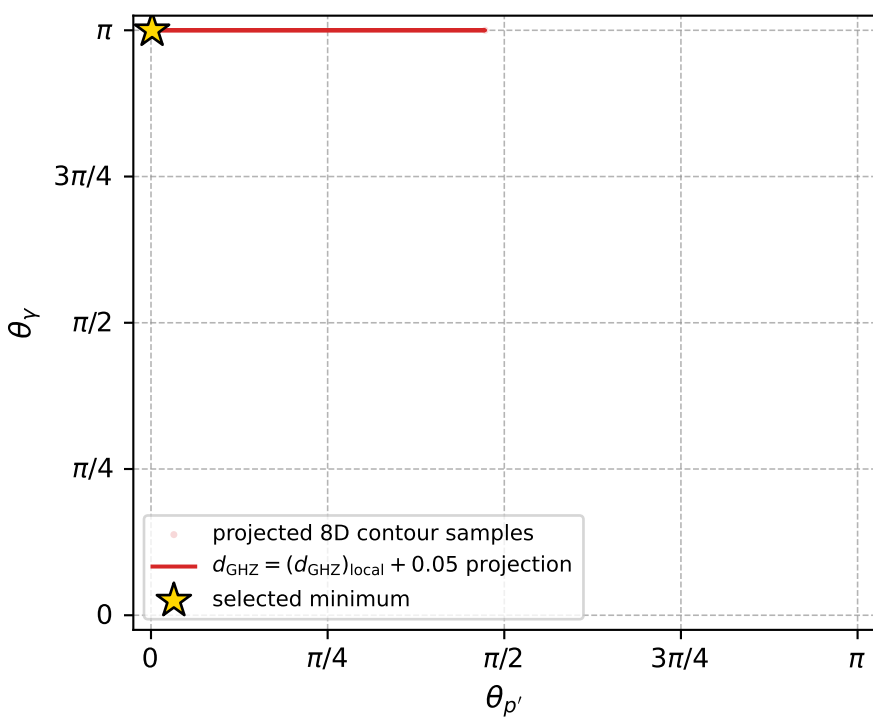
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 35; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.636361\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000026$
 above global minimum = $4.0801477\text{e-}09$
 8D contour samples = 123

selected phase-space point:

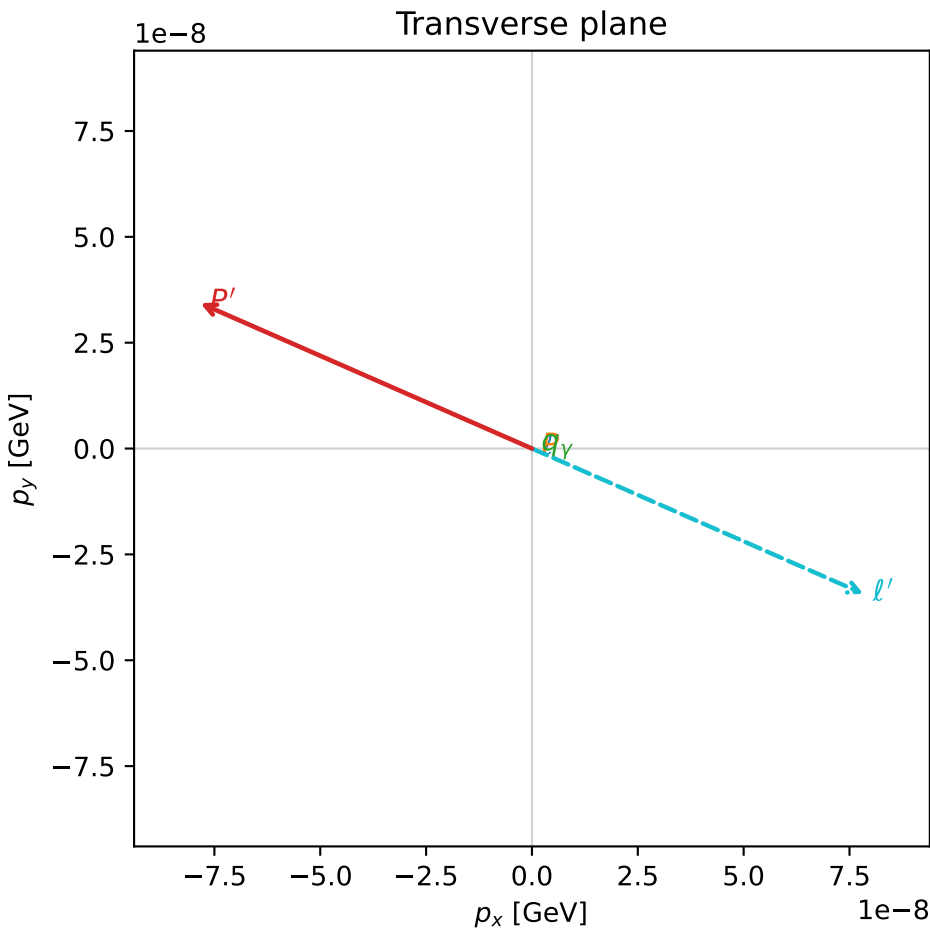
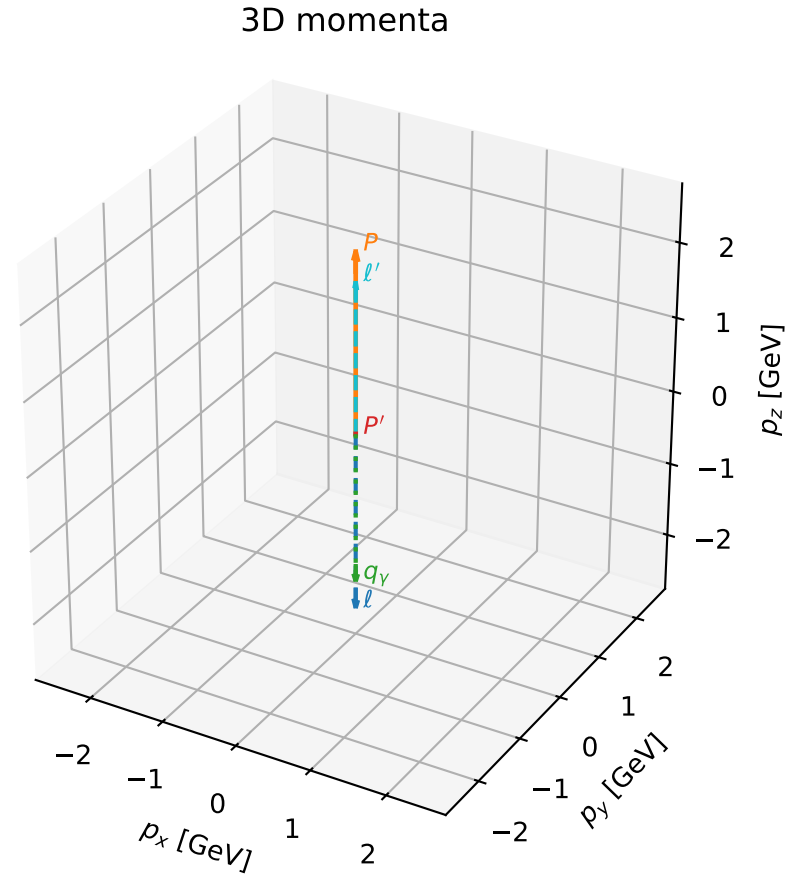
$\text{sqrt_s} = 4.999902$
 $\text{theta_p_out} = 0.004876729$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.032146$
 $\text{phi_p_out} = 6.282932$
 $\text{phi_gamma_out} = 3.506769$
 $\alpha_e = 0.8624705$
 $\alpha_p = 0.8624641$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_44 d_{\rm GHZ}=2.65428e-08
region: polarization_cluster_01_local_minimum_44
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.70477583$ rad
incoming lepton: $0.761757|+\rangle +0.647863|-\rangle$
proton mixing angle: $\alpha_p=0.70475776$ rad
incoming proton: $0.761769|+\rangle +0.647849|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00358496$
 $\theta_{P'}=2.38484\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=2.72855$, $\phi_Y=3.48858$
 $E_Y=2.03279$, $\theta_{\ell'}=4.21468\text{e-}08$, $\phi_{\ell'}=5.87014$
 $Q^2=19.5779$, $x_B=0.836447$, $t=-3.07811$, $W^2=4.70796$, $y=0.970392$

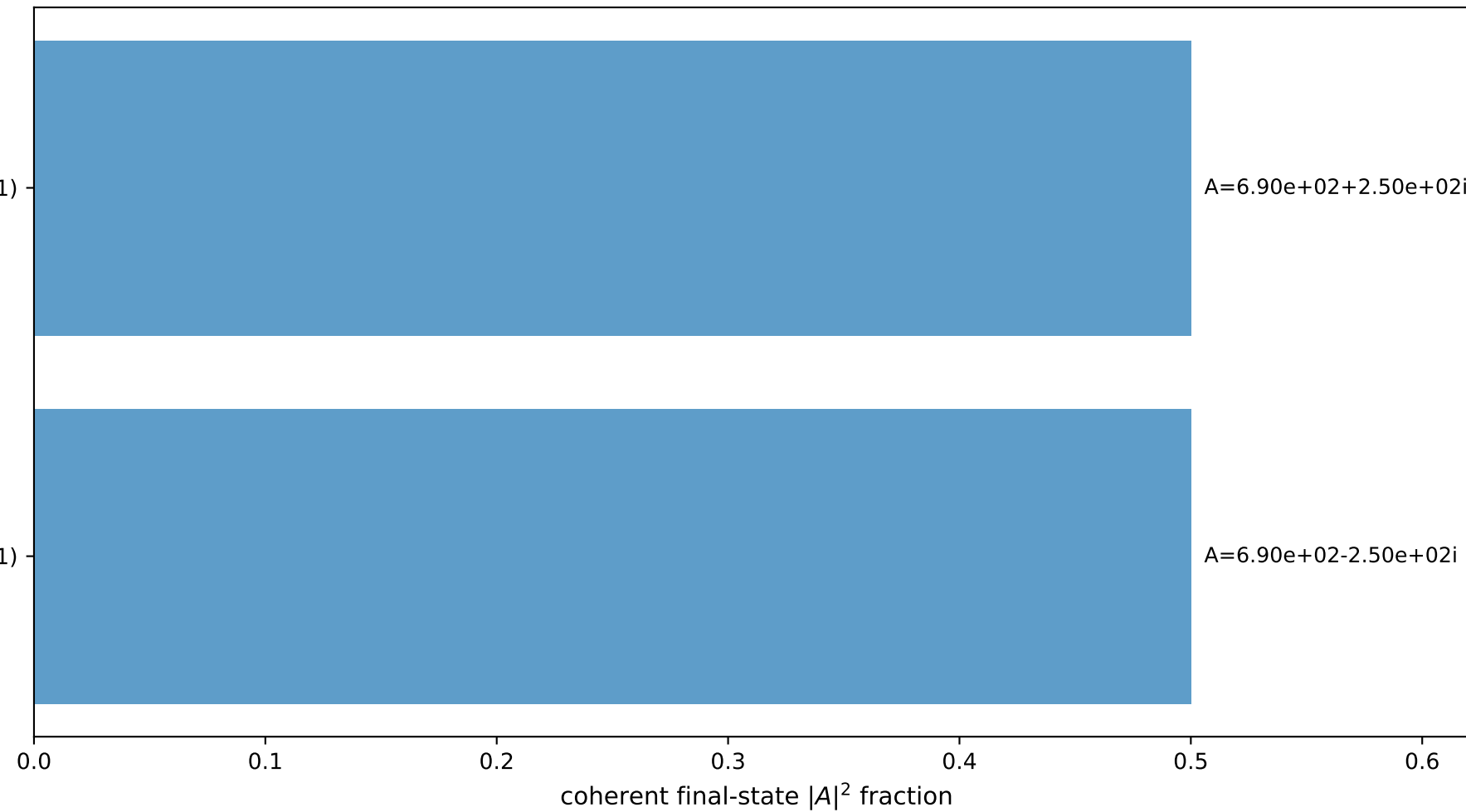
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0292$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0292$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0036$
 q_Y $E= 2.0328$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0328$



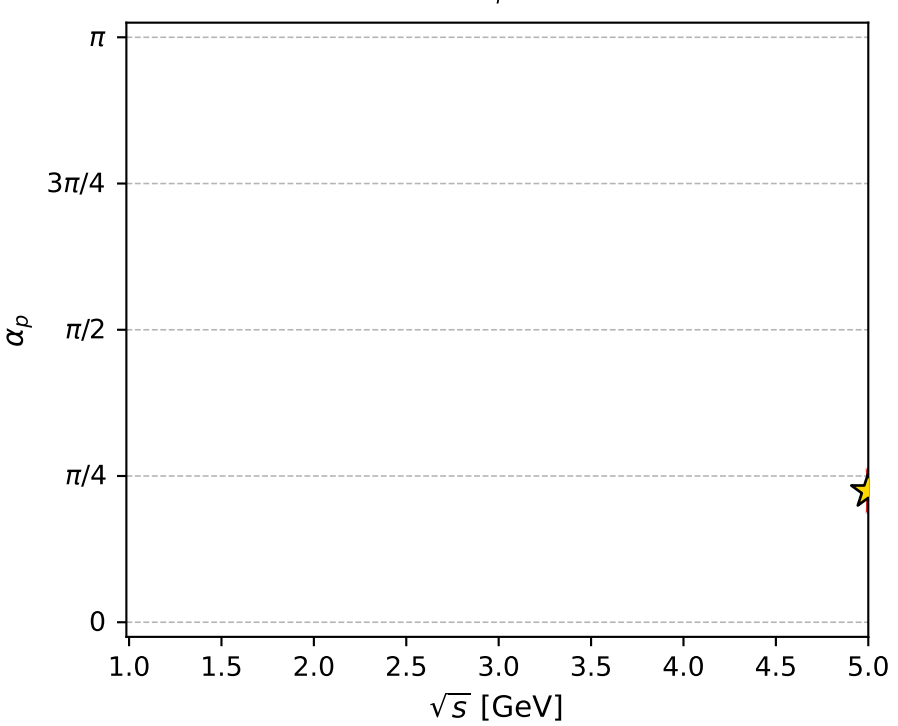
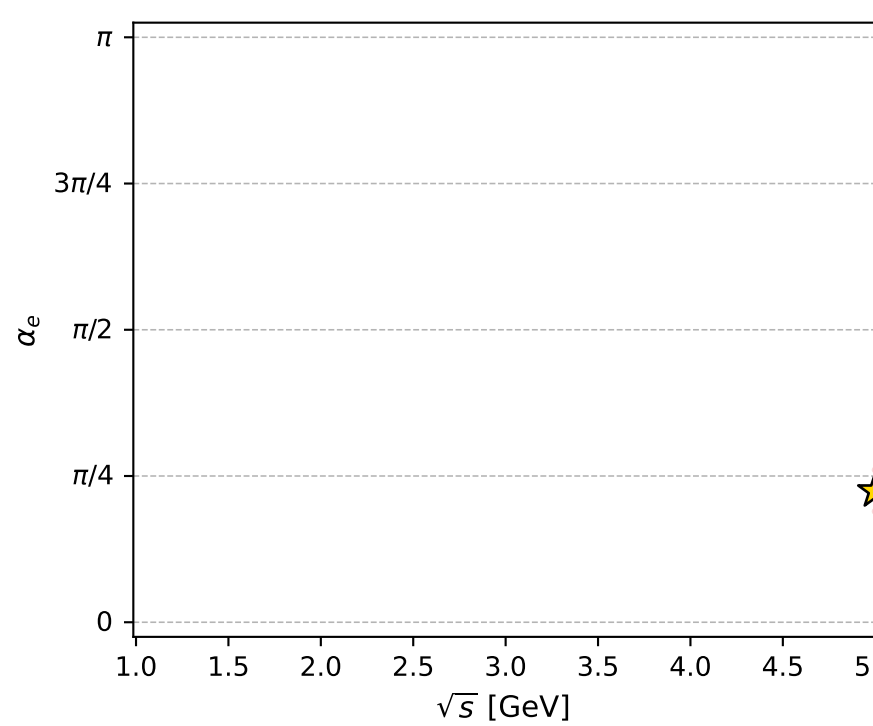
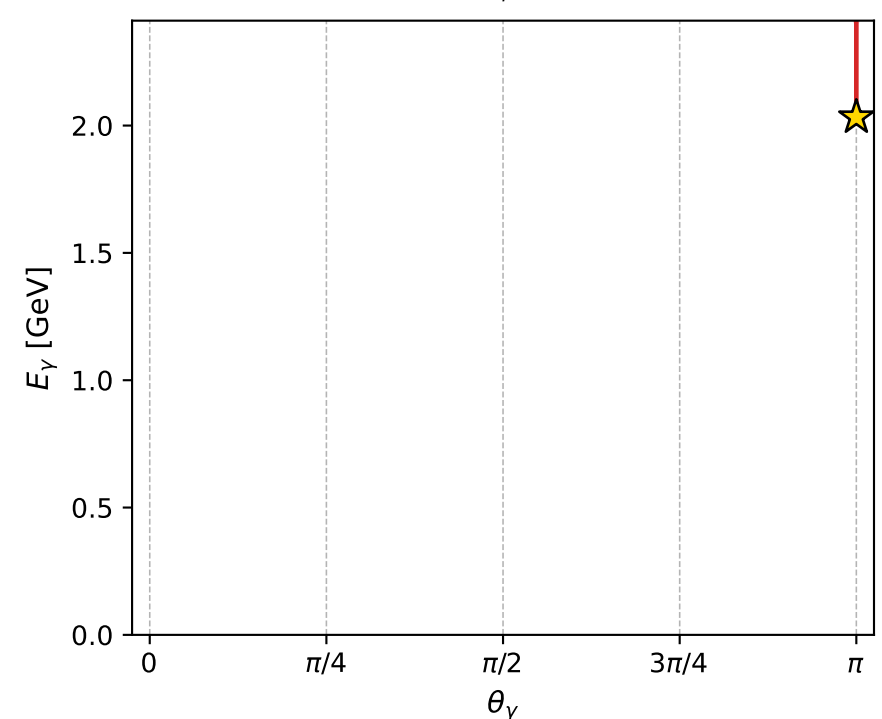
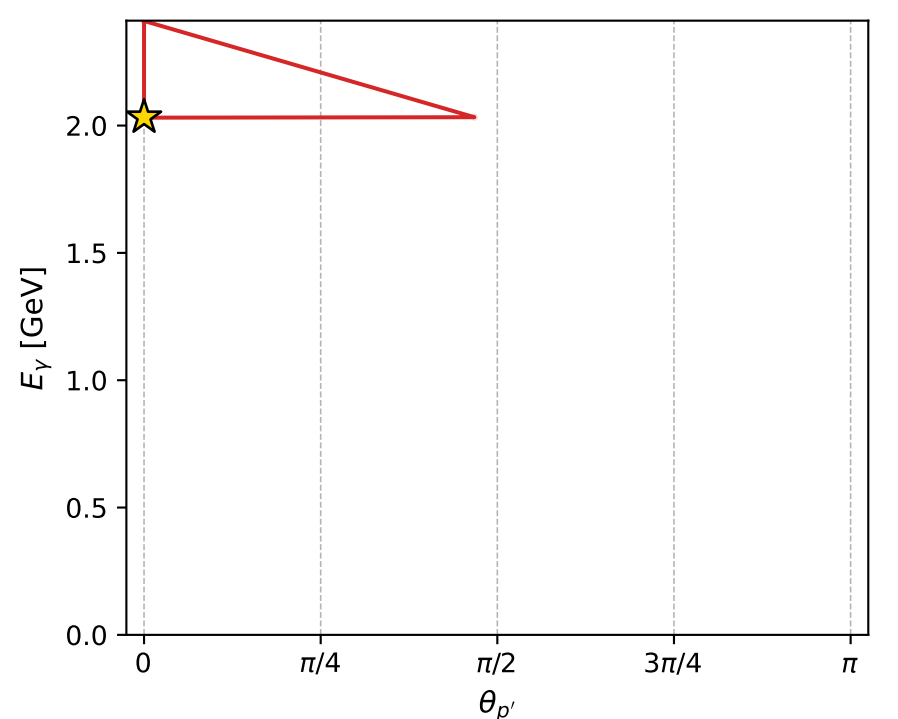
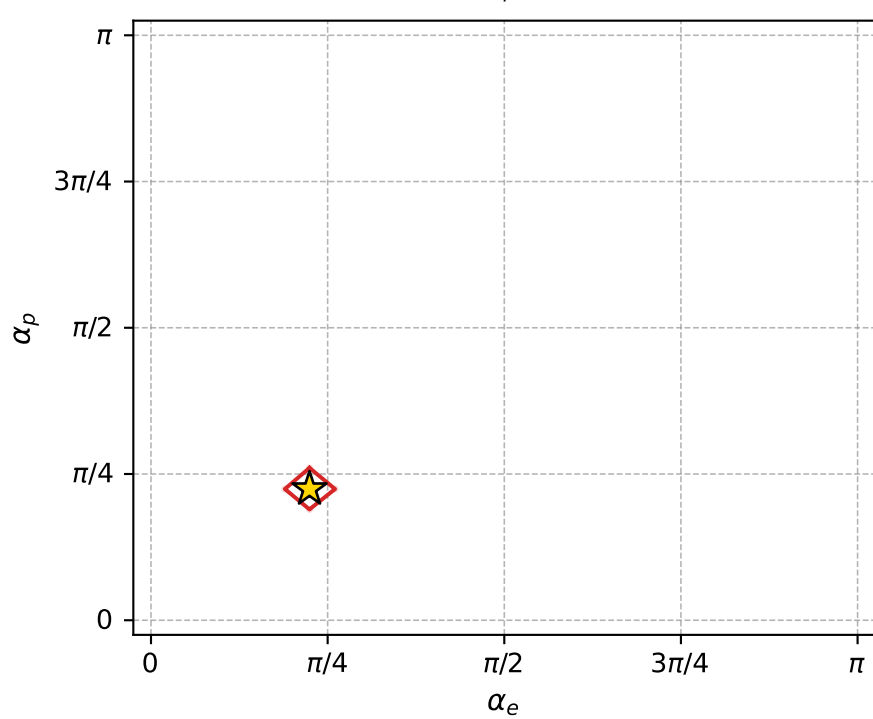
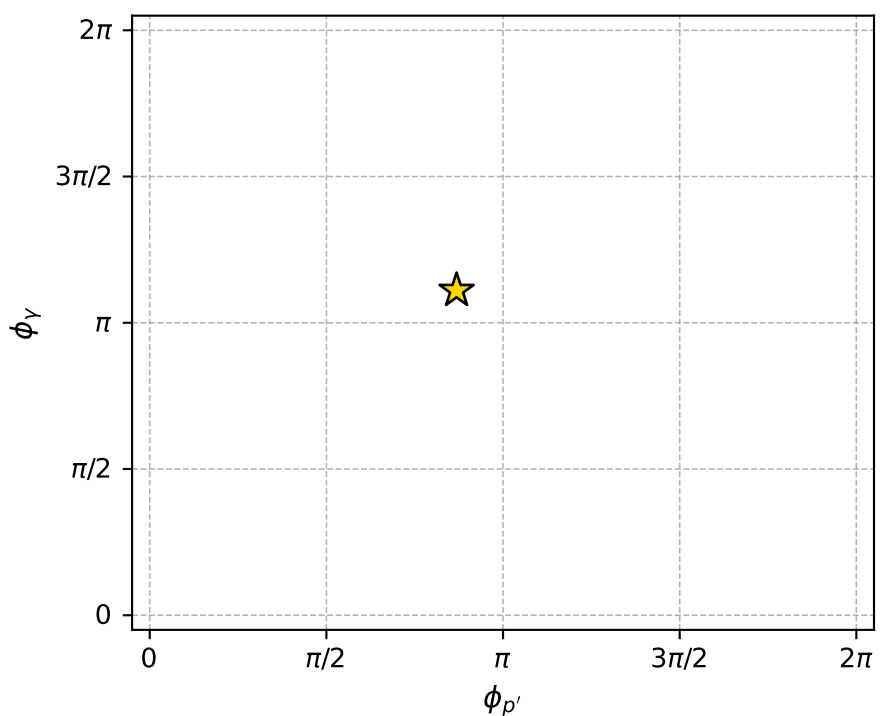
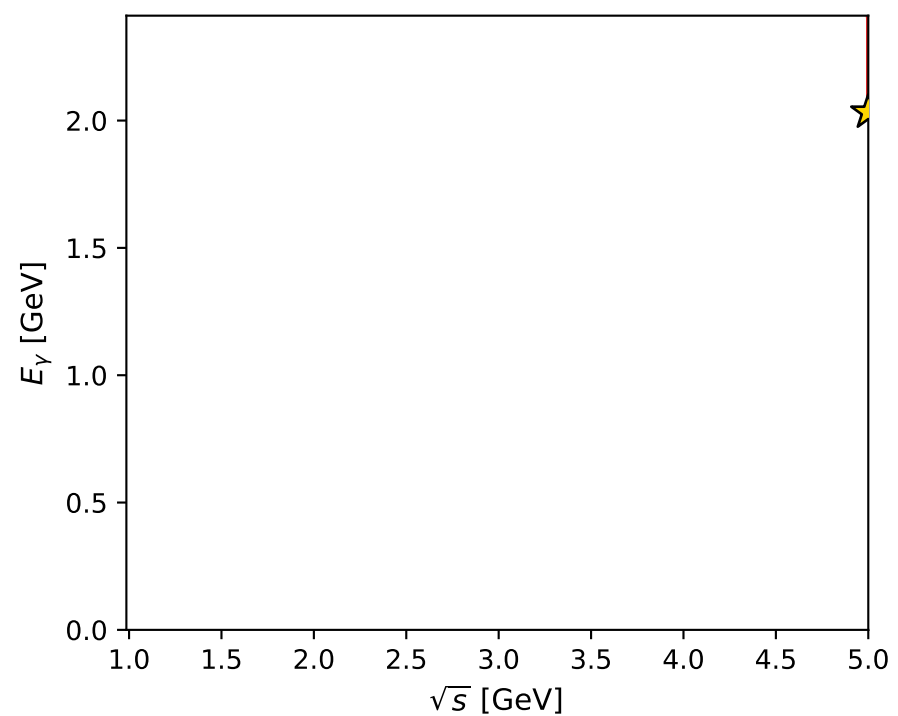
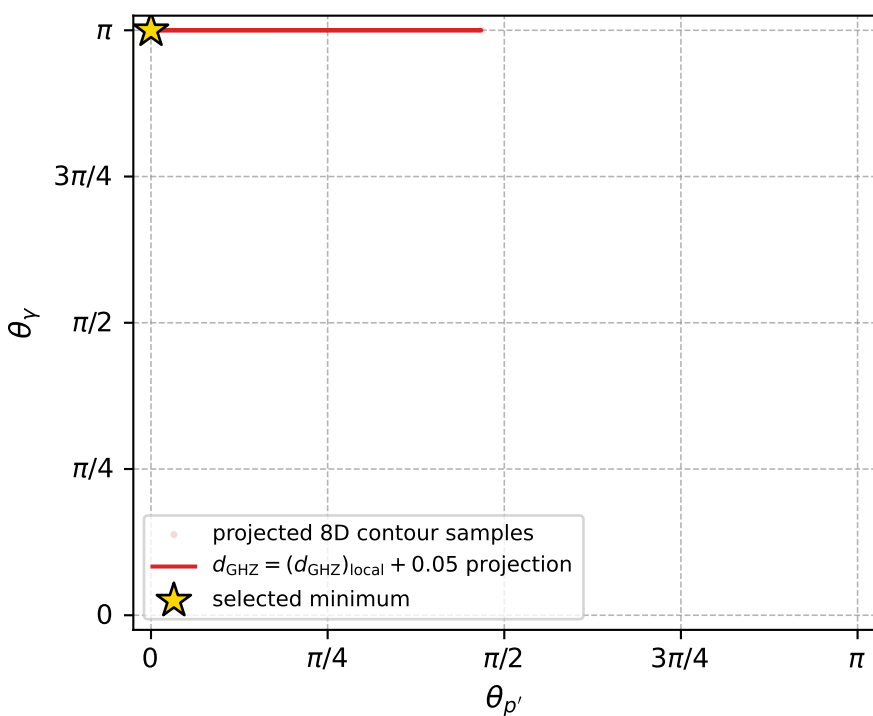
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 44; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6542832\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.2593703\text{e-}09$
 8D contour samples = 43

selected phase-space point:

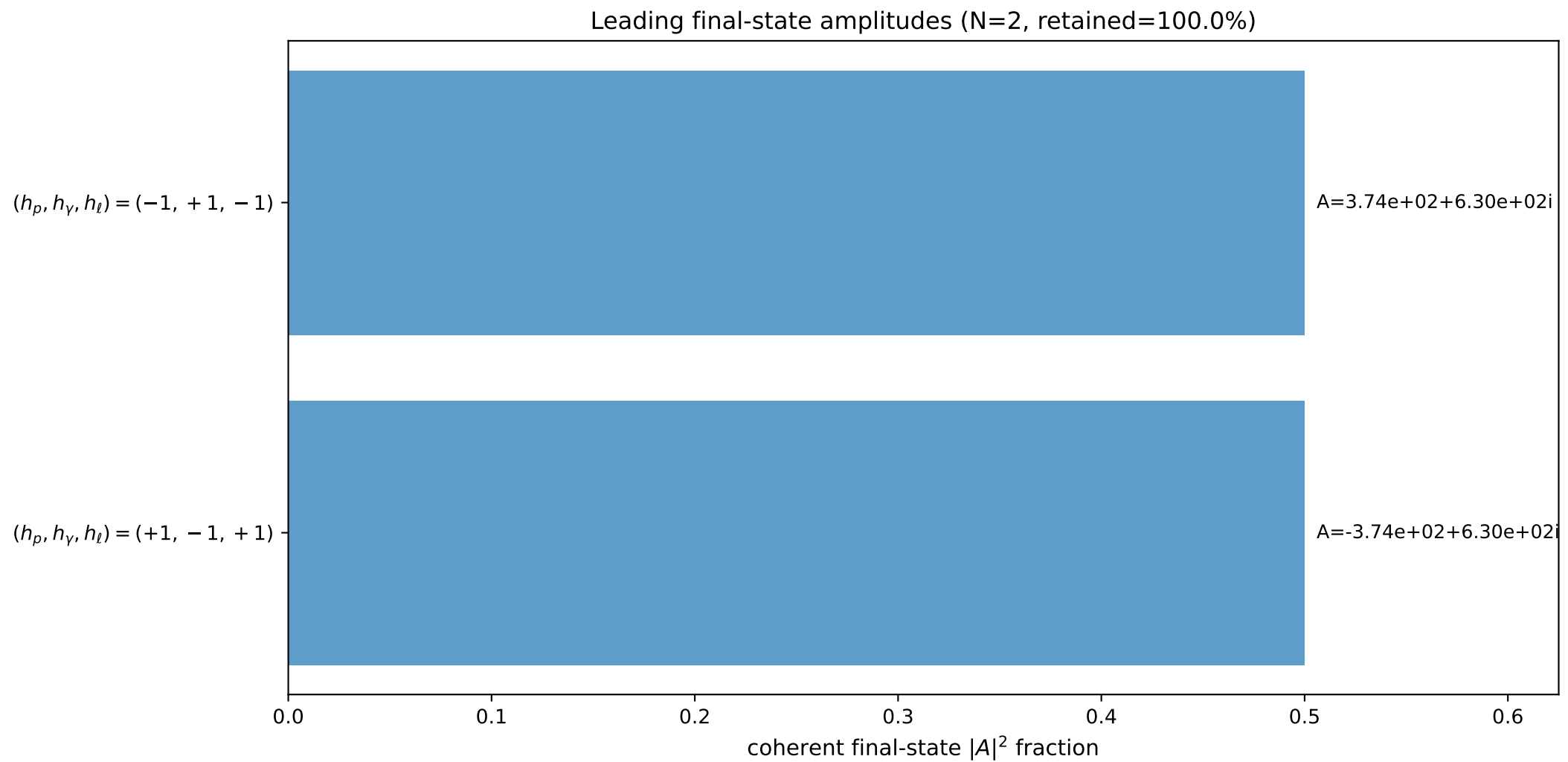
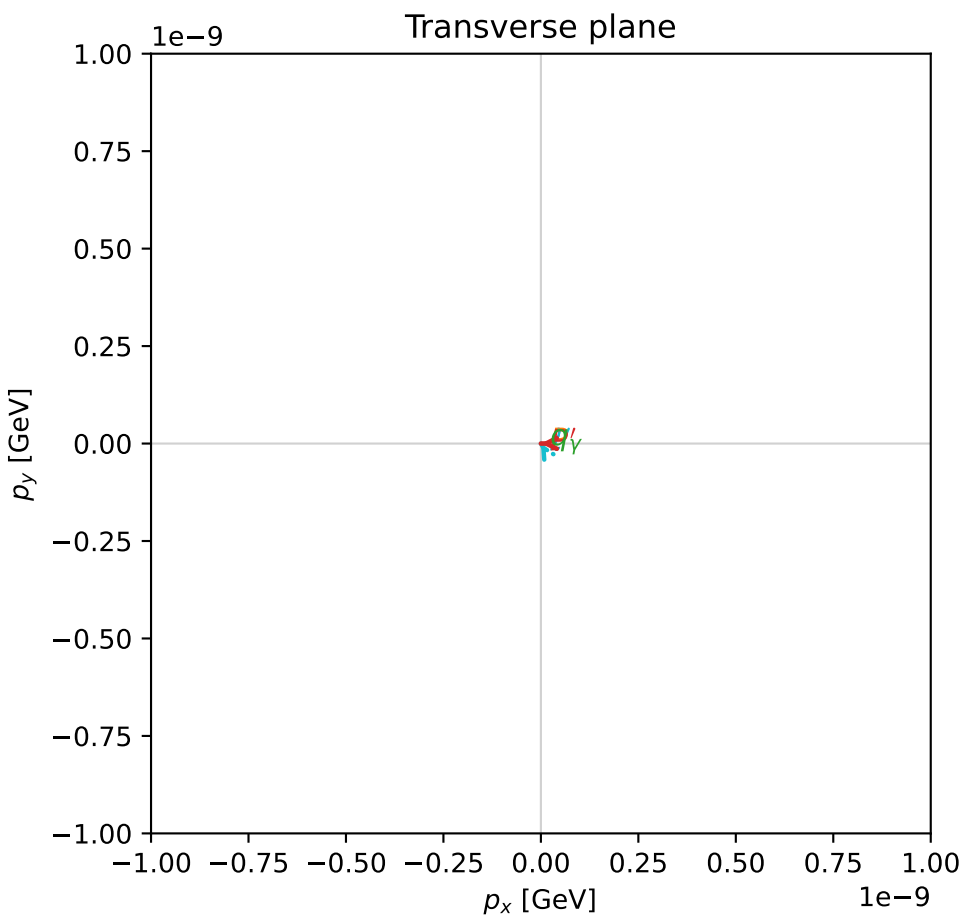
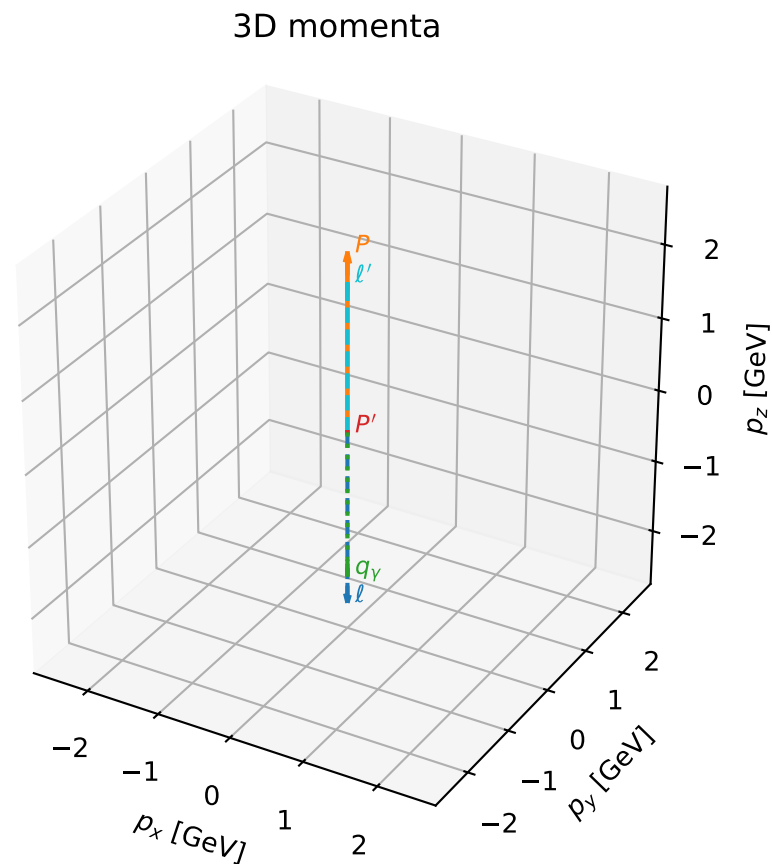
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 2.384839\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.032789$
 $\text{phi_p_out} = 2.728549$
 $\text{phi_gamma_out} = 3.488583$
 $\text{alpha_e} = 0.7047758$
 $\text{alpha_p} = 0.7047578$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

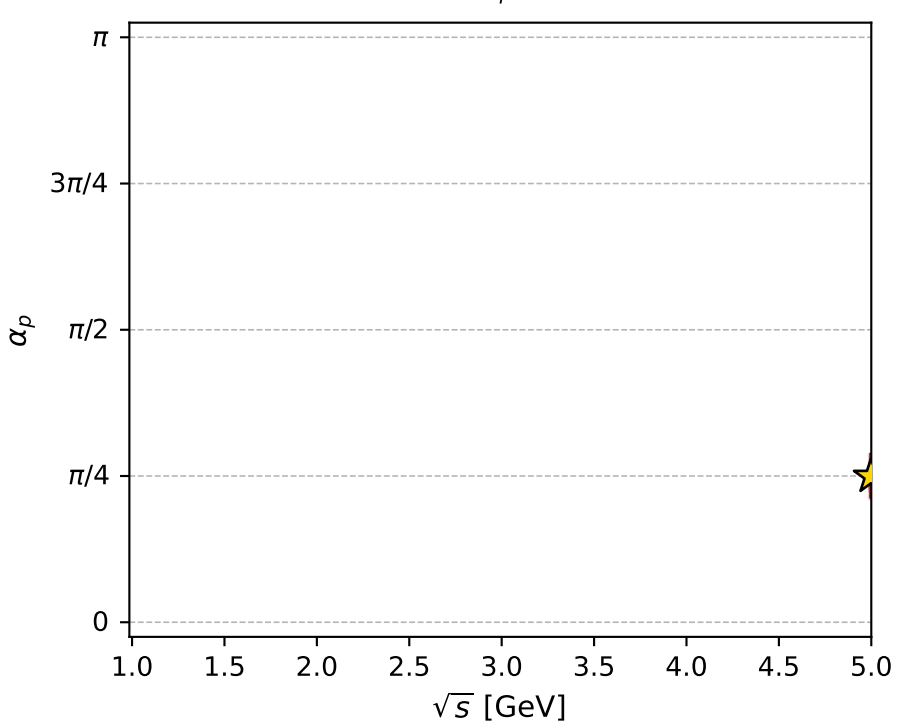
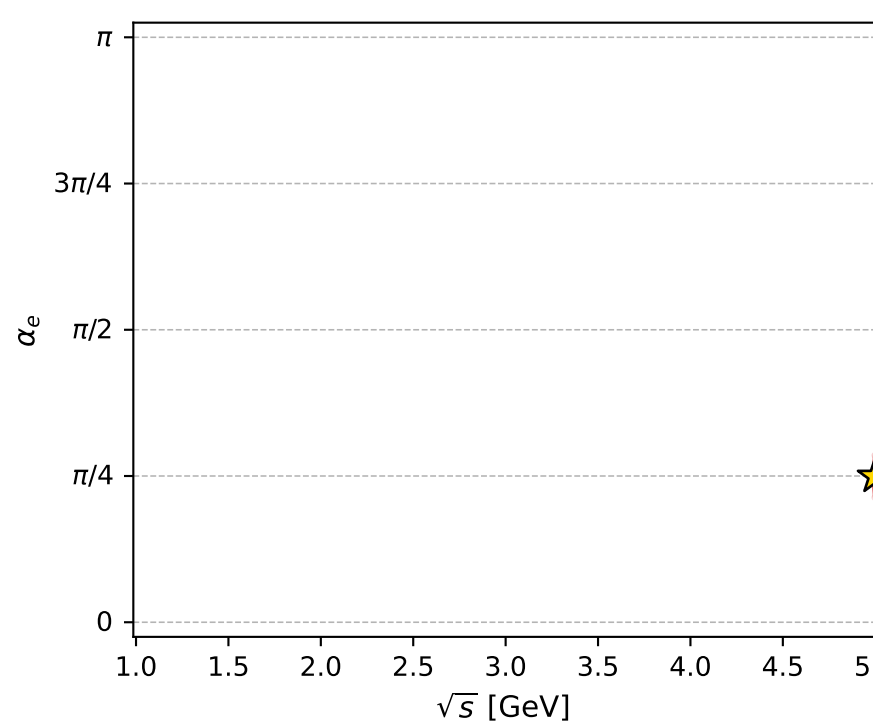
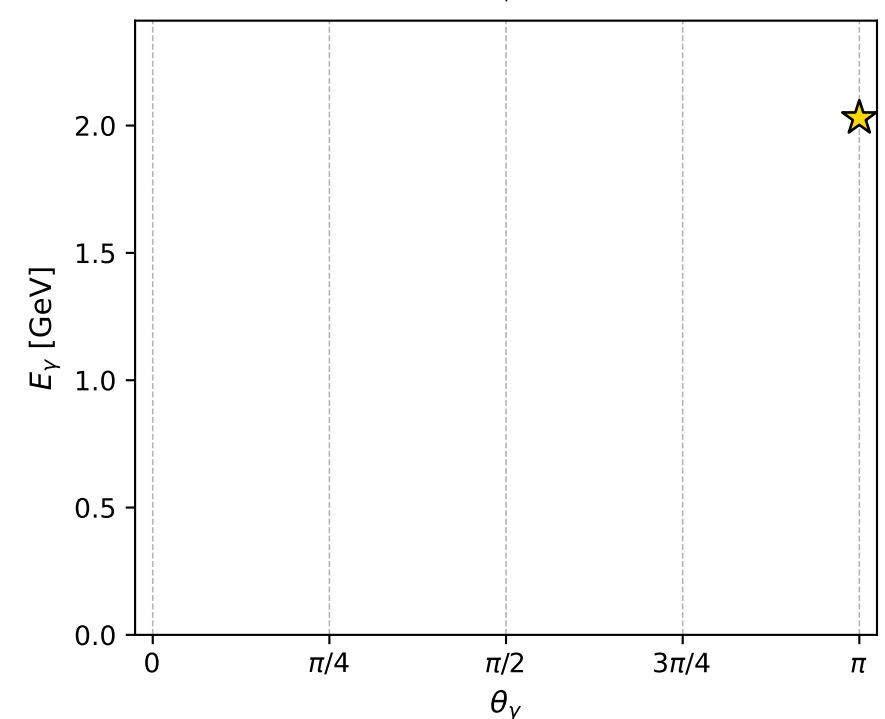
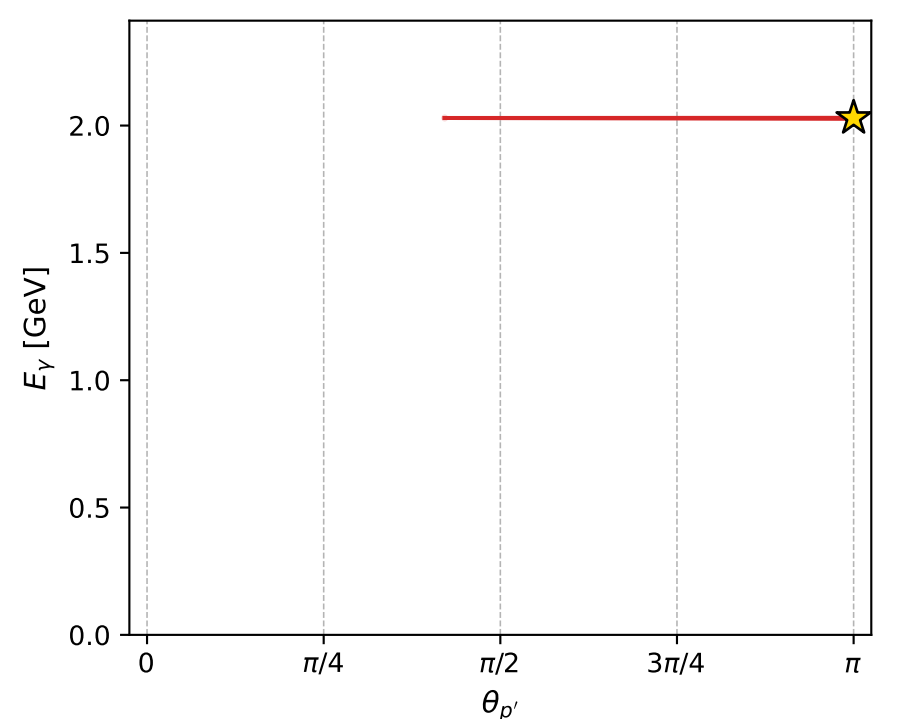
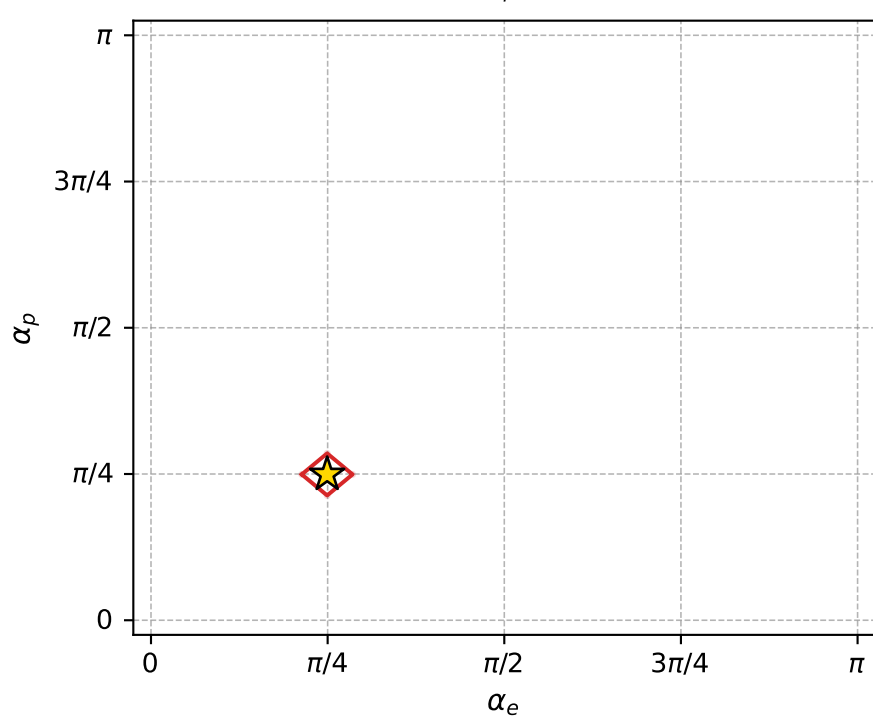
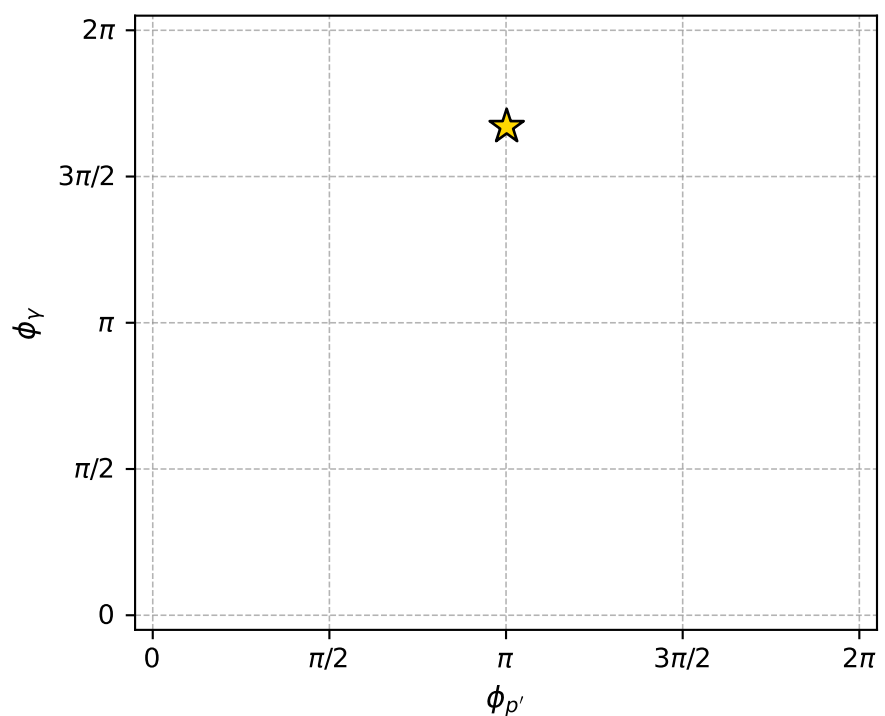
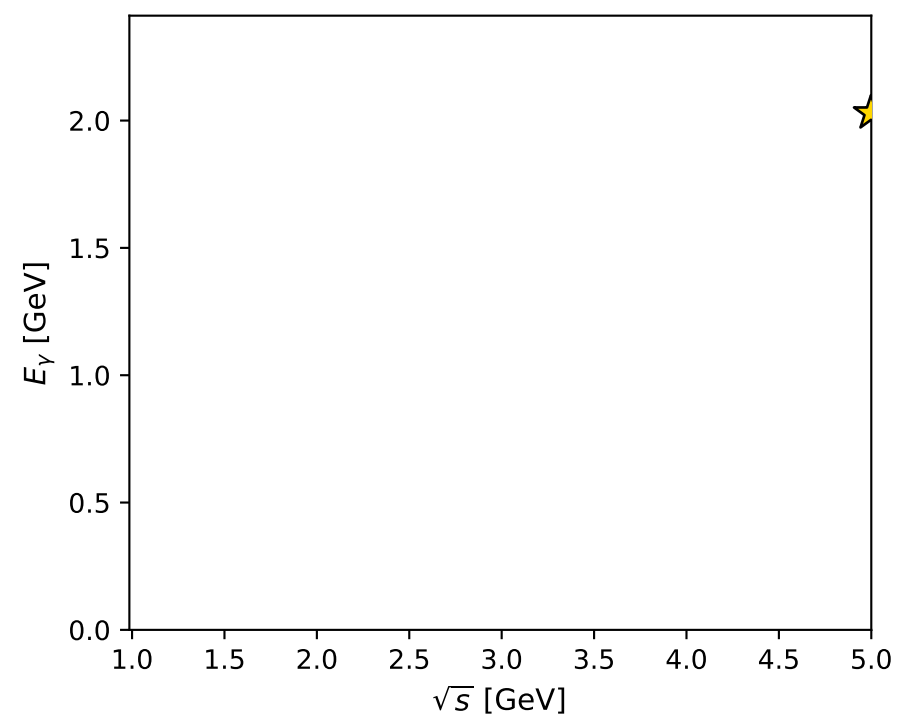
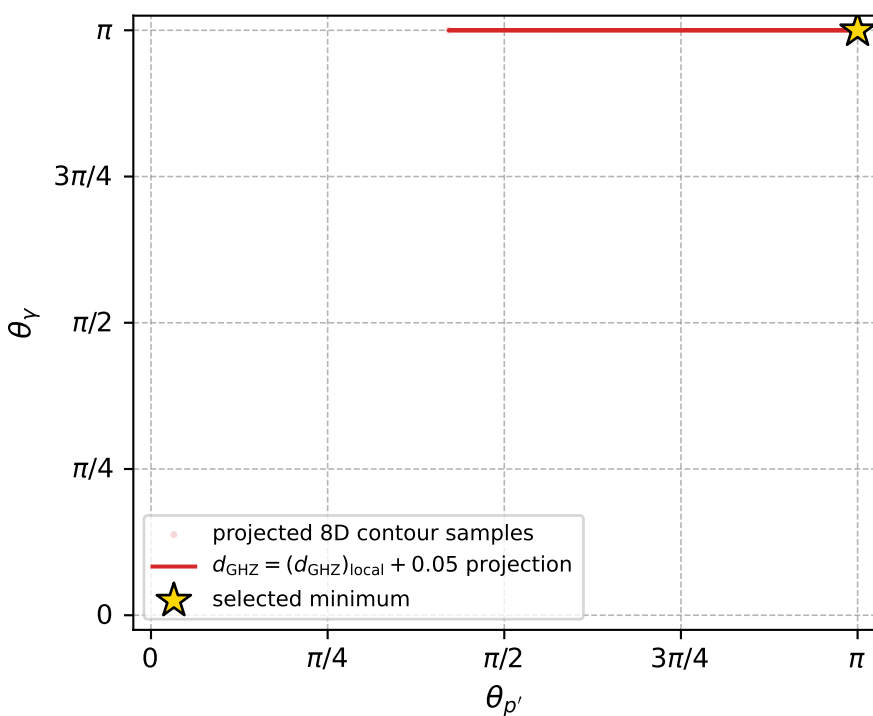
dghz_mixing_angles_polarization_01_local_minimum_46 $d_{\{\text{rm GHZ}\}}=2.66651\text{e-}08$
 region: polarization_cluster_01_local_minimum_46
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78323852$ rad
 incoming lepton: $0.708632|+\rangle +0.705578|-\rangle$
 proton mixing angle: $\alpha_p=0.78324505$ rad
 incoming proton: $0.708628|+\rangle +0.705583|-\rangle$

$s=24.9844$, $\sqrt{s}=4.99844$, $|\vec{P}|=2.41121$, $|\vec{P}'|=0.000949457$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=3.14724$, $\phi_Y=5.24835$
 $E_Y=2.02974$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.10635$
 $Q^2=19.5857$, $x_B=0.837366$, $t=-3.09854$, $W^2=4.68379$, $y=0.970342$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4112$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4112$
 P $E= 2.5872$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4112$
 ℓ' $E= 2.0307$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0307$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0009$
 q_Y $E= 2.0297$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0297$



electron: polarization cluster P1, local minimum 46; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6665119\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.3816573\text{e-}09$
 8D contour samples = 62

selected phase-space point:

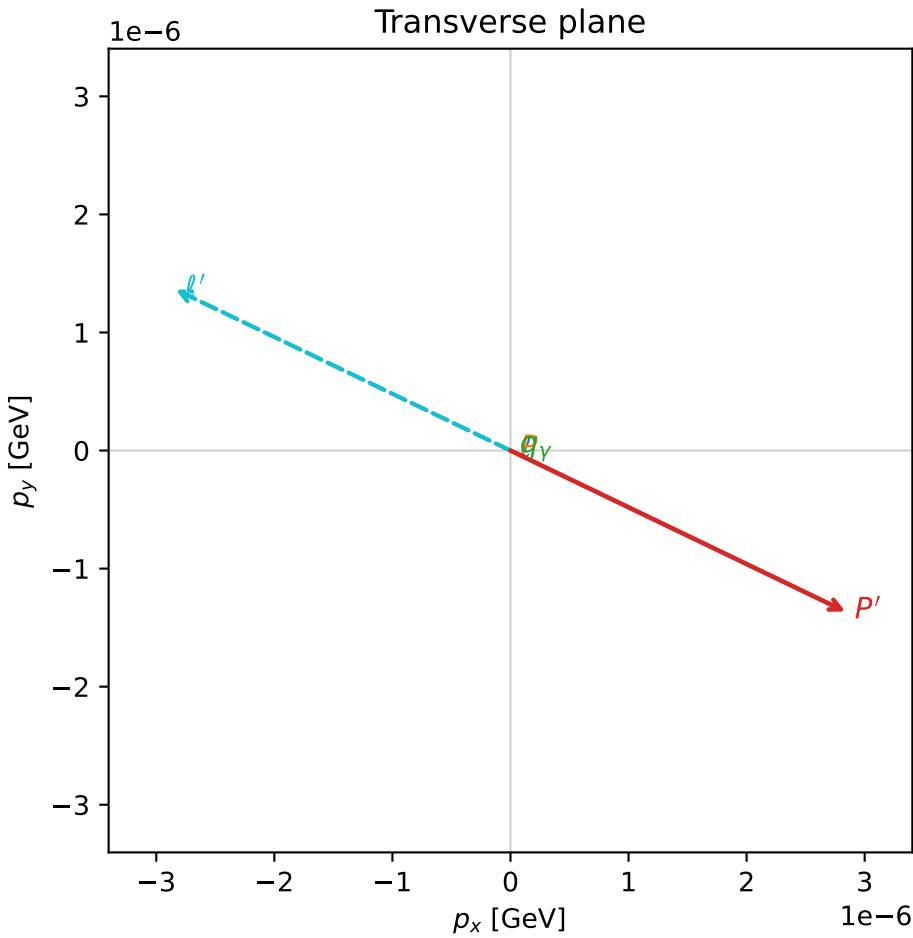
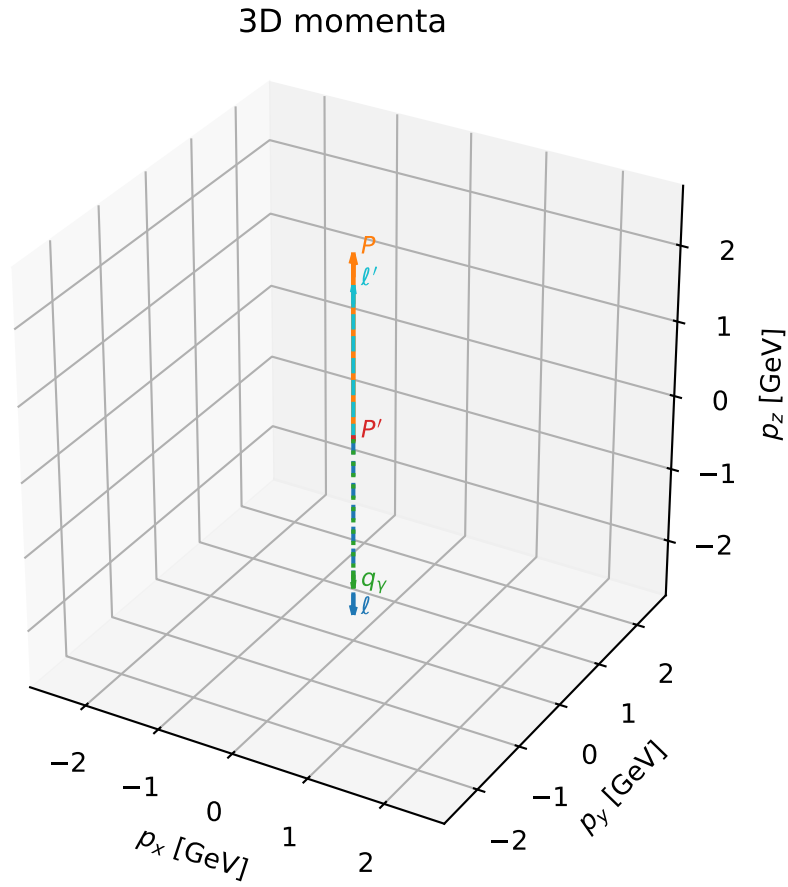
$\text{sqrt_s} = 4.998438$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.029744$
 $\text{phi_p_out} = 3.147244$
 $\text{phi_gamma_out} = 5.248346$
 $\text{alpha_e} = 0.7832385$
 $\text{alpha_p} = 0.7832451$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_47 $d_{\{\text{rm GHZ}\}}=2.6669\text{e-}08$
 region: polarization_cluster_01_local_minimum_47
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77596456$ rad
 incoming lepton: $0.713746|+\rangle +0.700405|-\rangle$
 proton mixing angle: $\alpha_p=0.77595102$ rad
 incoming proton: $0.713755|+\rangle +0.700395|-\rangle$

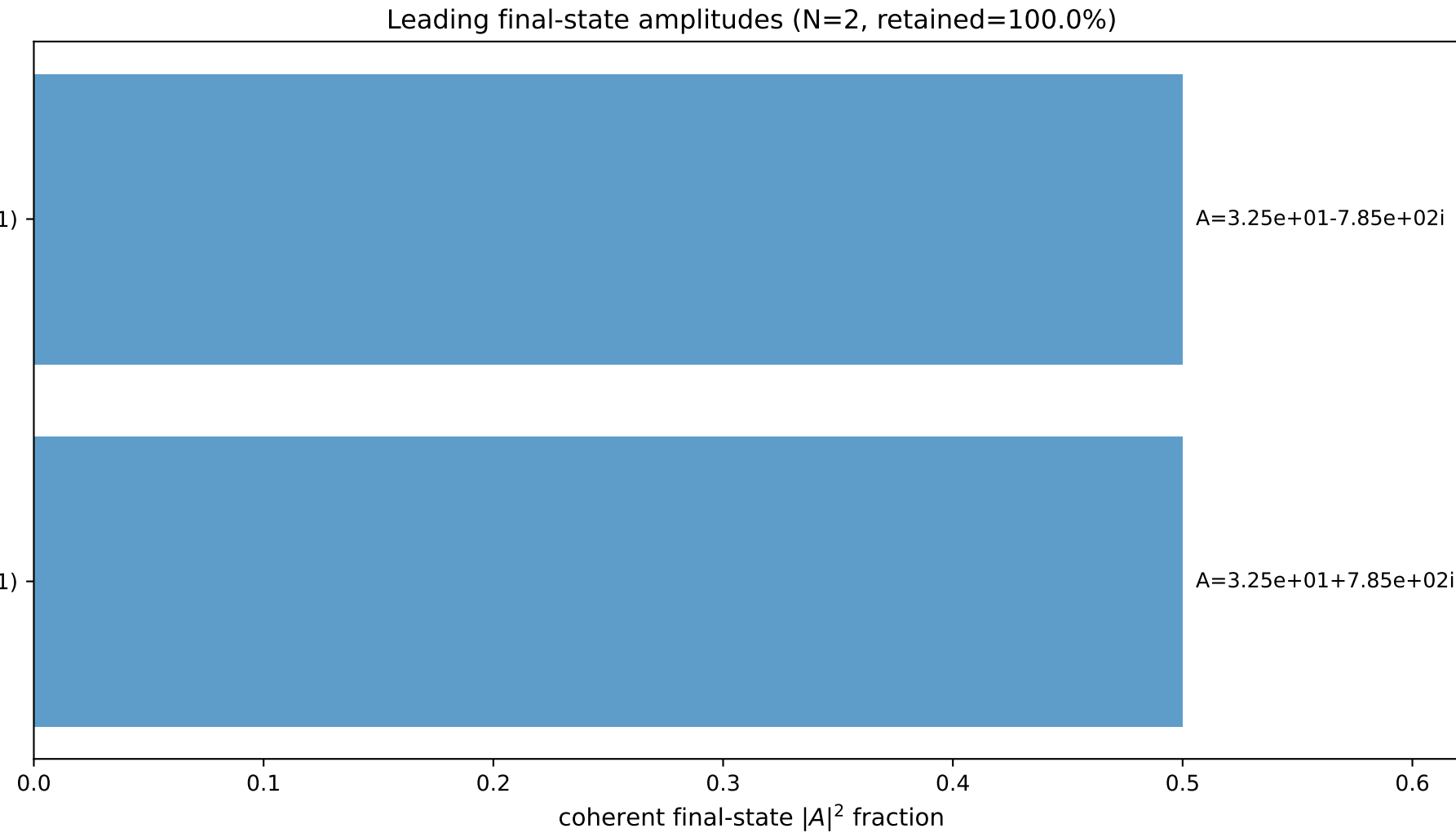
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0205162$
 $\theta_{P'}=0.000153398$, $\theta_Y=3.14159$, $\phi_{P'}=5.83508$, $\phi_Y=1.61218$
 $E_Y=2.04115$, $\theta_{\ell'}=1.55751\text{e-}06$, $\phi_{\ell'}=2.69349$
 $Q^2=19.4952$, $x_B=0.832806$, $t=-2.99756$, $W^2=4.7937$, $y=0.970517$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0206$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0205$
 q_Y $E= 2.0411$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0411$

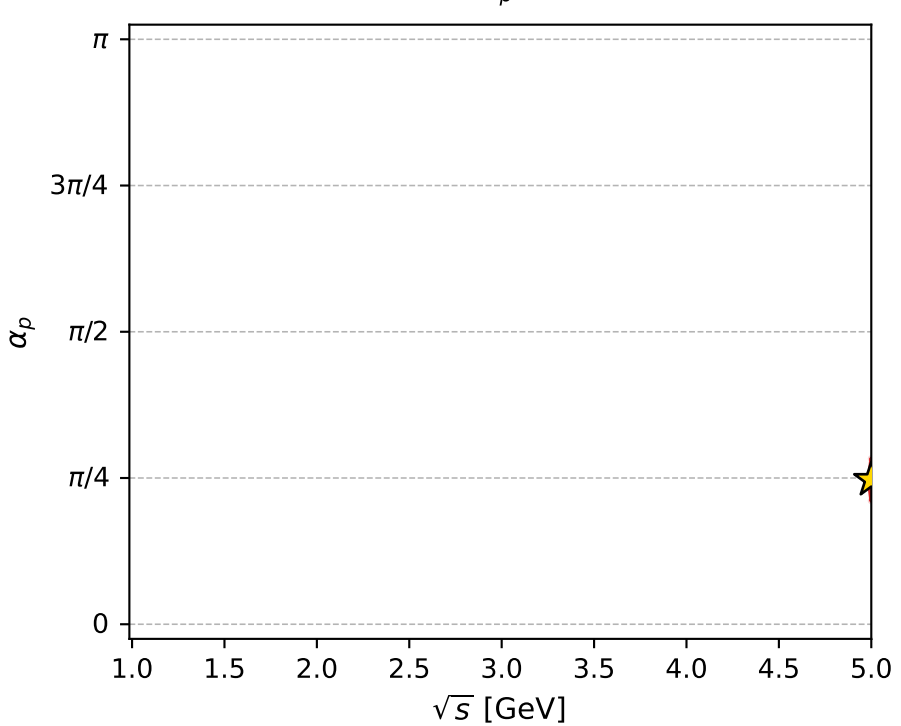
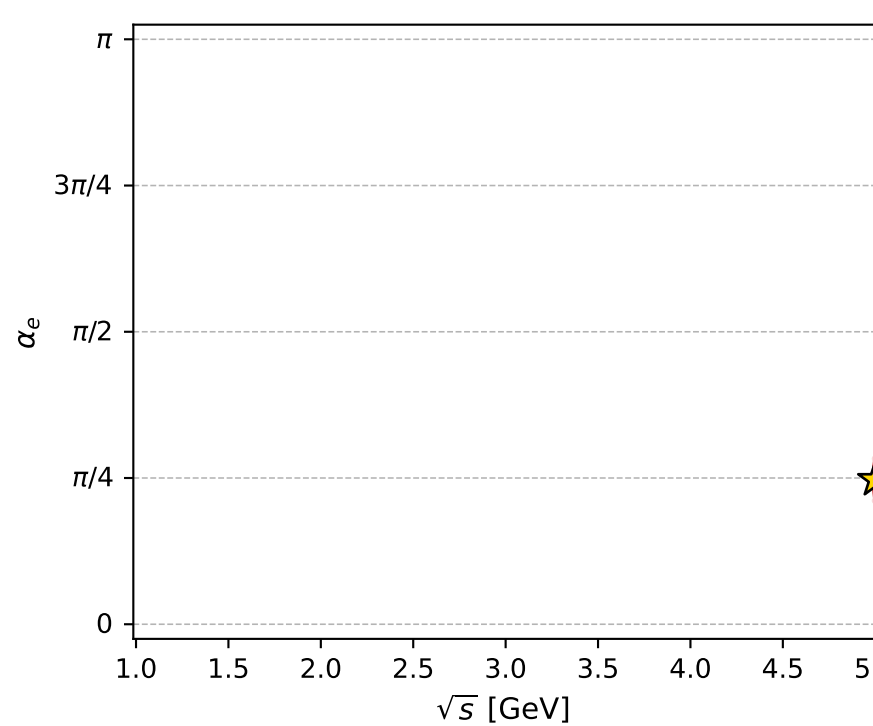
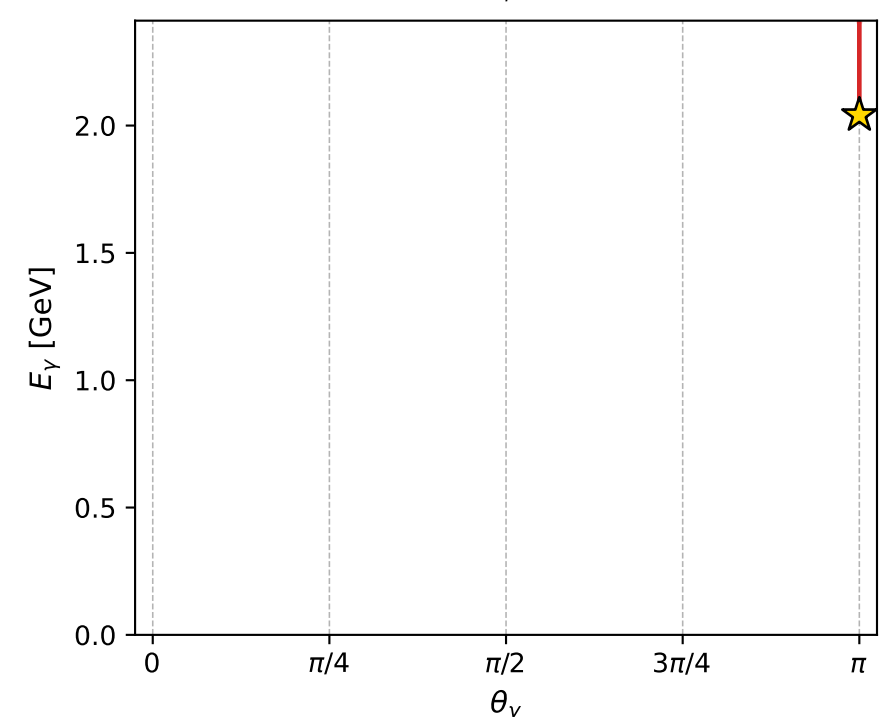
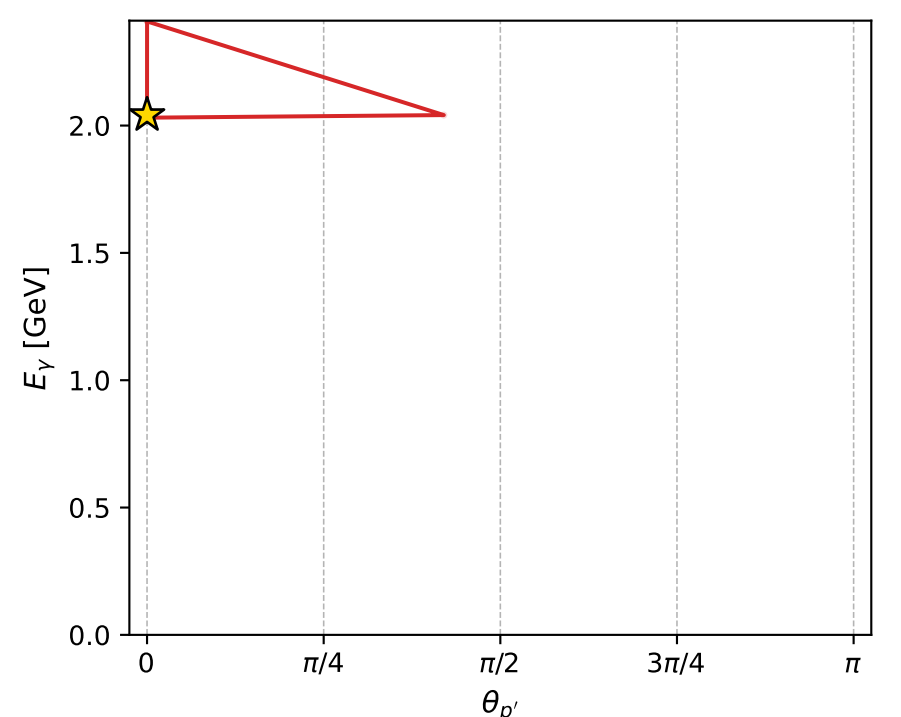
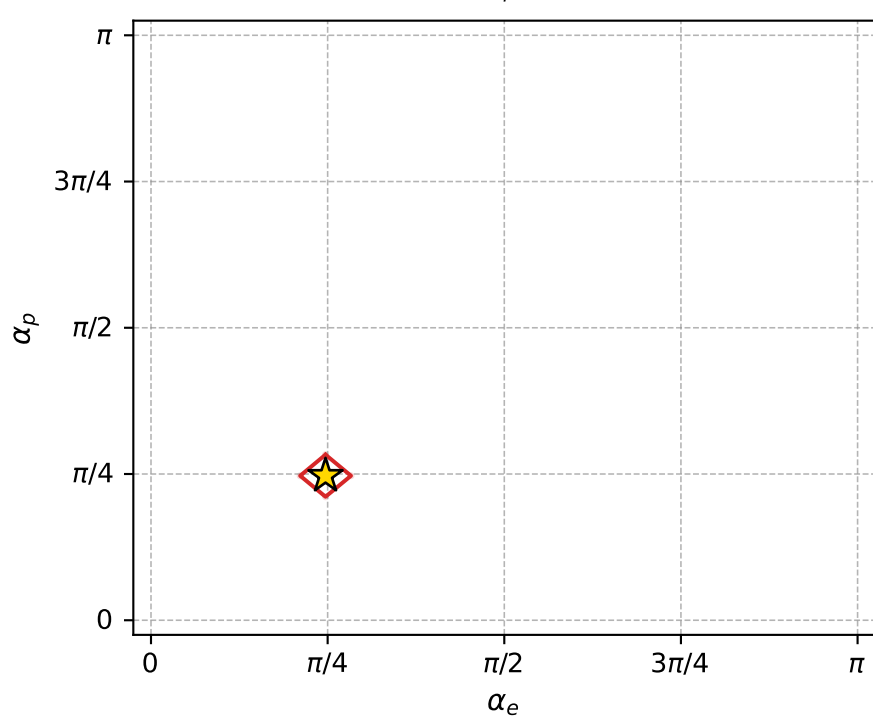
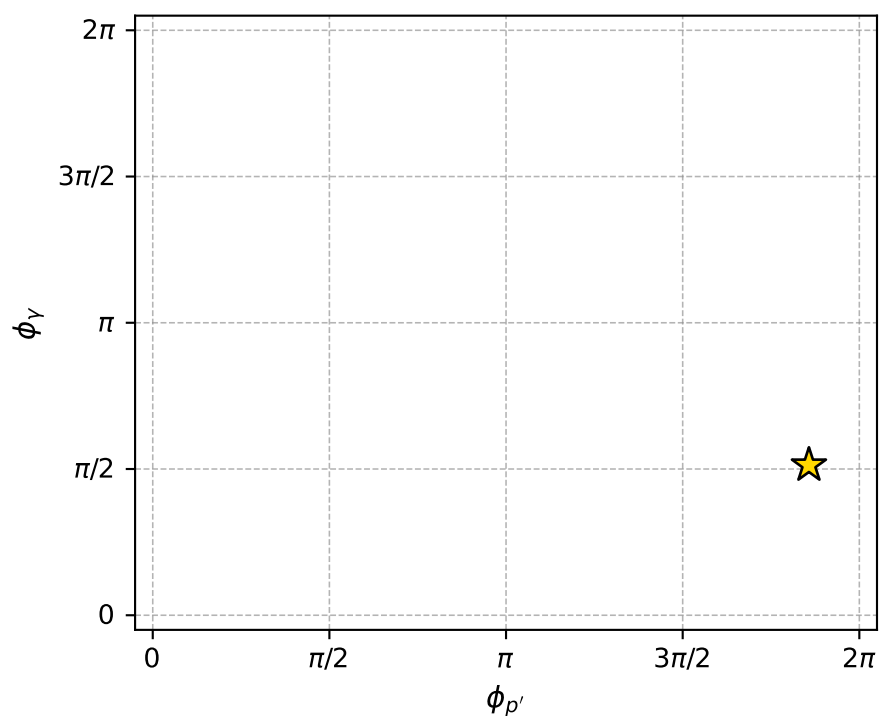
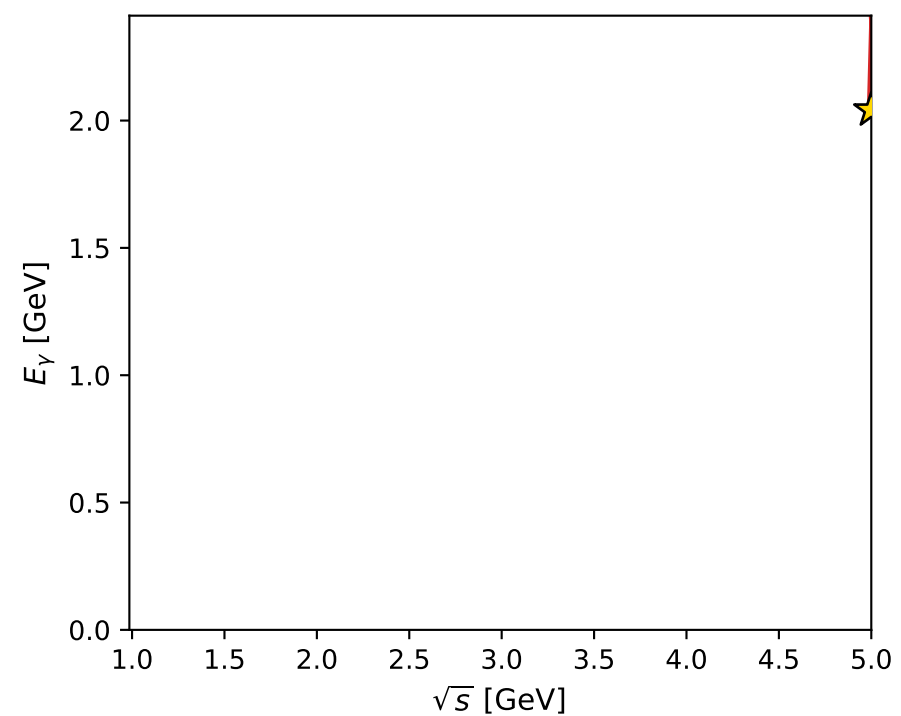
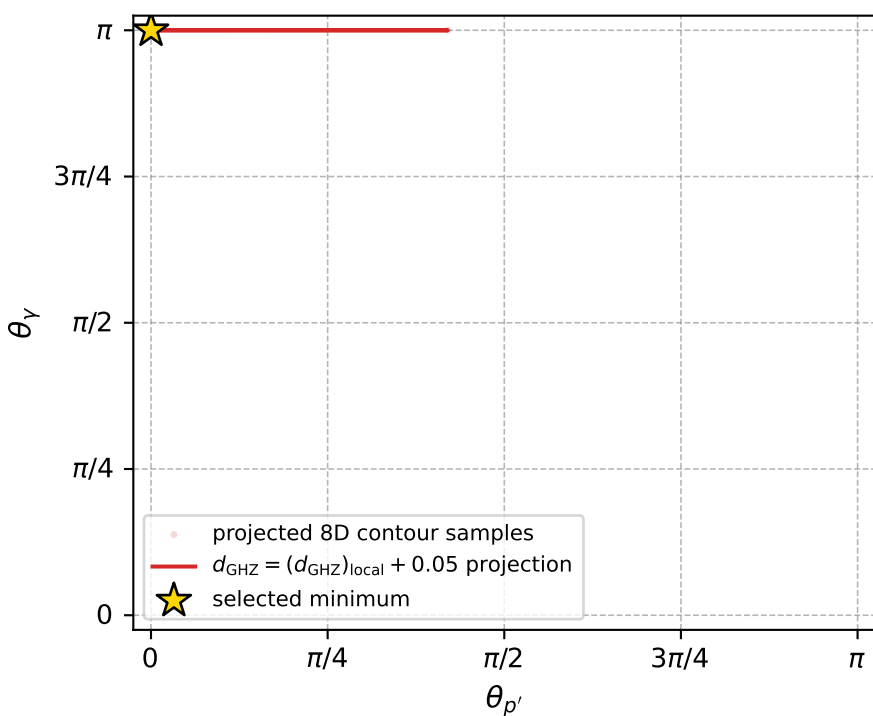


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P1, local minimum 47; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.6668984\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.3855221\text{e-}09$
 8D contour samples = 62

selected phase-space point:

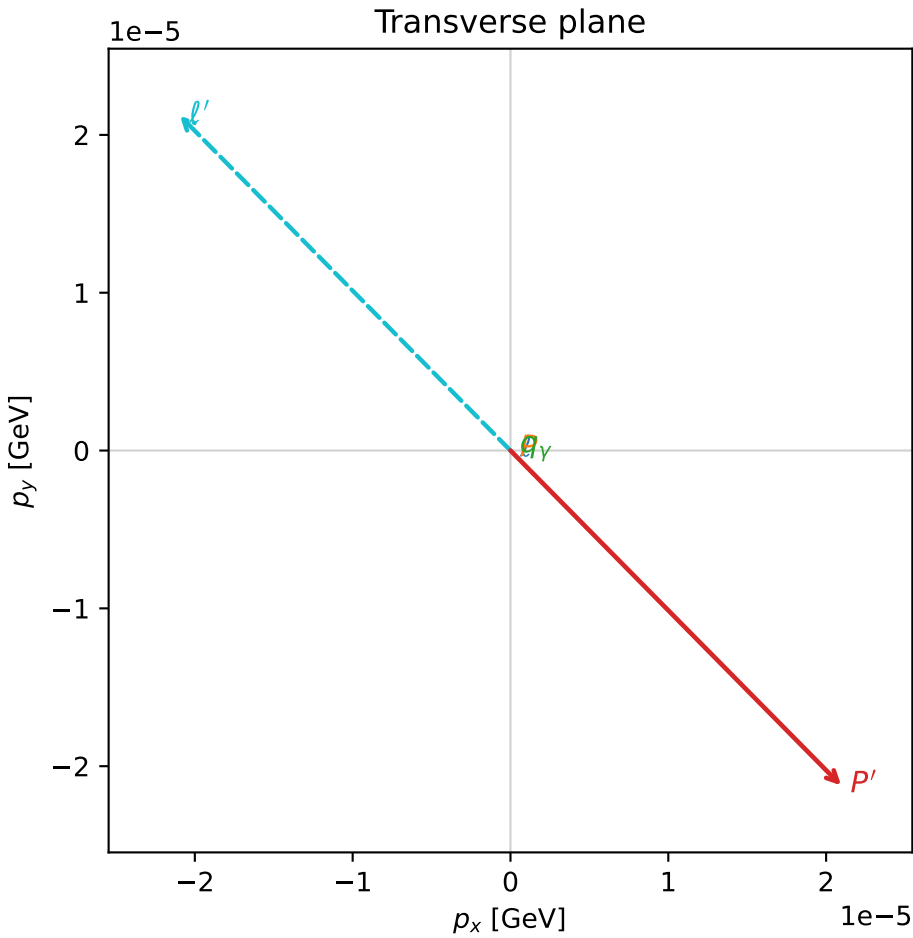
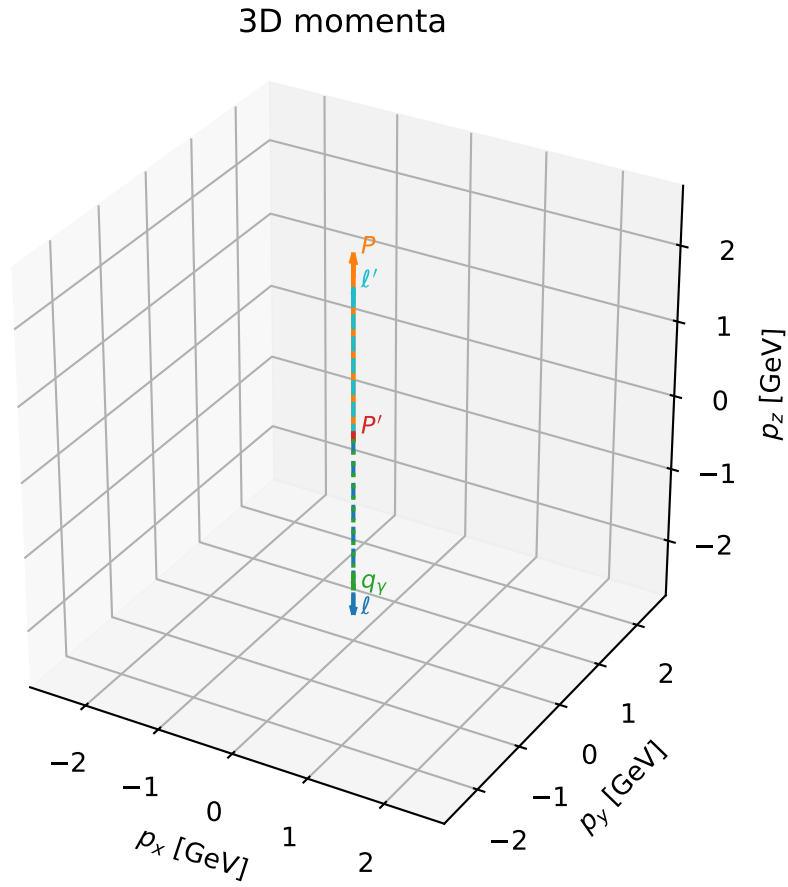
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.0001533981$
 $\text{theta_gamma_out} = 3.141593$
 $\text{qOut} = 2.041146$
 $\text{phi_p_out} = 5.835085$
 $\text{phi_gamma_out} = 1.612181$
 $\text{alpha_e} = 0.7759646$
 $\text{alpha_p} = 0.775951$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_57 $d_{\{\text{rm GHZ}\}}=2.72176\text{e-}08$
region: polarization_cluster_01_local_minimum_57
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.84620821$ rad
incoming lepton: $0.662827|+\rangle +0.748772|-\rangle$
proton mixing angle: $\alpha_p=0.84620257$ rad
incoming proton: $0.662831|+\rangle +0.748769|-\rangle$

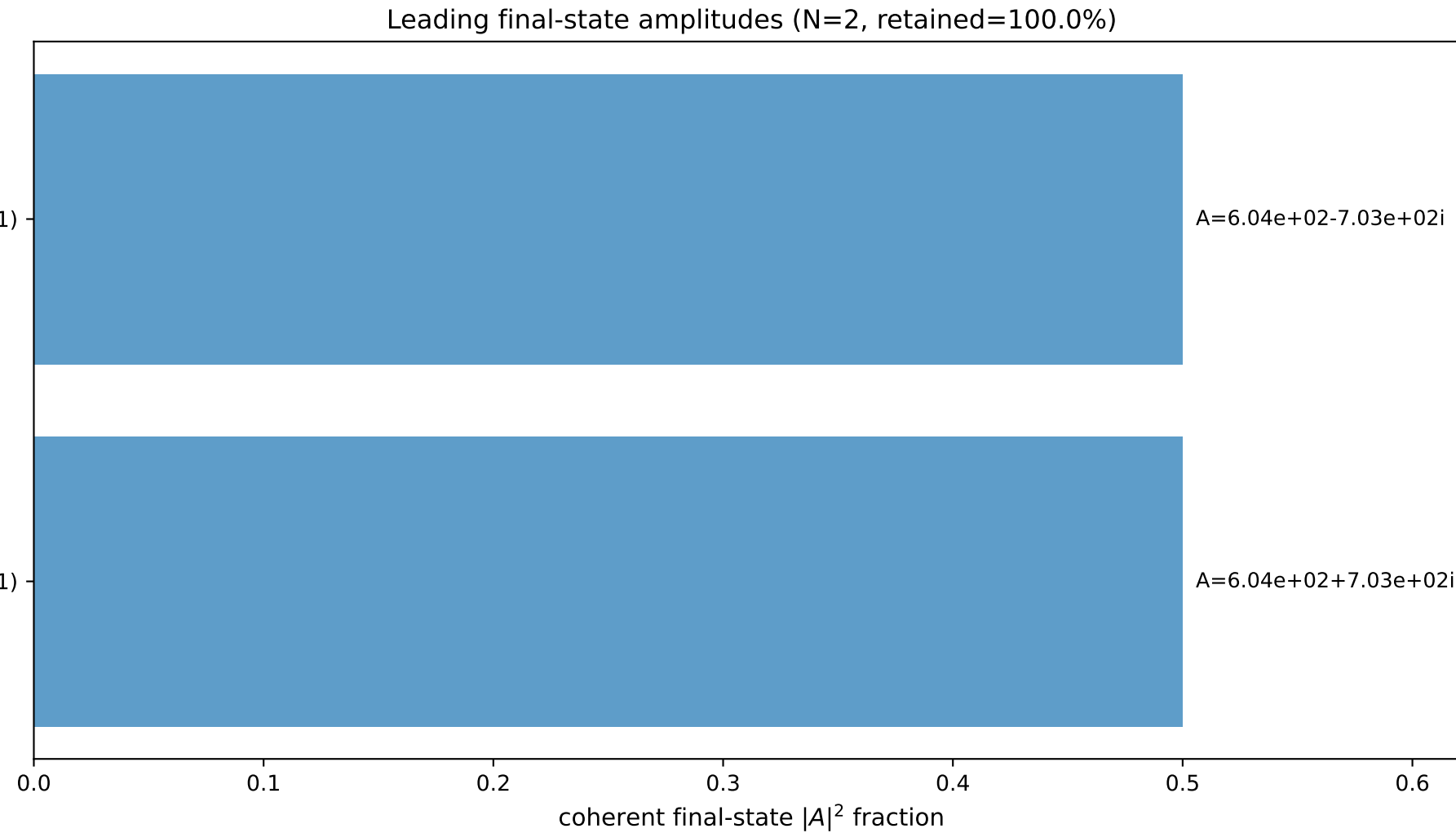
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0736114$
 $\theta_{P'}=0.000405073$, $\theta_Y=3.14159$, $\phi_{P'}=5.49139$, $\phi_Y=2.28109$
 $E_Y=2.06636$, $\theta_{\ell'}=1.49632\text{e-}05$, $\phi_{\ell'}=2.34979$
 $Q^2=19.2262$, $x_B=0.820972$, $t=-2.75519$, $W^2=5.07248$, $y=0.970924$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9928$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9928$
 P' $E= 0.9409$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0736$
 q_Y $E= 2.0664$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0664$

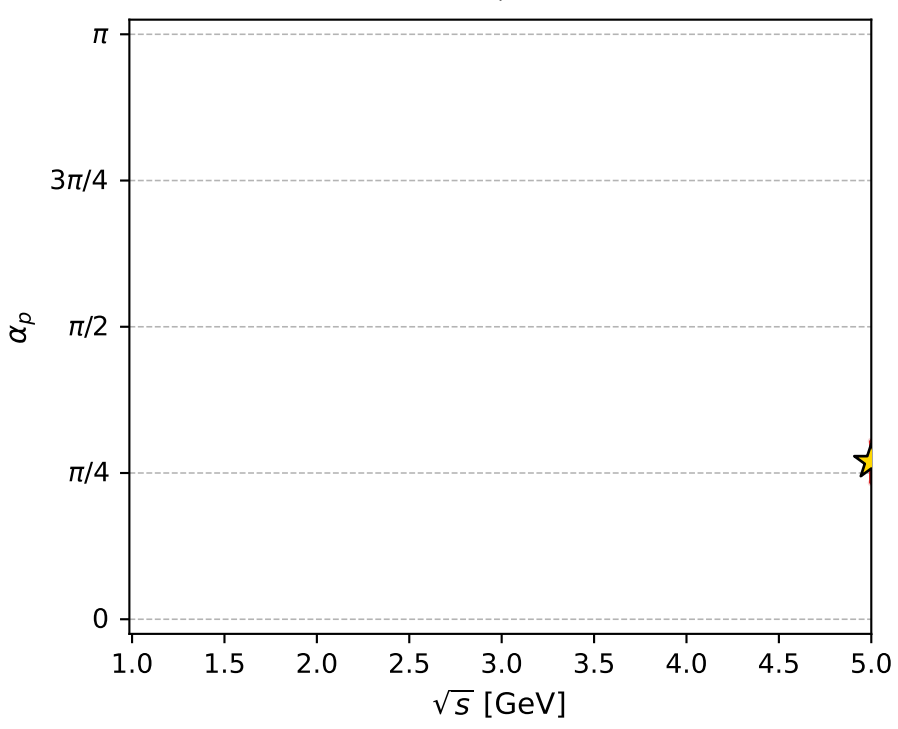
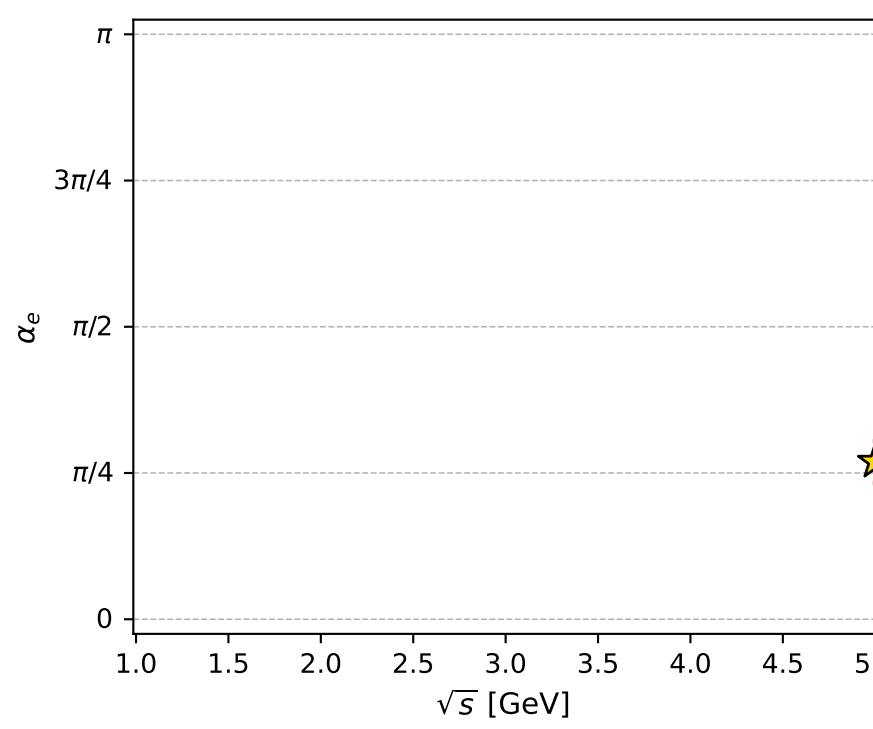
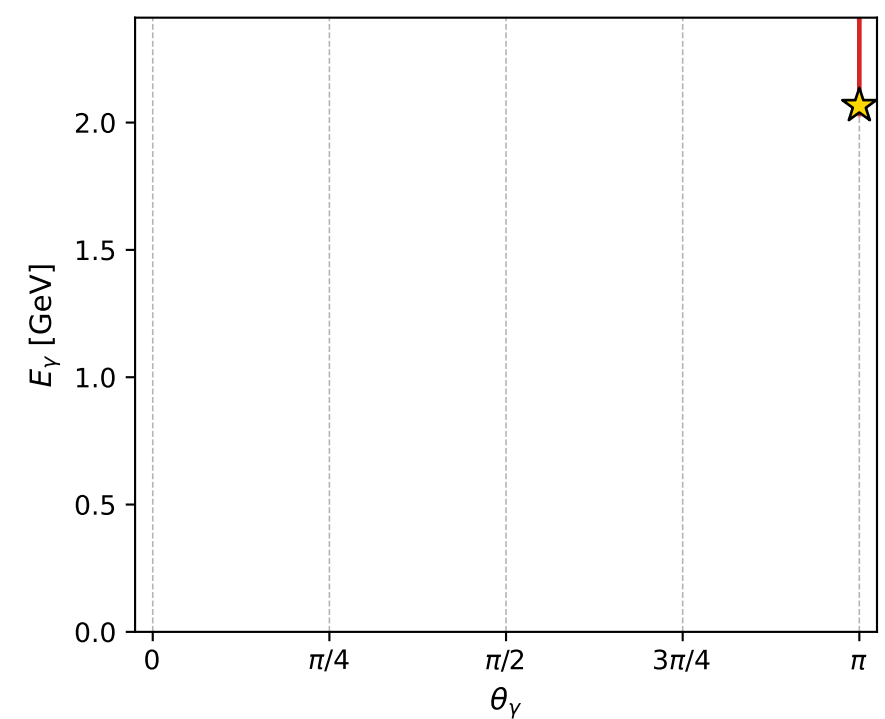
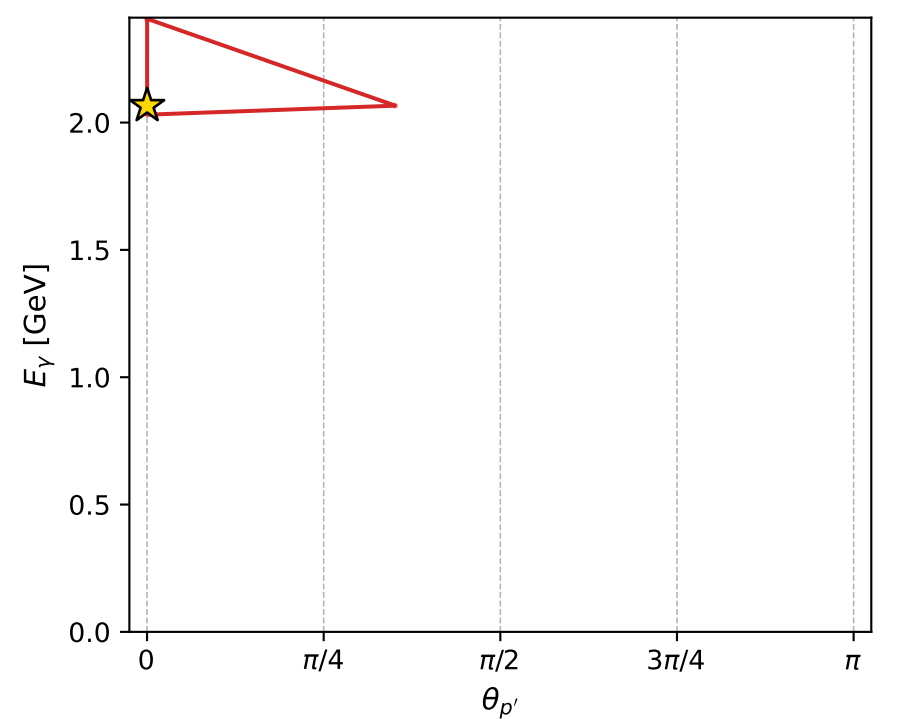
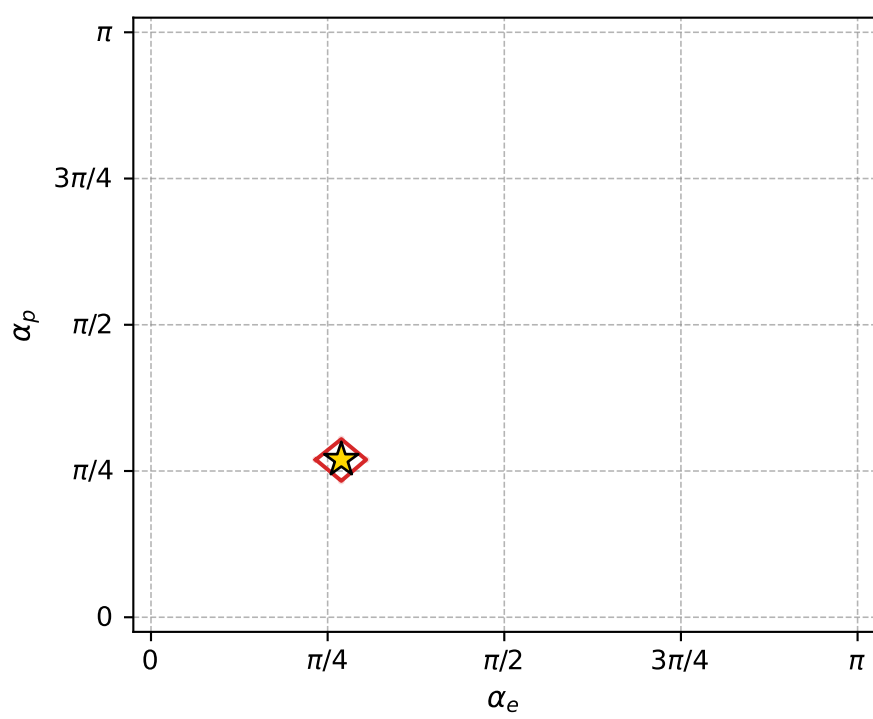
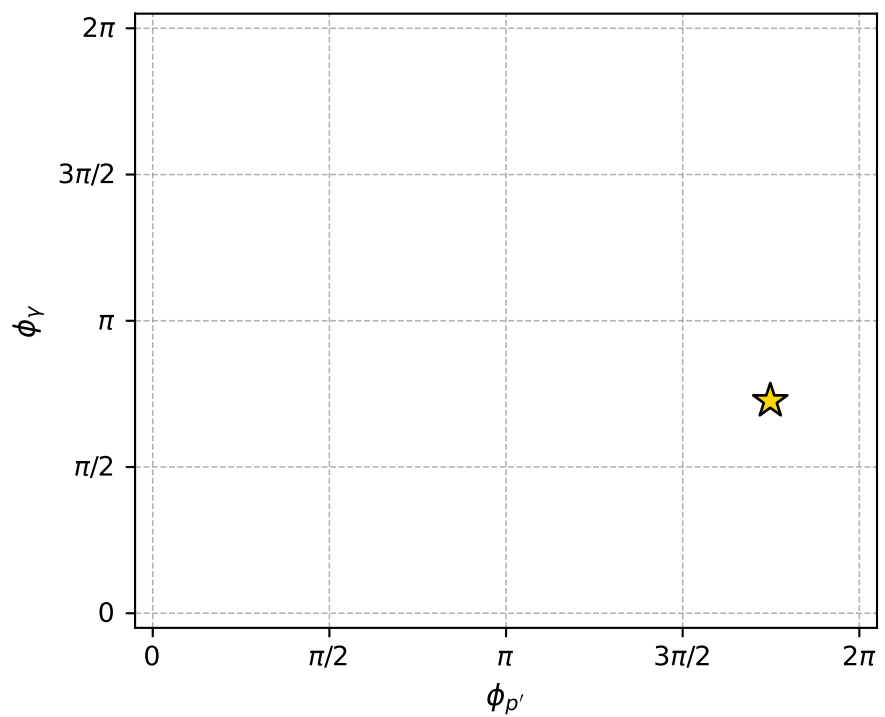
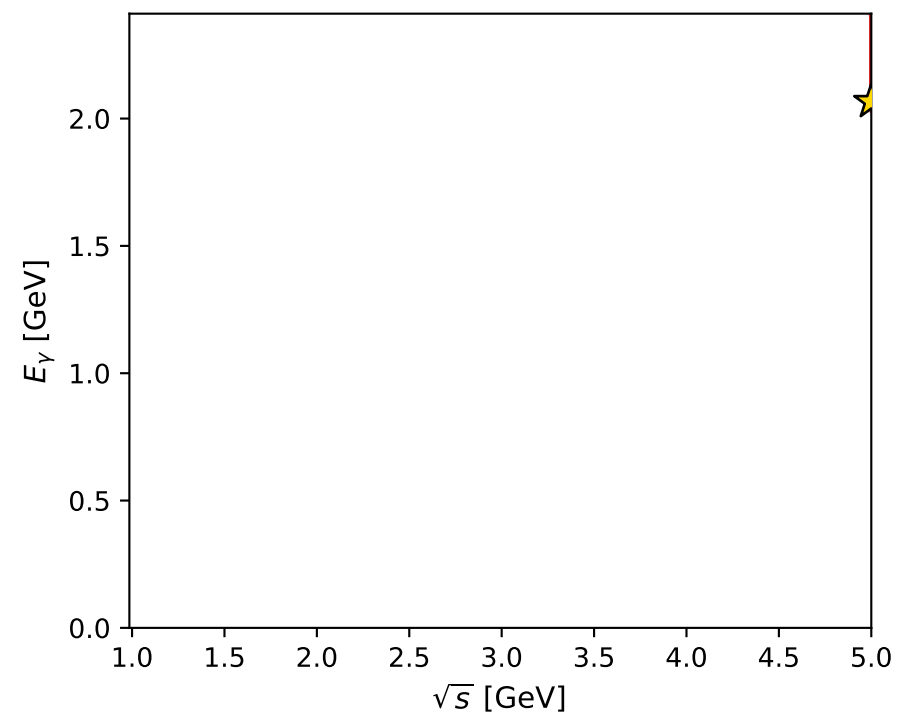
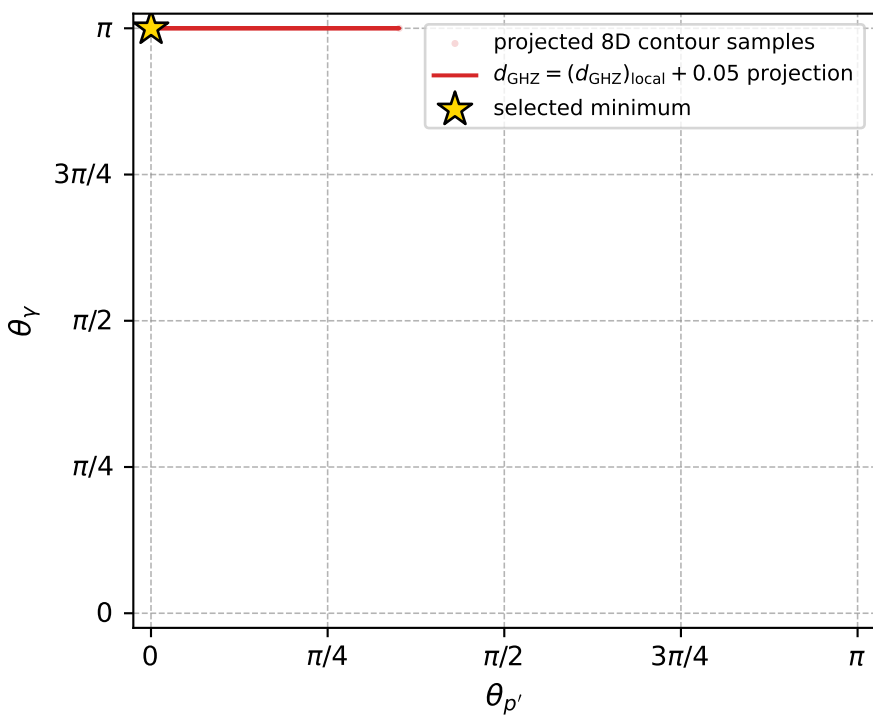


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P1, local minimum 57; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7217624\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.9341617\text{e-}09$
 8D contour samples = 63

selected phase-space point:

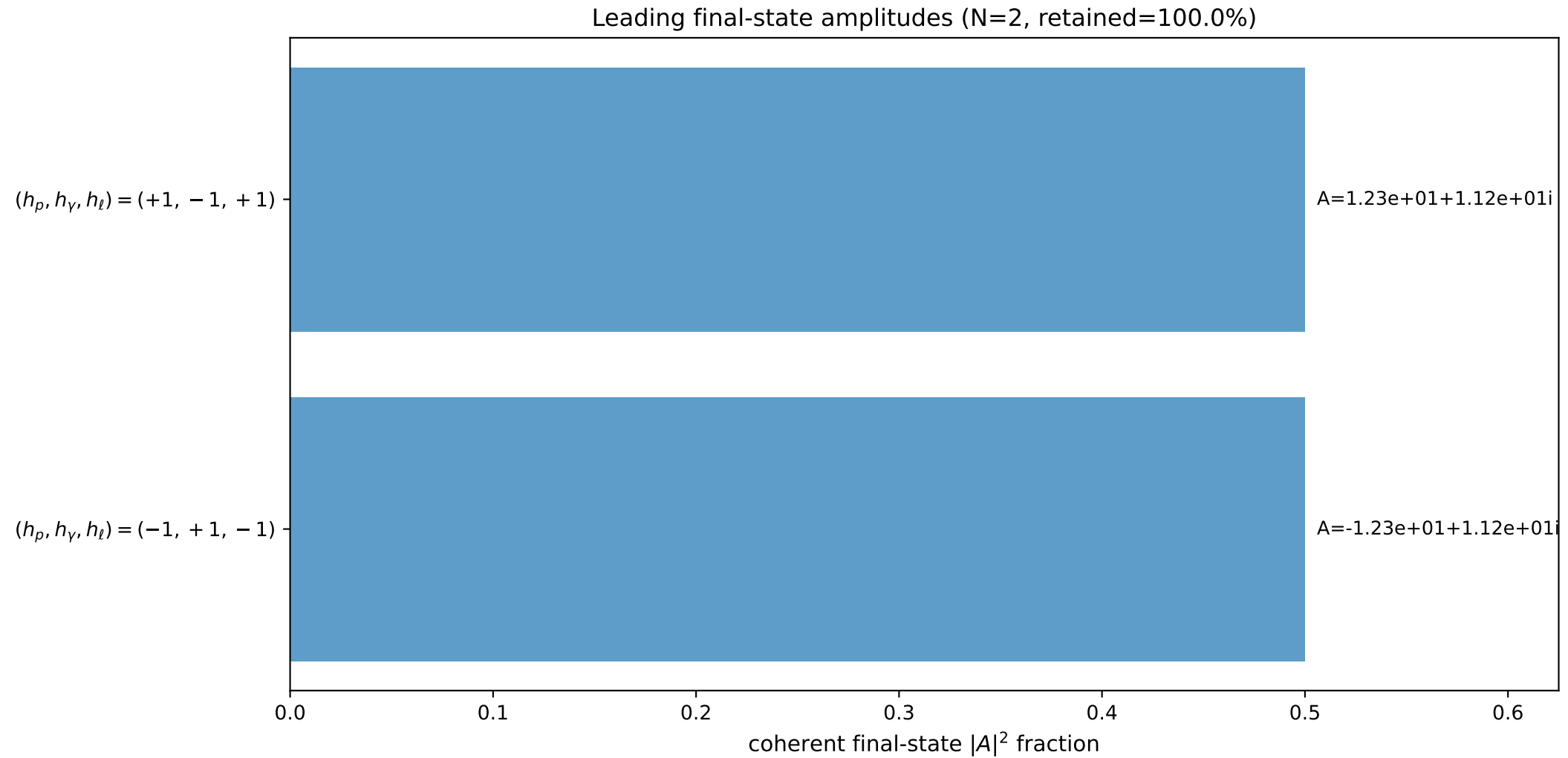
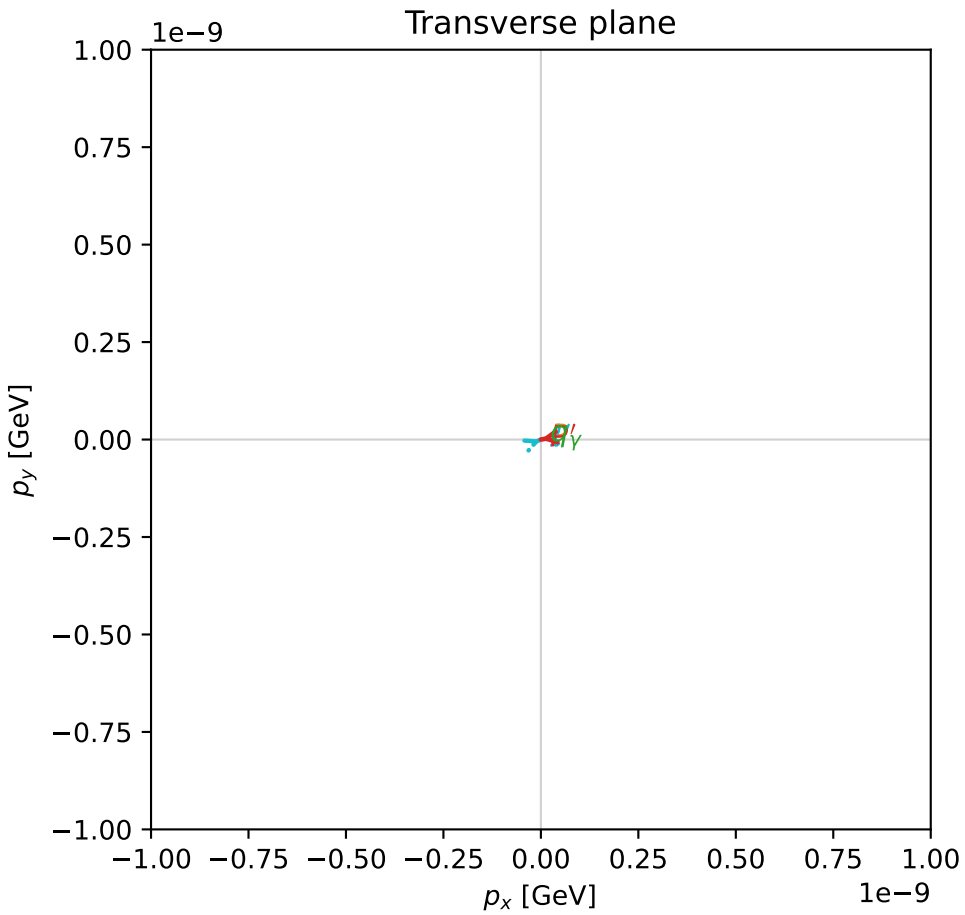
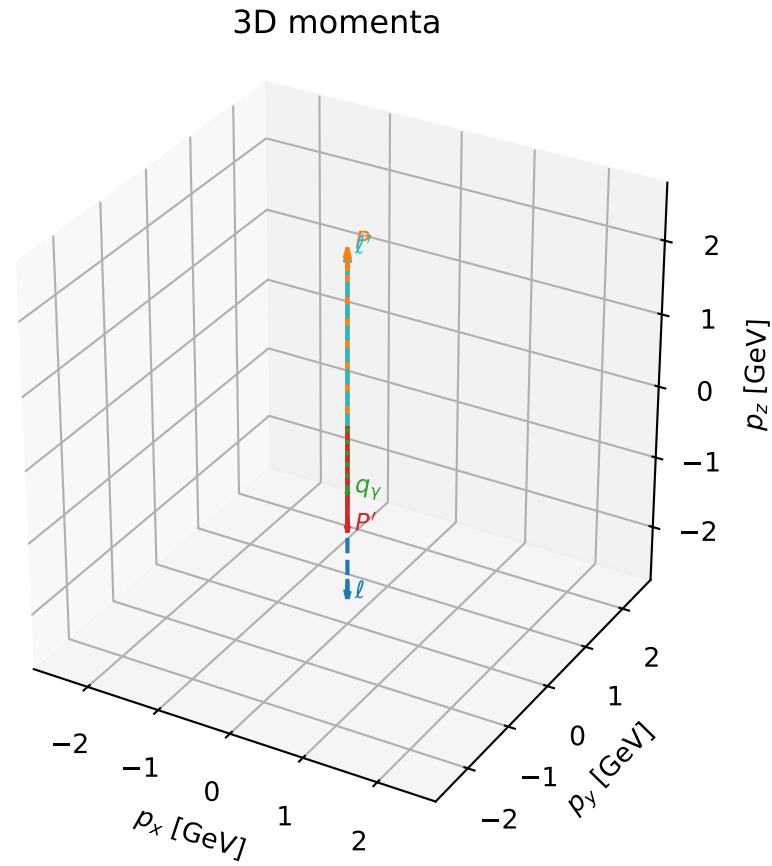
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.0004050734$
 $\text{theta_gamma_out} = 3.141593$
 $\text{qOut} = 2.066364$
 $\text{phi_p_out} = 5.491386$
 $\text{phi_gamma_out} = 2.28109$
 $\text{alpha_e} = 0.8462082$
 $\text{alpha_p} = 0.8462026$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

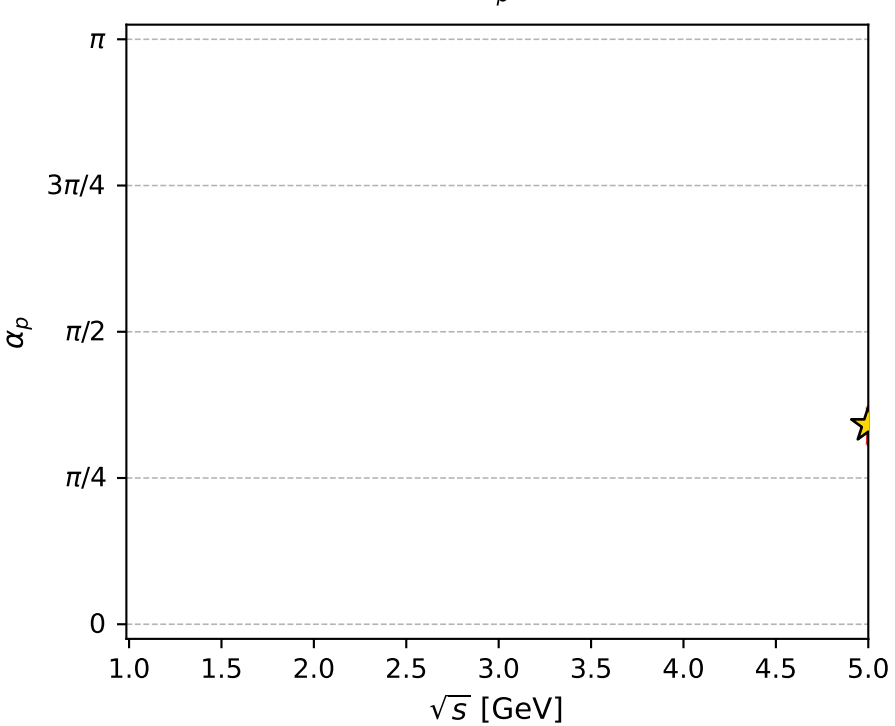
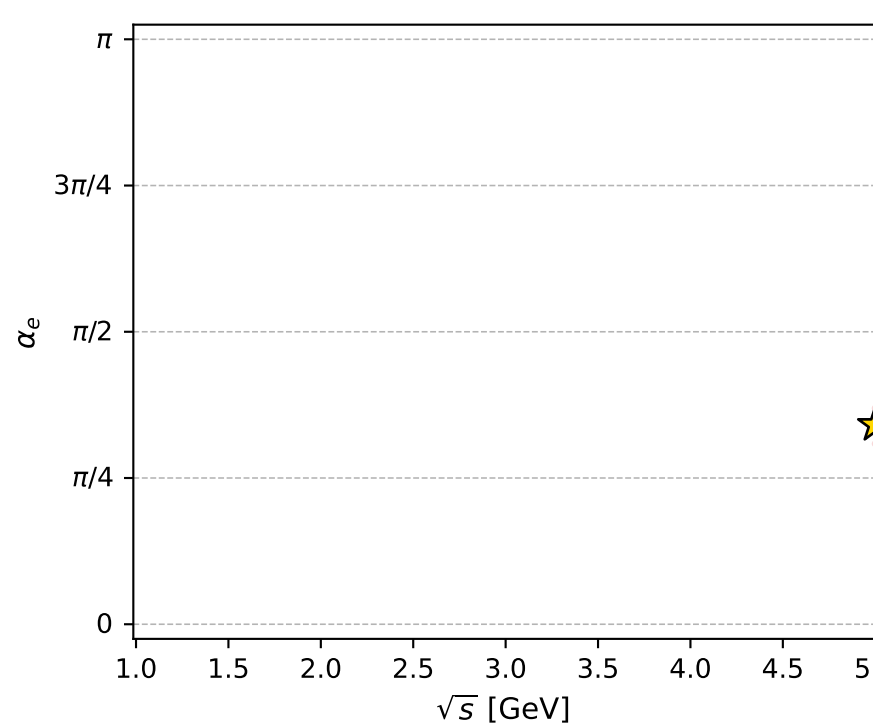
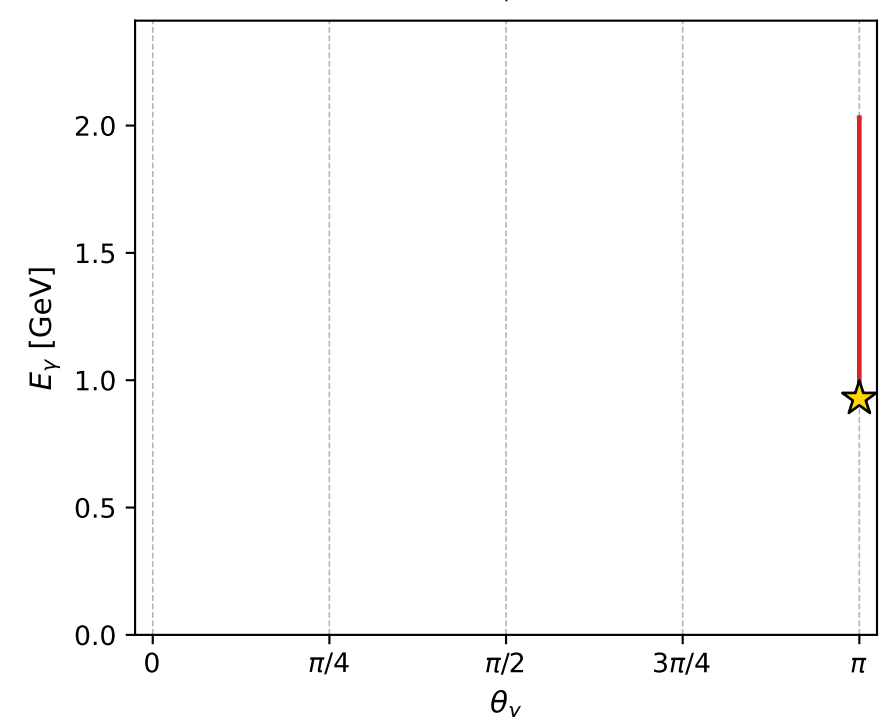
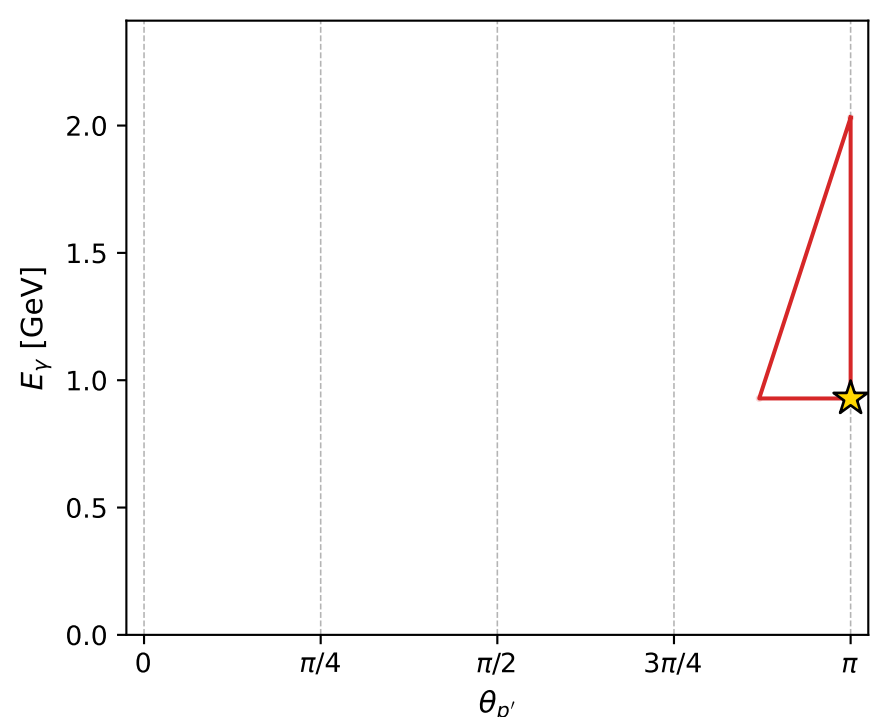
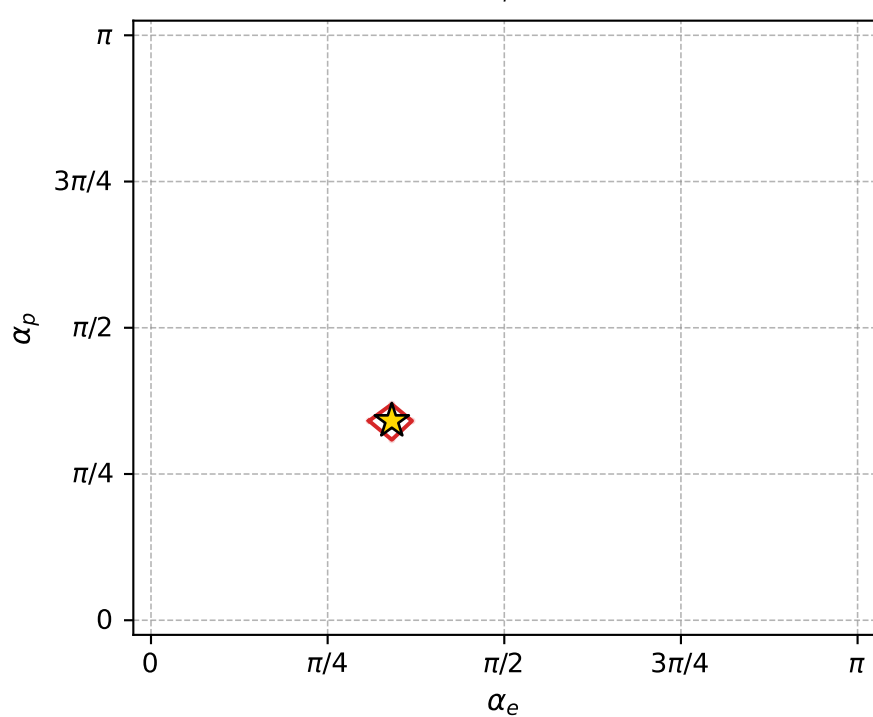
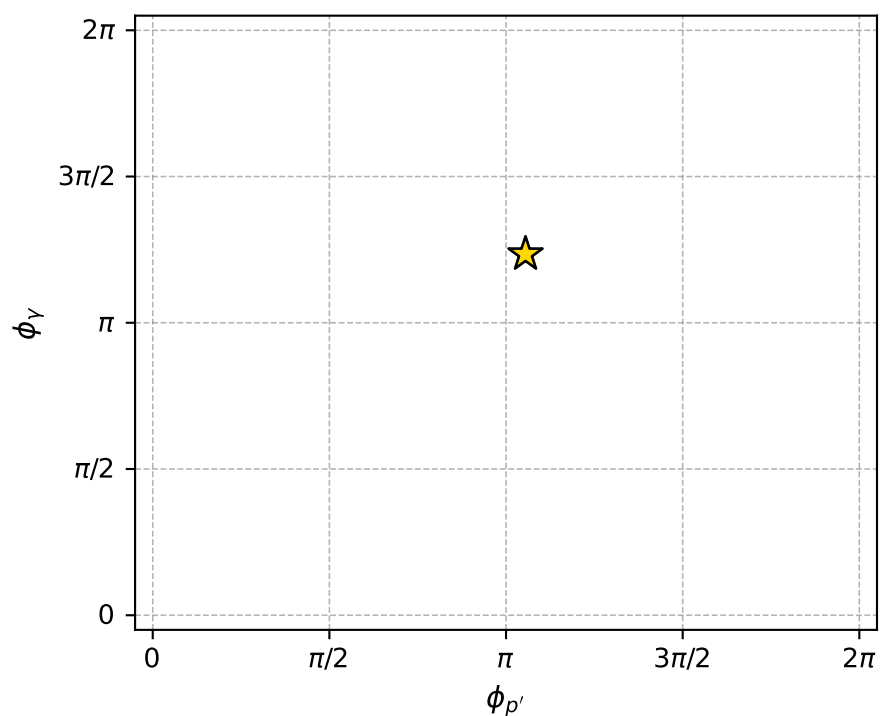
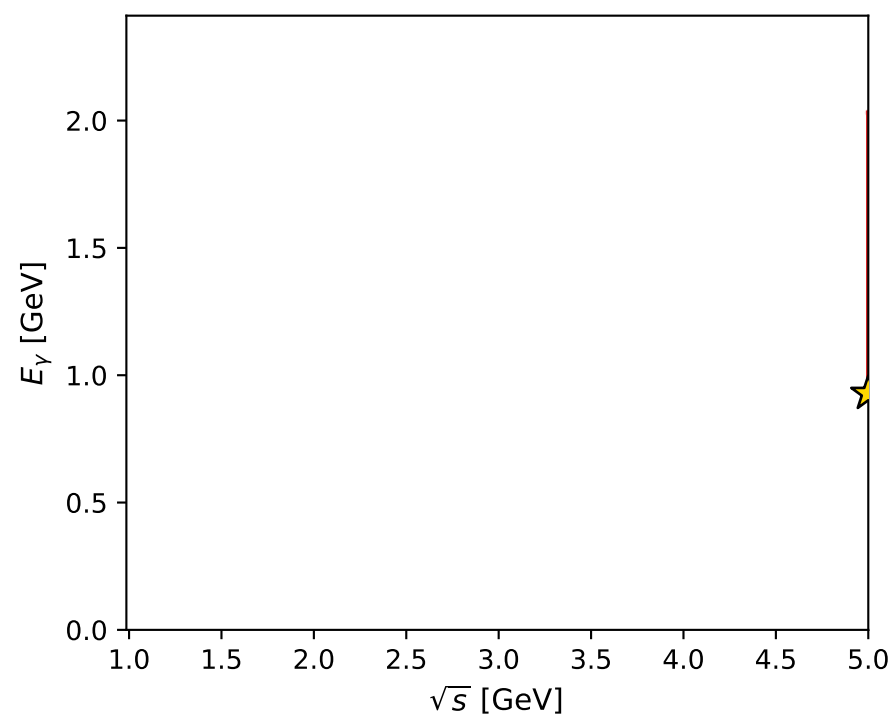
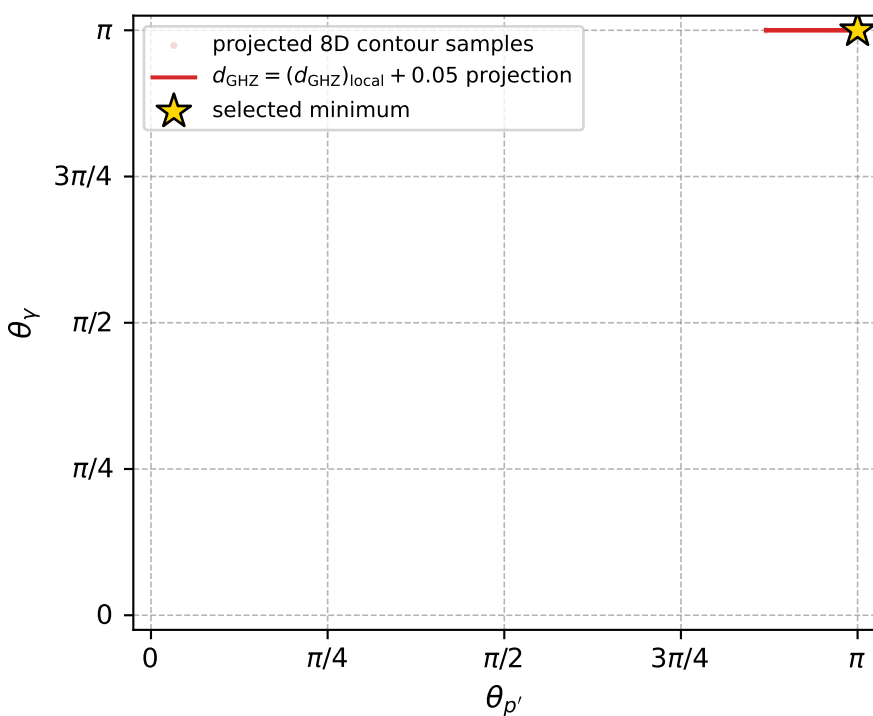
dghz_mixing_angles_polarization_01_local_minimum_58 $d_{\{\text{rm GHZ}\}}=2.7258\text{e-}08$
 region: polarization_cluster_01_local_minimum_58
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.0707816$ rad
 incoming lepton: $0.479438|+\rangle +0.877576|-\rangle$
 proton mixing angle: $\alpha_p=1.0707773$ rad
 incoming proton: $0.479442|+\rangle +0.877573|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.43163$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=3.31472$, $\phi_Y=3.87889$
 $E_Y=0.92841$, $\theta_{l'}=0$, $\phi_{l'}=0.393495$
 $Q^2=22.7698$, $x_B=0.977683$, $t=-14.0055$, $W^2=1.39961$, $y=0.965565$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.3600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3600$
 P' $E= 1.7116$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4316$
 q_Y $E= 0.9284$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.9284$



electron: polarization cluster P1, local minimum 58; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7258014\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000027$
 above global minimum = $4.974552\text{e-}09$
 8D contour samples = 46

selected phase-space point:

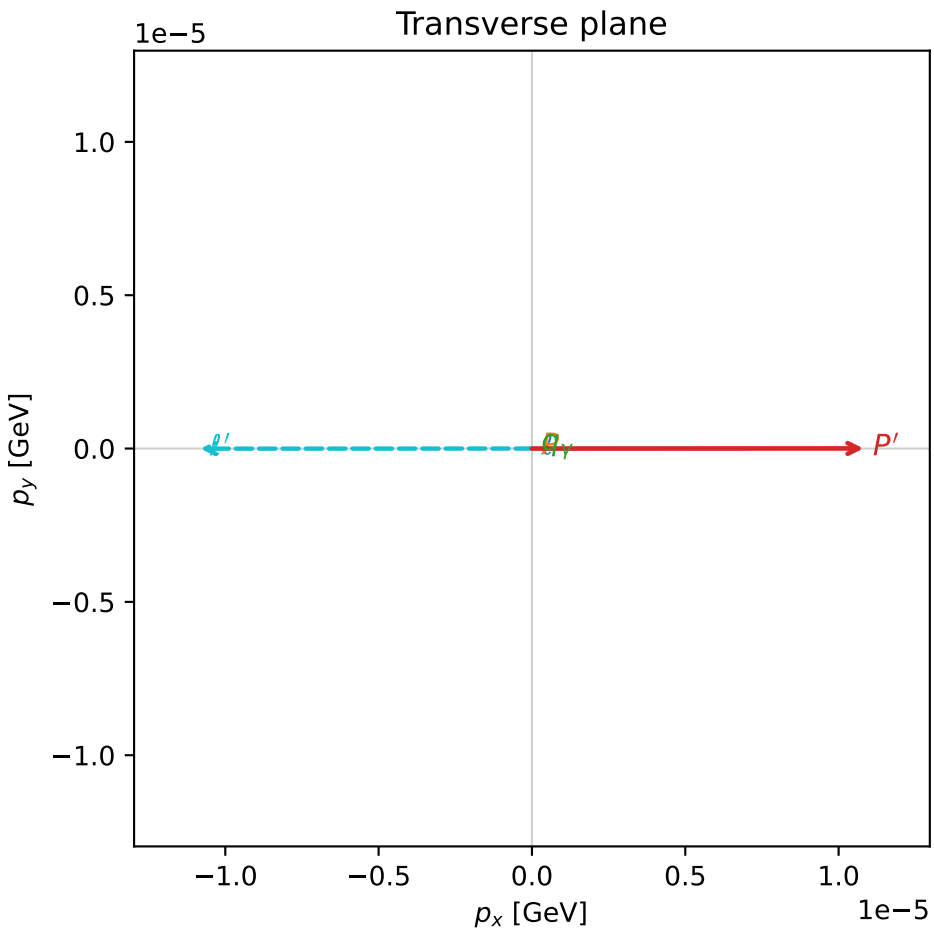
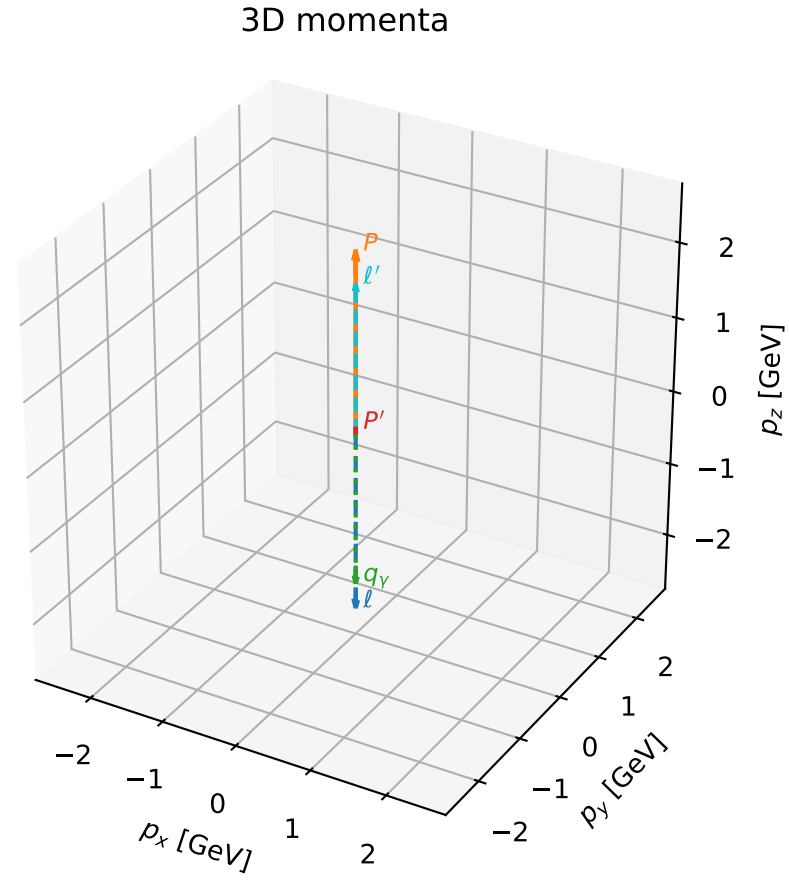
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 0.9284103$
 $\text{phi_p_out} = 3.31472$
 $\text{phi_gamma_out} = 3.878889$
 $\text{alpha_e} = 1.070782$
 $\text{alpha_p} = 1.070777$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_61 d_{\rm GHZ}=2.75163e-08
region: polarization_cluster_01_local_minimum_61
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.88093079$ rad
incoming lepton: $0.636433|+\rangle +0.771332|-\rangle$
proton mixing angle: $\alpha_p=0.88091145$ rad
incoming proton: $0.636448|+\rangle +0.771319|-\rangle$

$s=24.9904$, $\sqrt{s}=4.99903$, $|\vec{P}|=2.41152$, $|\vec{P}'|=0.0704794$
 $\theta_{P'}=0.000153398$, $\theta_Y=3.14159$, $\phi_{P'}=0.000314416$, $\phi_Y=3.07524$
 $E_Y=2.06444$, $\theta_{\ell'}=5.4221e-06$, $\phi_{\ell'}=3.14191$
 $Q^2=19.2338$, $x_B=0.821656$, $t=-2.76826$, $W^2=5.05464$, $y=0.970889$

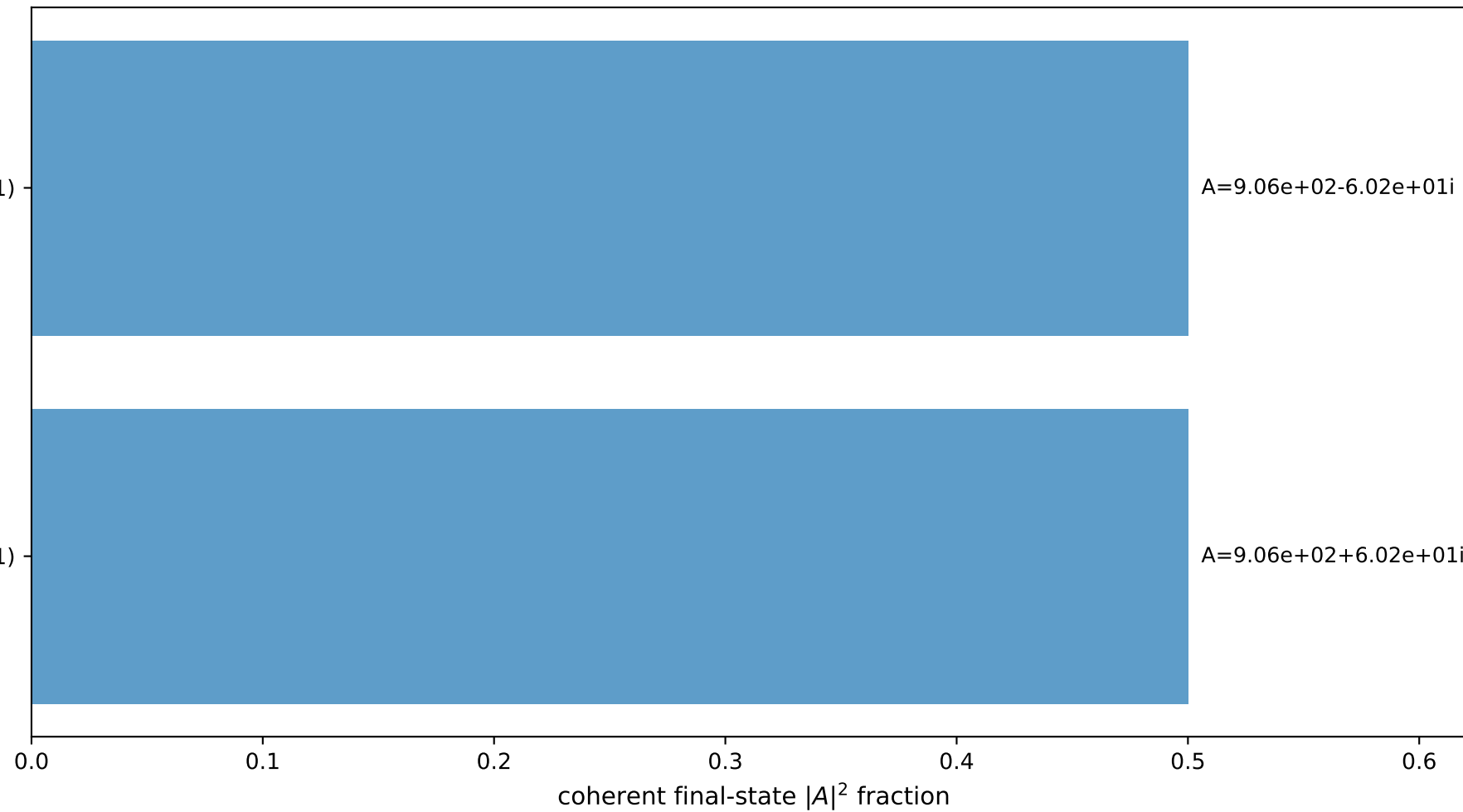
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4115$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4115$
 P $E= 2.5875$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4115$
 ℓ' $E= 1.9940$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9940$
 P' $E= 0.9406$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0705$
 q_Y $E= 2.0644$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0644$



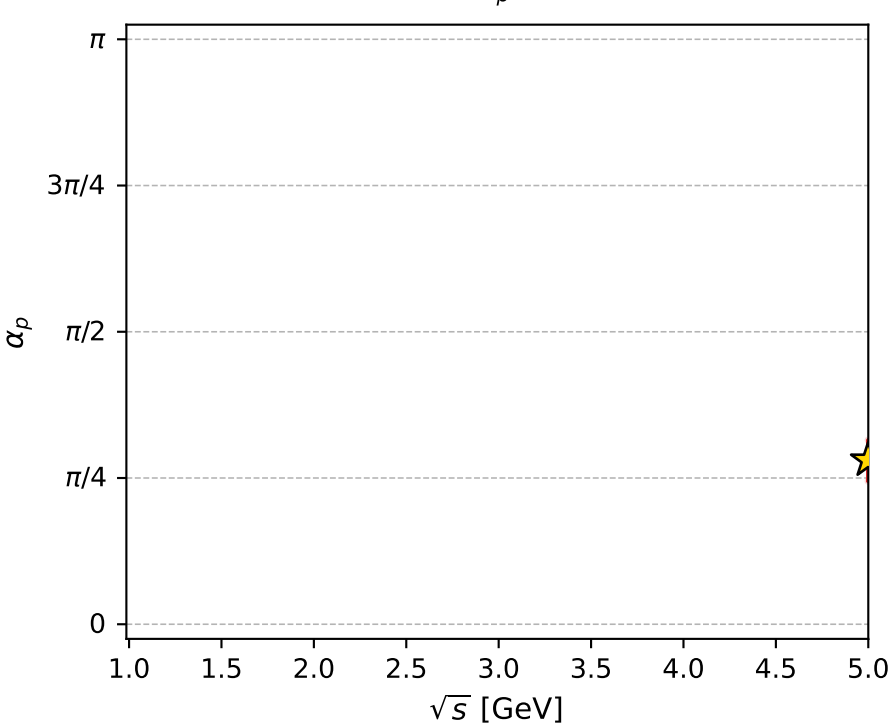
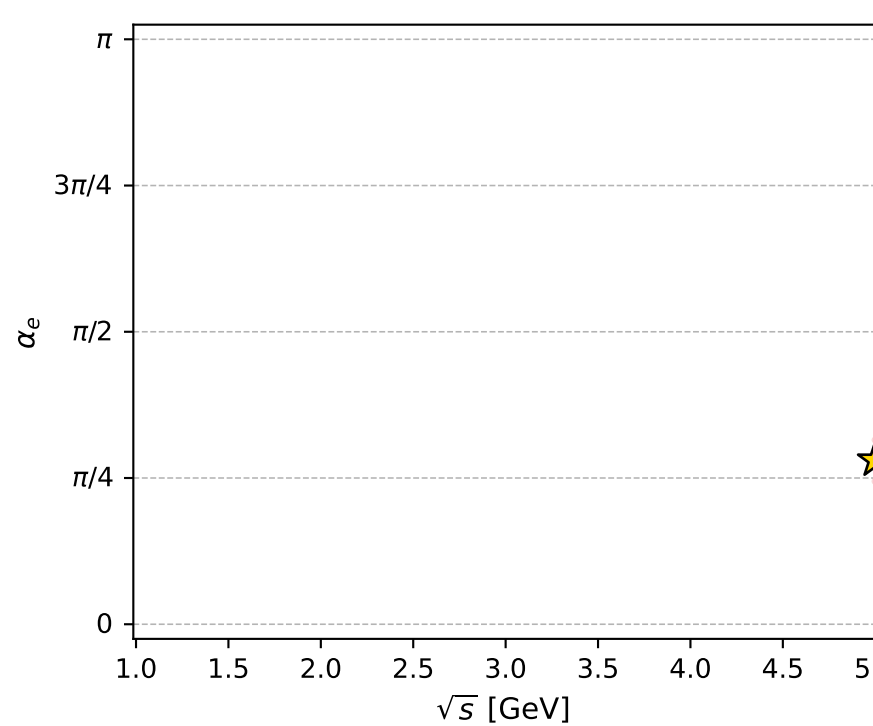
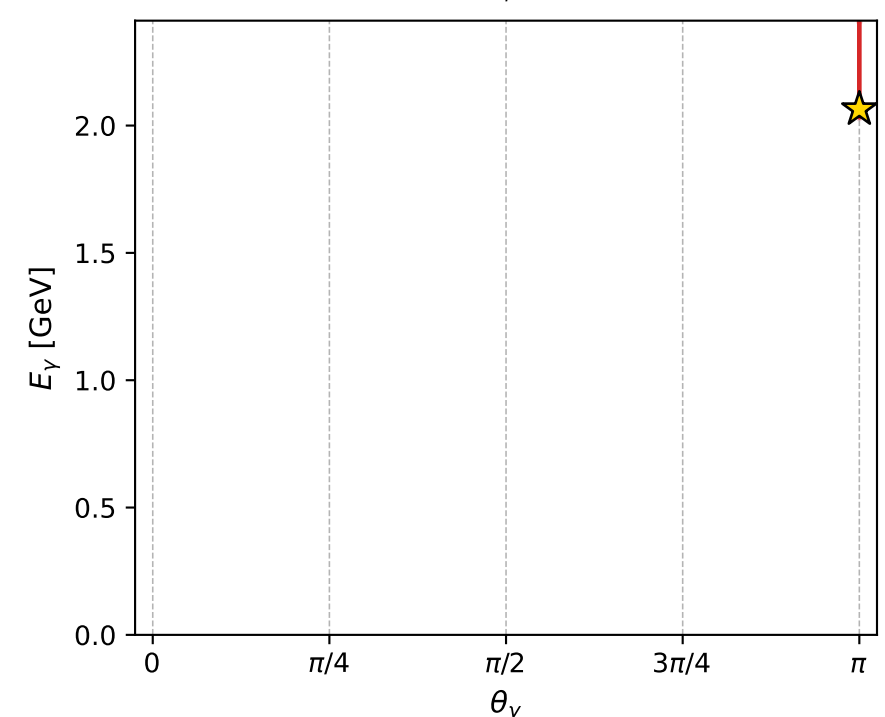
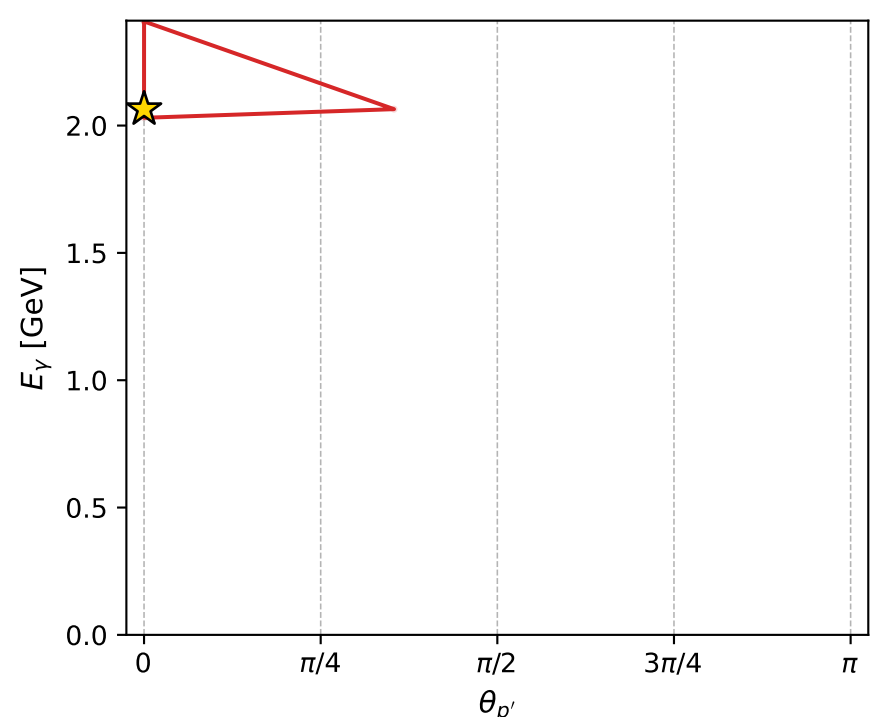
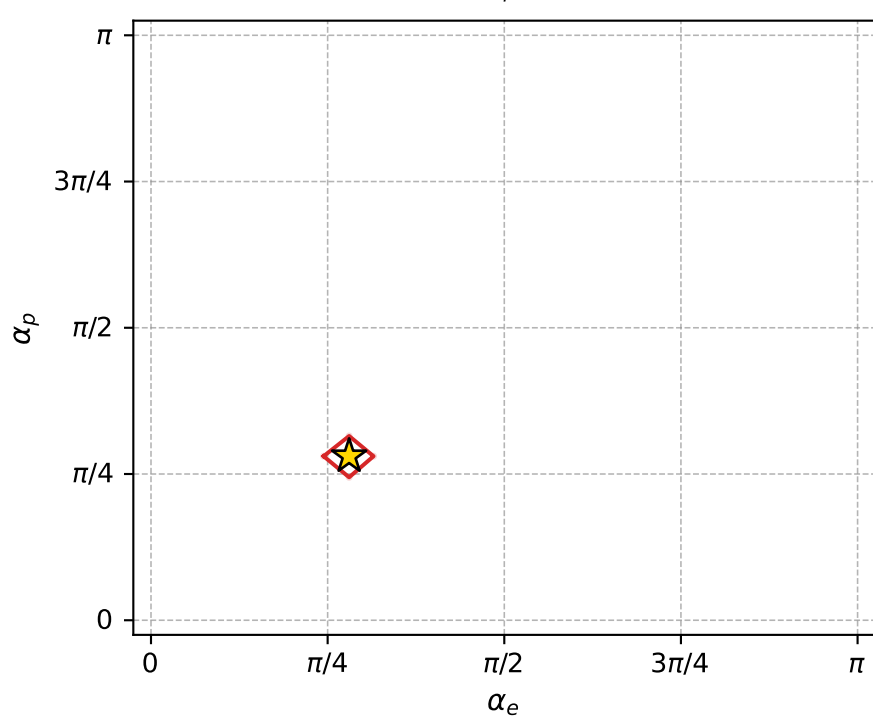
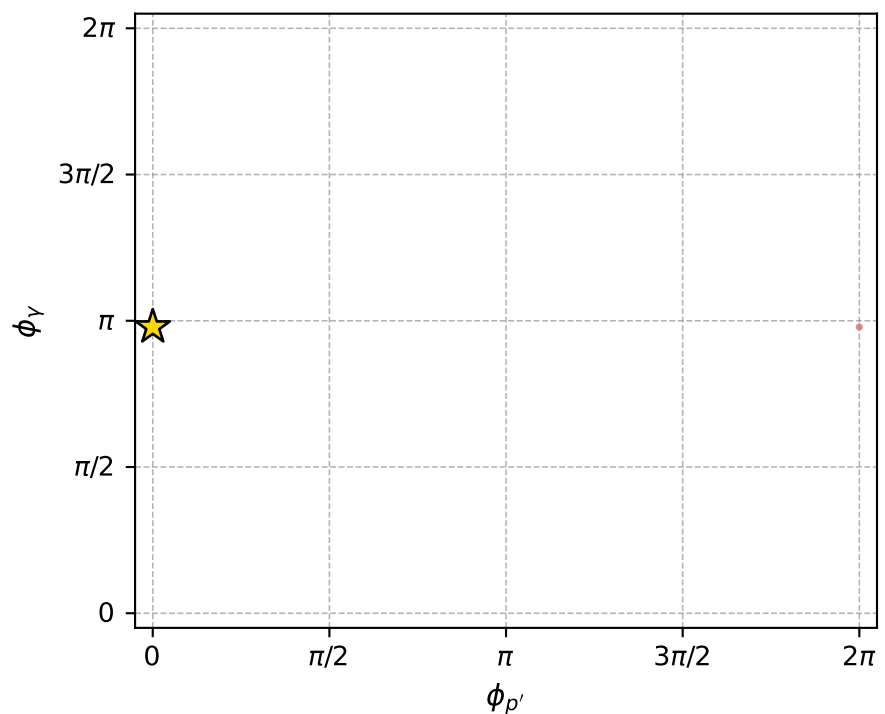
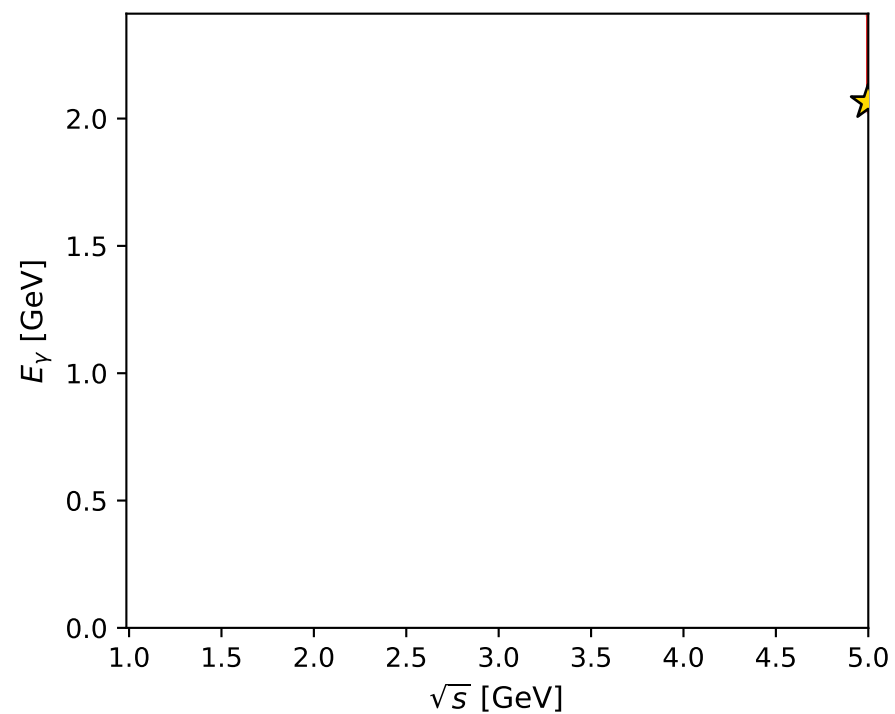
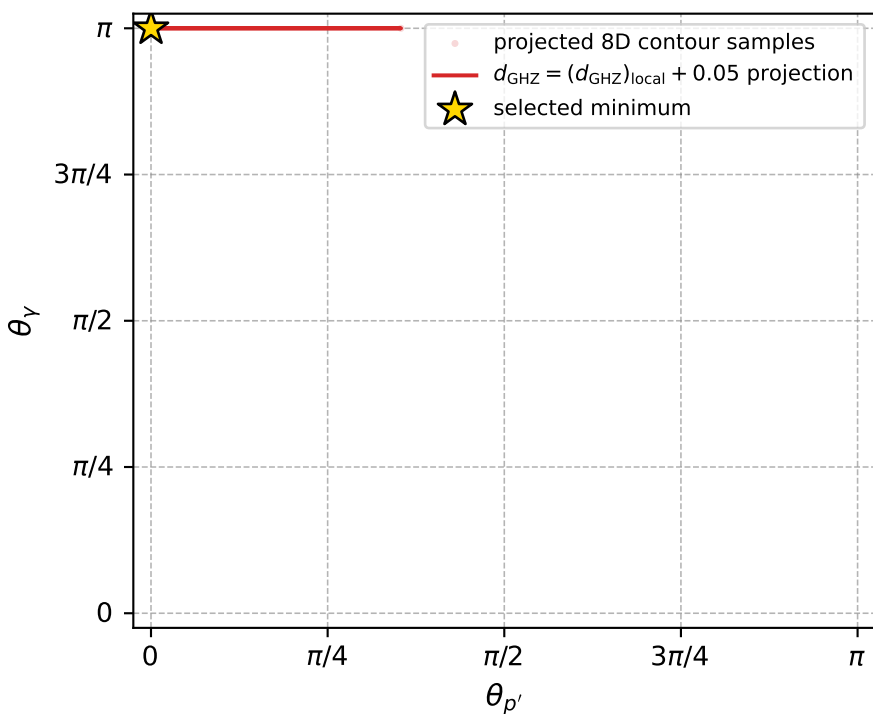
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 61; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7516278\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000028$
 above global minimum = $5.2328164\text{e-}09$
 8D contour samples = 126

selected phase-space point:

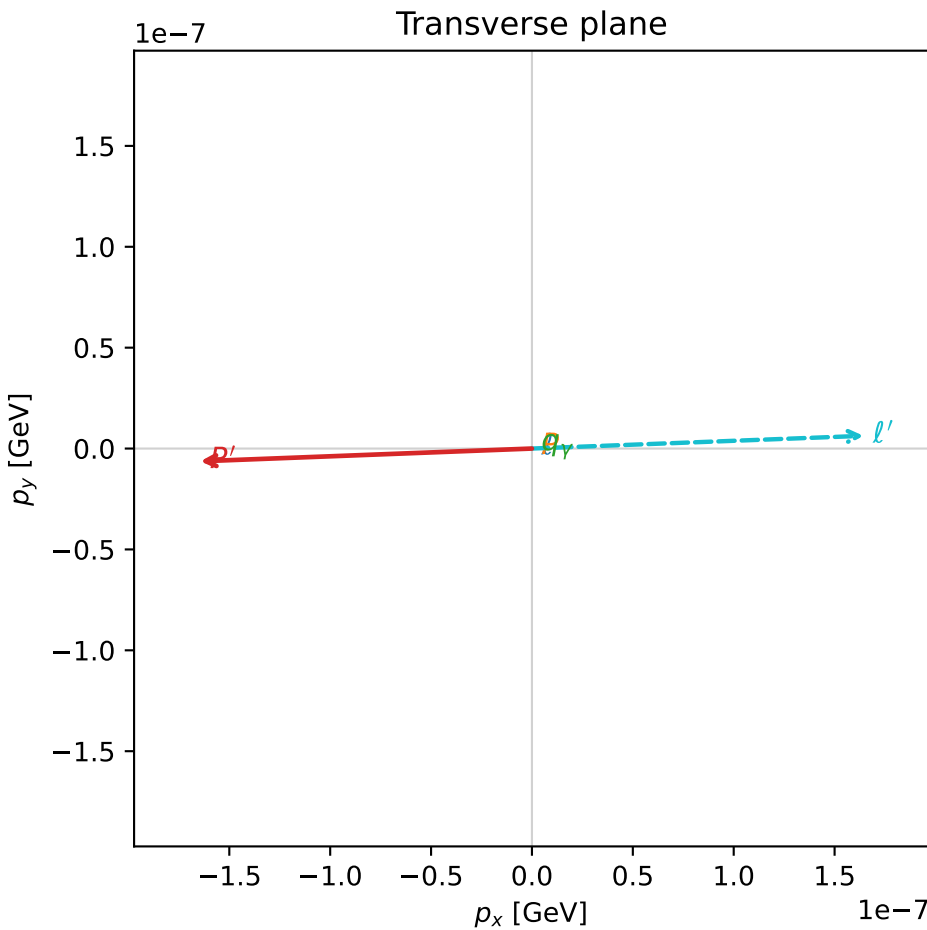
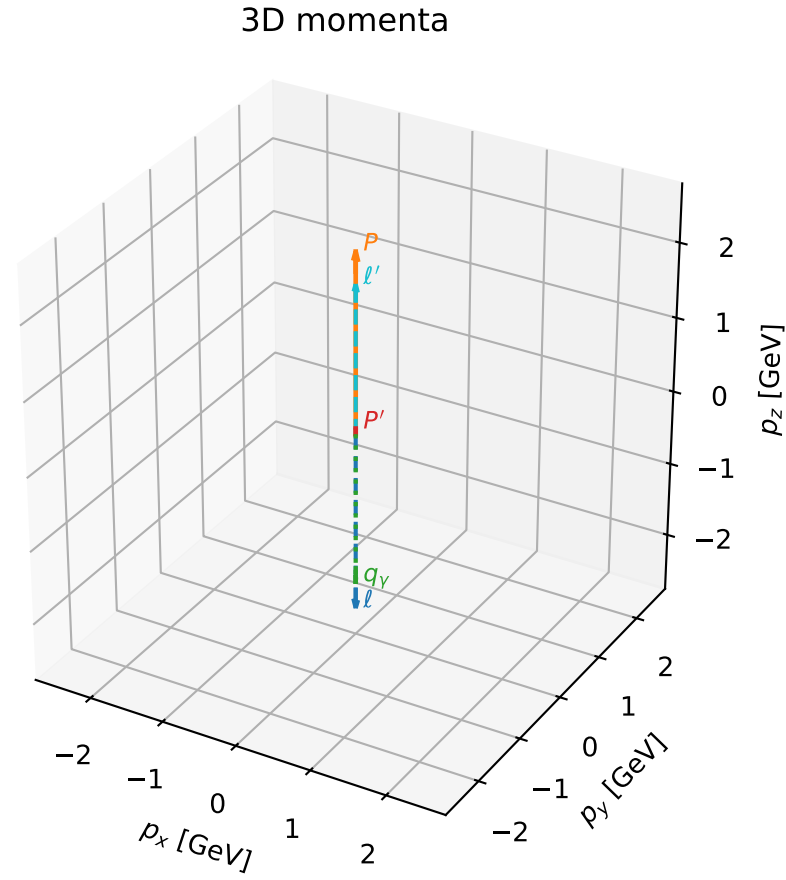
$\text{sqrt_s} = 4.999035$
 $\text{theta_p_out} = 0.0001533981$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.064435$
 $\text{phi_p_out} = 0.0003144157$
 $\text{phi_gamma_out} = 3.075242$
 $\text{alpha_e} = 0.8809308$
 $\text{alpha_p} = 0.8809114$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_65 $d_{\{\text{rm GHZ}\}}=2.78329\text{e-}08$
 region: polarization_cluster_01_local_minimum_65
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.5734809$ rad
 incoming lepton: $0.840017|+\rangle +0.542559|-\rangle$
 proton mixing angle: $\alpha_p=0.57346771$ rad
 incoming proton: $0.840025|+\rangle +0.542548|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0755206$
 $\theta_{P'}=2.17755\text{e-}06$, $\theta_Y=3.14159$, $\phi_{P'}=3.18$, $\phi_Y=2.24128$
 $E_Y=2.06724$, $\theta_{\ell'}=8.1617\text{e-}08$, $\phi_{\ell'}=0.0384033$
 $Q^2=19.2163$, $x_B=0.820535$, $t=-2.74677$, $W^2=5.08278$, $y=0.970939$

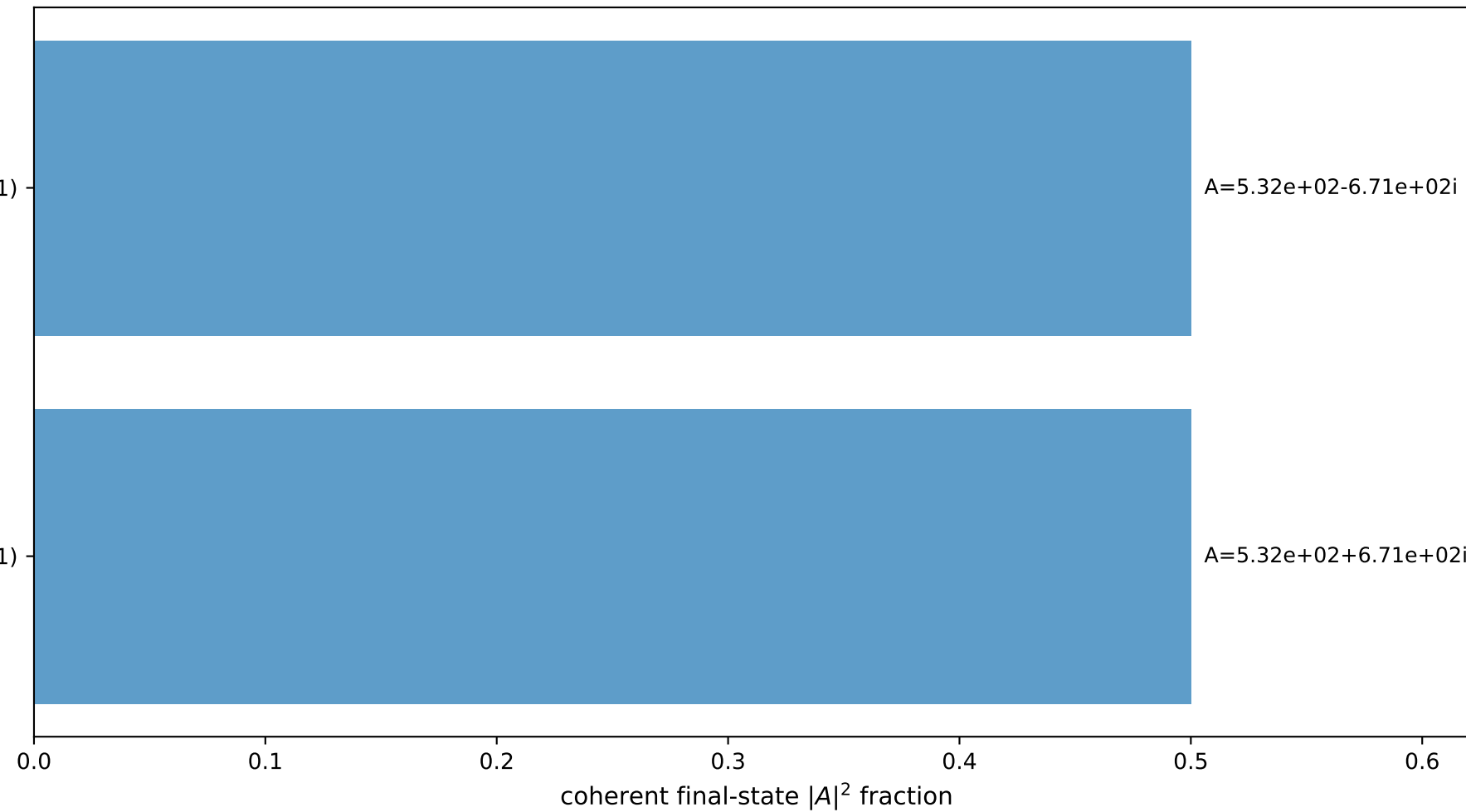
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9917$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9917$
 P' $E= 0.9410$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0755$
 q_Y $E= 2.0672$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0672$



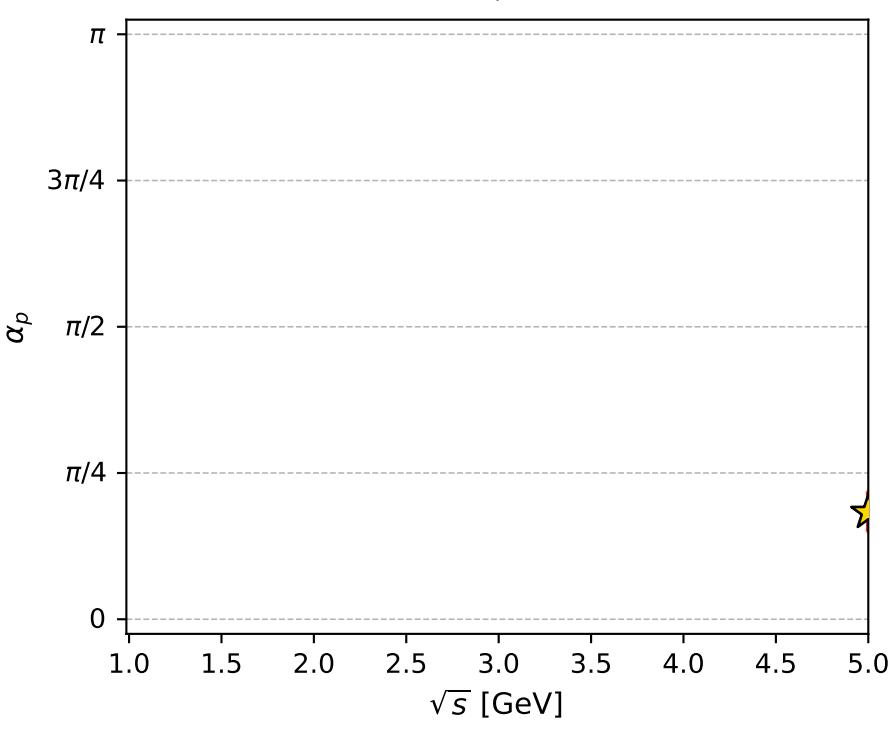
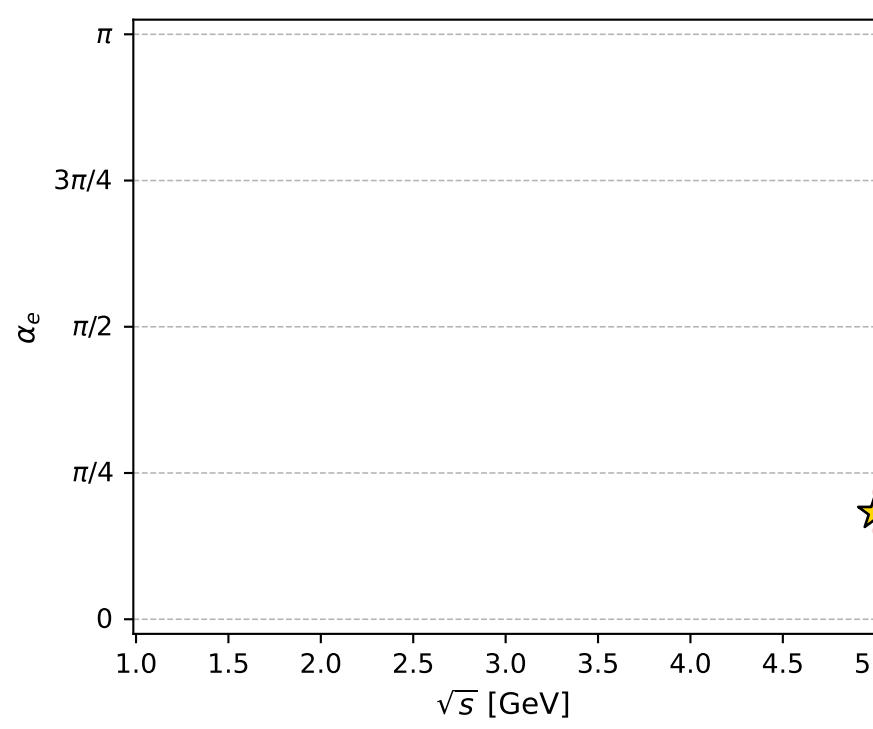
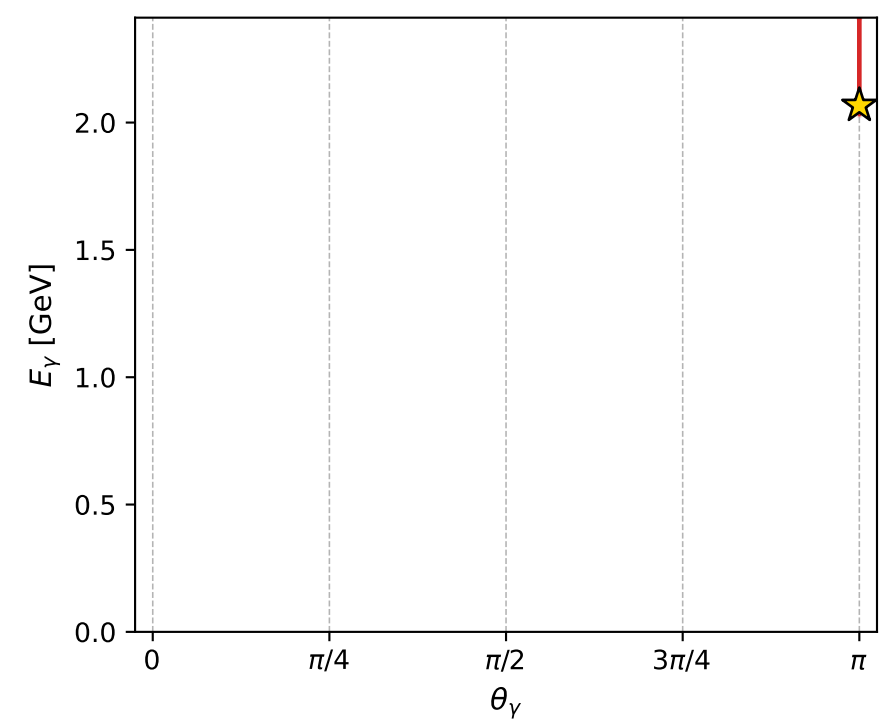
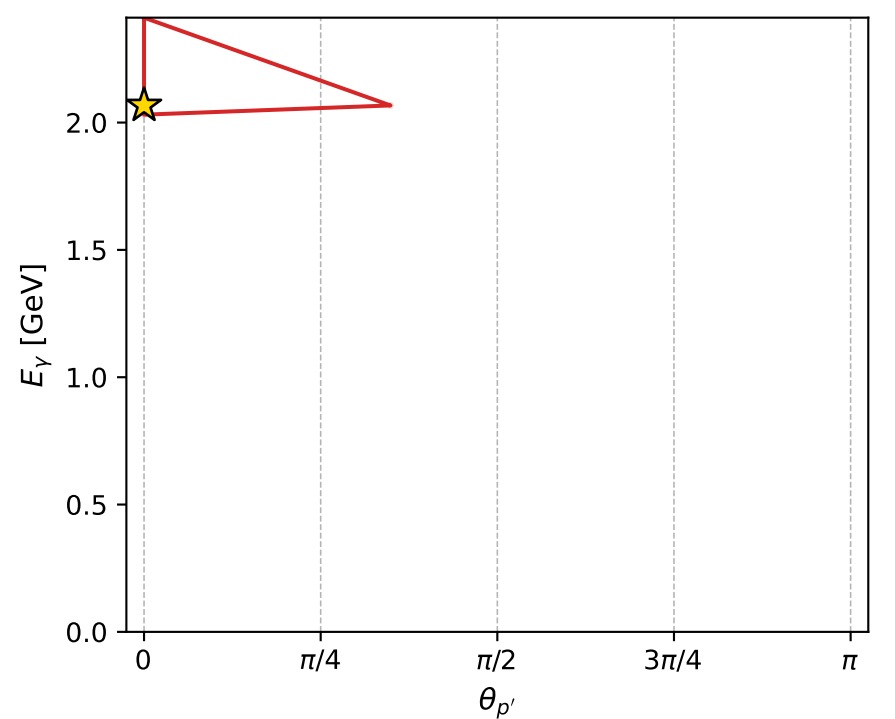
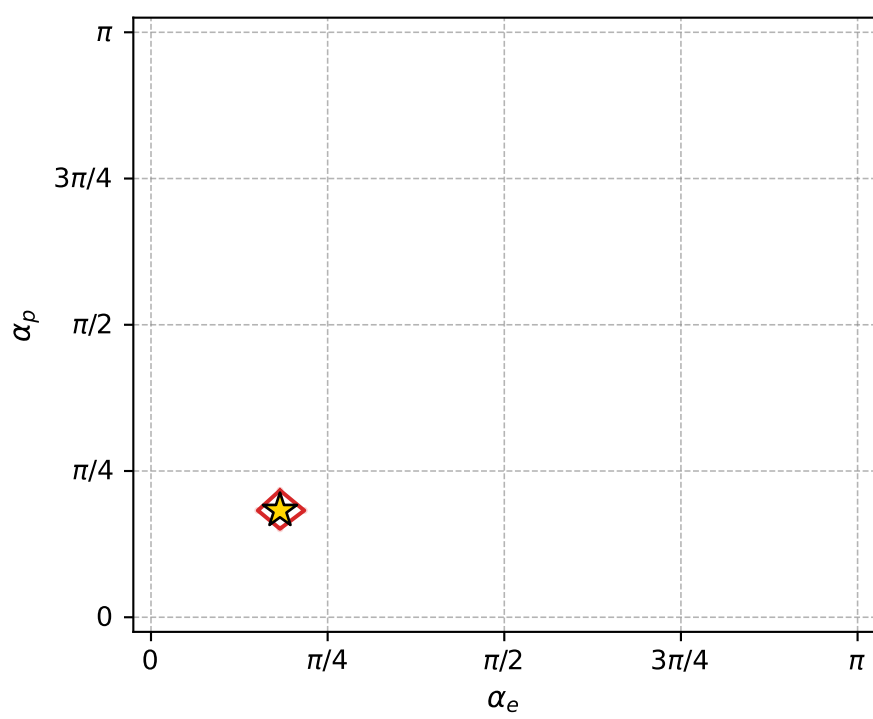
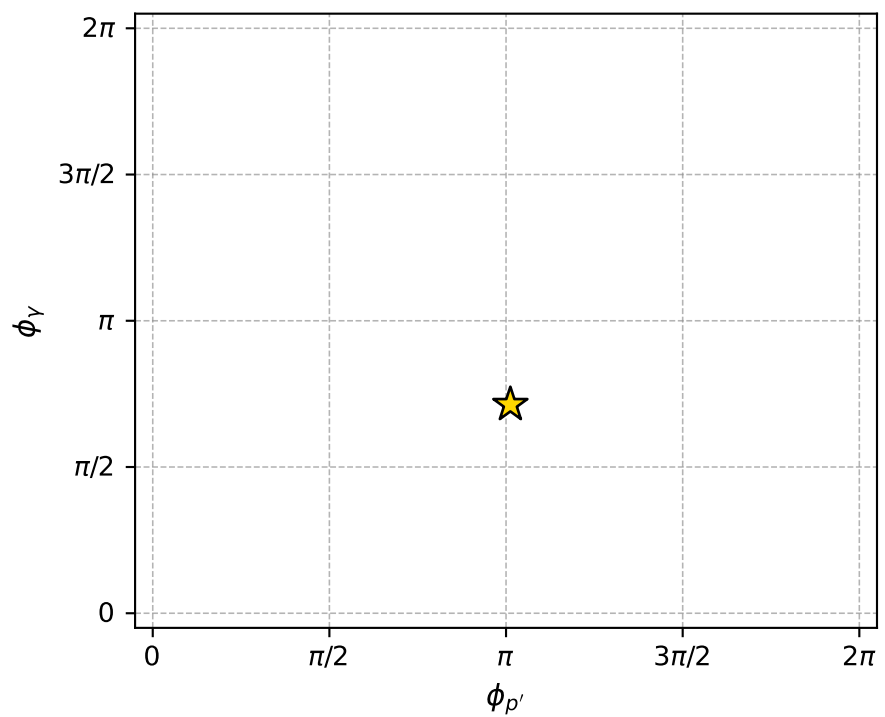
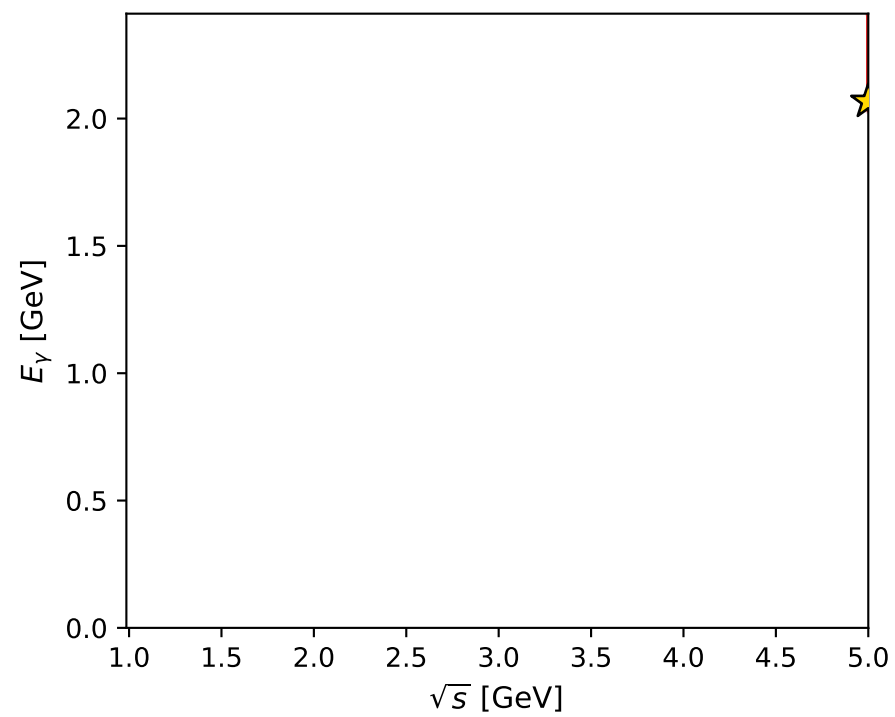
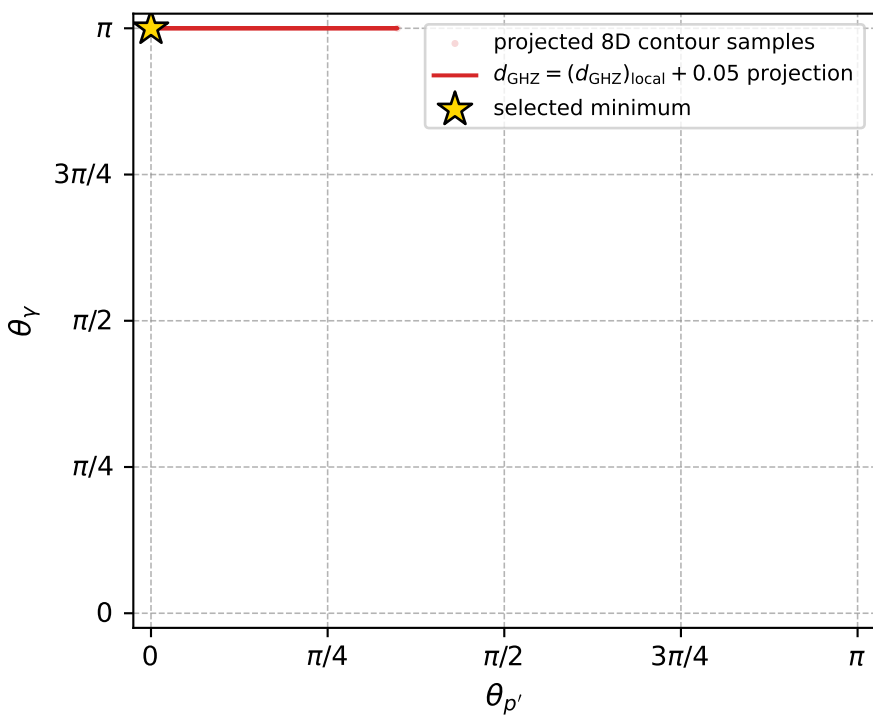
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 65; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.7832909\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000028$
 above global minimum = $5.5494468\text{e-}09$
 8D contour samples = 41

selected phase-space point:

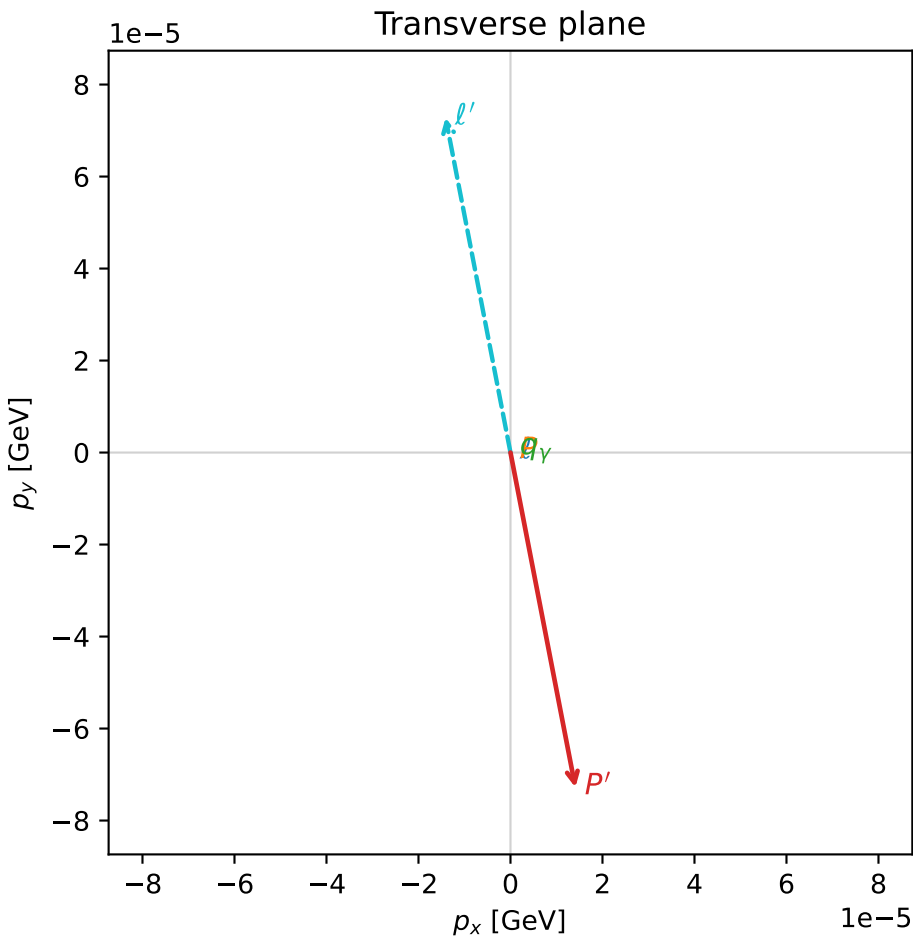
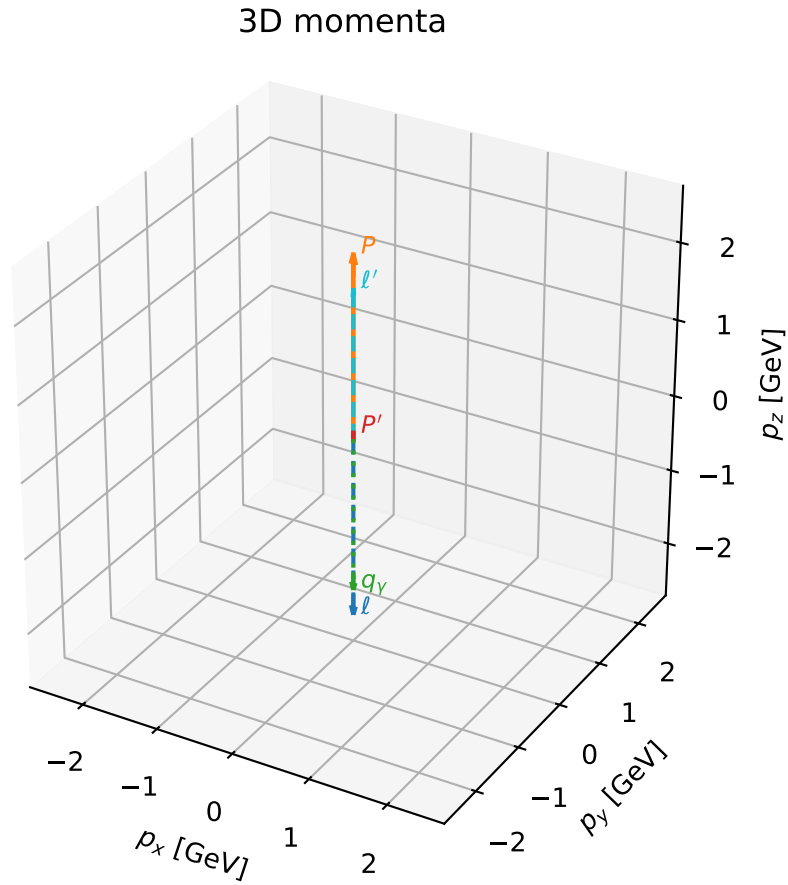
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 2.177553\text{e-}06$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.067242$
 $\text{phi_p_out} = 3.179996$
 $\text{phi_gamma_out} = 2.241283$
 $\text{alpha_e} = 0.5734809$
 $\text{alpha_p} = 0.5734677$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_69 $d_{\{\text{rm GHZ}\}}=2.83483\text{e-}08$
 region: polarization_cluster_01_local_minimum_69
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82158482$ rad
 incoming lepton: $0.681062|+\rangle +0.732226|-\rangle$
 proton mixing angle: $\alpha_p=0.82156325$ rad
 incoming proton: $0.681077|+\rangle +0.732211|-\rangle$

$s=24.1272$, $\sqrt{s}=4.91194$, $|\vec{P}|=2.36641$, $|\vec{P}'|=0.080731$
 $\theta_{P'}=0.00091855$, $\theta_Y=3.14159$, $\phi_{P'}=4.90382$, $\phi_Y=3.13744$
 $E_Y=2.0256$, $\theta_{\ell'}=3.81287\text{e-}05$, $\phi_{\ell'}=1.76223$
 $Q^2=18.4095$, $x_B=0.816362$, $t=-2.6513$, $W^2=5.02098$, $y=0.970029$

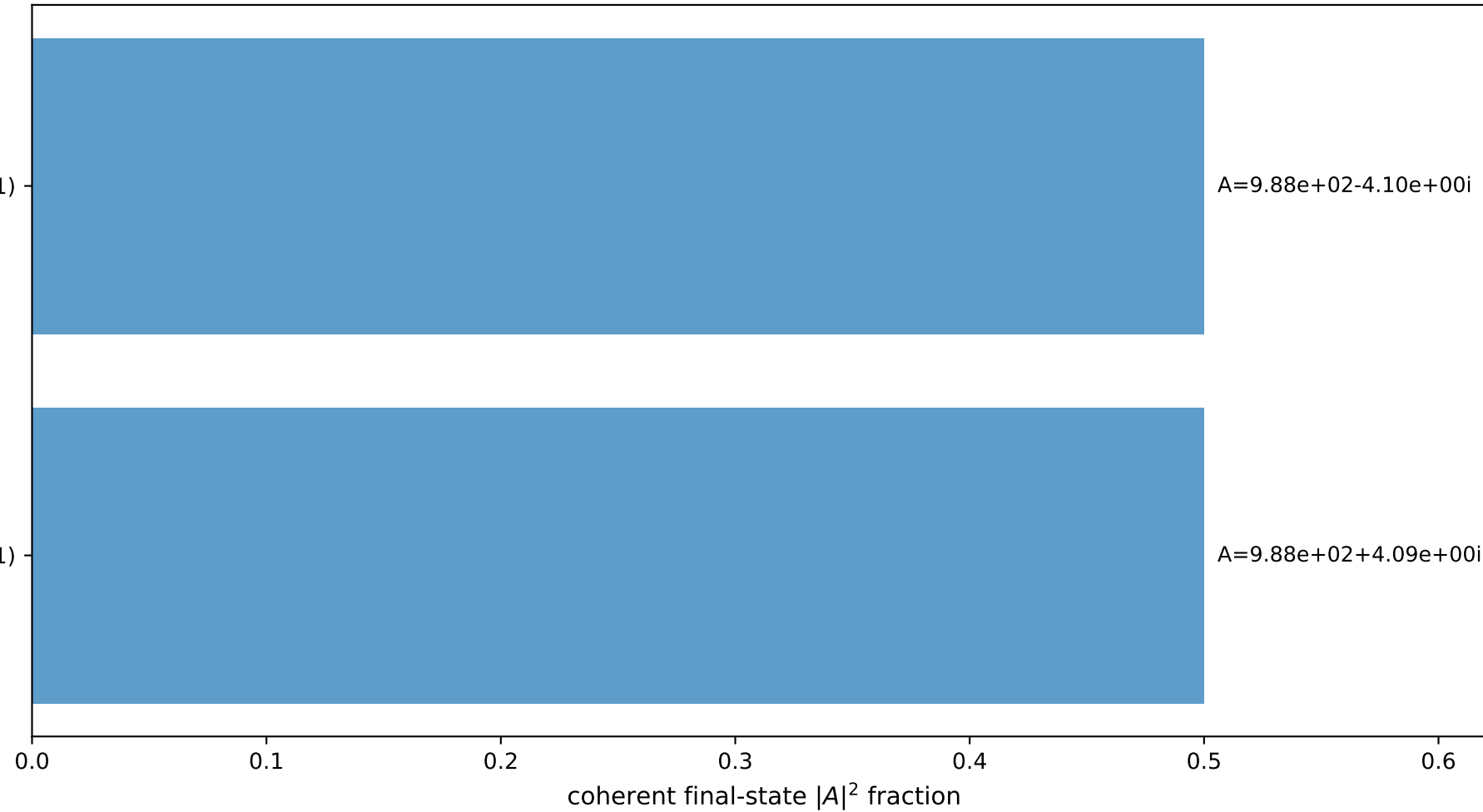
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3664$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3664$
 P $E= 2.5455$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3664$
 ℓ' $E= 1.9449$ $p_x= -0.0000$ $p_y= 0.0001$ $p_z= 1.9449$
 P' $E= 0.9415$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= 0.0807$
 q_Y $E= 2.0256$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0256$



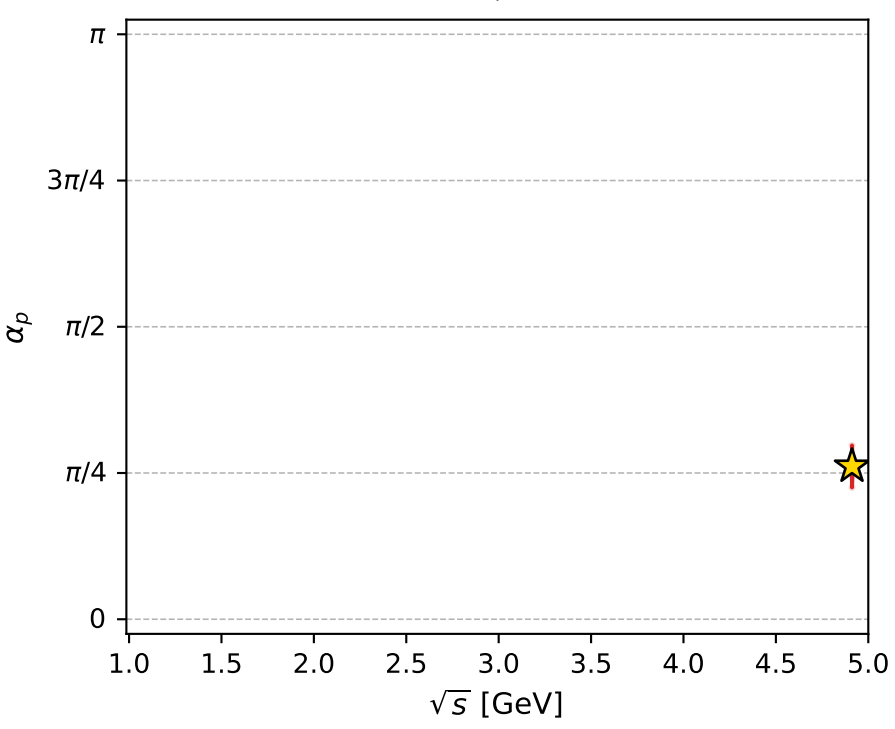
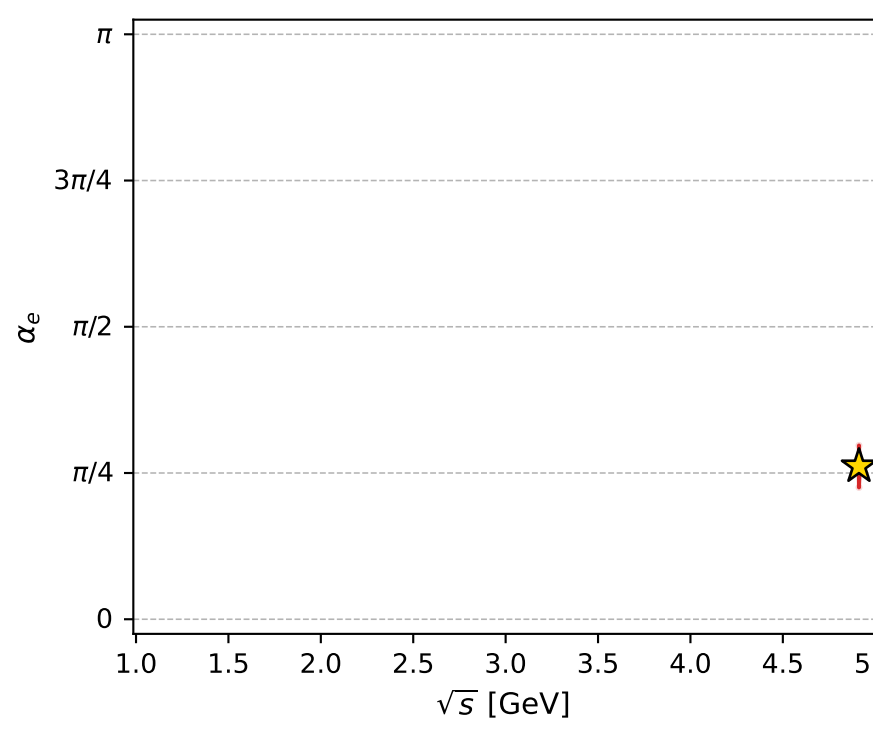
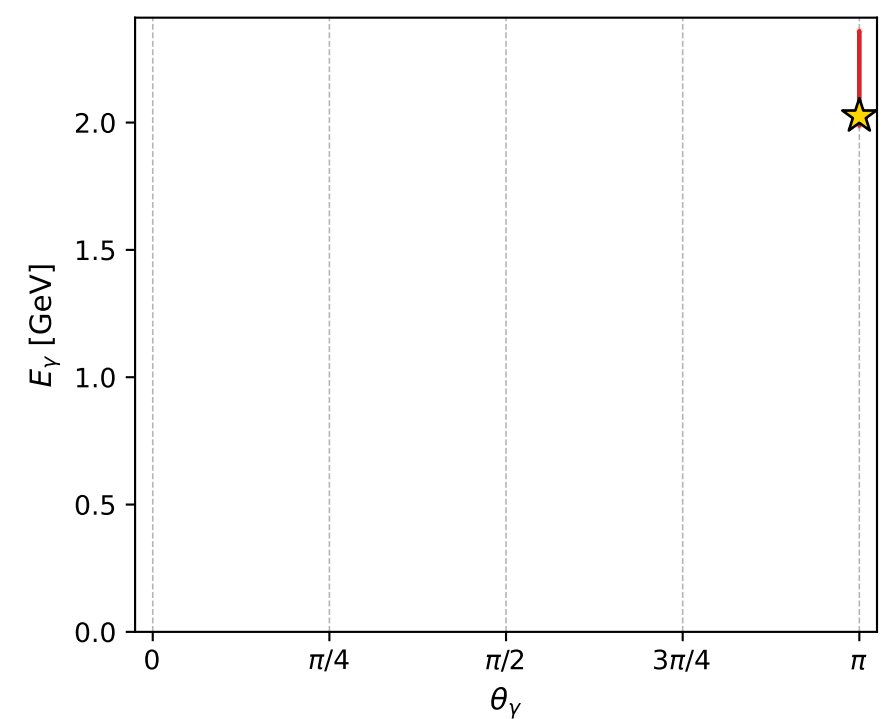
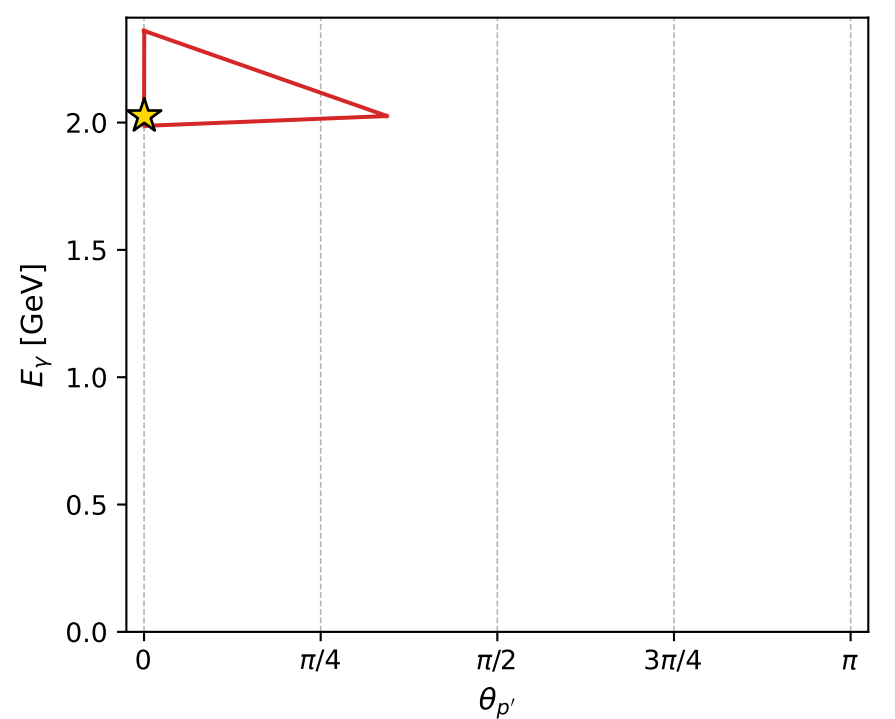
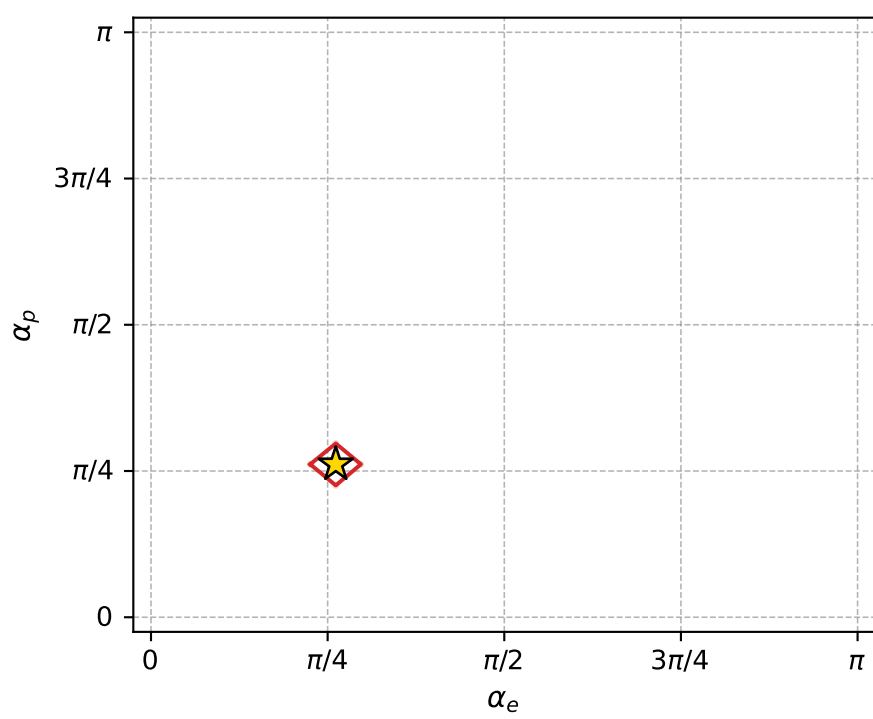
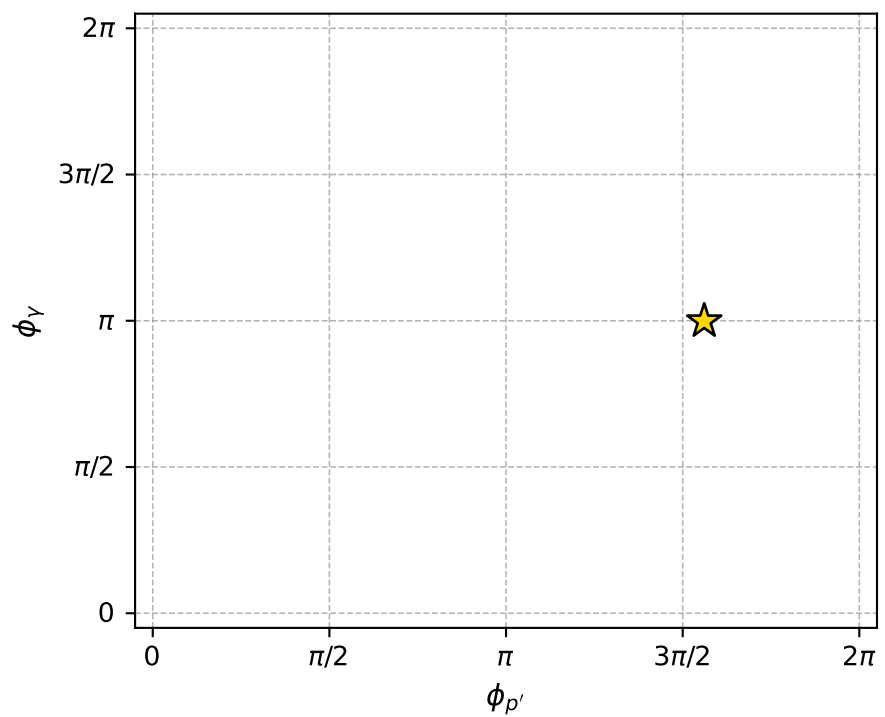
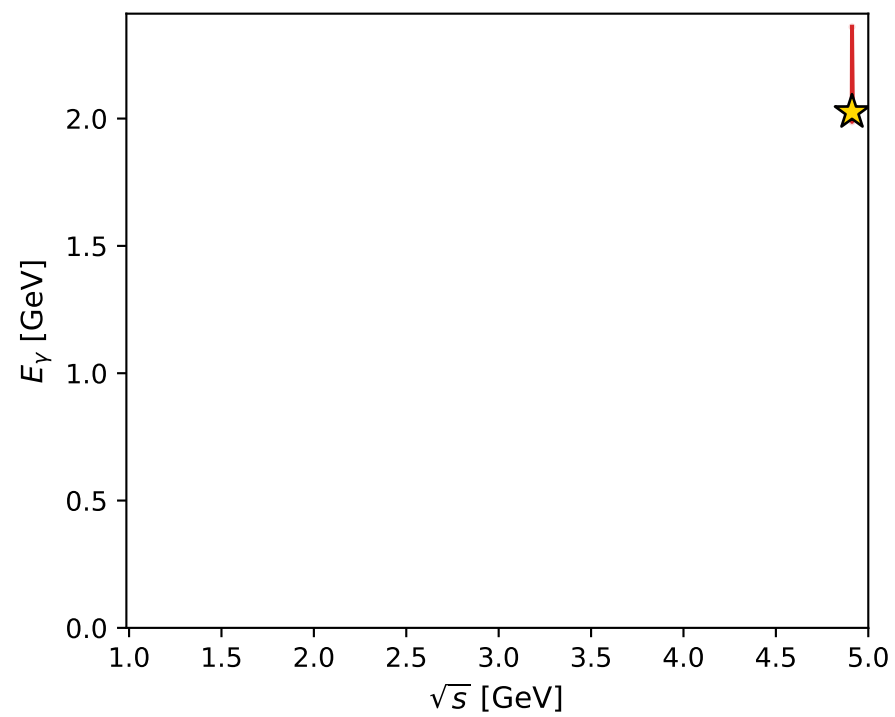
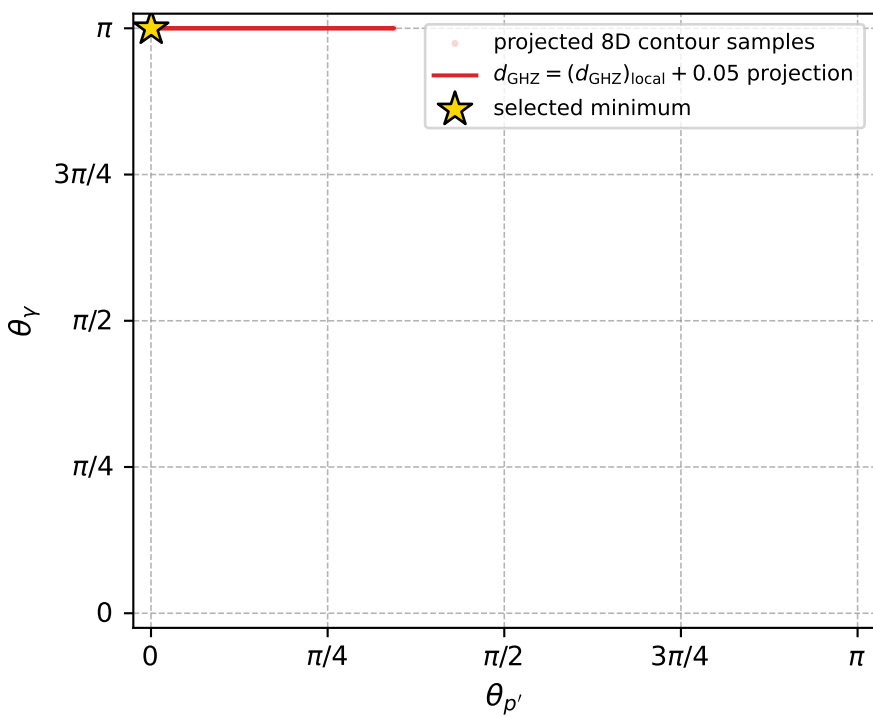
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 69; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.8348331\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000028$
 above global minimum = $6.0648693\text{e-}09$
 8D contour samples = 133

selected phase-space point:

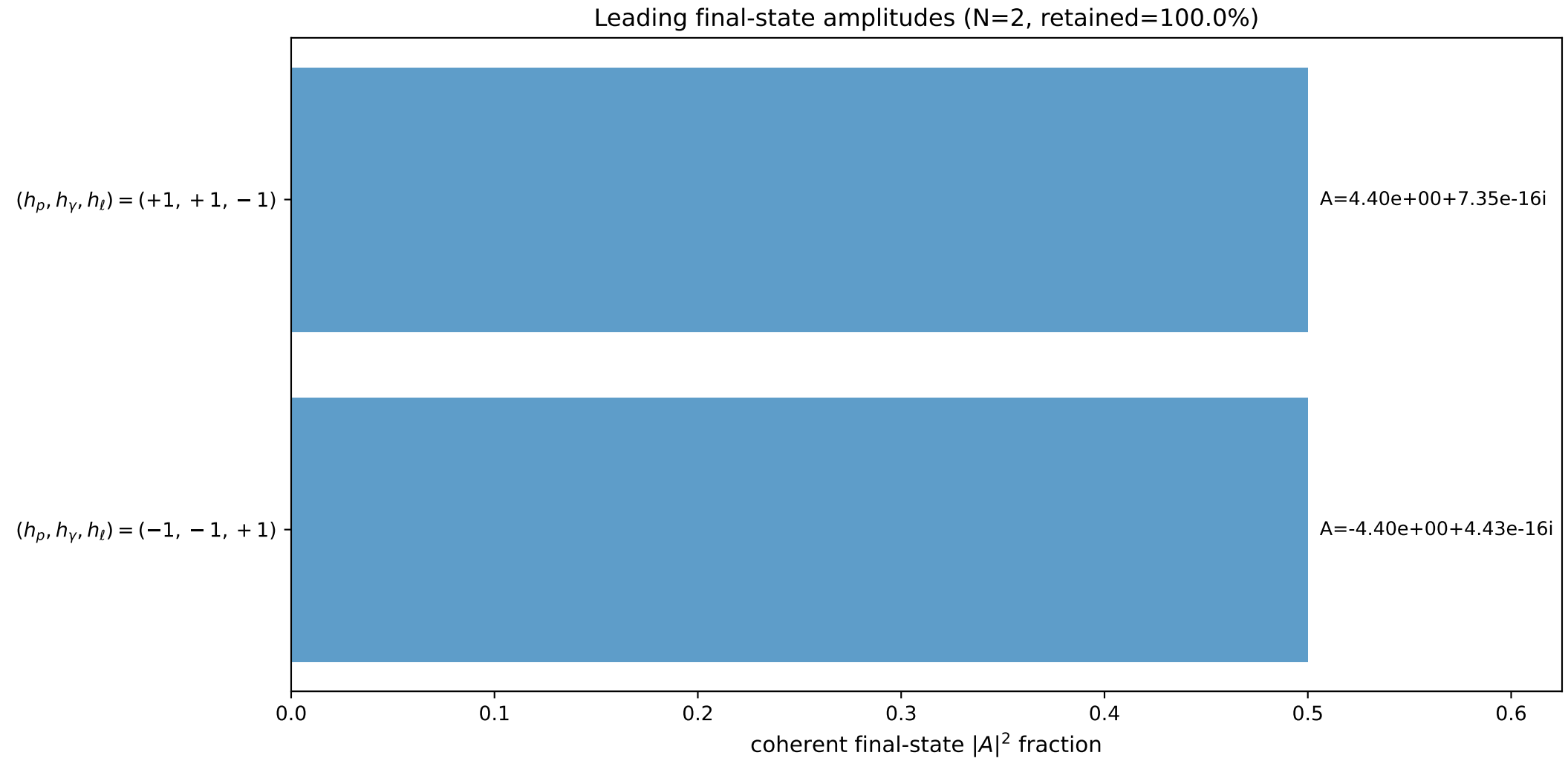
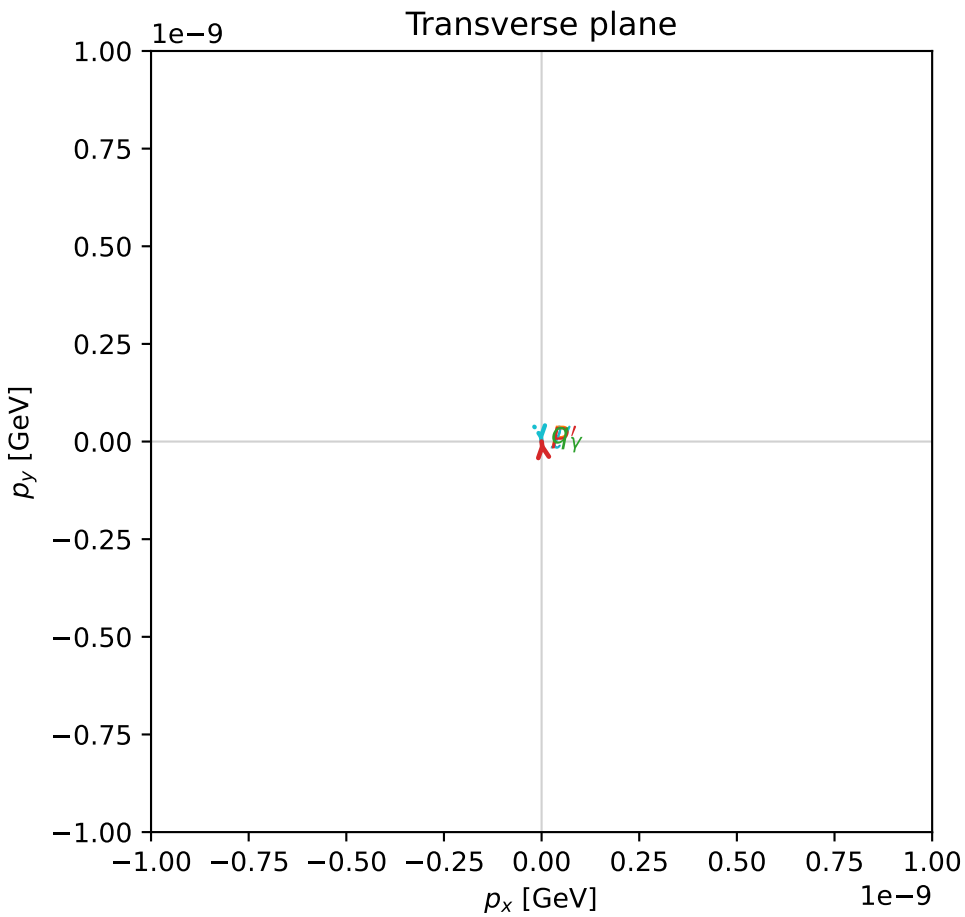
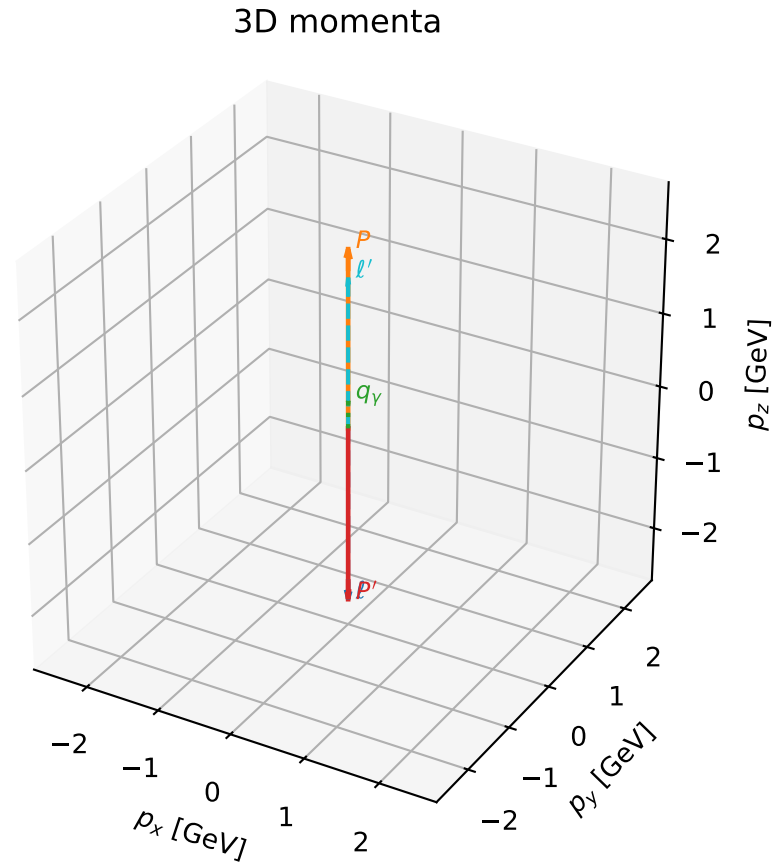
$\text{sqrt_s} = 4.91192$
 $\text{theta_p_out} = 0.0009185497$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.025603$
 $\text{phi_p_out} = 4.903824$
 $\text{phi_gamma_out} = 3.137445$
 $\text{alpha_e} = 0.8215848$
 $\text{alpha_p} = 0.8215632$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

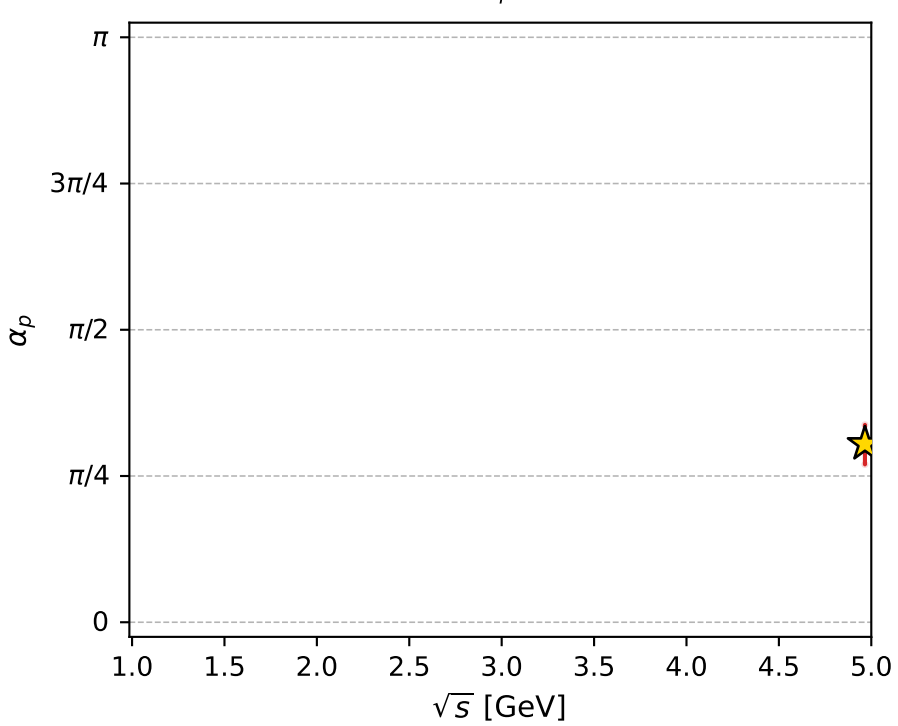
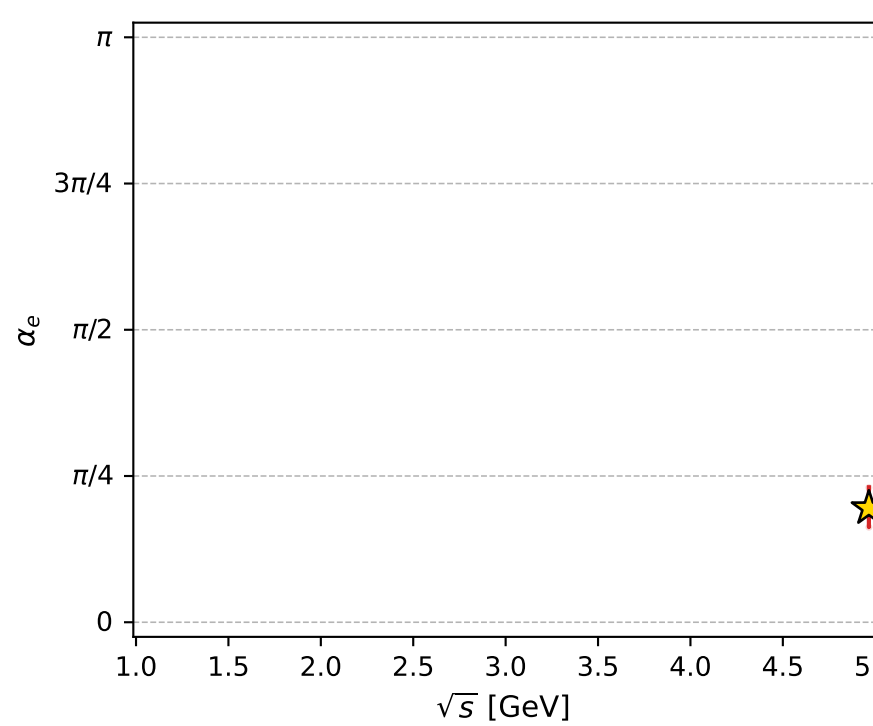
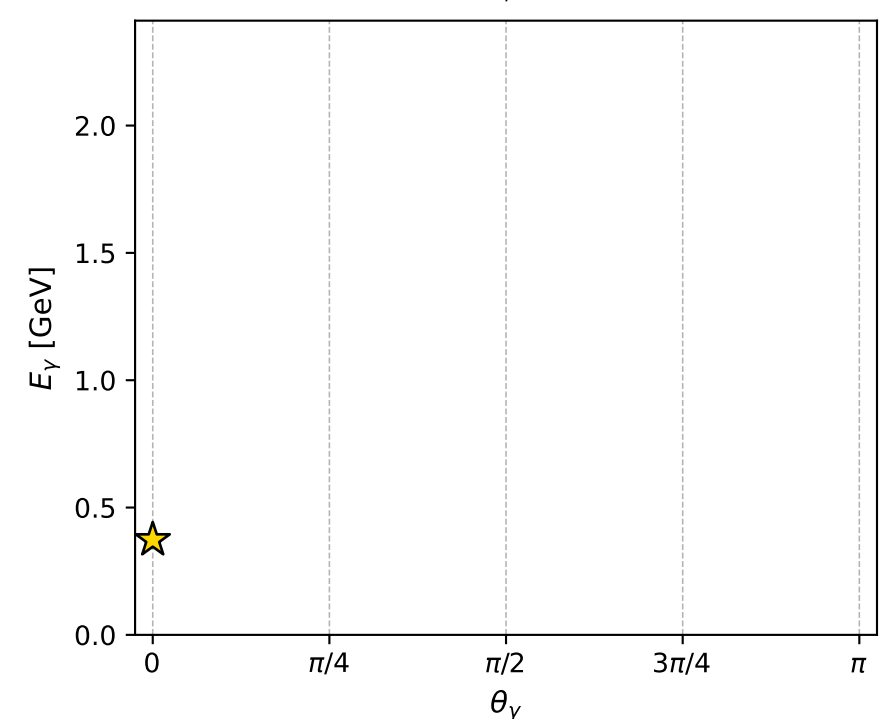
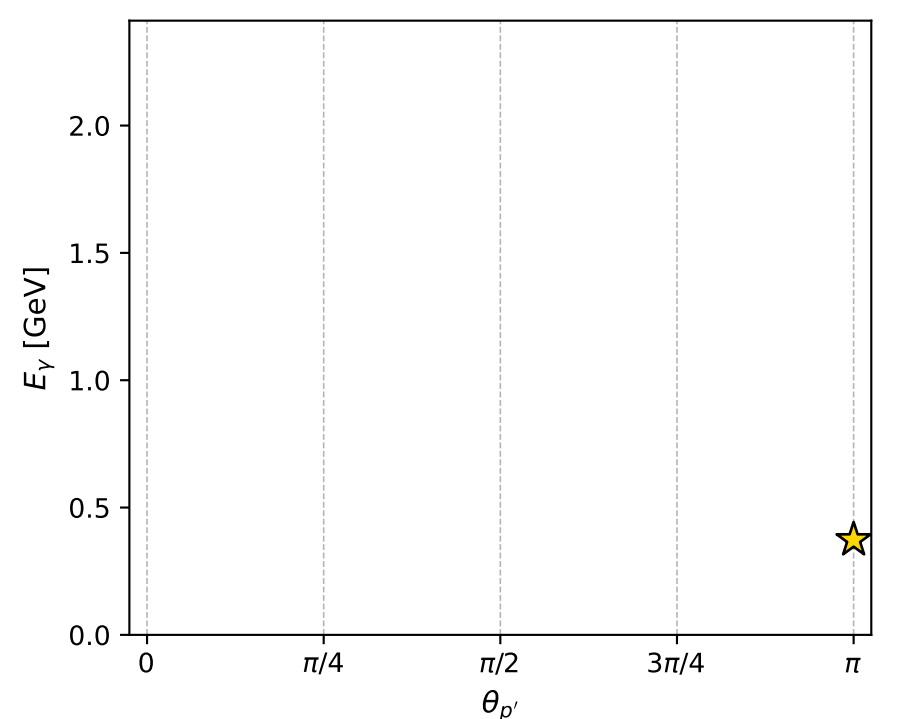
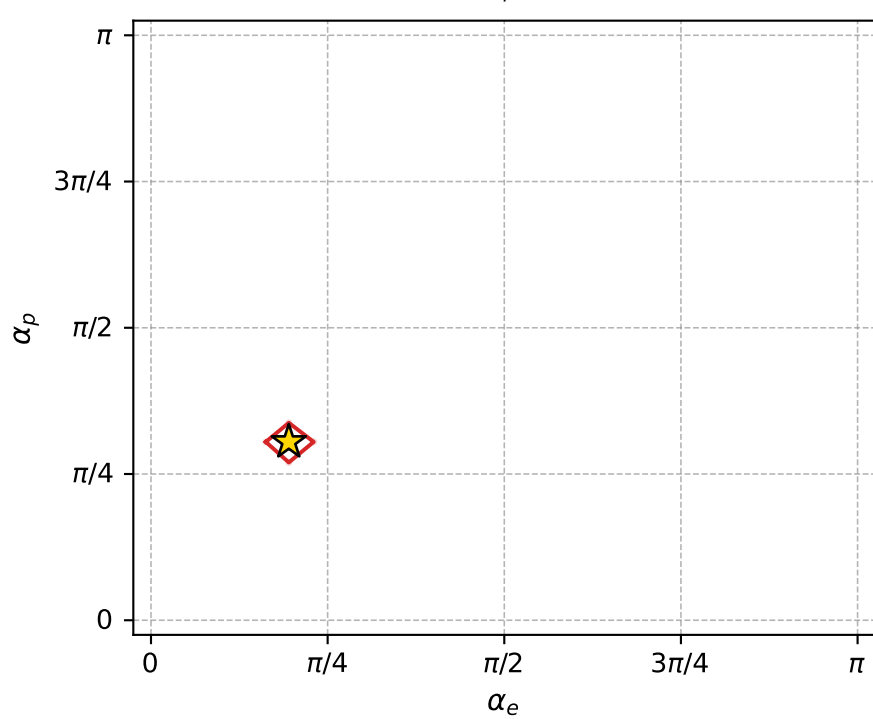
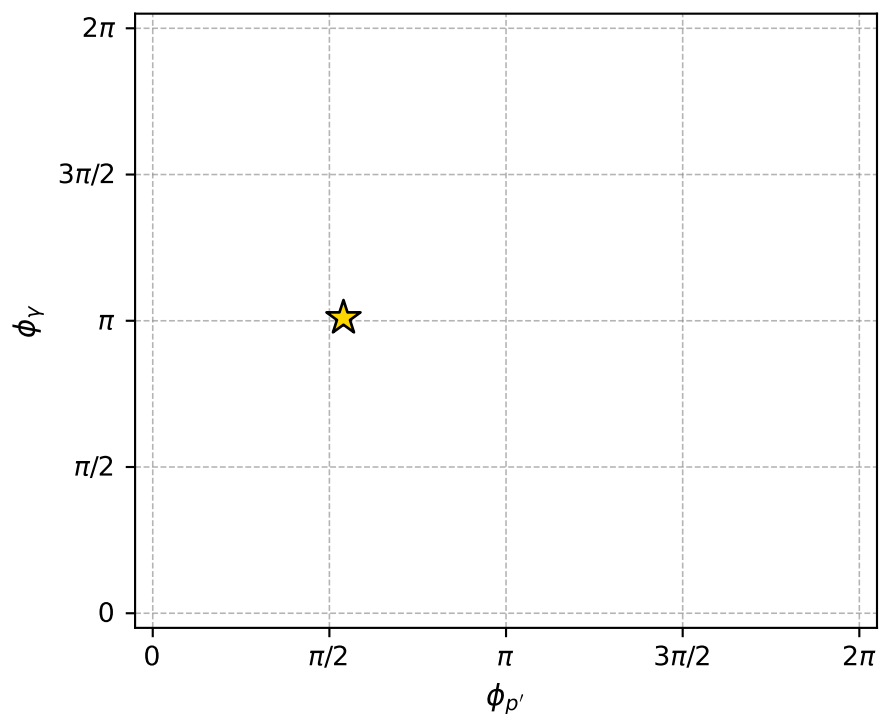
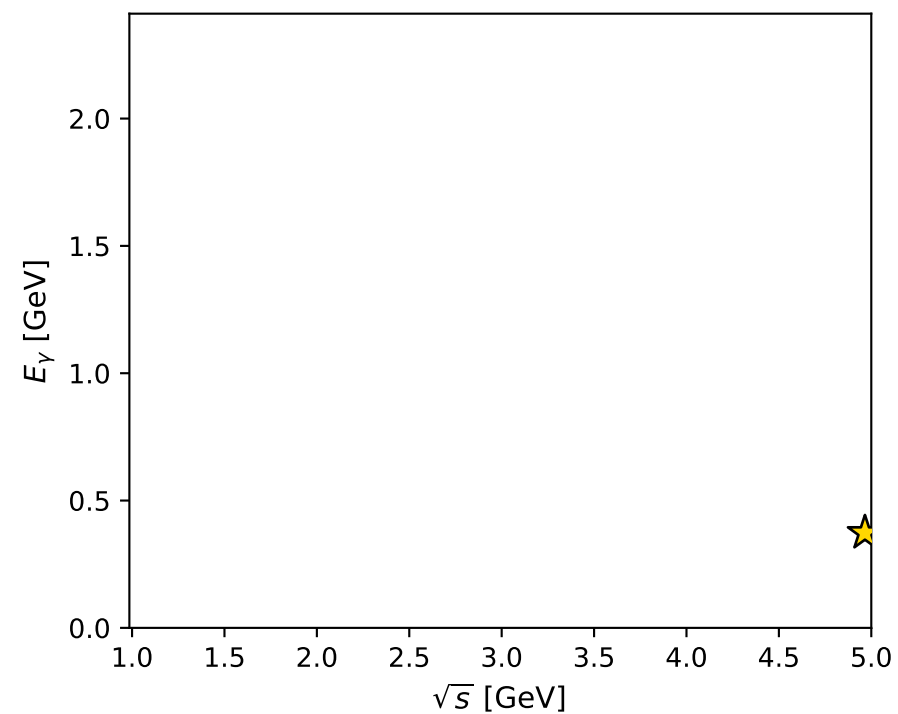
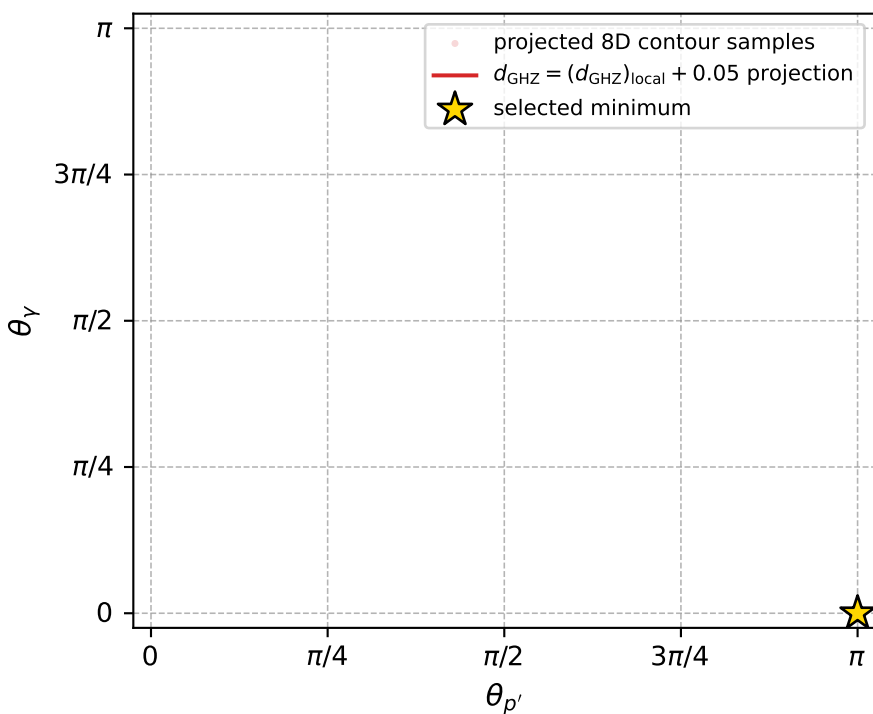
dghz_mixing_angles_polarization_01_local_minimum_71 d_{\rm GHZ}=2.87516e-08
region: polarization_cluster_01_local_minimum_71
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.61281827$ rad
incoming lepton: $0.81803|+\rangle \rightarrow +0.575175|-\rangle$
proton mixing angle: $\alpha_p=0.95795616$ rad
incoming proton: $0.575193|+\rangle \rightarrow +0.818018|-\rangle$

$s=24.6579$, $\sqrt{s}=4.96567$, $|\vec{P}|=2.39424$, $|\vec{P}'|=2.39424$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=1.69602$, $\phi_Y=3.17247$
 $E_Y=0.373172$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.83761$
 $Q^2=19.3557$, $x_B=0.839297$, $t=-22.9296$, $W^2=4.58594$, $y=0.969879$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3942$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3942$
 P $E= 2.5714$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3942$
 ℓ' $E= 2.0211$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0211$
 P' $E= 2.5714$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.3942$
 q_Y $E= 0.3732$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.3732$



electron: polarization cluster P1, local minimum 71; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.8751643\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000029$
 above global minimum = $6.4681812\text{e-}09$
 8D contour samples = 72

selected phase-space point:

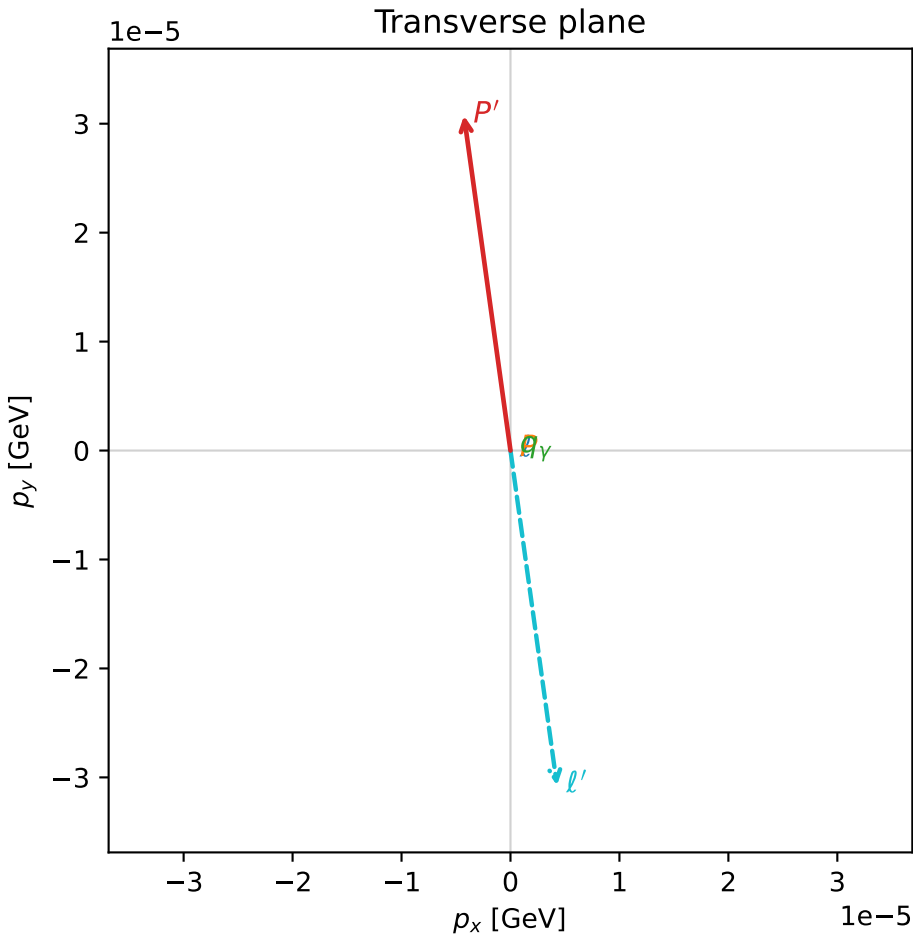
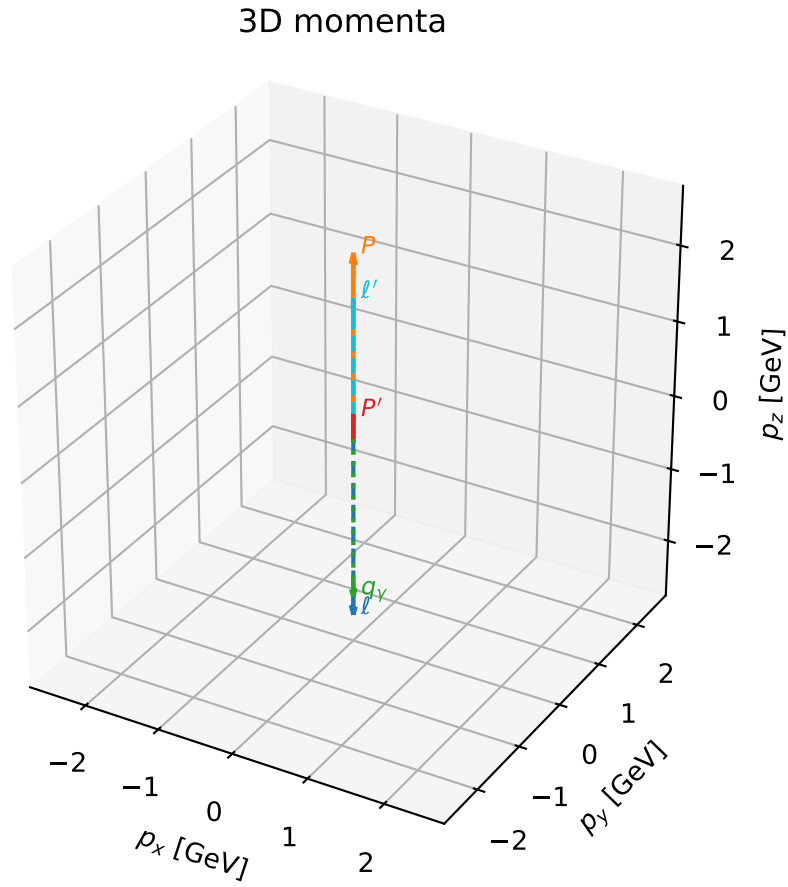
$\text{sqrt_s} = 4.965668$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 0.3731722$
 $\text{phi_p_out} = 1.69602$
 $\text{phi_gamma_out} = 3.172473$
 $\text{alpha_e} = 0.6128183$
 $\text{alpha_p} = 0.9579562$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_72 $d_{\{\text{rm GHZ}\}}=2.90848\text{e-}08$
region: polarization_cluster_01_local_minimum_72
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.79108284$ rad
incoming lepton: $0.703076|+\rangle +0.711115|-\rangle$
proton mixing angle: $\alpha_p=0.7910723$ rad
incoming proton: $0.703083|+\rangle +0.711108|-\rangle$

$s=24.9857$, $\sqrt{s}=4.99857$, $|\vec{P}|=2.41128$, $|\vec{P}'|=0.305039$
 $\theta_{P'}=0.000101732$, $\theta_Y=3.14159$, $\phi_{P'}=1.70875$, $\phi_Y=4.11801$
 $E_Y=2.15863$, $\theta_{\ell'}=1.67417\text{e-}05$, $\phi_{\ell'}=4.85035$
 $Q^2=17.8781$, $x_B=0.762282$, $t=-1.87322$, $W^2=6.45512$, $y=0.97293$

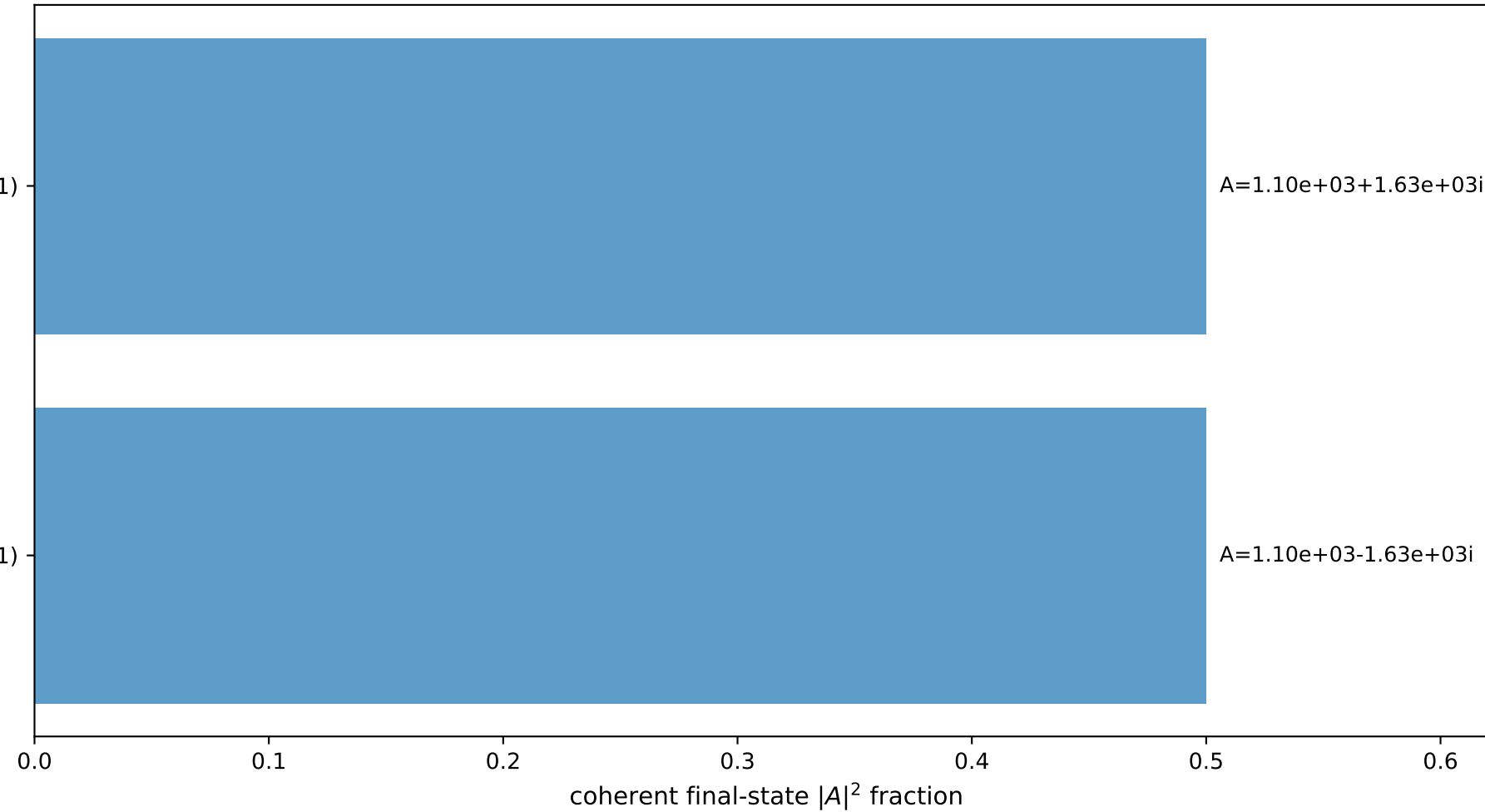
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 1.8536$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.8536$
 P' $E= 0.9864$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3050$
 q_Y $E= 2.1586$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1586$



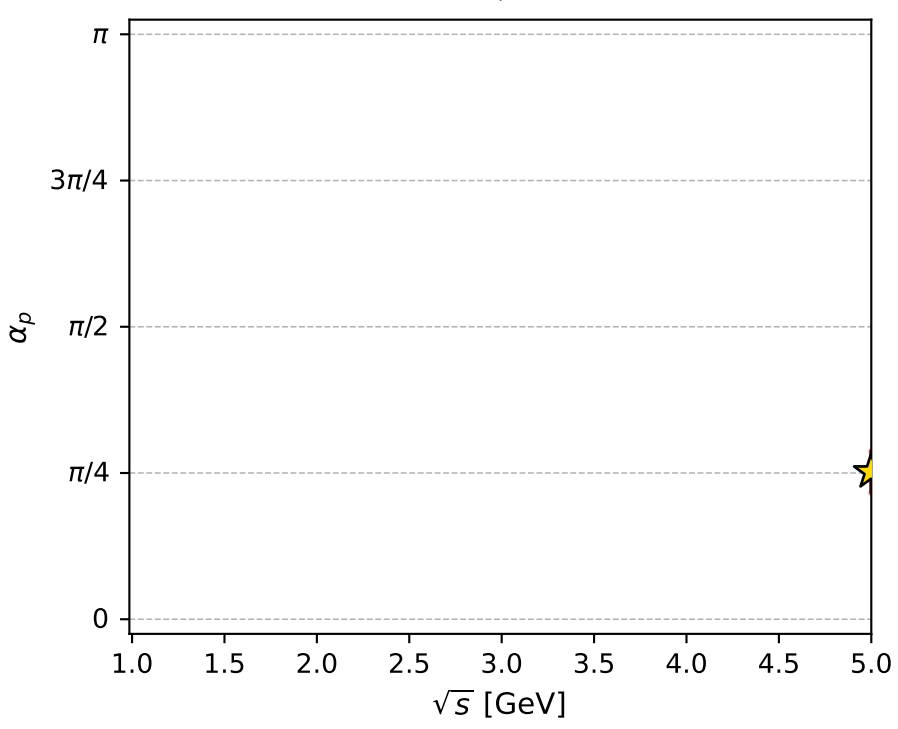
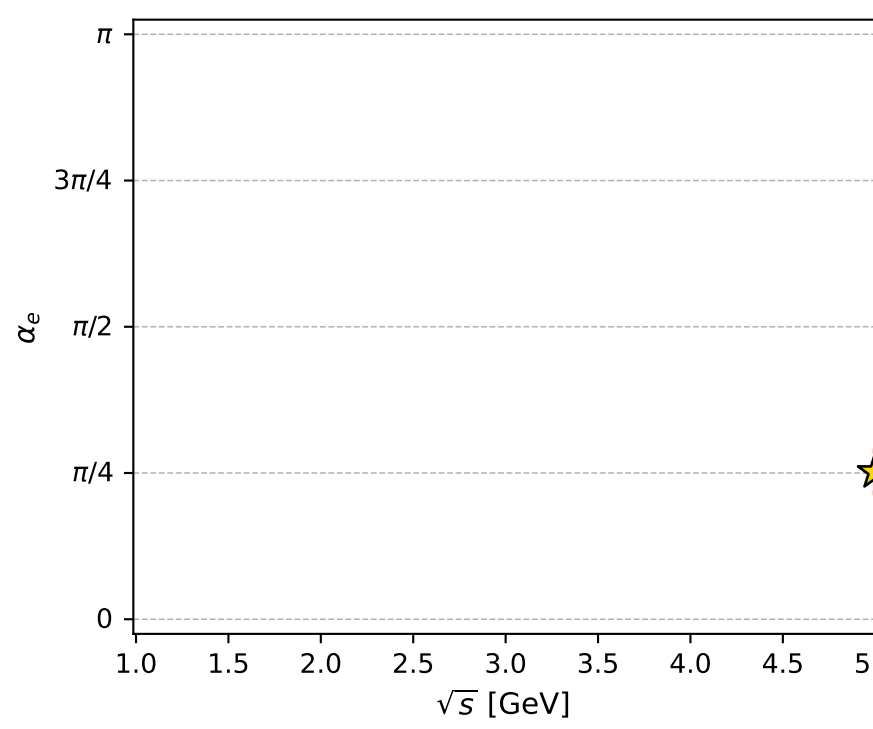
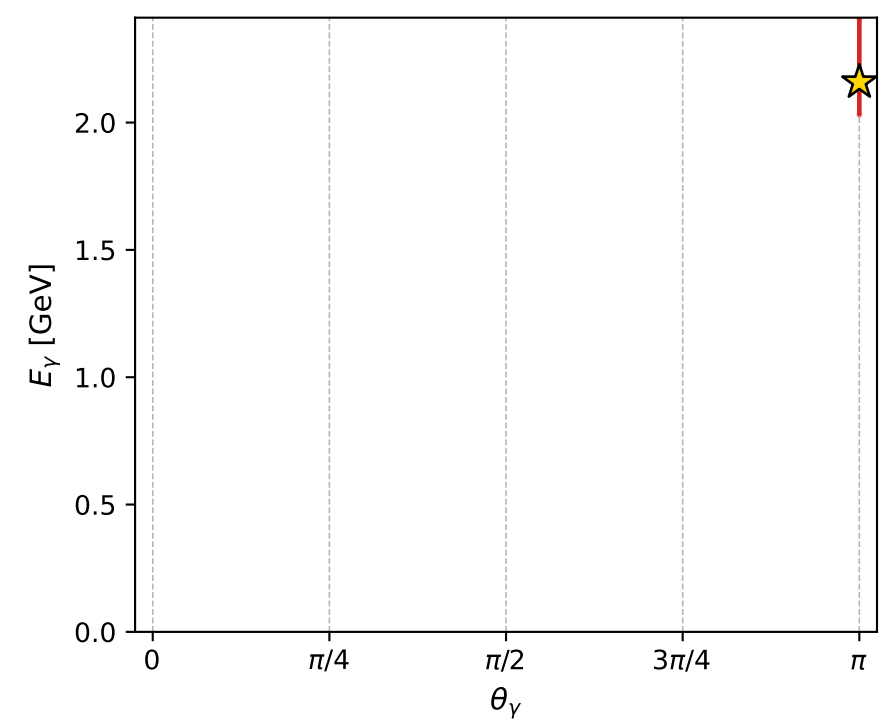
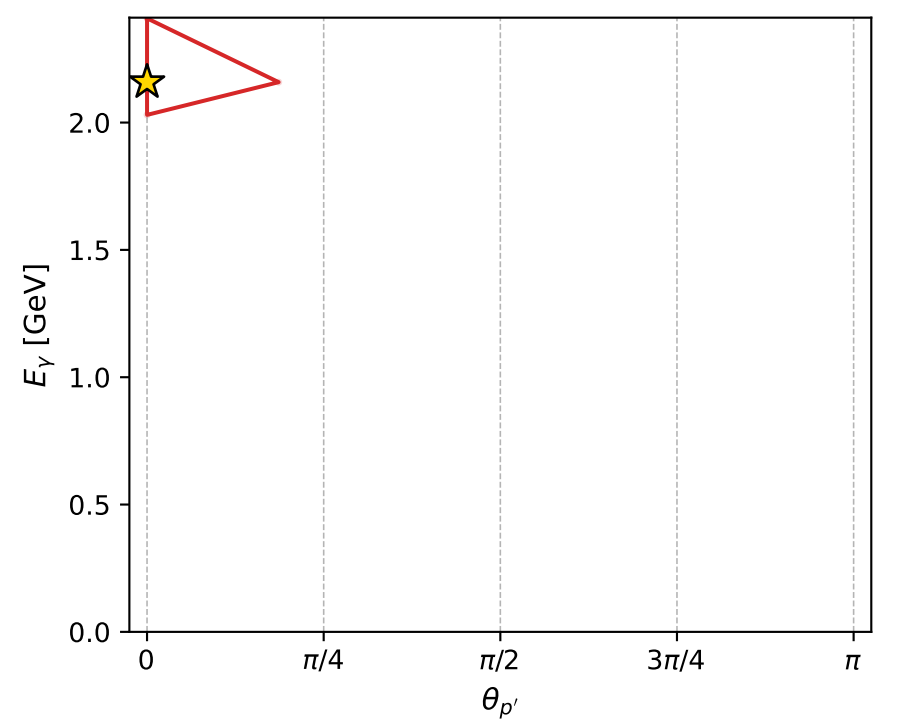
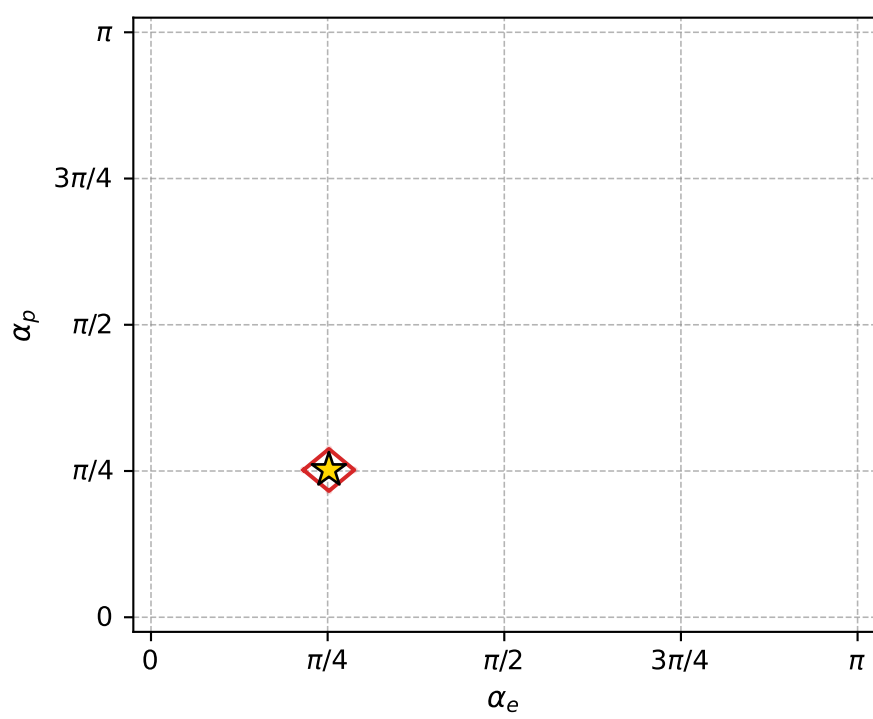
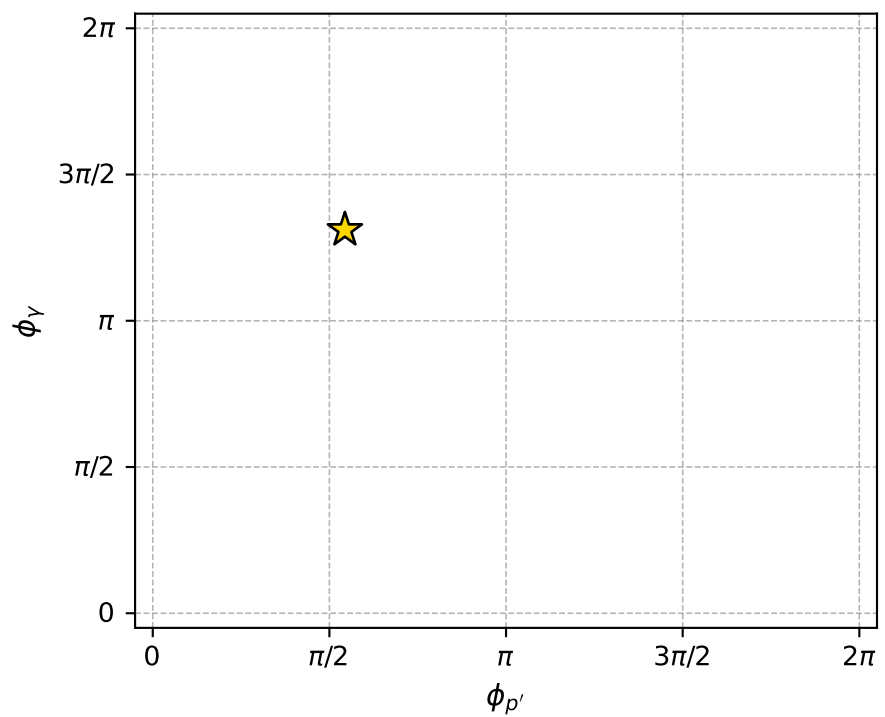
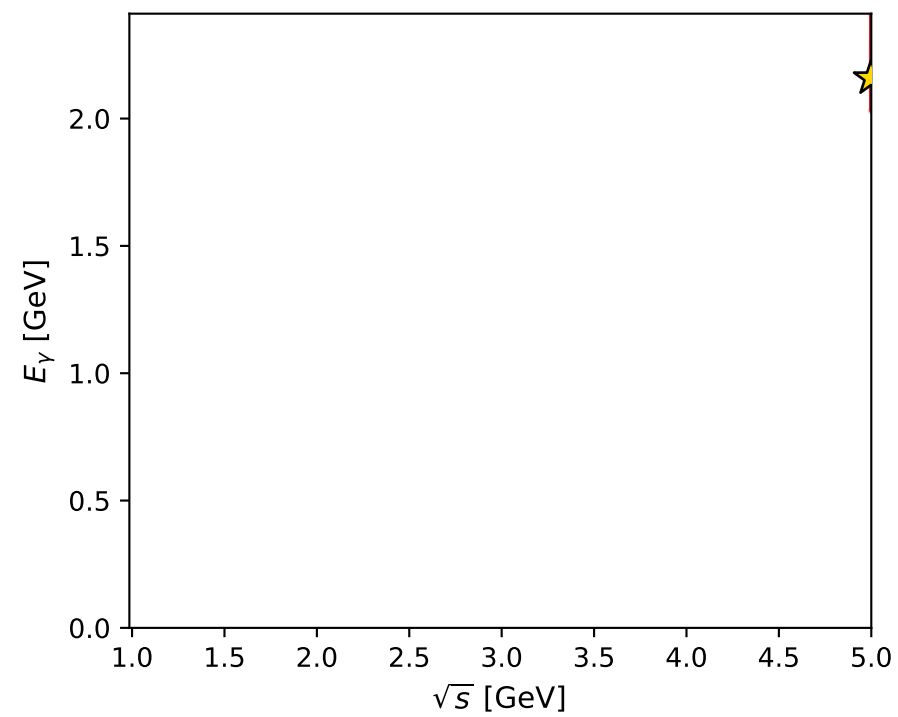
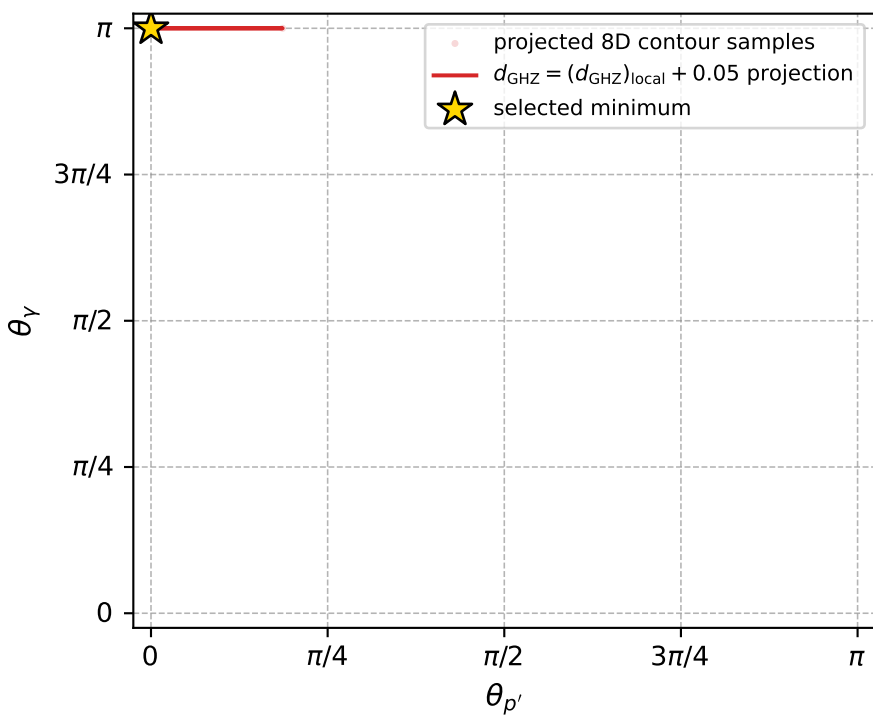
$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 72; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.9084813\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000029$
 above global minimum = $6.8013508\text{e-}09$
 8D contour samples = 79

selected phase-space point:

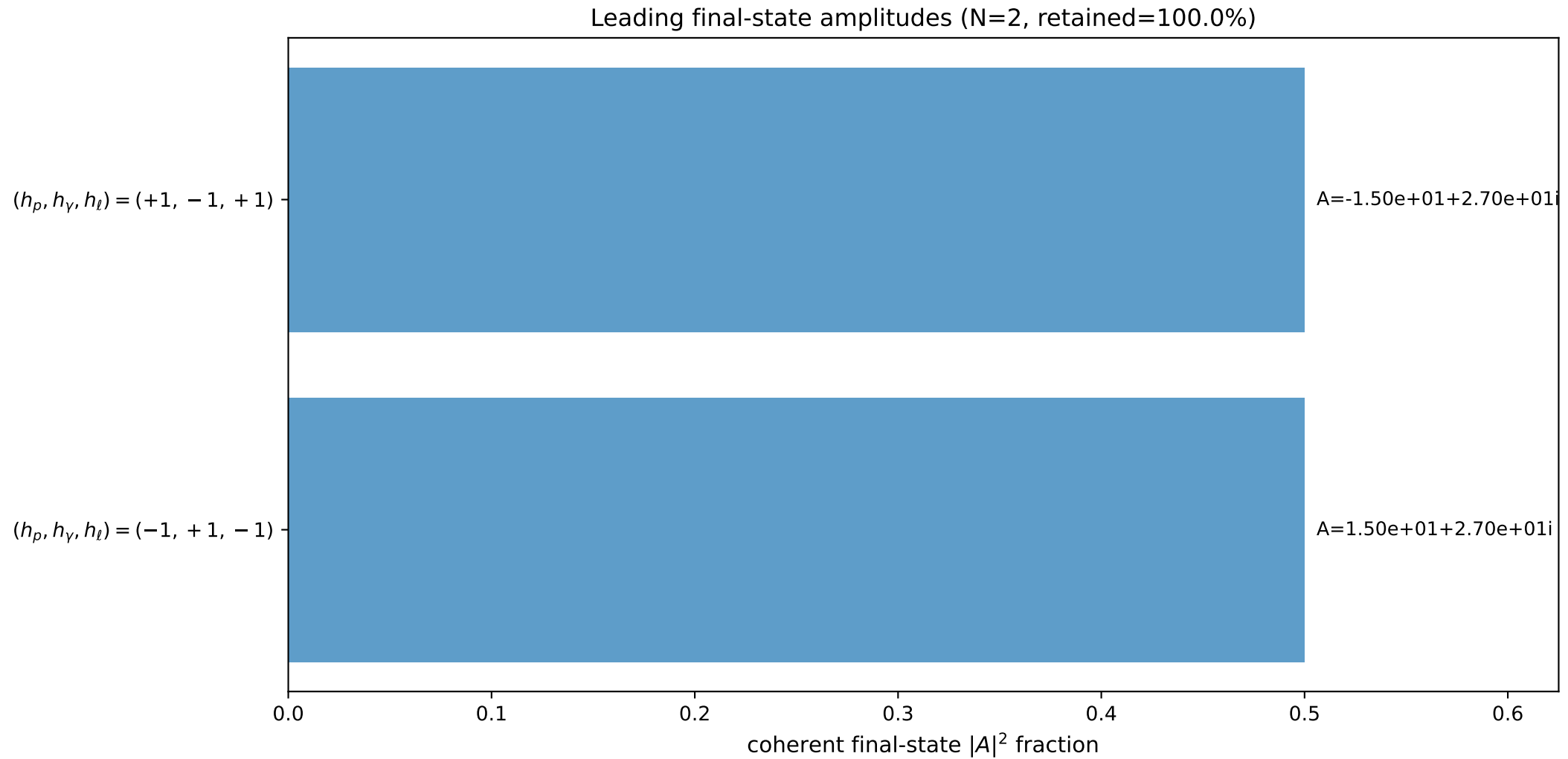
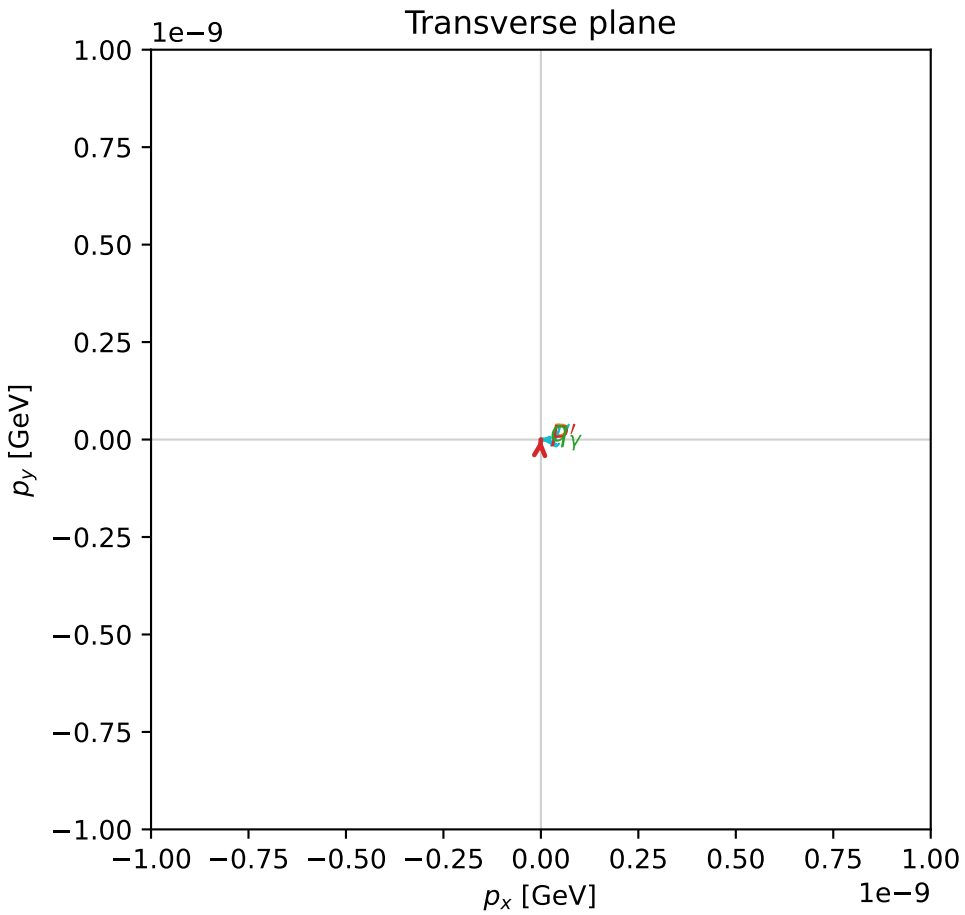
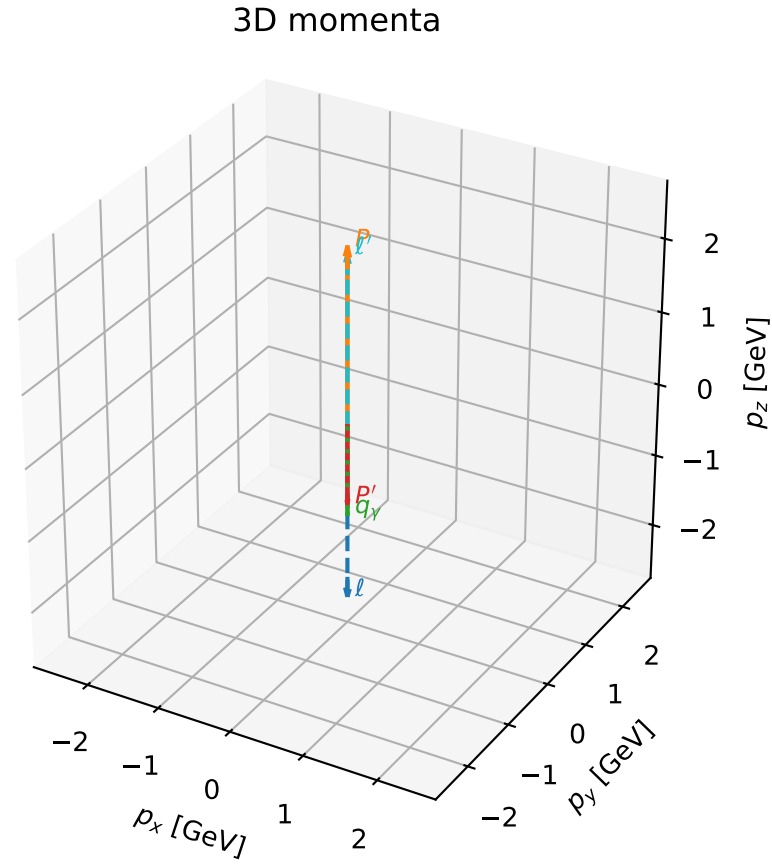
$\text{sqrt_s} = 4.998572$
 $\text{theta_p_out} = 0.000107319$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 2.158629$
 $\text{phi_p_out} = 1.708754$
 $\text{phi_gamma_out} = 4.118013$
 $\text{alpha_e} = 0.7910828$
 $\text{alpha_p} = 0.7910723$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

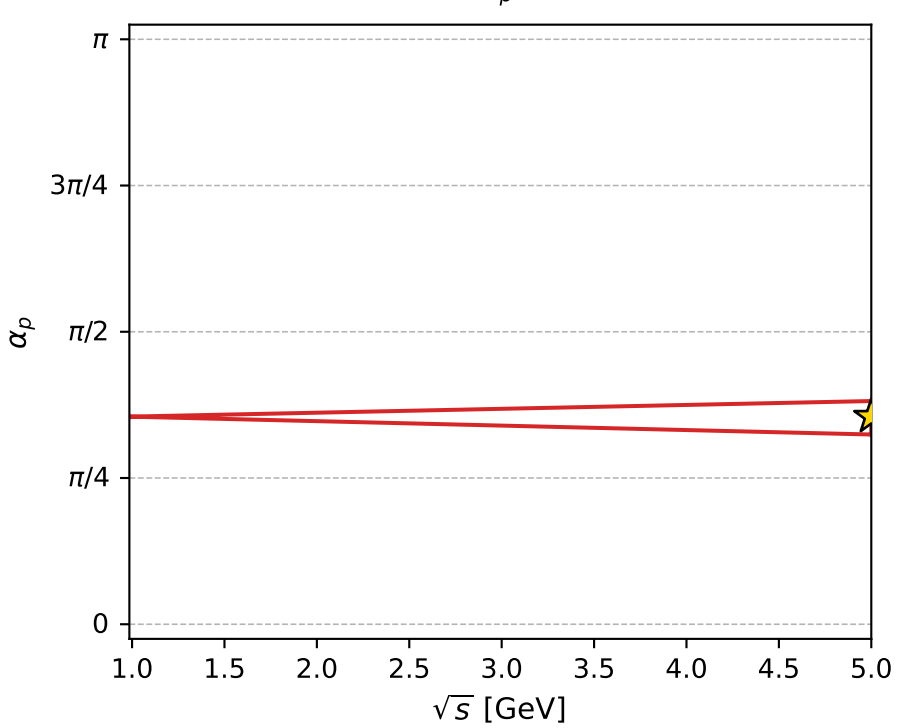
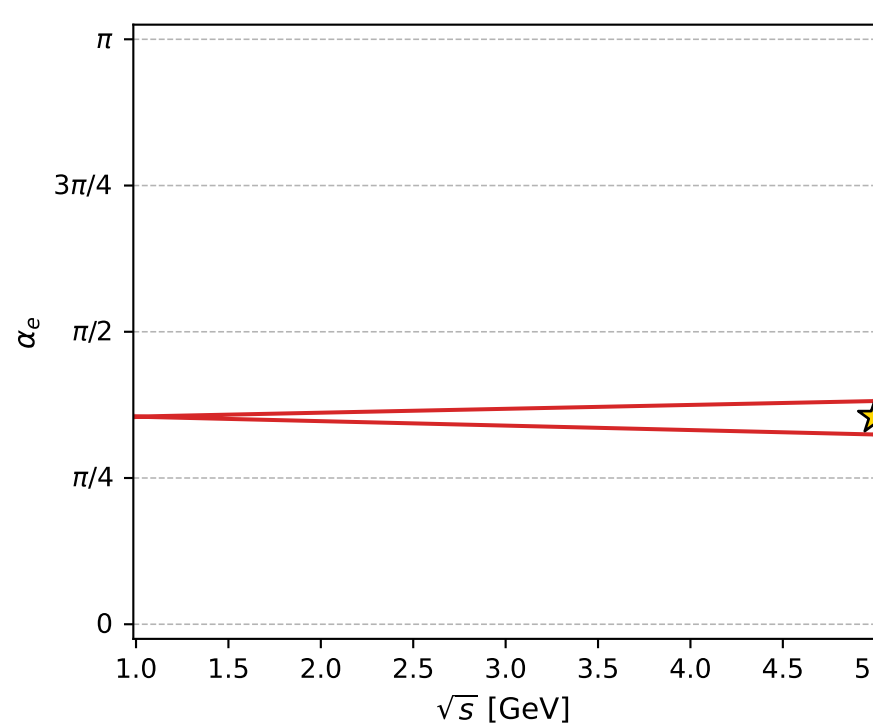
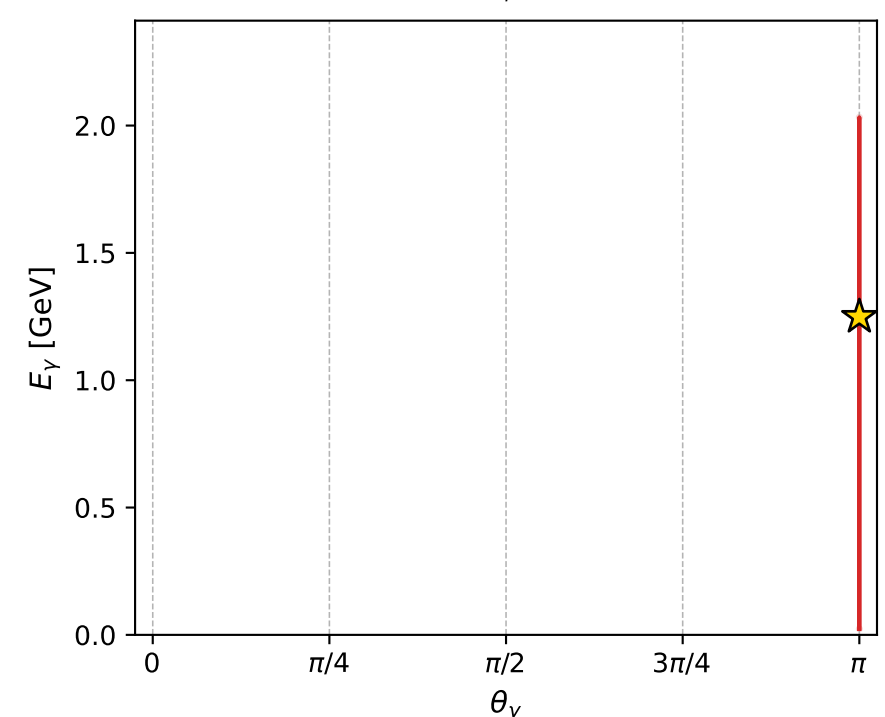
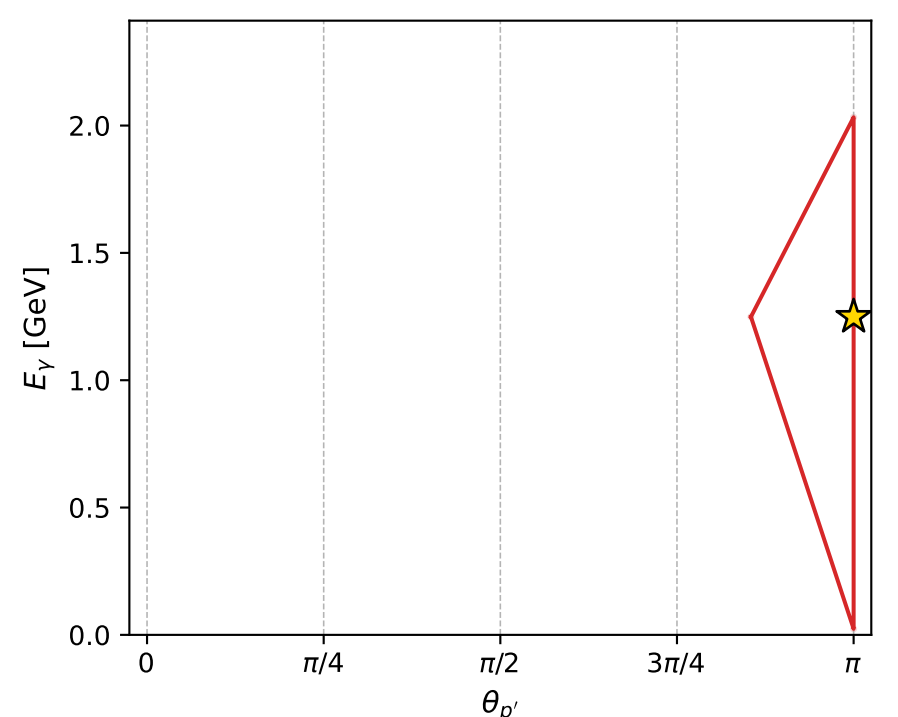
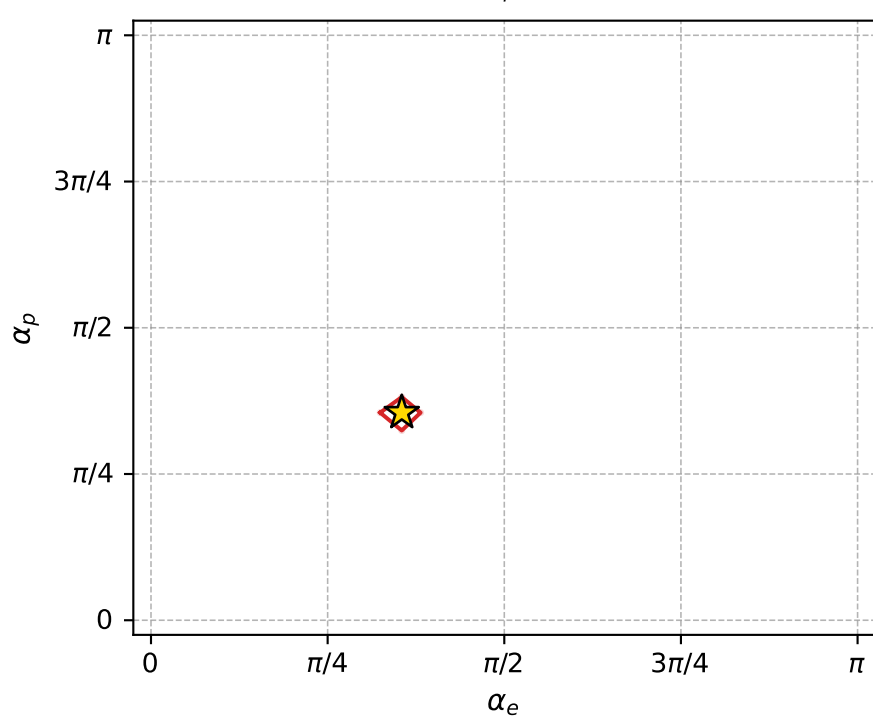
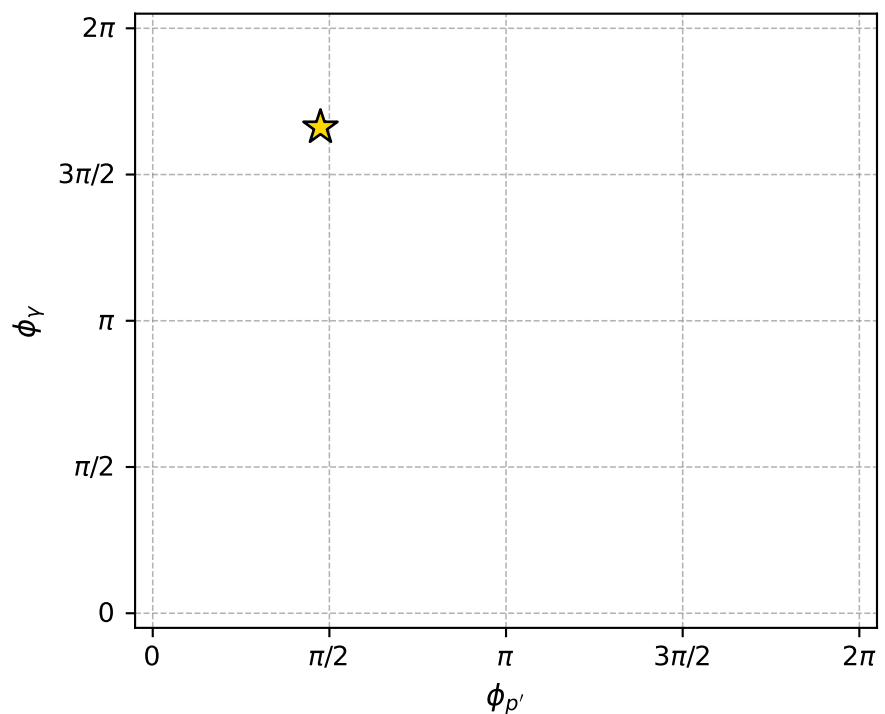
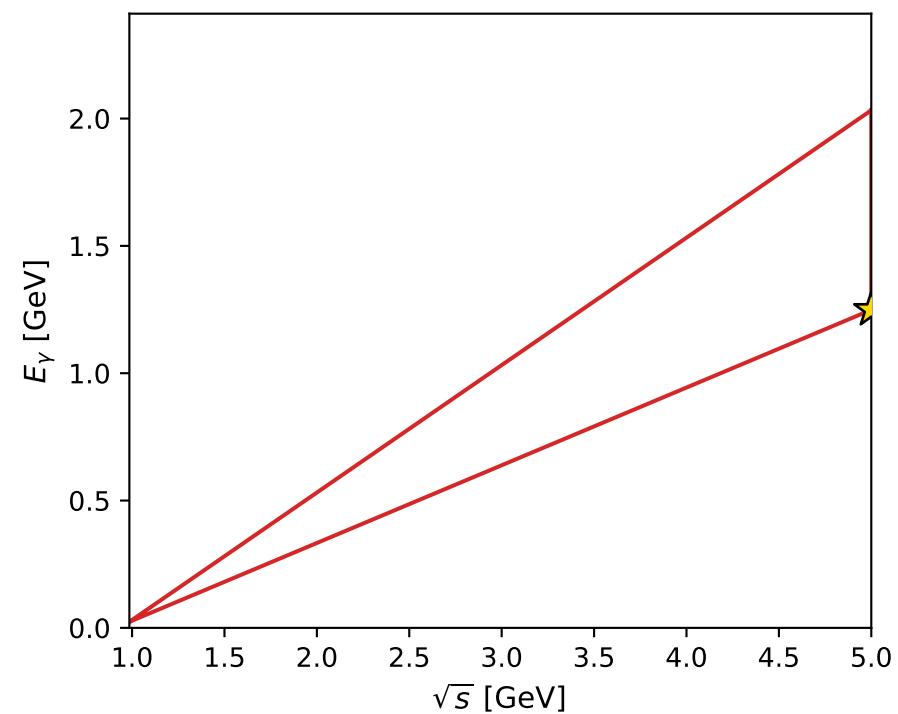
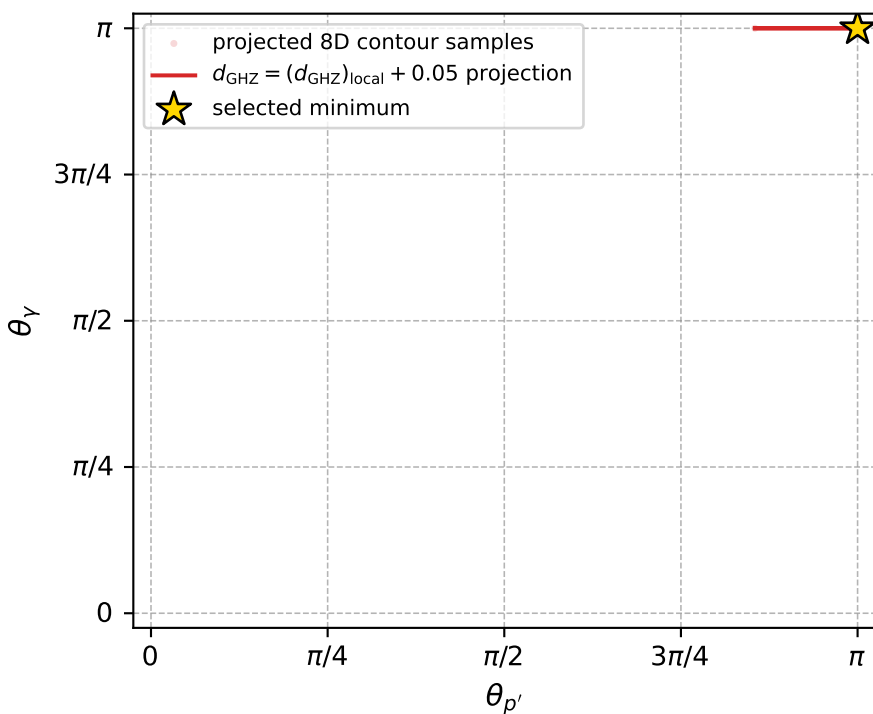
dghz_mixing_angles_polarization_01_local_minimum_75 $d_{\{\text{rm GHZ}\}}=2.94743\text{e-}08$
 region: polarization_cluster_01_local_minimum_75
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.1150181$ rad
 incoming lepton: $0.440161|+\rangle +0.897919|-\rangle$
 proton mixing angle: $\alpha_p=1.1150063$ rad
 incoming proton: $0.440172|+\rangle +0.897914|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.07529$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=1.49134$, $\phi_Y=5.21955$
 $E_Y=1.2489$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.11298$
 $Q^2=22.4239$, $x_B=0.962308$, $t=-10.8132$, $W^2=1.75814$, $y=0.966088$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3242$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.3242$
 P' $E= 1.4269$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.0753$
 q_Y $E= 1.2489$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.2489$



electron: polarization cluster P1, local minimum 75; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.9474276\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000029$
 above global minimum = $7.1908146\text{e-}09$
 8D contour samples = 41

selected phase-space point:

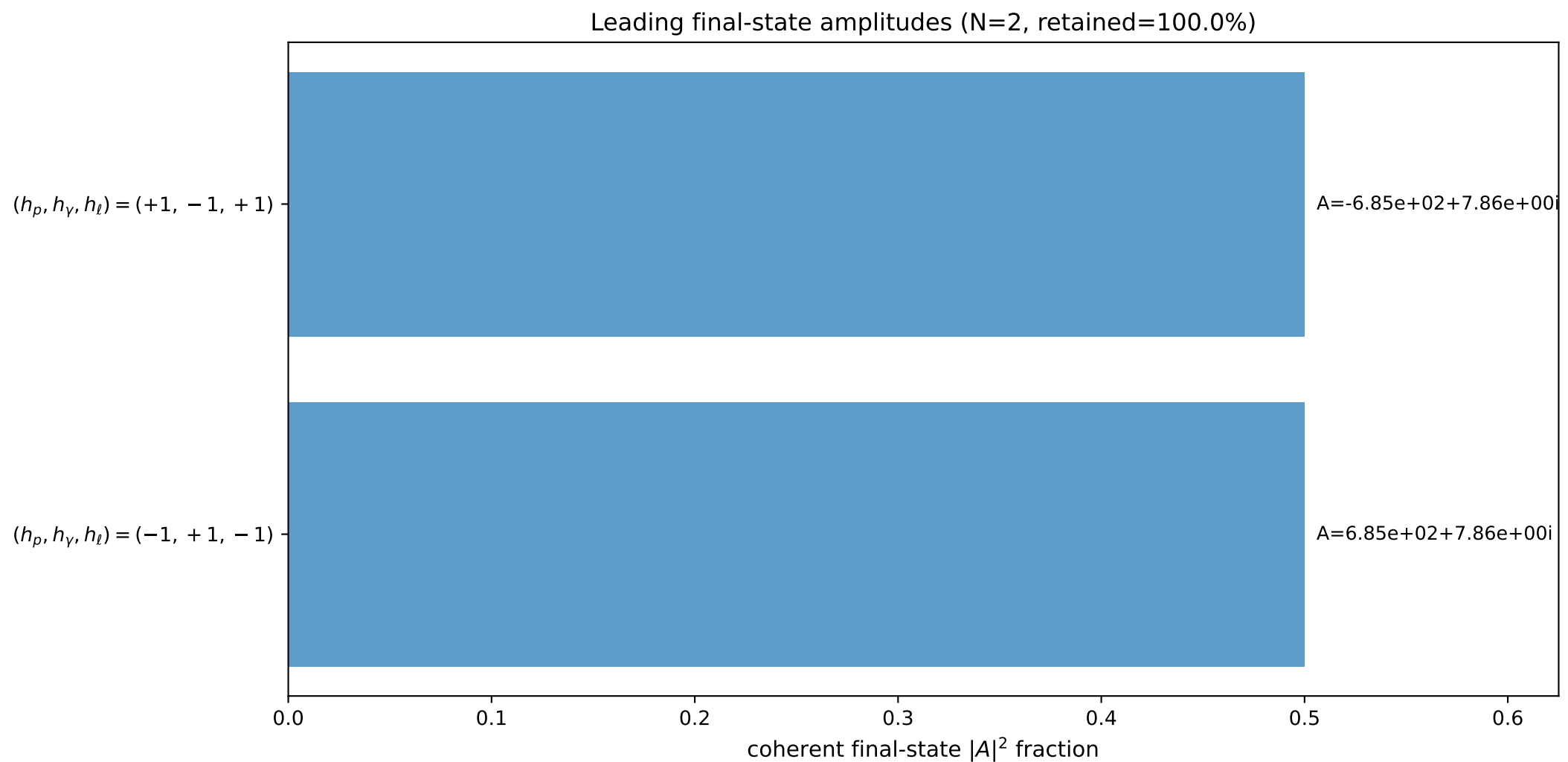
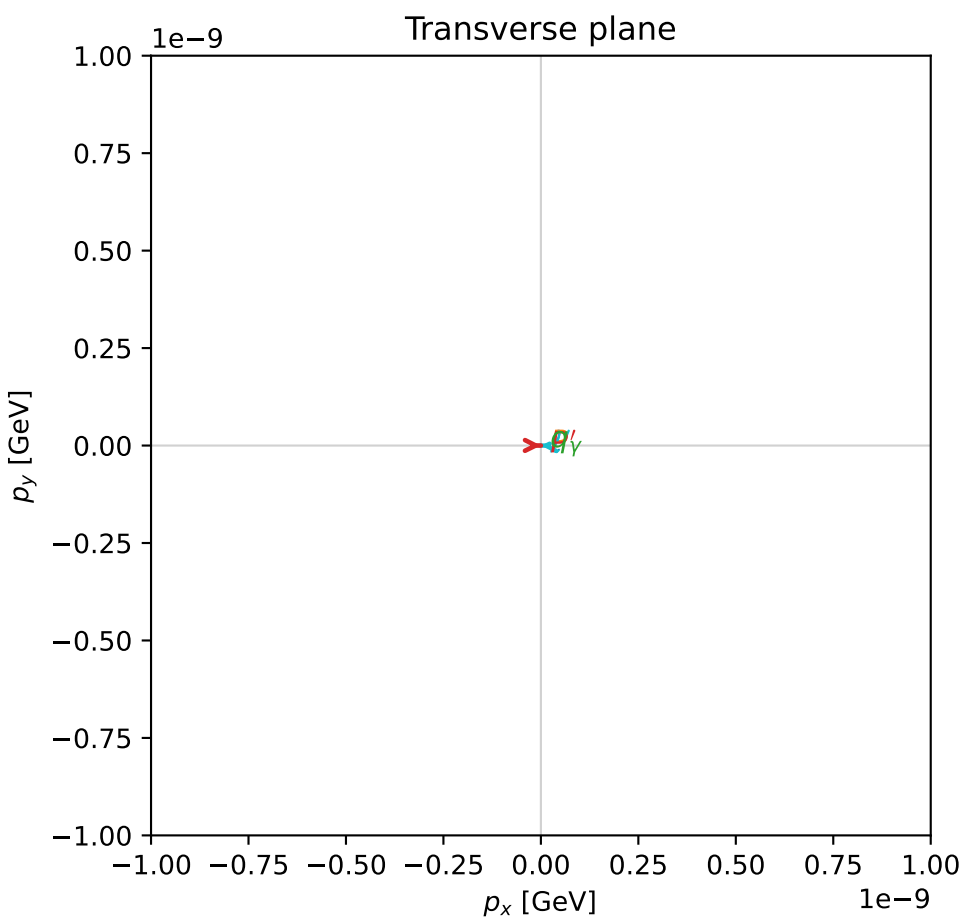
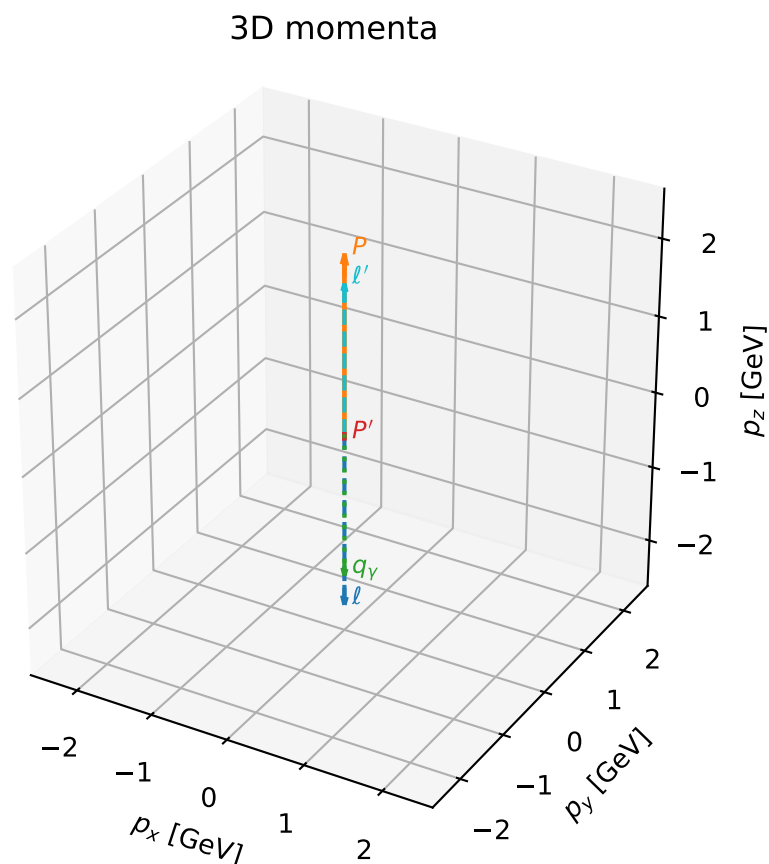
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.2489$
 $\text{phi_p_out} = 1.491339$
 $\text{phi_gamma_out} = 5.219546$
 $\text{alpha_e} = 1.115018$
 $\text{alpha_p} = 1.115006$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

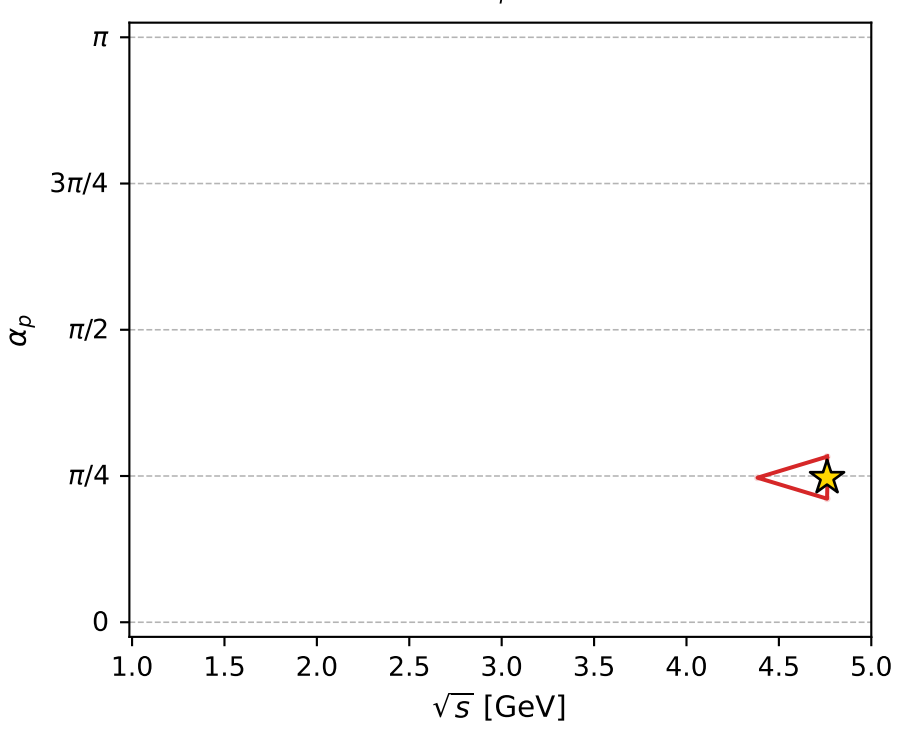
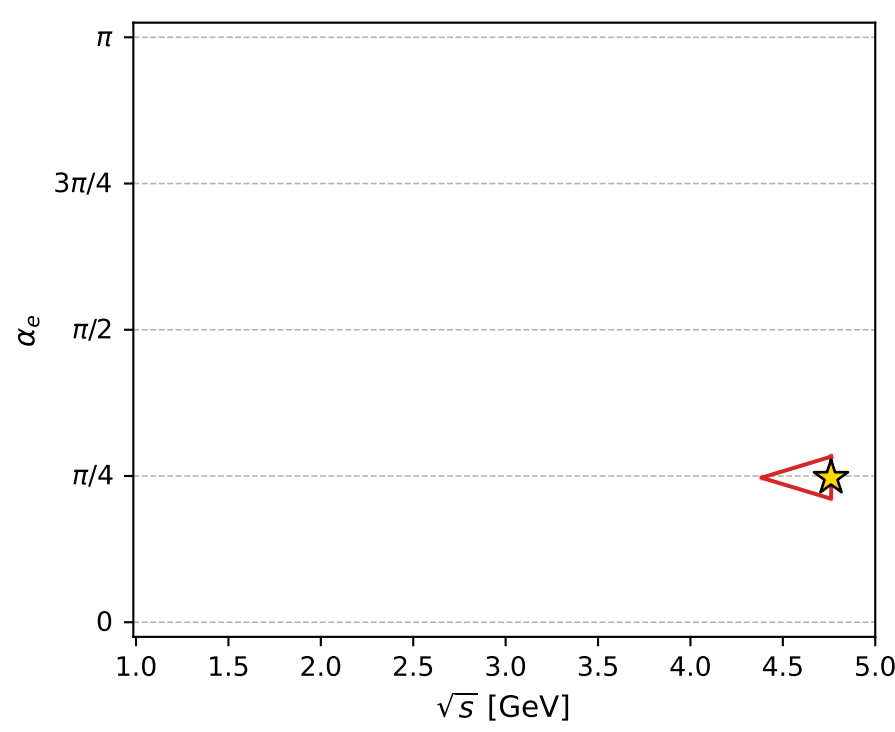
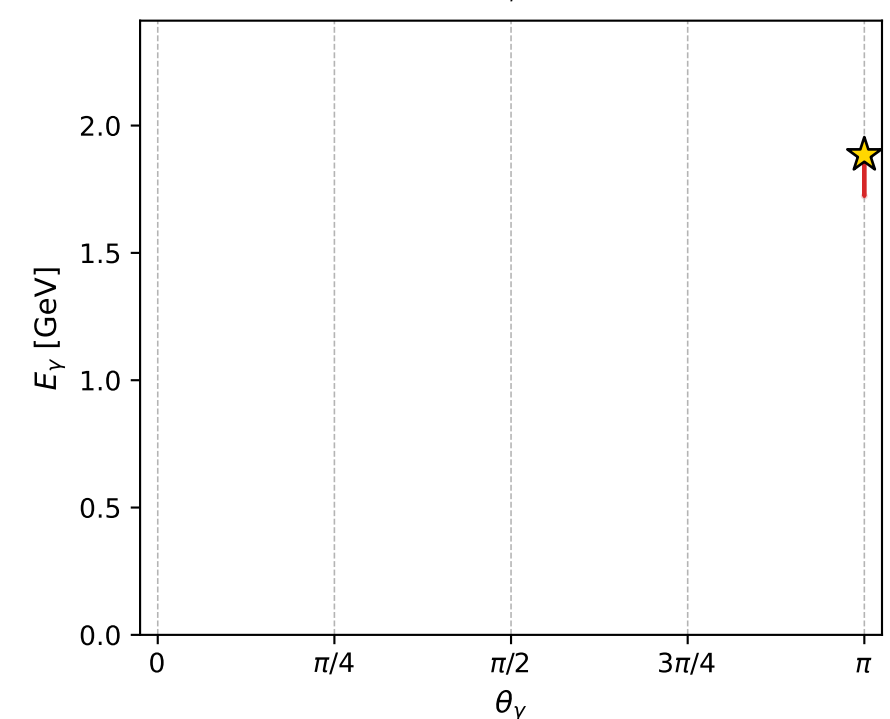
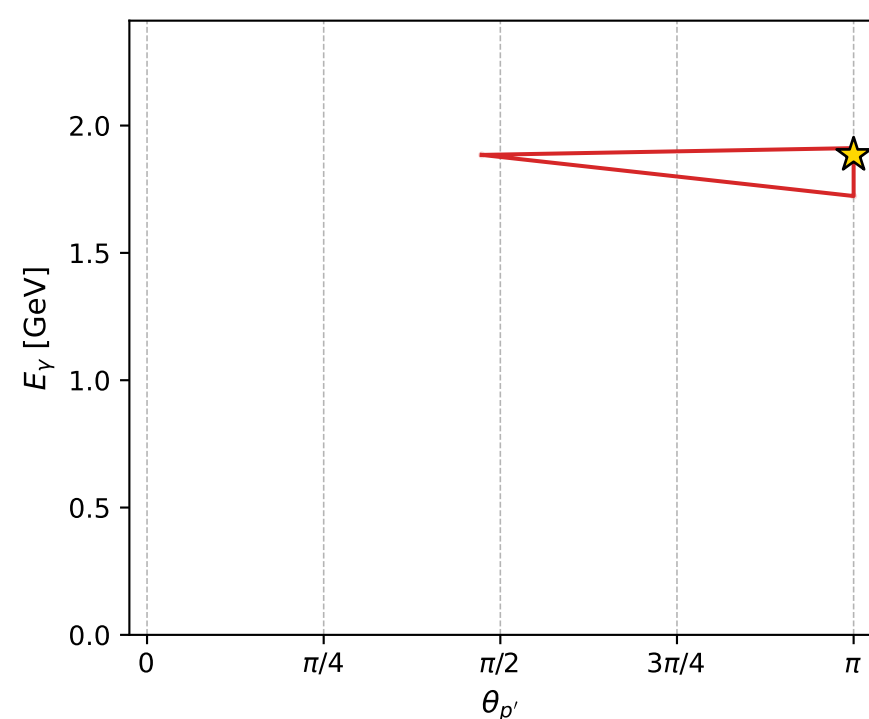
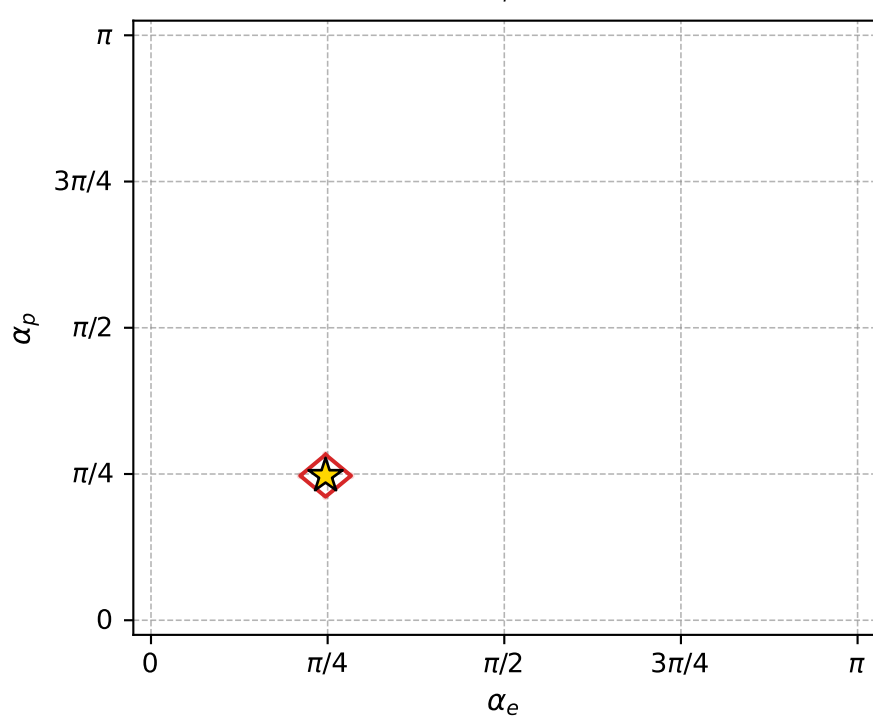
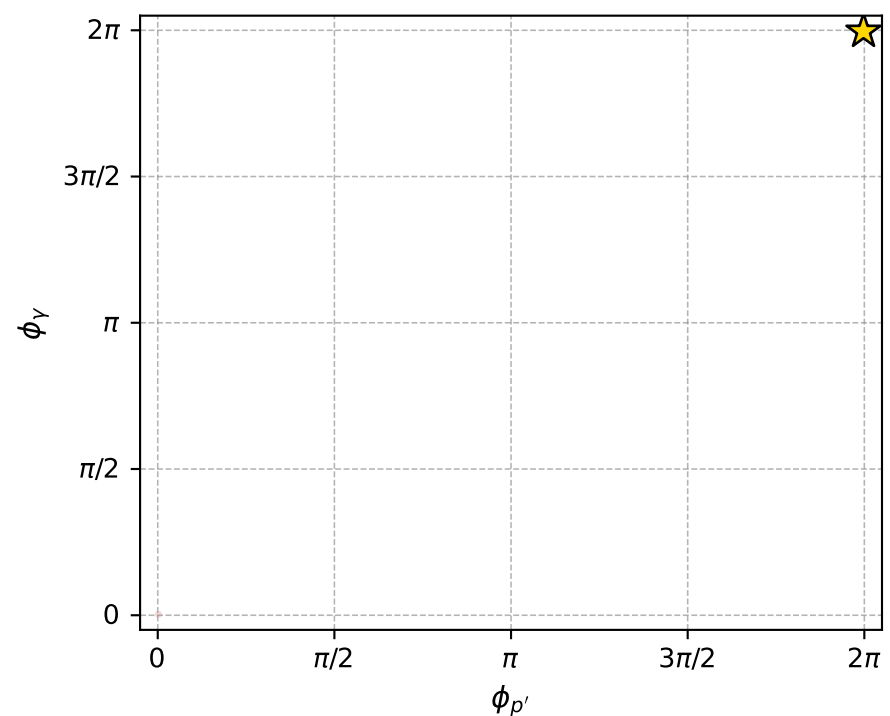
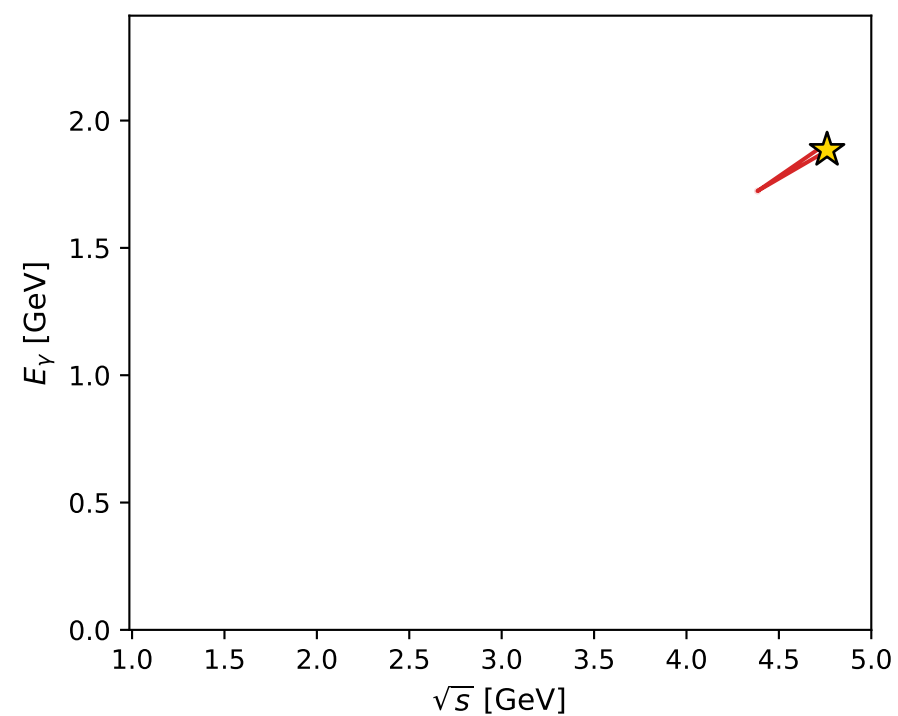
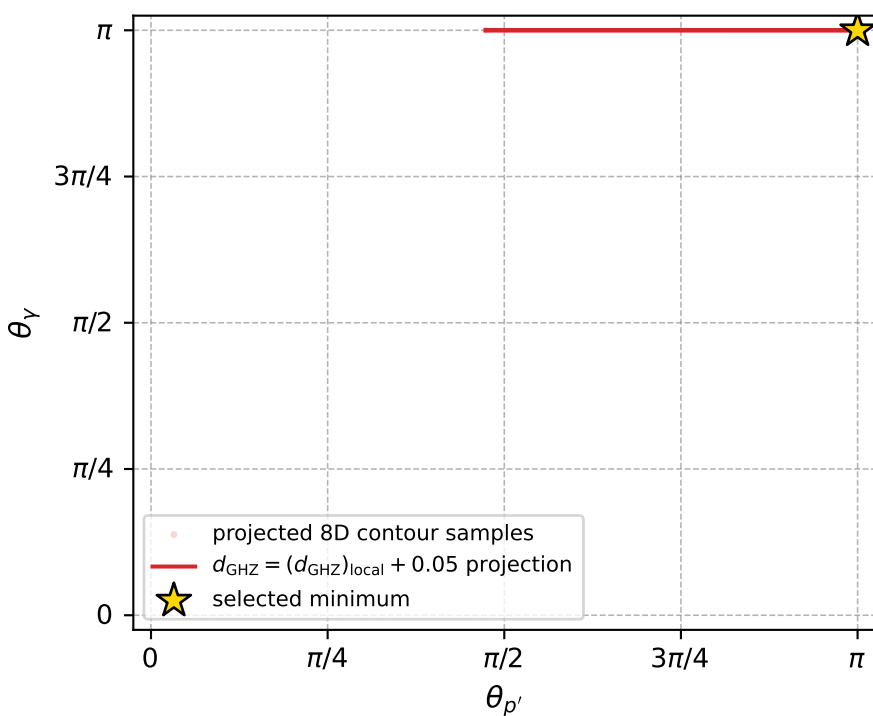
dghz_mixing_angles_polarization_01_local_minimum_76 $d_{\{\text{rm GHZ}\}}=2.95168\text{e-}08$
 region: polarization_cluster_01_local_minimum_76
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77617426$ rad
 incoming lepton: $0.713599|+\rangle +0.700555|-\rangle$
 proton mixing angle: $\alpha_p=0.77615415$ rad
 incoming proton: $0.713613|+\rangle +0.70054|-\rangle$

$s=22.6644$, $\sqrt{s}=4.76071$, $|\vec{P}|=2.28795$, $|\vec{P}'|=0.0517317$
 $\theta_{p'}=3.14159$, $\theta_Y=3.14159$, $\phi_{p'}=6.27601$, $\phi_Y=6.27171$
 $E_Y=1.88478$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.13024$
 $Q^2=17.7225$, $x_B=0.841177$, $t=-3.12298$, $W^2=4.22605$, $y=0.967142$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2879$
 P $E= 2.4728$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2879$
 ℓ' $E= 1.9365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9365$
 P' $E= 0.9394$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0517$
 q_Y $E= 1.8848$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8848$



electron: polarization cluster P1, local minimum 76; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.9516846\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.05000003$
 above global minimum = $7.2333837\text{e-}09$
 8D contour samples = 67

selected phase-space point:

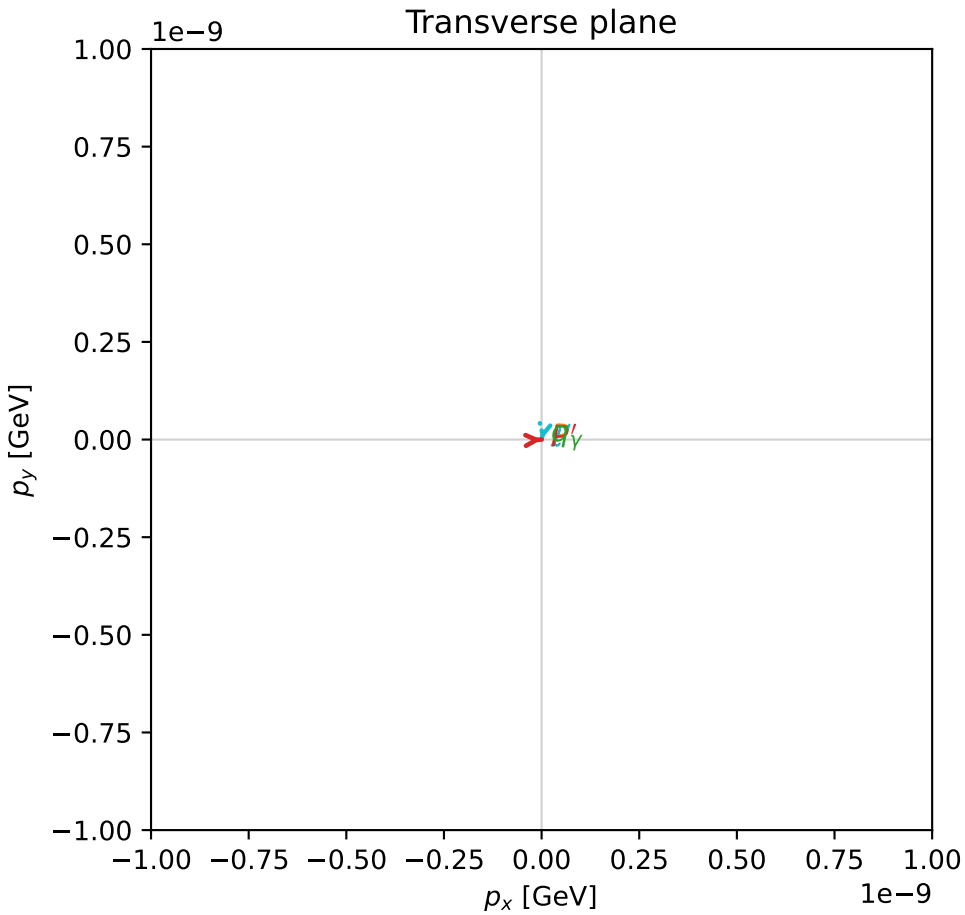
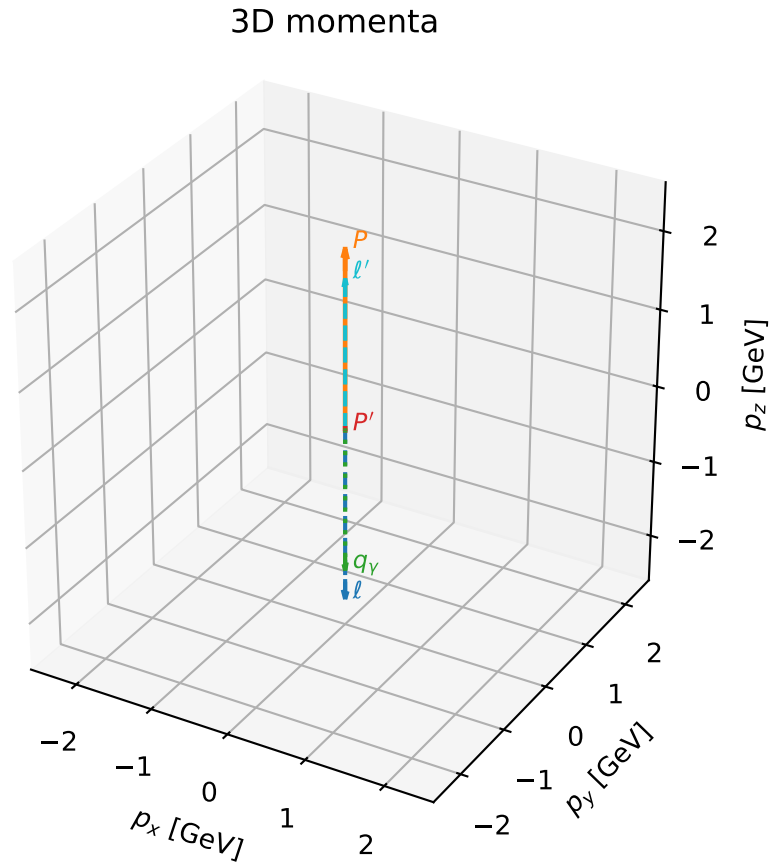
$\text{sqrt_s} = 4.760712$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.884777$
 $\text{phi_p_out} = 6.276008$
 $\text{phi_gamma_out} = 6.271714$
 $\alpha_e = 0.7761743$
 $\alpha_p = 0.7761542$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_79 $d_{\{\text{rm GHZ}\}}=3.01399\text{e-}08$
 region: polarization_cluster_01_local_minimum_79
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76405178$ rad
 incoming lepton: $0.722039|+\rangle +0.691853|-\rangle$
 proton mixing angle: $\alpha_p=0.76404422$ rad
 incoming proton: $0.722044|+\rangle +0.691847|-\rangle$

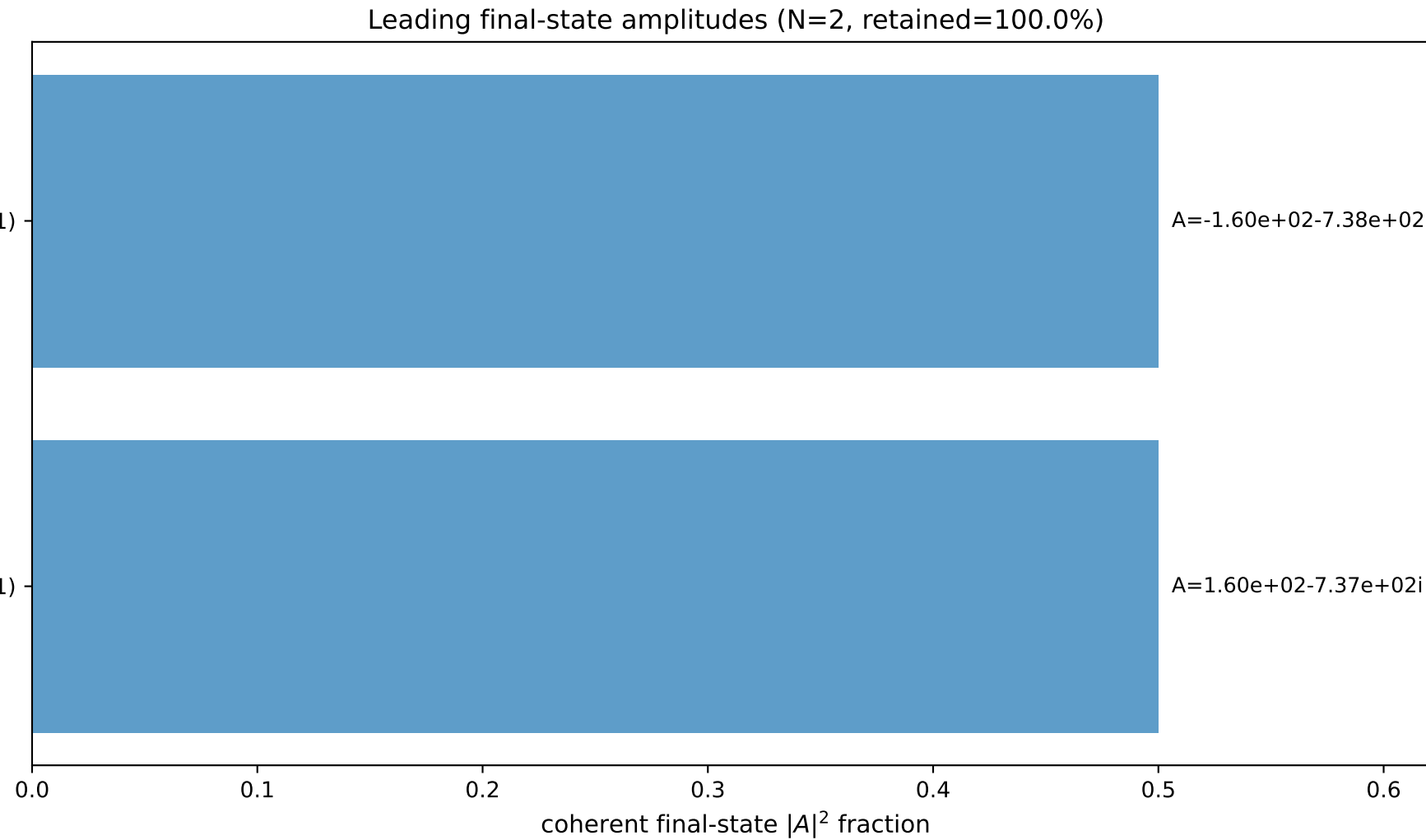
$s=22.3692$, $\sqrt{s}=4.72961$, $|\vec{P}|=2.27179$, $|\vec{P}'|=0.0259335$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=0.0568276$, $\phi_Y=1.35777$
 $E_Y=1.88266$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.48614$
 $Q^2=17.3437$, $x_B=0.834663$, $t=-2.97077$, $W^2=4.31541$, $y=0.966955$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2718$
 P $E= 2.4578$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2718$
 ℓ' $E= 1.9086$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9086$
 P' $E= 0.9384$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0259$
 q_Y $E= 1.8827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8827$

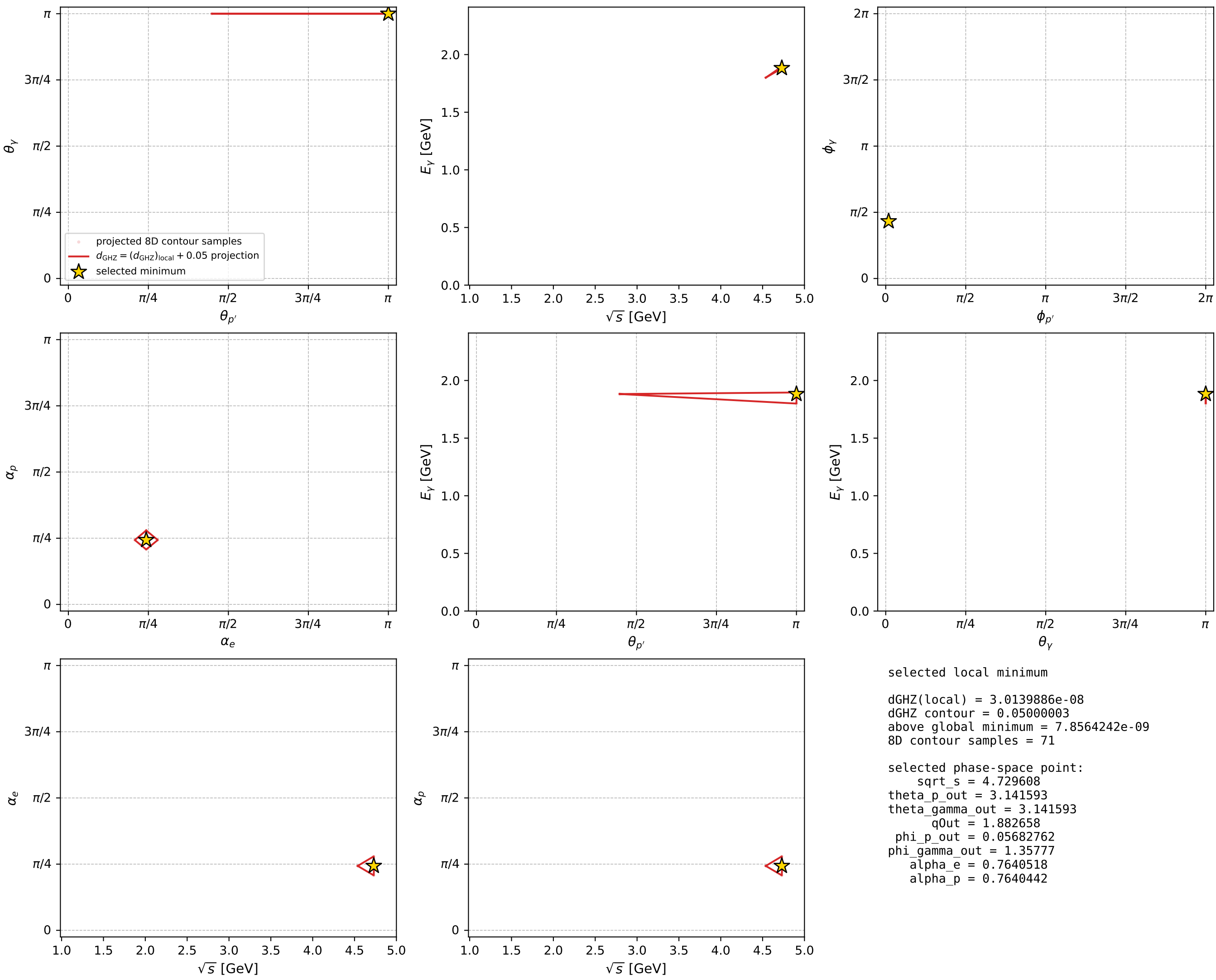


$(h_p, h_Y, h_\ell) = (+1, -1, +1)$

$(h_p, h_Y, h_\ell) = (-1, +1, -1)$



electron: polarization cluster P1, local minimum 79; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

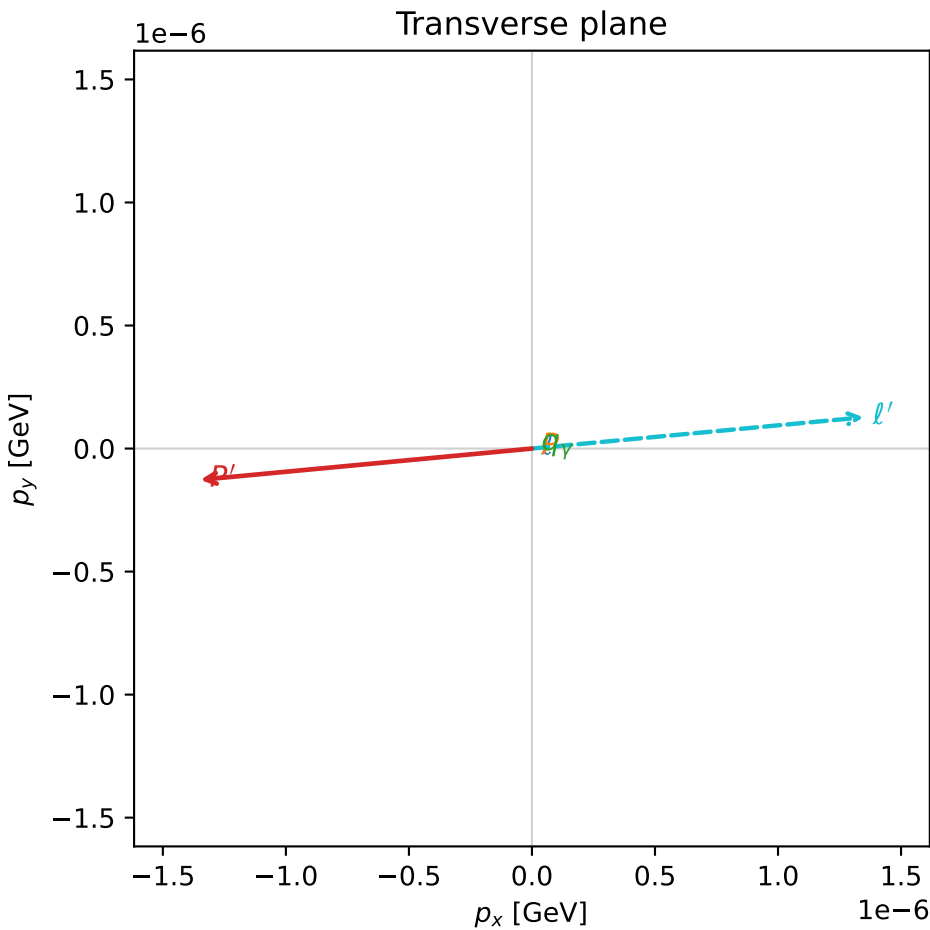
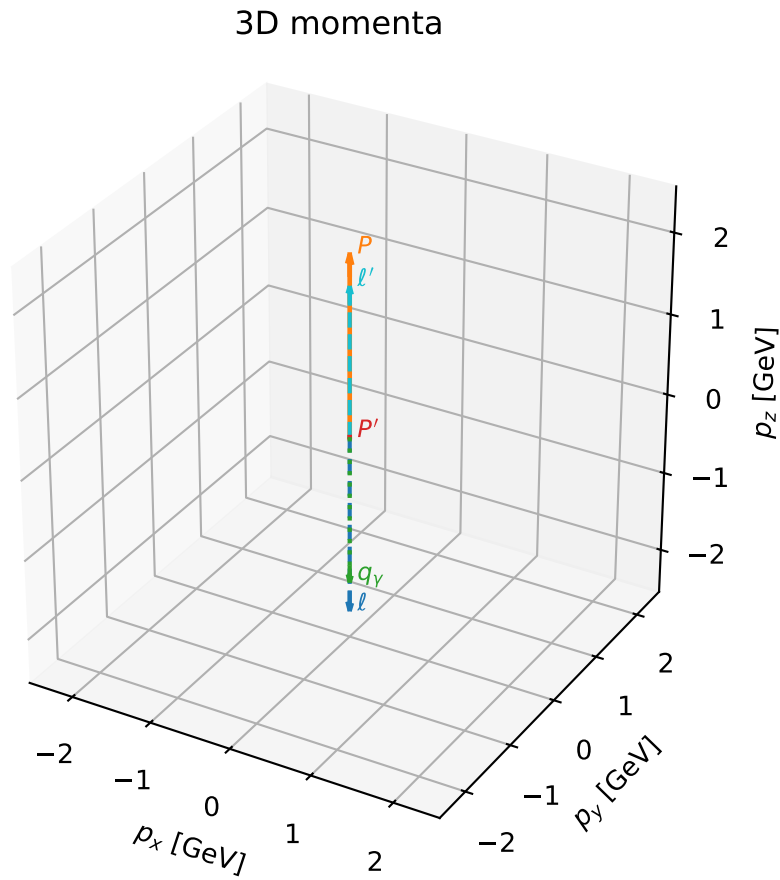


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_83 $d_{\{\text{rm GHZ}\}}=3.05691\text{e-}08$
 region: polarization_cluster_01_local_minimum_83
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.74952575$ rad
 incoming lepton: $0.732012|+\rangle +0.681292|-\rangle$
 proton mixing angle: $\alpha_p=0.74952252$ rad
 incoming proton: $0.732014|+\rangle +0.681289|-\rangle$

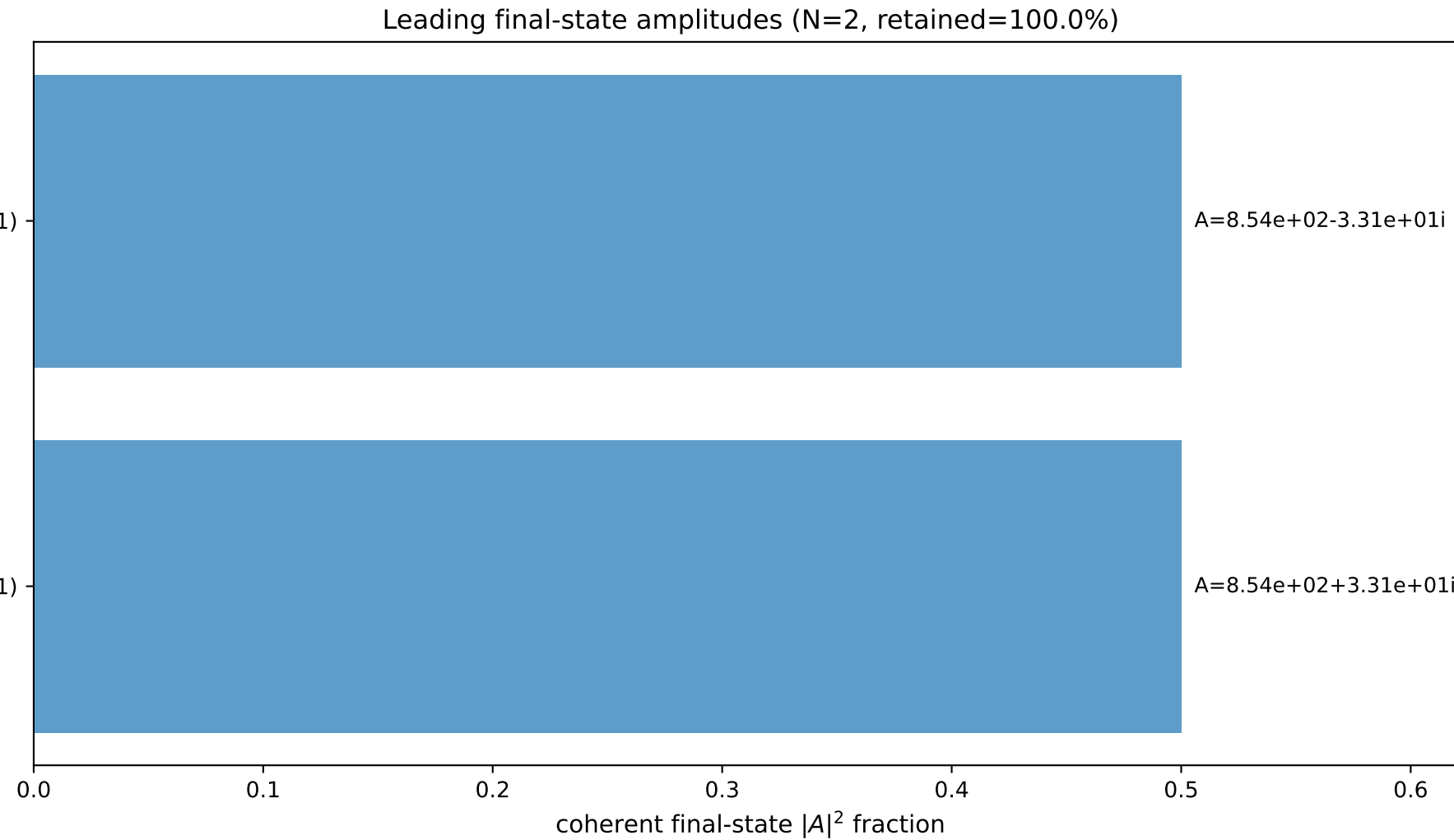
$s=21.4675$, $\sqrt{s}=4.63331$, $|\vec{P}|=2.22171$, $|\vec{P}'|=8.39622\text{e-}05$
 $\theta_{P'}=0.0161201$, $\theta_Y=3.14159$, $\phi_{P'}=3.23565$, $\phi_Y=3.1028$
 $E_Y=1.8477$, $\theta_{\ell'}=7.32586\text{e-}07$, $\phi_{\ell'}=0.0940553$
 $Q^2=16.4194$, $x_B=0.825677$, $t=-2.7641$, $W^2=4.34643$, $y=0.965916$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2217$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2217$
 P $E= 2.4116$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2217$
 ℓ' $E= 1.8476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8476$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0001$
 q_Y $E= 1.8477$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.8477$

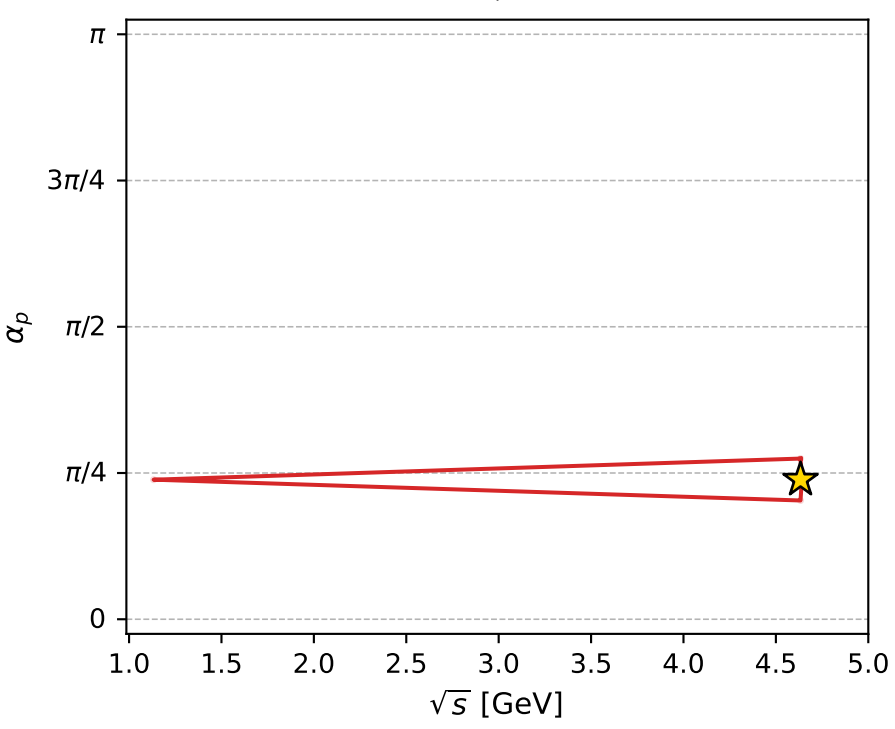
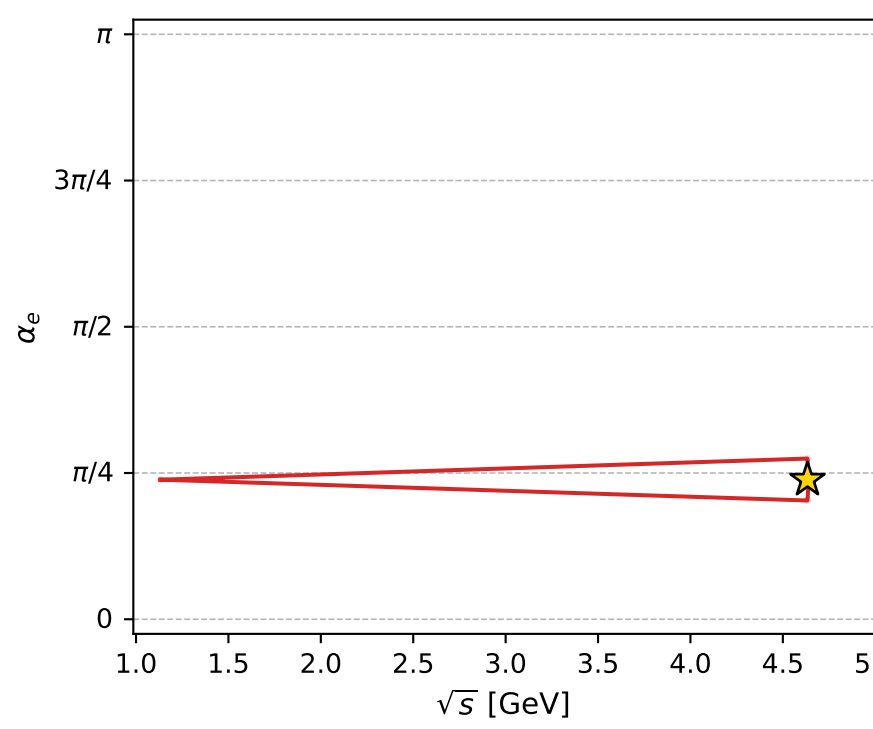
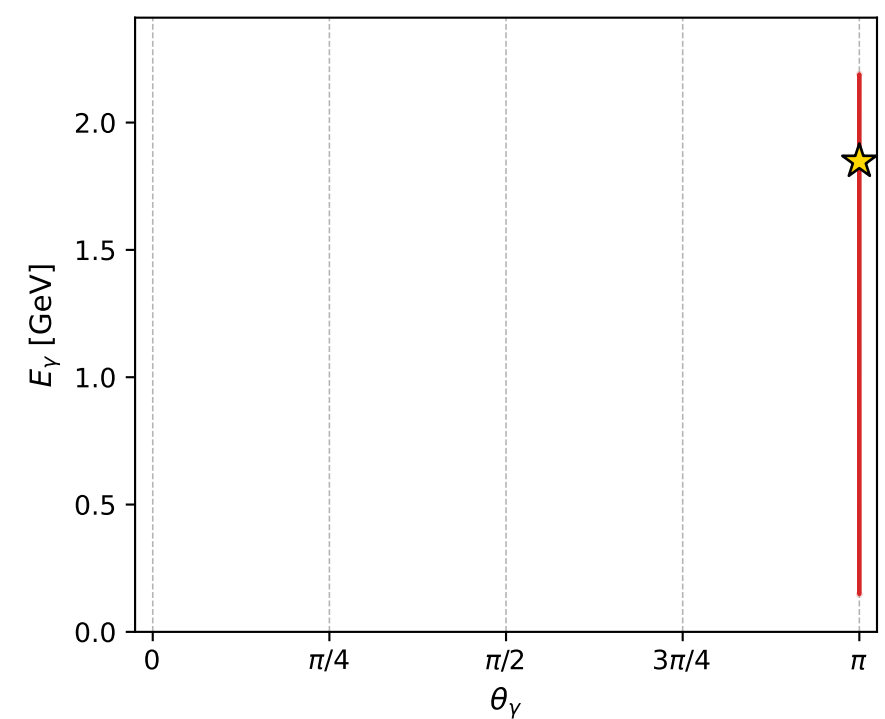
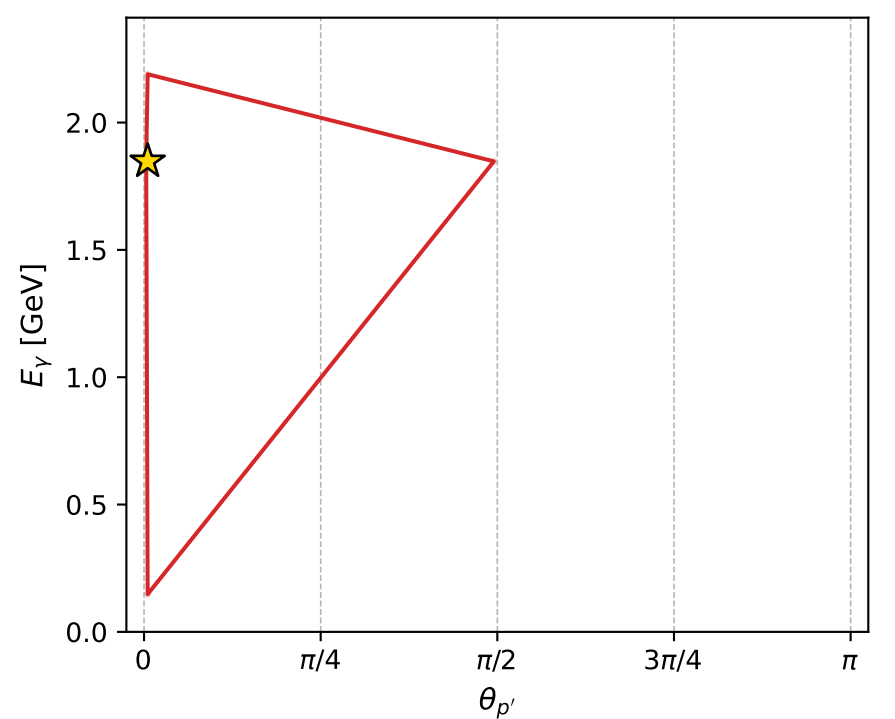
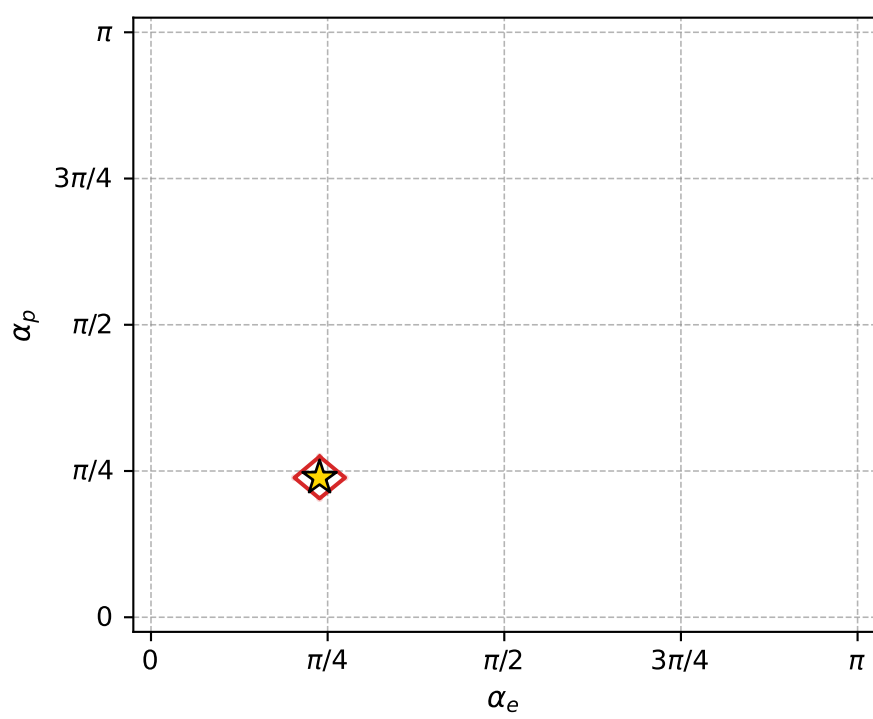
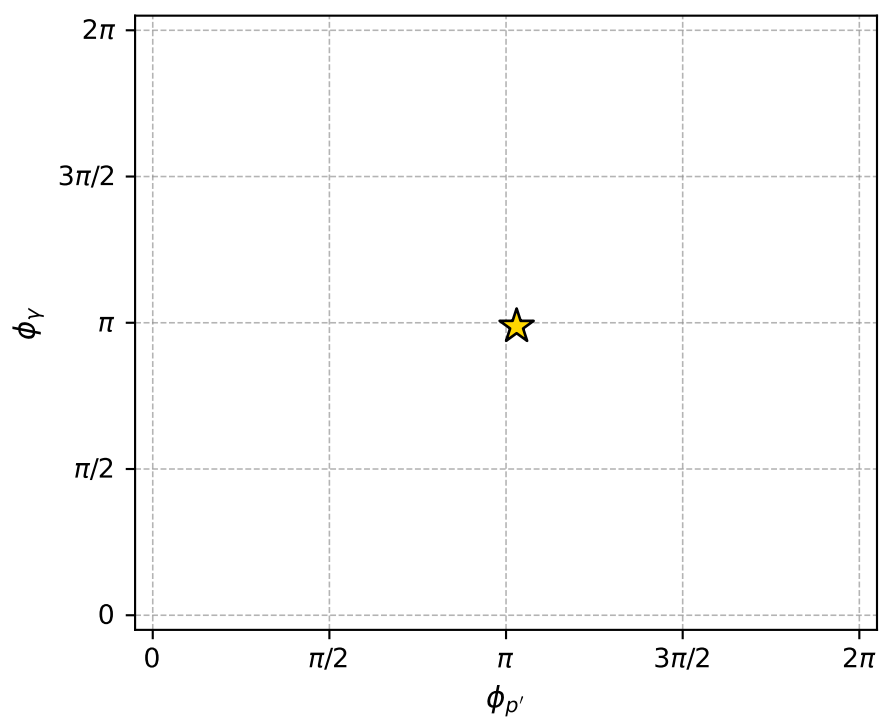
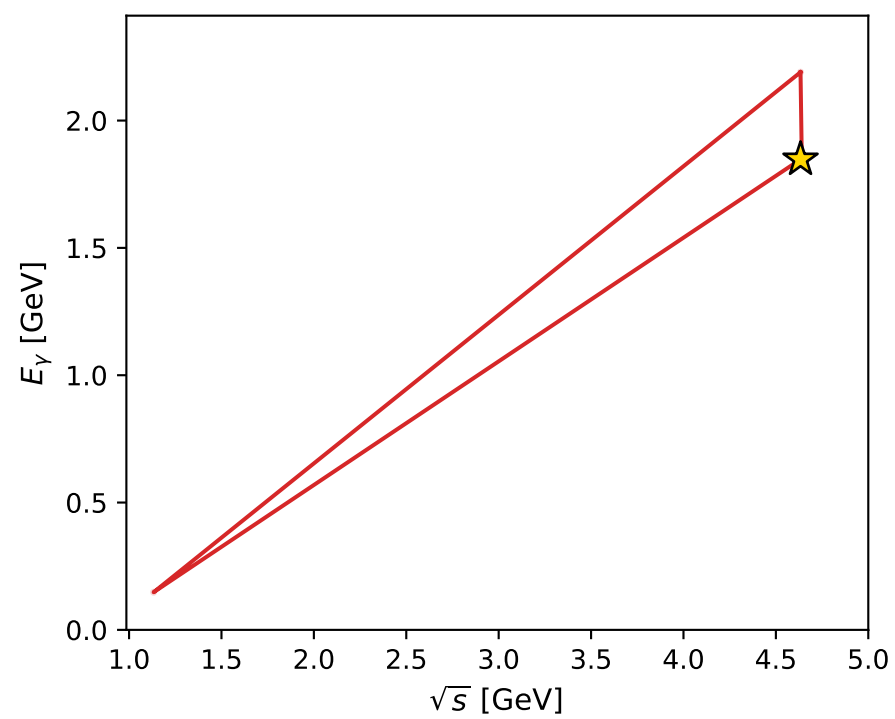
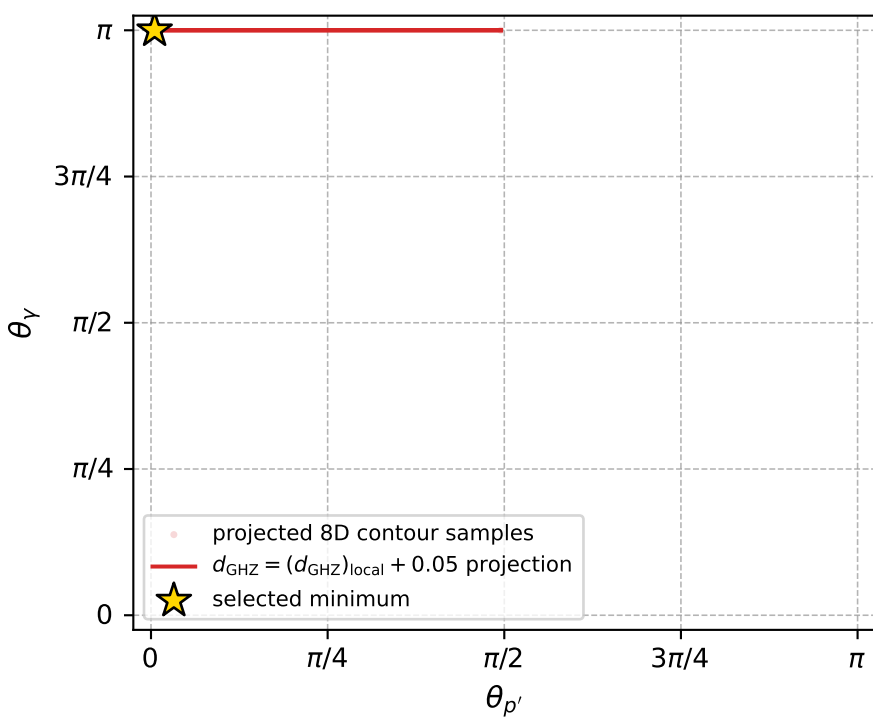


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$



electron: polarization cluster P1, local minimum 83; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.0569066\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000031$
 above global minimum = $8.285604\text{e-}09$
 8D contour samples = 120

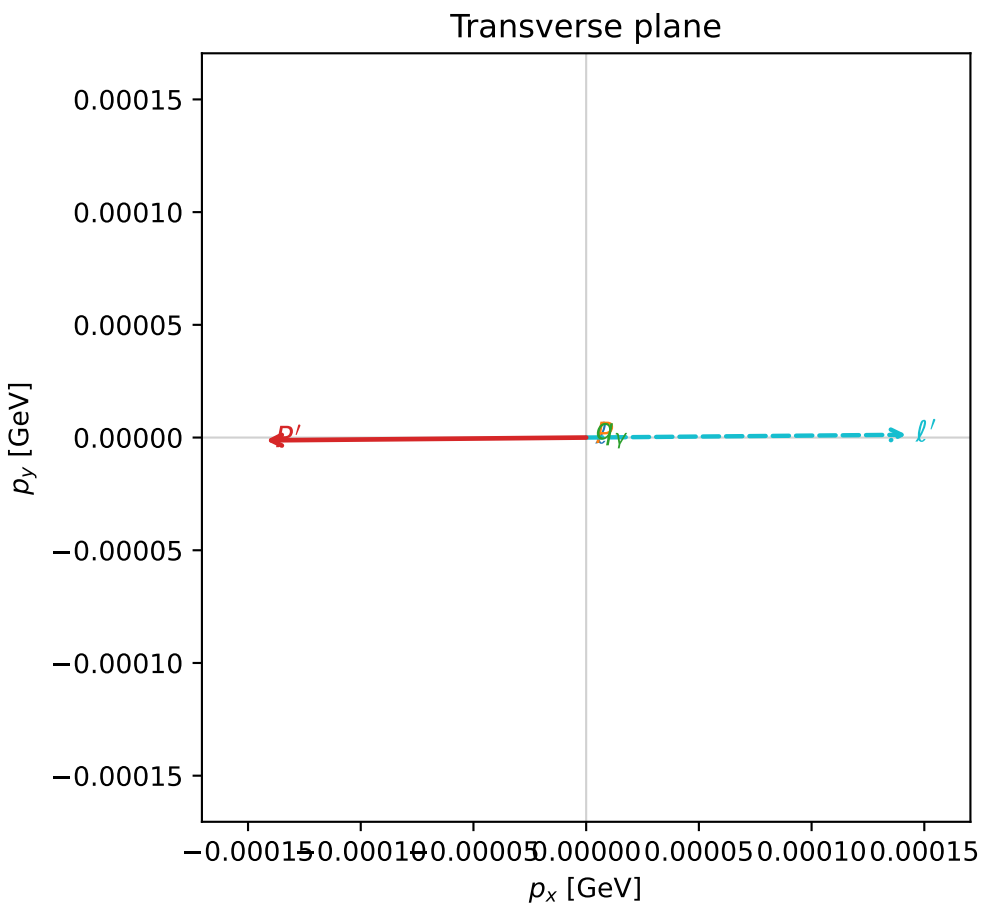
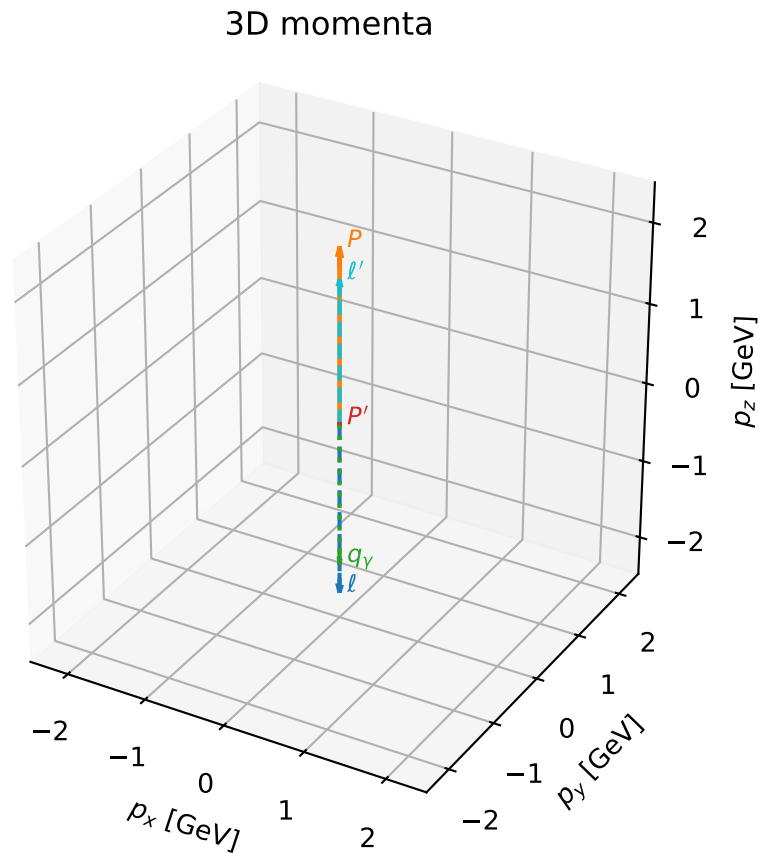
selected phase-space point:
 $\text{sqrt_s} = 4.633307$
 $\text{theta_p_out} = 0.01612007$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.847696$
 $\text{phi_p_out} = 3.235648$
 $\text{phi_gamma_out} = 3.102796$
 $\text{alpha_e} = 0.7495257$
 $\text{alpha_p} = 0.7495225$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_95 $d_{\{\text{rm GHZ}\}}=3.40442\text{e-}08$
region: polarization_cluster_01_local_minimum_95
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.83191558$ rad
incoming lepton: $0.673461|+\rangle +0.739223|-\rangle$
proton mixing angle: $\alpha_p=0.83190695$ rad
incoming proton: $0.673467|+\rangle +0.739217|-\rangle$

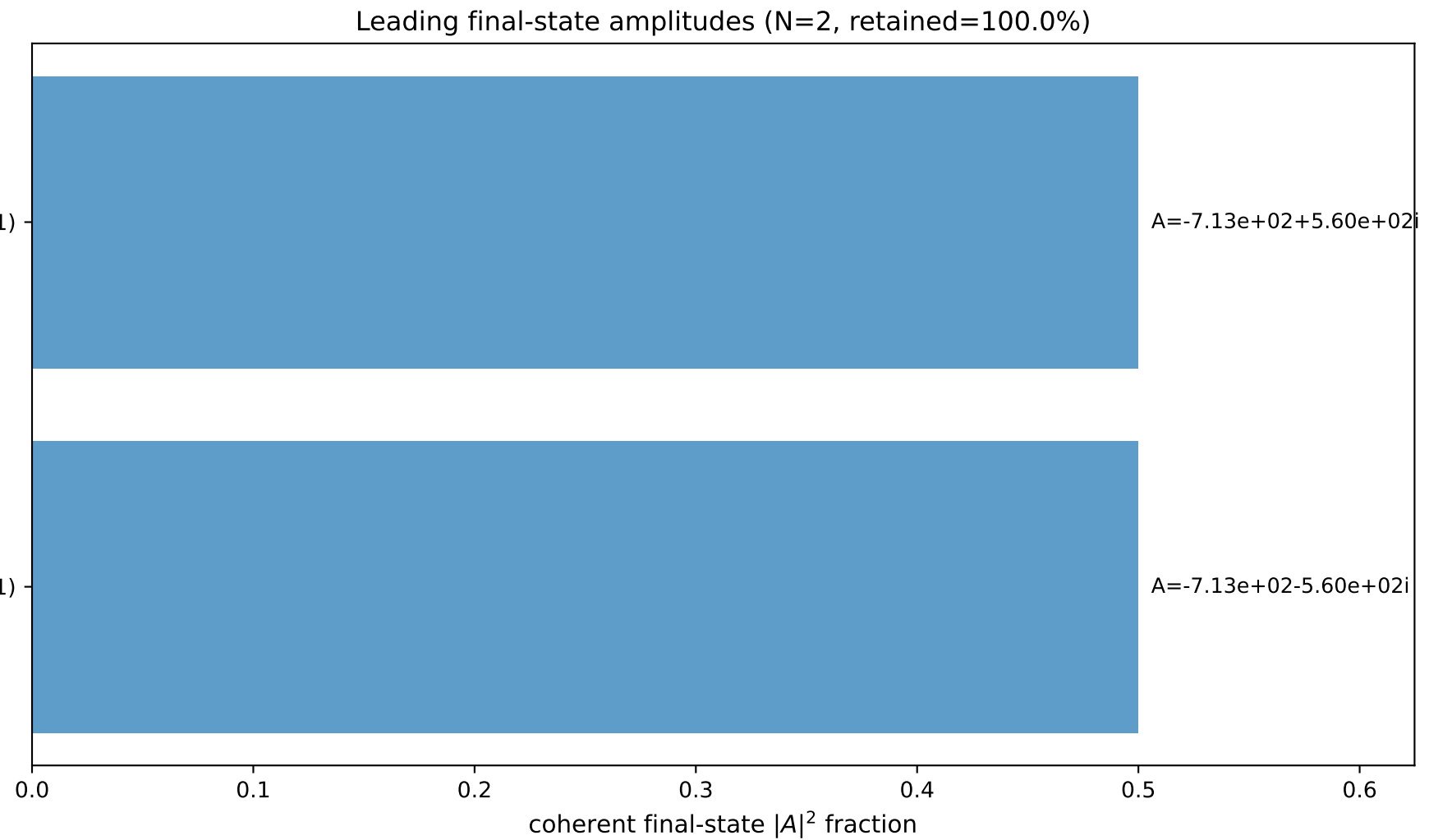
$s=20.4568$, $\sqrt{s}=4.52291$, $|\vec{P}|=2.16419$, $|\vec{P}'|=0.00379415$
 $\theta_{P'}=0.0374492$, $\theta_Y=3.14159$, $\phi_{P'}=3.15038$, $\phi_Y=5.61715$
 $E_Y=1.79435$, $\theta_{\ell'}=7.93354\text{e-}05$, $\phi_{\ell'}=0.00878615$
 $Q^2=15.5004$, $x_B=0.820986$, $t=-2.6489$, $W^2=4.25968$, $y=0.964415$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1642$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1642$
 P $E= 2.3587$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1642$
 ℓ' $E= 1.7906$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.7906$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 0.0038$
 q_Y $E= 1.7943$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.7943$

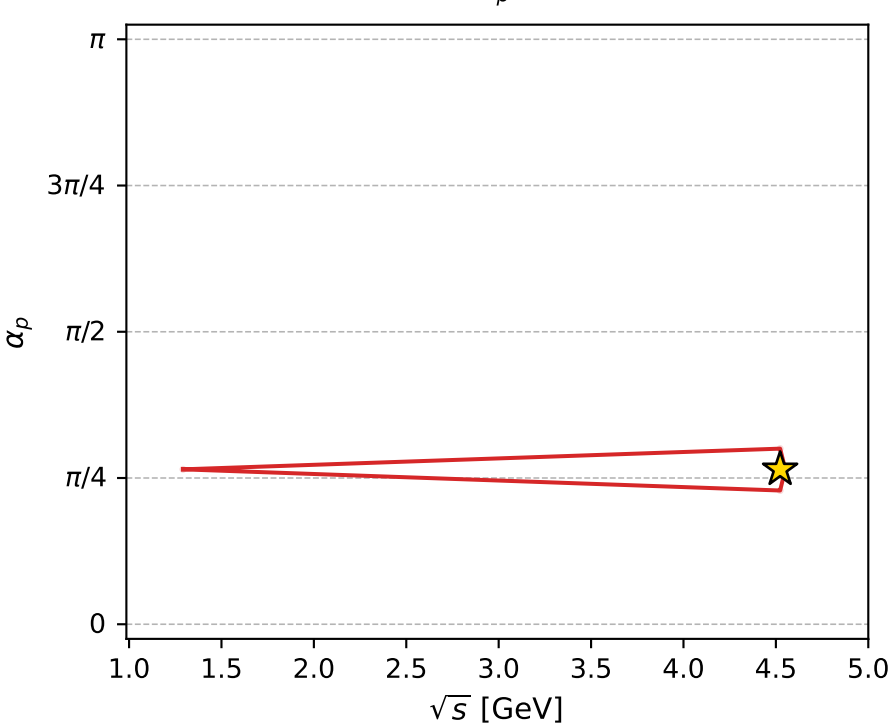
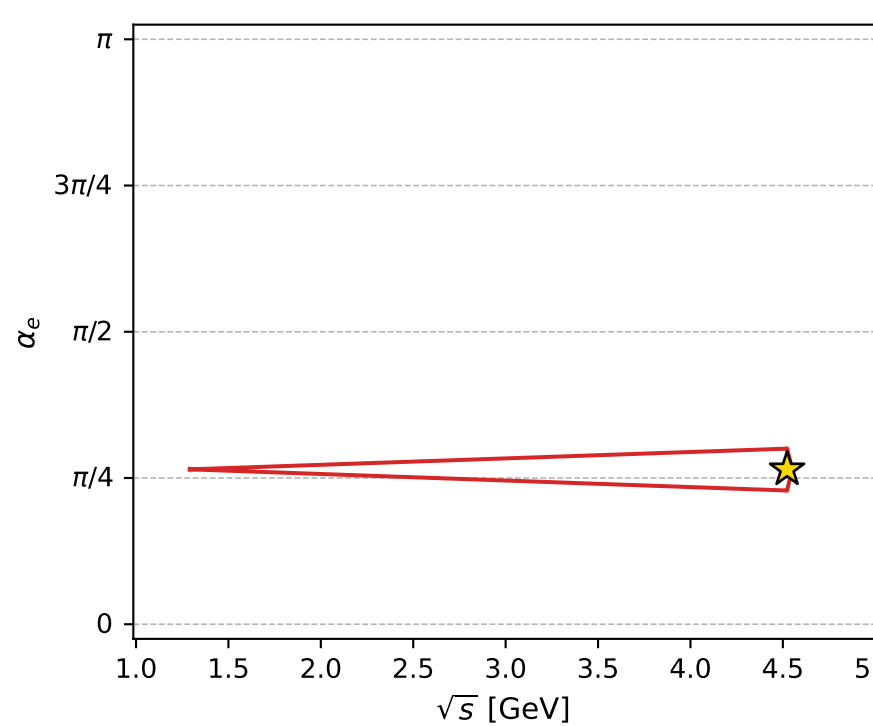
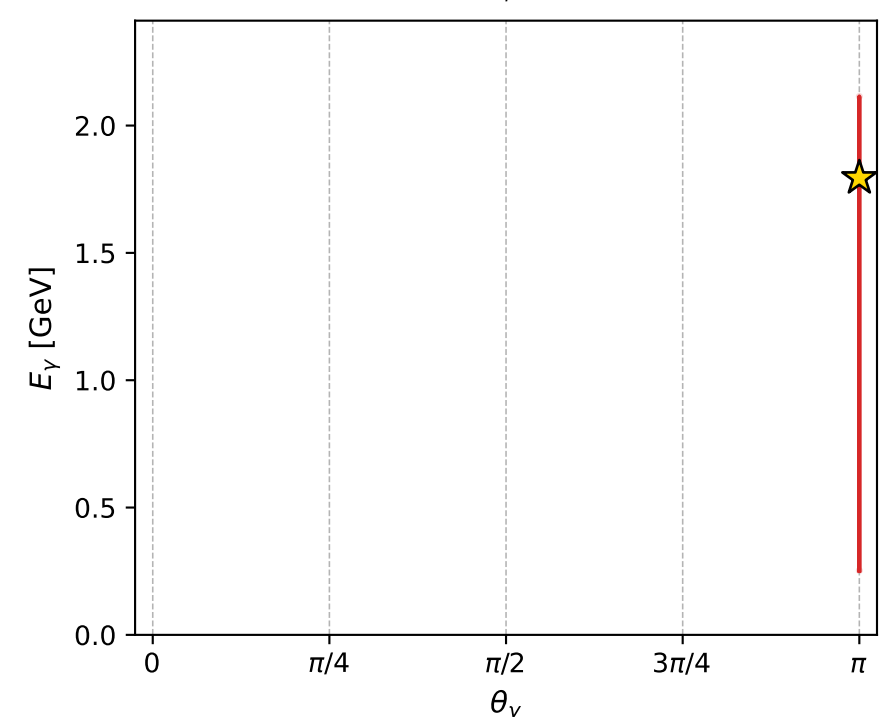
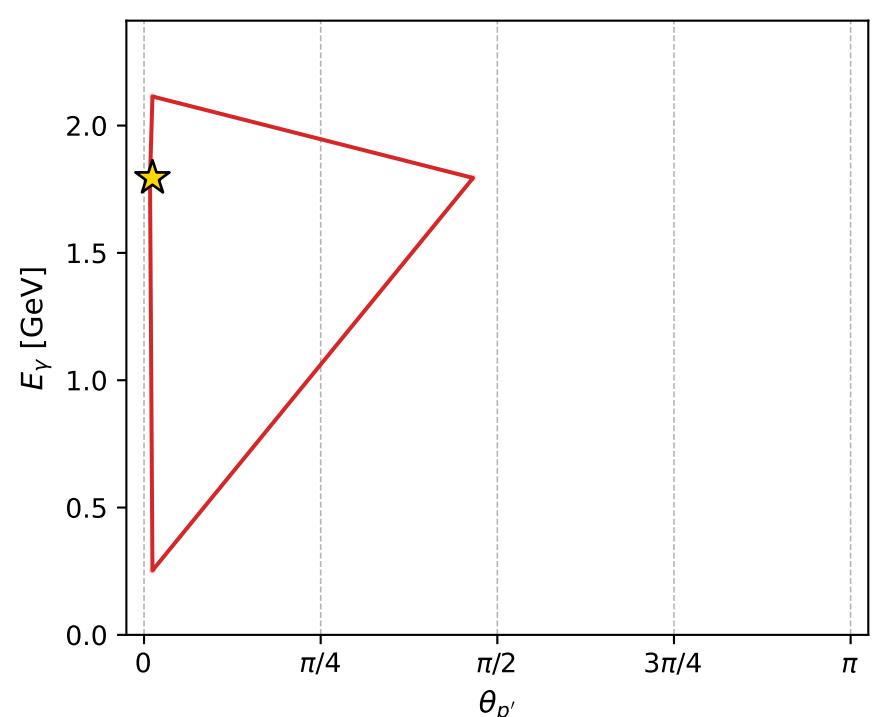
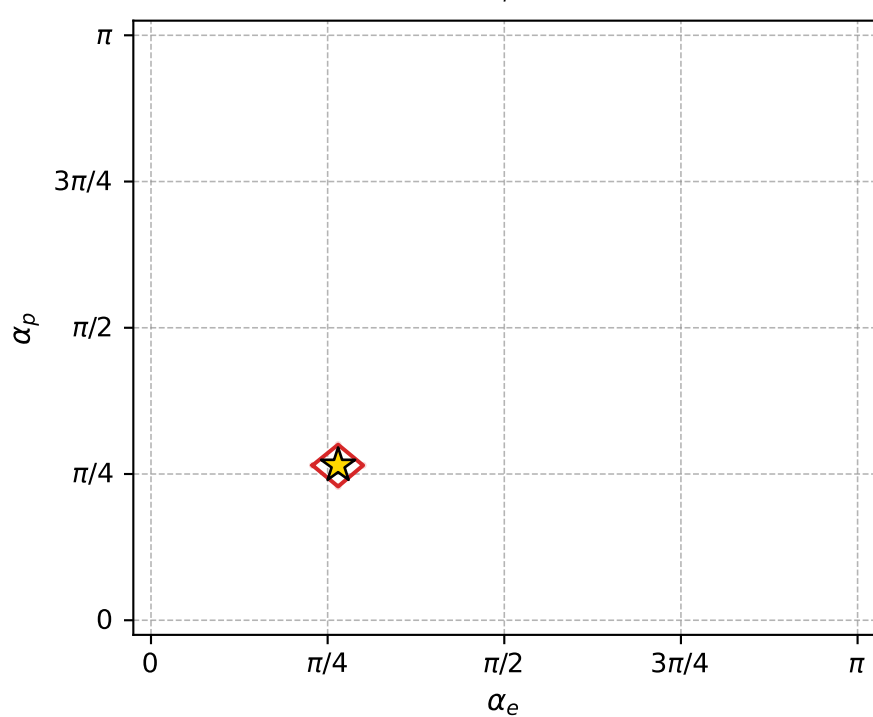
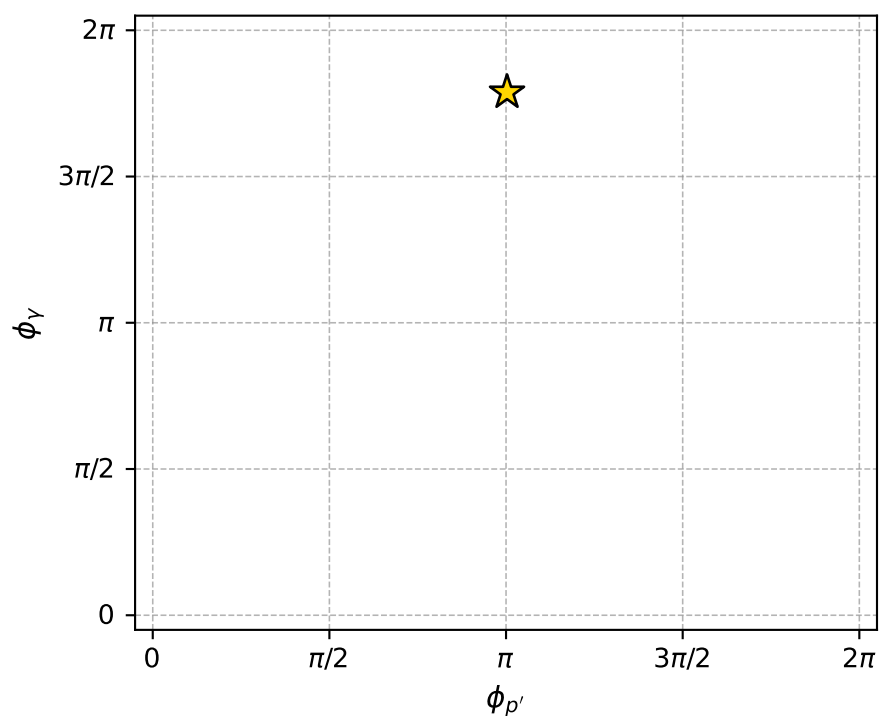
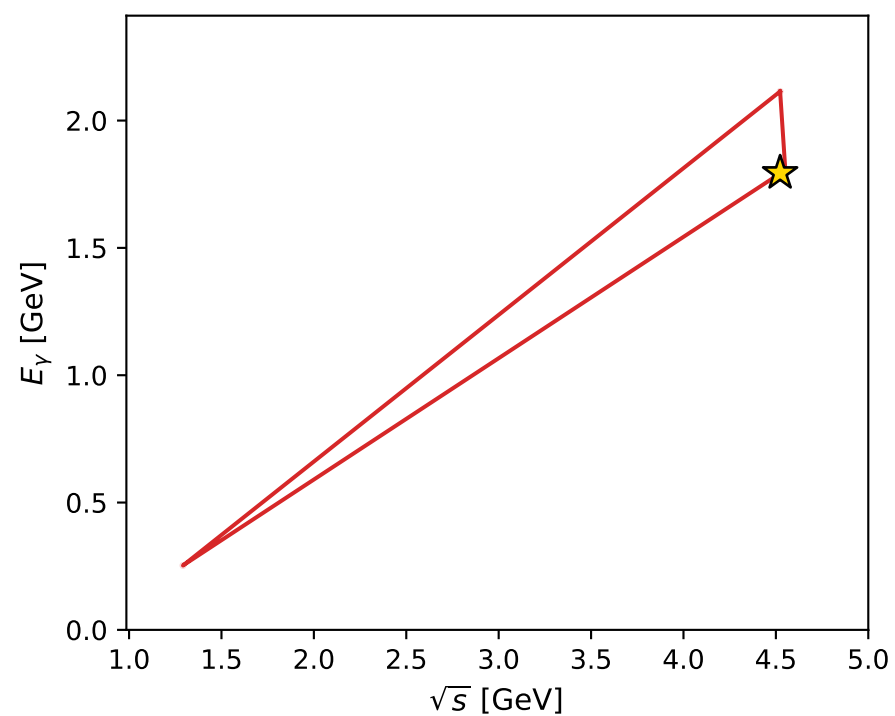
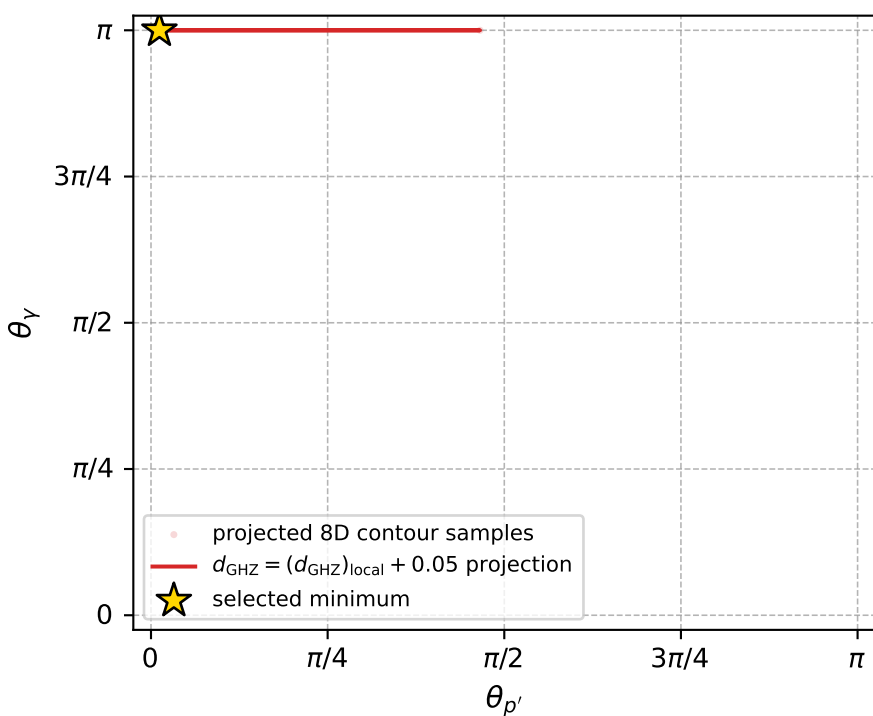


$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$



electron: polarization cluster P1, local minimum 95; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.4044151\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000034$
 above global minimum = $1.1760689\text{e-}08$
 8D contour samples = 130

selected phase-space point:

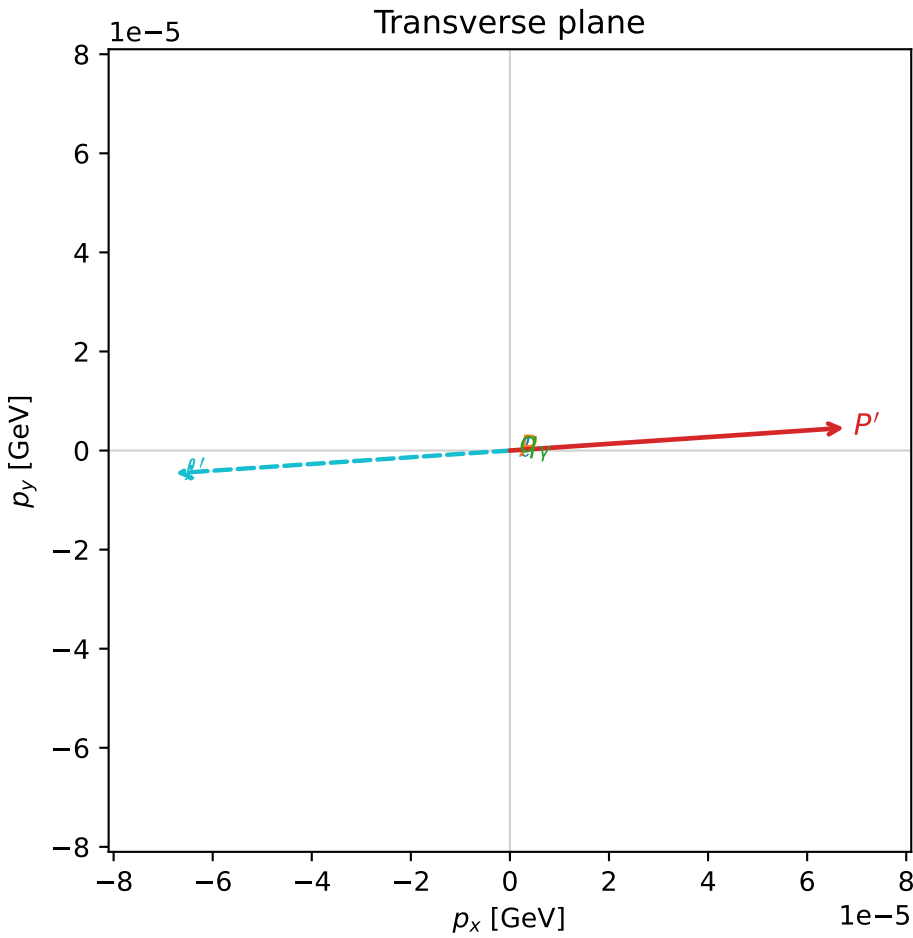
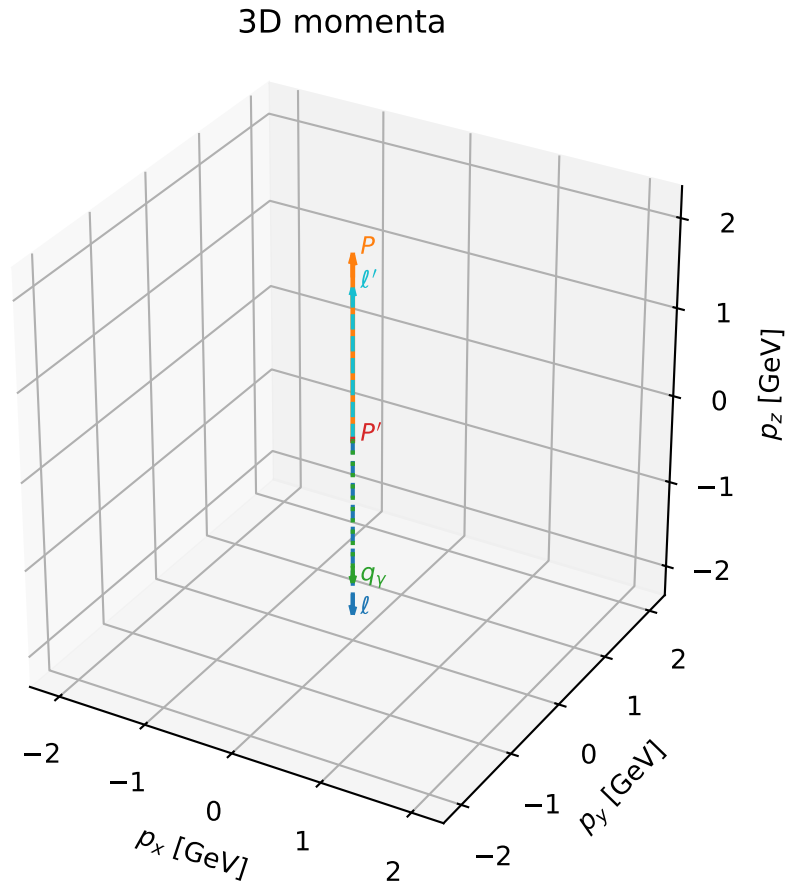
$\text{sqrt_s} = 4.522915$
 $\text{theta_p_out} = 0.03744918$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.794349$
 $\text{phi_p_out} = 3.150379$
 $\text{phi_gamma_out} = 5.617154$
 $\alpha_e = 0.8319156$
 $\alpha_p = 0.8319069$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_104 $d_{\{\text{rm GHZ}\}}=3.80072\text{e-}08$
region: polarization_cluster_01_local_minimum_104
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.79818758$ rad
incoming lepton: $0.698006|+\rangle +0.716092|-\rangle$
proton mixing angle: $\alpha_p=0.79817765$ rad
incoming proton: $0.698013|+\rangle +0.716085|-\rangle$

$s=18.4089$, $\sqrt{s}=4.29056$, $|\vec{P}|=2.04275$, $|\vec{P}'|=0.0189369$
 $\theta_{p'}=3.13802$, $\theta_Y=3.14159$, $\phi_{p'}=0.06771$, $\phi_Y=3.86678$
 $E_Y=1.66672$, $\theta_{\ell'}=4.01396\text{e-}05$, $\phi_{\ell'}=3.2093$
 $Q^2=13.7735$, $x_B=0.818012$, $t=-2.53544$, $W^2=3.94412$, $y=0.960561$

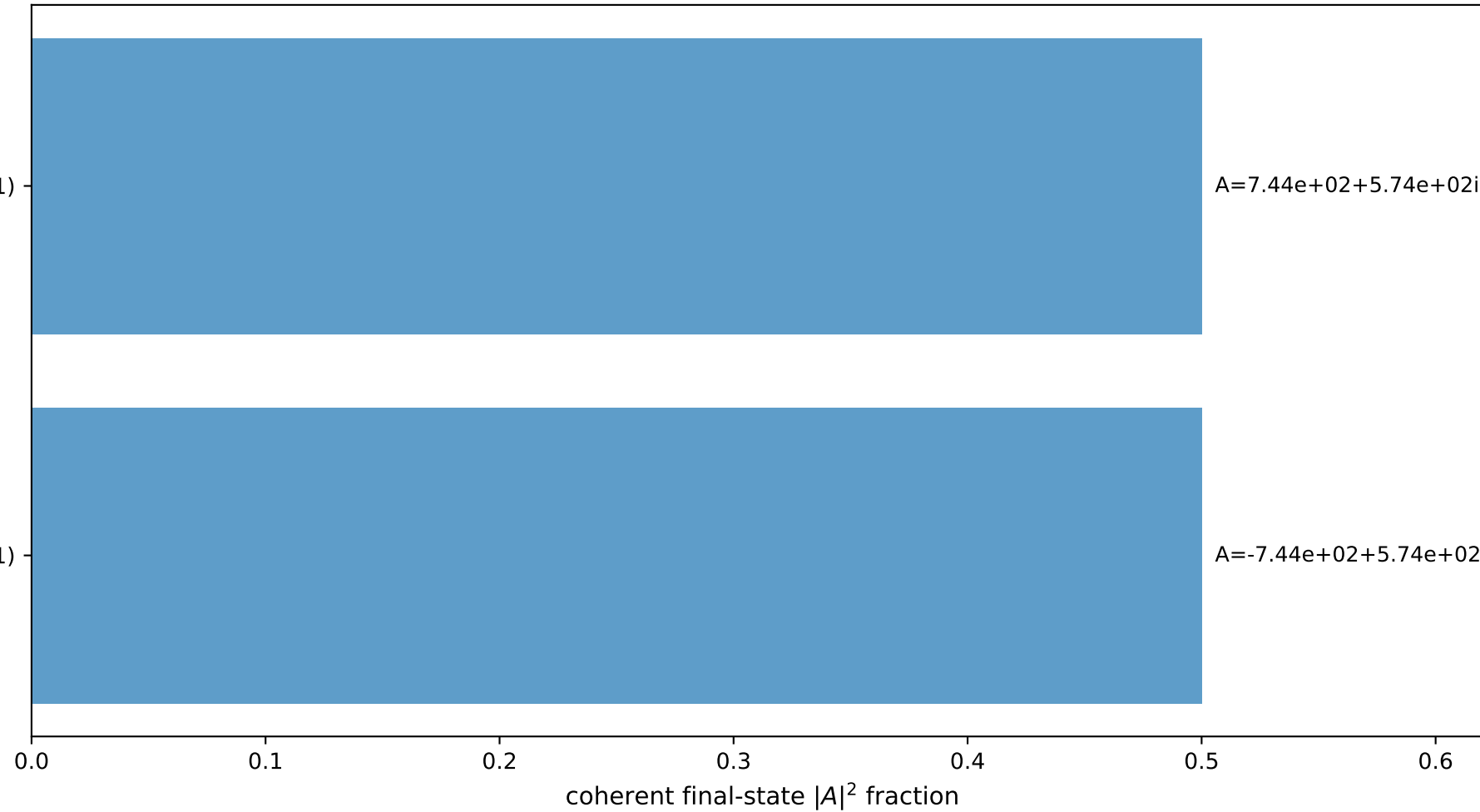
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0427$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0427$
 P $E= 2.2478$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0427$
 ℓ' $E= 1.6857$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 1.6857$
 P' $E= 0.9382$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.0189$
 q_Y $E= 1.6667$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.6667$



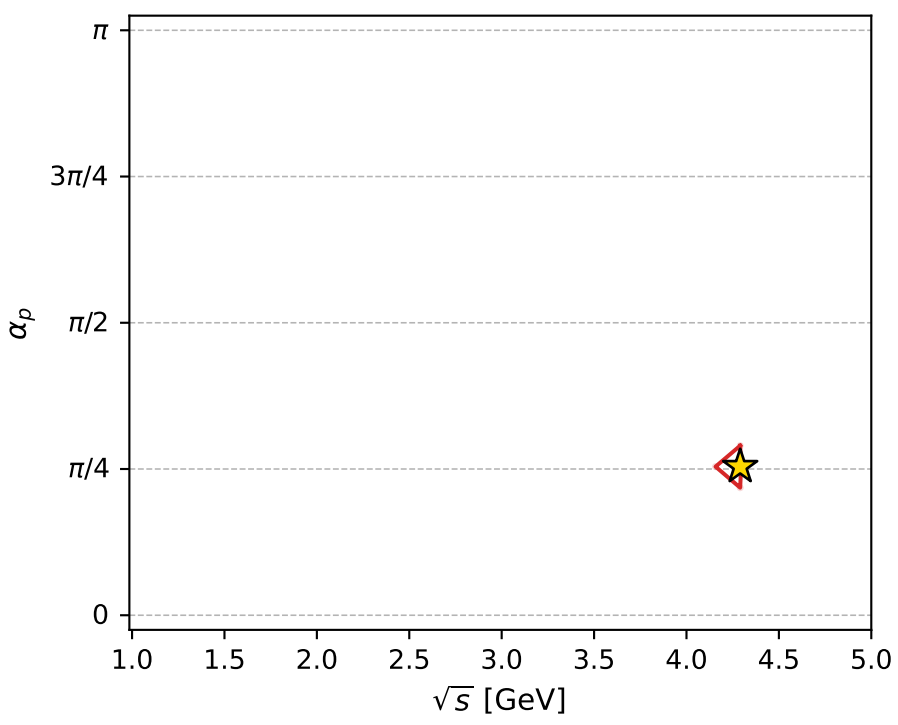
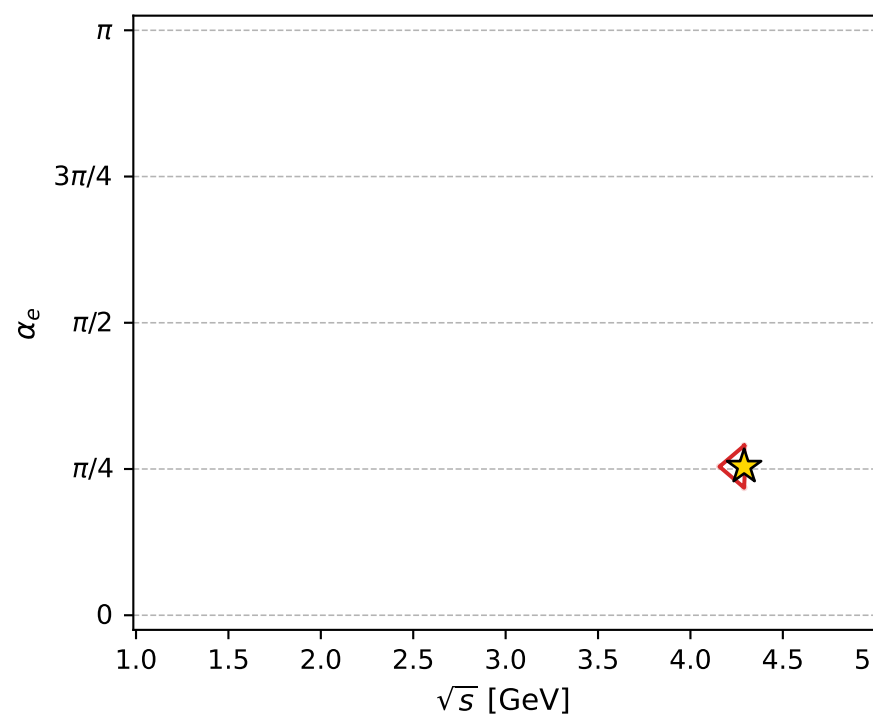
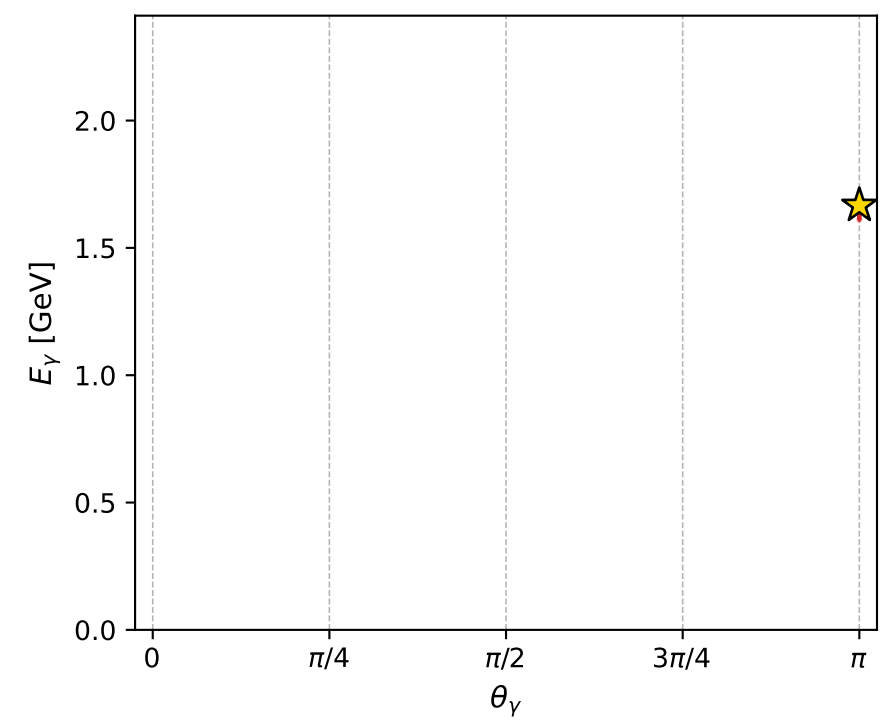
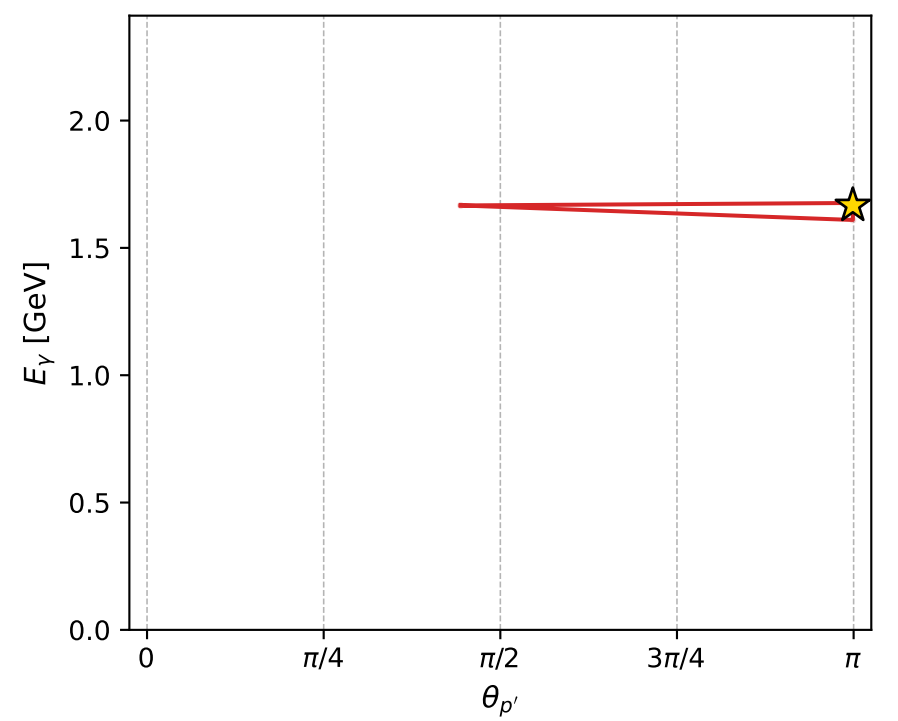
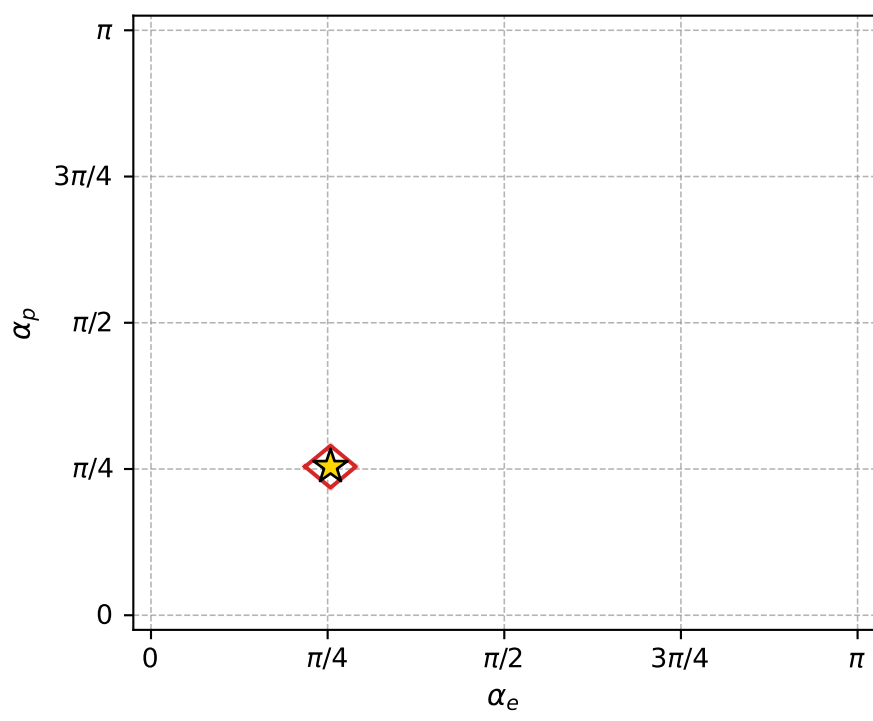
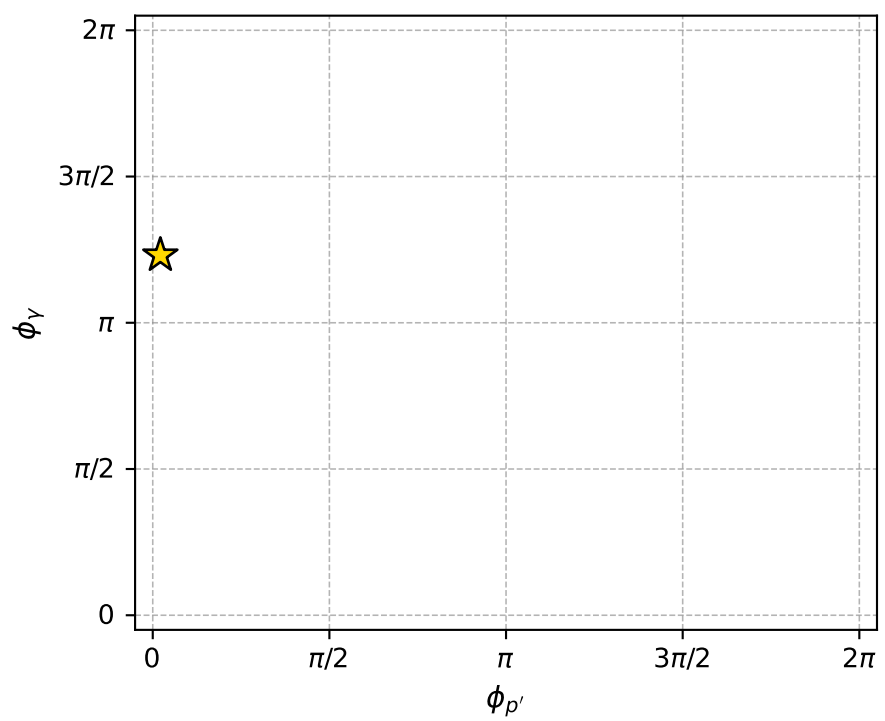
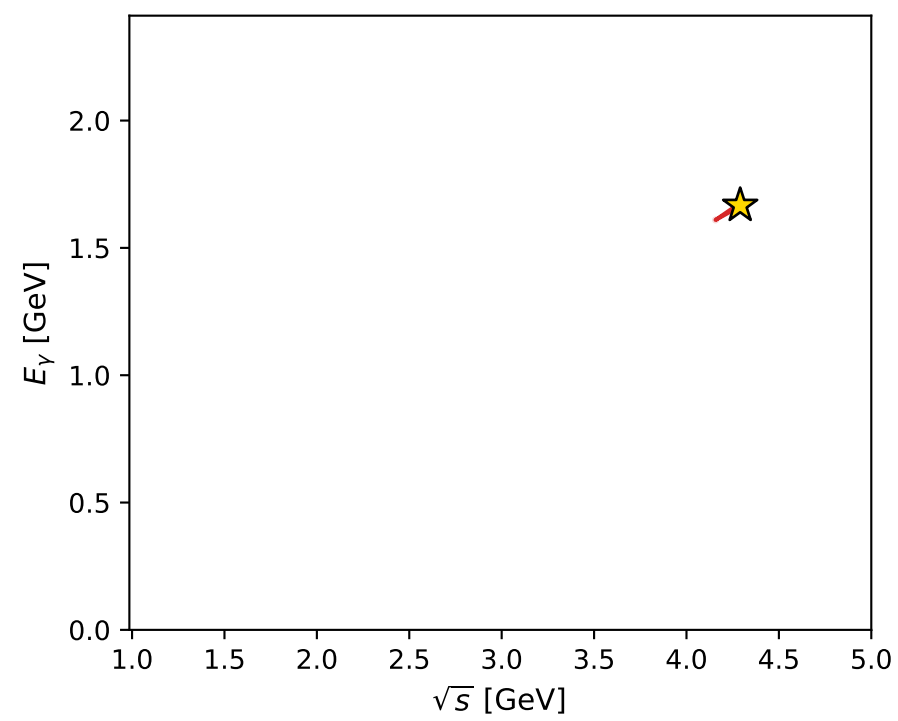
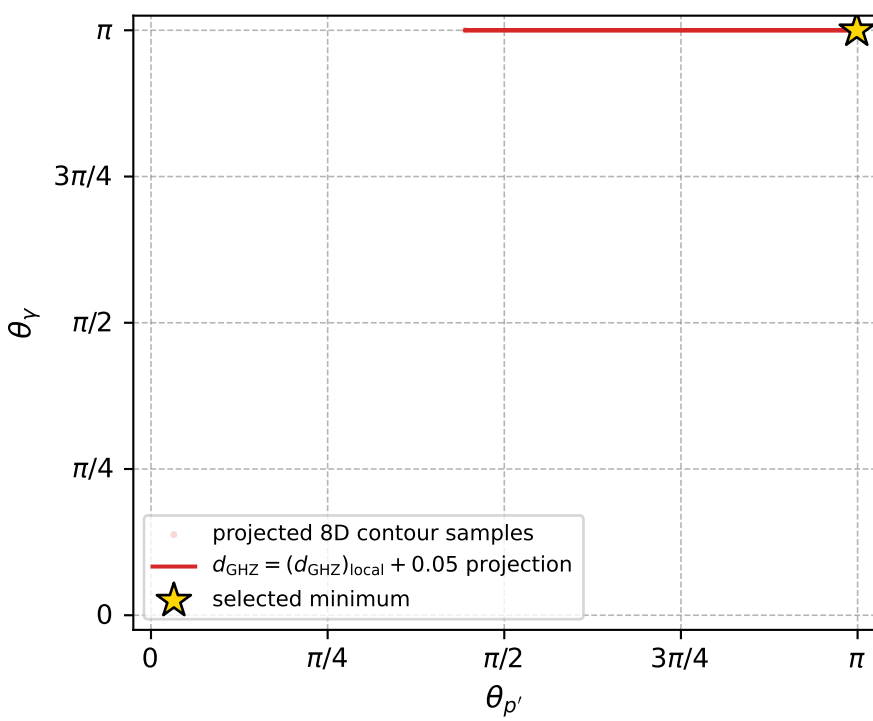
$(h_p, h_Y, h_\ell) = (+1, -1, +1)$

$(h_p, h_Y, h_\ell) = (-1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 104; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.8007186\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000038$
 above global minimum = $1.5723724\text{e-}08$
 8D contour samples = 134

selected phase-space point:

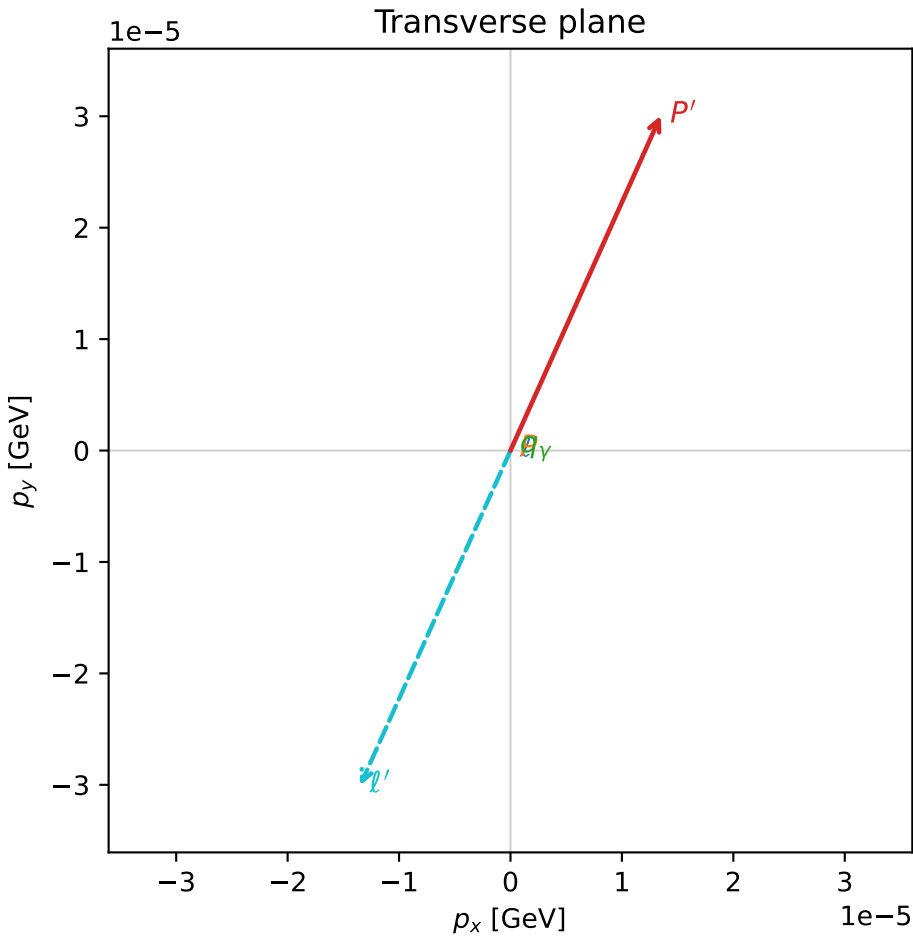
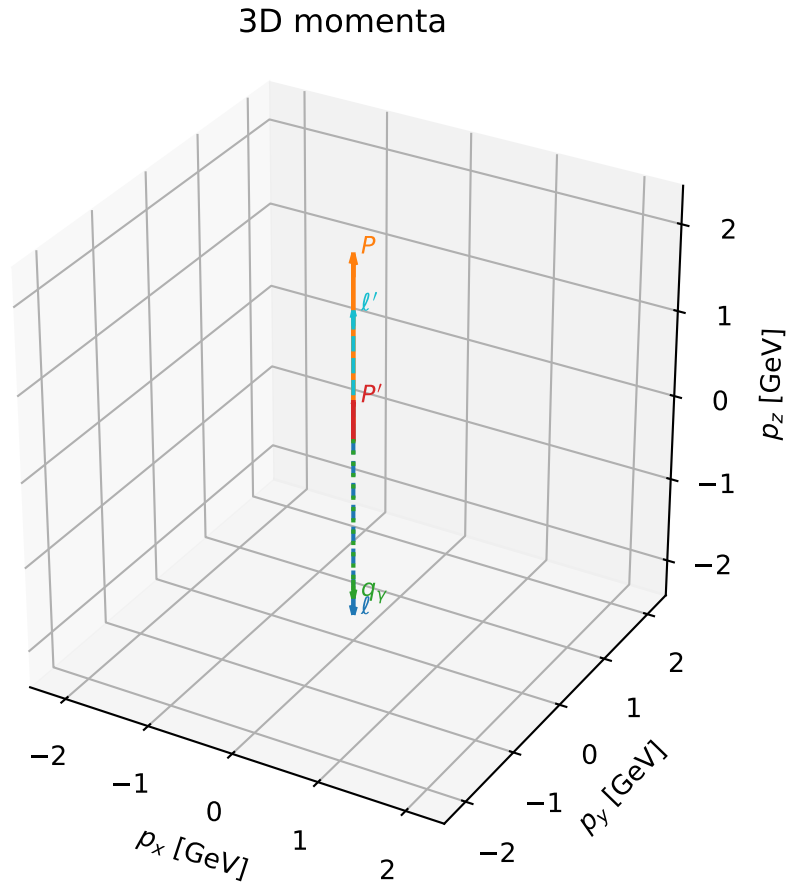
$\text{sqrt_s} = 4.290564$
 $\text{theta_p_out} = 3.13802$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.666718$
 $\text{phi_p_out} = 0.06770995$
 $\text{phi_gamma_out} = 3.866778$
 $\text{alpha_e} = 0.7981876$
 $\text{alpha_p} = 0.7981776$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_112 $d_{\{\text{rm GHZ}\}}=4.14029\text{e-}08$
region: polarization_cluster_01_local_minimum_112
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.7523389$ rad
incoming lepton: $0.730093|+\rangle +0.683348|-\rangle$
proton mixing angle: $\alpha_p=0.75235965$ rad
incoming proton: $0.730078|+\rangle +0.683363|-\rangle$

$s=19.6572$, $\sqrt{s}=4.43364$, $|\vec{P}|=2.1176$, $|\vec{P}'|=0.429483$
 $\theta_{P'}=7.6699\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=1.14894$, $\phi_Y=4.37324$
 $E_Y=1.91574$, $\theta_{\ell'}=2.21637\text{e-}05$, $\phi_{\ell'}=4.29053$
 $Q^2=12.5892$, $x_B=0.692189$, $t=-1.20006$, $W^2=6.47813$, $y=0.968585$

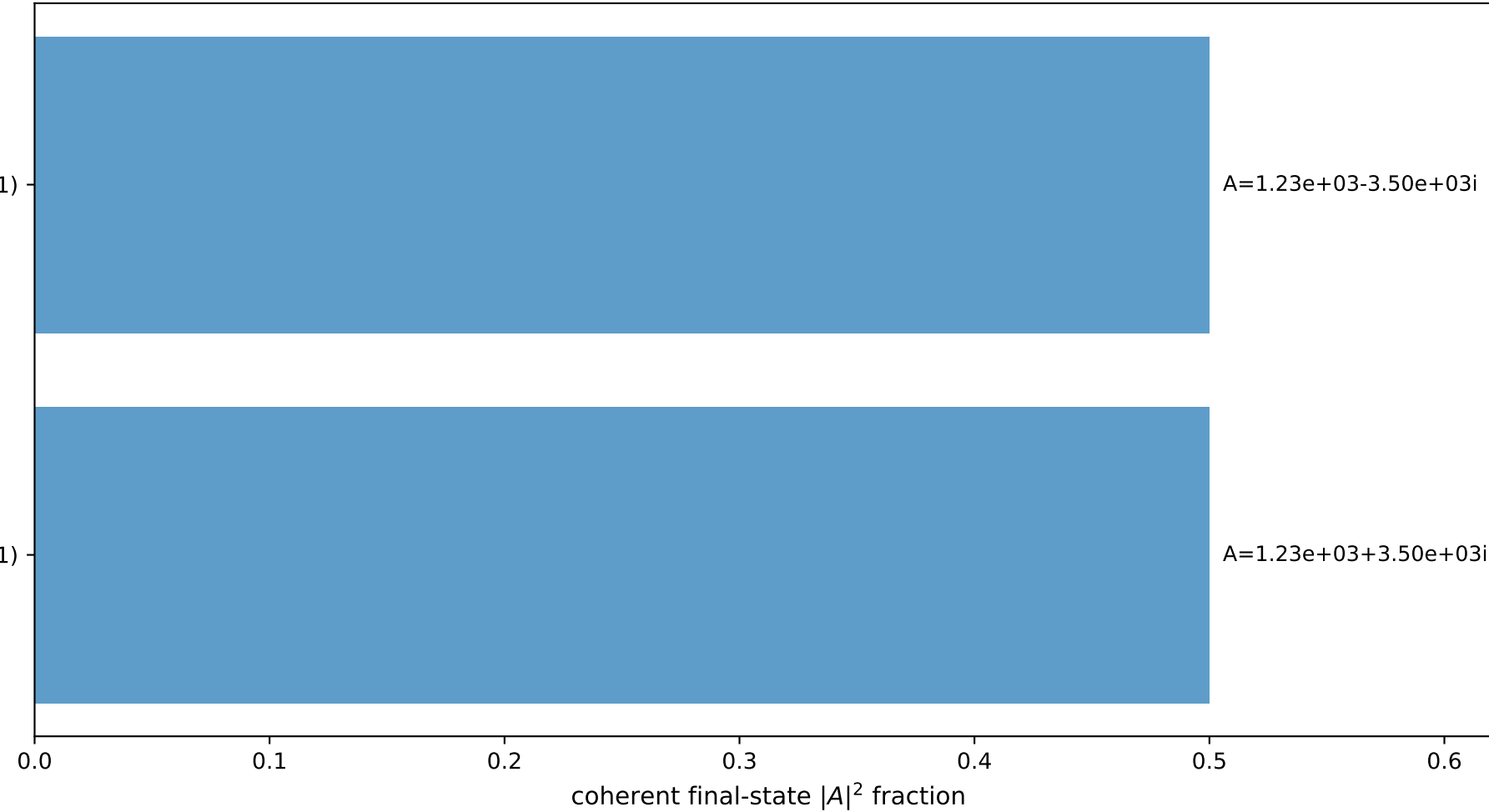
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1176$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1176$
 P $E= 2.3160$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1176$
 ℓ' $E= 1.4863$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4863$
 P' $E= 1.0316$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4295$
 q_Y $E= 1.9157$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9157$



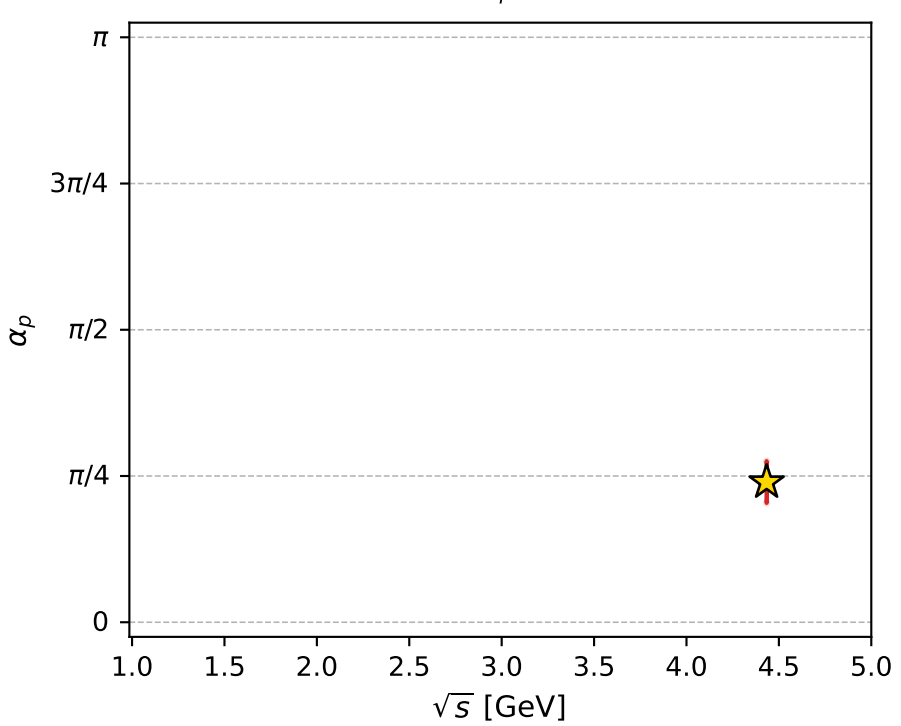
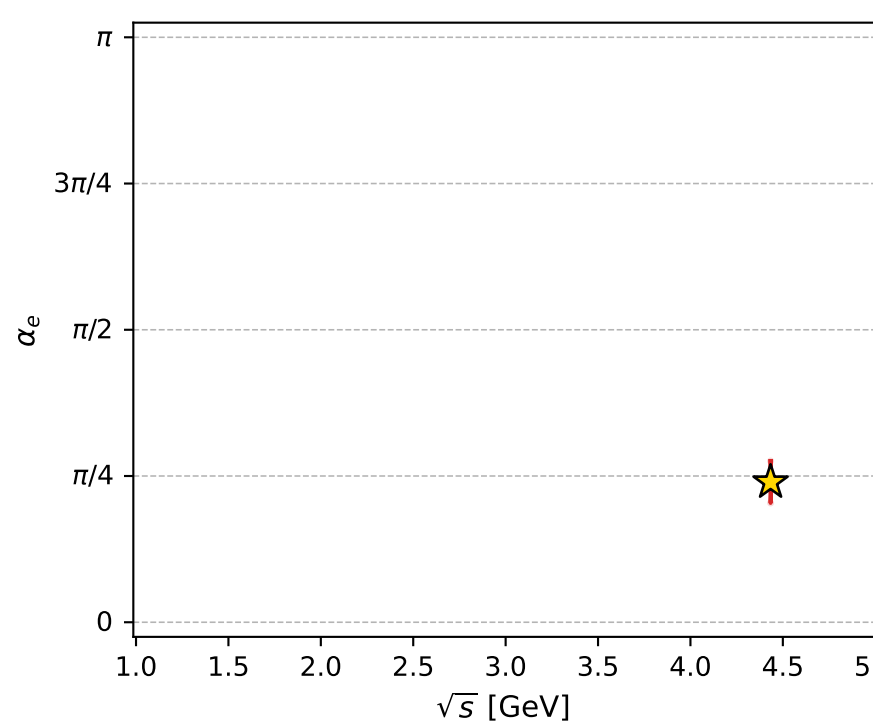
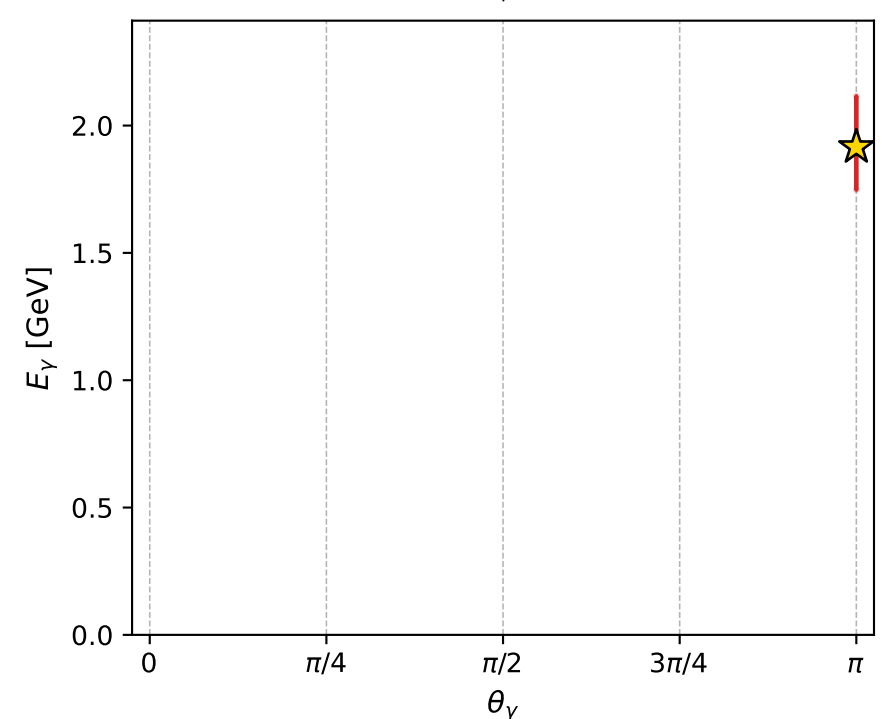
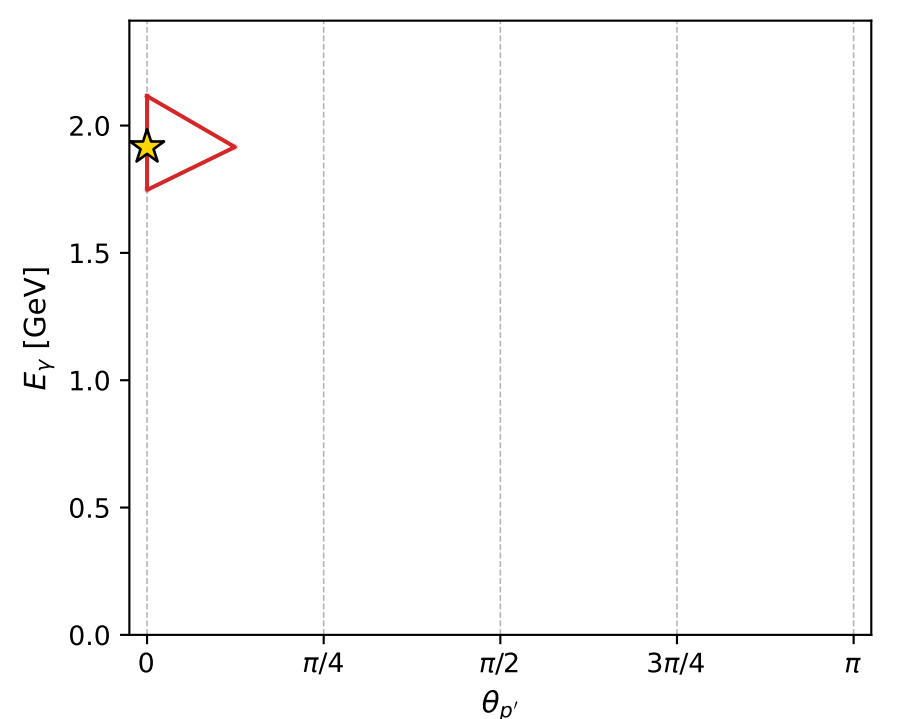
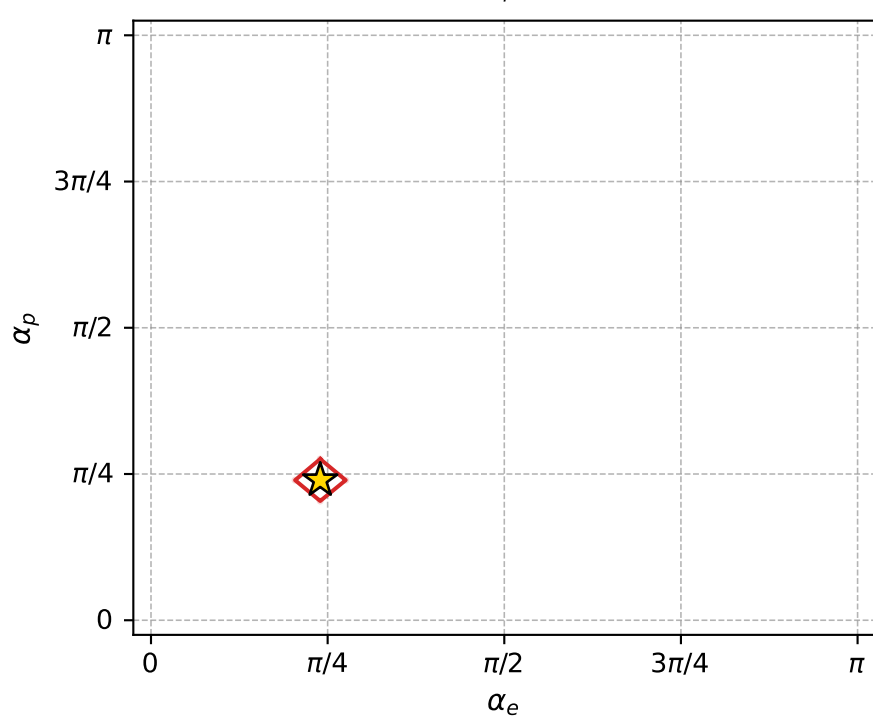
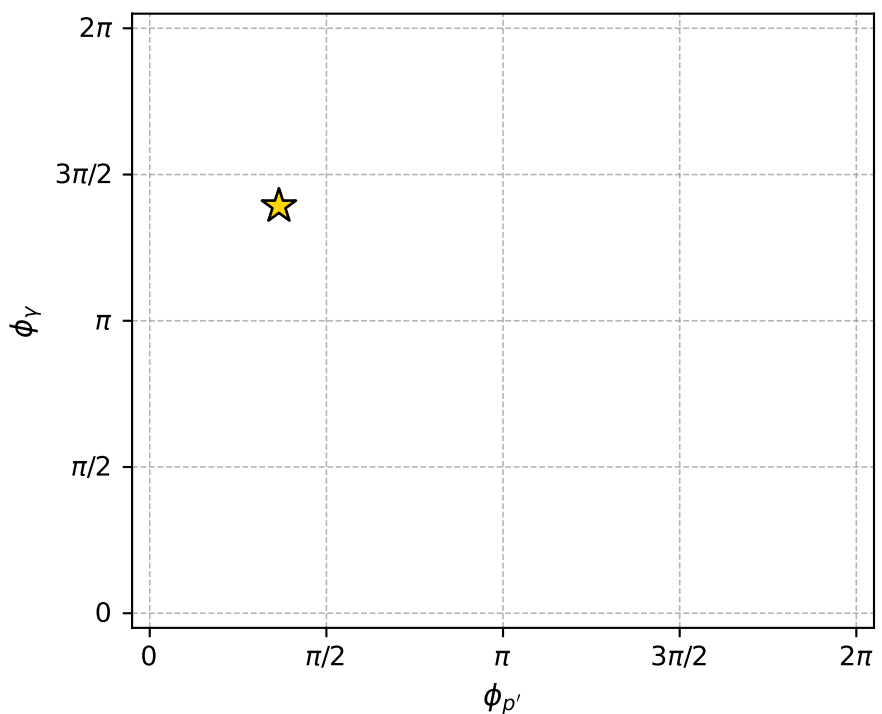
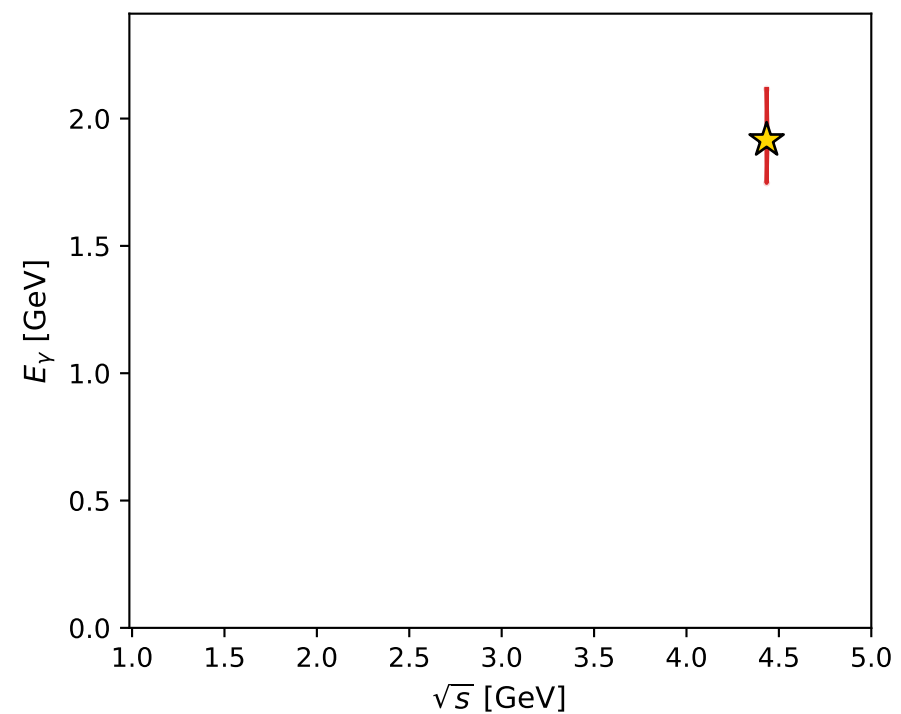
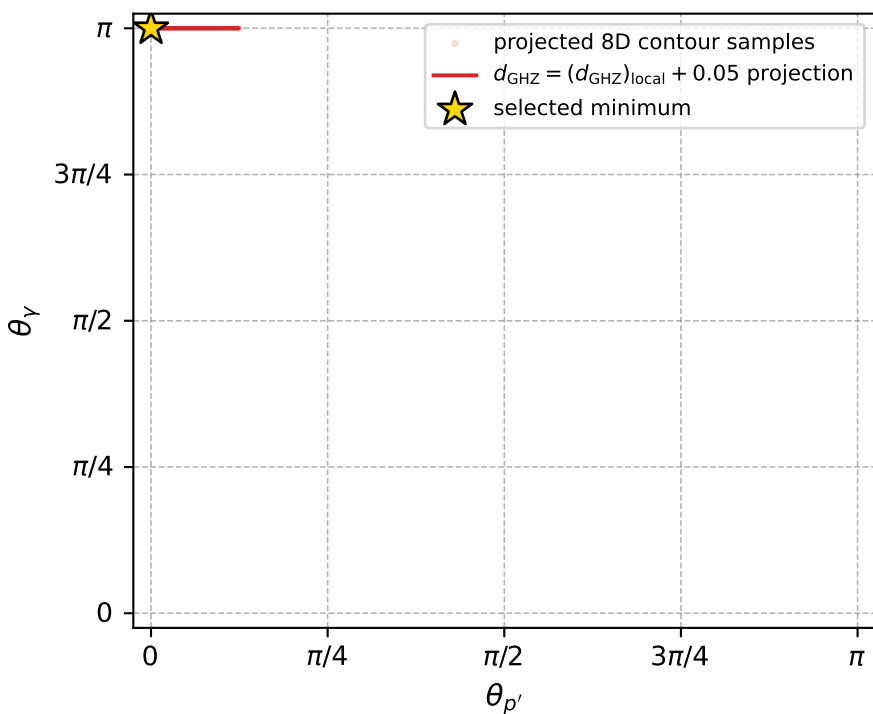
$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 112; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.140286\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000041$
 above global minimum = $1.9119398\text{e-}08$
 8D contour samples = 86

selected phase-space point:

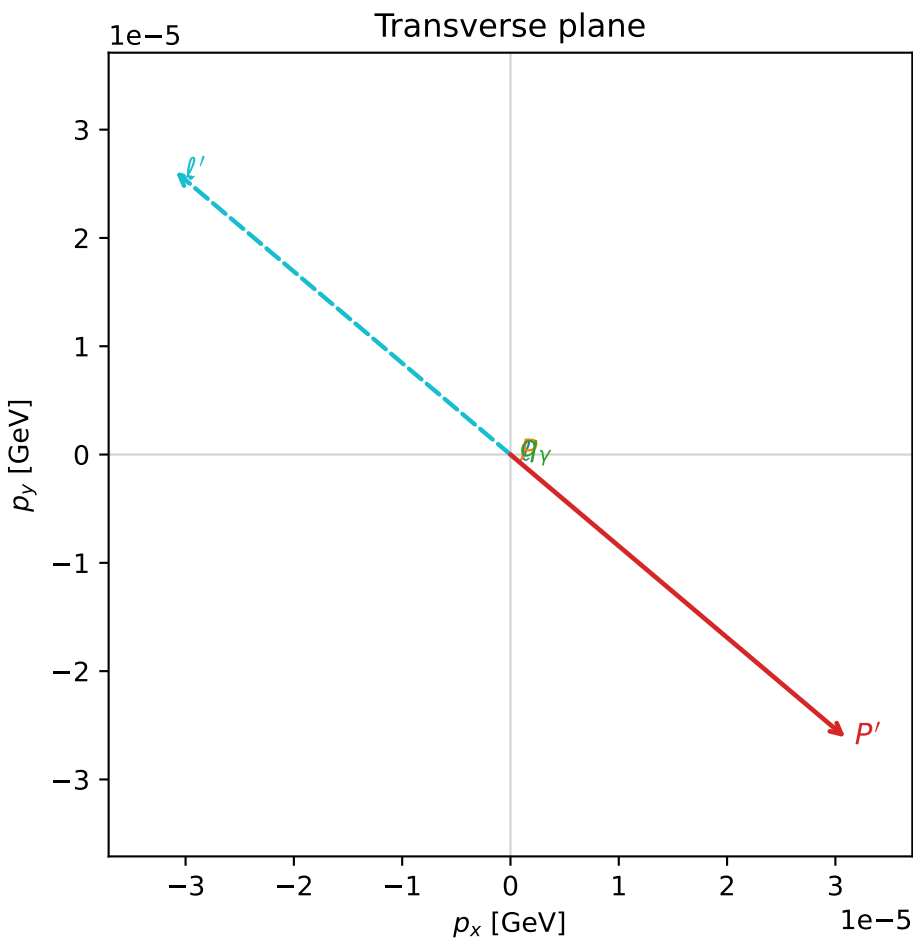
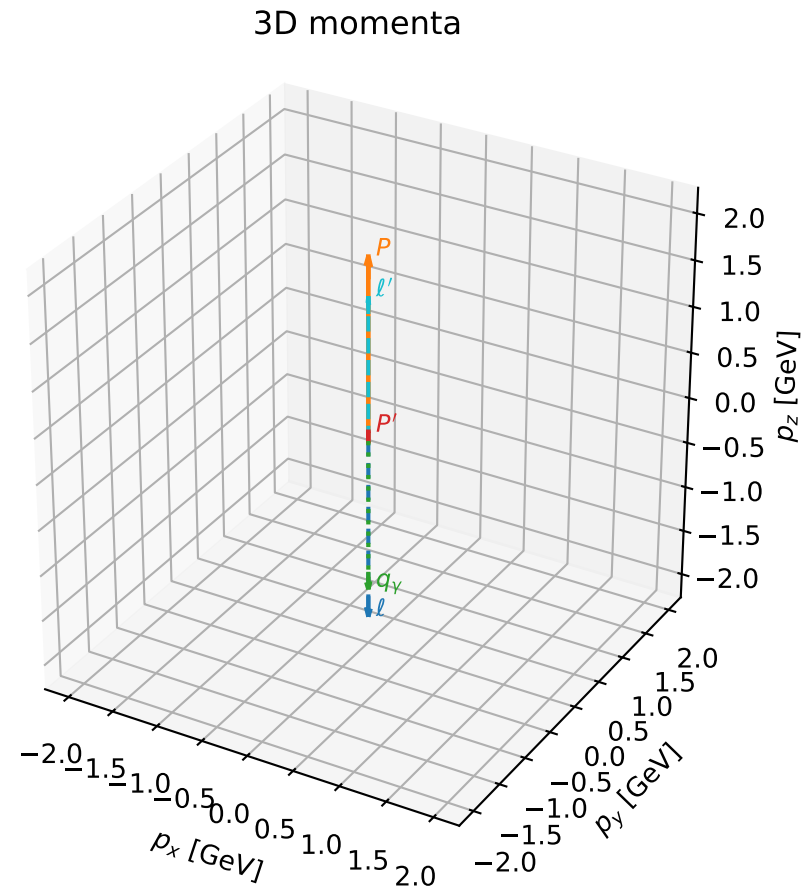
$\text{sqrt_s} = 4.433642$
 $\text{theta_p_out} = 7.669904\text{e-}05$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.915738$
 $\text{phi_p_out} = 1.14894$
 $\text{phi_gamma_out} = 4.373239$
 $\text{alpha_e} = 0.7523389$
 $\text{alpha_p} = 0.7523597$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_113 $d_{\{\text{rm GHZ}\}}=4.24901\text{e-}08$
region: polarization_cluster_01_local_minimum_113
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.83948165$ rad
incoming lepton: $0.667849|+\rangle +0.744297|-\rangle$
proton mixing angle: $\alpha_p=0.8394633$ rad
incoming proton: $0.667862|+\rangle +0.744285|-\rangle$

$s=17.2983$, $\sqrt{s}=4.15912$, $|\vec{P}|=1.97379$, $|\vec{P}'|=0.101055$
 $\theta_{P'}=0.000400551$, $\theta_Y=3.14159$, $\phi_{P'}=5.58175$, $\phi_Y=4.12102$
 $E_Y=1.65837$, $\theta_{\ell'}=2.59918\text{e-}05$, $\phi_{\ell'}=2.44016$
 $Q^2=12.2953$, $x_B=0.780179$, $t=-1.9648$, $W^2=4.34413$, $y=0.959869$

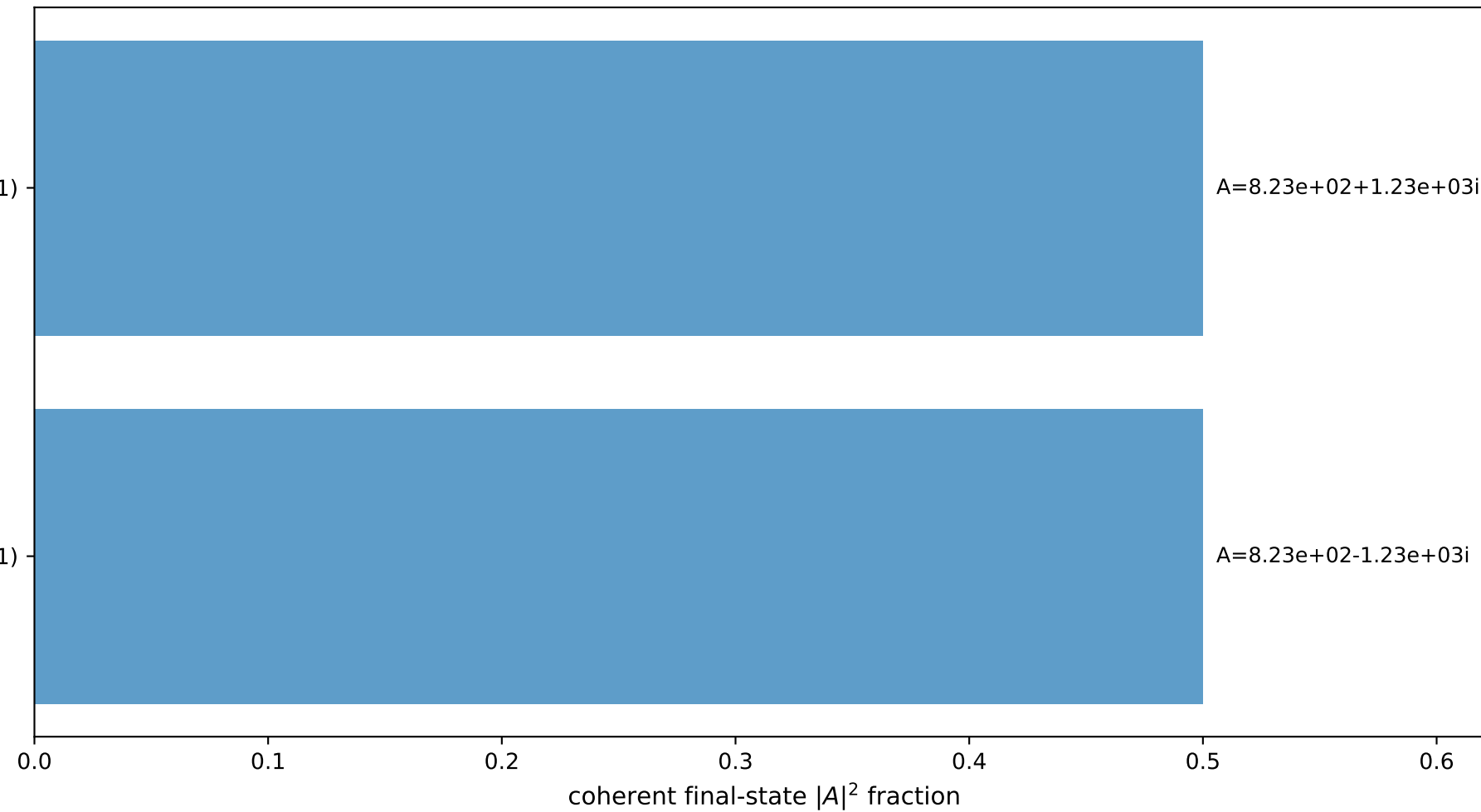
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9738$
 P $E= 2.1853$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9738$
 ℓ' $E= 1.5573$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5573$
 P' $E= 0.9434$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.1011$
 q_Y $E= 1.6584$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.6584$



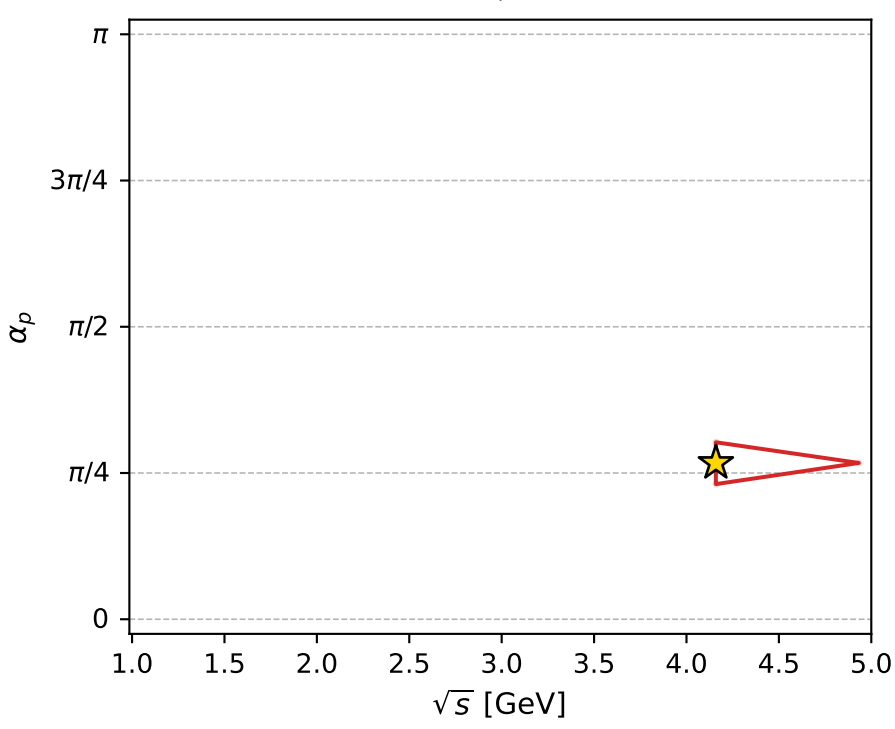
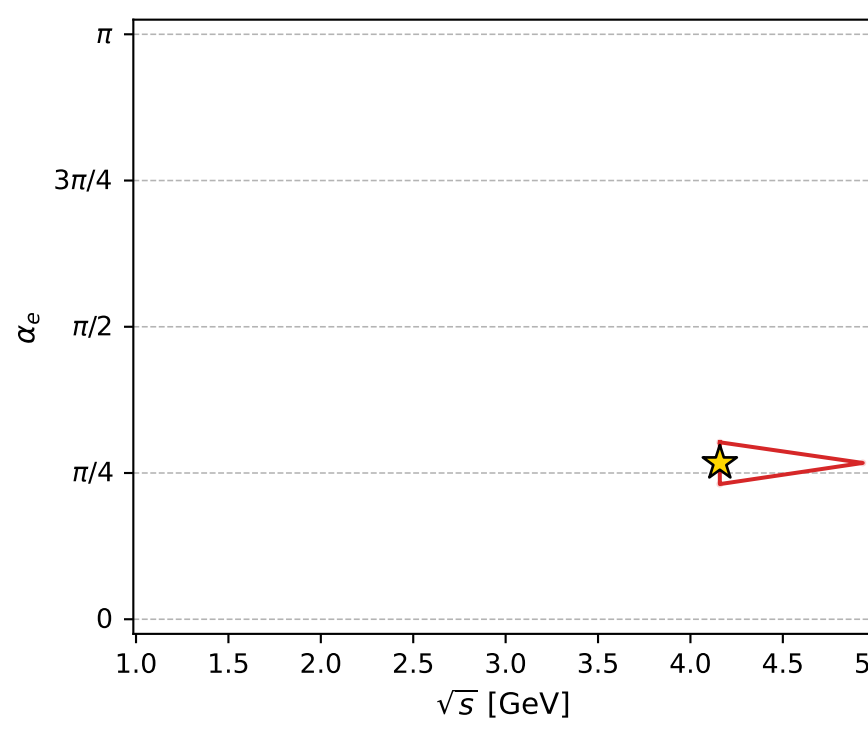
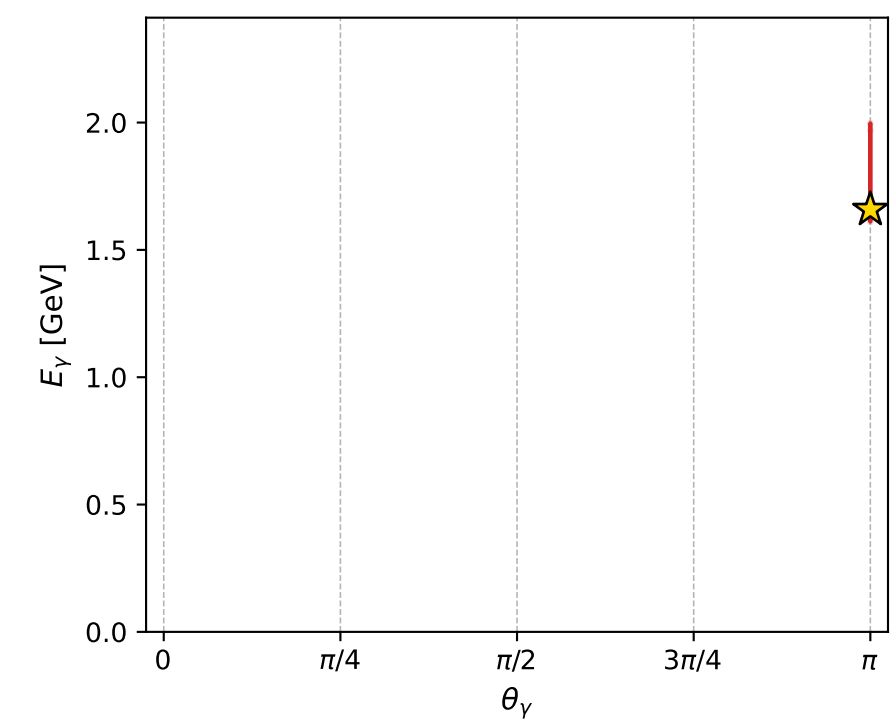
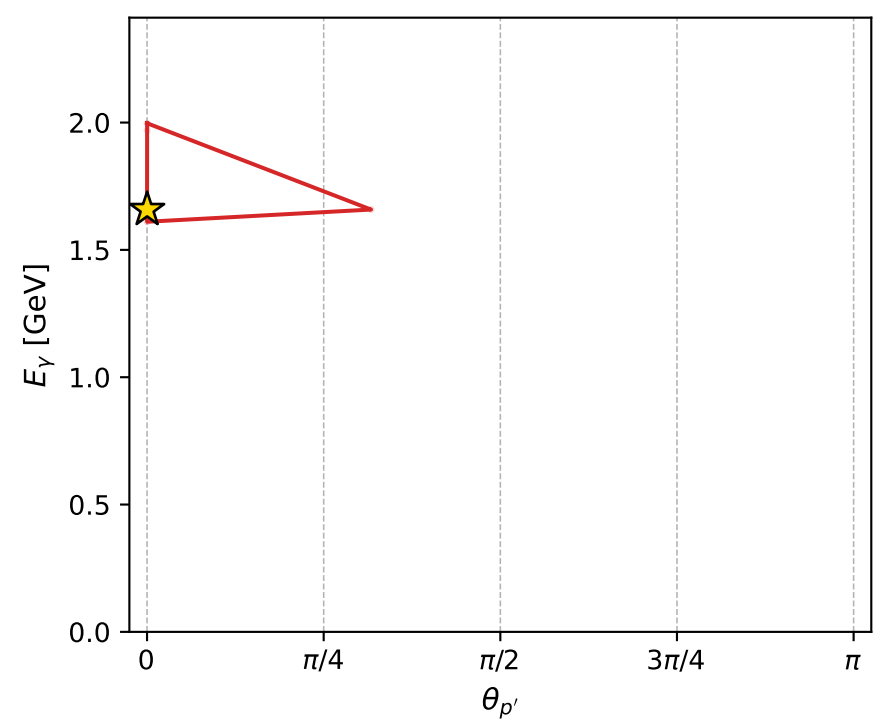
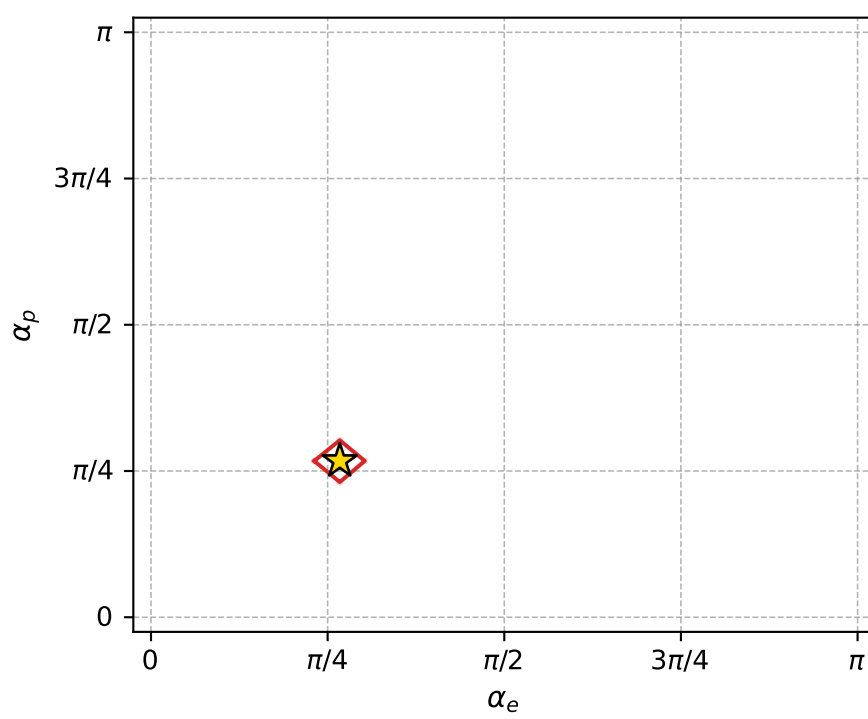
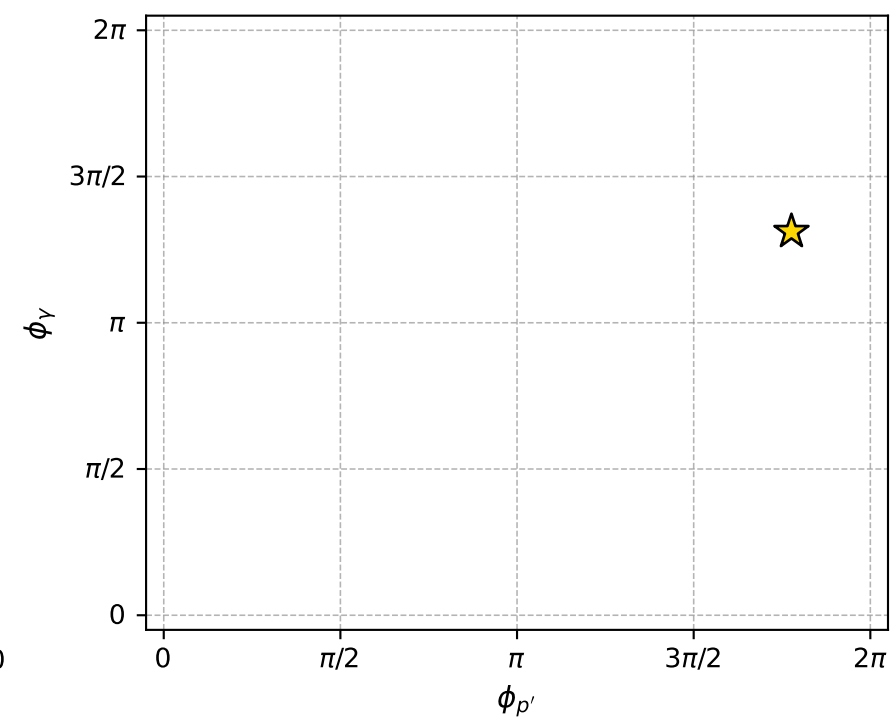
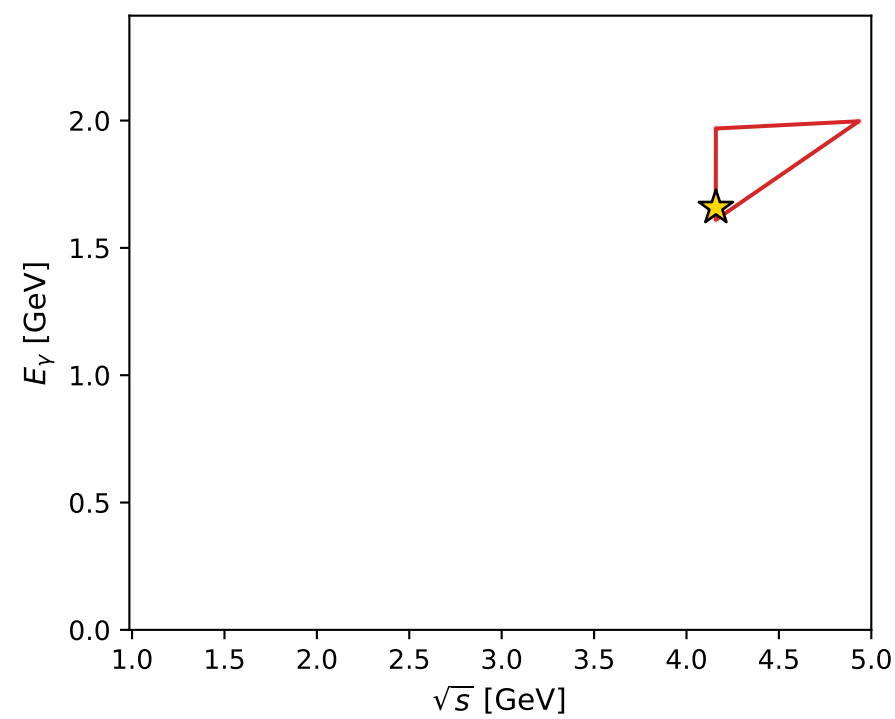
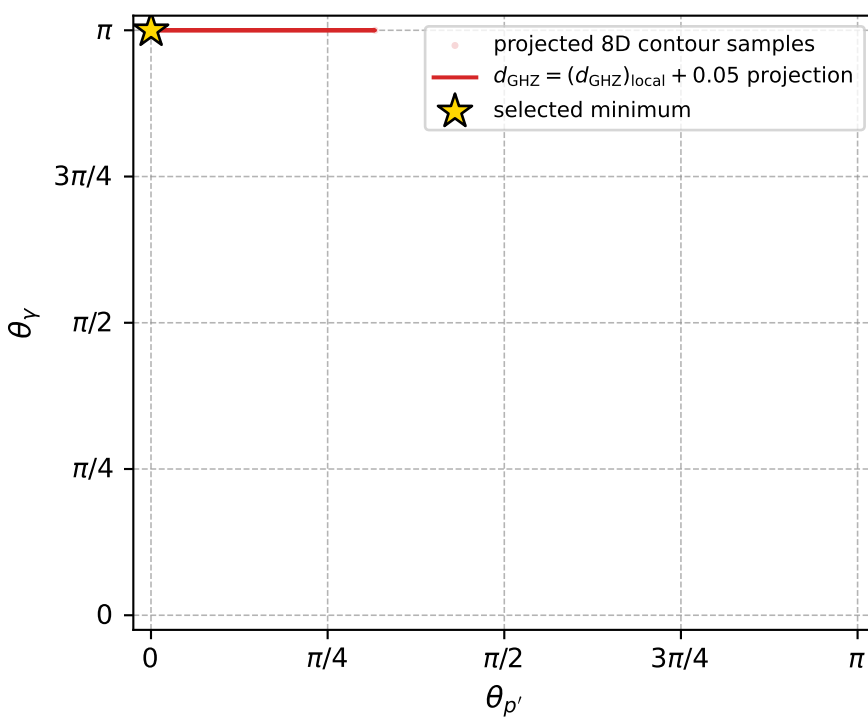
$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 113; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.2490077\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000042$
 above global minimum = $2.0206615\text{e-}08$
 8D contour samples = 125

selected phase-space point:

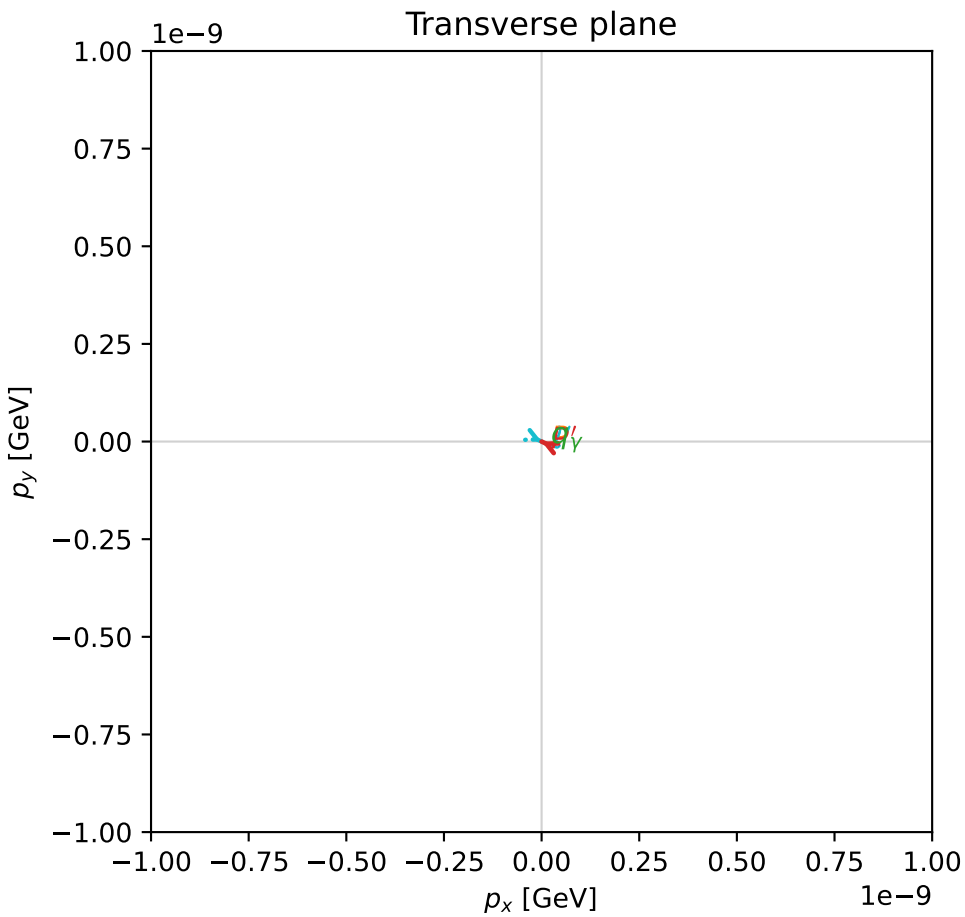
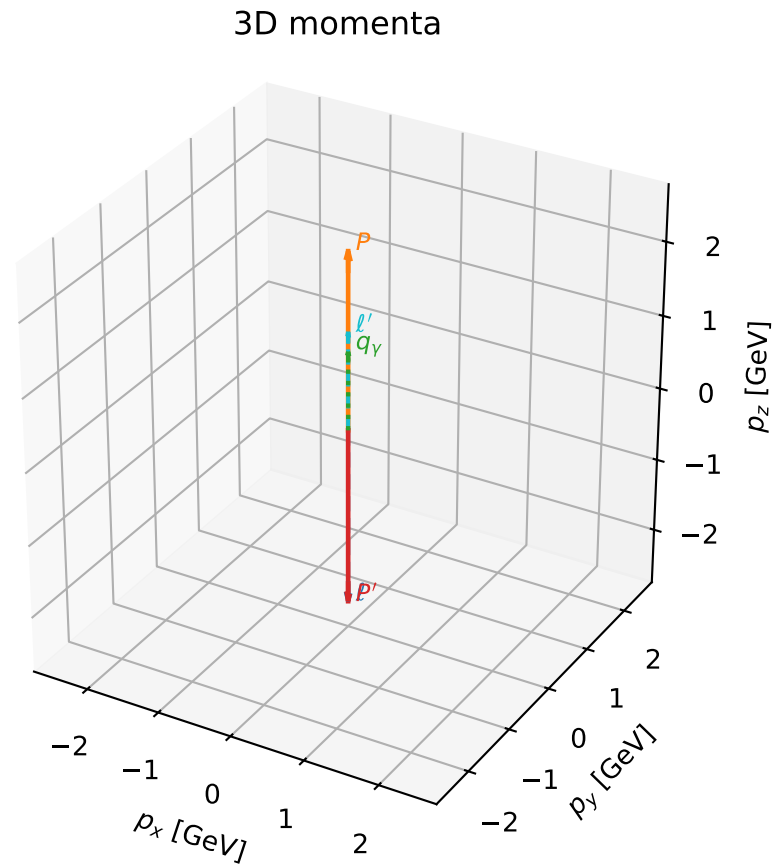
$\text{sqrt_s} = 4.15912$
 $\text{theta_p_out} = 0.0004005507$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.658373$
 $\text{phi_p_out} = 5.581748$
 $\text{phi_gamma_out} = 4.121016$
 $\text{alpha_e} = 0.8394817$
 $\text{alpha_p} = 0.8394633$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_114 $d_{\{\text{rm GHZ}\}}=4.29606\text{e-}08$
region: polarization_cluster_01_local_minimum_114
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.94445649$ rad
incoming lepton: $0.586183|+\rangle +0.810178|-\rangle$
proton mixing angle: $\alpha_p=0.62636419$ rad
incoming proton: $0.810164|+\rangle +0.586203|-\rangle$

$s=24.8065$, $\sqrt{s}=4.98061$, $|\vec{P}|=2.40198$, $|\vec{P}'|=2.40198$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.70525$, $\phi_Y=5.75553$
 $E_Y=1.06741$, $\theta_{l'}=0$, $\phi_{l'}=5.84684$
 $Q^2=12.8224$, $x_B=0.546678$, $t=-23.078$, $W^2=11.5126$, $y=0.980293$

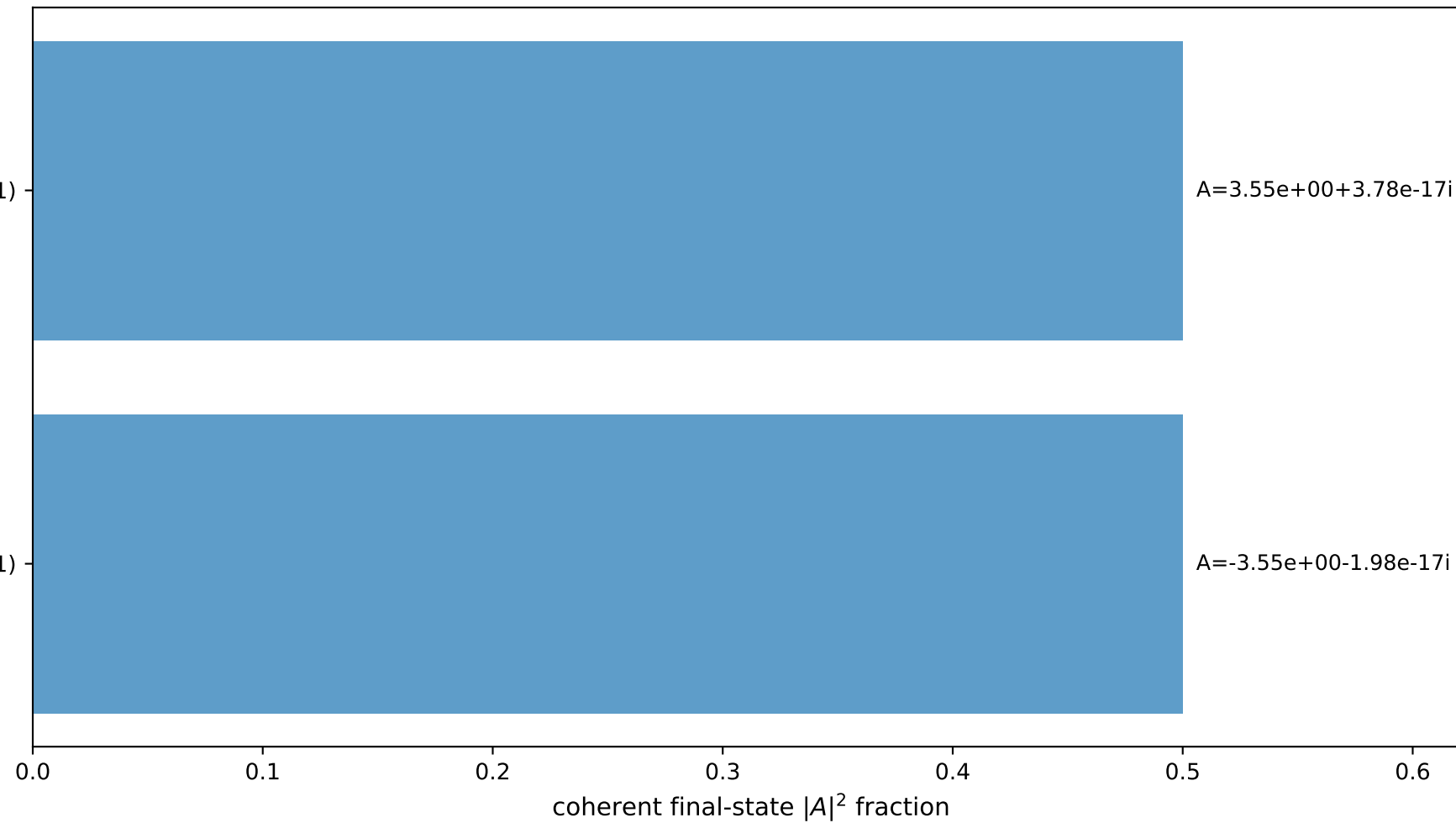
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4020$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4020$
 P $E= 2.5786$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4020$
 l' $E= 1.3346$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.3346$
 P' $E= 2.5786$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4020$
 q_Y $E= 1.0674$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0674$



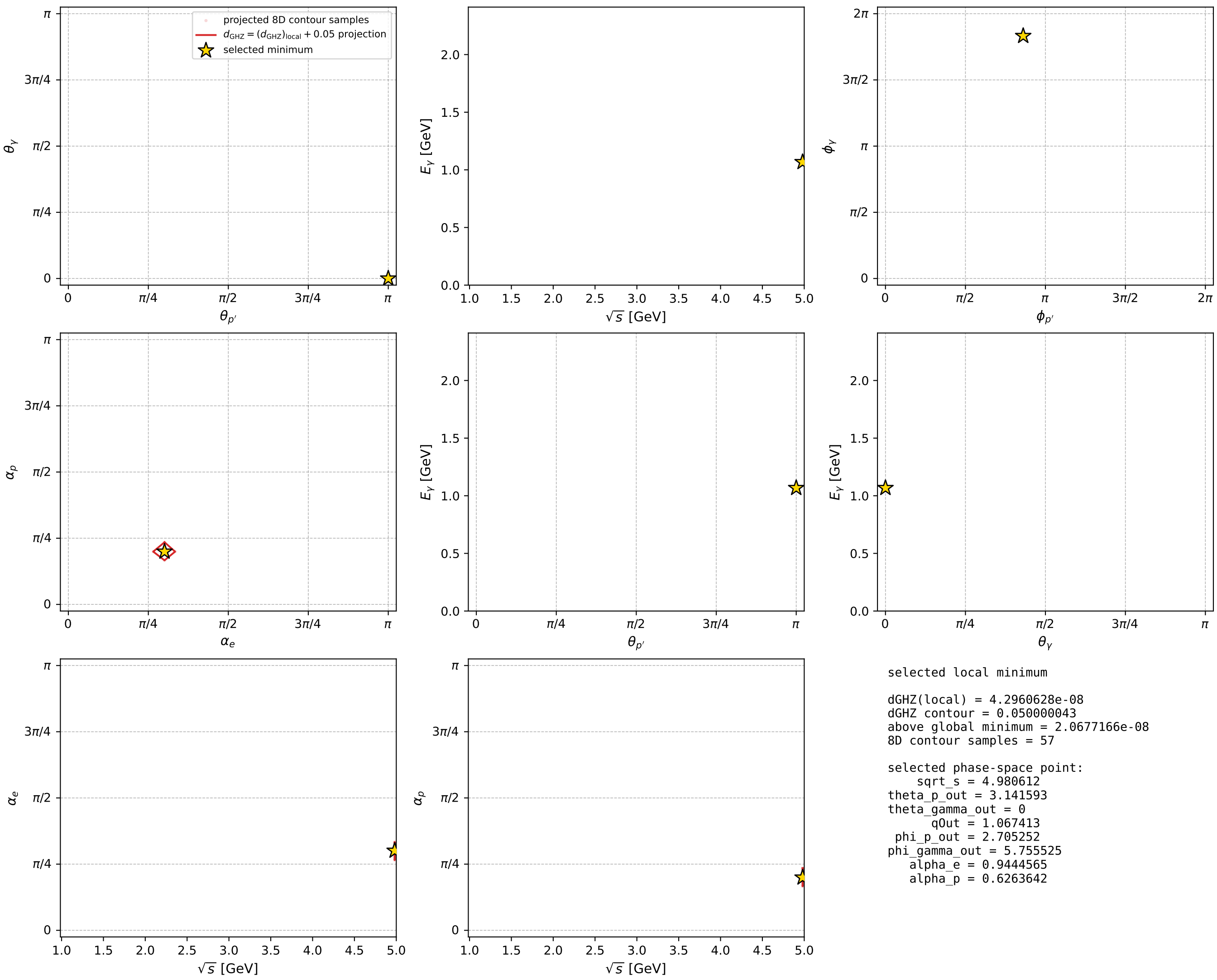
$(h_p, h_Y, h_l) = (-1, -1, +1)$

$(h_p, h_Y, h_l) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 114; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

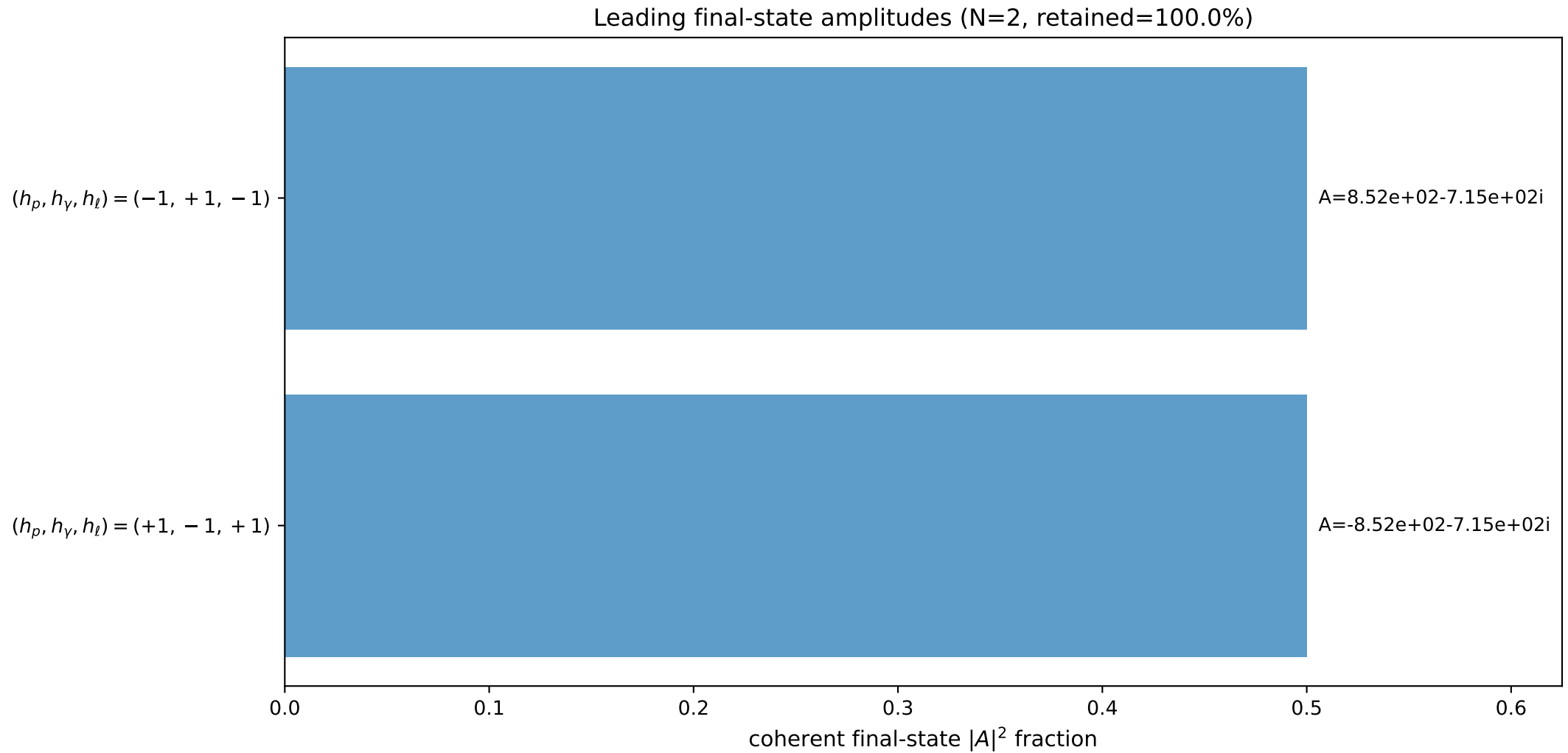
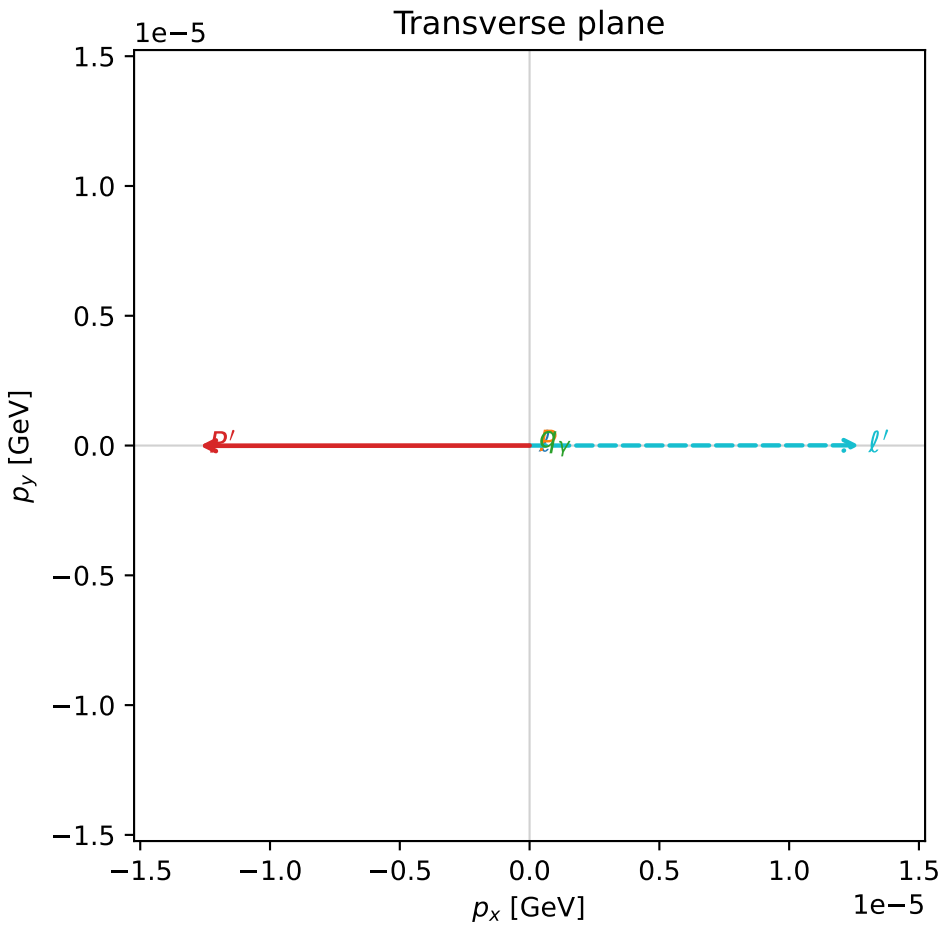
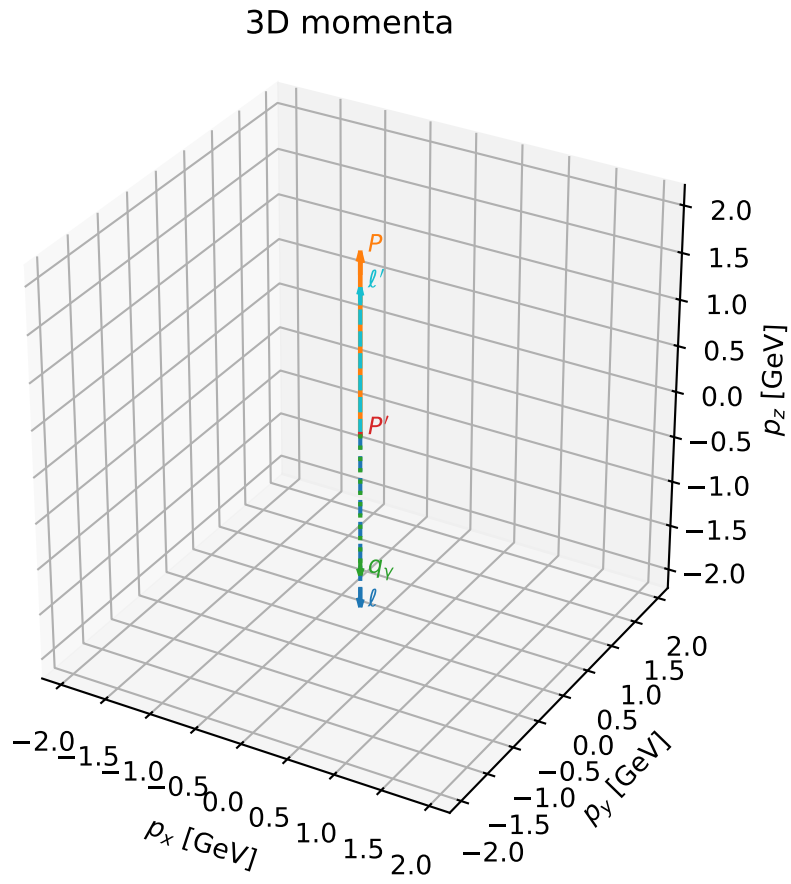


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

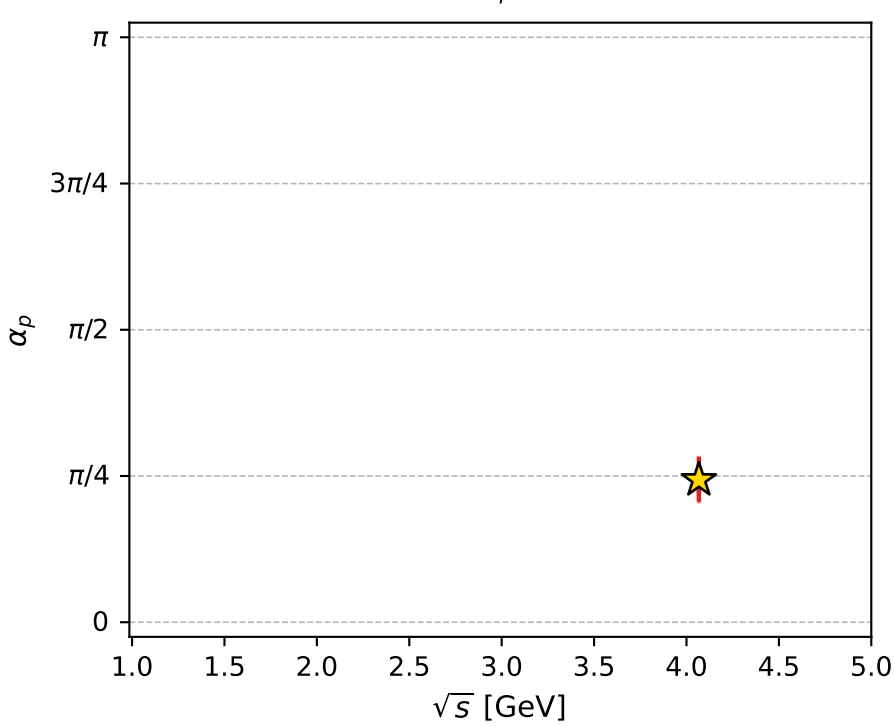
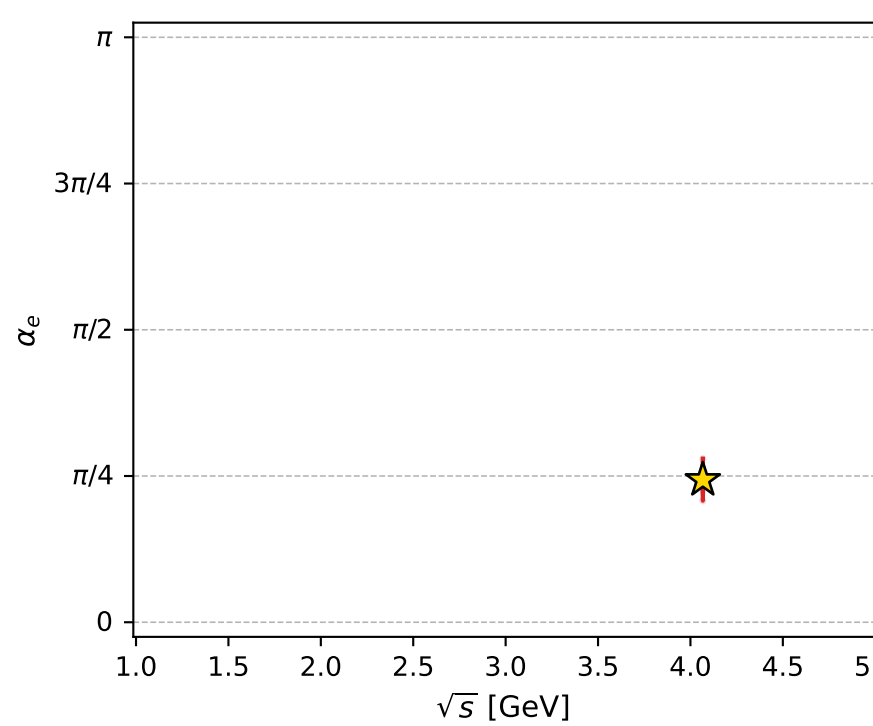
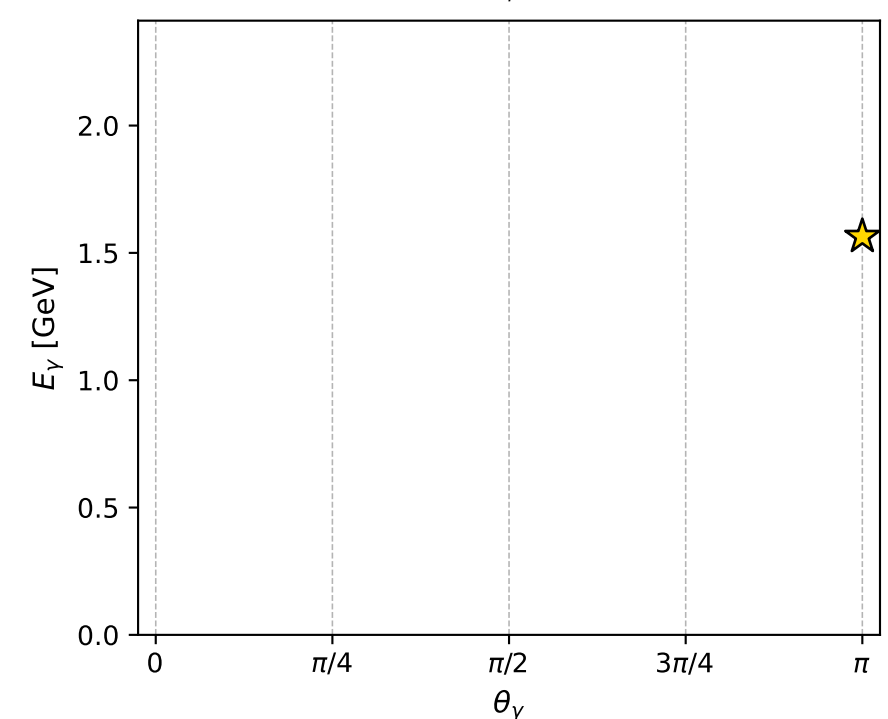
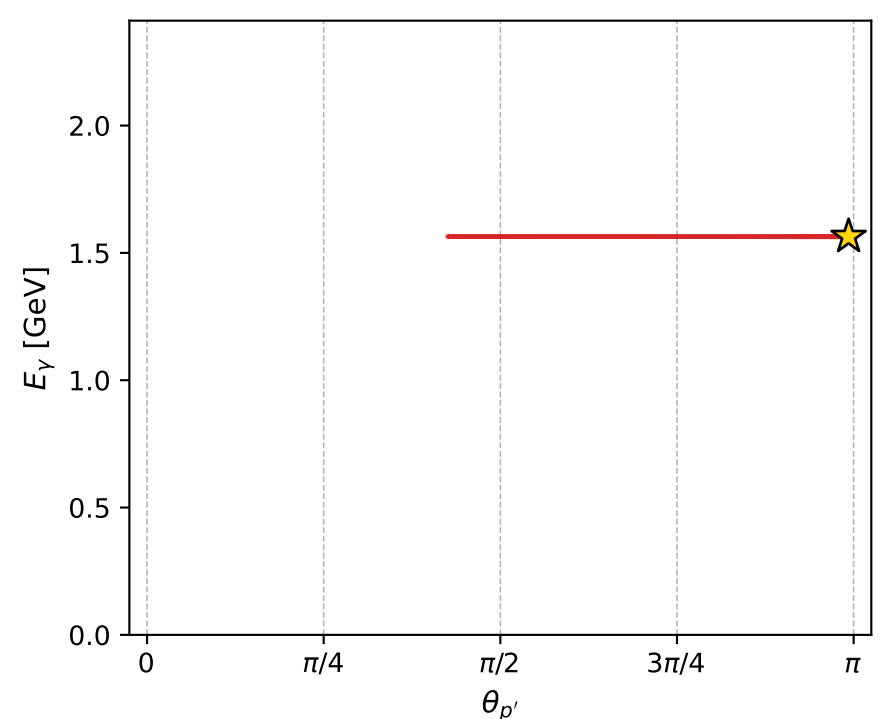
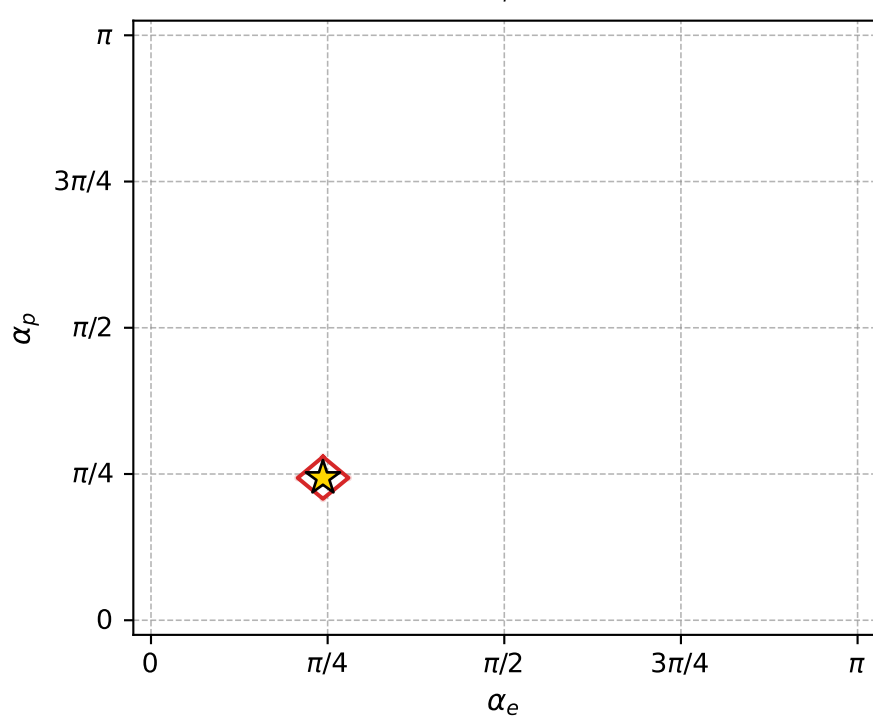
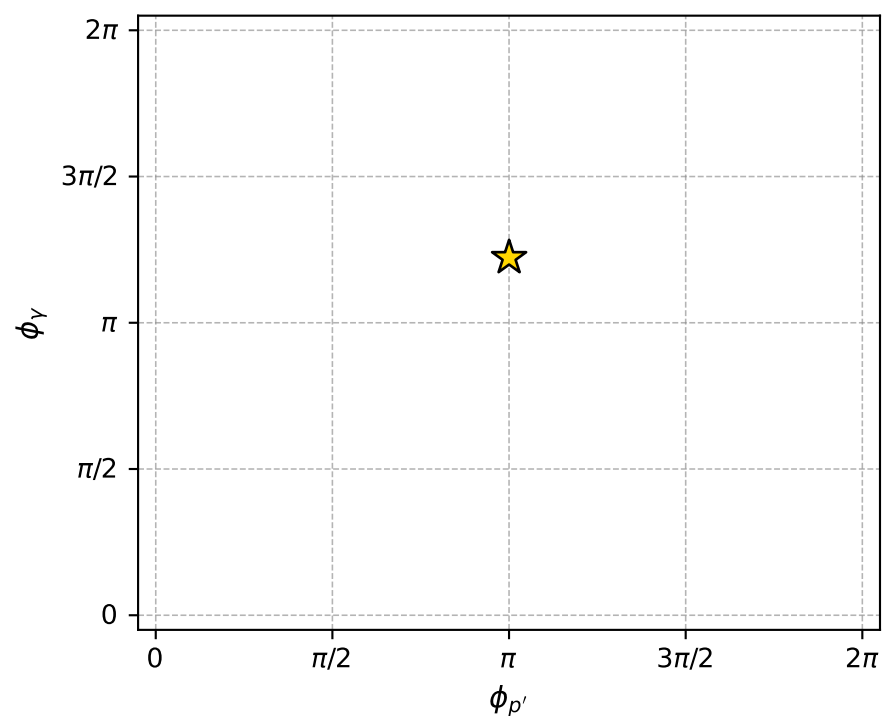
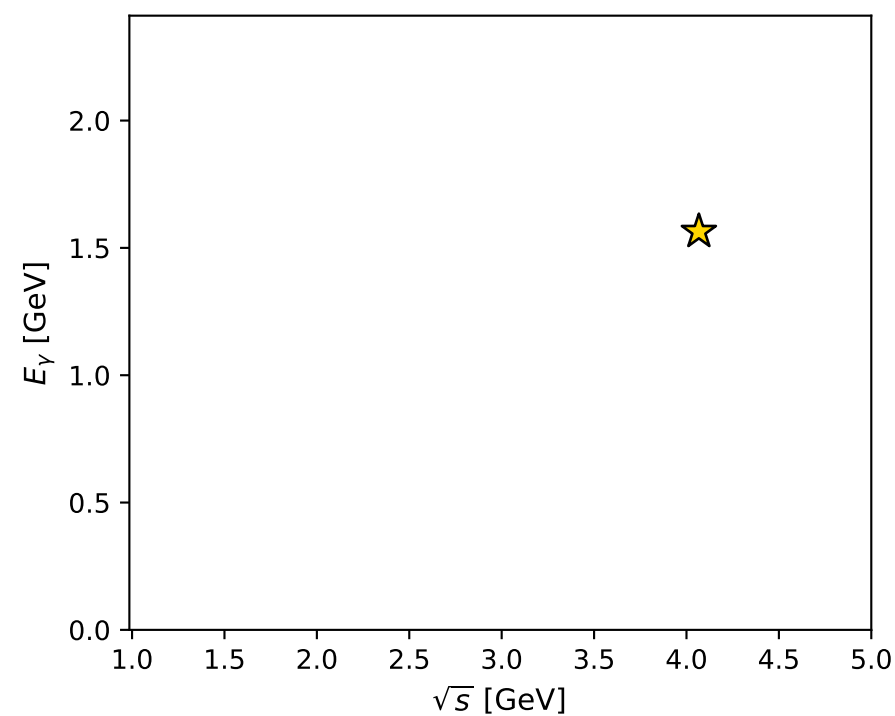
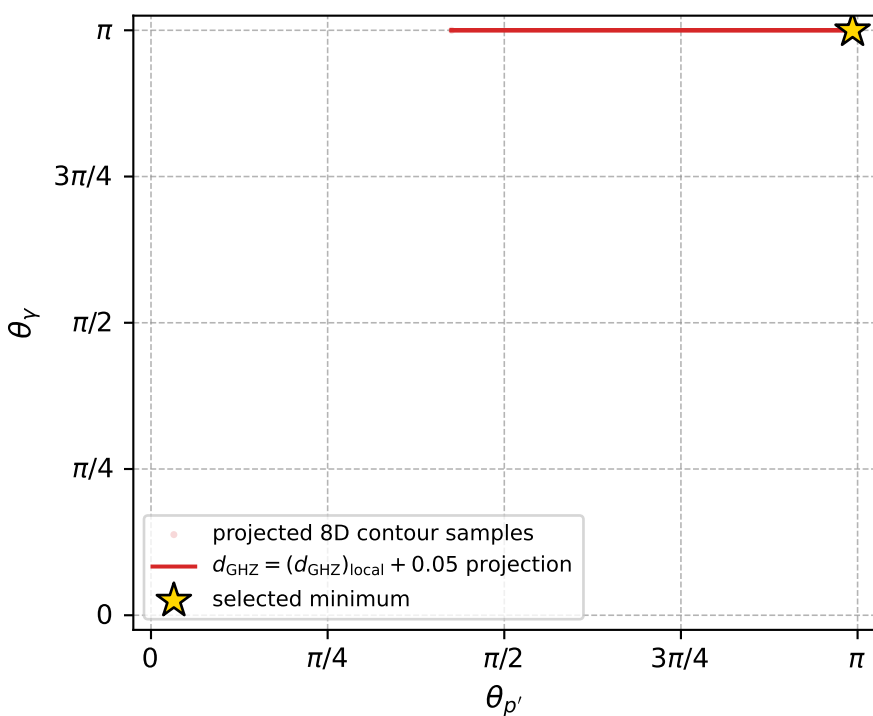
dghz_mixing_angles_polarization_01_local_minimum_115 d_{\rm GHZ}=4.30756e-08
region: polarization_cluster_01_local_minimum_115
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.76452518$ rad
incoming lepton: $0.721711|+\rangle +0.692194|-\rangle$
proton mixing angle: $\alpha_p=0.76454044$ rad
incoming proton: $0.721701|+\rangle +0.692205|-\rangle$

$s=16.5413$, $\sqrt{s}=4.0671$, $|\vec{P}|=1.92538$, $|\vec{P}'|=0.000570618$
 $\theta_{P'}=3.11934$, $\theta_Y=3.14159$, $\phi_{P'}=3.14223$, $\phi_Y=3.8405$
 $E_Y=1.56427$, $\theta_{\ell'}=8.11432\text{e-}06$, $\phi_{\ell'}=0.000639486$
 $Q^2=12.0516$, $x_B=0.804278$, $t=-2.26037$, $W^2=3.81262$, $y=0.95677$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9254$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9254$
 P $E= 2.1417$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9254$
 ℓ' $E= 1.5648$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5648$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0006$
 q_Y $E= 1.5643$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.5643$



electron: polarization cluster P1, local minimum 115; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.3075554\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000043$
 above global minimum = $2.0792092\text{e-}08$
 8D contour samples = 142

selected phase-space point:

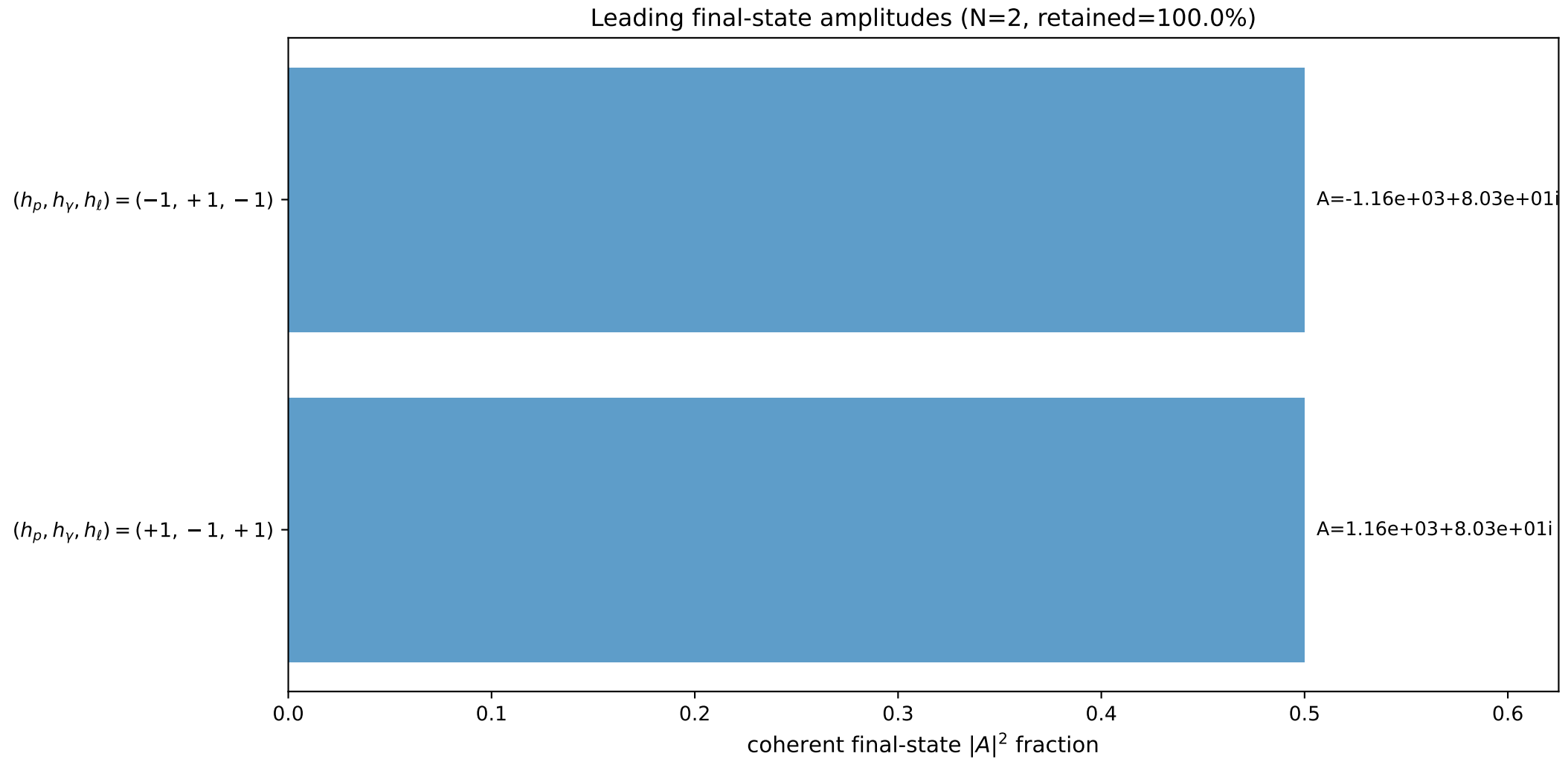
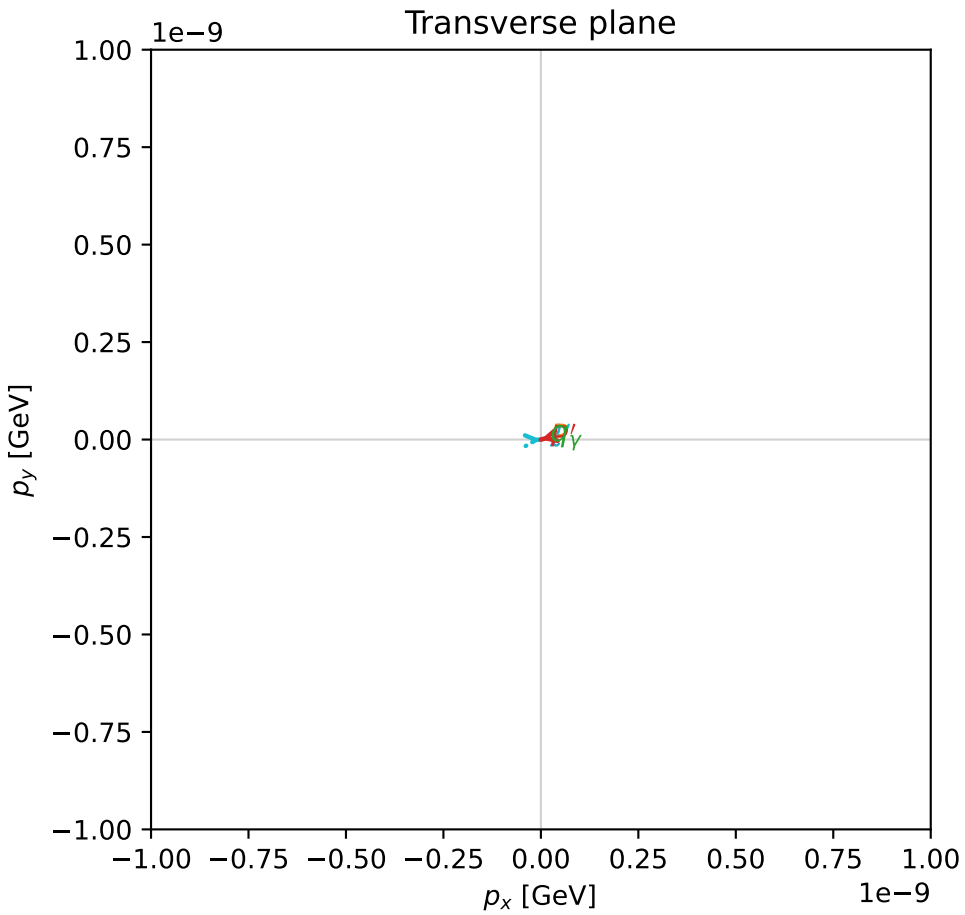
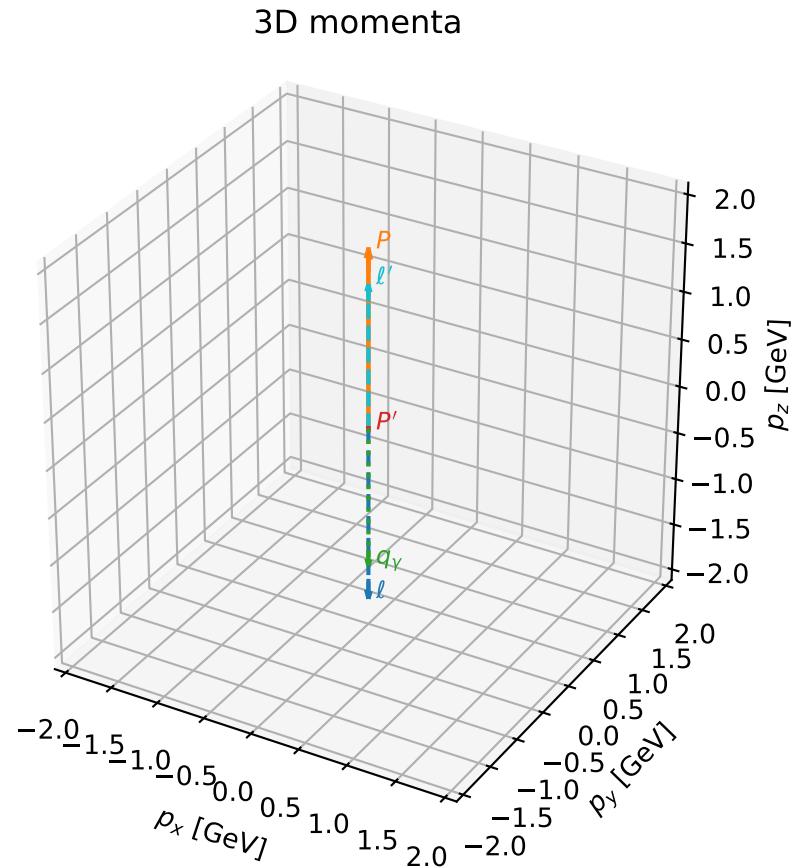
$\text{sqrt_s} = 4.067102$
 $\text{theta_p_out} = 3.119338$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.564265$
 $\text{phi_p_out} = 3.142232$
 $\text{phi_gamma_out} = 3.840503$
 $\text{alpha_e} = 0.7645252$
 $\text{alpha_p} = 0.7645404$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

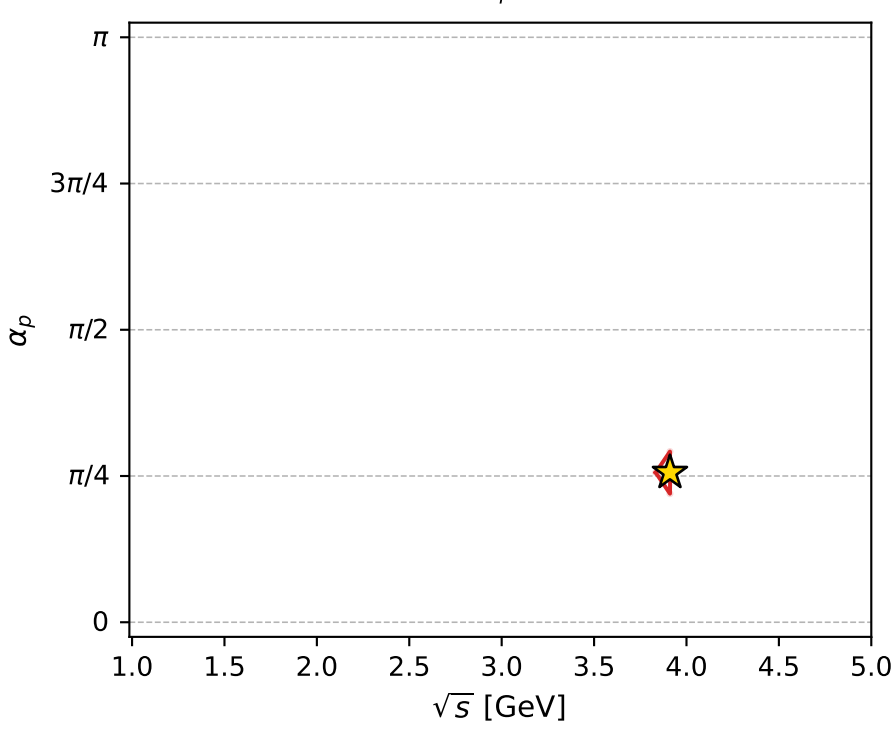
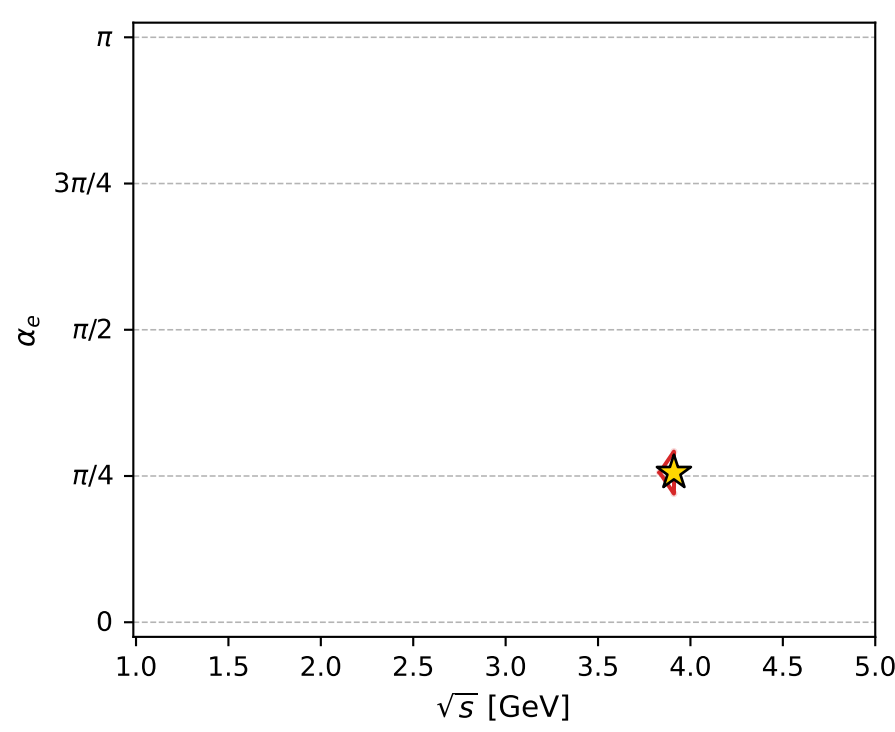
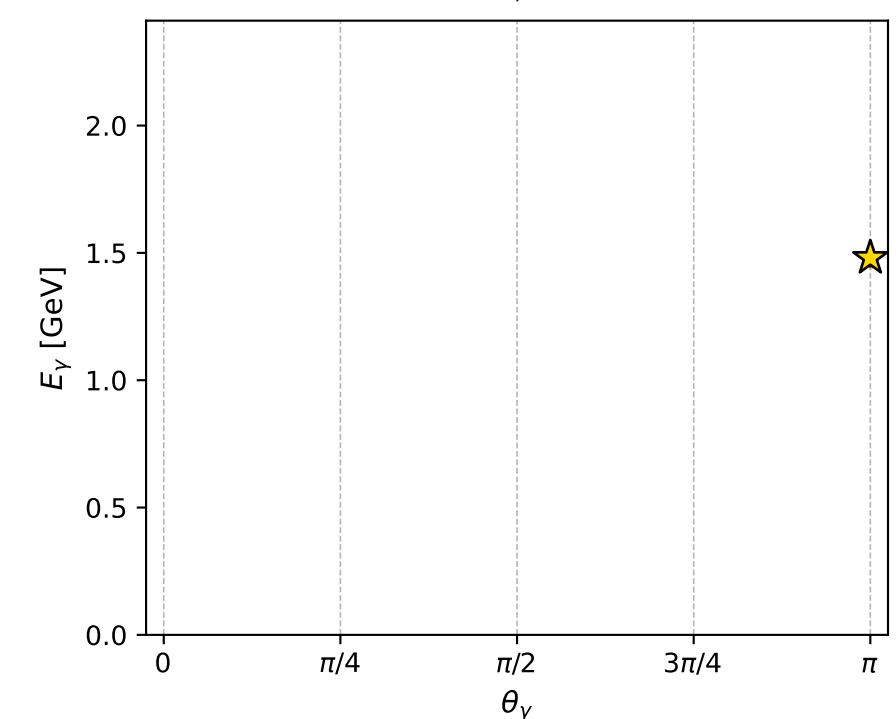
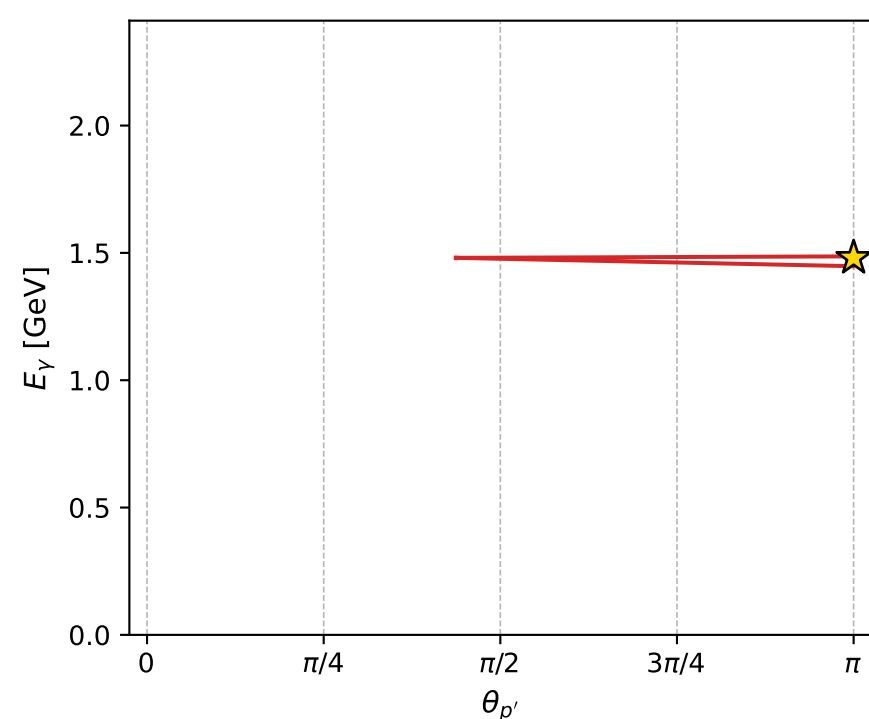
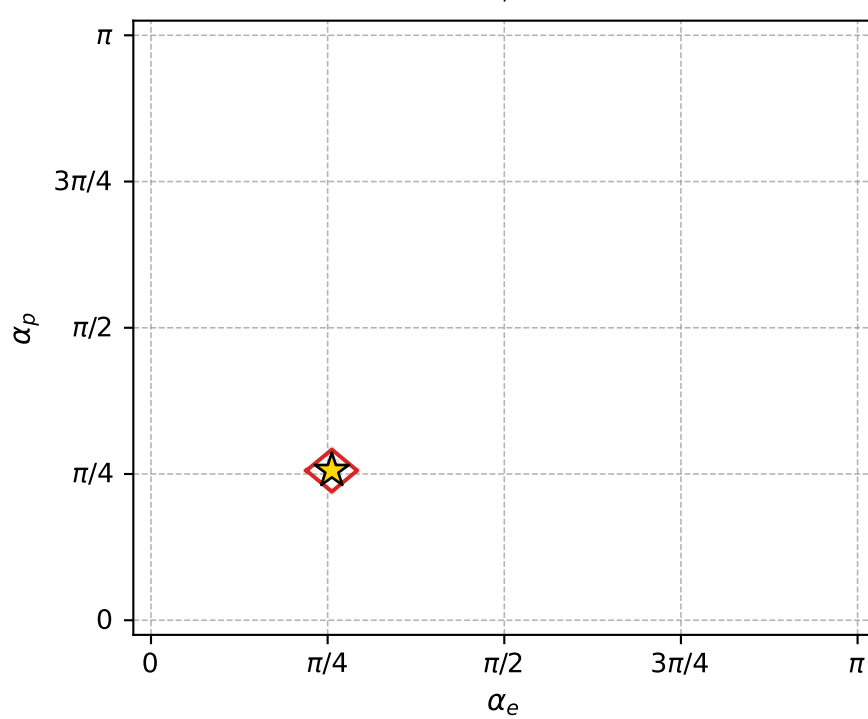
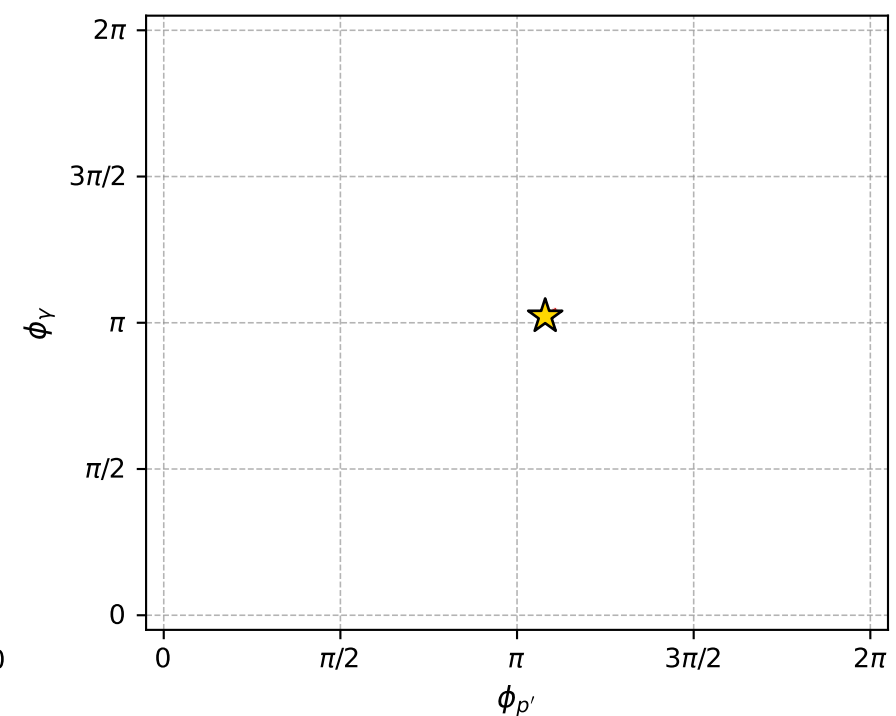
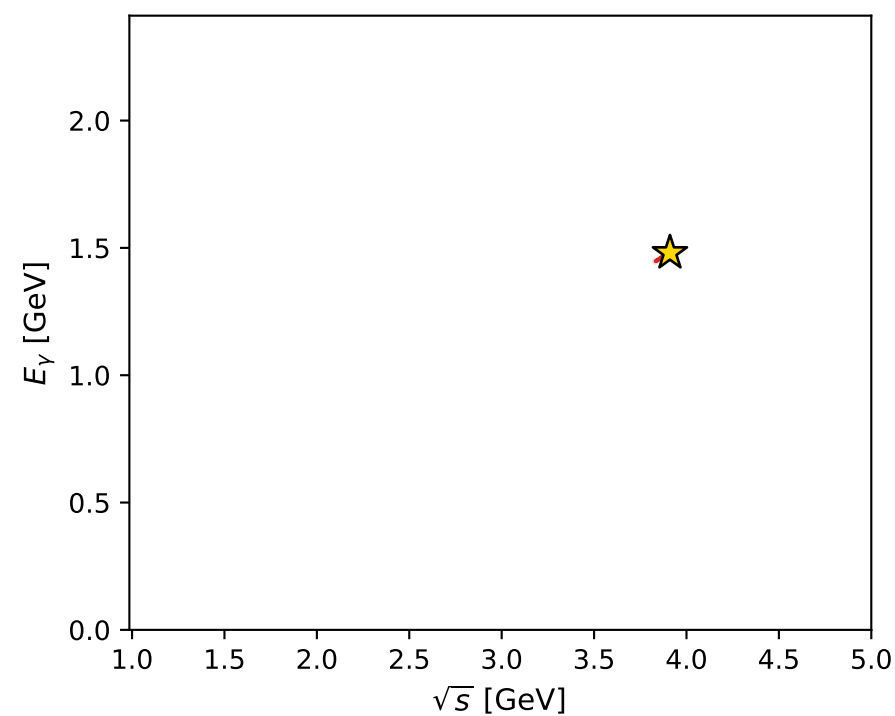
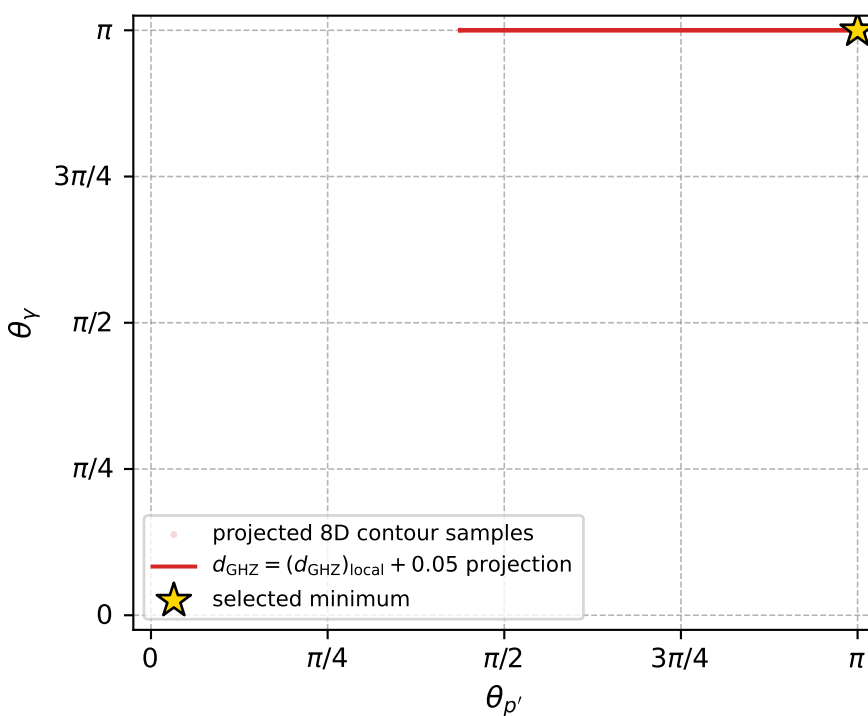
dghz_mixing_angles_polarization_01_local_minimum_121 $d_{\{\text{rm GHZ}\}}=4.75198\text{e-}08$
 region: polarization_cluster_01_local_minimum_121
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80409735$ rad
 incoming lepton: $0.693762|+\rangle +0.720205|-\rangle$
 proton mixing angle: $\alpha_p=0.80410559$ rad
 incoming proton: $0.693756|+\rangle +0.72021|-\rangle$

$s=15.2928$, $\sqrt{s}=3.9106$, $|\vec{P}|=1.8428$, $|\vec{P}'|=0.0114504$
 $\theta_{p'}=3.14159$, $\theta_V=3.14159$, $\phi_{p'}=3.39183$, $\phi_V=3.21079$
 $E_V=1.48054$, $\theta_{l'}=0$, $\phi_{l'}=0.0705831$
 $Q^2=10.9978$, $x_B=0.800329$, $t=-2.16199$, $W^2=3.62364$, $y=0.953419$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8428$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8428$
 P $E= 2.0678$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8428$
 l' $E= 1.4920$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4920$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0115$
 q_Y $E= 1.4805$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4805$



electron: polarization cluster P1, local minimum 121; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.7519806\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000048$
 above global minimum = $2.5236344\text{e-}08$
 8D contour samples = 54

selected phase-space point:

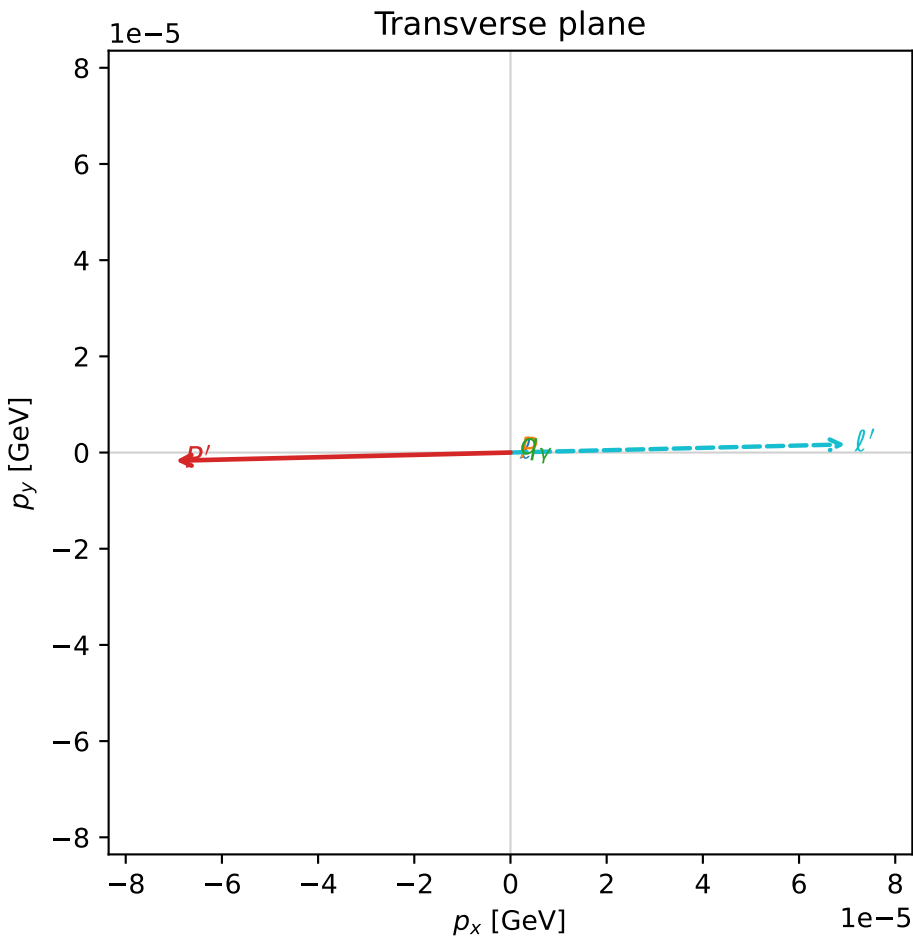
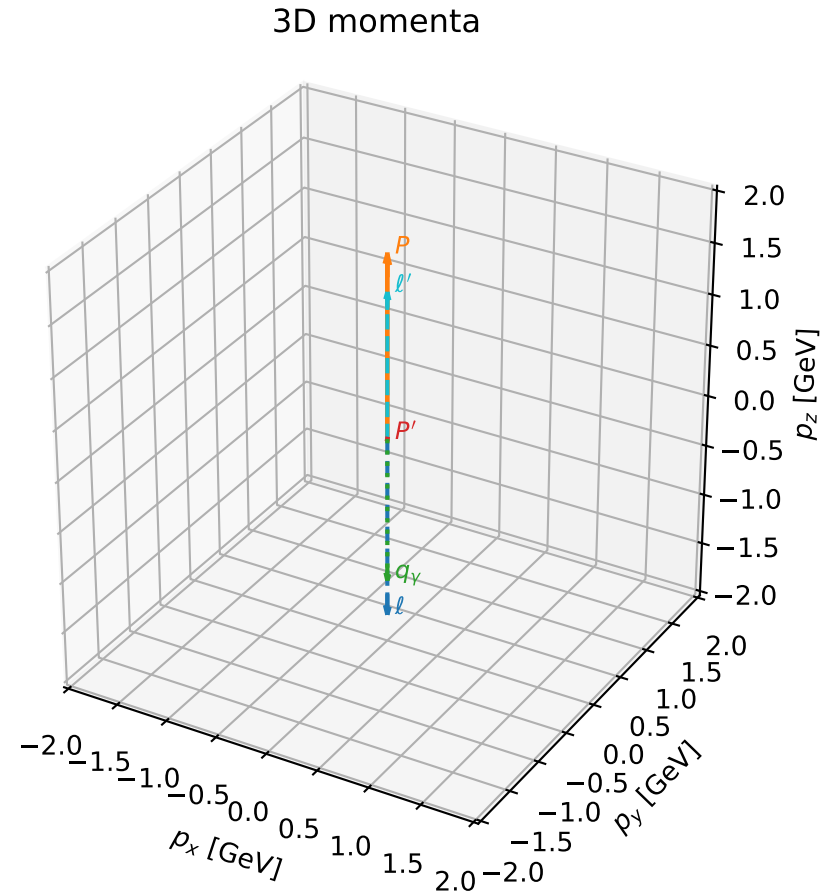
$\text{sqrt_s} = 3.910599$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.48054$
 $\text{phi_p_out} = 3.391832$
 $\text{phi_gamma_out} = 3.210794$
 $\alpha_e = 0.8040973$
 $\alpha_p = 0.8041056$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_126 $d_{\{\text{rm GHZ}\}}=4.95818\text{e-}08$
region: polarization_cluster_01_local_minimum_126
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.79324484$ rad
incoming lepton: $0.701537|+\rangle +0.712633|-\rangle$
proton mixing angle: $\alpha_p=0.79327627$ rad
incoming proton: $0.701514|+\rangle +0.712655|-\rangle$

$s=14.3543$, $\sqrt{s}=3.78871$, $|\vec{P}|=1.77824$, $|\vec{P}'|=0.000403116$
 $\theta_{p'}=2.96794$, $\theta_Y=3.14159$, $\phi_{p'}=3.16594$, $\phi_Y=3.0289$
 $E_Y=1.42516$, $\theta_{\ell'}=4.88583\text{e-}05$, $\phi_{\ell'}=0.0243458$
 $Q^2=10.1399$, $x_B=0.791416$, $t=-2.01337$, $W^2=3.55231$, $y=0.950862$

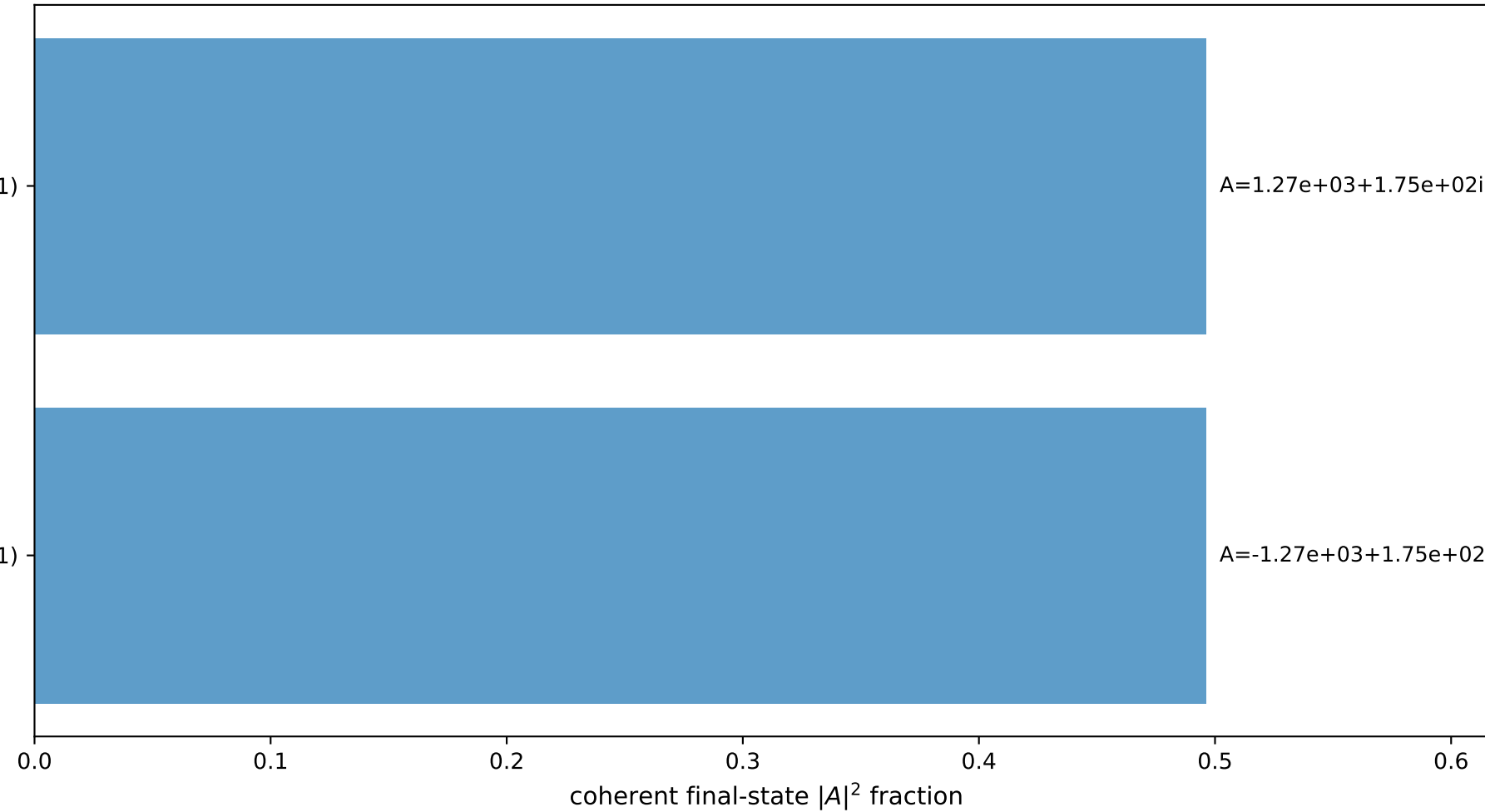
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7782$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7782$
 P $E= 2.0105$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7782$
 ℓ' $E= 1.4256$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.4256$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.0004$
 q_Y $E= 1.4252$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.4252$



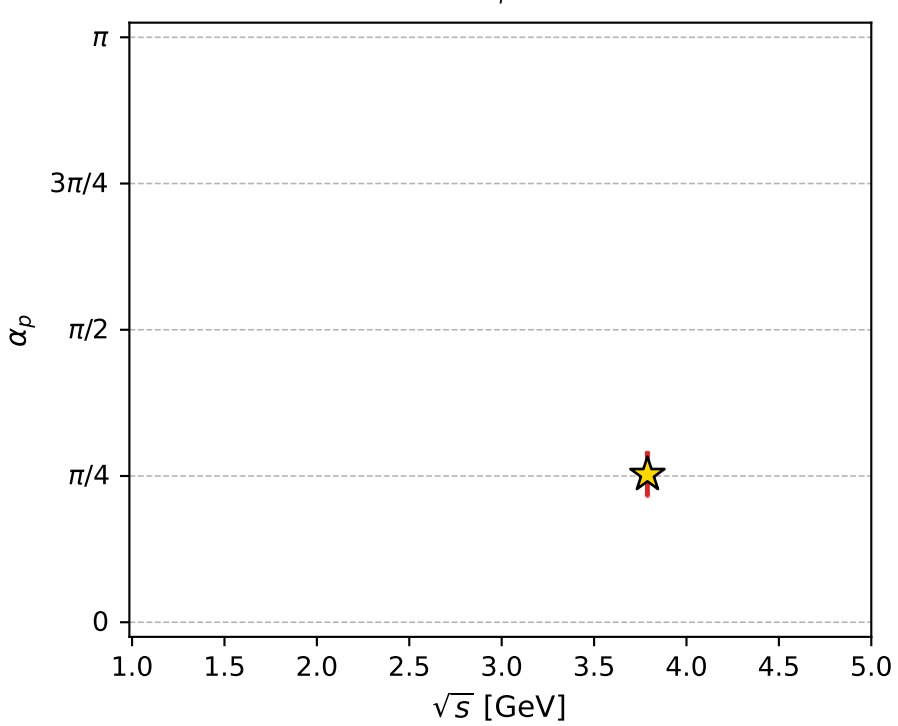
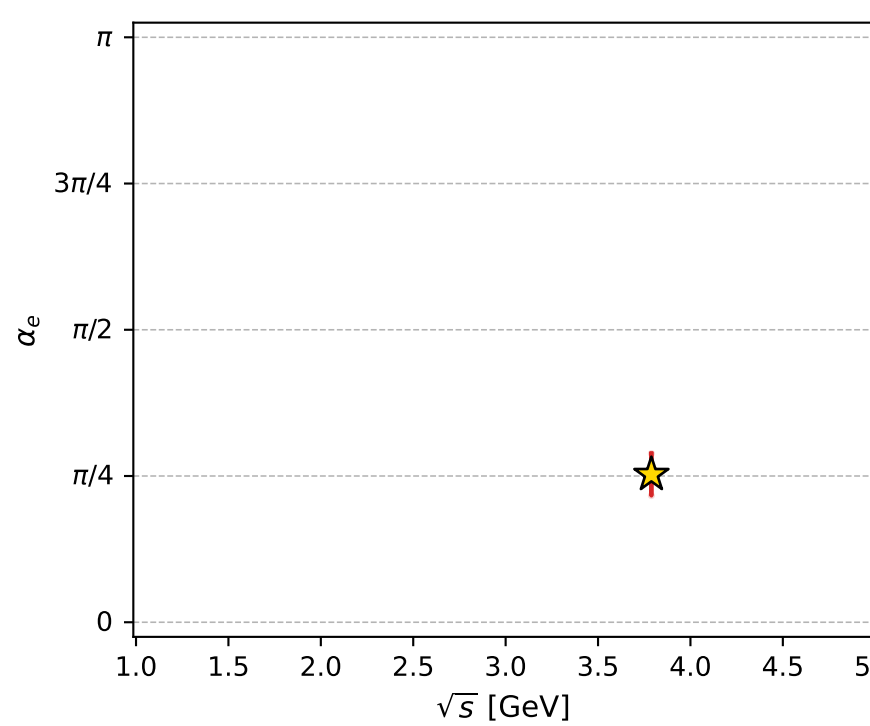
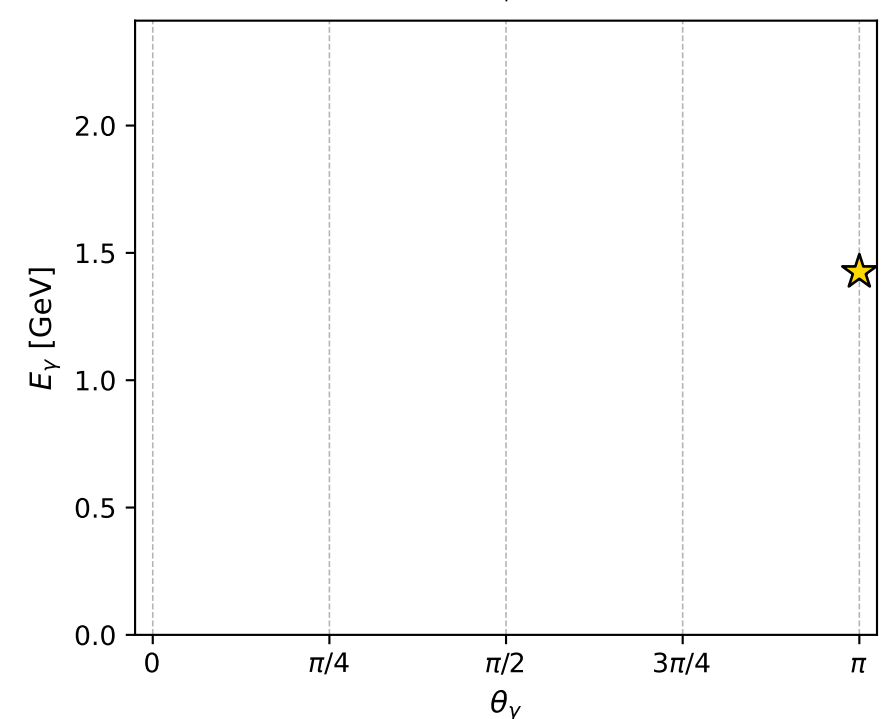
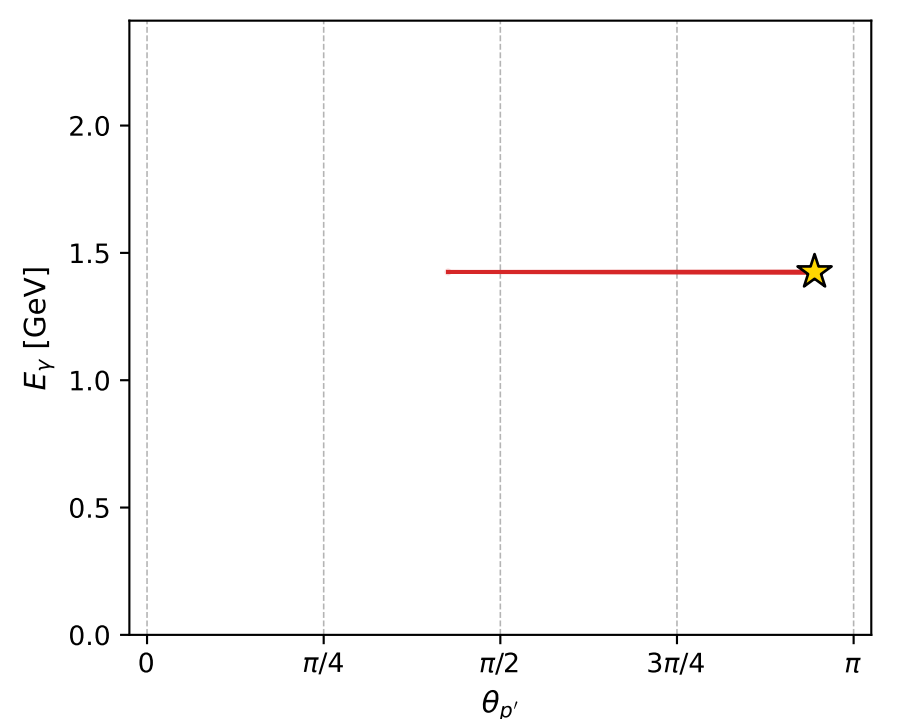
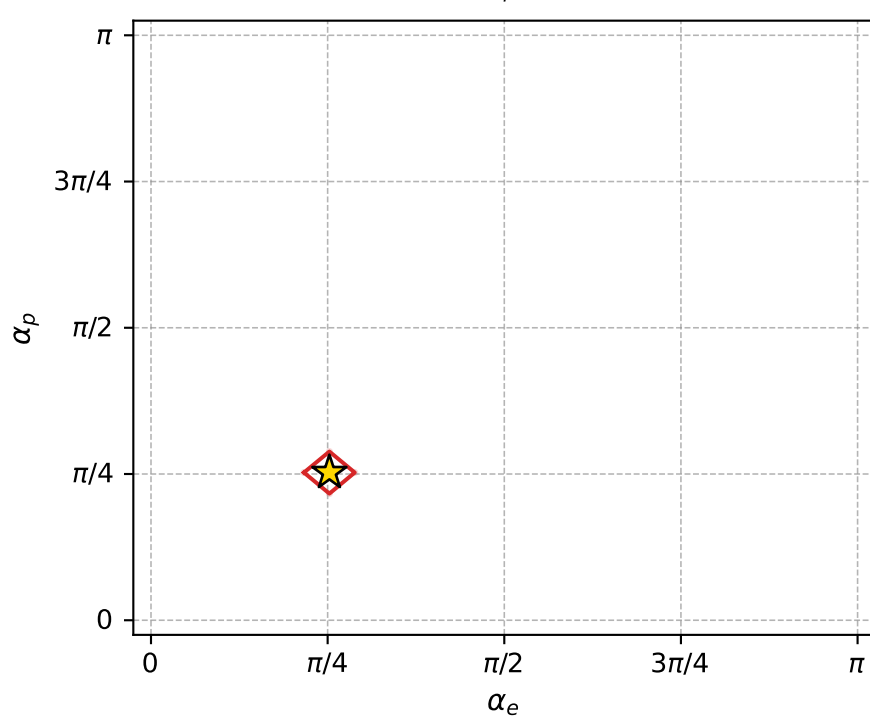
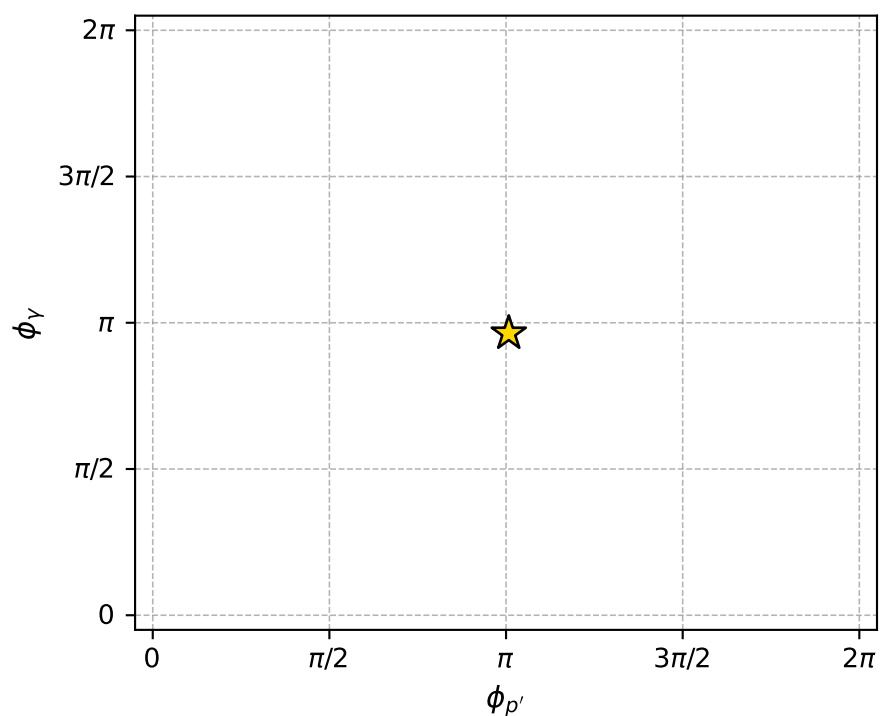
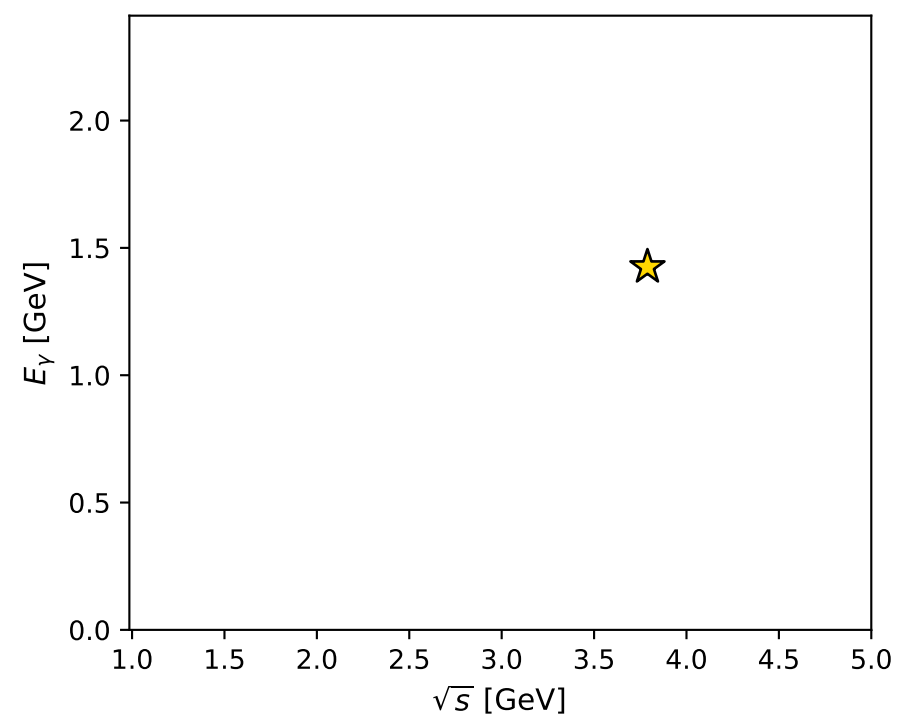
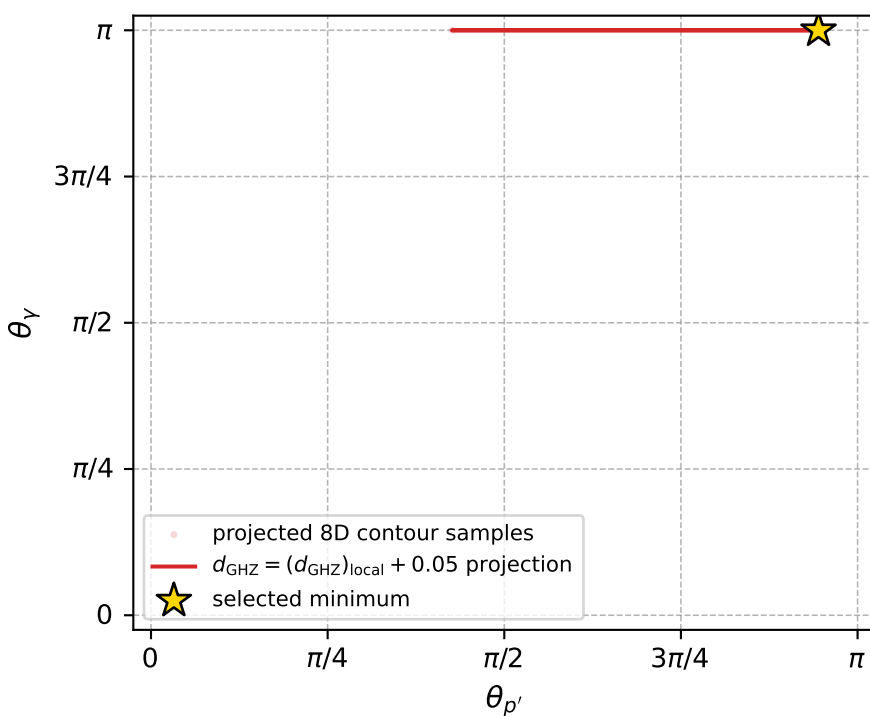
$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

Leading final-state amplitudes (N=2, retained=99.2%)



electron: polarization cluster P1, local minimum 126; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.9581823\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.05000005$
 above global minimum = $2.7298361\text{e-}08$
 8D contour samples = 119

selected phase-space point:

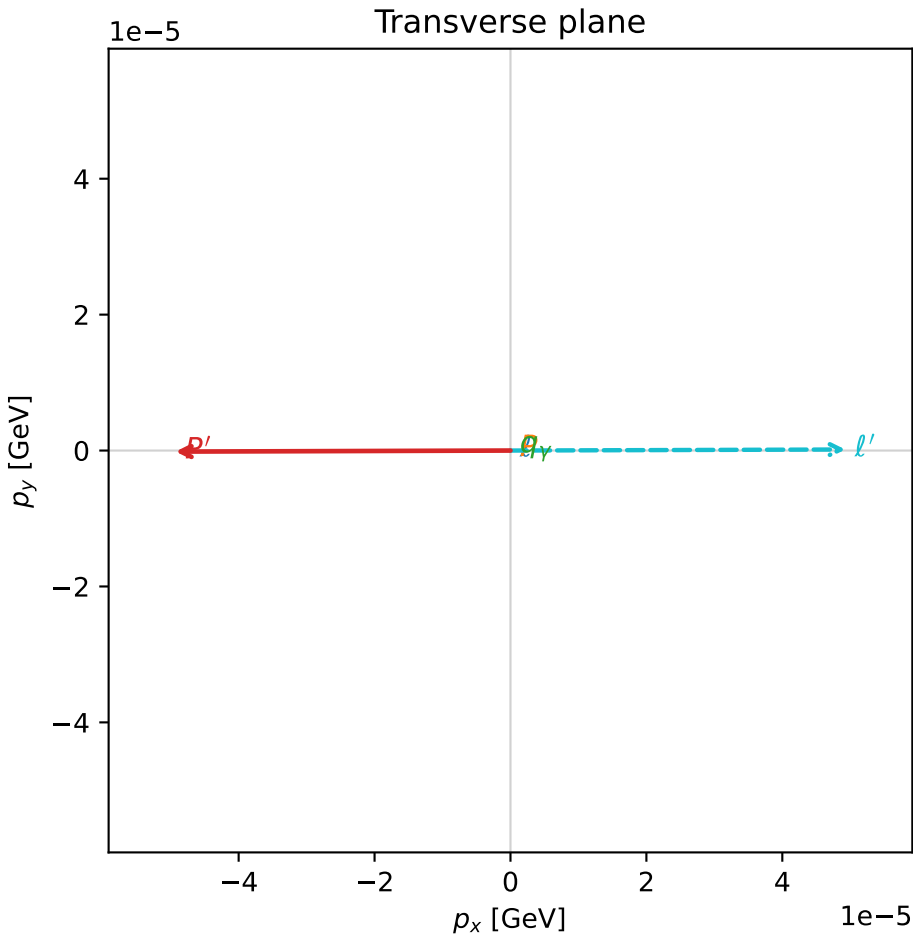
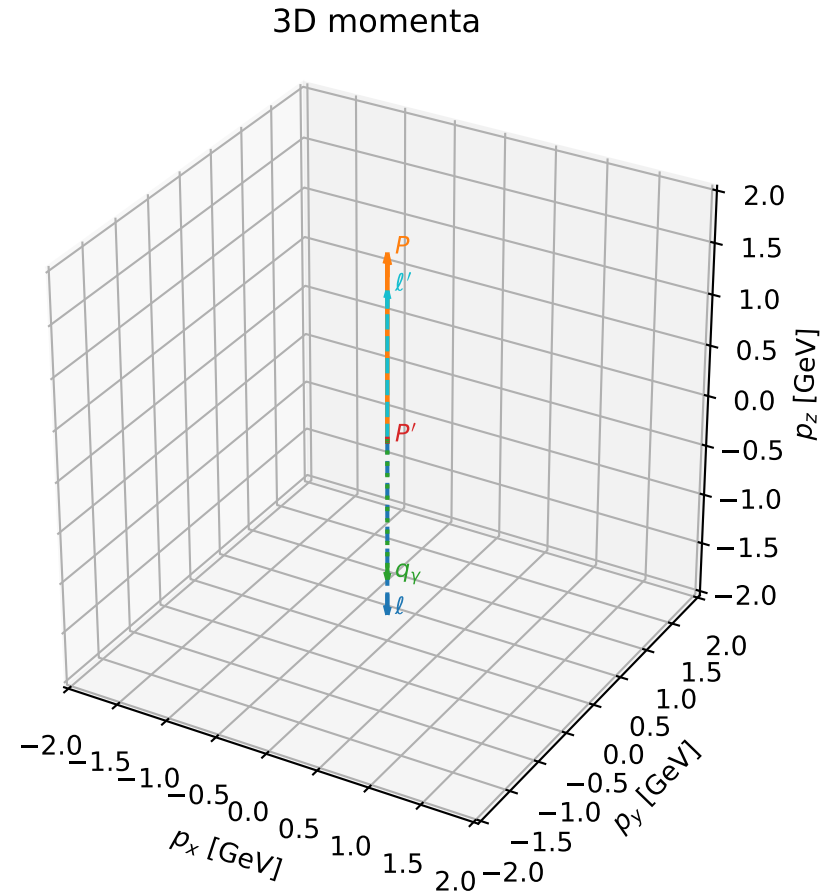
$\text{sqrt_s} = 3.788713$
 $\text{theta_p_out} = 2.967942$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.425158$
 $\text{phi_p_out} = 3.165938$
 $\text{phi_gamma_out} = 3.028902$
 $\alpha_e = 0.7932448$
 $\alpha_p = 0.7932763$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_129 d_{\rm GHZ}=5.2277e-08
region: polarization_cluster_01_local_minimum_129
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.89040334$ rad
incoming lepton: $0.629099|+\rangle +0.777326|-\rangle$
proton mixing angle: $\alpha_p=0.89039462$ rad
incoming proton: $0.629105|+\rangle +0.77732|-\rangle$

$s=14.3507$, $\sqrt{s}=3.78823$, $|\vec{P}|=1.77798$, $|\vec{P}'|=0.024118$
 $\theta_{P'}=3.13955$, $\theta_Y=3.14159$, $\phi_{P'}=3.14474$, $\phi_Y=3.30063$
 $E_Y=1.4129$, $\theta_{\ell'}=3.42918\text{e-}05$, $\phi_{\ell'}=0.00314395$
 $Q^2=10.22$, $x_B=0.79823$, $t=-2.09854$, $W^2=3.46317$, $y=0.950447$

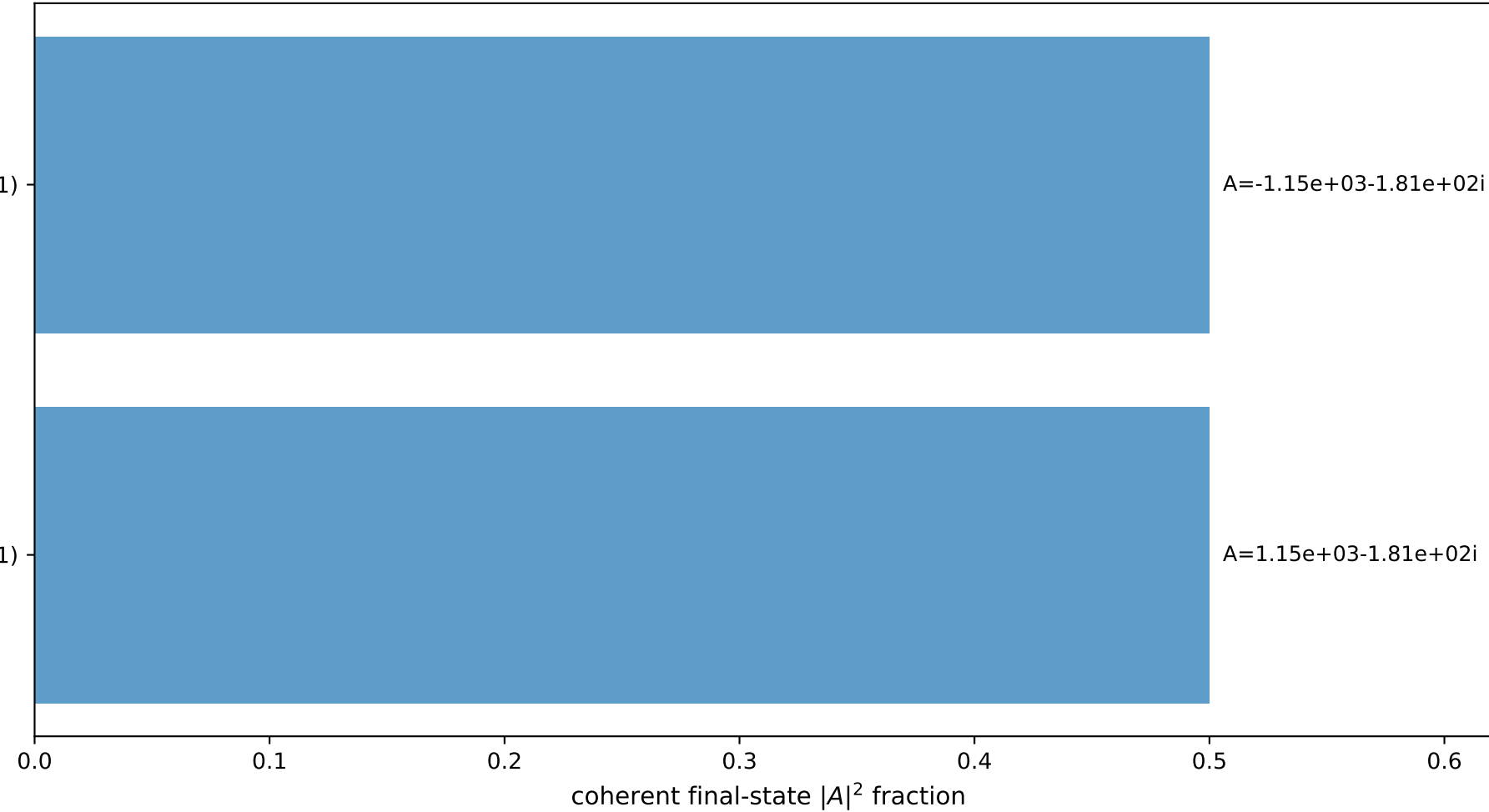
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7780$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7780$
 P $E= 2.0102$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7780$
 ℓ' $E= 1.4370$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4370$
 P' $E= 0.9383$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0241$
 q_Y $E= 1.4129$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4129$



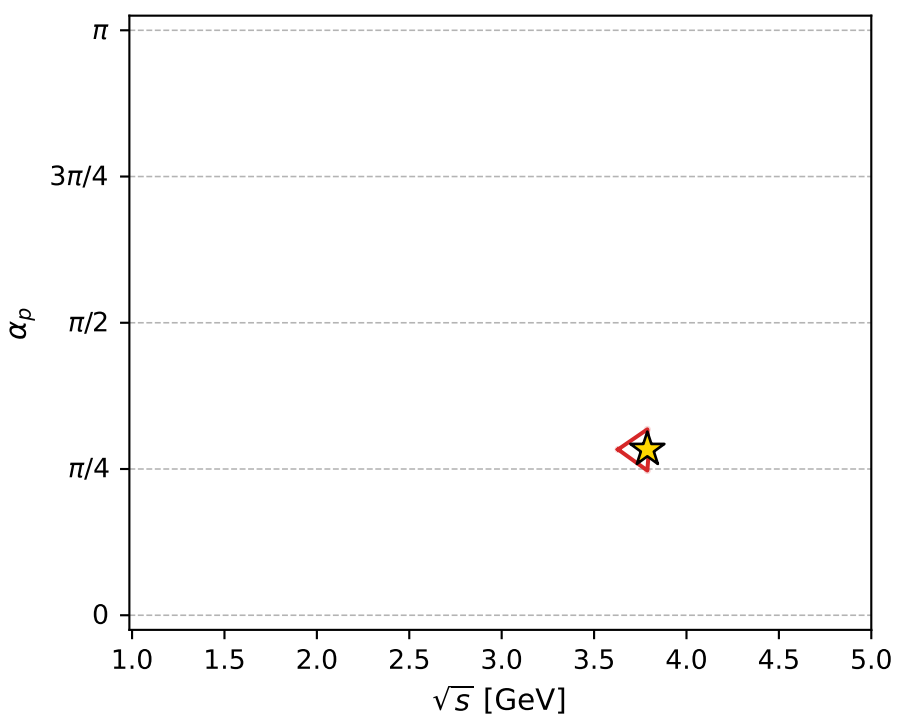
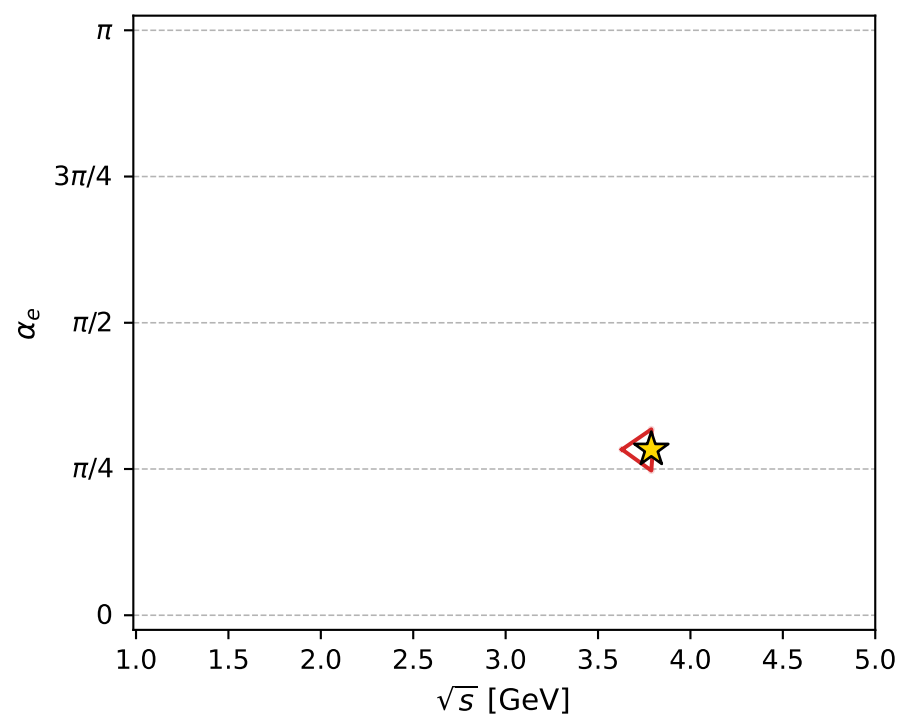
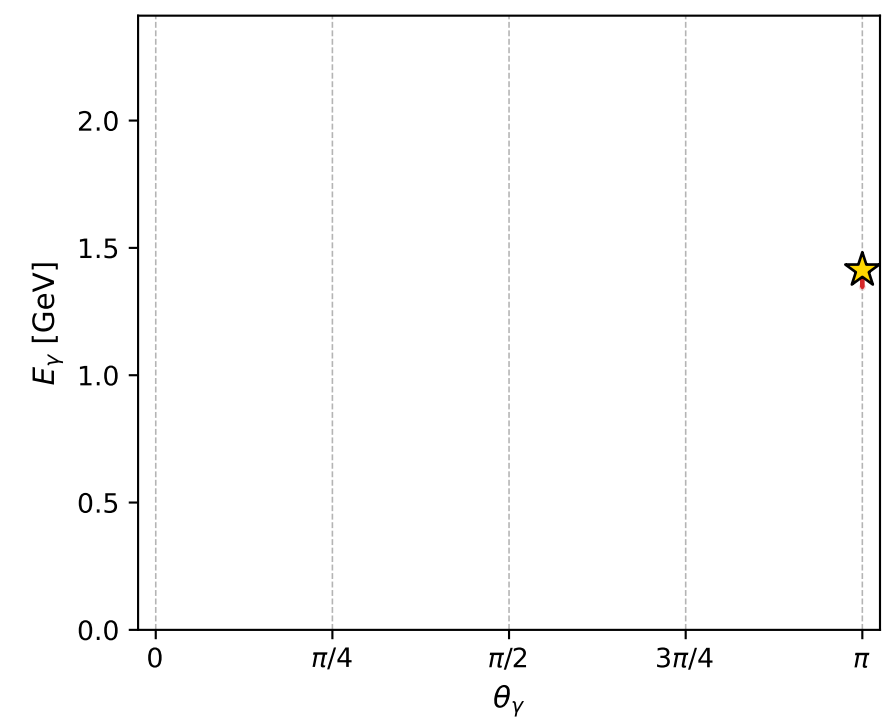
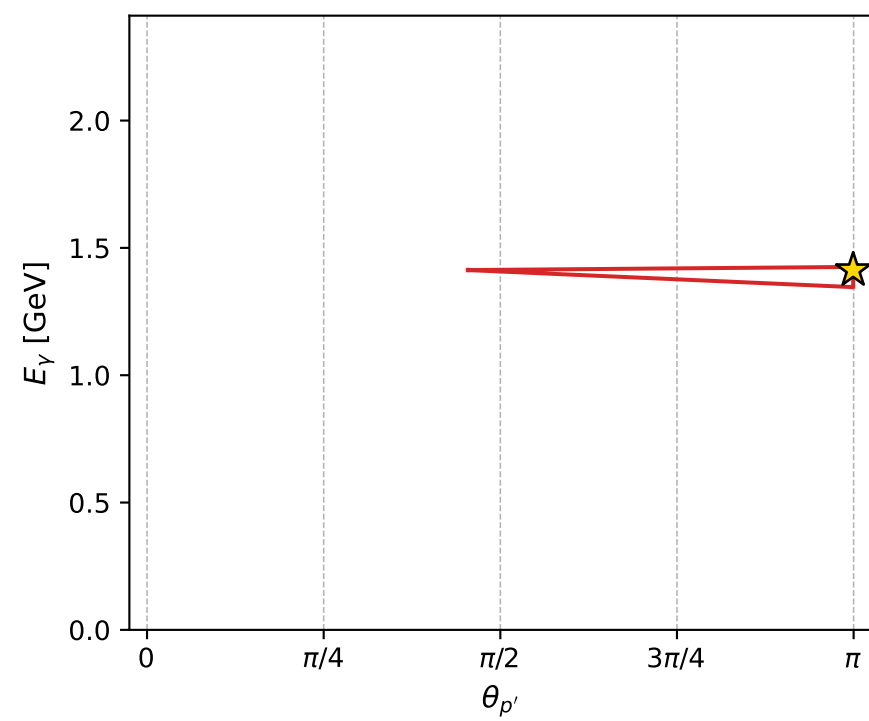
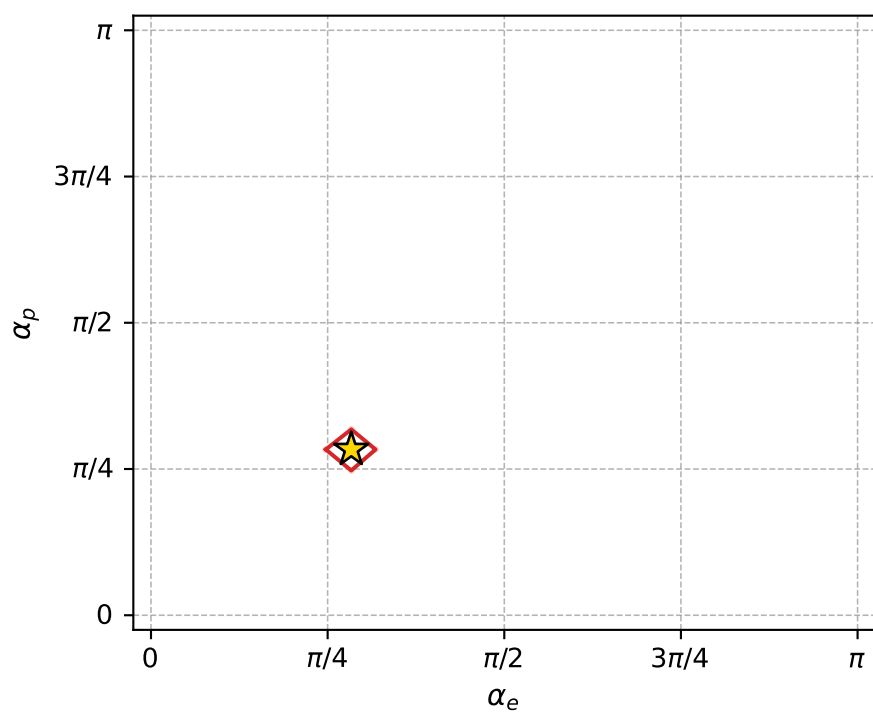
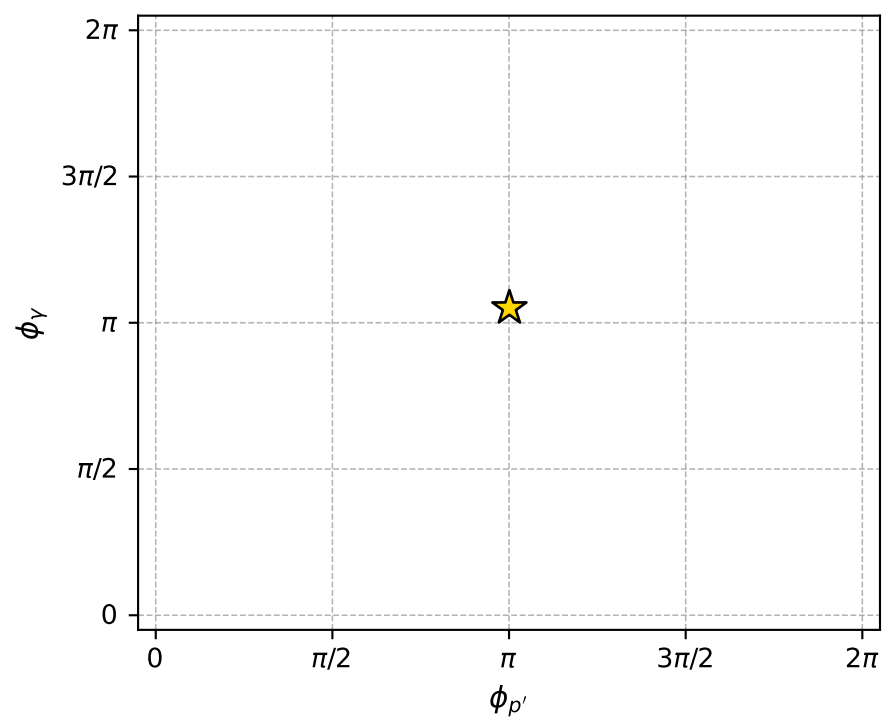
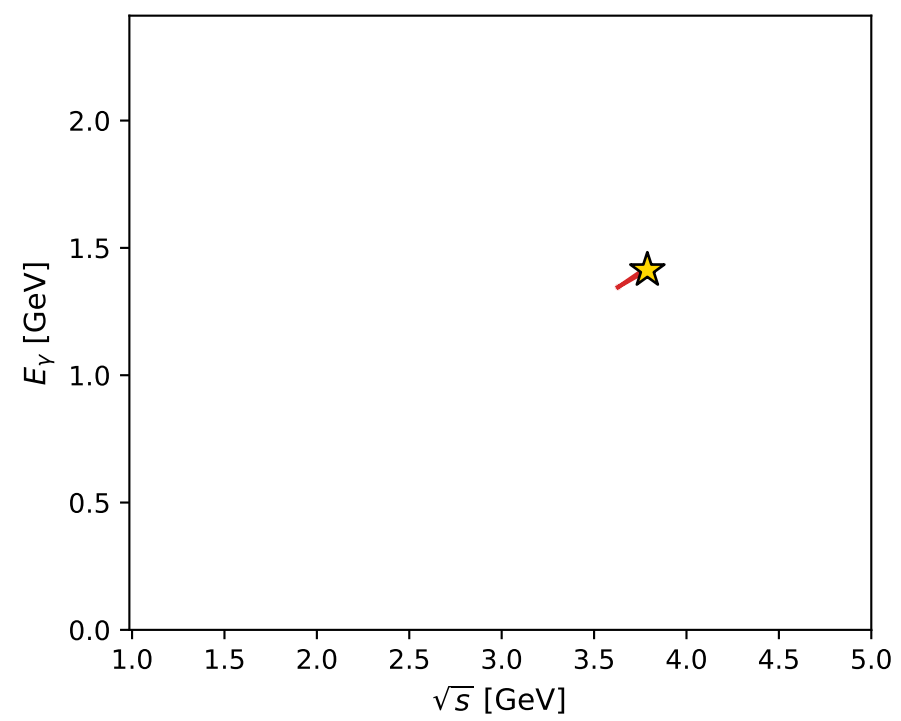
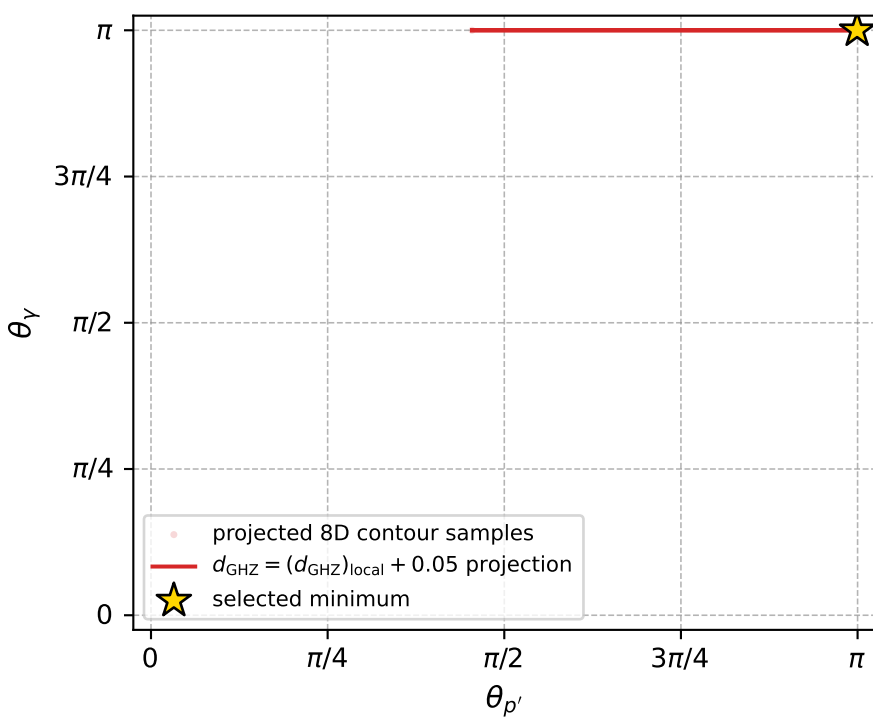
$(h_p, h_Y, h_\ell) = (+1, -1, +1)$

$(h_p, h_Y, h_\ell) = (-1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 129; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 5.2276992\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000052$
 above global minimum = $2.9993531\text{e-}08$
 8D contour samples = 137

selected phase-space point:

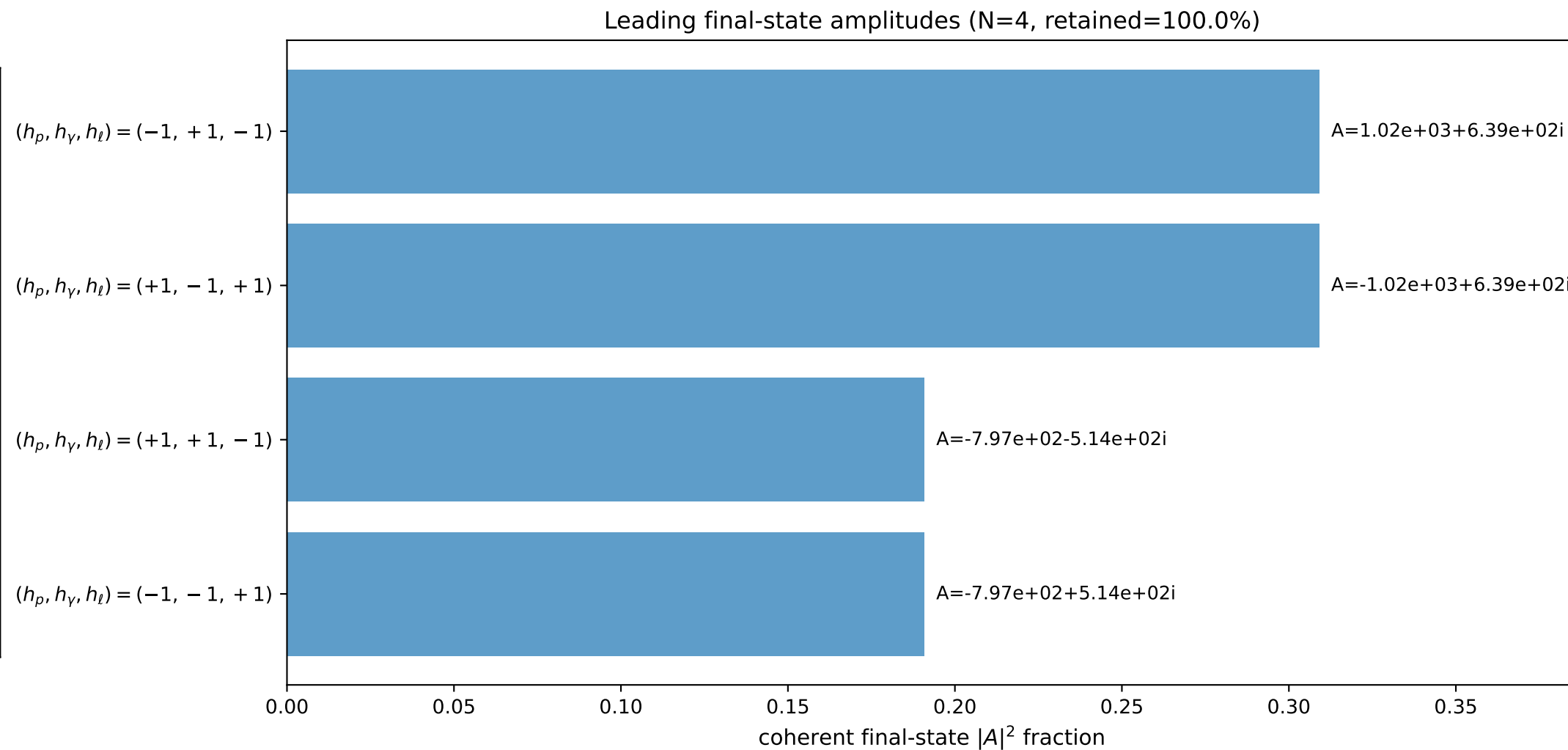
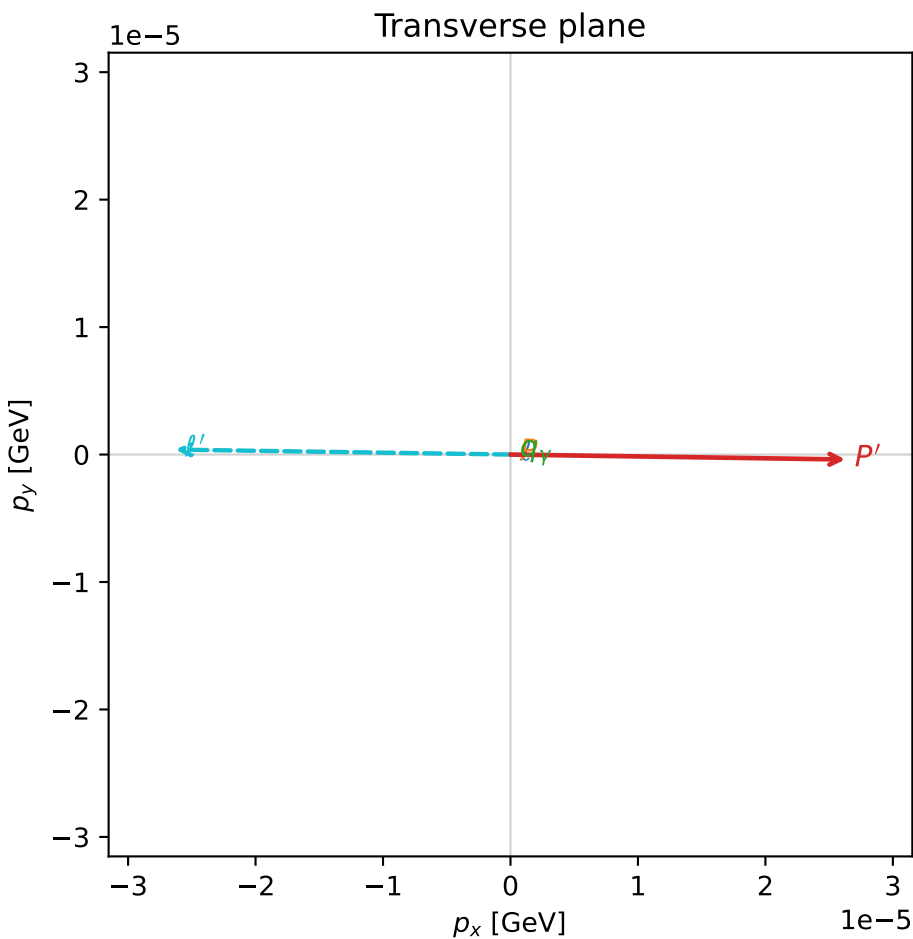
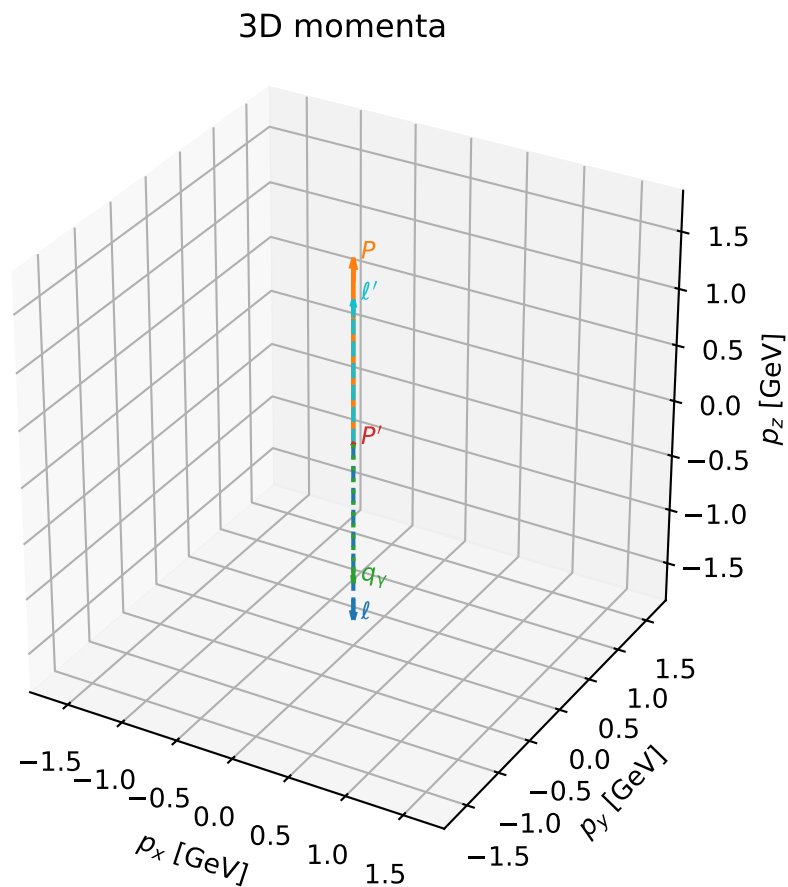
$\text{sqrt_s} = 3.788227$
 $\text{theta_p_out} = 3.139549$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.4129$
 $\text{phi_p_out} = 3.144737$
 $\text{phi_gamma_out} = 3.300629$
 $\alpha_e = 0.8904033$
 $\alpha_p = 0.8903946$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

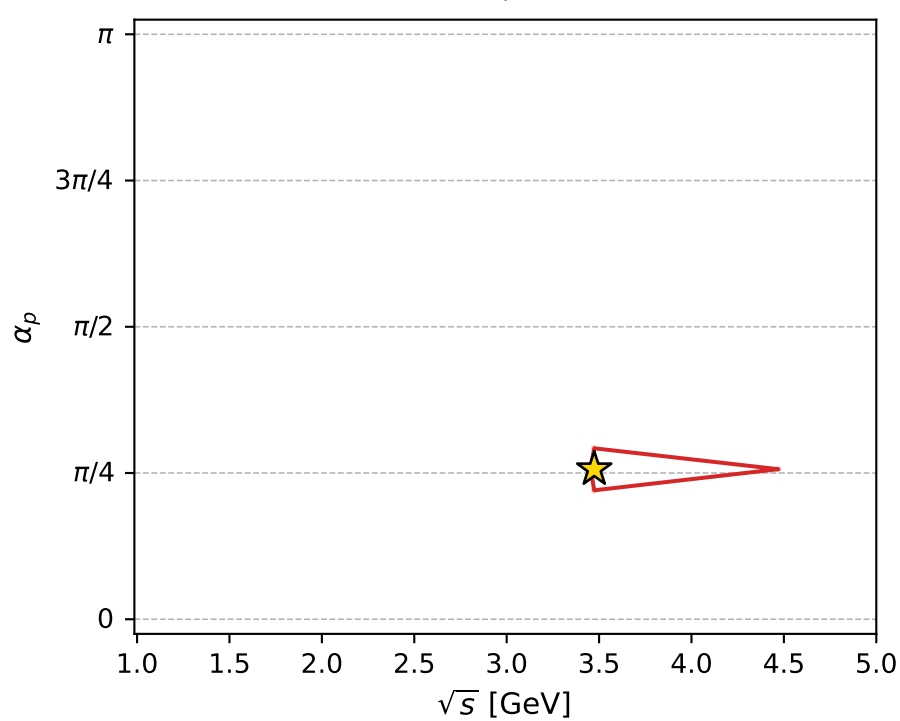
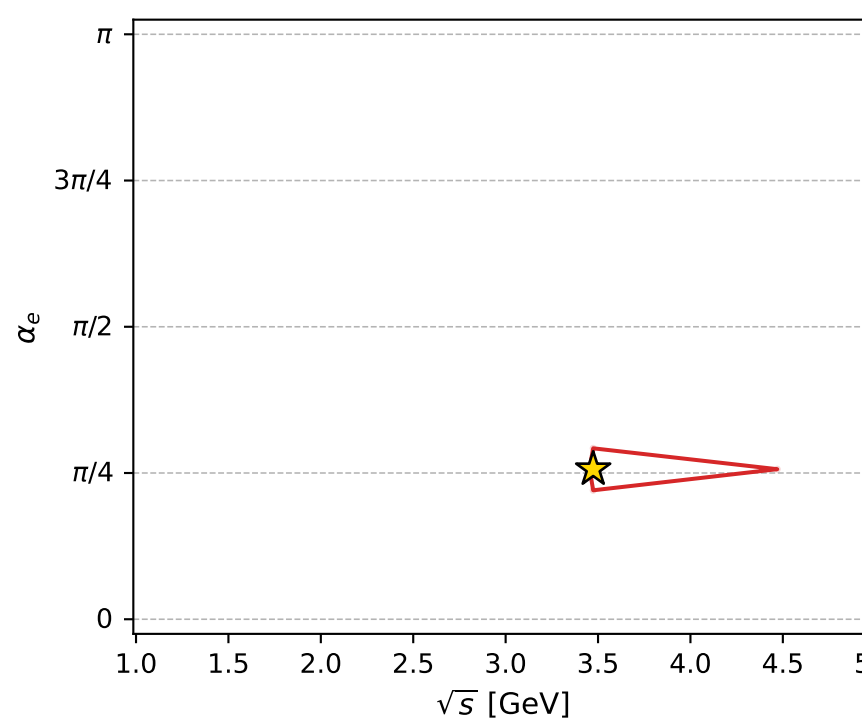
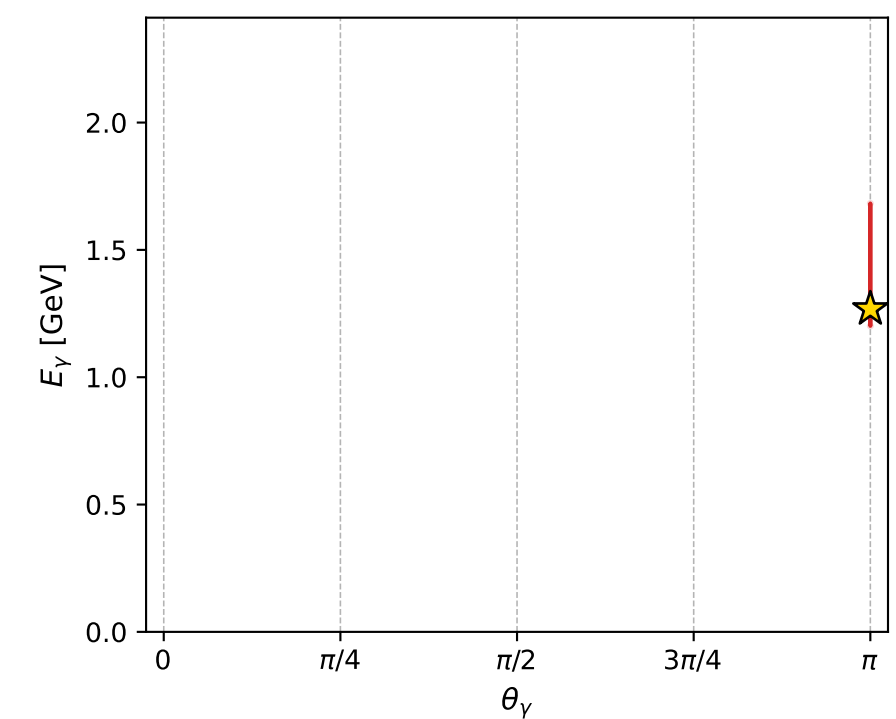
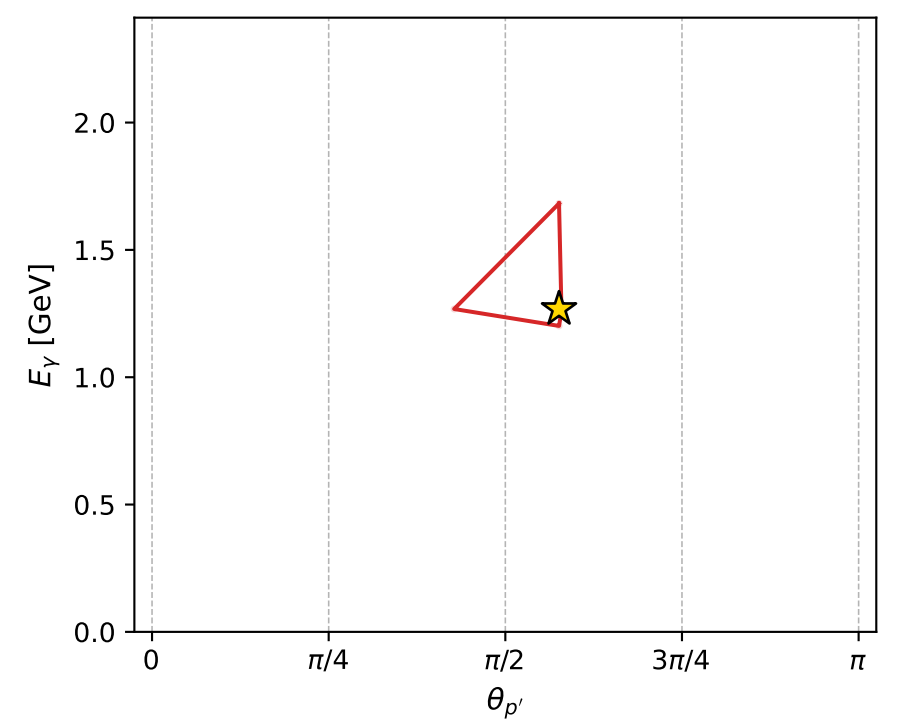
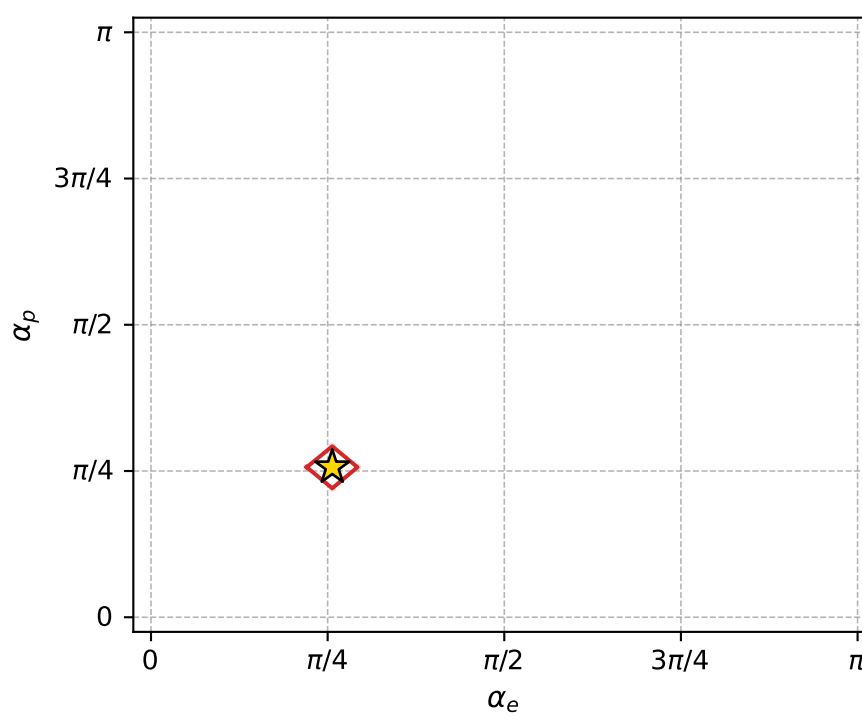
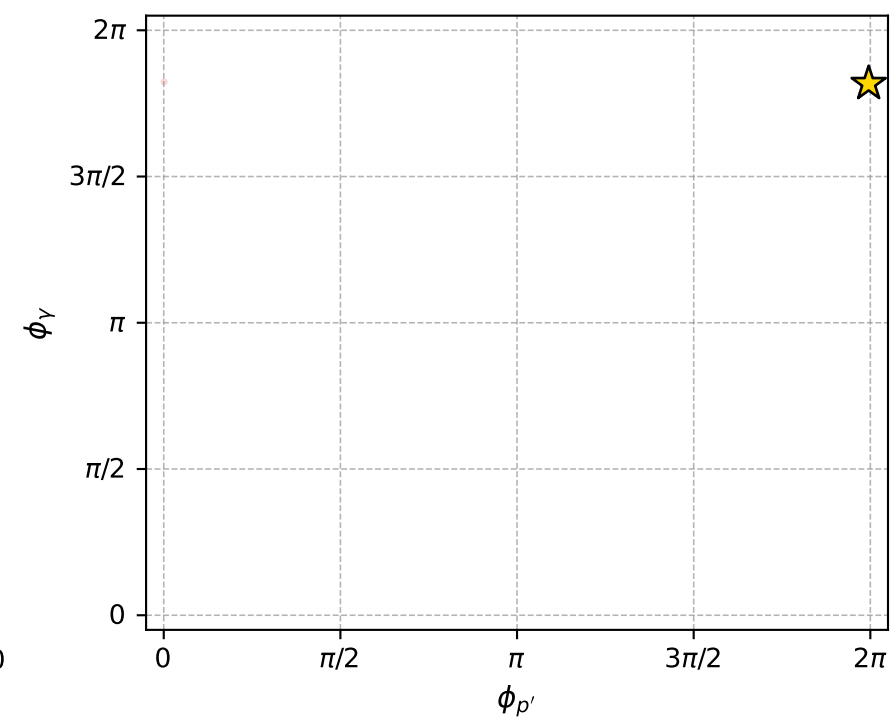
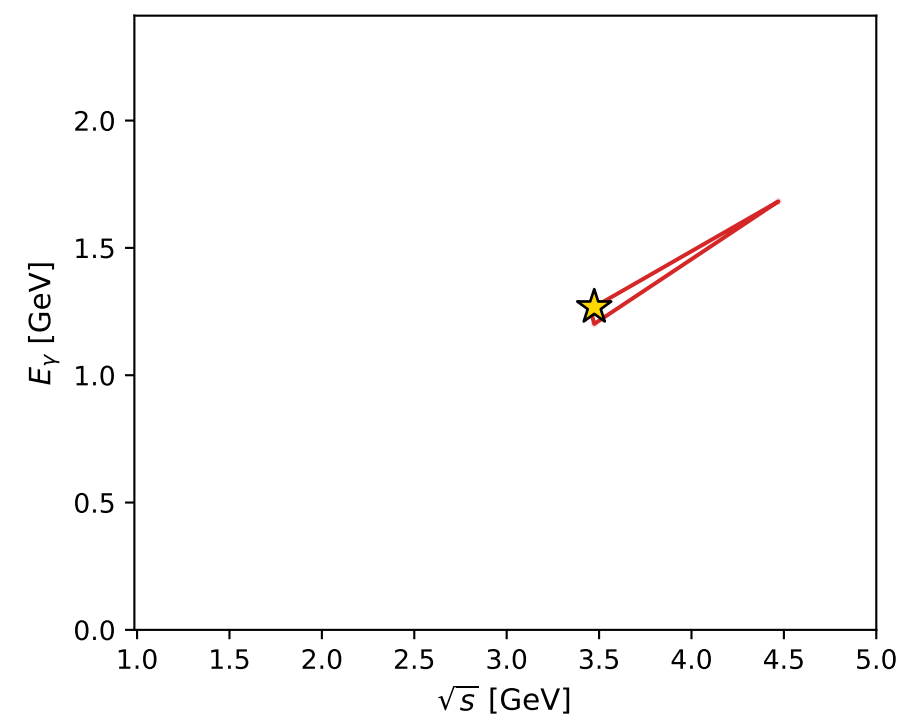
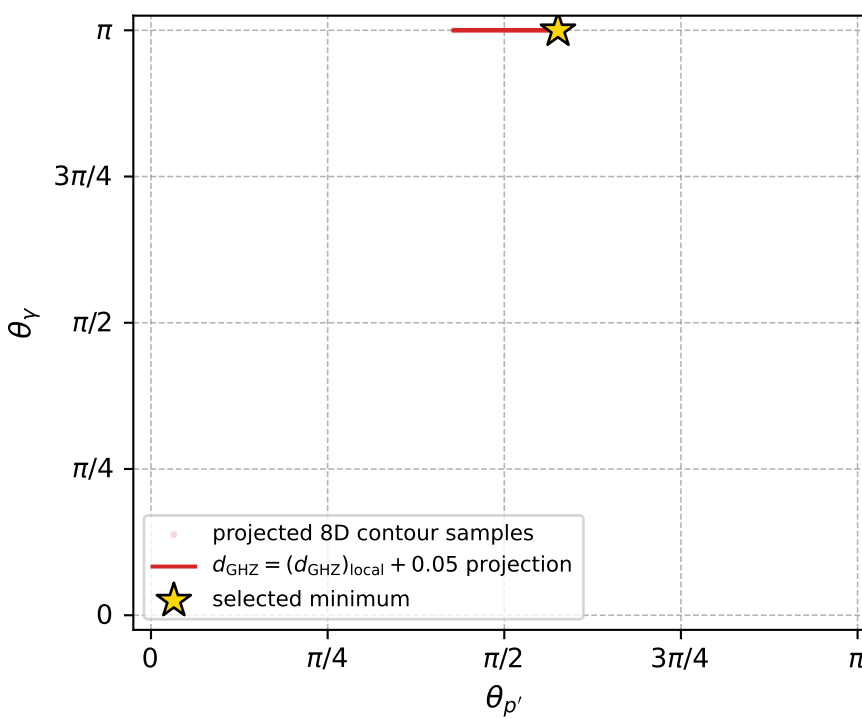
dghz_mixing_angles_polarization_01_local_minimum_130 d_{\rm GHZ}=5.25419e-08
region: polarization_cluster_01_local_minimum_130
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.8058868$ rad
incoming lepton: $0.692472|+\rangle +0.721445|-\rangle$
proton mixing angle: $\alpha_p=0.80590263$ rad
incoming proton: $0.69246|+\rangle +0.721456|-\rangle$

$s=12.0671$, $\sqrt{s}=3.47377$, $|\vec{P}|=1.61025$, $|\vec{P}'|=2.70429\text{e-}05$
 $\theta_{P'}=1.80954$, $\theta_Y=3.14159$, $\phi_{P'}=6.26842$, $\phi_Y=5.71067$
 $E_Y=1.26788$, $\theta_{\ell'}=2.07241\text{e-}05$, $\phi_{\ell'}=3.12683$
 $Q^2=8.16645$, $x_B=0.774439$, $t=-1.73631$, $W^2=3.25838$, $y=0.942589$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.6102$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6102$
 P $E= 1.8635$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6102$
 ℓ' $E= 1.2679$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.2679$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_Y $E= 1.2679$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.2679$



electron: polarization cluster P1, local minimum 130; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 5.2541893\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000053$
 above global minimum = $3.0258431\text{e-}08$
 8D contour samples = 135

selected phase-space point:

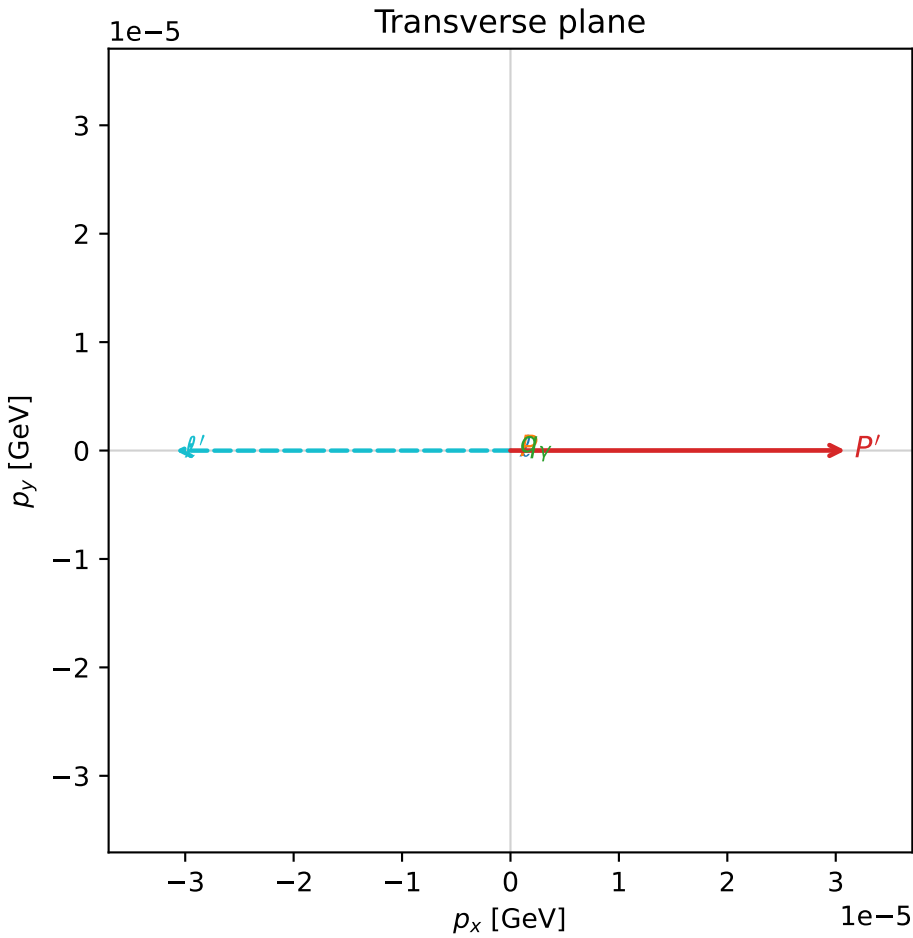
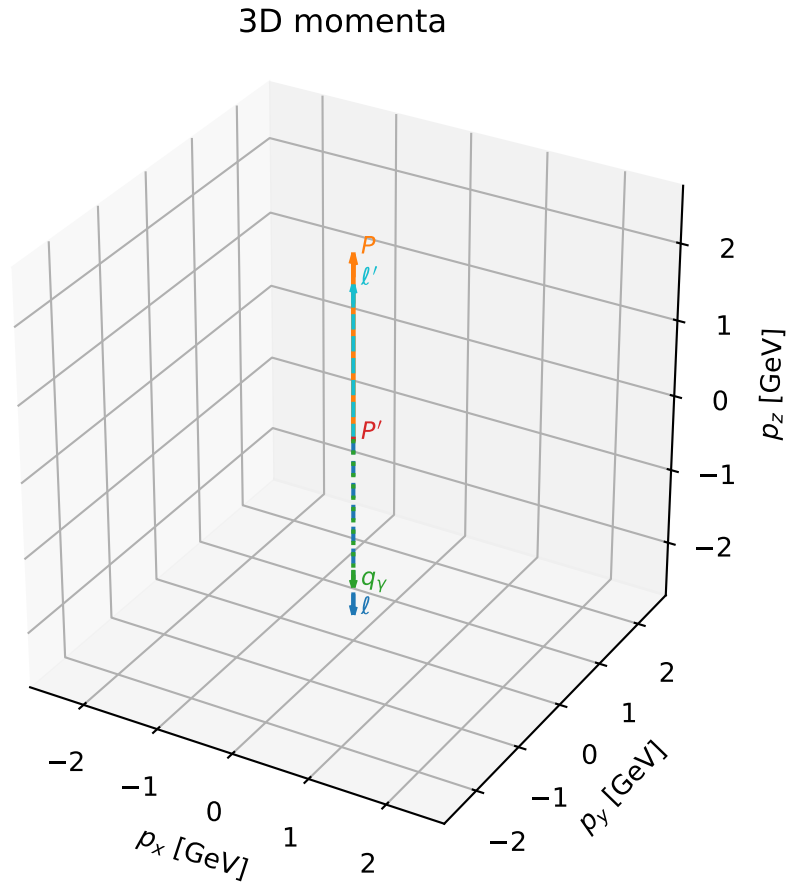
$\text{sqrt_s} = 3.473772$
 $\text{theta_p_out} = 1.809543$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.267883$
 $\text{phi_p_out} = 6.268421$
 $\text{phi_gamma_out} = 5.710674$
 $\text{alpha_e} = 0.8058868$
 $\text{alpha_p} = 0.8059026$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_135 $d_{\{\text{rm GHZ}\}}=5.61544\text{e-}08$
region: polarization_cluster_01_local_minimum_135
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.3218649$ rad
incoming lepton: $0.246369|+\rangle +0.969176|-\rangle$
proton mixing angle: $\alpha_p=1.3218678$ rad
incoming proton: $0.246366|+\rangle +0.969177|-\rangle$

$s=24.3709$, $\sqrt{s}=4.93669$, $|\vec{P}|=2.37923$, $|\vec{P}'|=0.000777917$
 $\theta_{P'}=0.0397154$, $\theta_Y=3.14159$, $\phi_{P'}=0.000309928$, $\phi_Y=3.64684$
 $E_Y=1.99973$, $\theta_{\ell'}=1.54516\text{e-}05$, $\phi_{\ell'}=3.1419$
 $Q^2=19.0239$, $x_B=0.835169$, $t=-3.0344$, $W^2=4.63445$, $y=0.969668$

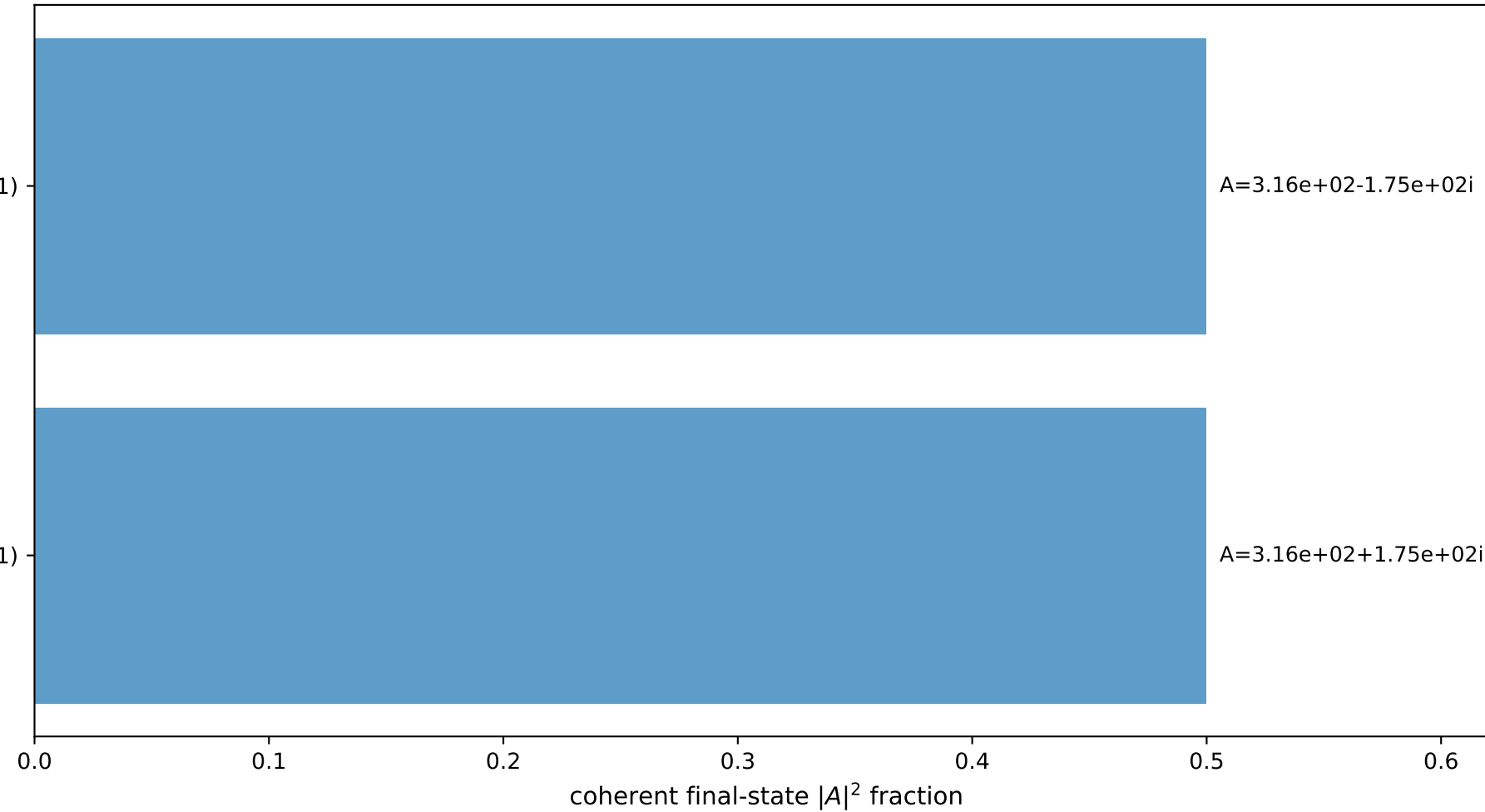
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3792$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3792$
 P $E= 2.5575$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3792$
 ℓ' $E= 1.9990$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9990$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0008$
 q_Y $E= 1.9997$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9997$



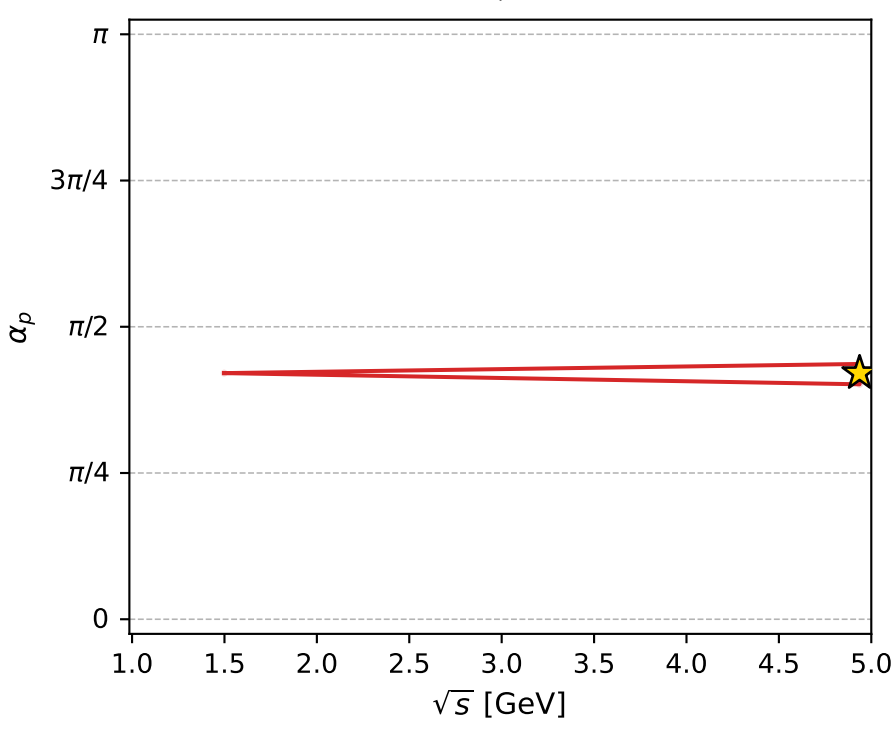
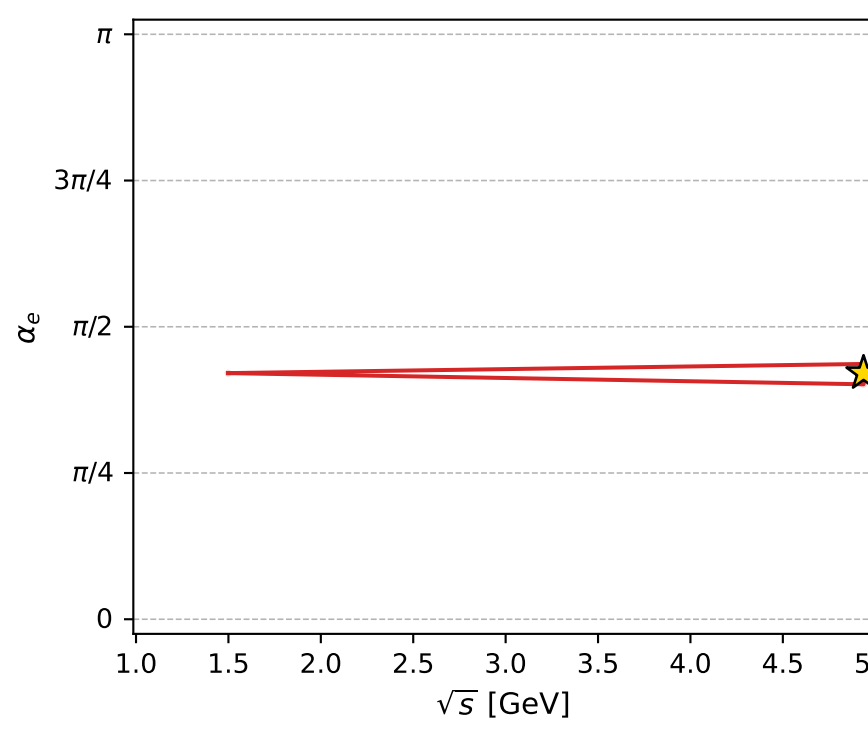
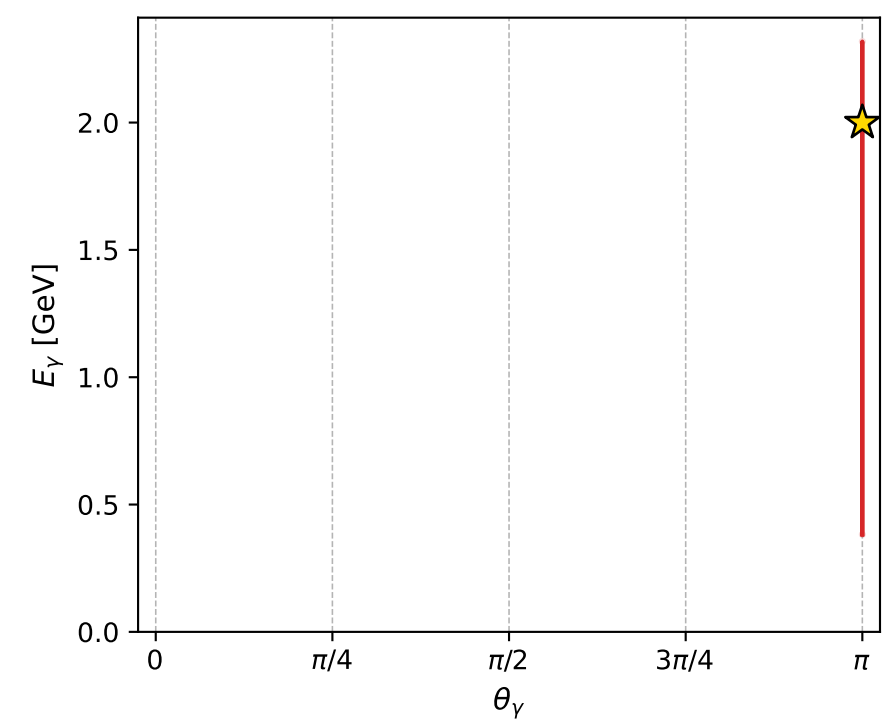
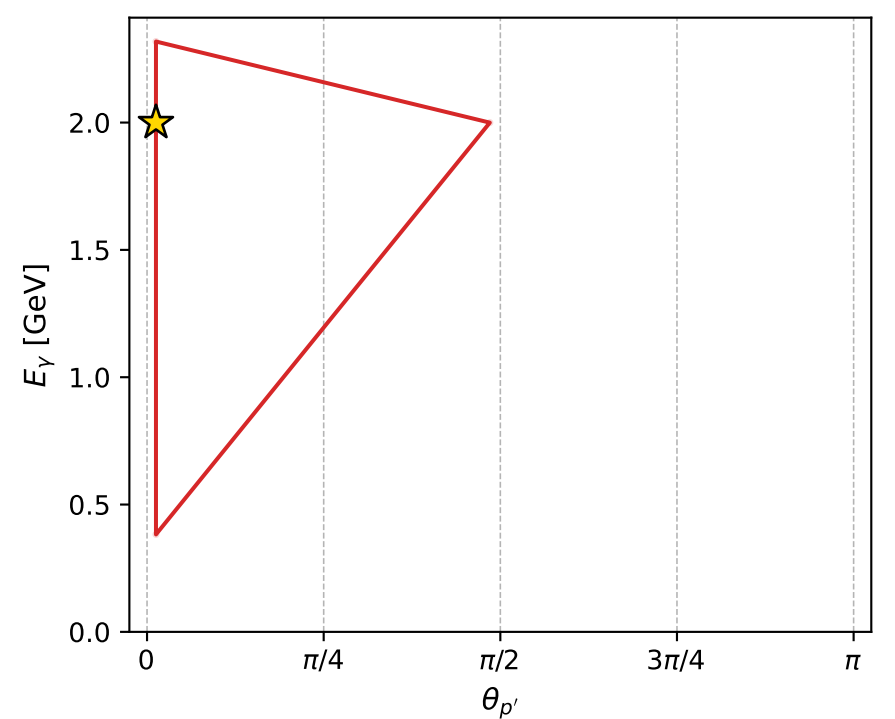
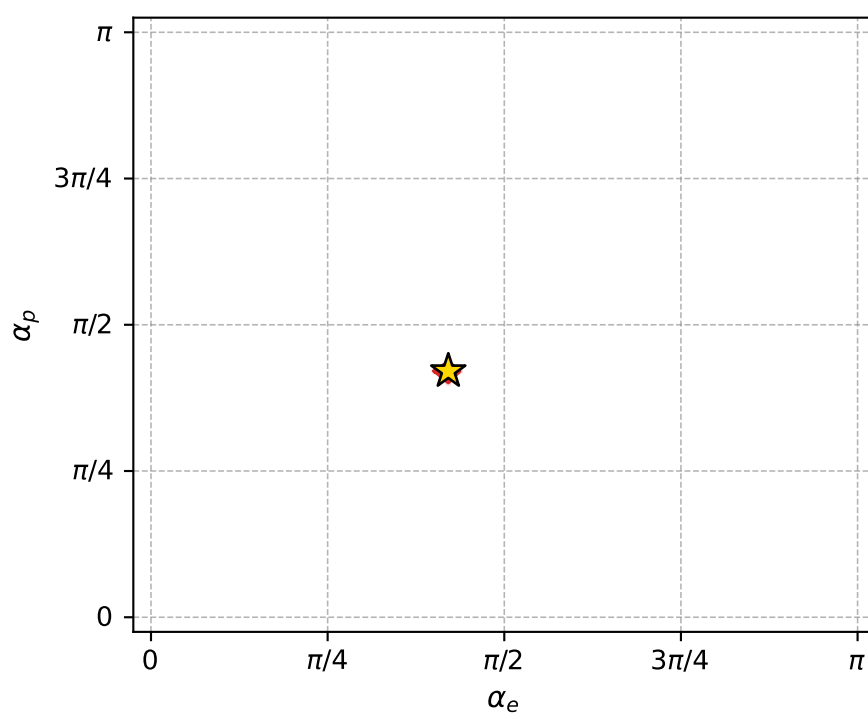
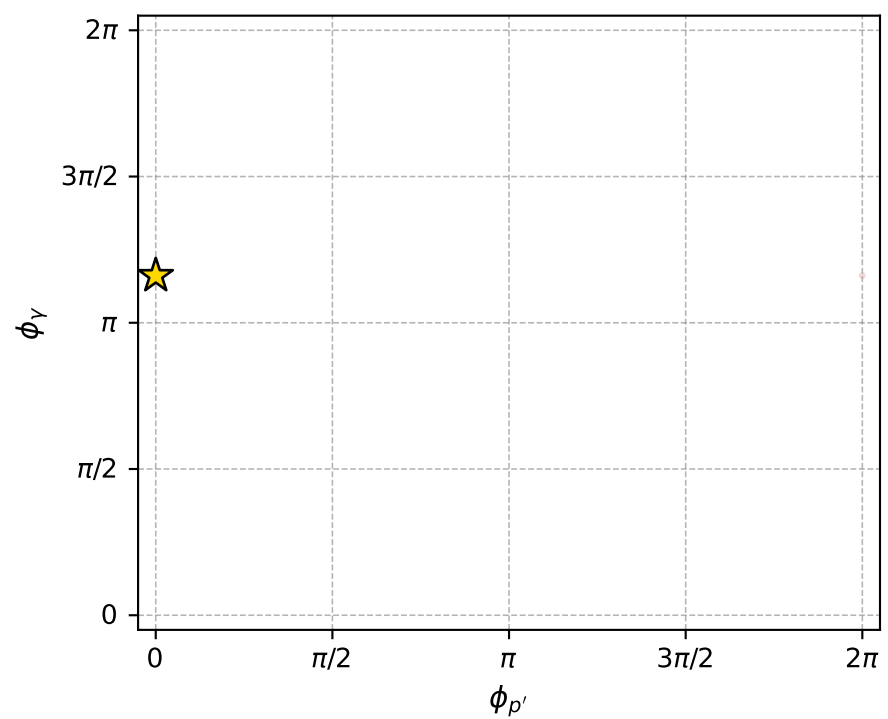
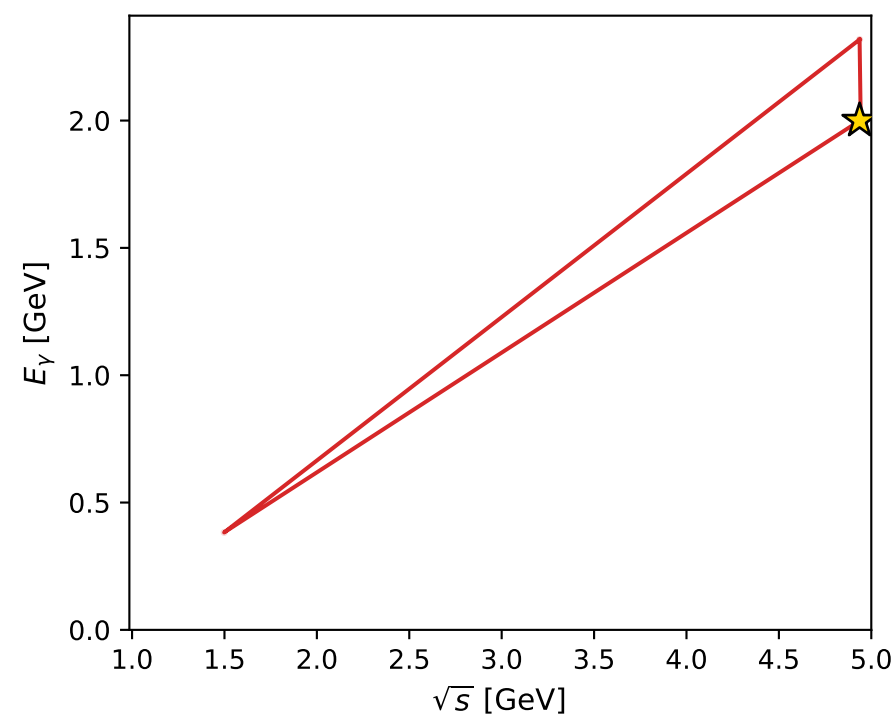
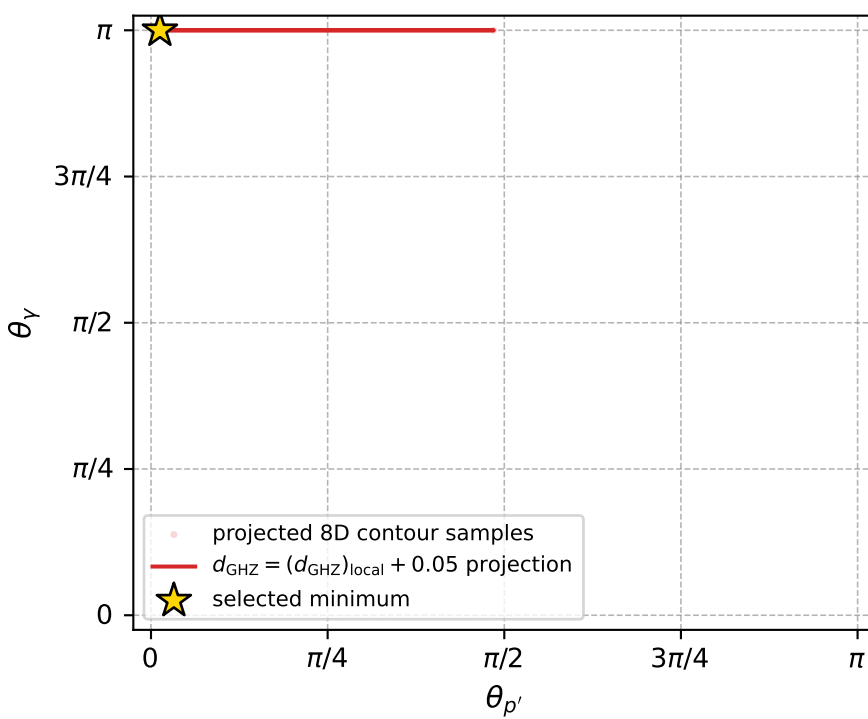
$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 135; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 5.6154366\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000056$
 above global minimum = $3.3870904\text{e-}08$
 8D contour samples = 139

selected phase-space point:

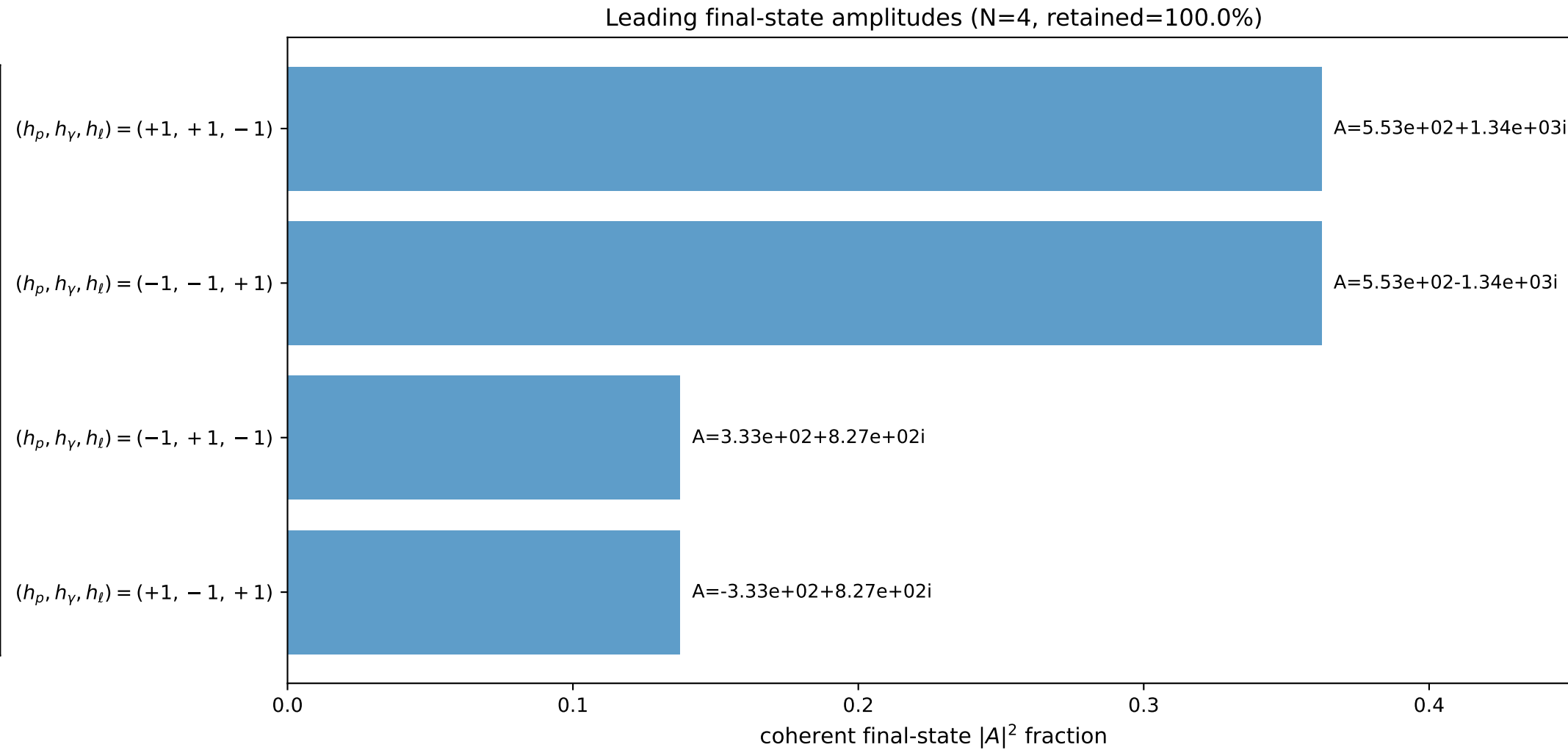
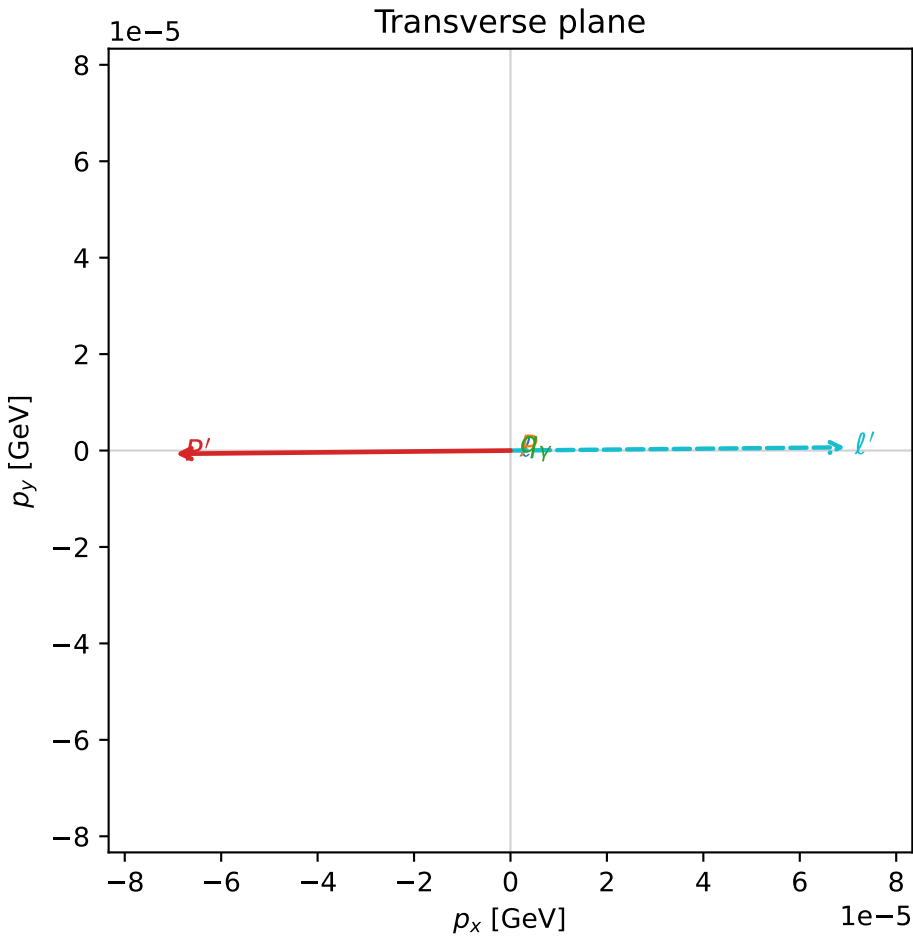
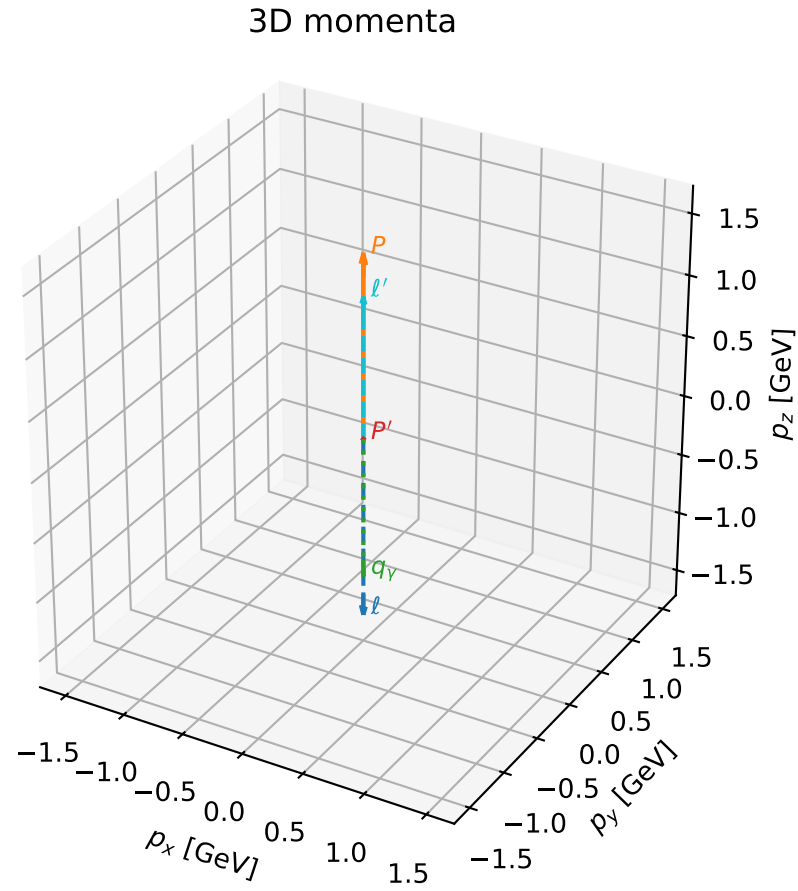
$\text{sqrt_s} = 4.936689$
 $\text{theta_p_out} = 0.03971537$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.999733$
 $\text{phi_p_out} = 0.0003099278$
 $\text{phi_gamma_out} = 3.646842$
 $\alpha_e = 1.321865$
 $\alpha_p = 1.321868$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

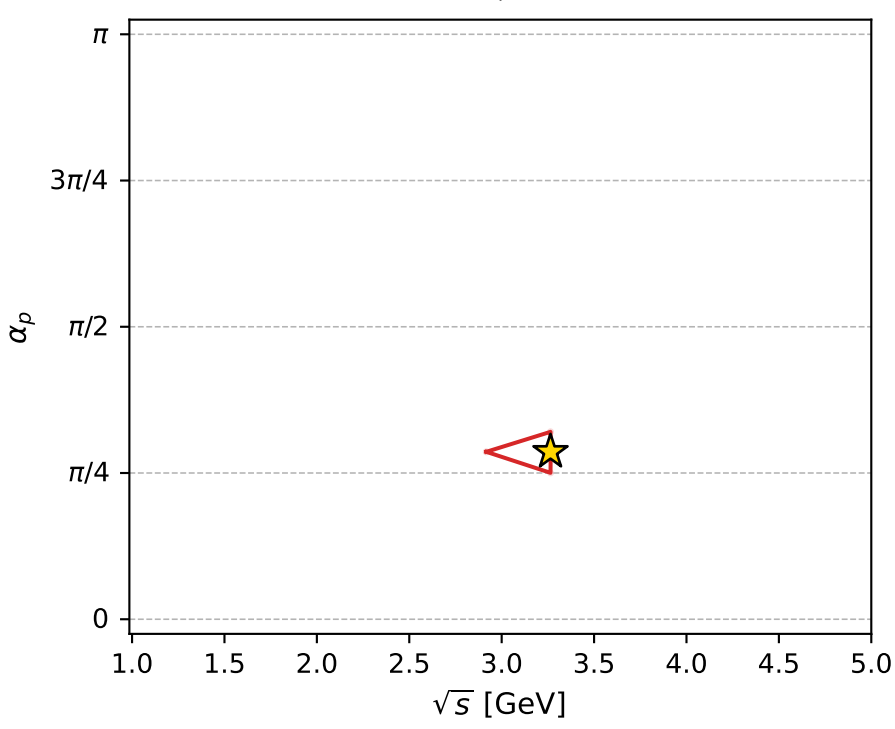
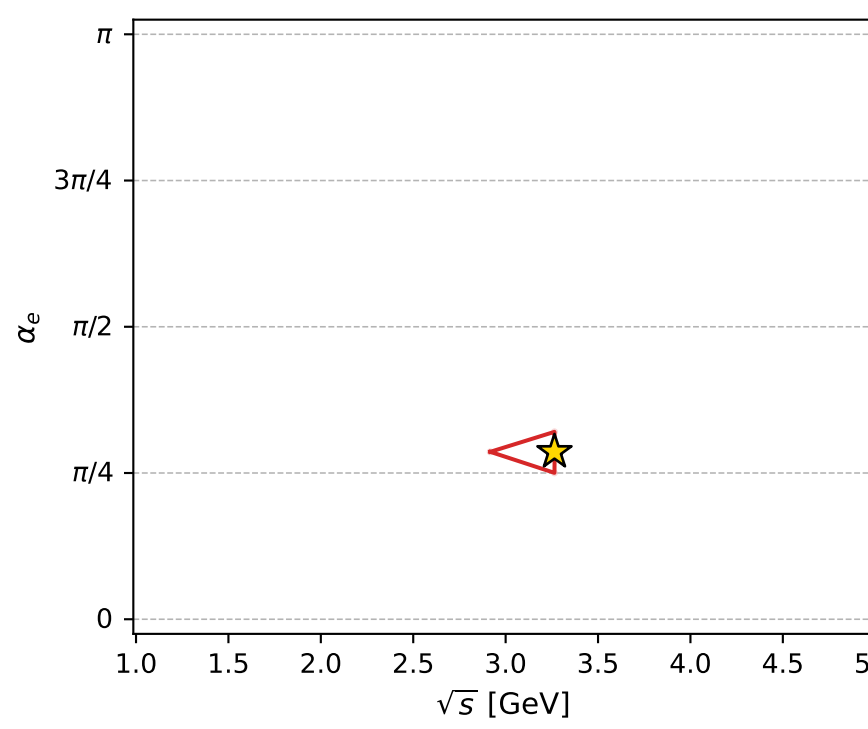
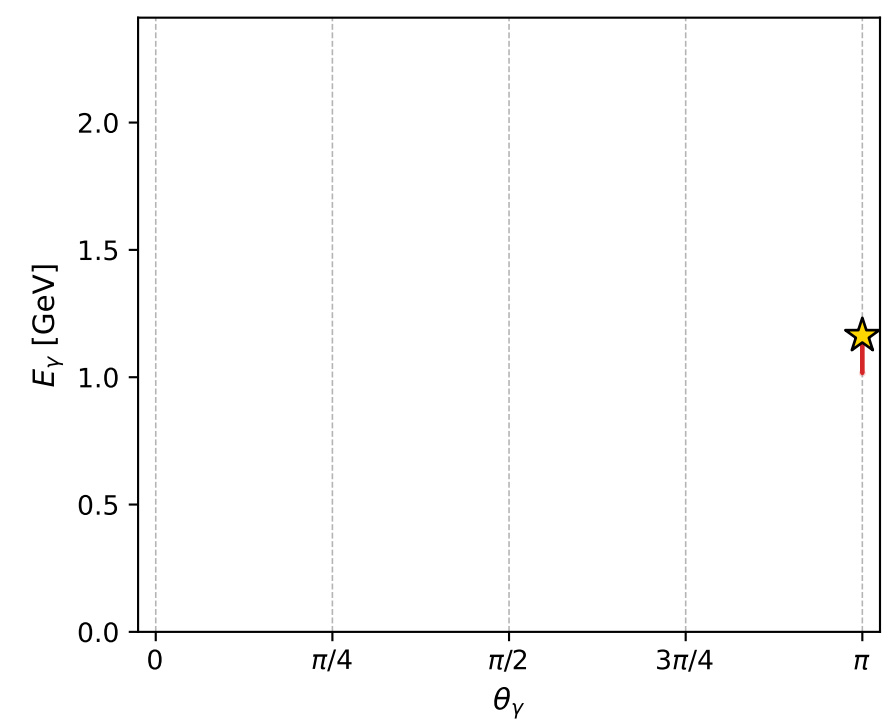
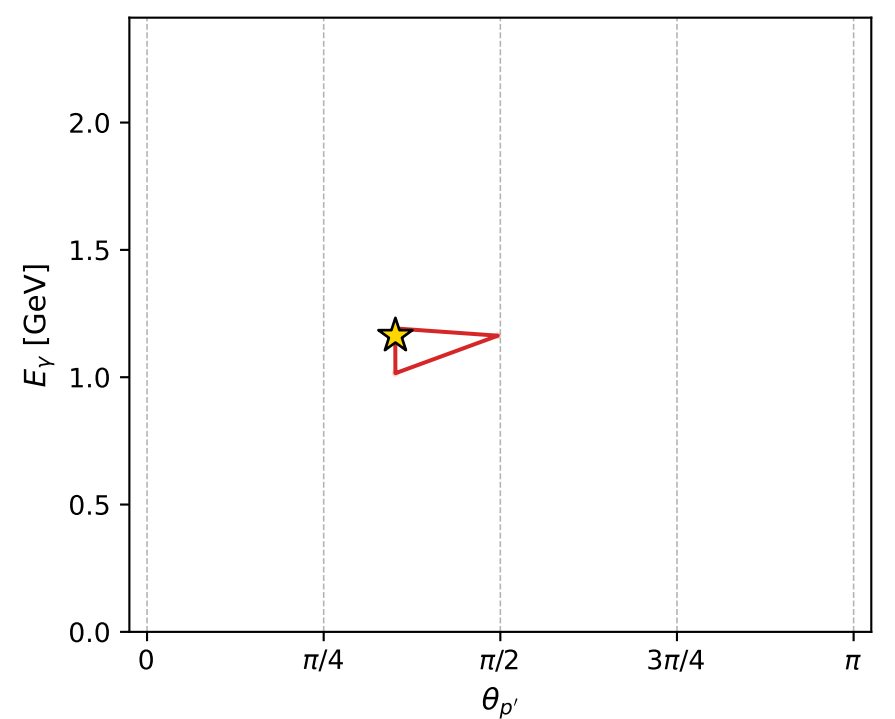
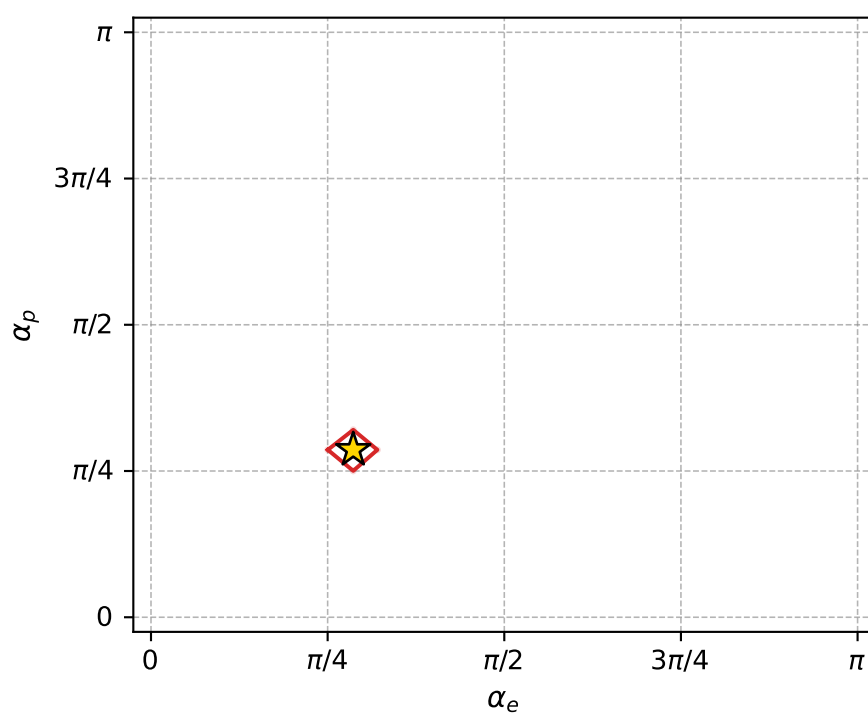
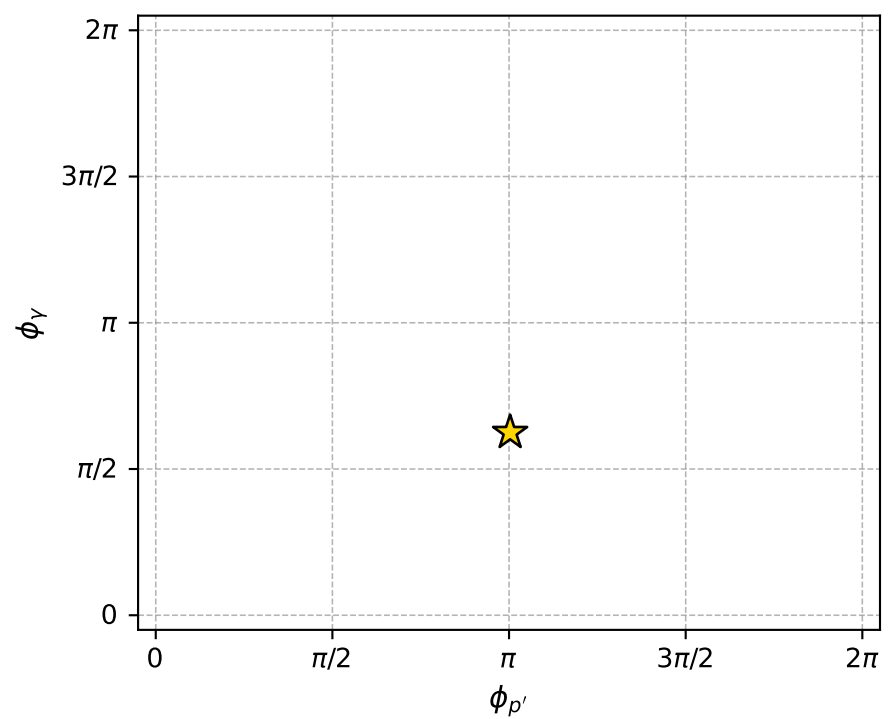
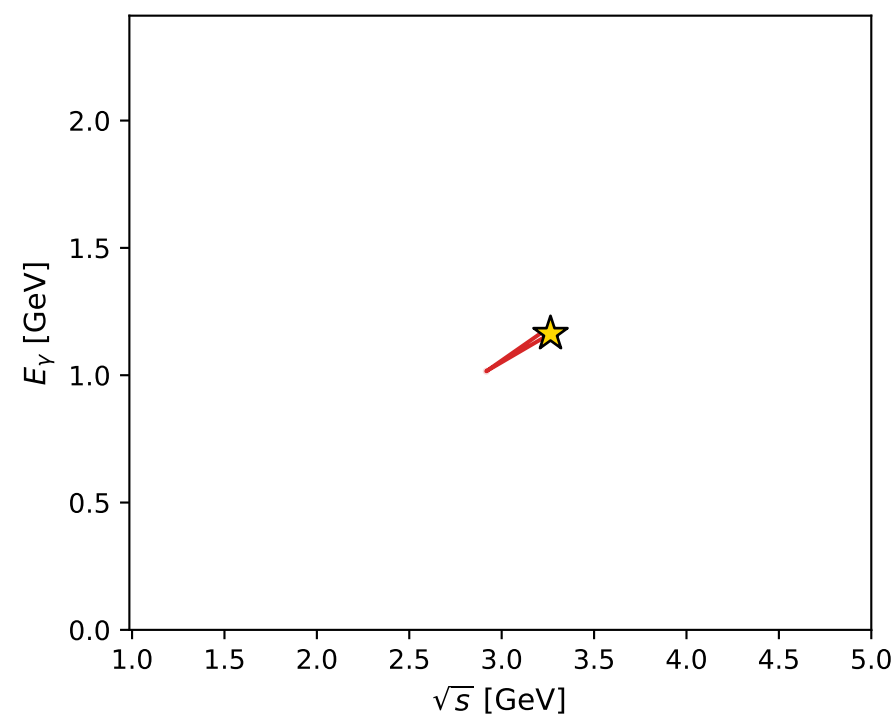
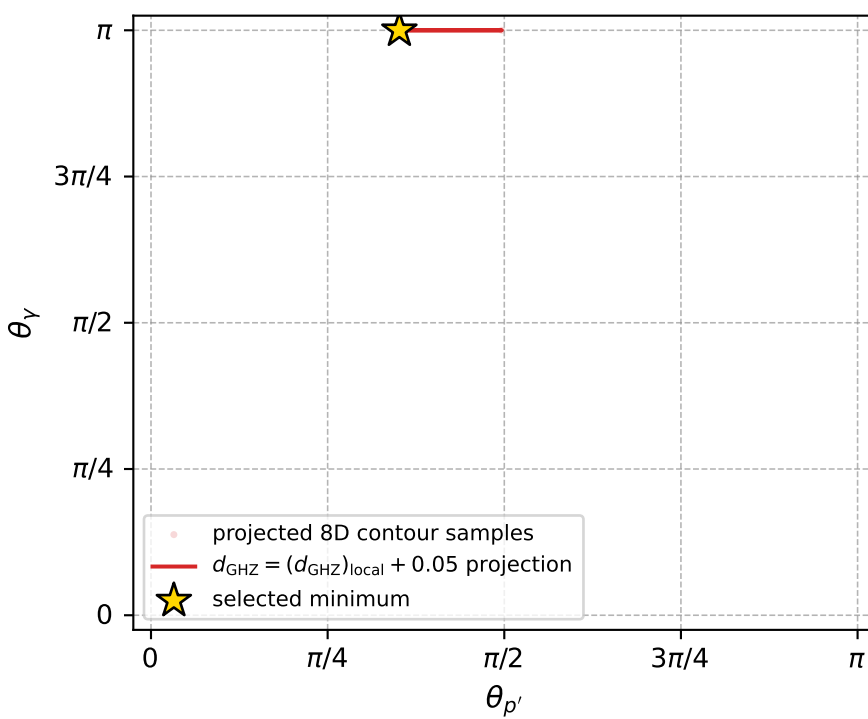
dghz_mixing_angles_polarization_01_local_minimum_137 $d_{\{\text{rm GHZ}\}}=6.5179\text{e-}08$
region: polarization_cluster_01_local_minimum_137
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.89915933$ rad
incoming lepton: $0.622268|+\rangle +0.782804|-\rangle$
proton mixing angle: $\alpha_p=0.89919378$ rad
incoming proton: $0.622241|+\rangle +0.782826|-\rangle$

$s=10.6551$, $\sqrt{s}=3.26422$, $|\vec{P}|=1.49734$, $|\vec{P}'|=7.77652\text{e-}05$
 $\theta_{p'}=1.10433$, $\theta_Y=3.14159$, $\phi_{p'}=3.15129$, $\phi_Y=1.96285$
 $E_Y=1.16313$, $\theta_{\ell'}=5.97176\text{e-}05$, $\phi_{\ell'}=0.00969301$
 $Q^2=6.96616$, $x_B=0.761473$, $t=-1.55487$, $W^2=3.06195$, $y=0.935858$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.4973$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4973$
 P $E= 1.7669$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4973$
 ℓ' $E= 1.1631$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.1631$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 0.0000$
 q_Y $E= 1.1631$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.1631$



electron: polarization cluster P1, local minimum 137; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 6.5178966\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000065$
 above global minimum = $4.2895504\text{e-}08$
 8D contour samples = 124

selected phase-space point:

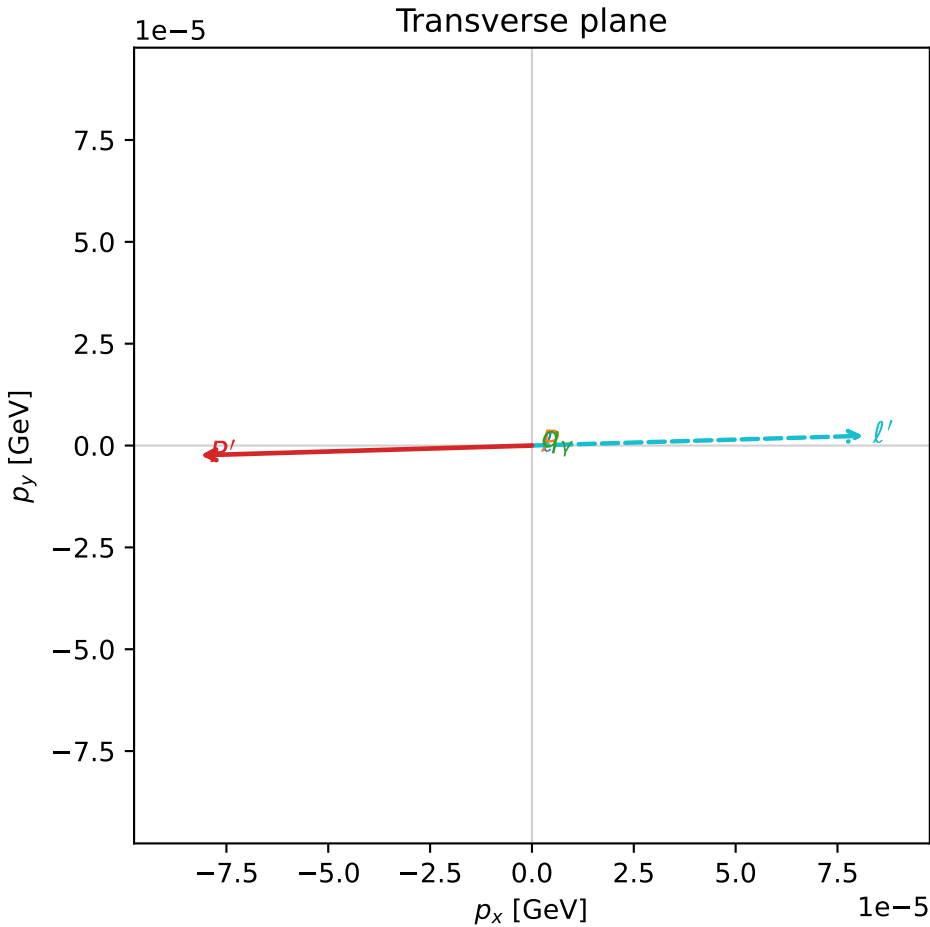
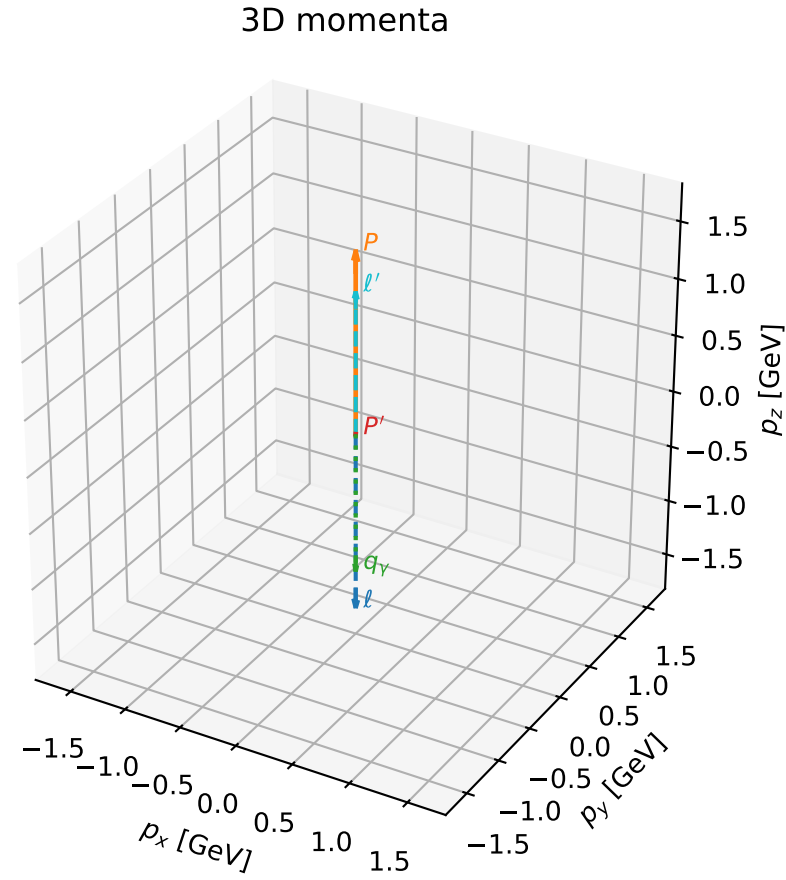
$\text{sqrt_s} = 3.264217$
 $\text{theta_p_out} = 1.104331$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.163126$
 $\text{phi_p_out} = 3.151286$
 $\text{phi_gamma_out} = 1.962849$
 $\text{alpha_e} = 0.8991593$
 $\text{alpha_p} = 0.8991938$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_138 $d_{\{\text{rm GHZ}\}}=6.56334\text{e-}08$
region: polarization_cluster_01_local_minimum_138
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.83474474$ rad
incoming lepton: $0.671367|+\rangle +0.741125|-\rangle$
proton mixing angle: $\alpha_p=0.83473256$ rad
incoming proton: $0.671376|+\rangle +0.741117|-\rangle$

$s=11.7789$, $\sqrt{s}=3.43203$, $|\vec{P}|=1.58784$, $|\vec{P}'|=0.00100941$
 $\theta_{p'}=3.06084$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.17093$, $\phi_\gamma=2.78545$
 $E_\gamma=1.24651$, $\theta_{\ell'}=6.52663\text{e-}05$, $\phi_{\ell'}=0.0293406$
 $Q^2=7.92343$, $x_B=0.772311$, $t=-1.70322$, $W^2=3.2158$, $y=0.941313$

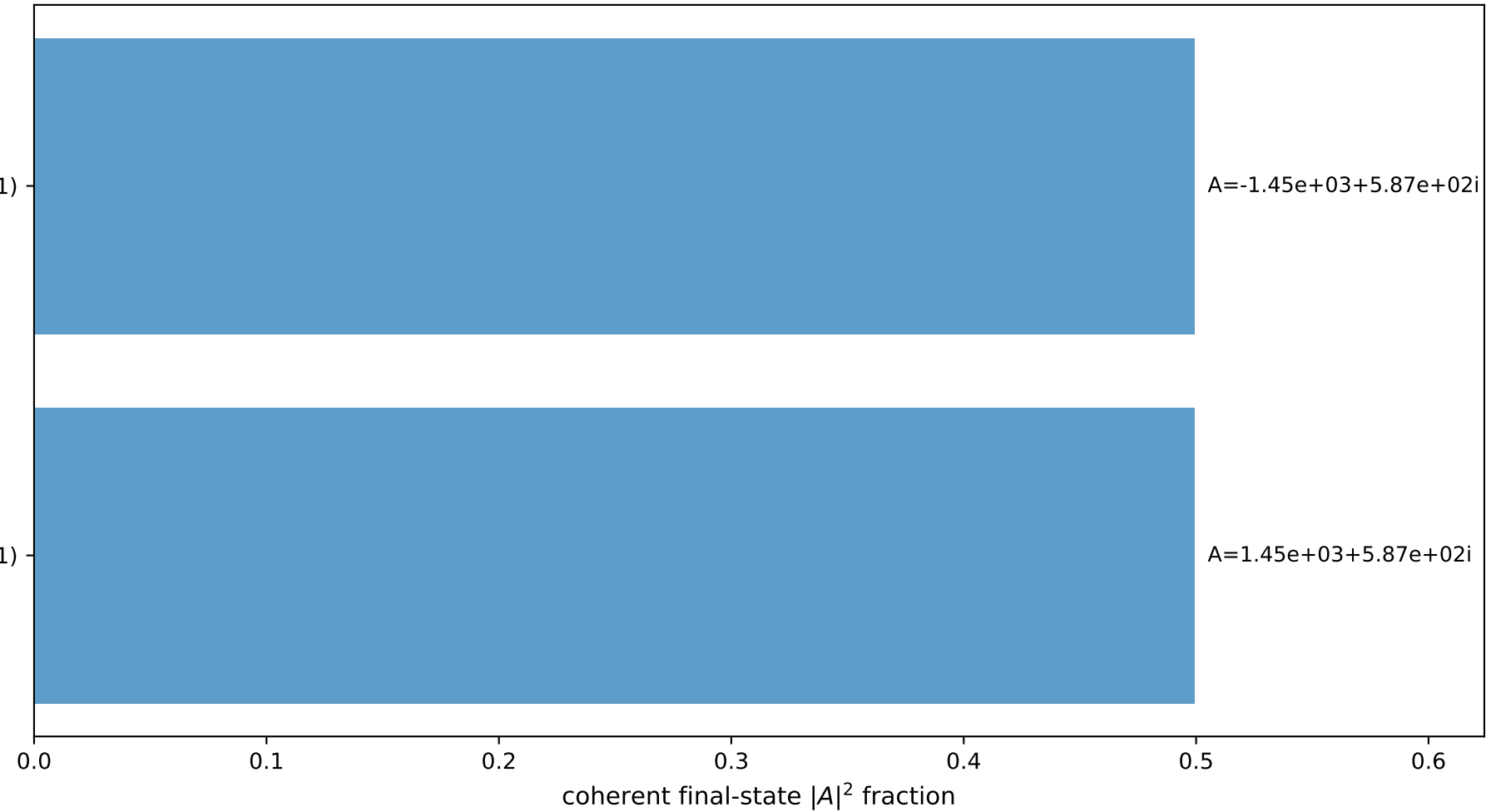
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5878$
 P $E= 1.8442$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5878$
 ℓ' $E= 1.2475$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.2475$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.0010$
 q_γ $E= 1.2465$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.2465$



$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$

$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$

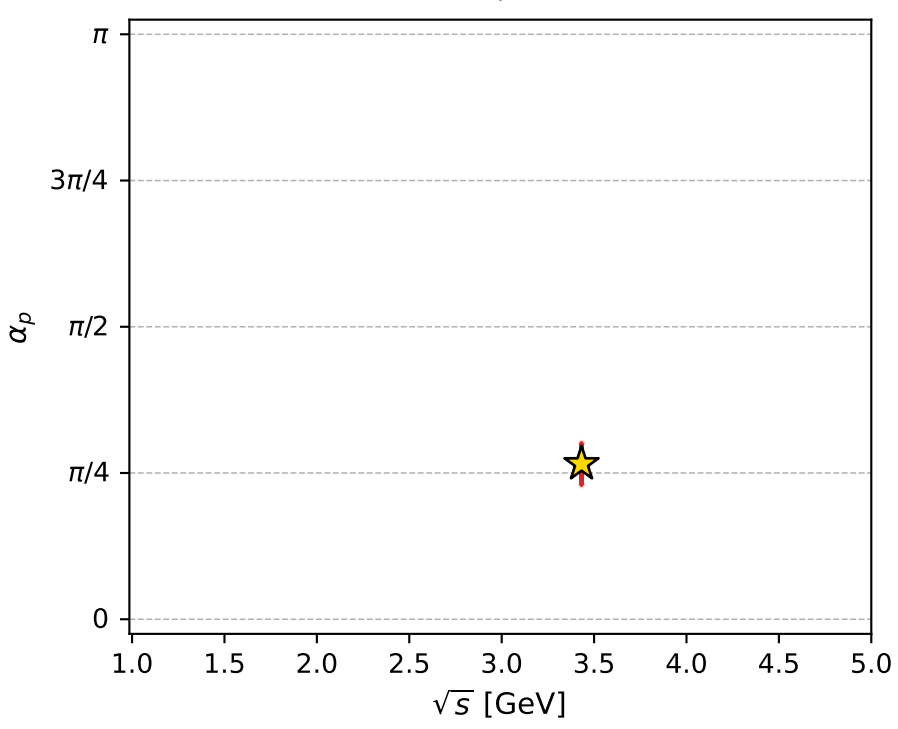
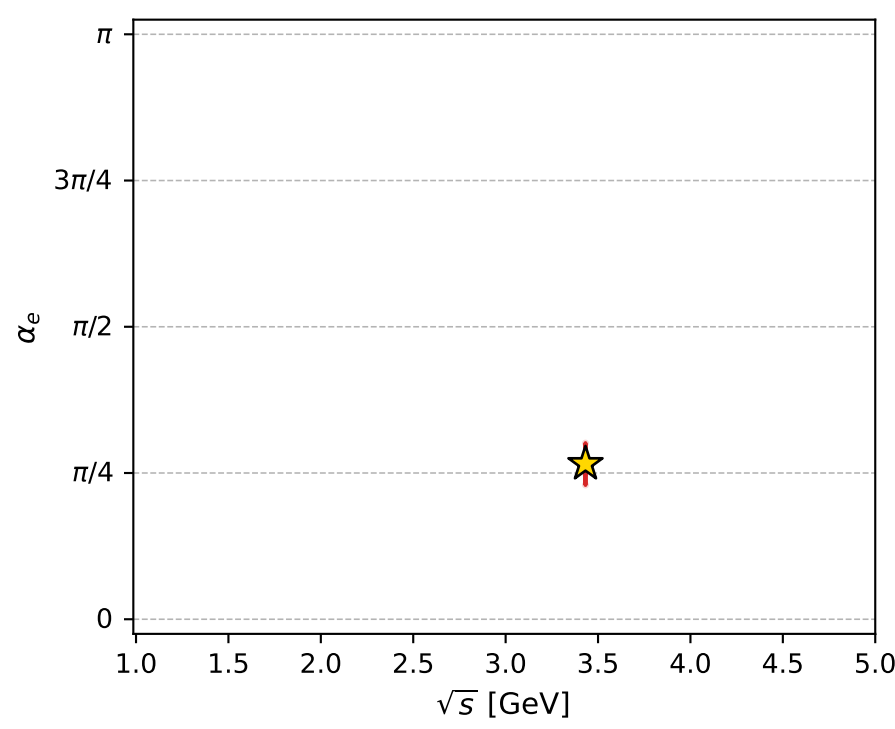
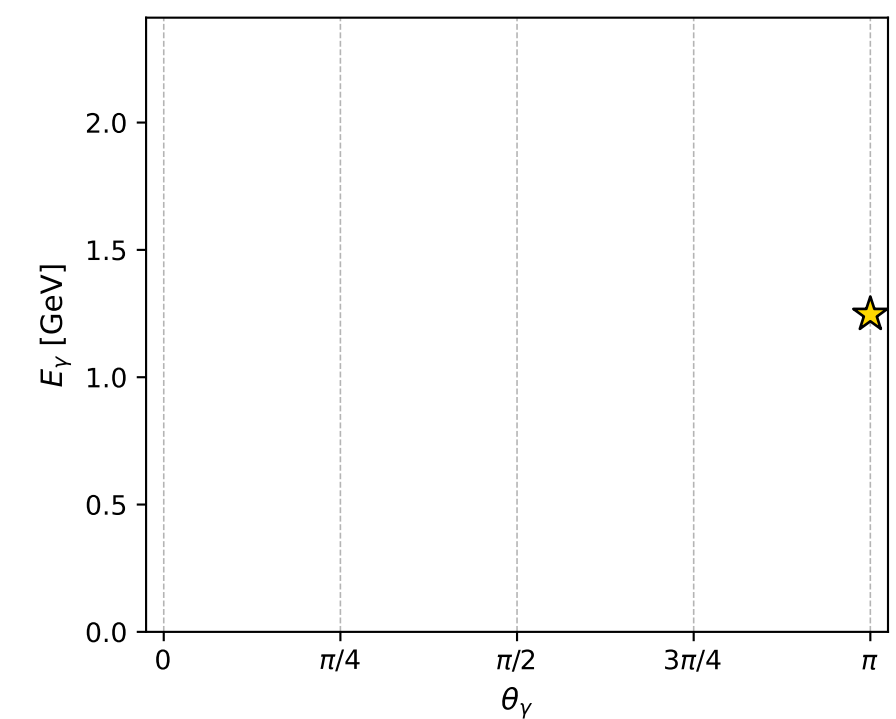
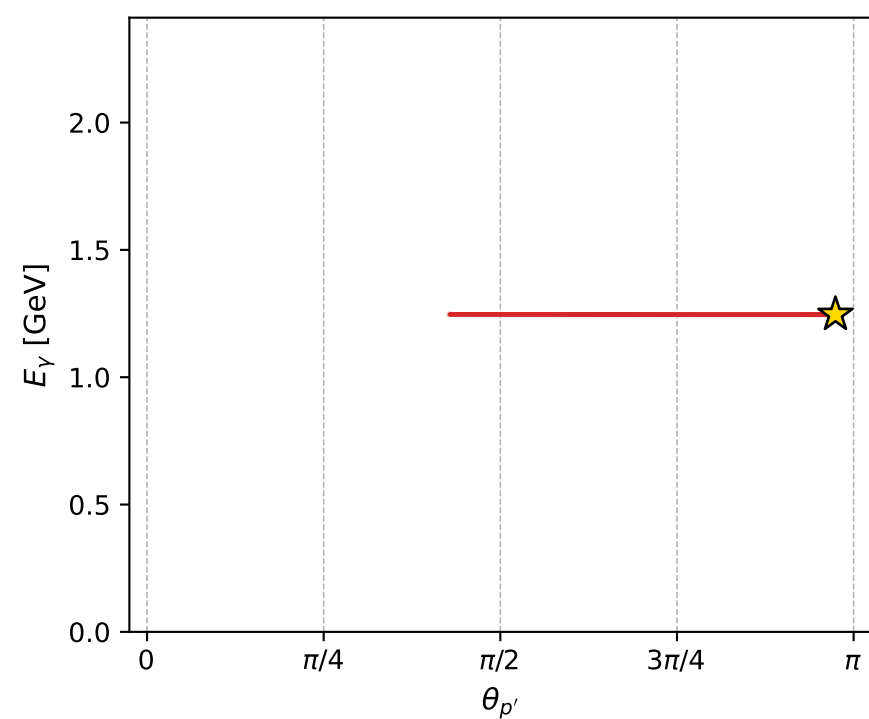
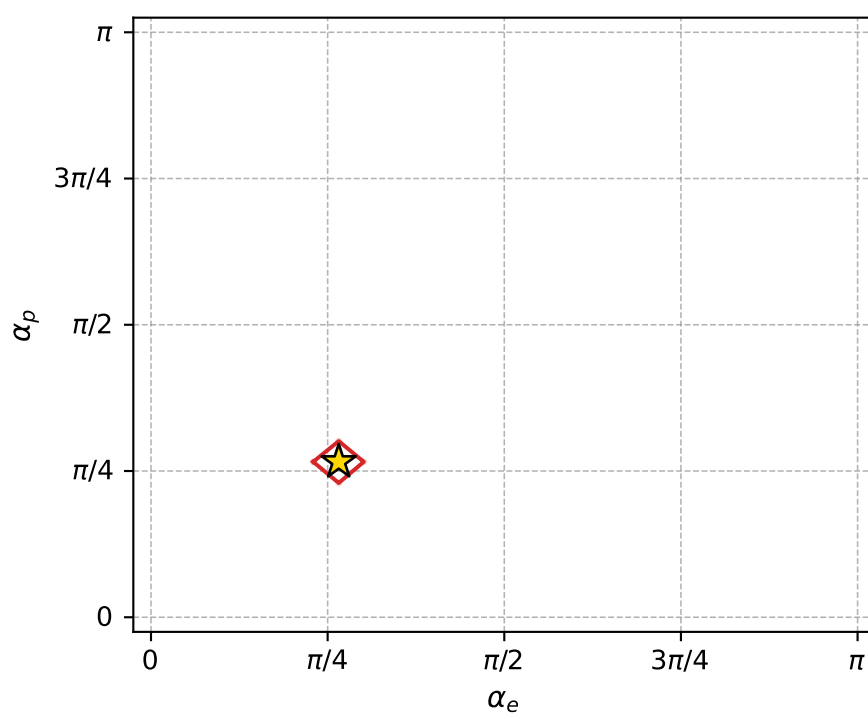
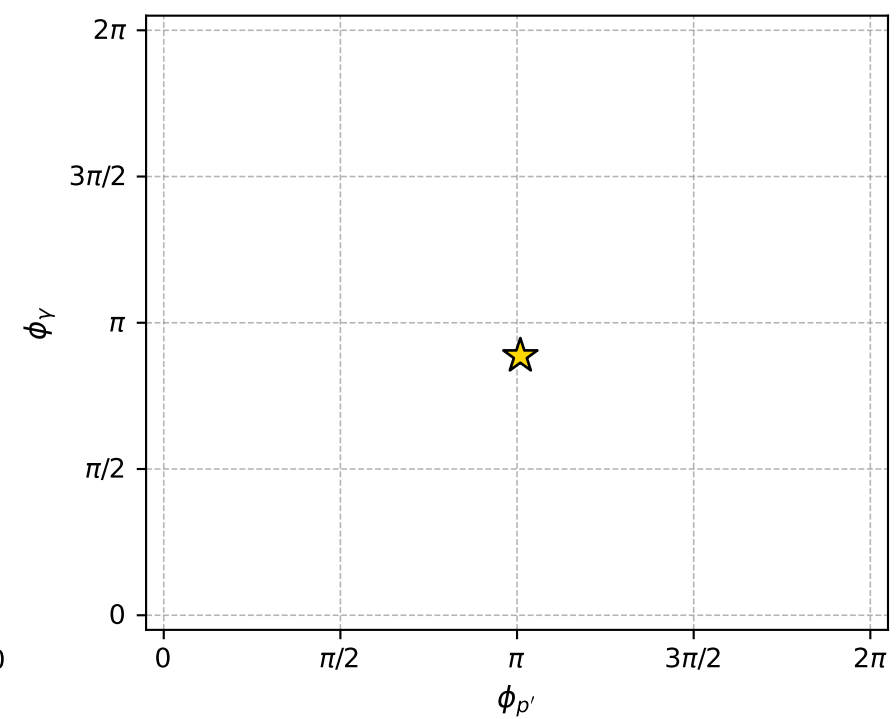
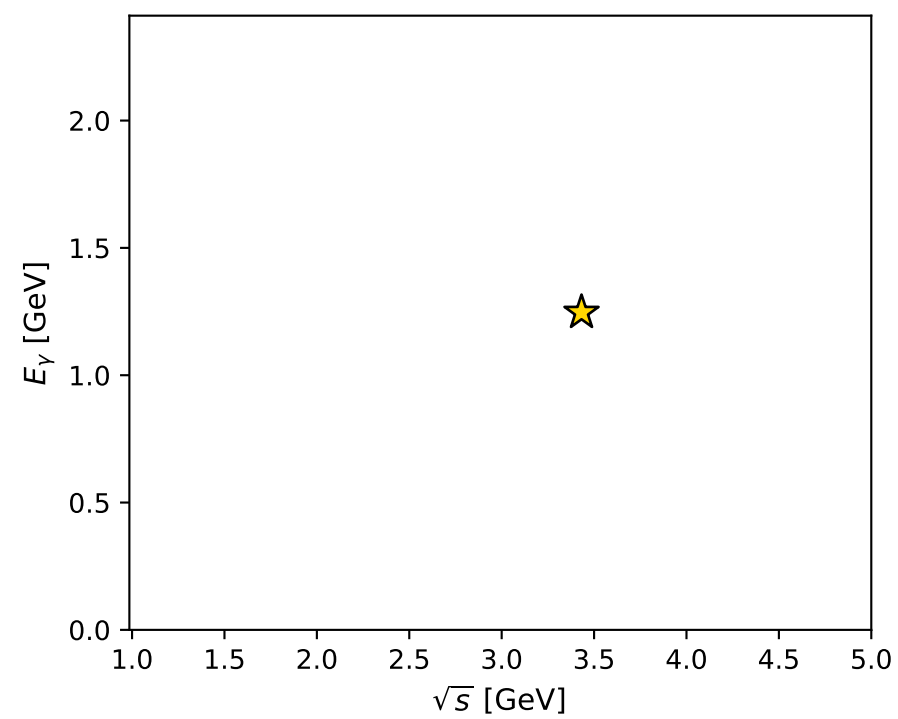
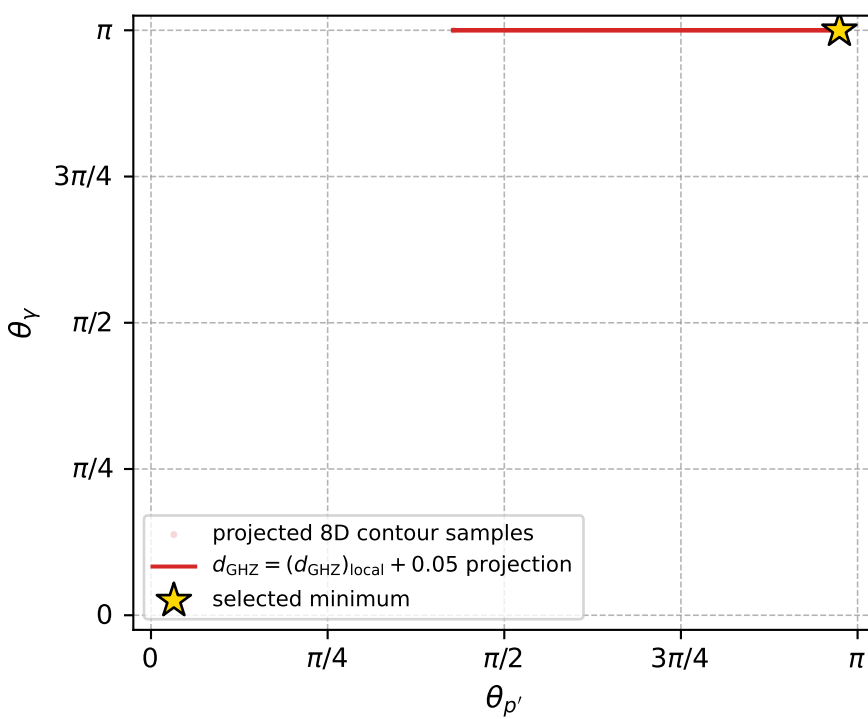
Leading final-state amplitudes (N=2, retained=99.8%)



$A=-1.45\text{e}+03+5.87\text{e}+02i$

$A=1.45\text{e}+03+5.87\text{e}+02i$

electron: polarization cluster P1, local minimum 138; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 6.563336\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000066$
 above global minimum = $4.3349898\text{e-}08$
 8D contour samples = 130

selected phase-space point:

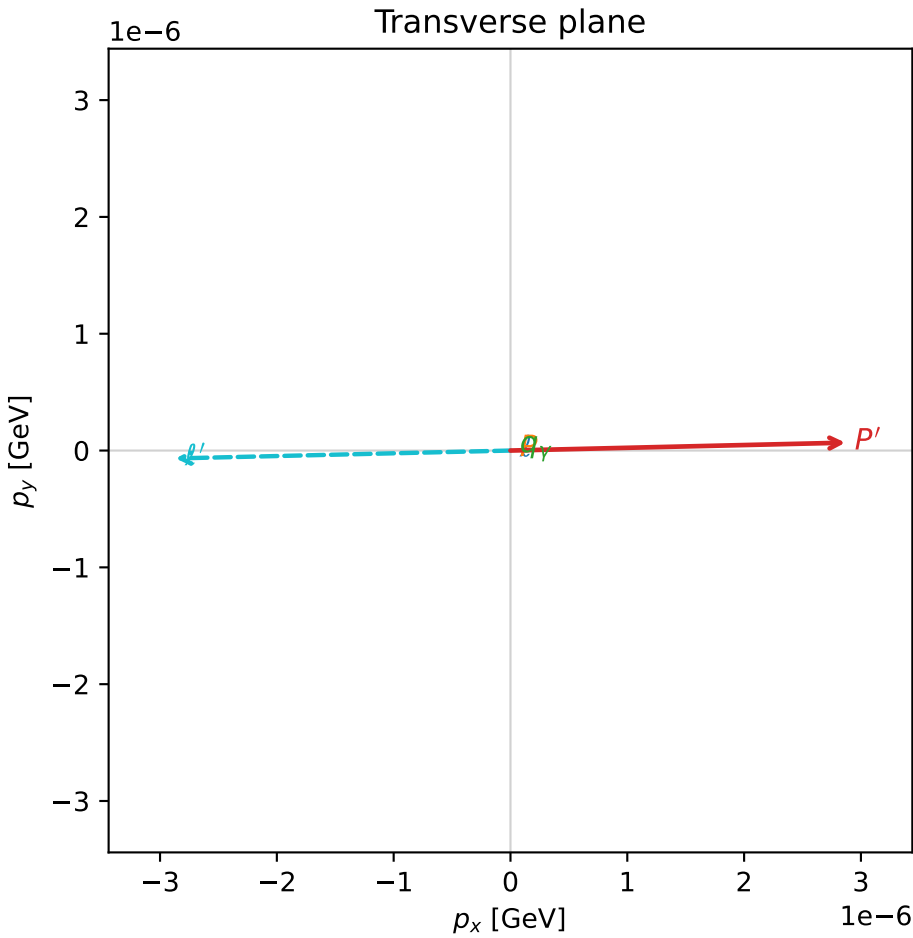
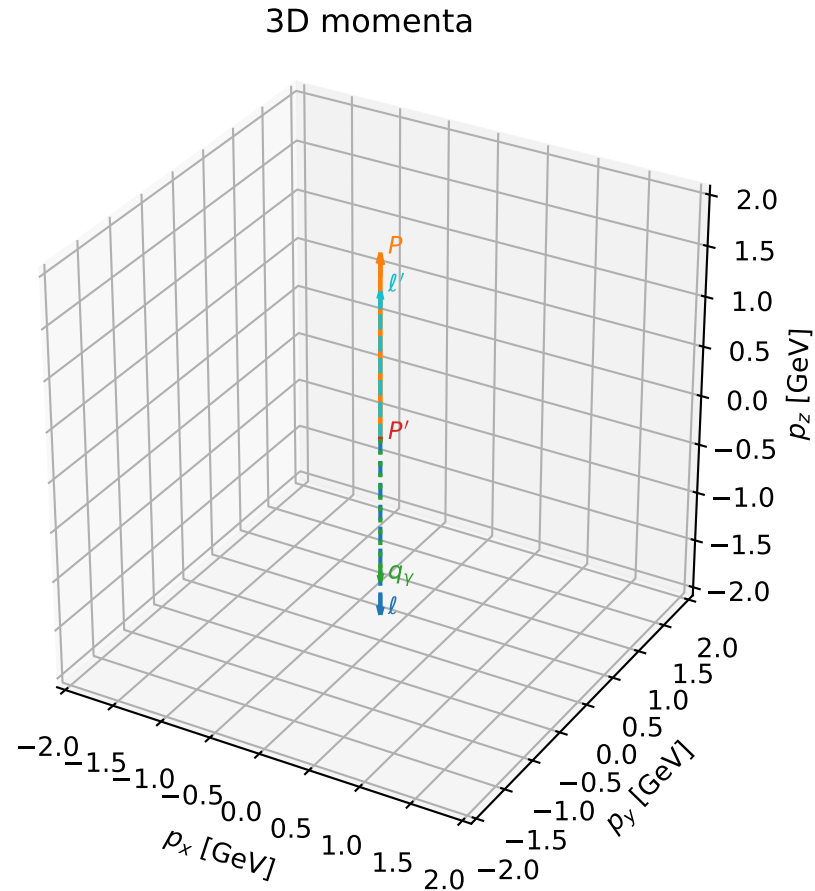
$\text{sqrt_s} = 3.432034$
 $\text{theta_p_out} = 3.060843$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.246514$
 $\text{phi_p_out} = 3.170933$
 $\text{phi_gamma_out} = 2.785449$
 $\alpha_e = 0.8347447$
 $\alpha_p = 0.8347326$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_139 $d_{\{\text{rm GHZ}\}}=6.7347\text{e-}08$
 region: polarization_cluster_01_local_minimum_139
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.40504113$ rad
 incoming lepton: $0.919086|+\rangle +0.394057|-\rangle$
 proton mixing angle: $\alpha_p=0.40503483$ rad
 incoming proton: $0.919089|+\rangle +0.394051|-\rangle$

$s=14.8742$, $\sqrt{s}=3.85671$, $|\vec{P}|=1.81429$, $|\vec{P}'|=0.000584182$
 $\theta_{P'}=0.00490874$, $\theta_V=3.14159$, $\phi_{P'}=0.0235349$, $\phi_V=1.27295$
 $E_V=1.45965$, $\theta_{\ell'}=1.96537\text{e-}06$, $\phi_{\ell'}=3.16513$
 $Q^2=10.5886$, $x_B=0.794427$, $t=-2.06977$, $W^2=3.61985$, $y=0.952429$

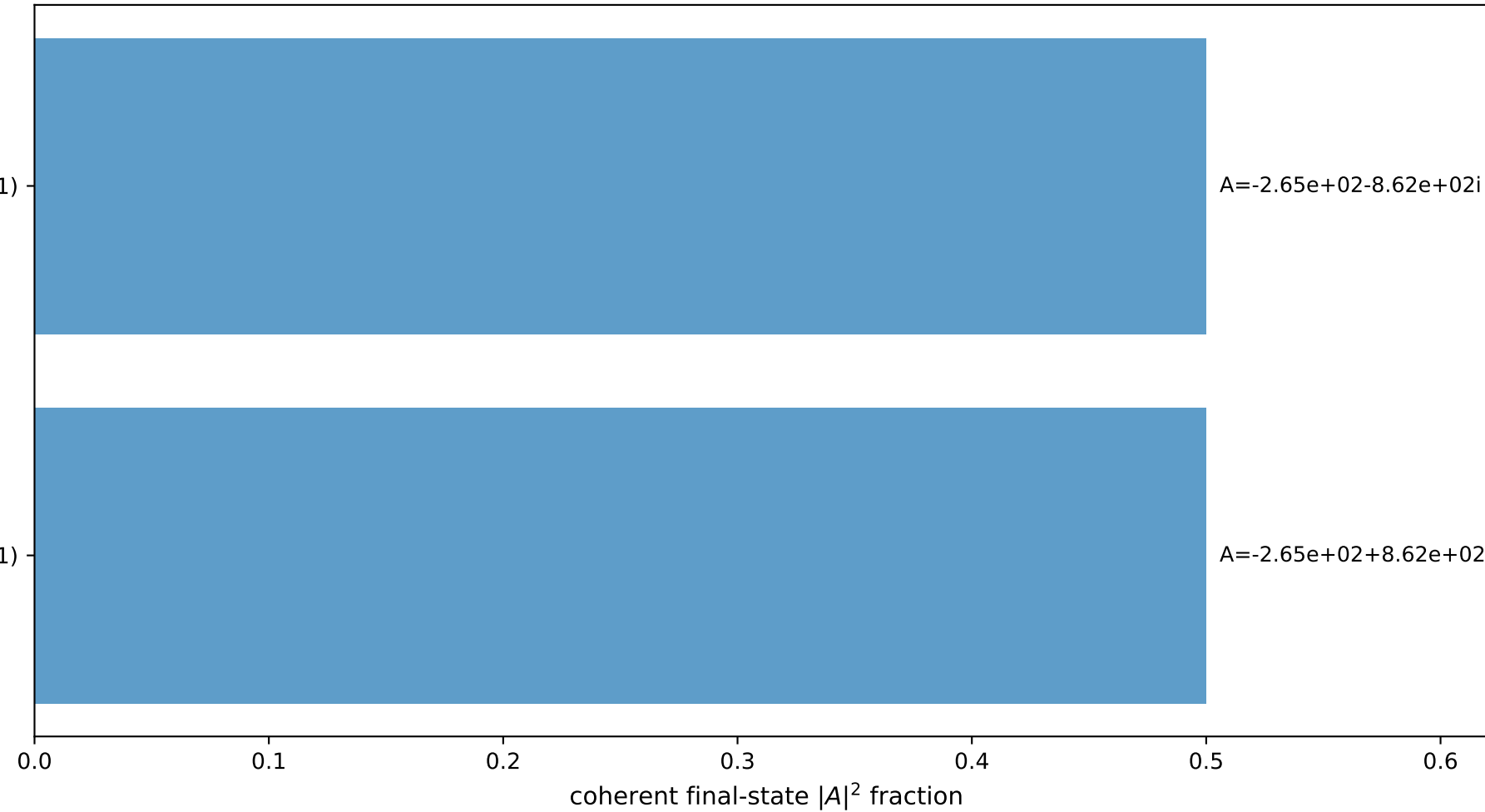
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8143$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8143$
 P $E= 2.0424$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8143$
 ℓ' $E= 1.4591$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4591$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0006$
 q_Y $E= 1.4596$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4596$



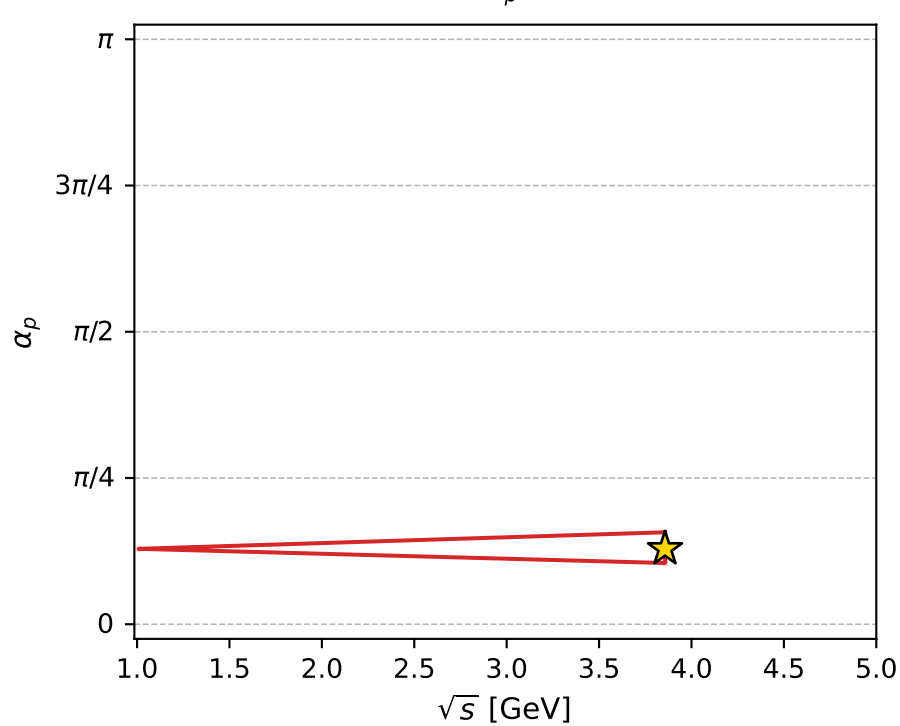
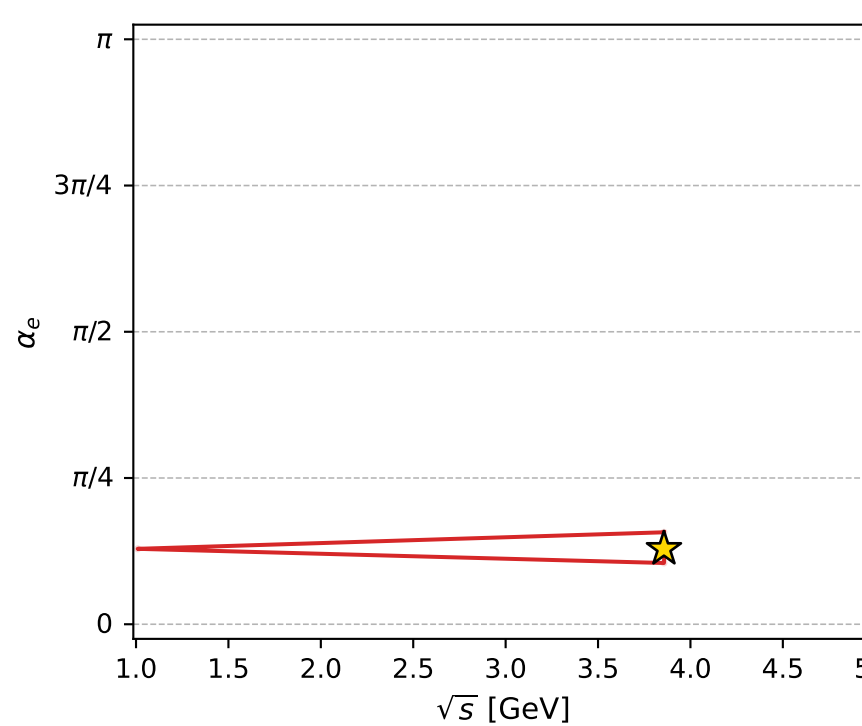
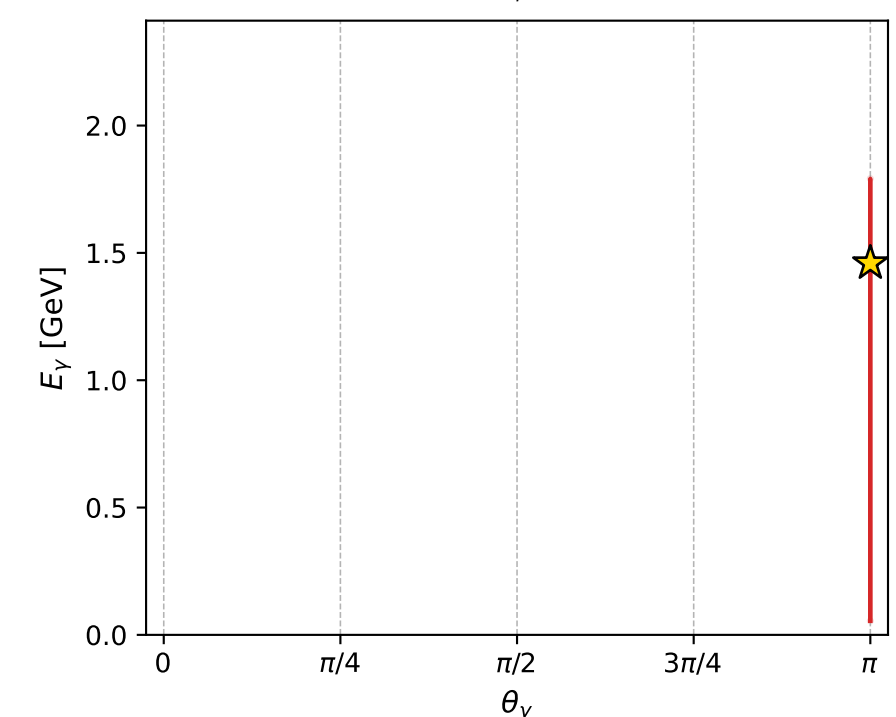
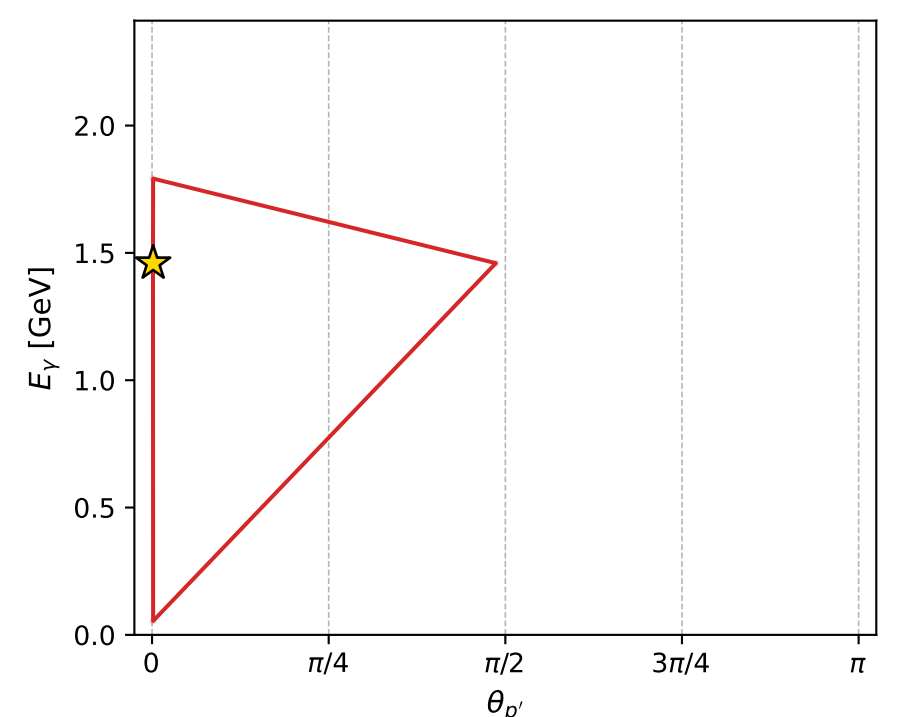
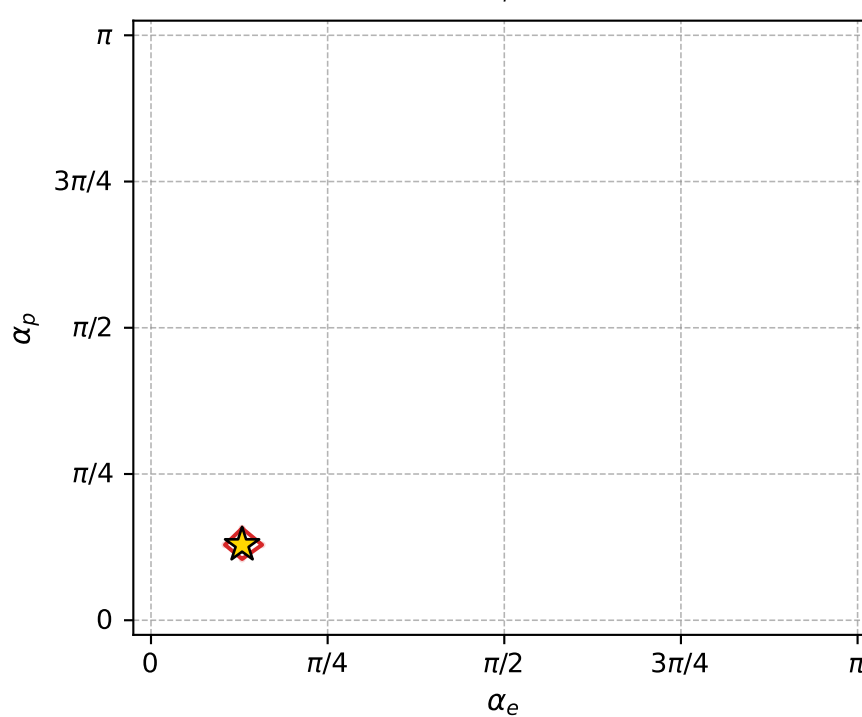
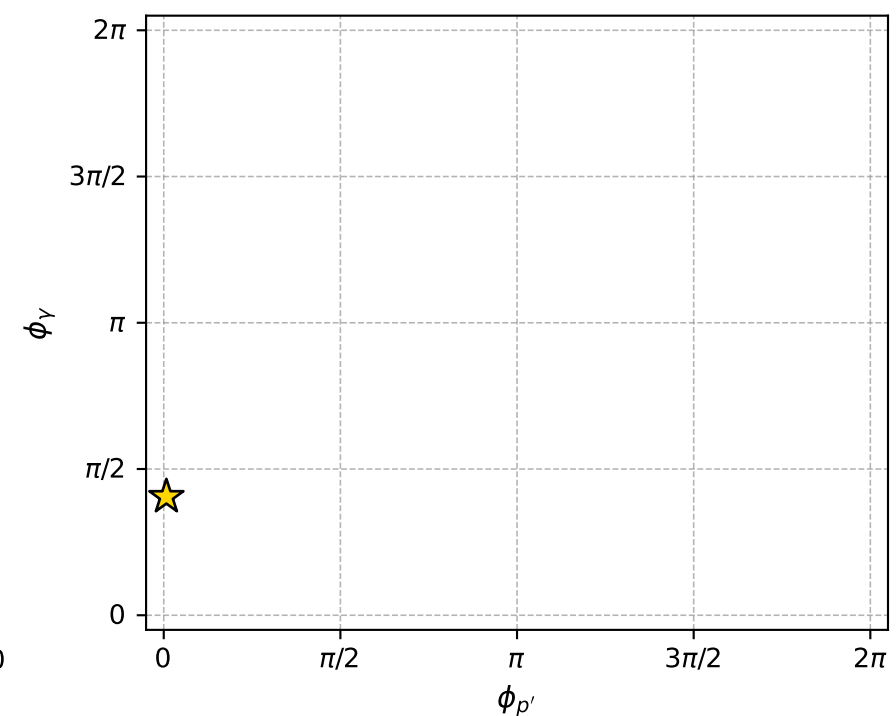
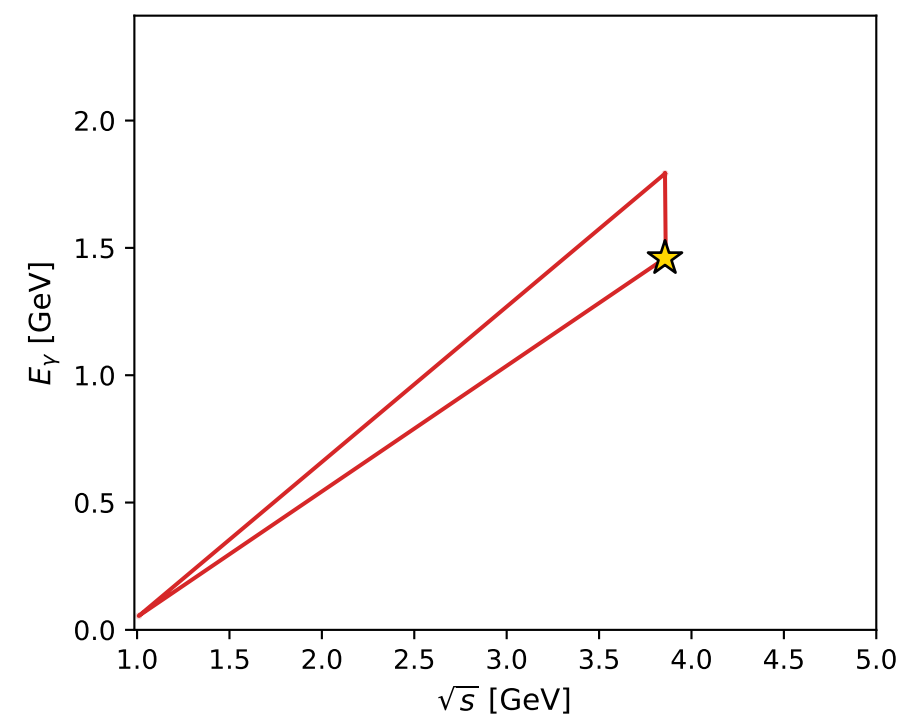
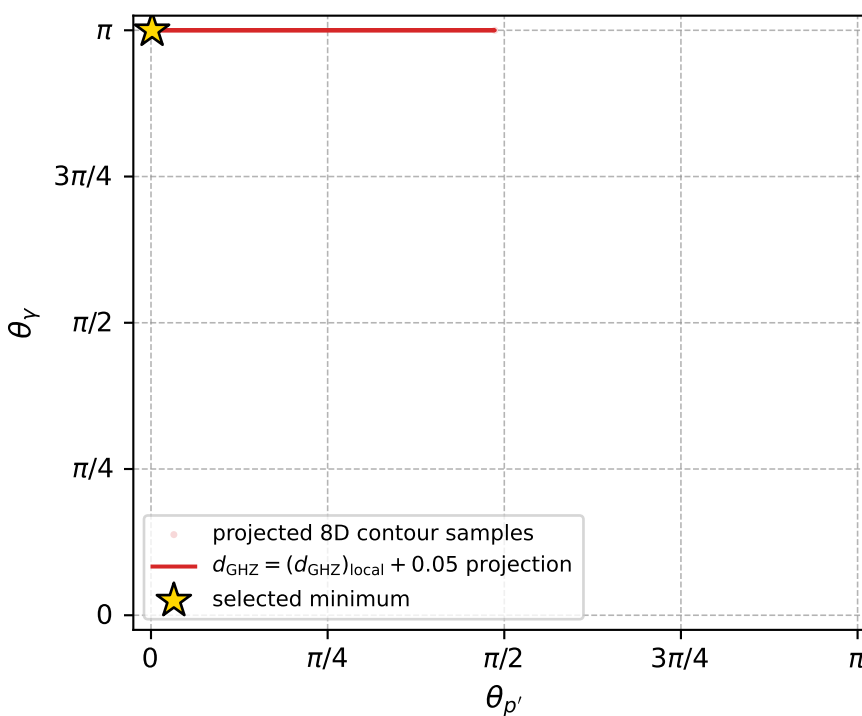
$(h_p, h_Y, h_\ell) = (-1, -1, +1)$

$(h_p, h_Y, h_\ell) = (+1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 139; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 6.7347026\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000067$
 above global minimum = $4.5063564\text{e-}08$
 8D contour samples = 130

selected phase-space point:

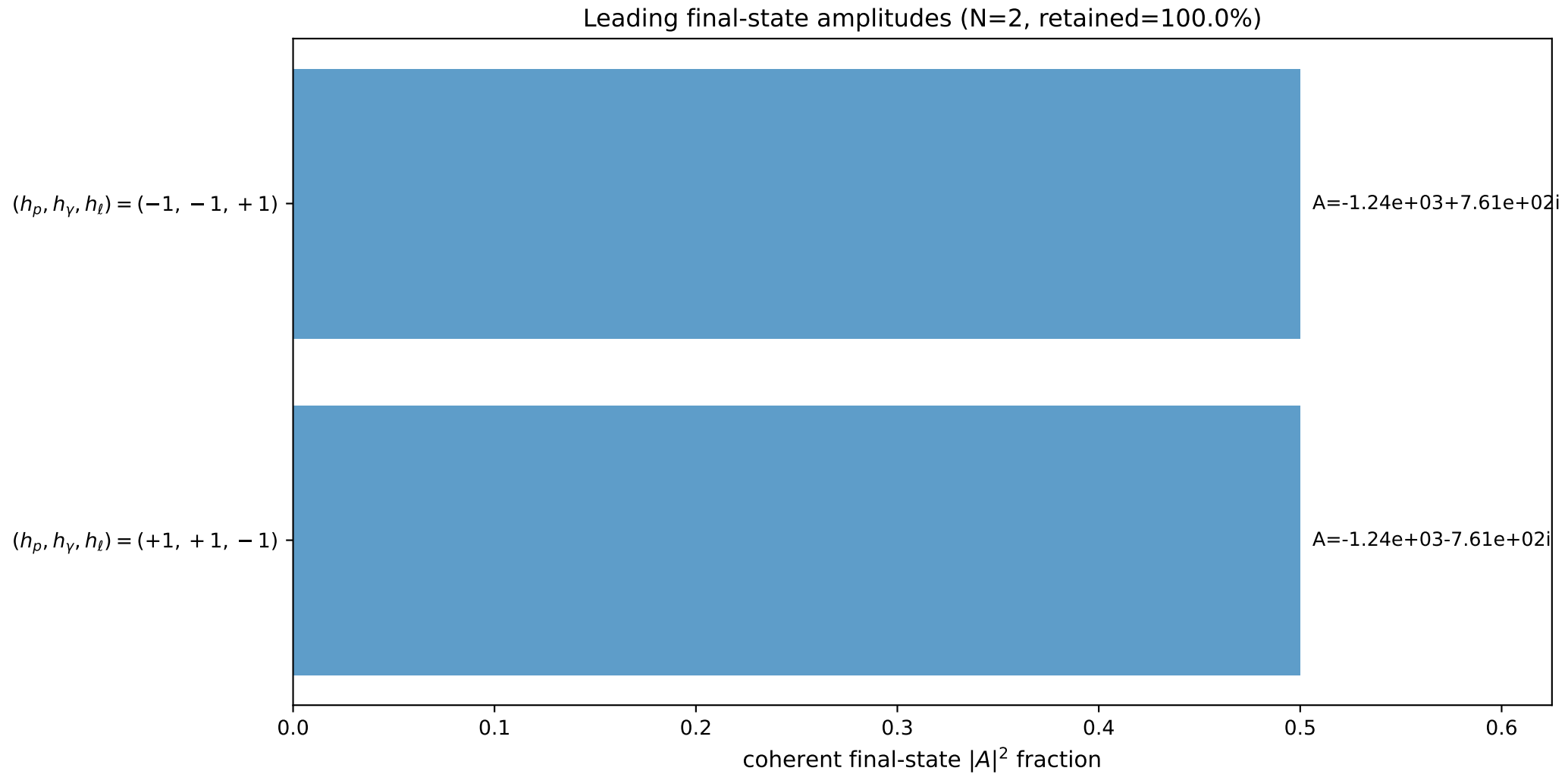
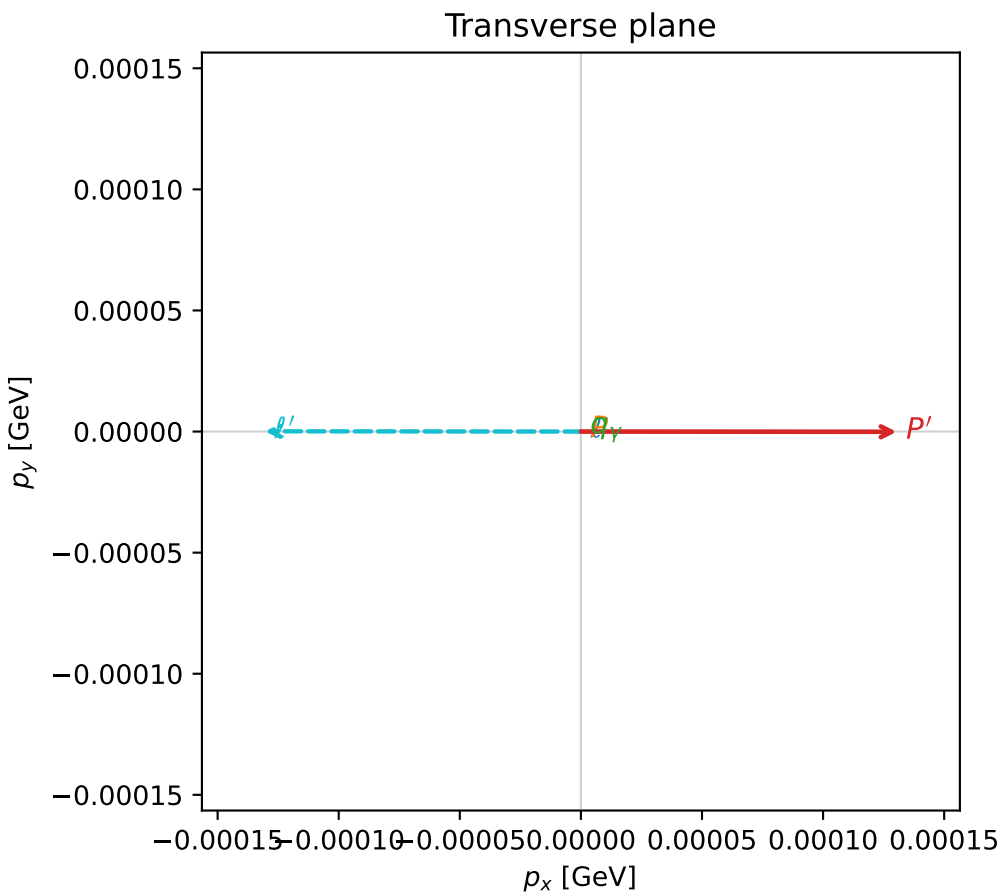
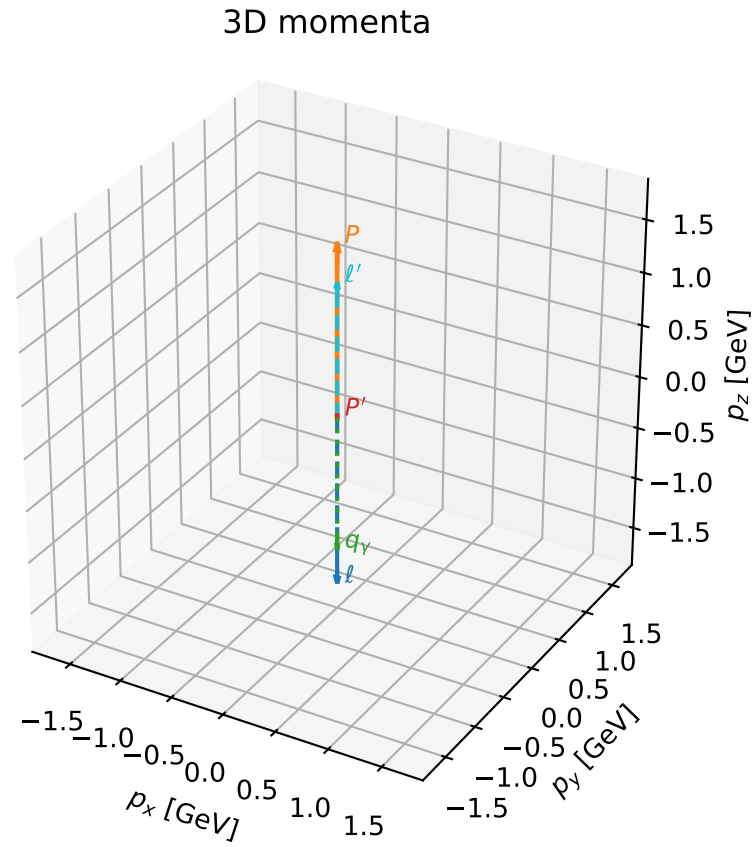
$\text{sqrt_s} = 3.856707$
 $\text{theta_p_out} = 0.004908739$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0ut} = 1.459646$
 $\text{phi_p_out} = 0.02353493$
 $\text{phi_gamma_out} = 1.272947$
 $\alpha_e = 0.4050411$
 $\alpha_p = 0.4050348$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

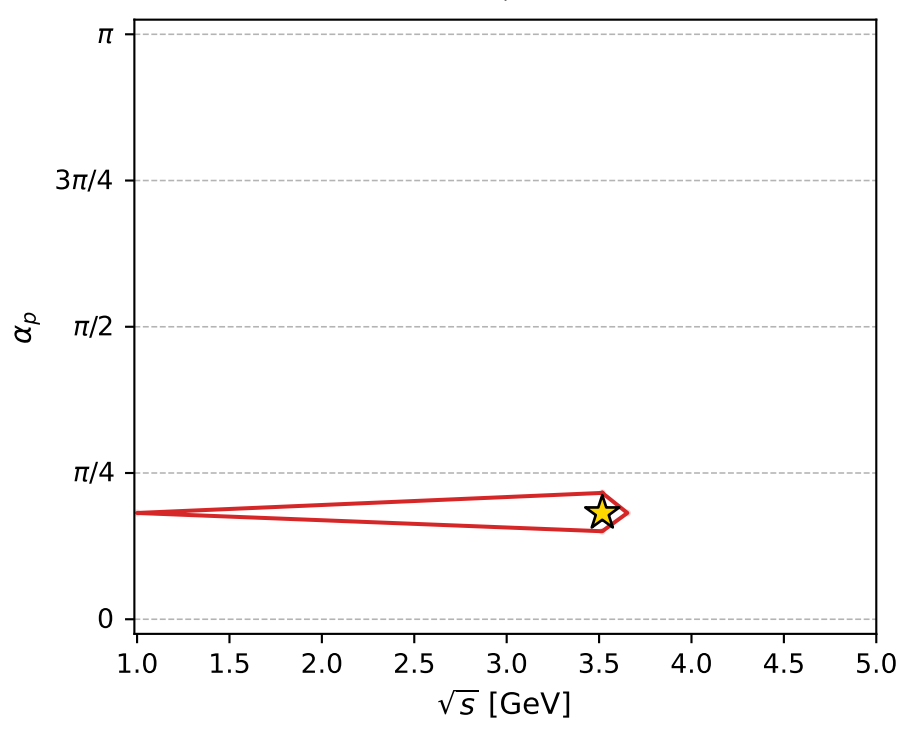
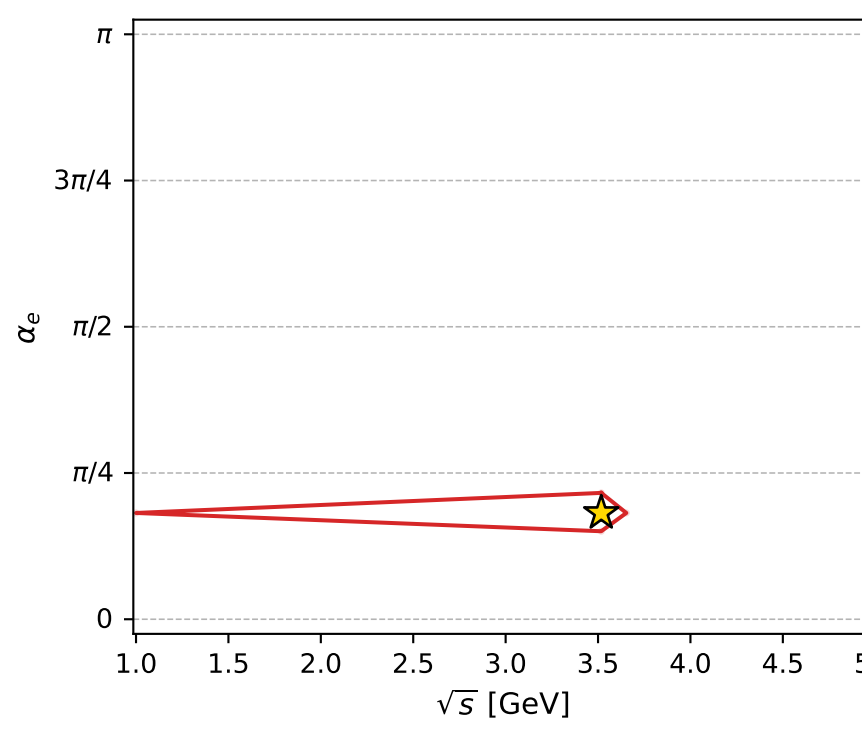
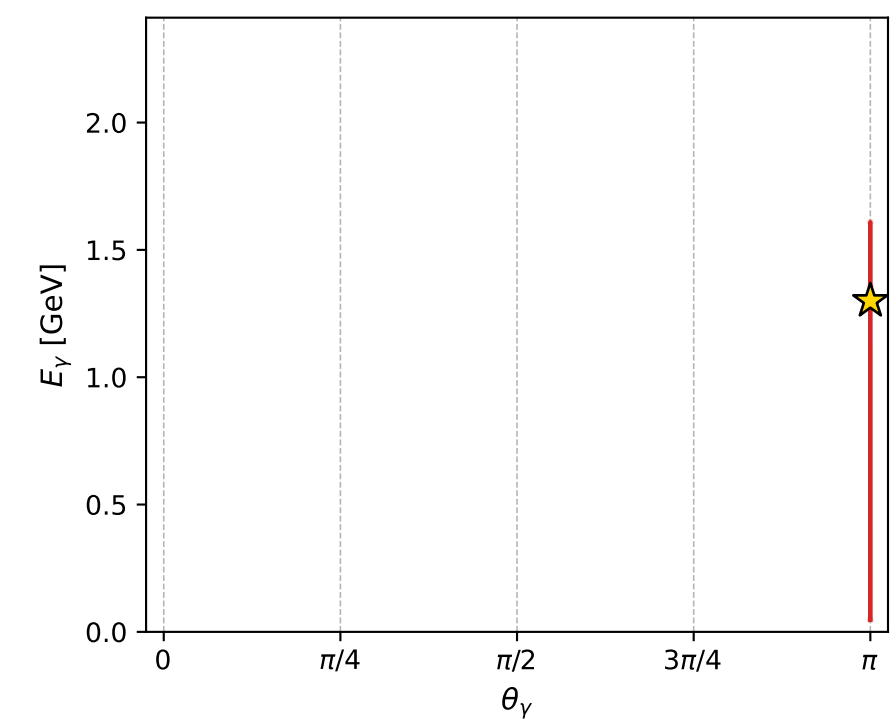
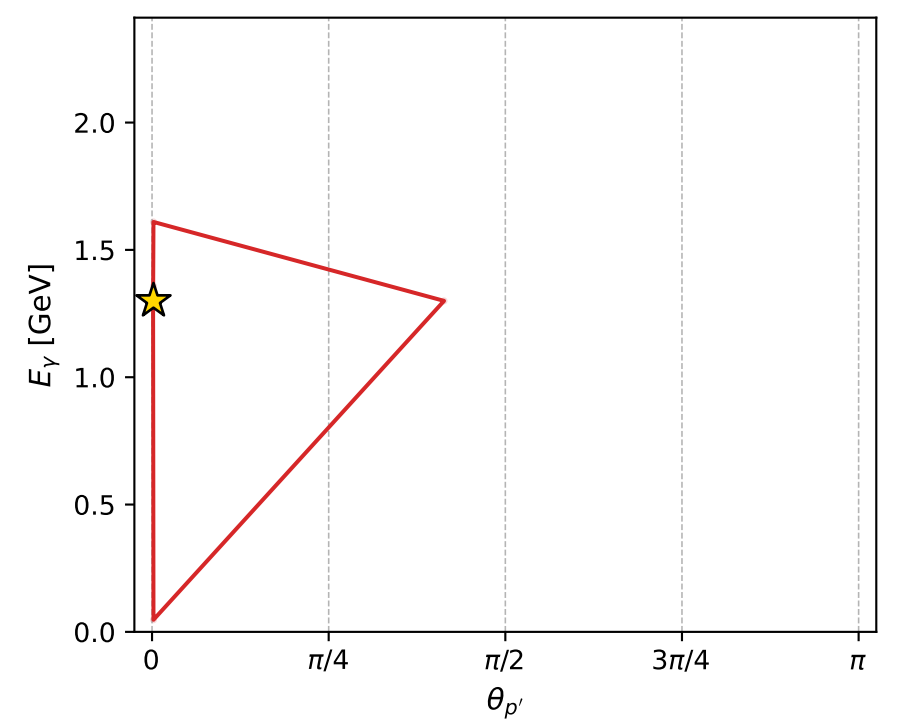
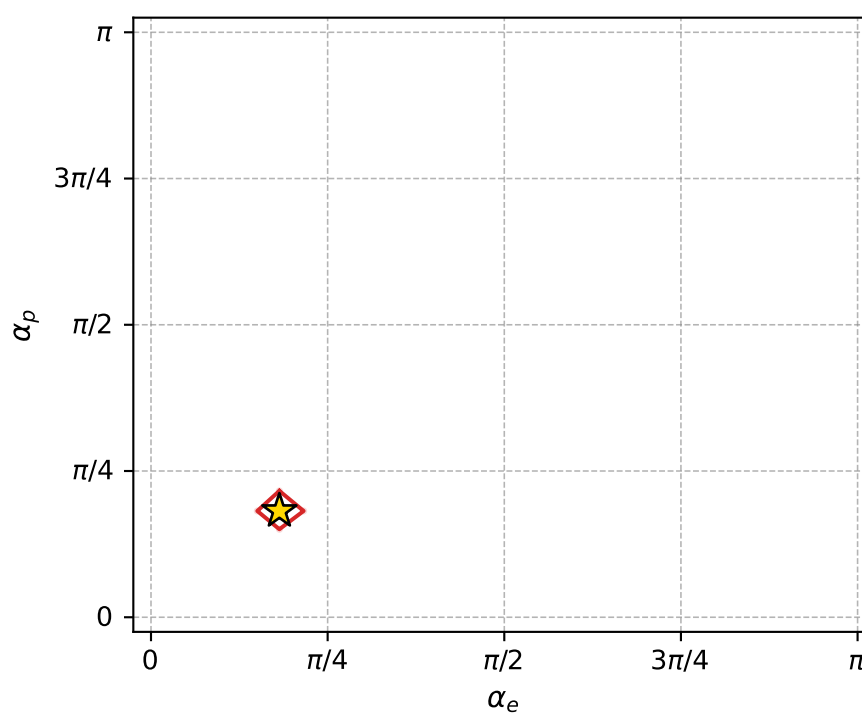
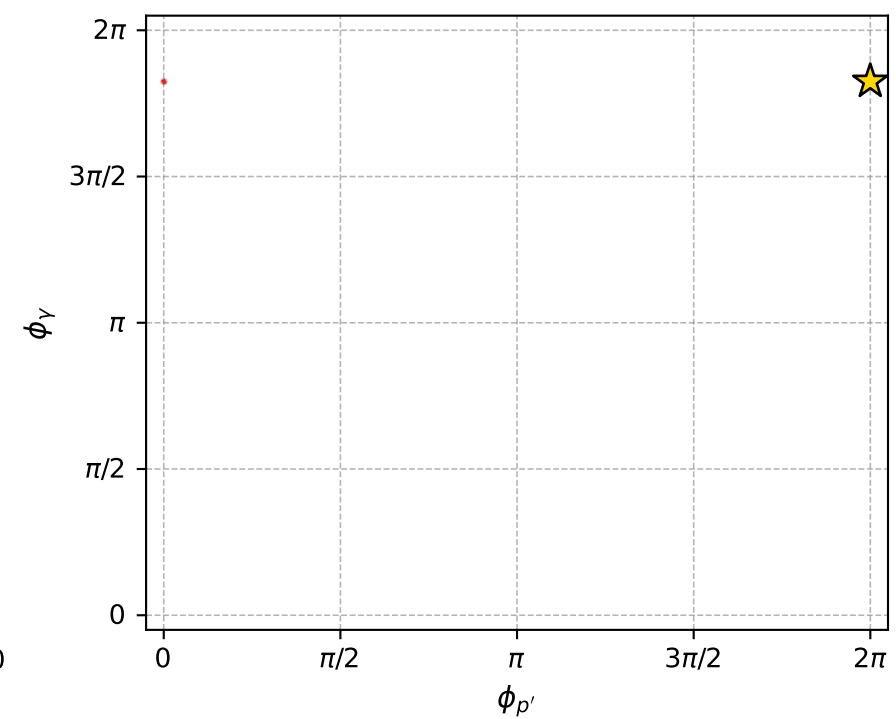
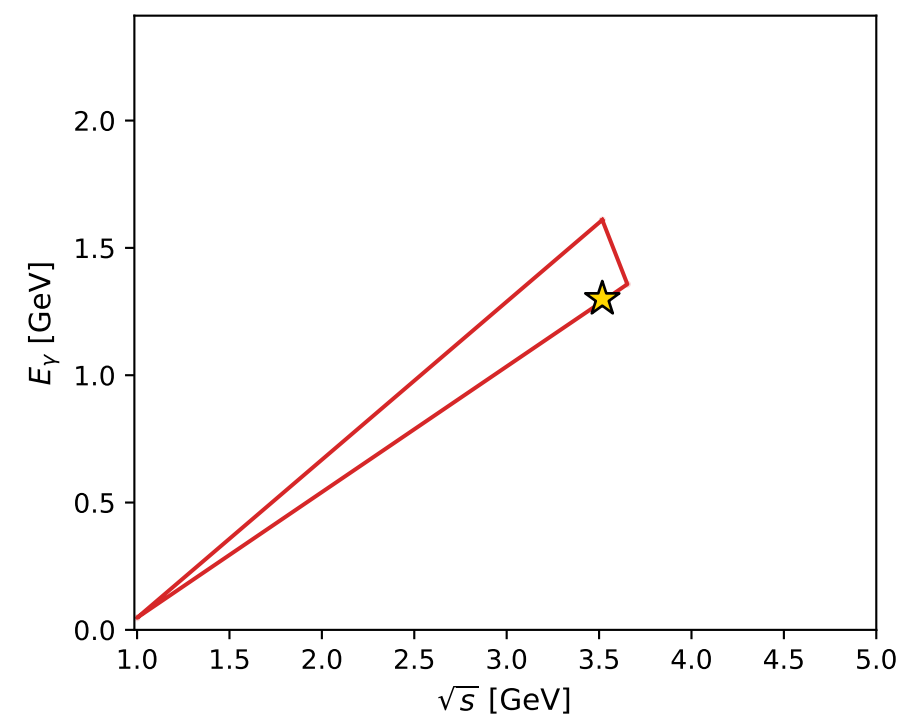
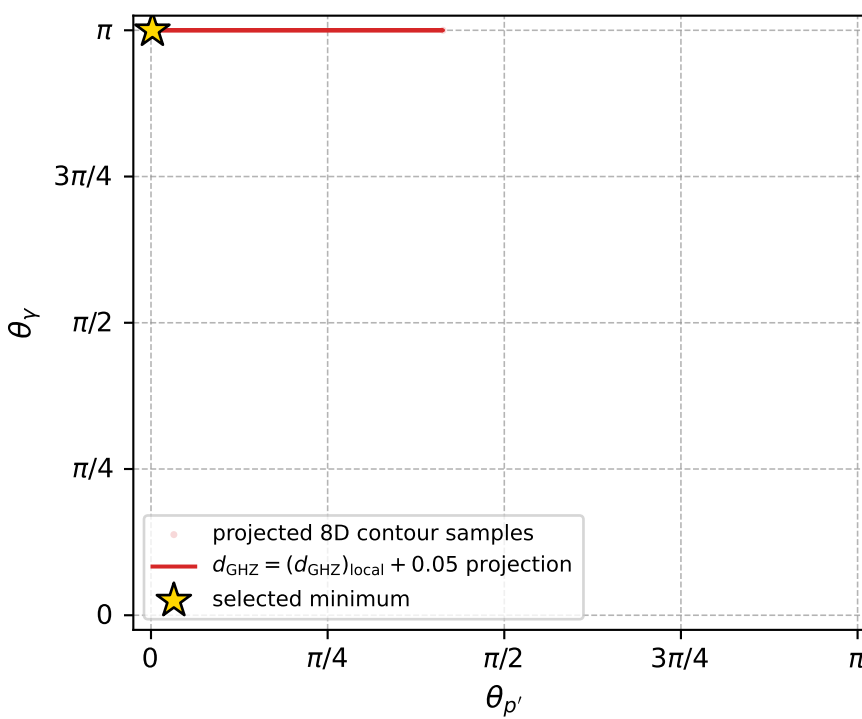
dghz_mixing_angles_polarization_01_local_minimum_140 d_{\rm GHZ}=7.0516e-08
region: polarization_cluster_01_local_minimum_140
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.57080929$ rad
incoming lepton: $0.841464|+\rangle +0.540313|-\rangle$
proton mixing angle: $\alpha_p=0.57079935$ rad
incoming proton: $0.841469|+\rangle +0.540305|-\rangle$

$s=12.3715$, $\sqrt{s}=3.51731$, $|\vec{P}|=1.63358$, $|\vec{P}'|=0.0205126$
 $\theta_{p'}=0.00635735$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.28239$, $\phi_\gamma=5.7342$
 $E_\gamma=1.2998$, $\theta_{\ell'}=0.000101936$, $\phi_{\ell'}=3.1408$
 $Q^2=8.35928$, $x_B=0.770326$, $t=-1.70802$, $W^2=3.37217$, $y=0.944306$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.6336$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6336$
 P $E= 1.8837$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6336$
 ℓ' $E= 1.2793$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 1.2793$
 P' $E= 0.9382$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0205$
 q_γ $E= 1.2998$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.2998$



electron: polarization cluster P1, local minimum 140; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 7.0516005\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000071$
 above global minimum = $4.8232543\text{e-}08$
 8D contour samples = 122

selected phase-space point:

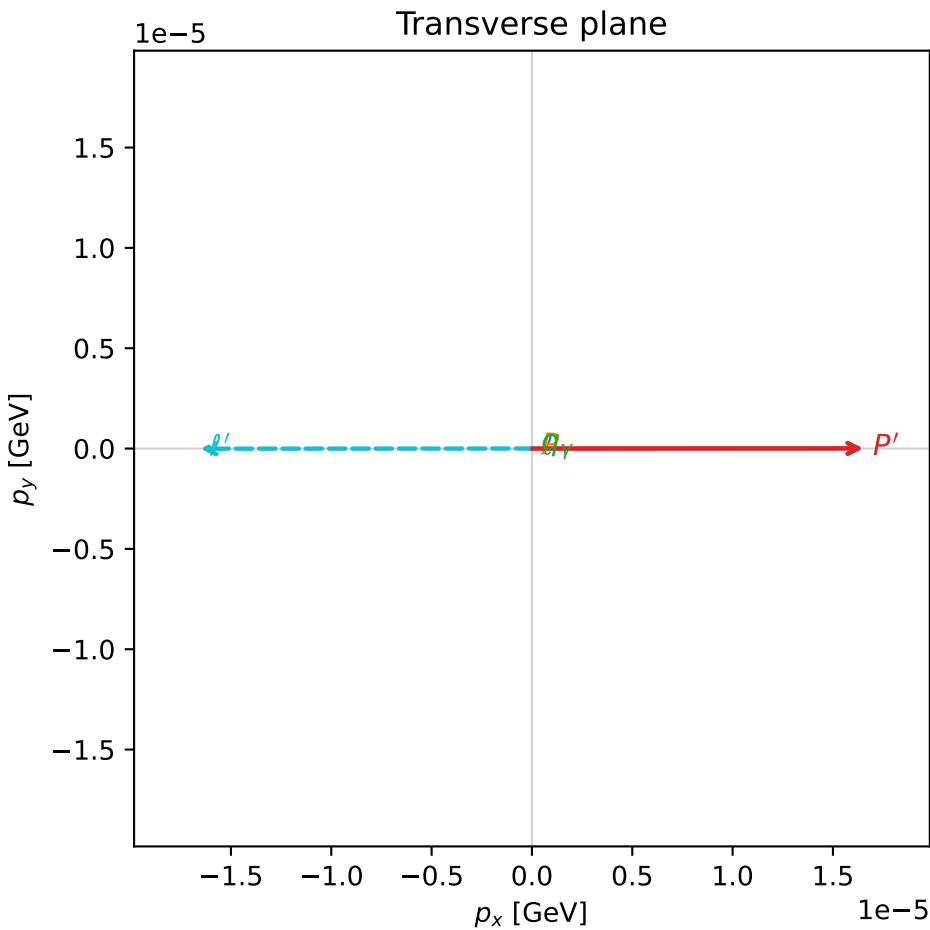
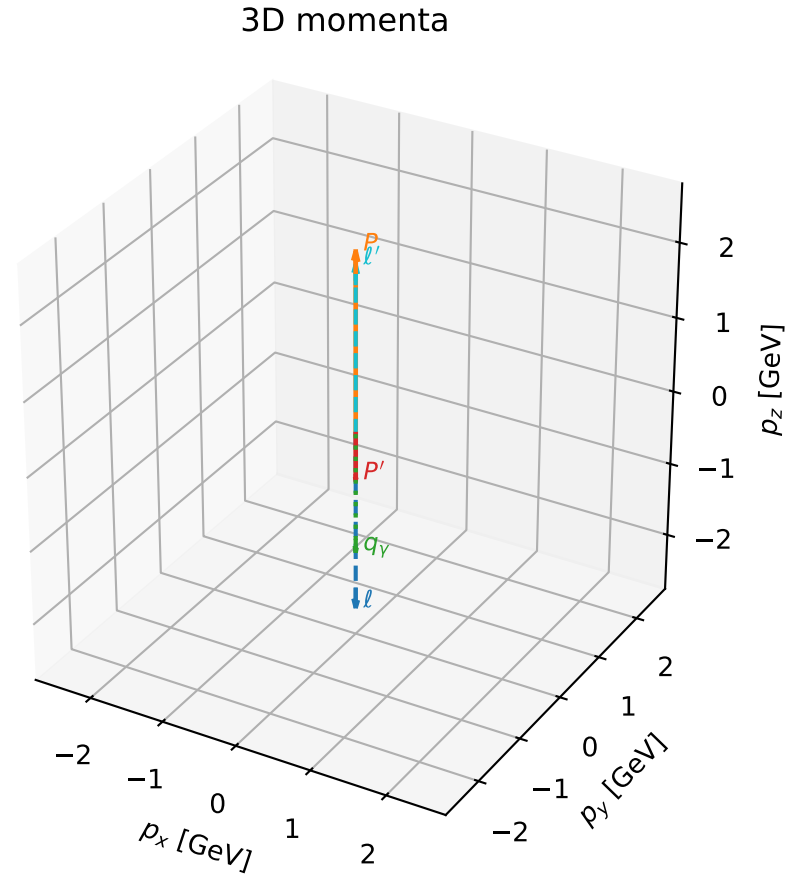
$\text{sqrt_s} = 3.517311$
 $\text{theta_p_out} = 0.006357351$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.299799$
 $\text{phi_p_out} = 6.282388$
 $\text{phi_gamma_out} = 5.734202$
 $\alpha_e = 0.5708093$
 $\alpha_p = 0.5707994$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_146 $d_{\{\text{rm GHZ}\}}=7.9425\text{e-}08$
 region: polarization_cluster_01_local_minimum_146
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.15440538$ rad
 incoming lepton: $0.988103|+\rangle +0.153793|-\rangle$
 proton mixing angle: $\alpha_p=0.15439576$ rad
 incoming proton: $0.988105|+\rangle +0.153783|-\rangle$

$s=24.9939$, $\sqrt{s}=4.99939$, $|\vec{P}|=2.4117$, $|\vec{P}'|=0.615551$
 $\theta_{P'}=3.14157$, $\theta_Y=3.14159$, $\phi_{P'}=0.000416712$, $\phi_Y=2.69483$
 $E_Y=1.63095$, $\theta_{\ell'}=7.35473\text{e-}06$, $\phi_{\ell'}=3.14201$
 $Q^2=21.6715$, $x_B=0.929179$, $t=-7.01582$, $W^2=2.53163$, $y=0.967209$

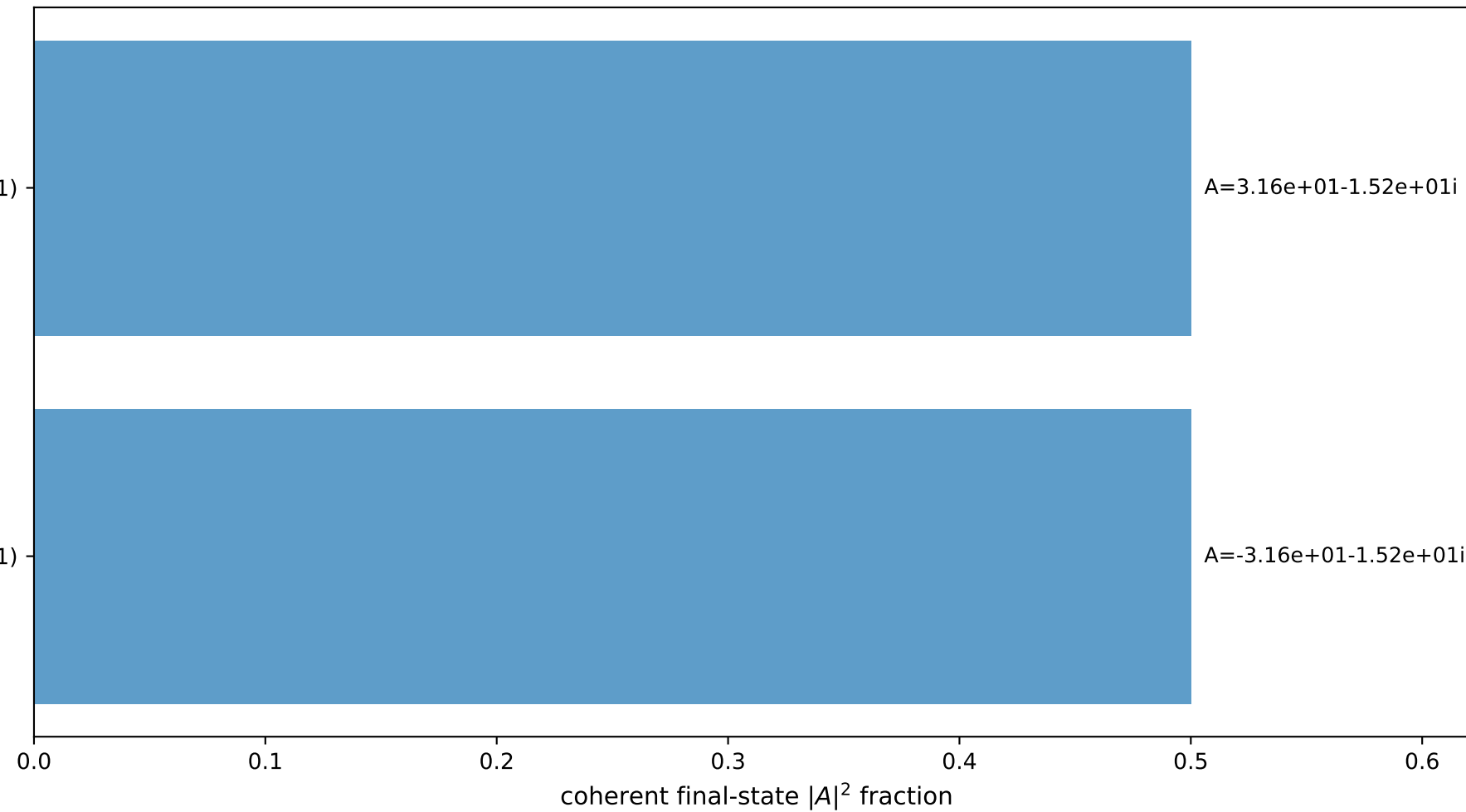
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4117$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4117$
 P $E= 2.5877$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4117$
 ℓ' $E= 2.2465$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2465$
 P' $E= 1.1219$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6156$
 q_Y $E= 1.6309$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.6309$



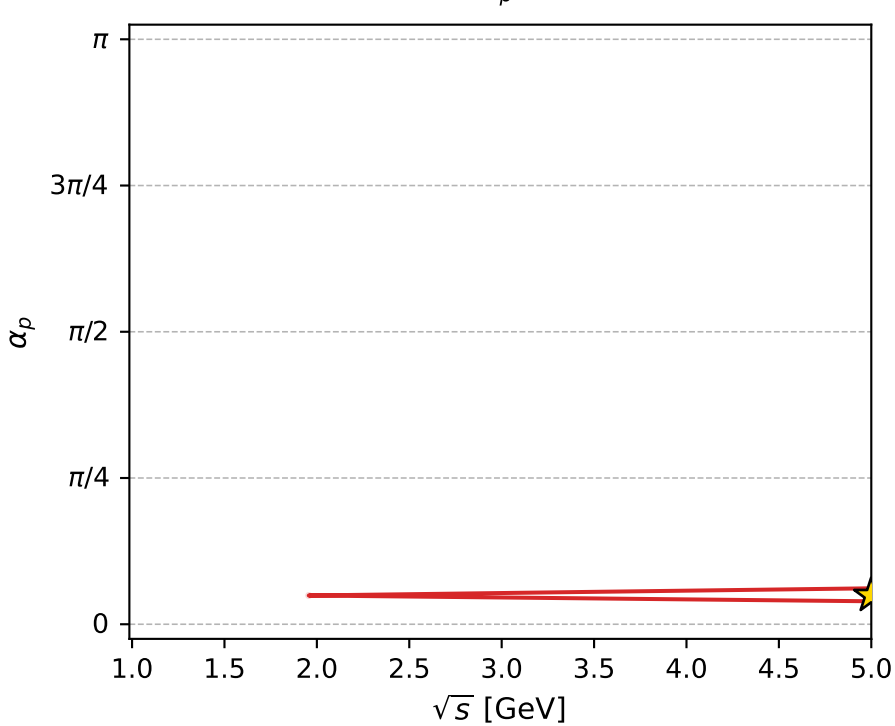
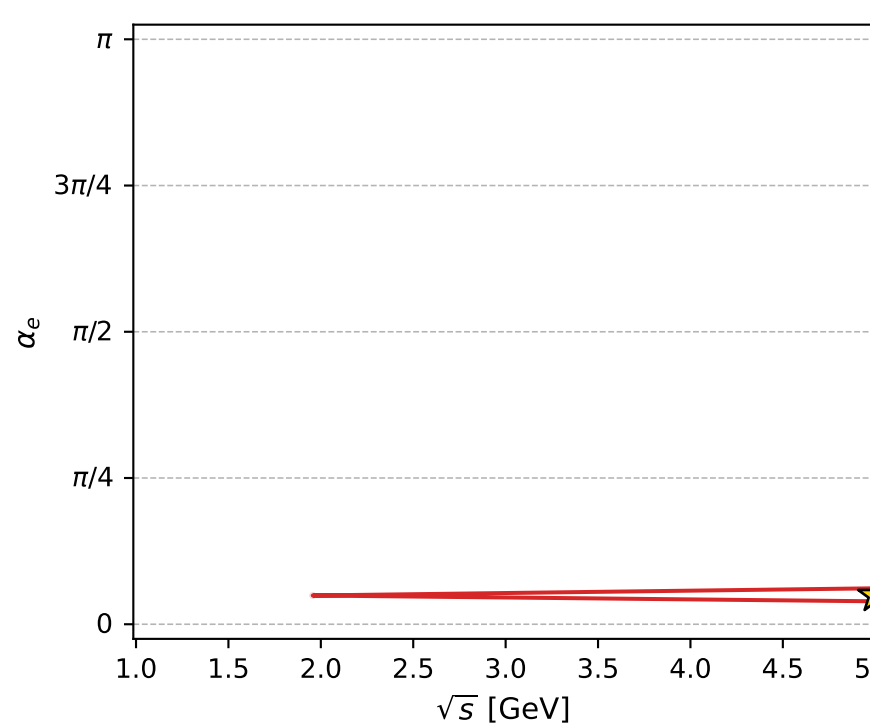
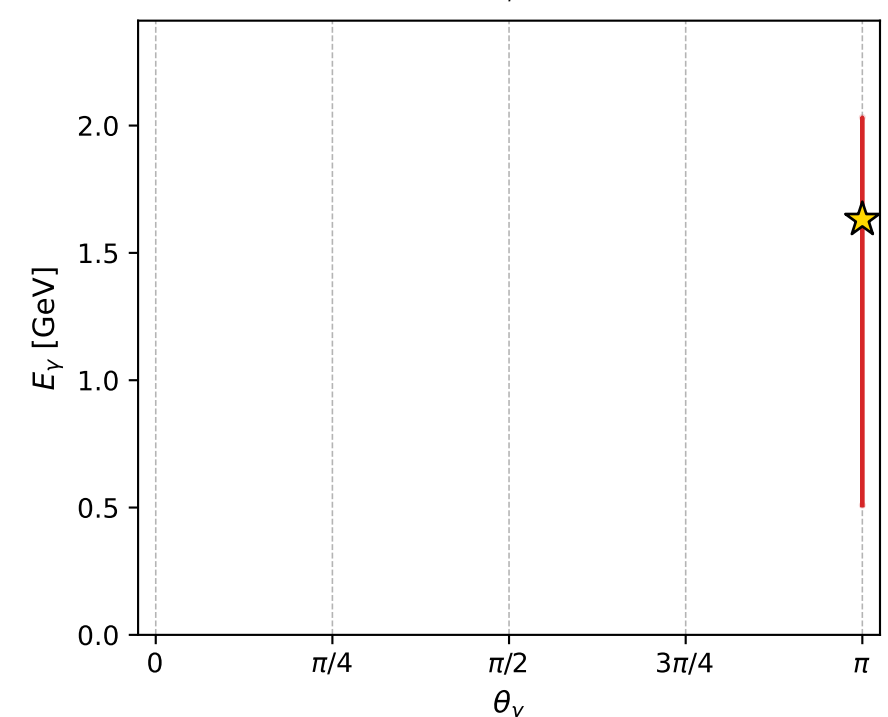
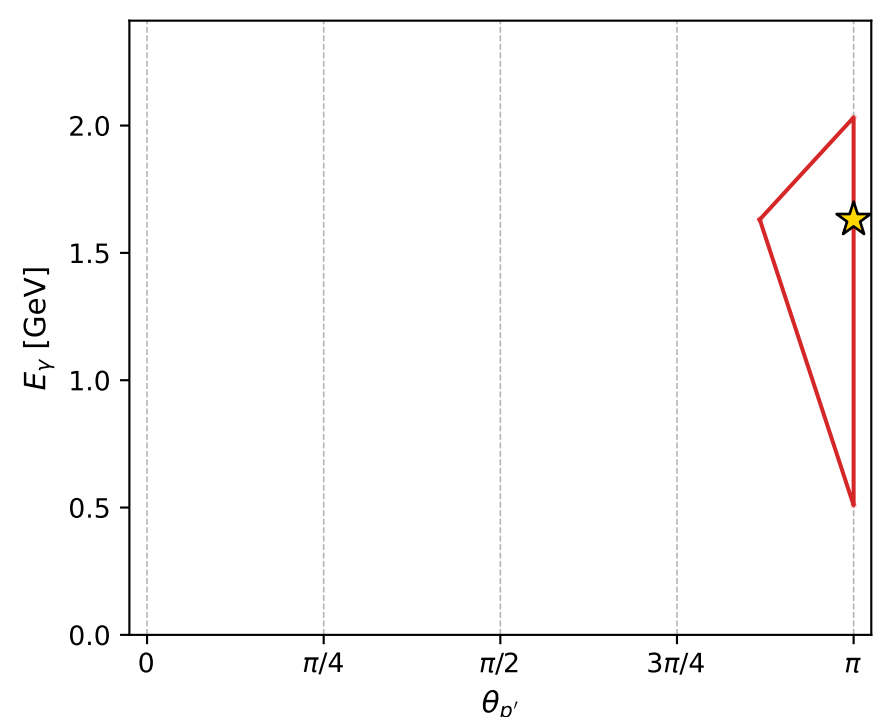
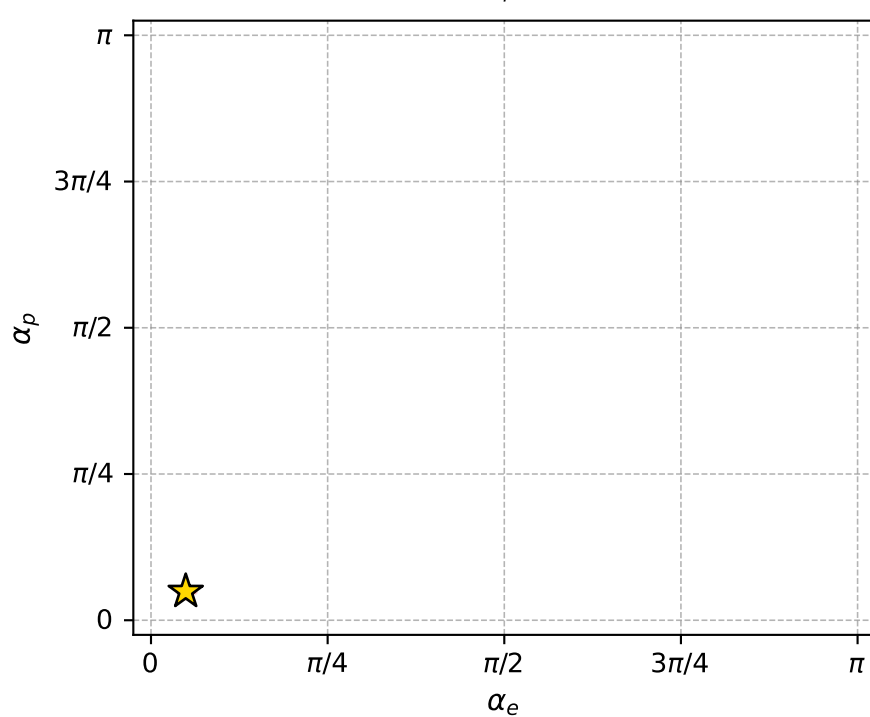
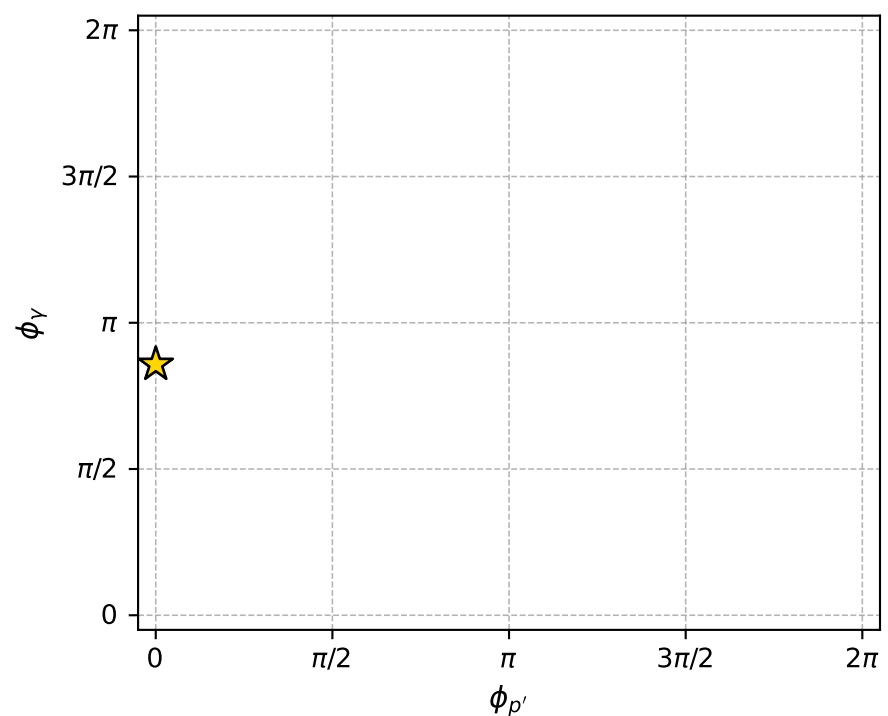
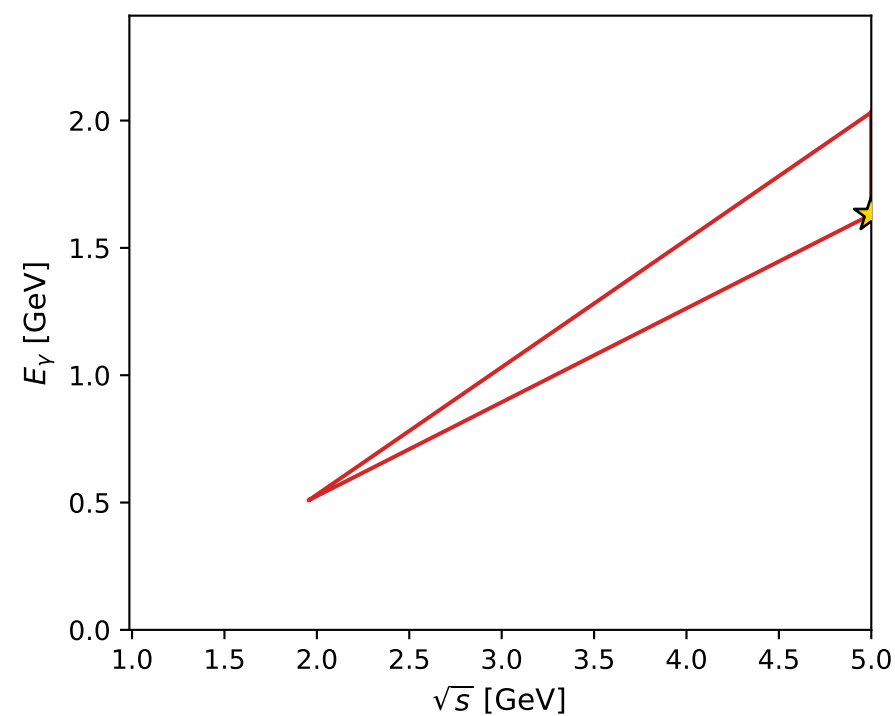
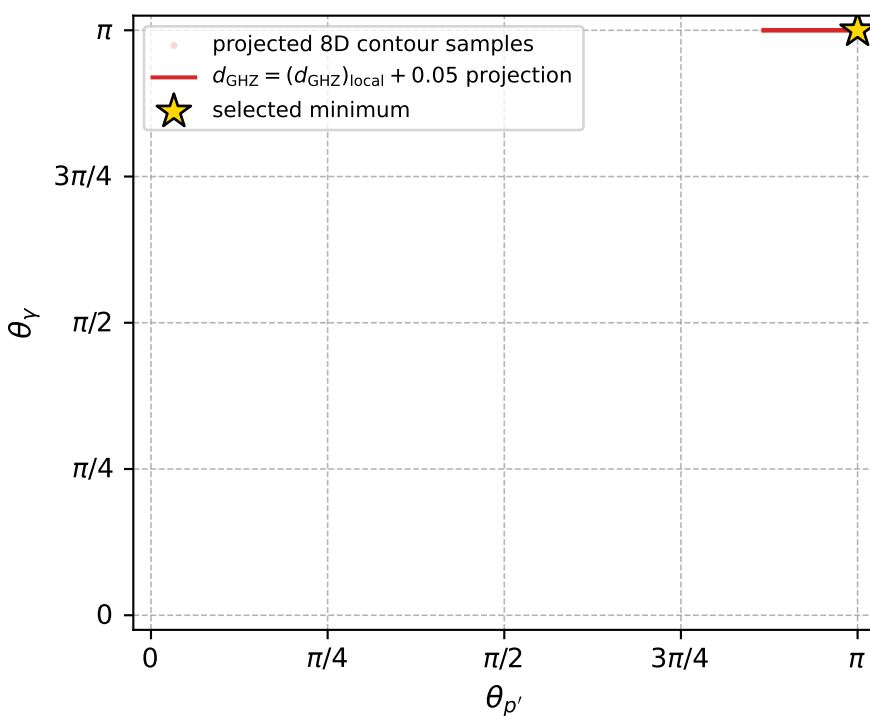
$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 146; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 7.9424972\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000079$
 above global minimum = $5.714151\text{e-}08$
 8D contour samples = 129

selected phase-space point:
 $\text{sqrt_s} = 4.99939$
 $\text{theta_p_out} = 3.141566$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.63095$
 $\text{phi_p_out} = 0.0004167123$
 $\text{phi_gamma_out} = 2.694826$
 $\alpha_e = 0.1544054$
 $\alpha_p = 0.1543958$

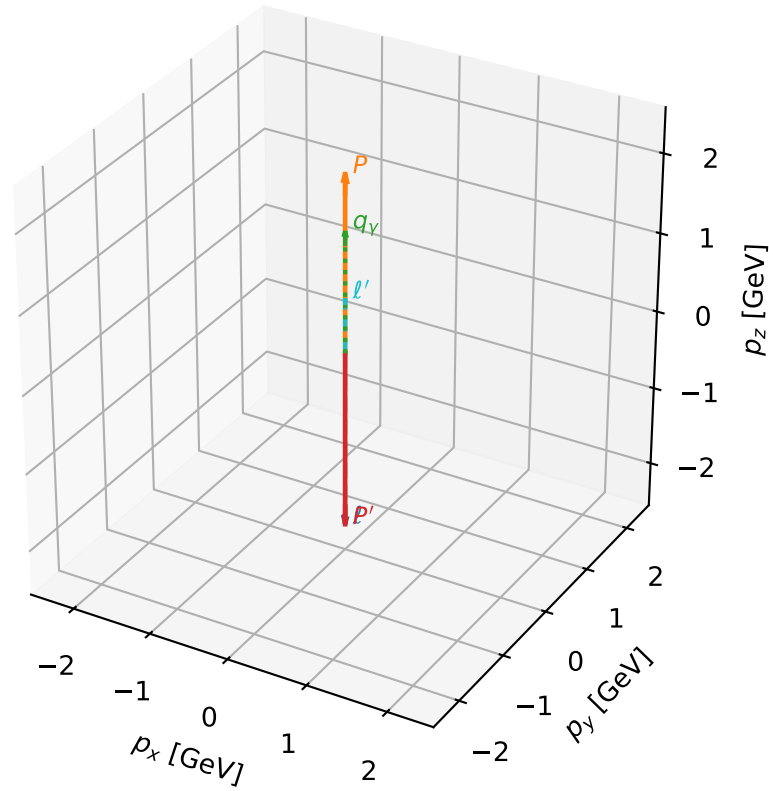
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_149 $d_{\{\text{rm GHZ}\}}=8.40053\text{e-}08$
 region: polarization_cluster_01_local_minimum_149
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.79062427$ rad
 incoming lepton: $0.703402|+\rangle +0.710793|-\rangle$
 proton mixing angle: $\alpha_p=0.78016465$ rad
 incoming proton: $0.710798|+\rangle +0.703396|-\rangle$

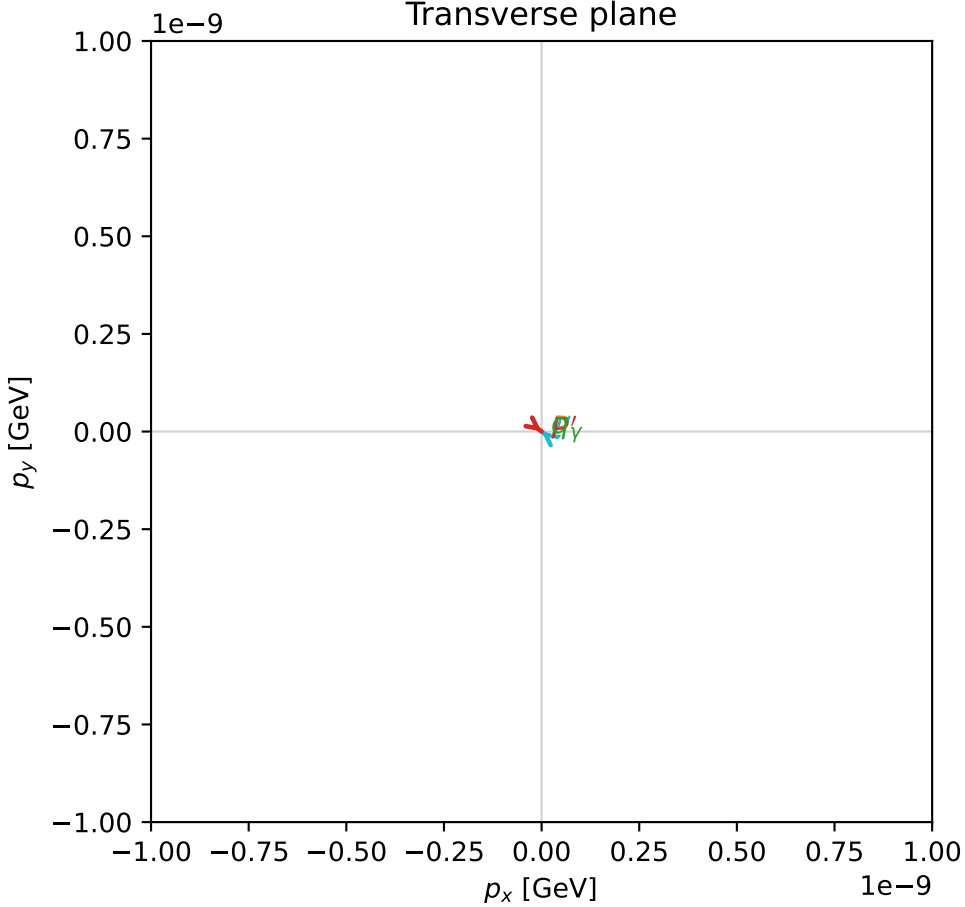
$s=21.6733$, $\sqrt{s}=4.65546$, $|\vec{P}|=2.23324$, $|\vec{P}'|=2.23324$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=5.62947$, $\phi_Y=0.0541682$
 $E_Y=1.53726$, $\theta_{l'}=0$, $\phi_{l'}=2.48787$
 $Q^2=6.2171$, $x_B=0.302824$, $t=-19.9494$, $W^2=15.1932$, $y=0.987349$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2332$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2332$
 P $E= 2.4222$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2332$
 l' $E= 0.6960$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6960$
 P' $E= 2.4222$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.2332$
 q_Y $E= 1.5373$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5373$

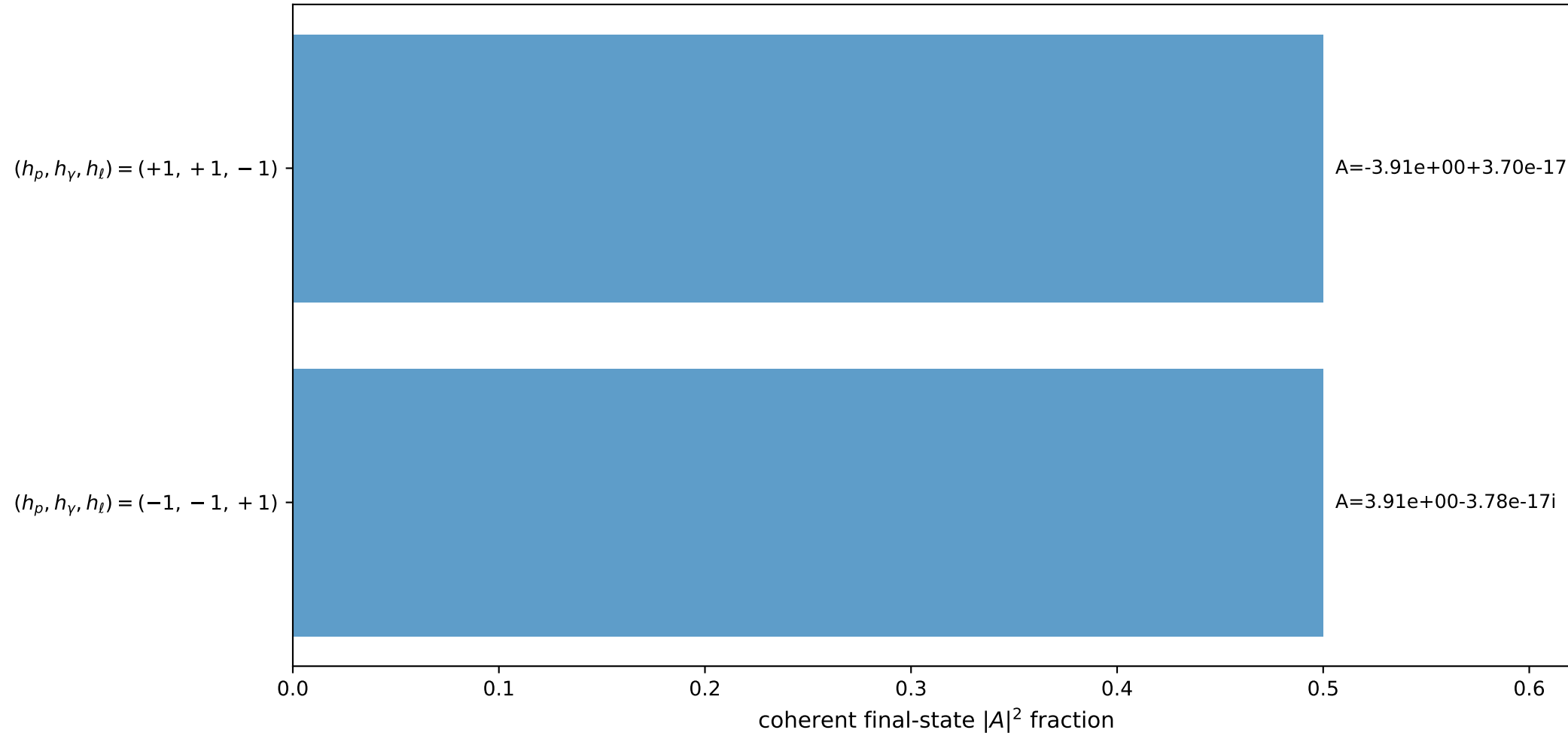
3D momenta



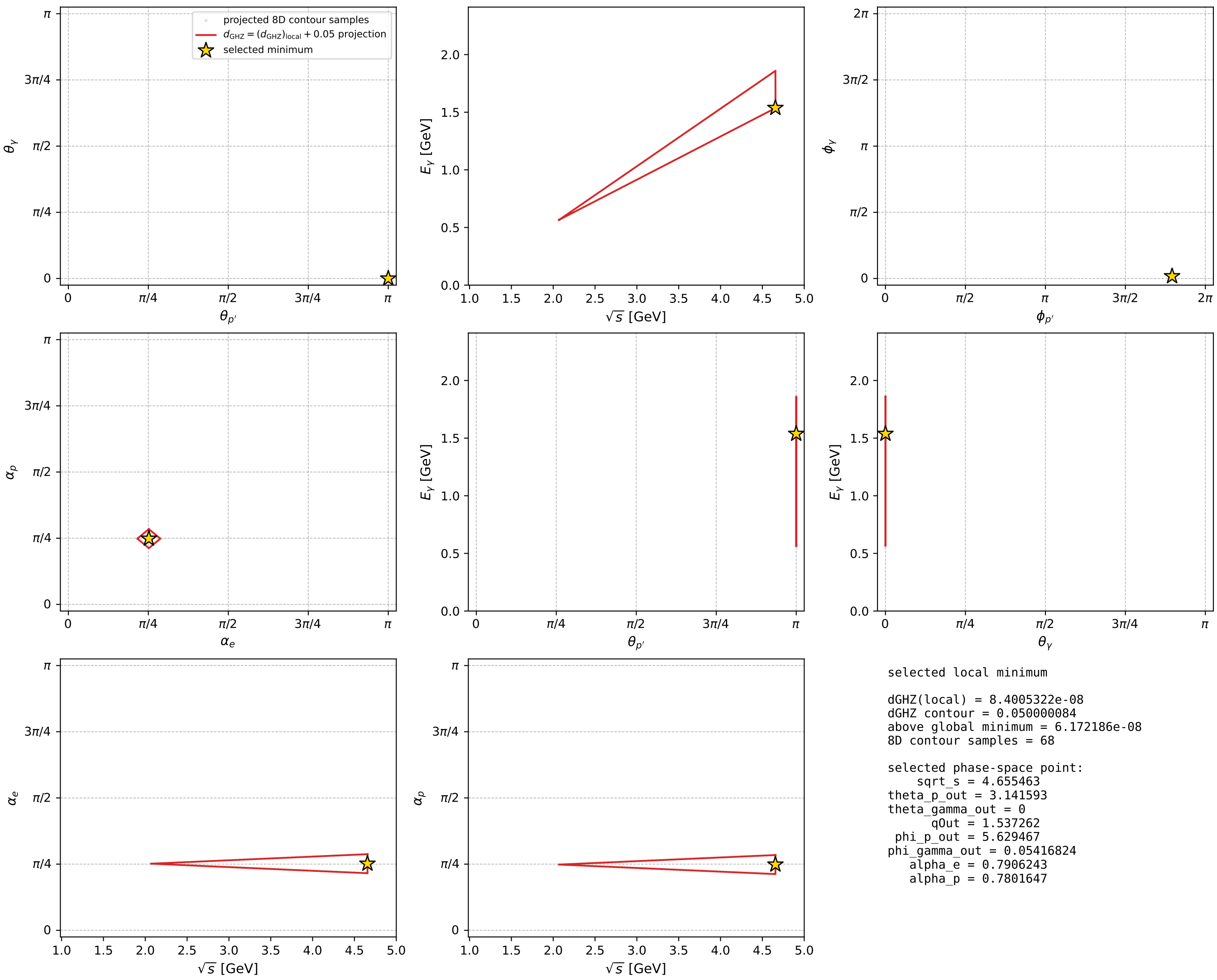
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 149; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

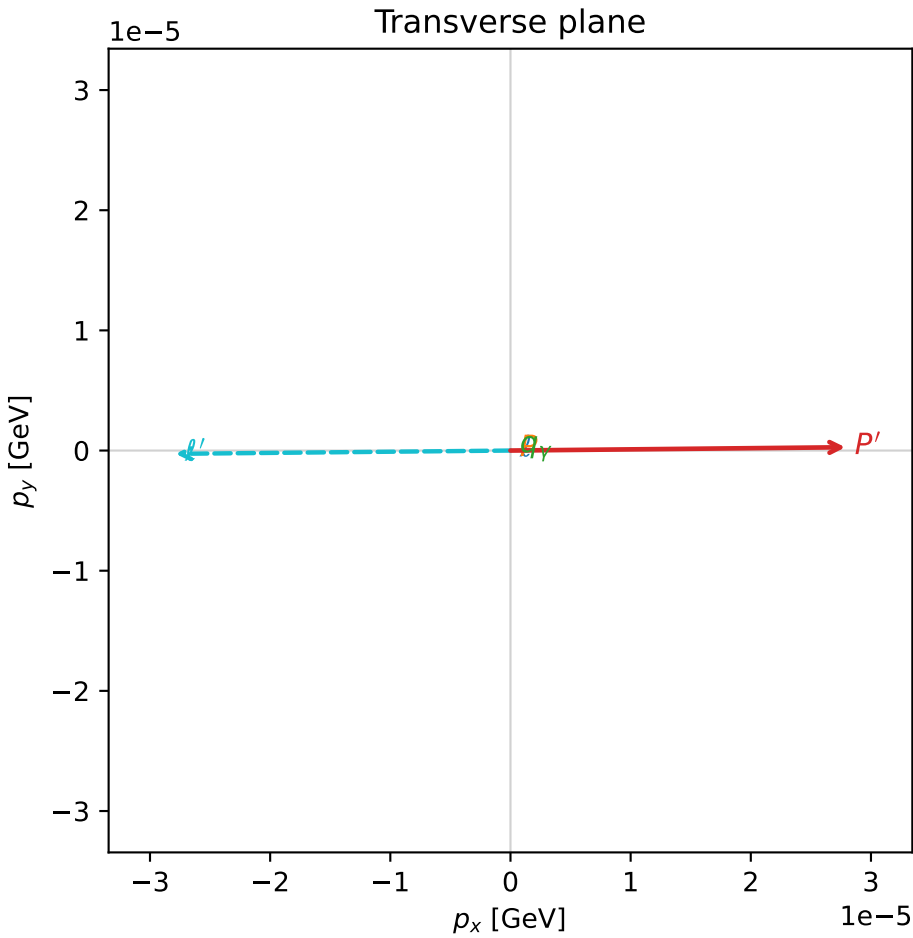
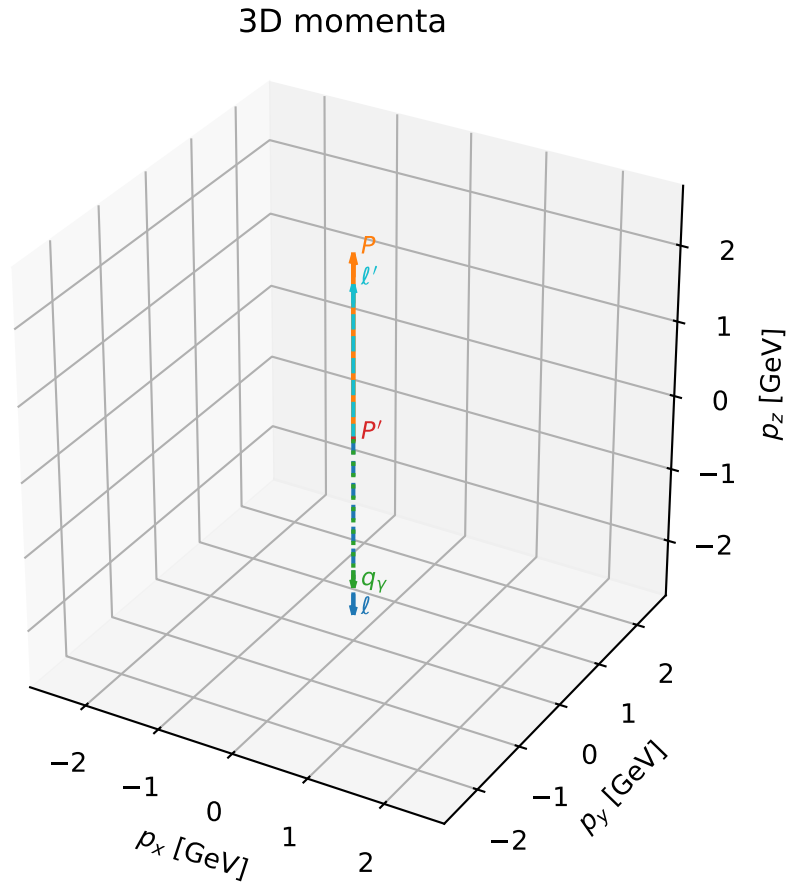


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_151 $d_{\{\text{rm GHZ}\}}=8.54757\text{e-}08$
region: polarization_cluster_01_local_minimum_151
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.15707963$ rad
incoming lepton: $0.987688|+\rangle +0.156434|-\rangle$
proton mixing angle: $\alpha_p=0.15707963$ rad
incoming proton: $0.987688|+\rangle +0.156434|-\rangle$

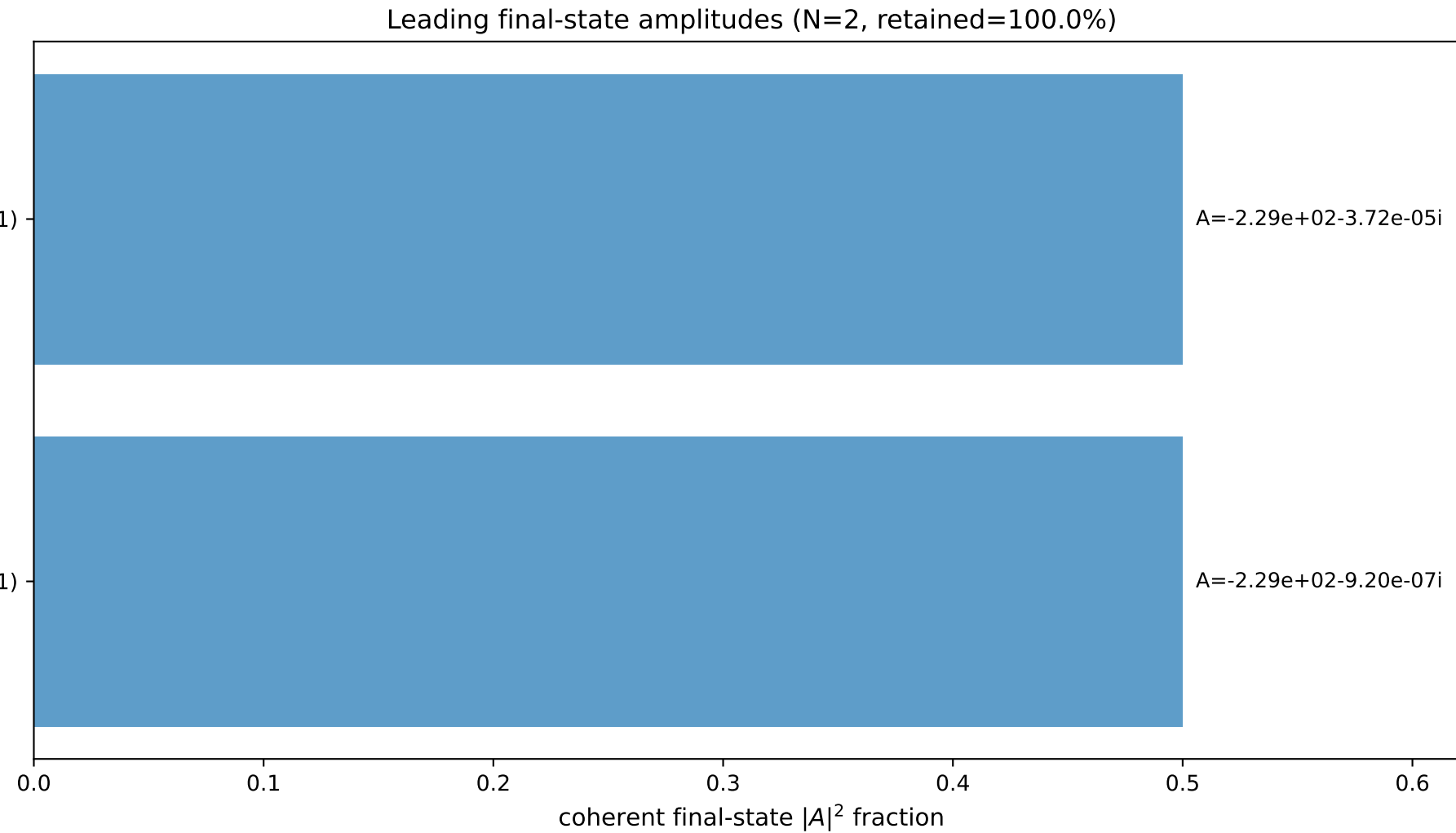
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00283891$
 $\theta_{P'}=0.00981748$, $\theta_Y=3.14159$, $\phi_{P'}=0.00981748$, $\phi_Y=0$
 $E_Y=2.03242$, $\theta_{\ell'}=1.37321\text{e-}05$, $\phi_{\ell'}=3.15141$
 $Q^2=19.5815$, $x_B=0.836606$, $t=-3.0817$, $W^2=4.70422$, $y=0.970386$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0296$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0296$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0028$
 q_Y $E= 2.0324$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0324$

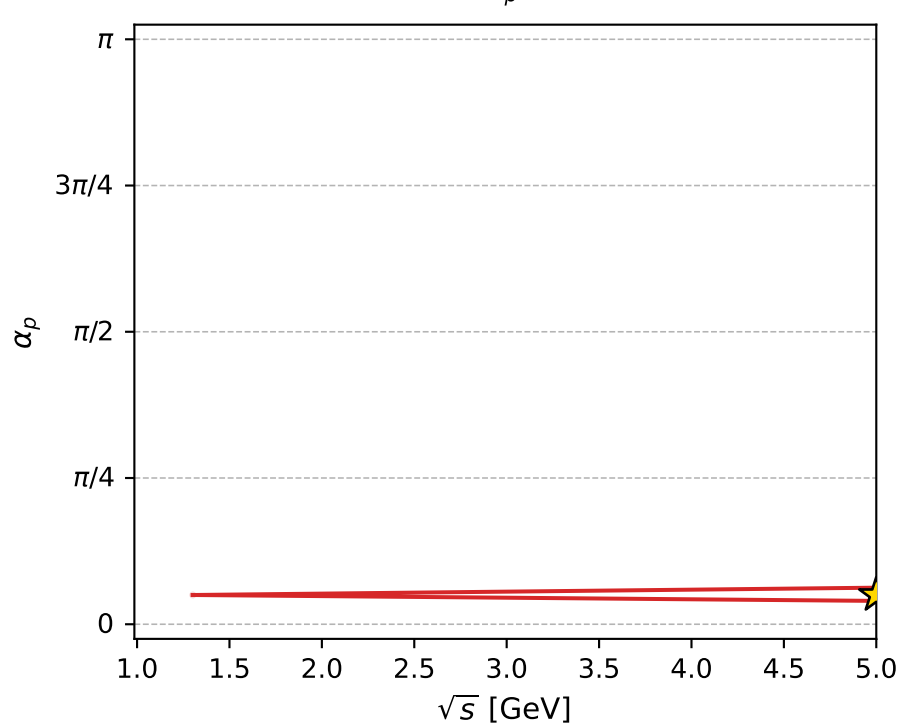
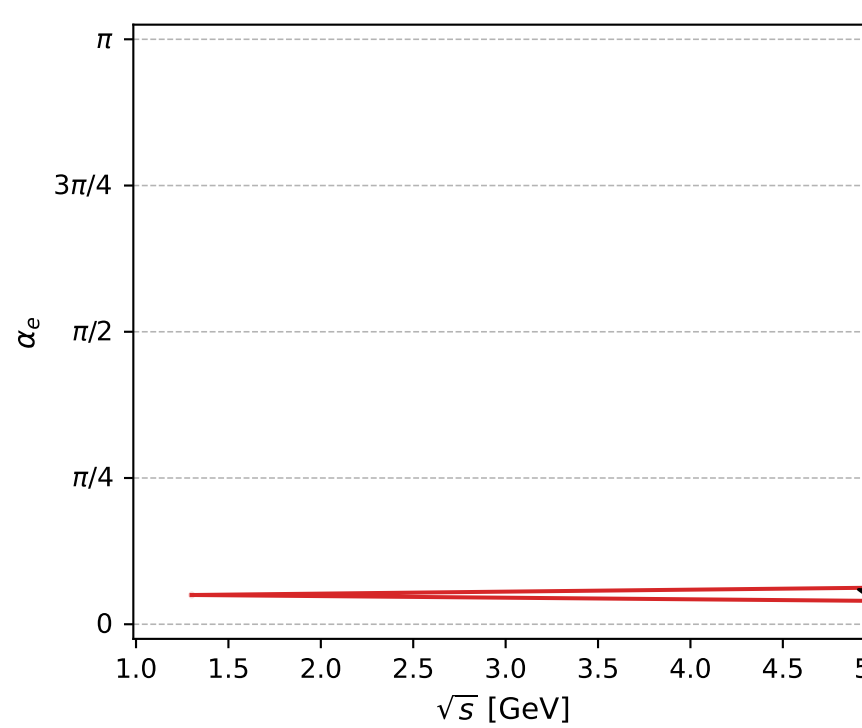
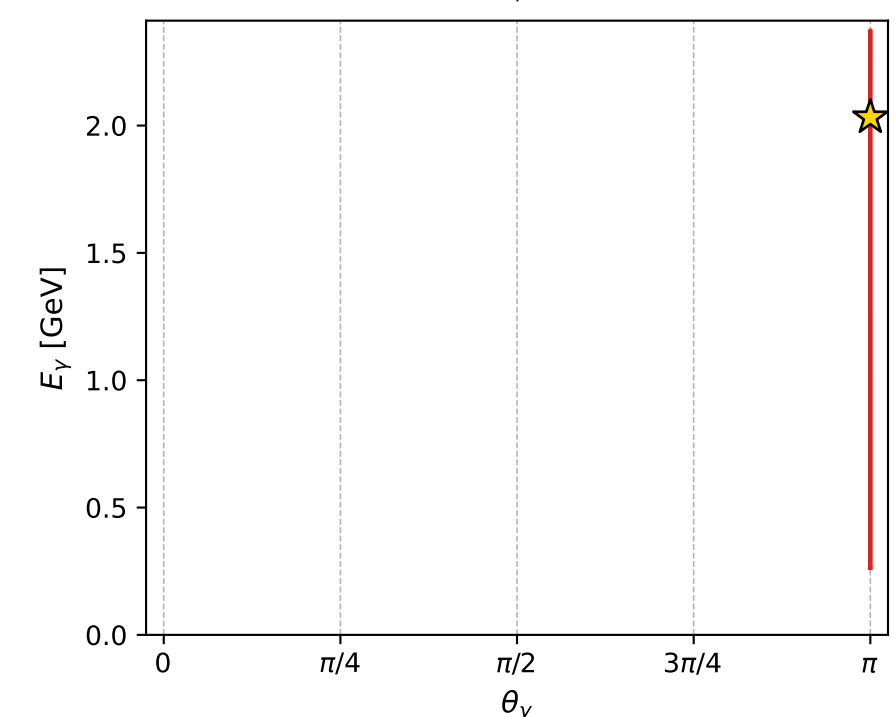
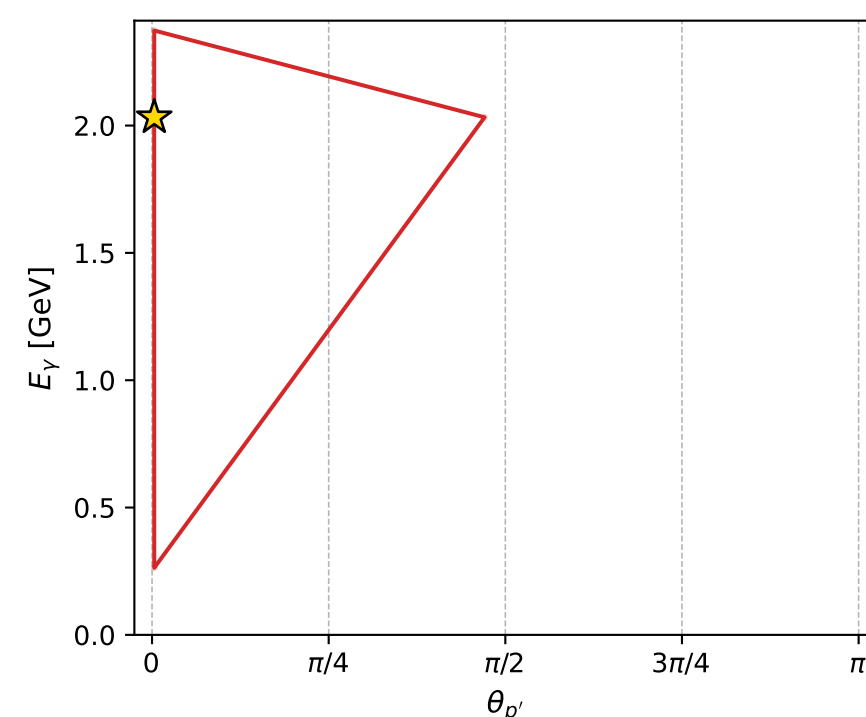
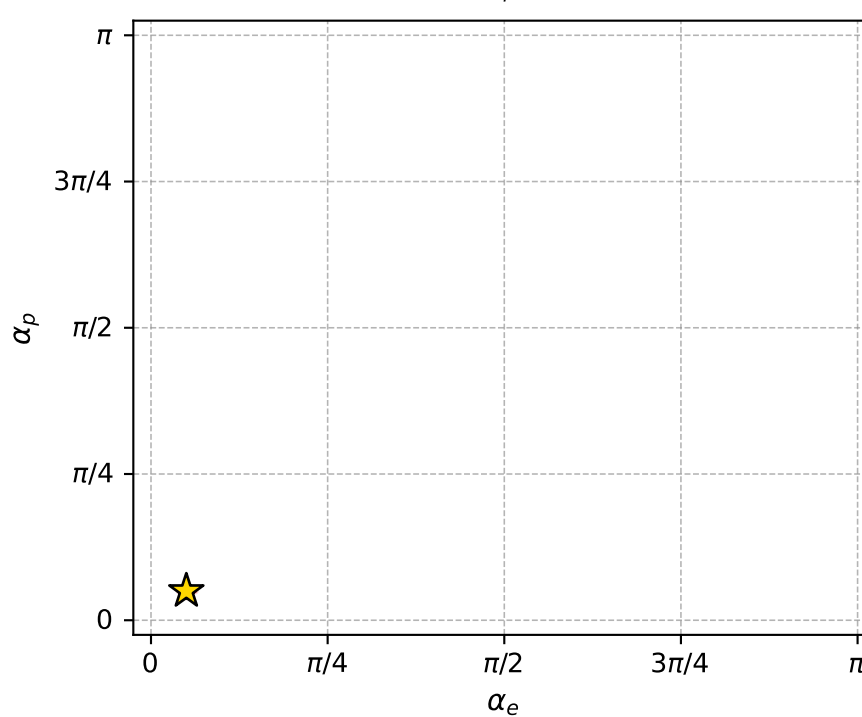
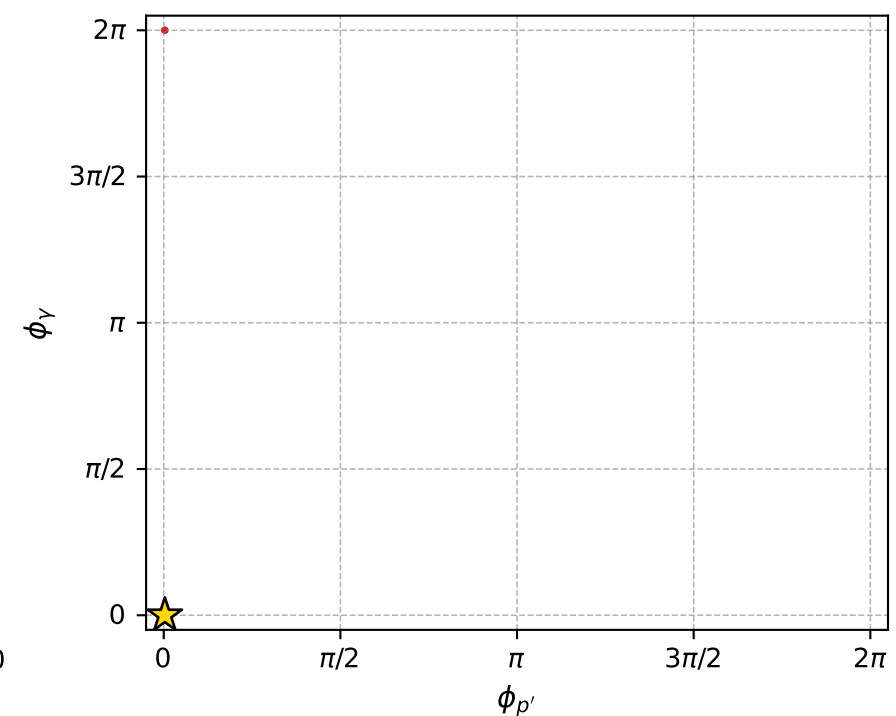
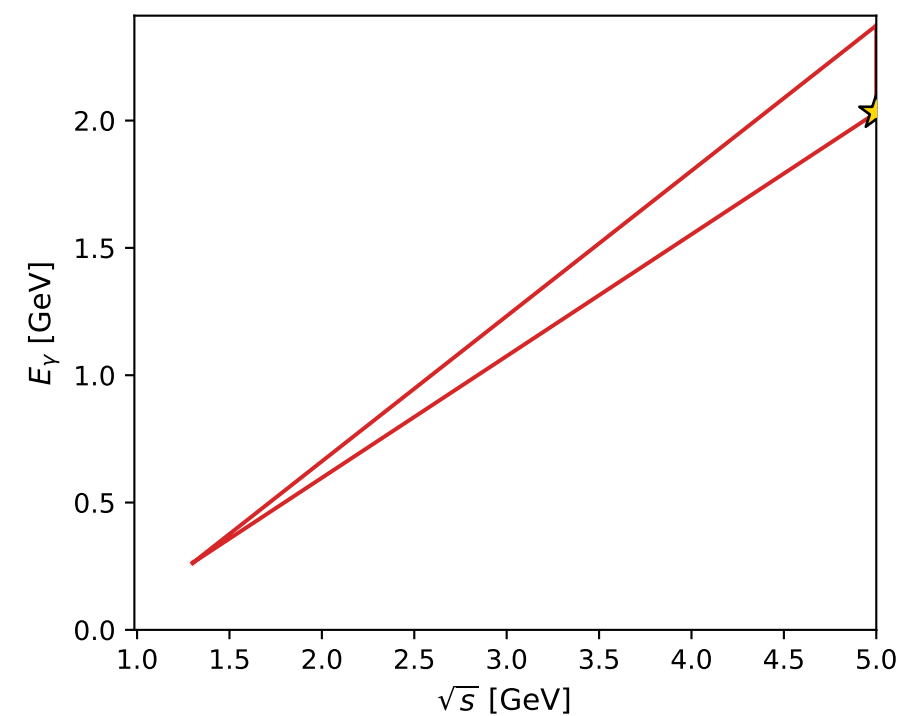
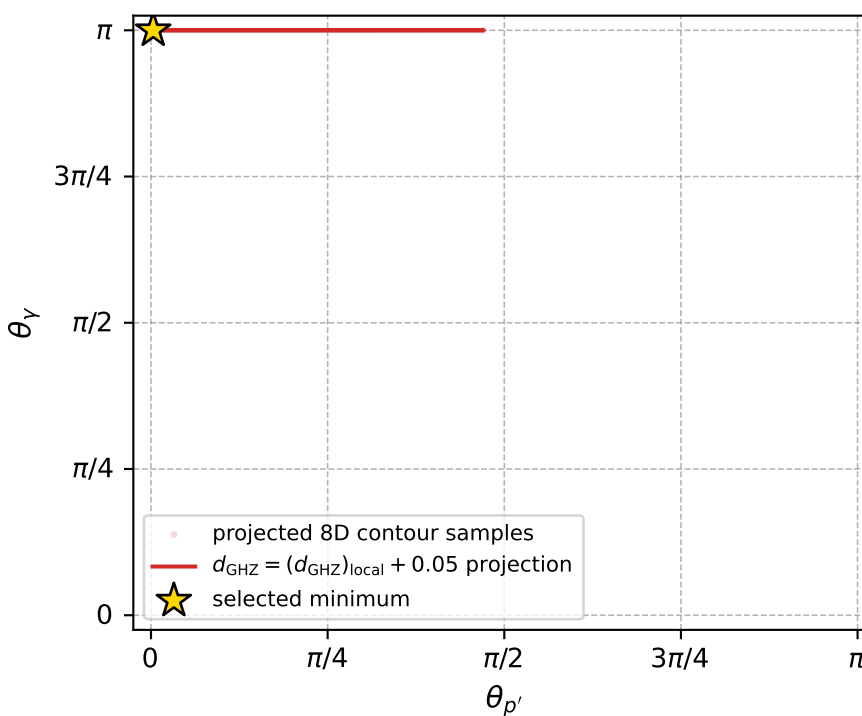


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P1, local minimum 151; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 8.5475667\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000085$
 above global minimum = $6.3192206\text{e-}08$
 8D contour samples = 64

selected phase-space point:

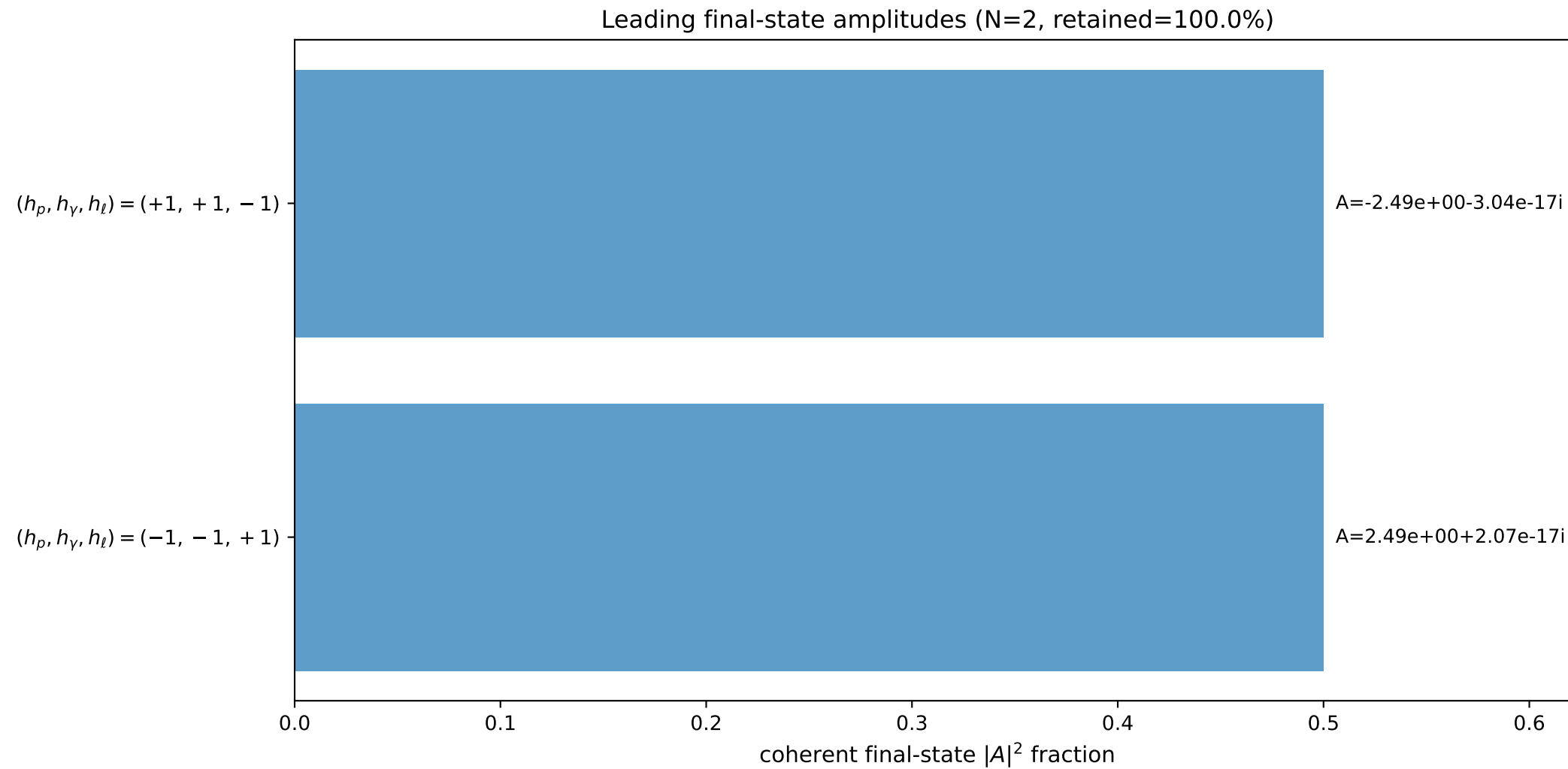
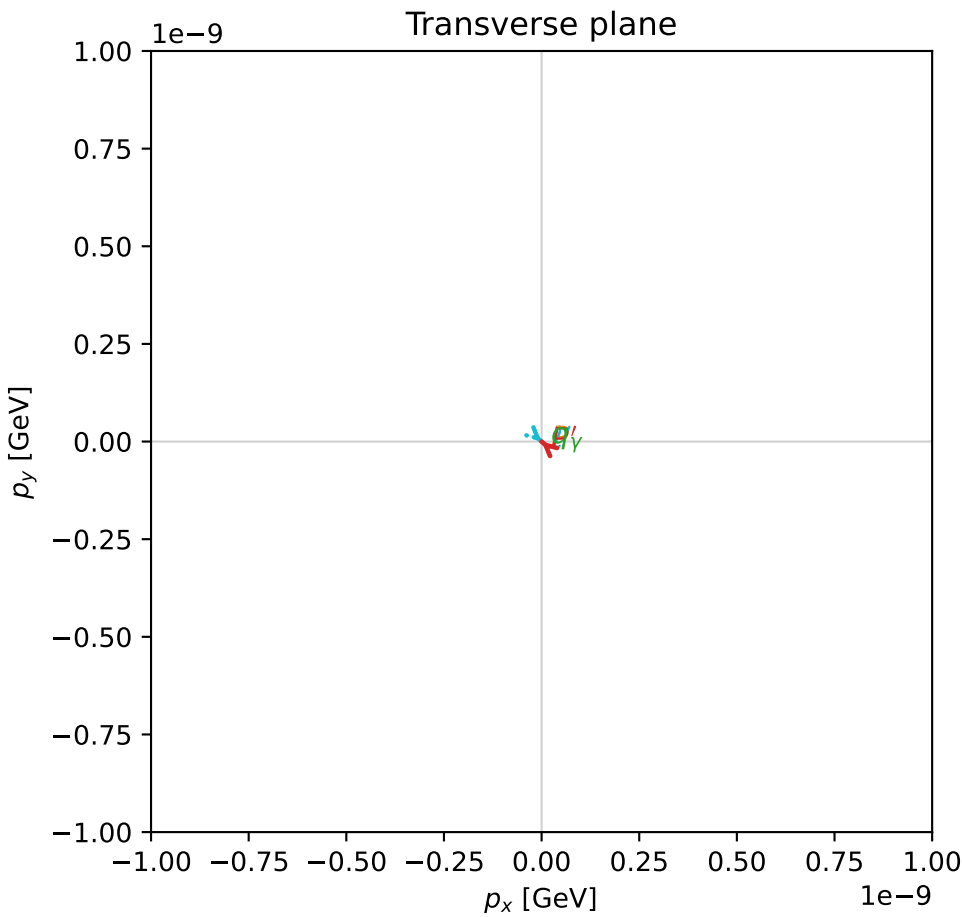
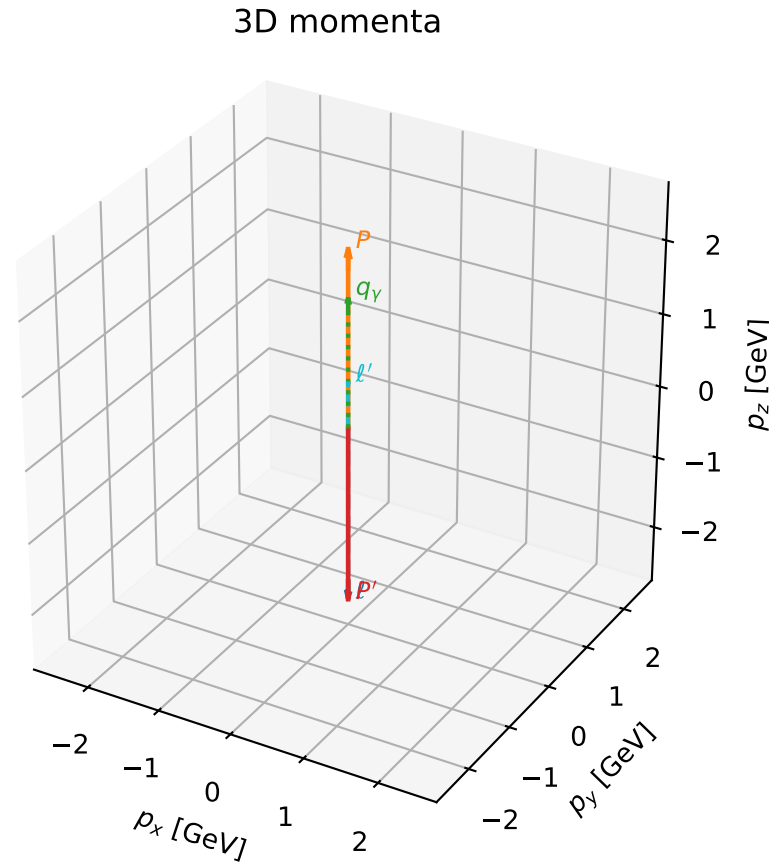
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 0.009817477$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 2.032417$
 $\text{phi_p_out} = 0.009817477$
 $\text{phi_gamma_out} = 0$
 $\alpha_e = 0.1570796$
 $\alpha_p = 0.1570796$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

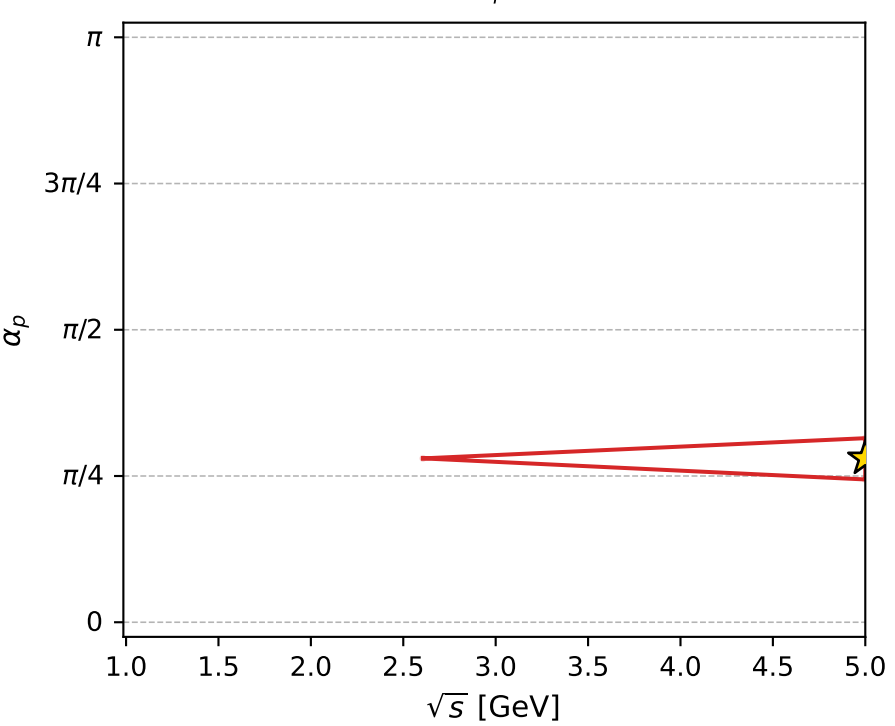
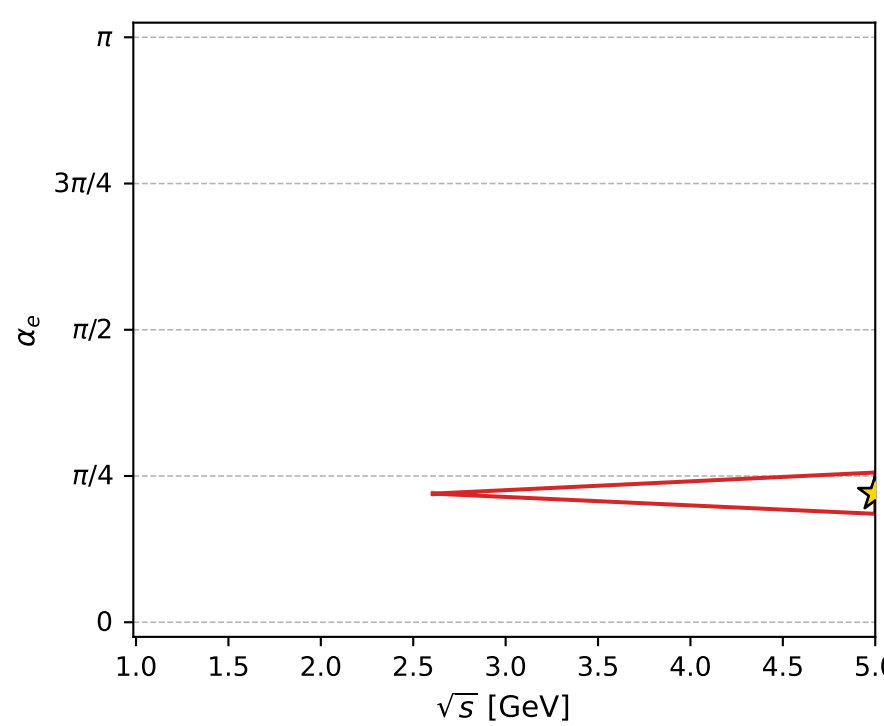
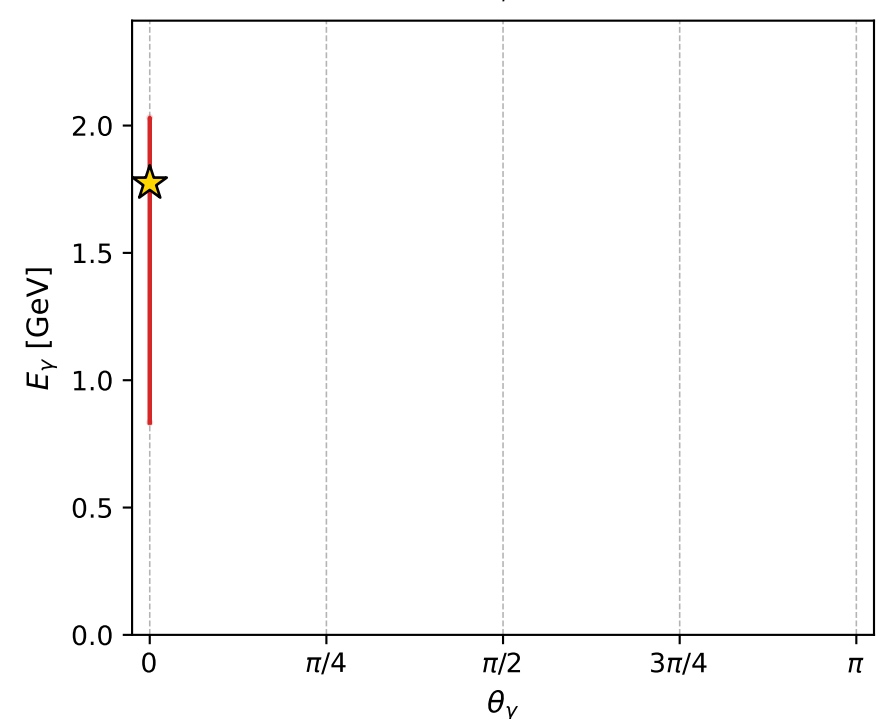
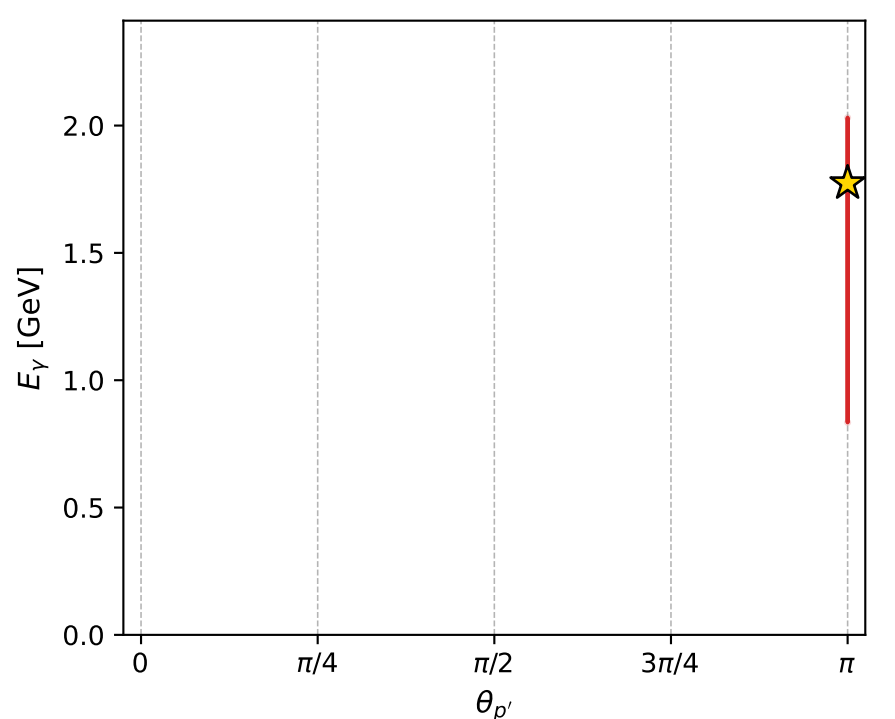
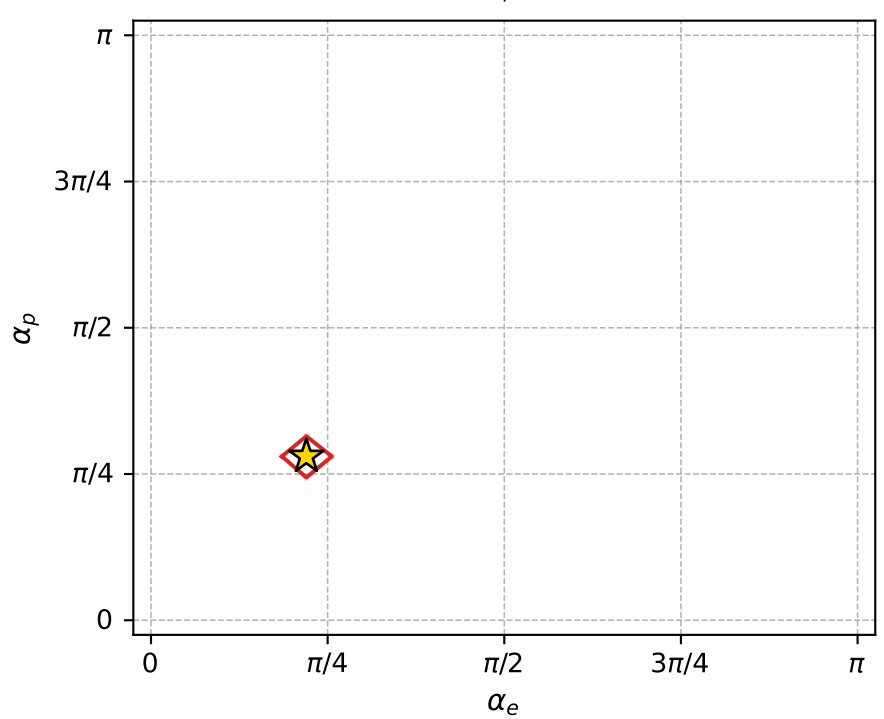
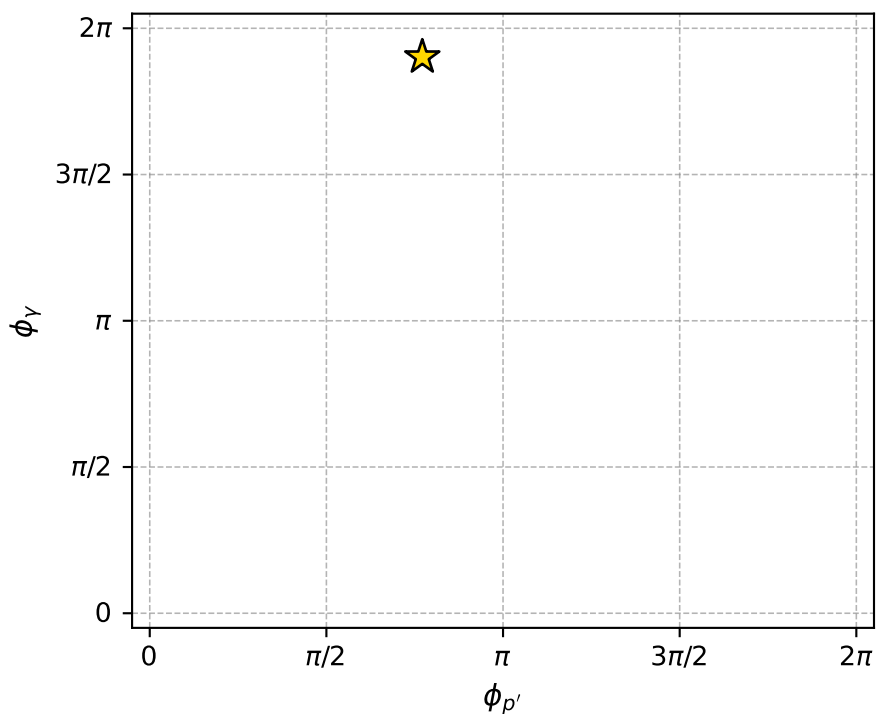
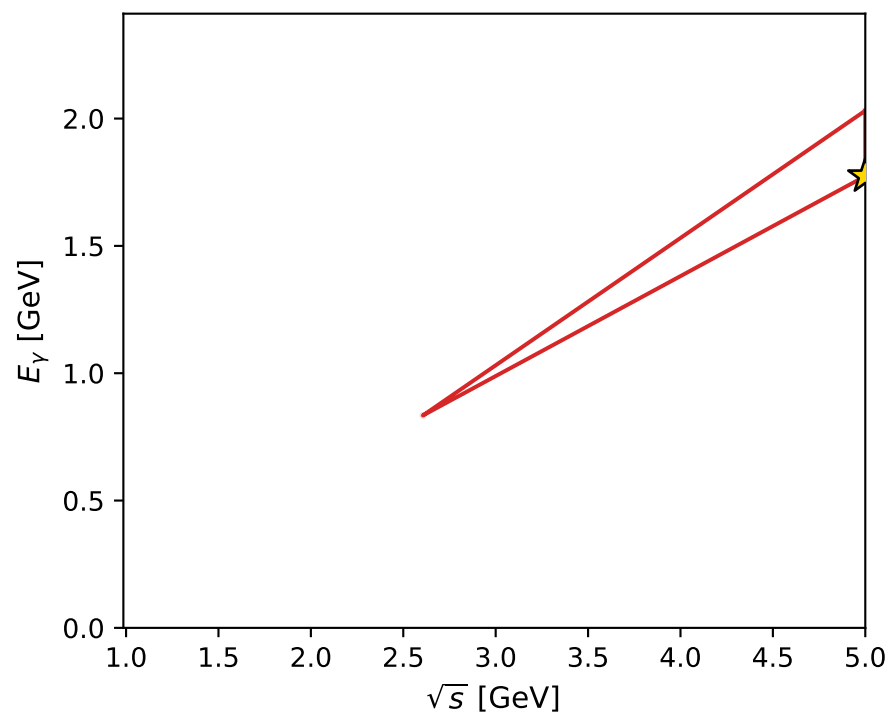
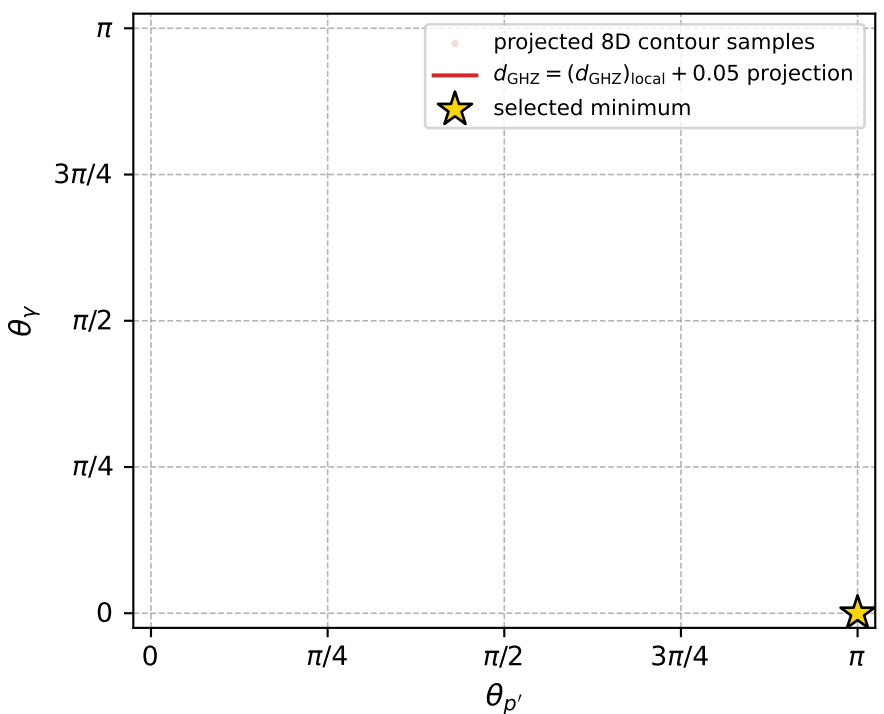
dghz_mixing_angles_polarization_01_local_minimum_152 $d_{\{\text{rm GHZ}\}}=8.63787\text{e-}08$
region: polarization_cluster_01_local_minimum_152
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.69075328$ rad
incoming lepton: $0.770766|+\rangle +0.637118|-\rangle$
proton mixing angle: $\alpha_p=0.88003373$ rad
incoming proton: $0.637125|+\rangle +0.77076|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.42356$, $\phi_Y=5.97423$
 $E_Y=1.77397$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.56515$
 $Q^2=6.15594$, $x_B=0.257618$, $t=-23.2713$, $W^2=18.6195$, $y=0.99069$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 0.6380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6380$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_Y $E= 1.7740$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7740$



electron: polarization cluster P1, local minimum 152; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 8.6378686\text{e-}08$
 $d_{\text{GHZ}} \text{ contour} = 0.050000086$
 above global minimum = $6.4095224\text{e-}08$
 8D contour samples = 39

selected phase-space point:

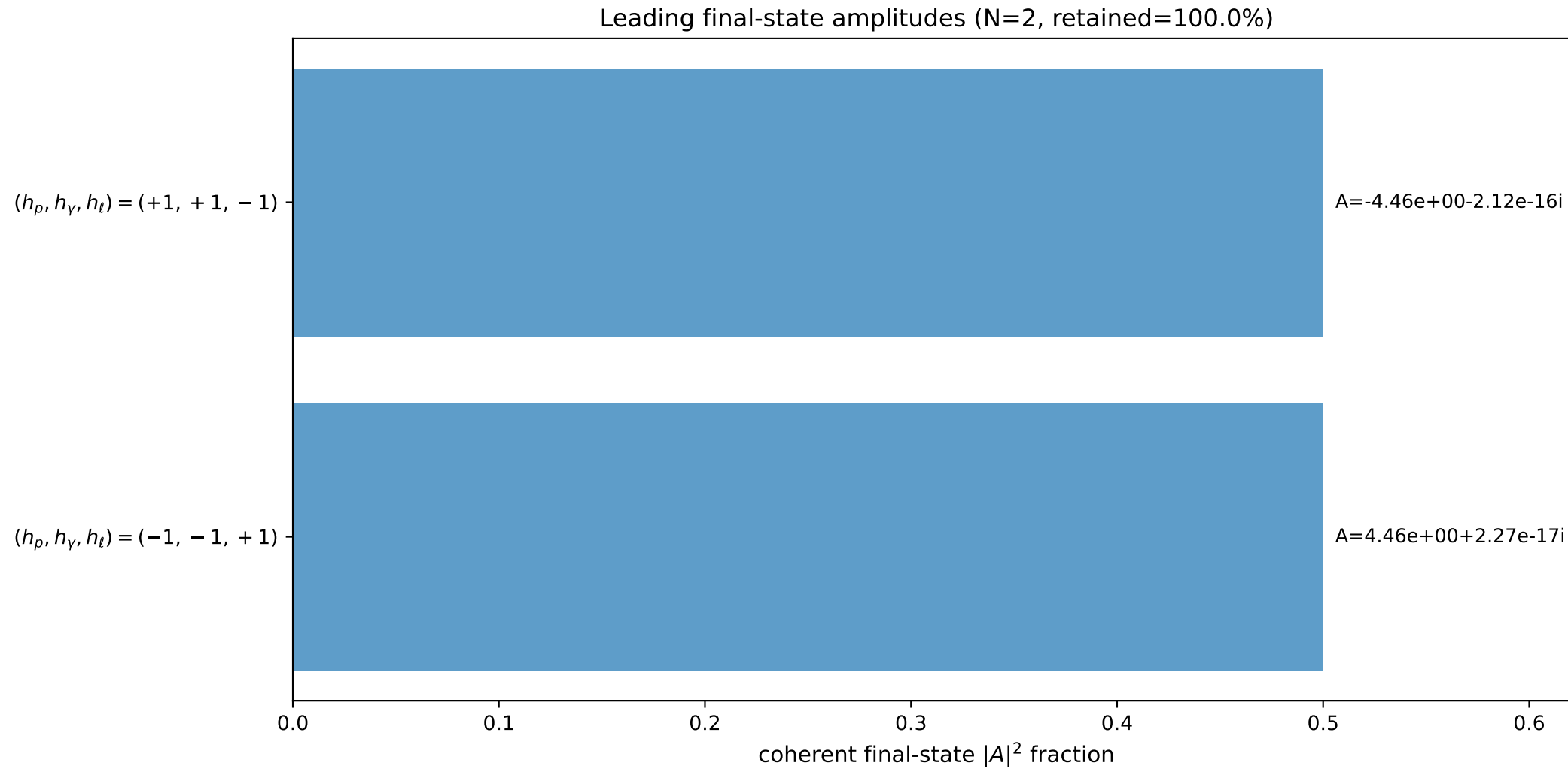
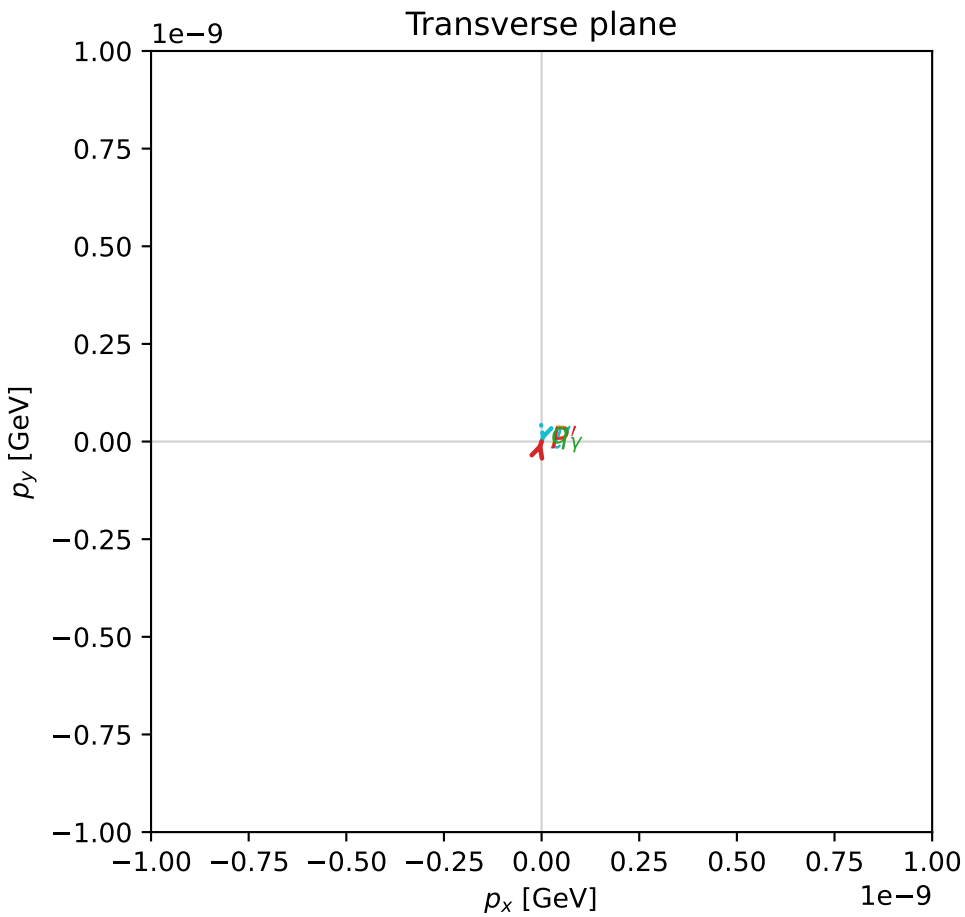
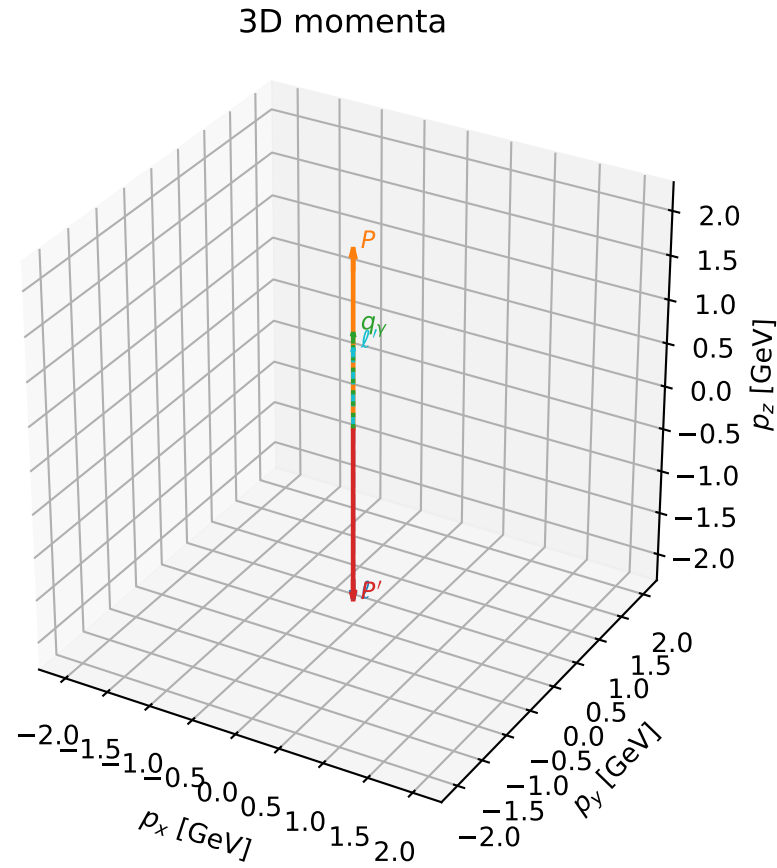
$\text{sqrt_s} = 5$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0ut} = 1.773966$
 $\text{phi_p_out} = 2.423558$
 $\text{phi_gamma_out} = 5.974228$
 $\alpha_e = 0.6907533$
 $\alpha_p = 0.8800337$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

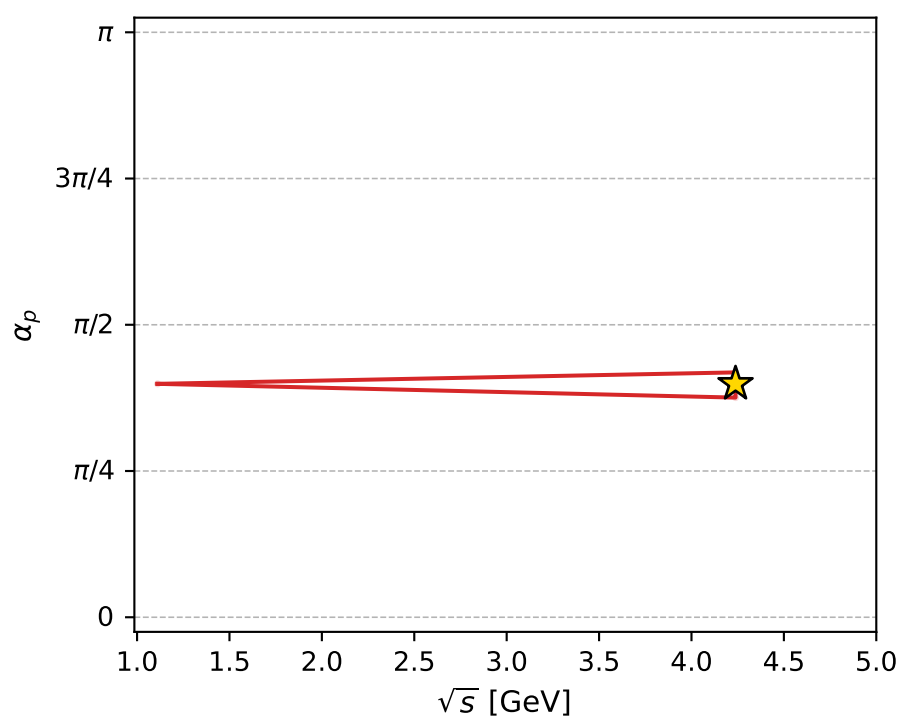
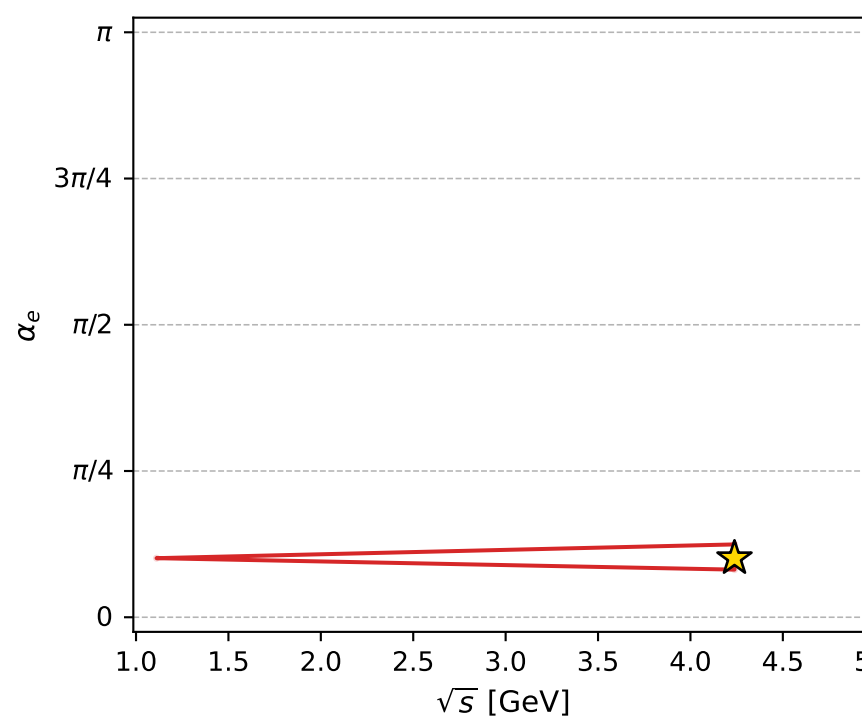
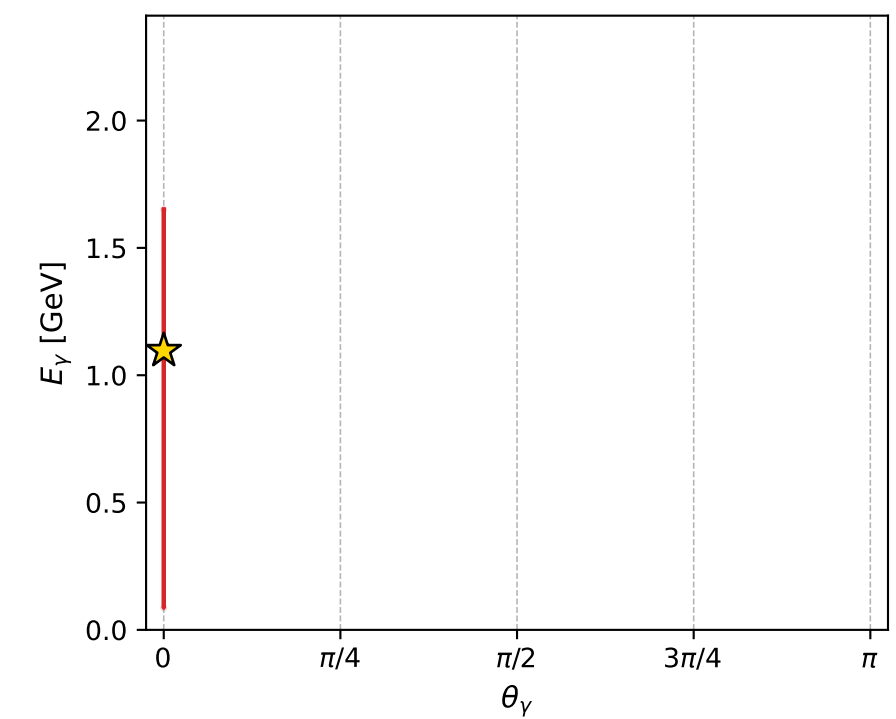
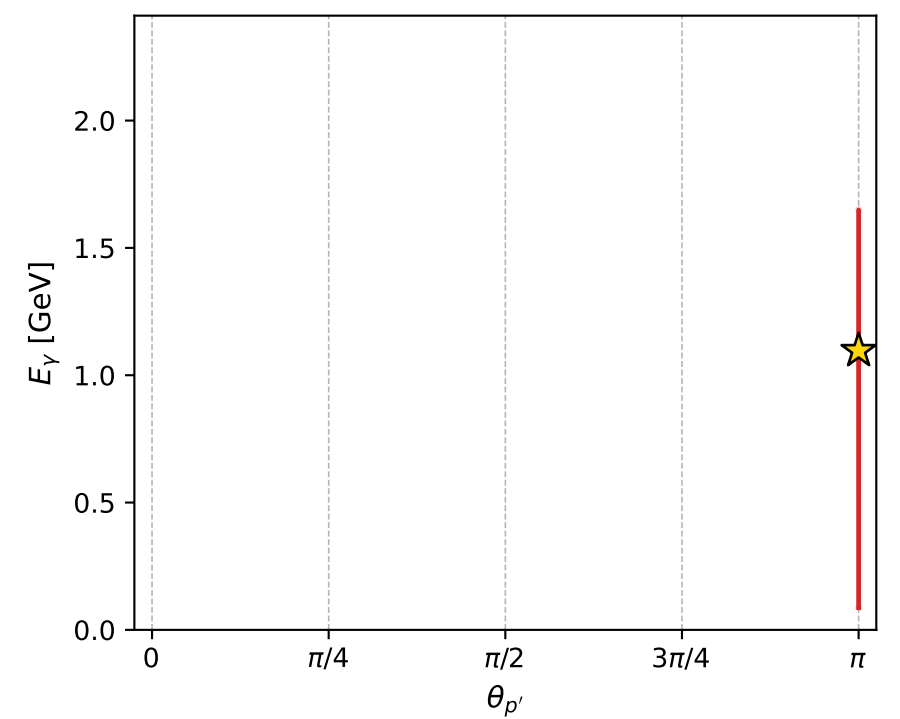
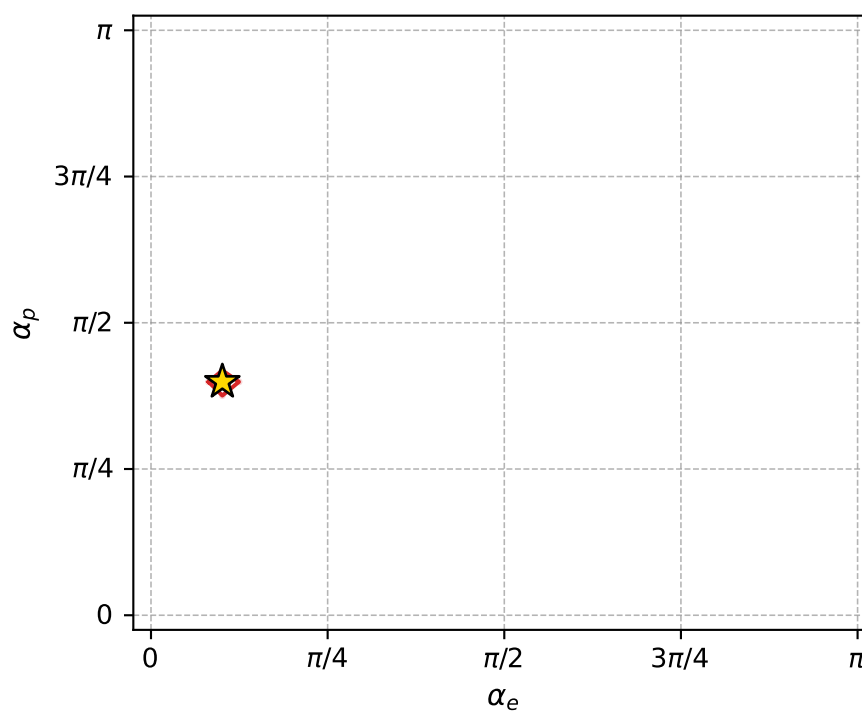
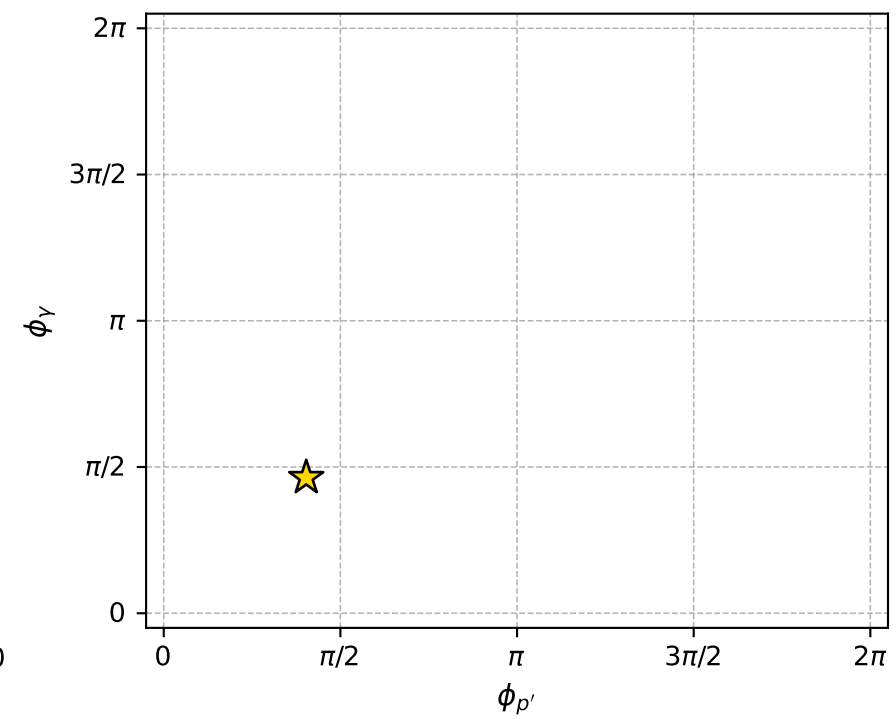
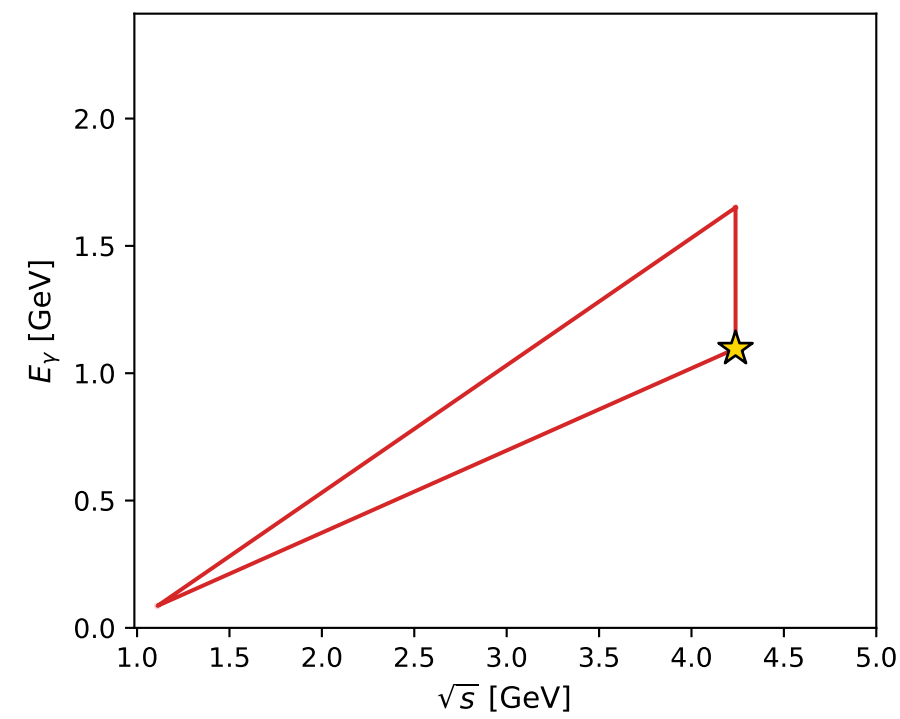
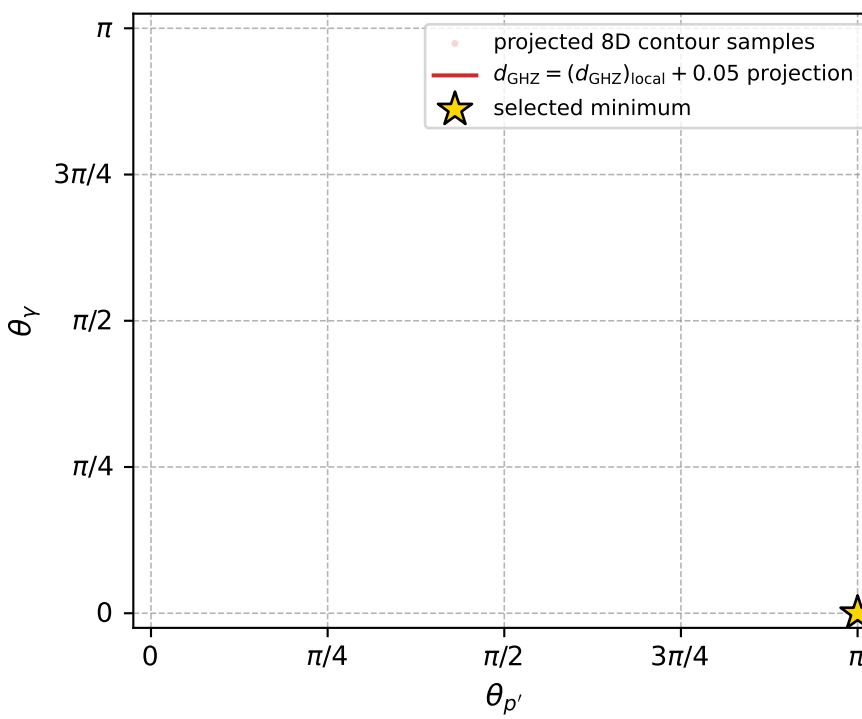
dghz_mixing_angles_polarization_01_local_minimum_163 $d_{\{\text{rm GHZ}\}}=1.18968\text{e-}07$
region: polarization_cluster_01_local_minimum_163
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.31723973$ rad
incoming lepton: $0.9501|+\rangle +0.311945|-\rangle$
proton mixing angle: $\alpha_p=1.2535406$ rad
incoming proton: $0.31196|+\rangle +0.950095|-\rangle$

$s=17.9637$, $\sqrt{s}=4.23836$, $|\vec{P}|=2.01539$, $|\vec{P}'|=2.01539$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=1.26708$, $\phi_Y=1.45652$
 $E_Y=1.09647$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.40868$
 $Q^2=7.40786$, $x_B=0.443522$, $t=-16.2471$, $W^2=10.1743$, $y=0.977668$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0154$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0154$
 P $E= 2.2230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0154$
 ℓ' $E= 0.9189$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.9189$
 P' $E= 2.2230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0154$
 q_Y $E= 1.0965$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0965$



electron: polarization cluster P1, local minimum 163; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.1896791\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000119$
 above global minimum = $9.668445\text{e-}08$
 8D contour samples = 68

selected phase-space point:

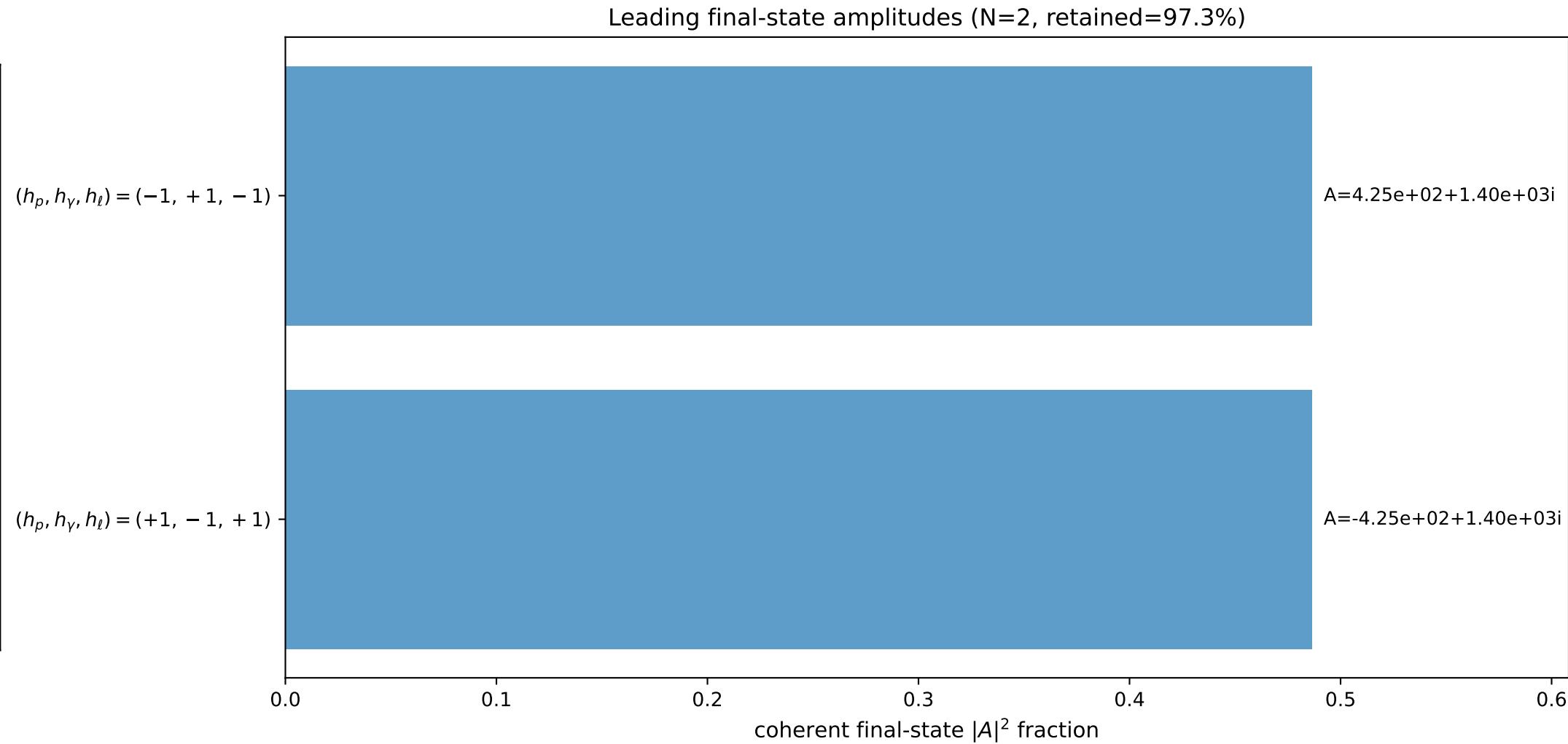
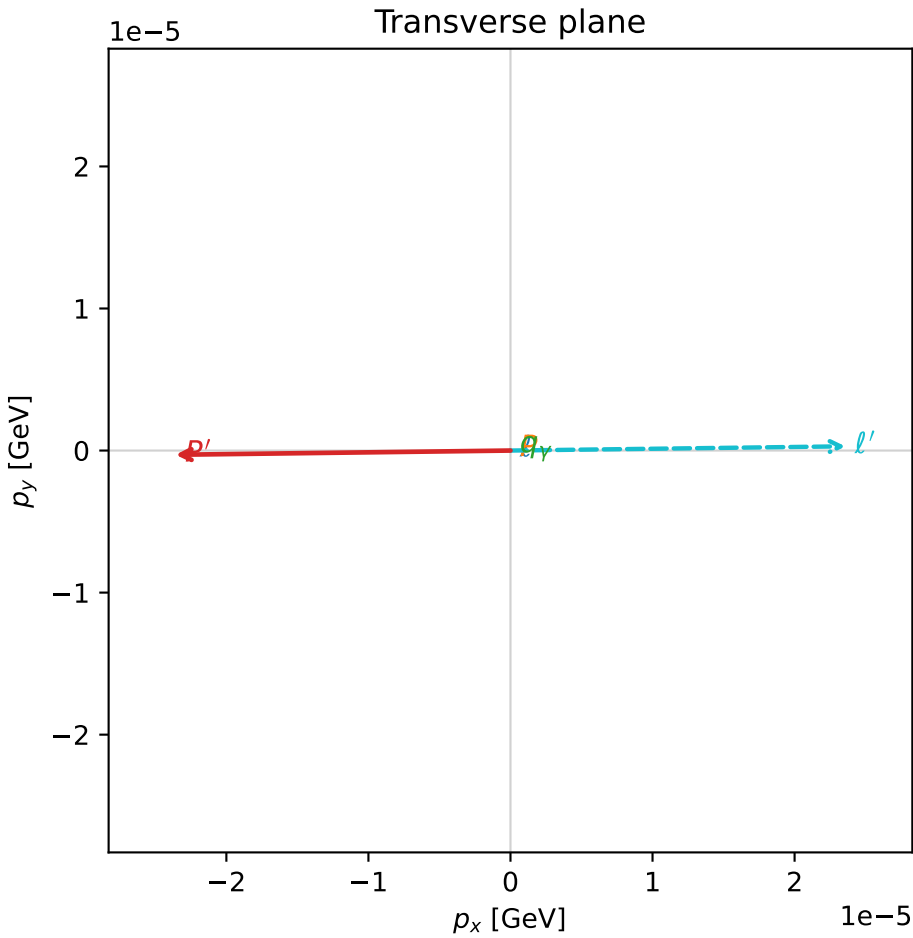
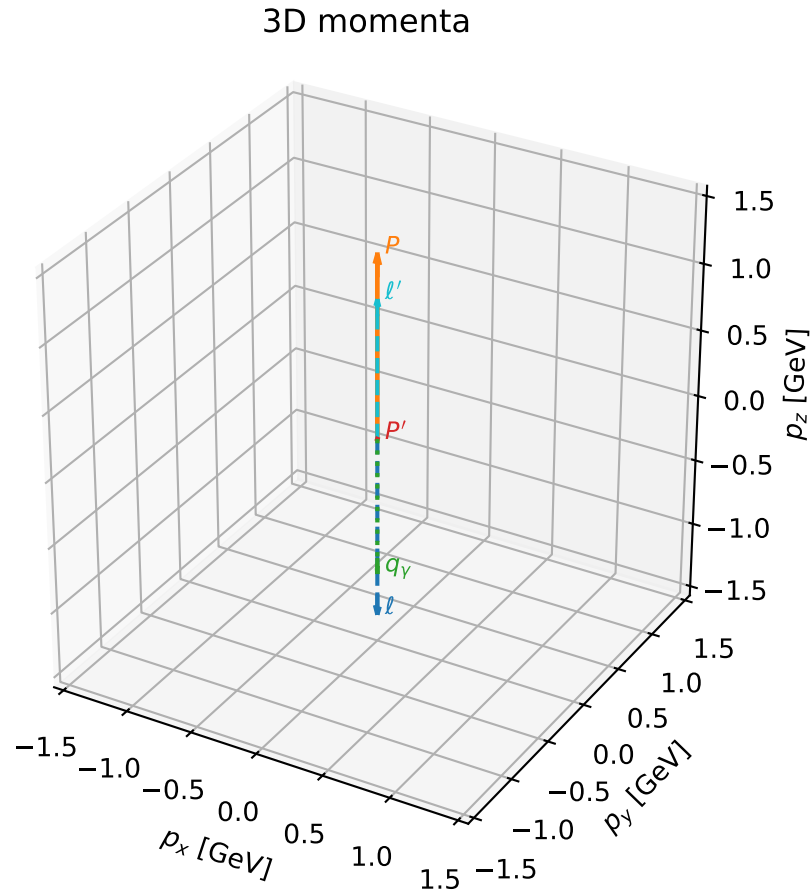
$\text{sqrt_s} = 4.238363$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0$
 $\text{q0out} = 1.096473$
 $\text{phi_p_out} = 1.267083$
 $\text{phi_gamma_out} = 1.456517$
 $\text{alpha_e} = 0.3172397$
 $\text{alpha_p} = 1.253541$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

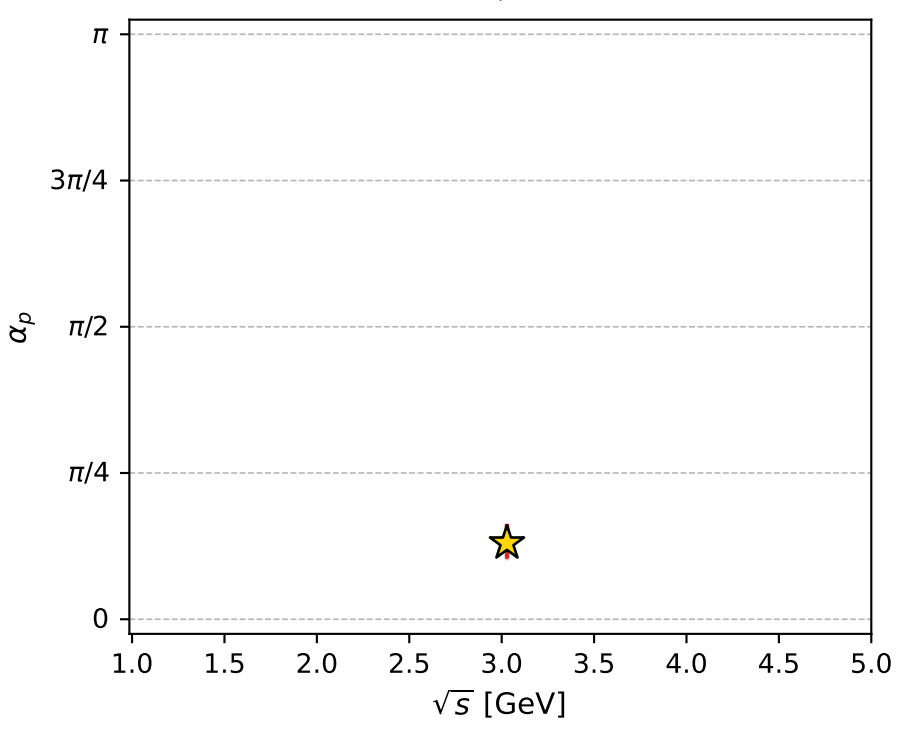
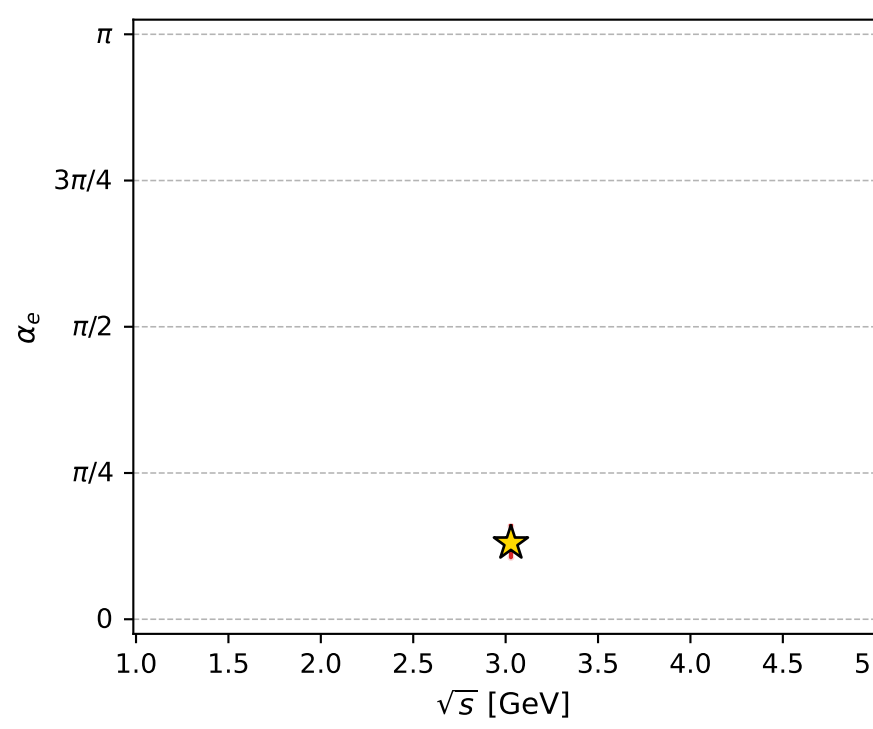
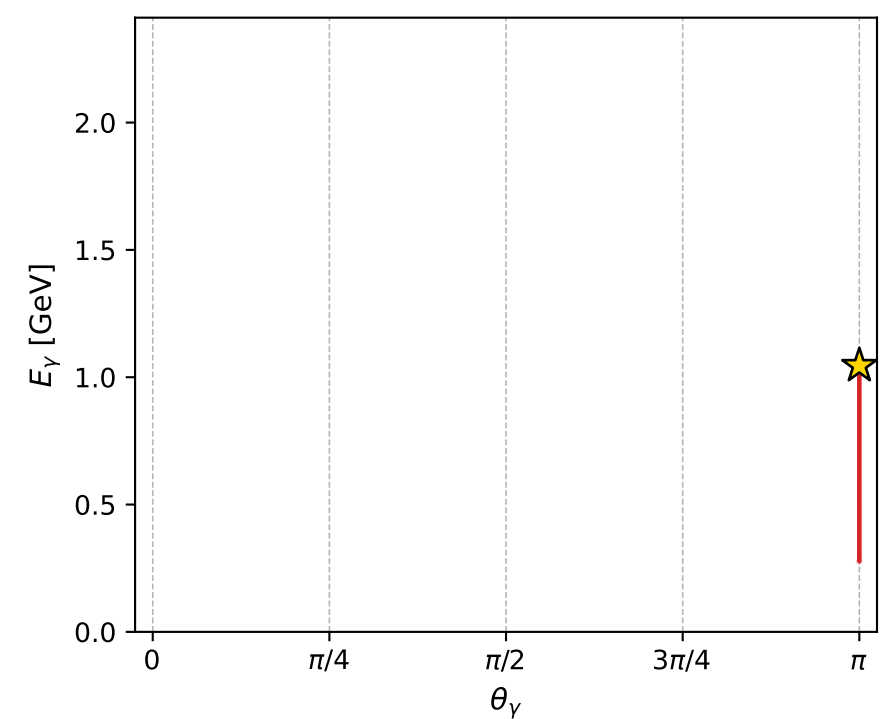
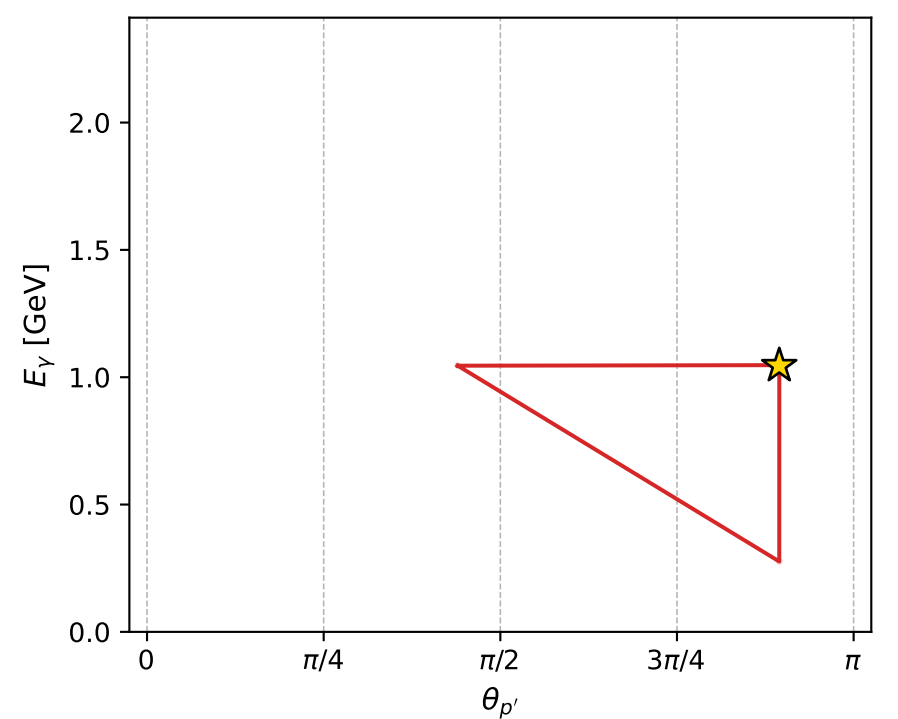
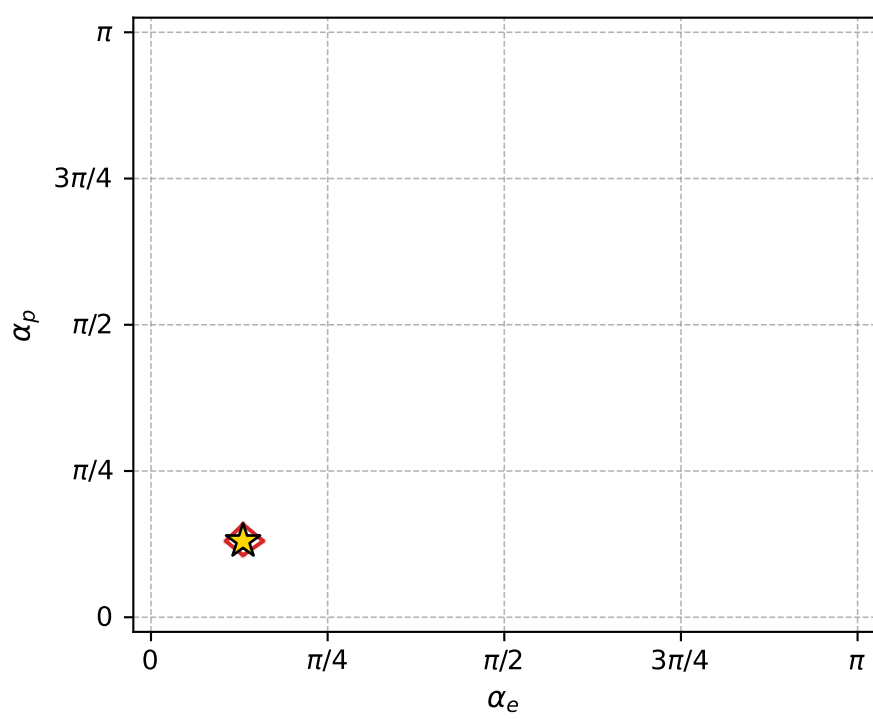
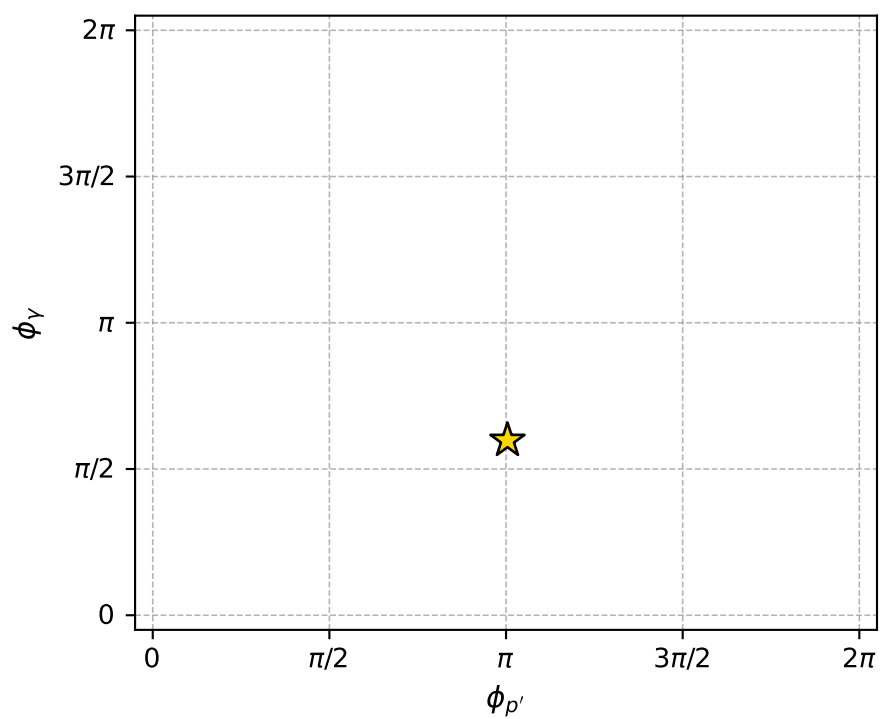
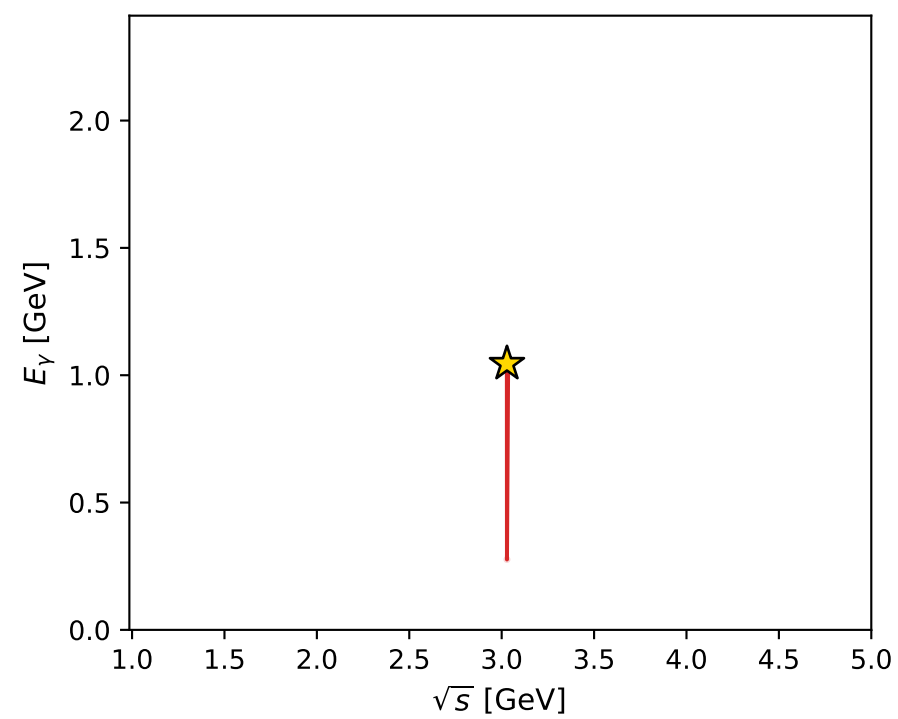
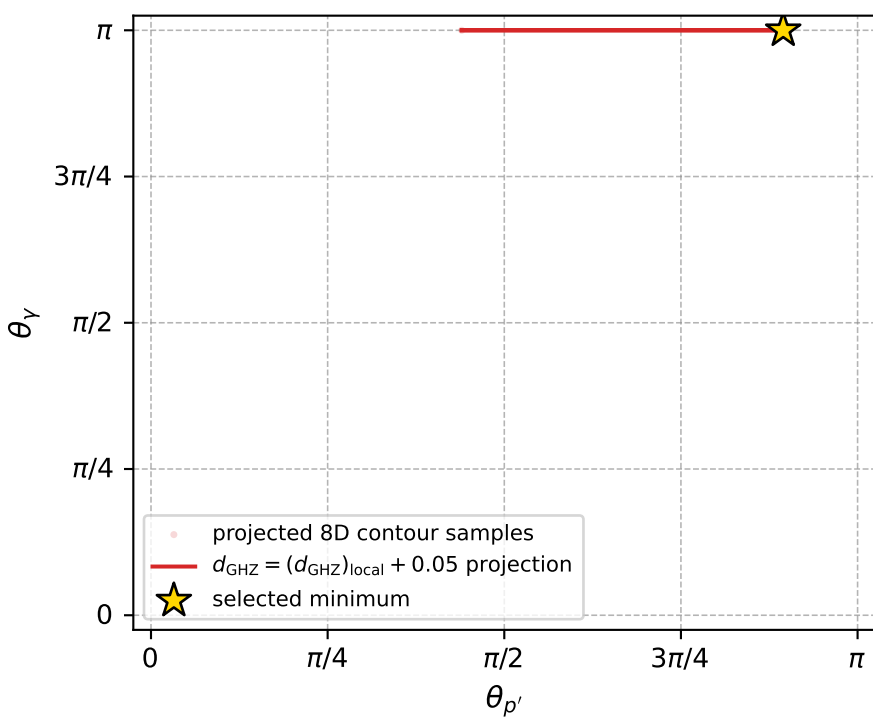
dghz_mixing_angles_polarization_01_local_minimum_165 $d_{\{\text{rm GHZ}\}}=1.20835\text{e-}07$
region: polarization_cluster_01_local_minimum_165
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.40930724$ rad
incoming lepton: $0.917397|+\rangle +0.397974|-\rangle$
proton mixing angle: $\alpha_p=0.40933059$ rad
incoming proton: $0.917387|+\rangle +0.397995|-\rangle$

$s=9.17245$, $\sqrt{s}=3.0286$, $|\vec{P}|=1.36905$, $|\vec{P}'|=7.26171\text{e-}05$
 $\theta_{p'}=2.81094$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.1542$, $\phi_\gamma=1.87819$
 $E_\gamma=1.04527$, $\theta_{\ell'}=2.25534\text{e-}05$, $\phi_{\ell'}=0.0126103$
 $Q^2=5.72446$, $x_B=0.744864$, $t=-1.35383$, $W^2=2.84062$, $y=0.926758$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3690$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3690$
 P $E= 1.6596$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3690$
 ℓ' $E= 1.0453$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0453$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0001$
 q_γ $E= 1.0453$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.0453$



electron: polarization cluster P1, local minimum 165; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.2083544\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000121$
 above global minimum = $9.8551979\text{e-}08$
 8D contour samples = 134

selected phase-space point:

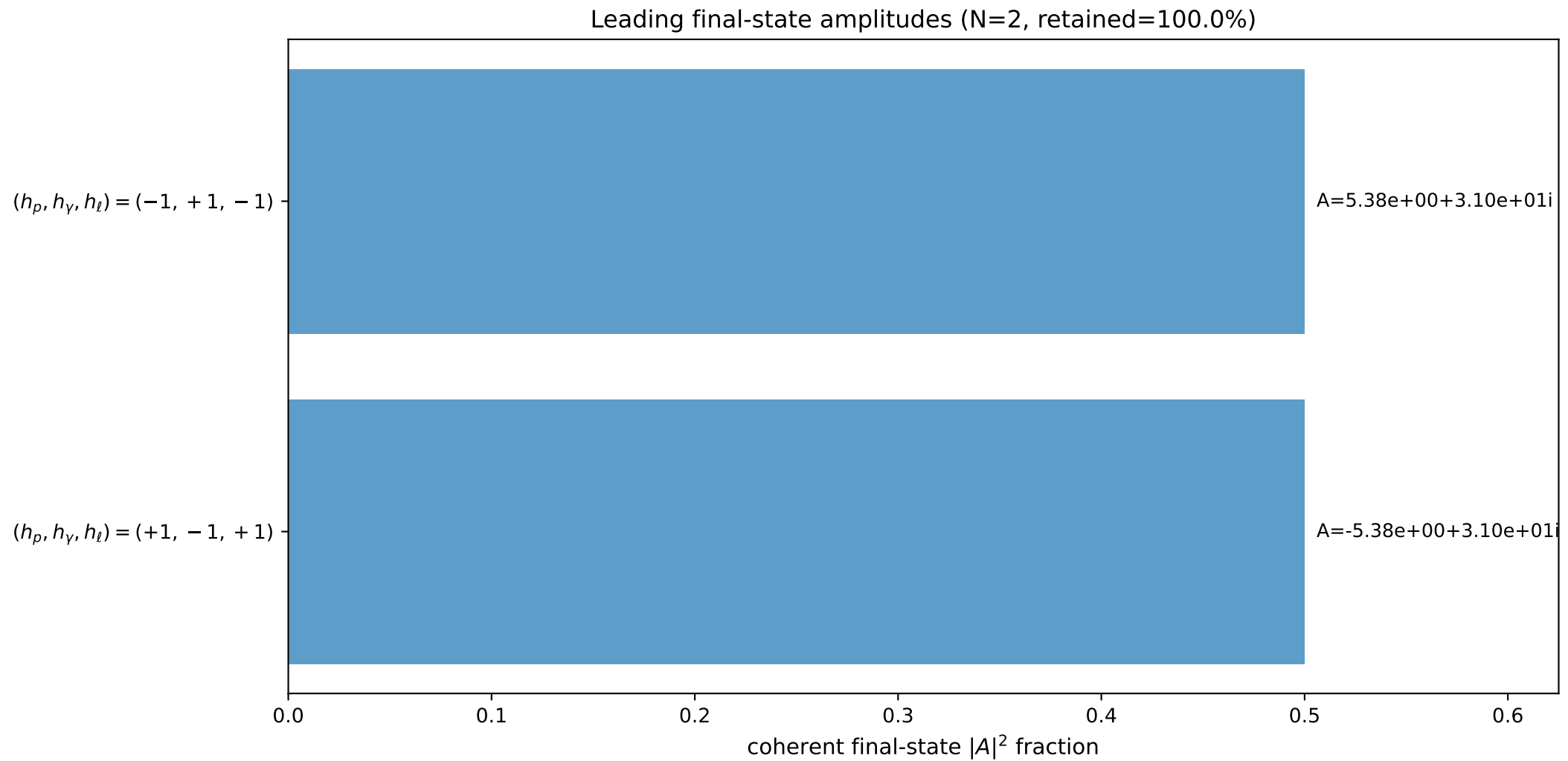
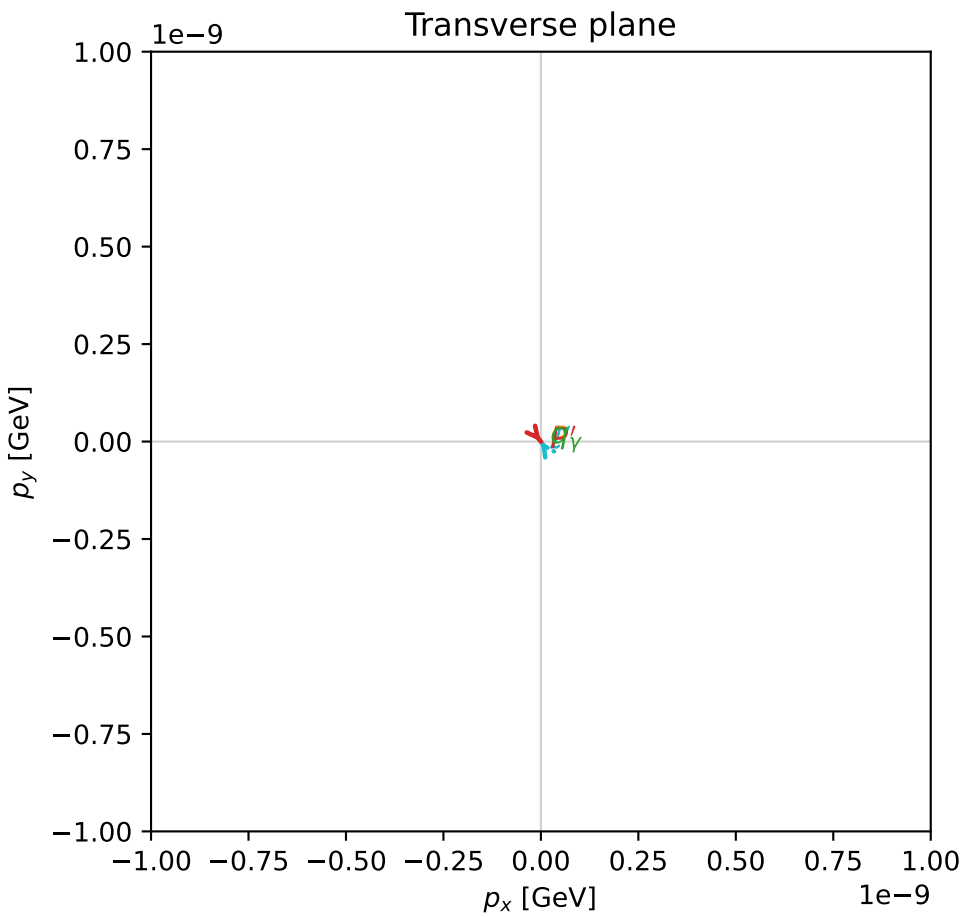
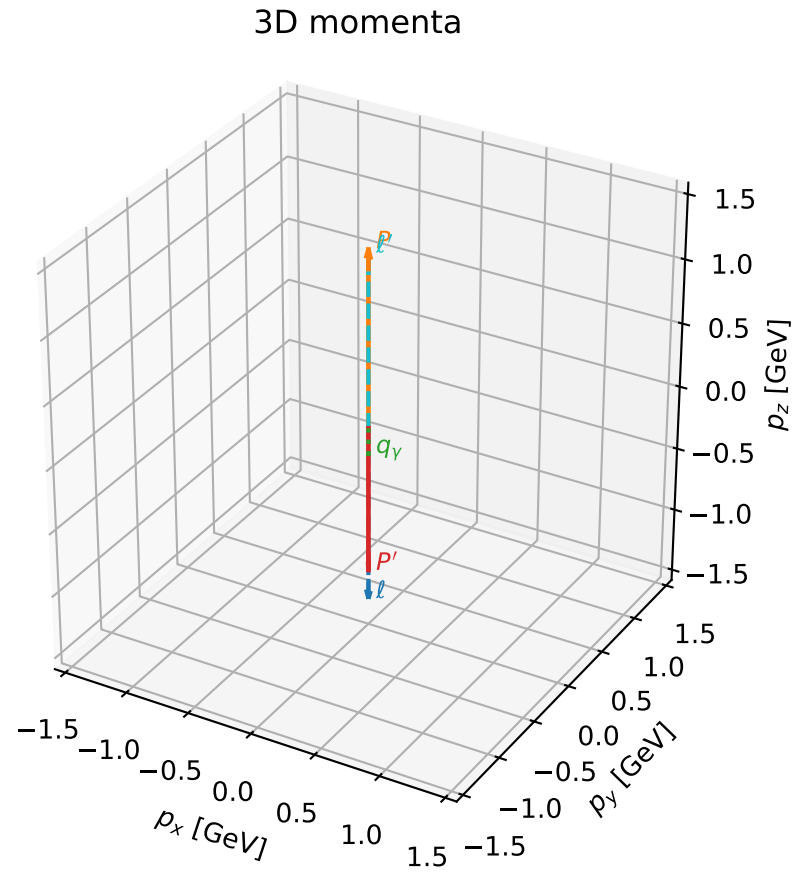
$\text{sqrt_s} = 3.028605$
 $\text{theta_p_out} = 2.81094$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 1.045268$
 $\text{phi_p_out} = 3.154203$
 $\text{phi_gamma_out} = 1.878185$
 $\text{alpha_e} = 0.4093072$
 $\text{alpha_p} = 0.4093306$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

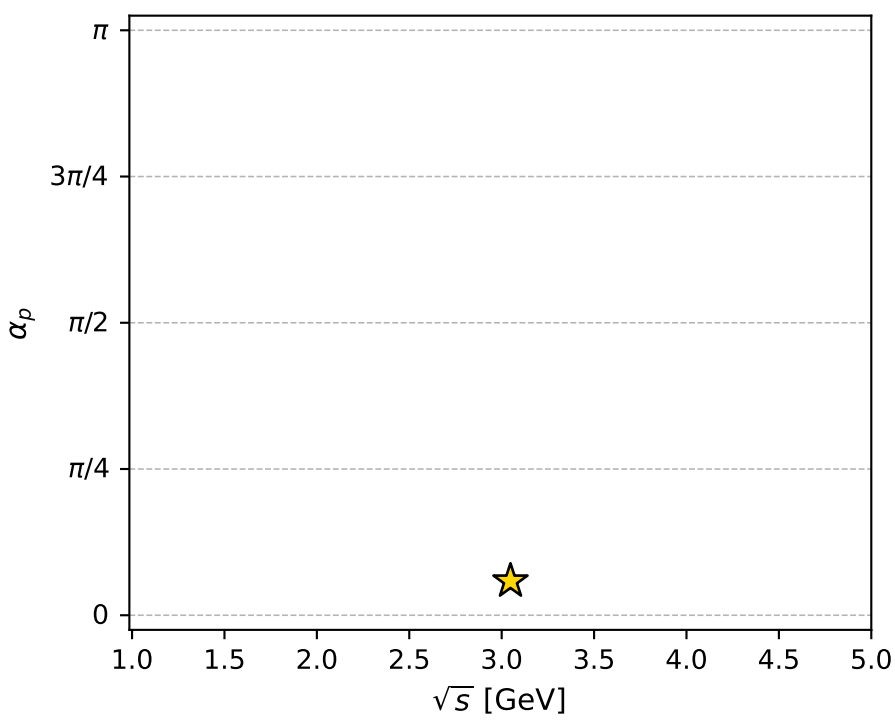
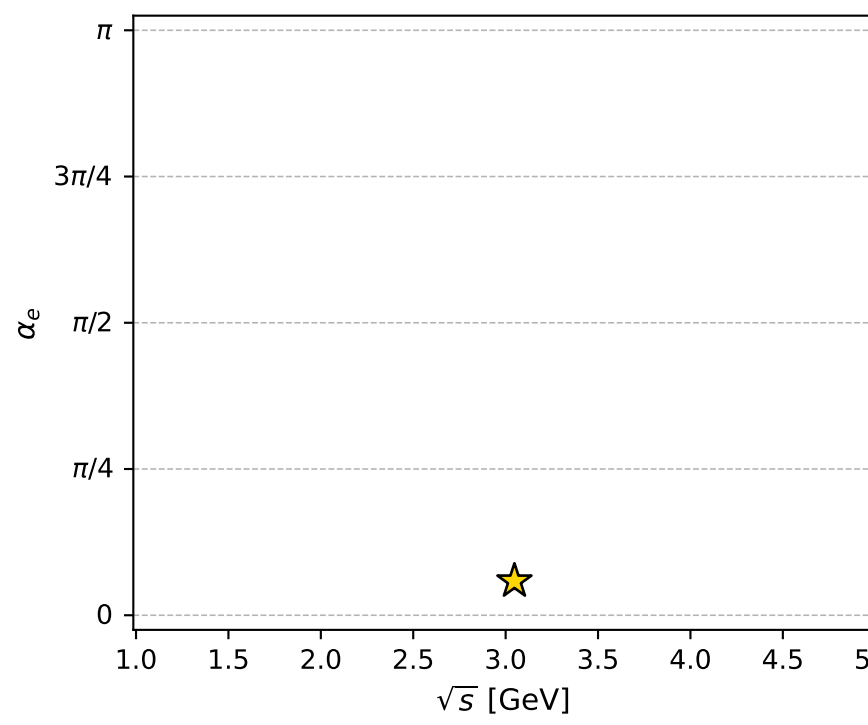
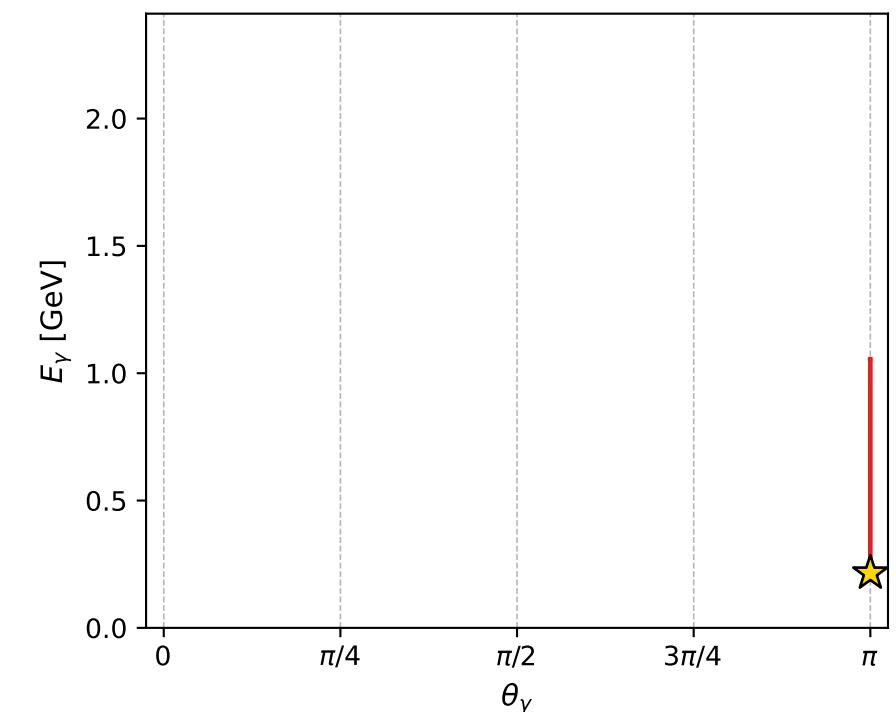
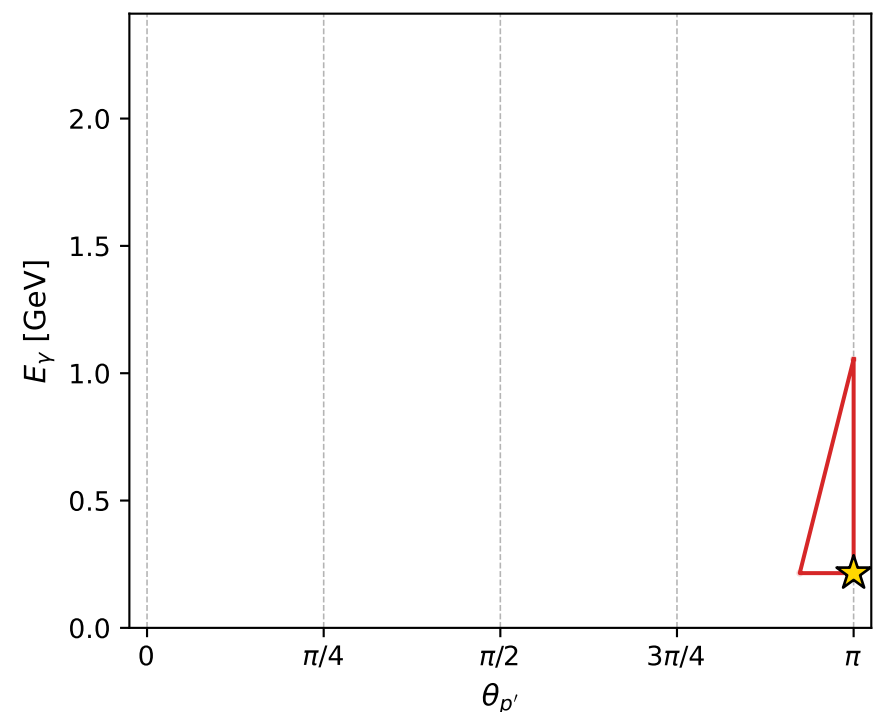
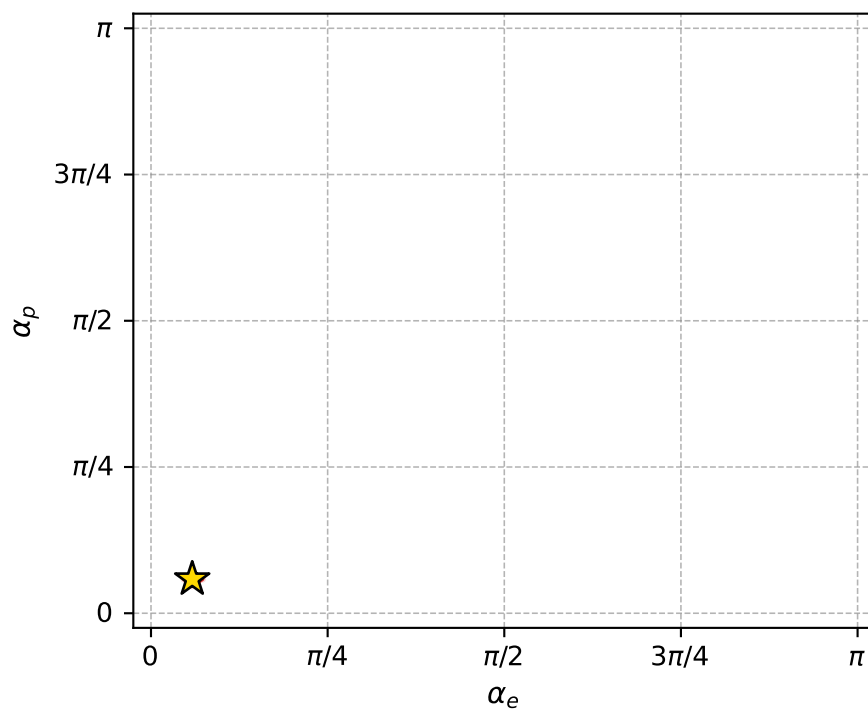
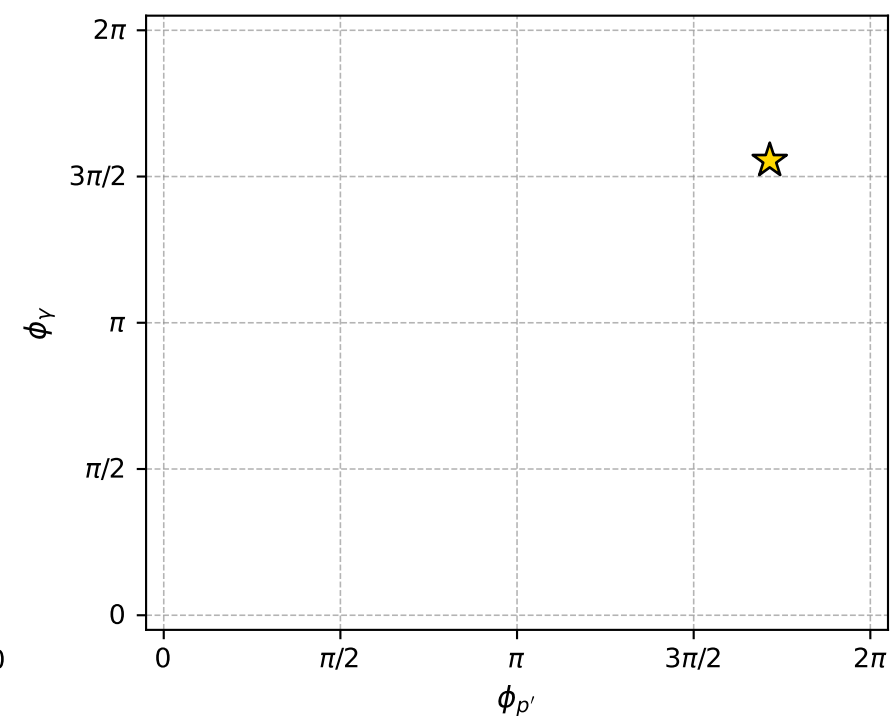
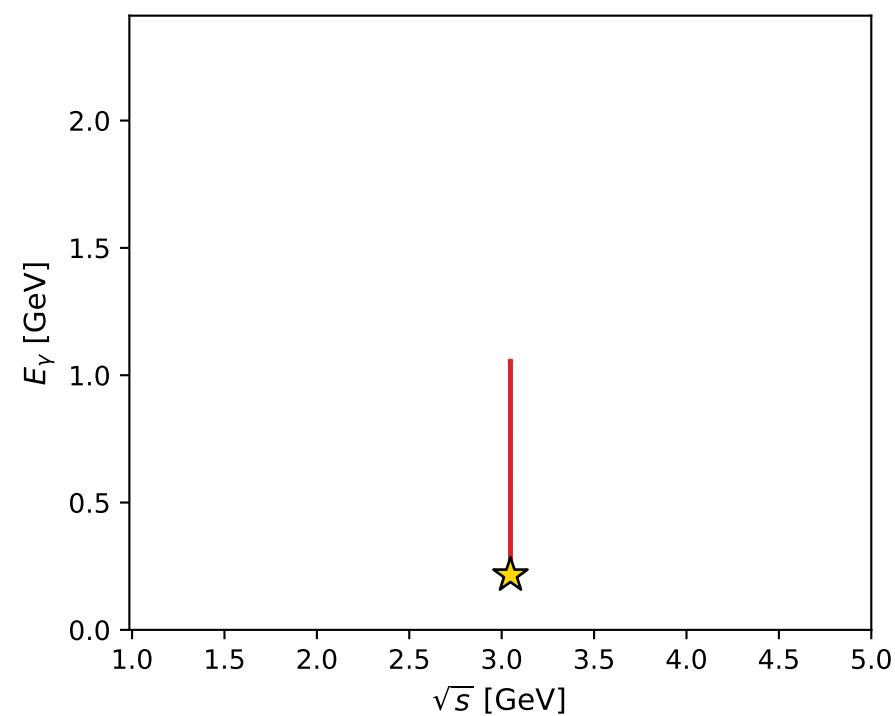
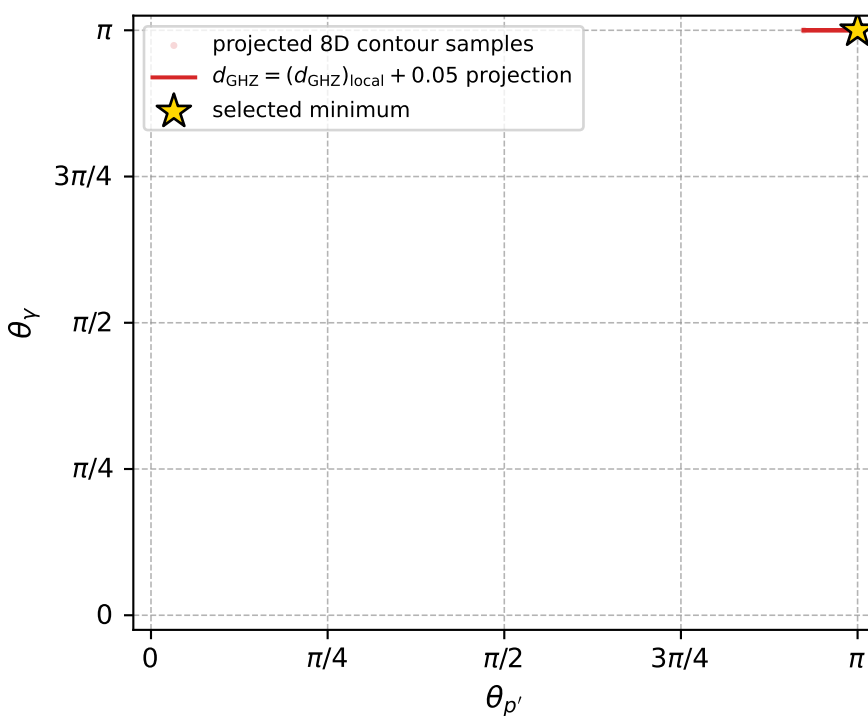
dghz_mixing_angles_polarization_01_local_minimum_170 d_{\rm GHZ}=1.94761e-07
 region: polarization_cluster_01_local_minimum_170
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.18329382$ rad
 incoming lepton: $0.983249|+\rangle +0.182269|-\rangle$
 proton mixing angle: $\alpha_p=0.18329532$ rad
 incoming proton: $0.983248|+\rangle +0.182271|-\rangle$

$s=9.28794$, $\sqrt{s}=3.04761$, $|\vec{P}|=1.37946$, $|\vec{P}'|=1.14077$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=5.38865$, $\phi_Y=4.88405$
 $E_Y=0.214975$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.16901$
 $Q^2=7.48078$, $x_B=0.981048$, $t=-6.31497$, $W^2=1.02436$, $y=0.906898$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3795$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3795$
 P $E= 1.6682$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3795$
 ℓ' $E= 1.3557$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.3557$
 P' $E= 1.4769$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.1408$
 q_Y $E= 0.2150$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.2150$



electron: polarization cluster P1, local minimum 170; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.9476071\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000195$
 above global minimum = $1.7247725\text{e-}07$
 8D contour samples = 80

selected phase-space point:

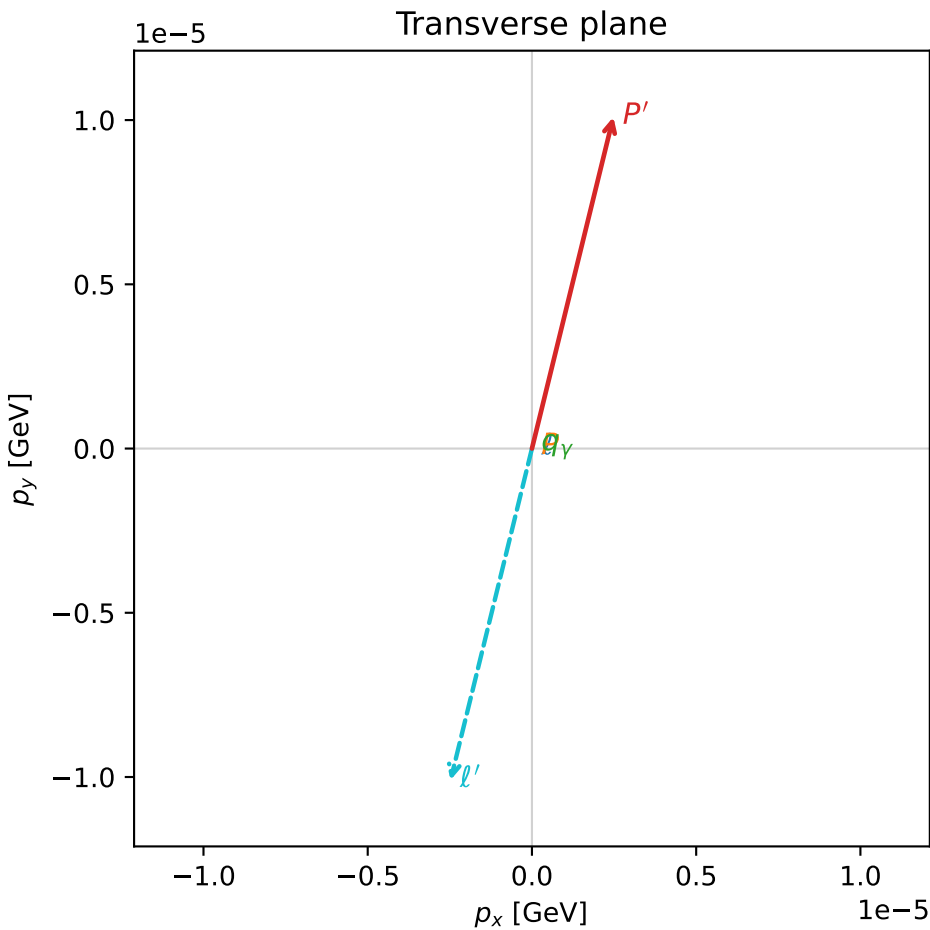
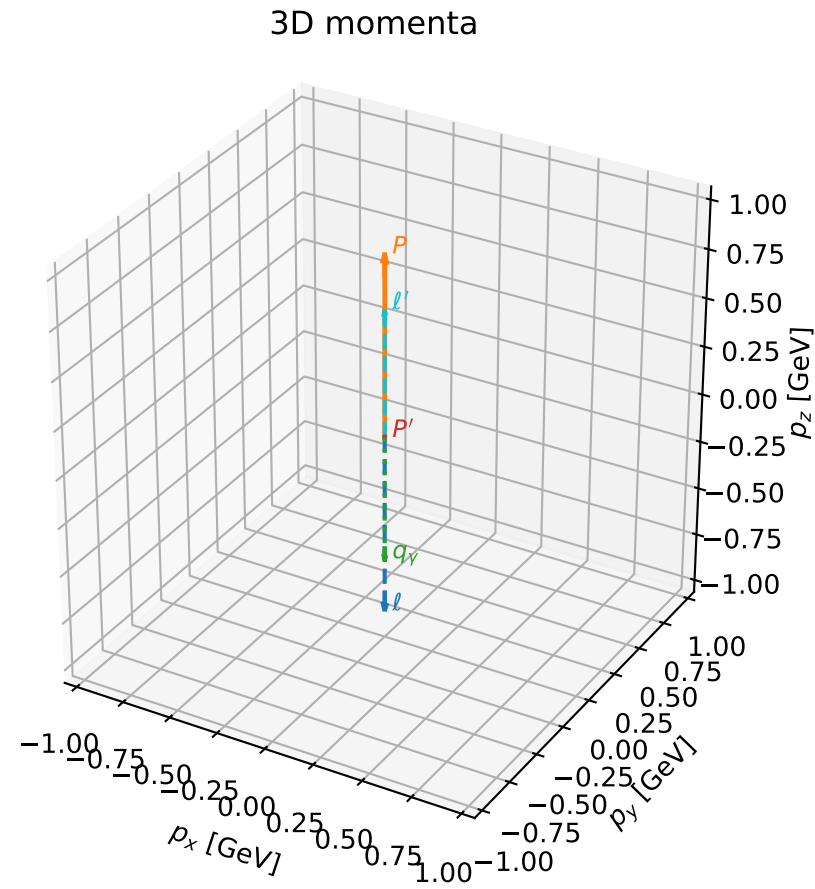
$\text{sqrt_s} = 3.047613$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.2149748$
 $\text{phi_p_out} = 5.388652$
 $\text{phi_gamma_out} = 4.884047$
 $\text{alpha_e} = 0.1832938$
 $\text{alpha_p} = 0.1832953$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_172 $d_{\{\text{rm GHZ}\}}=2.15034\text{e-}07$
 region: polarization_cluster_01_local_minimum_172
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.74567333$ rad
 incoming lepton: $0.734631|+\rangle +0.678467|-\rangle$
 proton mixing angle: $\alpha_p=0.74572493$ rad
 incoming proton: $0.734596|+\rangle +0.678505|-\rangle$

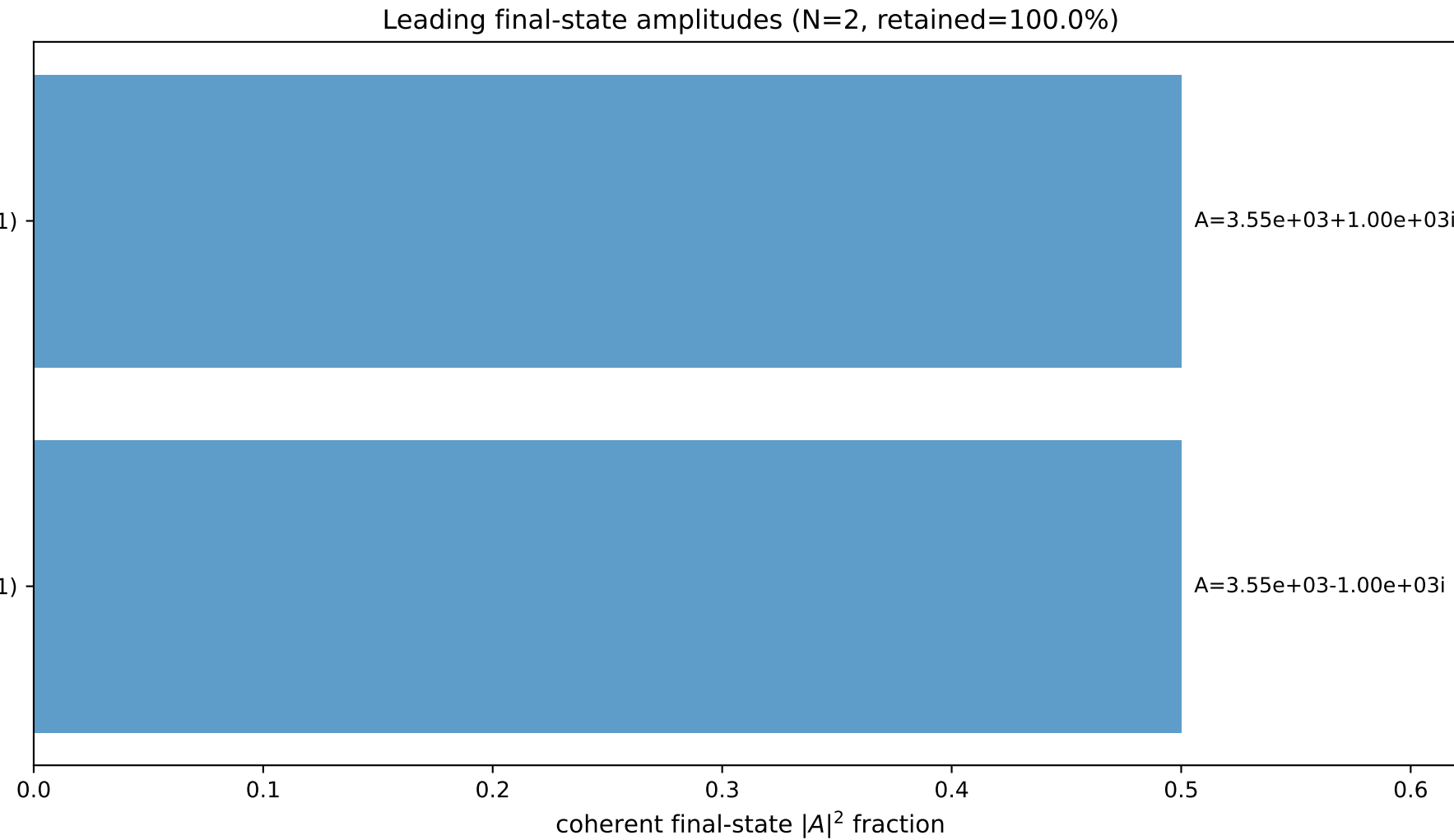
$s=5.04216$, $\sqrt{s}=2.24548$, $|\vec{P}|=0.926823$, $|\vec{P}'|=0.000424397$
 $\theta_{P'}=0.0244936$, $\theta_Y=3.14159$, $\phi_{P'}=1.33005$, $\phi_Y=2.86593$
 $E_Y=0.65395$, $\theta_{\ell'}=1.59044\text{e-}05$, $\phi_{\ell'}=4.47164$
 $Q^2=2.42281$, $x_B=0.663752$, $t=-0.713318$, $W^2=2.10721$, $y=0.876957$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.9268$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9268$
 P $E= 1.3187$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9268$
 ℓ' $E= 0.6535$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.6535$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0004$
 q_Y $E= 0.6539$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.6539$

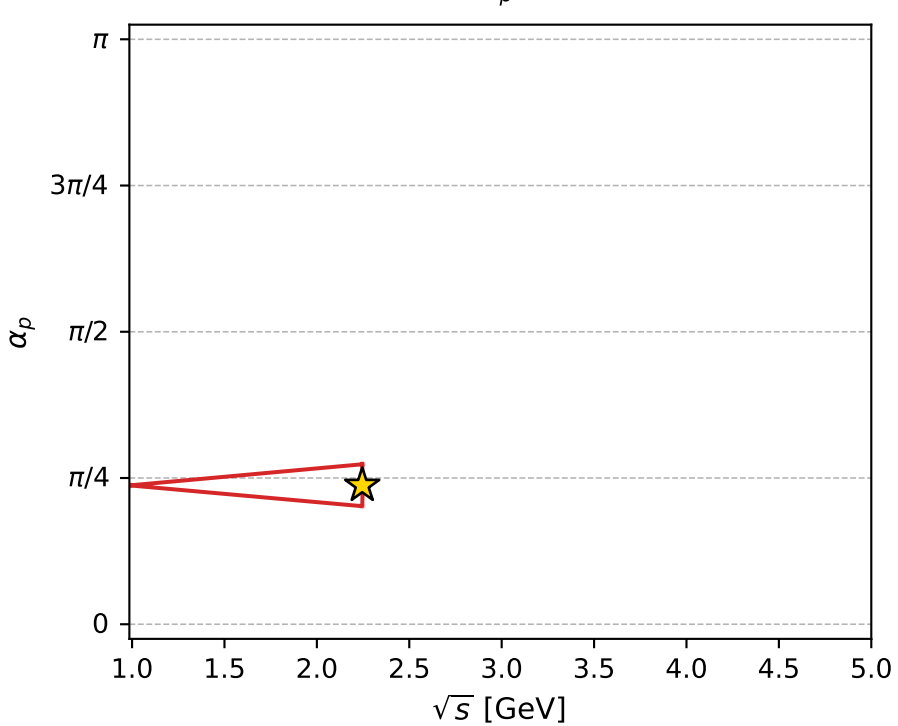
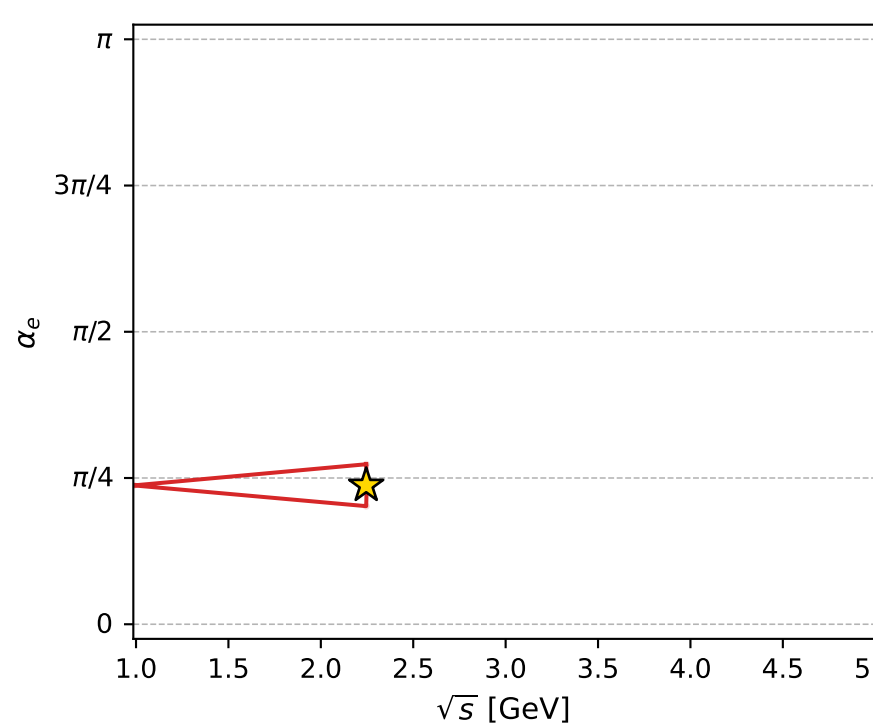
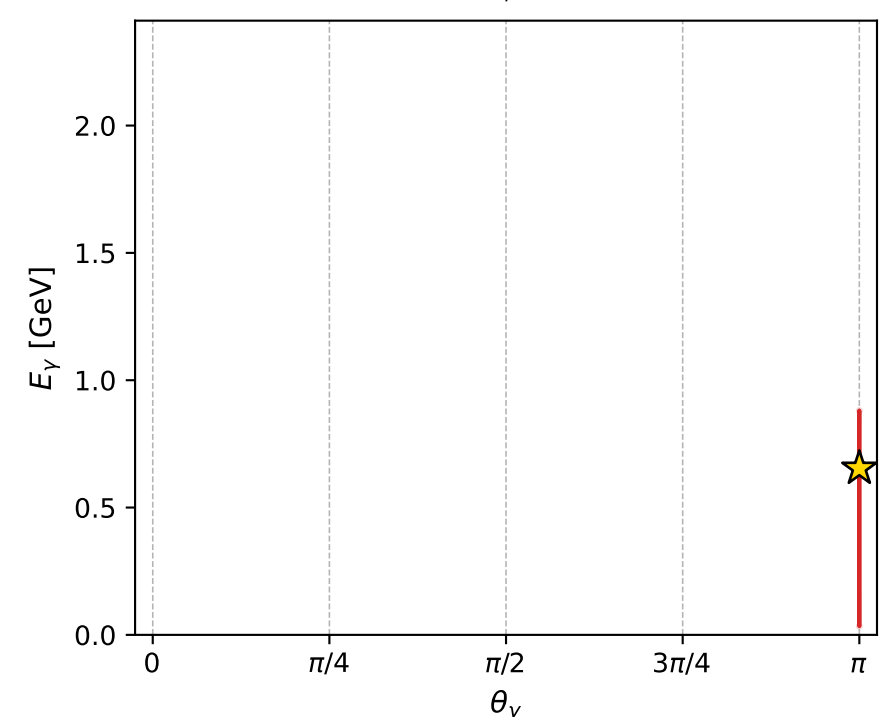
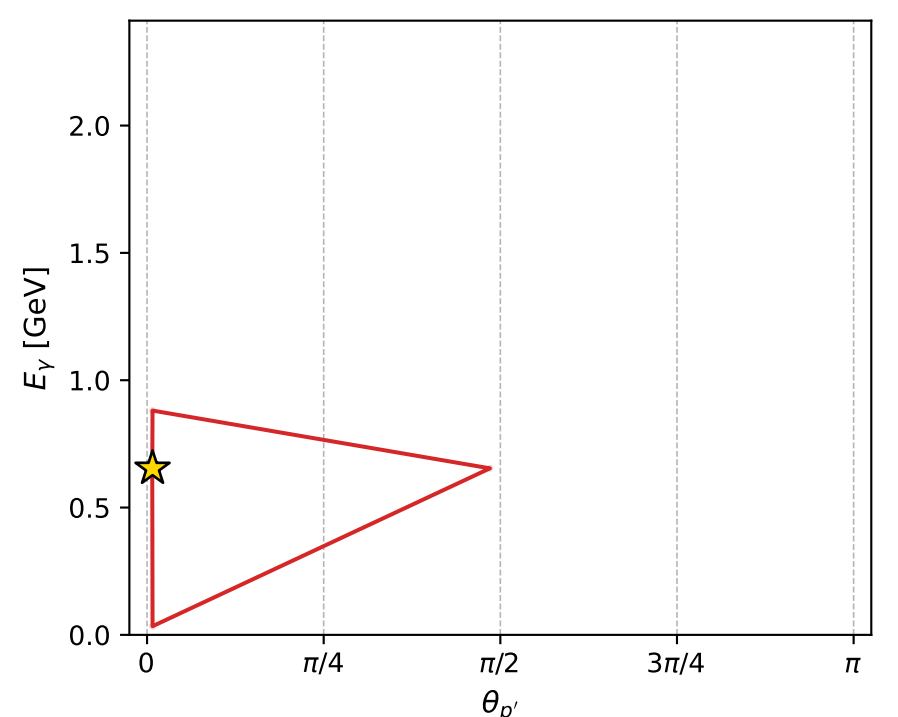
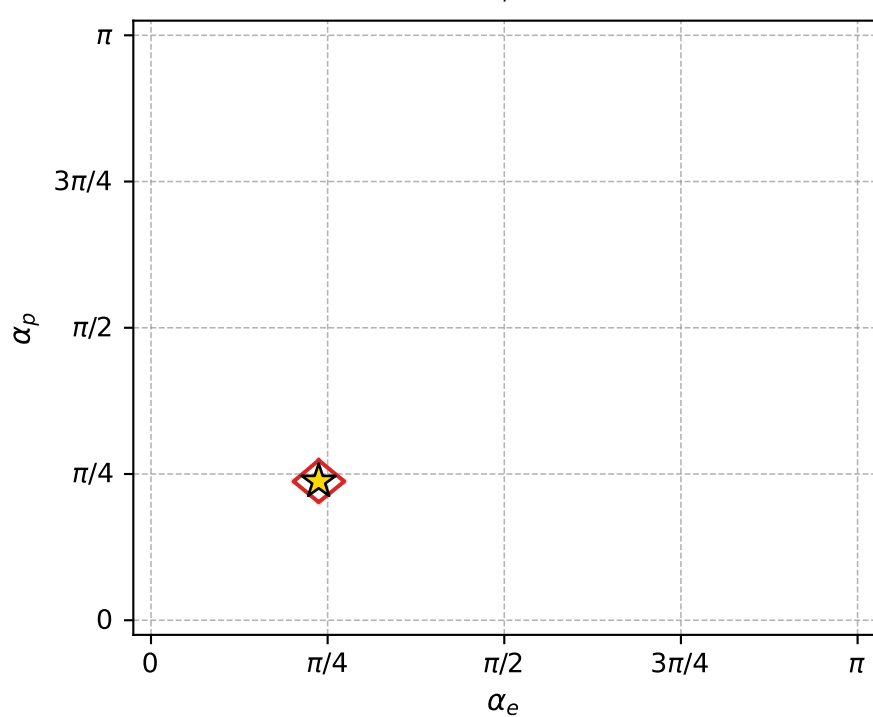
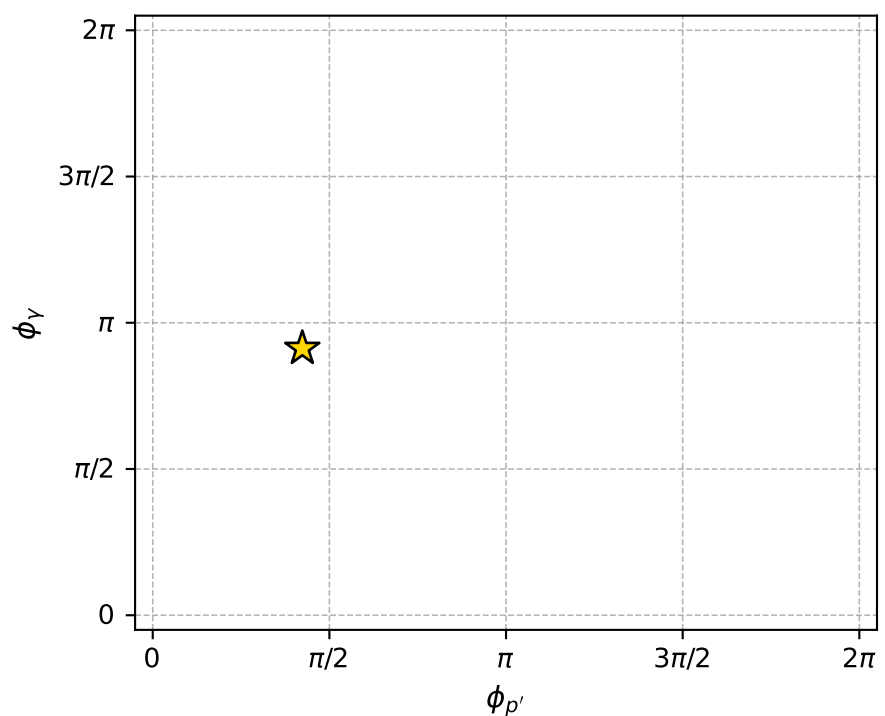
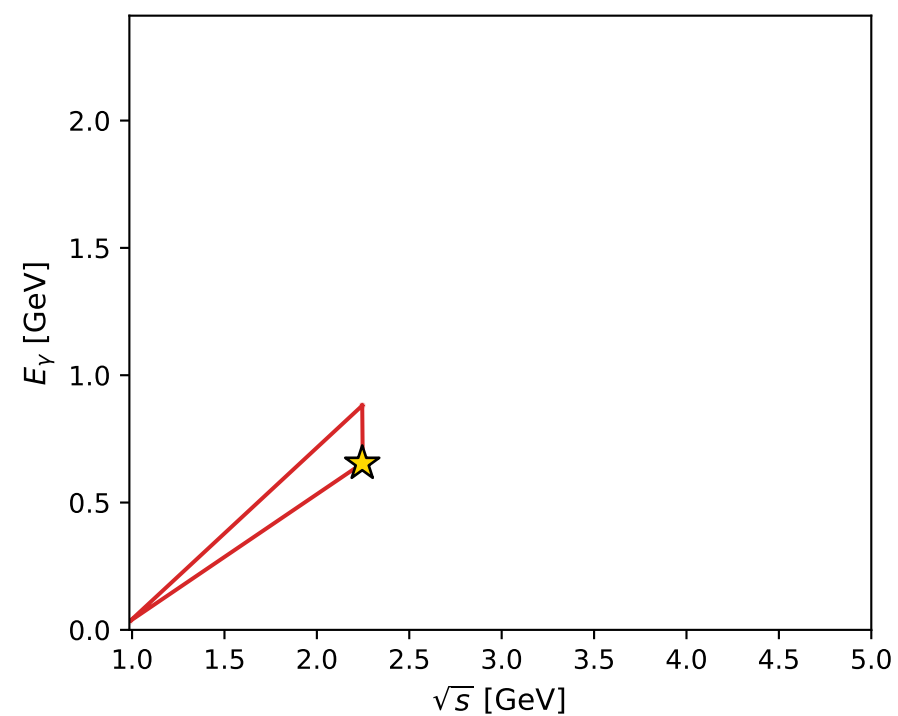
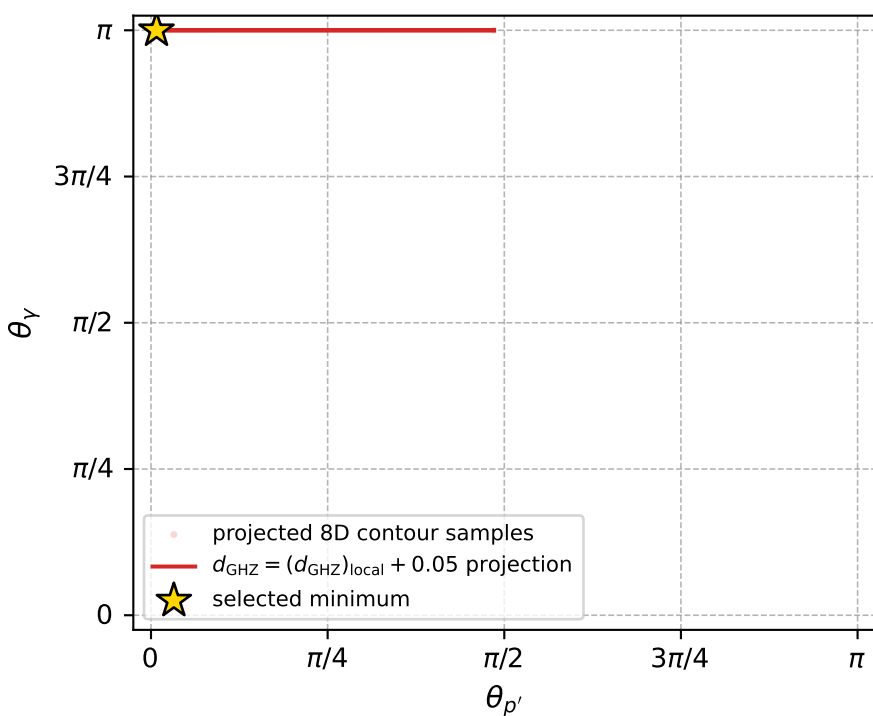


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$



electron: polarization cluster P1, local minimum 172; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.1503375\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000215$
 above global minimum = $1.9275028\text{e-}07$
 8D contour samples = 134

selected phase-space point:

$\text{sqrt_s} = 2.245476$
 $\text{theta_p_out} = 0.02449358$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.6539498$
 $\text{phi_p_out} = 1.330048$
 $\text{phi_gamma_out} = 2.865926$
 $\text{alpha_e} = 0.7456733$
 $\text{alpha_p} = 0.7457249$

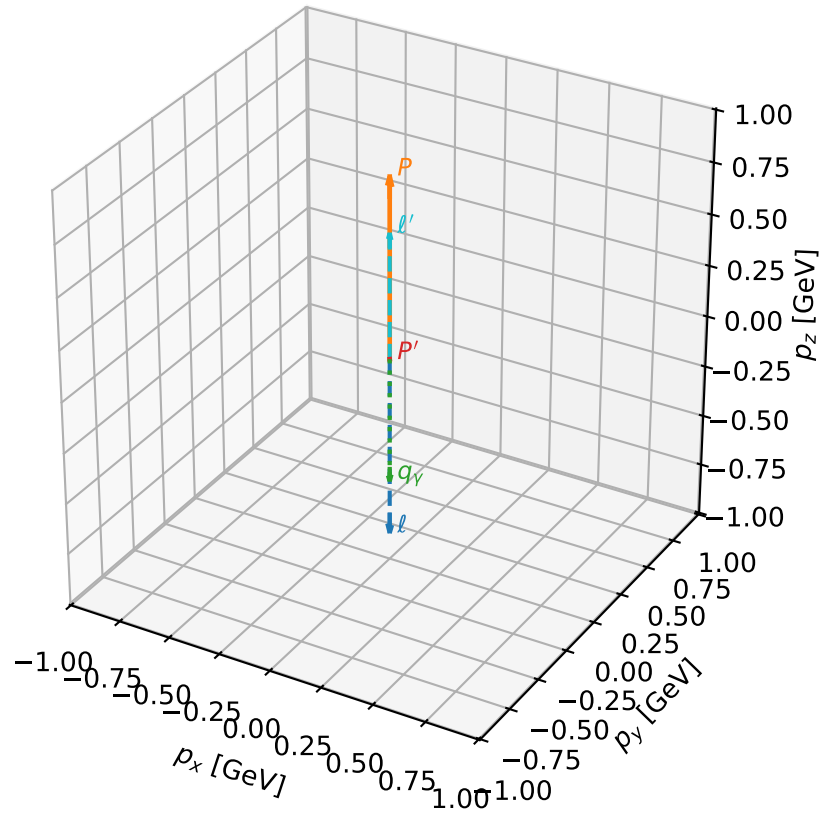
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_173 $d_{\{\text{rm GHZ}\}}=2.33712\text{e-}07$
 region: polarization_cluster_01_local_minimum_173
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.74702707$ rad
 incoming lepton: $0.733712|+\rangle +0.67946|-\rangle$
 proton mixing angle: $\alpha_p=0.7470235$ rad
 incoming proton: $0.733715|+\rangle +0.679458|-\rangle$

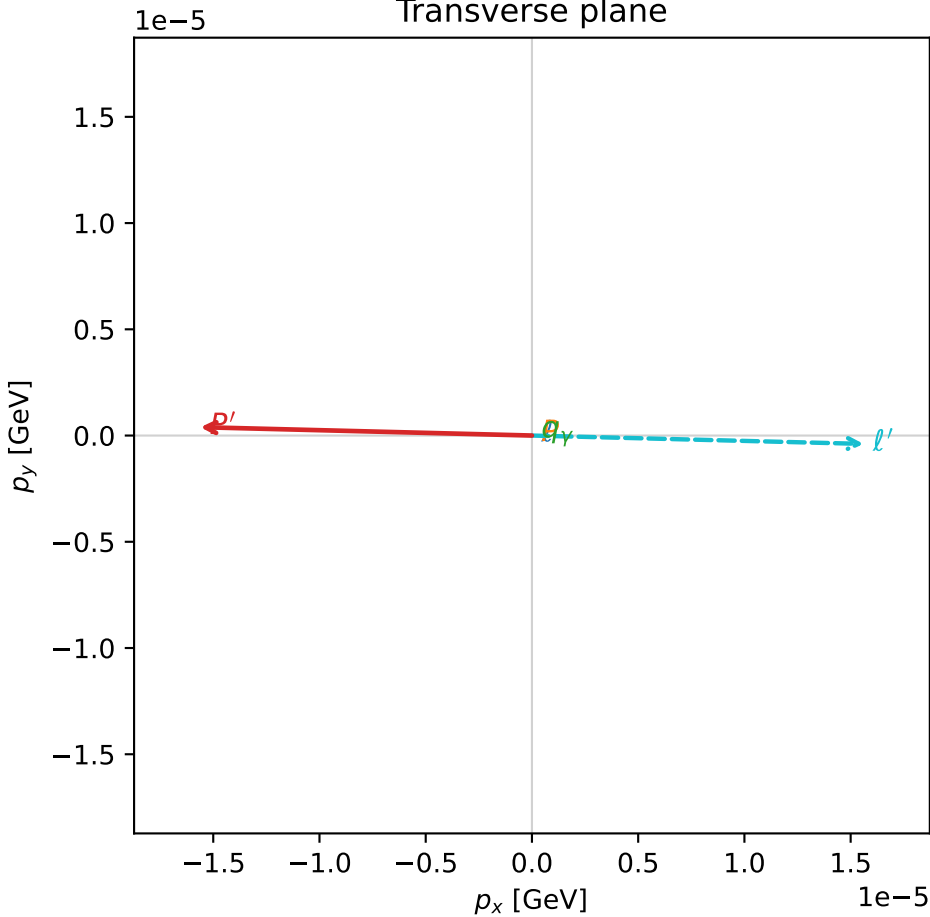
$s=4.65491$, $\sqrt{s}=2.15752$, $|\vec{P}|=0.874861$, $|\vec{P}'|=1.68118\text{e-}05$
 $\theta_{p'}=1.95121$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.11639$, $\phi_\gamma=0.976409$
 $E_\gamma=0.609759$, $\theta_{\ell'}=2.55999\text{e-}05$, $\phi_{\ell'}=6.25798$
 $Q^2=2.13384$, $x_B=0.651009$, $t=-0.646599$, $W^2=2.02374$, $y=0.86826$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.8749$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.8749$
 P $E= 1.2827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8749$
 ℓ' $E= 0.6098$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6098$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_γ $E= 0.6098$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6098$

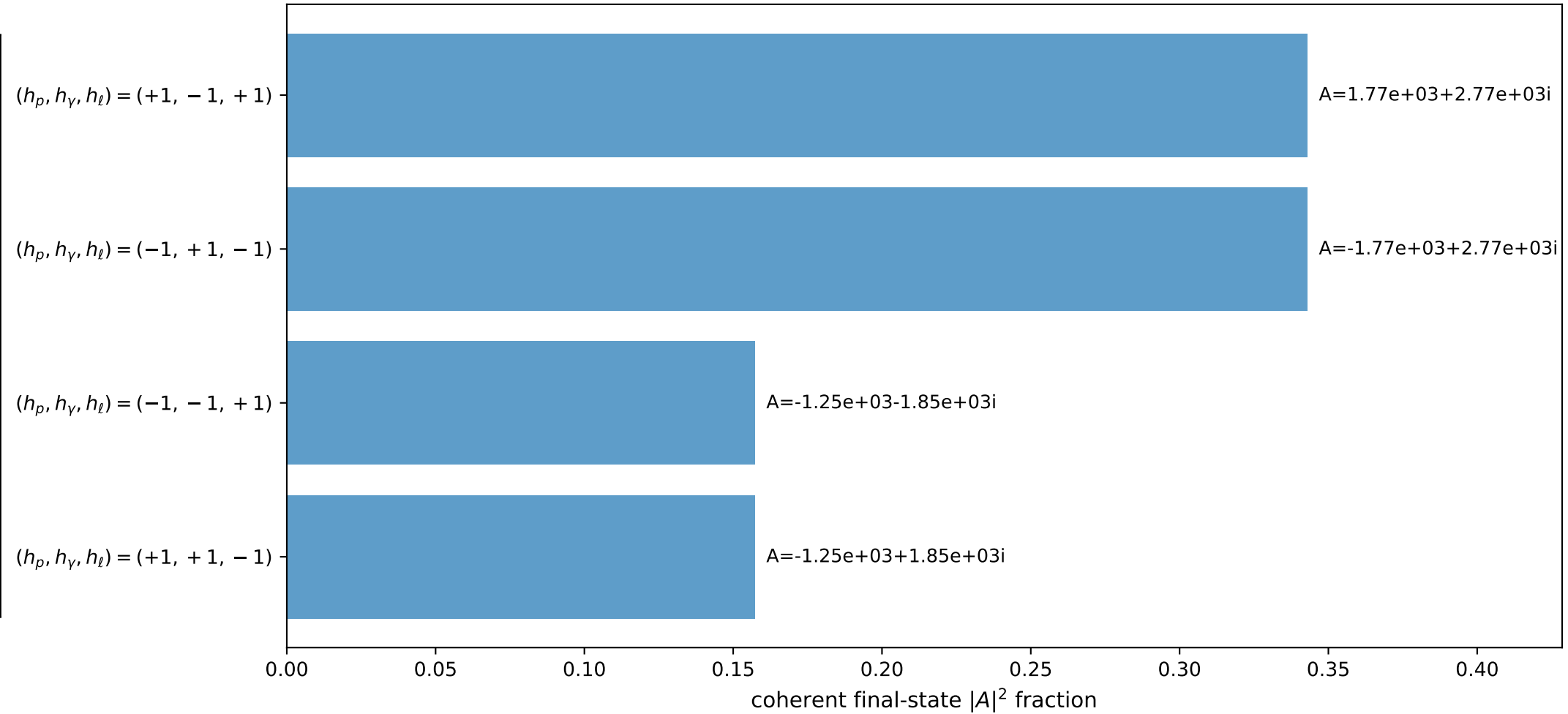
3D momenta



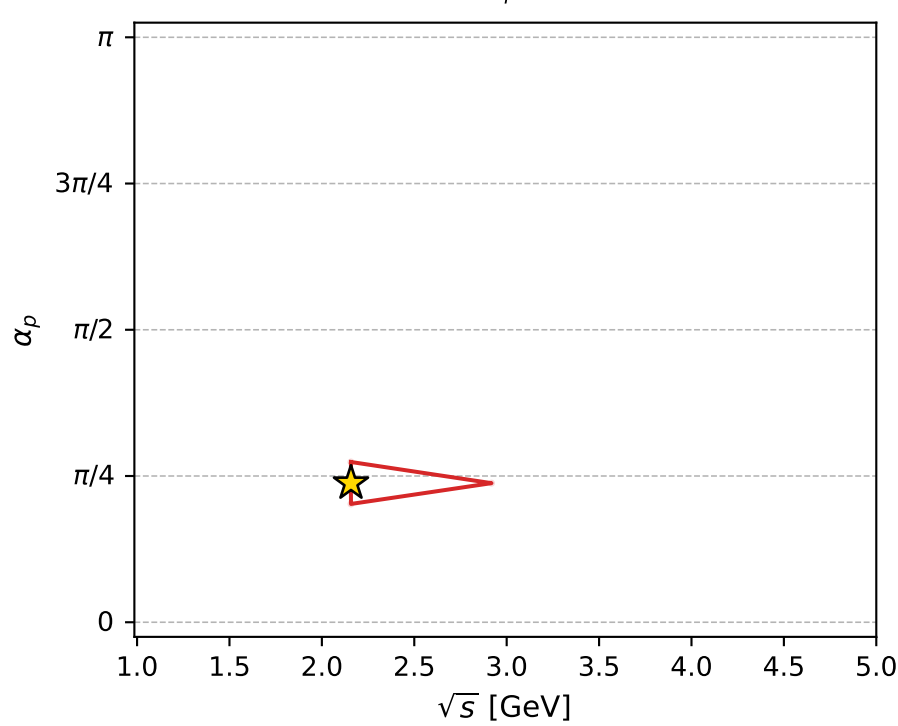
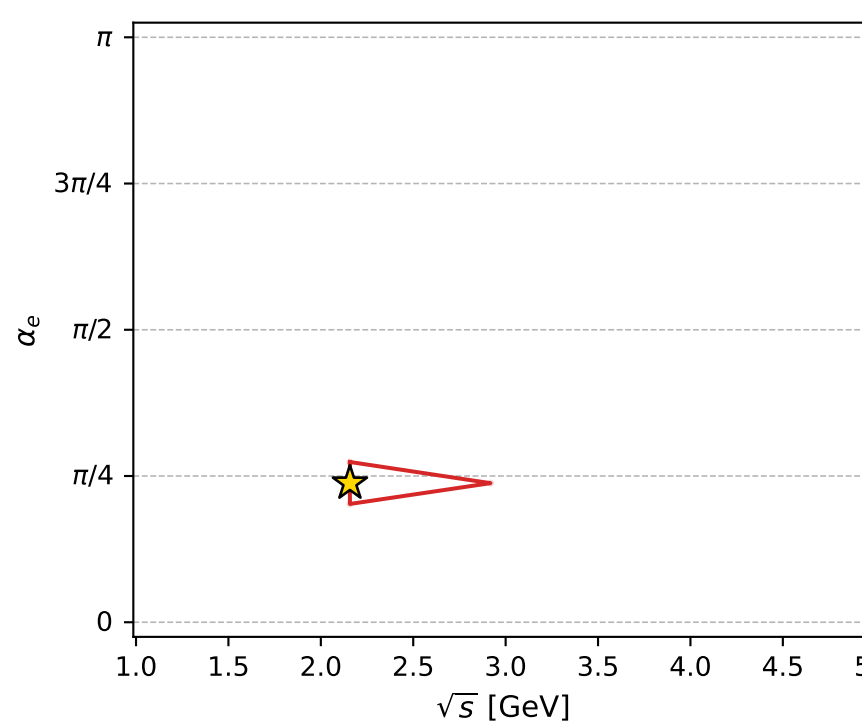
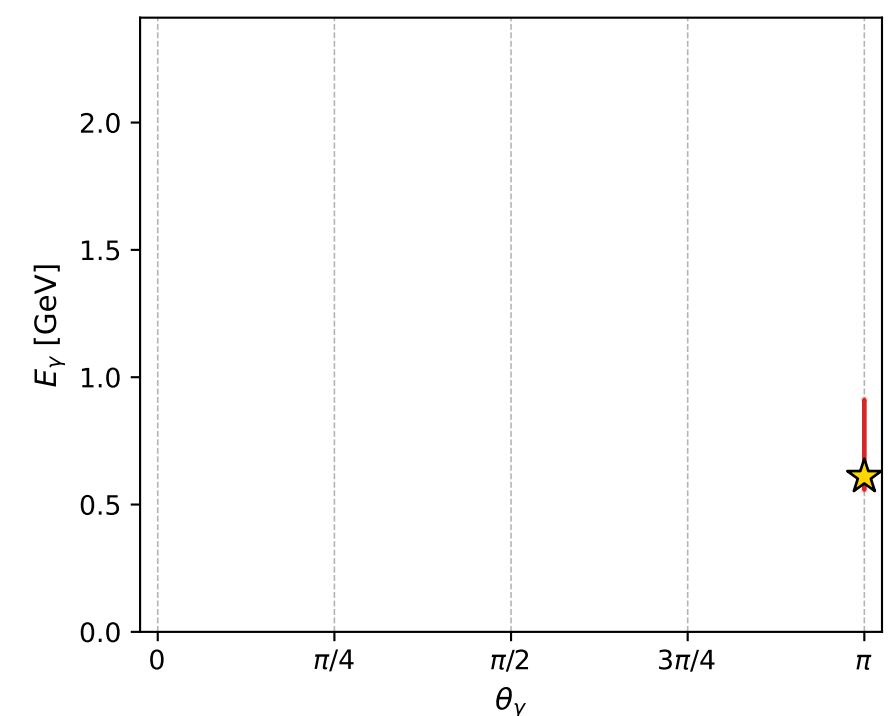
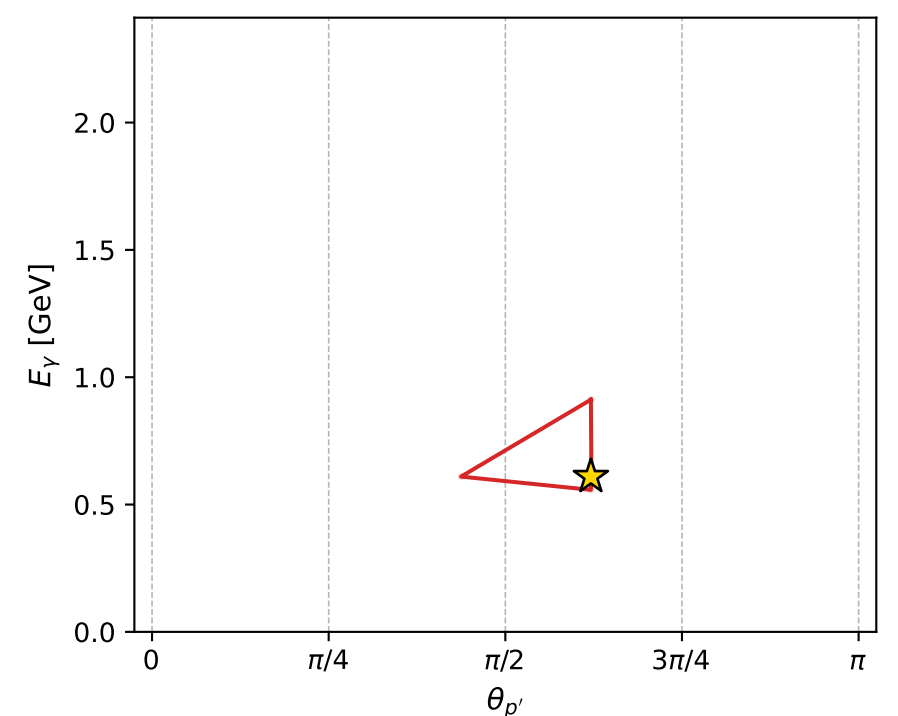
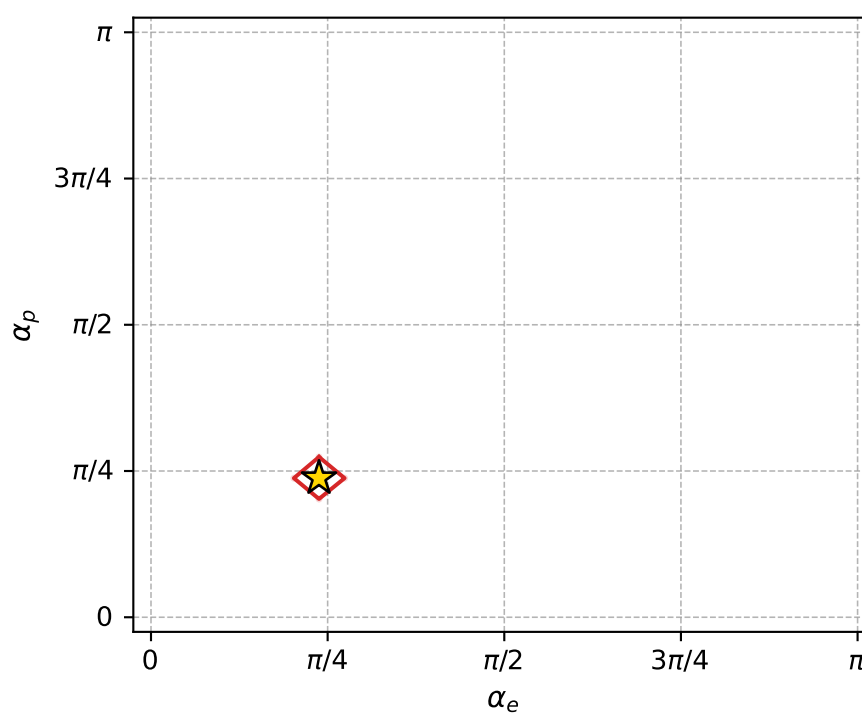
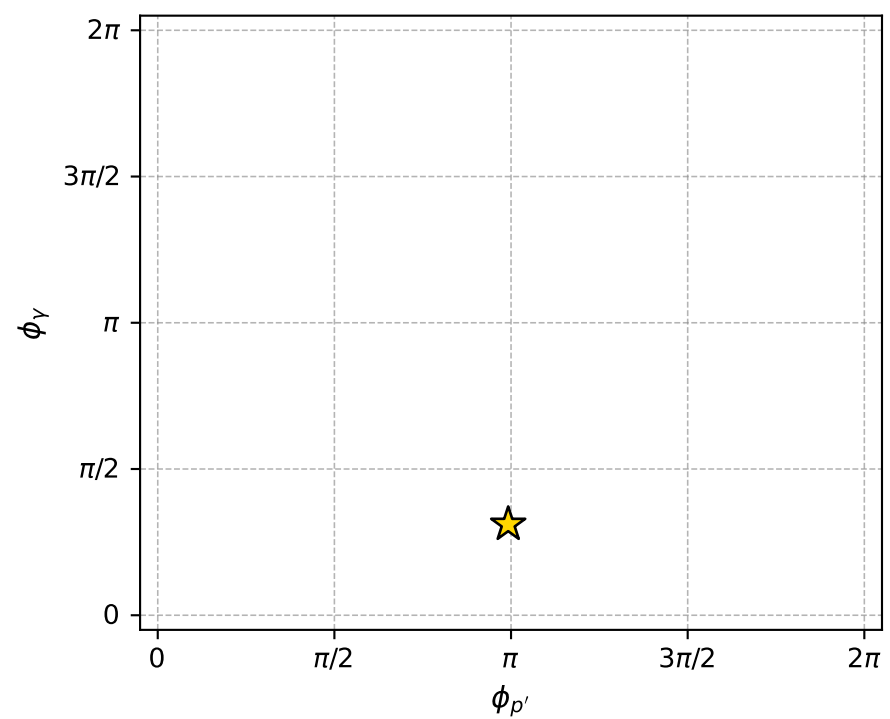
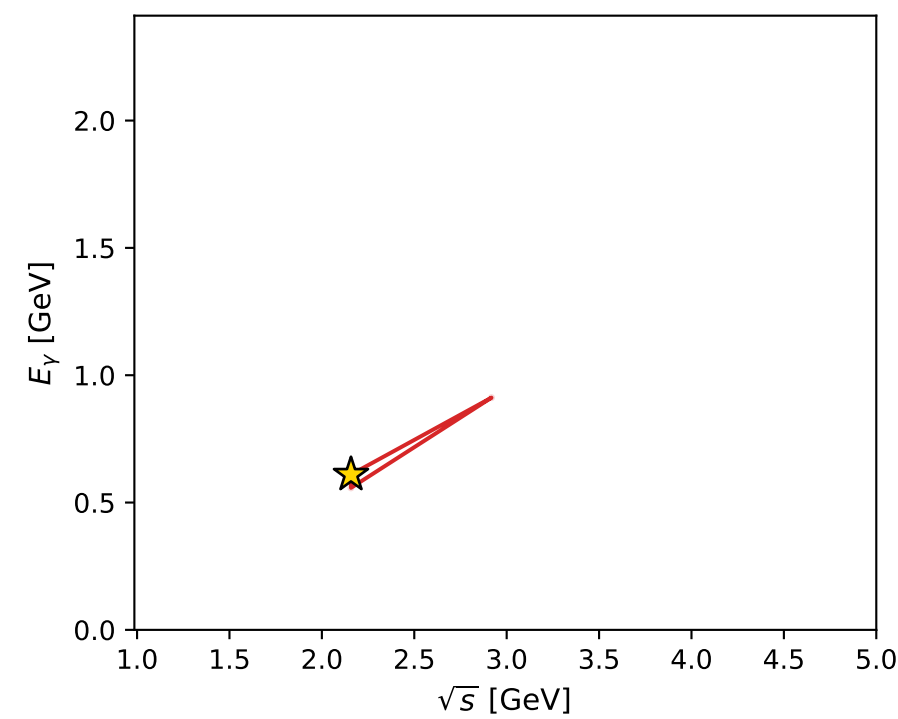
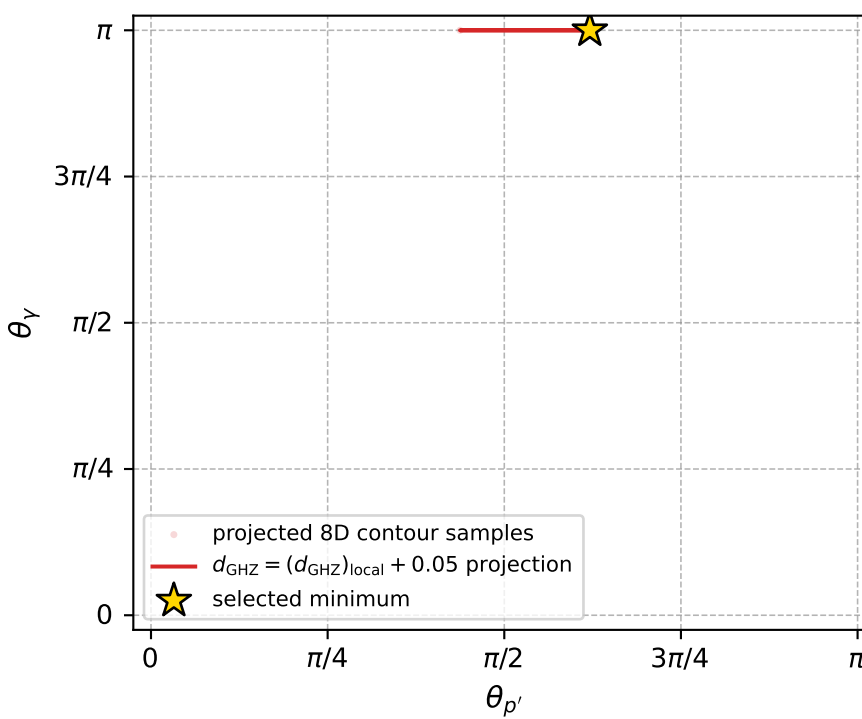
Transverse plane



Leading final-state amplitudes (N=4, retained=100.0%)



electron: polarization cluster P1, local minimum 173; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.3371198\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000234$
 above global minimum = $2.1142851\text{e-}07$
 8D contour samples = 135

selected phase-space point:

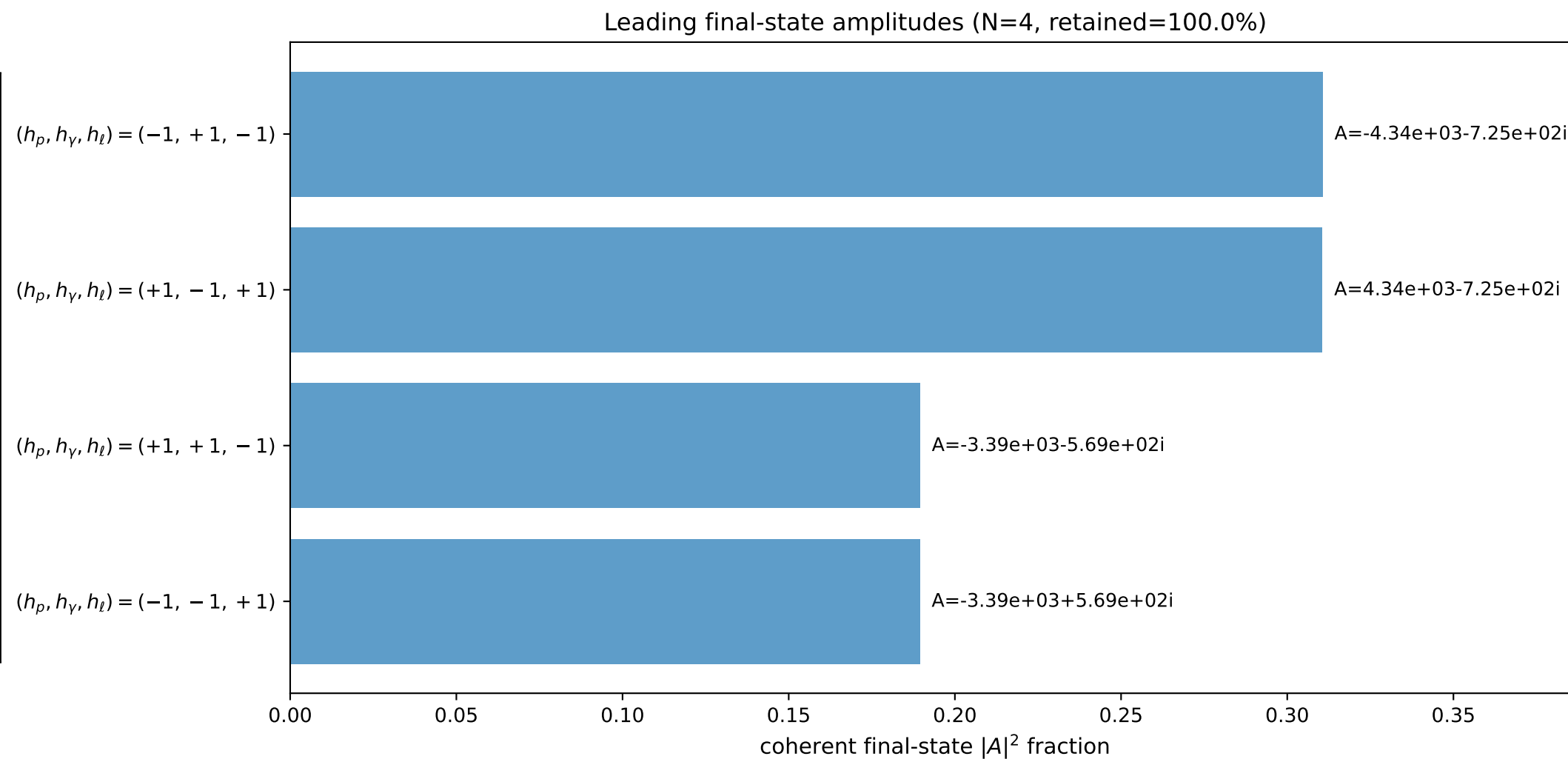
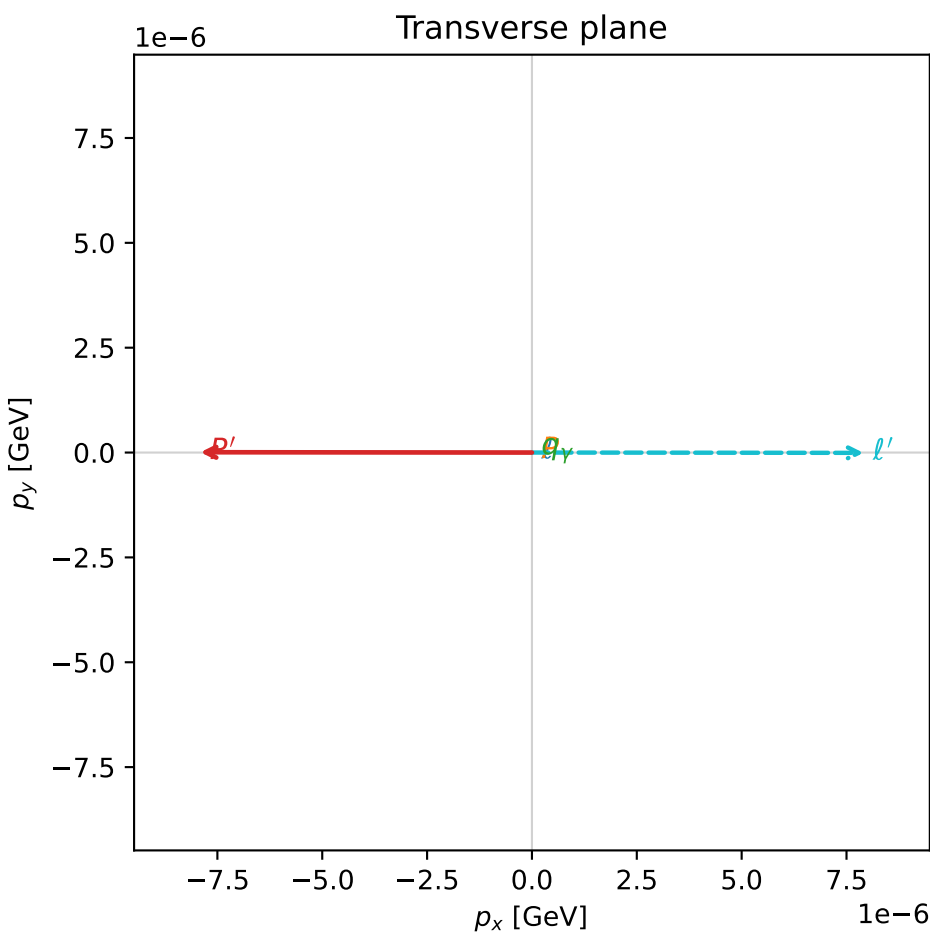
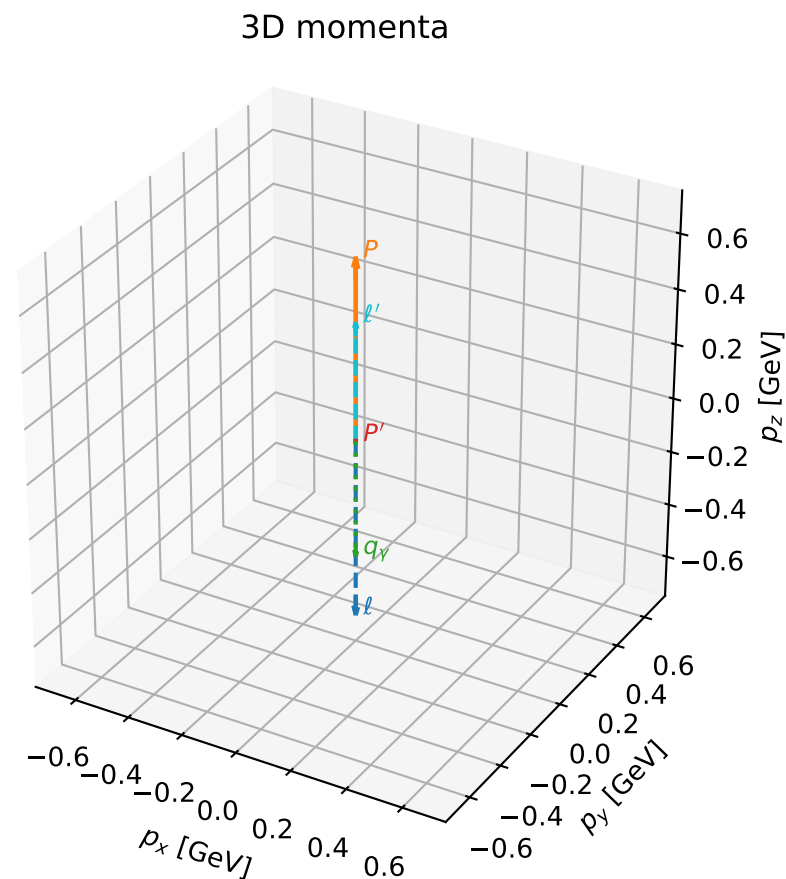
$\text{sqrt_s} = 2.157524$
 $\text{theta_p_out} = 1.951209$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.6097589$
 $\text{phi_p_out} = 3.11639$
 $\text{phi_gamma_out} = 0.9764086$
 $\text{alpha_e} = 0.7470271$
 $\text{alpha_p} = 0.7470235$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

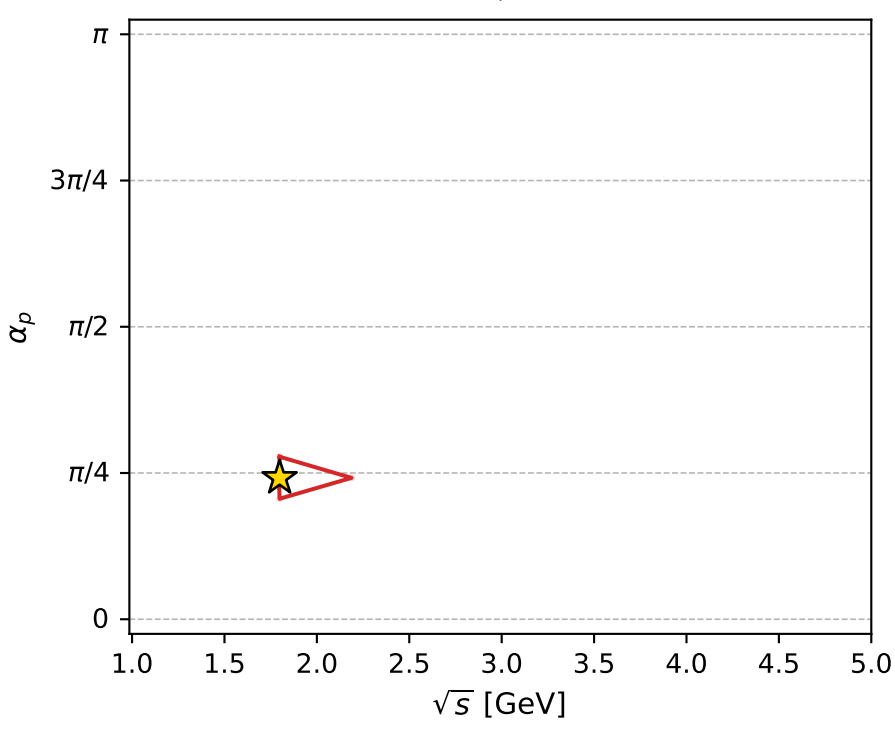
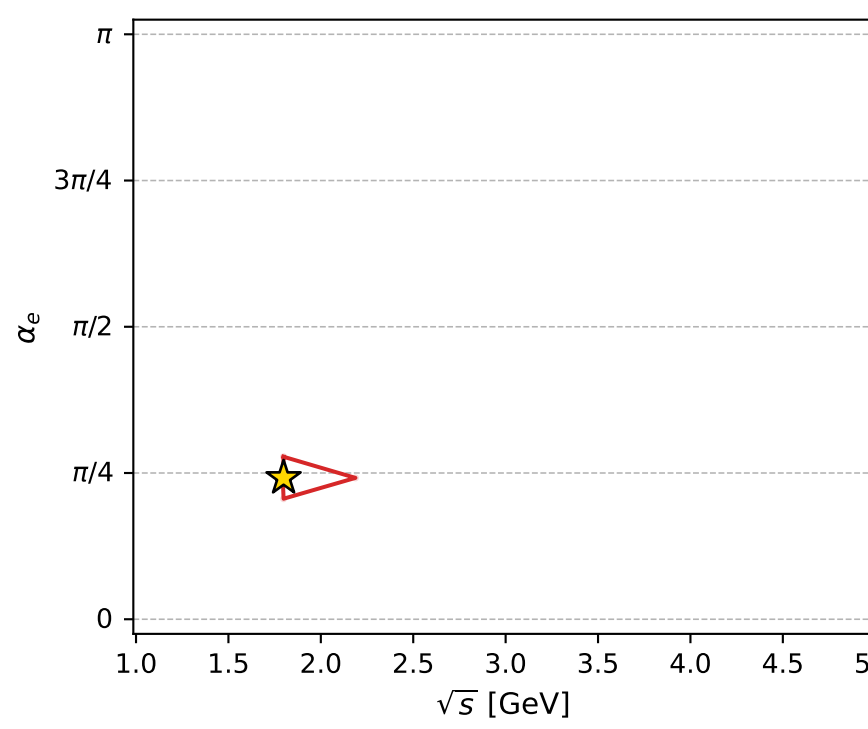
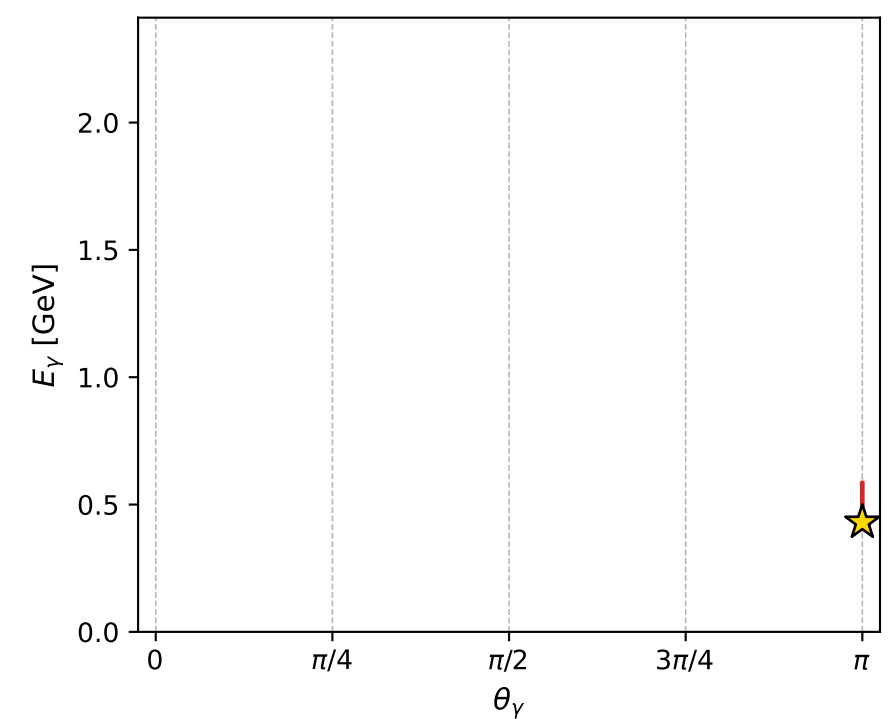
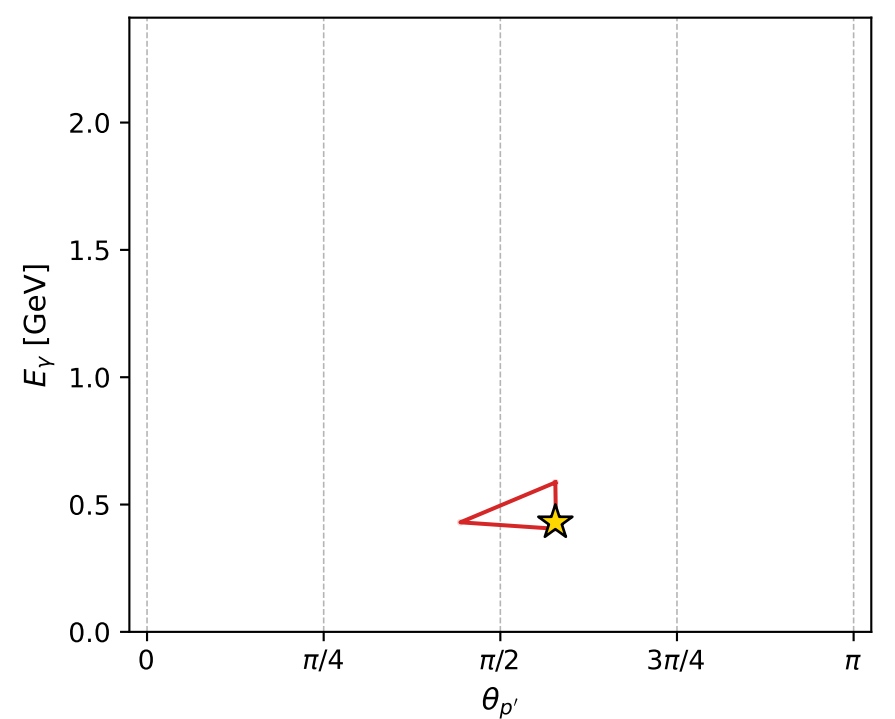
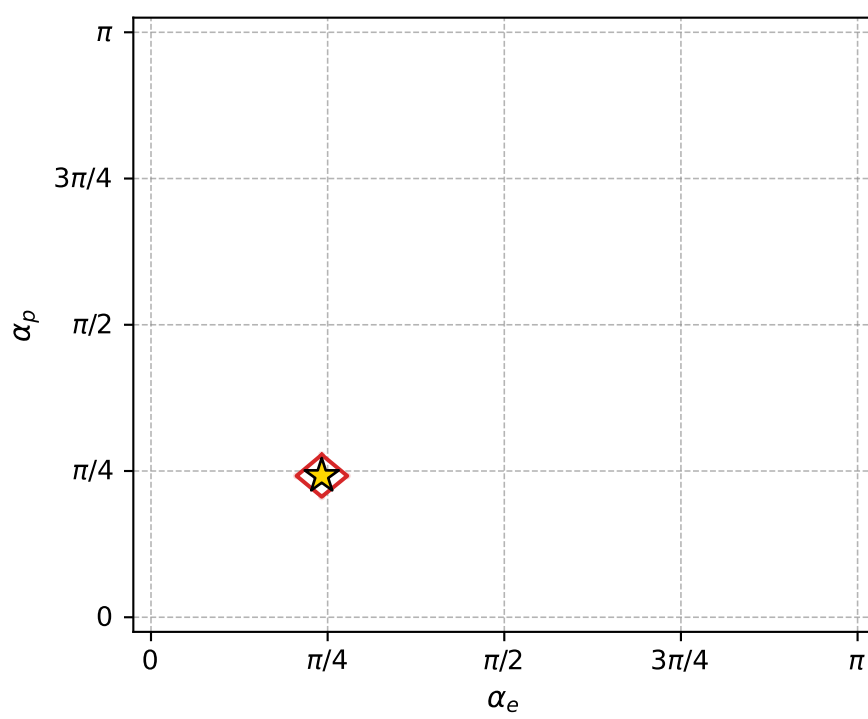
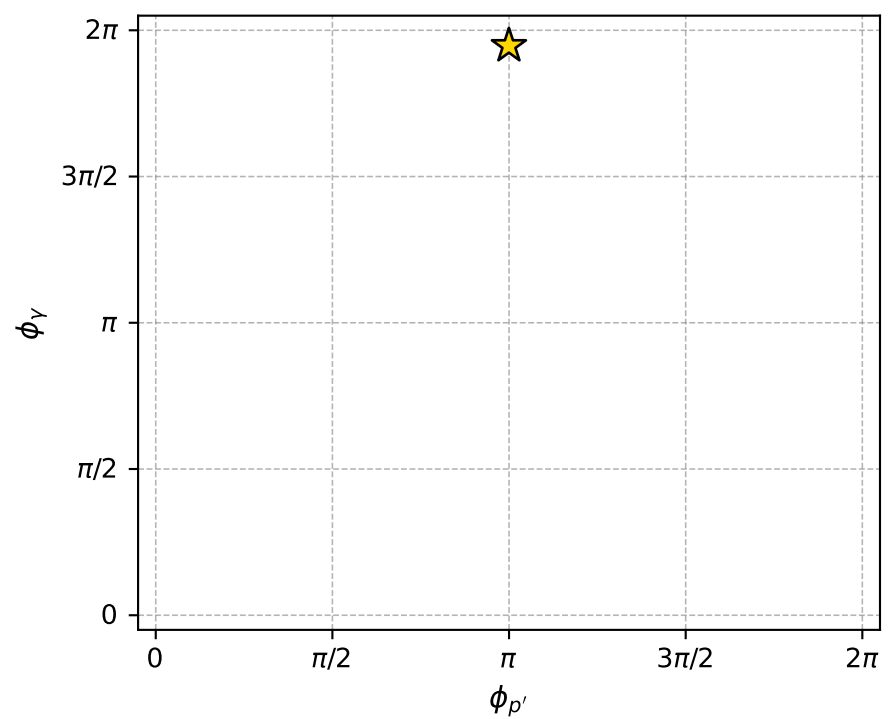
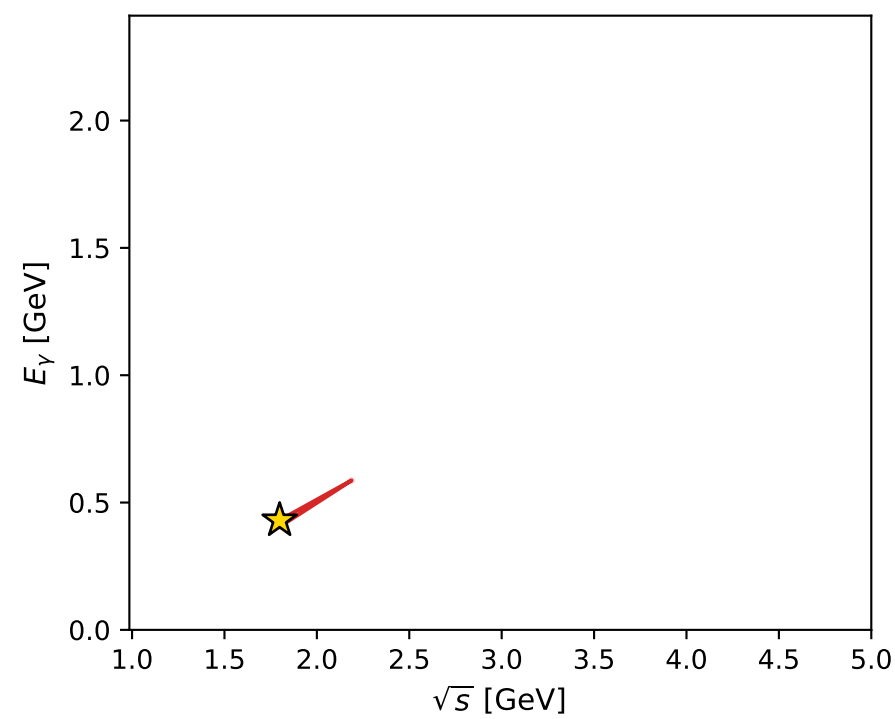
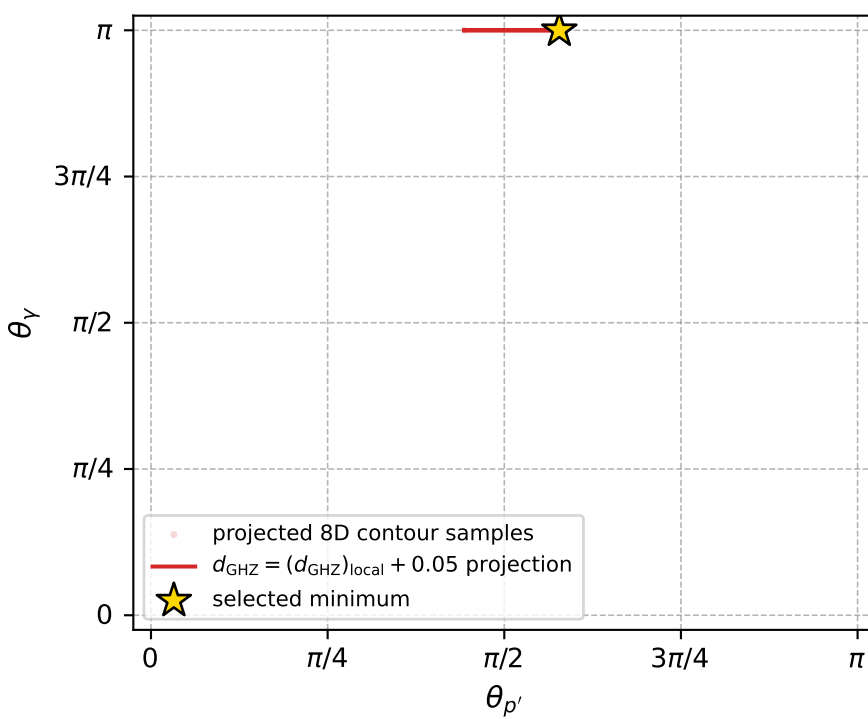
dghz_mixing_angles_polarization_01_local_minimum_179 d_{\rm GHZ}=4.53824e-07
region: polarization_cluster_01_local_minimum_179
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.75918142$ rad
incoming lepton: $0.7254|+\rangle +0.688328|-\rangle$
proton mixing angle: $\alpha_p=0.75924297$ rad
incoming proton: $0.725357|+\rangle +0.688373|-\rangle$

$s=3.23451$, $\sqrt{s}=1.79847$, $|\vec{P}|=0.654628$, $|\vec{P}'|=8.14786e-06$
 $\theta_{p'}=1.81546$, $\theta_Y=3.14159$, $\phi_{p'}=3.14064$, $\phi_Y=6.11665$
 $E_Y=0.430236$, $\theta_{l'}=1.8374e-05$, $\phi_{l'}=6.28224$
 $Q^2=1.12658$, $x_B=0.582604$, $t=-0.386168$, $W^2=1.68696$, $y=0.821223$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.6546$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6546$
 P $E= 1.1438$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6546$
 l' $E= 0.4302$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.4302$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_Y $E= 0.4302$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.4302$



electron: polarization cluster P1, local minimum 179; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.5382447\text{e-}07$
 $d_{\text{GHZ}} \text{ contour} = 0.050000454$
 above global minimum = $4.3154101\text{e-}07$
 8D contour samples = 124

selected phase-space point:

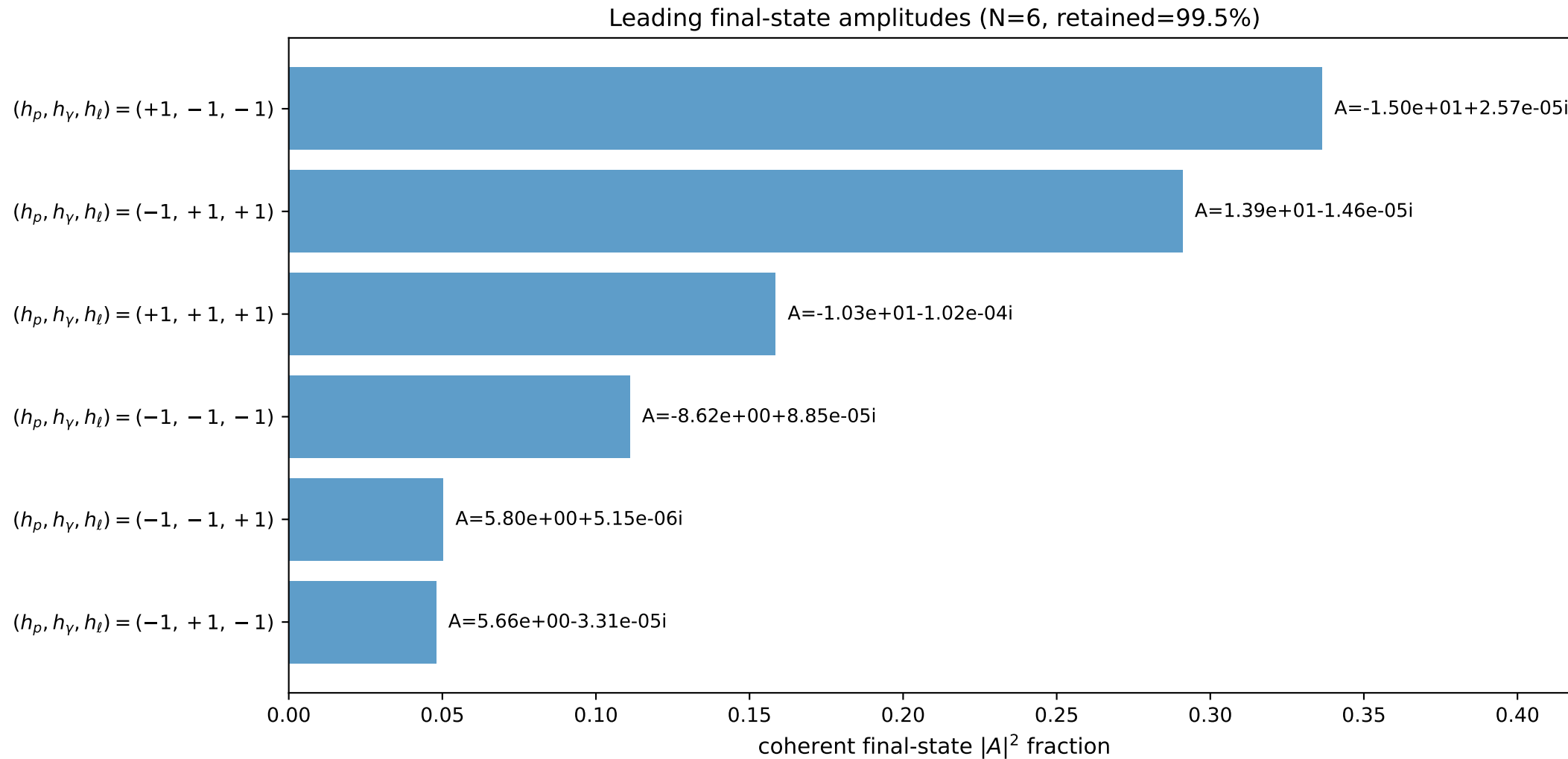
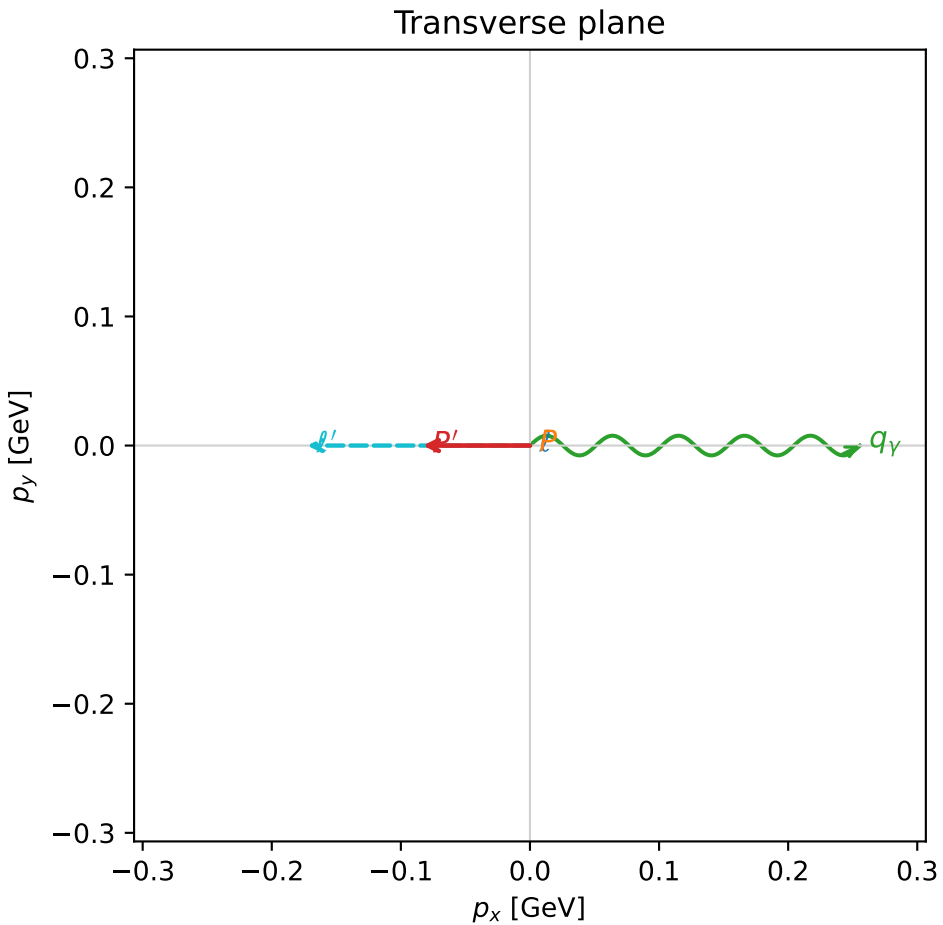
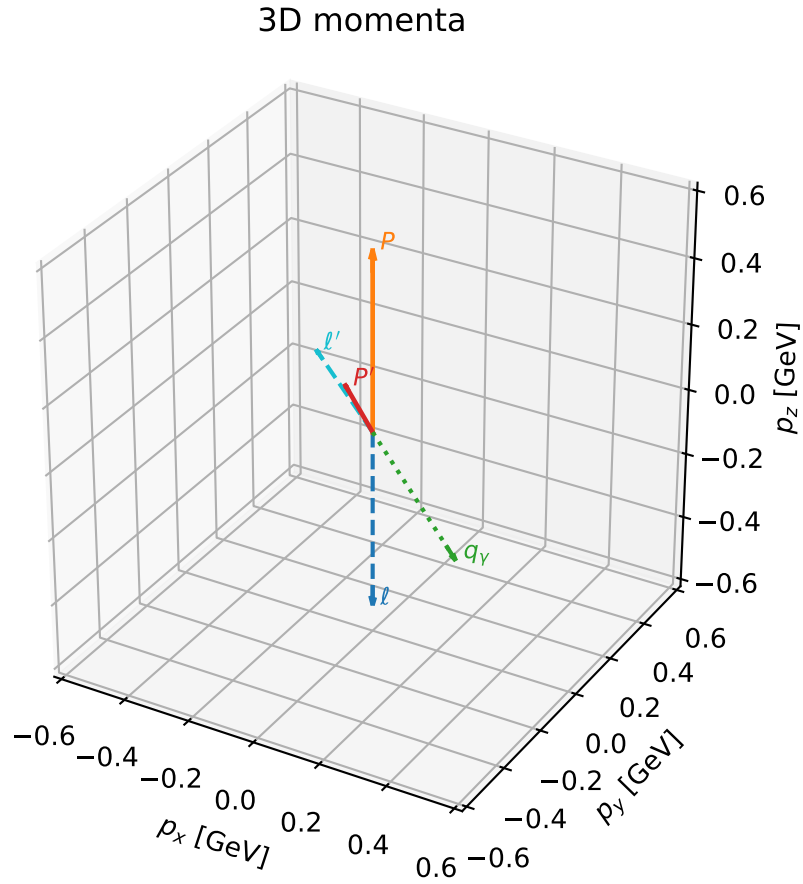
$\text{sqrt_s} = 1.798473$
 $\text{theta_p_out} = 1.815463$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.4302356$
 $\text{phi_p_out} = 3.140642$
 $\text{phi_gamma_out} = 6.116649$
 $\text{alpha_e} = 0.7591814$
 $\text{alpha_p} = 0.759243$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

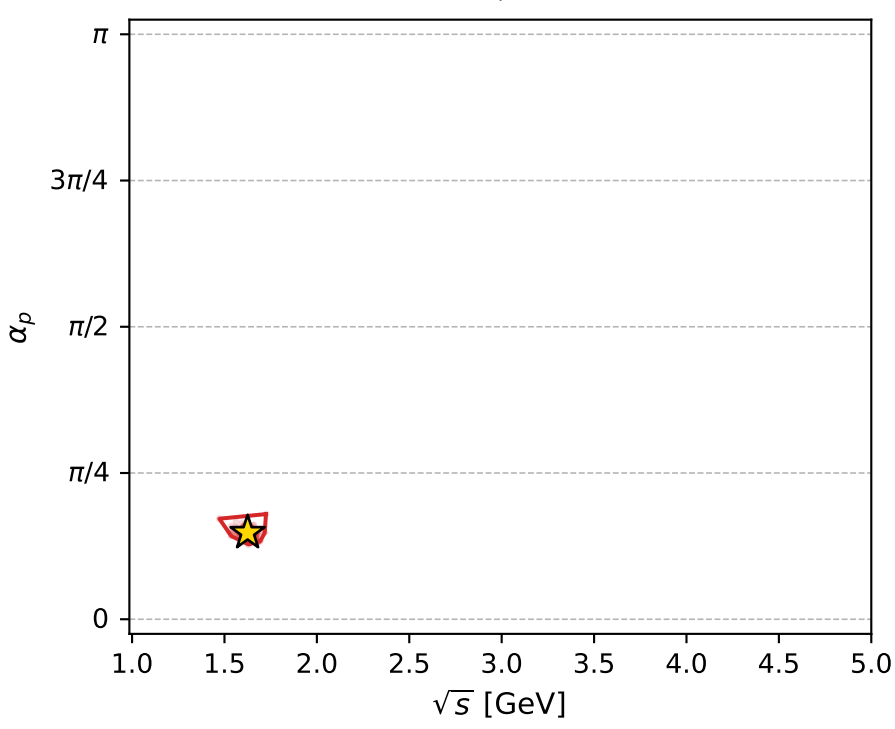
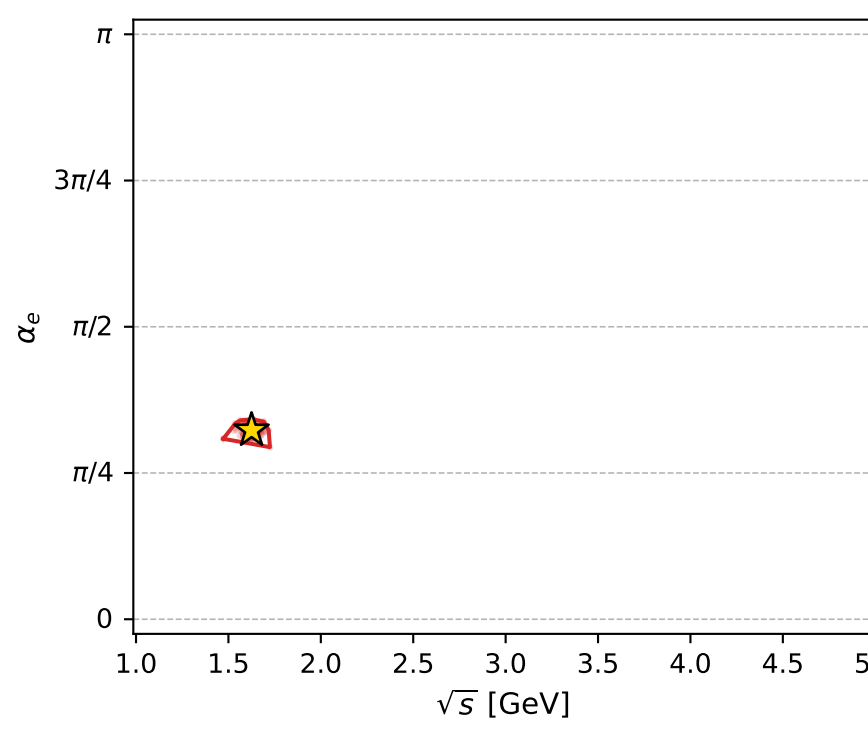
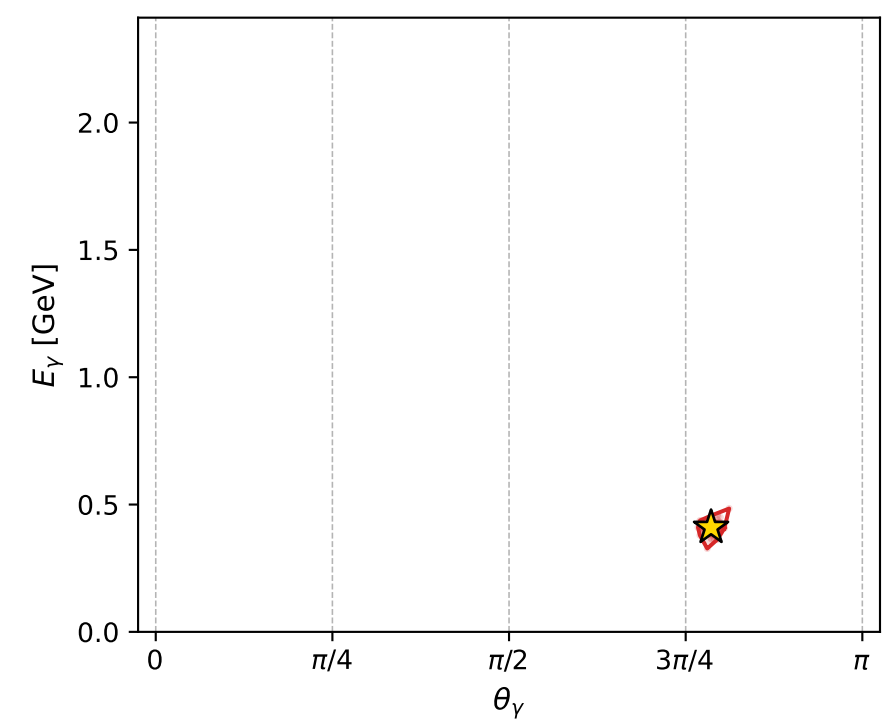
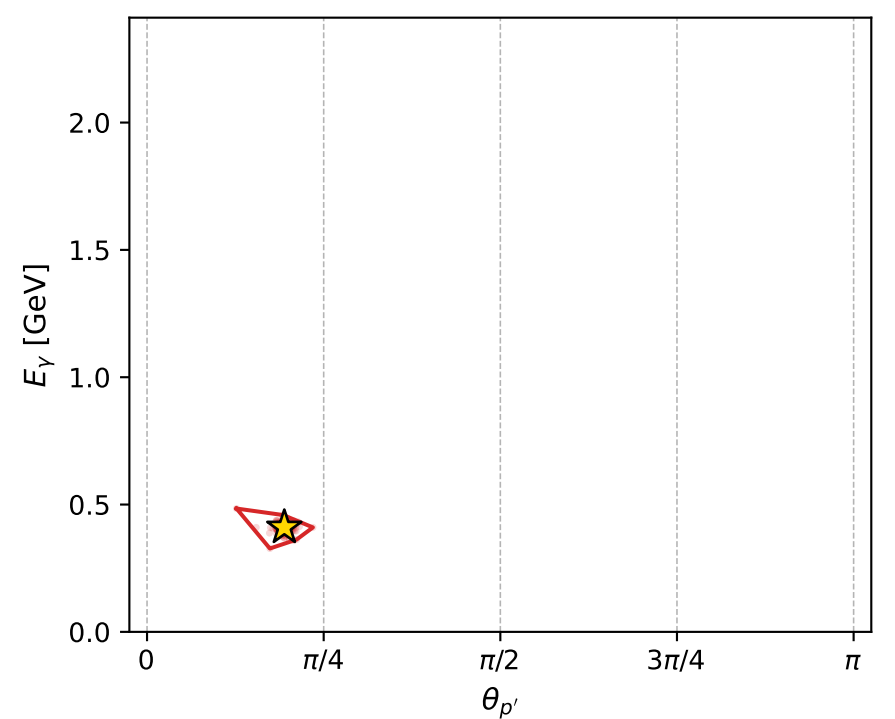
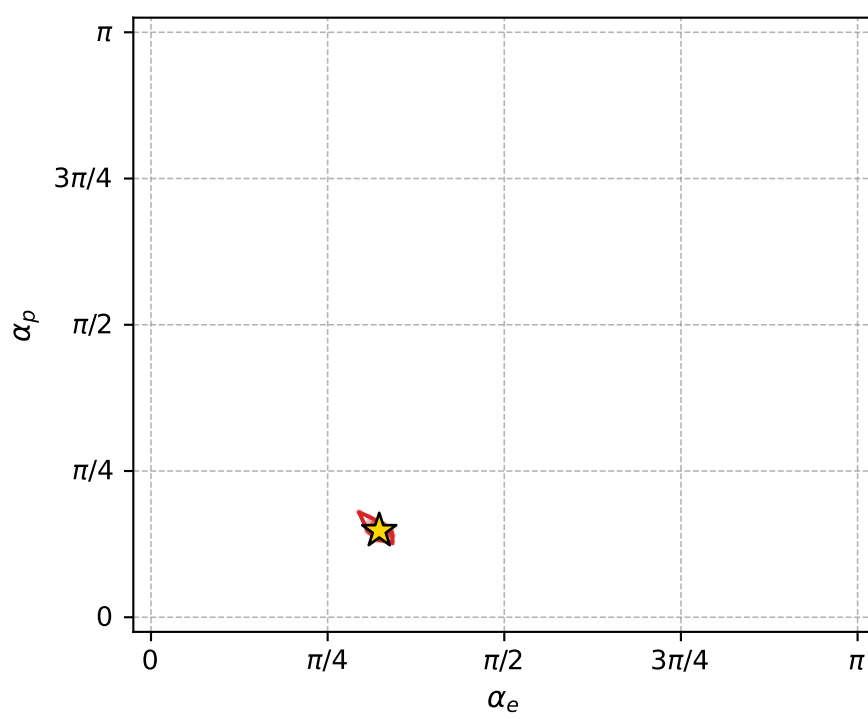
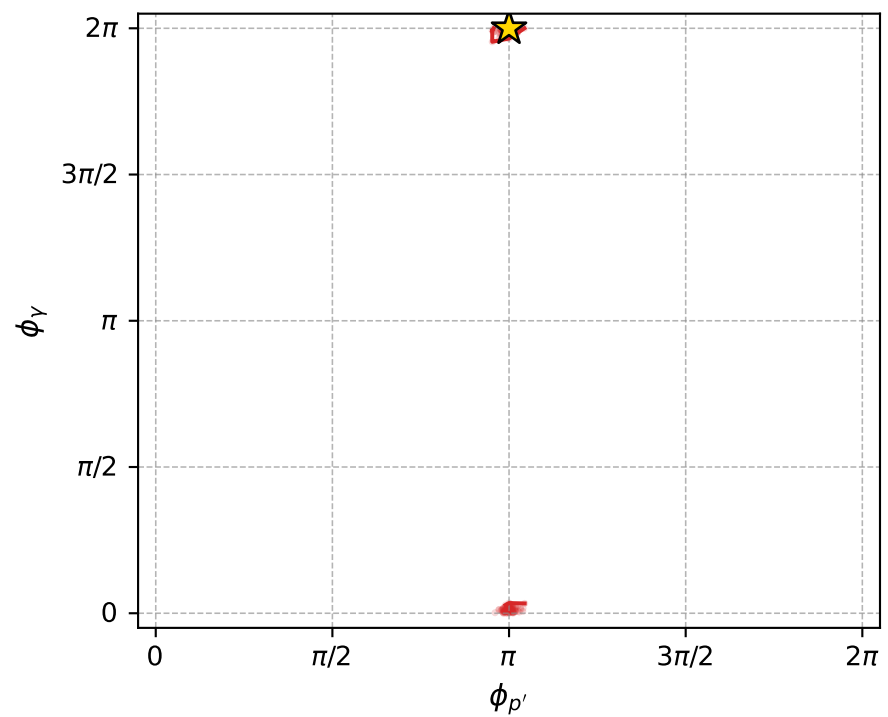
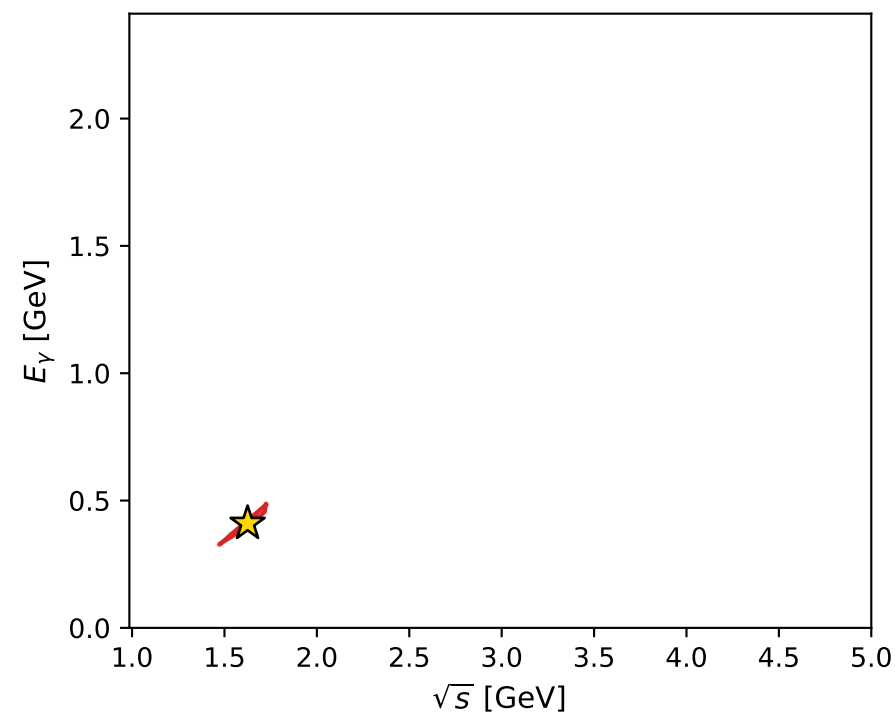
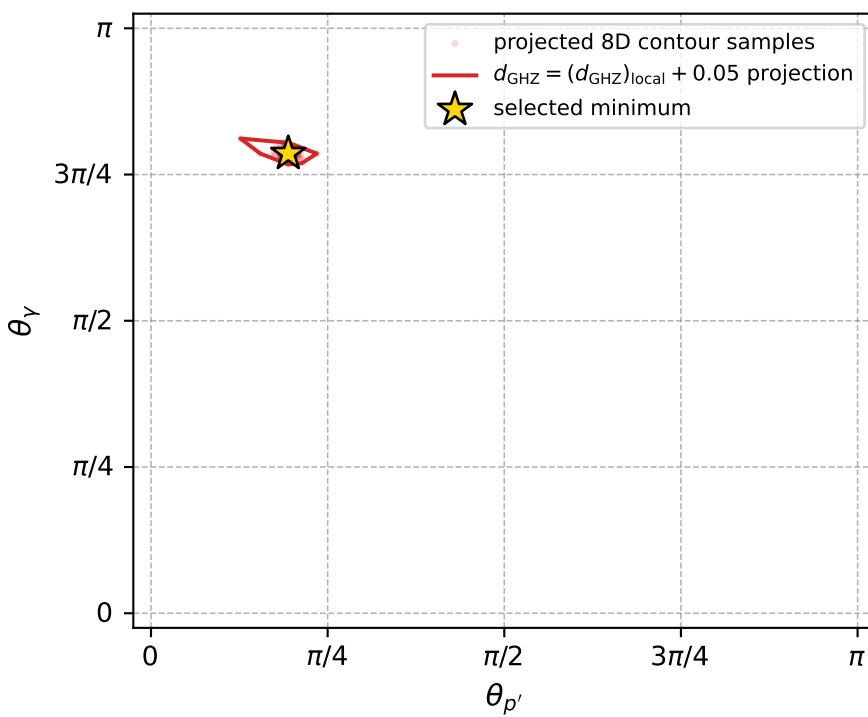
dghz_mixing_angles_polarization_01_local_minimum_187 $d_{\{\text{r m GHZ}\}}=1.10771\text{e-}05$
 region: polarization_cluster_01_local_minimum_187
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.0148851$ rad
 incoming lepton: $0.527718|+\rangle +0.84942|-\rangle$
 proton mixing angle: $\alpha_p=0.46484743$ rad
 incoming proton: $0.89389|+\rangle +0.448286|-\rangle$

$s=2.64114$, $\sqrt{s}=1.62516$, $|\vec{P}|=0.541884$, $|\vec{P}'|=0.144684$
 $\theta_{P'}=0.610365$, $\theta_Y=2.46897$, $\phi_{P'}=3.14158$, $\phi_Y=6.28318$
 $E_Y=0.410158$, $\theta_{\ell'}=0.706488$, $\phi_{\ell'}=3.14159$
 $Q^2=0.507384$, $x_B=0.361283$, $t=-0.168076$, $W^2=1.77686$, $y=0.797367$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5419$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5419$
 P $E= 1.0833$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5419$
 ℓ' $E= 0.2659$ $p_x= -0.1726$ $p_y= 0.0000$ $p_z= 0.2023$
 P' $E= 0.9491$ $p_x= -0.0829$ $p_y= 0.0000$ $p_z= 0.1186$
 q_Y $E= 0.4102$ $p_x= 0.2555$ $p_y= -0.0000$ $p_z= -0.3208$



electron: polarization cluster P1, local minimum 187; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.1077138\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050011077$
 above global minimum = $1.1054855\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.625157$
 $\text{theta_p_out} = 0.6103654$
 $\text{theta_gamma_out} = 2.468967$
 $\text{q0out} = 0.4101575$
 $\text{phi_p_out} = 3.141582$
 $\text{phi_gamma_out} = 6.283181$
 $\text{alpha_e} = 1.014885$
 $\text{alpha_p} = 0.4648474$

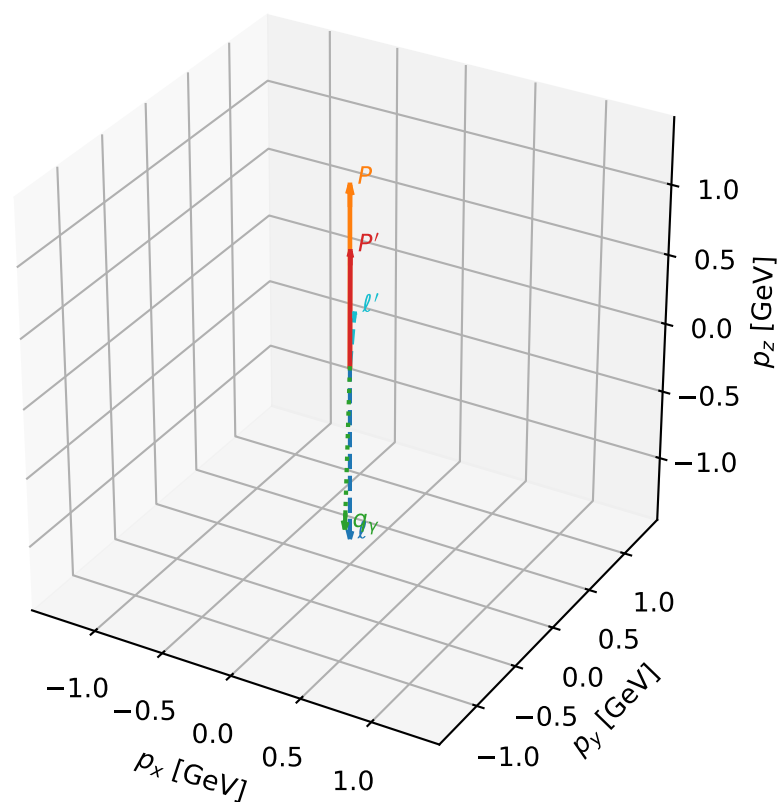
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_193 d_{\rm GHZ}=1.77496e-05
 region: polarization_cluster_01_local_minimum_193
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.24273564$ rad
 incoming lepton: $0.970684|+\rangle + 0.240359|-\rangle$
 proton mixing angle: $\alpha_p=1.3360225$ rad
 incoming proton: $0.232623|+\rangle + 0.972567|-\rangle$

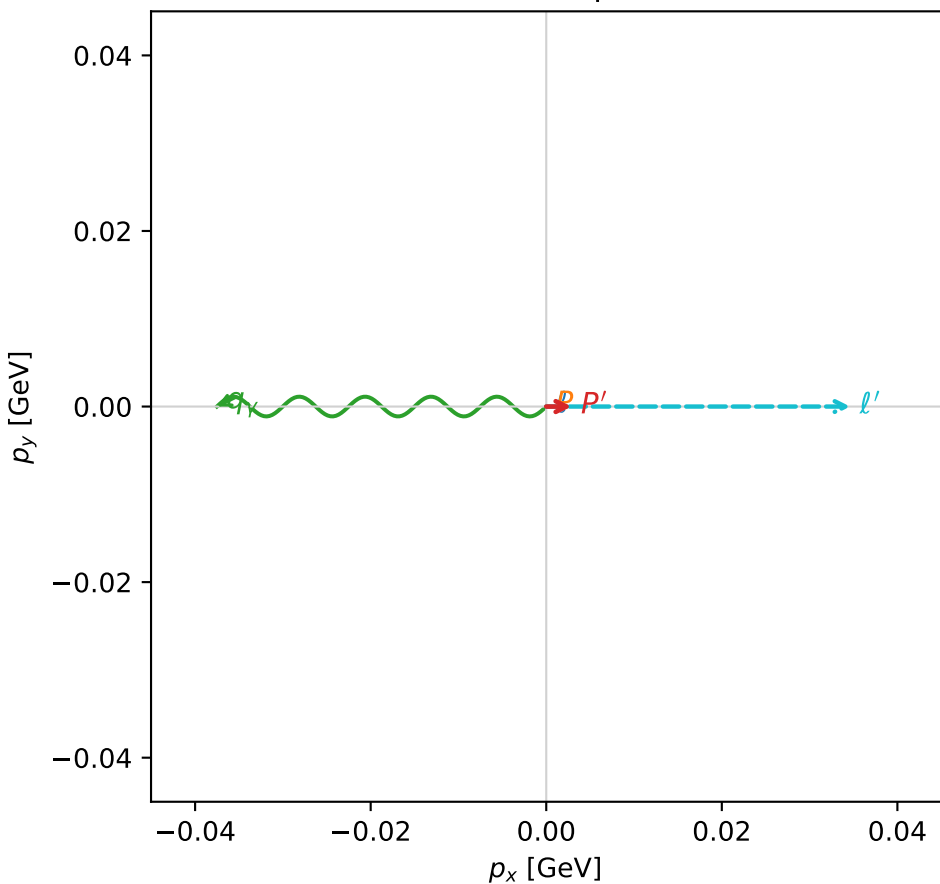
$s=8.22294$, $\sqrt{s}=2.86757$, $|\vec{P}|=1.28037$, $|\vec{P}'|=0.818827$
 $\theta_{P'}=0.00352053$, $\theta_Y=3.11084$, $\phi_{P'}=6.28317$, $\phi_Y=3.14158$
 $E_Y=1.22018$, $\theta_{\ell'}=0.086202$, $\phi_{\ell'}=6.28317$
 $Q^2=2.0564$, $x_B=0.289943$, $t=-0.0960184$, $W^2=5.91586$, $y=0.965862$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2804$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2804$
 P $E= 1.5872$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2804$
 ℓ' $E= 0.4023$ $p_x= 0.0346$ $p_y= -0.0000$ $p_z= 0.4008$
 P' $E= 1.2451$ $p_x= 0.0029$ $p_y= -0.0000$ $p_z= 0.8188$
 q_Y $E= 1.2202$ $p_x= -0.0375$ $p_y= 0.0000$ $p_z= -1.2196$

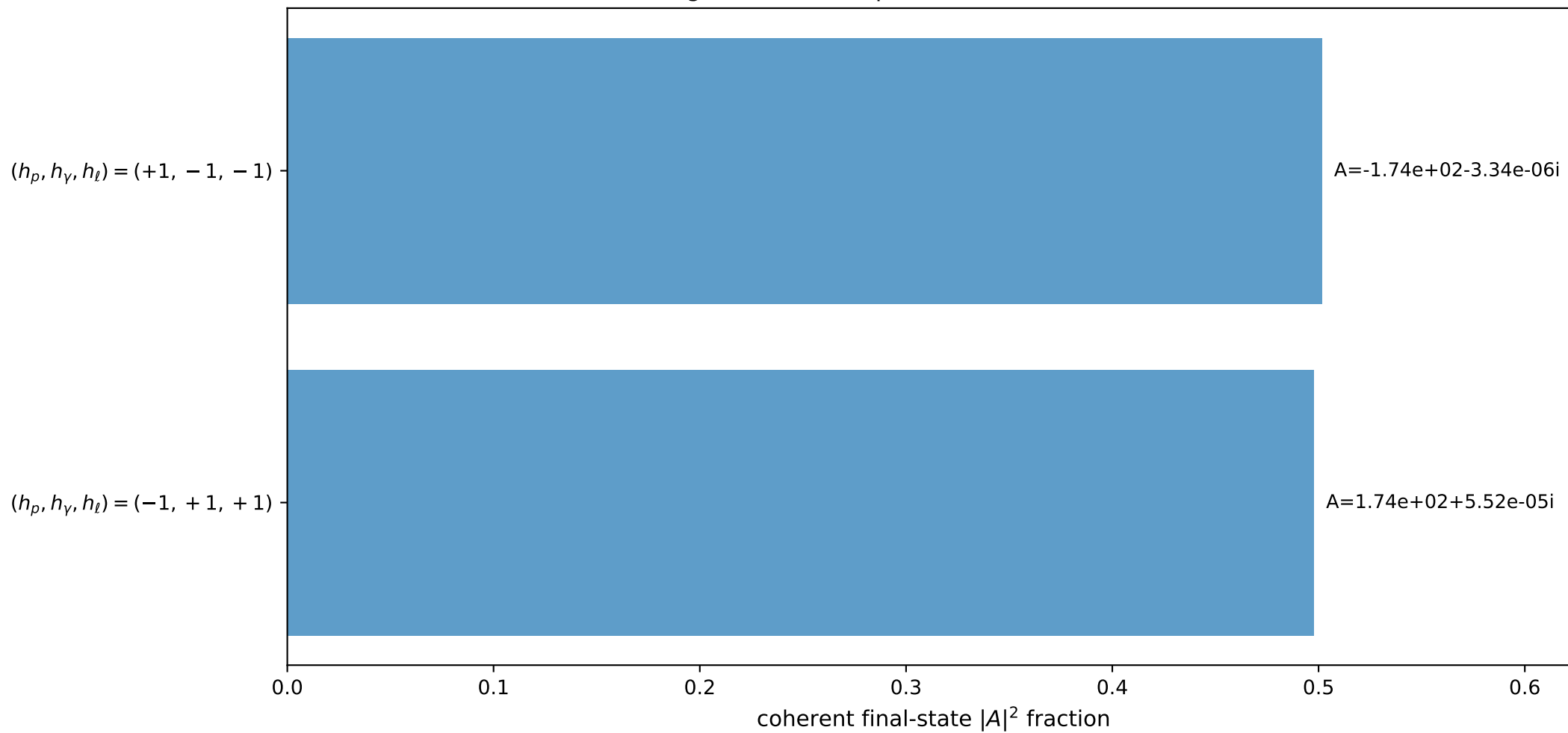
3D momenta



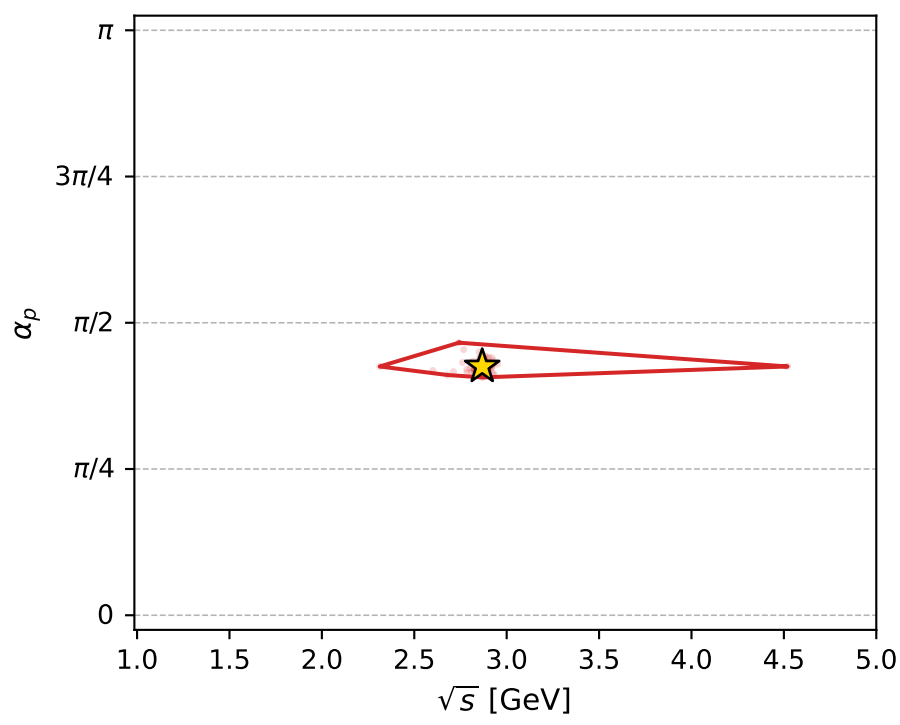
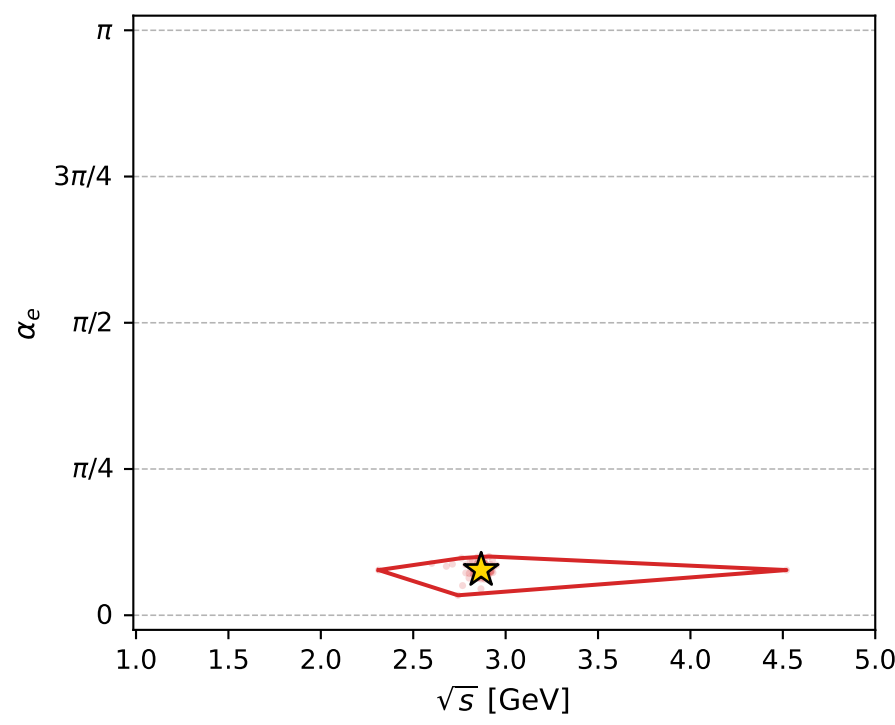
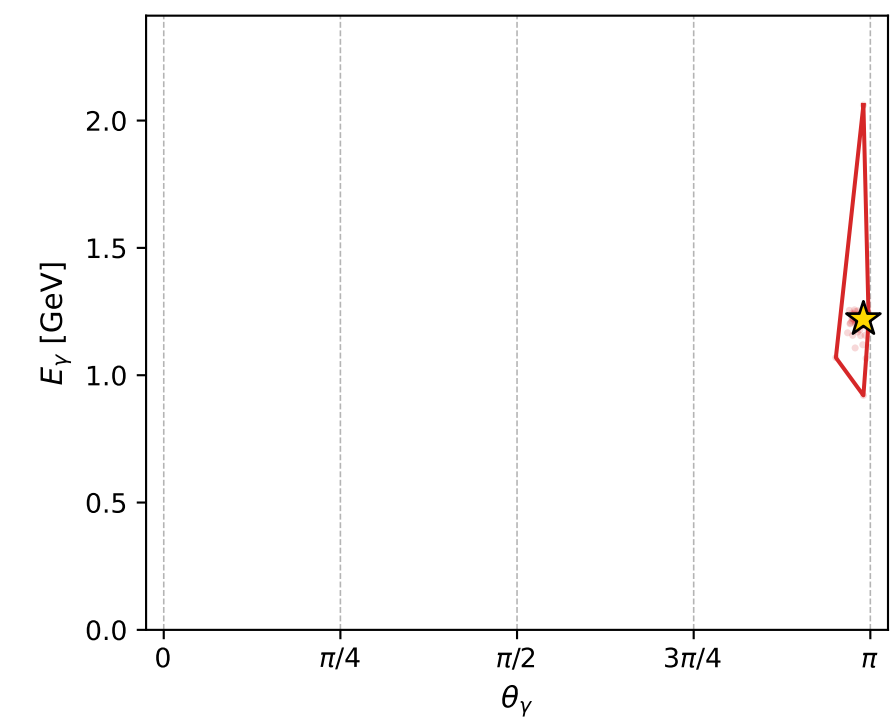
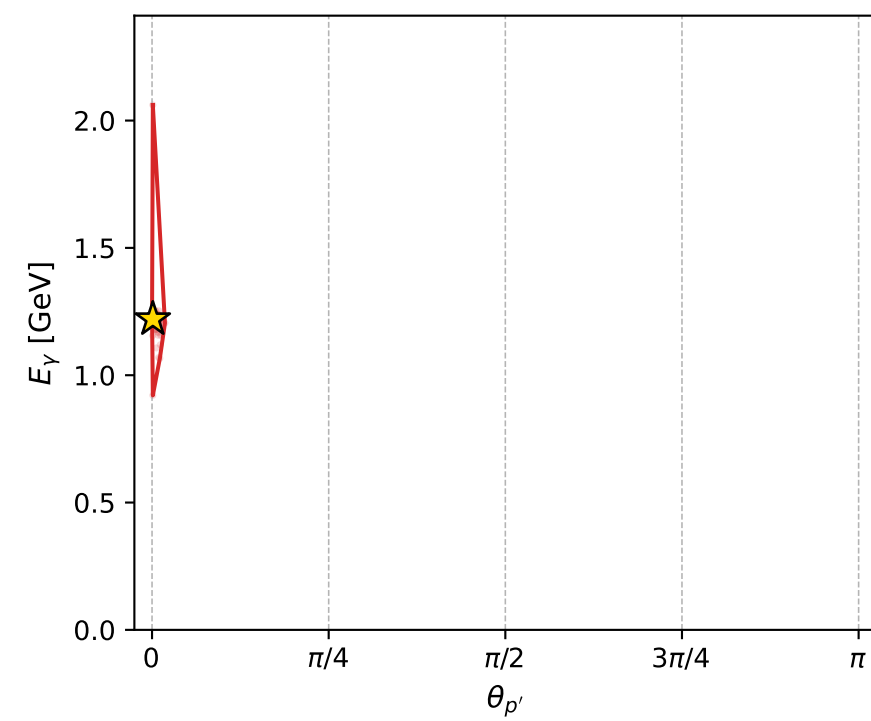
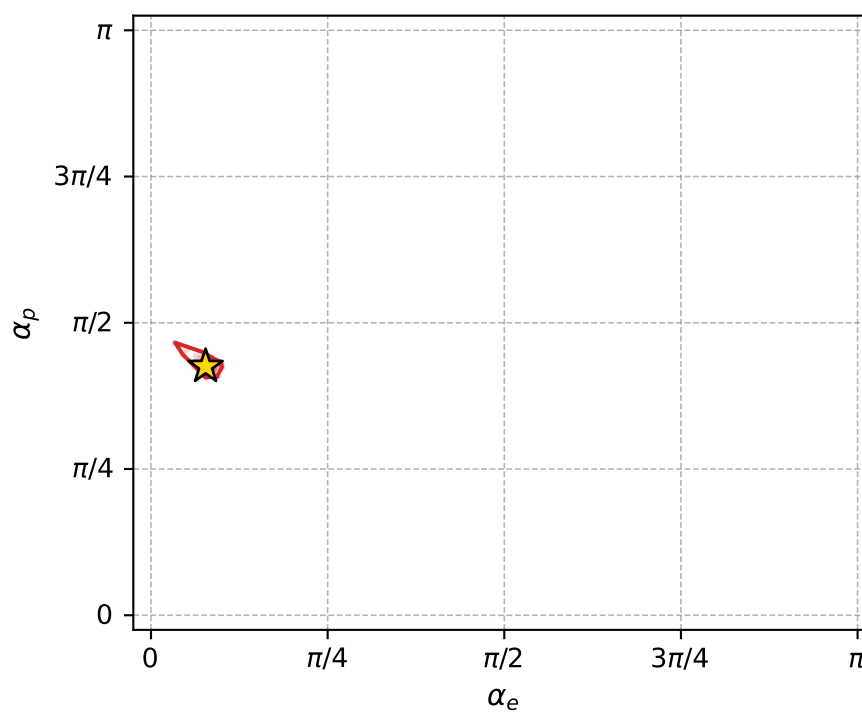
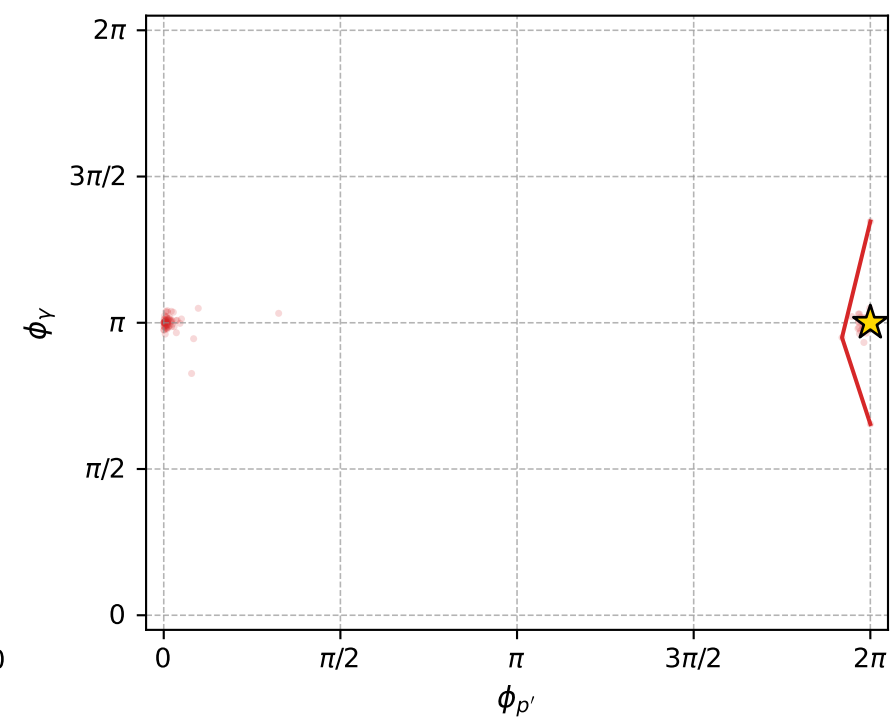
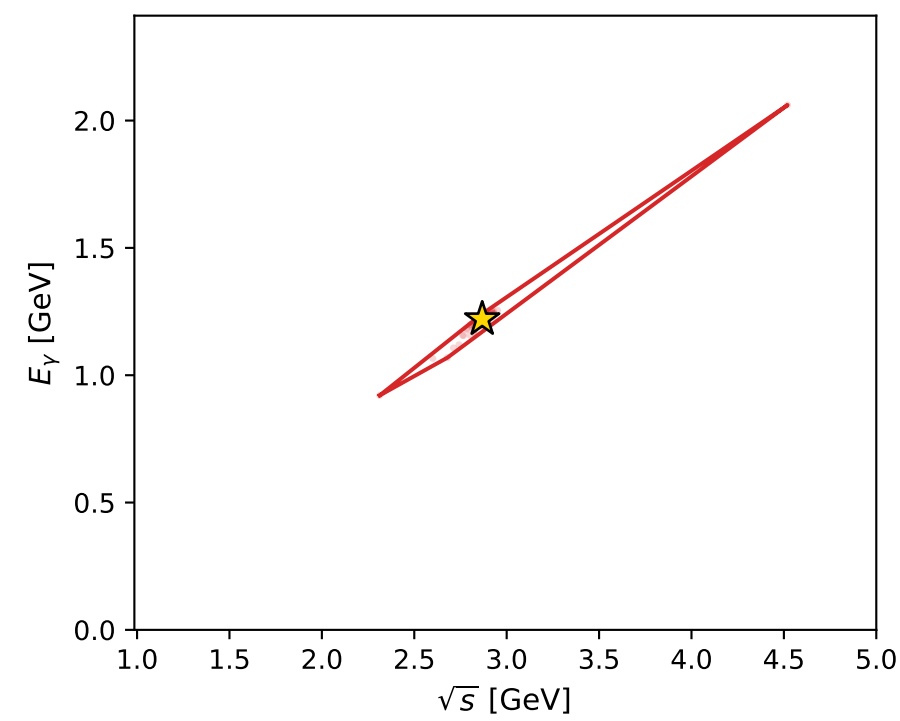
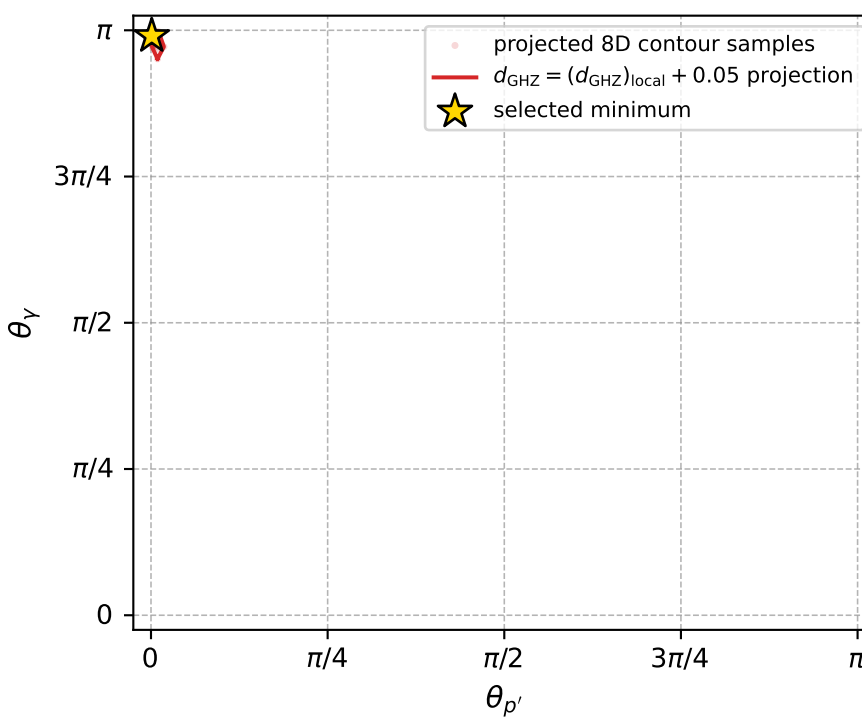
Transverse plane



Leading final-state amplitudes (N=2, retained=99.9%)



electron: polarization cluster P1, local minimum 193; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.7749646\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.05001775$
 above global minimum = $1.7727362\text{e-}05$
 8D contour samples = 163

selected phase-space point:

$\text{sqrt_s} = 2.867567$
 $\text{theta_p_out} = 0.003520534$
 $\text{theta_gamma_out} = 3.110841$
 $\text{q0ut} = 1.220176$
 $\text{phi_p_out} = 6.283174$
 $\text{phi_gamma_out} = 3.141579$
 $\text{alpha_e} = 0.2427356$
 $\text{alpha_p} = 1.336023$

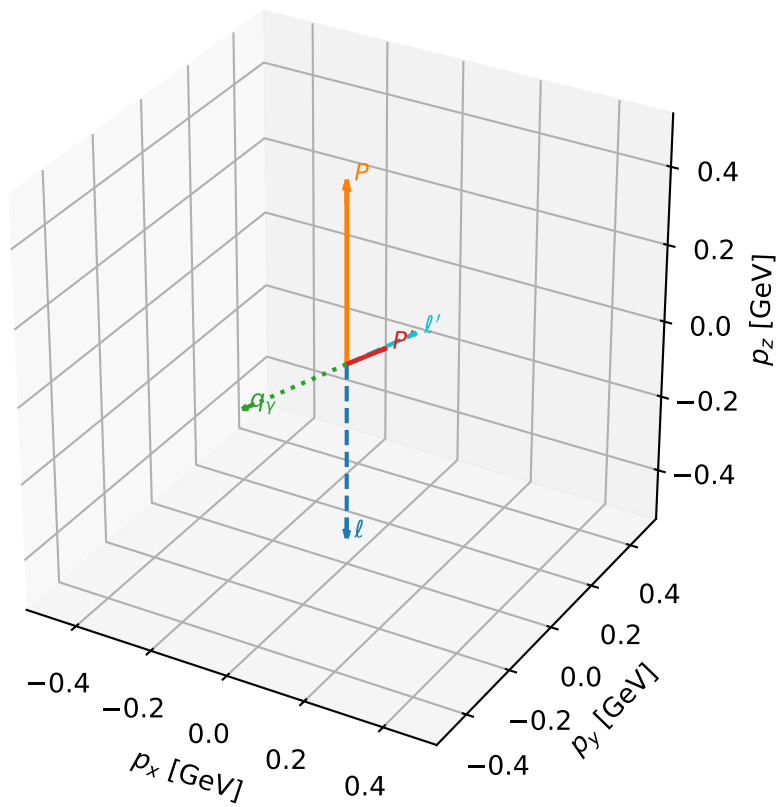
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_196 $d_{\{\text{rm GHZ}\}}=1.90562\text{e-}05$
 region: polarization_cluster_01_local_minimum_196
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.69896132$ rad
 incoming lepton: $0.765511|+\rangle +0.643423|-\rangle$
 proton mixing angle: $\alpha_p=0.98358756$ rad
 incoming proton: $0.55404|+\rangle +0.83249|-\rangle$

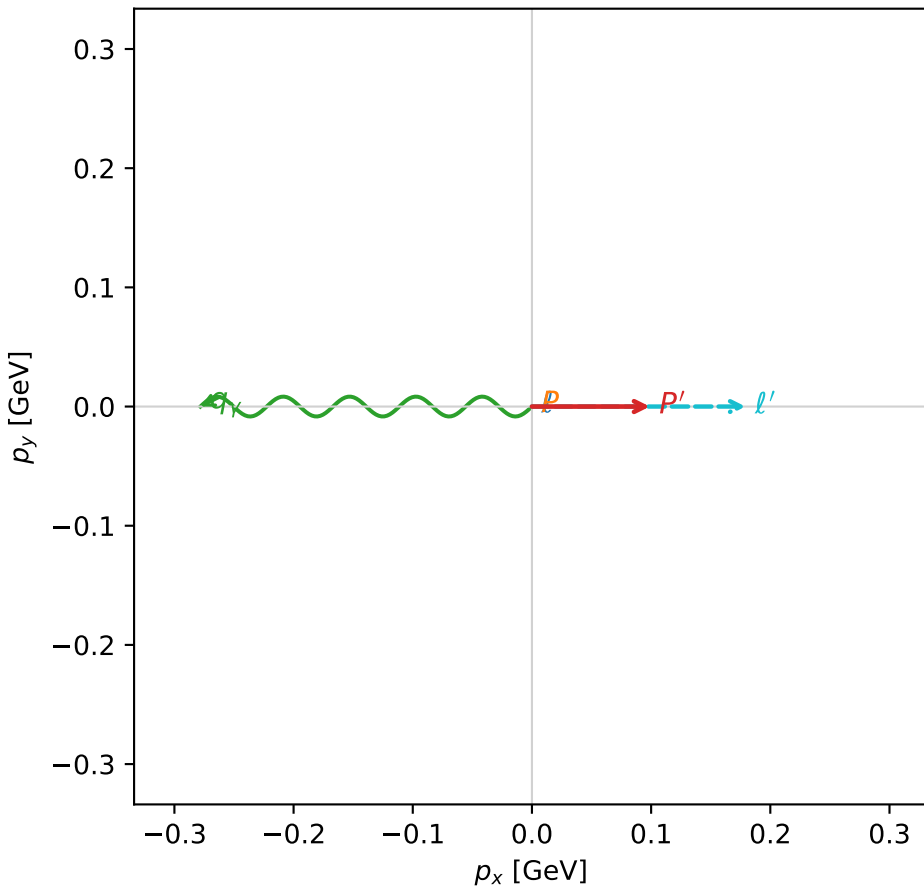
$s=2.28397$, $\sqrt{s}=1.51128$, $|\vec{P}|=0.464547$, $|\vec{P}'|=0.120367$
 $\theta_{P'}=0.964747$, $\theta_Y=2.19546$, $\phi_{P'}=3.02427\text{e-}05$, $\phi_Y=3.14163$
 $E_Y=0.342961$, $\theta_{\ell'}=0.936071$, $\phi_{\ell'}=4.22327\text{e-}05$
 $Q^2=0.329489$, $x_B=0.310631$, $t=-0.156381$, $W^2=1.61106$, $y=0.755424$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4645$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4645$
 P $E= 1.0467$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4645$
 ℓ' $E= 0.2226$ $p_x= 0.1793$ $p_y= 0.0000$ $p_z= 0.1320$
 P' $E= 0.9457$ $p_x= 0.0989$ $p_y= 0.0000$ $p_z= 0.0686$
 q_Y $E= 0.3430$ $p_x= -0.2782$ $p_y= -0.0000$ $p_z= -0.2006$

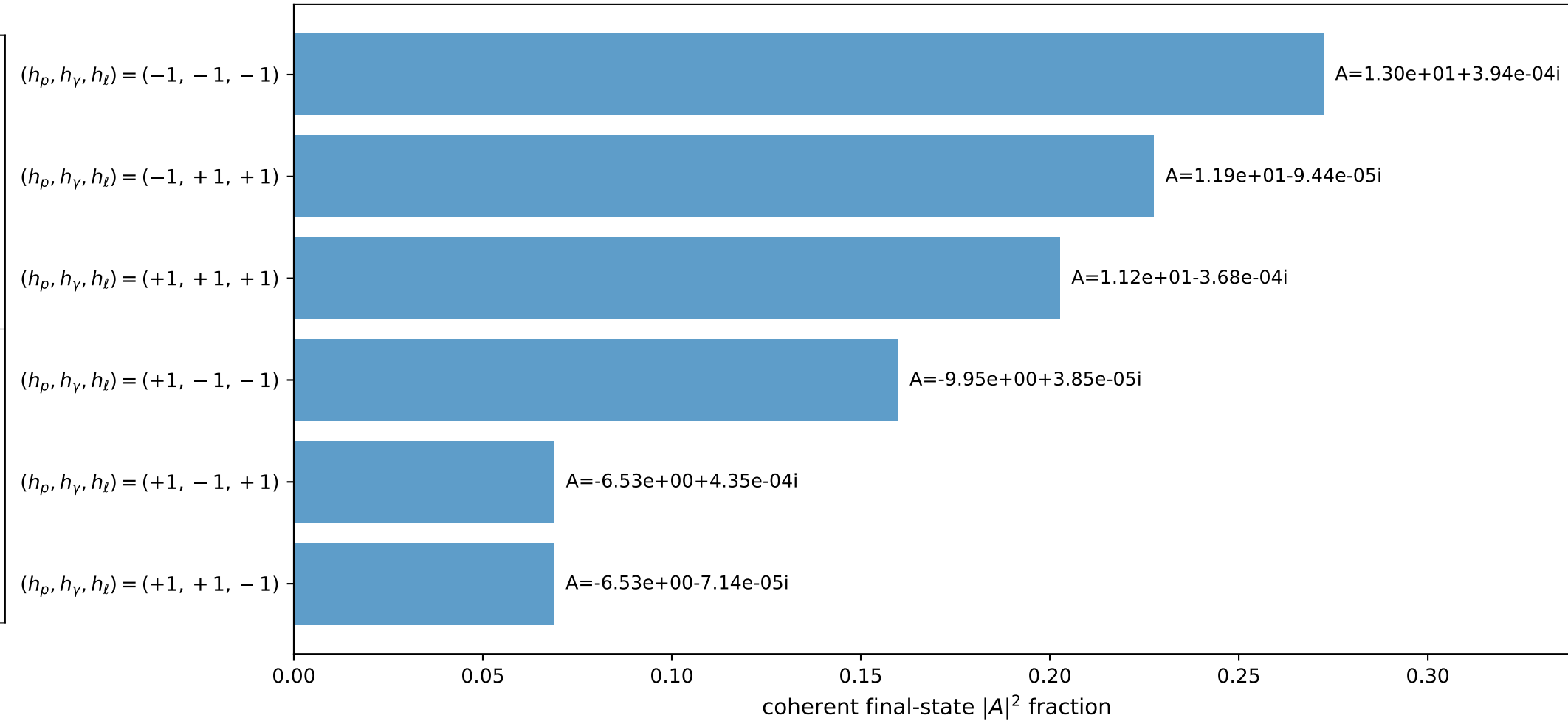
3D momenta



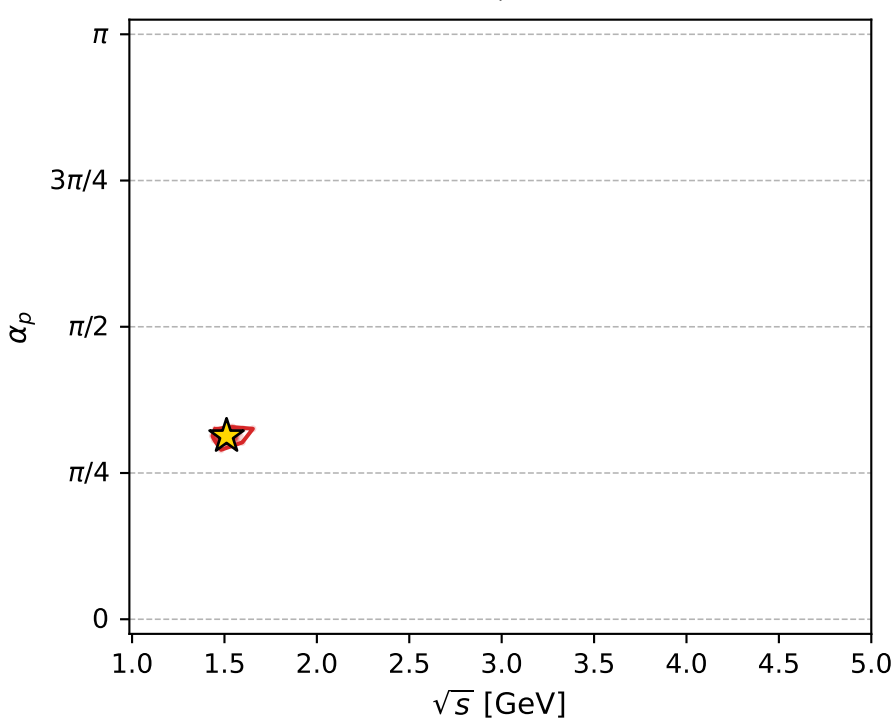
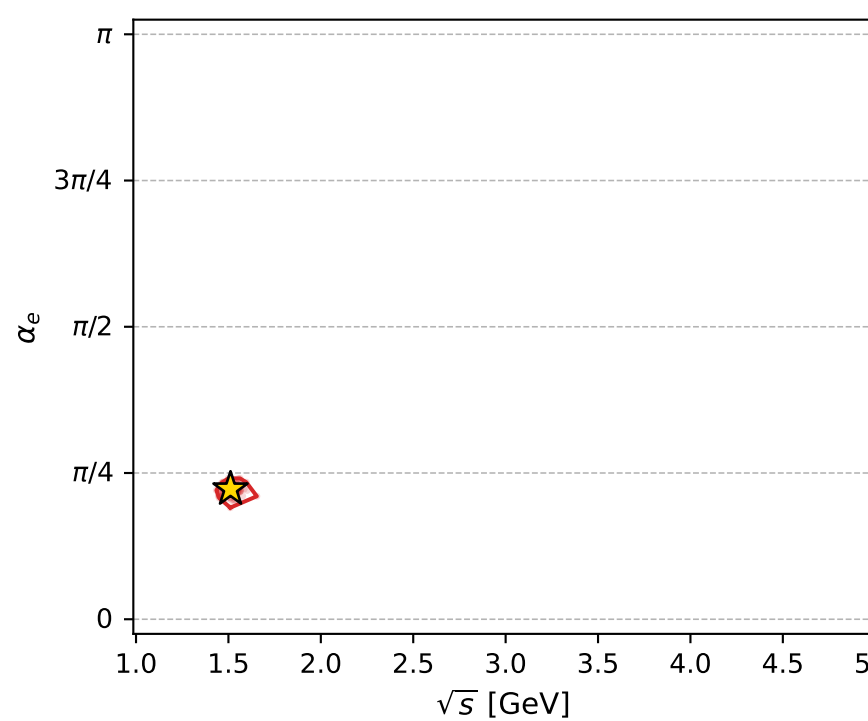
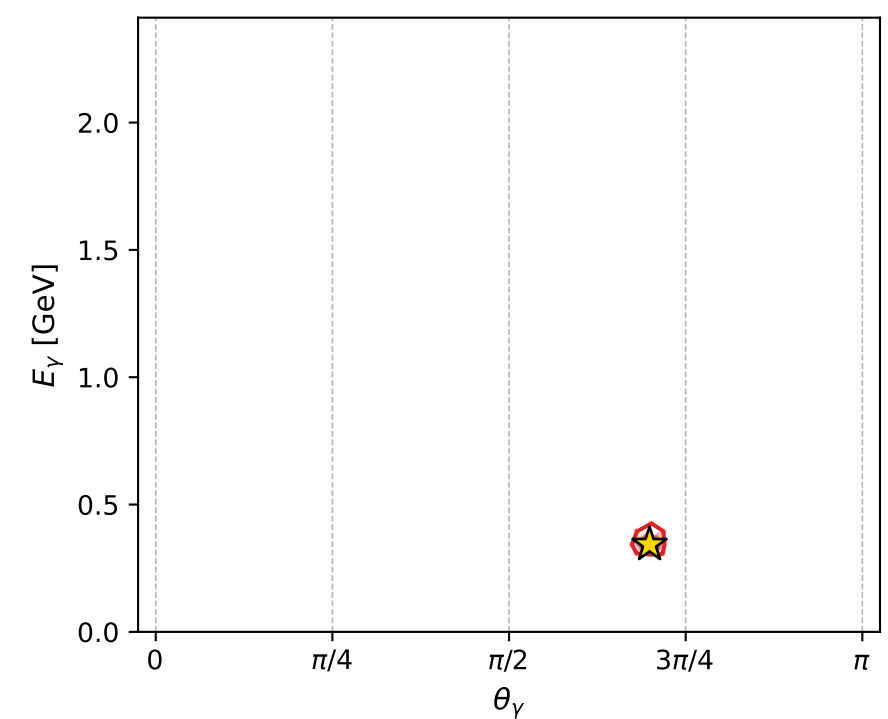
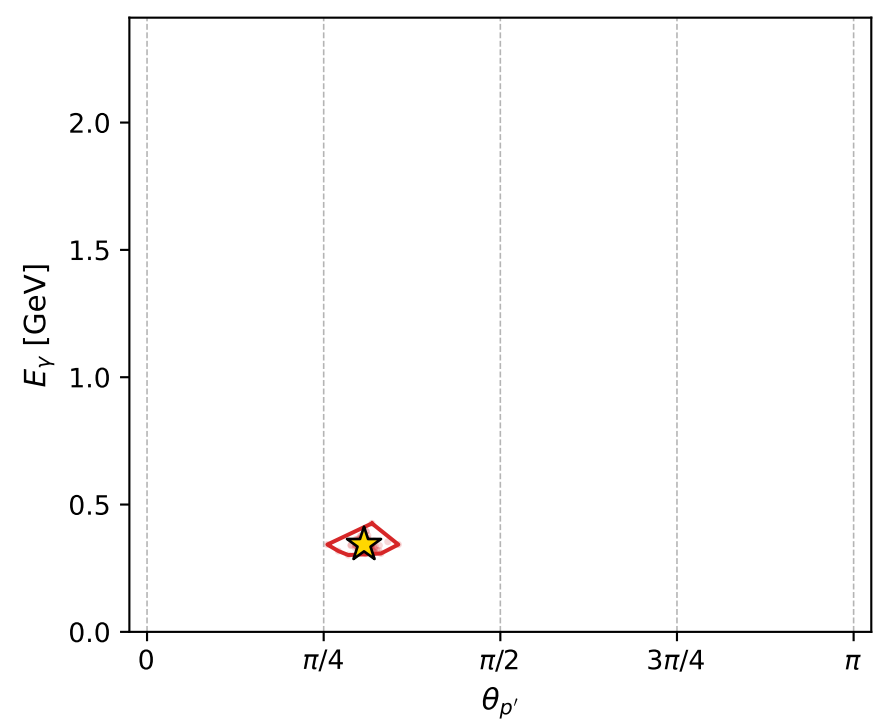
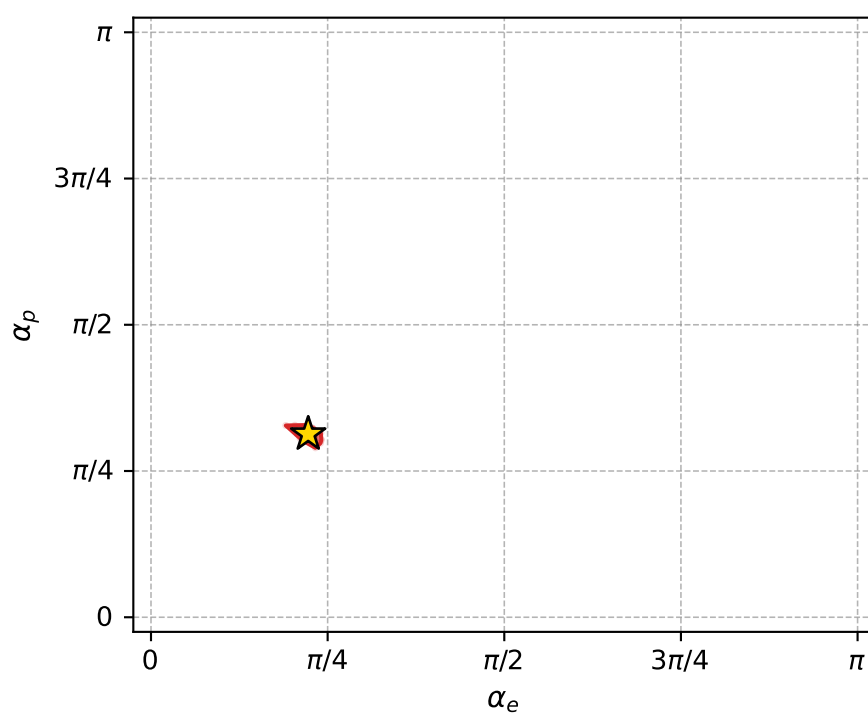
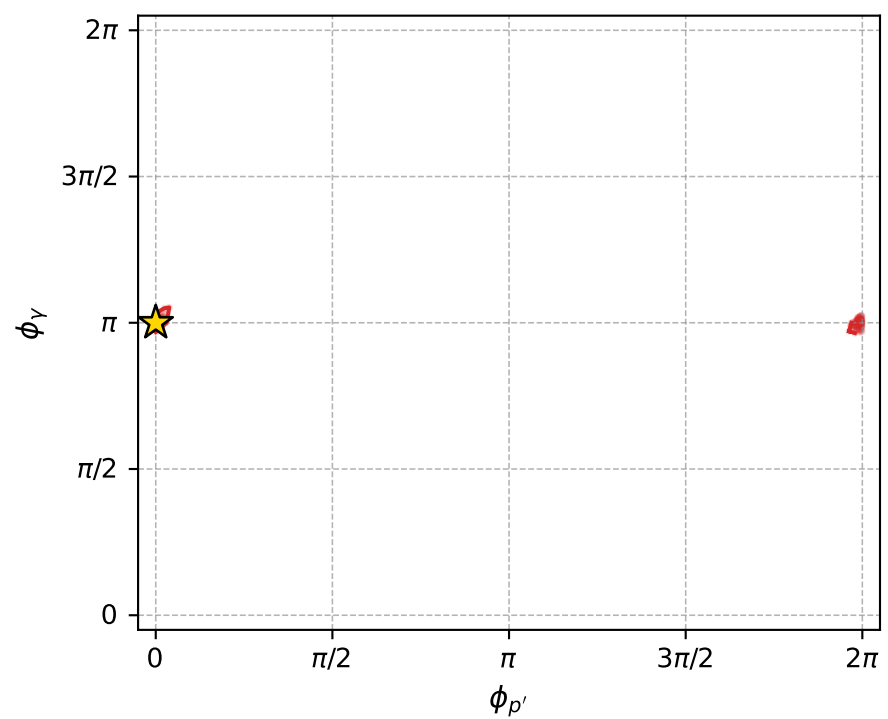
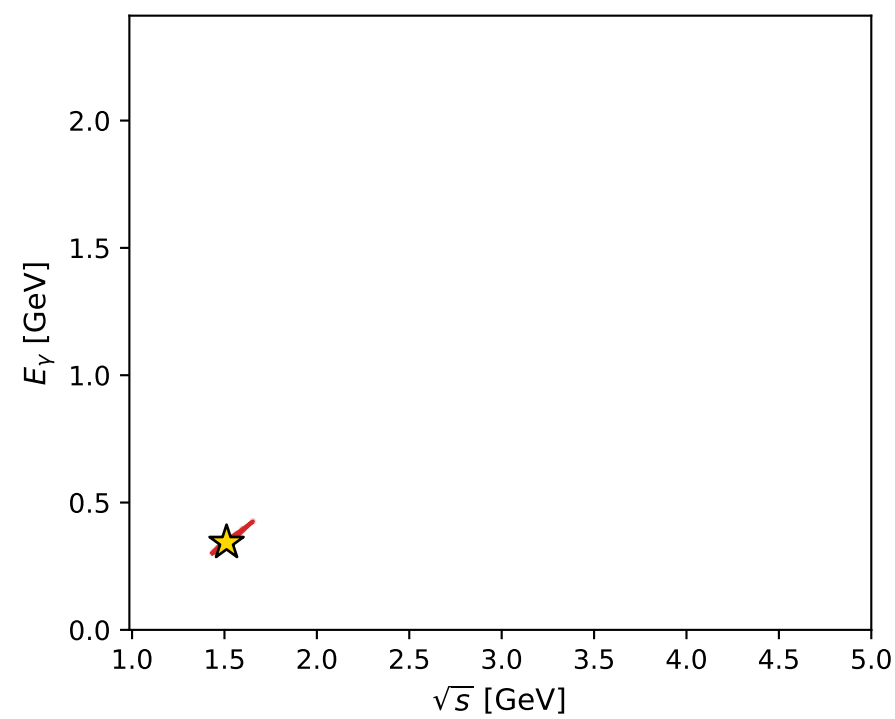
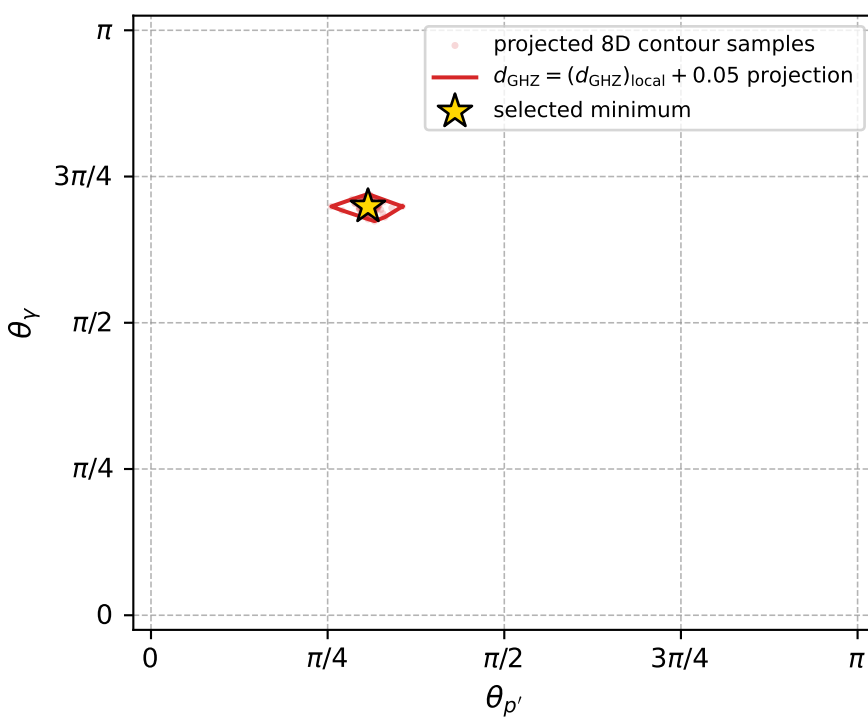
Transverse plane



Leading final-state amplitudes (N=6, retained=100.0%)



electron: polarization cluster P1, local minimum 196; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 1.9056247\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050019056$
 above global minimum = $1.9033963\text{e-}05$
 8D contour samples = 256

selected phase-space point:

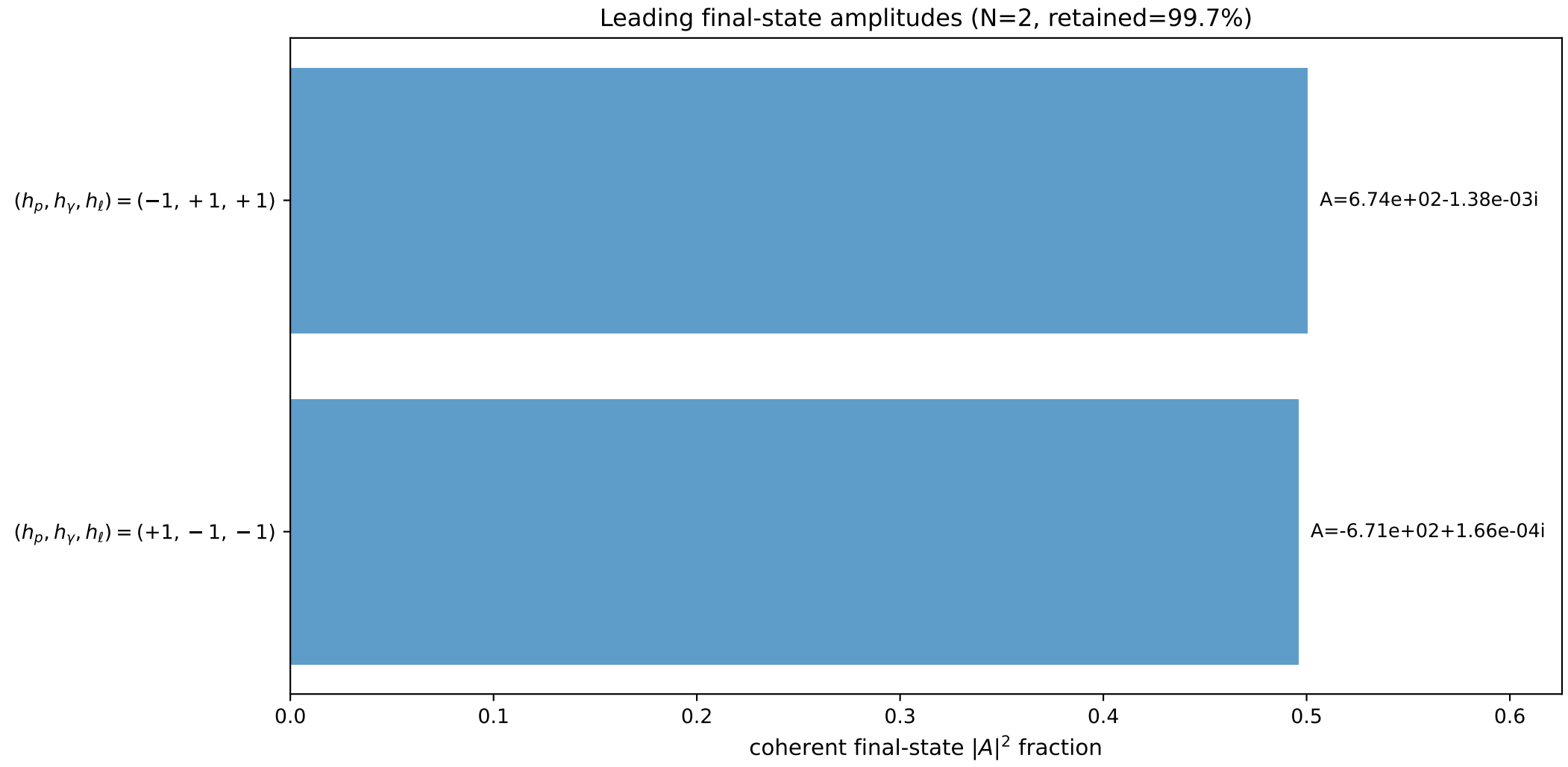
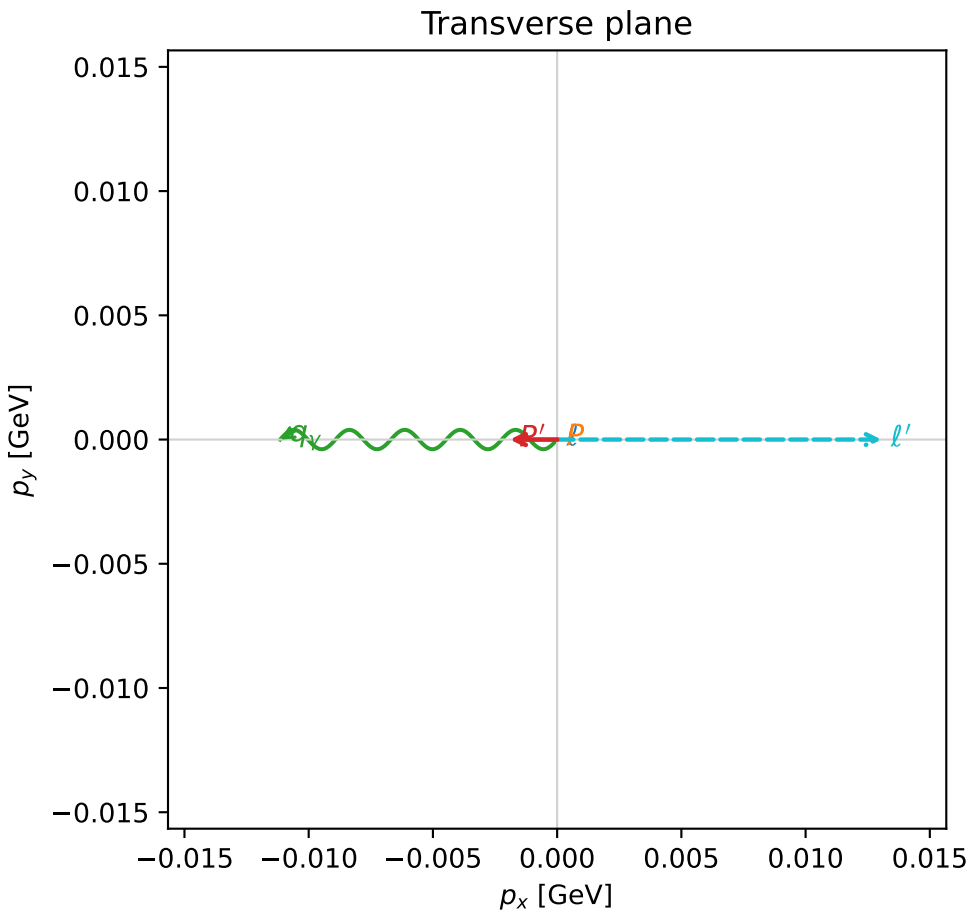
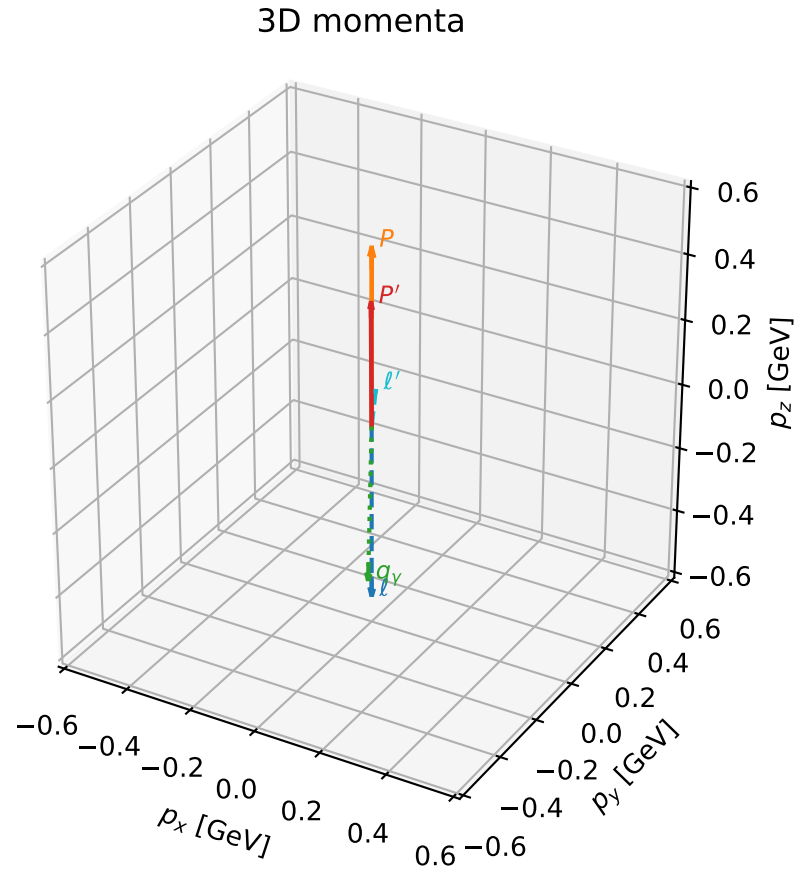
$\text{sqrt_s} = 1.51128$
 $\text{theta_p_out} = 0.9647466$
 $\text{theta_gamma_out} = 2.195459$
 $\text{q0ut} = 0.3429614$
 $\text{phi_p_out} = 3.024269\text{e-}05$
 $\text{phi_gamma_out} = 3.141631$
 $\text{alpha_e} = 0.6989613$
 $\text{alpha_p} = 0.9835876$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

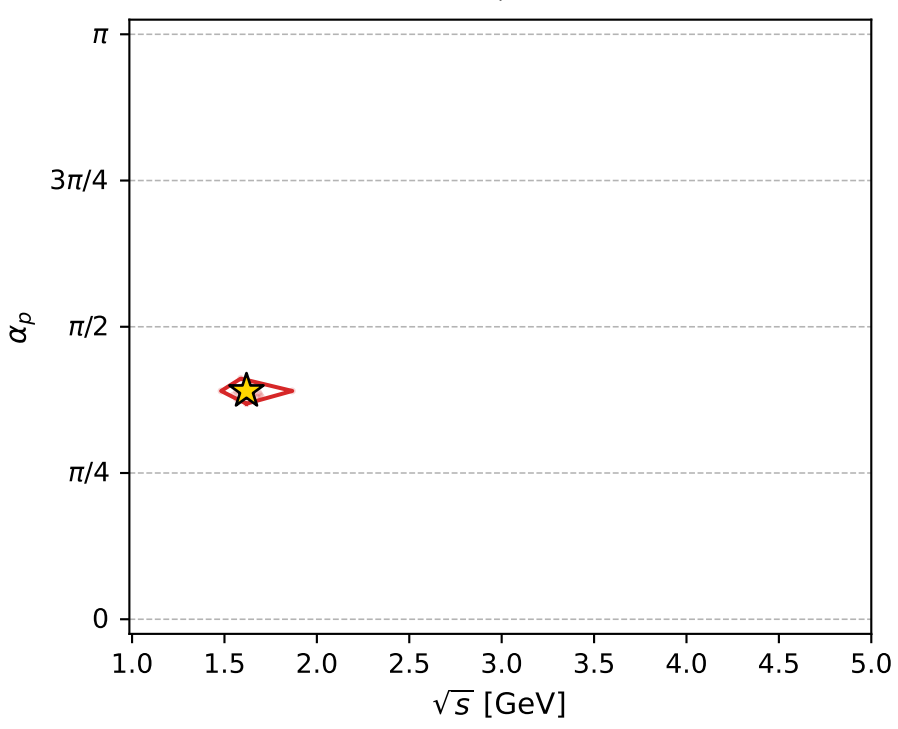
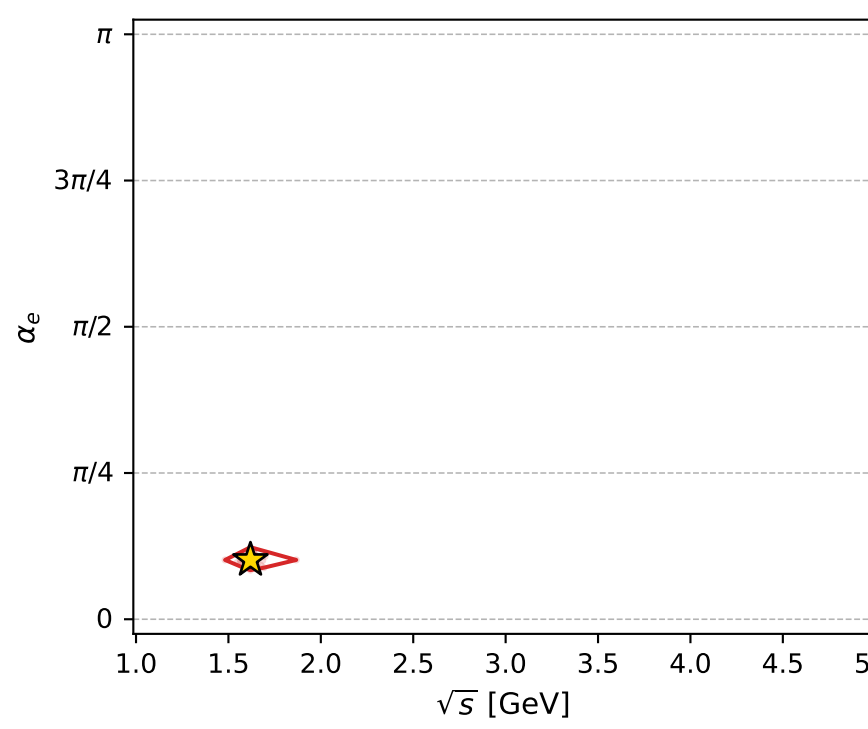
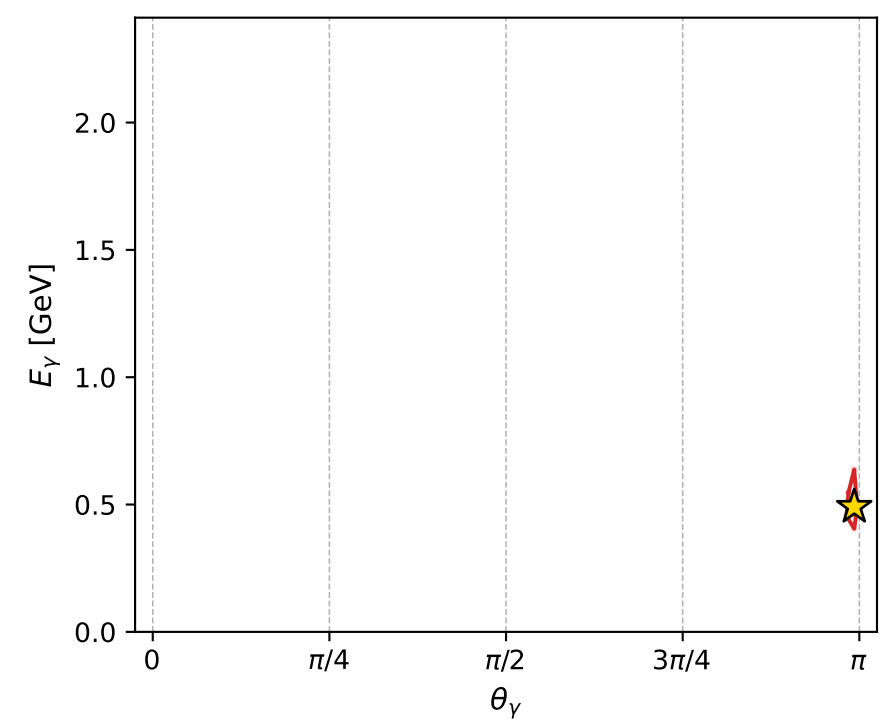
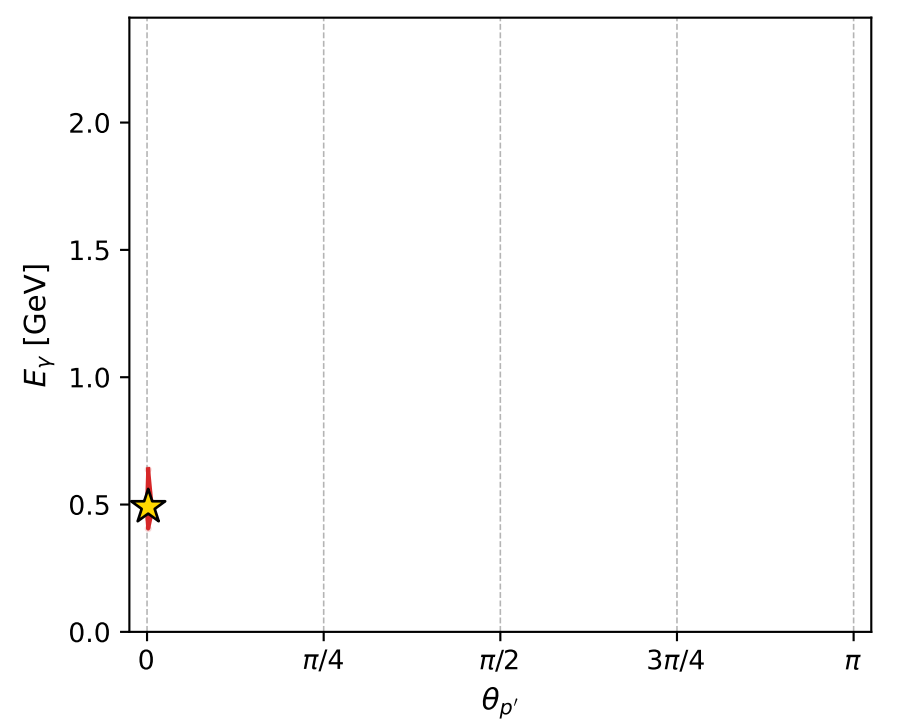
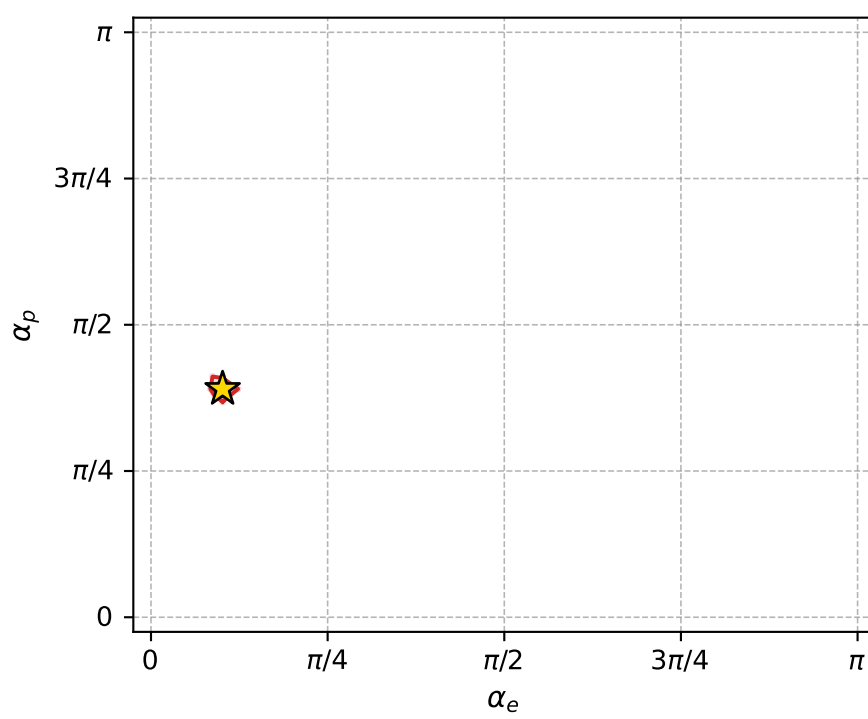
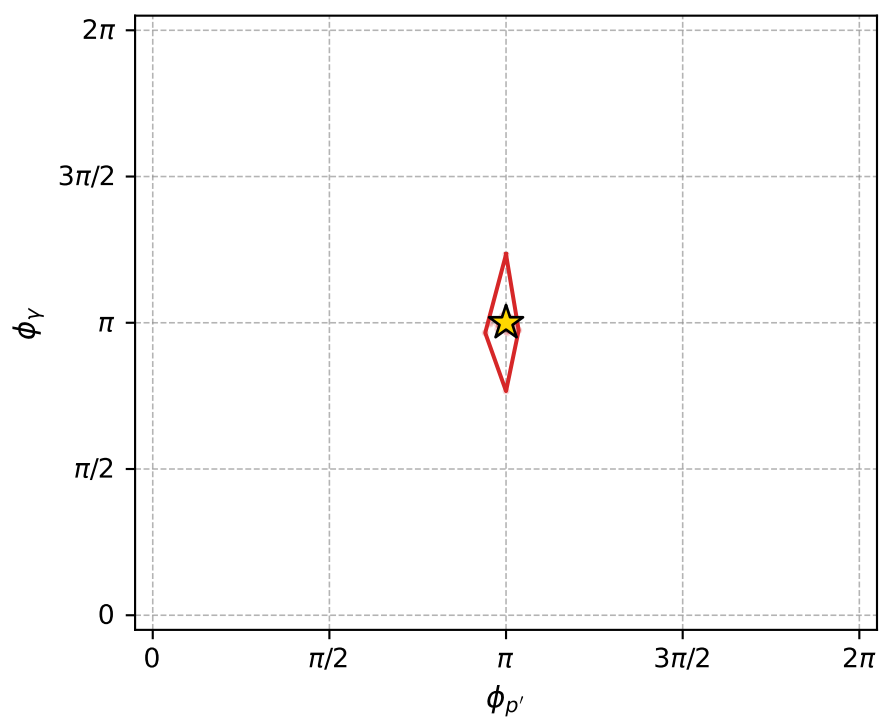
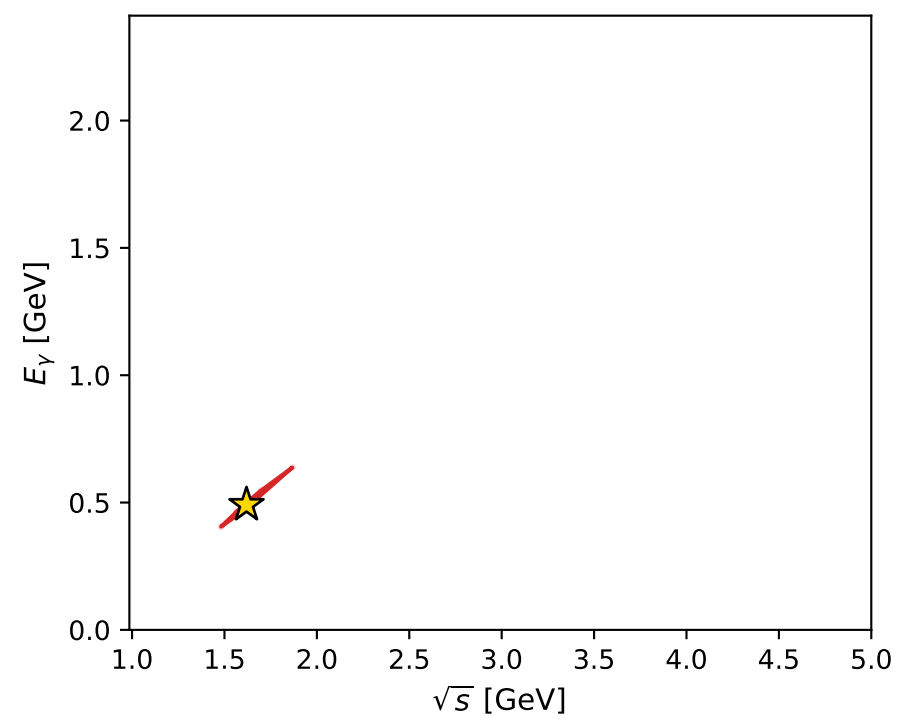
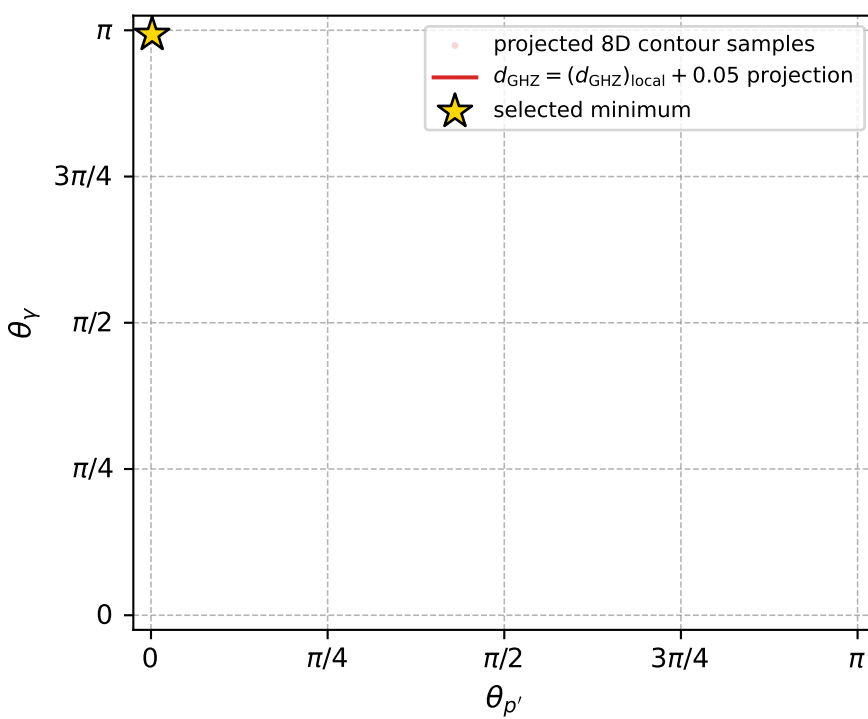
dghz_mixing_angles_polarization_01_local_minimum_198 $d_{\{\text{rm GHZ}\}}=2.02825\text{e-}05$
 region: polarization_cluster_01_local_minimum_198
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.31842078$ rad
 incoming lepton: $0.949731|+\rangle +0.313067|-\rangle$
 proton mixing angle: $\alpha_p=1.2260476$ rad
 incoming proton: $0.33796|+\rangle +0.94116|-\rangle$

$s=2.62187$, $\sqrt{s}=1.61922$, $|\vec{P}|=0.537922$, $|\vec{P}'|=0.373454$
 $\theta_{P'}=0.00509094$, $\theta_V=3.11889$, $\phi_{P'}=3.14157$, $\phi_V=3.14159$
 $E_V=0.491231$, $\theta_{l'}=0.110485$, $\phi_{l'}=6.28318$
 $Q^2=0.253934$, $x_B=0.157469$, $t=-0.0219156$, $W^2=2.23851$, $y=0.925704$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.5379$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5379$
 P $E= 1.0813$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5379$
 l' $E= 0.1184$ $p_x= 0.0131$ $p_y= -0.0000$ $p_z= 0.1177$
 P' $E= 1.0096$ $p_x= -0.0019$ $p_y= 0.0000$ $p_z= 0.3734$
 q_V $E= 0.4912$ $p_x= -0.0112$ $p_y= 0.0000$ $p_z= -0.4911$



electron: polarization cluster P1, local minimum 198; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.0282517\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050020283$
 above global minimum = $2.0260233\text{e-}05$
 8D contour samples = 156

selected phase-space point:

$\text{sqrt_s} = 1.619219$
 $\text{theta_p_out} = 0.005090941$
 $\text{theta_gamma_out} = 3.11889$
 $\text{q0out} = 0.4912312$
 $\text{phi_p_out} = 3.141565$
 $\text{phi_gamma_out} = 3.141592$
 $\text{alpha_e} = 0.3184208$
 $\text{alpha_p} = 1.226048$

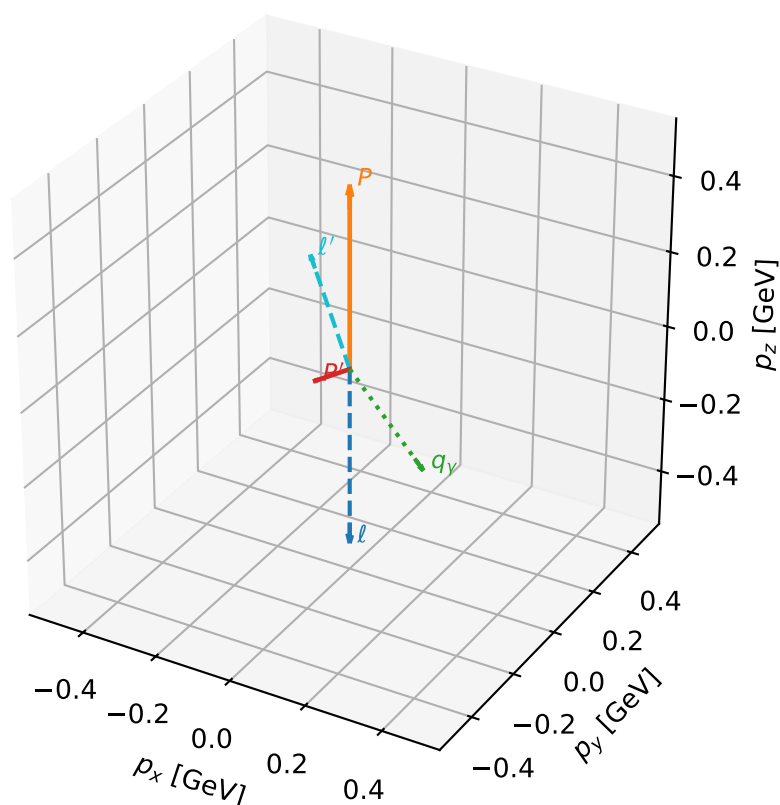
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_200 $d_{\{\text{rm GHZ}\}}=2.18346\text{e-}05$
 region: polarization_cluster_01_local_minimum_200
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.3199569$ rad
 incoming lepton: $0.248217|+\rangle +0.968704|-\rangle$
 proton mixing angle: $\alpha_p=0.20130687$ rad
 incoming proton: $0.979806|+\rangle +0.19995|-\rangle$

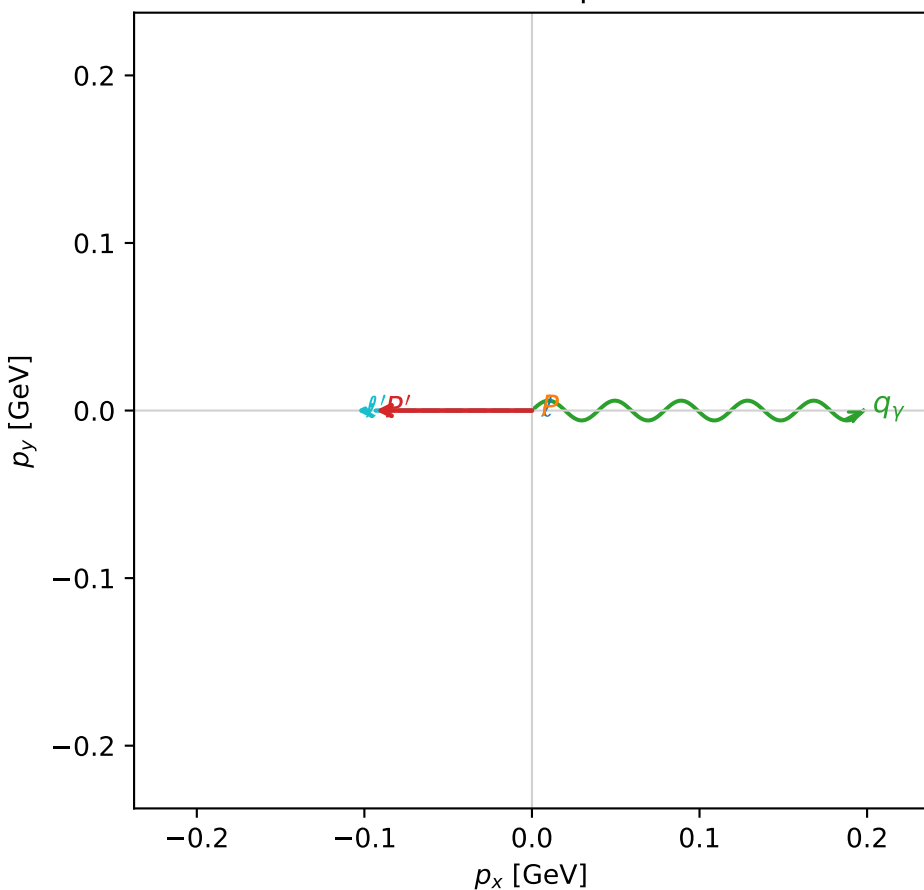
$s=2.34006$, $\sqrt{s}=1.52972$, $|\vec{P}|=0.477279$, $|\vec{P}'|=0.11$
 $\theta_{p'}=2.13041$, $\theta_\gamma=2.39819$, $\phi_{p'}=3.14155$, $\phi_\gamma=6.28317$
 $E_\gamma=0.292352$, $\theta_{\ell'}=0.365278$, $\phi_{\ell'}=3.1416$
 $Q^2=0.540814$, $x_B=0.489522$, $t=-0.283969$, $W^2=1.44381$, $y=0.756589$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4773$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4773$
 P $E= 1.0524$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4773$
 ℓ' $E= 0.2929$ $p_x= -0.1046$ $p_y= -0.0000$ $p_z= 0.2736$
 P' $E= 0.9444$ $p_x= -0.0932$ $p_y= 0.0000$ $p_z= -0.0584$
 q_γ $E= 0.2924$ $p_x= 0.1979$ $p_y= -0.0000$ $p_z= -0.2152$

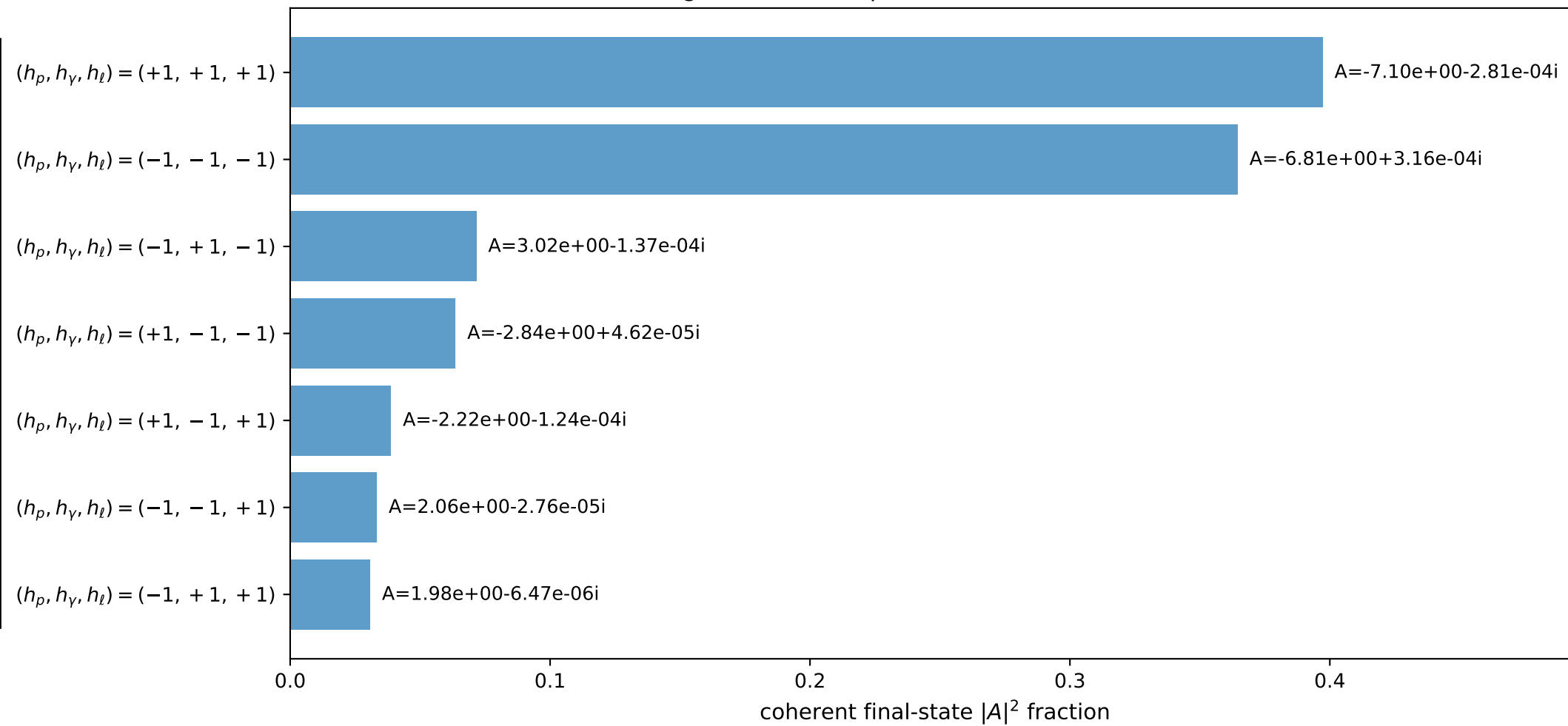
3D momenta



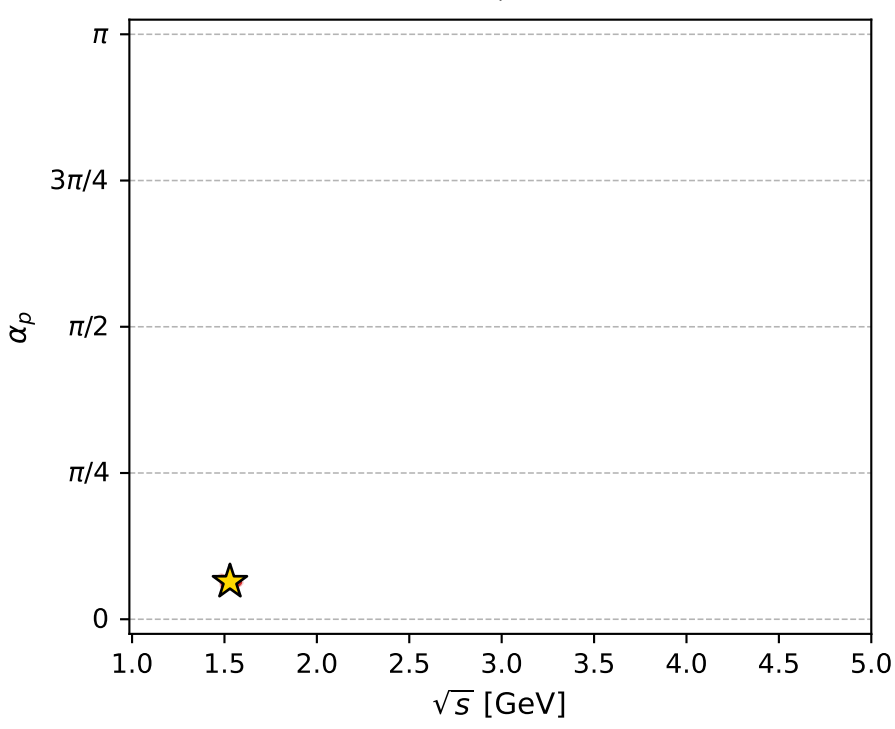
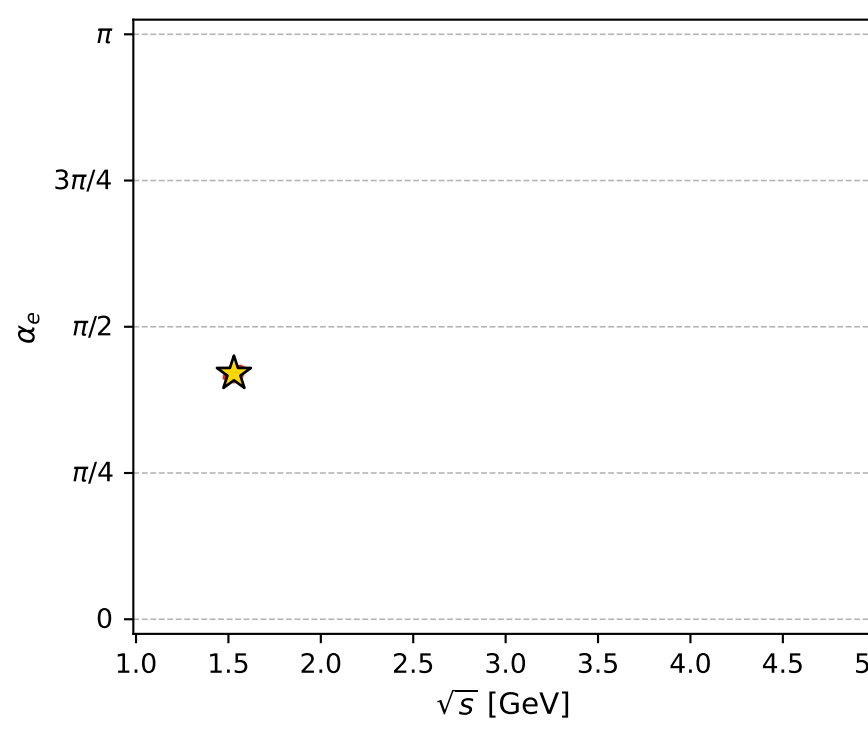
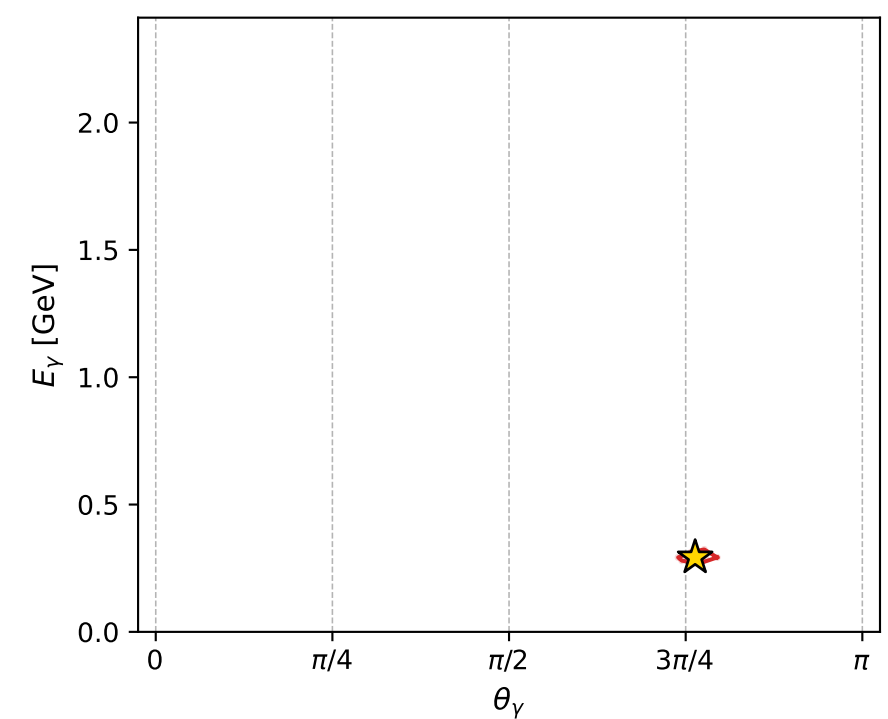
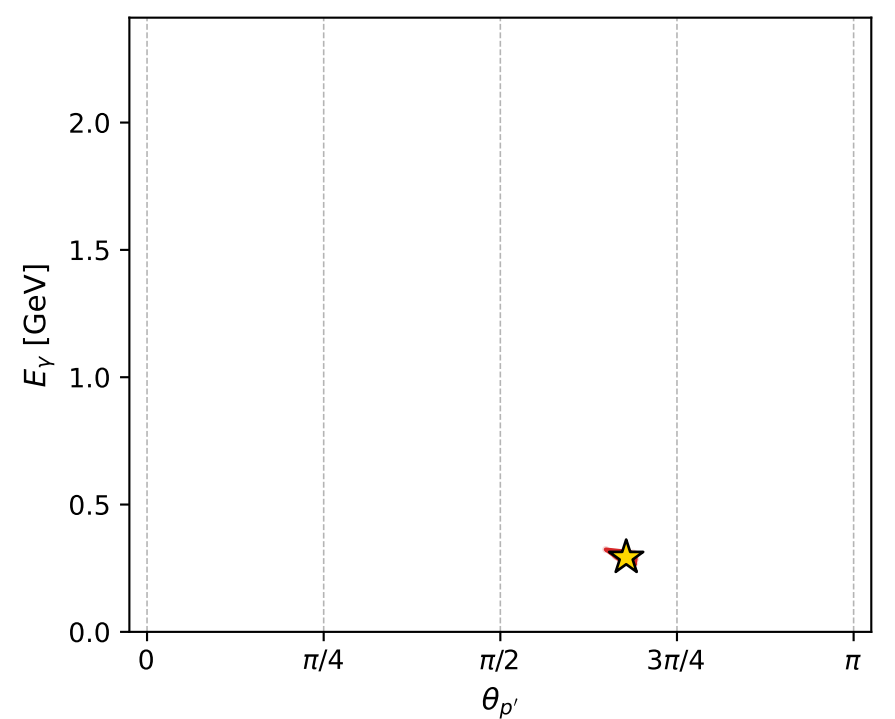
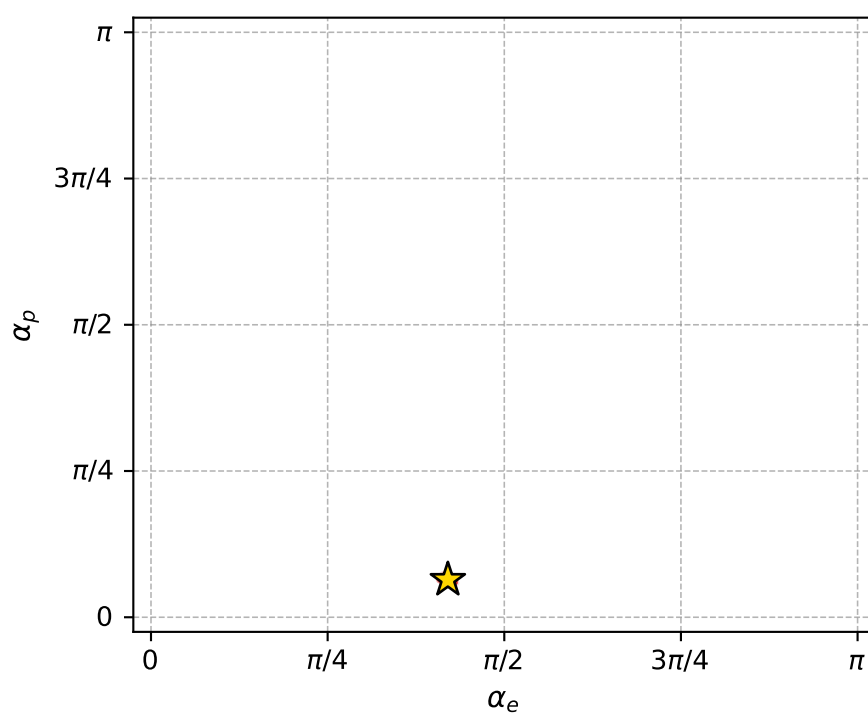
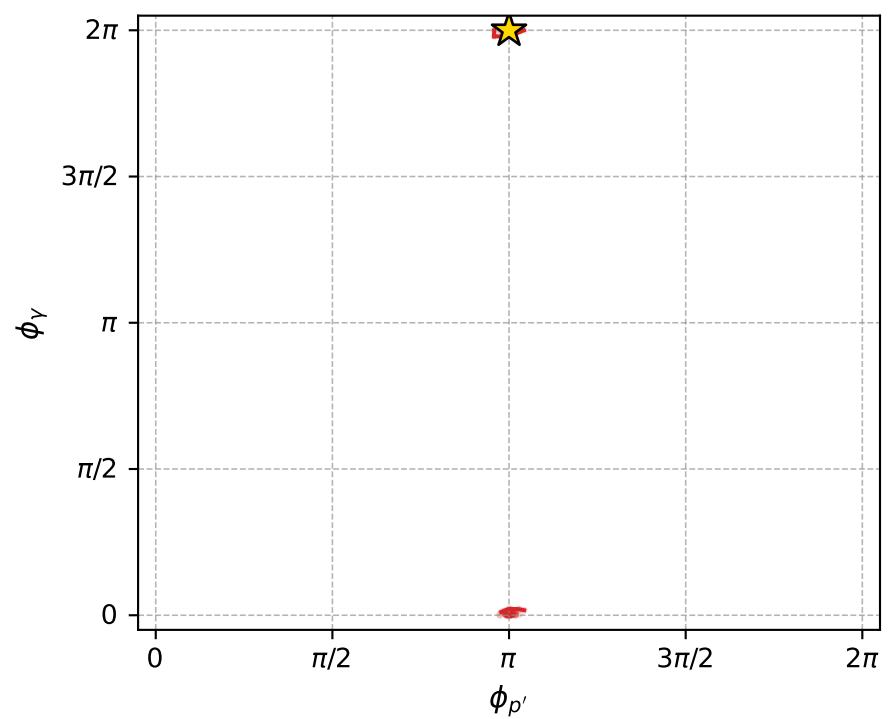
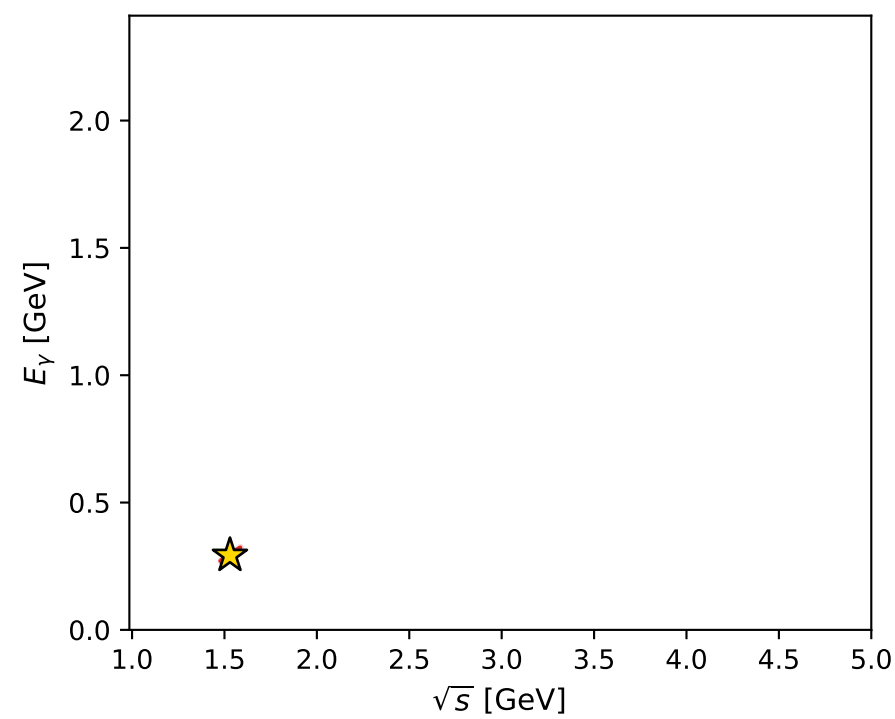
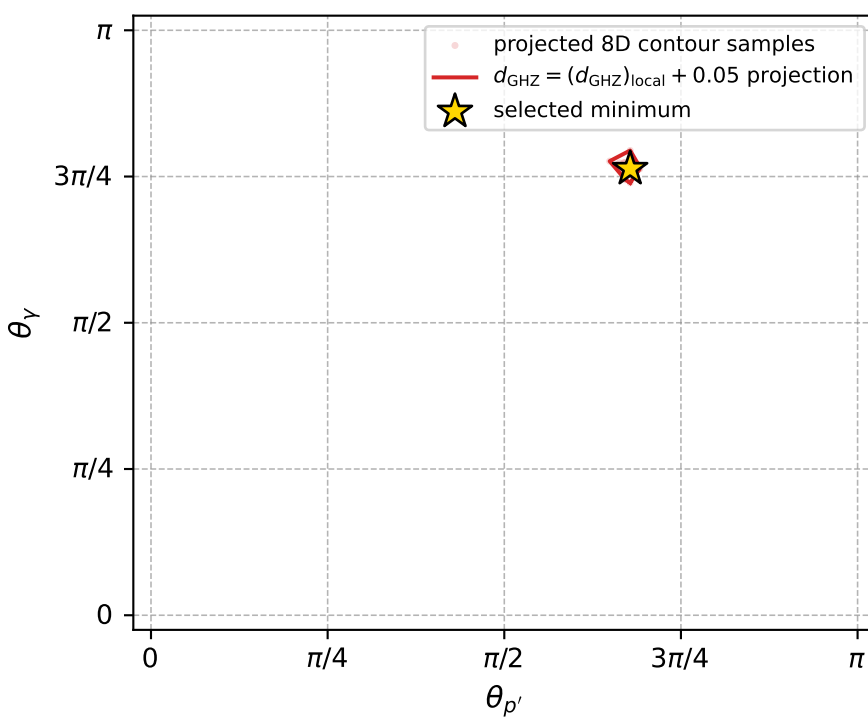
Transverse plane



Leading final-state amplitudes (N=7, retained=100.0%)



electron: polarization cluster P1, local minimum 200; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.1834627\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050021835$
 above global minimum = $2.1812343\text{e-}05$
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.529724$
 $\text{theta_p_out} = 2.130413$
 $\text{theta_gamma_out} = 2.398191$
 $\text{q0out} = 0.2923523$
 $\text{phi_p_out} = 3.141551$
 $\text{phi_gamma_out} = 6.283172$
 $\text{alpha_e} = 1.319957$
 $\text{alpha_p} = 0.2013069$

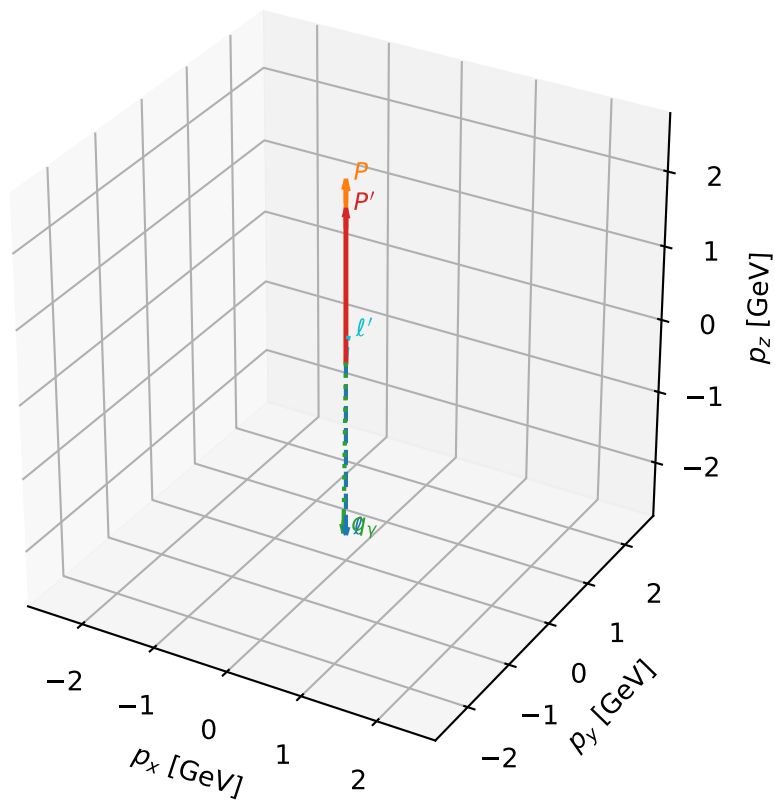
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_203 $d_{\{\text{rm GHZ}\}}=2.34128\text{e-}05$
 region: polarization_cluster_01_local_minimum_203
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.64267685$ rad
 incoming lepton: $0.800494|+\rangle +0.59934|-\rangle$
 proton mixing angle: $\alpha_p=0.93529282$ rad
 incoming proton: $0.593583|+\rangle +0.804773|-\rangle$

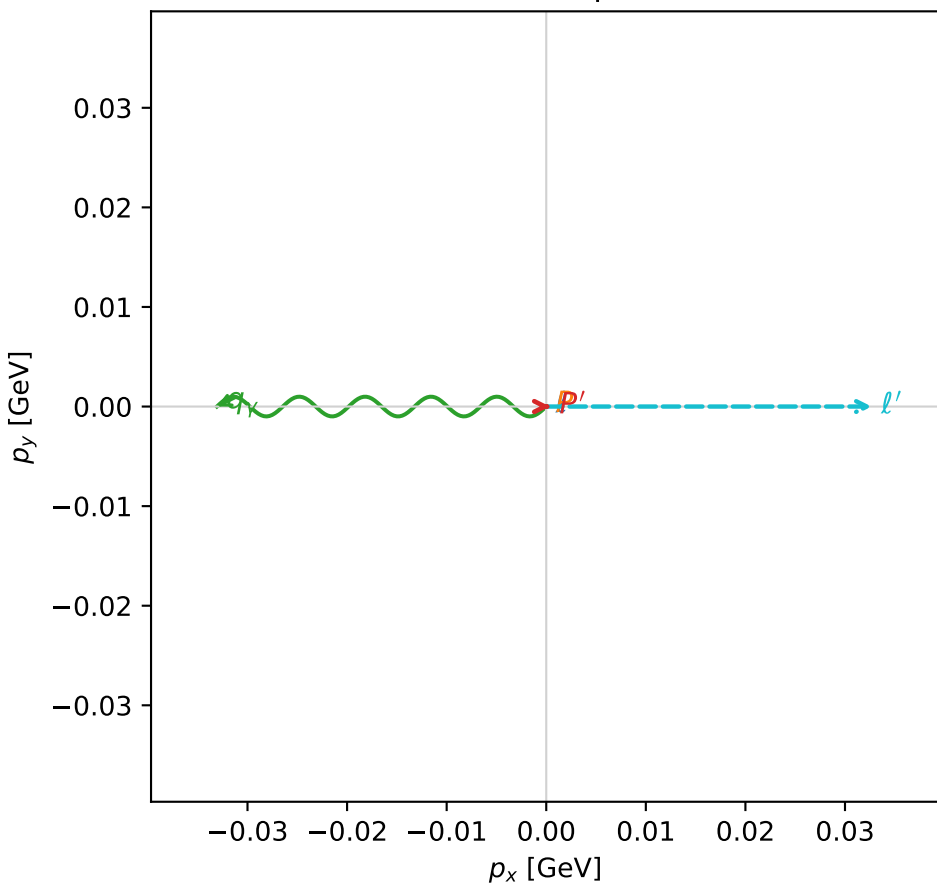
$s=24.8566$, $\sqrt{s}=4.98564$, $|\vec{P}|=2.40458$, $|\vec{P}'|=2.02645$
 $\theta_{P'}=0.000191071$, $\theta_V=3.12775$, $\phi_{P'}=6.28238$, $\phi_V=3.14161$
 $E_V=2.38892$, $\theta_{\ell'}=0.0899866$, $\phi_{\ell'}=2.49864\text{e-}05$
 $Q^2=3.49121$, $x_B=0.146436$, $t=-0.0218485$, $W^2=21.2299$, $y=0.994351$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4046$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4046$
 P $E= 2.5811$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4046$
 ℓ' $E= 0.3637$ $p_x= 0.0327$ $p_y= 0.0000$ $p_z= 0.3622$
 P' $E= 2.2330$ $p_x= 0.0004$ $p_y= -0.0000$ $p_z= 2.0264$
 q_V $E= 2.3889$ $p_x= -0.0331$ $p_y= -0.0000$ $p_z= -2.3887$

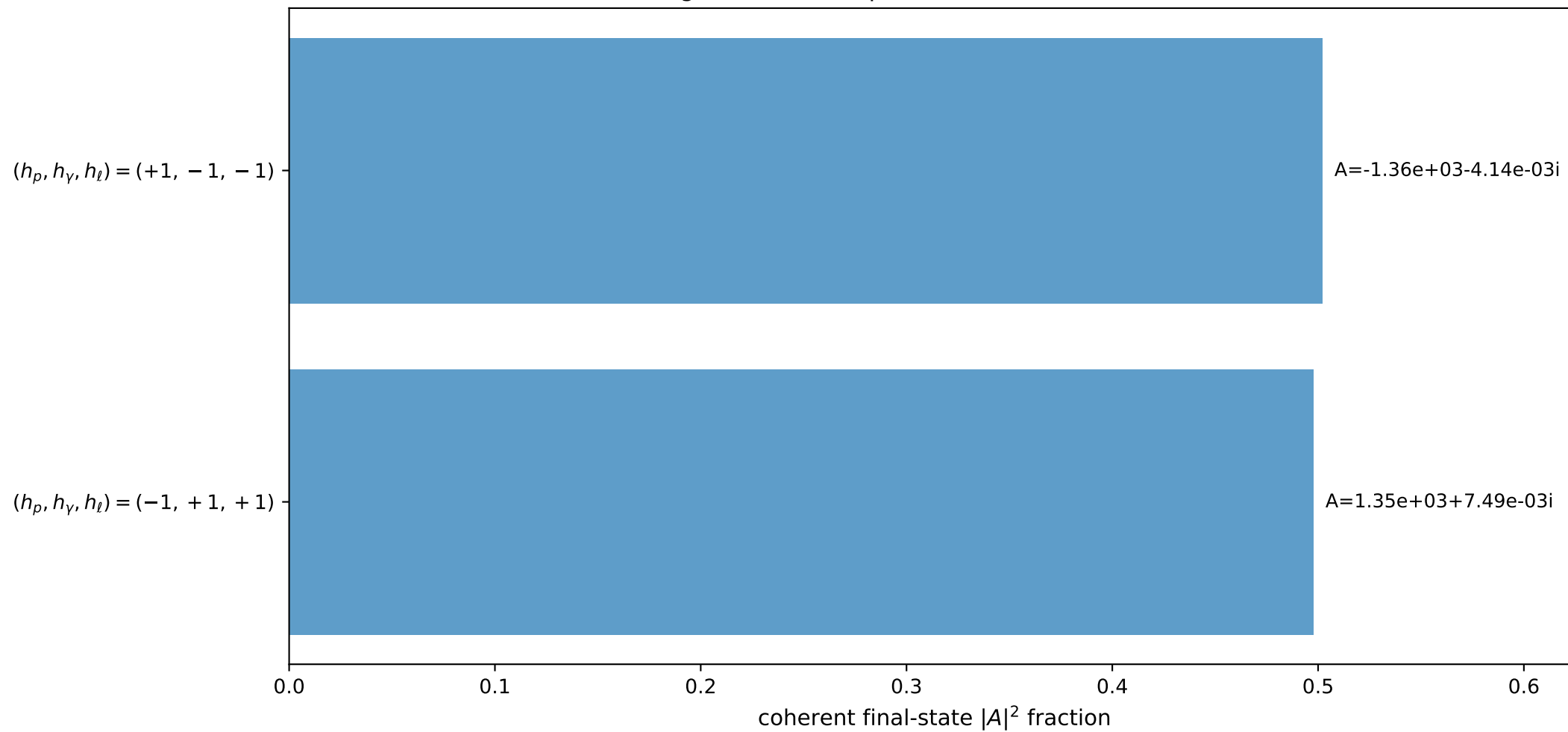
3D momenta



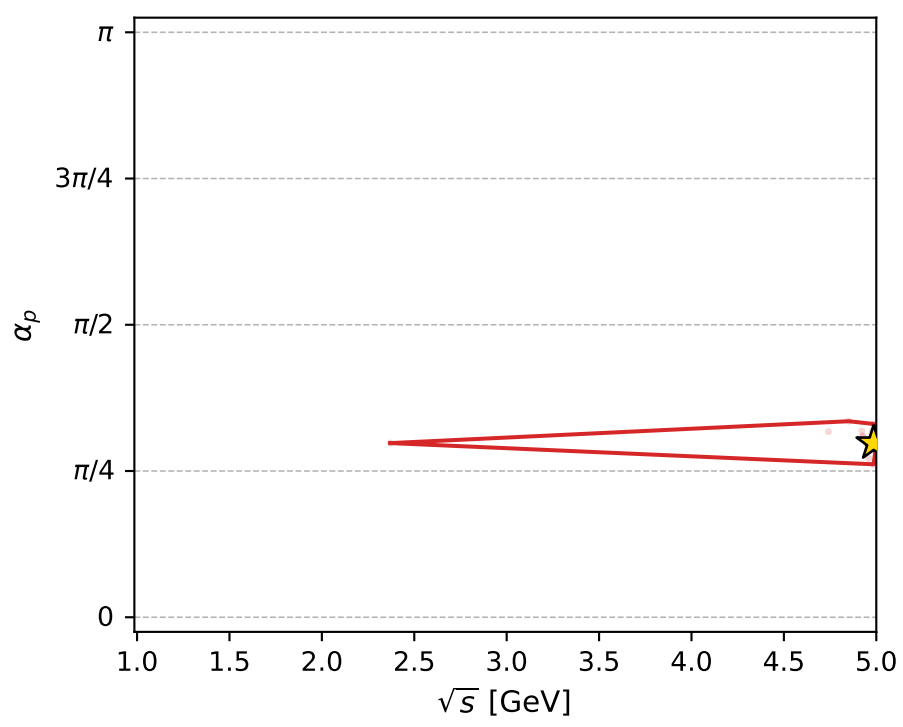
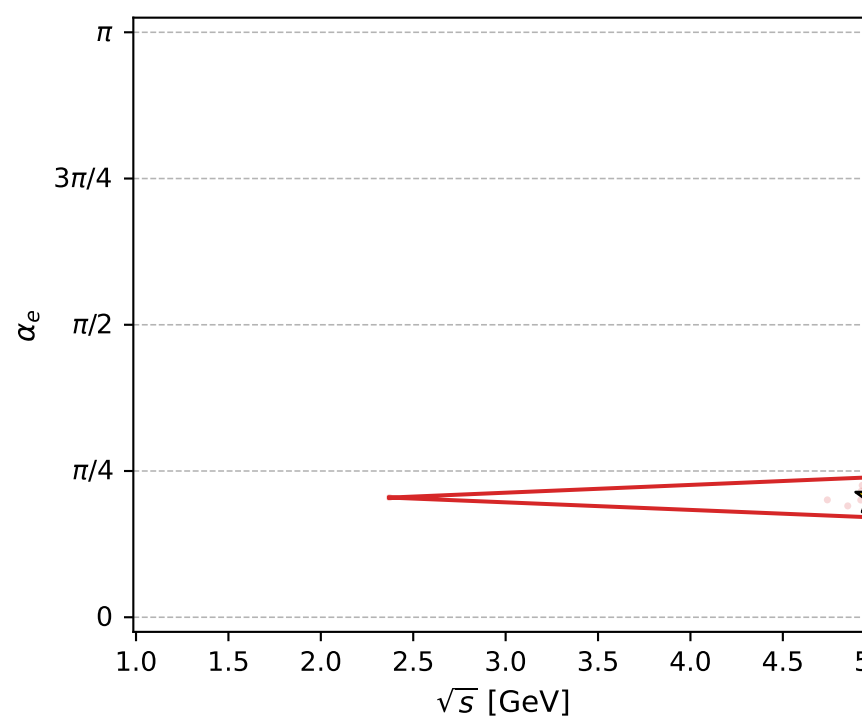
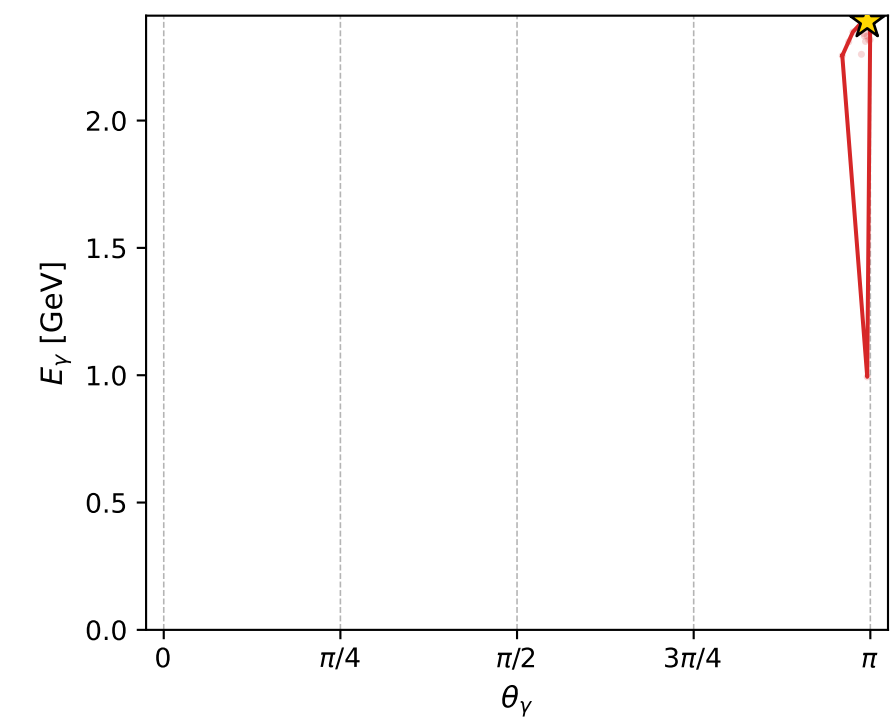
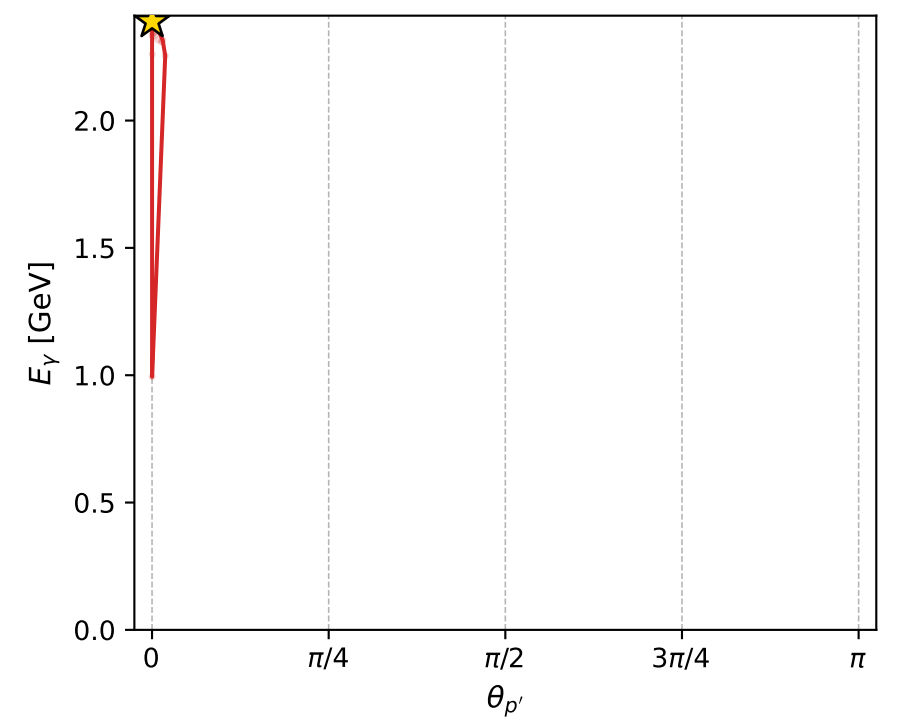
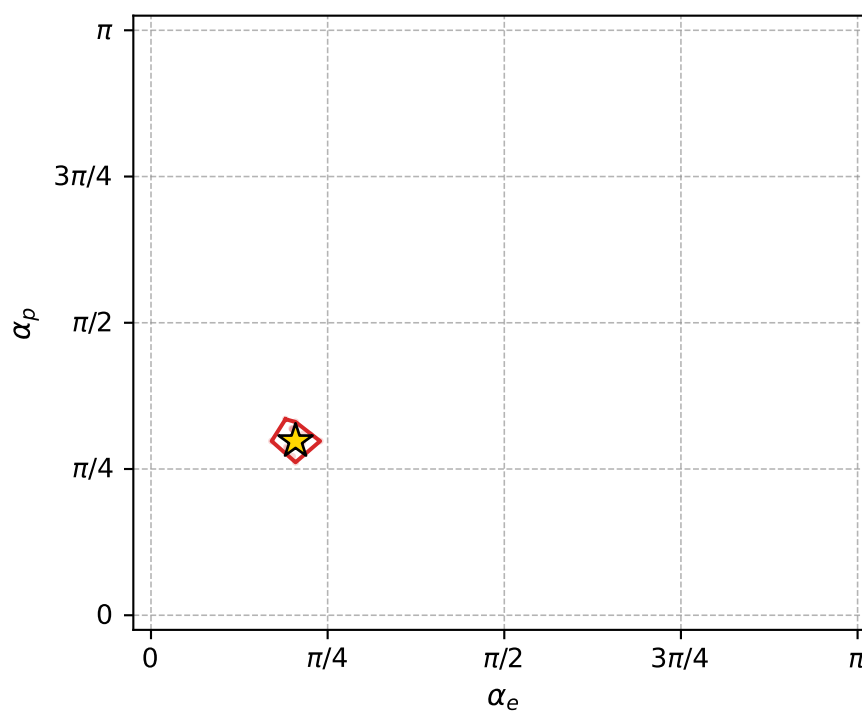
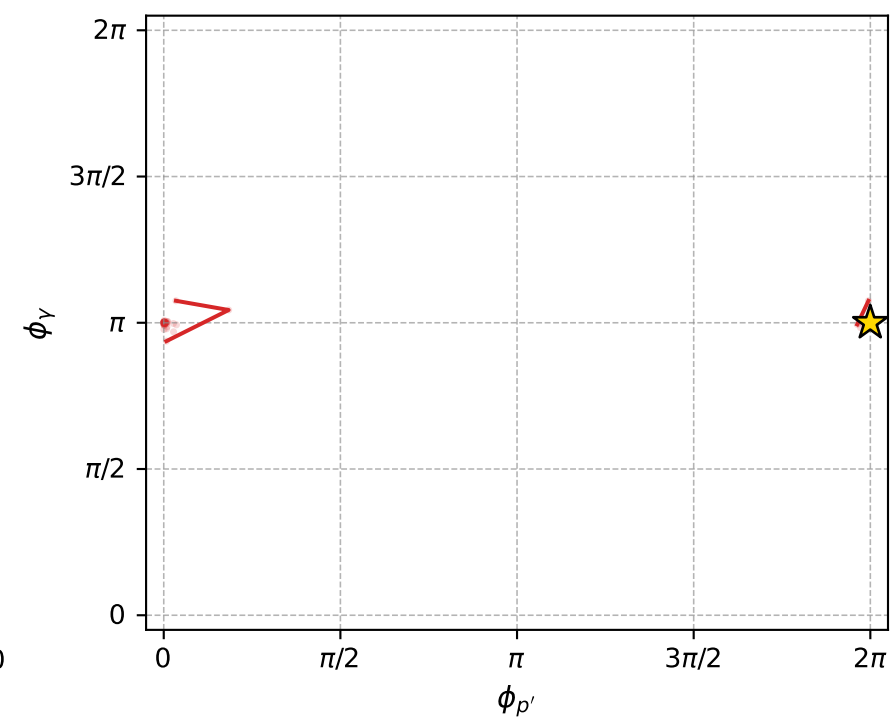
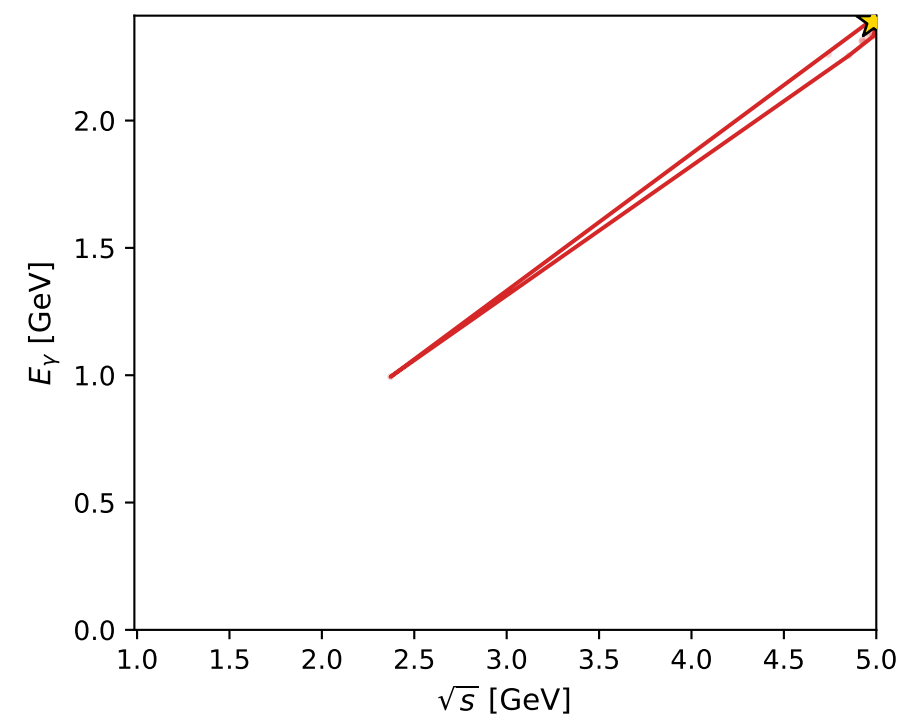
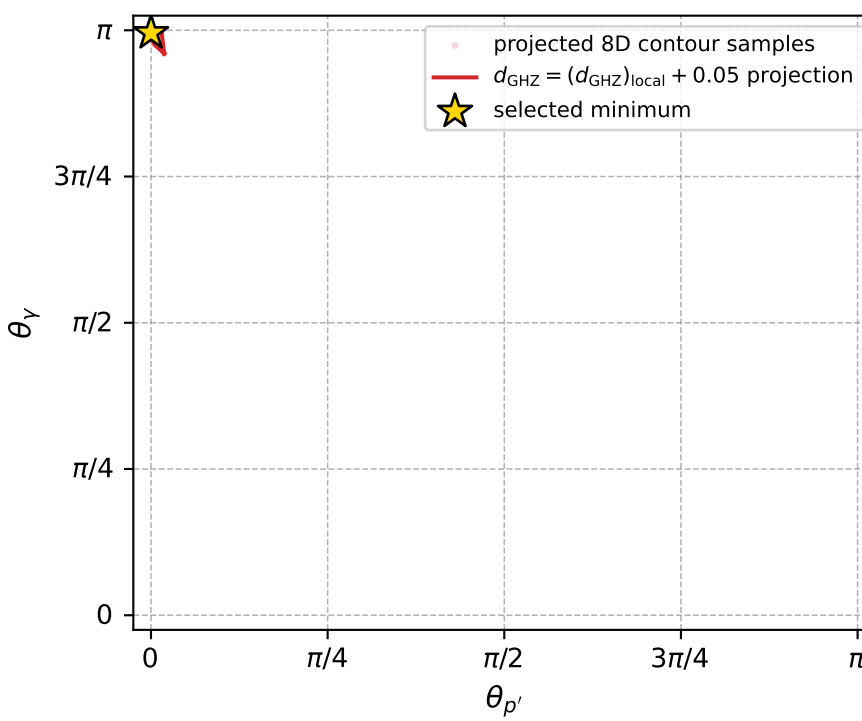
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 203; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.3412804\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050023413$
 above global minimum = $2.3390521\text{e-}05$
 8D contour samples = 130

selected phase-space point:

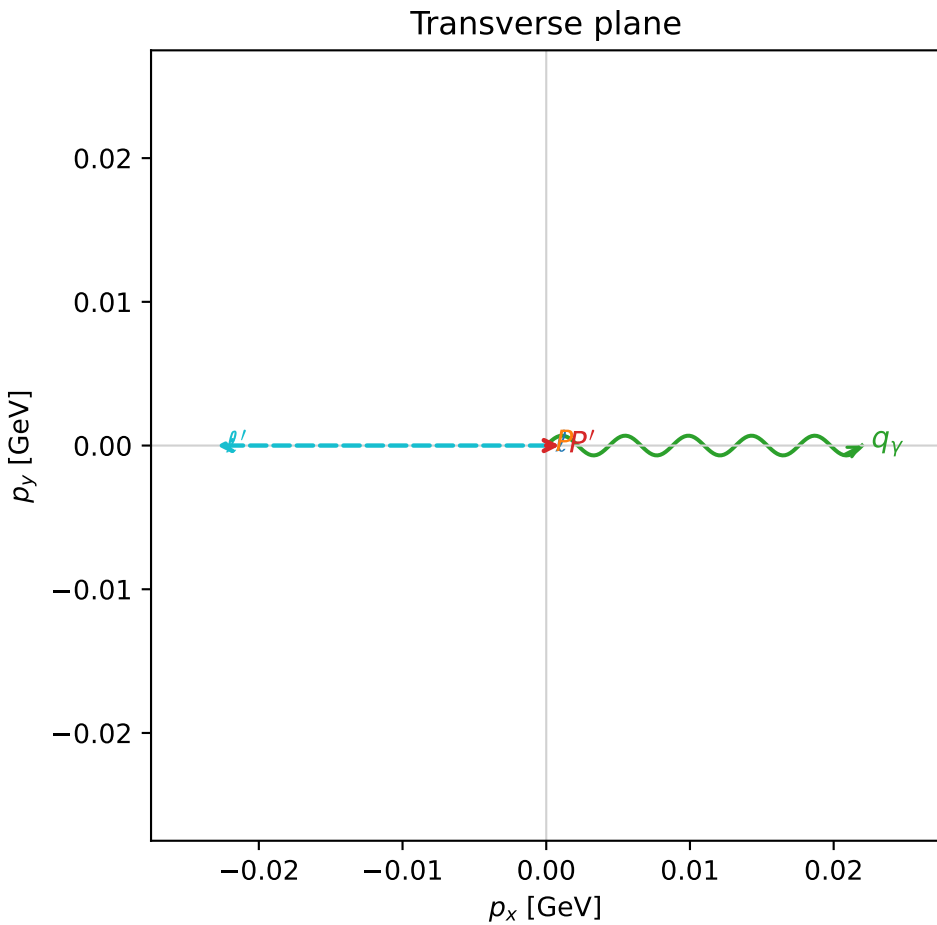
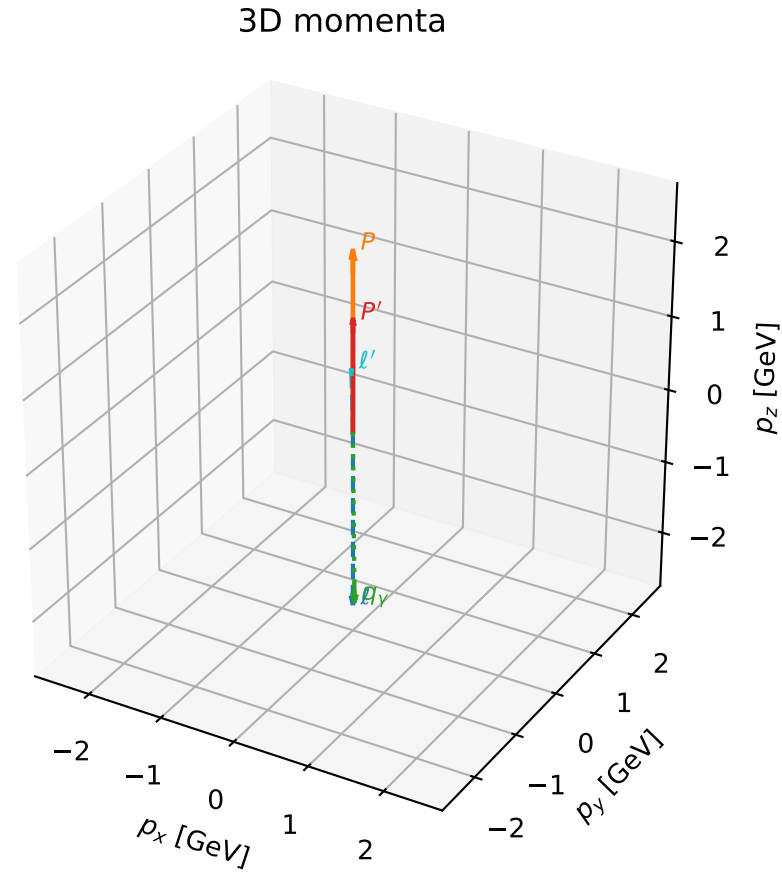
$\text{sqrt_s} = 4.985638$
 $\text{theta_p_out} = 0.0001910708$
 $\text{theta_gamma_out} = 3.127748$
 $\text{q0out} = 2.388916$
 $\text{phi_p_out} = 6.282377$
 $\text{phi_gamma_out} = 3.141608$
 $\alpha_e = 0.6426769$
 $\alpha_p = 0.9352928$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_206 $d_{\{\text{rm GHZ}\}}=2.59585\text{e-}05$
region: polarization_cluster_01_local_minimum_206
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.1210285$ rad
incoming lepton: $0.434756|+\rangle +0.900548|-\rangle$
proton mixing angle: $\alpha_p=0.45230063$ rad
incoming proton: $0.899444|+\rangle +0.437036|-\rangle$

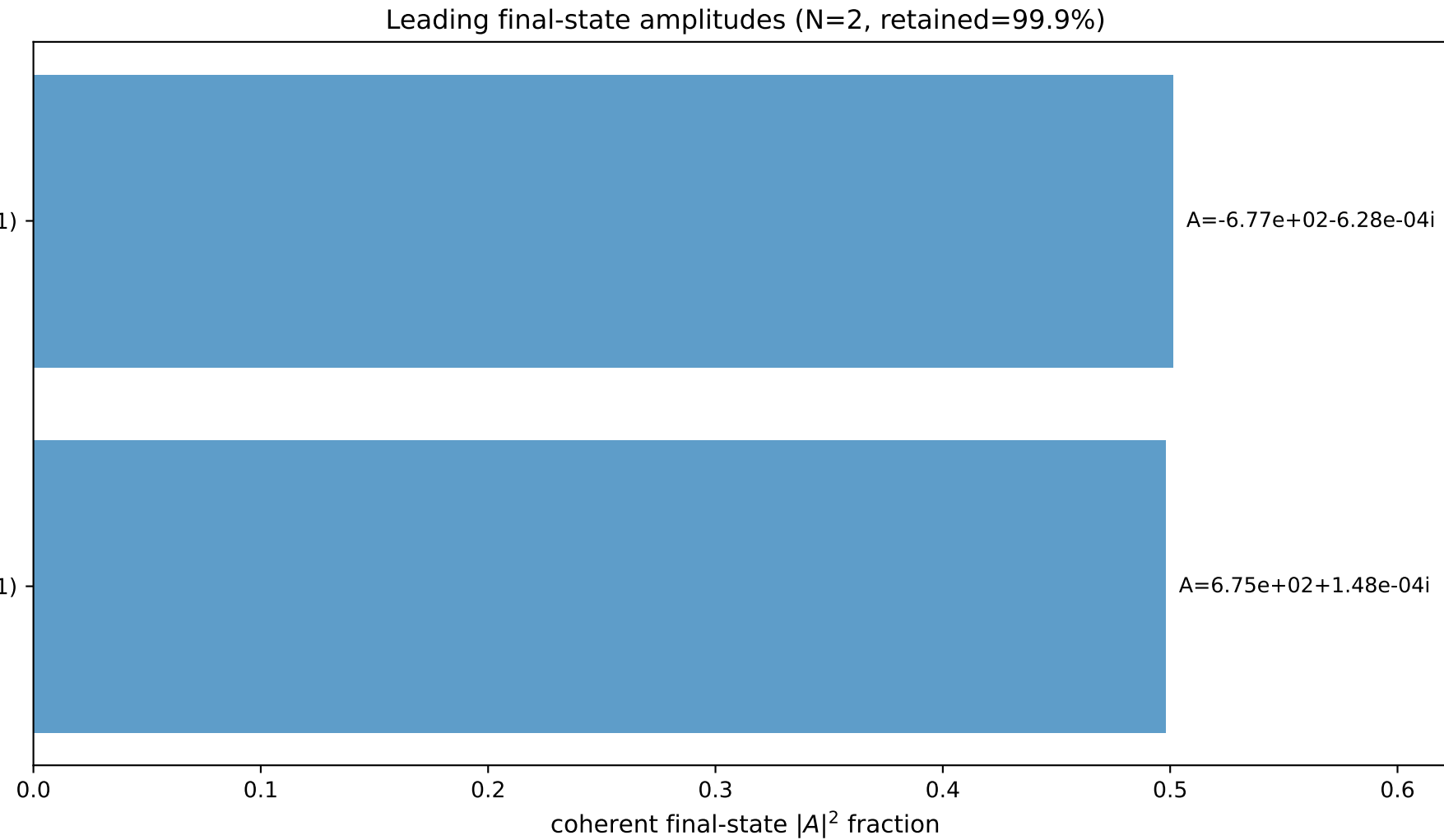
$s=24.8343$, $\sqrt{s}=4.9834$, $|\vec{P}|=2.40342$, $|\vec{P}'|=1.50557$
 $\theta_{P'}=0.000624821$, $\theta_V=3.13227$, $\phi_{P'}=6.28282$, $\phi_V=0$
 $E_V=2.35745$, $\theta_{\ell'}=0.0268985$, $\phi_{\ell'}=3.14158$
 $Q^2=8.19023$, $x_B=0.346279$, $t=-0.156318$, $W^2=16.3417$, $y=0.987378$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4034$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4034$
 P $E= 2.5800$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4034$
 ℓ' $E= 0.8521$ $p_x= -0.0229$ $p_y= 0.0000$ $p_z= 0.8518$
 P' $E= 1.7739$ $p_x= 0.0009$ $p_y= -0.0000$ $p_z= 1.5056$
 q_Y $E= 2.3575$ $p_x= 0.0220$ $p_y= 0.0000$ $p_z= -2.3574$

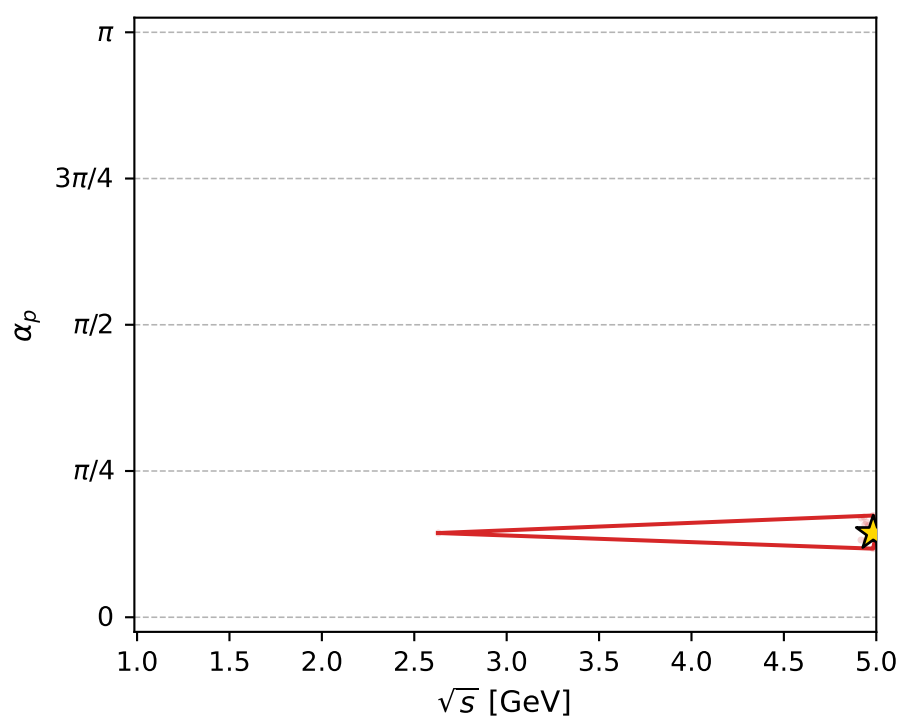
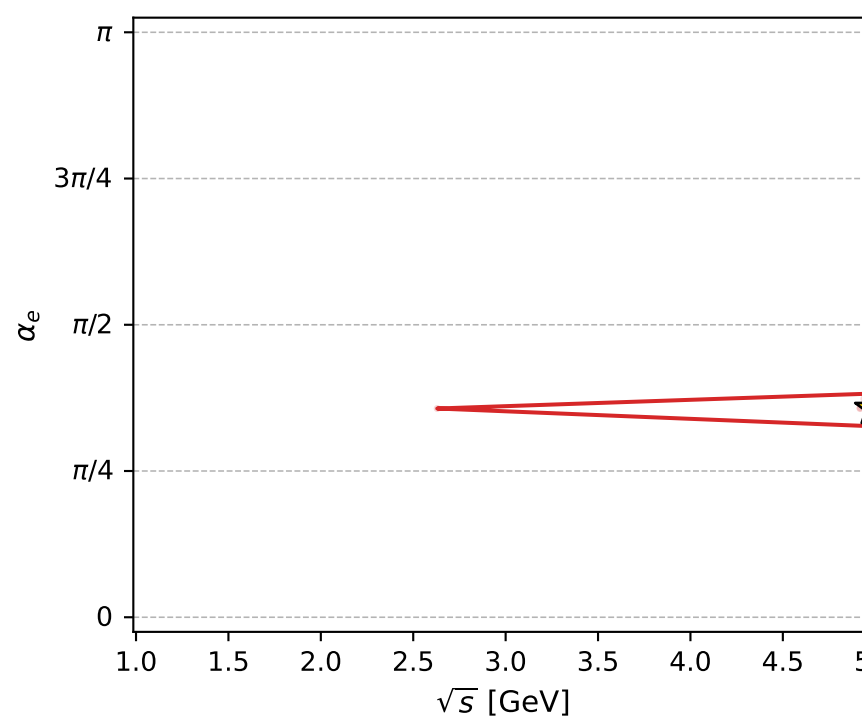
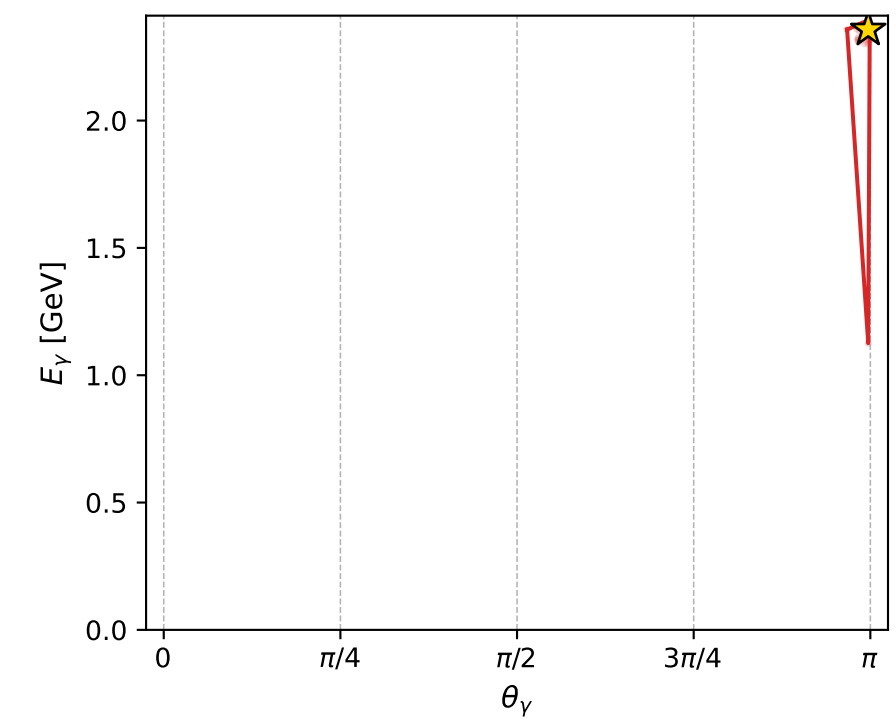
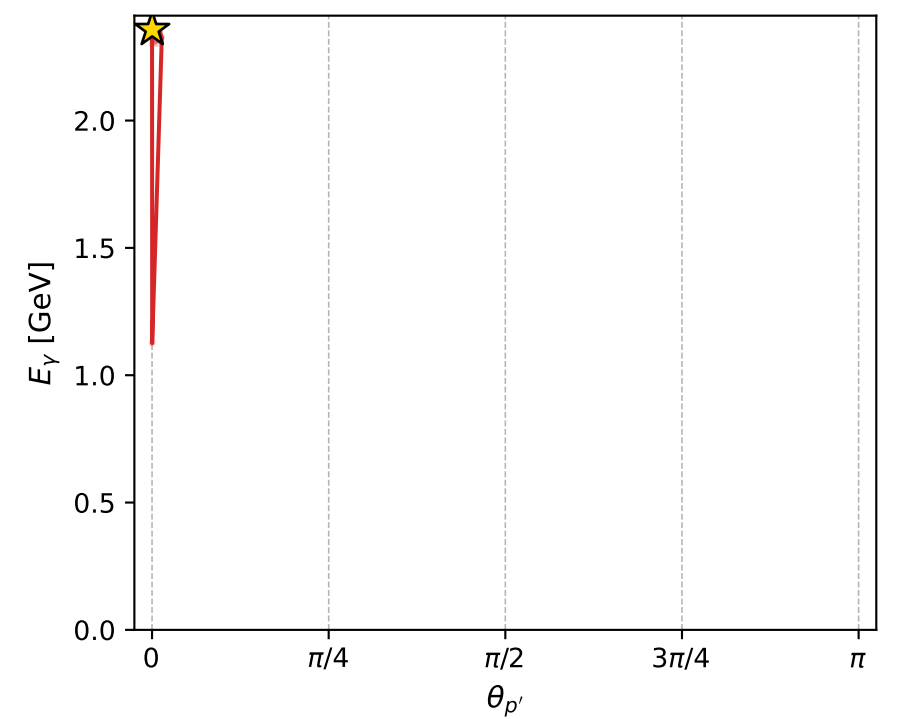
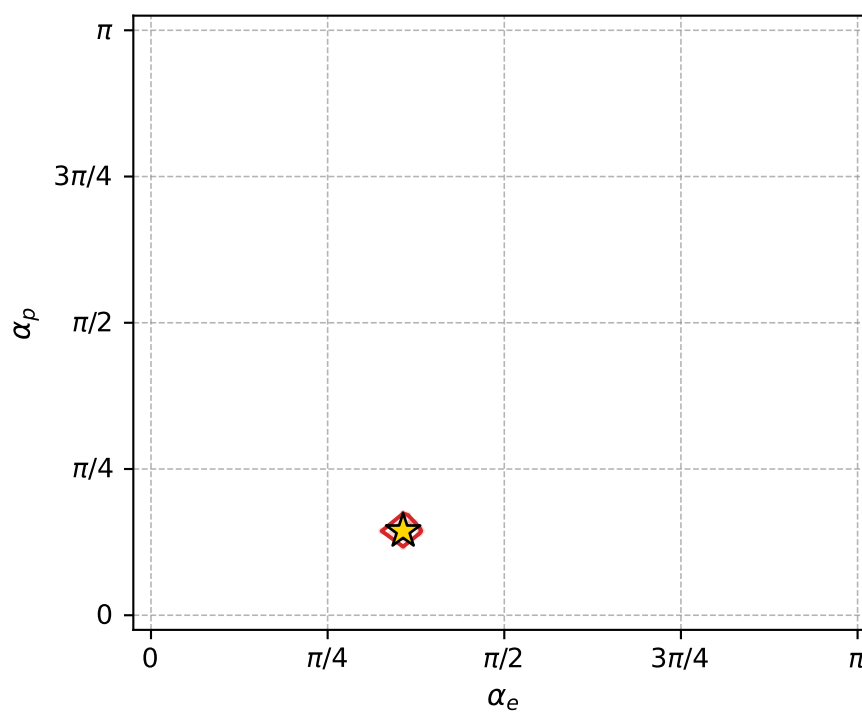
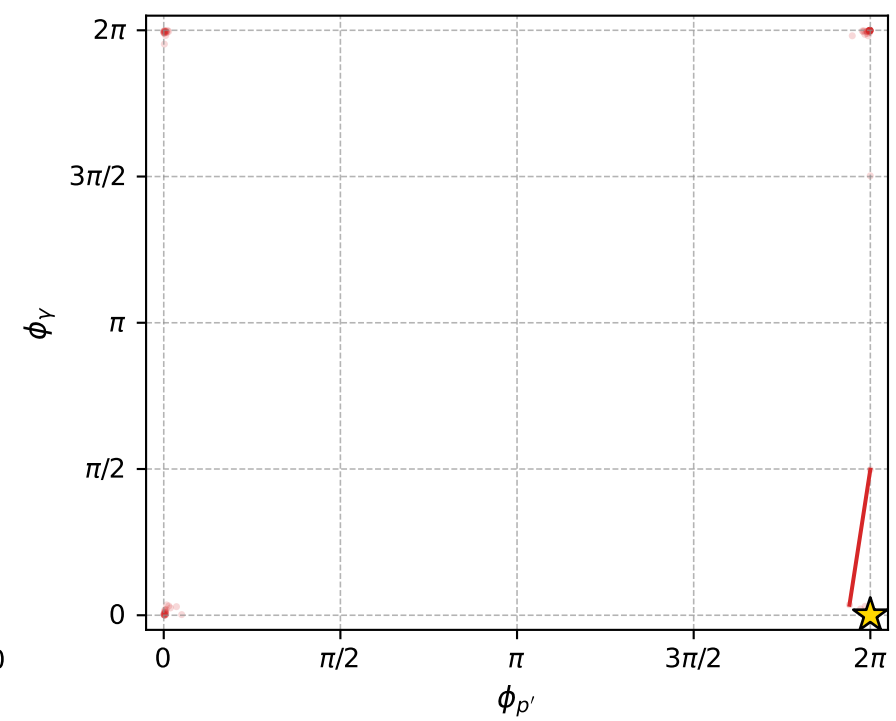
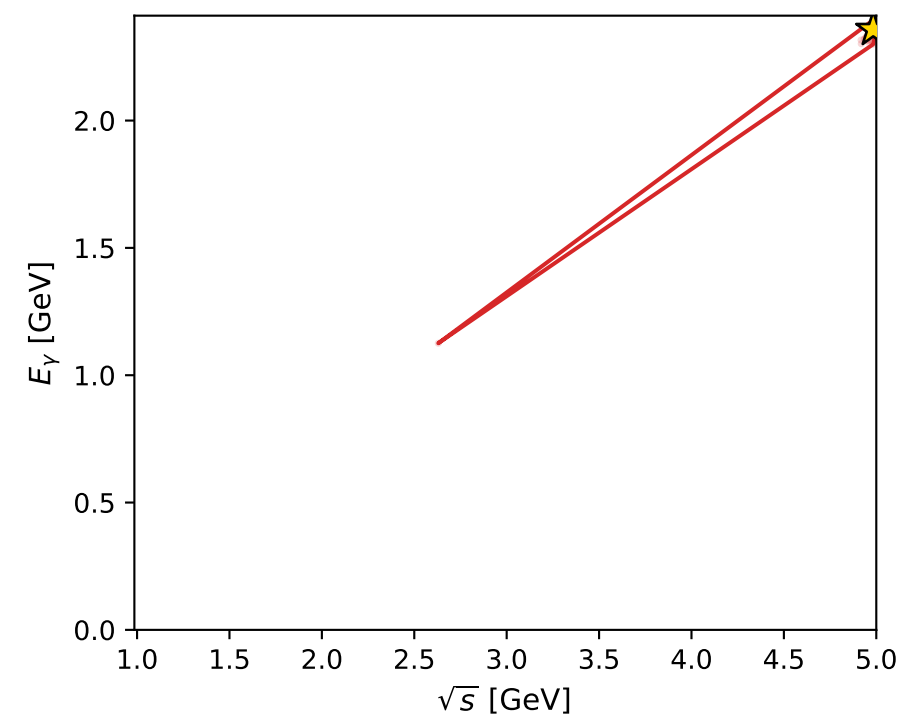
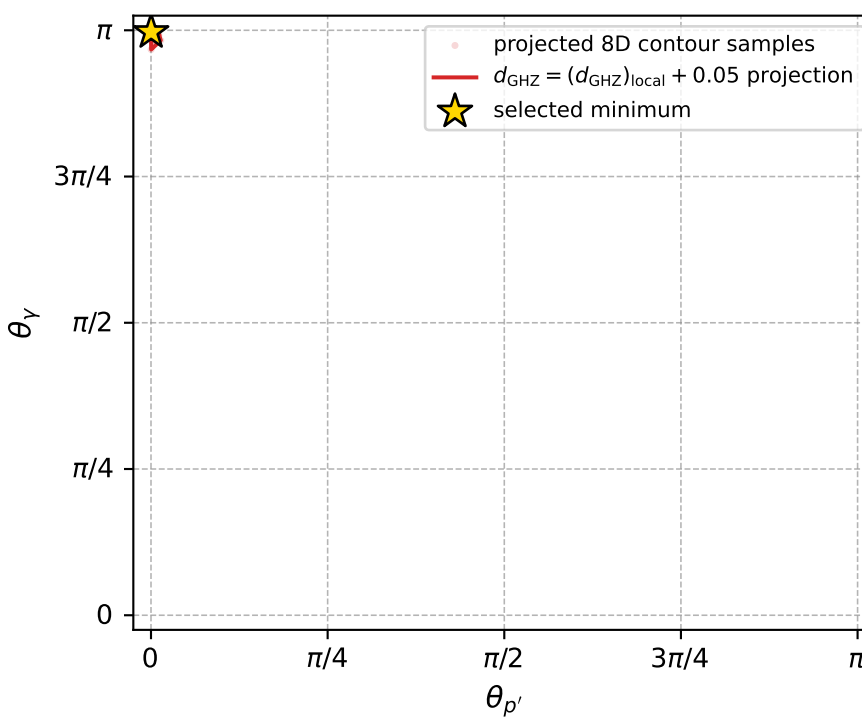


$(h_p, h_Y, h_\ell) = (+1, -1, -1)$

$(h_p, h_Y, h_\ell) = (-1, +1, +1)$



electron: polarization cluster P1, local minimum 206; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.5958529\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050025959$
 above global minimum = $2.5936245\text{e-}05$
 8D contour samples = 109

selected phase-space point:

$\text{sqrt_s} = 4.983404$
 $\text{theta_p_out} = 0.0006248208$
 $\text{theta_gamma_out} = 3.13227$
 $\text{q0out} = 2.357453$
 $\text{phi_p_out} = 6.282825$
 $\text{phi_gamma_out} = 0$
 $\text{alpha_e} = 1.121028$
 $\text{alpha_p} = 0.4523006$

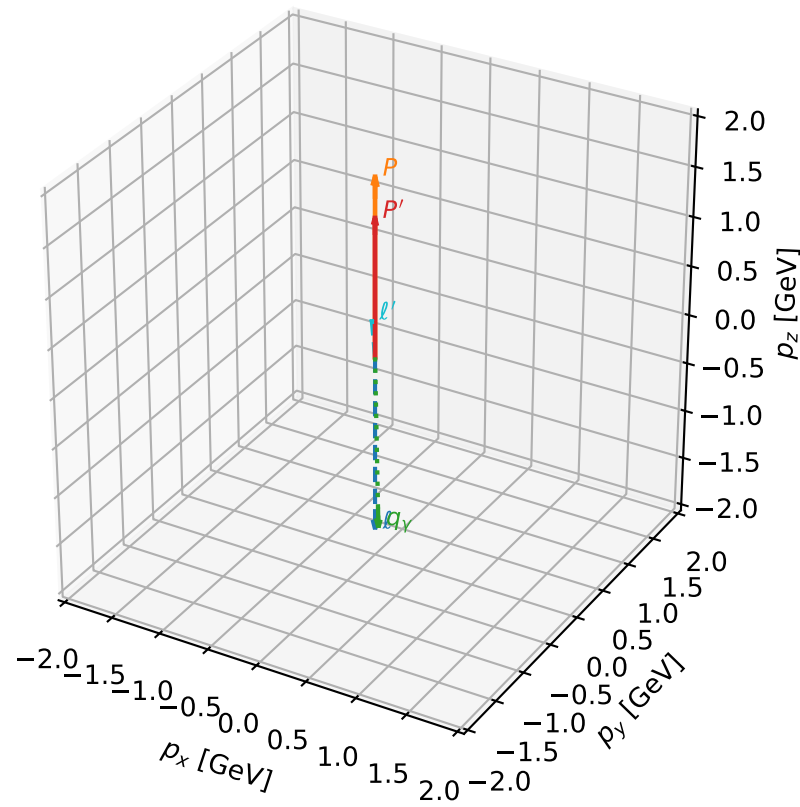
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_207 $d_{\{\text{rm GHZ}\}}=2.62949\text{e-}05$
 region: polarization_cluster_01_local_minimum_207
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.0858098$ rad
 incoming lepton: $0.466196|+\rangle + 0.884681|-\rangle$
 proton mixing angle: $\alpha_p=0.47680636$ rad
 incoming proton: $0.888465|+\rangle + 0.458944|-\rangle$

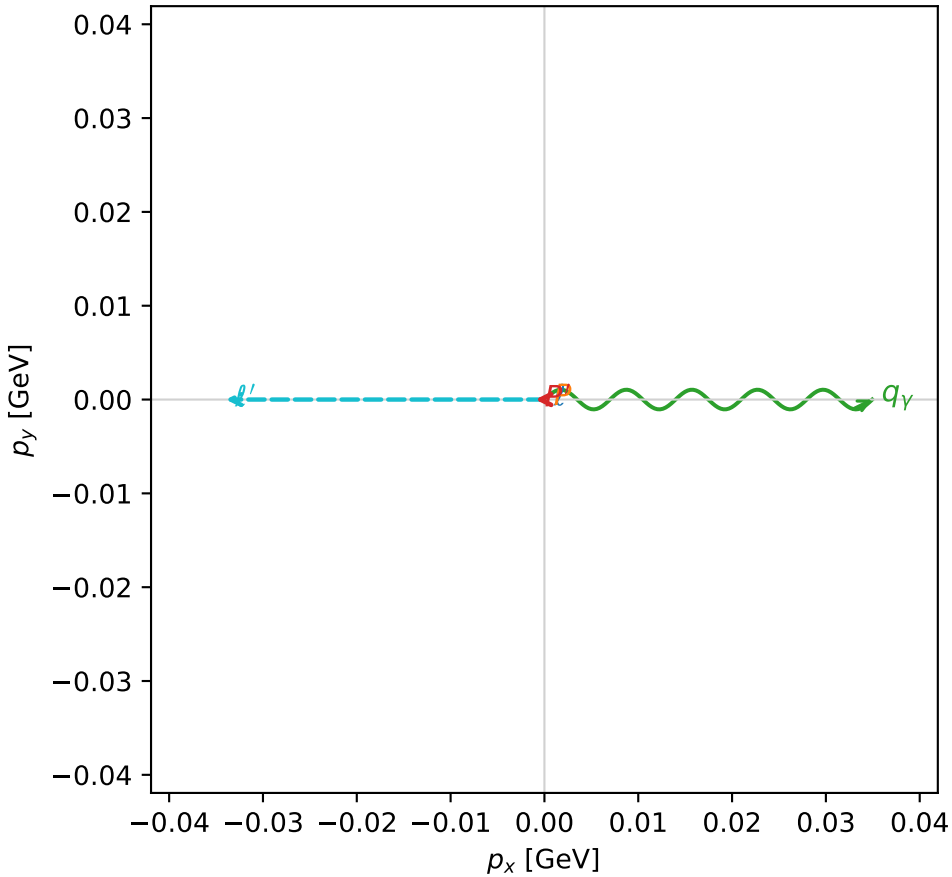
$s=14.6057$, $\sqrt{s}=3.82175$, $|\vec{P}|=1.79576$, $|\vec{P}'|=1.39318$
 $\theta_{p'}=0.000700268$, $\theta_V=3.12182$, $\phi_{p'}=3.14146$, $\phi_V=1.35121\text{e-}05$
 $E_V=1.7671$, $\theta_{\ell'}=0.0906712$, $\phi_{\ell'}=3.14161$
 $Q^2=2.68895$, $x_B=0.198481$, $t=-0.0420382$, $W^2=11.7385$, $y=0.987013$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7958$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7958$
 P $E= 2.0260$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7958$
 ℓ' $E= 0.3751$ $p_x= -0.0340$ $p_y= -0.0000$ $p_z= 0.3736$
 P' $E= 1.6795$ $p_x= -0.0010$ $p_y= 0.0000$ $p_z= 1.3932$
 q_V $E= 1.7671$ $p_x= 0.0349$ $p_y= 0.0000$ $p_z= -1.7668$

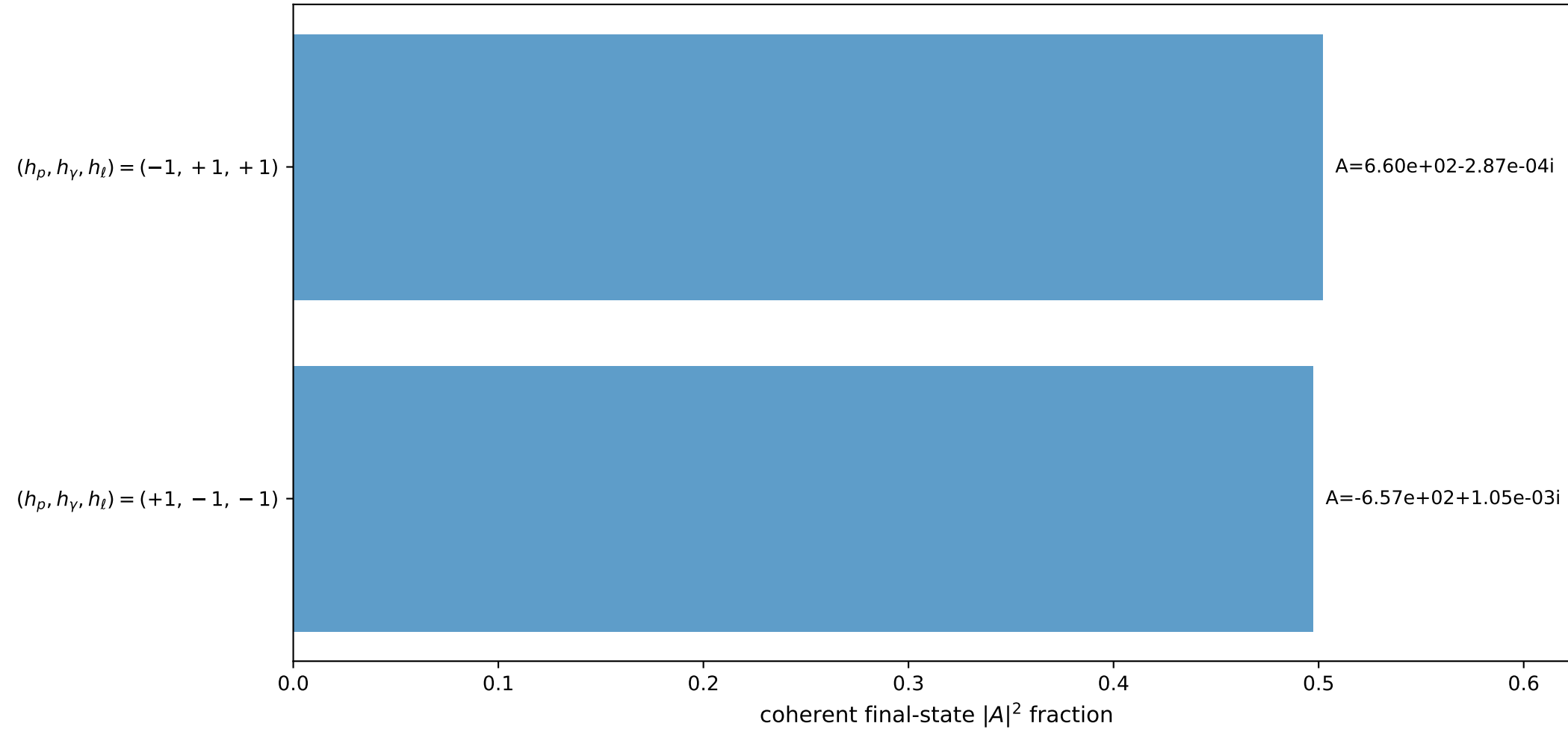
3D momenta



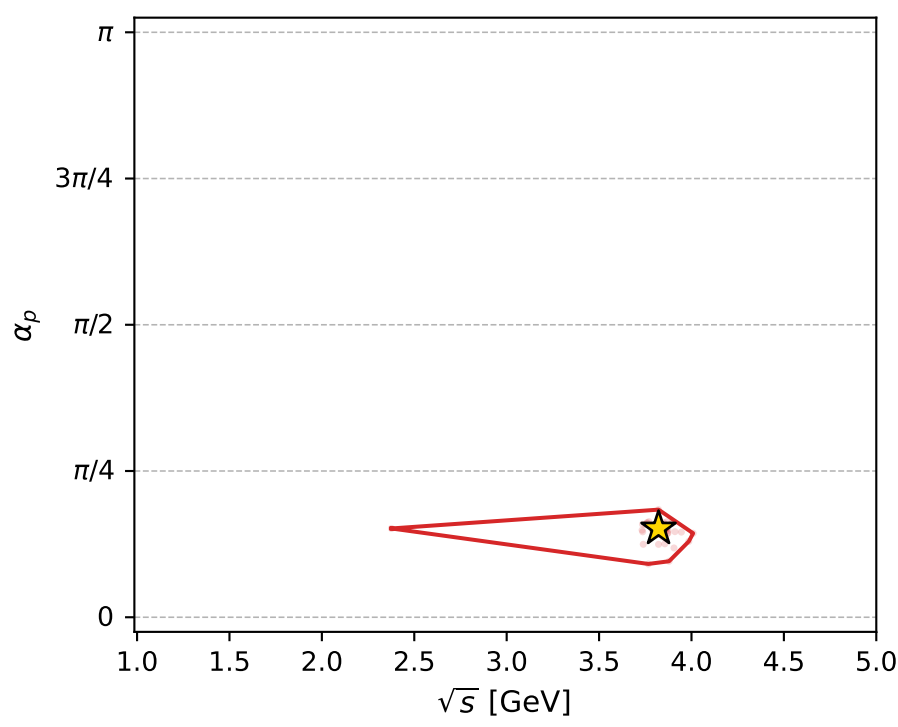
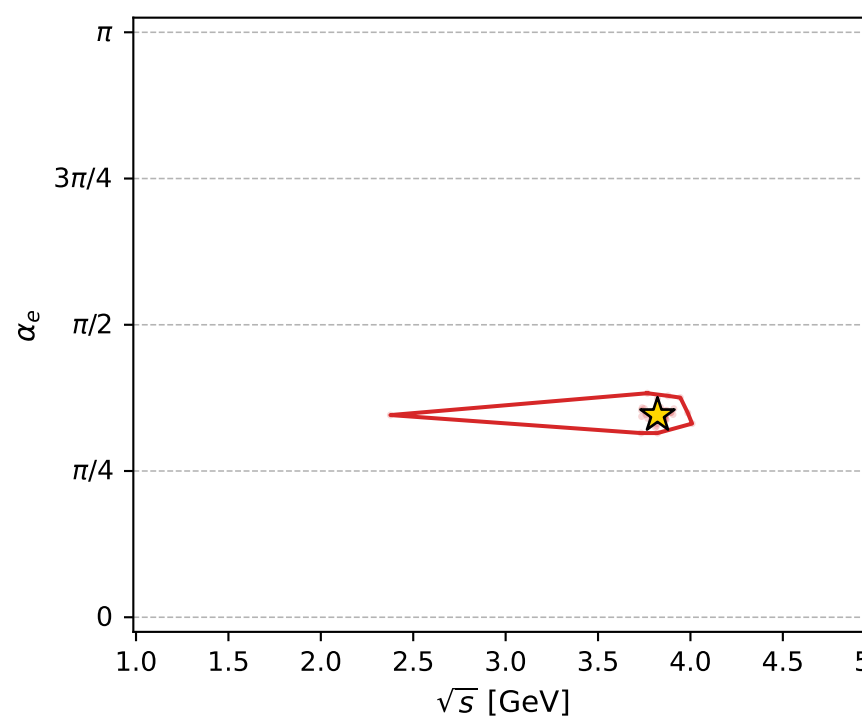
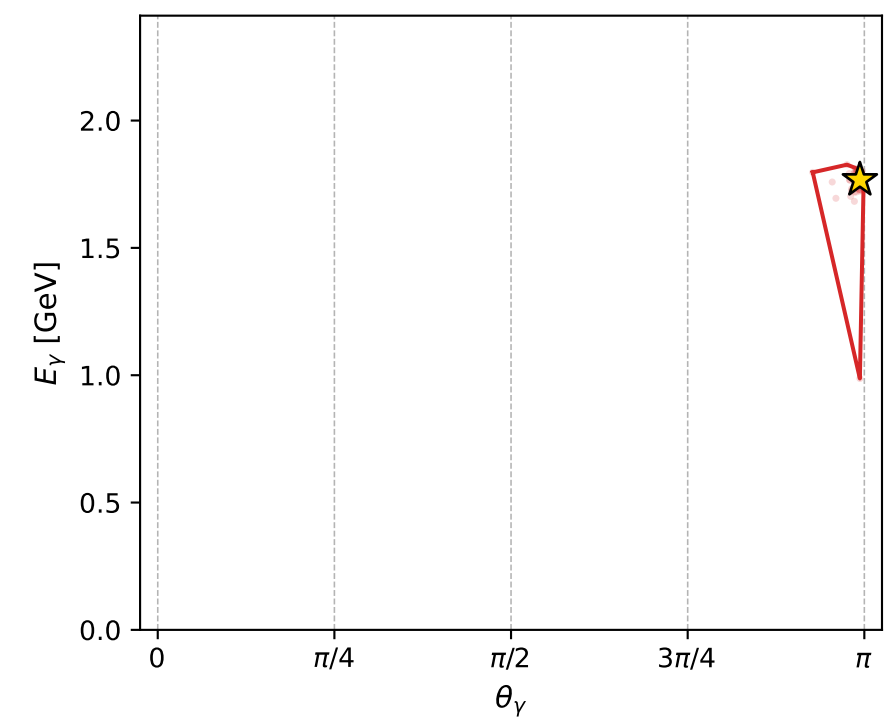
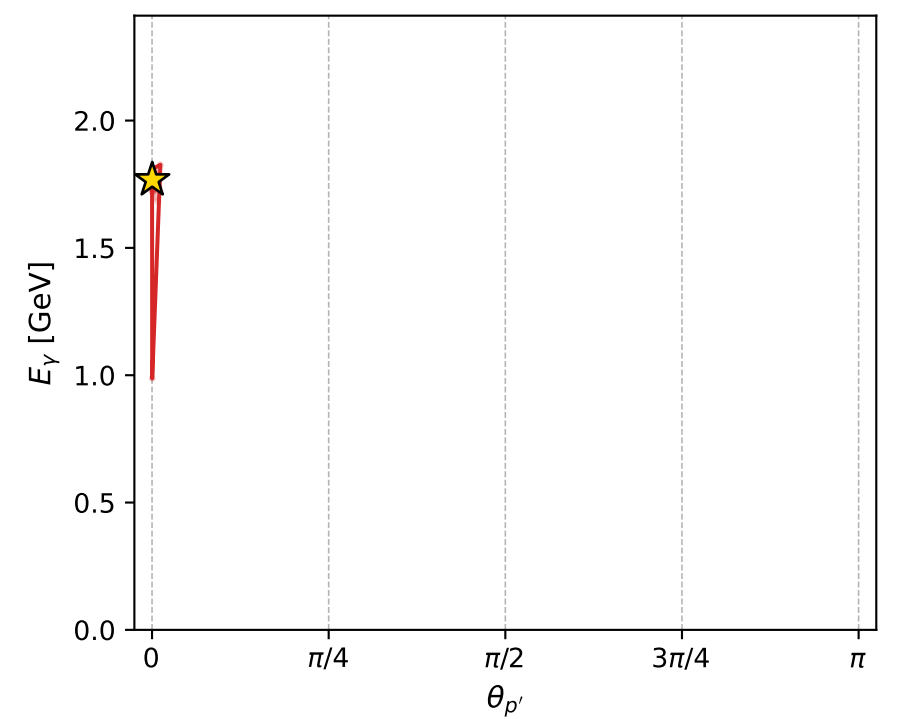
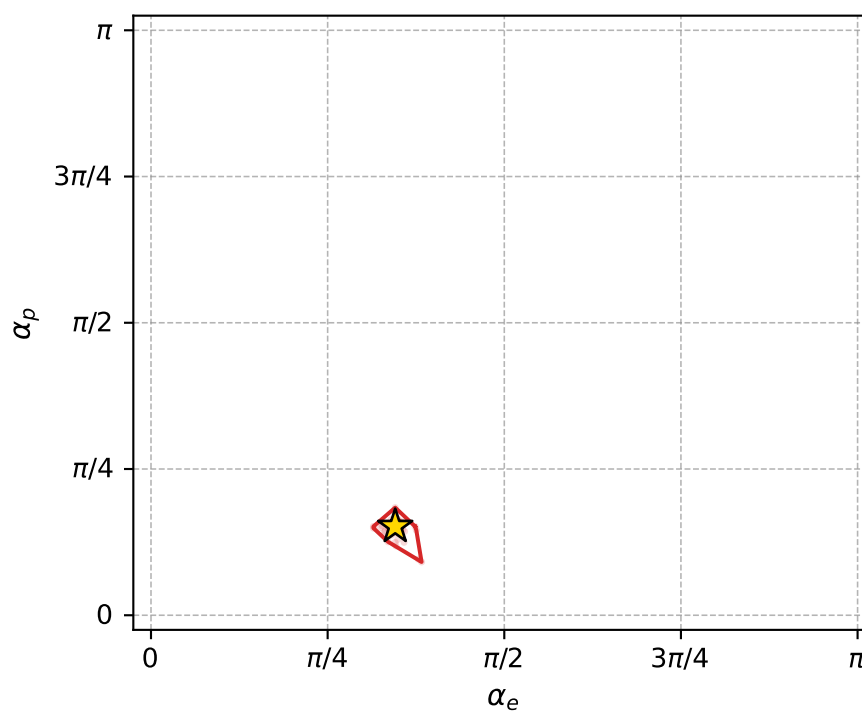
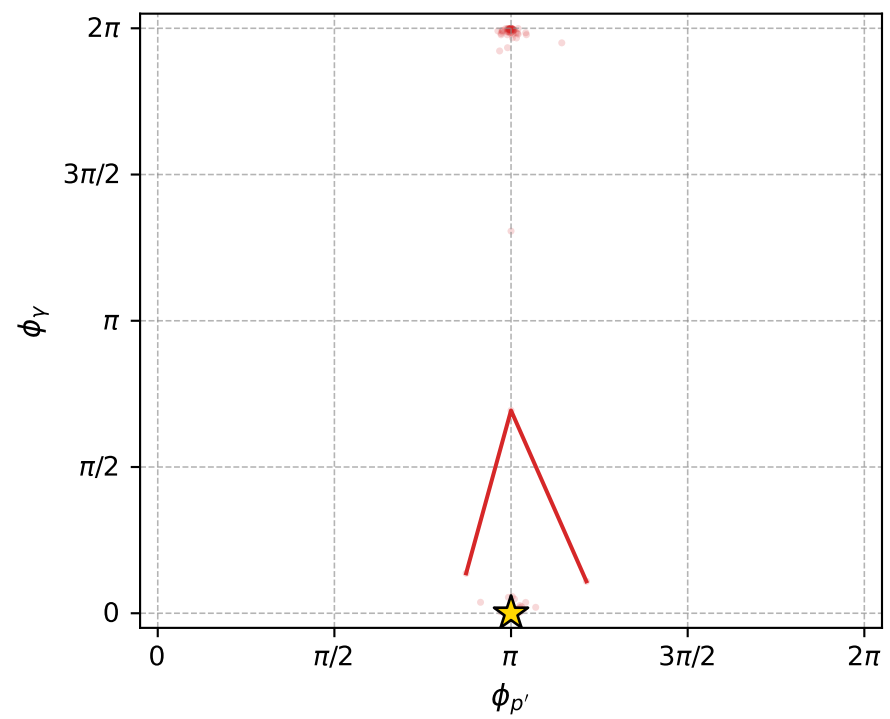
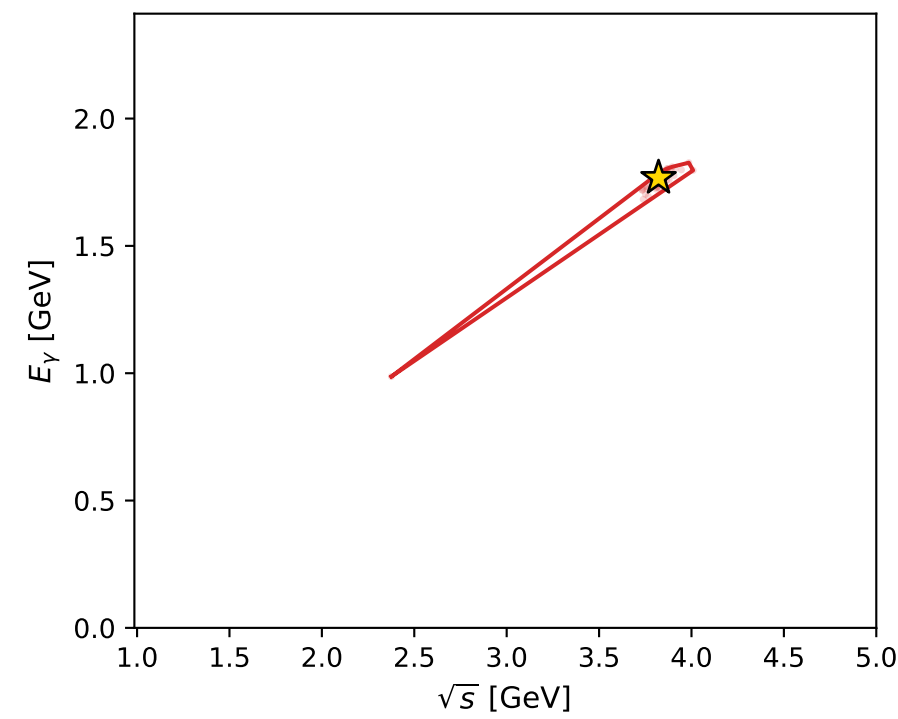
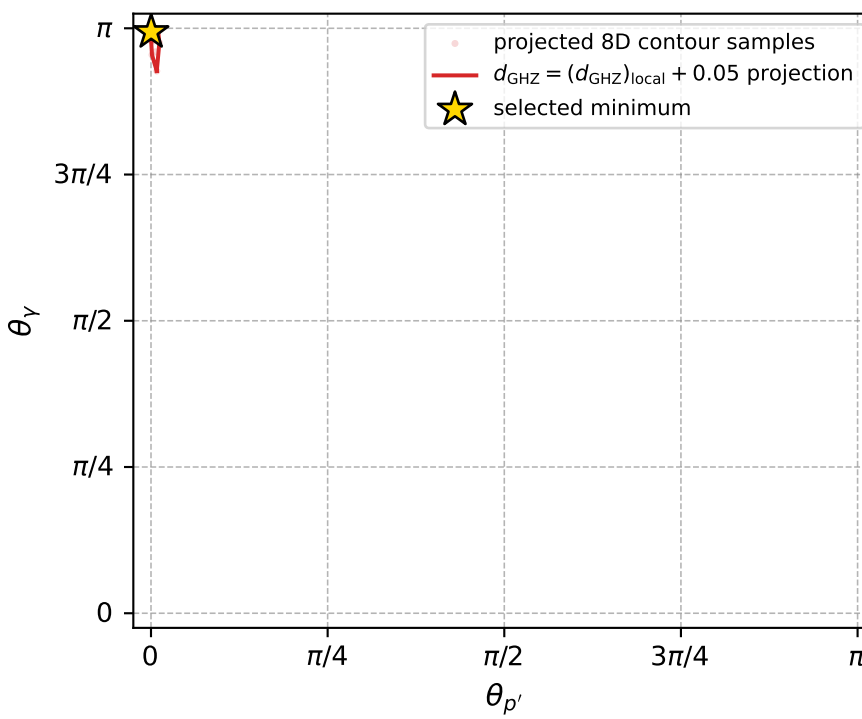
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 207; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 2.629491\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050026295$
 above global minimum = $2.6272626\text{e-}05$
 8D contour samples = 138

selected phase-space point:

$\text{sqrt_s} = 3.821746$
 $\text{theta_p_out} = 0.0007002678$
 $\text{theta_gamma_out} = 3.121818$
 $\text{q0ut} = 1.767104$
 $\text{phi_p_out} = 3.141459$
 $\text{phi_gamma_out} = 1.351213\text{e-}05$
 $\text{alpha_e} = 1.08581$
 $\text{alpha_p} = 0.4768064$

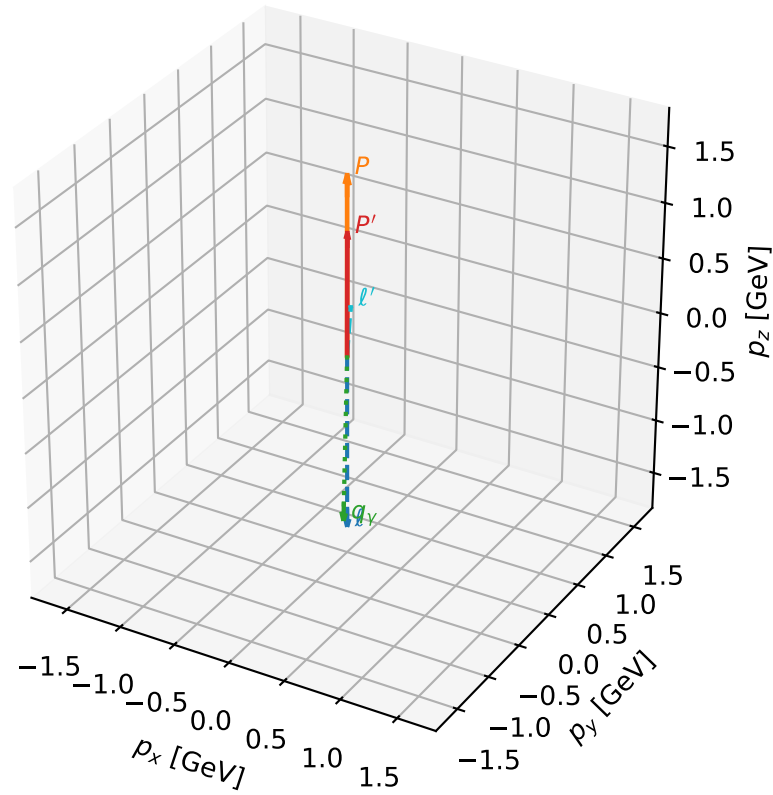
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_216 $d_{\text{GHZ}}=3.98745\text{e-}05$
 region: polarization_cluster_01_local_minimum_216
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.32329631$ rad
 incoming lepton: $0.948193|+\rangle +0.317694|-\rangle$
 proton mixing angle: $\alpha_p=1.2539312$ rad
 incoming proton: $0.311589|+\rangle +0.950217|-\rangle$

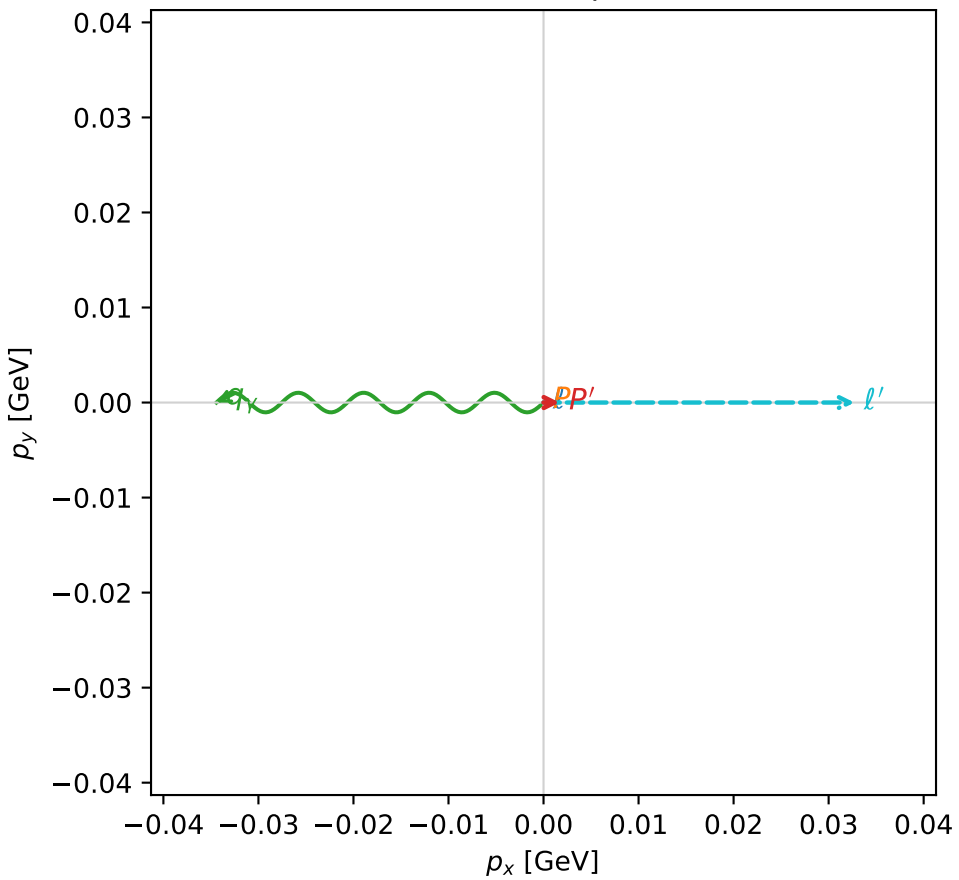
$s=11.9072$, $\sqrt{s}=3.45069$, $|\vec{P}|=1.59785$, $|\vec{P}'|=1.09272$
 $\theta_{p'}=0.00155077$, $\theta_Y=3.1194$, $\phi_{p'}=6.28293$, $\phi_Y=3.14161$
 $E_Y=1.55126$, $\theta_{\ell'}=0.0712991$, $\phi_{\ell'}=2.66731\text{e-}05$
 $Q^2=2.93205$, $x_B=0.271753$, $t=-0.0848181$, $W^2=8.73722$, $y=0.97842$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5979$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5979$
 P $E= 1.8528$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5979$
 ℓ' $E= 0.4593$ $p_x= 0.0327$ $p_y= 0.0000$ $p_z= 0.4582$
 P' $E= 1.4401$ $p_x= 0.0017$ $p_y= -0.0000$ $p_z= 1.0927$
 q_Y $E= 1.5513$ $p_x= -0.0344$ $p_y= -0.0000$ $p_z= -1.5509$

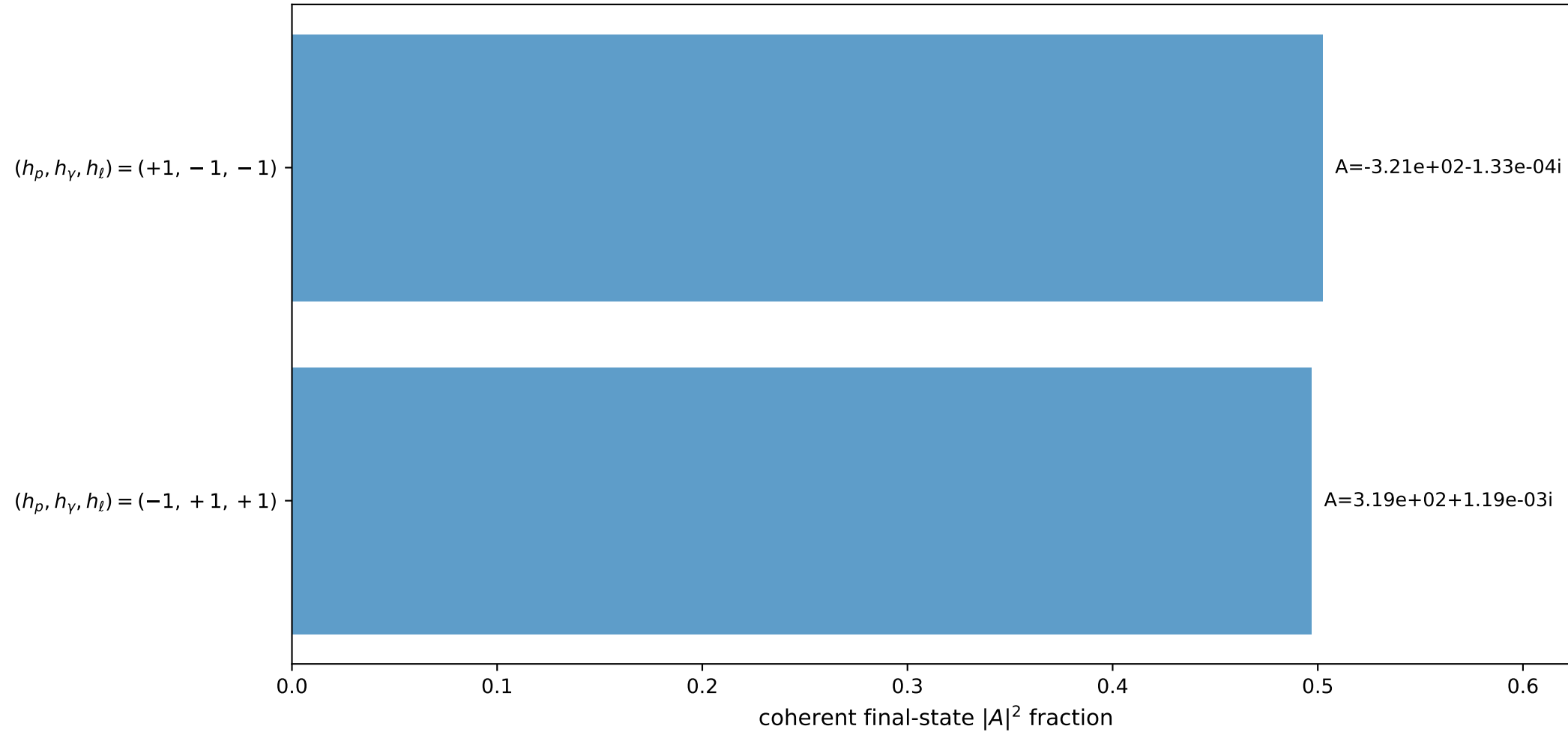
3D momenta



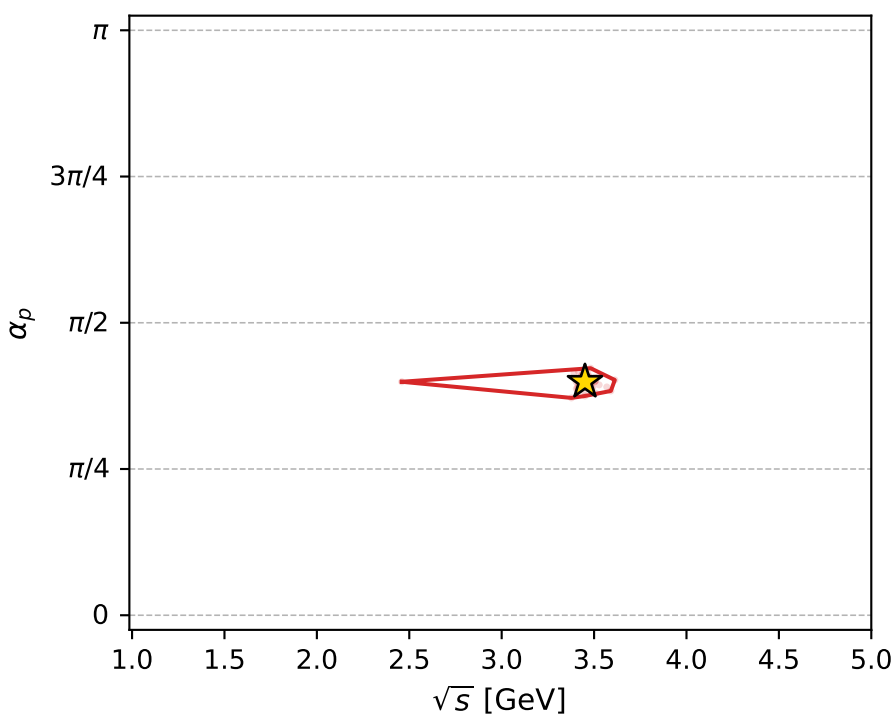
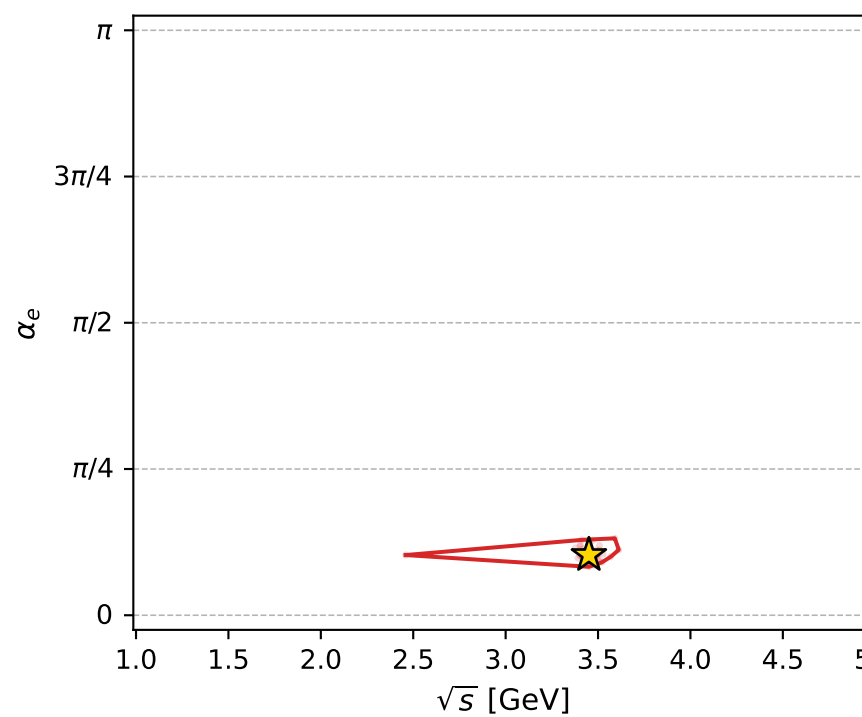
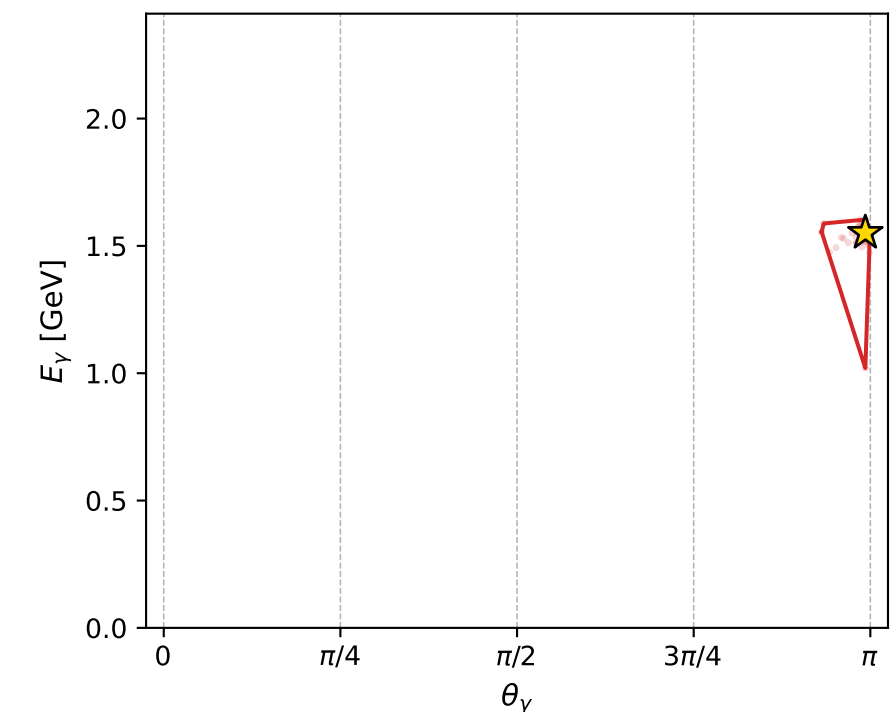
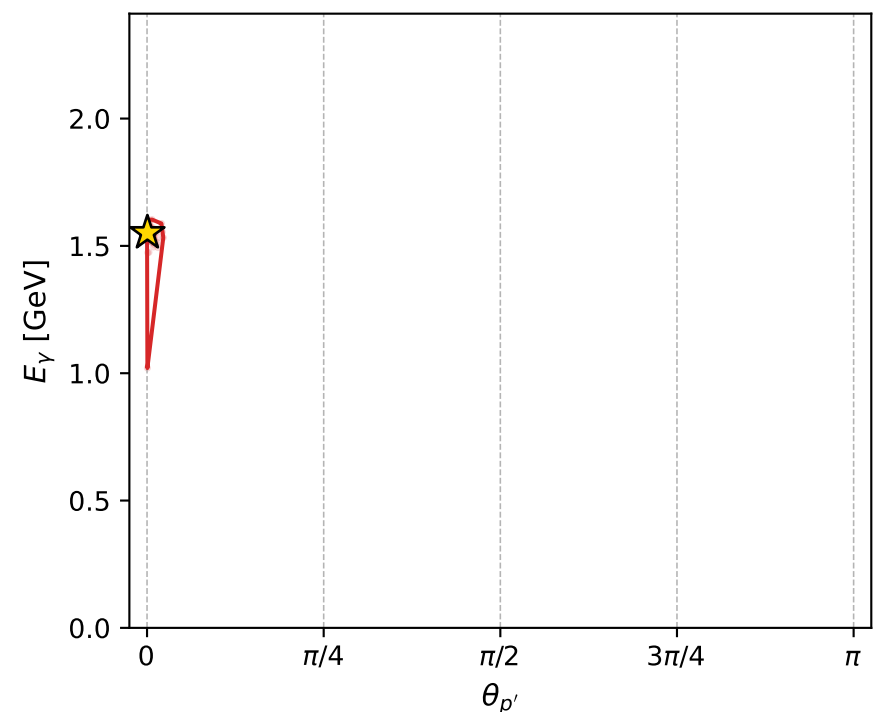
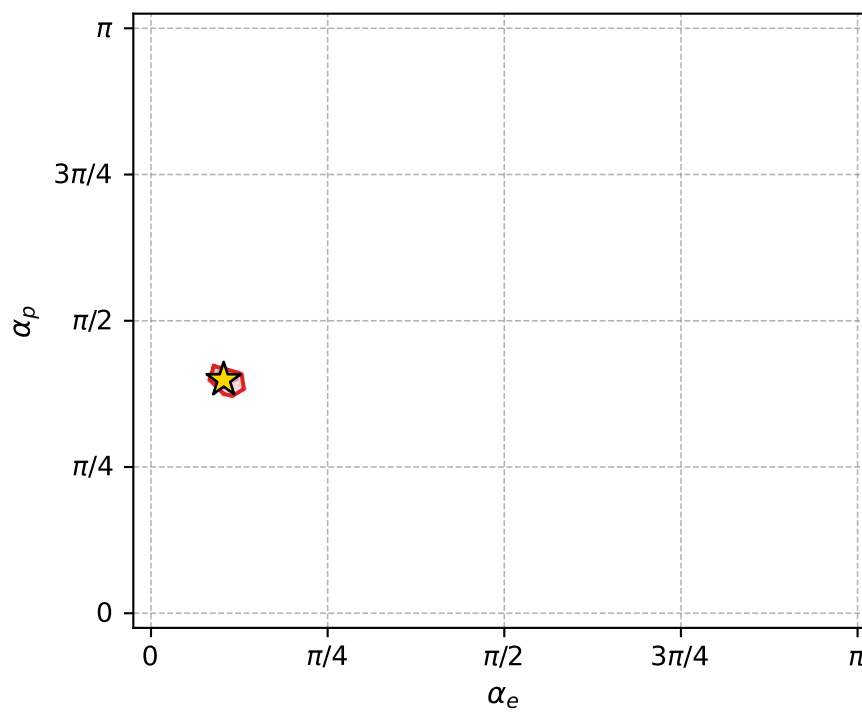
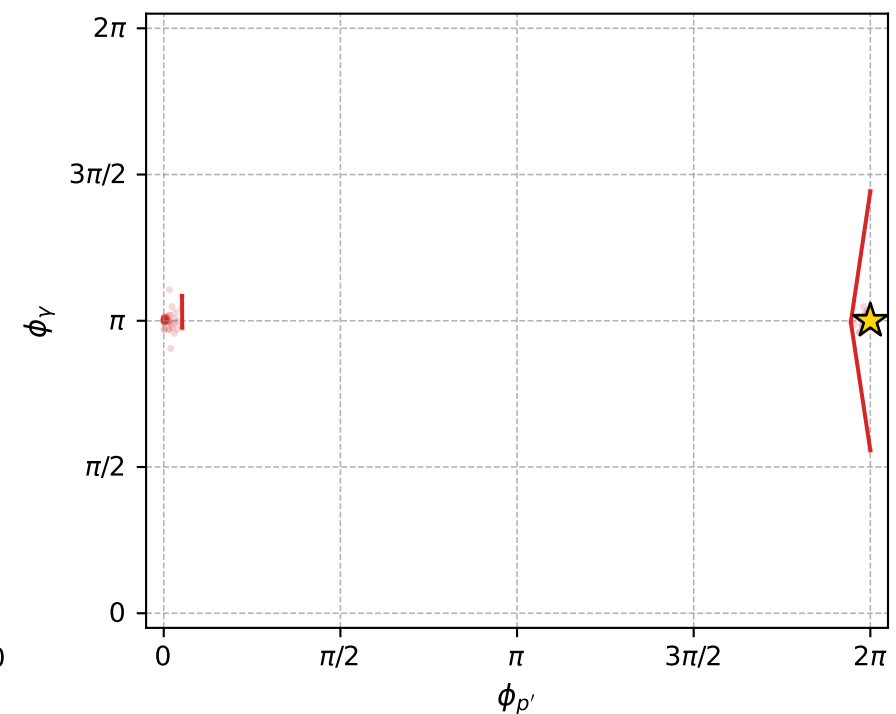
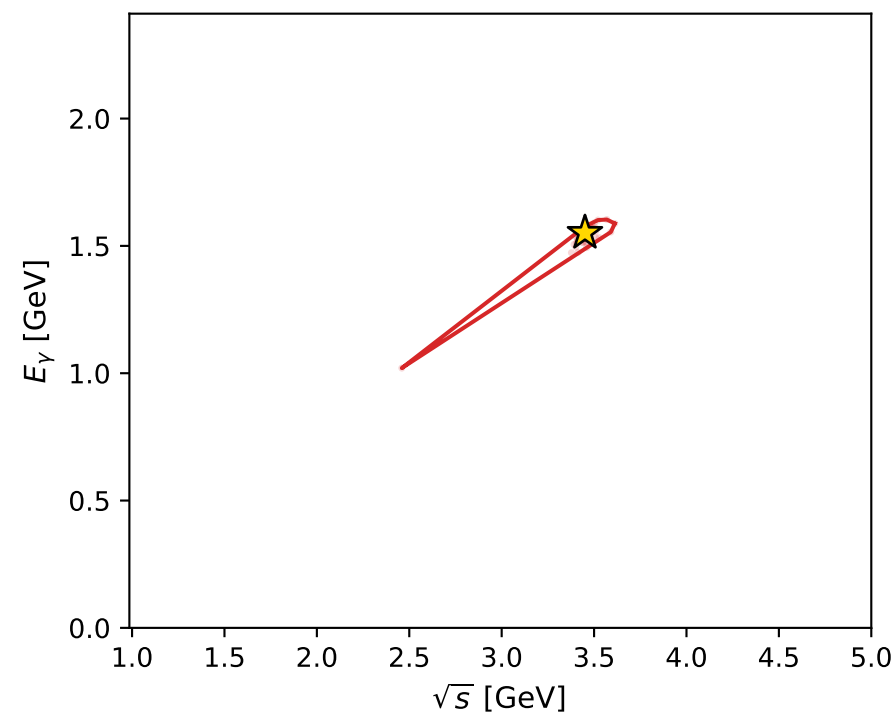
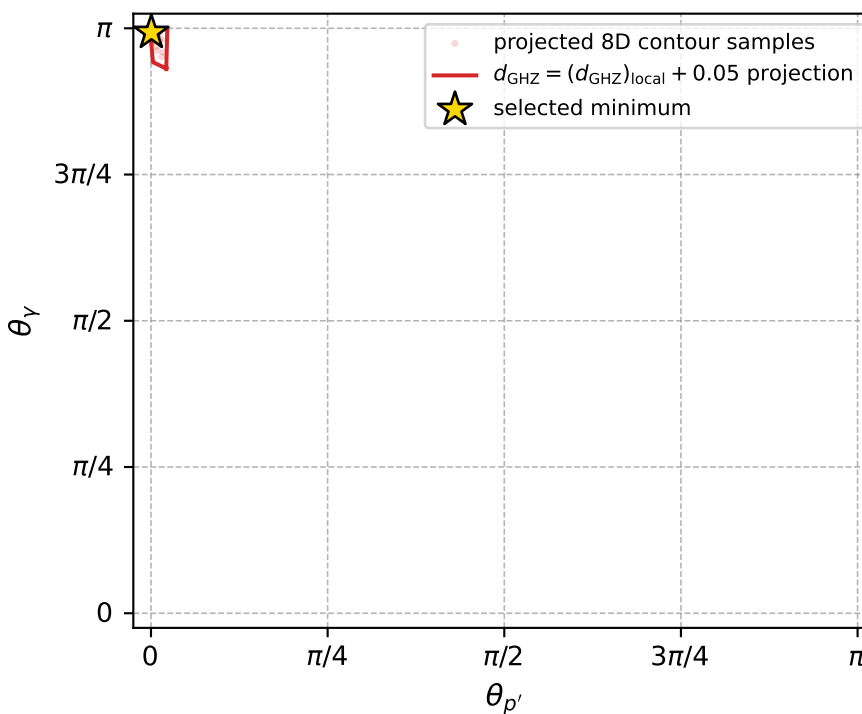
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 216; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 3.9874454\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050039874$
 above global minimum = $3.9852171\text{e-}05$
 8D contour samples = 119

selected phase-space point:

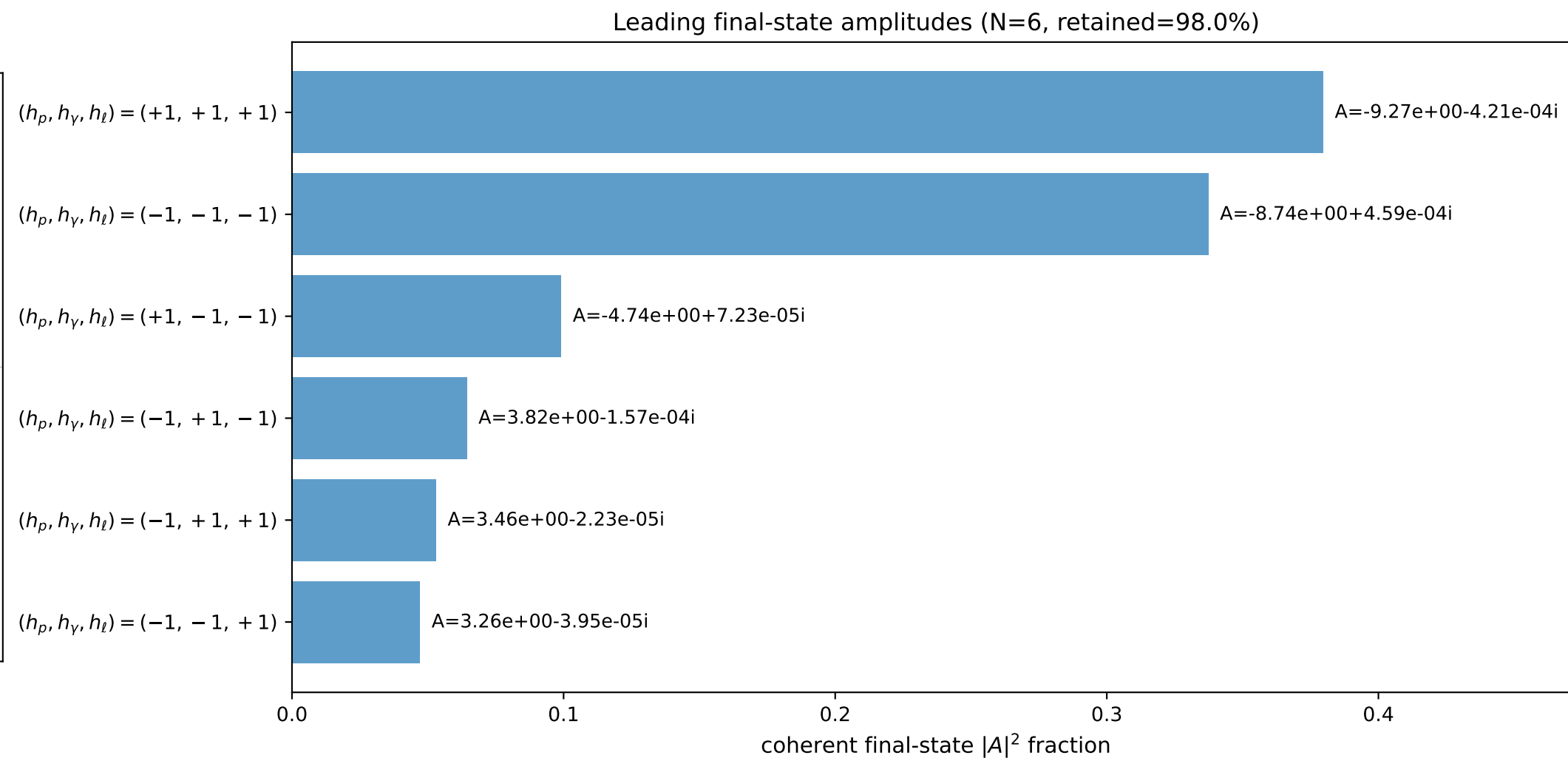
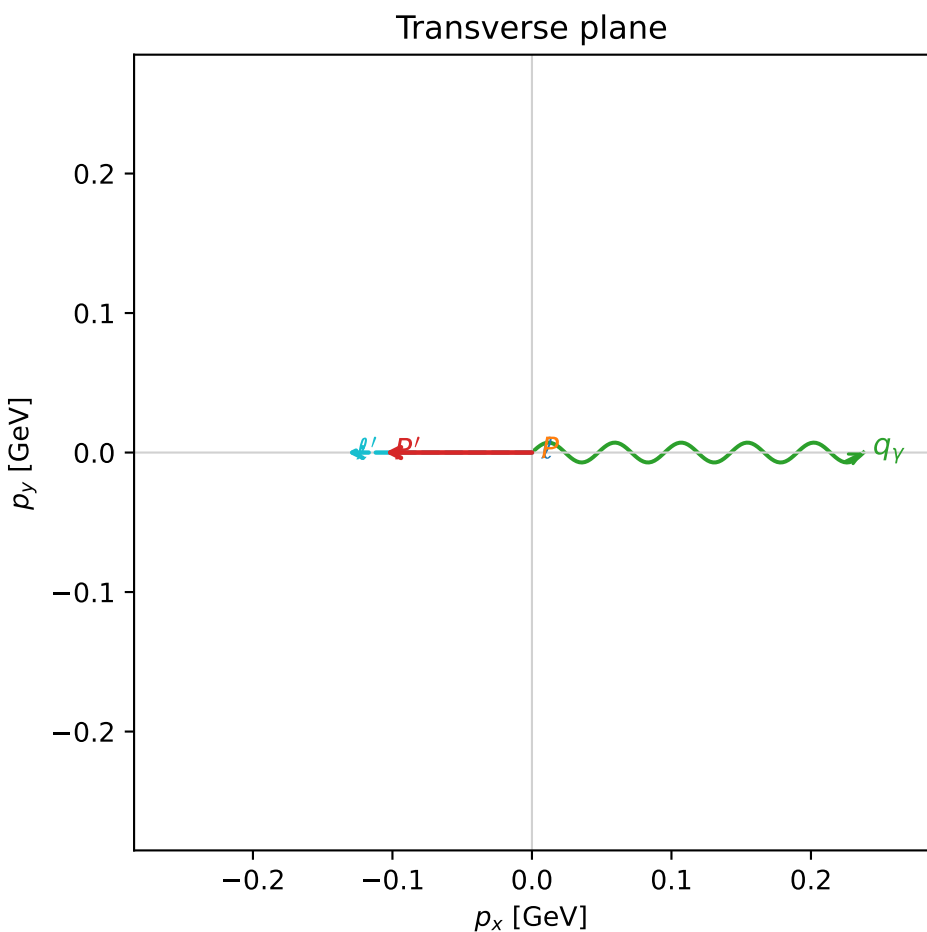
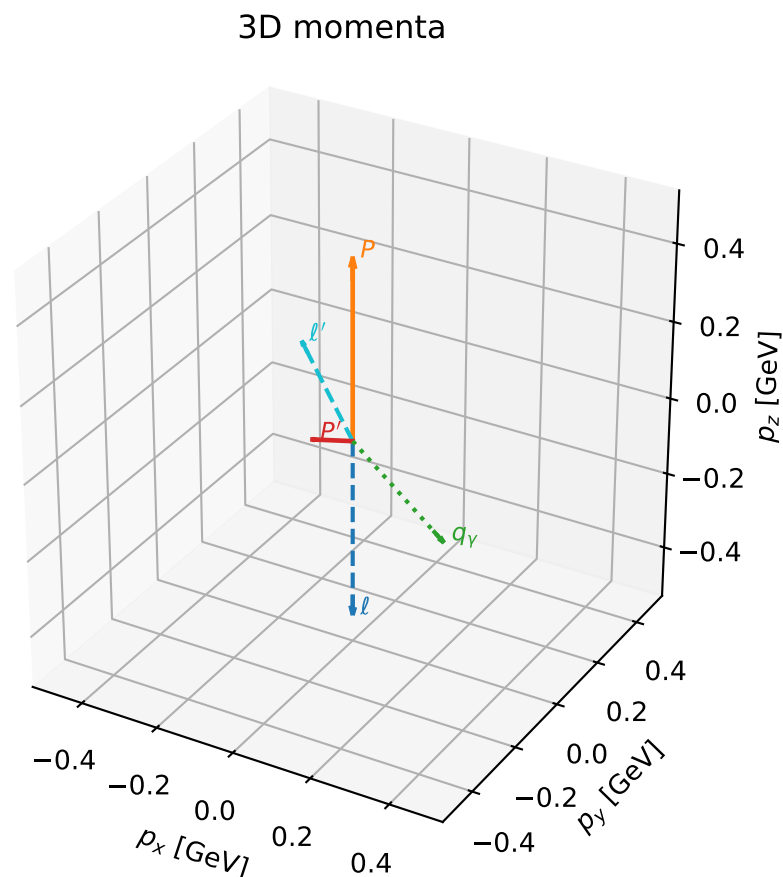
$\text{sqrt_s} = 3.450686$
 $\text{theta_p_out} = 0.00155077$
 $\text{theta_gamma_out} = 3.119405$
 $\text{q0ut} = 1.55126$
 $\text{phi_p_out} = 6.282927$
 $\text{phi_gamma_out} = 3.141605$
 $\text{alpha_e} = 0.3232963$
 $\text{alpha_p} = 1.253931$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

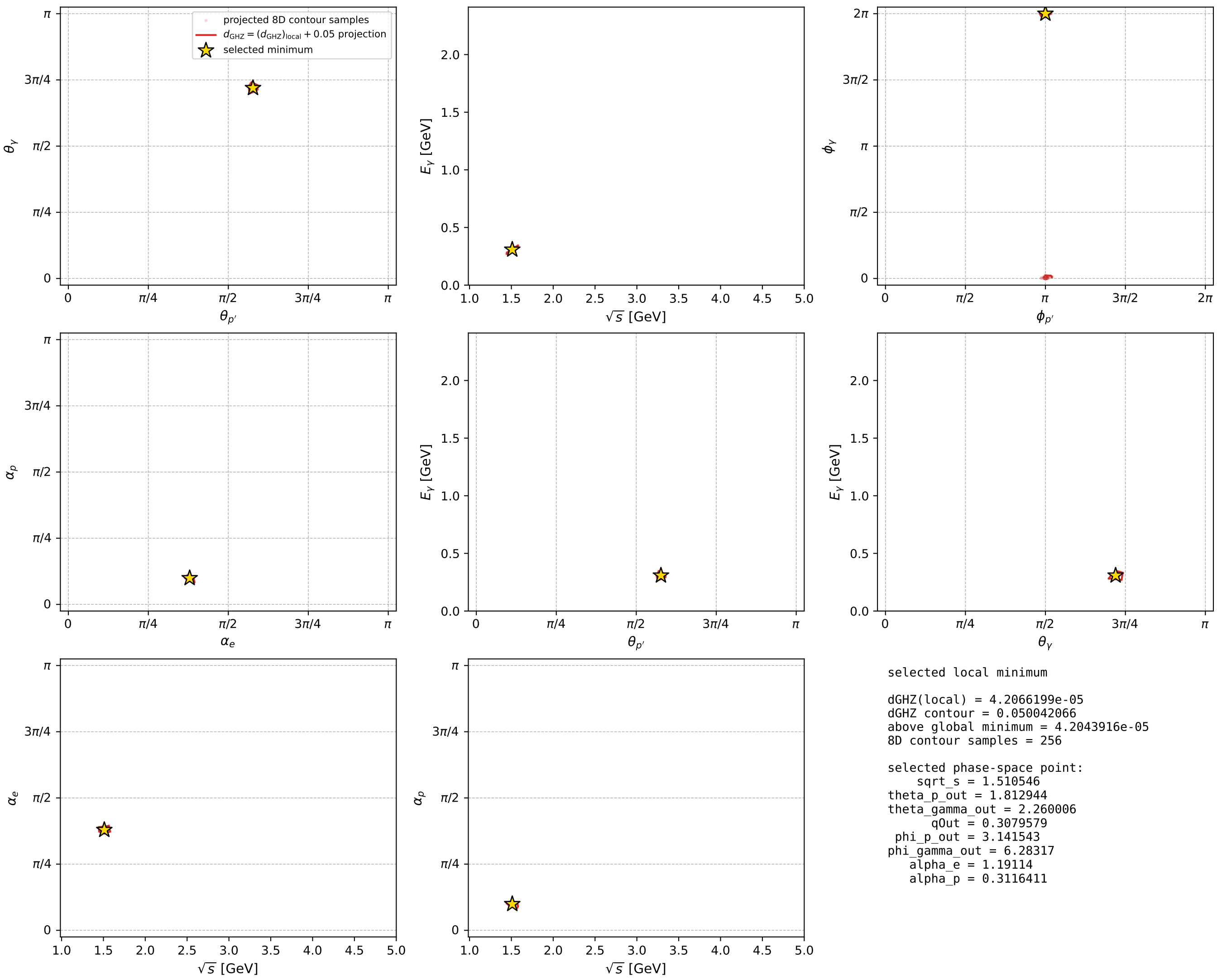
dghz_mixing_angles_polarization_01_local_minimum_218 $d_{\{\text{rm GHZ}\}}=4.20662\text{e-}05$
 region: polarization_cluster_01_local_minimum_218
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.1911401$ rad
 incoming lepton: $0.370601|+\rangle +0.928792|-\rangle$
 proton mixing angle: $\alpha_p=0.31164112$ rad
 incoming proton: $0.951832|+\rangle +0.306621|-\rangle$

$s=2.28175$, $\sqrt{s}=1.51055$, $|\vec{P}|=0.464039$, $|\vec{P}'|=0.108451$
 $\theta_{p'}=1.81294$, $\theta_\gamma=2.26001$, $\phi_{p'}=3.14154$, $\phi_\gamma=6.28317$
 $E_\gamma=0.307958$, $\theta_\ell=0.538005$, $\phi_\ell=3.1416$
 $Q^2=0.445647$, $x_B=0.41763$, $t=-0.240772$, $W^2=1.50128$, $y=0.761168$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4640$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4640$
 P $E= 1.0465$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4640$
 ℓ' $E= 0.2583$ $p_x= -0.1324$ $p_y= -0.0000$ $p_z= 0.2218$
 P' $E= 0.9442$ $p_x= -0.1053$ $p_y= 0.0000$ $p_z= -0.0260$
 q_γ $E= 0.3080$ $p_x= 0.2377$ $p_y= -0.0000$ $p_z= -0.1958$



electron: polarization cluster P1, local minimum 218; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

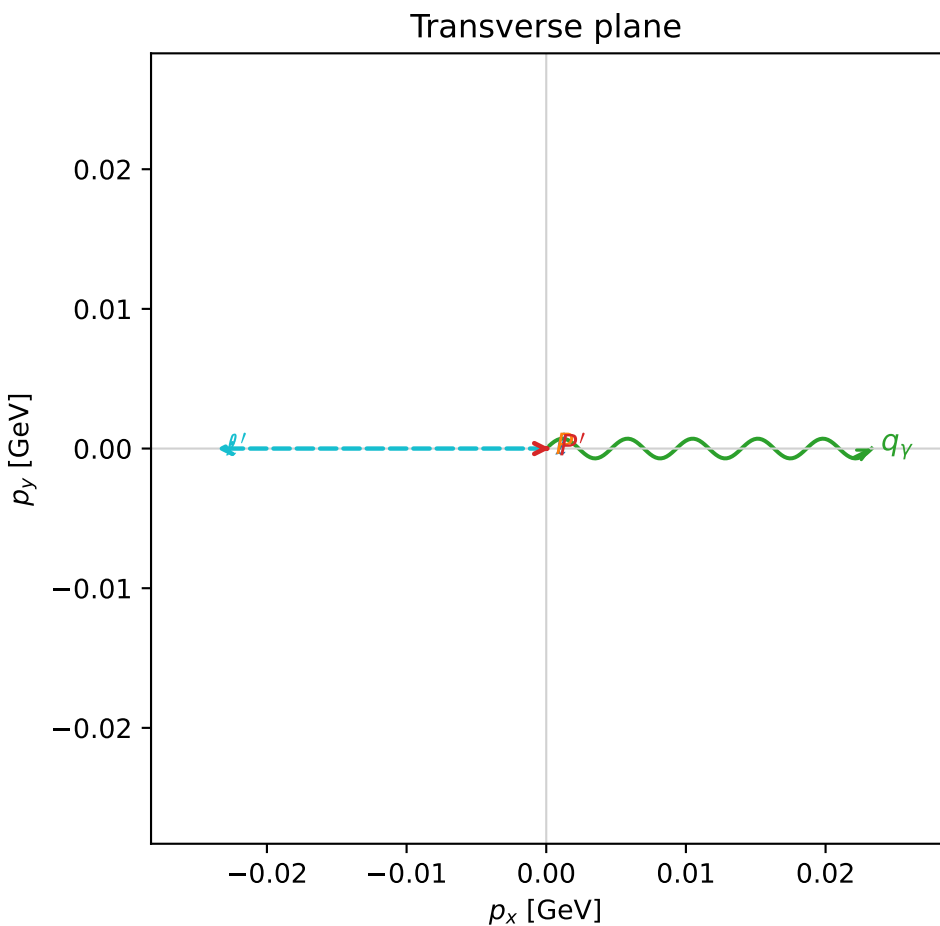
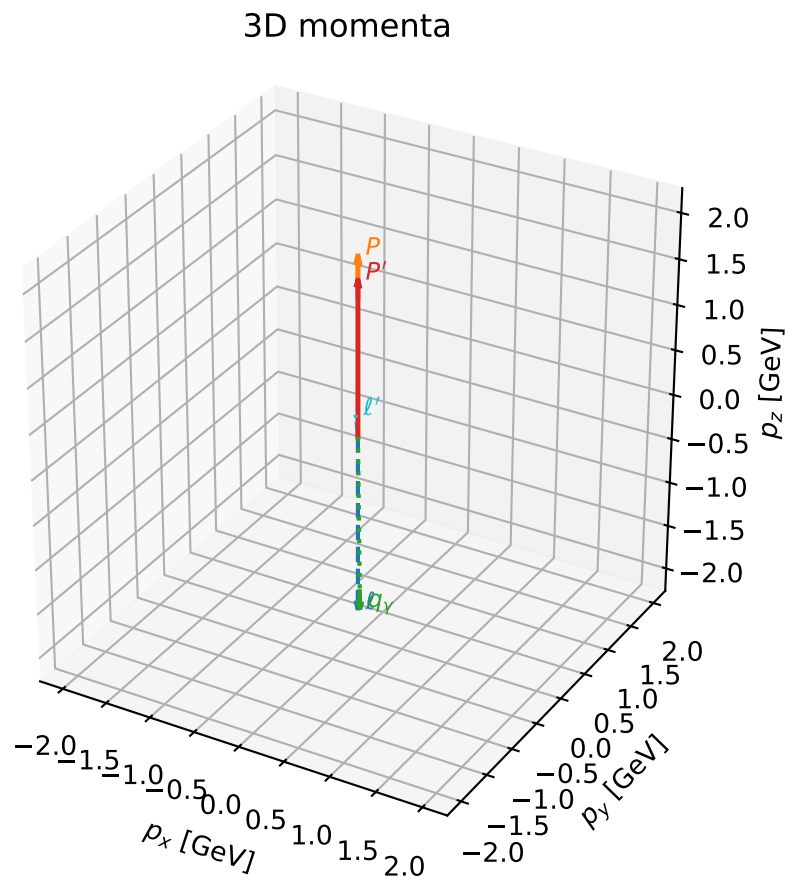


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_219 d_{\rm GHZ}=4.26109e-05
region: polarization_cluster_01_local_minimum_219
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.90729952$ rad
incoming lepton: $0.615876|+\rangle +0.787843|-\rangle$
proton mixing angle: $\alpha_p=0.66718354$ rad
incoming proton: $0.785568|+\rangle +0.618776|-\rangle$

$s=17.2279$, $\sqrt{s}=4.15065$, $|\vec{P}|=1.96934$, $|\vec{P}'|=1.71682$
 $\theta_{P'}=0.00016275$, $\theta_Y=3.12967$, $\phi_{P'}=0.000585085$, $\phi_Y=1.82782e-05$
 $E_Y=1.95504$, $\theta_{\ell'}=0.0987232$, $\phi_{\ell'}=3.14162$
 $Q^2=1.88008$, $x_B=0.115754$, $t=-0.0131571$, $W^2=15.2418$, $y=0.993515$

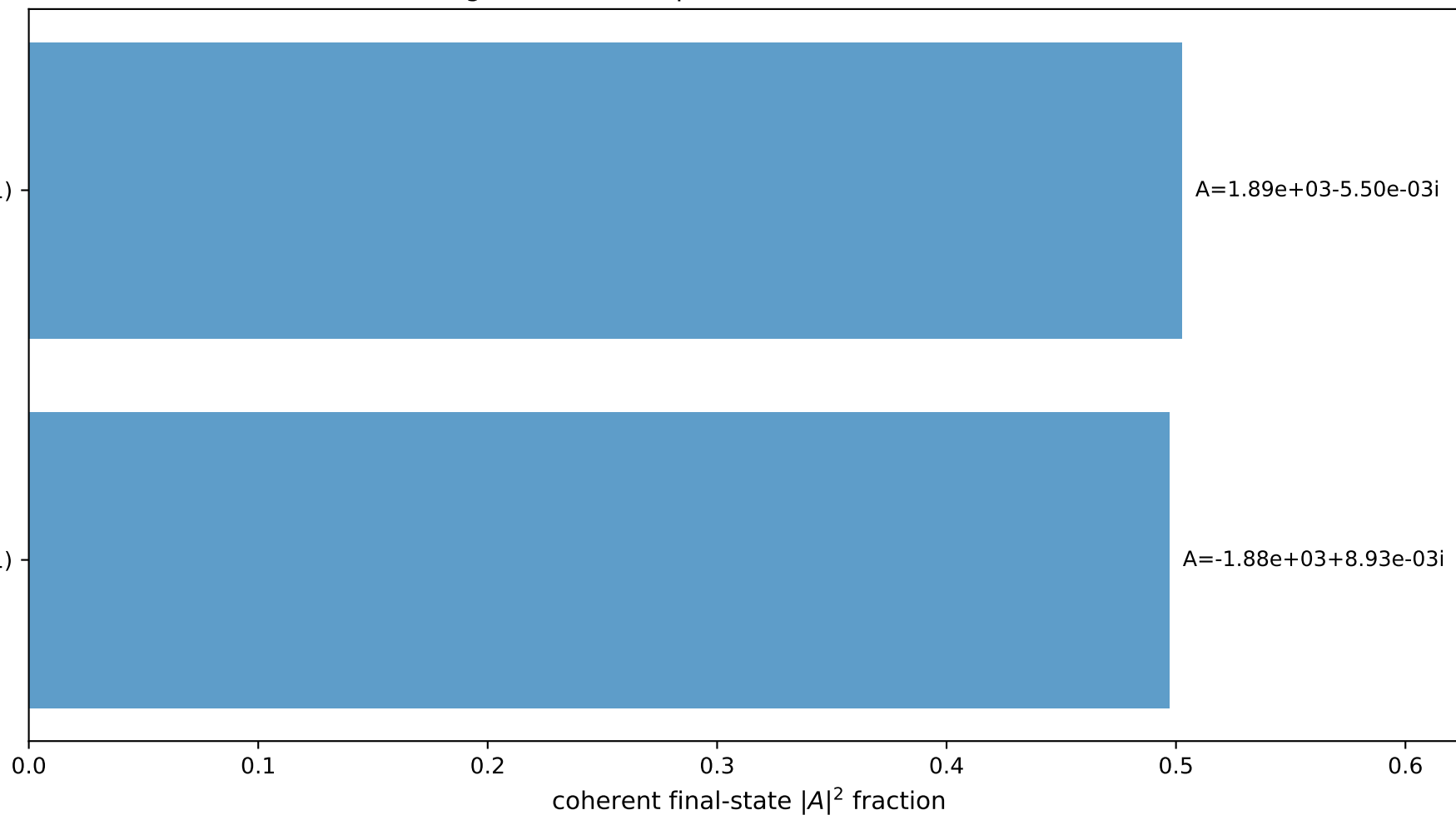
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9693$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9693$
 P $E= 2.1813$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9693$
 ℓ' $E= 0.2393$ $p_x= -0.0236$ $p_y= -0.0000$ $p_z= 0.2381$
 P' $E= 1.9564$ $p_x= 0.0003$ $p_y= 0.0000$ $p_z= 1.7168$
 q_Y $E= 1.9550$ $p_x= 0.0233$ $p_y= 0.0000$ $p_z= -1.9549$



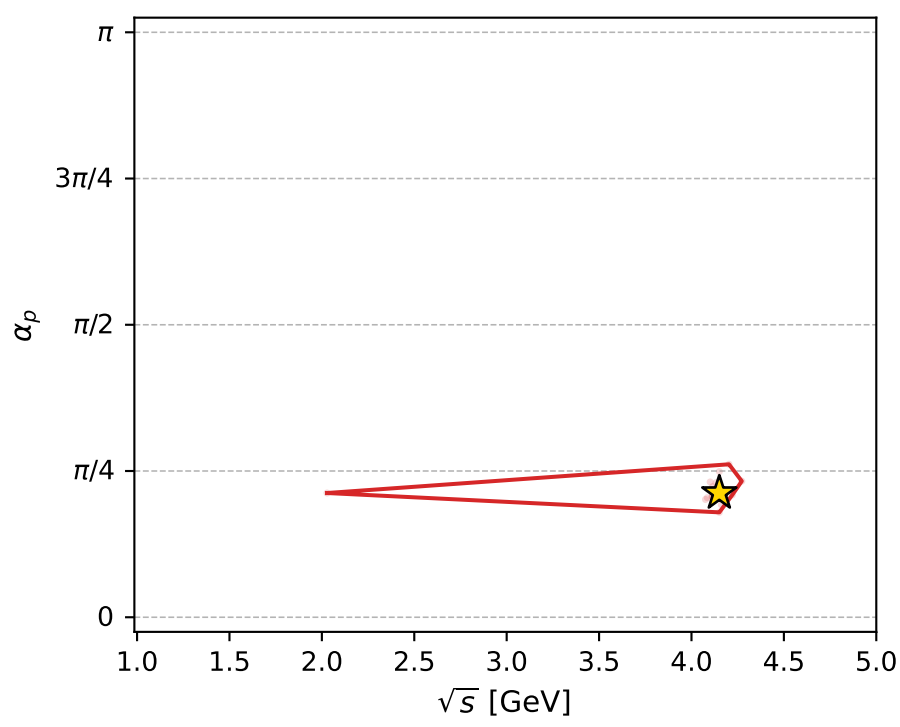
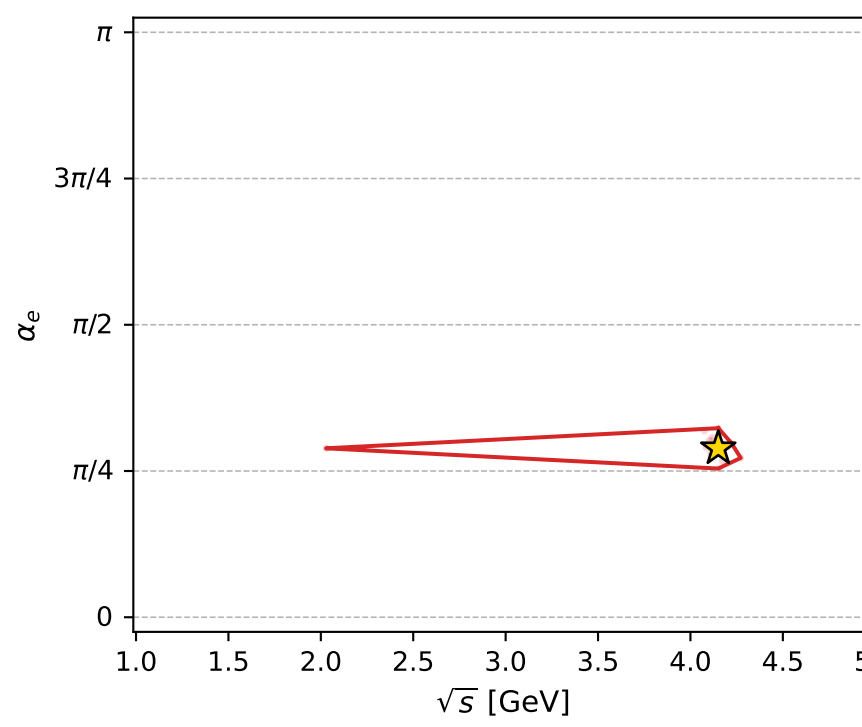
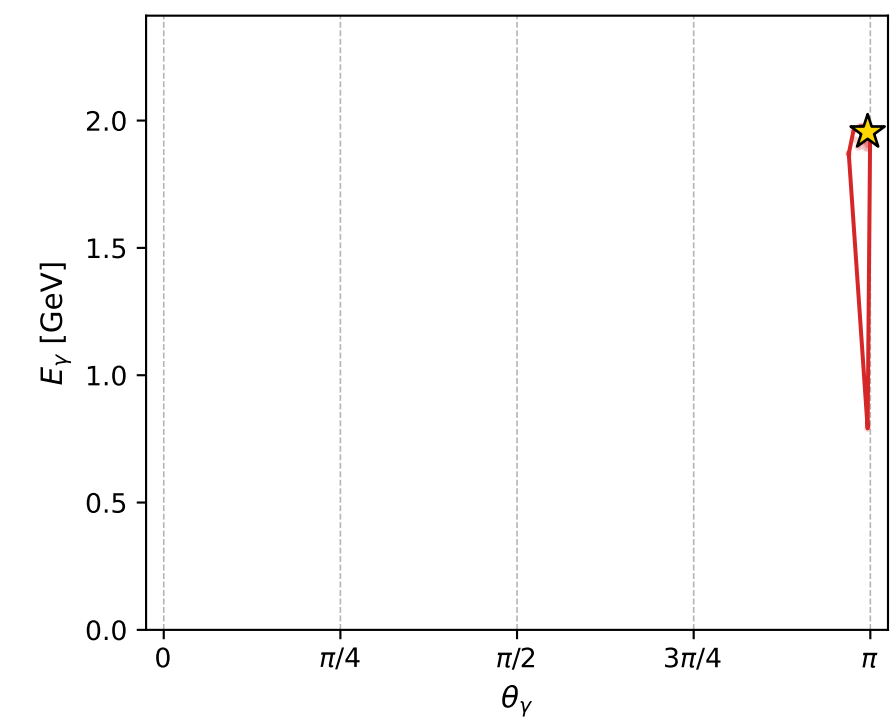
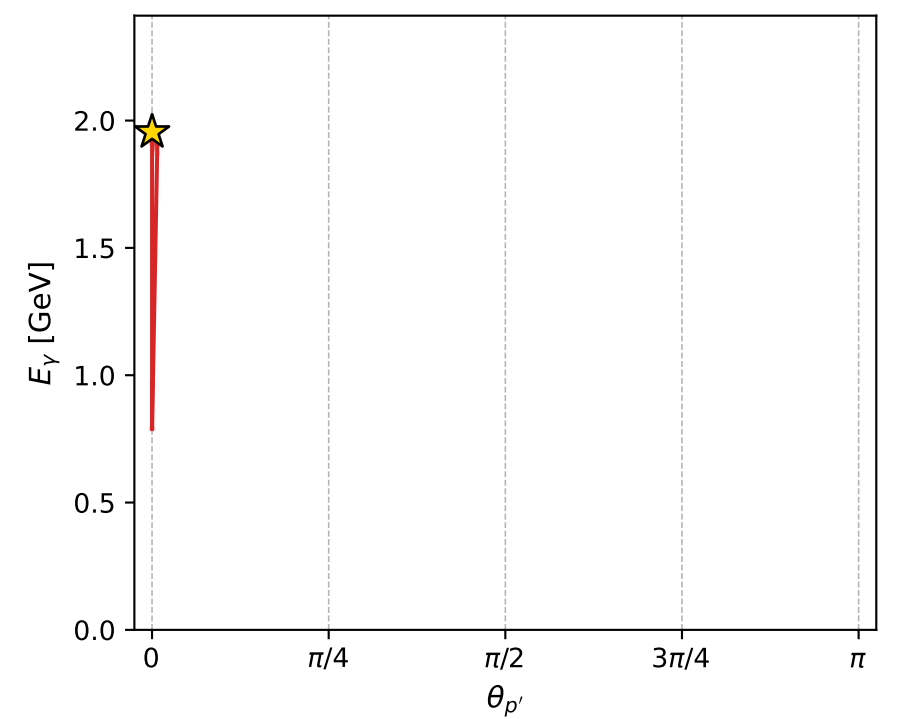
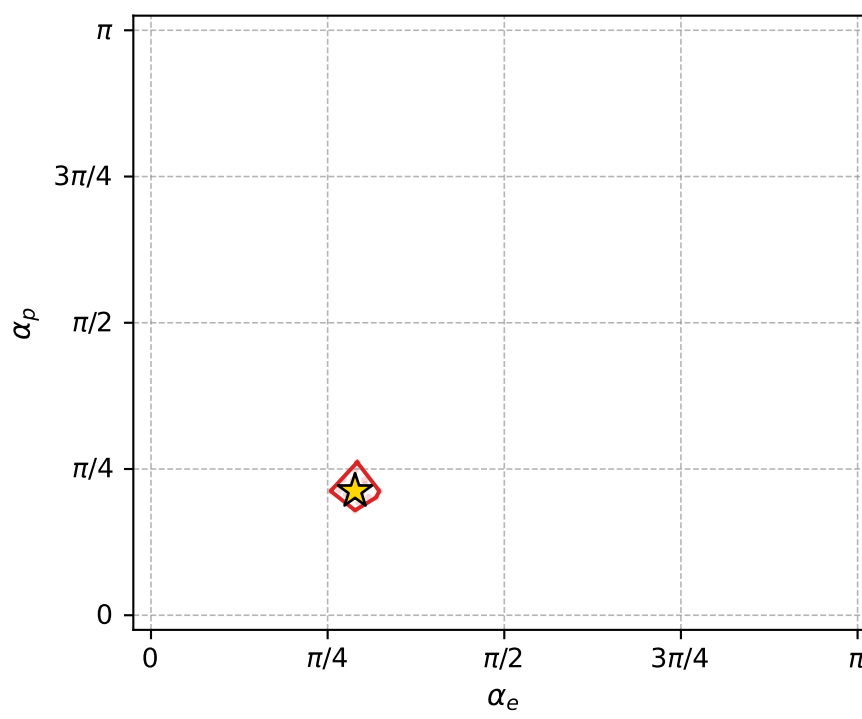
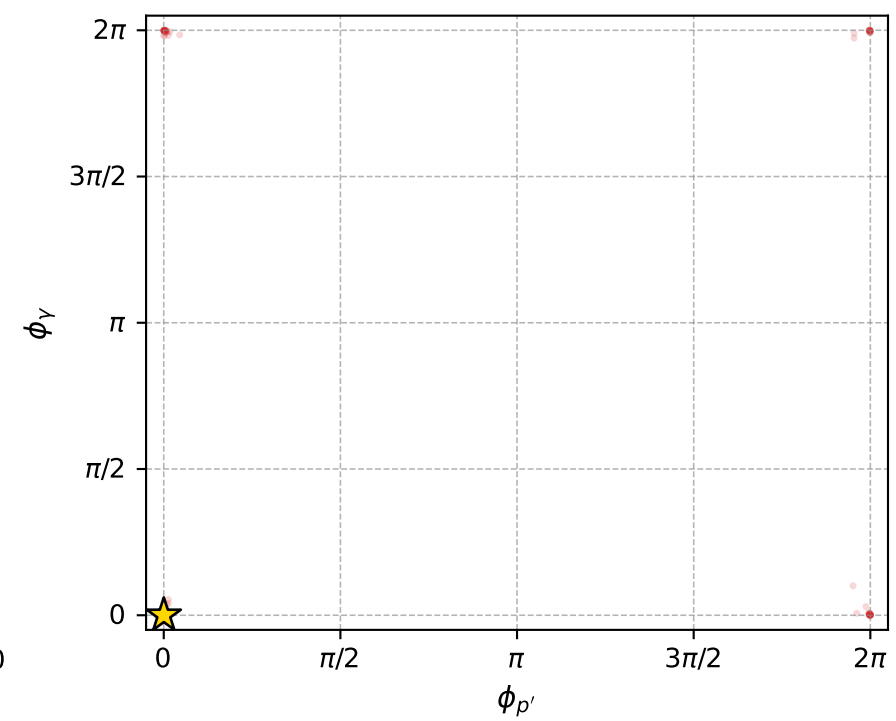
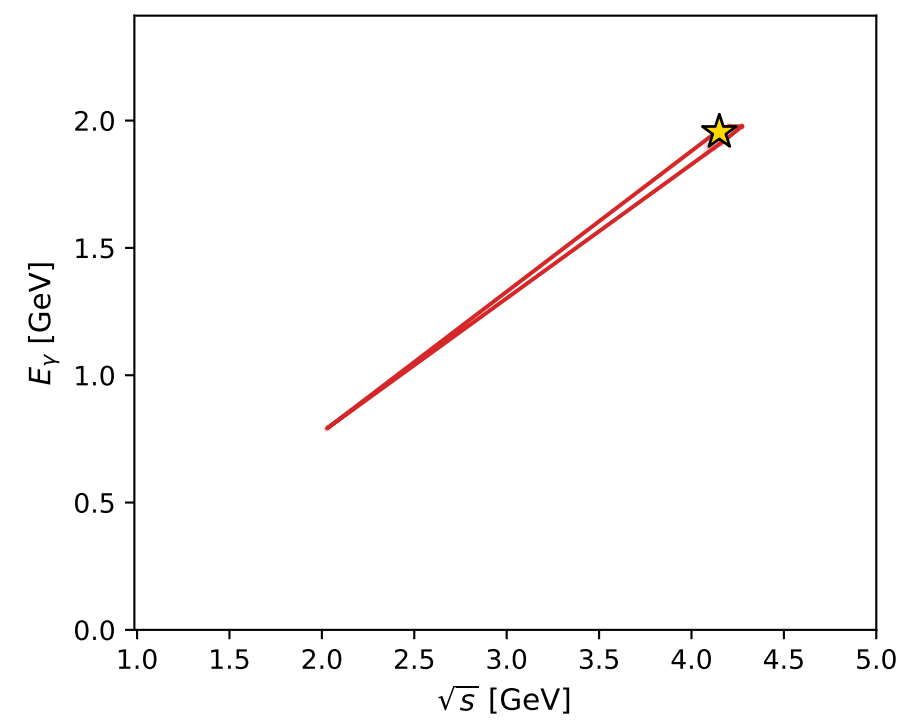
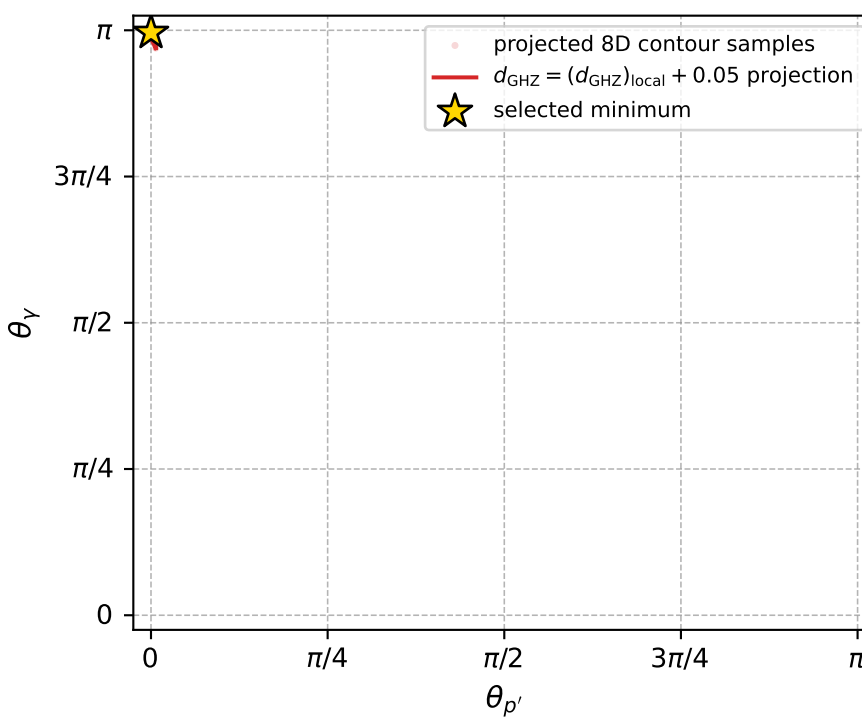
$(h_p, h_Y, h_{\ell}) = (-1, +1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, -1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 219; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.2610917\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050042611$
 above global minimum = $4.2588633\text{e-}05$
 8D contour samples = 127

selected phase-space point:

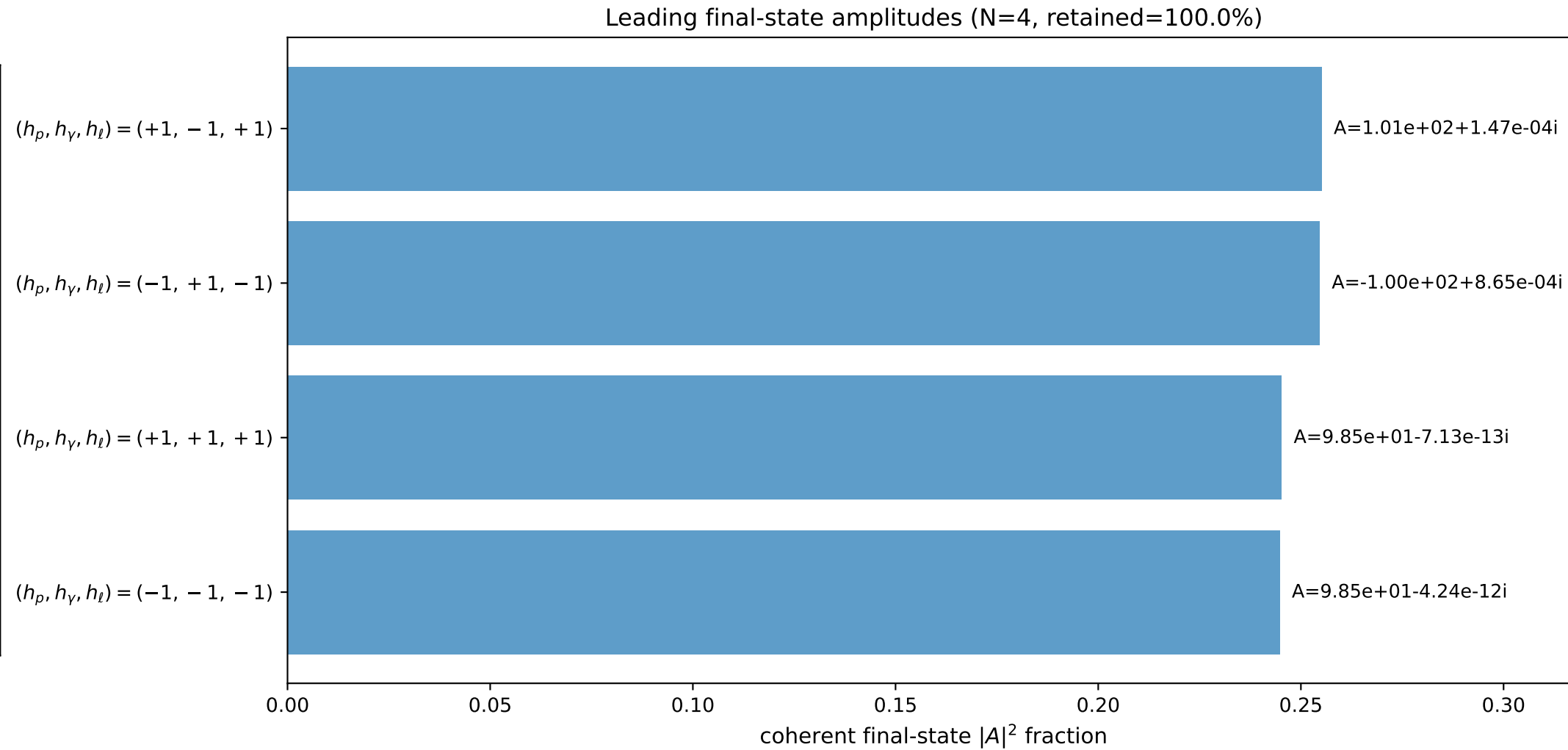
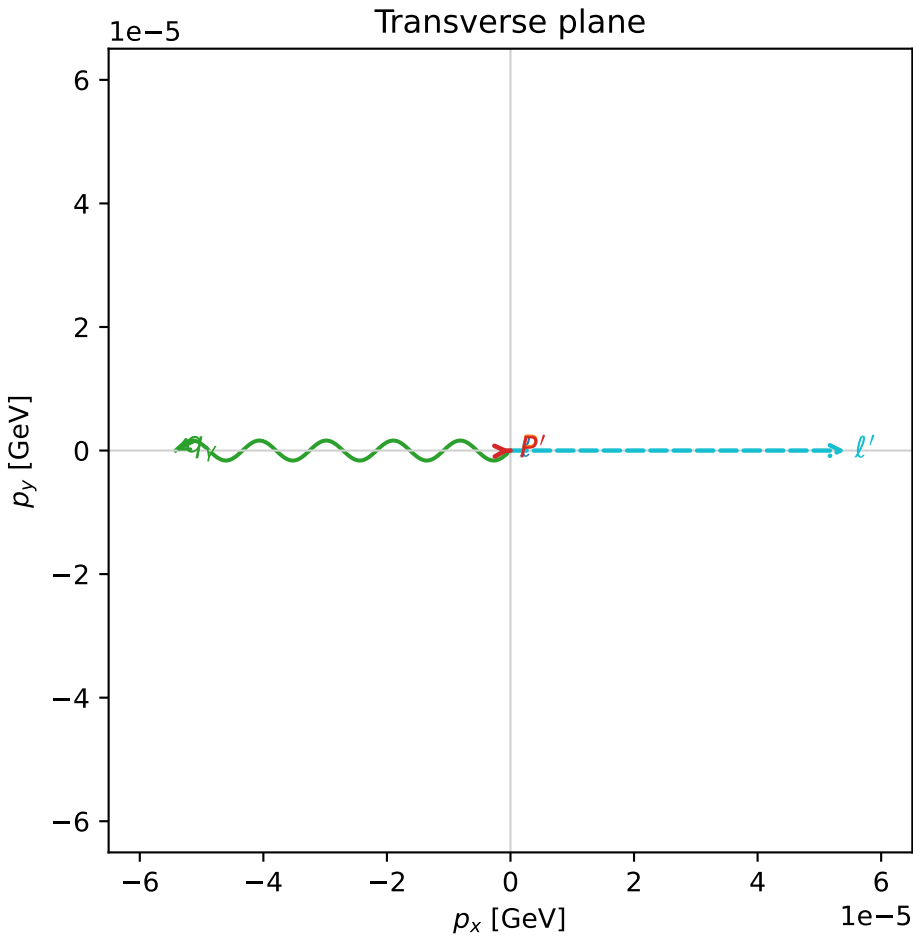
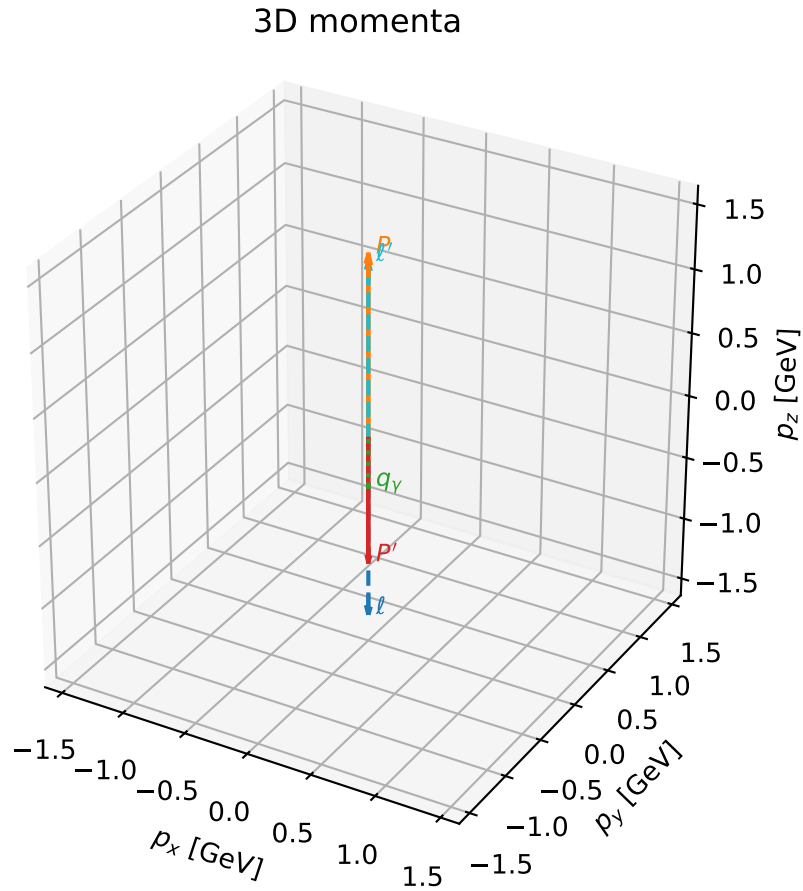
$\text{sqrt_s} = 4.150648$
 $\text{theta_p_out} = 0.0001627498$
 $\text{theta_gamma_out} = 3.129673$
 $\text{q0out} = 1.955045$
 $\text{phi_p_out} = 0.000585085$
 $\text{phi_gamma_out} = 1.827822\text{e-}05$
 $\alpha_e = 0.9072995$
 $\alpha_p = 0.6671835$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

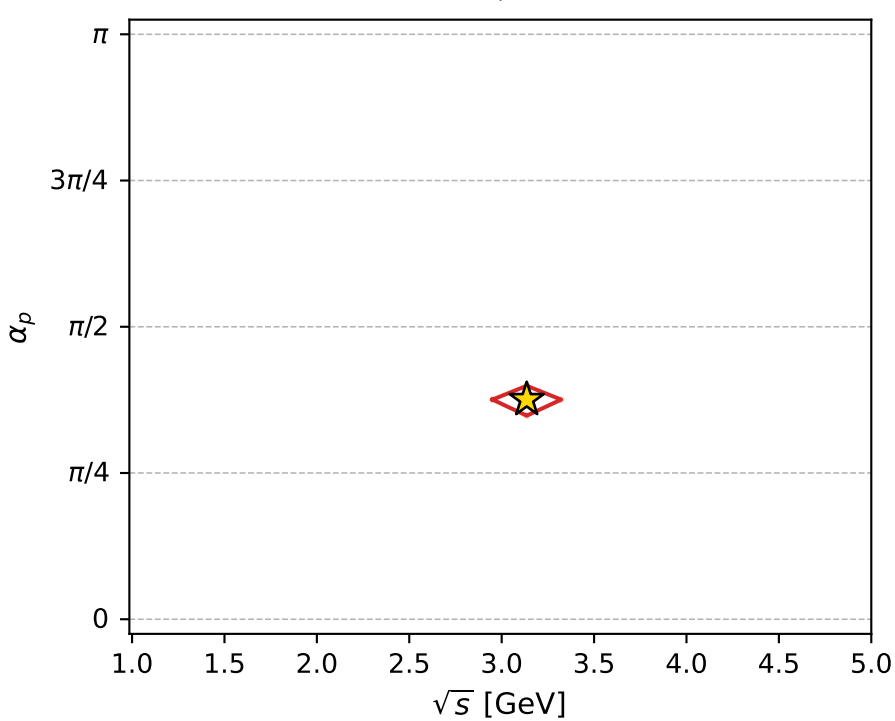
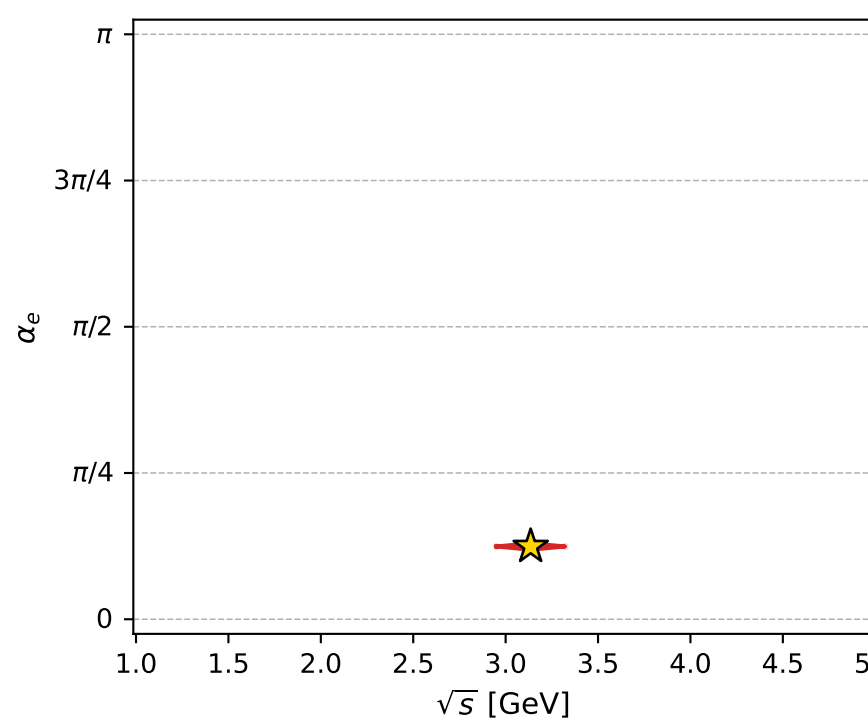
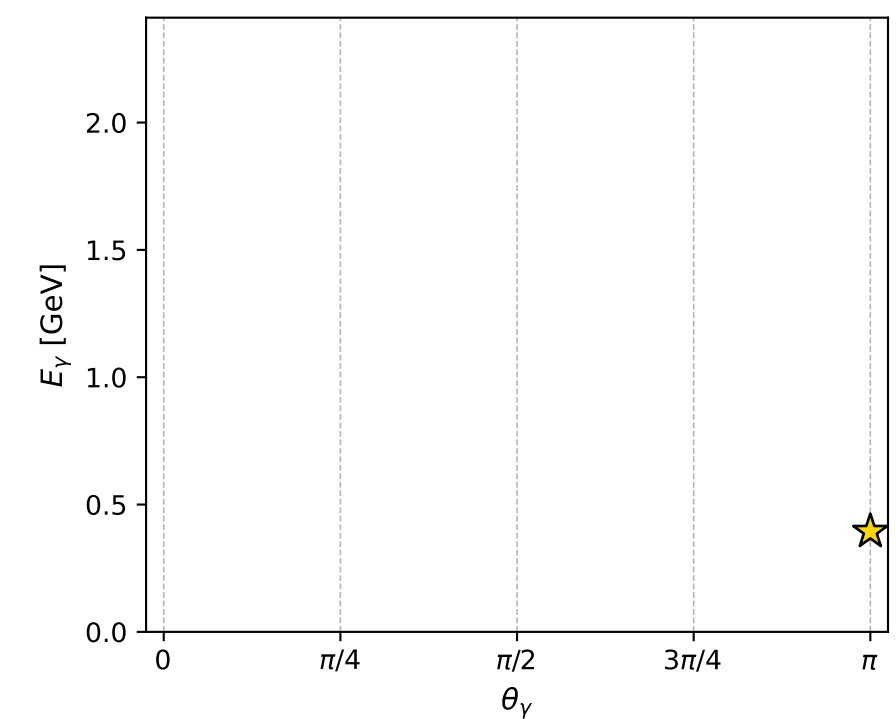
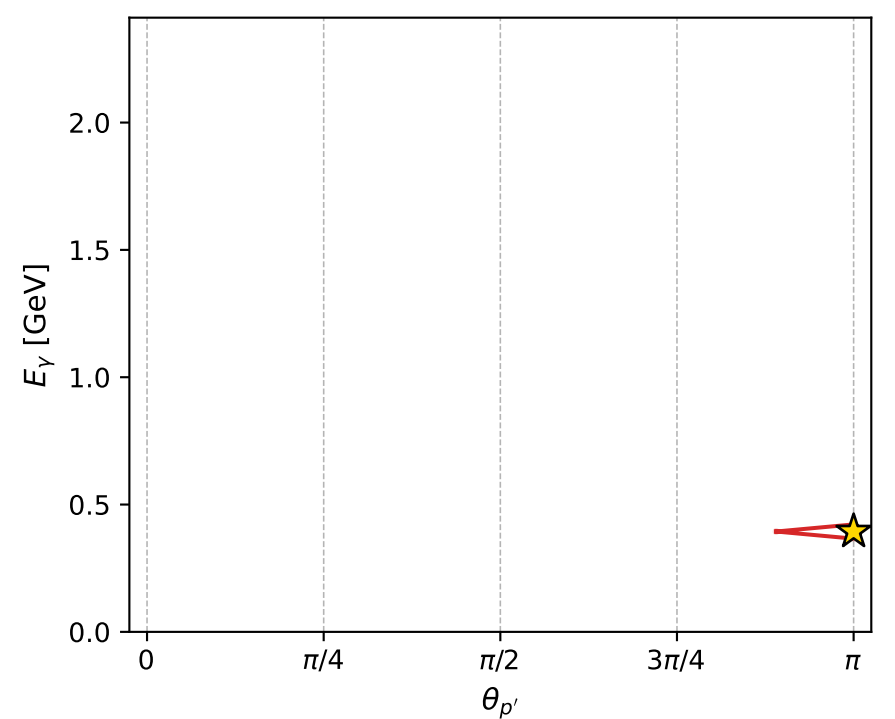
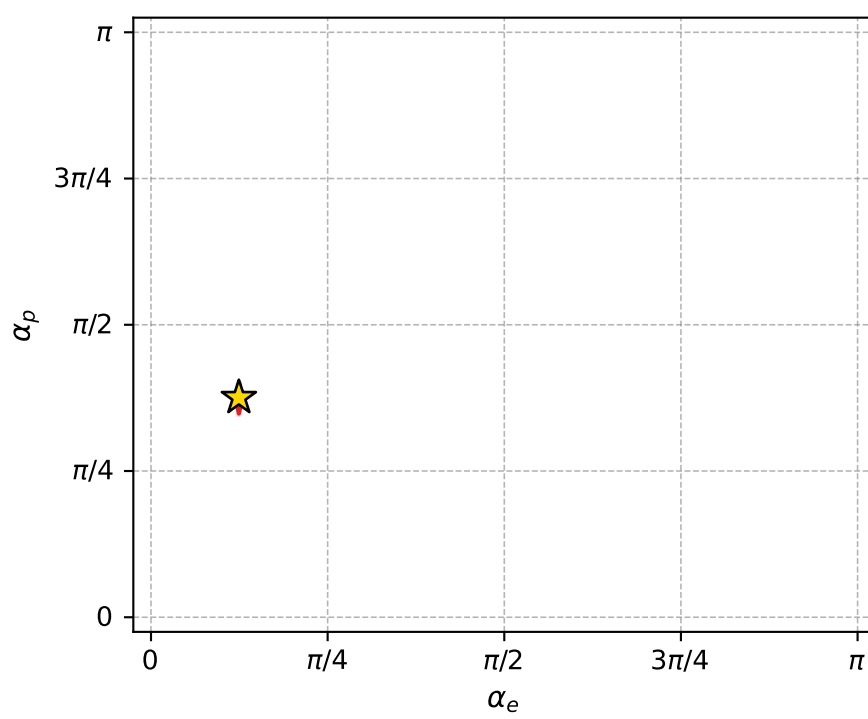
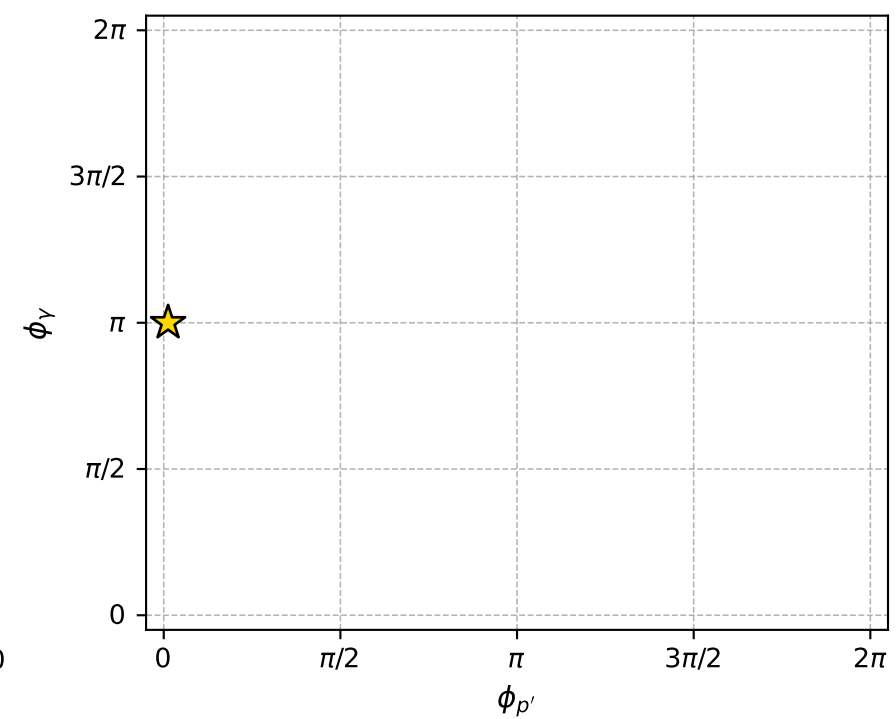
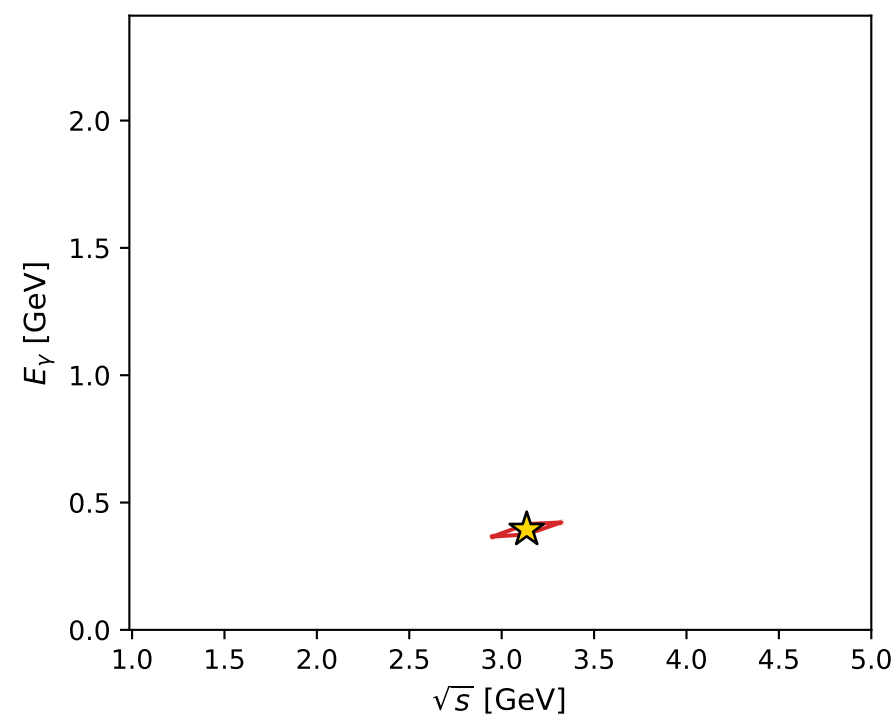
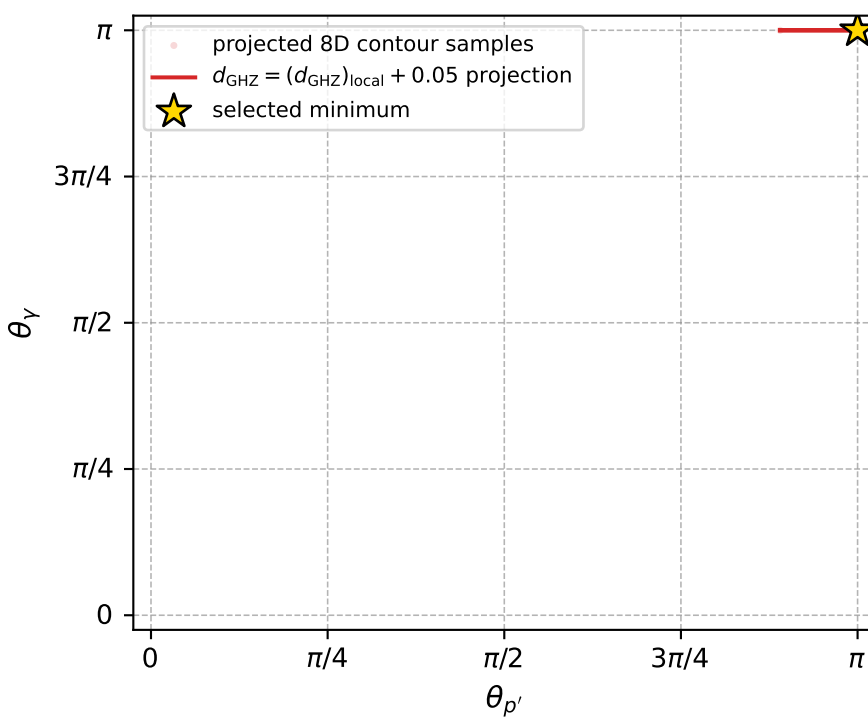
dghz_mixing_angles_polarization_01_local_minimum_220 d_{\rm GHZ}=4.51609e-05
region: polarization_cluster_01_local_minimum_220
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.39079798$ rad
incoming lepton: $0.924605|+\rangle +0.380926|-\rangle$
proton mixing angle: $\alpha_p=1.1796625$ rad
incoming proton: $0.381237|+\rangle +0.924477|-\rangle$

$s=9.83123$, $\sqrt{s}=3.13548$, $|\vec{P}|=1.42744$, $|\vec{P}'|=0.986259$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14146$, $\phi_{P'}=0.0385553$, $\phi_Y=3.1416$
 $E_Y=0.394068$, $\theta_{\ell'}=3.92703e-05$, $\phi_{\ell'}=5.03873e-06$
 $Q^2=7.88131$, $x_B=0.963871$, $t=-5.70554$, $W^2=1.17526$, $y=0.913459$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.4274$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4274$
 P $E= 1.7080$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4274$
 ℓ' $E= 1.3803$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.3803$
 P' $E= 1.3611$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9863$
 q_Y $E= 0.3941$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.3941$



electron: polarization cluster P1, local minimum 220; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 4.5160906\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050045161$
 above global minimum = $4.5138623\text{e-}05$
 8D contour samples = 131

selected phase-space point:

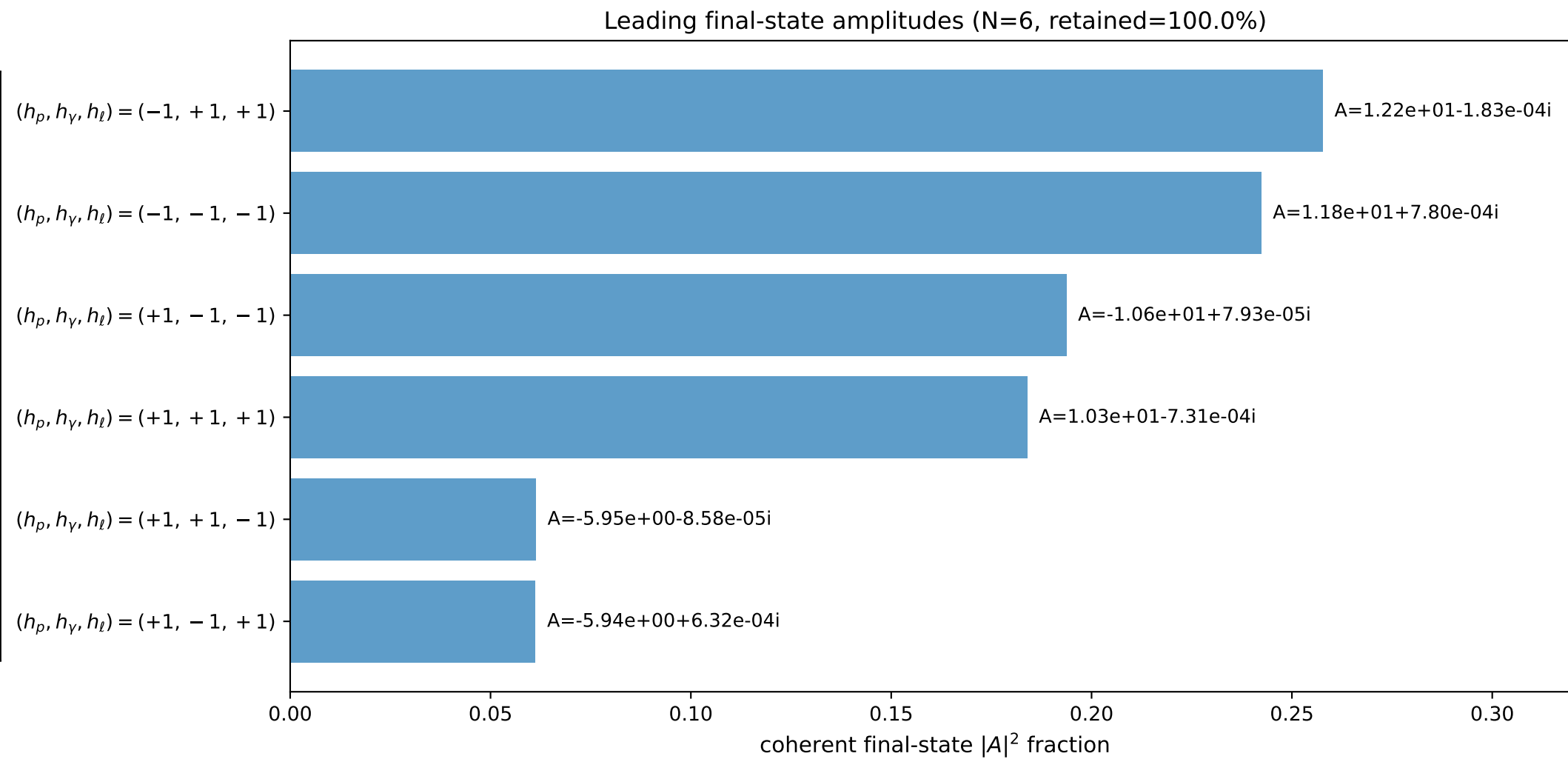
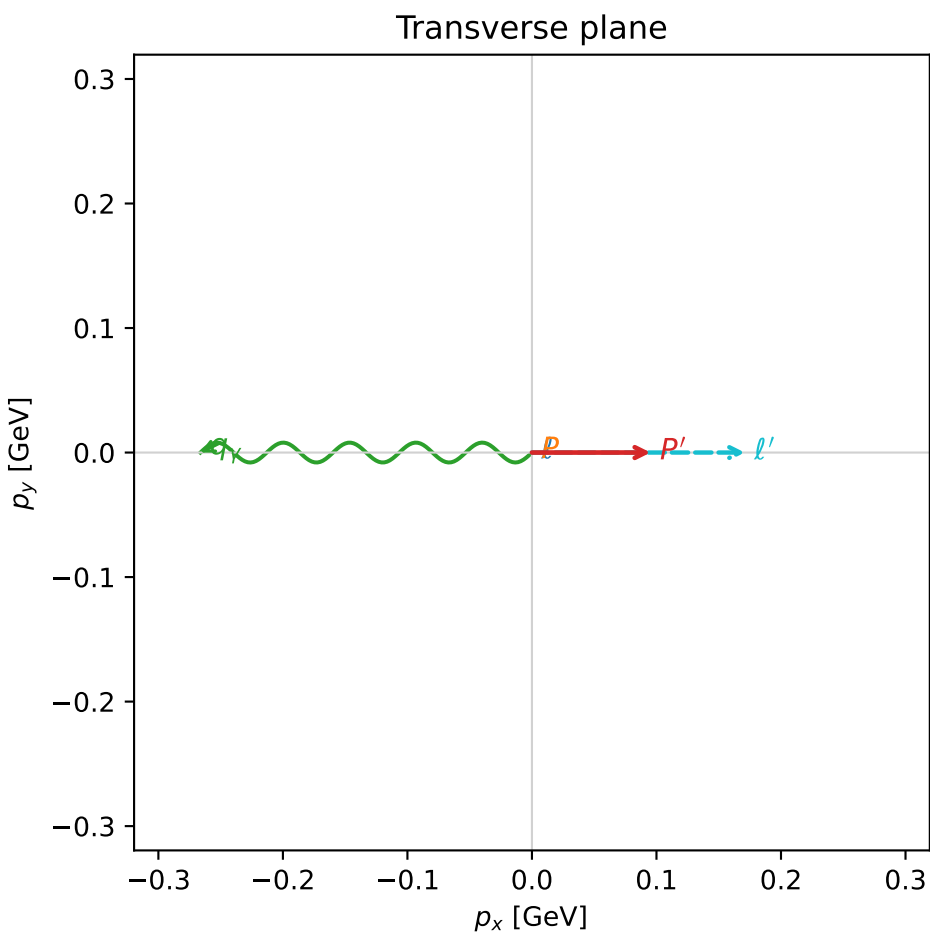
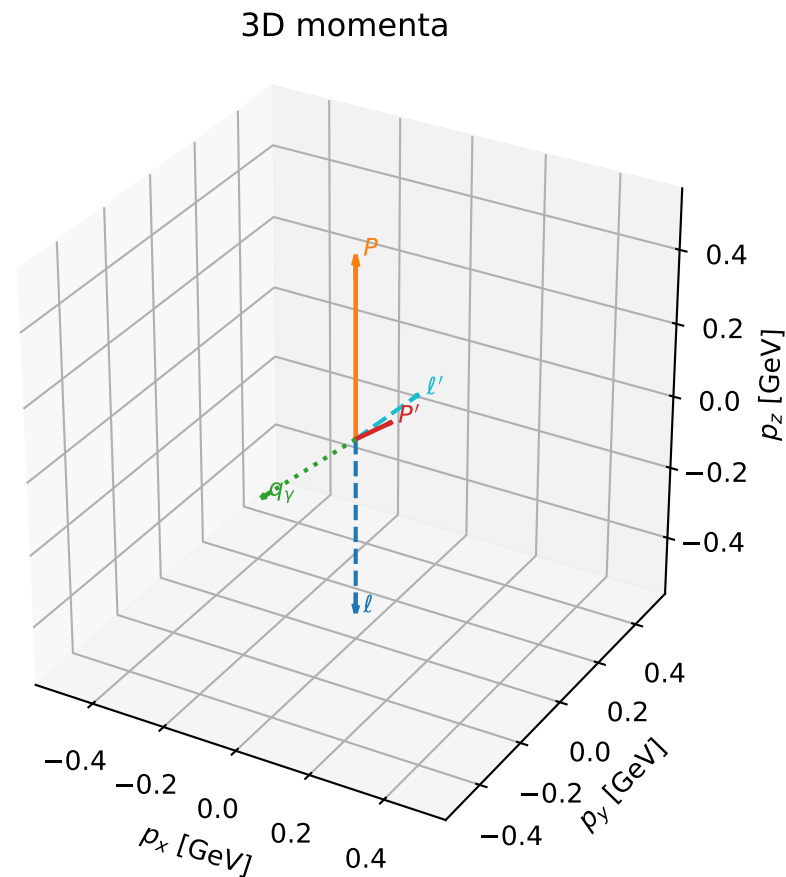
$\text{sqrt_s} = 3.135479$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.141455$
 $\text{q0out} = 0.3940682$
 $\text{phi_p_out} = 0.03855534$
 $\text{phi_gamma_out} = 3.141598$
 $\text{alpha_e} = 0.390798$
 $\text{alpha_p} = 1.179662$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

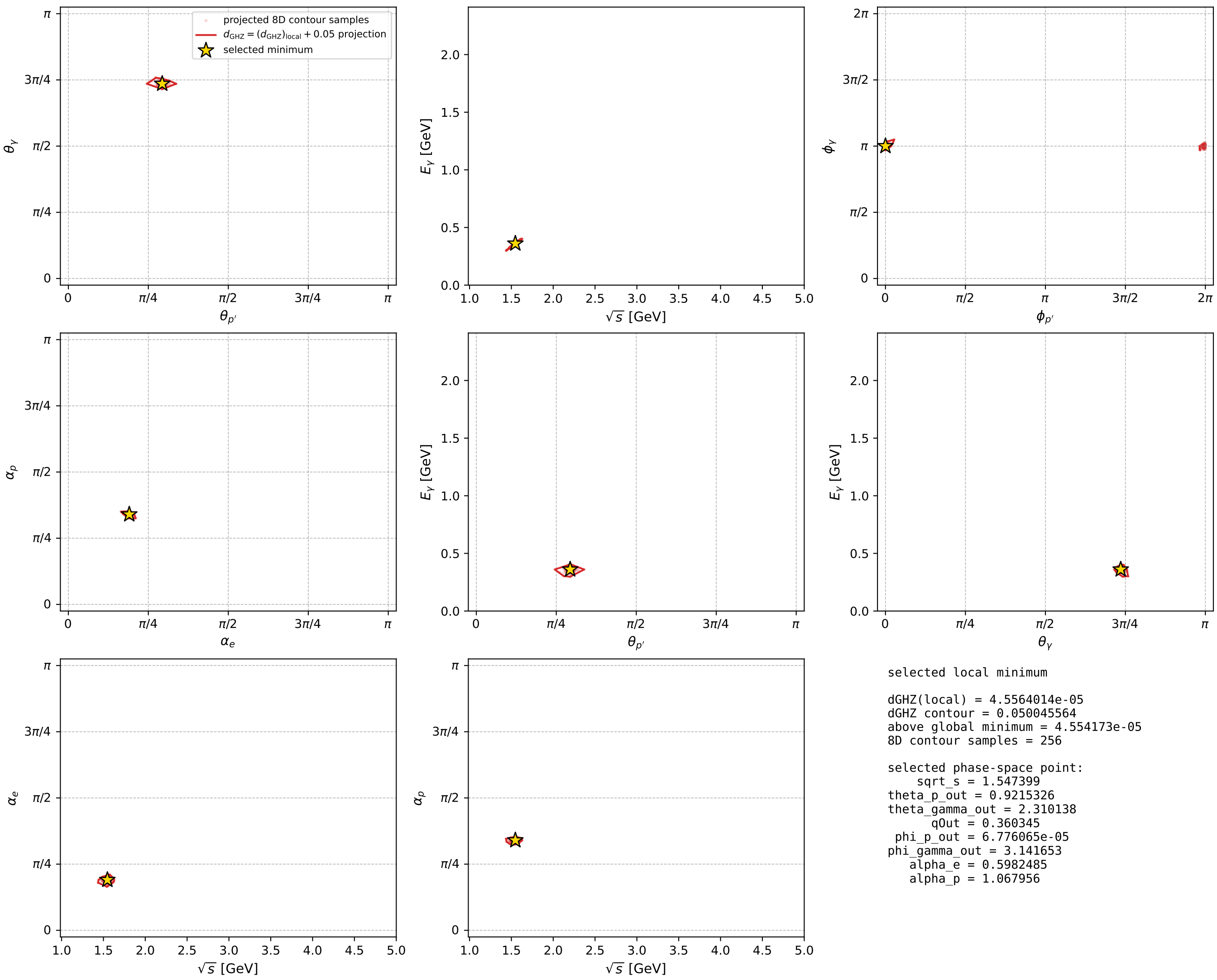
dghz_mixing_angles_polarization_01_local_minimum_223 $d_{\{\text{rm GHZ}\}}=4.5564\text{e-}05$
 region: polarization_cluster_01_local_minimum_223
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.59824849$ rad
 incoming lepton: $0.826323|+\rangle +0.563196|-\rangle$
 proton mixing angle: $\alpha_p=1.0679556$ rad
 incoming proton: $0.481917|+\rangle +0.876217|-\rangle$

$s=2.39444$, $\sqrt{s}=1.5474$, $|\vec{P}|=0.489402$, $|\vec{P}'|=0.119612$
 $\theta_{P'}=0.921533$, $\theta_Y=2.31014$, $\phi_{P'}=6.77606\text{e-}05$, $\phi_Y=3.14165$
 $E_Y=0.360345$, $\theta_{\ell'}=0.786878$, $\phi_{\ell'}=5.56911\text{e-}05$
 $Q^2=0.40321$, $x_B=0.344463$, $t=-0.170402$, $W^2=1.64718$, $y=0.772841$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4894$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4894$
 P $E= 1.0580$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4894$
 ℓ' $E= 0.2415$ $p_x= 0.1710$ $p_y= 0.0000$ $p_z= 0.1705$
 P' $E= 0.9456$ $p_x= 0.0953$ $p_y= 0.0000$ $p_z= 0.0723$
 q_Y $E= 0.3603$ $p_x= -0.2663$ $p_y= -0.0000$ $p_z= -0.2428$



electron: polarization cluster P1, local minimum 223; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



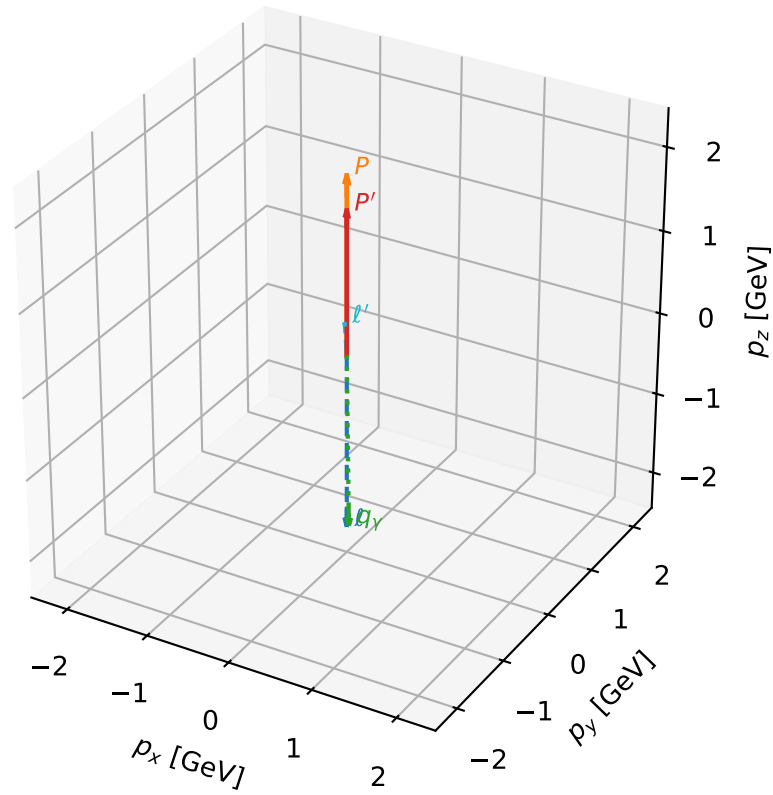
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_231 $d_{\{\text{rm GHZ}\}}=5.7484\text{e-}05$
 region: polarization_cluster_01_local_minimum_231
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.97671817$ rad
 incoming lepton: $0.559745|+\rangle +0.828665|-\rangle$
 proton mixing angle: $\alpha_p=0.59468799$ rad
 incoming proton: $0.828323|+\rangle +0.56025|-\rangle$

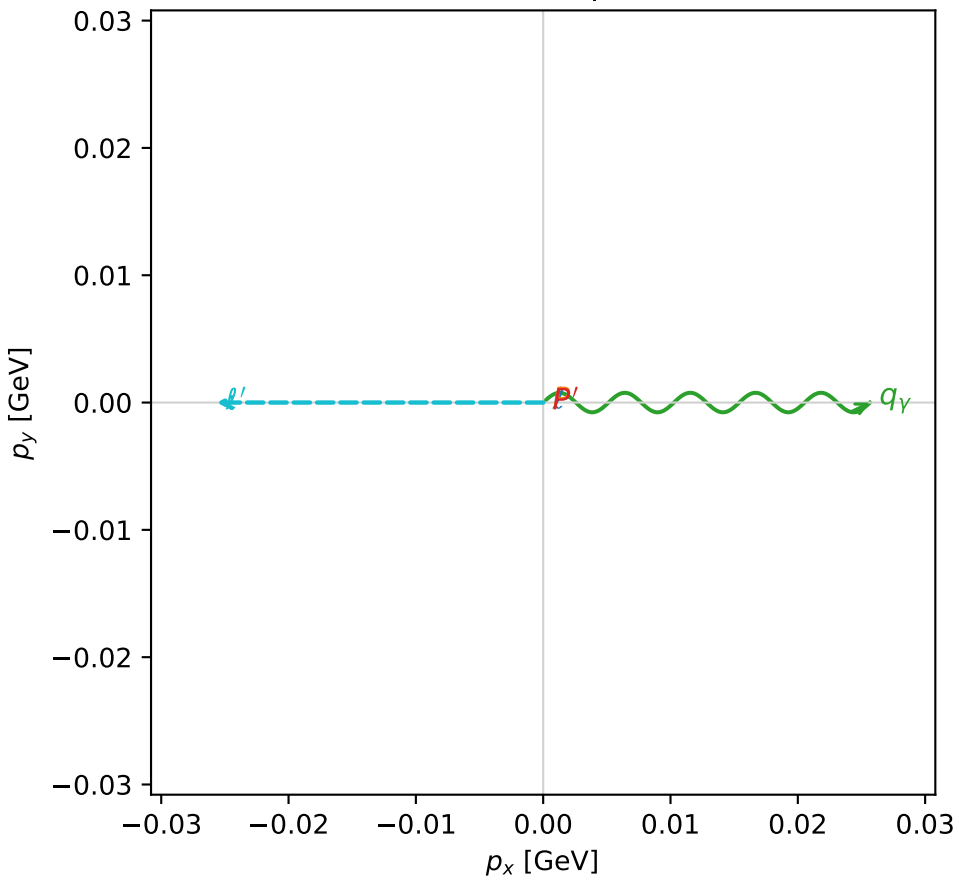
$s=19.9347$, $\sqrt{s}=4.46483$, $|\vec{P}|=2.13388$, $|\vec{P}'|=1.72462$
 $\theta_{P'}=0$, $\theta_Y=3.12944$, $\phi_{P'}=1.59257$, $\phi_Y=6.28318$
 $E_Y=2.11278$, $\theta_{\ell'}=0.066056$, $\phi_{\ell'}=3.14158$
 $Q^2=3.31544$, $x_B=0.175439$, $t=-0.0322616$, $W^2=16.4624$, $y=0.991767$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1339$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1339$
 P $E= 2.3309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1339$
 ℓ' $E= 0.3889$ $p_x= -0.0257$ $p_y= 0.0000$ $p_z= 0.3880$
 P' $E= 1.9632$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7246$
 q_Y $E= 2.1128$ $p_x= 0.0257$ $p_y= -0.0000$ $p_z= -2.1126$

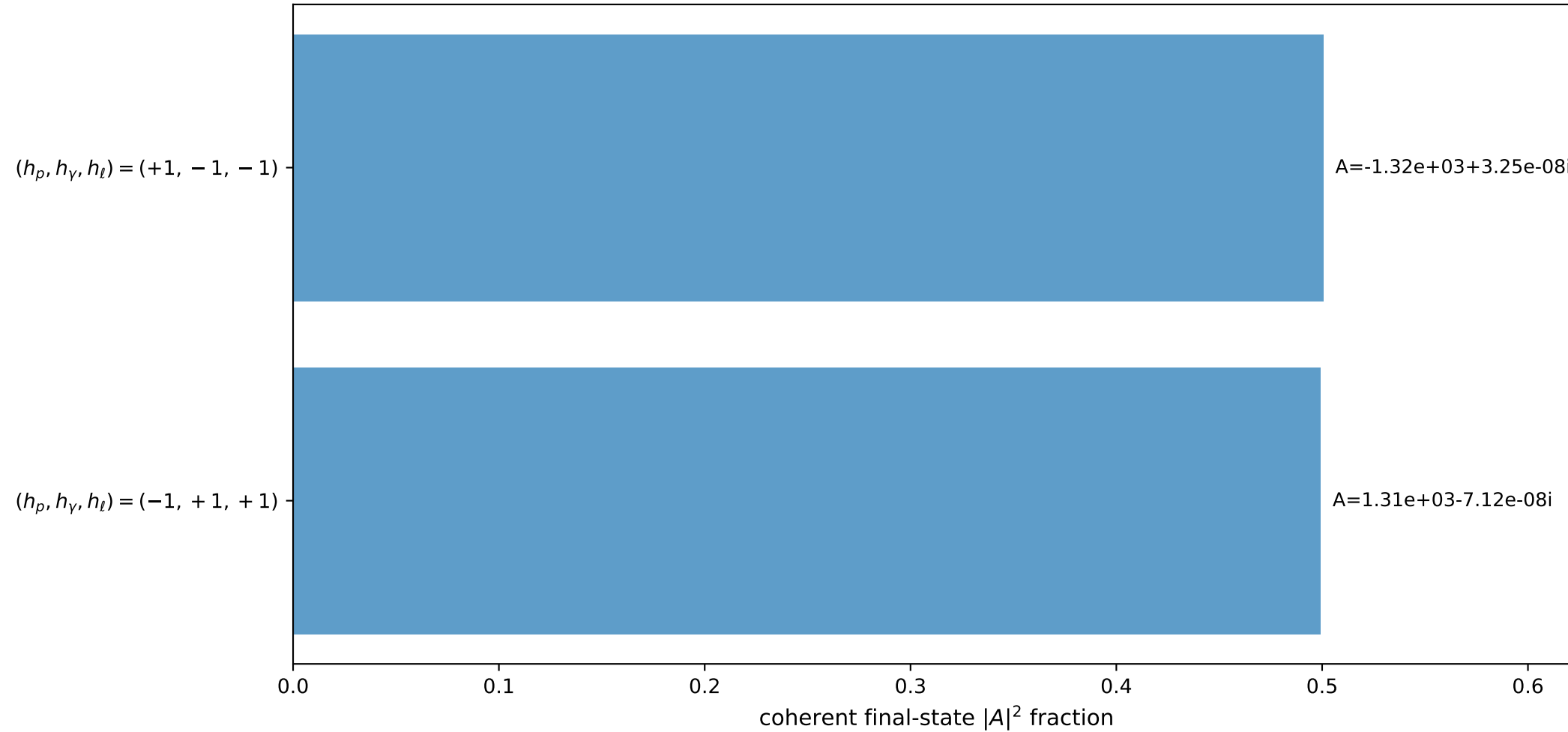
3D momenta



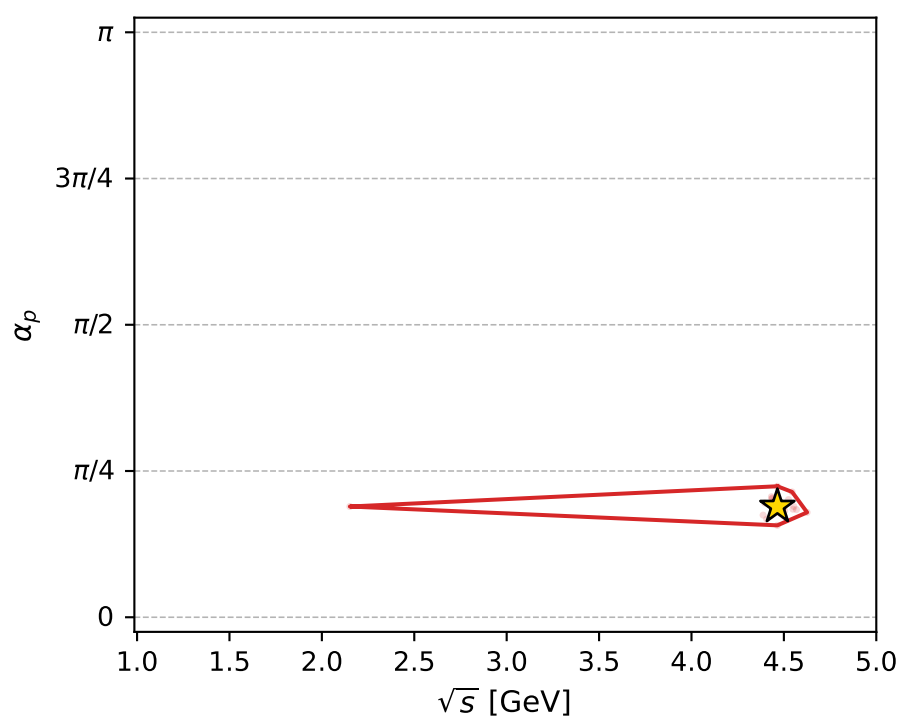
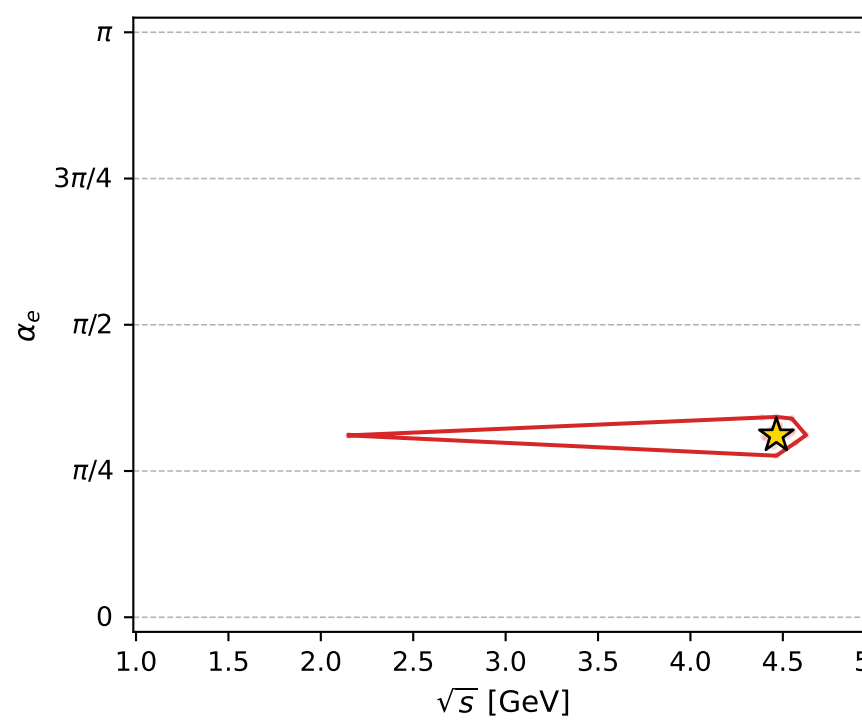
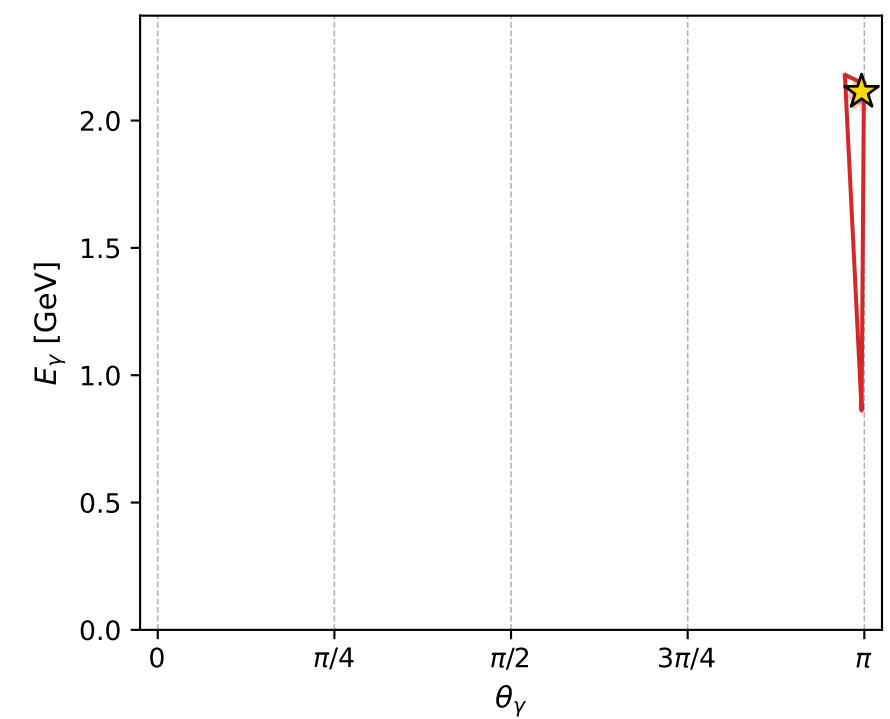
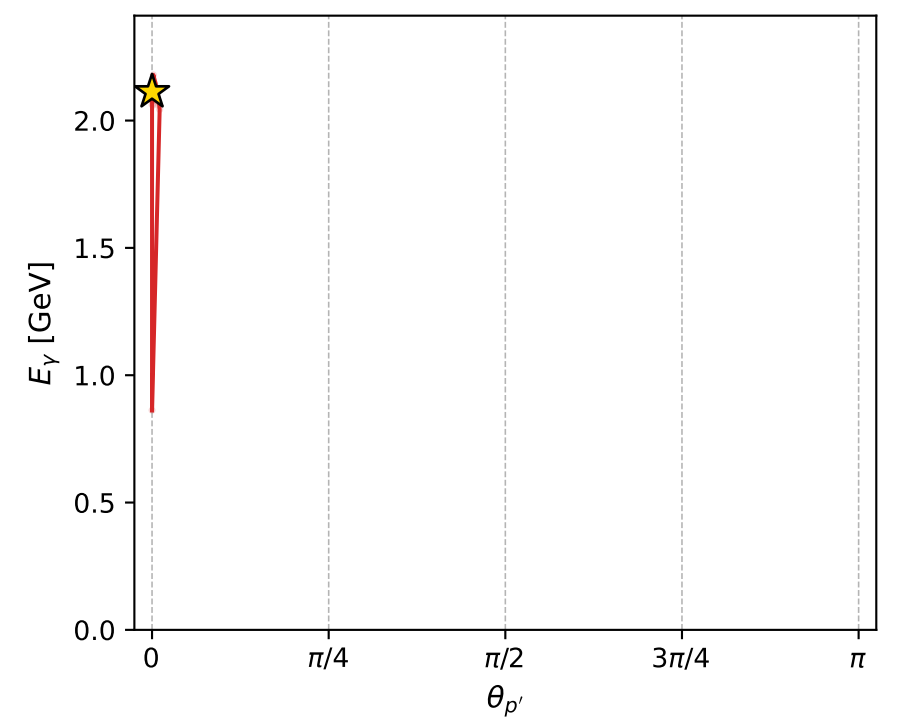
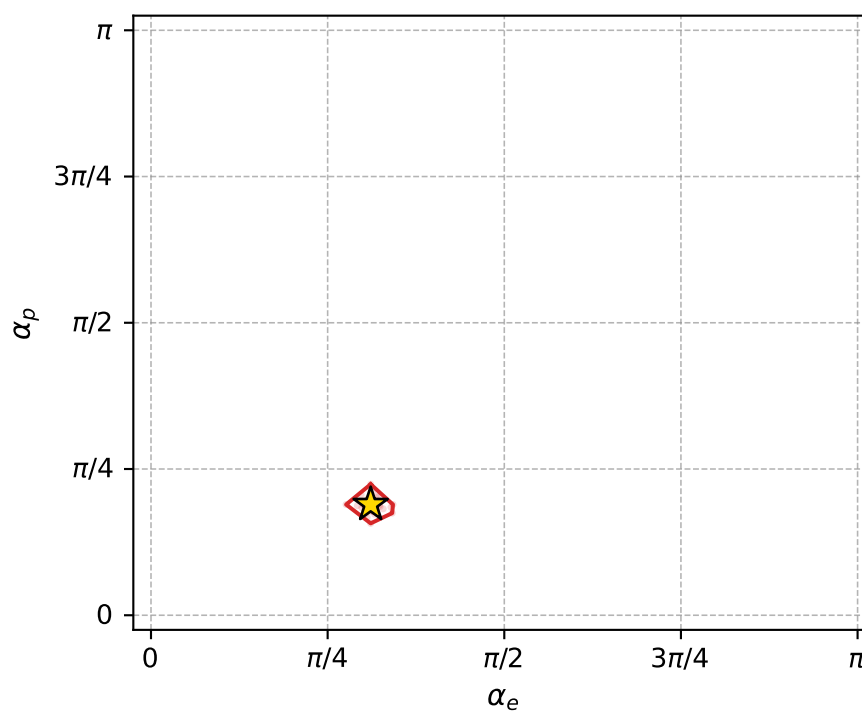
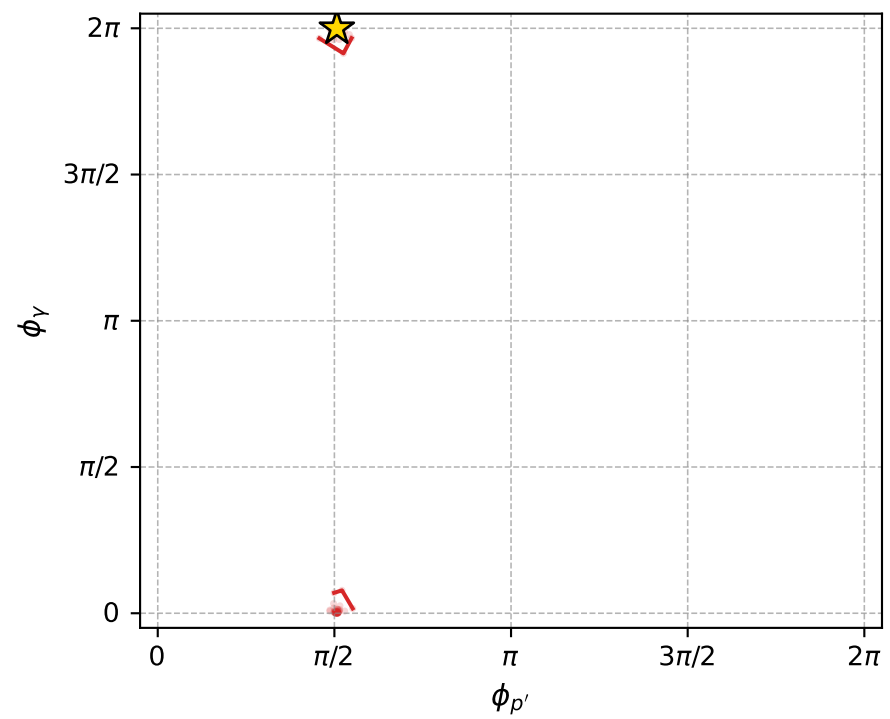
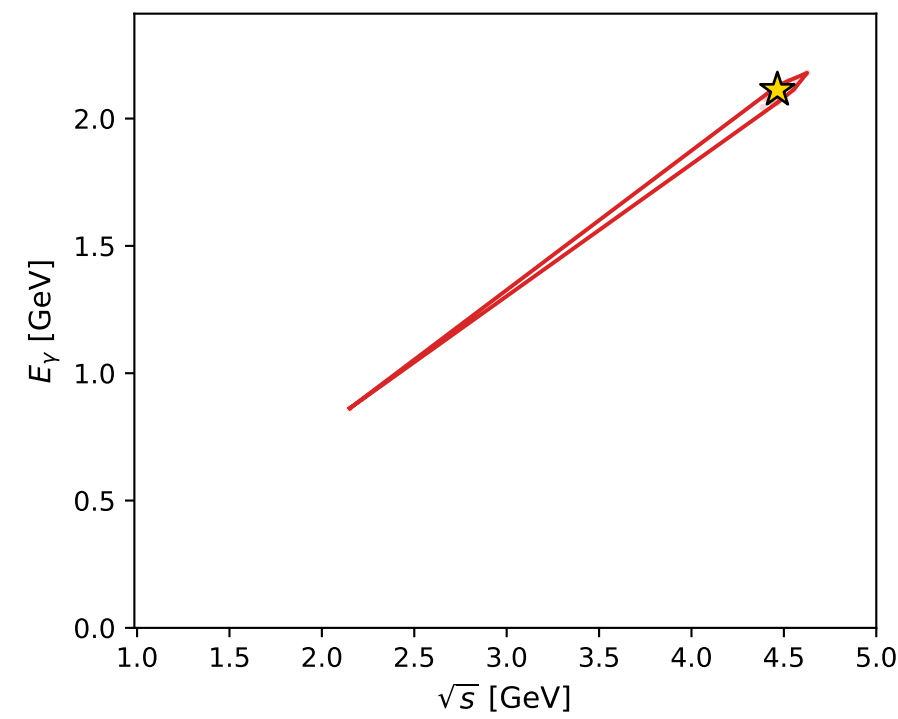
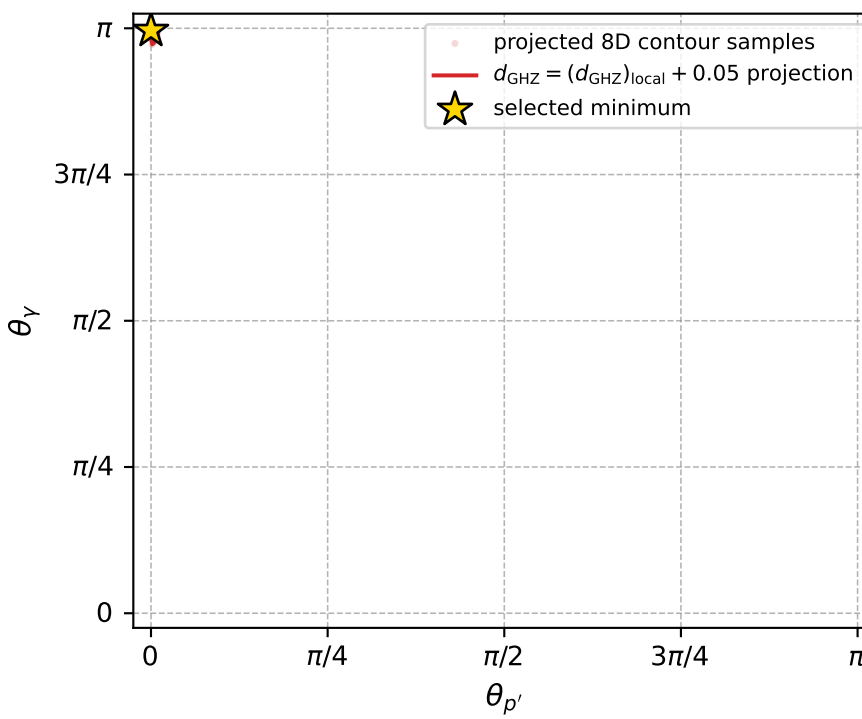
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 231; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 5.7483976\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050057484$
 above global minimum = $5.7461693\text{e-}05$
 8D contour samples = 129

selected phase-space point:

$\text{sqrt_s} = 4.46483$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 3.129444$
 $\text{q0out} = 2.112778$
 $\text{phi_p_out} = 1.592566$
 $\text{phi_gamma_out} = 6.283176$
 $\text{alpha_e} = 0.9767182$
 $\text{alpha_p} = 0.594688$

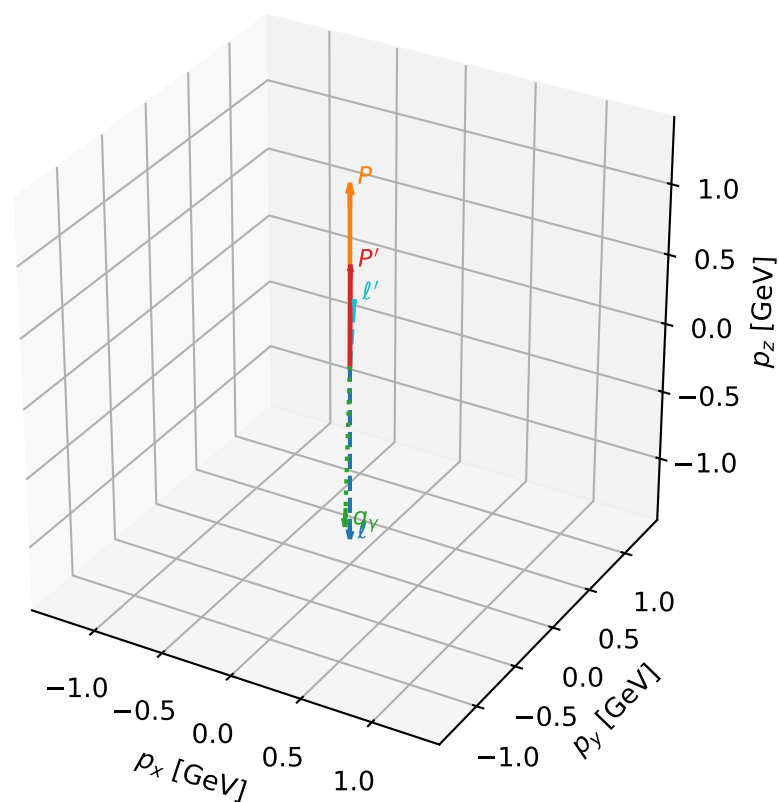
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_236 $d_{\{\text{rm GHZ}\}}=6.73836\text{e-}05$
 region: polarization_cluster_01_local_minimum_236
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.17076605$ rad
 incoming lepton: $0.985455|+\rangle +0.169937|-\rangle$
 proton mixing angle: $\alpha_p=1.4059166$ rad
 incoming proton: $0.164134|+\rangle +0.986438|-\rangle$

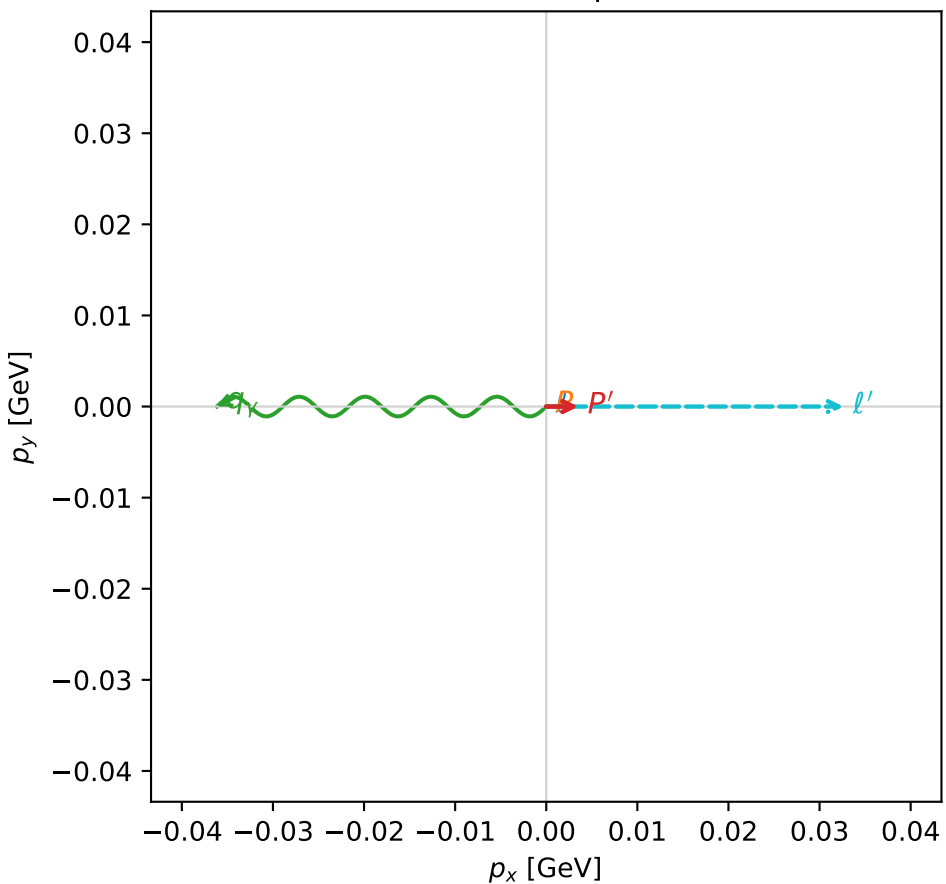
$s=8.17465$, $\sqrt{s}=2.85913$, $|\vec{P}|=1.2757$, $|\vec{P}'|=0.709145$
 $\theta_{P'}=0.00496652$, $\theta_Y=3.11136$, $\phi_{P'}=6.28315$, $\phi_Y=3.14159$
 $E_Y=1.19591$, $\theta_{\ell'}=0.0670059$, $\phi_{\ell'}=4.64873\text{e-}06$
 $Q^2=2.48393$, $x_B=0.355249$, $t=-0.154922$, $W^2=5.38799$, $y=0.958502$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2757$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2757$
 P $E= 1.5834$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2757$
 ℓ' $E= 0.4873$ $p_x= 0.0326$ $p_y= 0.0000$ $p_z= 0.4862$
 P' $E= 1.1759$ $p_x= 0.0035$ $p_y= -0.0000$ $p_z= 0.7091$
 q_Y $E= 1.1959$ $p_x= -0.0362$ $p_y= -0.0000$ $p_z= -1.1954$

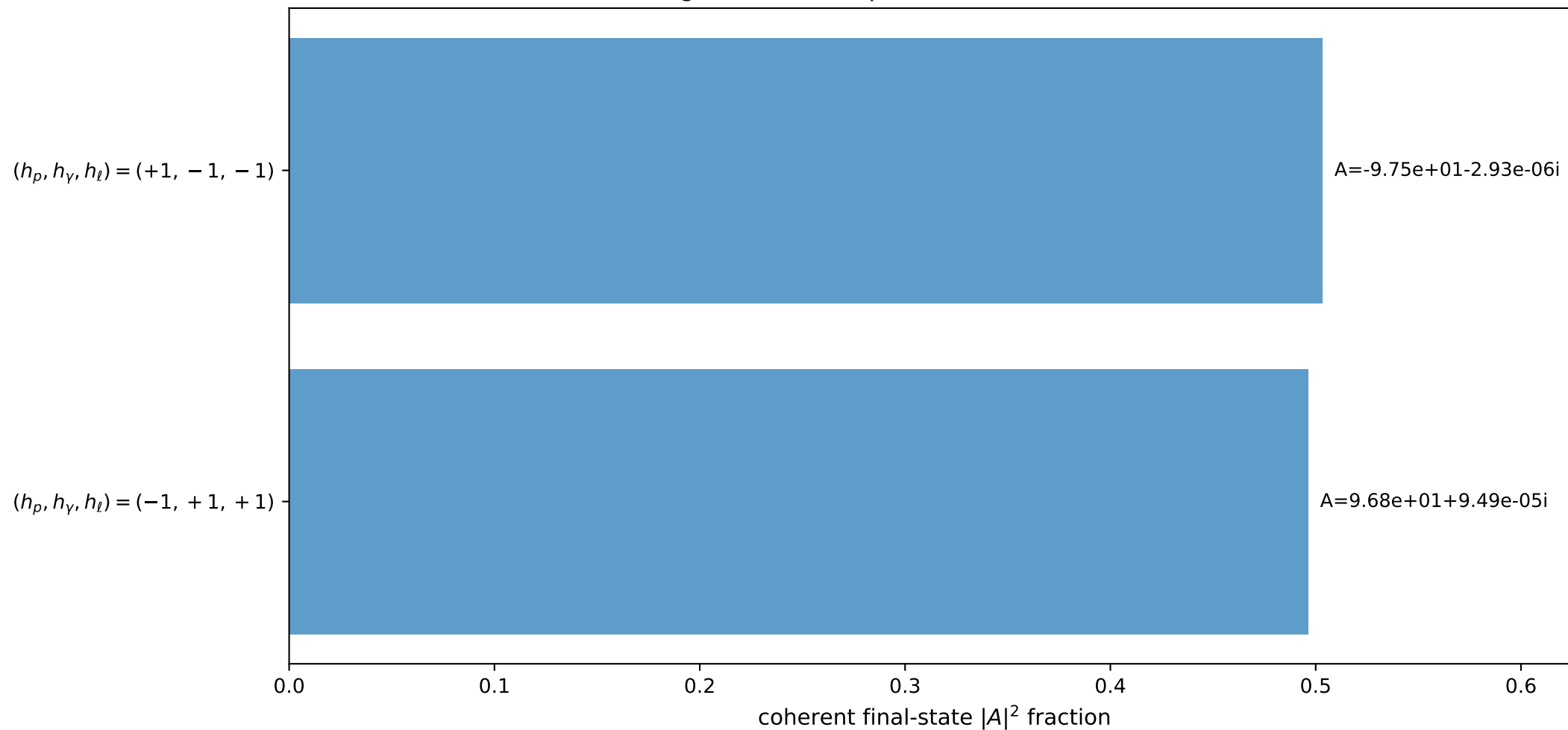
3D momenta



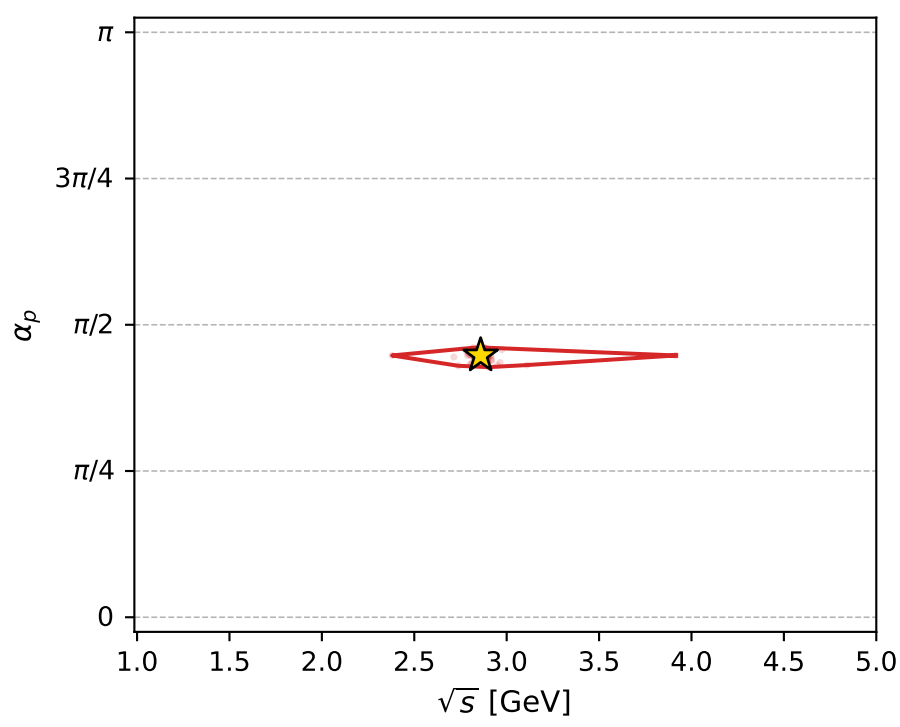
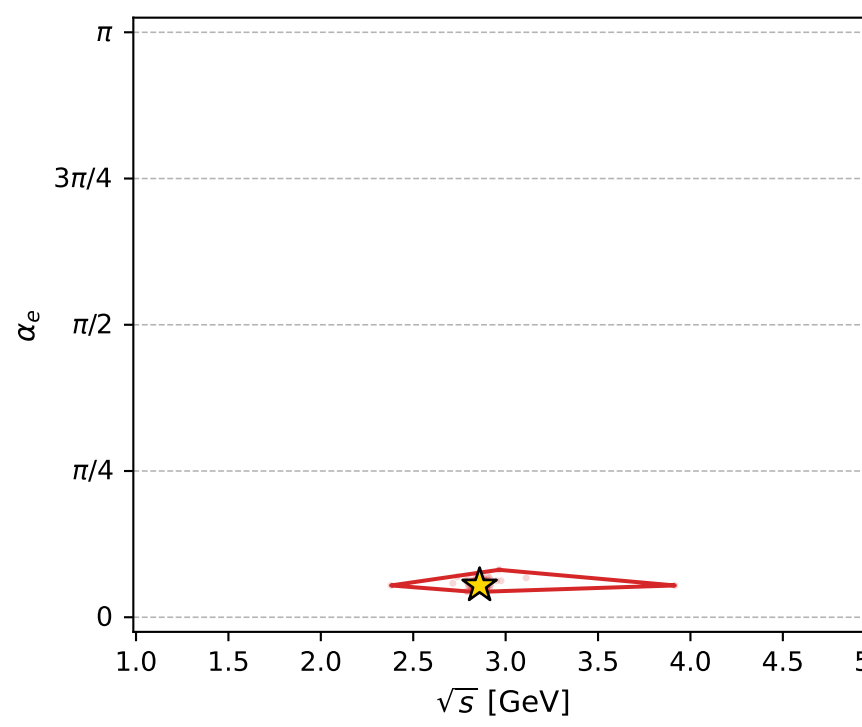
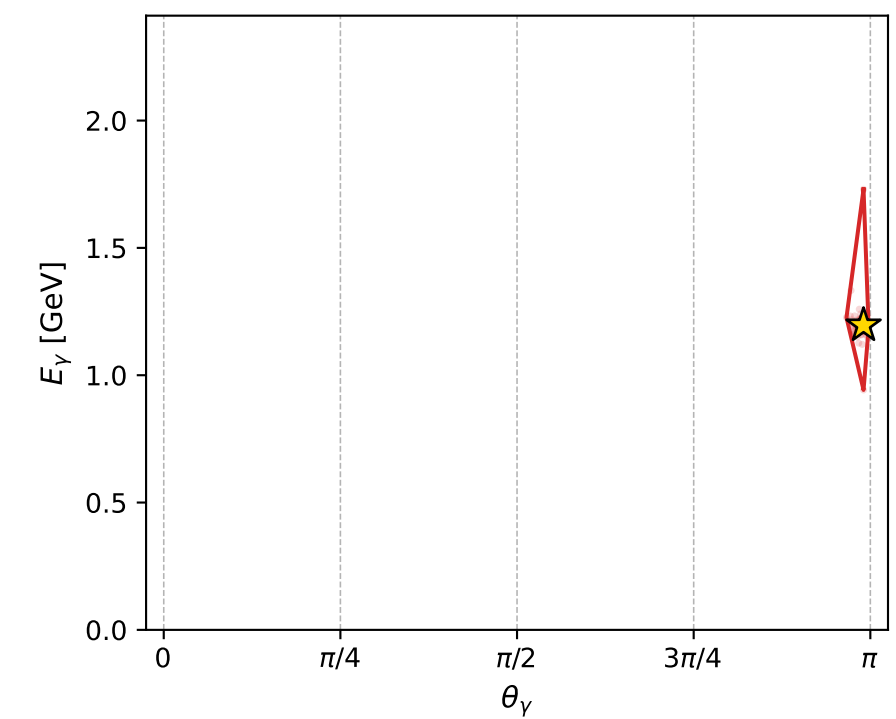
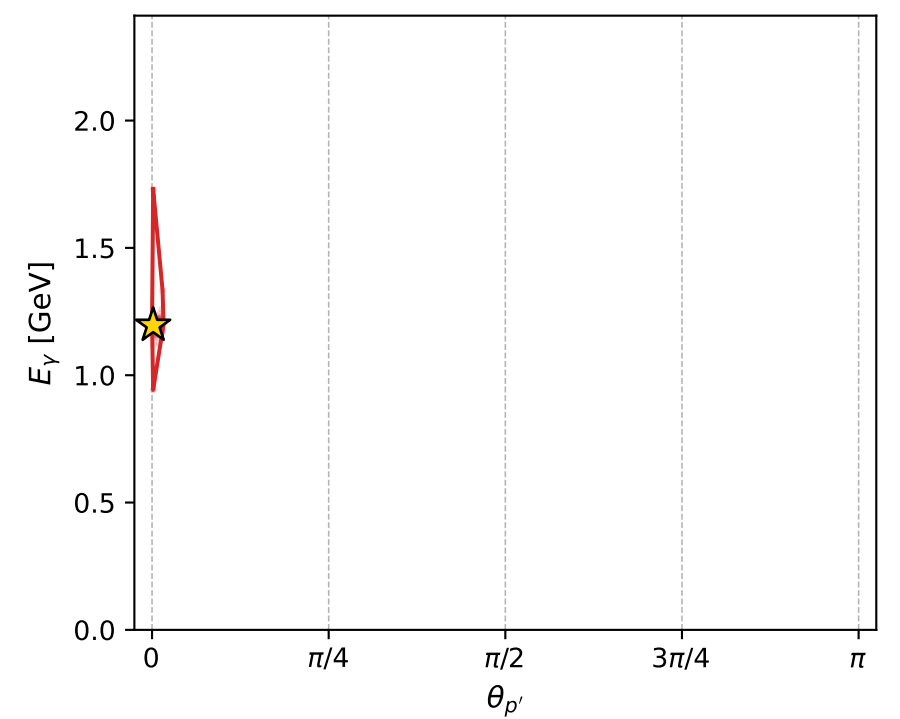
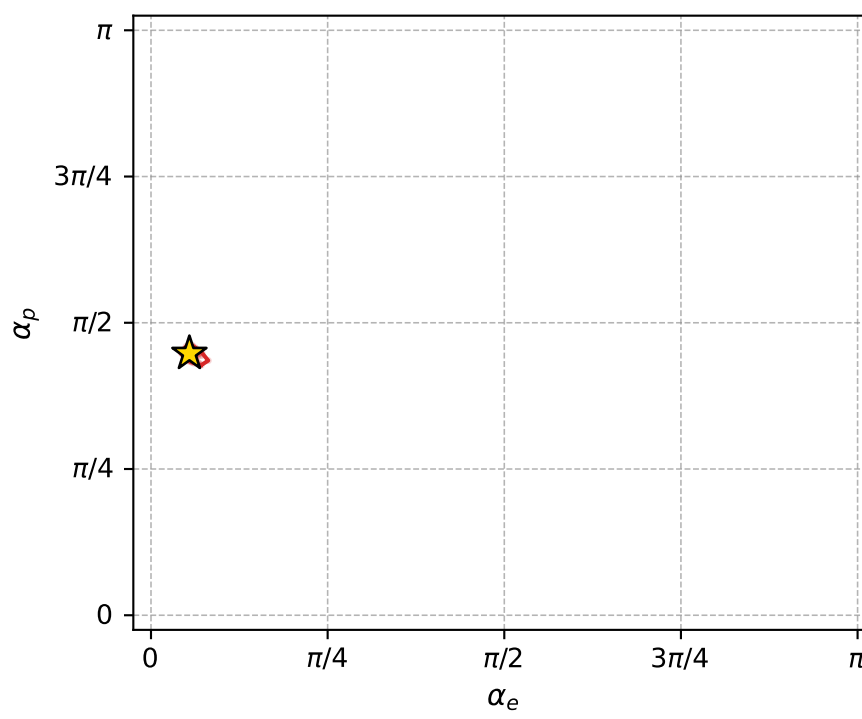
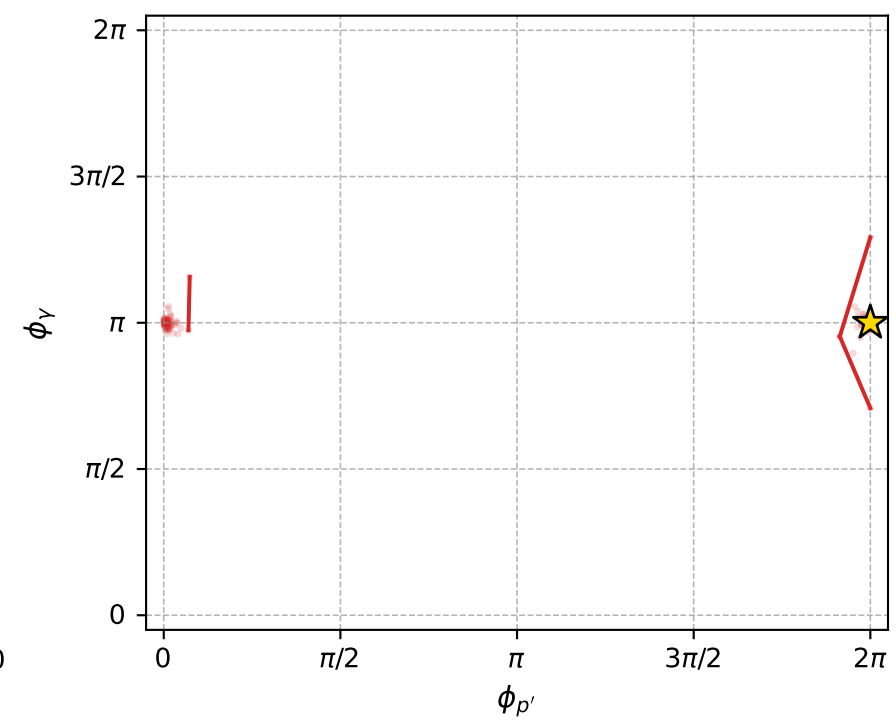
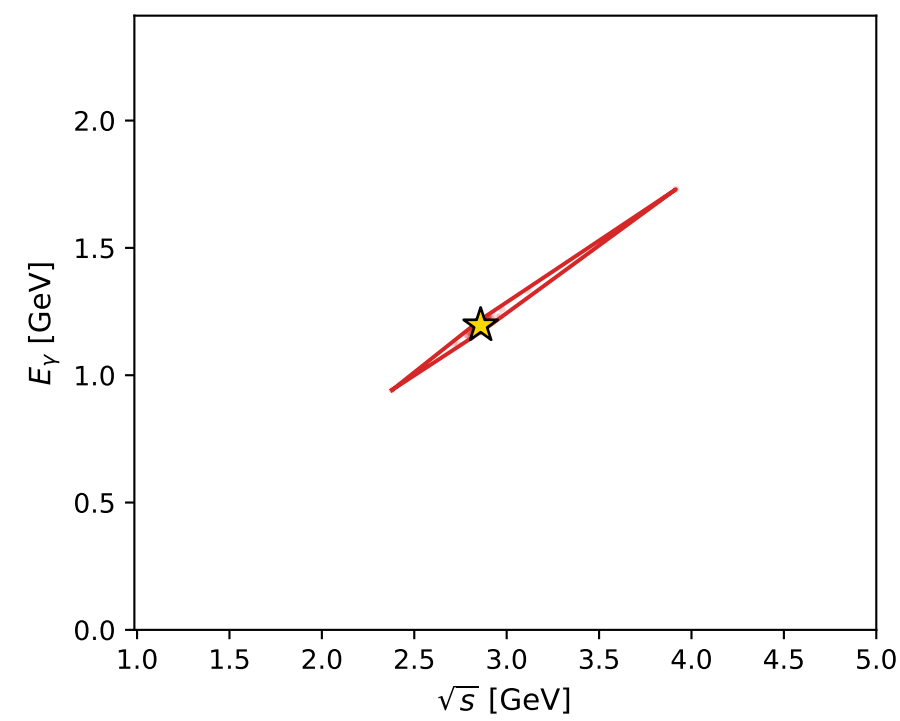
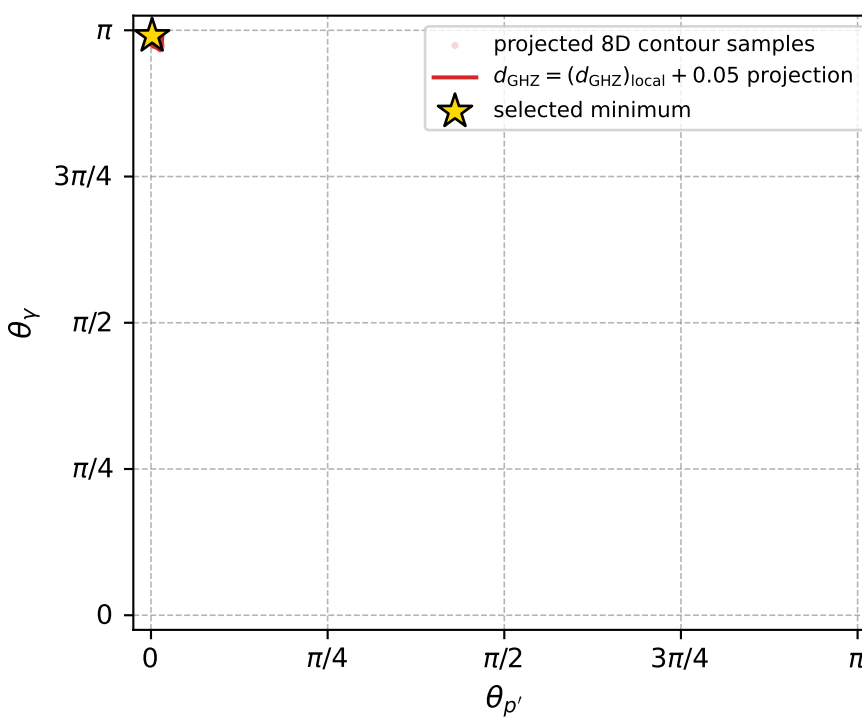
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 236; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 6.7383592\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050067384$
 above global minimum = $6.7361308\text{e-}05$
 8D contour samples = 156

selected phase-space point:

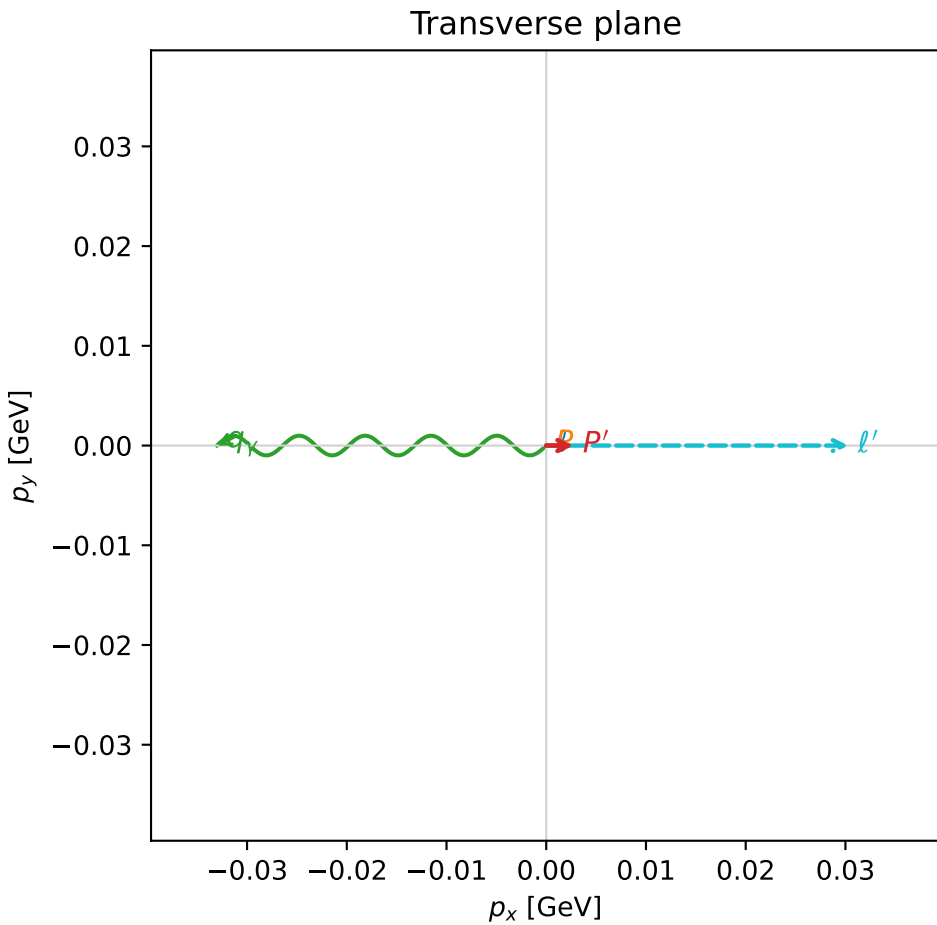
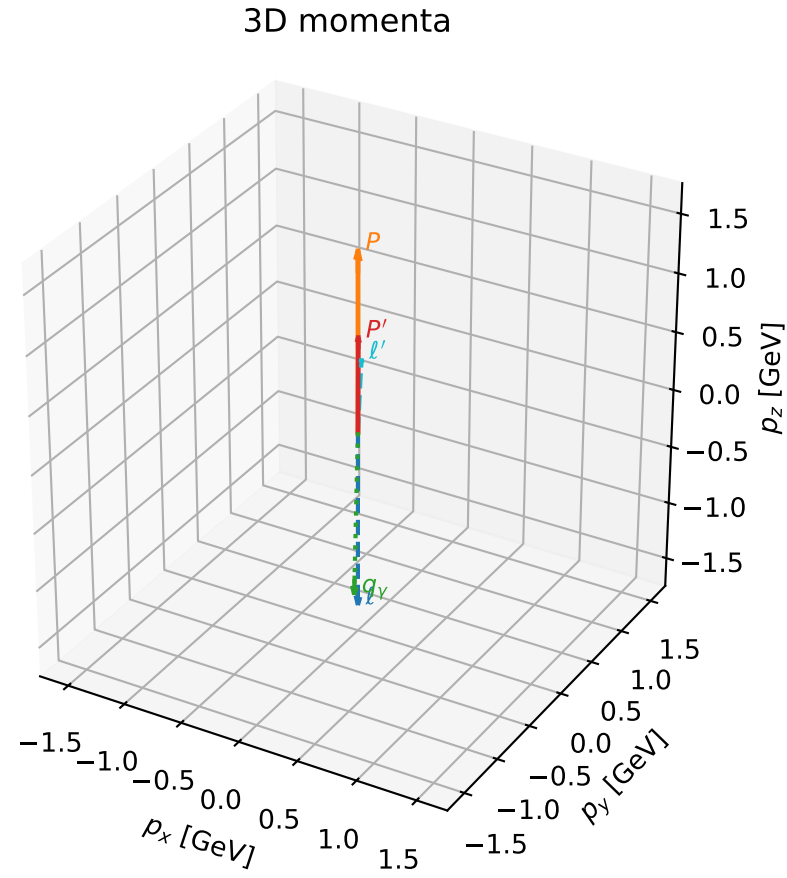
$\text{sqrt_s} = 2.859134$
 $\text{theta_p_out} = 0.004966517$
 $\text{theta_gamma_out} = 3.111359$
 $\text{q0ut} = 1.195914$
 $\text{phi_p_out} = 6.28315$
 $\text{phi_gamma_out} = 3.141593$
 $\text{alpha_e} = 0.1707661$
 $\text{alpha_p} = 1.405917$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_241 d_{\rm GHZ}=8.98184e-05
region: polarization_cluster_01_local_minimum_241
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.17233494$ rad
incoming lepton: $0.985187|+\rangle +0.171483|-\rangle$
proton mixing angle: $\alpha_p=1.4019813$ rad
incoming proton: $0.168014|+\rangle +0.985785|-\rangle$

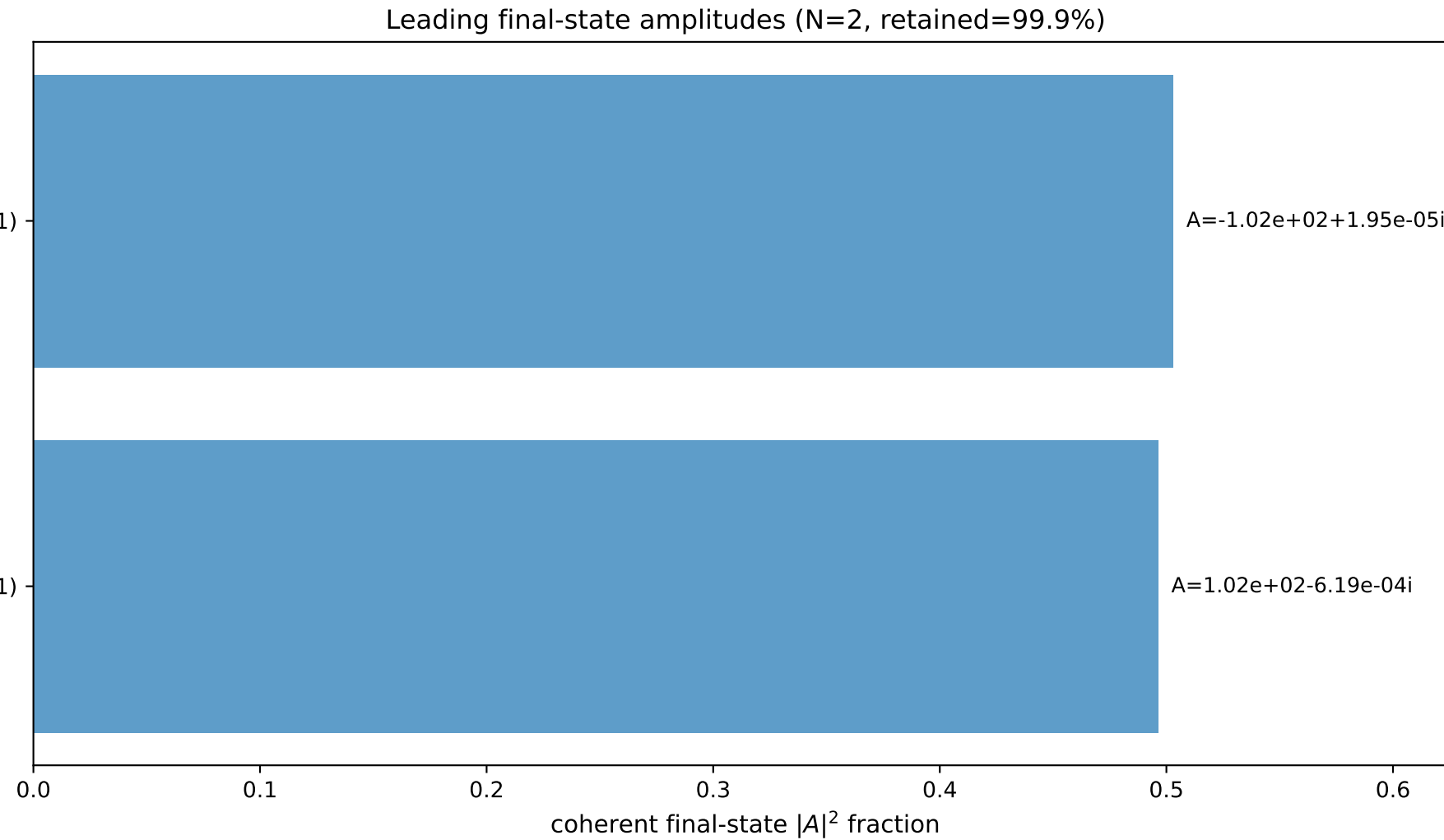
$s=10.9892$, $\sqrt{s}=3.31499$, $|\vec{P}|=1.52479$, $|\vec{P}'|=0.805866$
 $\theta_{P'}=0.0033848$, $\theta_V=3.11869$, $\phi_{P'}=0.000438679$, $\phi_V=3.14159$
 $E_V=1.44194$, $\theta_{\ell'}=0.0476231$, $\phi_{\ell'}=6.28315$
 $Q^2=3.87942$, $x_B=0.397103$, $t=-0.210426$, $W^2=6.76972$, $y=0.966365$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5248$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5248$
 P $E= 1.7902$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5248$
 ℓ' $E= 0.6364$ $p_x= 0.0303$ $p_y= -0.0000$ $p_z= 0.6357$
 P' $E= 1.2366$ $p_x= 0.0027$ $p_y= 0.0000$ $p_z= 0.8059$
 q_V $E= 1.4419$ $p_x= -0.0330$ $p_y= -0.0000$ $p_z= -1.4416$

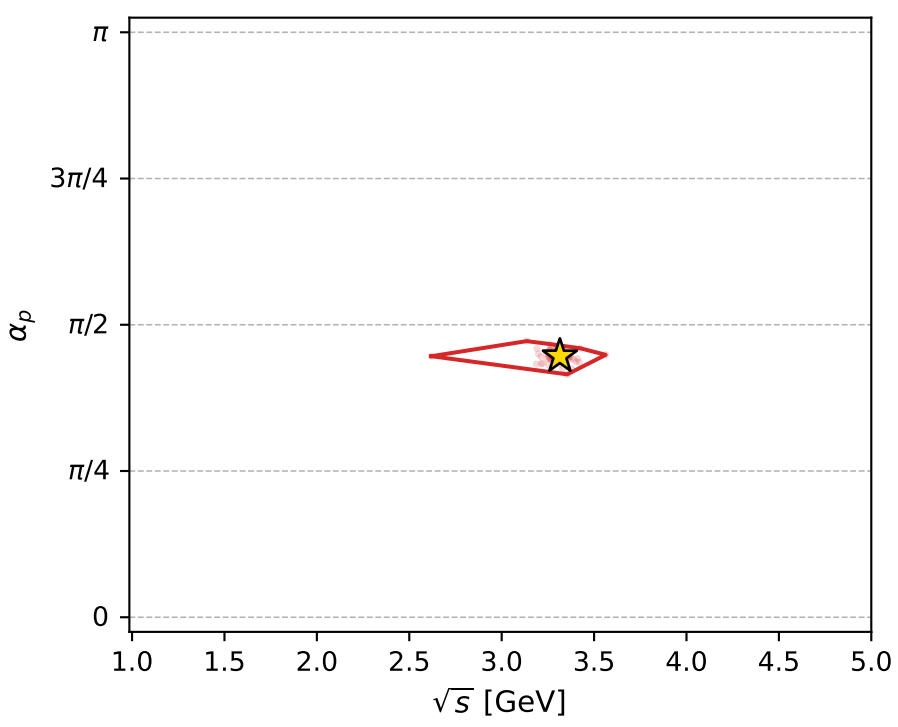
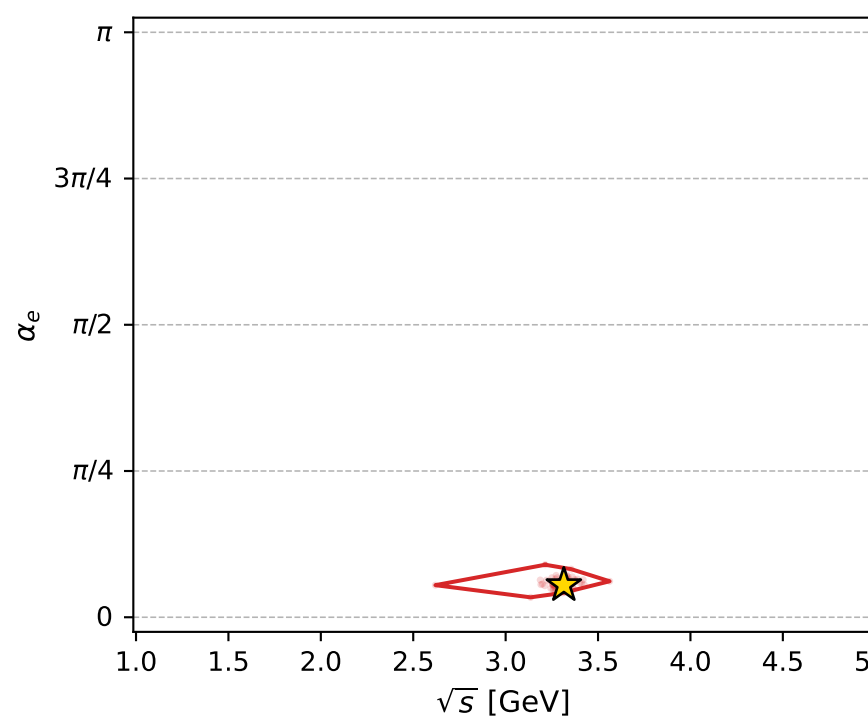
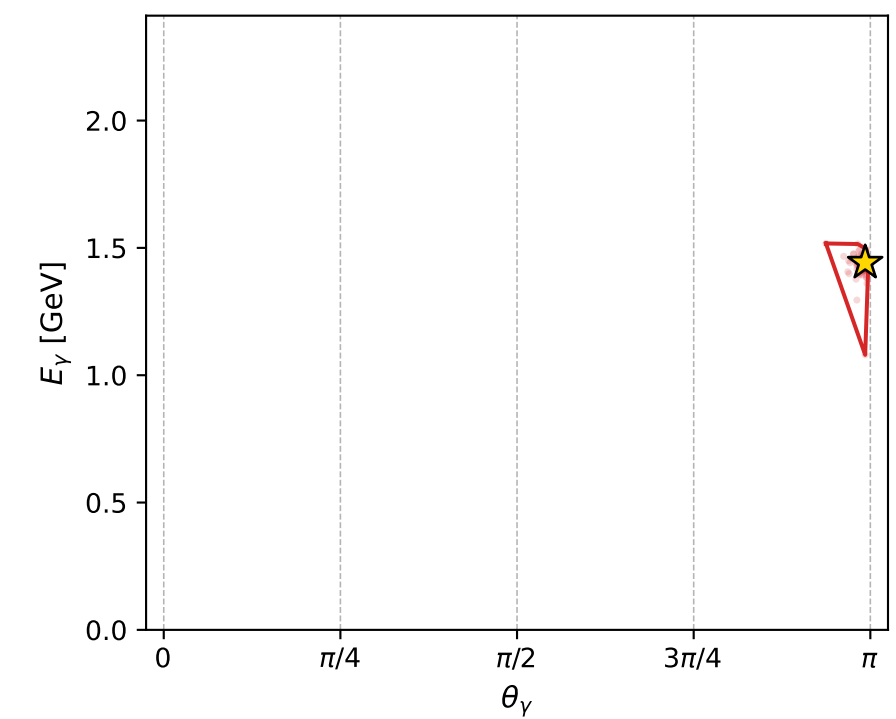
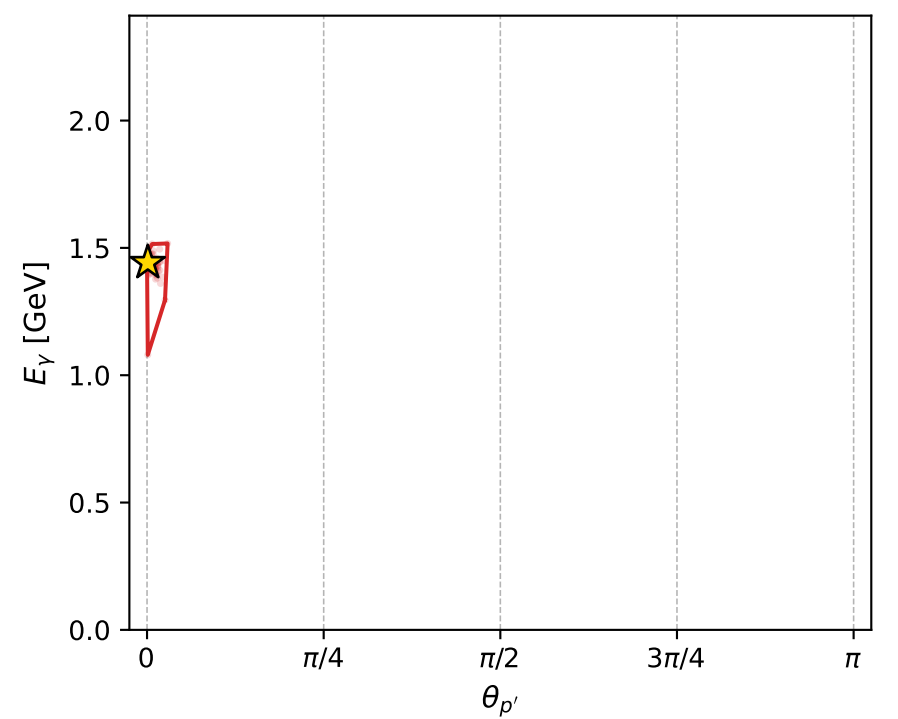
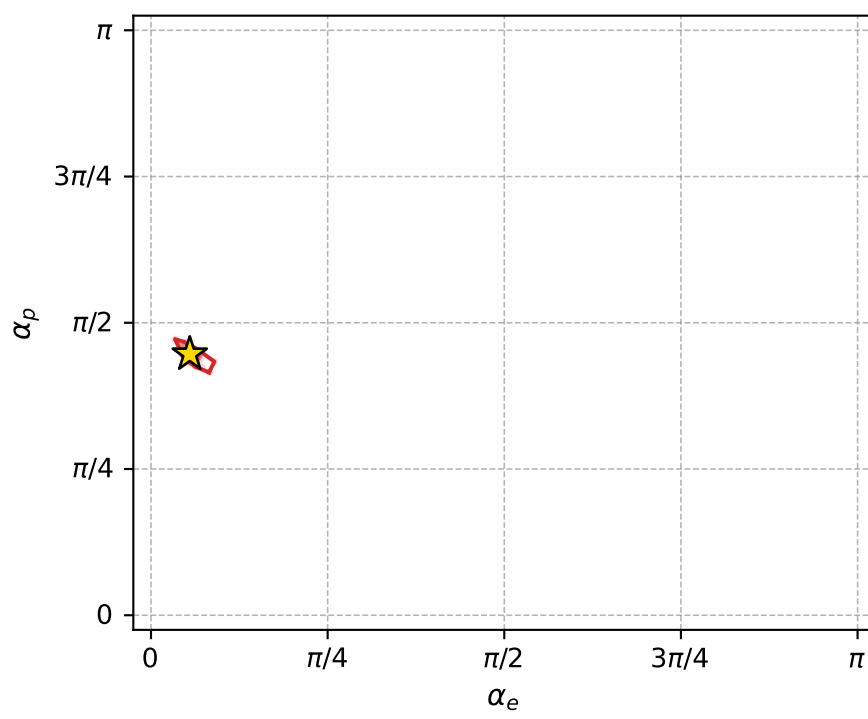
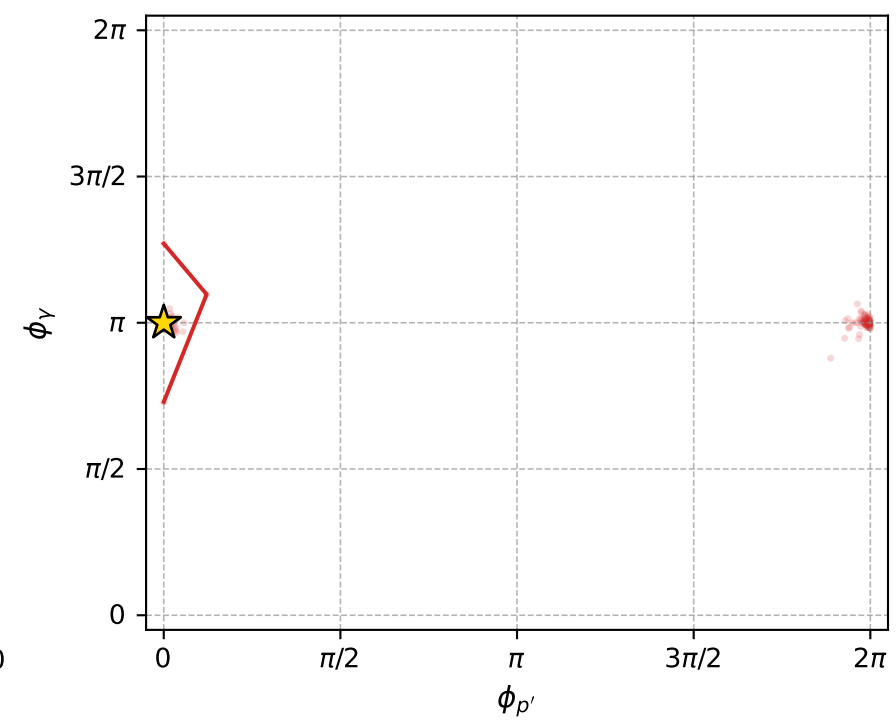
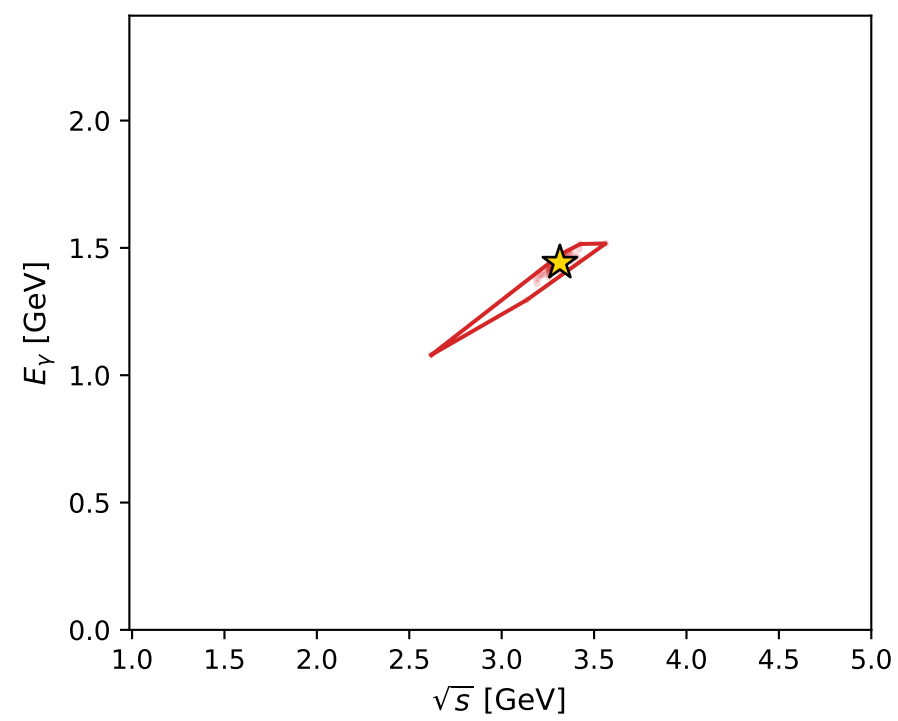
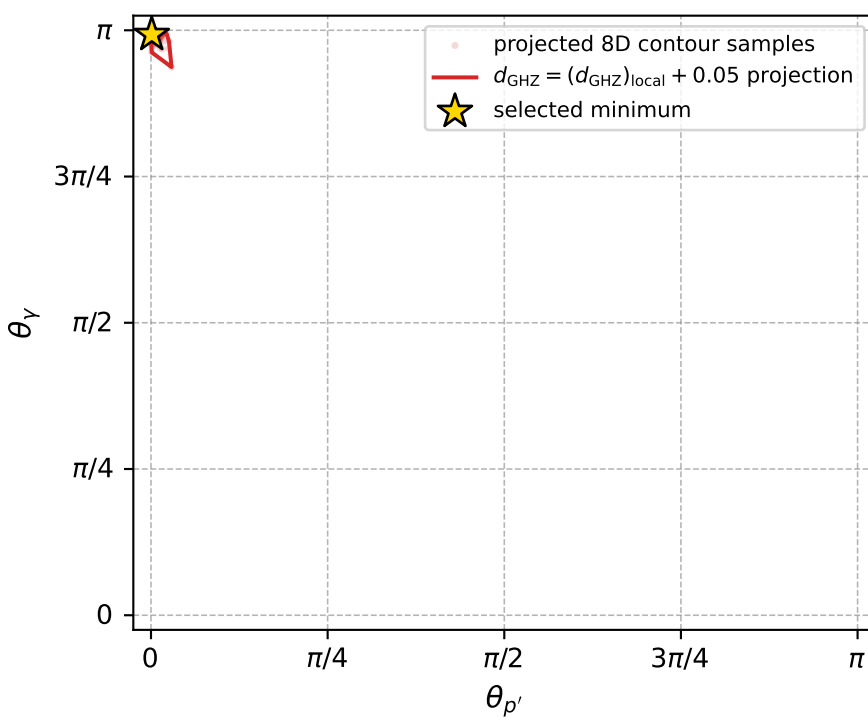


$(h_p, h_V, h_\ell) = (+1, -1, -1)$

$(h_p, h_V, h_\ell) = (-1, +1, +1)$



electron: polarization cluster P1, local minimum 241; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 8.9818412\text{e-}05$
 $d_{\text{GHZ}} \text{ contour} = 0.050089818$
 above global minimum = $8.9796128\text{e-}05$
 8D contour samples = 154

selected phase-space point:

$\text{sqrt_s} = 3.314991$
 $\text{theta_p_out} = 0.003384802$
 $\text{theta_gamma_out} = 3.118688$
 $\text{q0out} = 1.441938$
 $\text{phi_p_out} = 0.0004386794$
 $\text{phi_gamma_out} = 3.141594$
 $\text{alpha_e} = 0.1723349$
 $\text{alpha_p} = 1.401981$

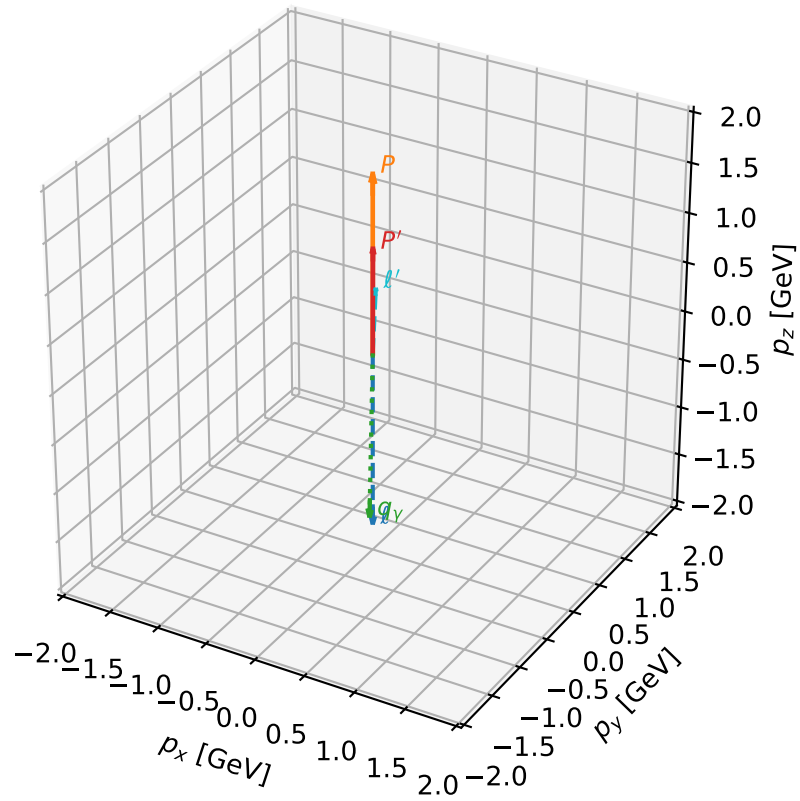
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_242 $d_{\{\text{rm GHZ}\}}=0.000102039$
 region: polarization_cluster_01_local_minimum_242
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.24367421$ rad
 incoming lepton: $0.970458|+\rangle +0.24127|-\rangle$
 proton mixing angle: $\alpha_p=1.3312341$ rad
 incoming proton: $0.237277|+\rangle +0.971442|-\rangle$

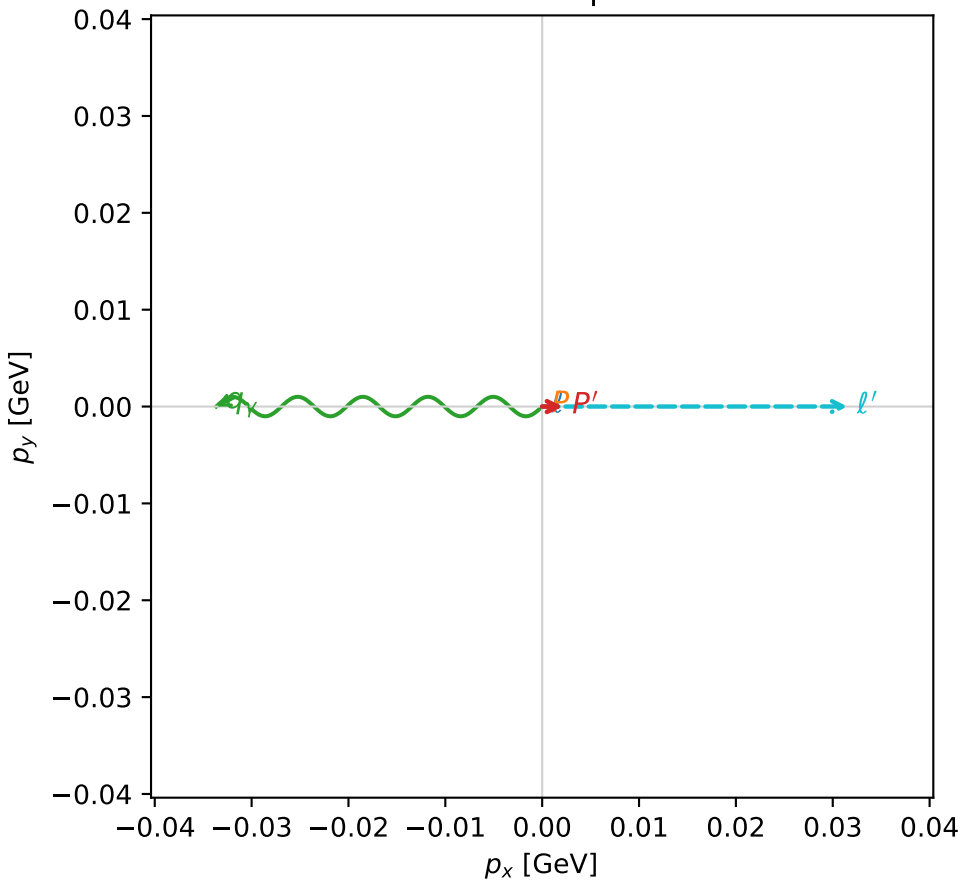
$s=14.4155$, $\sqrt{s}=3.79678$, $|\vec{P}|=1.78252$, $|\vec{P}'|=1.0492$
 $\theta_{P'}=0.00202721$, $\theta_Y=3.12202$, $\phi_{P'}=6.28087$, $\phi_Y=3.14159$
 $E_Y=1.7191$, $\theta_{l'}=0.0470478$, $\phi_{l'}=0.000148497$
 $Q^2=4.77679$, $x_B=0.361266$, $t=-0.169449$, $W^2=9.32542$, $y=0.976852$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7825$
 P $E= 2.0143$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7825$
 l' $E= 0.6703$ $p_x= 0.0315$ $p_y= 0.0000$ $p_z= 0.6696$
 P' $E= 1.4074$ $p_x= 0.0021$ $p_y= -0.0000$ $p_z= 1.0492$
 q_Y $E= 1.7191$ $p_x= -0.0337$ $p_y= 0.0000$ $p_z= -1.7188$

3D momenta



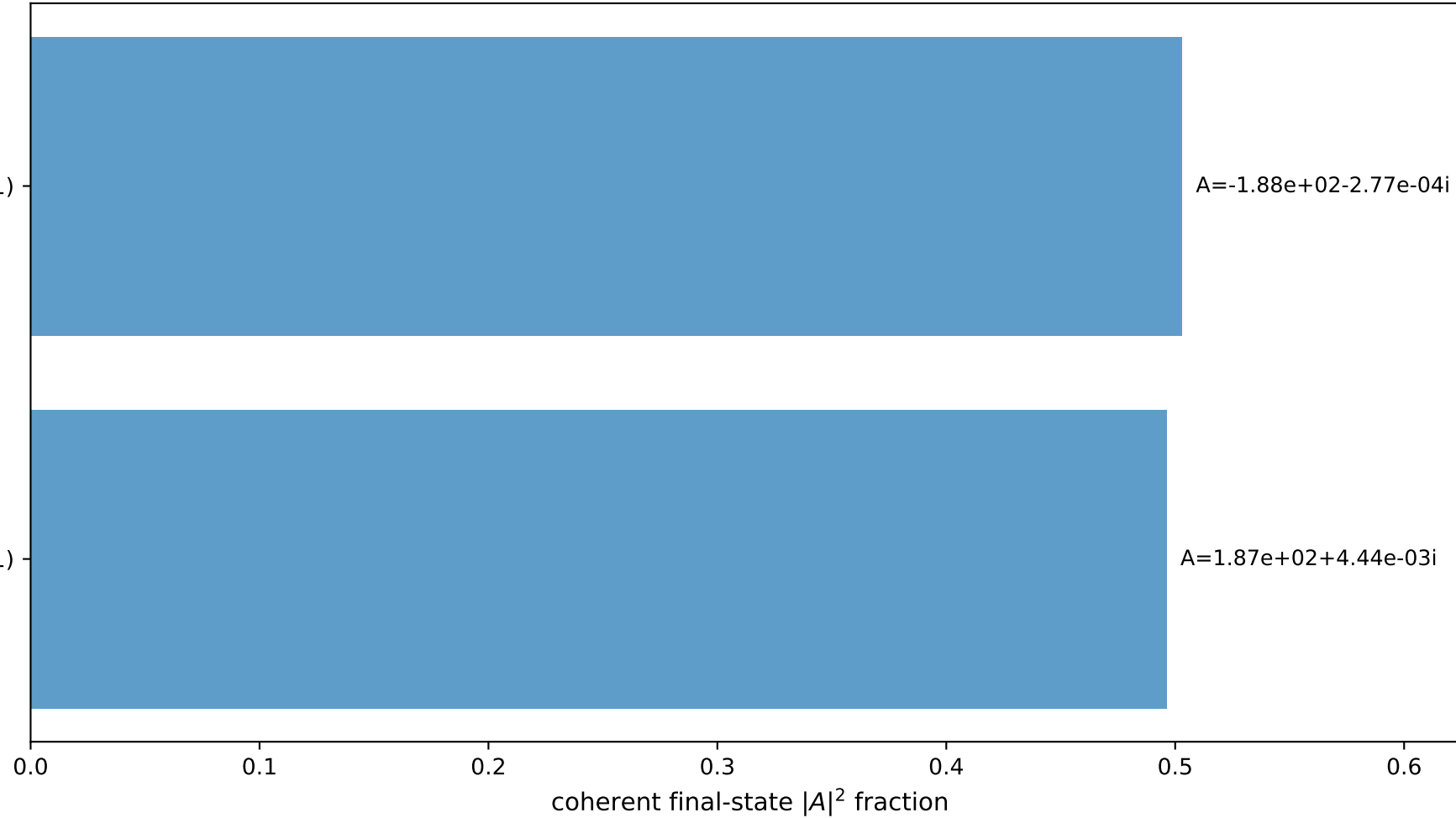
Transverse plane



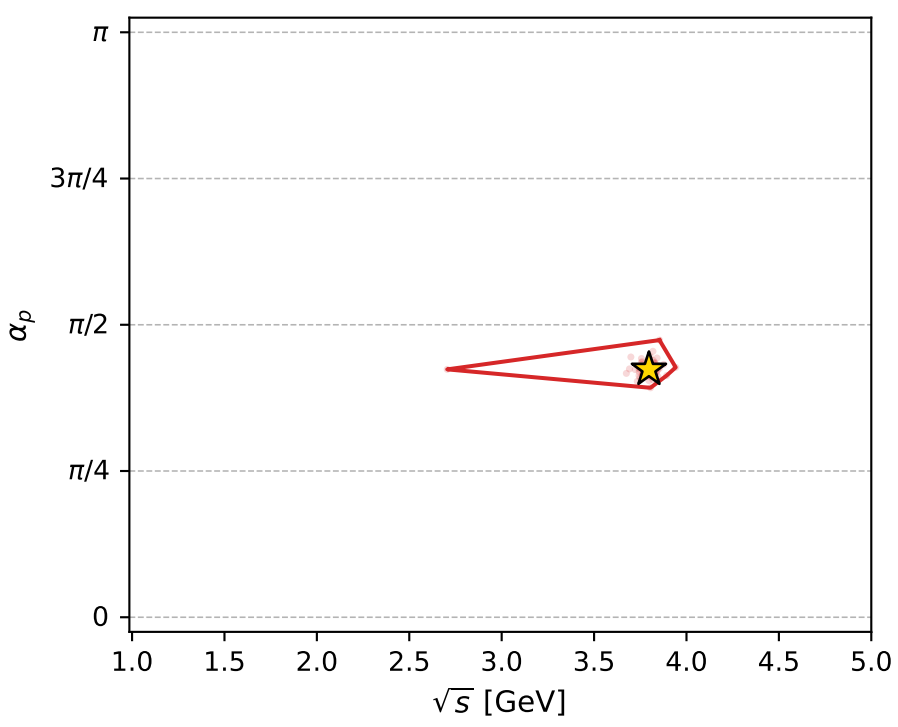
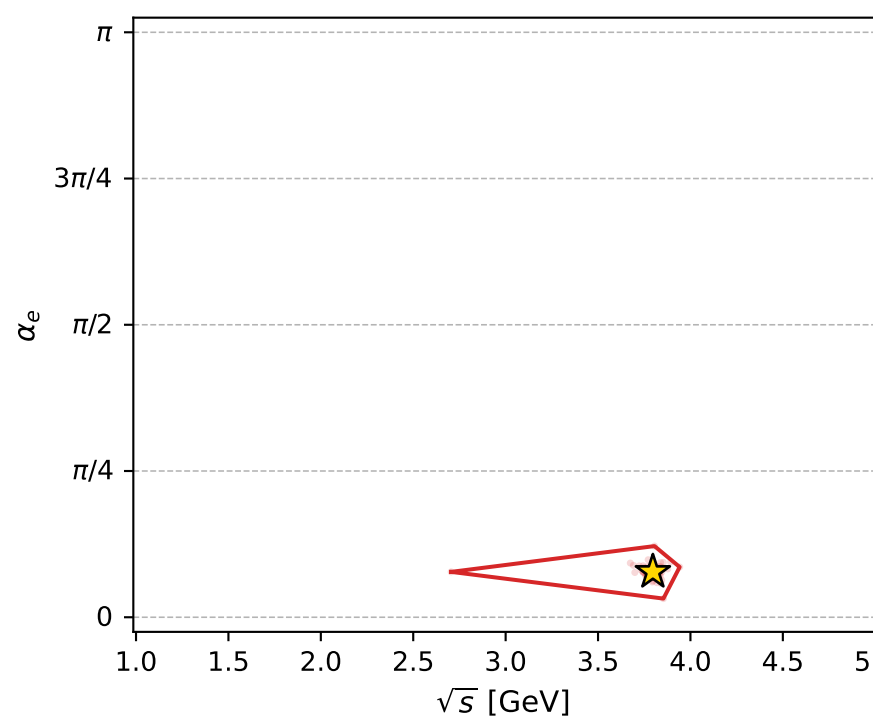
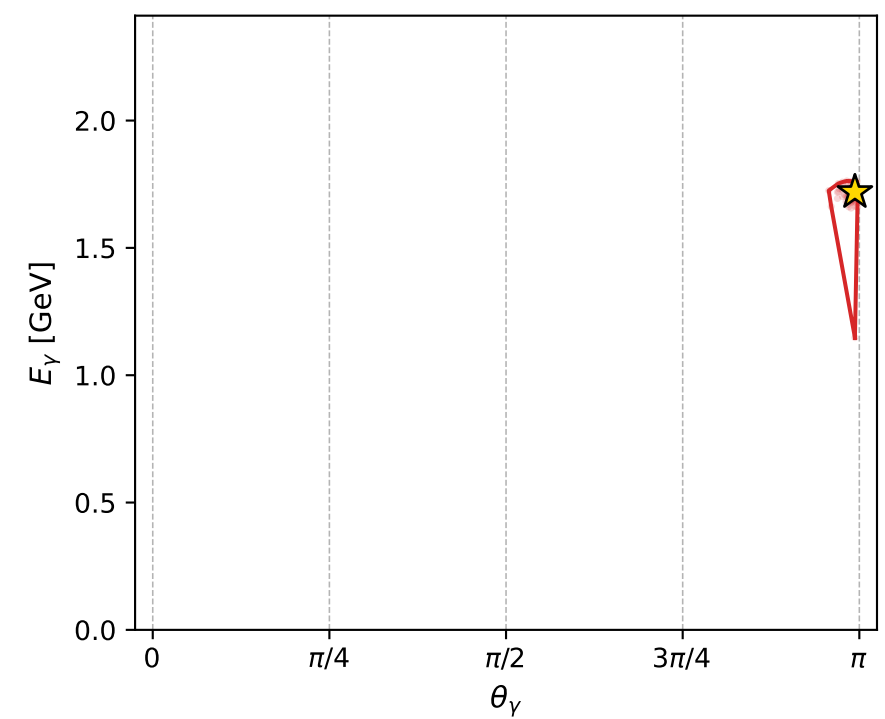
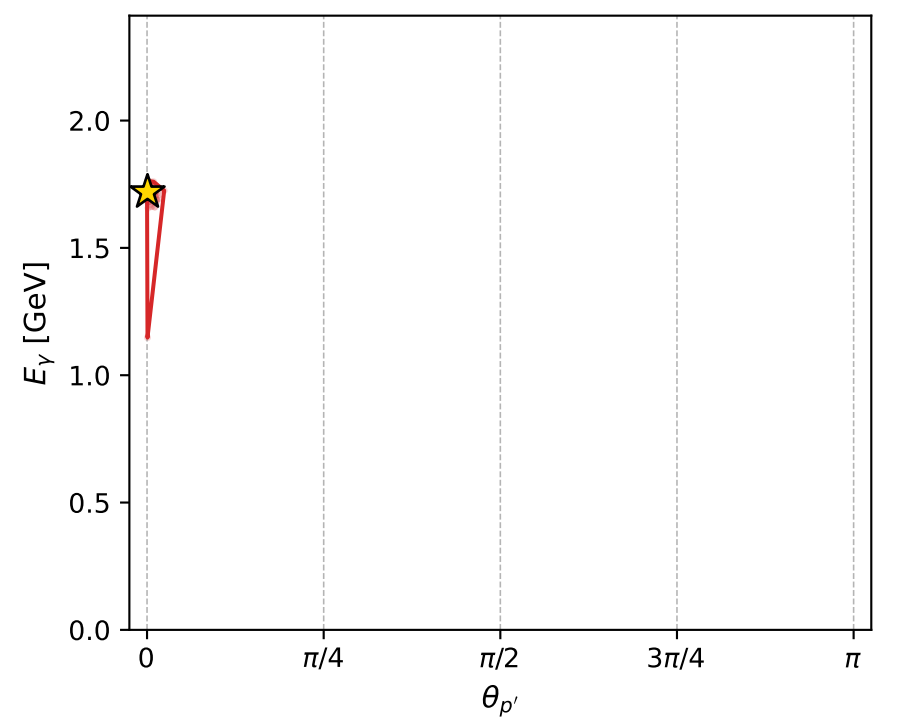
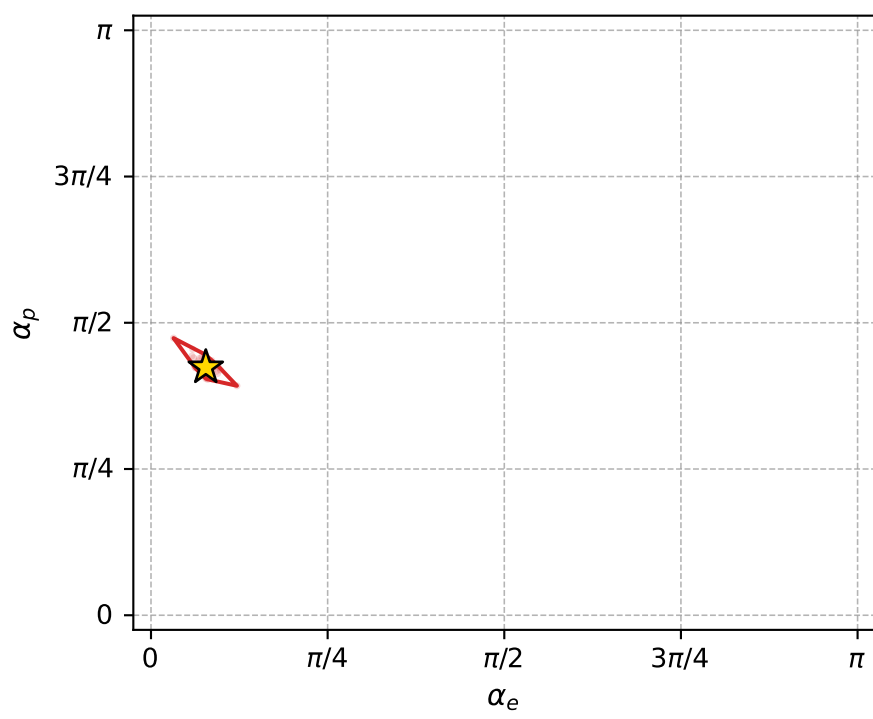
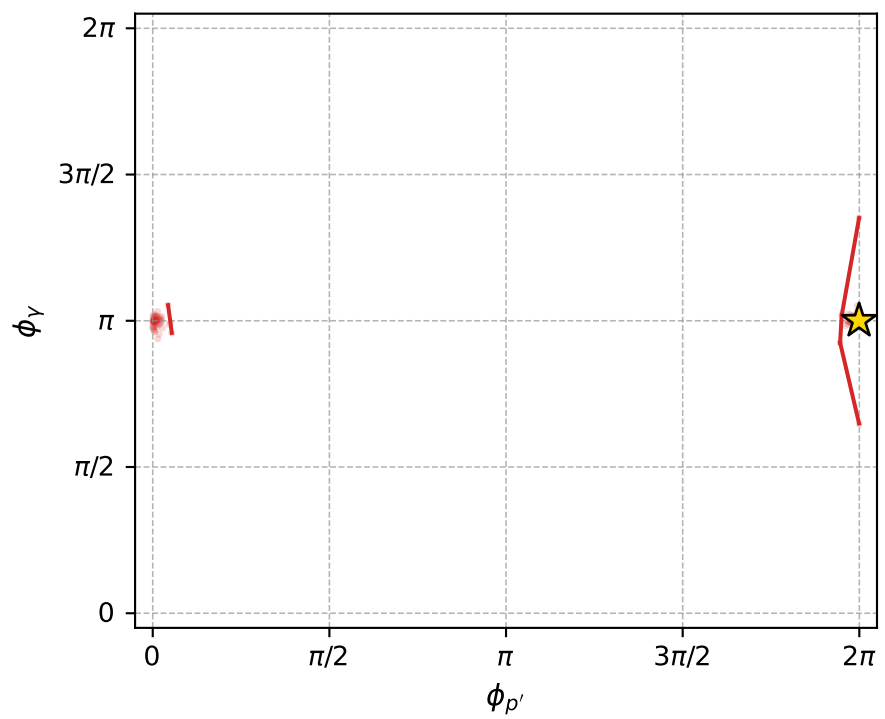
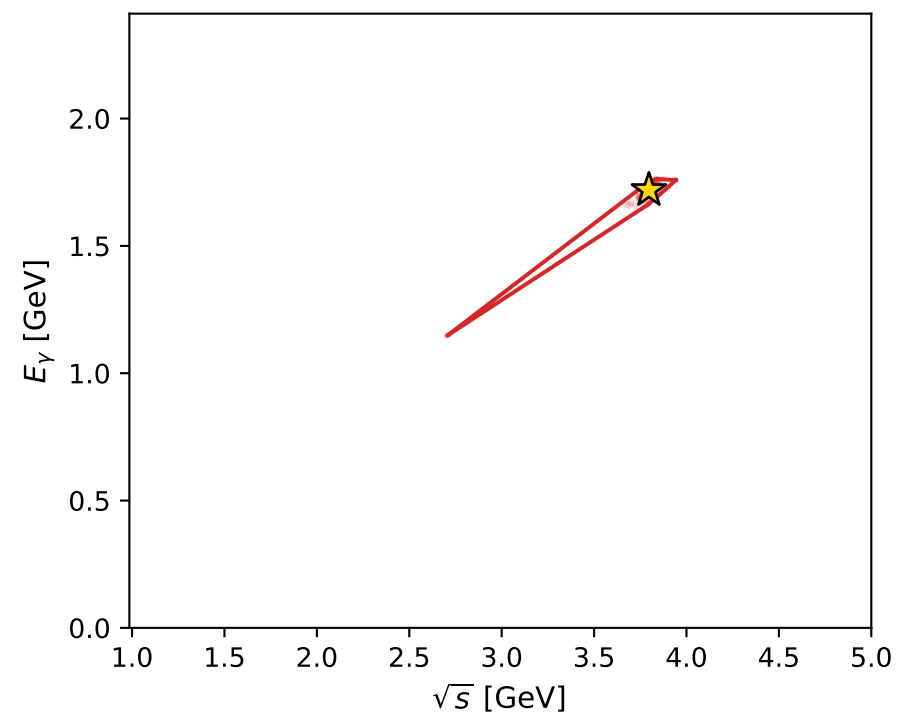
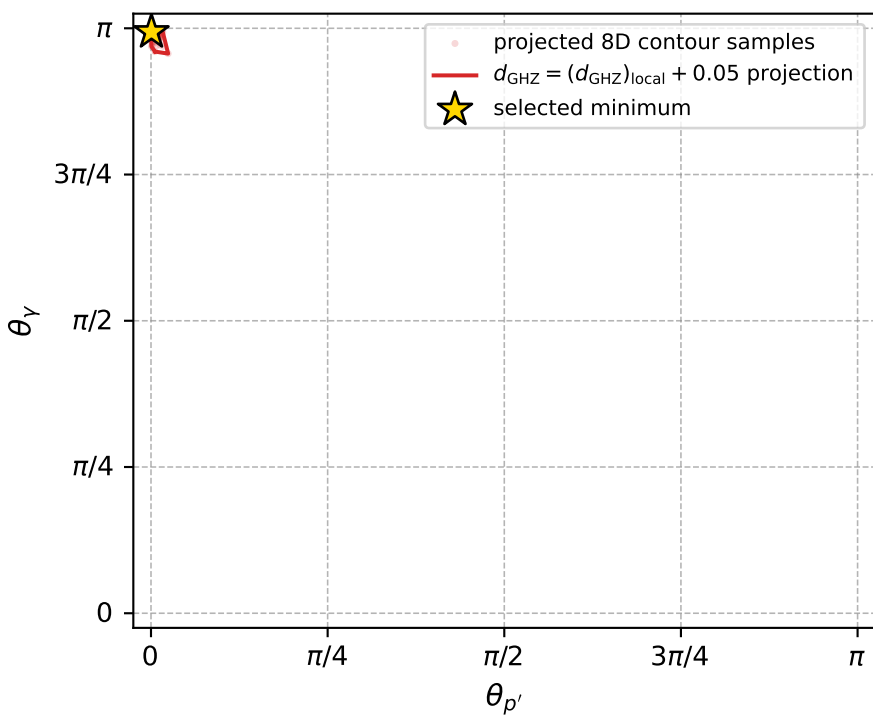
$(h_p, h_Y, h_l) = (+1, -1, -1)$

$(h_p, h_Y, h_l) = (-1, +1, +1)$

Leading final-state amplitudes (N=2, retained=99.9%)



electron: polarization cluster P1, local minimum 242; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00010203879$
 $d_{\text{GHZ}} \text{ contour} = 0.050102039$
 above global minimum = 0.00010201651
 8D contour samples = 139

selected phase-space point:

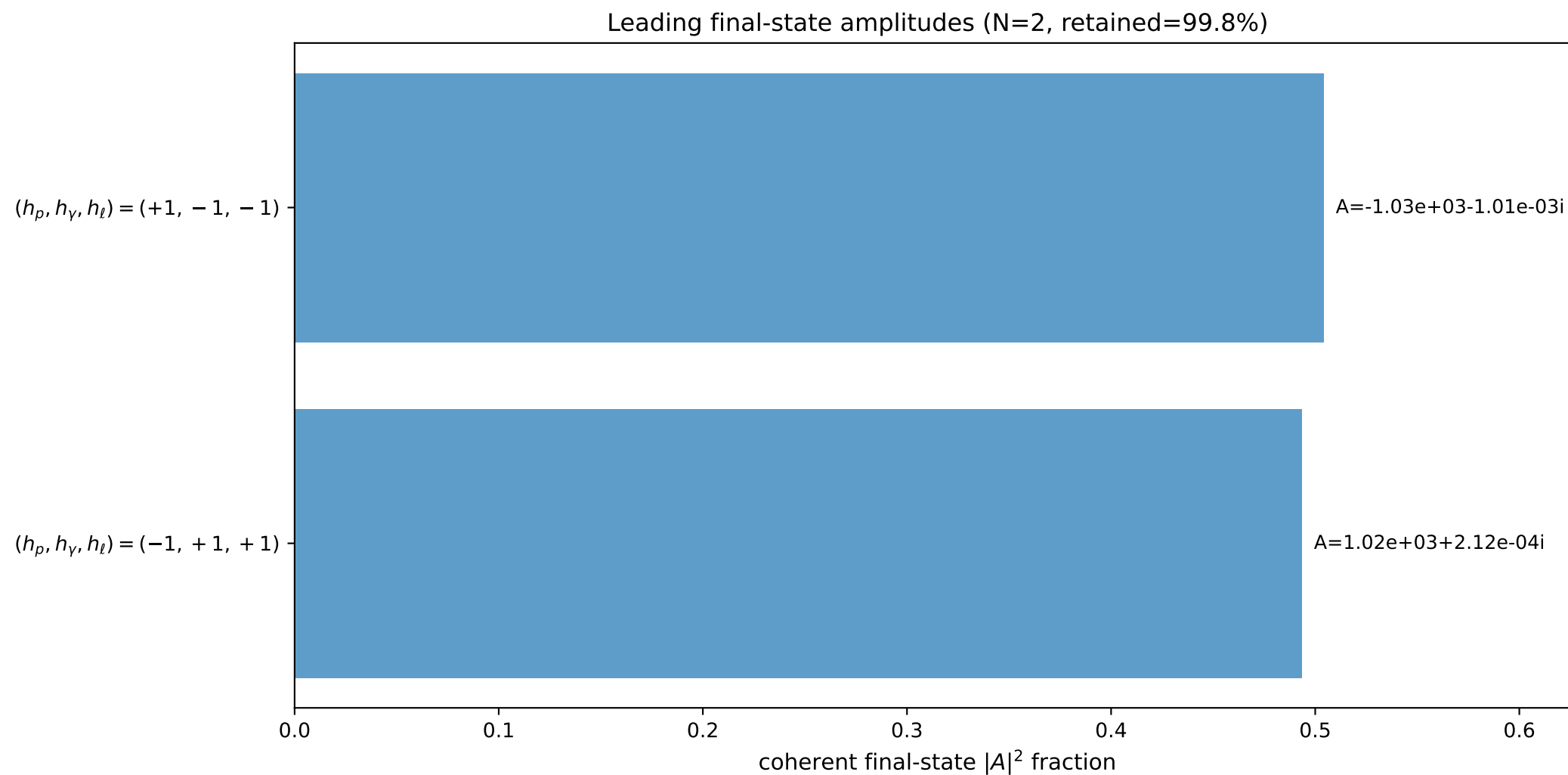
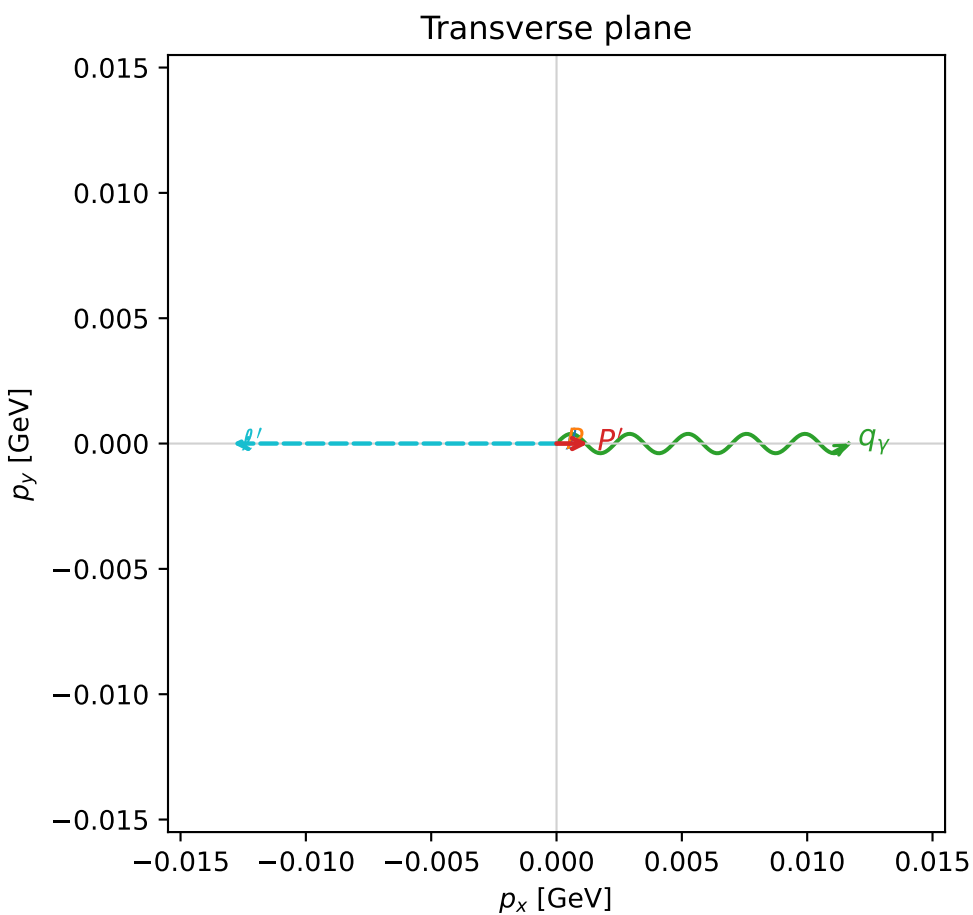
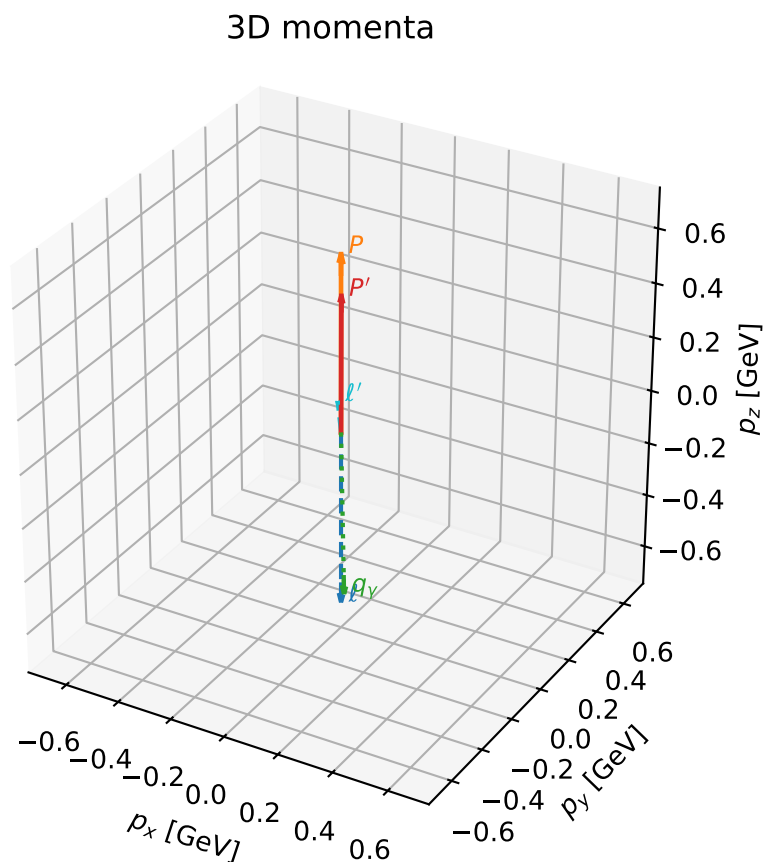
$\text{sqrt_s} = 3.796778$
 $\text{theta_p_out} = 0.002027211$
 $\text{theta_gamma_out} = 3.122016$
 $\text{q0out} = 1.719101$
 $\text{phi_p_out} = 6.280865$
 $\text{phi_gamma_out} = 3.141585$
 $\text{alpha_e} = 0.2436742$
 $\text{alpha_p} = 1.331234$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

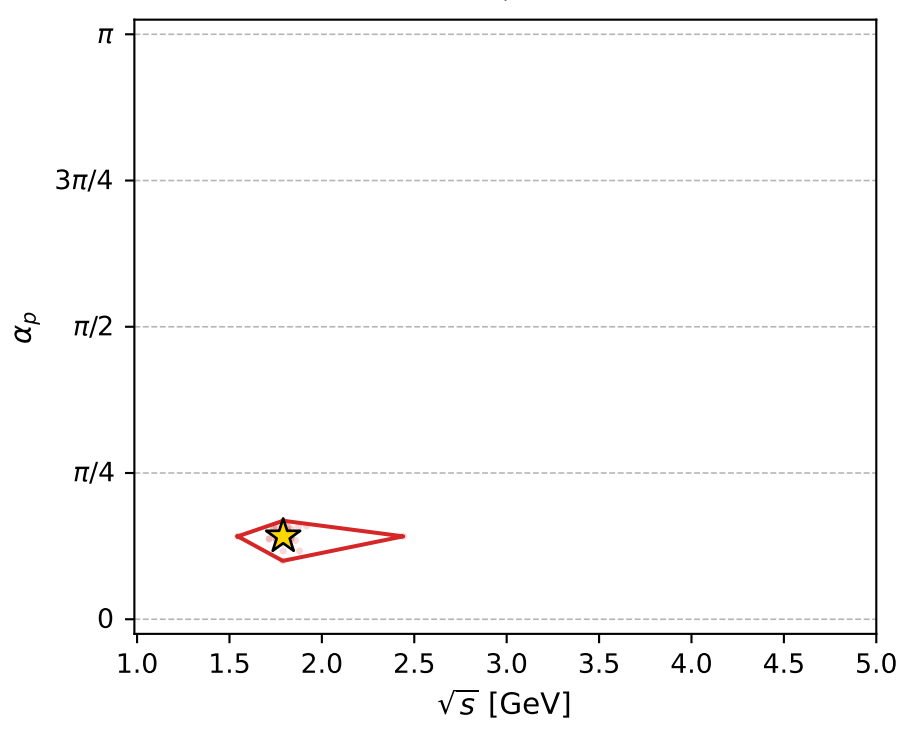
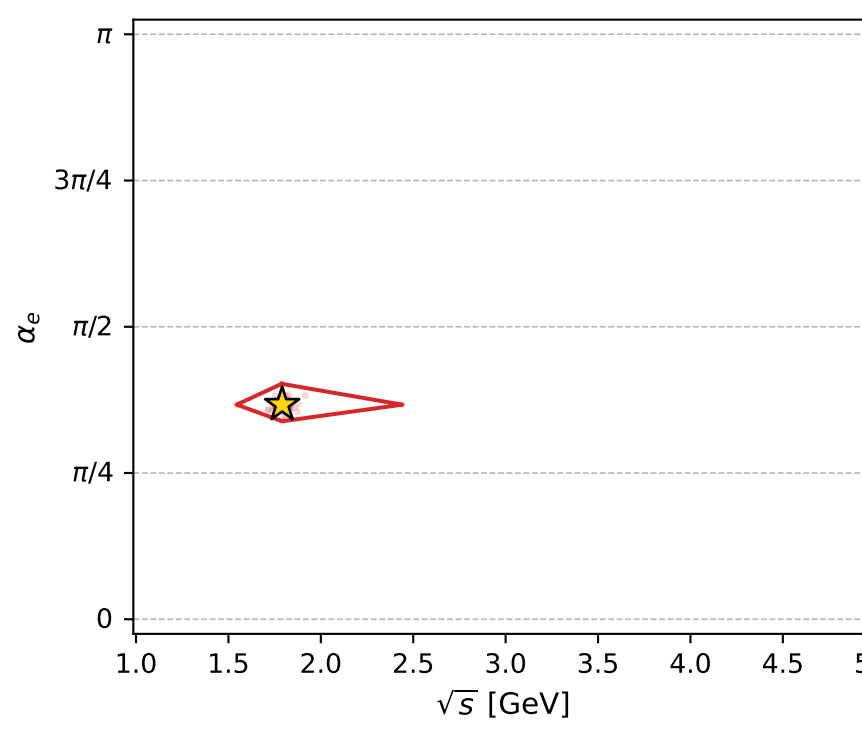
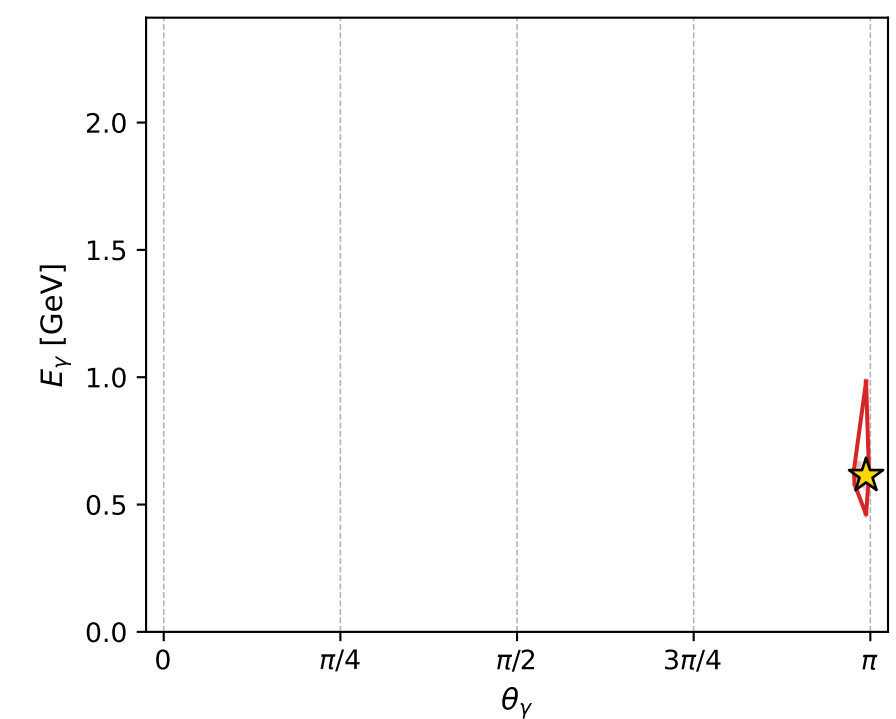
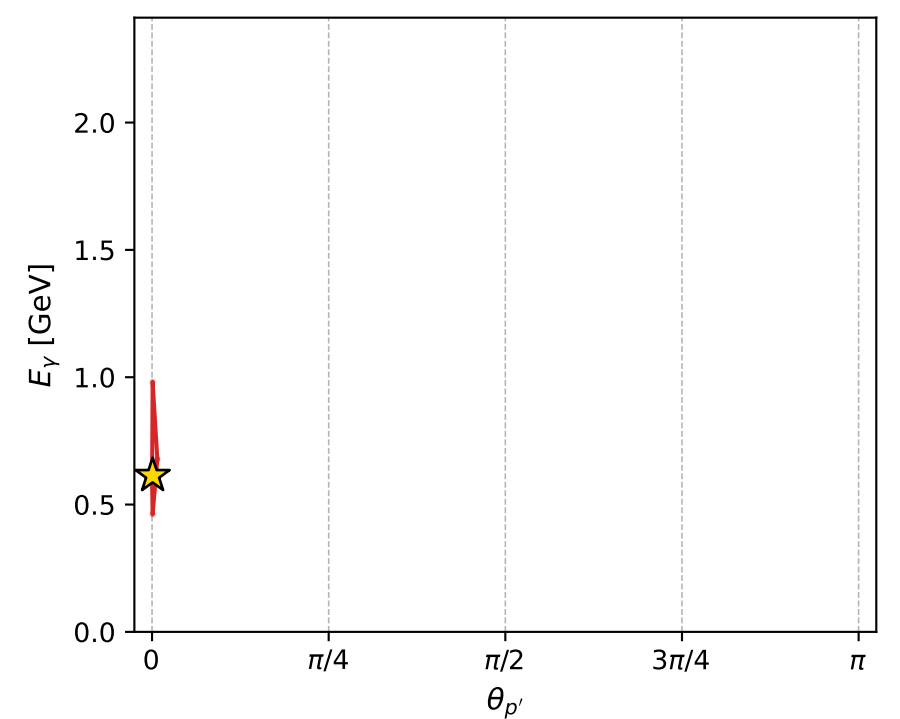
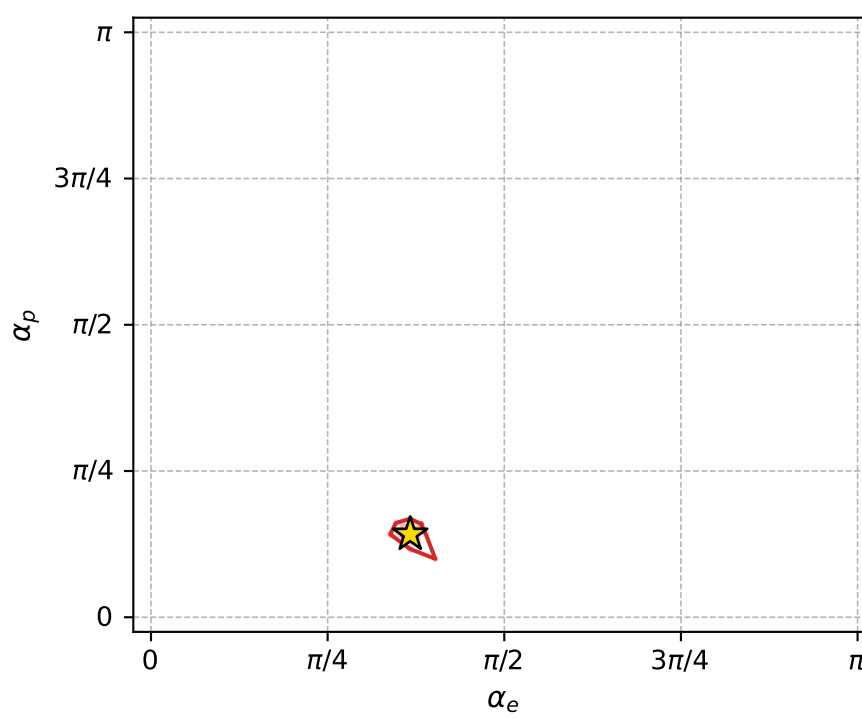
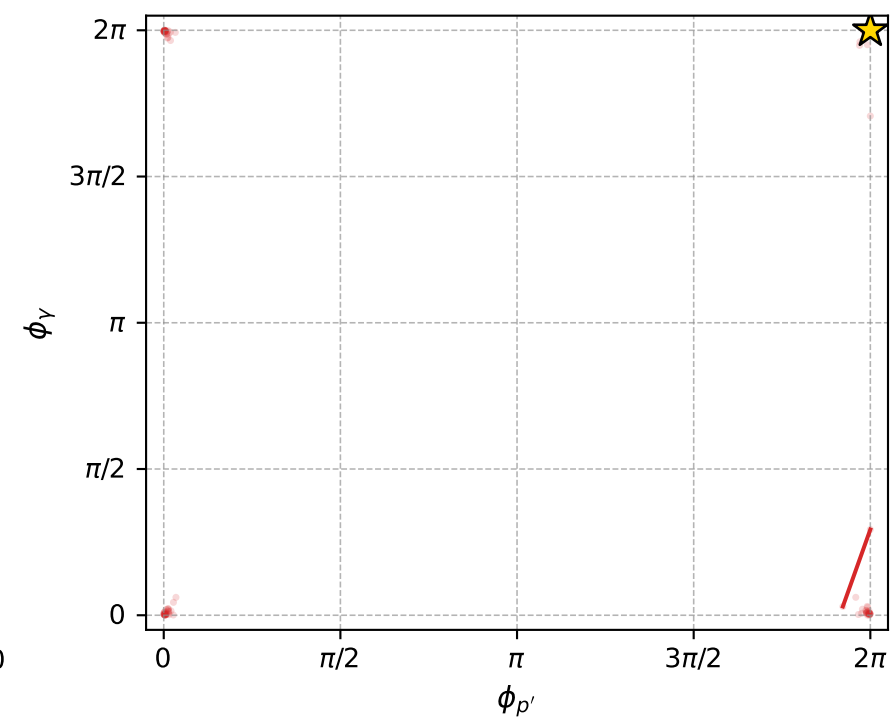
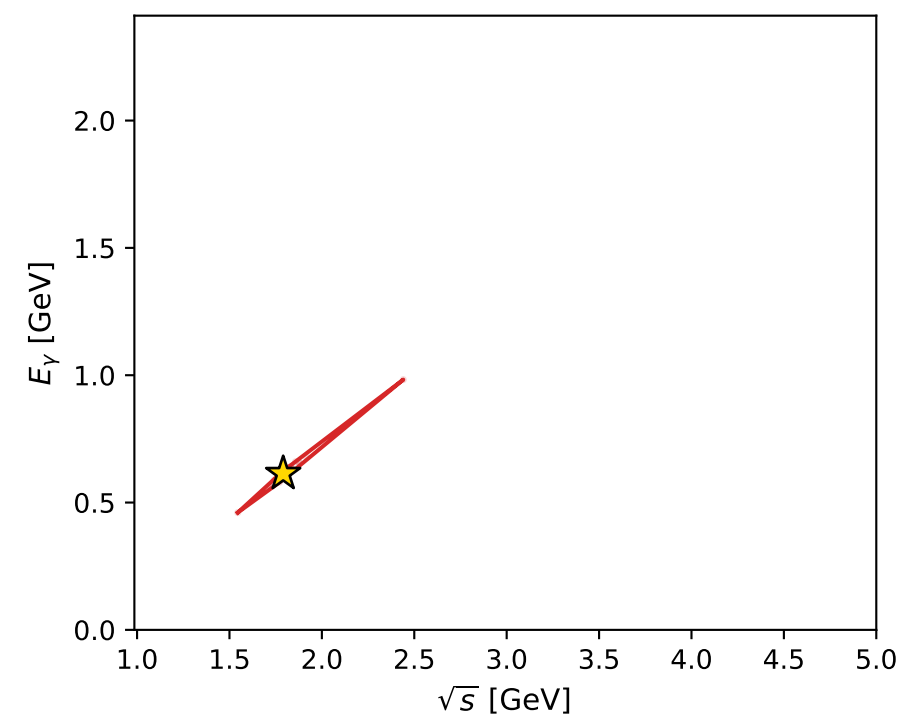
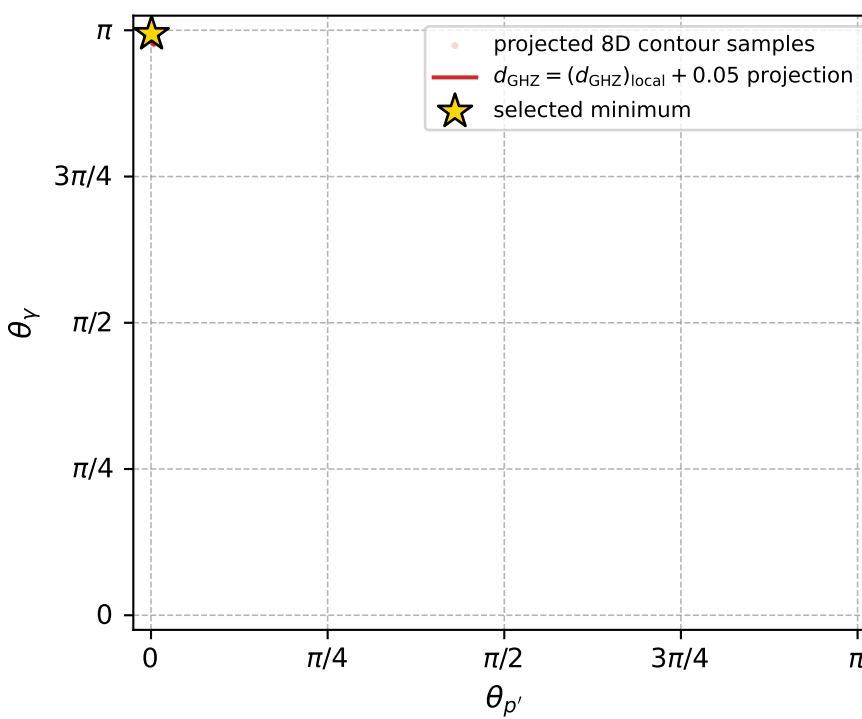
dghz_mixing_angles_polarization_01_local_minimum_249 $d_{\text{GHZ}}=0.000132871$
region: polarization_cluster_01_local_minimum_249
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.1524667$ rad
incoming lepton: $0.406235|+\rangle +0.913769|-\rangle$
proton mixing angle: $\alpha_p=0.44527808$ rad
incoming proton: $0.902491|+\rangle +0.430709|-\rangle$

$s=3.20651$, $\sqrt{s}=1.79067$, $|\vec{P}|=0.649662$, $|\vec{P}'|=0.500629$
 $\theta_{P'}=0.0025126$, $\theta_Y=3.12259$, $\phi_{P'}=6.28317$, $\phi_Y=6.28316$
 $E_Y=0.613719$, $\theta_{\ell'}=0.113834$, $\phi_{\ell'}=3.14157$
 $Q^2=0.294553$, $x_B=0.133043$, $t=-0.0161642$, $W^2=2.79925$, $y=0.951558$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6497$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6497$
 P $E= 1.1410$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6497$
 ℓ' $E= 0.1137$ $p_x= -0.0129$ $p_y= 0.0000$ $p_z= 0.1130$
 P' $E= 1.0632$ $p_x= 0.0013$ $p_y= -0.0000$ $p_z= 0.5006$
 q_Y $E= 0.6137$ $p_x= 0.0117$ $p_y= -0.0000$ $p_z= -0.6136$



electron: polarization cluster P1, local minimum 249; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00013287056$
 $d_{\text{GHZ}} \text{ contour} = 0.050132871$
 above global minimum = 0.00013284828
 8D contour samples = 140

selected phase-space point:

$\text{sqrt_s} = 1.790673$
 $\text{theta_p_out} = 0.002512602$
 $\text{theta_gamma_out} = 3.122594$
 $\text{q0out} = 0.6137187$
 $\text{phi_p_out} = 6.283166$
 $\text{phi_gamma_out} = 6.283162$
 $\text{alpha_e} = 1.152467$
 $\text{alpha_p} = 0.4452781$

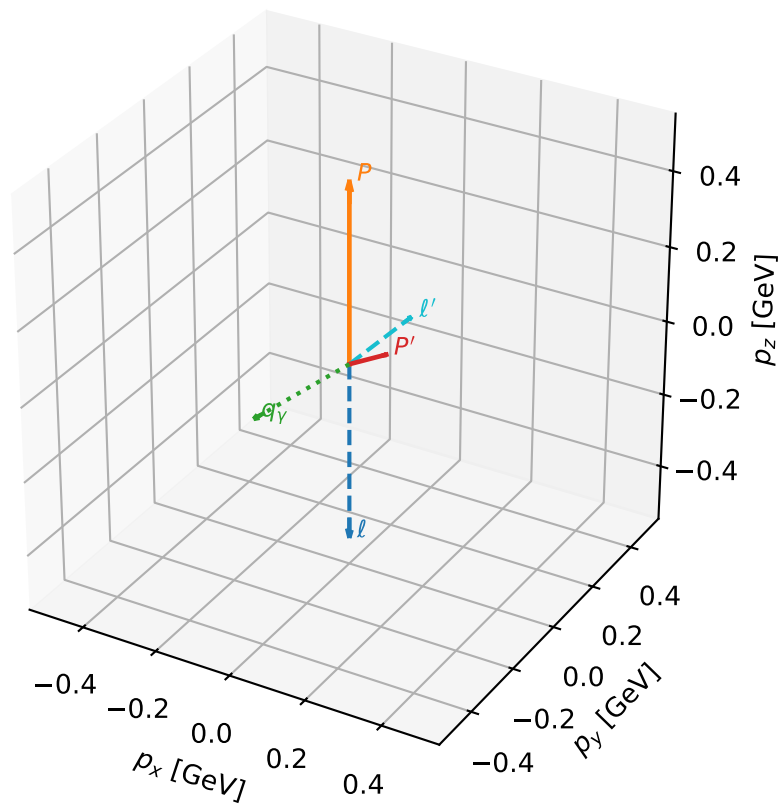
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_250 $d_{\{\text{rm GHZ}\}}=0.000133867$
 region: polarization_cluster_01_local_minimum_250
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.57284141$ rad
 incoming lepton: $0.840364|+\rangle +0.542022|-\rangle$
 proton mixing angle: $\alpha_p=1.0888305$ rad
 incoming proton: $0.463522|+\rangle +0.886085|-\rangle$

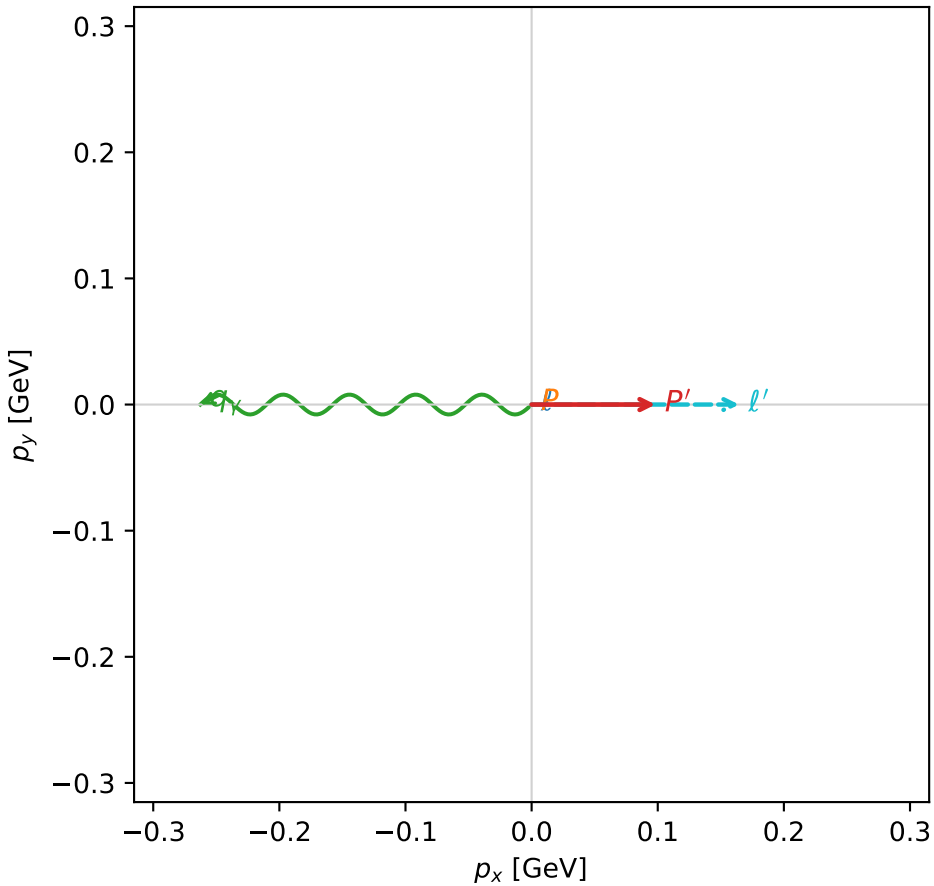
$s=2.34024$, $\sqrt{s}=1.52978$, $|\vec{P}|=0.477321$, $|\vec{P}'|=0.111952$
 $\theta_{p'}=1.06993$, $\theta_V=2.2821$, $\phi_{p'}=6.28316$, $\phi_V=3.14157$
 $E_V=0.34673$, $\theta_{\ell'}=0.761244$, $\phi_{\ell'}=6.28317$
 $Q^2=0.39235$, $x_B=0.349267$, $t=-0.177427$, $W^2=1.61085$, $y=0.769211$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4773$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4773$
 P $E= 1.0525$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4773$
 ℓ' $E= 0.2384$ $p_x= 0.1645$ $p_y= -0.0000$ $p_z= 0.1726$
 P' $E= 0.9447$ $p_x= 0.0982$ $p_y= -0.0000$ $p_z= 0.0538$
 q_V $E= 0.3467$ $p_x= -0.2627$ $p_y= 0.0000$ $p_z= -0.2264$

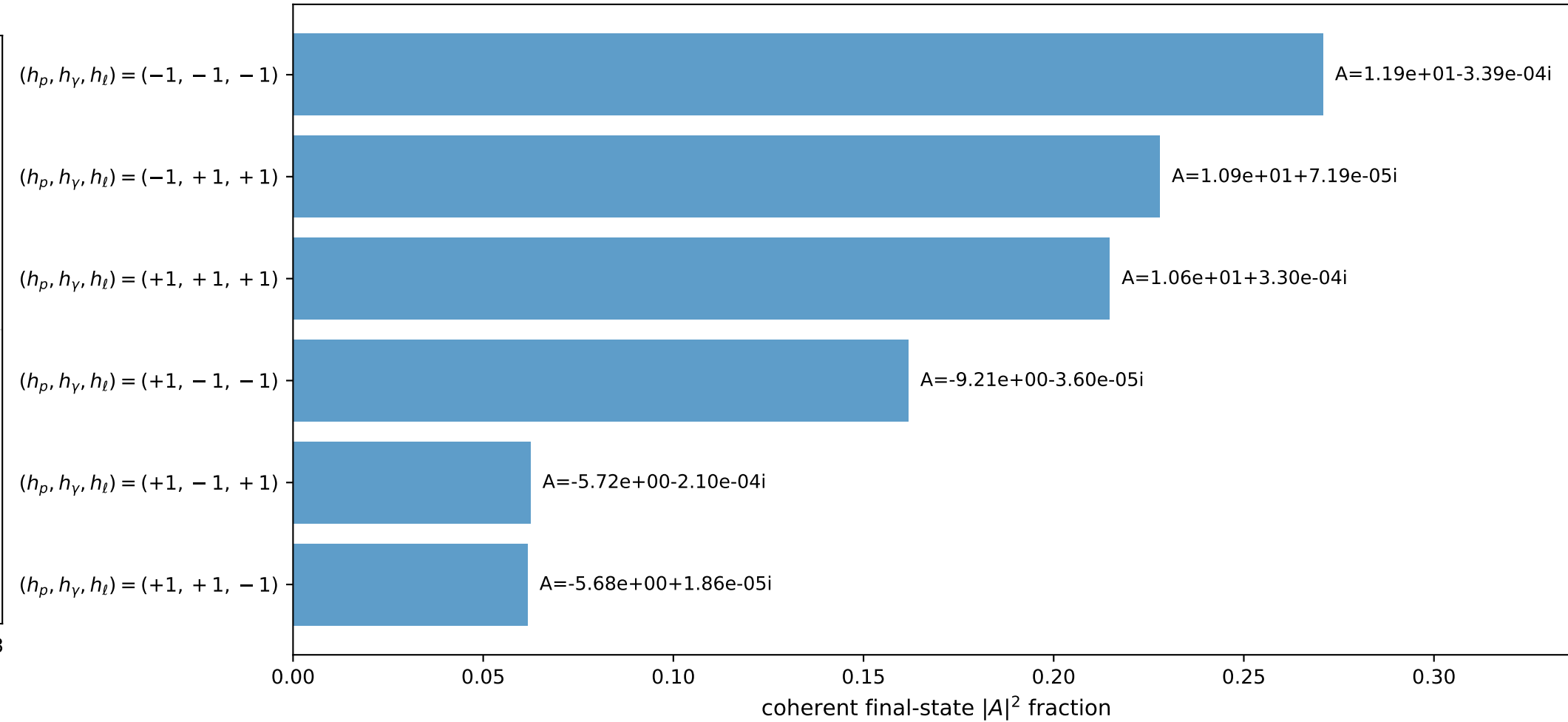
3D momenta



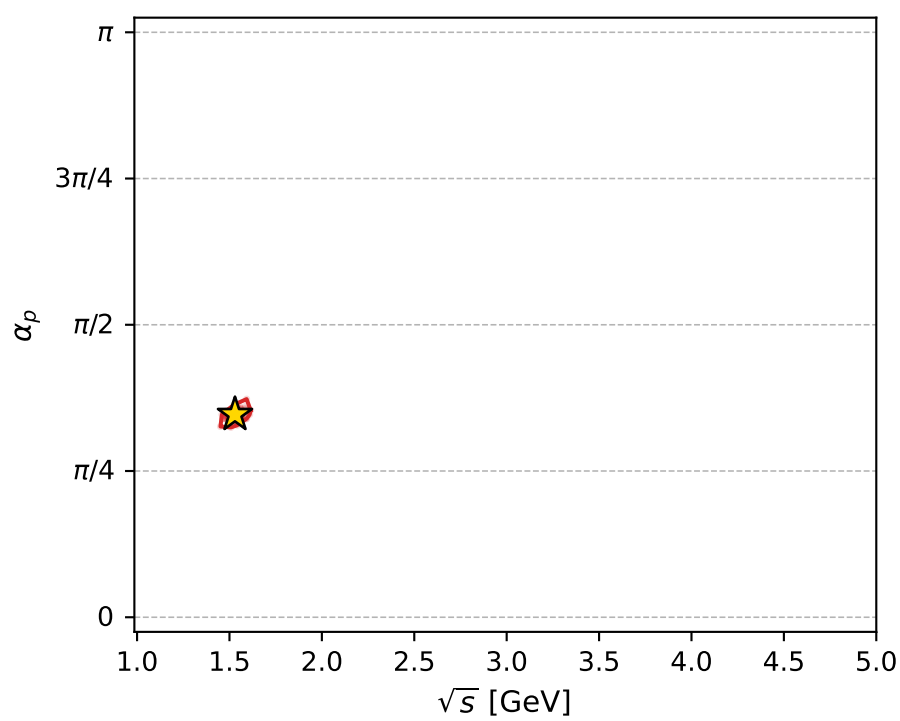
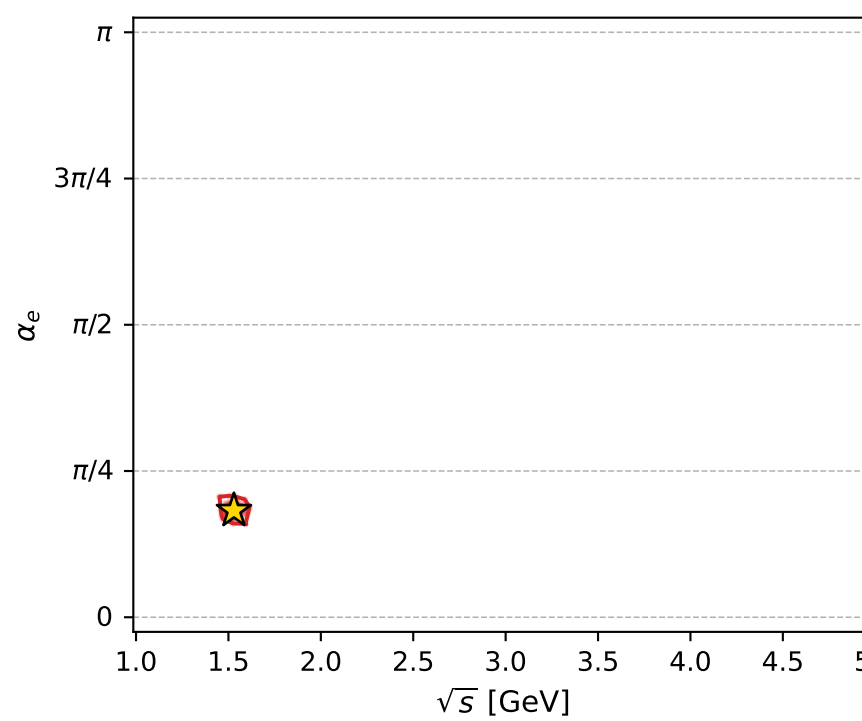
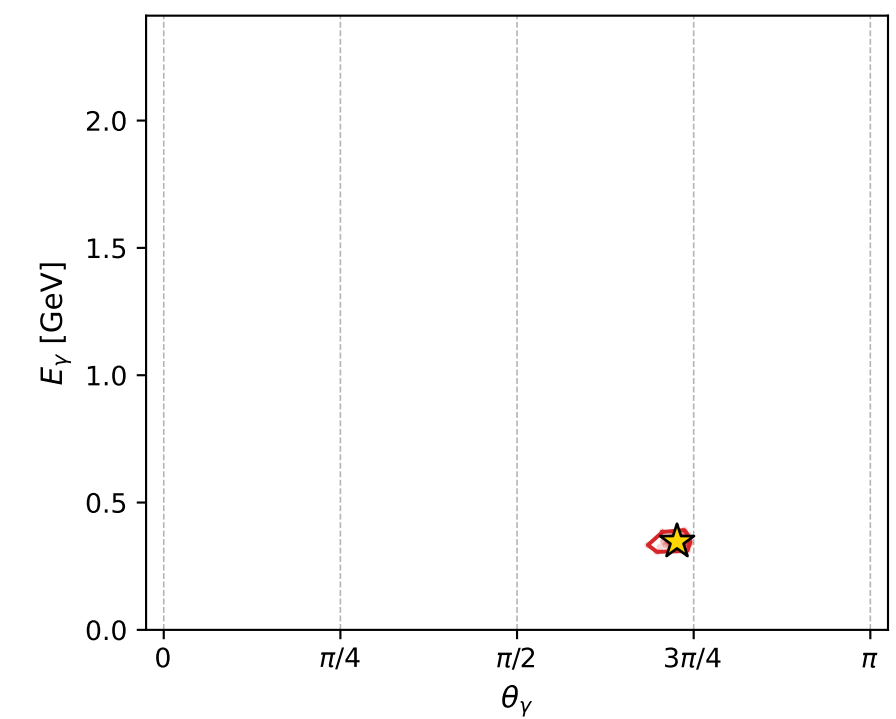
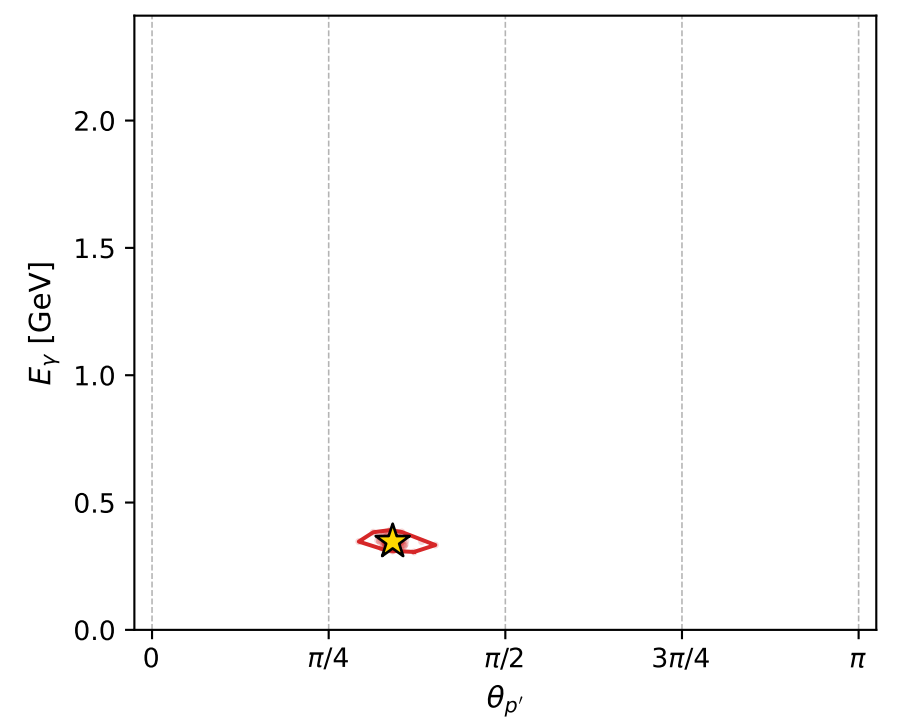
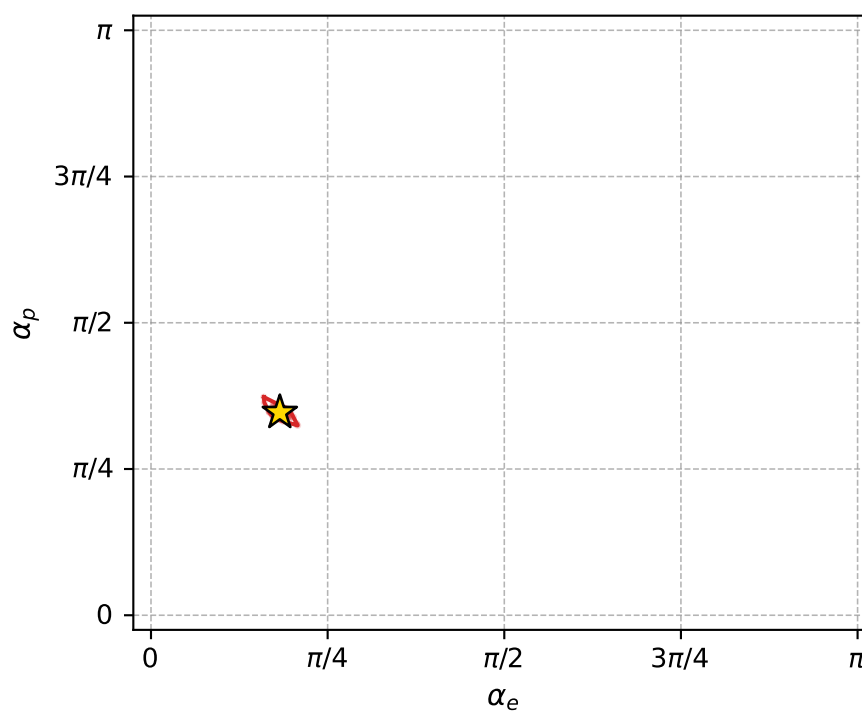
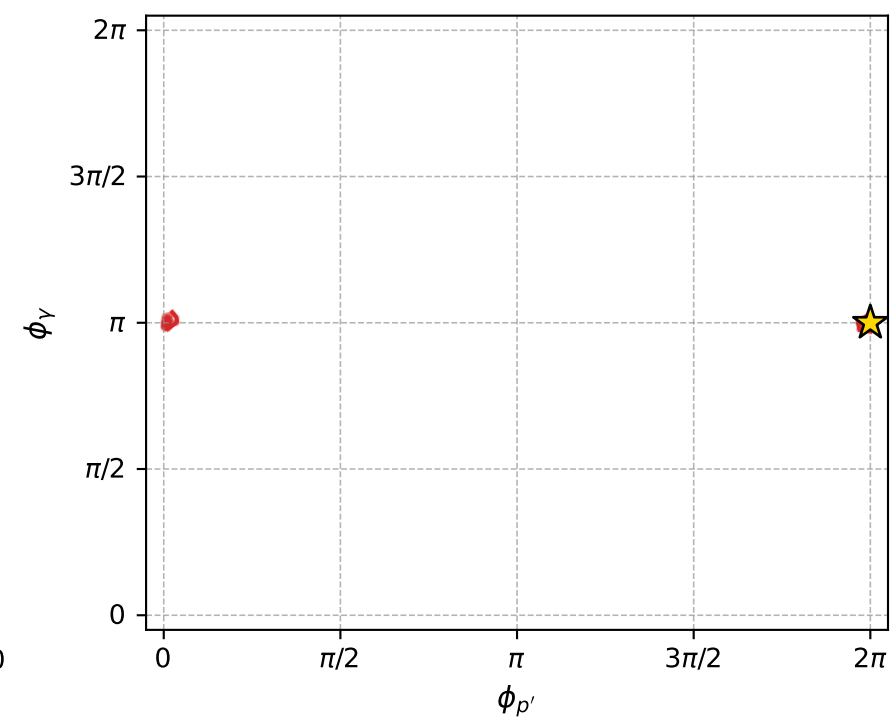
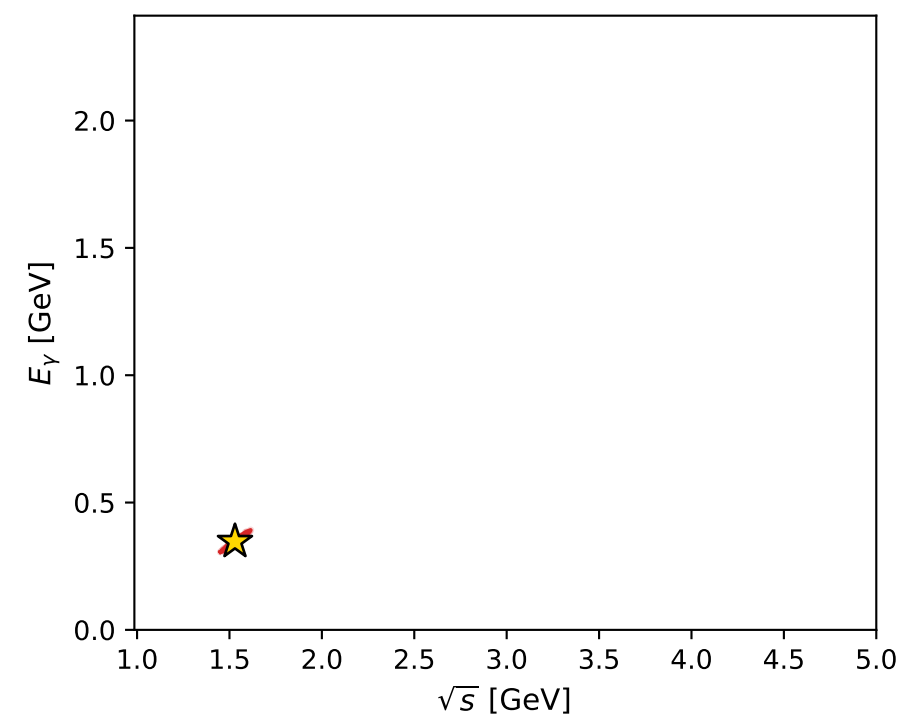
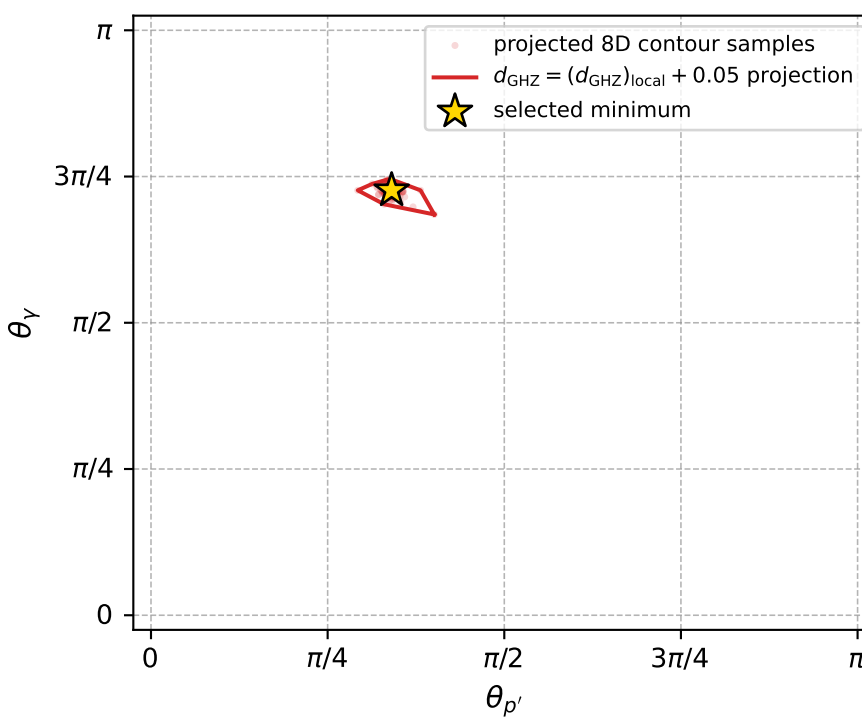
Transverse plane



Leading final-state amplitudes (N=6, retained=99.9%)



electron: polarization cluster P1, local minimum 250; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0001338669$
 $d_{\text{GHZ}} \text{ contour} = 0.050133867$
 above global minimum = 0.00013384462
 8D contour samples = 256

selected phase-space point:

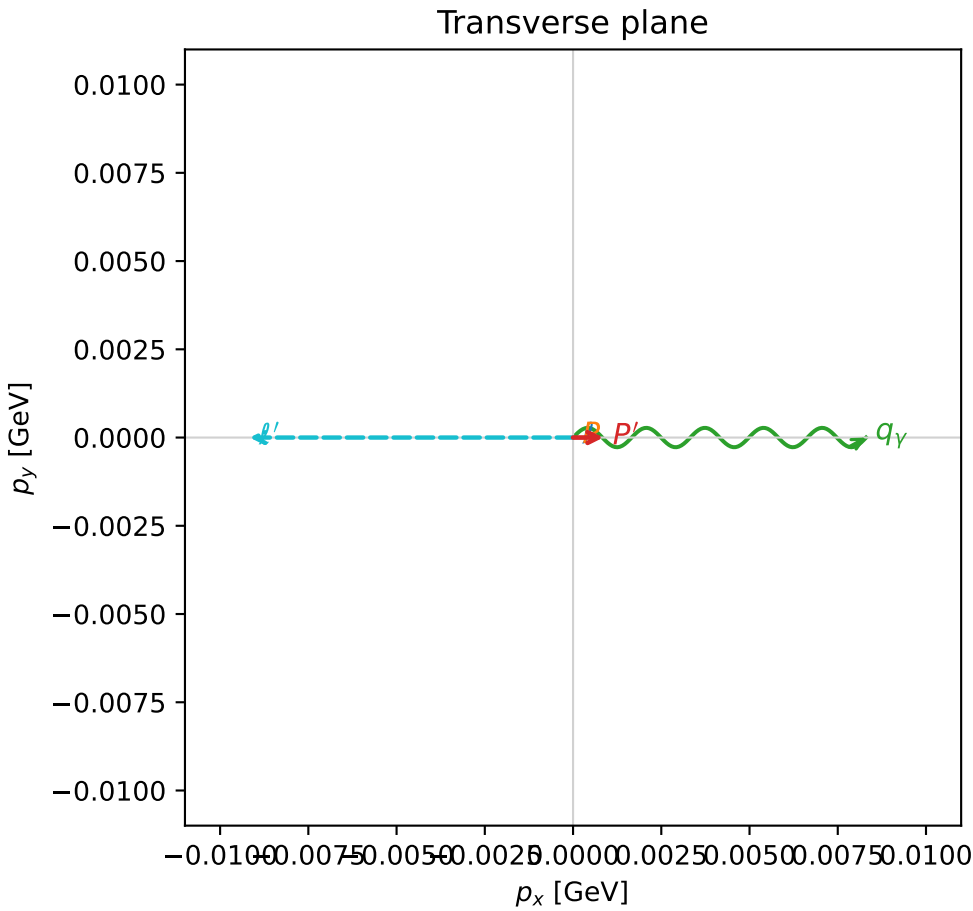
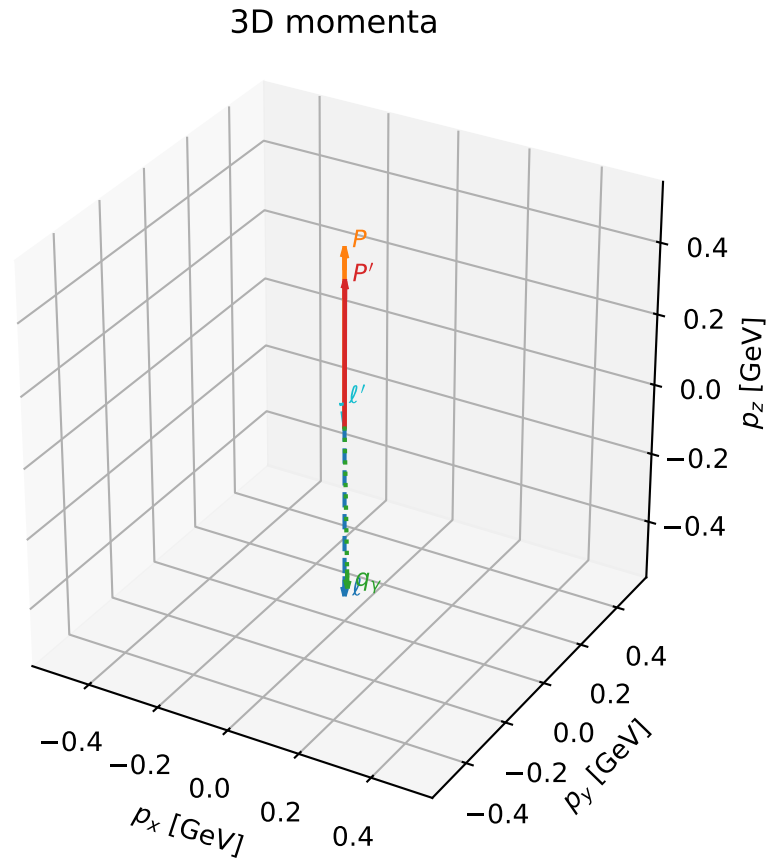
$\text{sqrt_s} = 1.529785$
 $\text{theta_p_out} = 1.069933$
 $\text{theta_gamma_out} = 2.282097$
 $\text{q0ut} = 0.3467296$
 $\text{phi_p_out} = 6.283155$
 $\text{phi_gamma_out} = 3.141572$
 $\text{alpha_e} = 0.5728414$
 $\text{alpha_p} = 1.08883$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_253 $d_{\text{GHZ}}=0.000141867$
region: polarization_cluster_01_local_minimum_253
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.0645582$ rad
incoming lepton: $0.484891|+\rangle +0.874575|-\rangle$
proton mixing angle: $\alpha_p=0.55250545$ rad
incoming proton: $0.851212|+\rangle +0.524822|-\rangle$

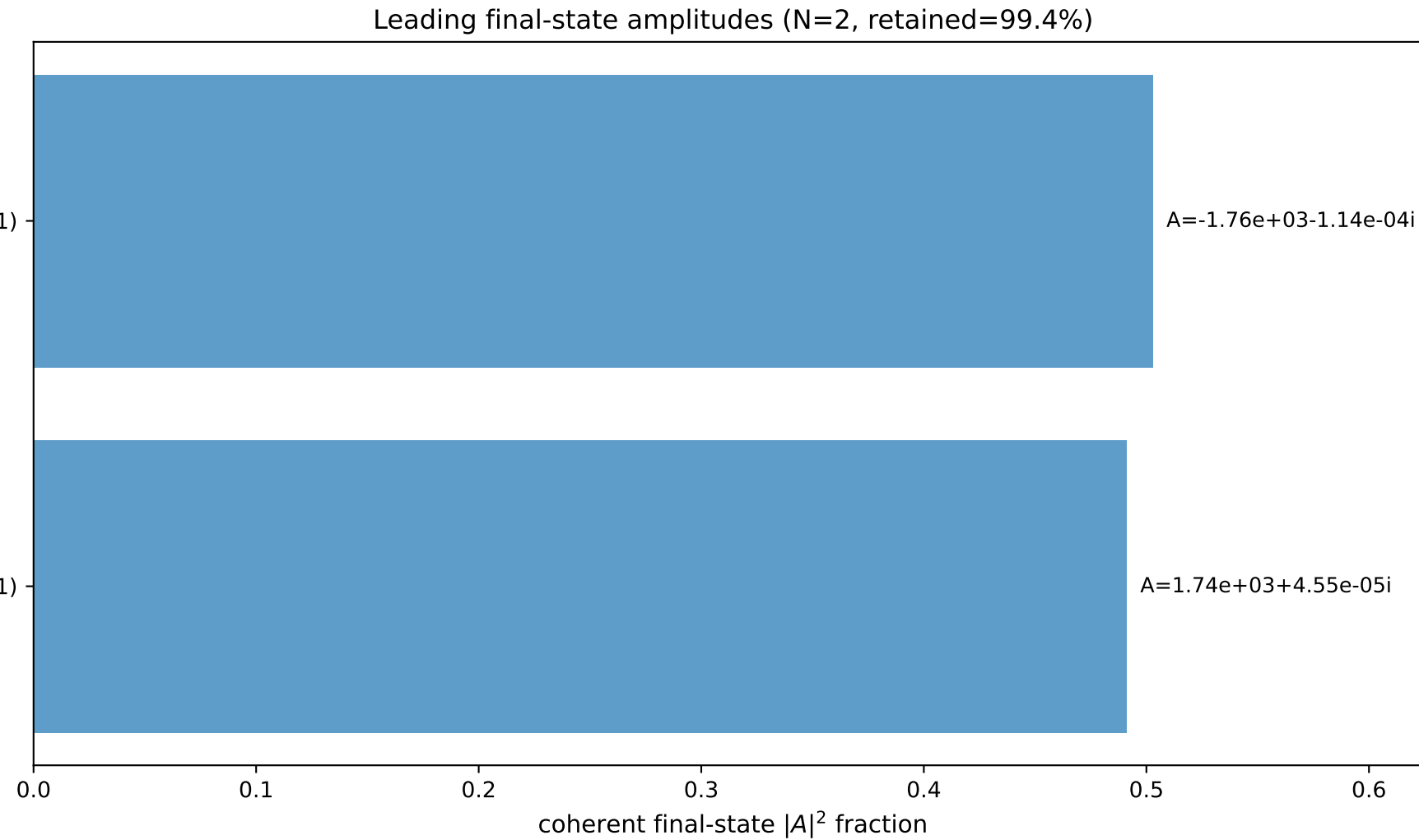
$s=2.40938$, $\sqrt{s}=1.55222$, $|\vec{P}|=0.492693$, $|\vec{P}'|=0.403492$
 $\theta_{P'}=0.00213162$, $\theta_Y=3.12381$, $\phi_{P'}=6.28318$, $\phi_Y=2.67404e-05$
 $E_Y=0.46701$, $\theta_{\ell'}=0.143434$, $\phi_{\ell'}=3.14162$
 $Q^2=0.125686$, $x_B=0.0863101$, $t=-0.00648154$, $W^2=2.21037$, $y=0.952061$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4927$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4927$
 P $E= 1.0595$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4927$
 ℓ' $E= 0.0641$ $p_x= -0.0092$ $p_y= -0.0000$ $p_z= 0.0634$
 P' $E= 1.0211$ $p_x= 0.0009$ $p_y= -0.0000$ $p_z= 0.4035$
 q_Y $E= 0.4670$ $p_x= 0.0083$ $p_y= 0.0000$ $p_z= -0.4669$

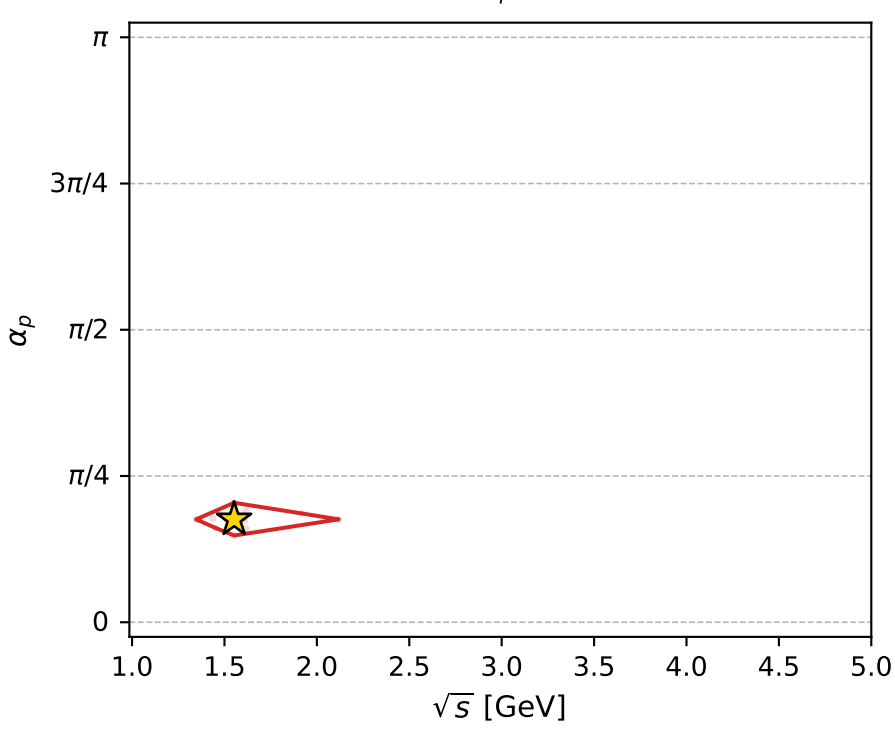
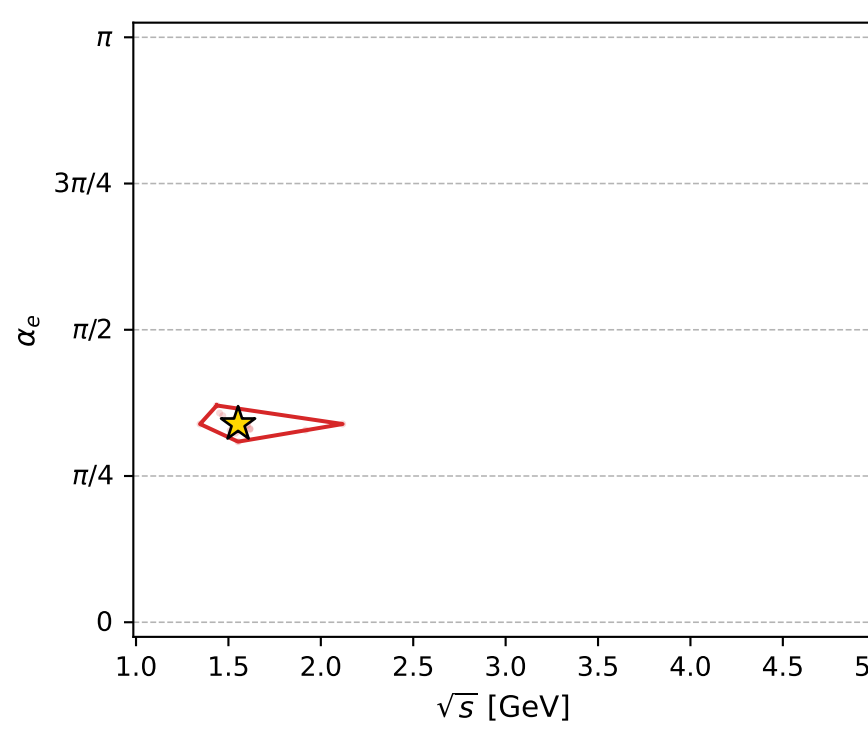
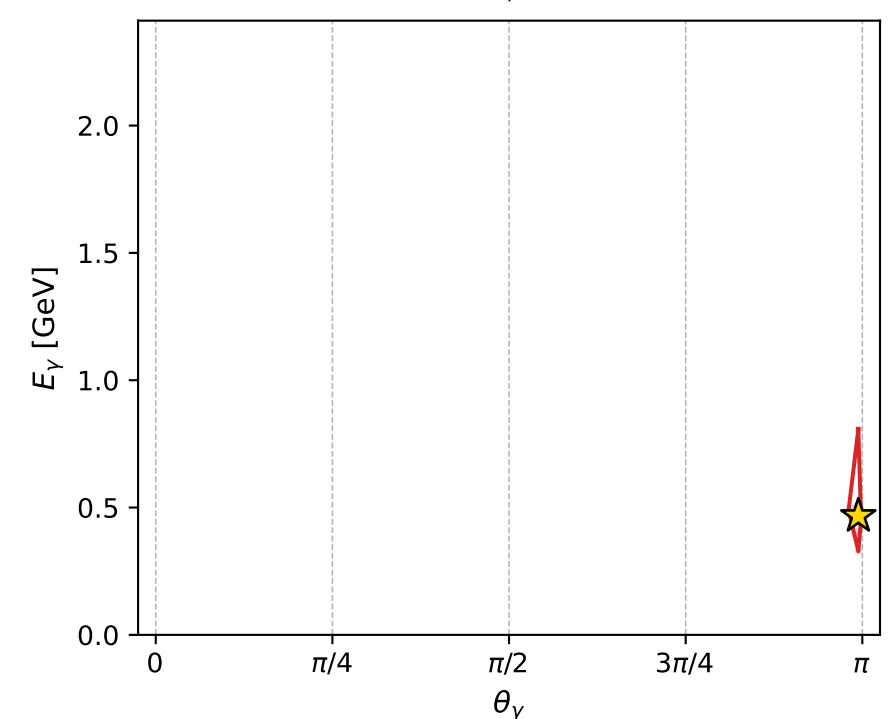
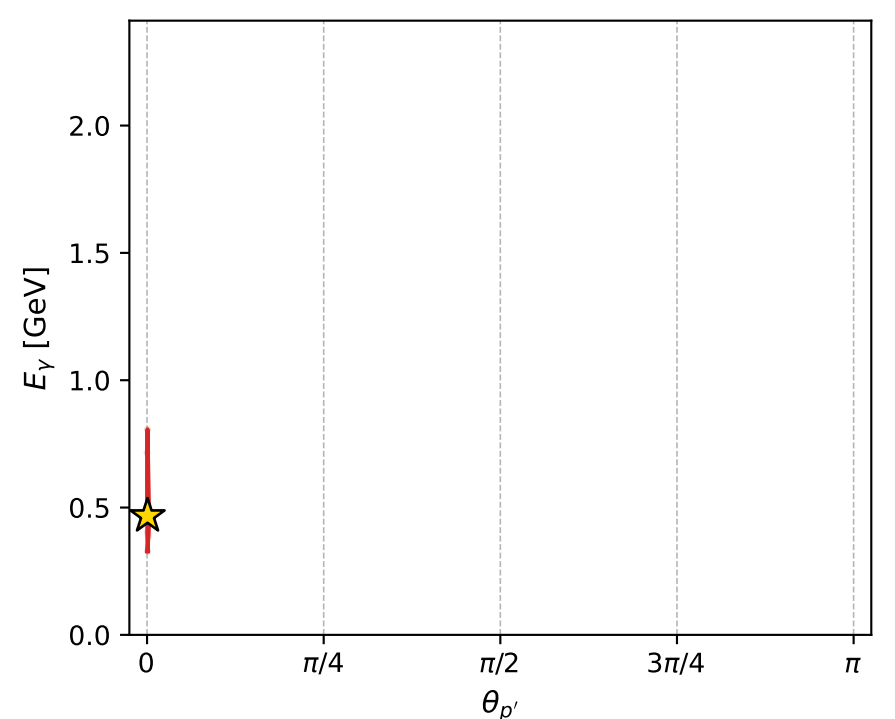
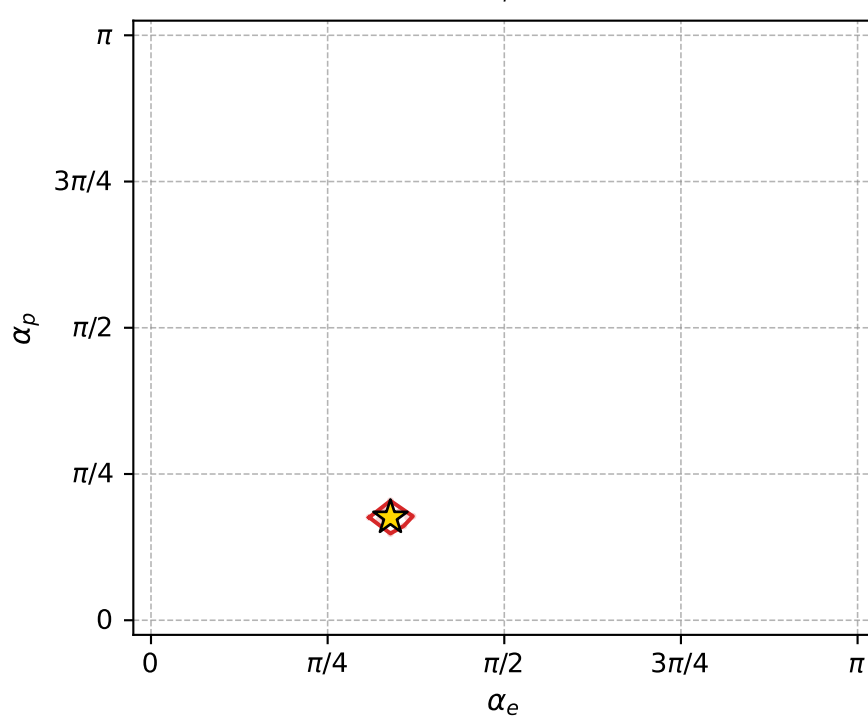
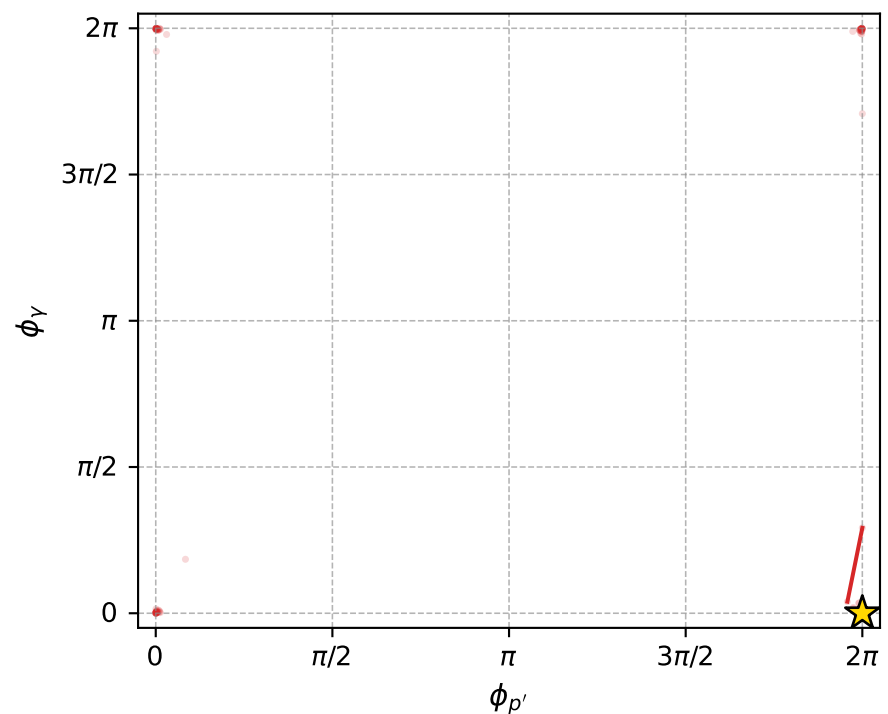
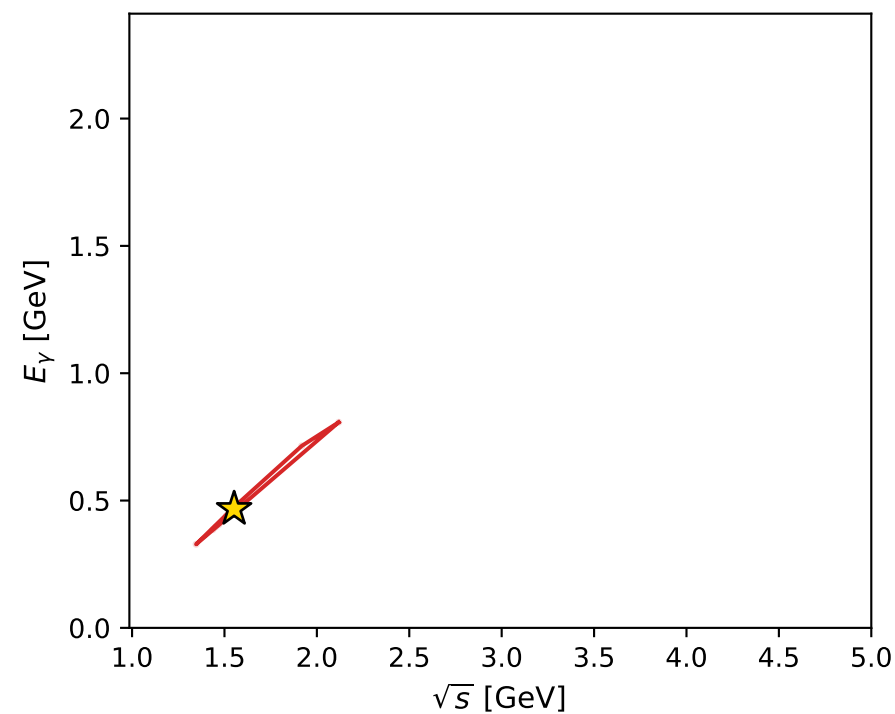
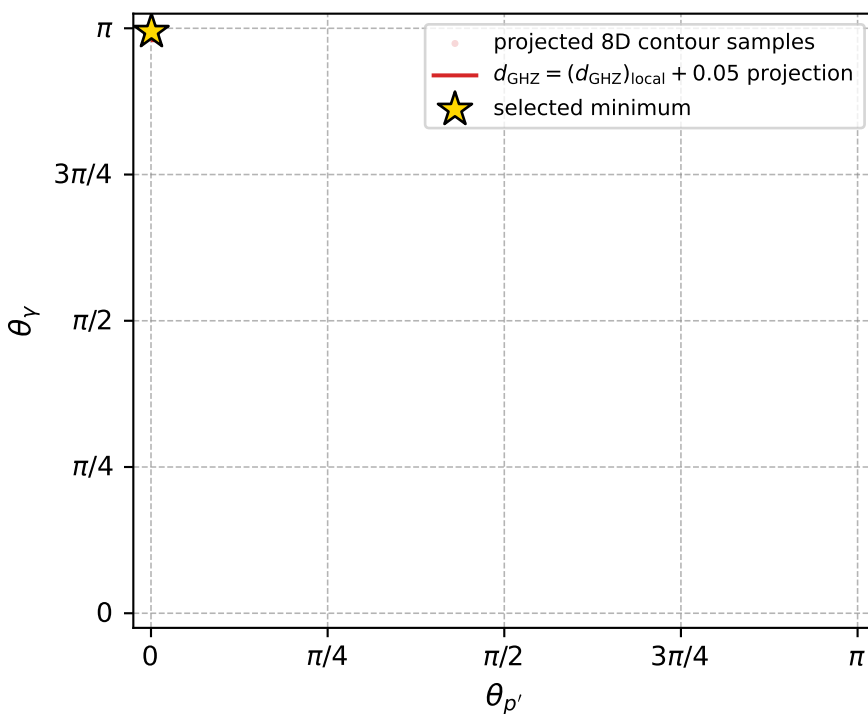


$(h_p, h_Y, h_{\ell}) = (+1, -1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, +1)$



electron: polarization cluster P1, local minimum 253; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00014186705$
 $d_{\text{GHZ}} \text{ contour} = 0.050141867$
 above global minimum = 0.00014184476
 8D contour samples = 145

selected phase-space point:

$\text{sqrt_s} = 1.552217$
 $\text{theta_p_out} = 0.002131624$
 $\text{theta_gamma_out} = 3.123813$
 $\text{q0ut} = 0.4670097$
 $\text{phi_p_out} = 6.283184$
 $\text{phi_gamma_out} = 2.674042\text{e-}05$
 $\text{alpha_e} = 1.064558$
 $\text{alpha_p} = 0.5525054$

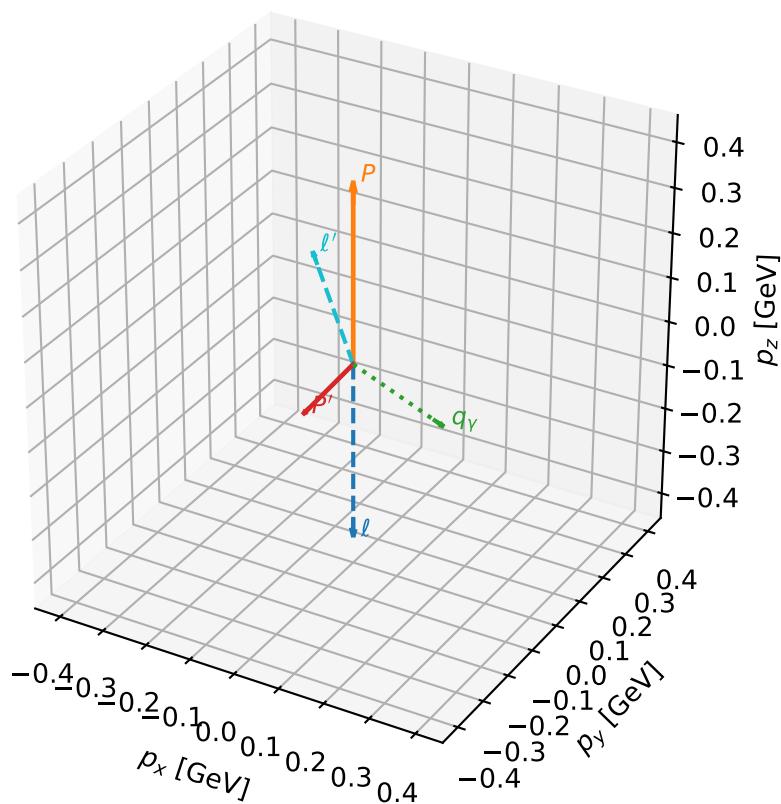
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_260 d_{\rm GHZ}=0.000169426
 region: polarization_cluster_01_local_minimum_260
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.3199313$ rad
 incoming lepton: $0.248242|+\rangle +0.968698|-\rangle$
 proton mixing angle: $\alpha_p=0.20111088$ rad
 incoming proton: $0.979845|+\rangle +0.199758|-\rangle$

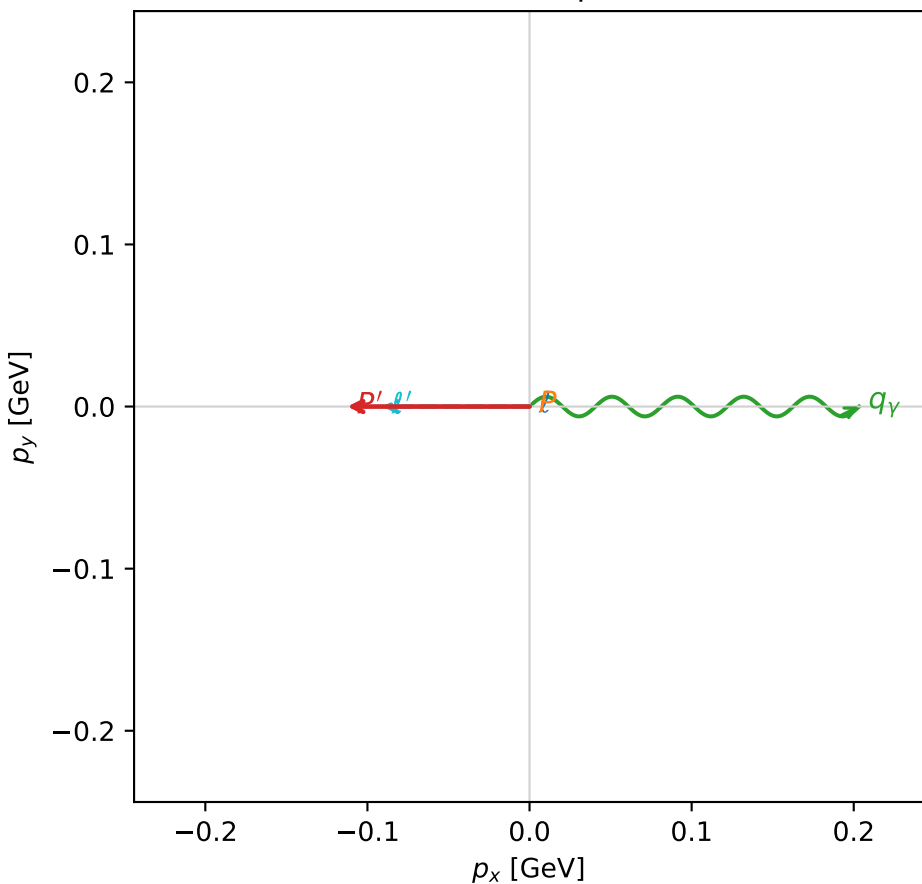
$s=2.01163$, $\sqrt{s}=1.41832$, $|\vec{P}|=0.398987$, $|\vec{P}'|=0.183112$
 $\theta_{P'}=2.48057$, $\theta_Y=1.95196$, $\phi_{P'}=3.14161$, $\phi_Y=1.10887\text{e-}05$
 $E_Y=0.219011$, $\theta_{\ell'}=0.382303$, $\phi_{\ell'}=3.14159$
 $Q^2=0.37474$, $x_B=0.459514$, $t=-0.304013$, $W^2=1.32062$, $y=0.720559$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3990$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3990$
 P $E= 1.0193$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3990$
 ℓ' $E= 0.2436$ $p_x= -0.0909$ $p_y= -0.0000$ $p_z= 0.2260$
 P' $E= 0.9557$ $p_x= -0.1124$ $p_y= -0.0000$ $p_z= -0.1445$
 q_Y $E= 0.2190$ $p_x= 0.2033$ $p_y= 0.0000$ $p_z= -0.0815$

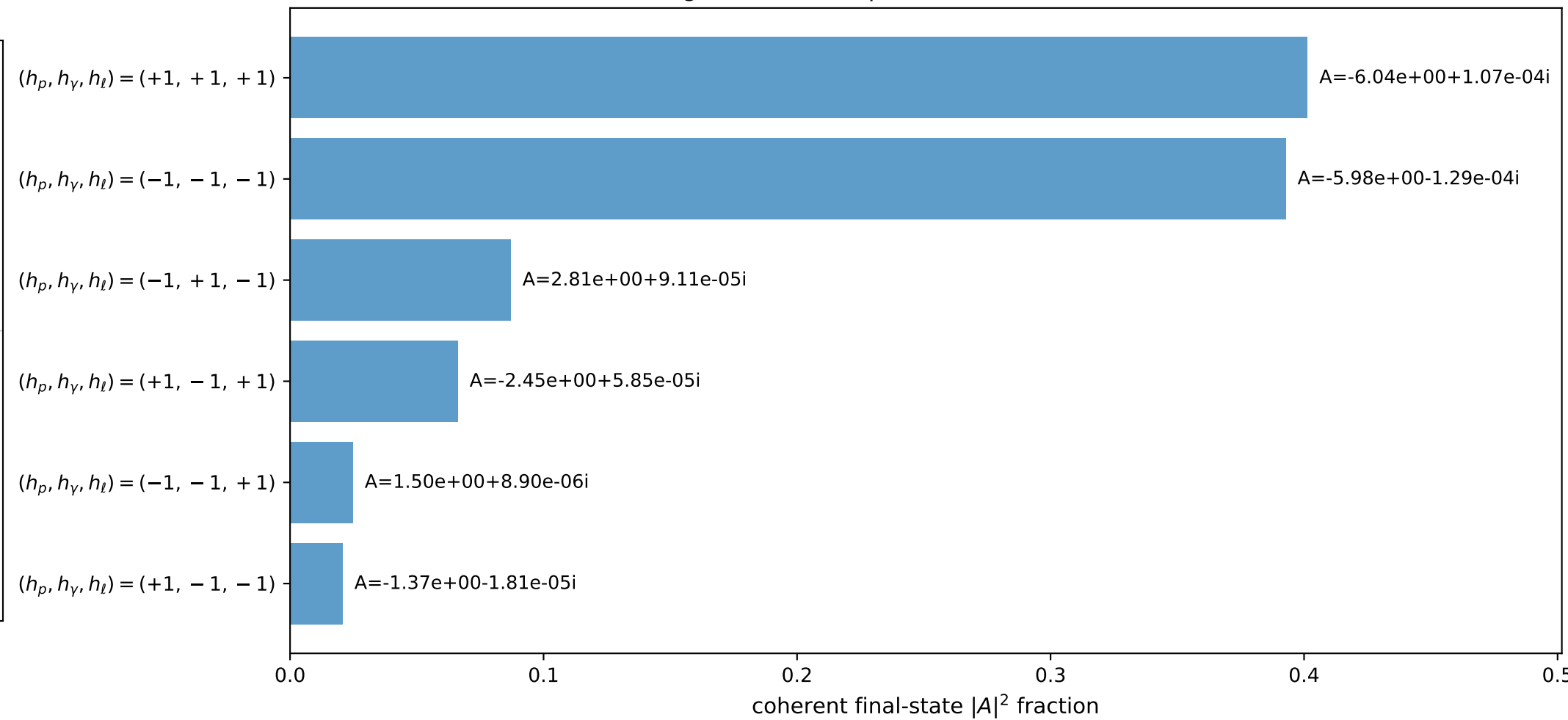
3D momenta



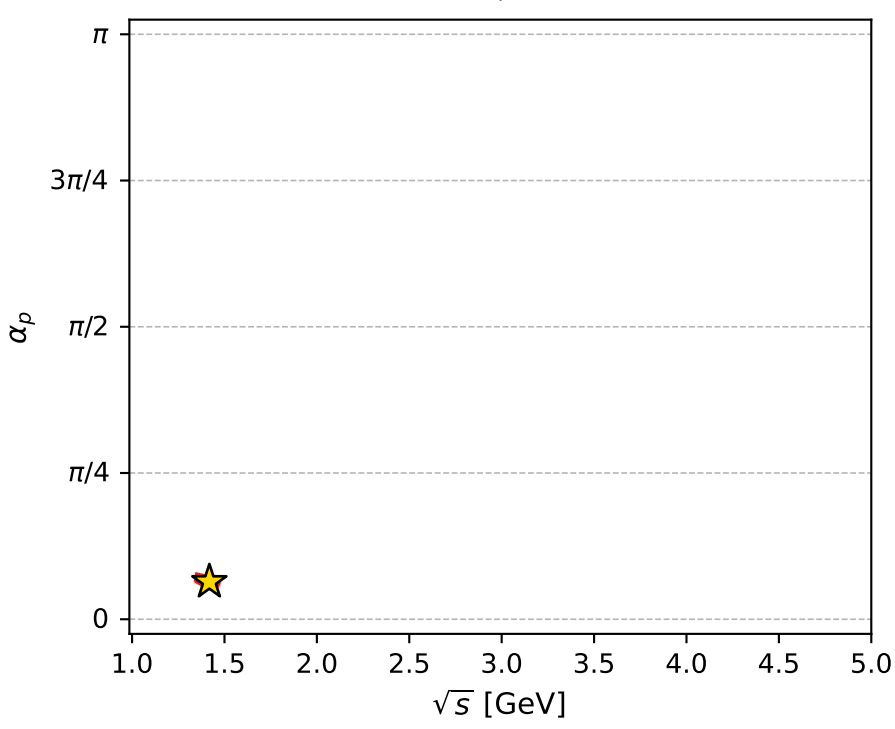
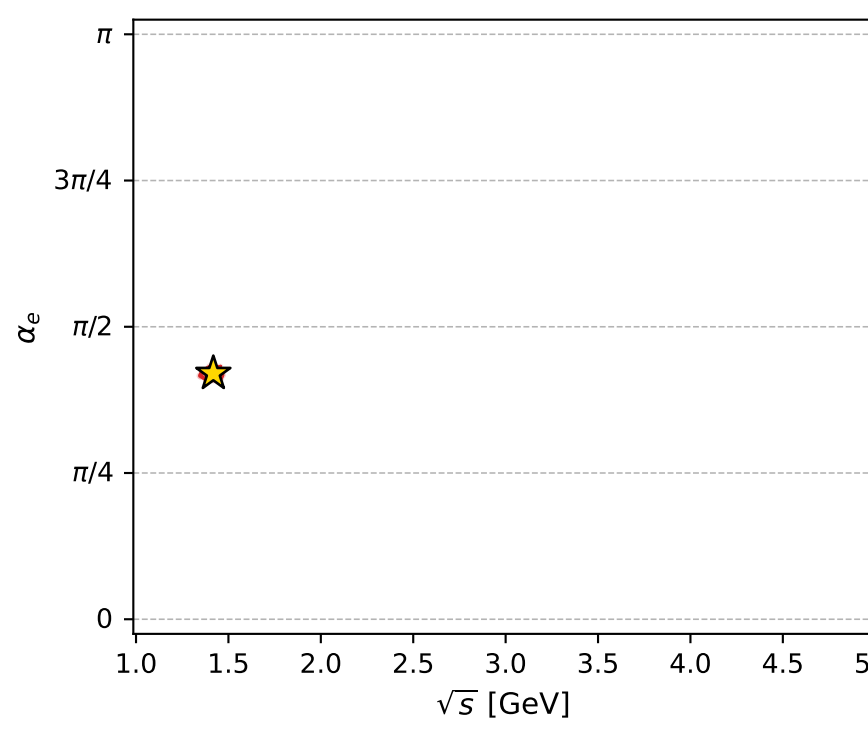
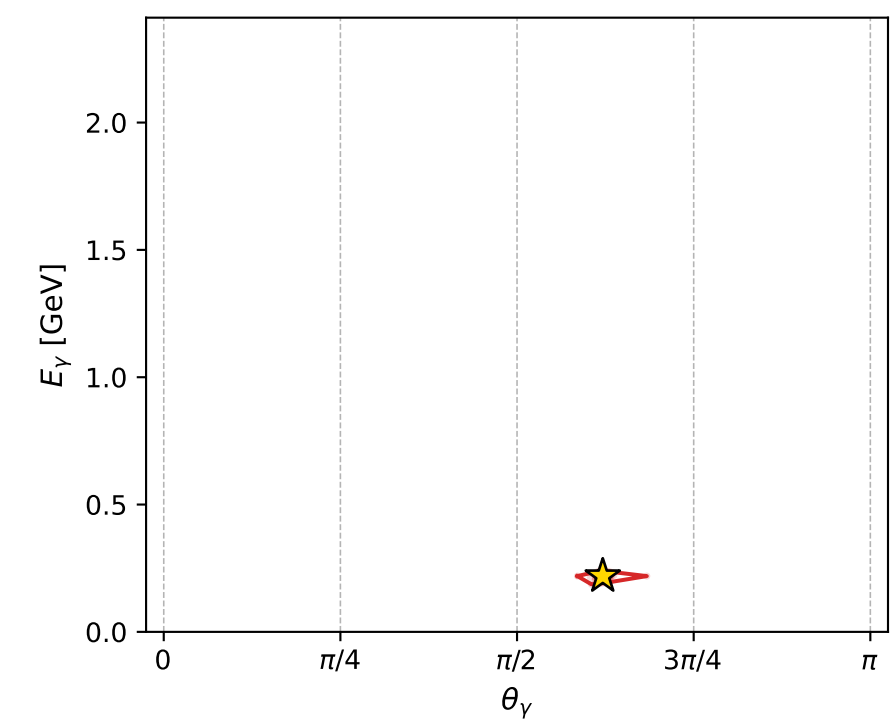
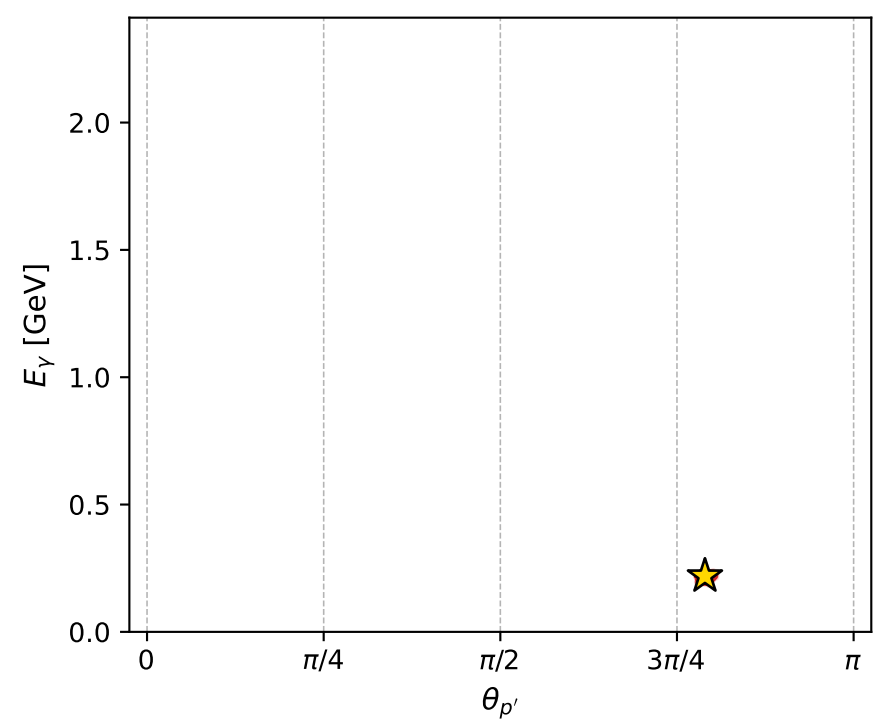
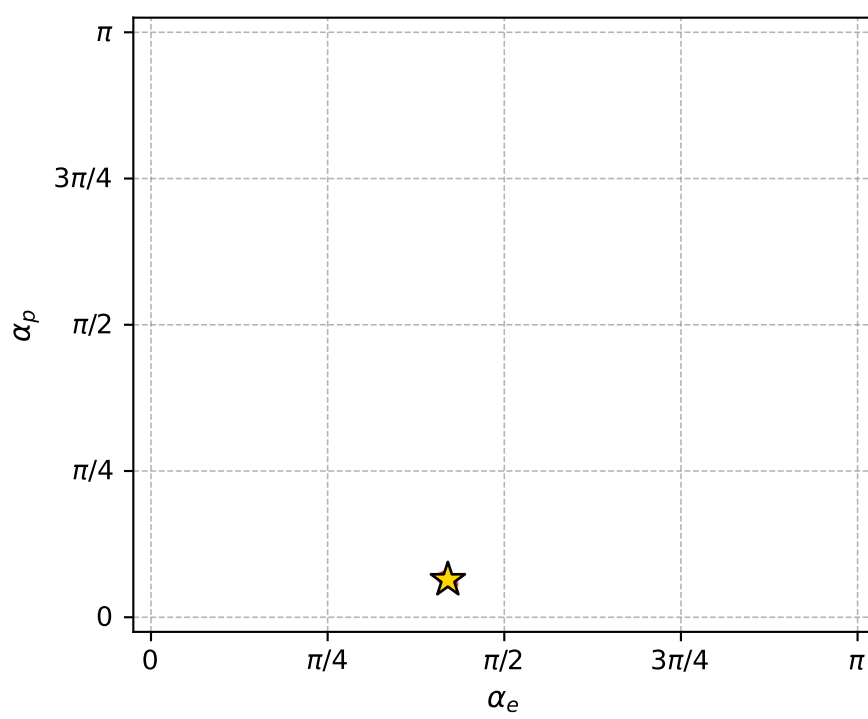
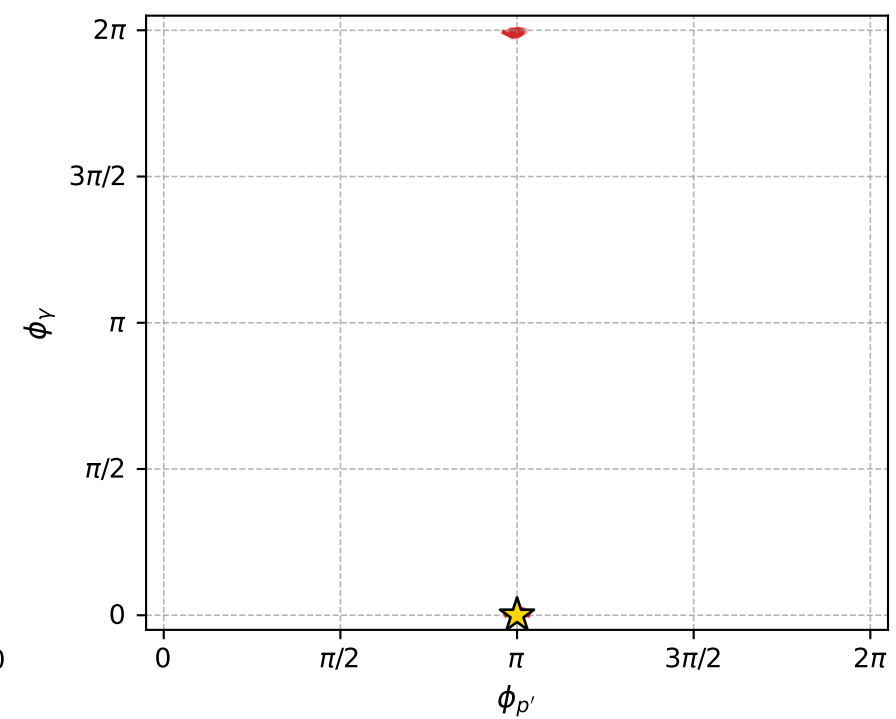
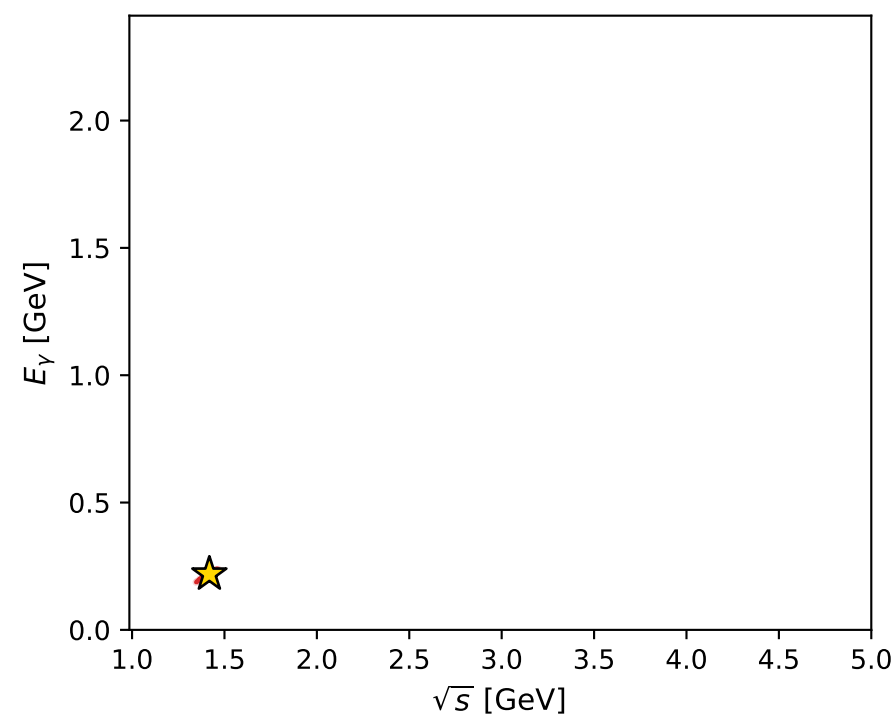
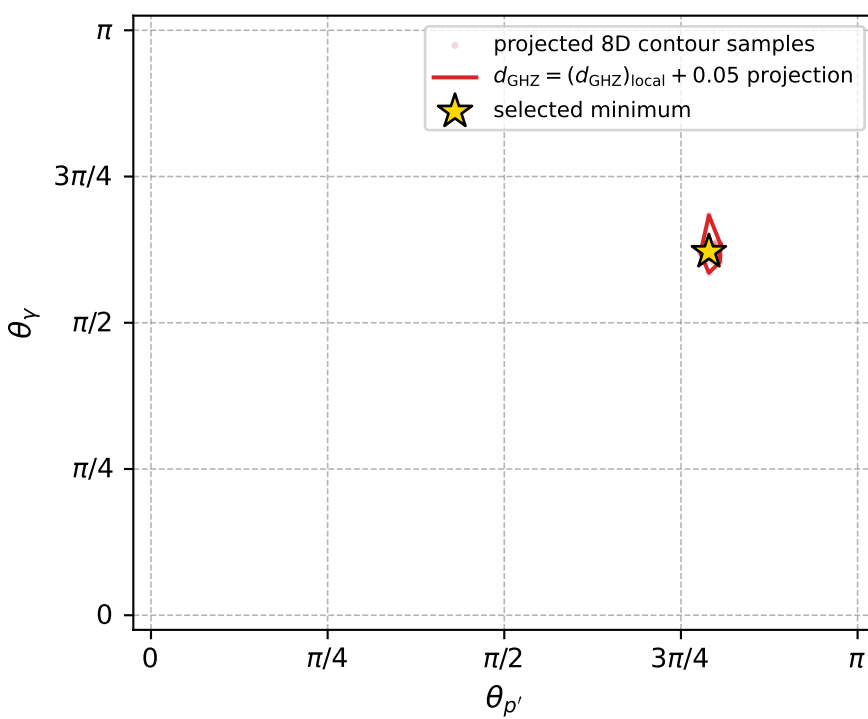
Transverse plane



Leading final-state amplitudes (N=6, retained=99.3%)



electron: polarization cluster P1, local minimum 260; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00016942645$
 $d_{\text{GHZ}} \text{ contour} = 0.050169426$
 above global minimum = 0.00016940417
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.418318$
 $\text{theta_p_out} = 2.480574$
 $\text{theta_gamma_out} = 1.951958$
 $\text{qOut} = 0.219011$
 $\text{phi_p_out} = 3.141612$
 $\text{phi_gamma_out} = 1.108872\text{e-}05$
 $\text{alpha_e} = 1.319931$
 $\text{alpha_p} = 0.2011109$

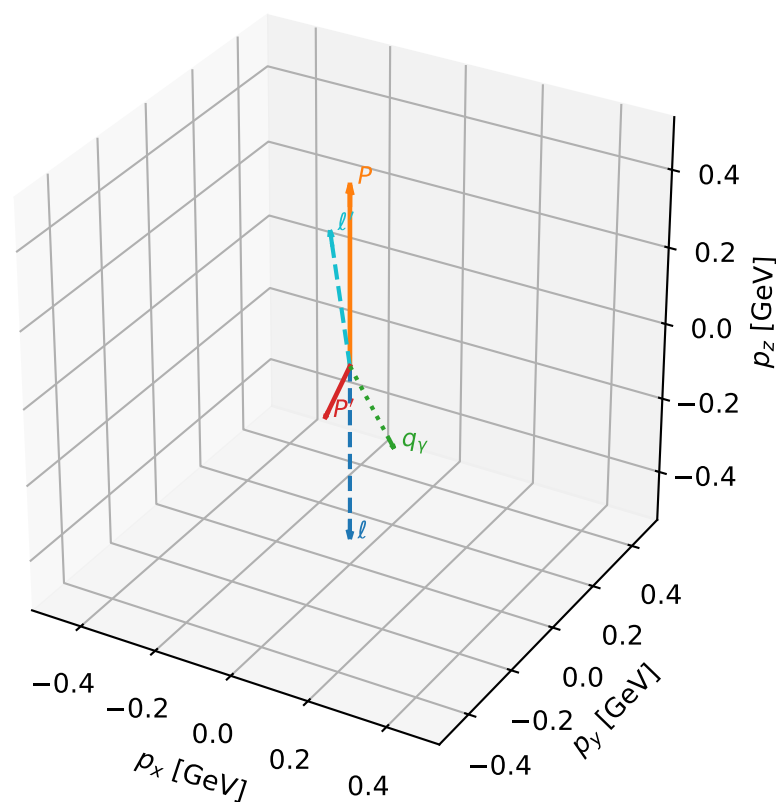
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_264 d_{\rm GHZ}=0.000183604
region: polarization_cluster_01_local_minimum_264
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.4665237$ rad
incoming lepton: $0.104084|+\rangle +0.994569|-\rangle$
proton mixing angle: $\alpha_p=0.080853583$ rad
incoming proton: $0.996733|+\rangle +0.0807655|-\rangle$

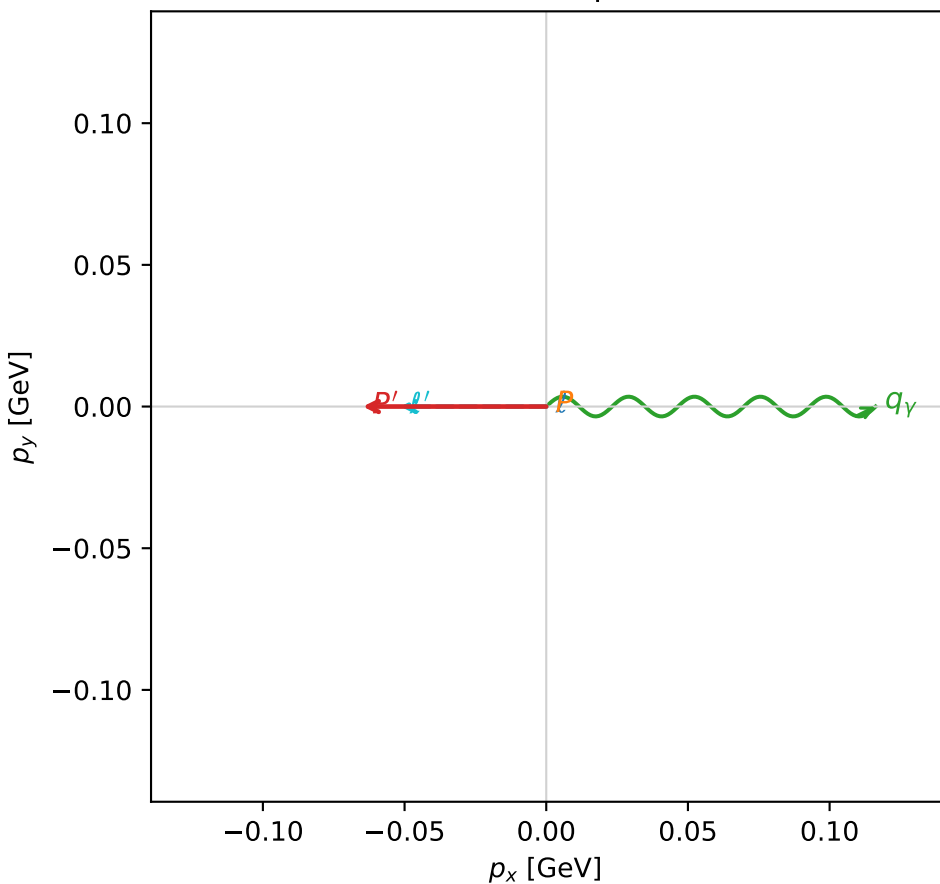
$s=2.27913$, $\sqrt{s}=1.50968$, $|\vec{P}|=0.463438$, $|\vec{P}'|=0.166332$
 $\theta_{P'}=2.74182$, $\theta_Y=2.57591$, $\phi_{P'}=3.14154$, $\phi_Y=6.28315$
 $E_Y=0.216844$, $\theta_{\ell'}=0.151931$, $\phi_{\ell'}=3.14158$
 $Q^2=0.627016$, $x_B=0.627572$, $t=-0.375692$, $W^2=1.25194$, $y=0.714017$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4634$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4634$
 P $E= 1.0462$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4634$
 ℓ' $E= 0.3402$ $p_x= -0.0515$ $p_y= 0.0000$ $p_z= 0.3363$
 P' $E= 0.9526$ $p_x= -0.0647$ $p_y= 0.0000$ $p_z= -0.1532$
 q_Y $E= 0.2168$ $p_x= 0.1162$ $p_y= -0.0000$ $p_z= -0.1831$

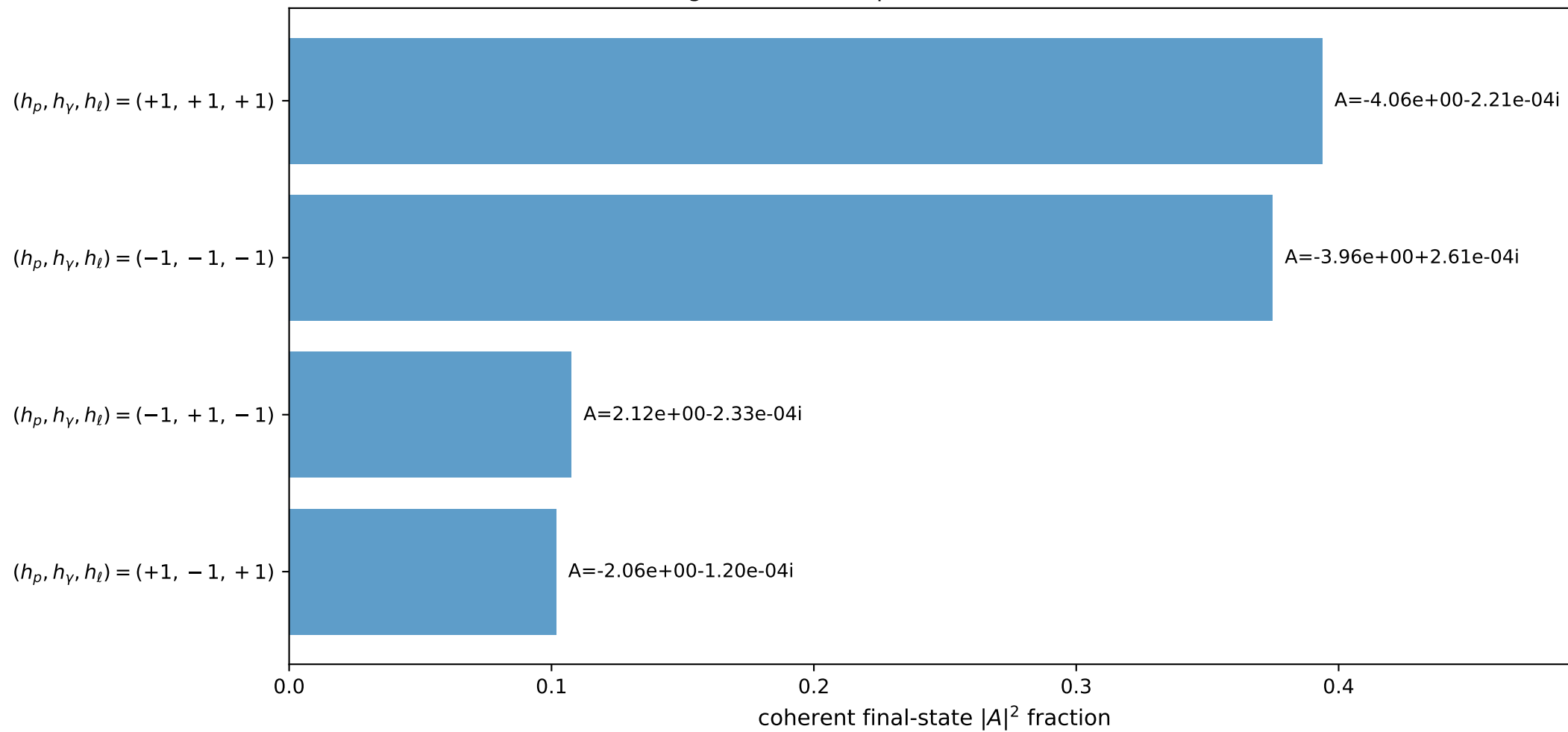
3D momenta



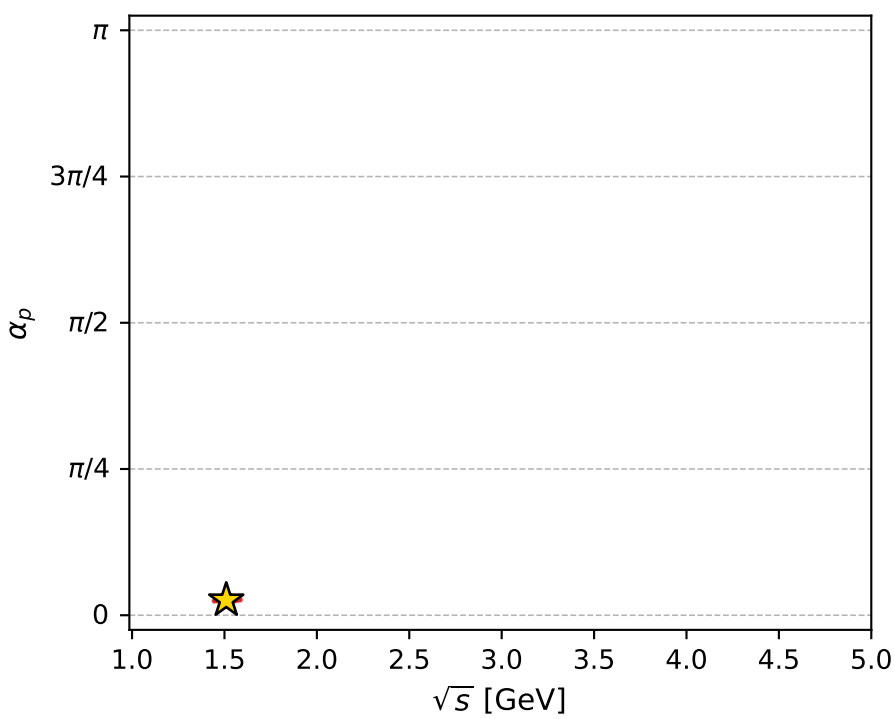
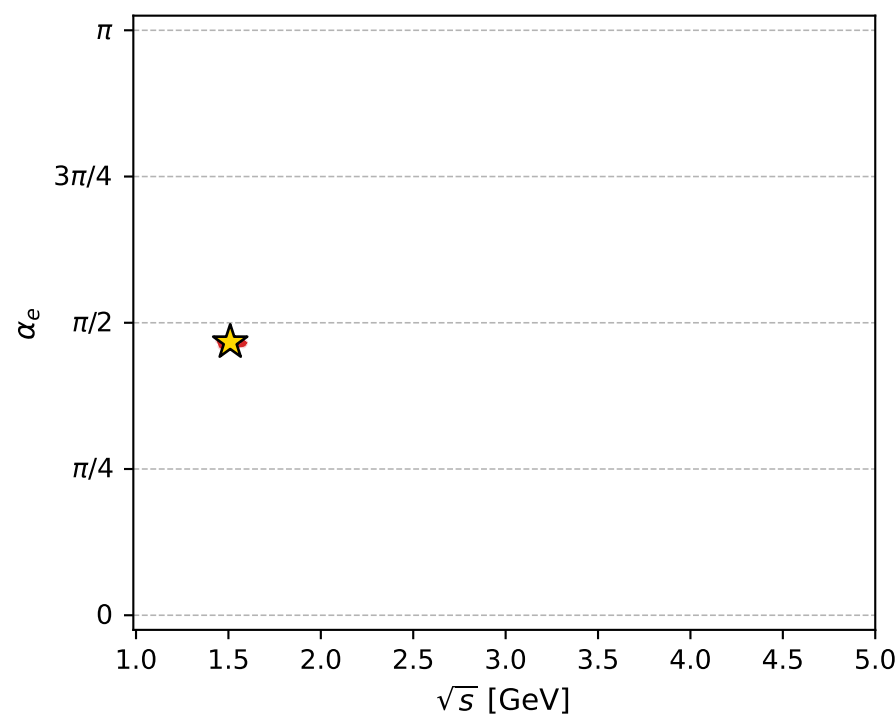
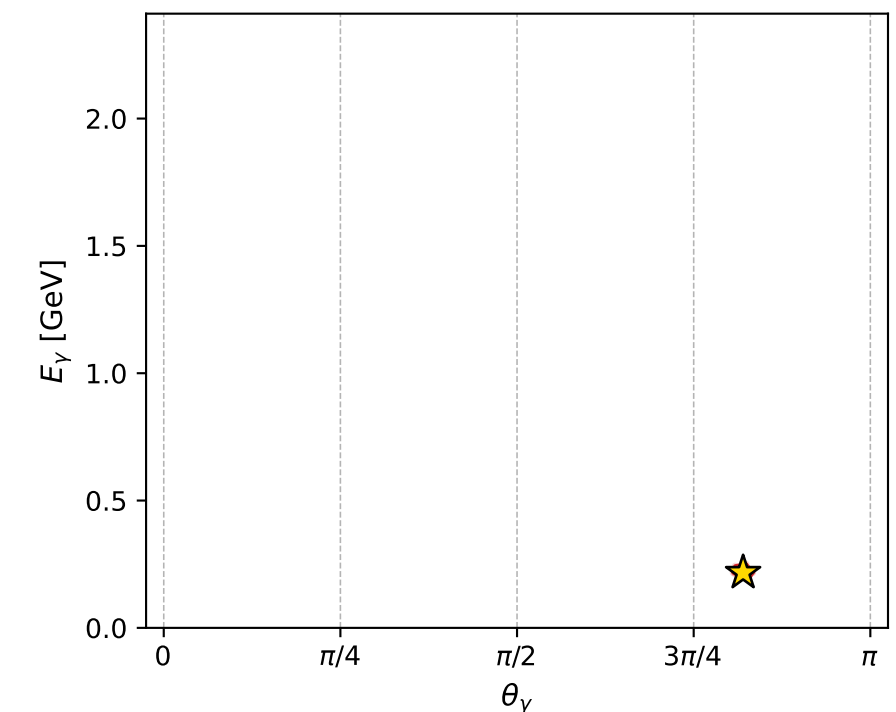
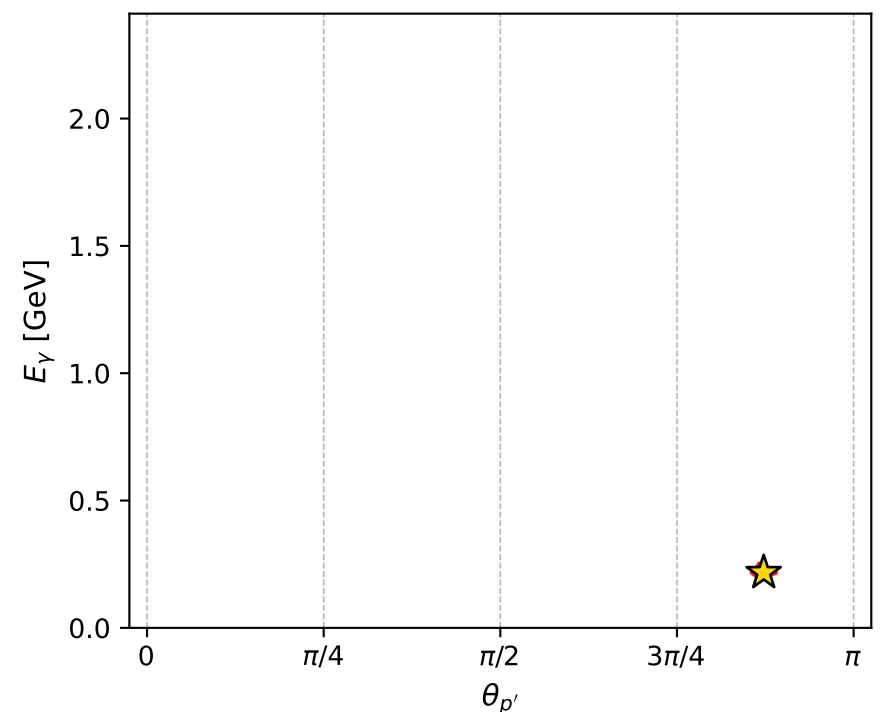
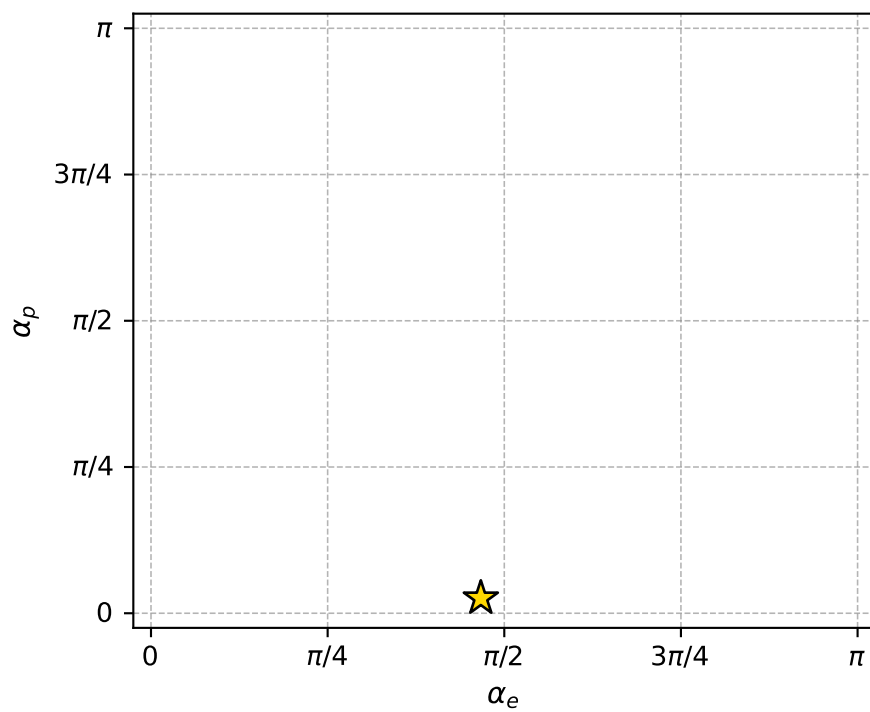
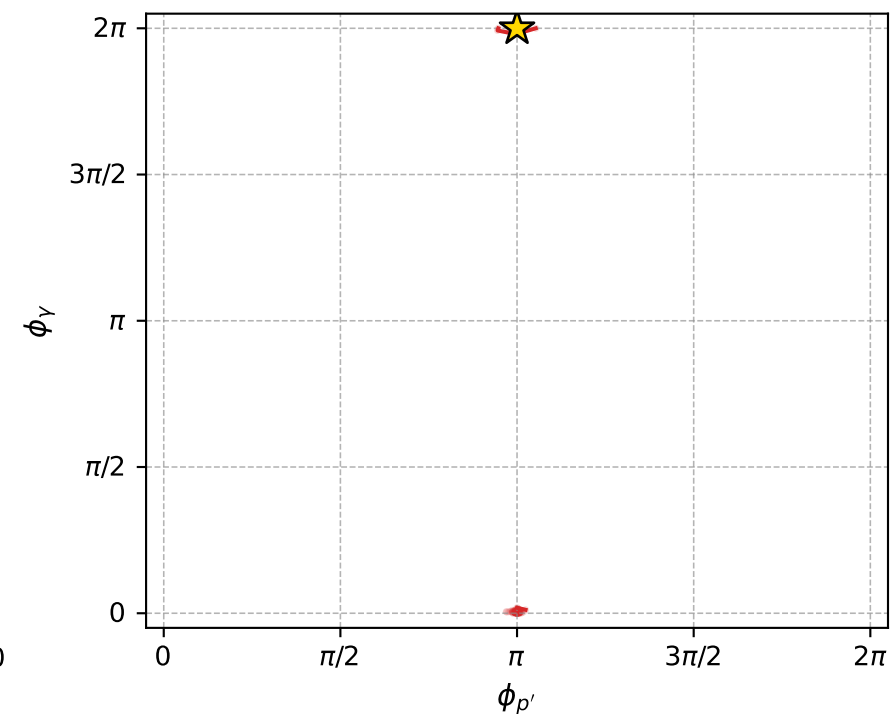
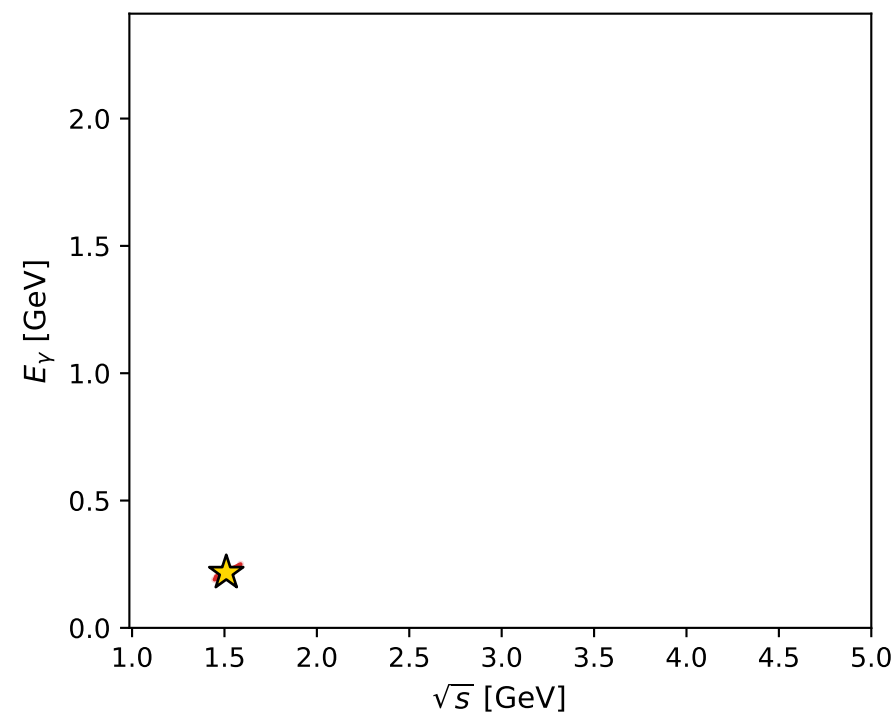
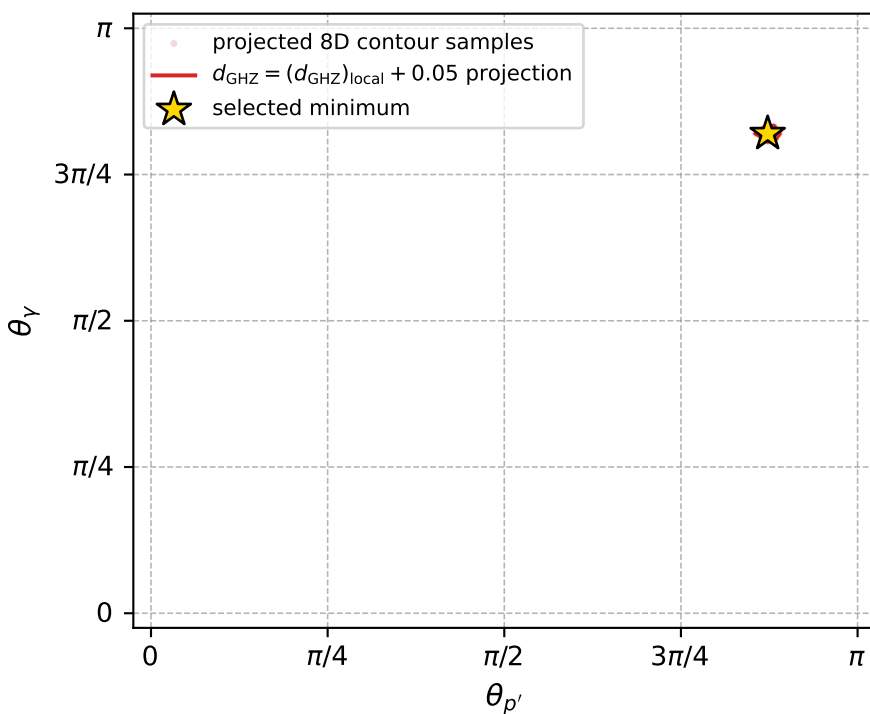
Transverse plane



Leading final-state amplitudes (N=4, retained=97.8%)



electron: polarization cluster P1, local minimum 264; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00018360365$
 $d_{\text{GHZ}} \text{ contour} = 0.050183604$
 above global minimum = 0.00018358137
 8D contour samples = 256

selected phase-space point:

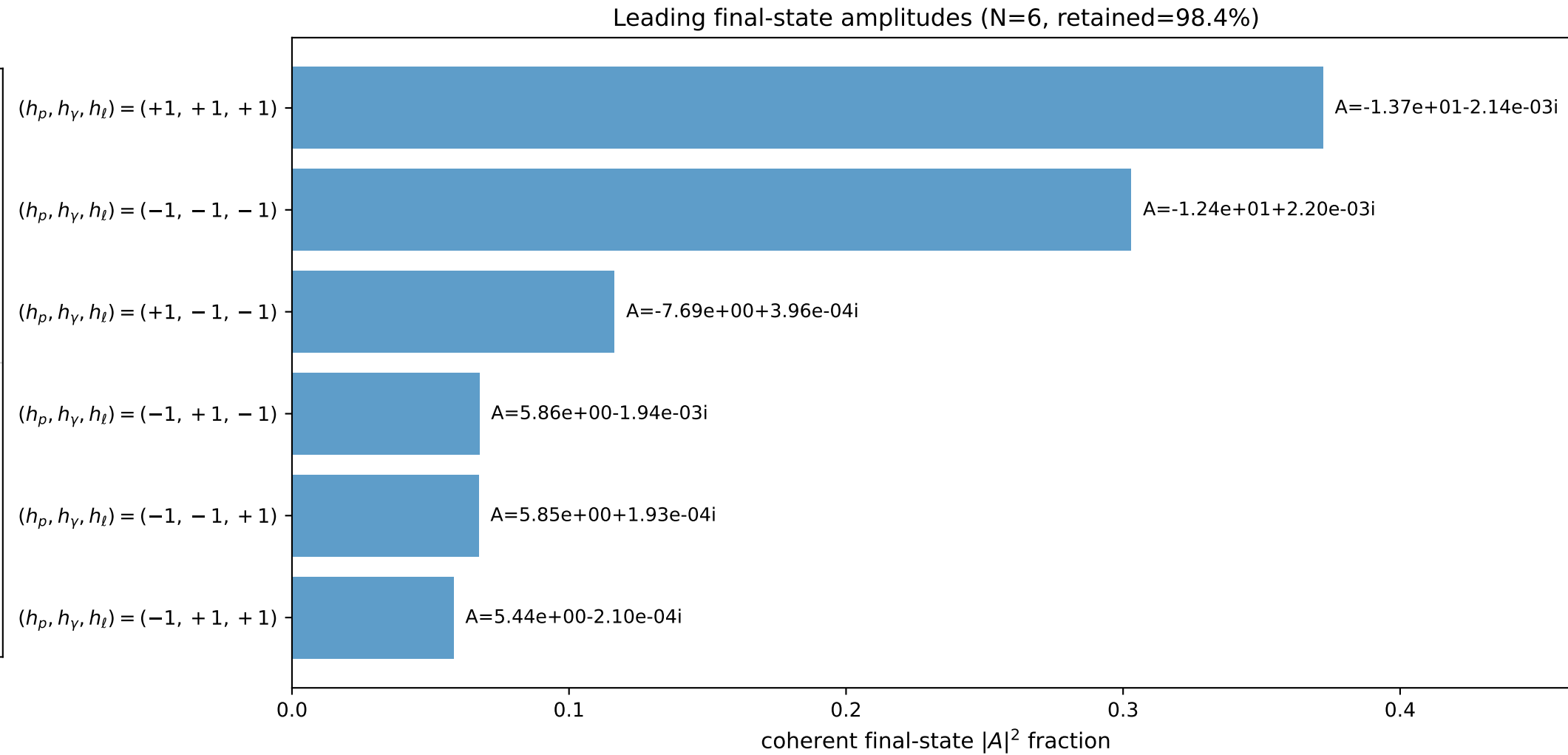
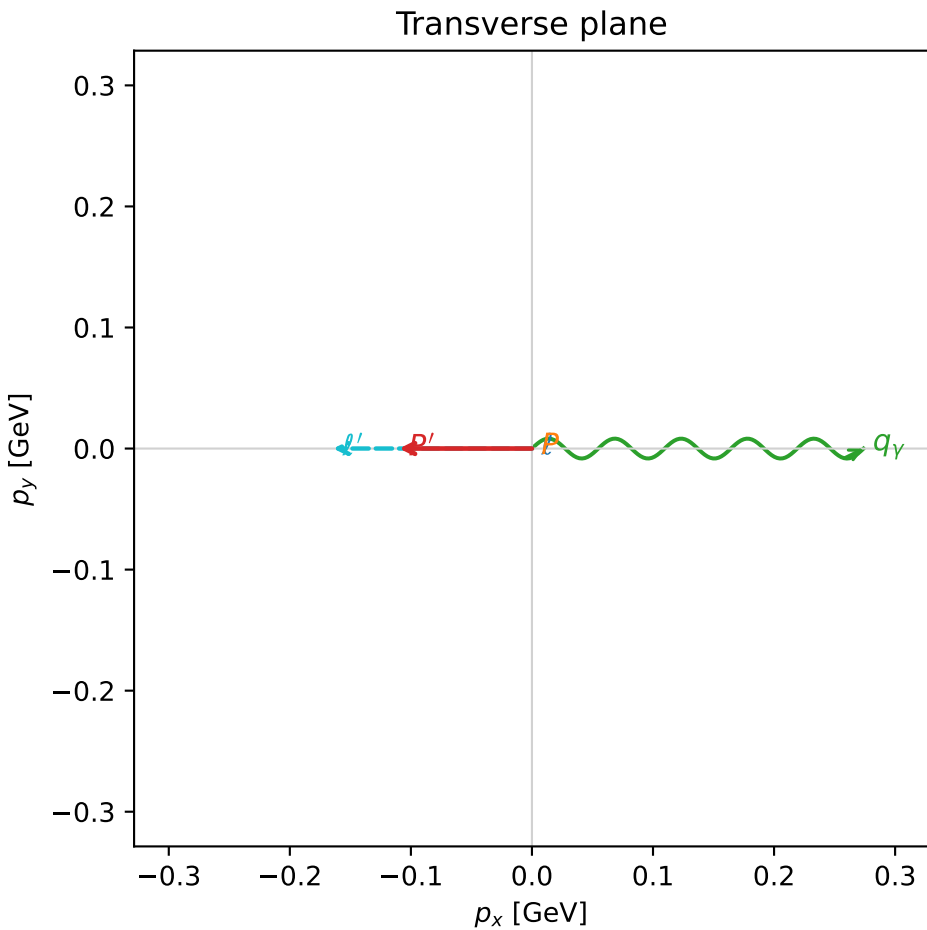
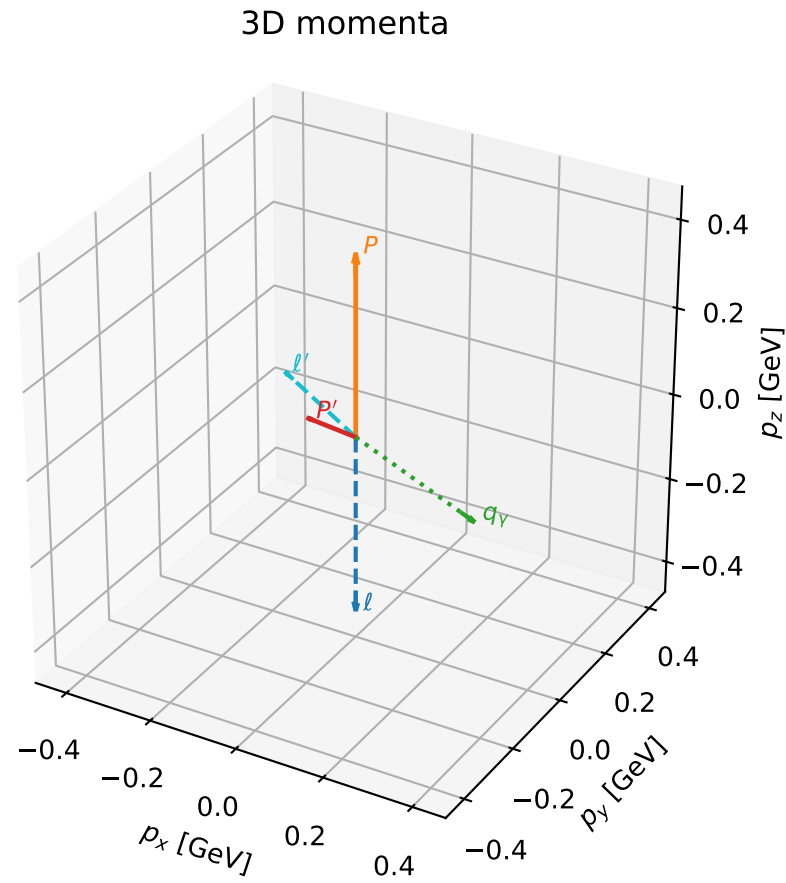
$\text{sqrt_s} = 1.509679$
 $\text{theta_p_out} = 2.741818$
 $\text{theta_gamma_out} = 2.575911$
 $\text{qOut} = 0.2168444$
 $\text{phi_p_out} = 3.141537$
 $\text{phi_gamma_out} = 6.28315$
 $\text{alpha_e} = 1.466524$
 $\text{alpha_p} = 0.08085358$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

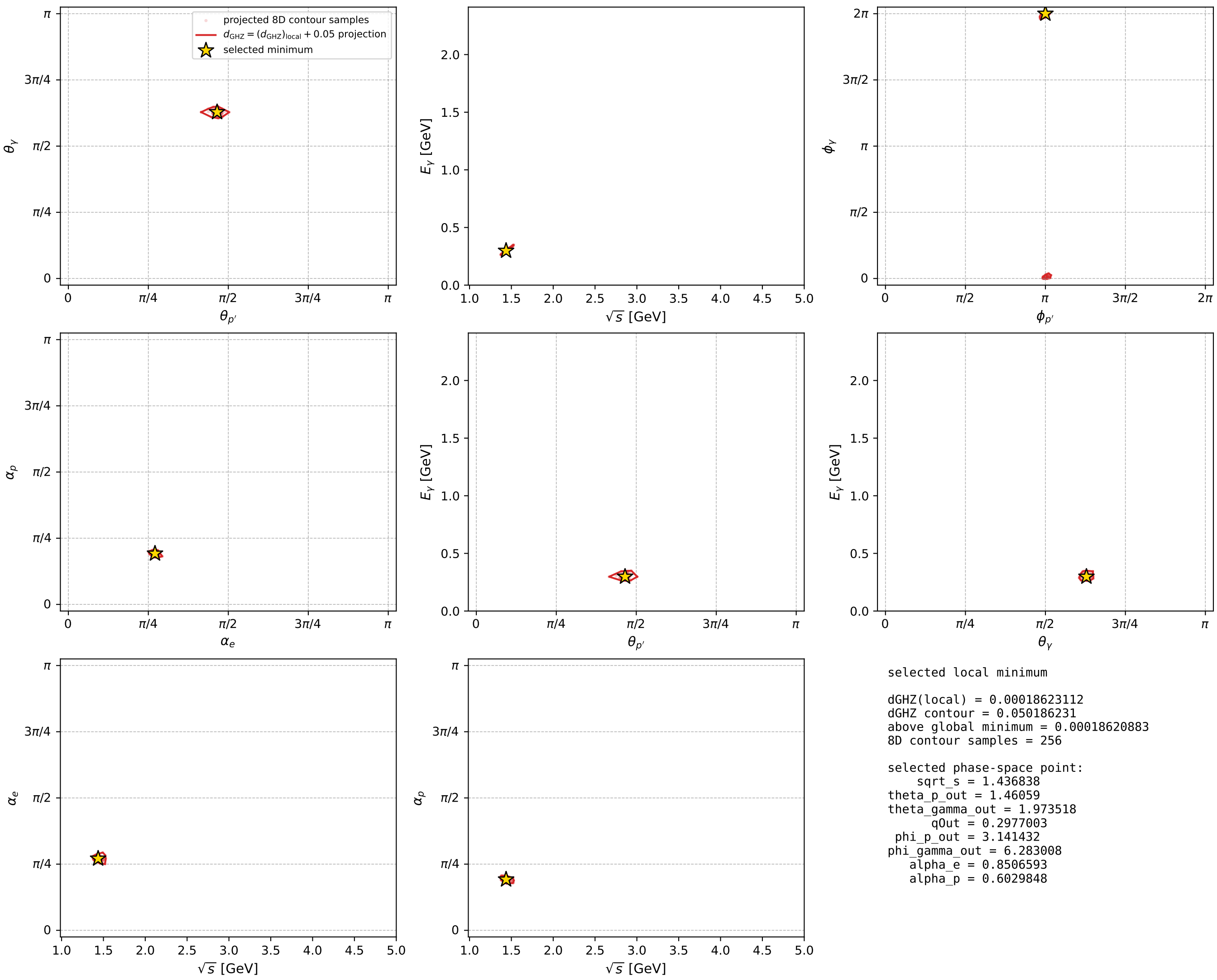
dghz_mixing_angles_polarization_01_local_minimum_265 $d_{\{\text{rm GHZ}\}}=0.000186231$
 region: polarization_cluster_01_local_minimum_265
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.8506593$ rad
 incoming lepton: $0.659488|+\rangle +0.751715|-\rangle$
 proton mixing angle: $\alpha_p=0.60298482$ rad
 incoming proton: $0.823647|+\rangle +0.567103|-\rangle$

$s=2.0645$, $\sqrt{s}=1.43684$, $|\vec{P}|=0.412245$, $|\vec{P}'|=0.110335$
 $\theta_{P'}=1.46059$, $\theta_Y=1.97352$, $\phi_{P'}=3.14143$, $\phi_Y=6.28301$
 $E_Y=0.2977$, $\theta_{l'}=1.0039$, $\phi_{l'}=3.1414$
 $Q^2=0.246696$, $x_B=0.282929$, $t=-0.165694$, $W^2=1.50508$, $y=0.736023$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.4122$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4122$
 P $E= 1.0246$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4122$
 l' $E= 0.1947$ $p_x= -0.1642$ $p_y= 0.0000$ $p_z= 0.1045$
 P' $E= 0.9445$ $p_x= -0.1097$ $p_y= 0.0000$ $p_z= 0.0121$
 q_Y $E= 0.2977$ $p_x= 0.2739$ $p_y= -0.0000$ $p_z= -0.1167$



electron: polarization cluster P1, local minimum 265; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



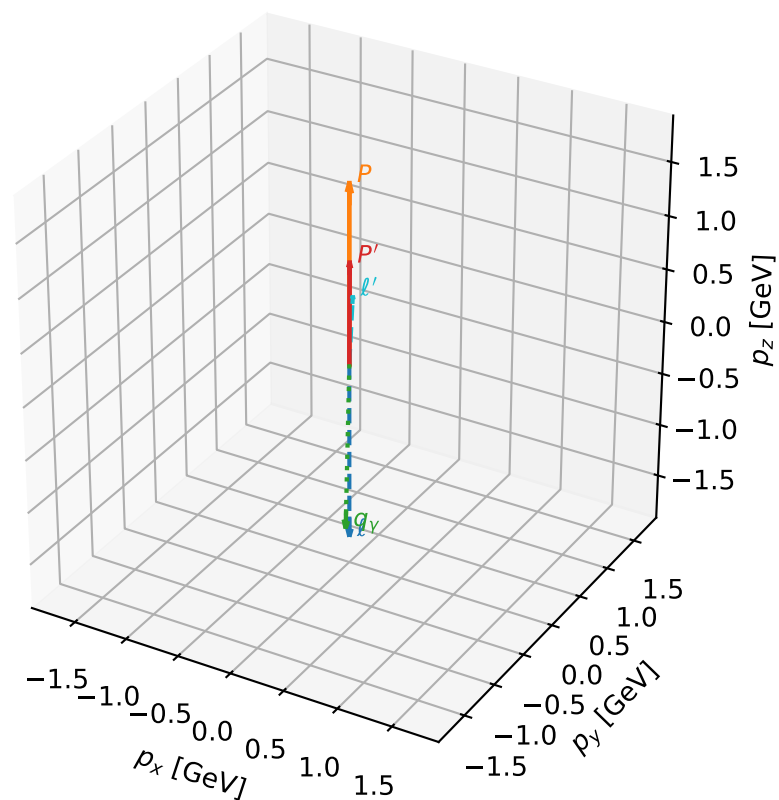
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_266 $d_{\{\text{rm GHZ}\}}=0.000189044$
 region: polarization_cluster_01_local_minimum_266
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.20844456$ rad
 incoming lepton: $0.978354|+\rangle +0.206938|-\rangle$
 proton mixing angle: $\alpha_p=1.3671115$ rad
 incoming proton: $0.202279|+\rangle +0.979328|-\rangle$

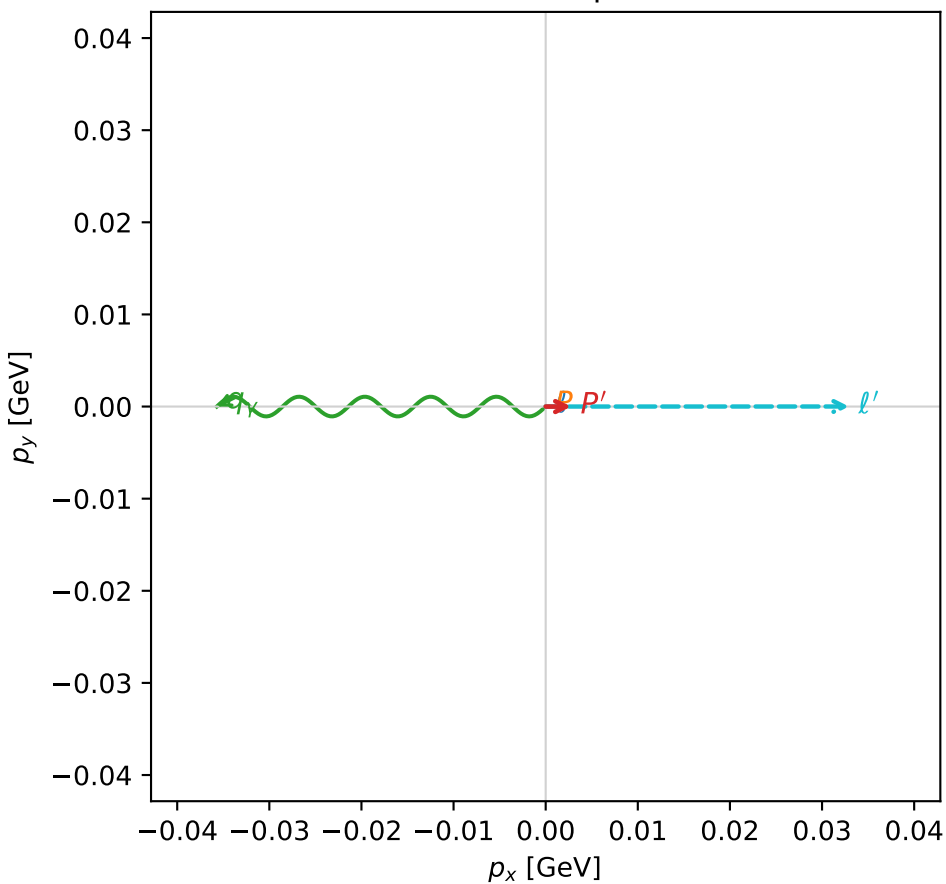
$s=12.8901$, $\sqrt{s}=3.59027$, $|\vec{P}|=1.67261$, $|\vec{P}'|=0.948936$
 $\theta_{P'}=0.00291456$, $\theta_V=3.11931$, $\phi_{P'}=6.2819$, $\phi_V=3.14164$
 $E_V=1.60224$, $\theta_{l'}=0.0504001$, $\phi_{l'}=0.000160702$
 $Q^2=4.37105$, $x_B=0.374008$, $t=-0.183378$, $W^2=8.19583$, $y=0.97309$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.6726$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6726$
 P $E= 1.9177$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6726$
 l' $E= 0.6537$ $p_x= 0.0329$ $p_y= 0.0000$ $p_z= 0.6529$
 P' $E= 1.3343$ $p_x= 0.0028$ $p_y= -0.0000$ $p_z= 0.9489$
 q_Y $E= 1.6022$ $p_x= -0.0357$ $p_y= -0.0000$ $p_z= -1.6018$

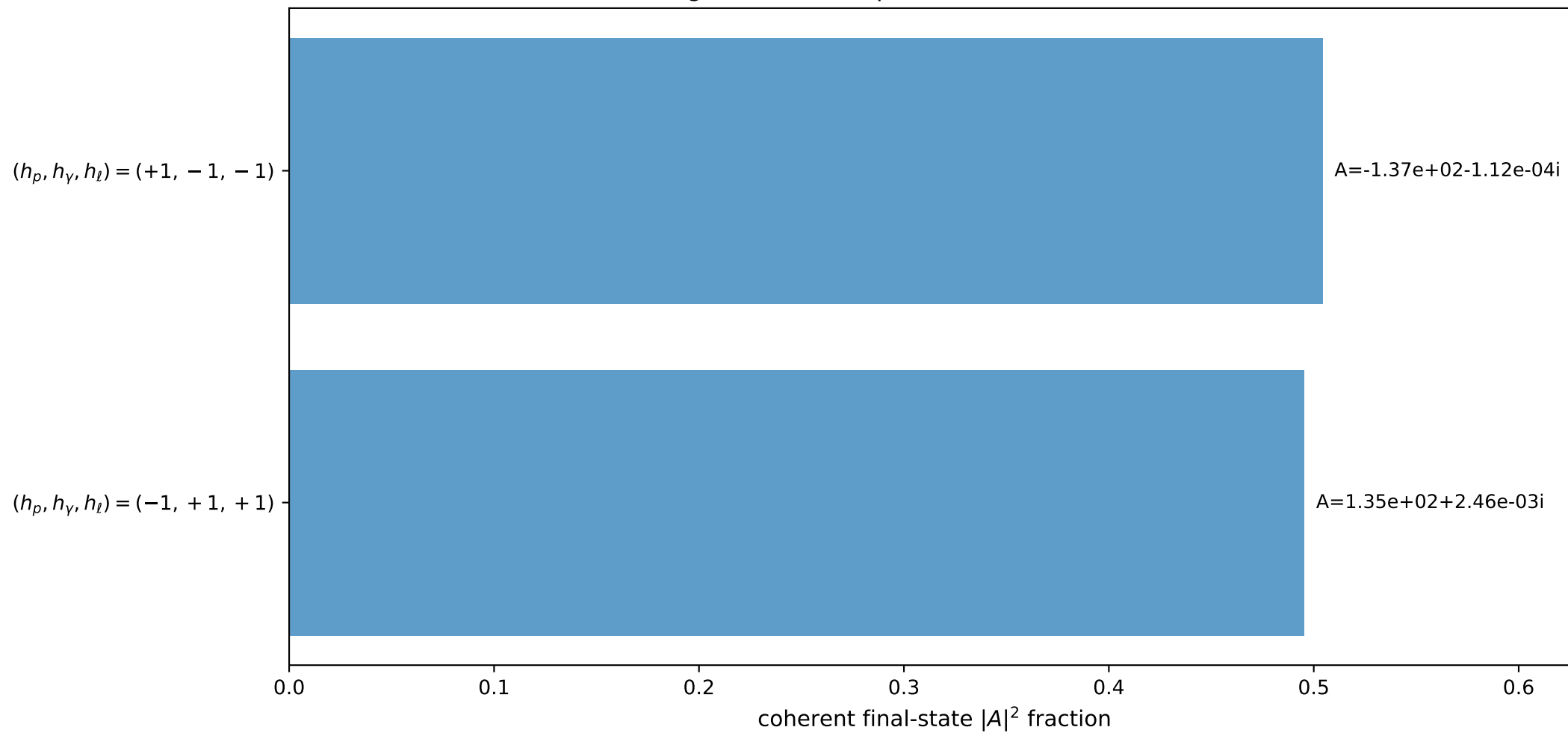
3D momenta



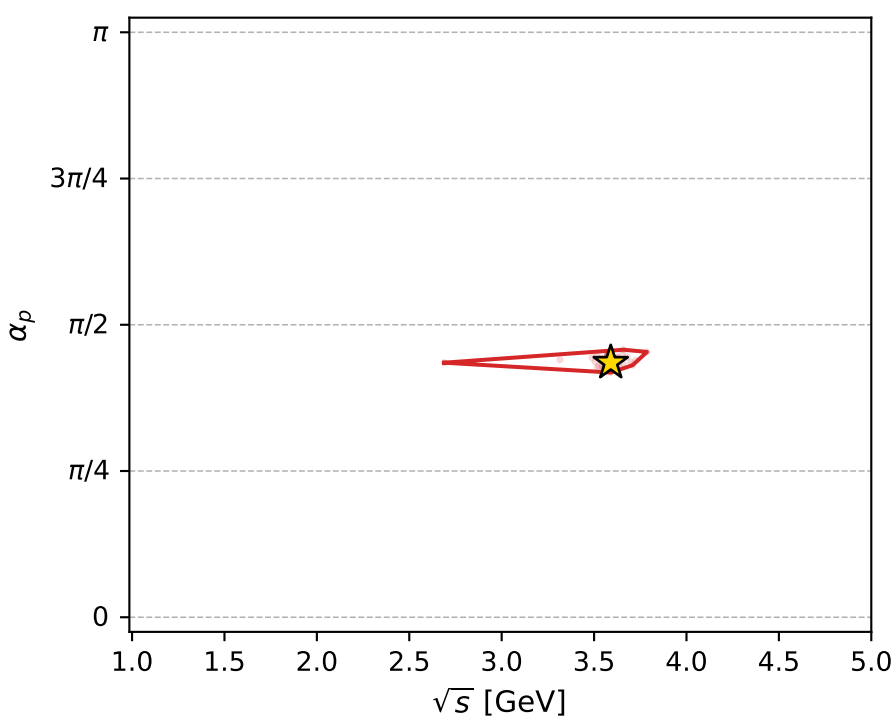
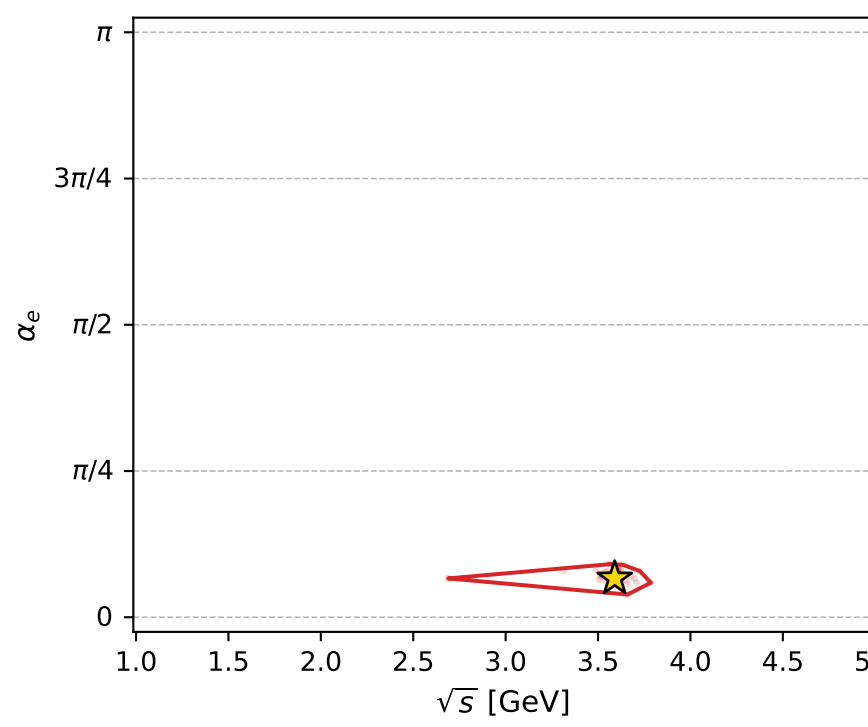
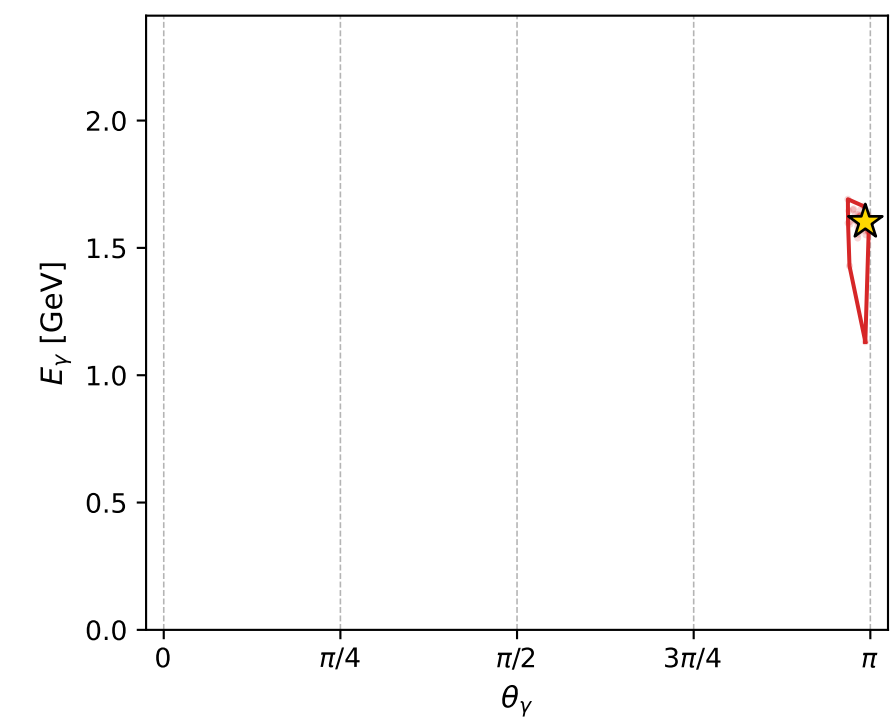
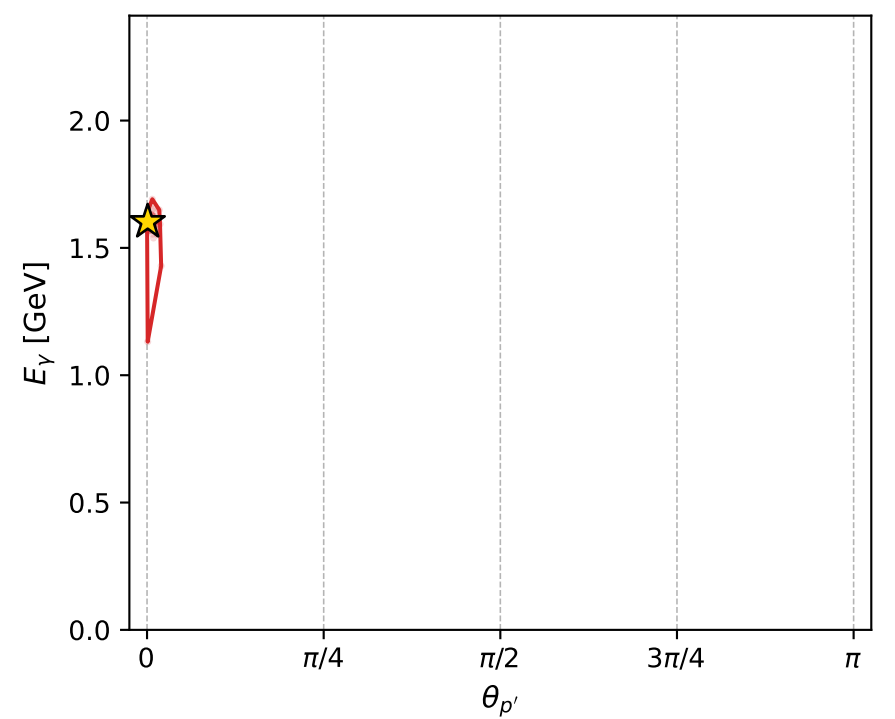
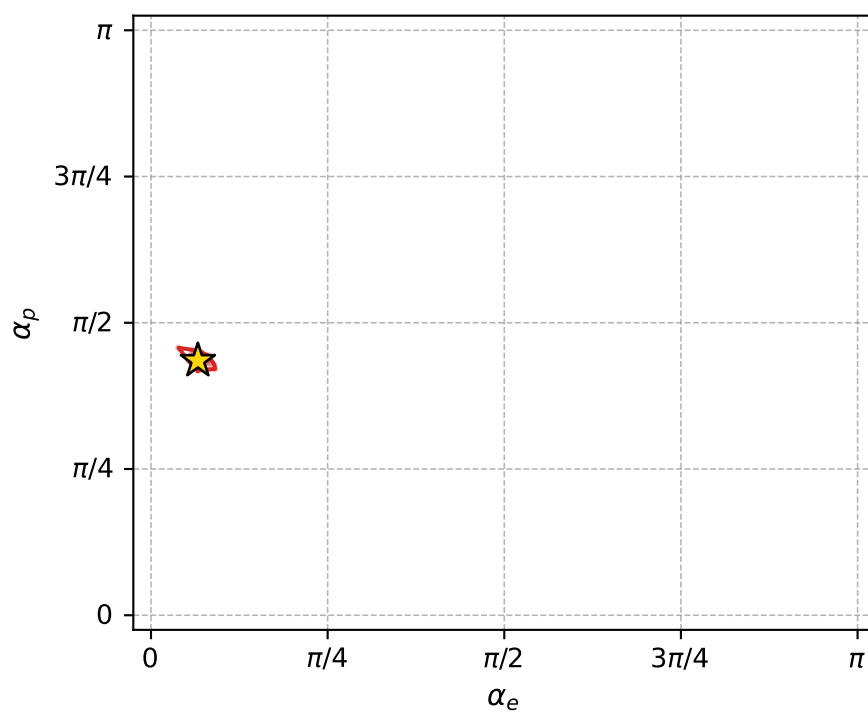
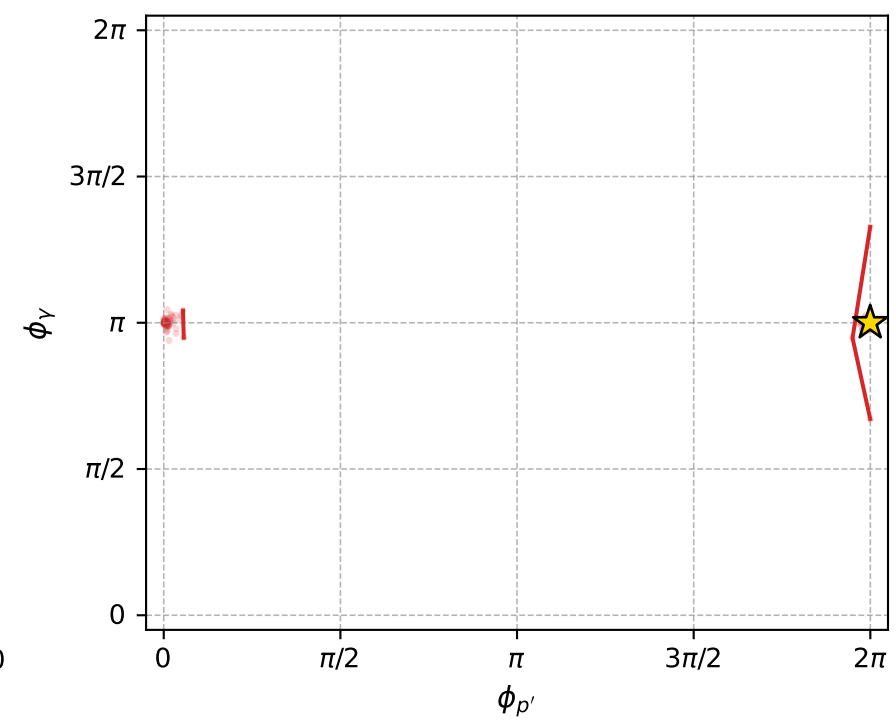
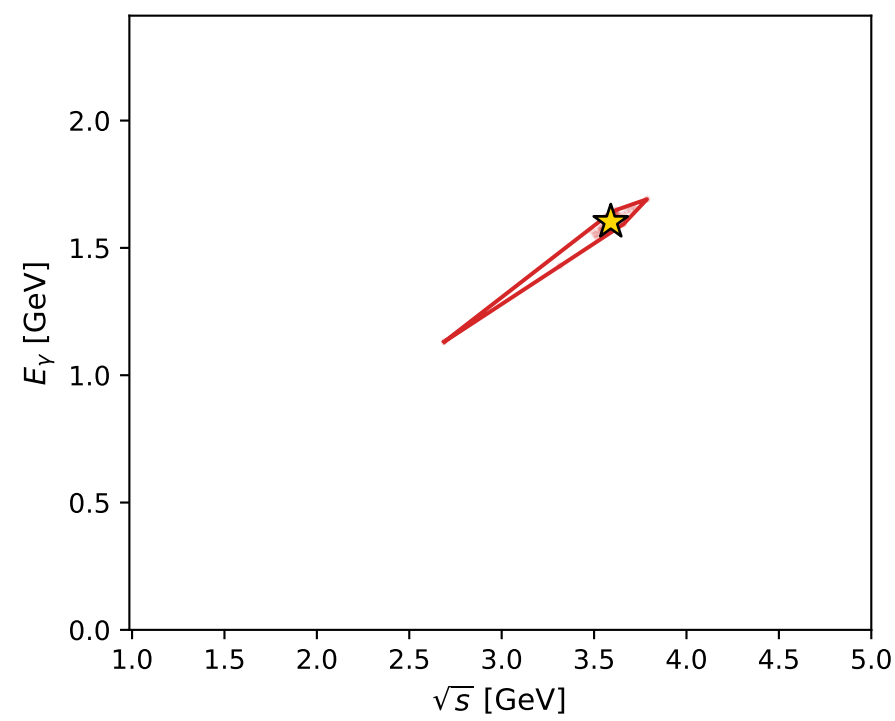
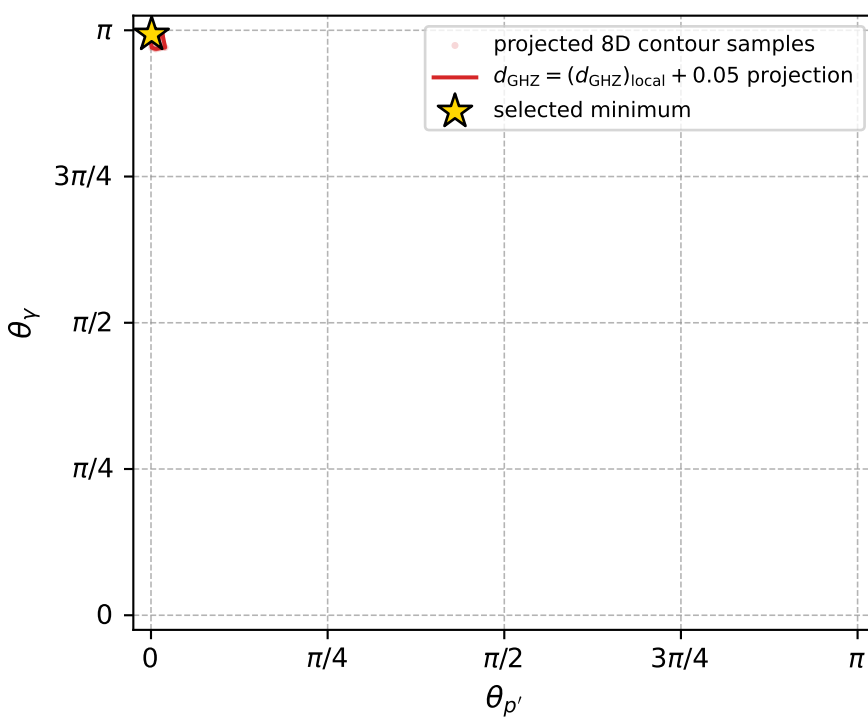
Transverse plane



Leading final-state amplitudes (N=2, retained=99.9%)



electron: polarization cluster P1, local minimum 266; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00018904359$
 $d_{\text{GHZ}} \text{ contour} = 0.050189044$
 above global minimum = 0.00018902131
 8D contour samples = 141

selected phase-space point:

$\text{sqrt_s} = 3.590275$
 $\text{theta_p_out} = 0.002914563$
 $\text{theta_gamma_out} = 3.119309$
 $\text{q0out} = 1.602243$
 $\text{phi_p_out} = 6.281897$
 $\text{phi_gamma_out} = 3.141641$
 $\text{alpha_e} = 0.2084446$
 $\text{alpha_p} = 1.367111$

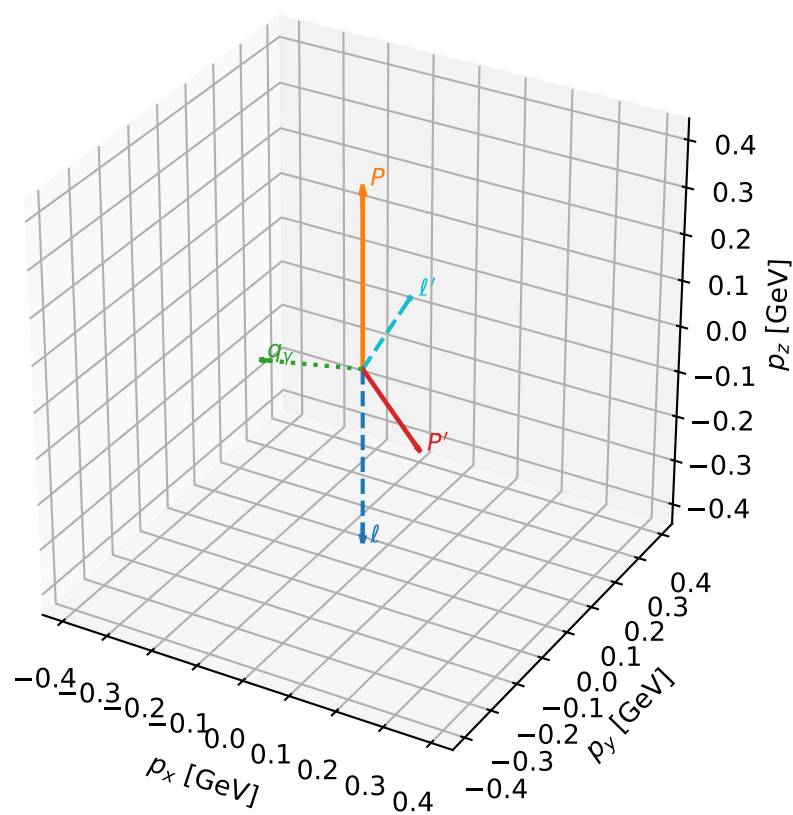
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_268 $d_{\{\text{rm GHZ}\}}=0.000192794$
 region: polarization_cluster_01_local_minimum_268
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.32833034$ rad
 incoming lepton: $0.946582|+\rangle +0.322463|-\rangle$
 proton mixing angle: $\alpha_p=1.3073254$ rad
 incoming proton: $0.260433|+\rangle +0.965492|-\rangle$

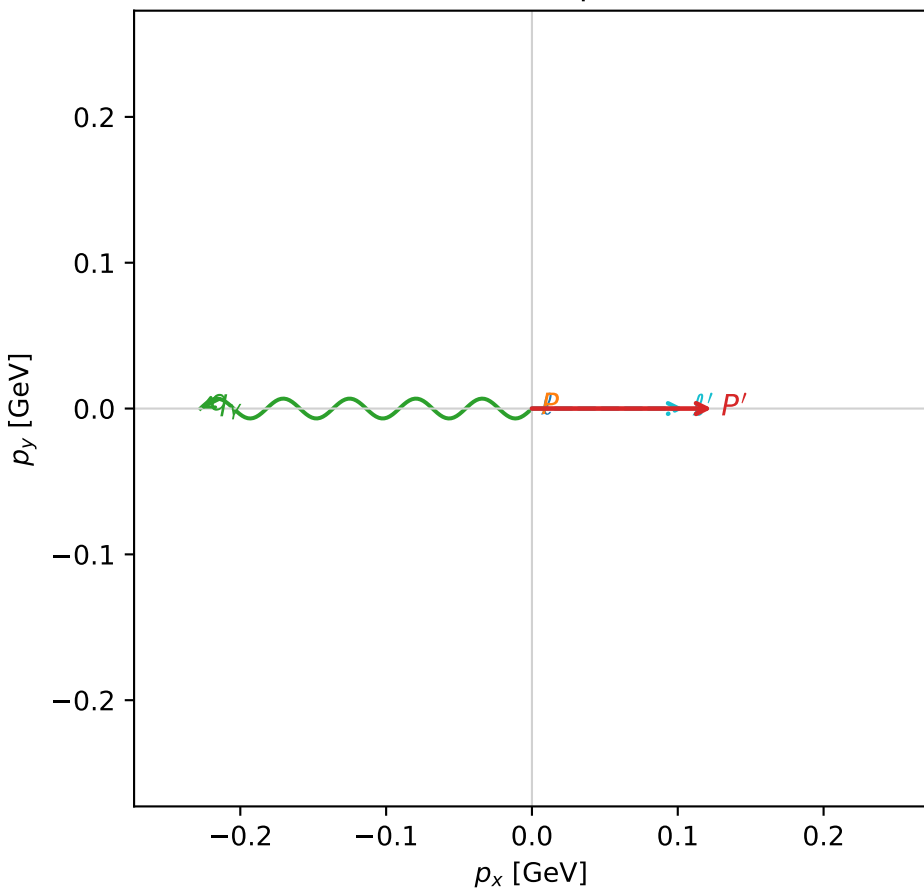
$s=1.96434$, $\sqrt{s}=1.40155$, $|\vec{P}|=0.386891$, $|\vec{P}'|=0.187552$
 $\theta_{P'}=2.42343$, $\theta_Y=1.76586$, $\phi_{P'}=6.28311$, $\phi_Y=3.14155$
 $E_Y=0.231766$, $\theta_{\ell'}=0.509329$, $\phi_{\ell'}=6.28317$
 $Q^2=0.309023$, $x_B=0.388291$, $t=-0.290767$, $W^2=1.36667$, $y=0.733848$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3869$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3869$
 P $E= 1.0147$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3869$
 ℓ' $E= 0.2132$ $p_x= 0.1040$ $p_y= -0.0000$ $p_z= 0.1862$
 P' $E= 0.9566$ $p_x= 0.1234$ $p_y= -0.0000$ $p_z= -0.1412$
 q_Y $E= 0.2318$ $p_x= -0.2274$ $p_y= 0.0000$ $p_z= -0.0449$

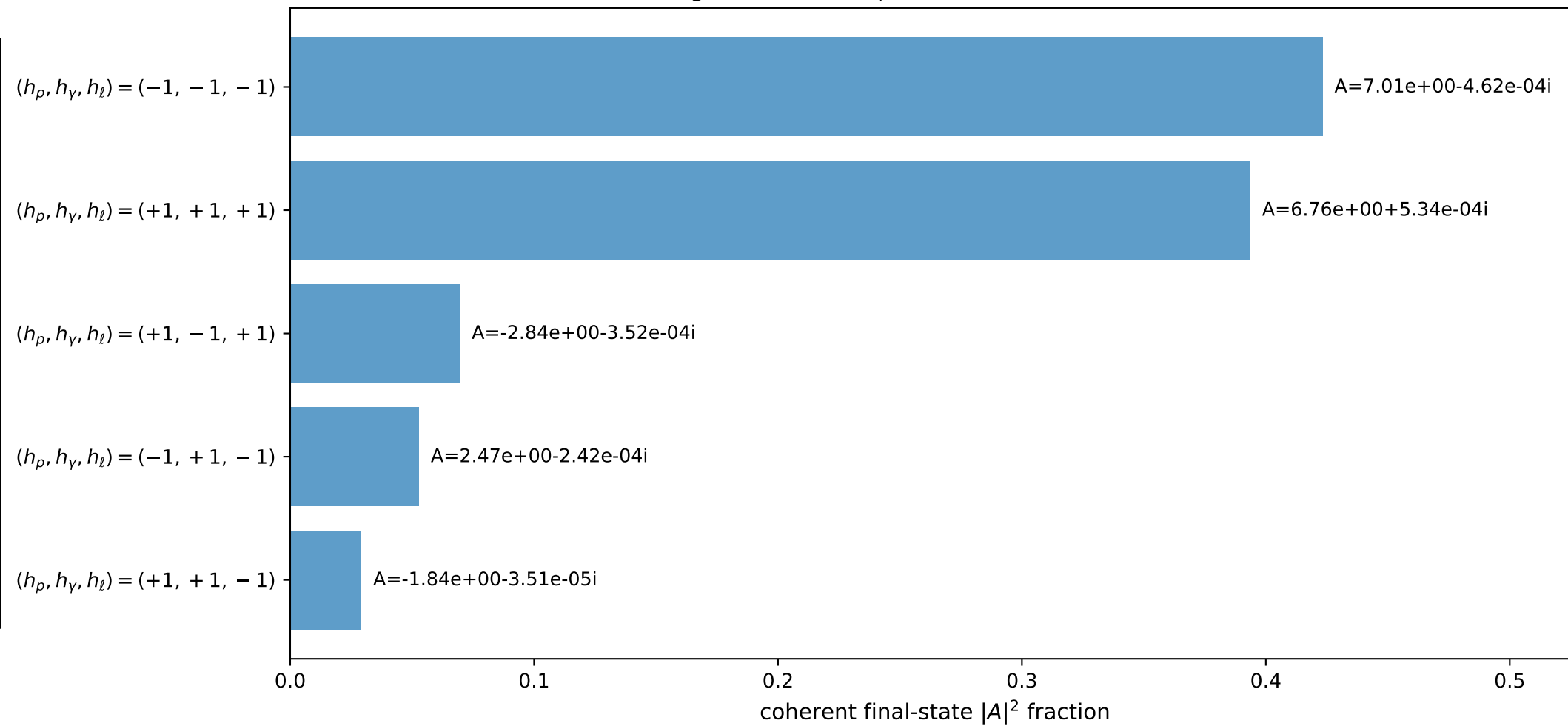
3D momenta



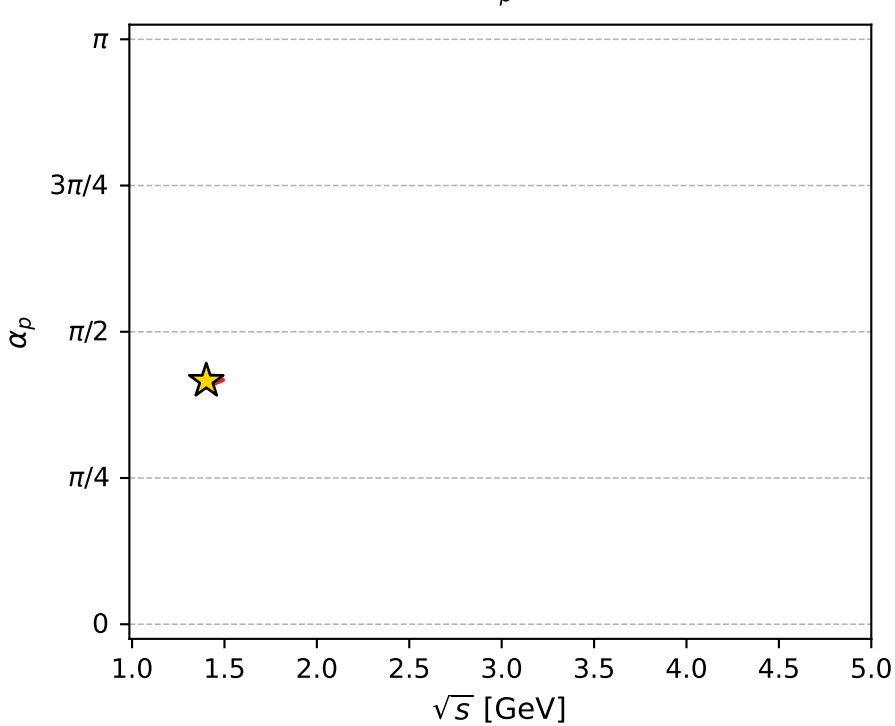
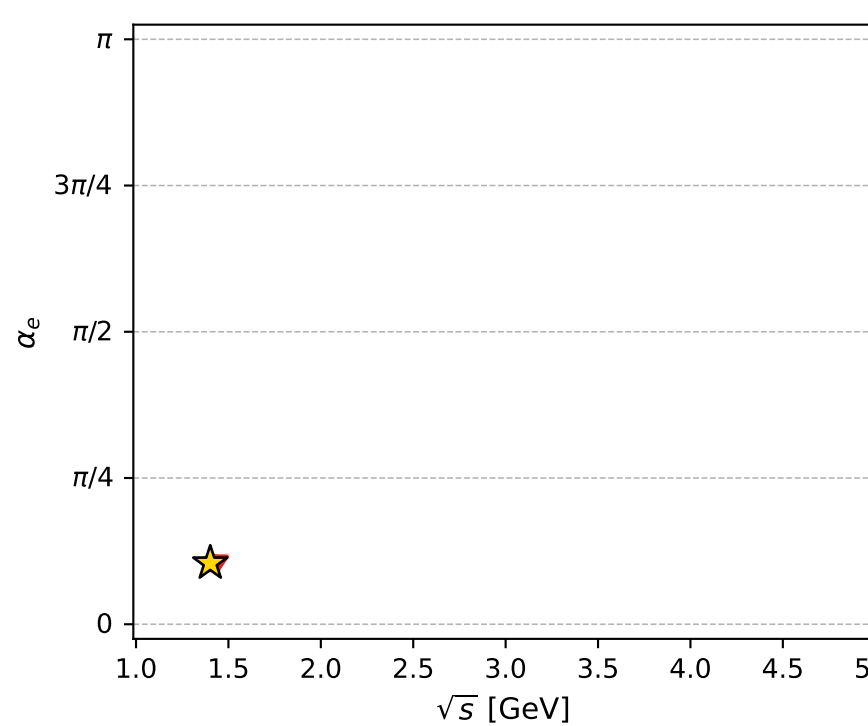
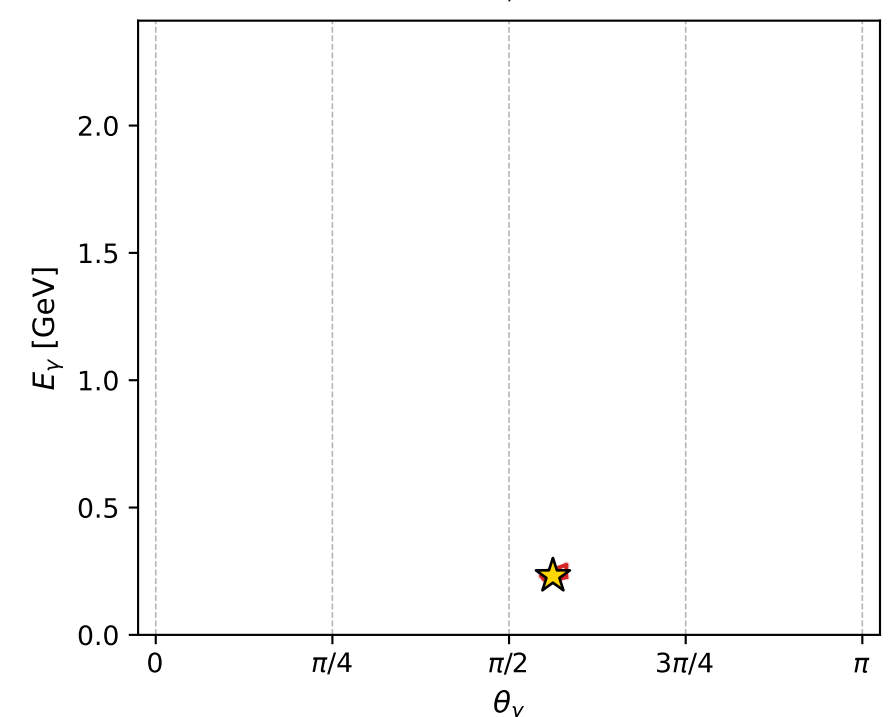
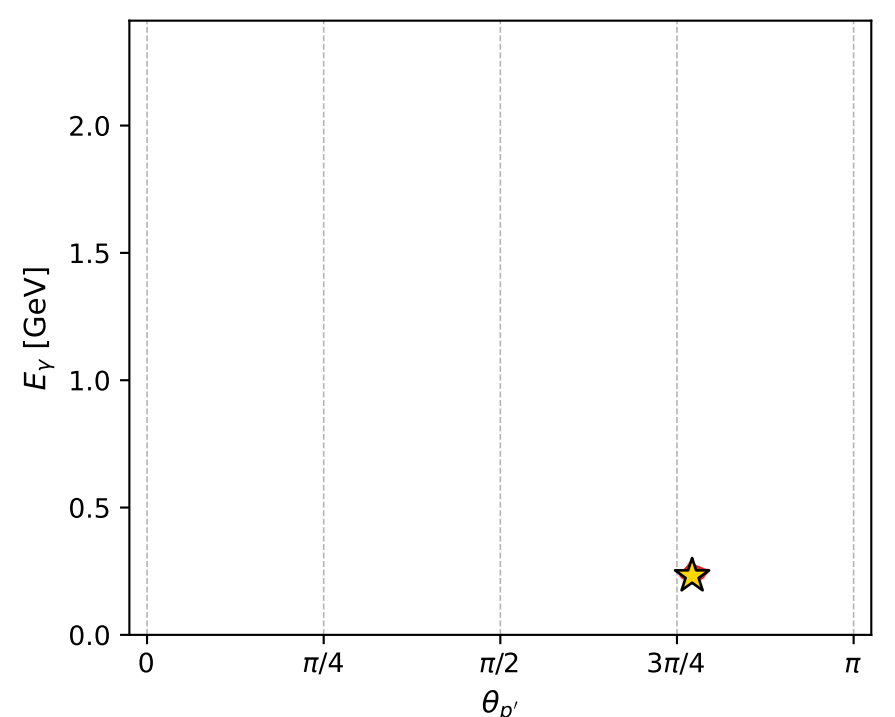
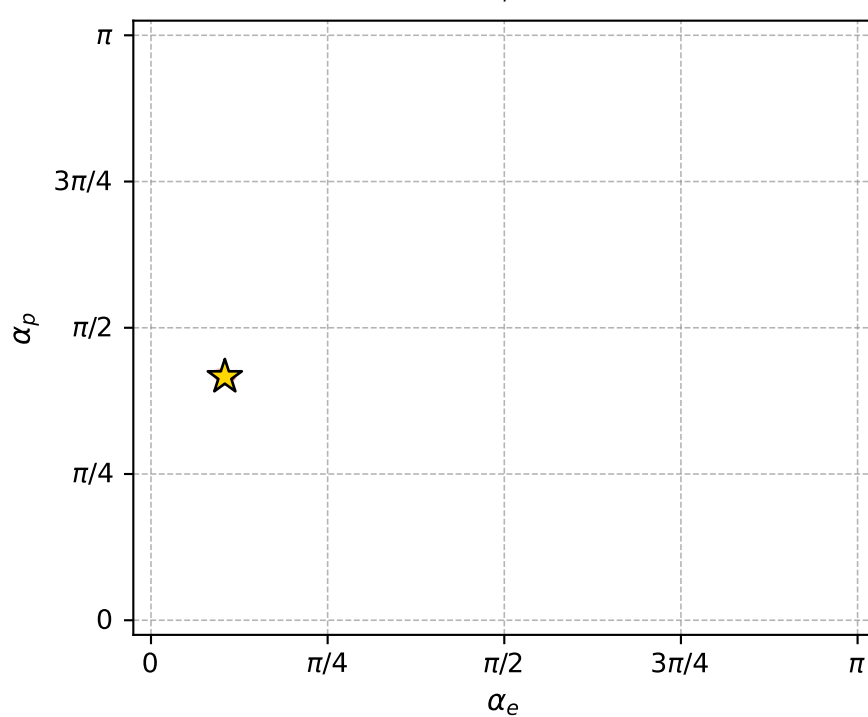
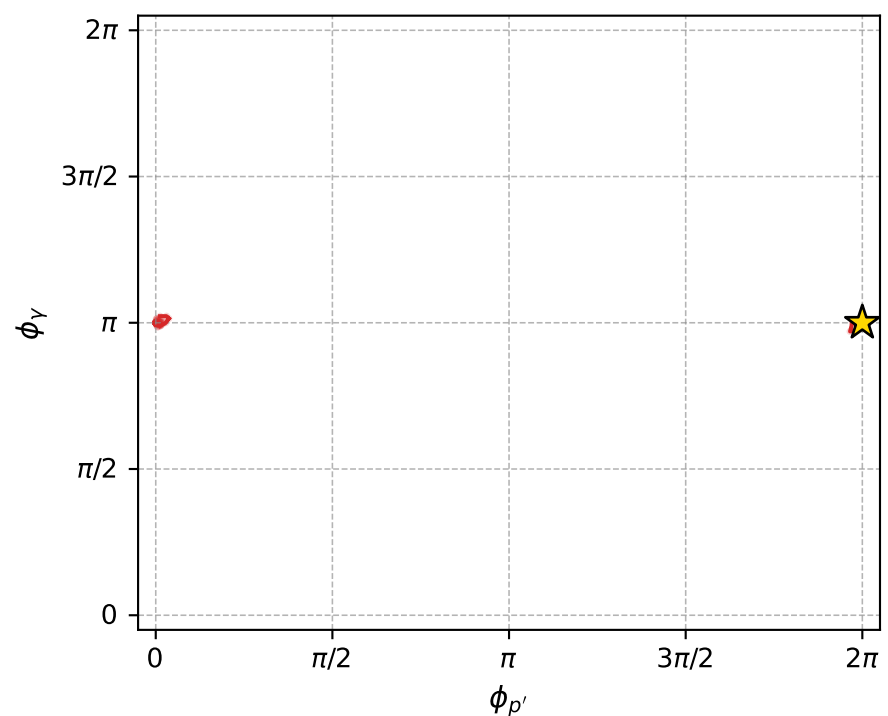
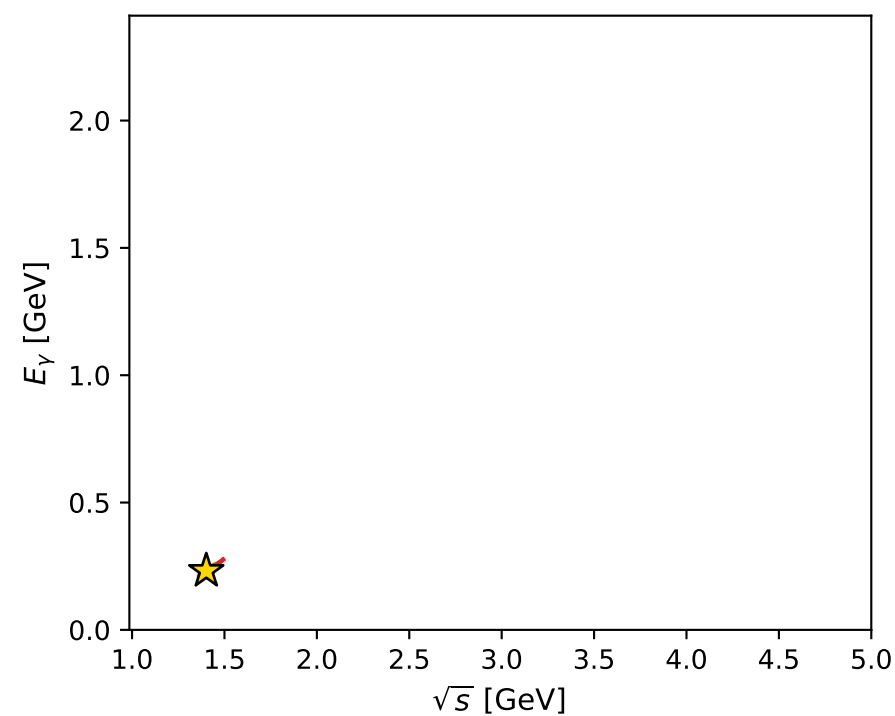
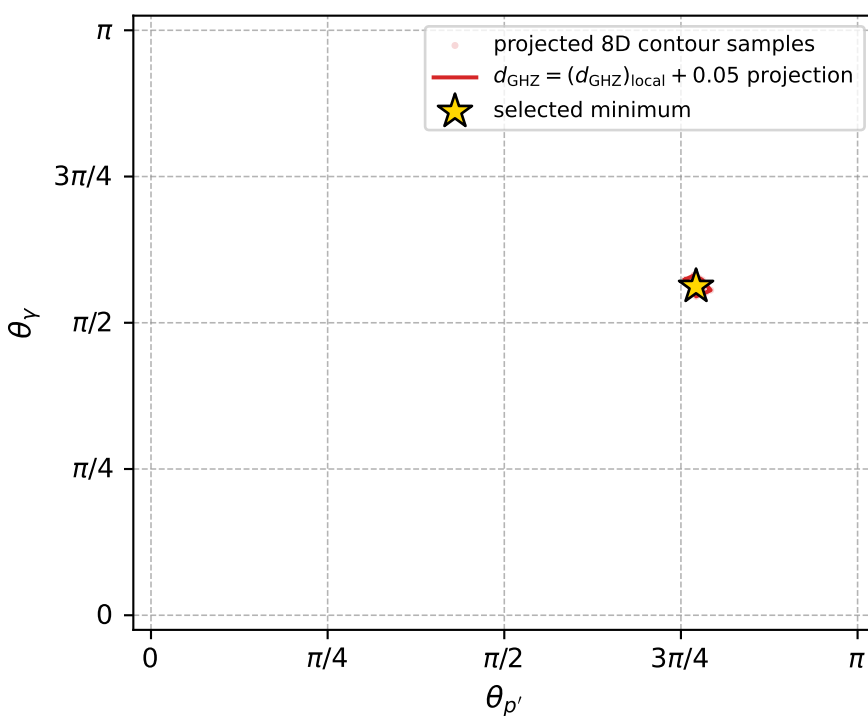
Transverse plane



Leading final-state amplitudes (N=5, retained=96.8%)



electron: polarization cluster P1, local minimum 268; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00019279408$
 $d_{\text{GHZ}} \text{ contour} = 0.050192794$
 above global minimum = 0.0001927718
 8D contour samples = 256

selected phase-space point:

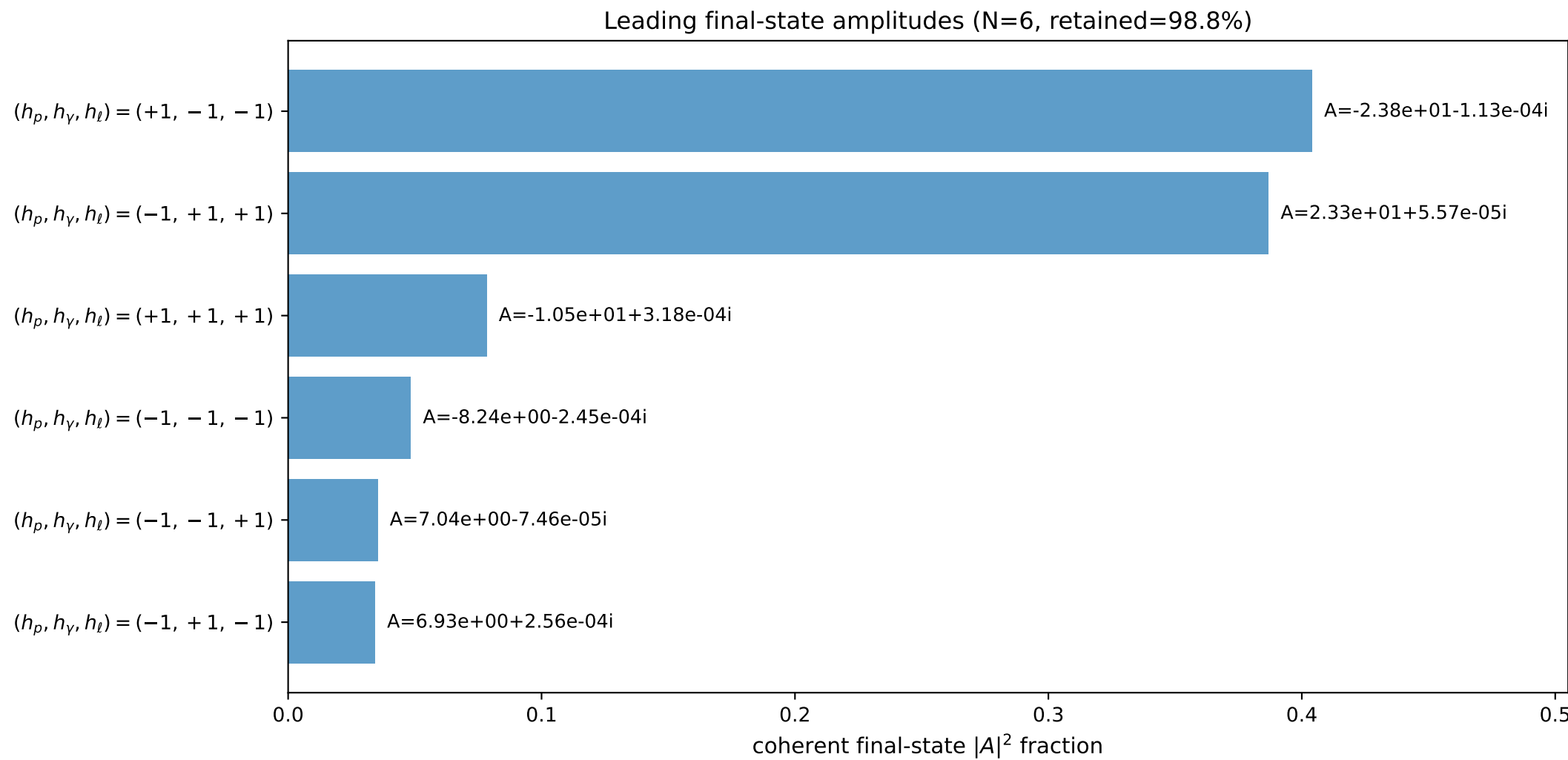
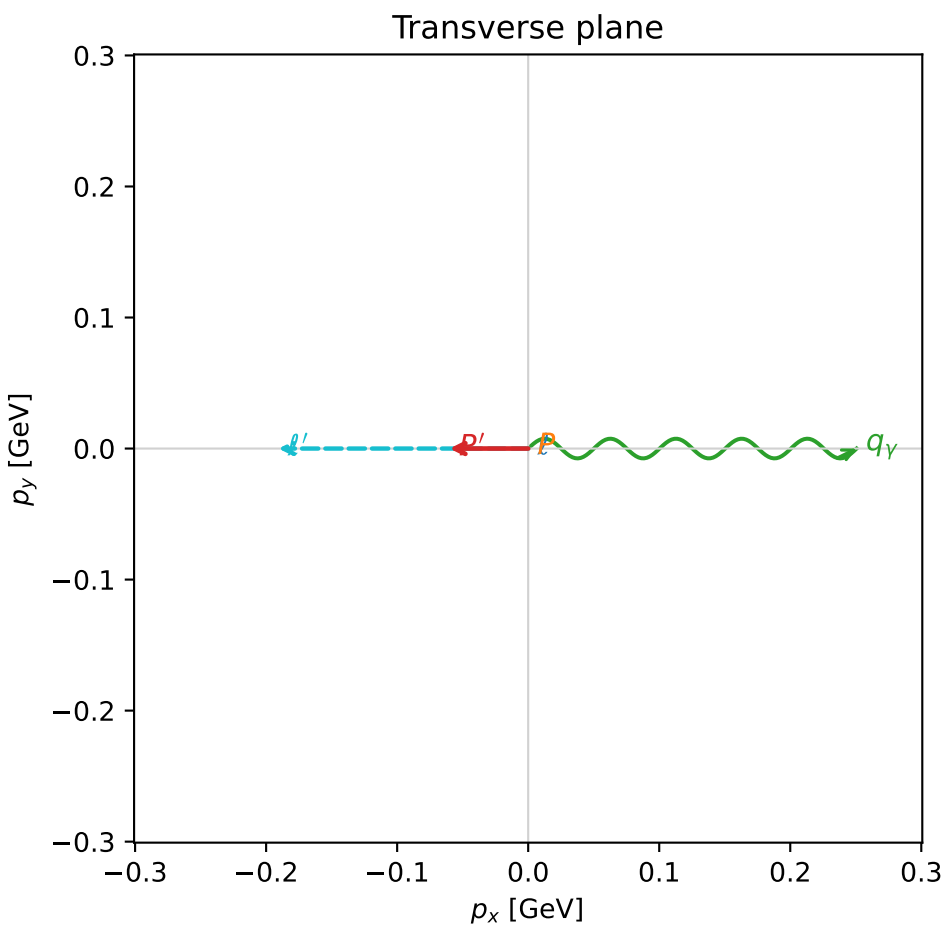
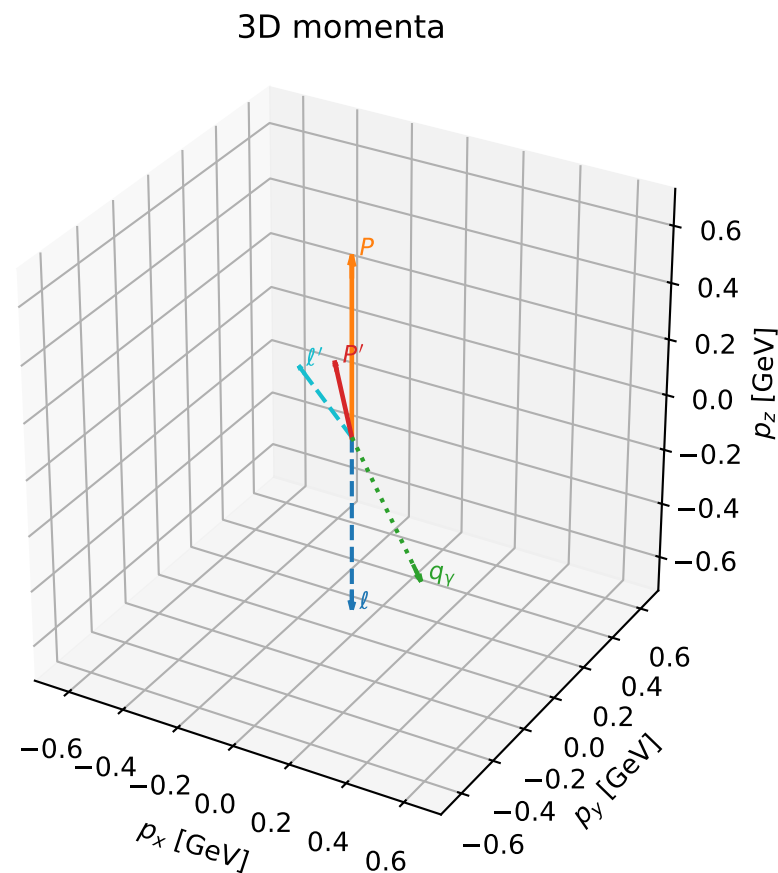
$\text{sqrt_s} = 1.401549$
 $\text{theta_p_out} = 2.42343$
 $\text{theta_gamma_out} = 1.765857$
 $\text{q0out} = 0.2317664$
 $\text{phi_p_out} = 6.283113$
 $\text{phi_gamma_out} = 3.141547$
 $\text{alpha_e} = 0.3283303$
 $\text{alpha_p} = 1.307325$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

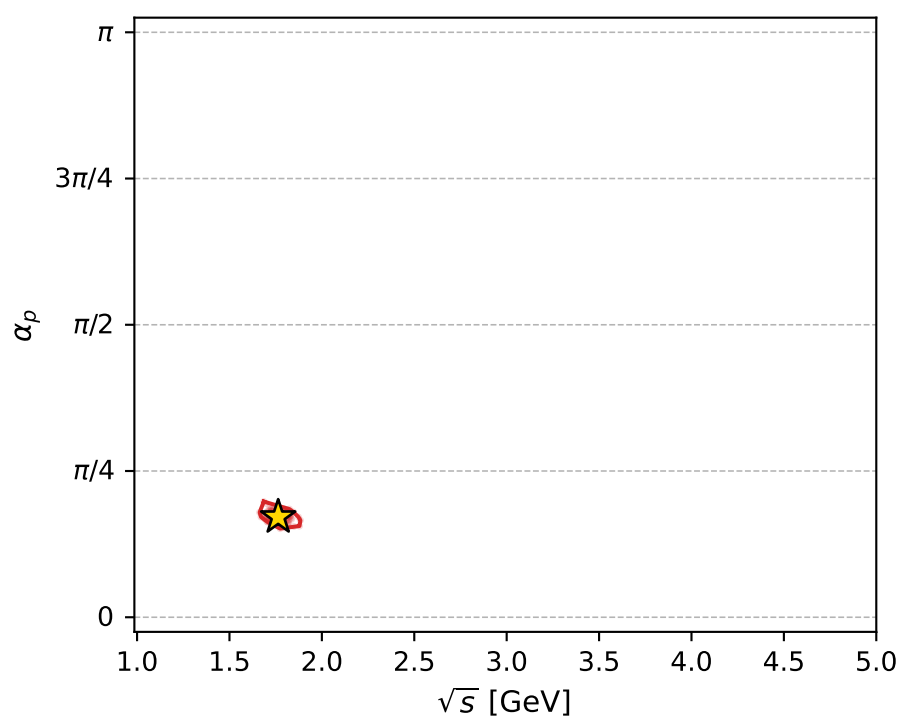
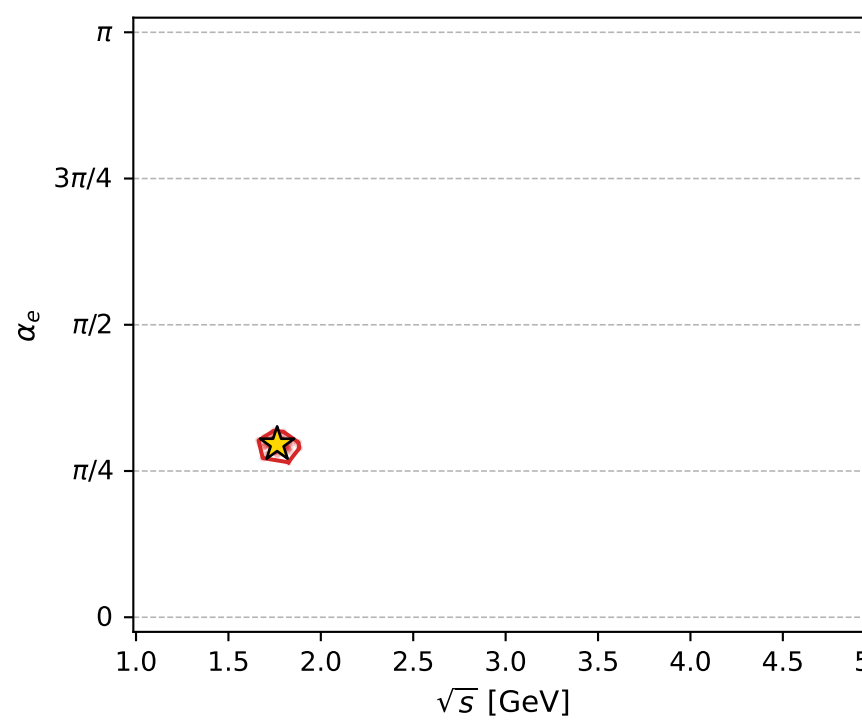
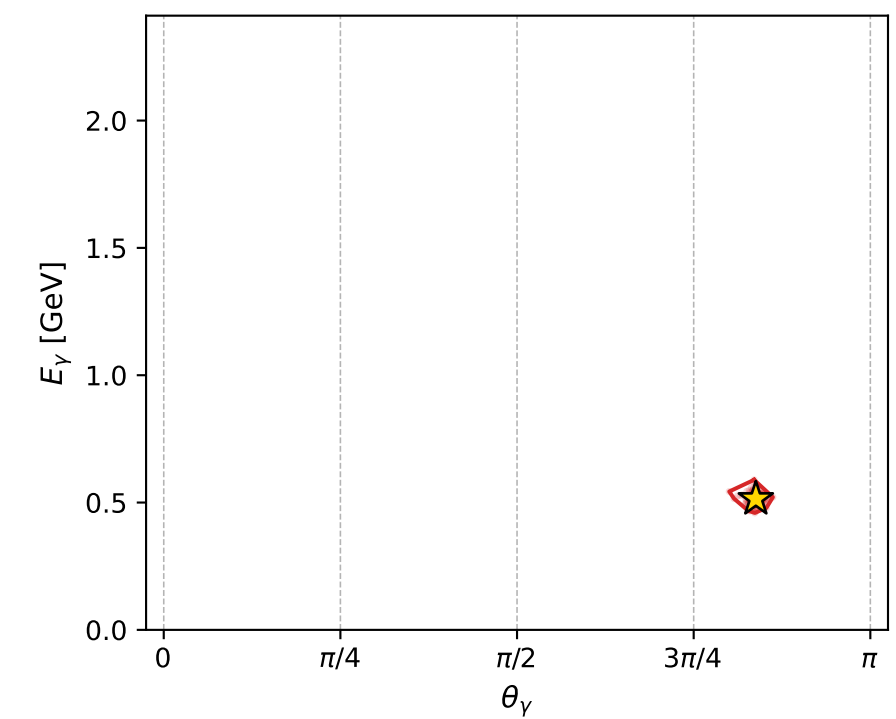
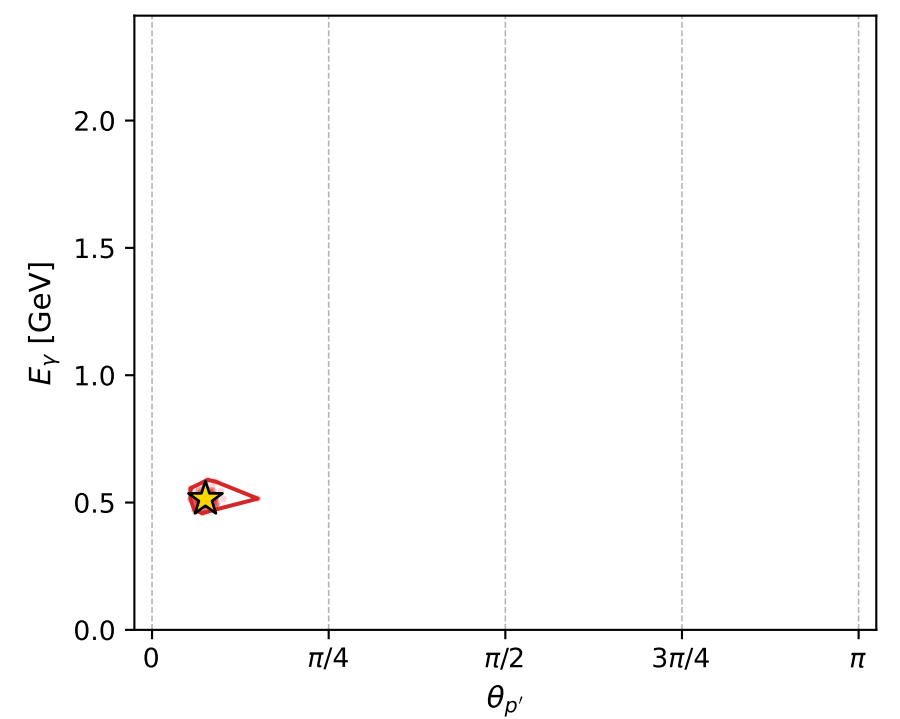
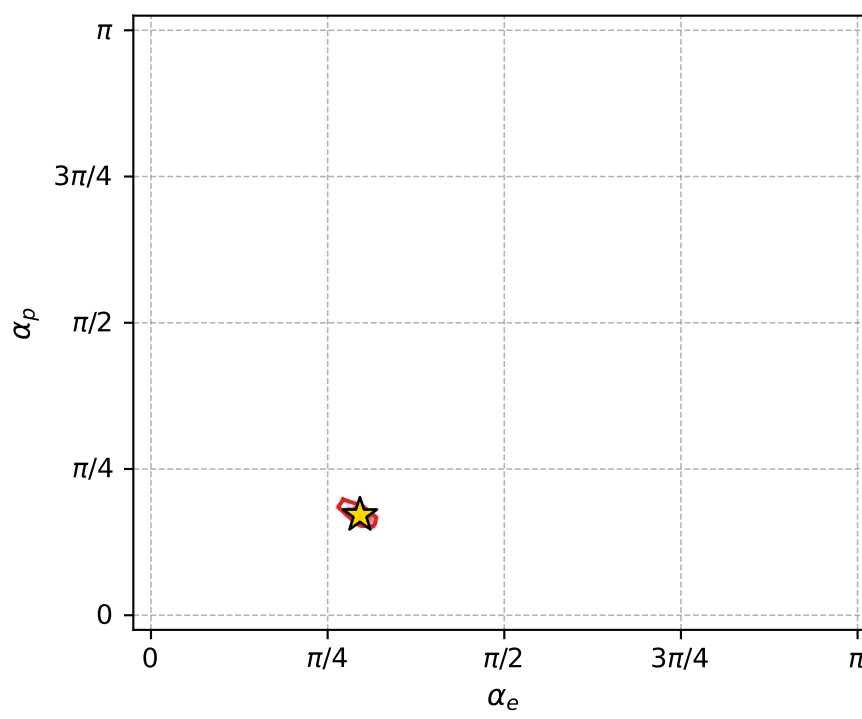
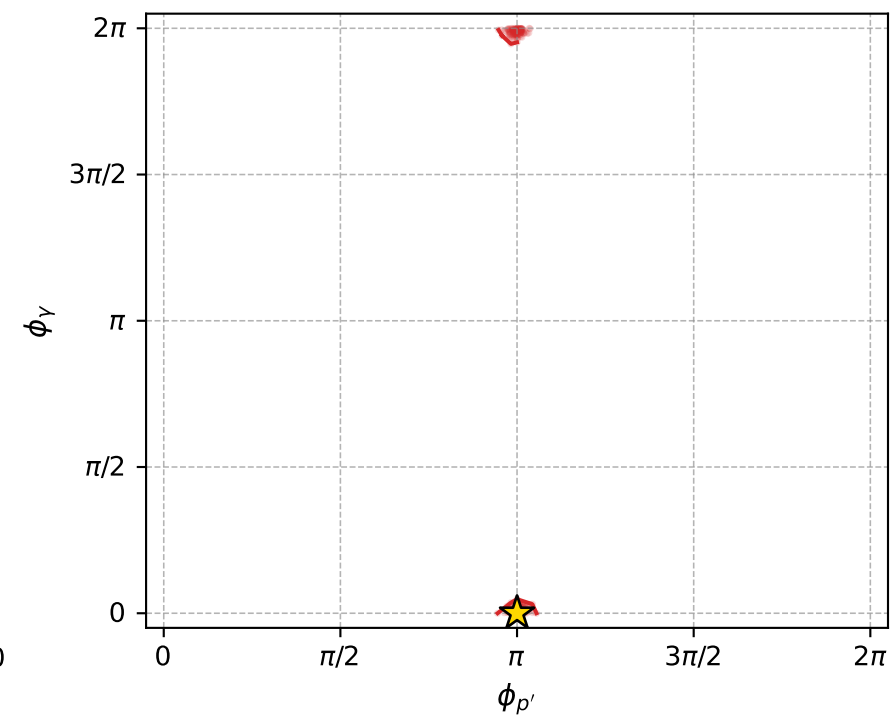
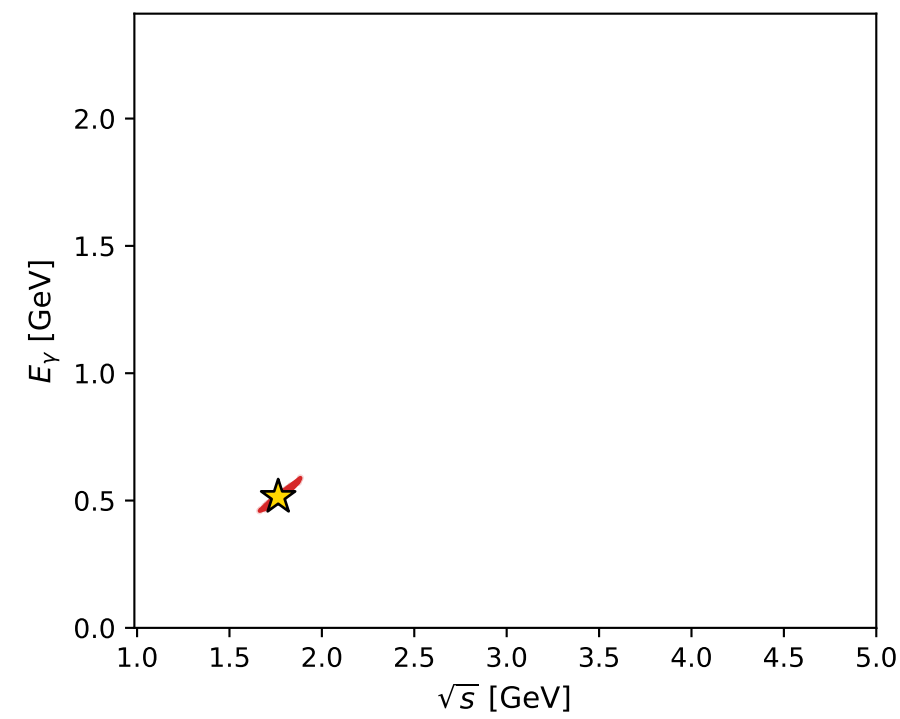
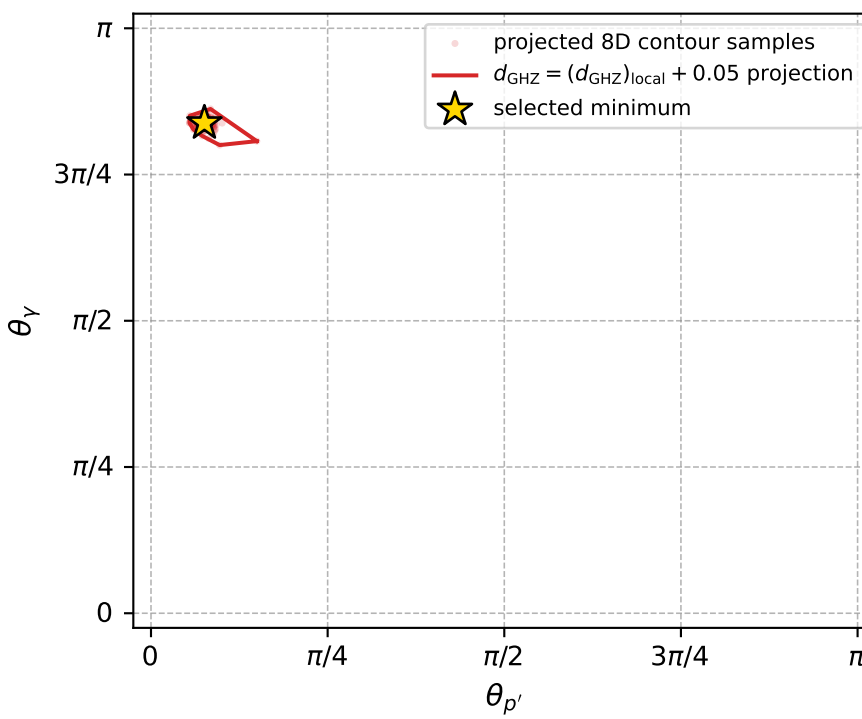
dghz_mixing_angles_polarization_01_local_minimum_269 d_{\rm GHZ}=0.000196339
 region: polarization_cluster_01_local_minimum_269
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.92911125$ rad
 incoming lepton: $0.598546|+\rangle +0.801088|-\rangle$
 proton mixing angle: $\alpha_p=0.53828034$ rad
 incoming proton: $0.858592|+\rangle +0.51266|-\rangle$

$s=3.10995$, $\sqrt{s}=1.76351$, $|\vec{P}|=0.632294$, $|\vec{P}'|=0.25525$
 $\theta_{P'}=0.237413$, $\theta_V=2.6326$, $\phi_{P'}=3.14162$, $\phi_V=2.50608\text{e-}05$
 $E_V=0.514348$, $\theta_{l'}=0.75873$, $\phi_{l'}=3.14162$
 $Q^2=0.604606$, $x_B=0.325484$, $t=-0.125903$, $W^2=2.1328$, $y=0.832947$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.6323$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6323$
 P $E= 1.1312$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6323$
 l' $E= 0.2770$ $p_x= -0.1906$ $p_y= -0.0000$ $p_z= 0.2011$
 P' $E= 0.9721$ $p_x= -0.0600$ $p_y= -0.0000$ $p_z= 0.2481$
 q_V $E= 0.5143$ $p_x= 0.2506$ $p_y= 0.0000$ $p_z= -0.4491$



electron: polarization cluster P1, local minimum 269; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00019633925$
 $d_{\text{GHZ}} \text{ contour} = 0.050196339$
 above global minimum = 0.00019631696
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.763506$
 $\text{theta_p_out} = 0.2374128$
 $\text{theta_gamma_out} = 2.632599$
 $\text{qOut} = 0.5143482$
 $\text{phi_p_out} = 3.141623$
 $\text{phi_gamma_out} = 2.506082\text{e-}05$
 $\text{alpha_e} = 0.9291113$
 $\text{alpha_p} = 0.5382803$

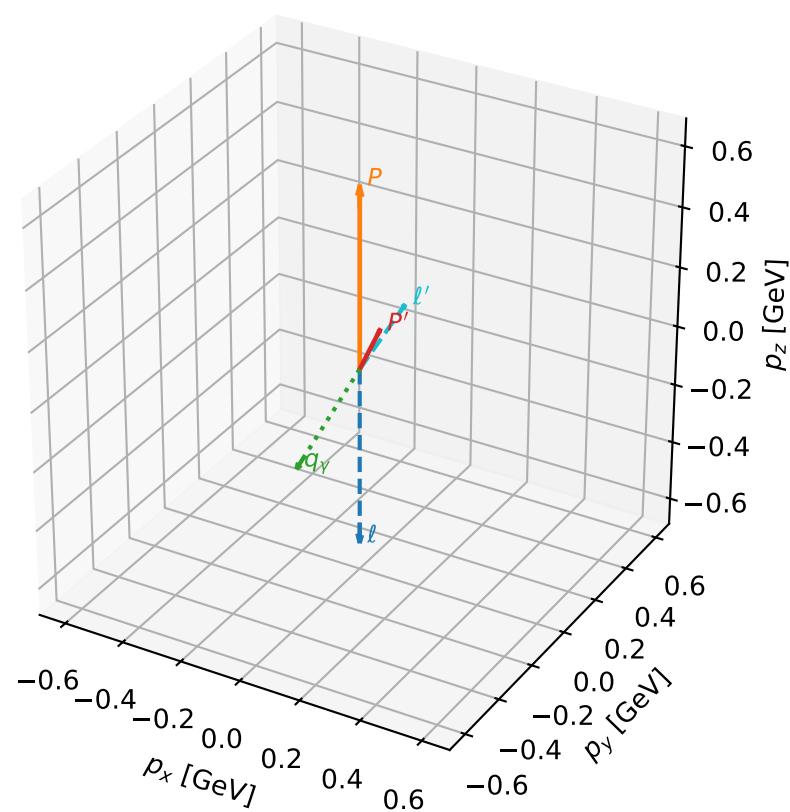
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_277 $d_{\{\text{rm GHZ}\}}=0.00022324$
 region: polarization_cluster_01_local_minimum_277
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.42740399$ rad
 incoming lepton: $0.910045|+\rangle +0.41451|-\rangle$
 proton mixing angle: $\alpha_p=1.2172573$ rad
 incoming proton: $0.34622|+\rangle +0.938153|-\rangle$

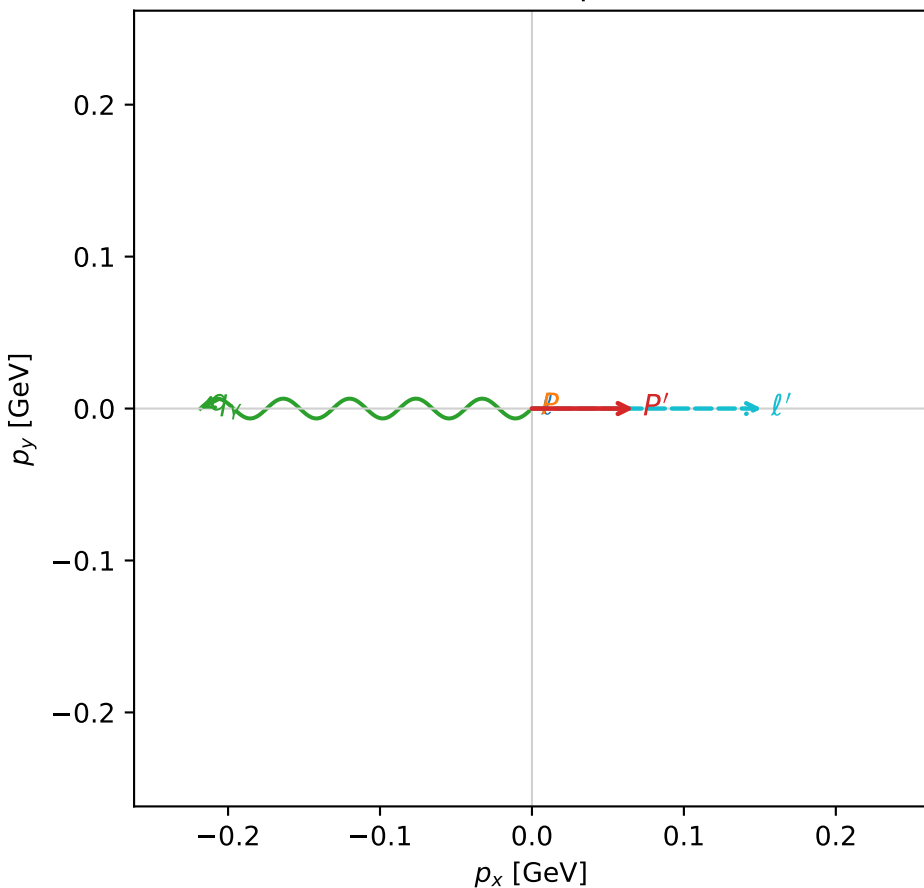
$s=2.93673$, $\sqrt{s}=1.71369$, $|\vec{P}|=0.600133$, $|\vec{P}'|=0.163303$
 $\theta_{P'}=0.422621$, $\theta_Y=2.64982$, $\phi_{P'}=7.11256\text{e-}05$, $\phi_Y=3.14164$
 $E_Y=0.462139$, $\theta_{\ell'}=0.529467$, $\phi_{\ell'}=3.10587\text{e-}05$
 $Q^2=0.669602$, $x_B=0.393839$, $t=-0.182001$, $W^2=1.91043$, $y=0.826587$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6001$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6001$
 P $E= 1.1136$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6001$
 ℓ' $E= 0.2994$ $p_x= 0.1512$ $p_y= 0.0000$ $p_z= 0.2584$
 P' $E= 0.9521$ $p_x= 0.0670$ $p_y= 0.0000$ $p_z= 0.1489$
 q_Y $E= 0.4621$ $p_x= -0.2182$ $p_y= -0.0000$ $p_z= -0.4074$

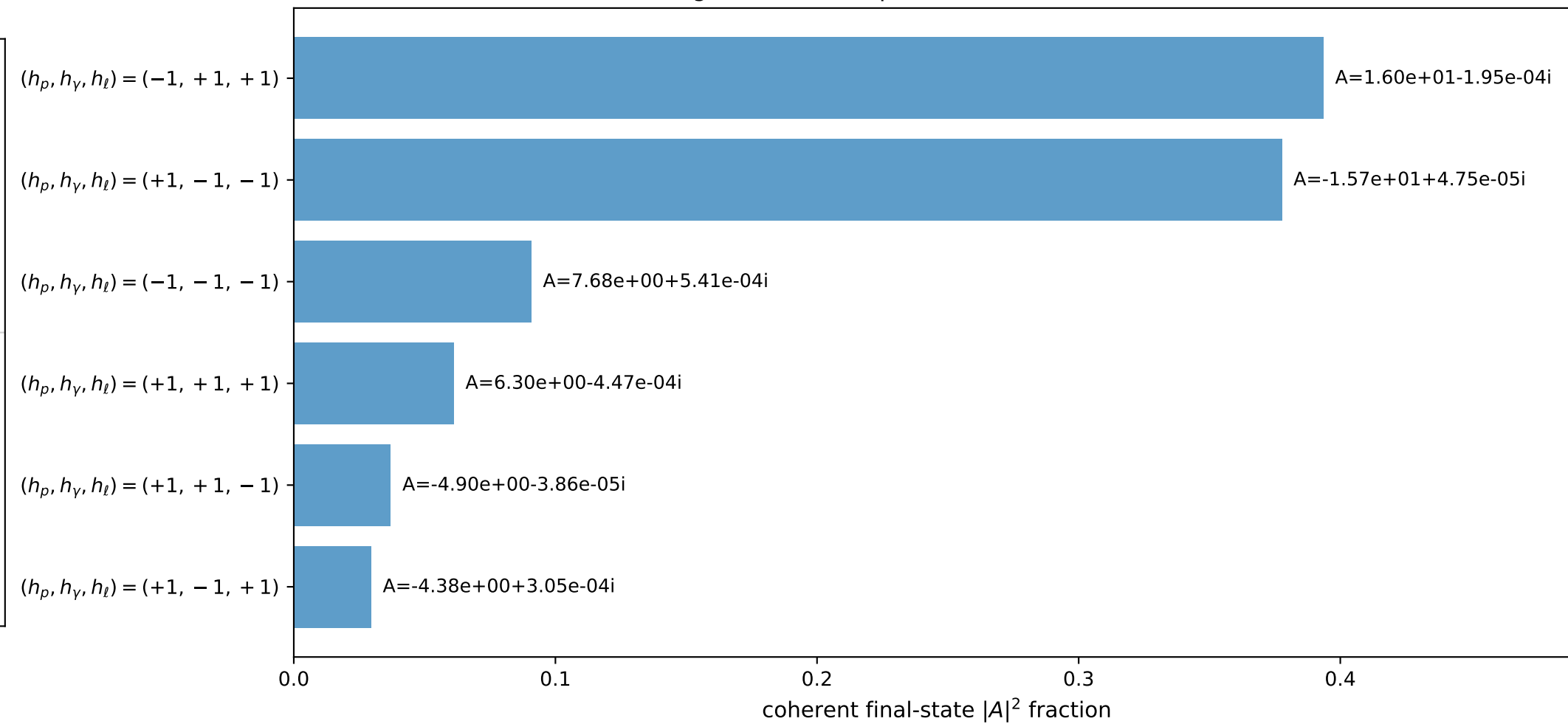
3D momenta



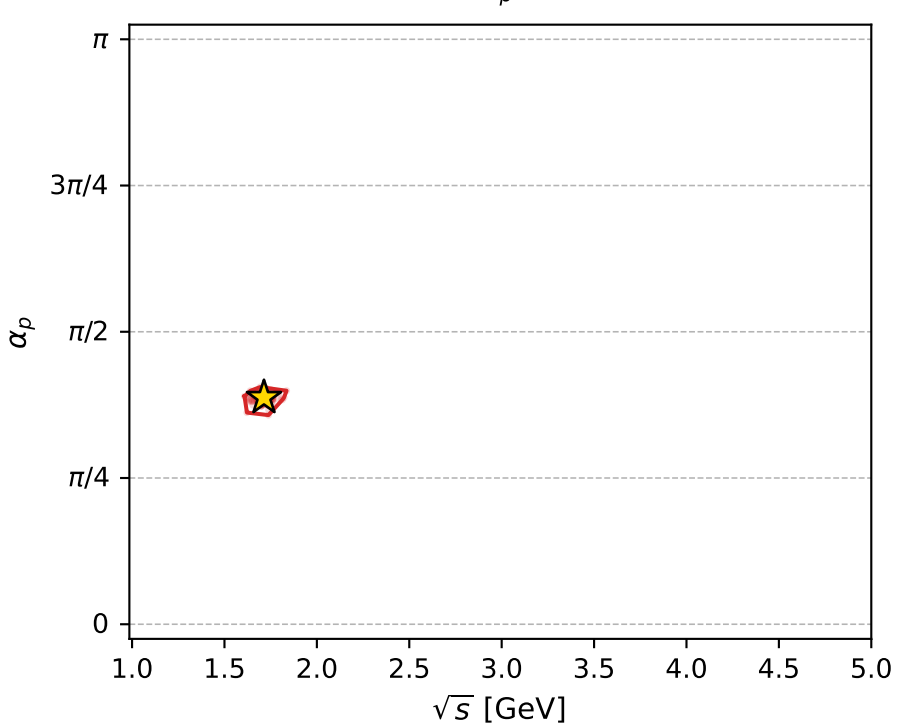
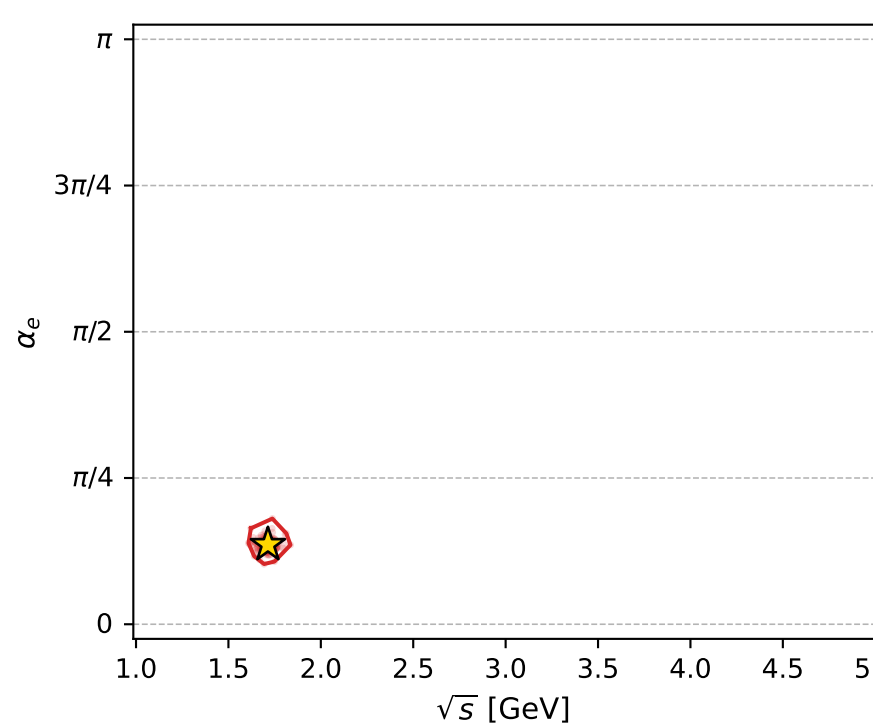
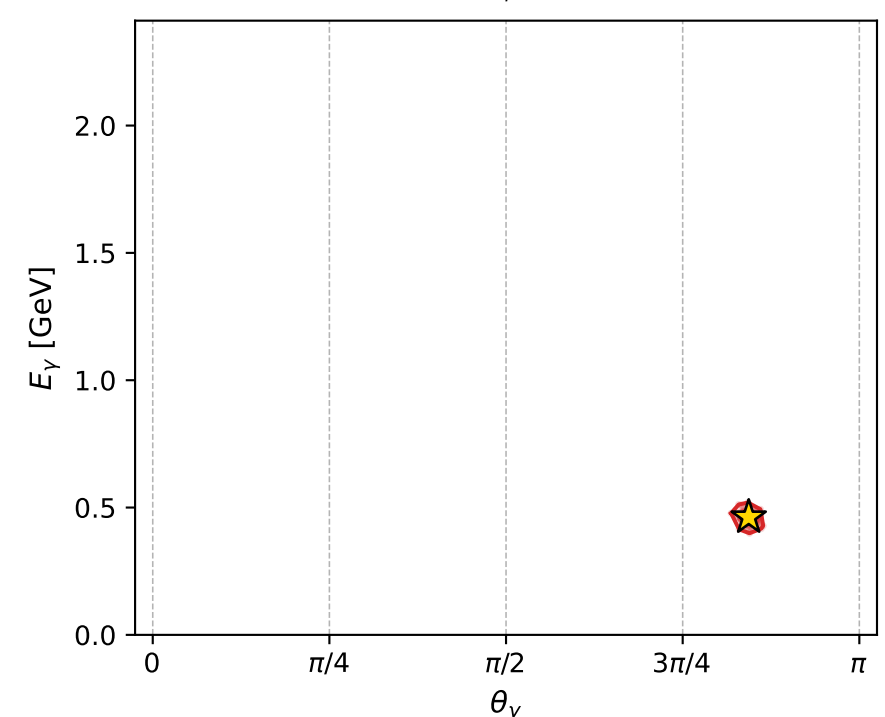
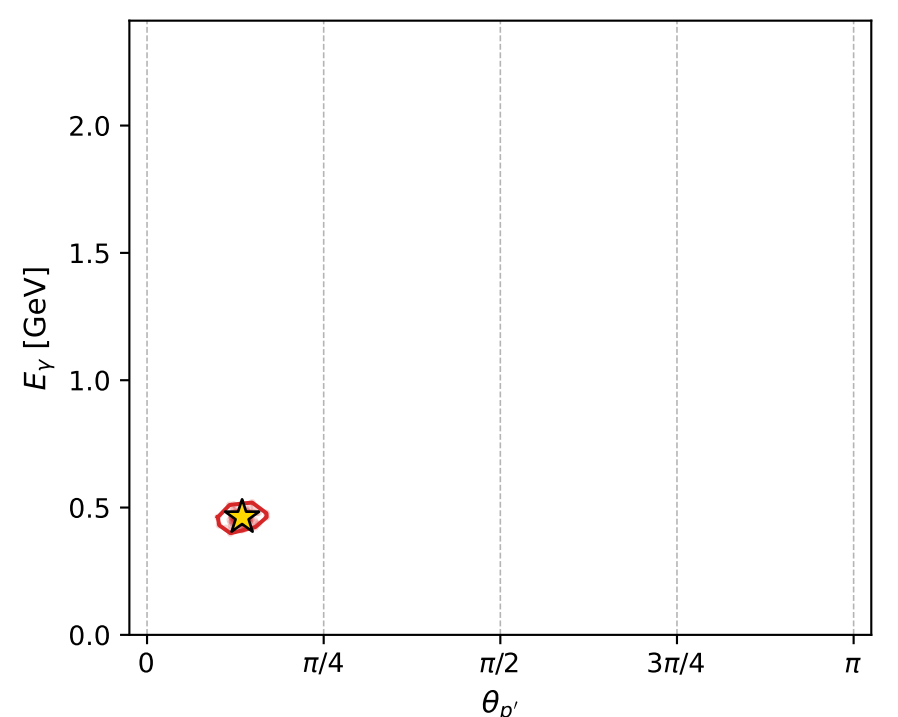
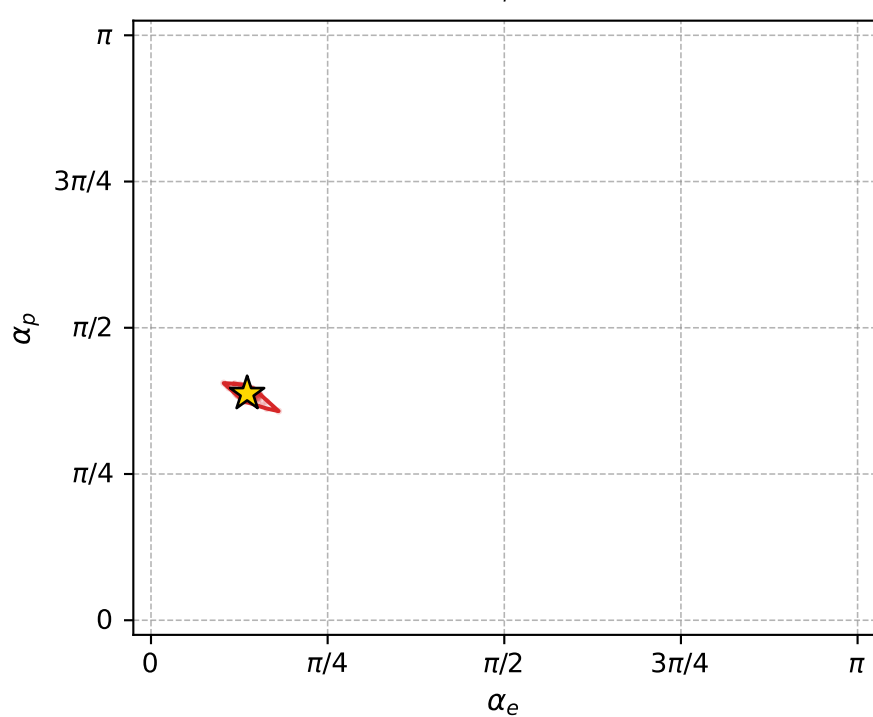
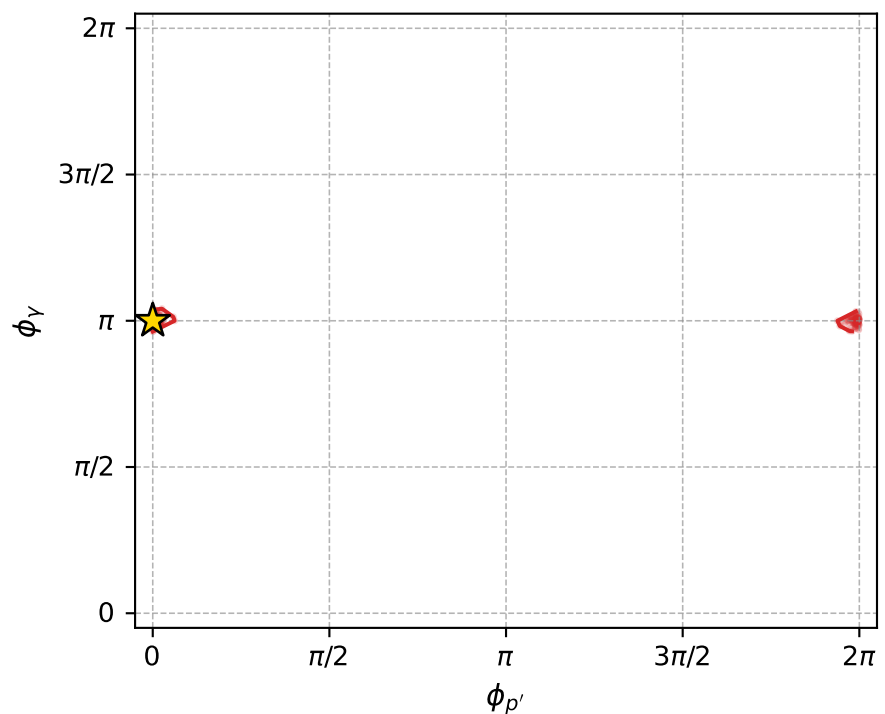
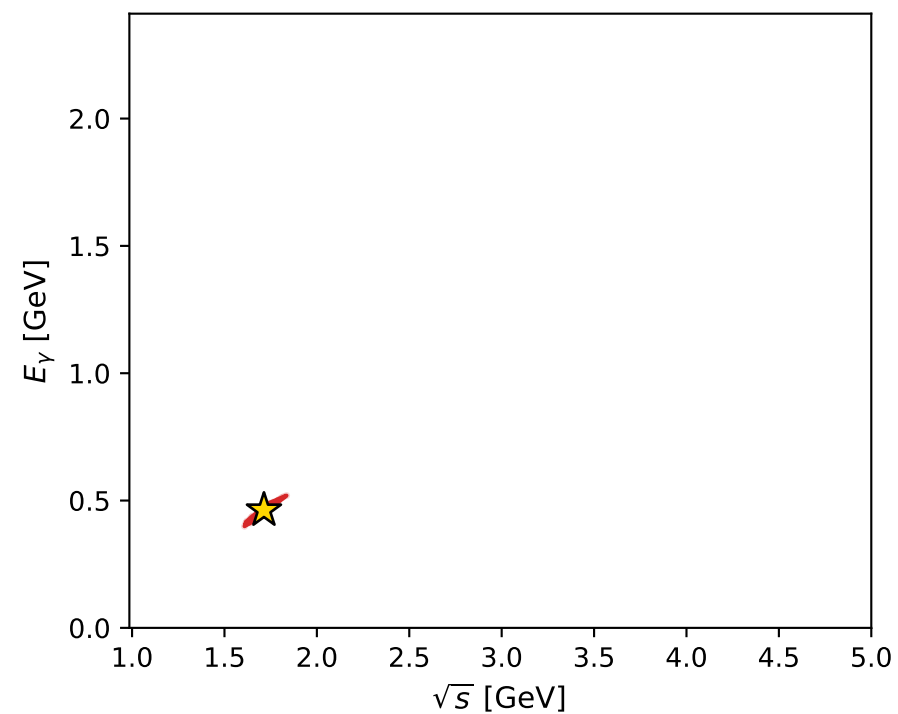
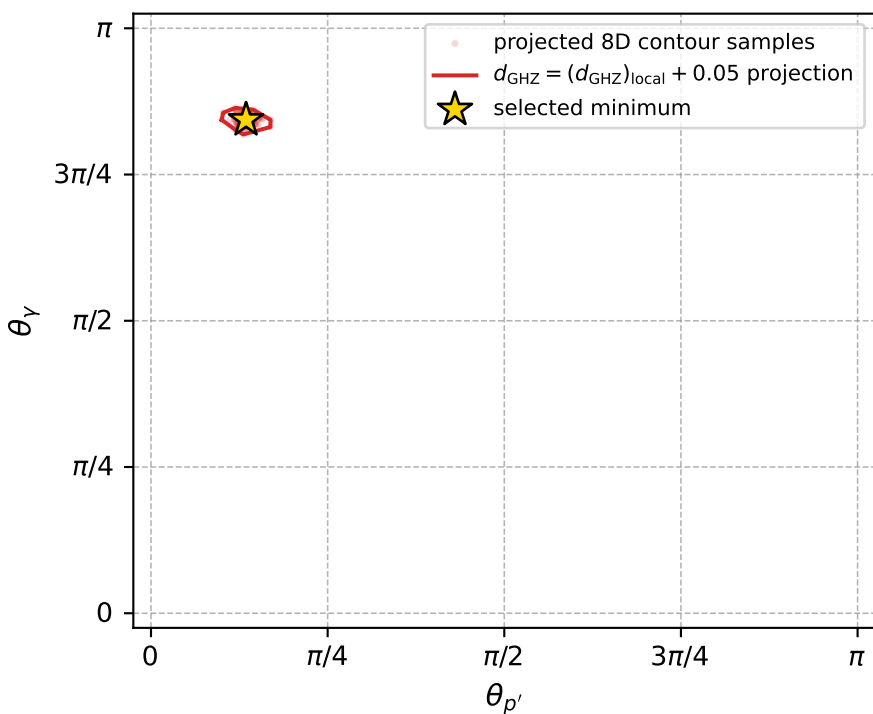
Transverse plane



Leading final-state amplitudes (N=6, retained=98.9%)



electron: polarization cluster P1, local minimum 277; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00022323994$
 $d_{\text{GHZ}} \text{ contour} = 0.05022324$
 above global minimum = 0.00022321765
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.713688$
 $\text{theta_p_out} = 0.4226211$
 $\text{theta_gamma_out} = 2.649818$
 $\text{q0ut} = 0.4621391$
 $\text{phi_p_out} = 7.112565\text{e-}05$
 $\text{phi_gamma_out} = 3.141636$
 $\text{alpha_e} = 0.427404$
 $\text{alpha_p} = 1.217257$

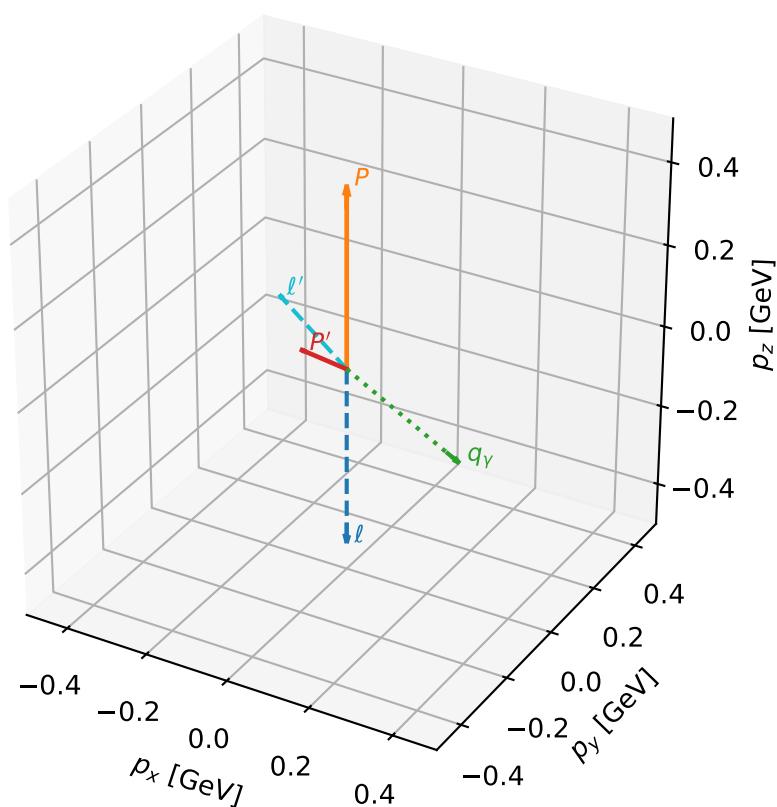
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_282 $d_{\{\text{rm GHZ}\}}=0.000234223$
 region: polarization_cluster_01_local_minimum_282
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.92579382$ rad
 incoming lepton: $0.6012|+\rangle + 0.799098|-\rangle$
 proton mixing angle: $\alpha_p=0.53762923$ rad
 incoming proton: $0.858925|+\rangle + 0.512101|-\rangle$

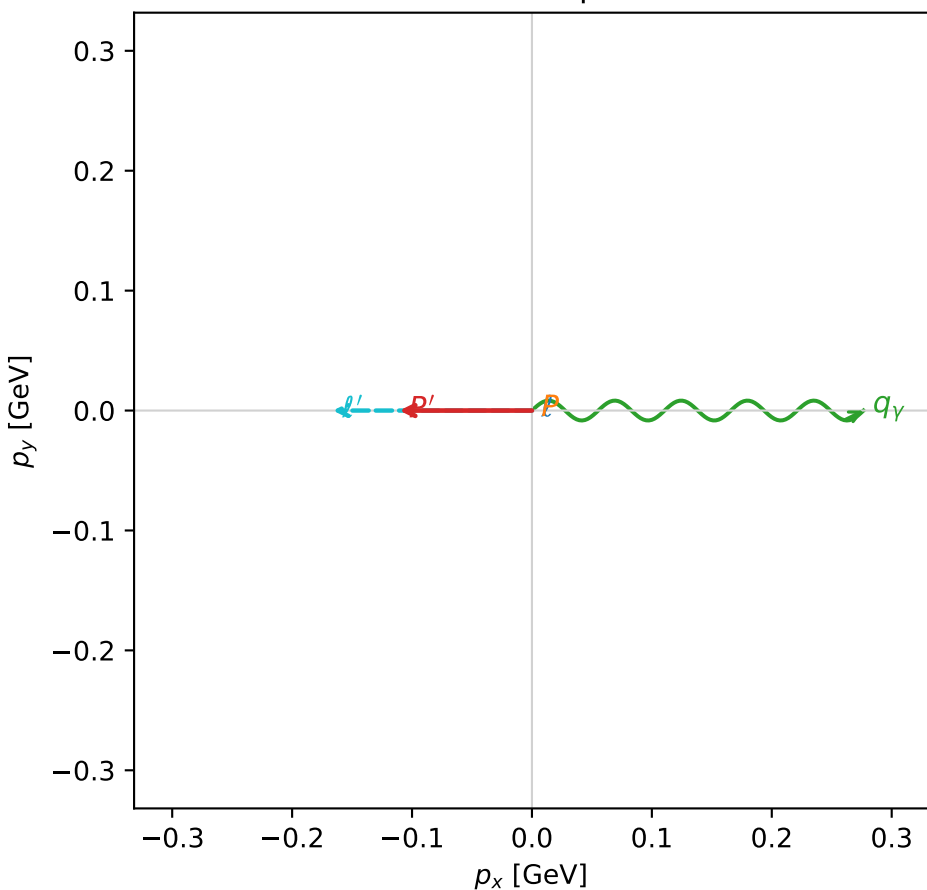
$s=2.17181$, $\sqrt{s}=1.47371$, $|\vec{P}|=0.43834$, $|\vec{P}'|=0.111547$
 $\theta_{p'}=1.43397$, $\theta_Y=2.06968$, $\phi_{p'}=3.14164$, $\phi_Y=2.10331\text{e-}05$
 $E_Y=0.314869$, $\theta_{\ell'}=0.88642$, $\phi_{\ell'}=3.14159$
 $Q^2=0.306542$, $x_B=0.316974$, $t=-0.183009$, $W^2=1.54039$, $y=0.748539$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4383$
 P $E= 1.0354$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4383$
 ℓ' $E= 0.2142$ $p_x= -0.1660$ $p_y= -0.0000$ $p_z= 0.1354$
 P' $E= 0.9446$ $p_x= -0.1105$ $p_y= -0.0000$ $p_z= 0.0152$
 q_Y $E= 0.3149$ $p_x= 0.2765$ $p_y= 0.0000$ $p_z= -0.1506$

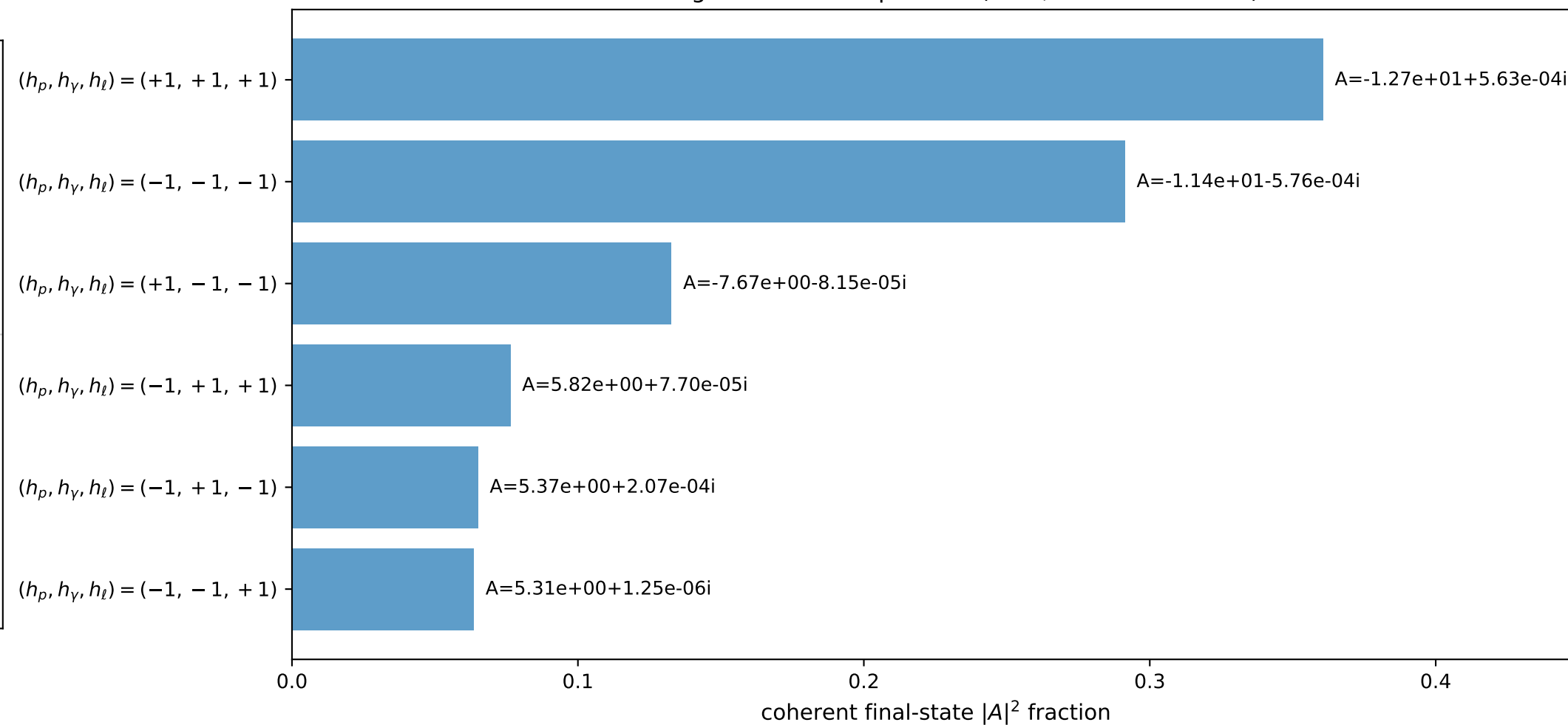
3D momenta



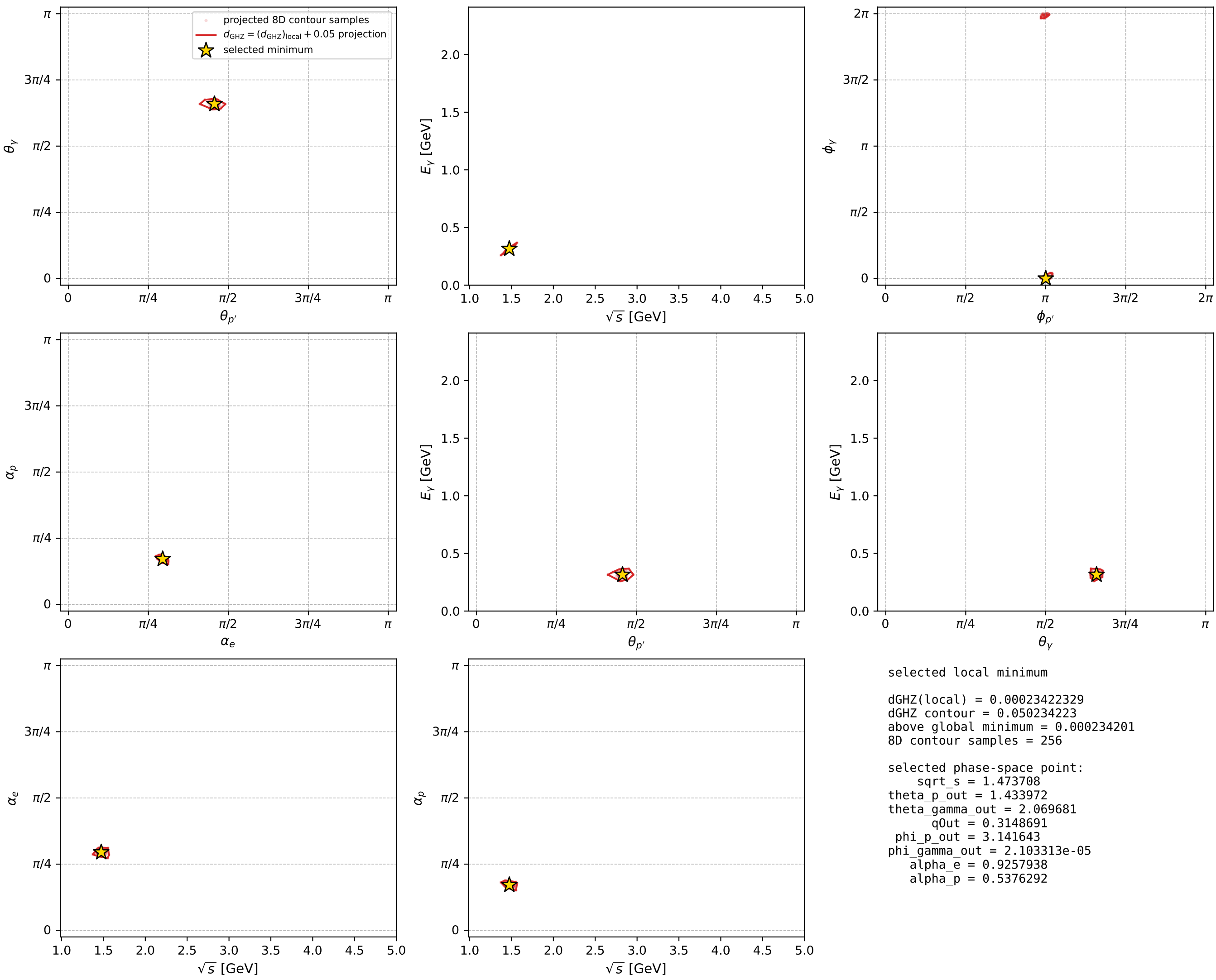
Transverse plane



Leading final-state amplitudes (N=6, retained=98.9%)



electron: polarization cluster P1, local minimum 282; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



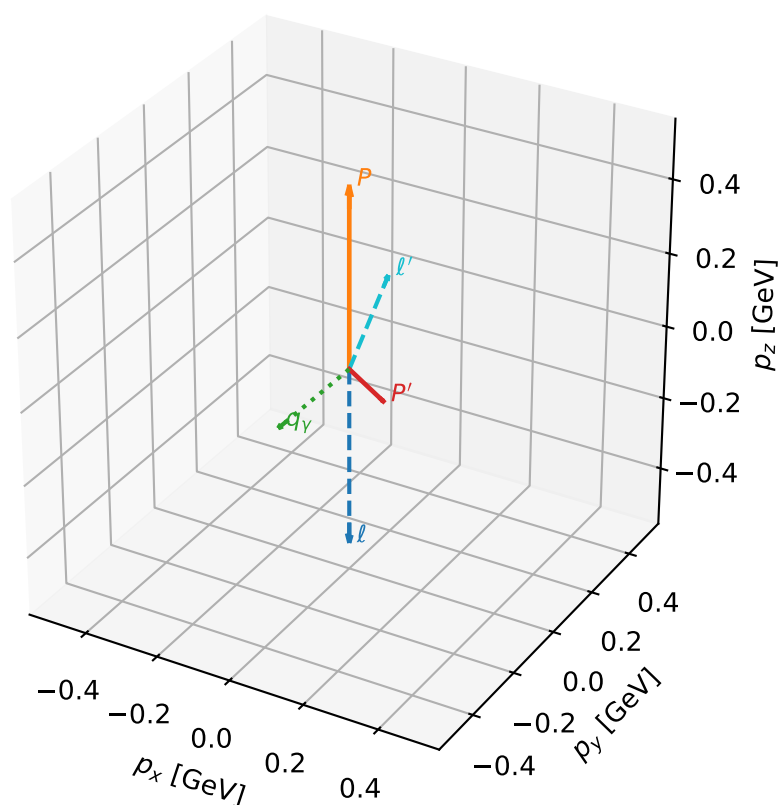
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_289 $d_{\{\text{rm GHZ}\}}=0.000275715$
 region: polarization_cluster_01_local_minimum_289
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.24333635$ rad
 incoming lepton: $0.97054|+\rangle +0.240942|-\rangle$
 proton mixing angle: $\alpha_p=1.3775684$ rad
 incoming proton: $0.192028|+\rangle +0.98139|-\rangle$

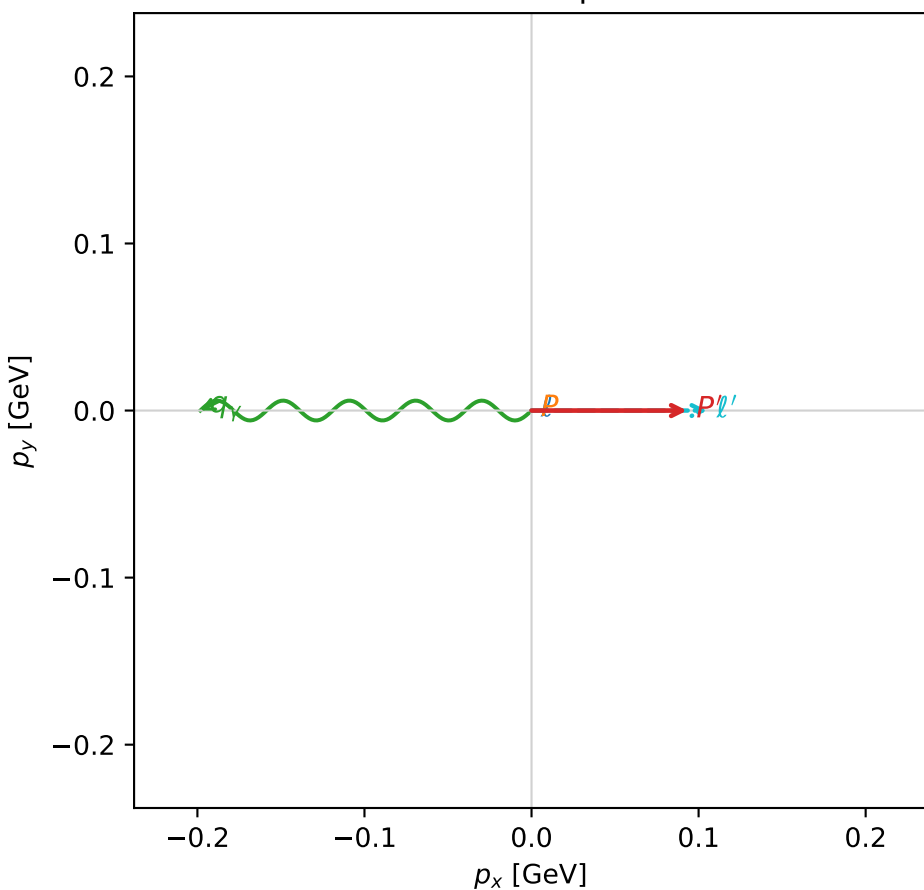
$s=2.38391$, $\sqrt{s}=1.54399$, $|\vec{P}|=0.48707$, $|\vec{P}'|=0.111271$
 $\theta_{P'}=2.15082$, $\theta_Y=2.41248$, $\phi_{P'}=6.28307$, $\phi_Y=3.14153$
 $E_Y=0.297584$, $\theta_{\ell'}=0.355942$, $\phi_{\ell'}=6.28317$
 $Q^2=0.569619$, $x_B=0.498949$, $t=-0.296402$, $W^2=1.45186$, $y=0.759035$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4871$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4871$
 P $E= 1.0569$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4871$
 ℓ' $E= 0.3018$ $p_x= 0.1052$ $p_y= -0.0000$ $p_z= 0.2829$
 P' $E= 0.9446$ $p_x= 0.0931$ $p_y= -0.0000$ $p_z= -0.0610$
 q_Y $E= 0.2976$ $p_x= -0.1983$ $p_y= 0.0000$ $p_z= -0.2219$

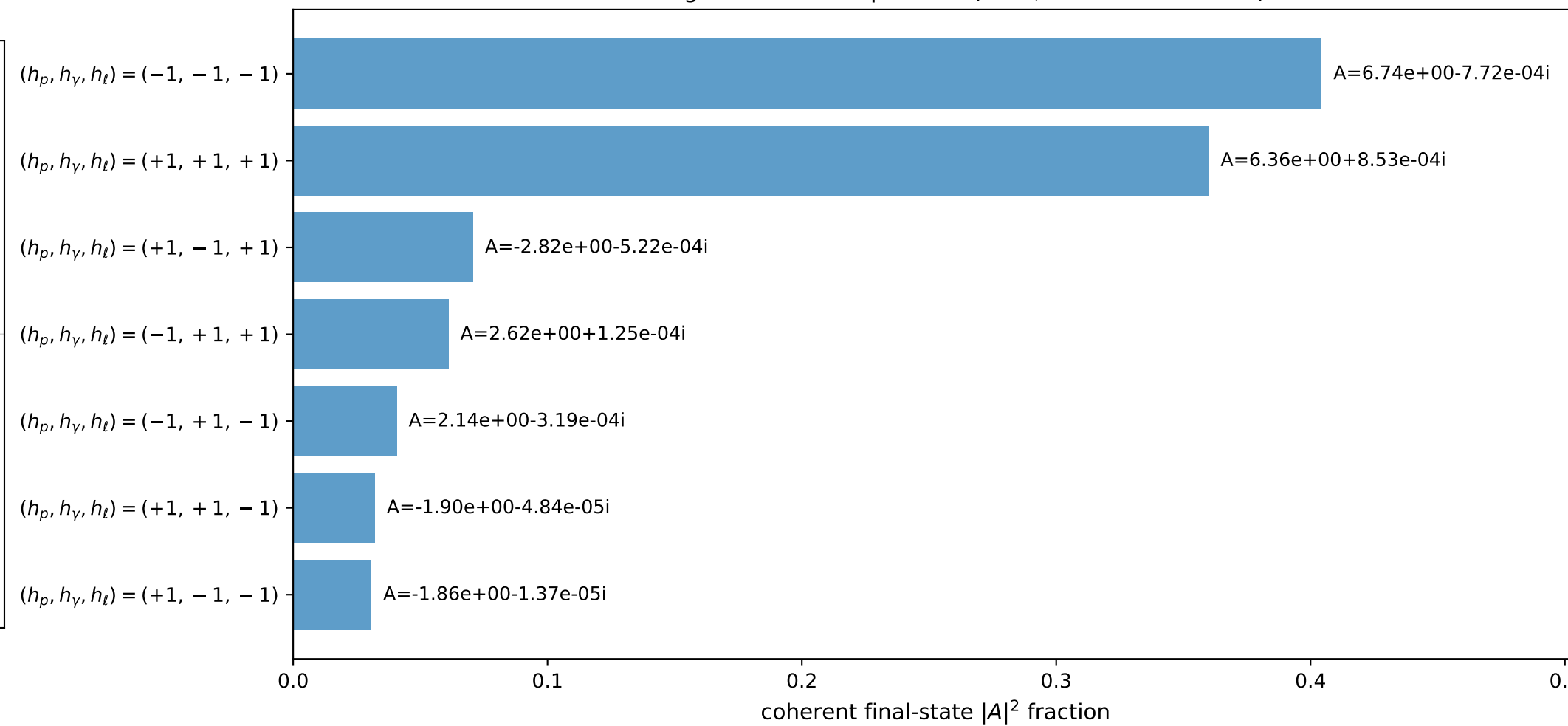
3D momenta



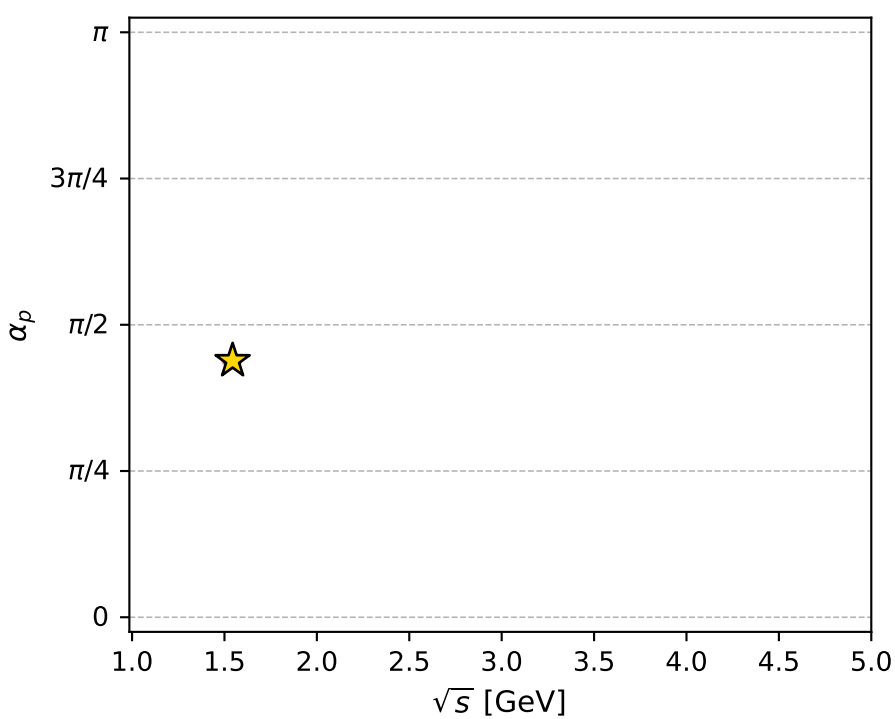
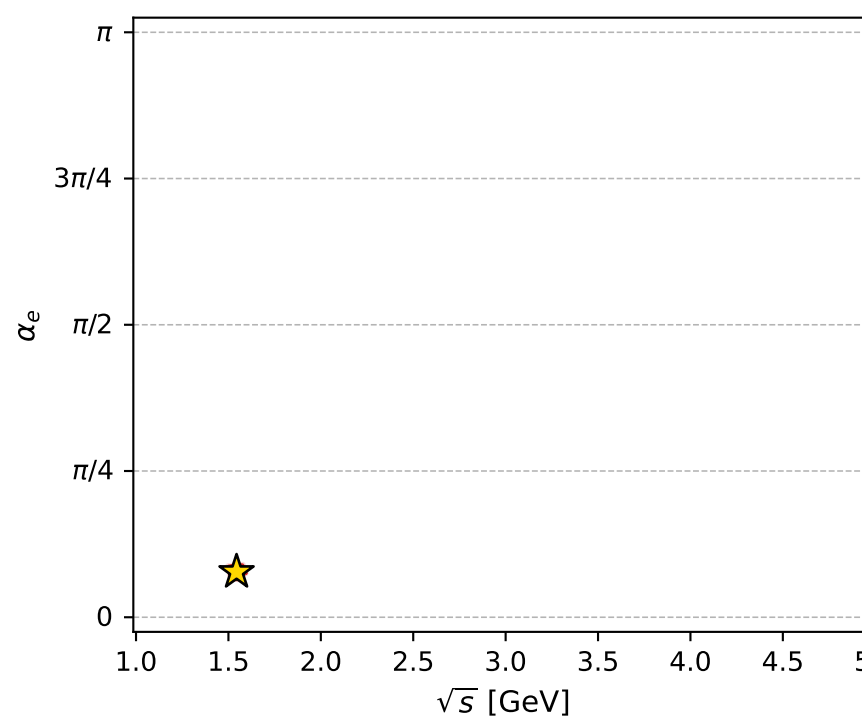
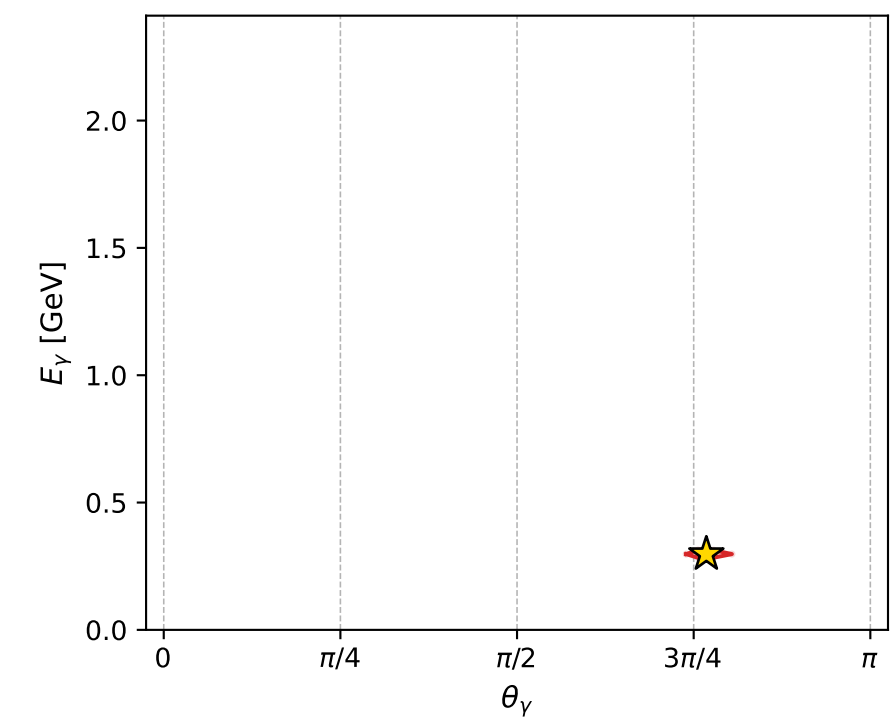
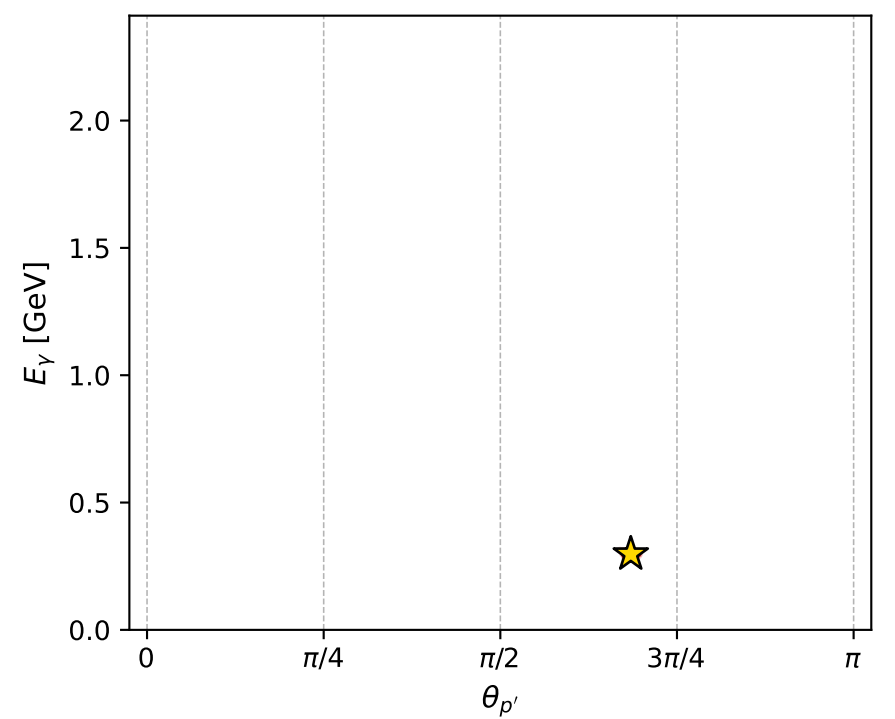
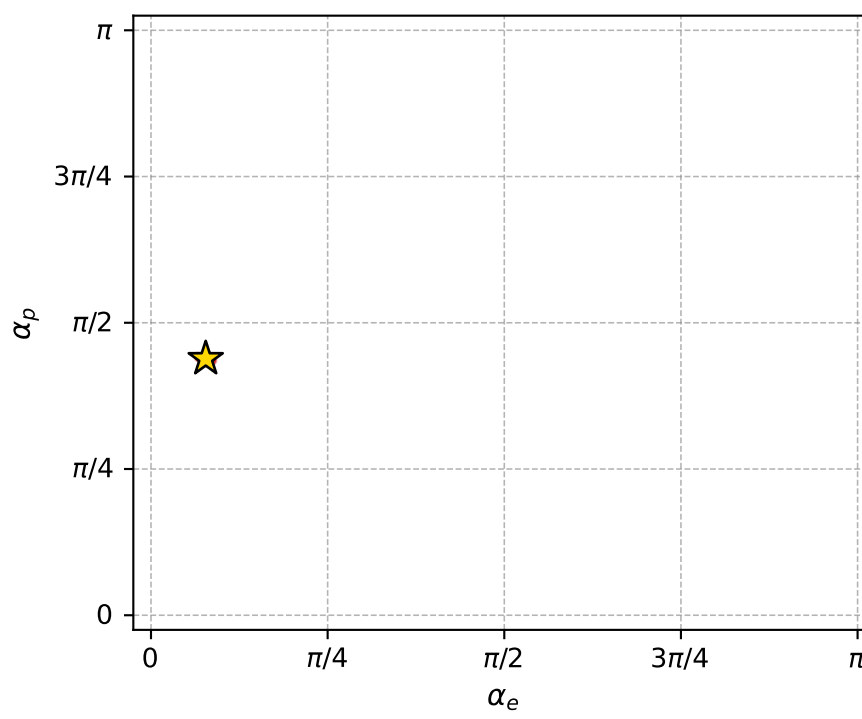
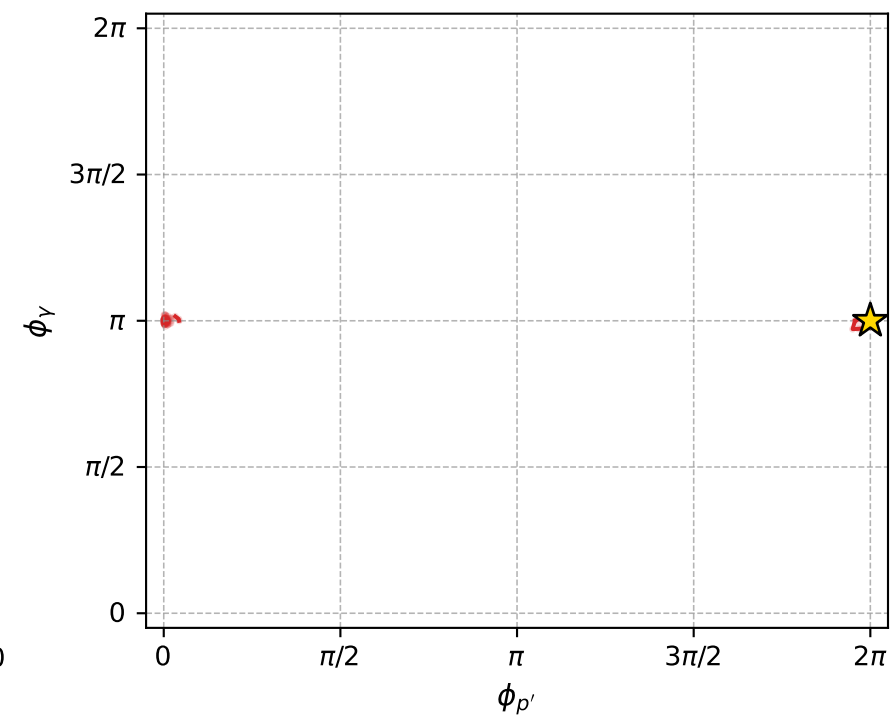
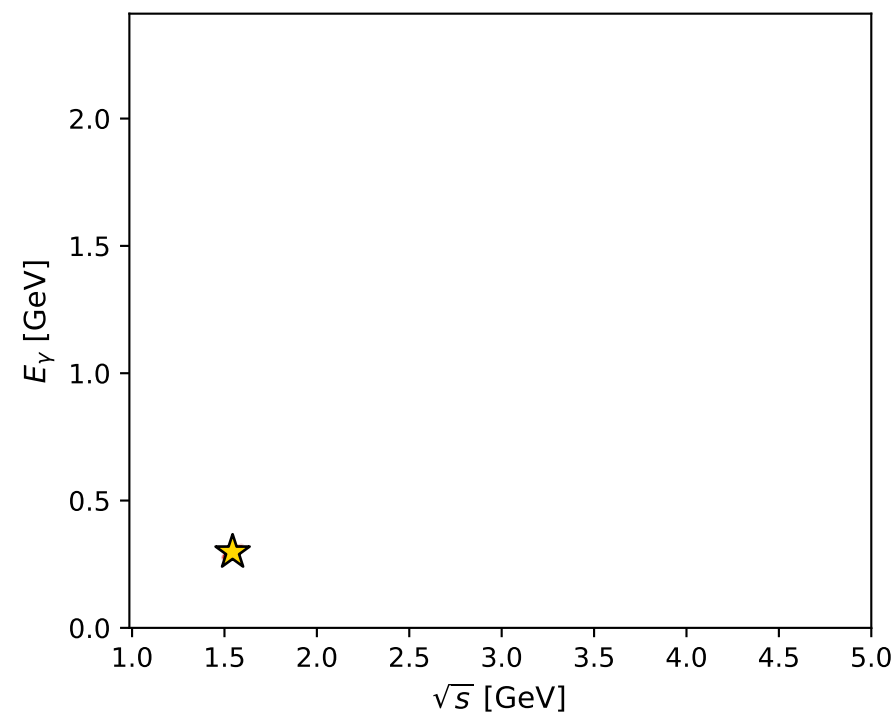
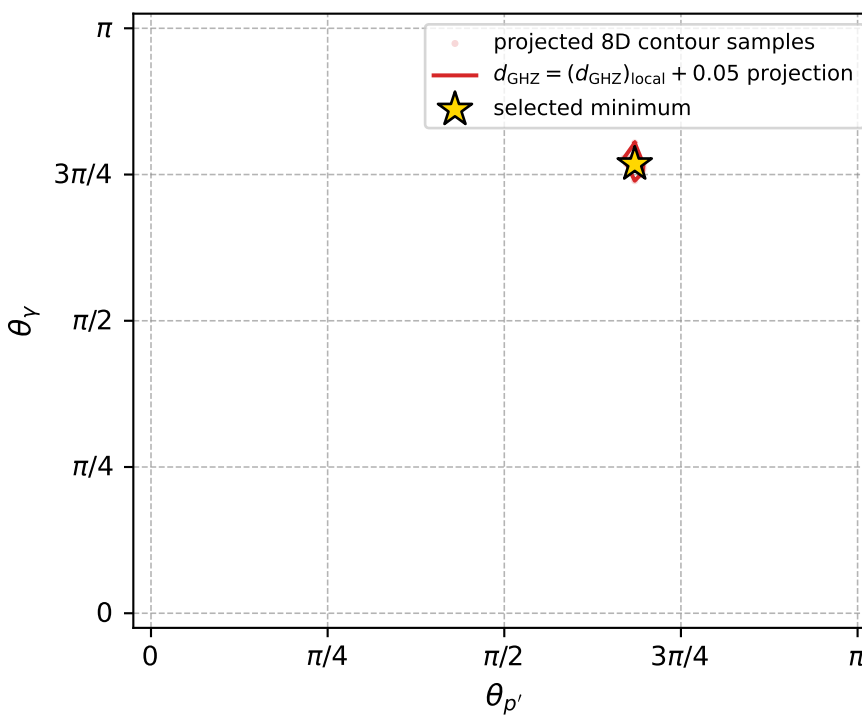
Transverse plane



Leading final-state amplitudes (N=7, retained=100.0%)



electron: polarization cluster P1, local minimum 289; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00027571491$
 $d_{\text{GHZ}} \text{ contour} = 0.050275715$
 above global minimum = 0.00027569263
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.543991$
 $\text{theta_p_out} = 2.150821$
 $\text{theta_gamma_out} = 2.412484$
 $\text{q0ut} = 0.2975843$
 $\text{phi_p_out} = 6.283067$
 $\text{phi_gamma_out} = 3.141529$
 $\text{alpha_e} = 0.2433364$
 $\text{alpha_p} = 1.377568$

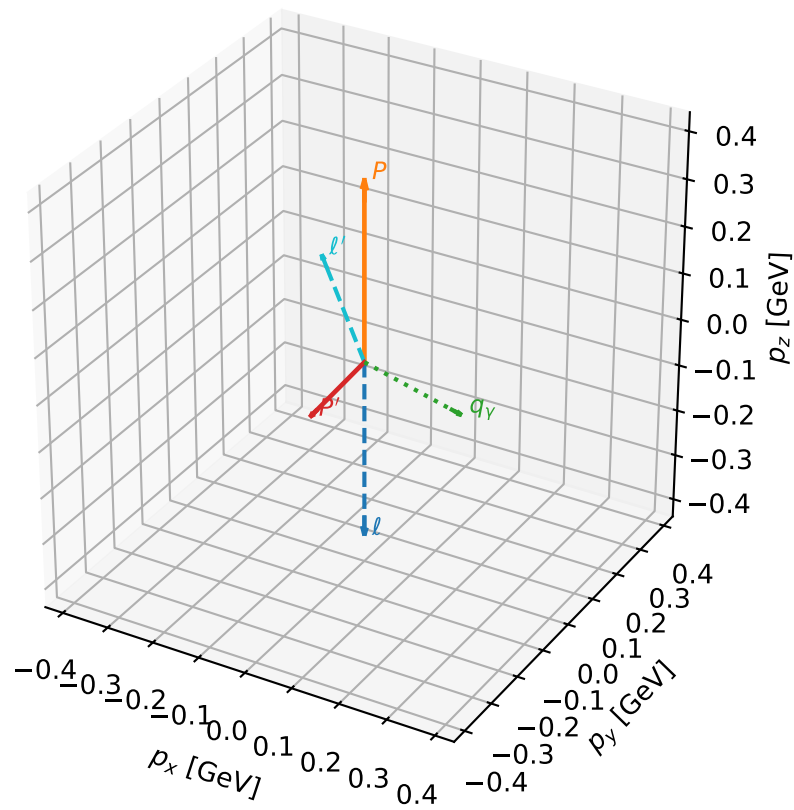
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_291 $d_{\{\text{rm GHZ}\}}=0.00027839$
 region: polarization_cluster_01_local_minimum_291
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.296893$ rad
 incoming lepton: $0.270491|+\rangle +0.962722|-\rangle$
 proton mixing angle: $\alpha_p=0.22100604$ rad
 incoming proton: $0.975677|+\rangle +0.219211|-\rangle$

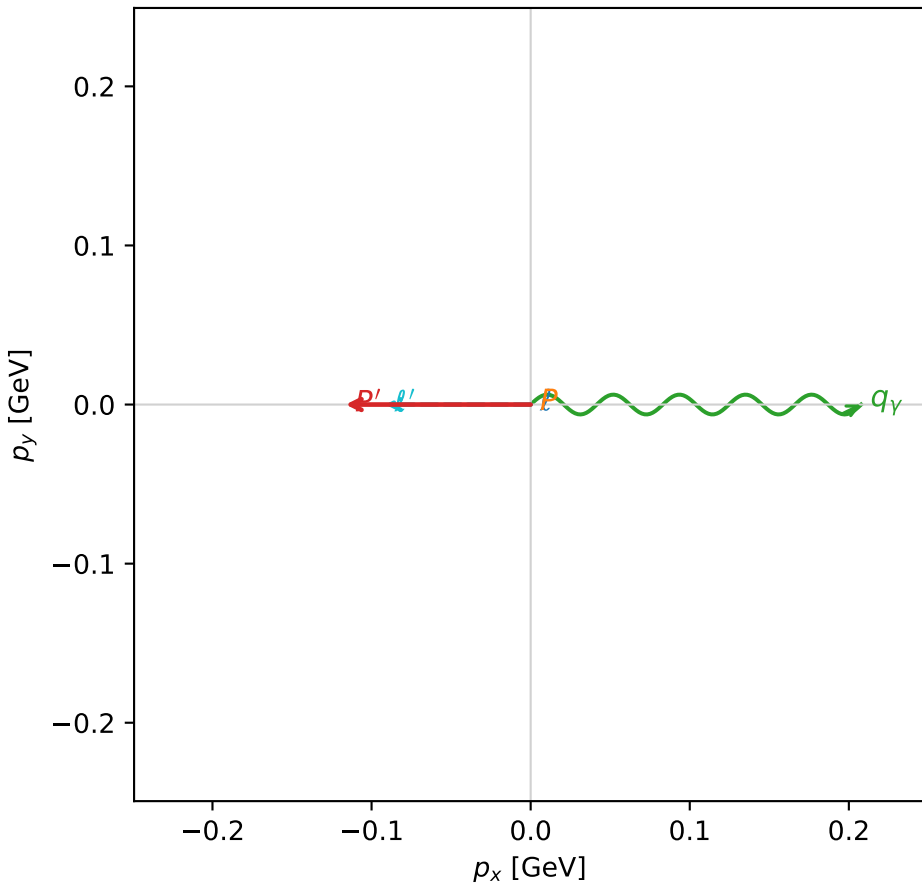
$s=1.94522$, $\sqrt{s}=1.39471$, $|\vec{P}|=0.381933$, $|\vec{P}'|=0.190351$
 $\theta_{P'}=2.48415$, $\theta_Y=1.82045$, $\phi_{P'}=3.1416$, $\phi_Y=2.70014\text{e-}05$
 $E_Y=0.214376$, $\theta_{\ell'}=0.421913$, $\phi_{\ell'}=3.14164$
 $Q^2=0.32606$, $x_B=0.424119$, $t=-0.294103$, $W^2=1.32258$, $y=0.721618$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3819$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3819$
 P $E= 1.0128$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3819$
 ℓ' $E= 0.2232$ $p_x= -0.0914$ $p_y= -0.0000$ $p_z= 0.2036$
 P' $E= 0.9571$ $p_x= -0.1163$ $p_y= -0.0000$ $p_z= -0.1507$
 q_Y $E= 0.2144$ $p_x= 0.2077$ $p_y= 0.0000$ $p_z= -0.0530$

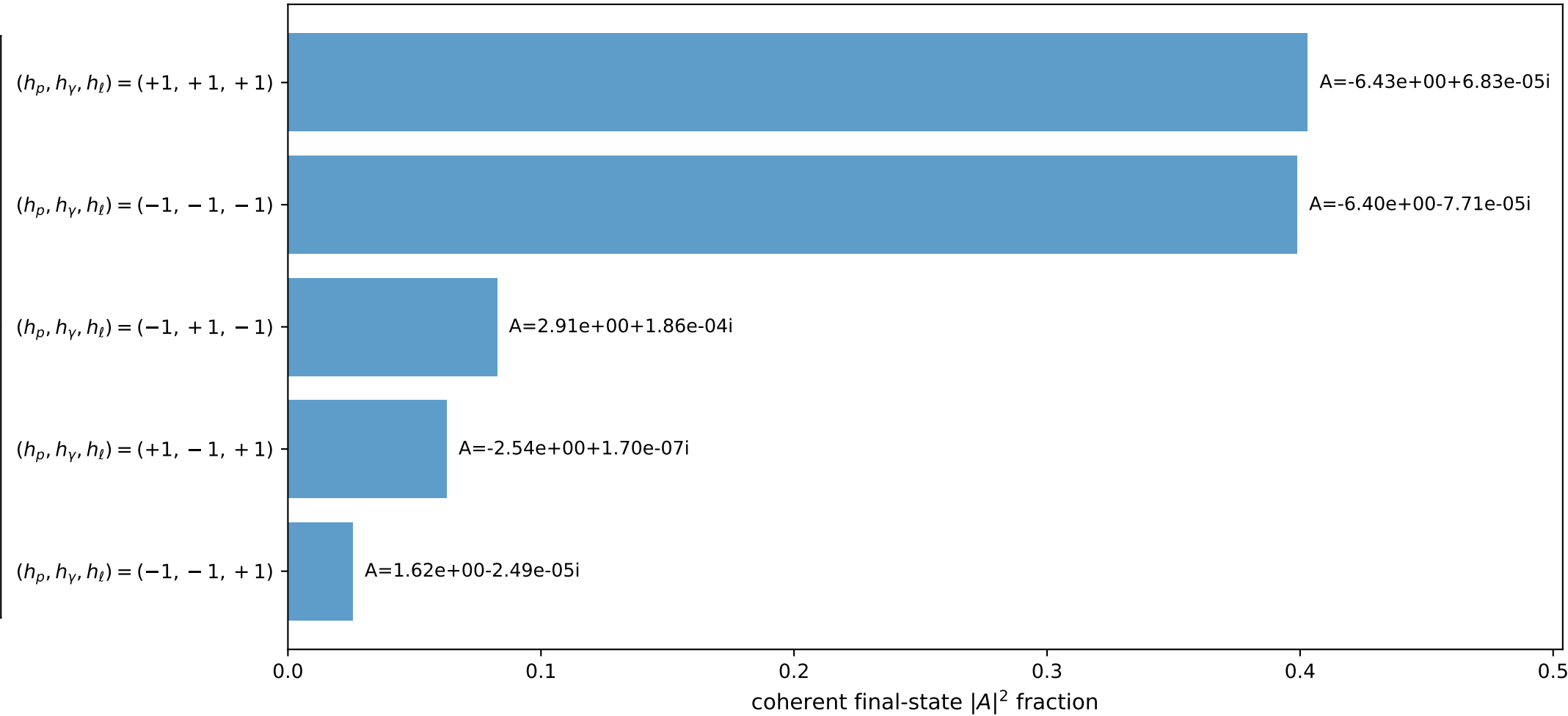
3D momenta



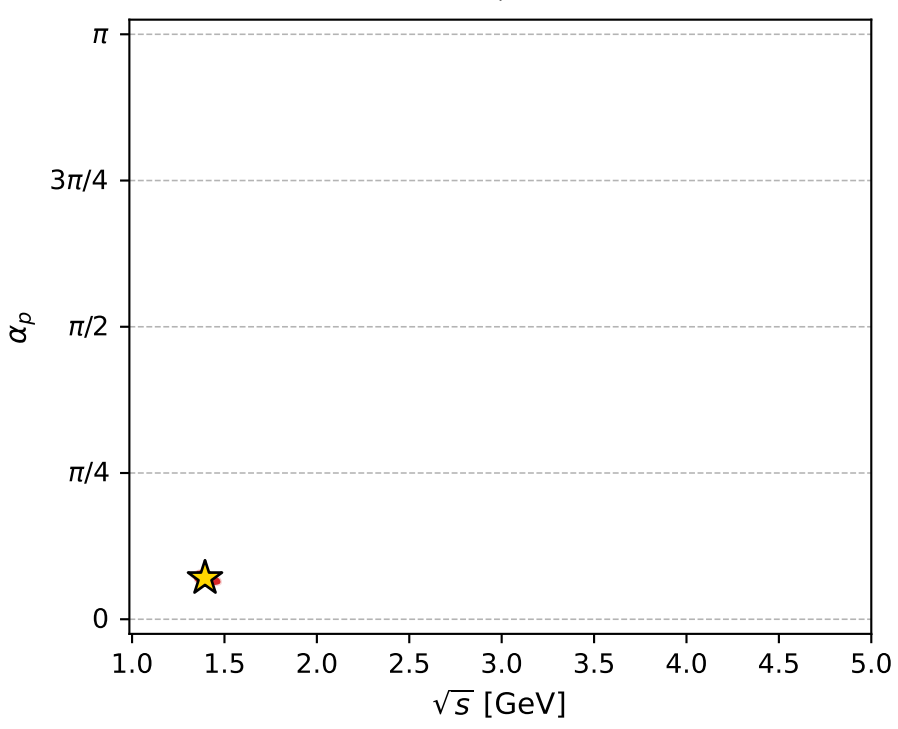
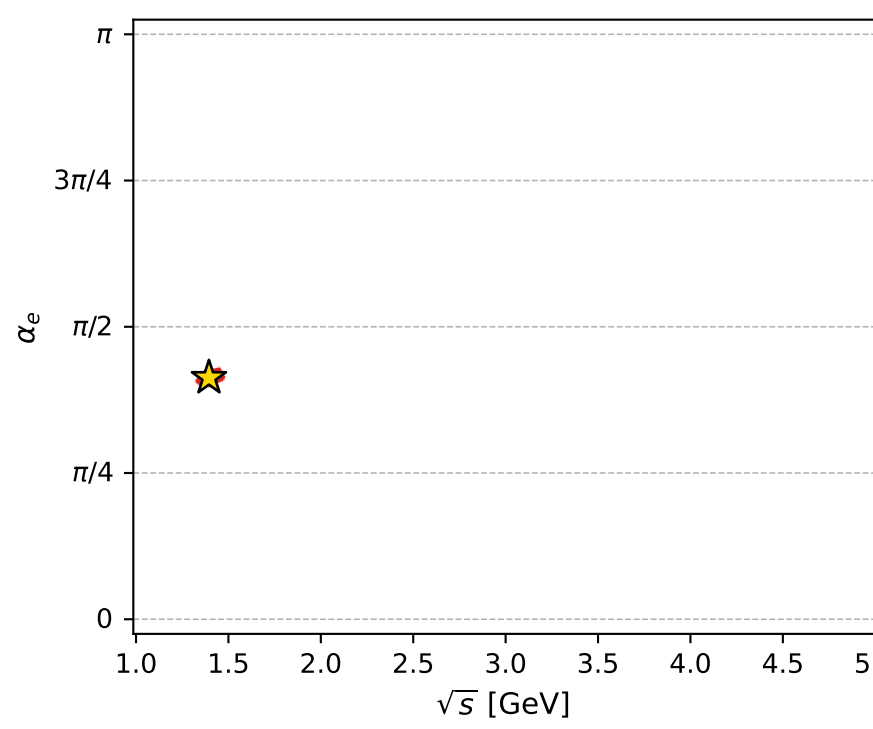
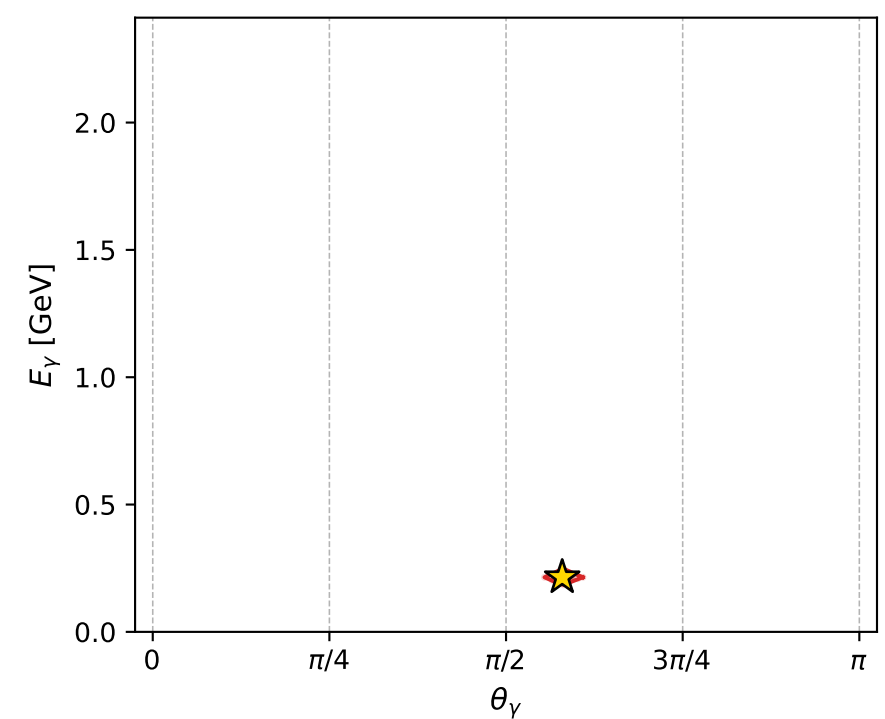
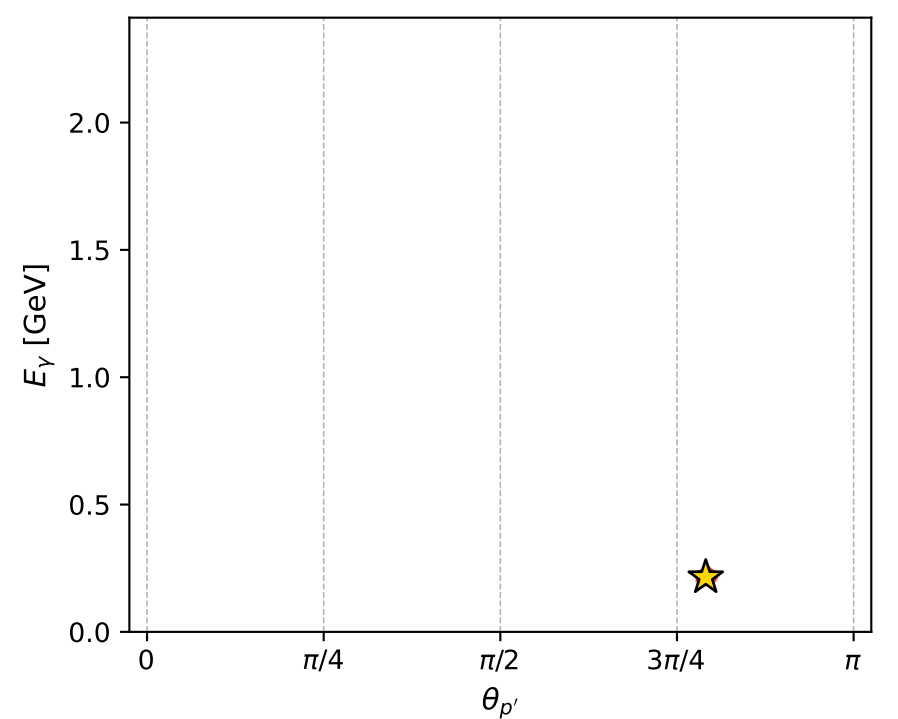
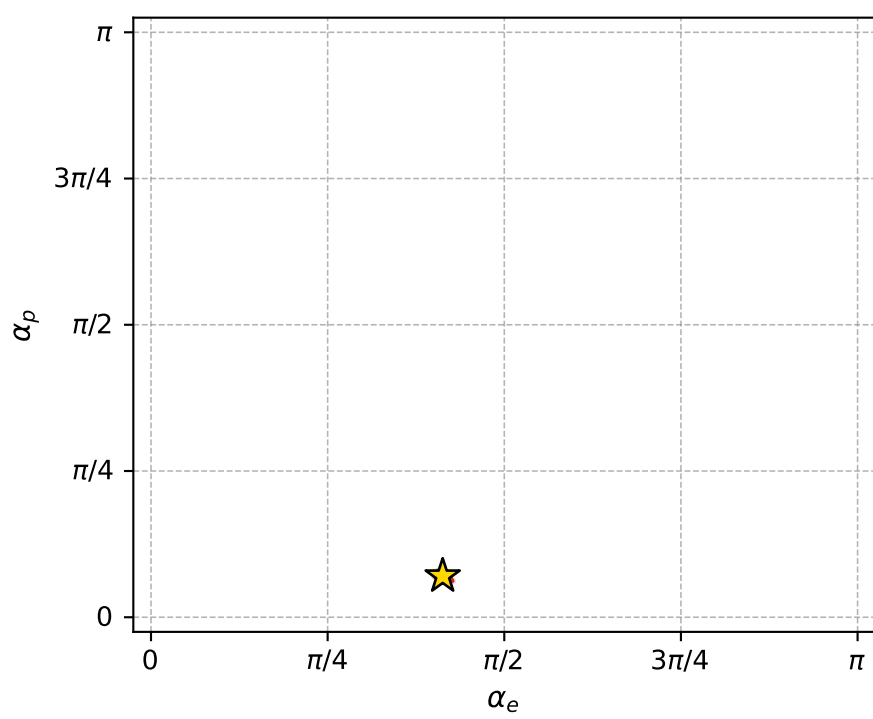
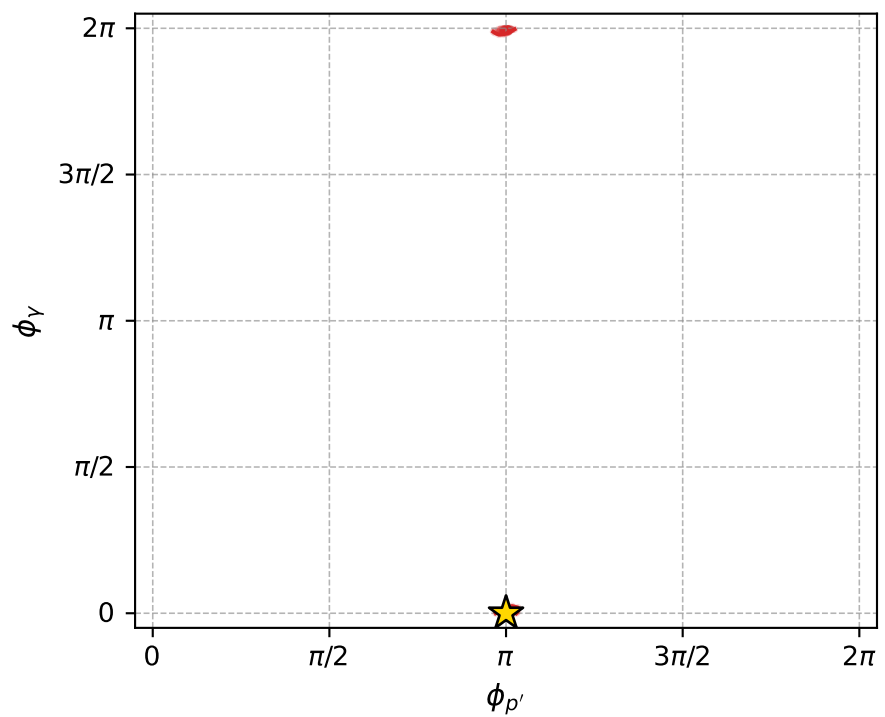
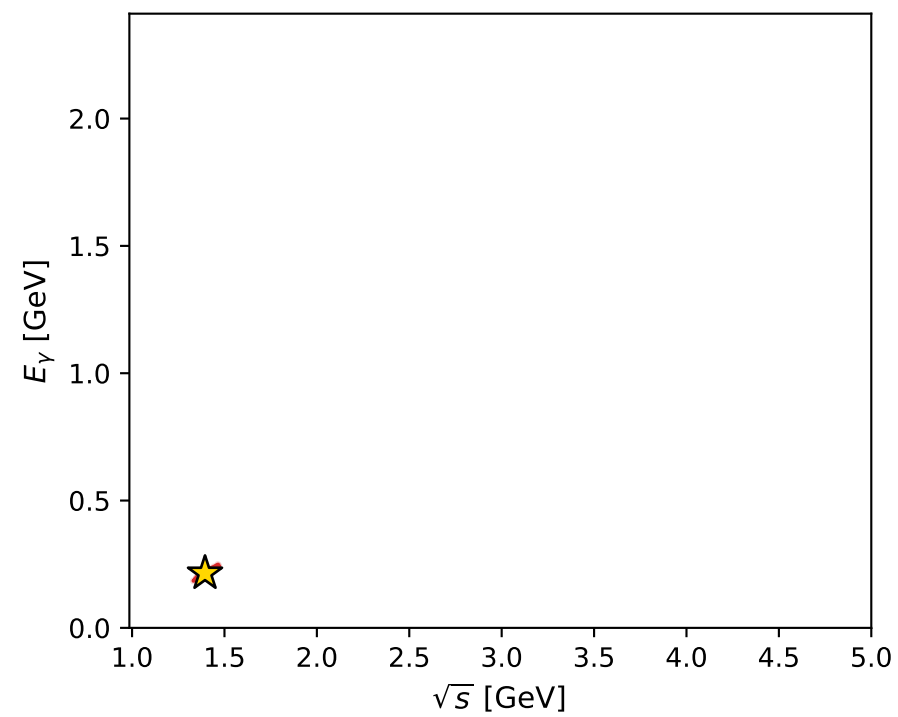
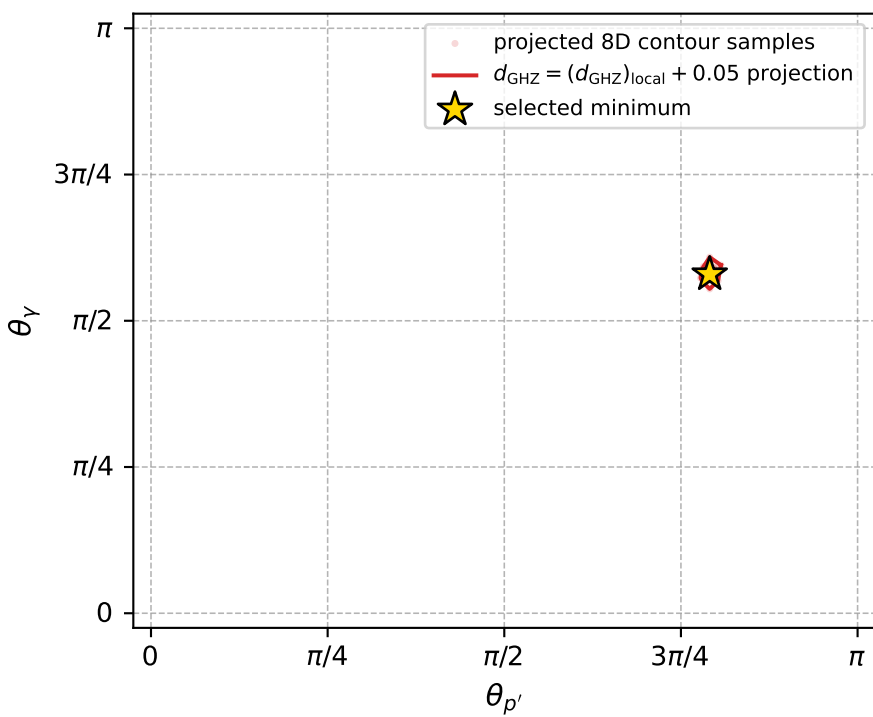
Transverse plane



Leading final-state amplitudes (N=5, retained=97.3%)



electron: polarization cluster P1, local minimum 291; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00027839028$
 $d_{\text{GHZ}} \text{ contour} = 0.05027839$
 above global minimum = 0.000278368
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.39471$
 $\text{theta_p_out} = 2.484152$
 $\text{theta_gamma_out} = 1.820453$
 $\text{q0ut} = 0.2143761$
 $\text{phi_p_out} = 3.1416$
 $\text{phi_gamma_out} = 2.700137\text{e-}05$
 $\text{alpha_e} = 1.296893$
 $\text{alpha_p} = 0.221006$

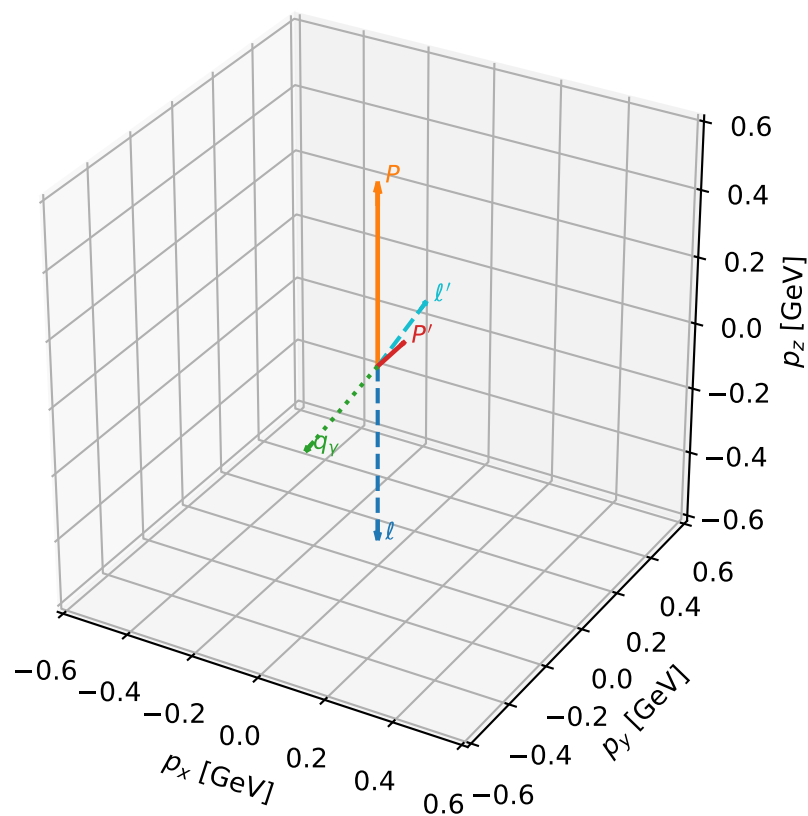
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_298 $d_{\{\text{rm GHZ}\}}=0.000313993$
 region: polarization_cluster_01_local_minimum_298
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.43020075$ rad
 incoming lepton: $0.908882|+\rangle +0.417053|-\rangle$
 proton mixing angle: $\alpha_p=1.2101725$ rad
 incoming proton: $0.352858|+\rangle +0.935677|-\rangle$

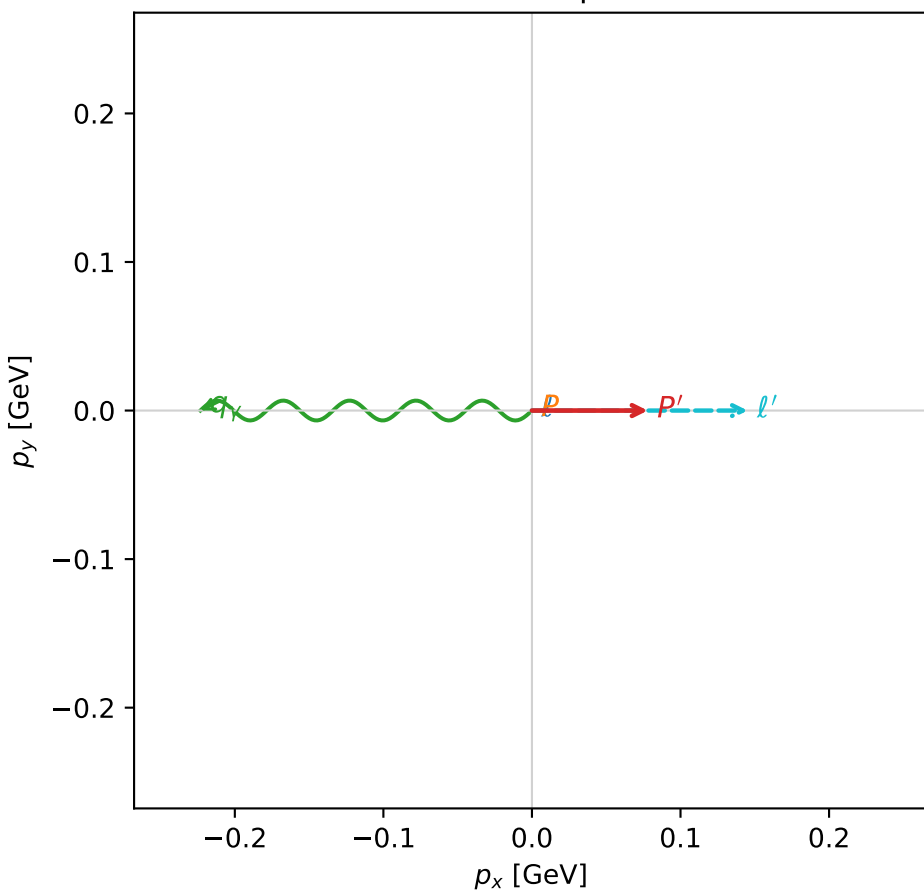
$s=2.61699$, $\sqrt{s}=1.61771$, $|\vec{P}|=0.536914$, $|\vec{P}'|=0.121116$
 $\theta_{P'}=0.699389$, $\theta_Y=2.543$, $\phi_{P'}=6.28307$, $\phi_Y=3.14151$
 $E_Y=0.396078$, $\theta_{\ell'}=0.554398$, $\phi_{\ell'}=6.28312$
 $Q^2=0.548055$, $x_B=0.393514$, $t=-0.185194$, $W^2=1.72451$, $y=0.80173$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5369$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5369$
 P $E= 1.0808$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5369$
 ℓ' $E= 0.2758$ $p_x= 0.1452$ $p_y= -0.0000$ $p_z= 0.2345$
 P' $E= 0.9458$ $p_x= 0.0780$ $p_y= -0.0000$ $p_z= 0.0927$
 q_Y $E= 0.3961$ $p_x= -0.2232$ $p_y= 0.0000$ $p_z= -0.3272$

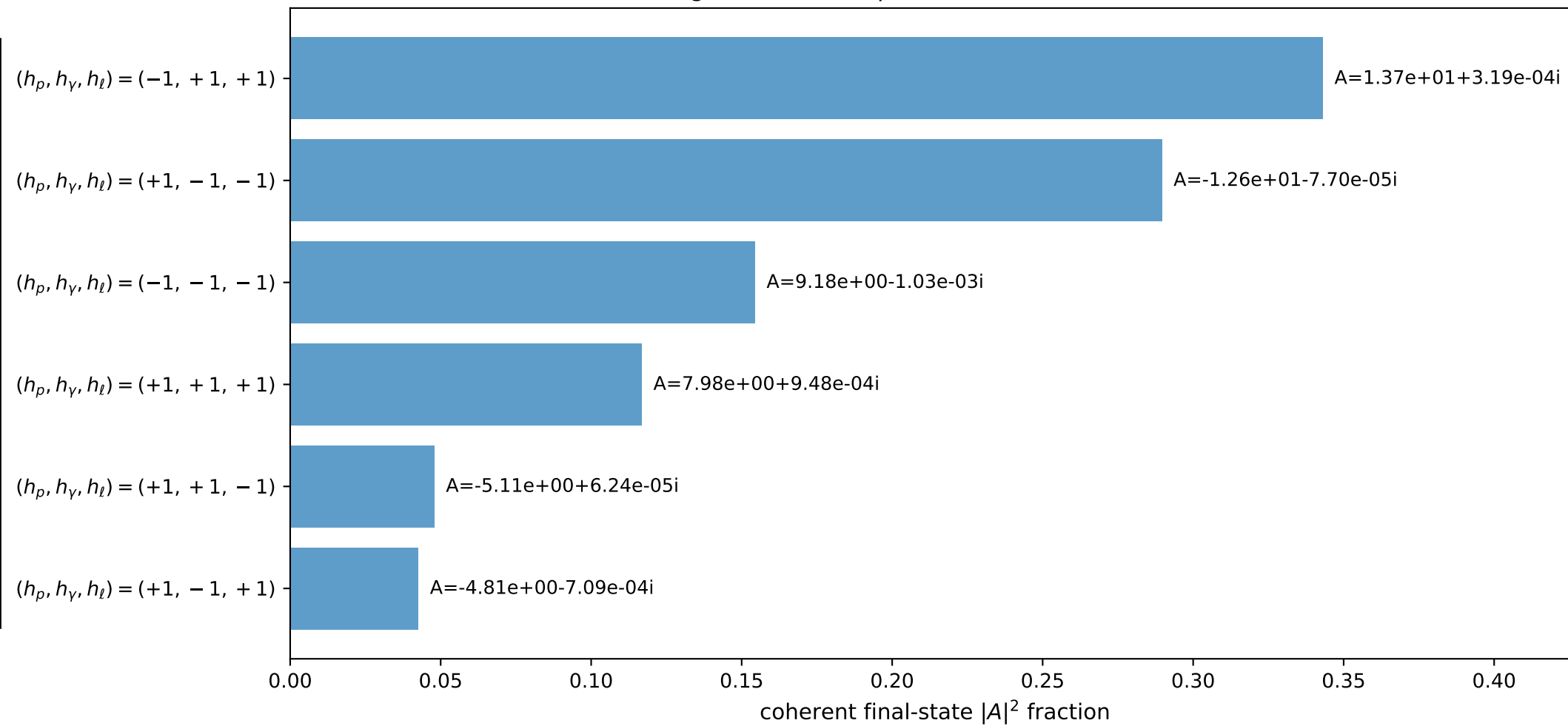
3D momenta



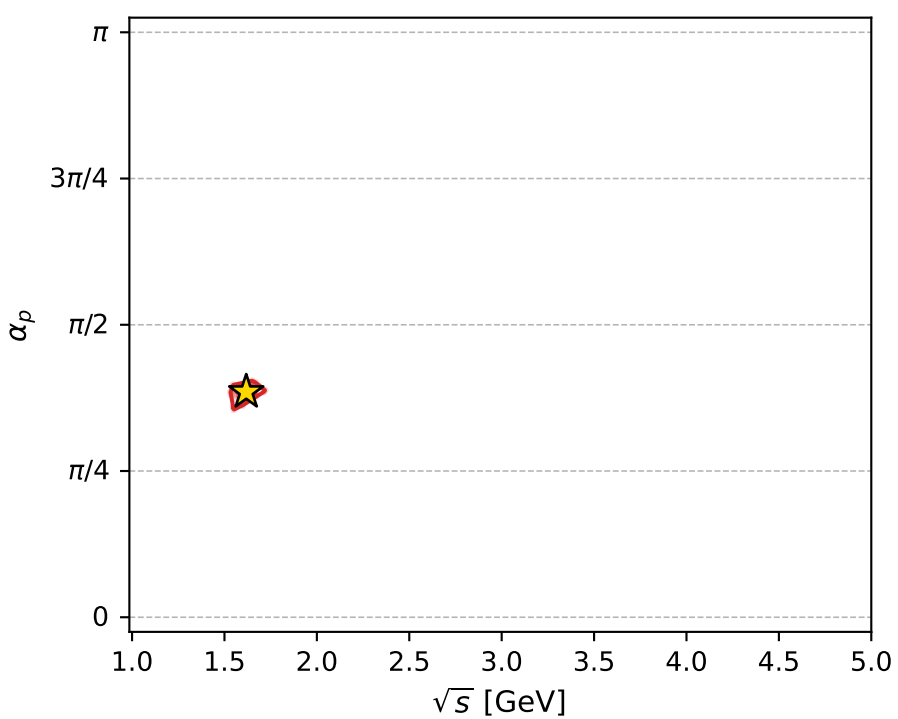
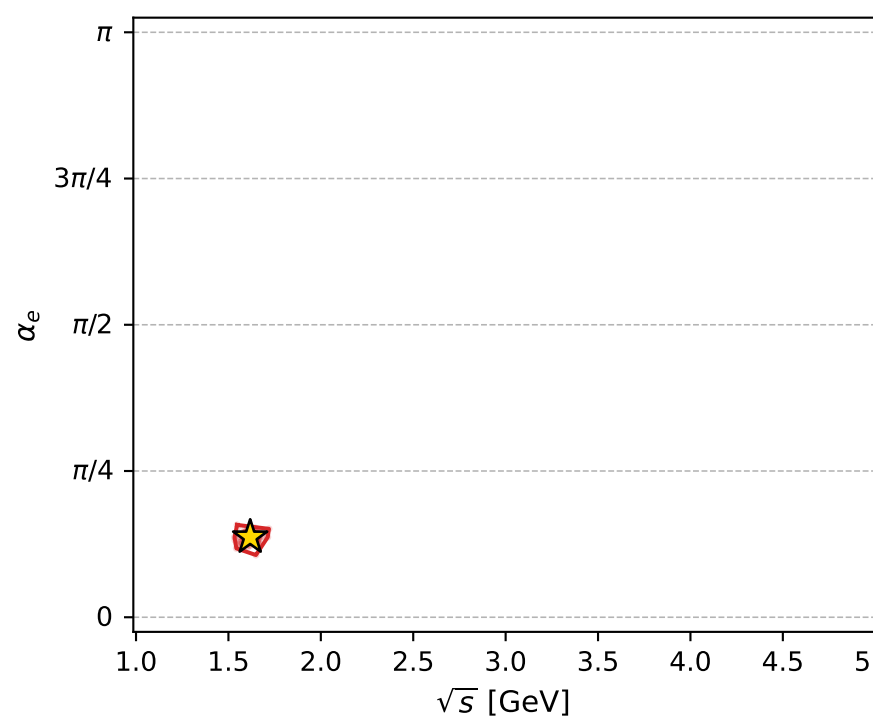
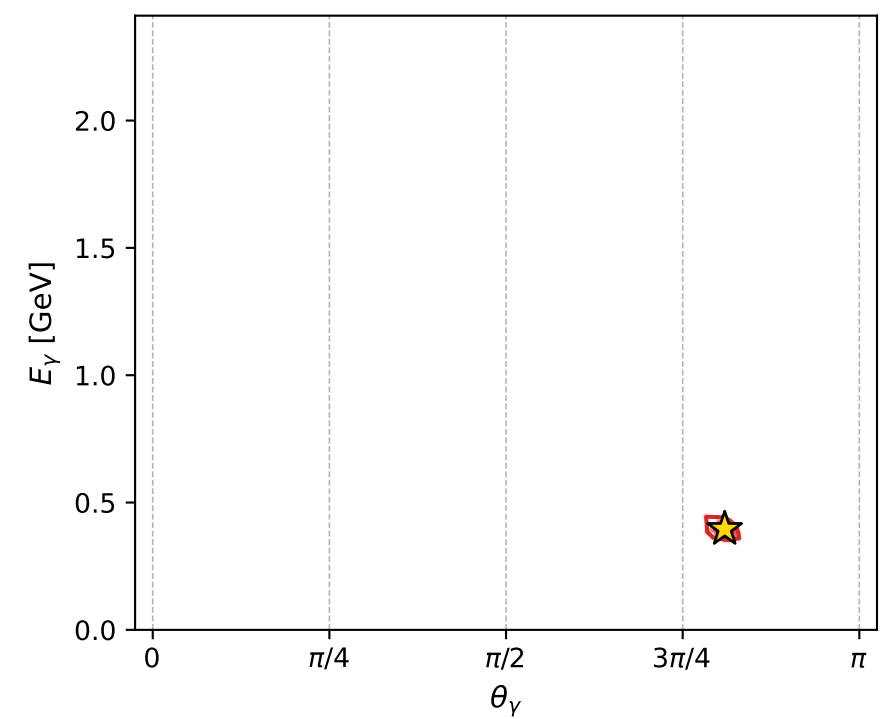
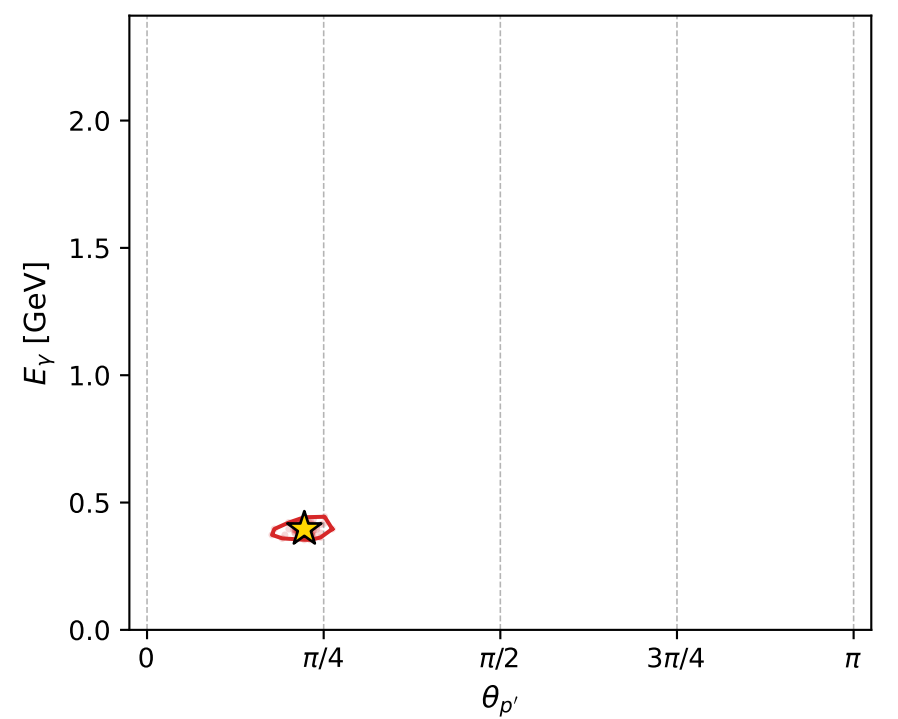
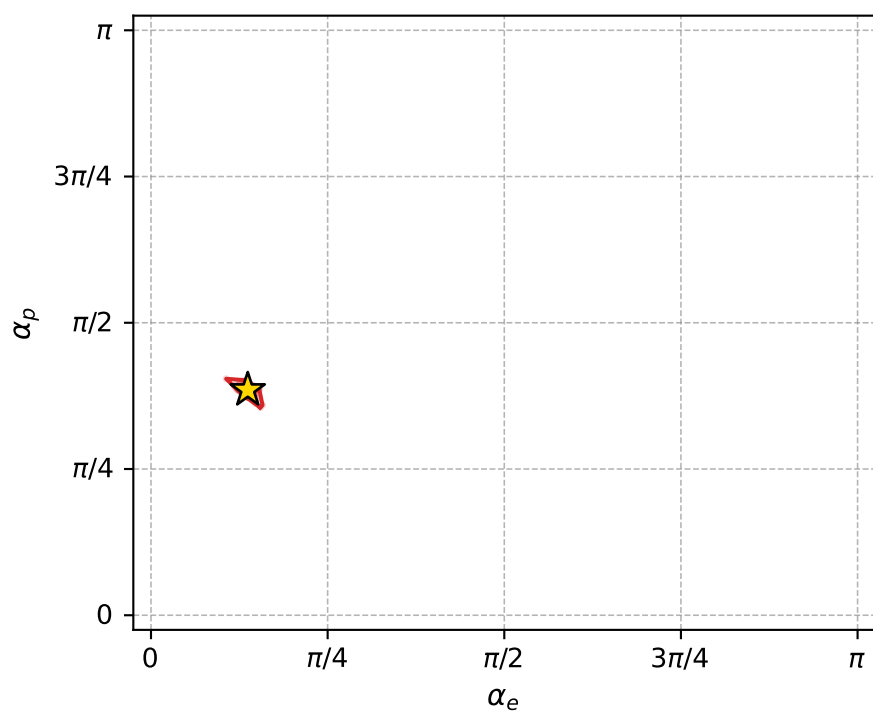
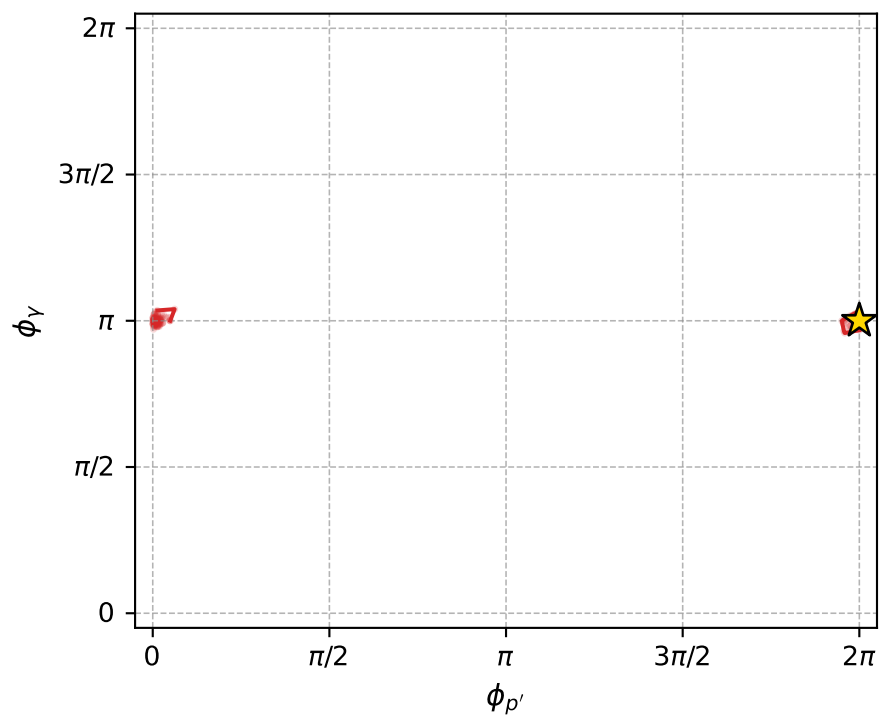
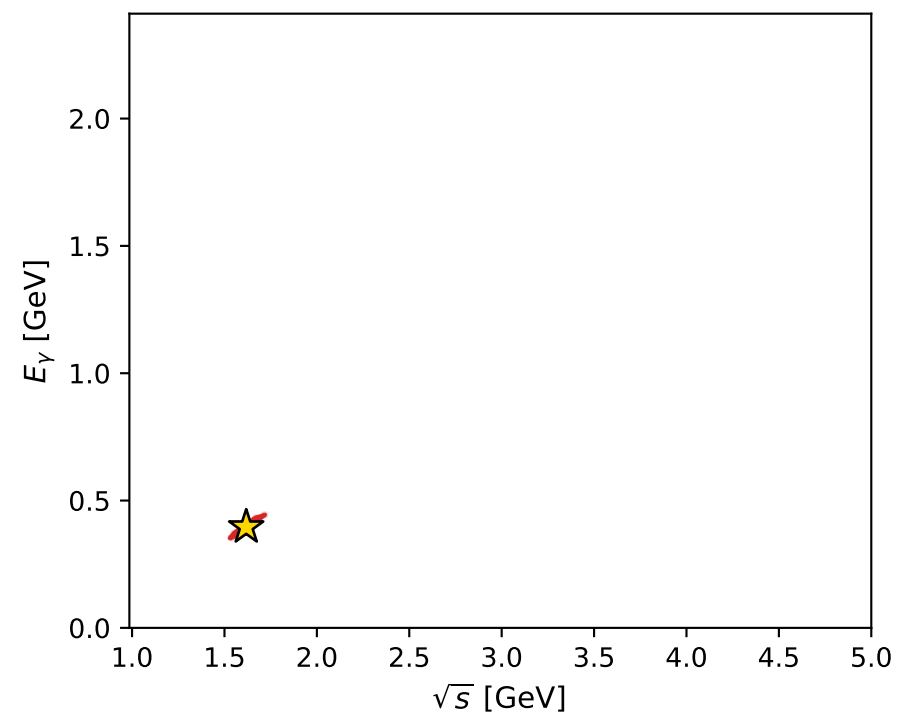
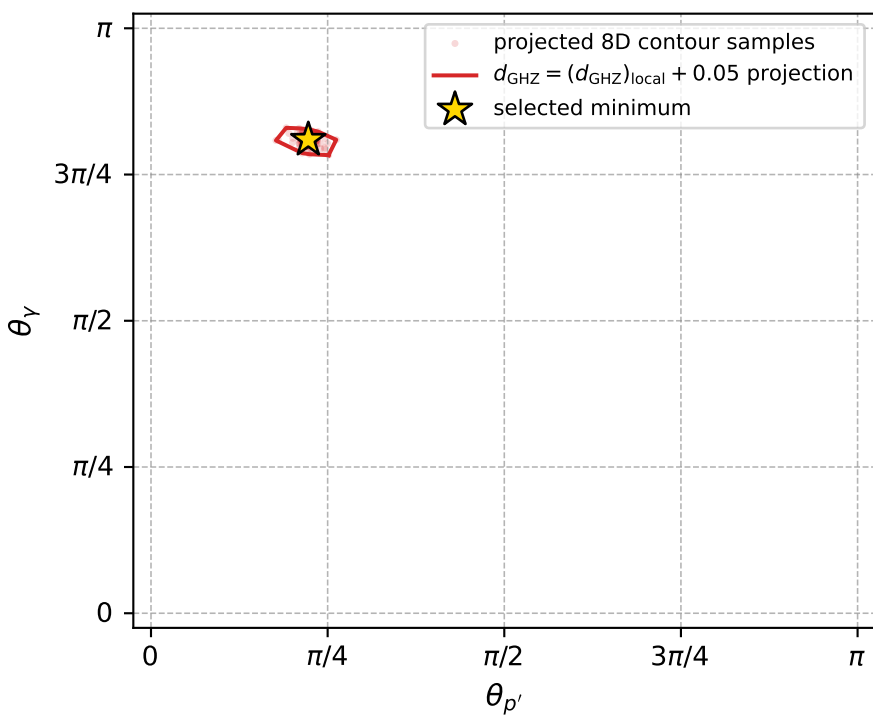
Transverse plane



Leading final-state amplitudes (N=6, retained=99.4%)



electron: polarization cluster P1, local minimum 298; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00031399258$
 $d_{\text{GHZ}} \text{ contour} = 0.050313993$
 above global minimum = 0.00031397029
 8D contour samples = 256

selected phase-space point:

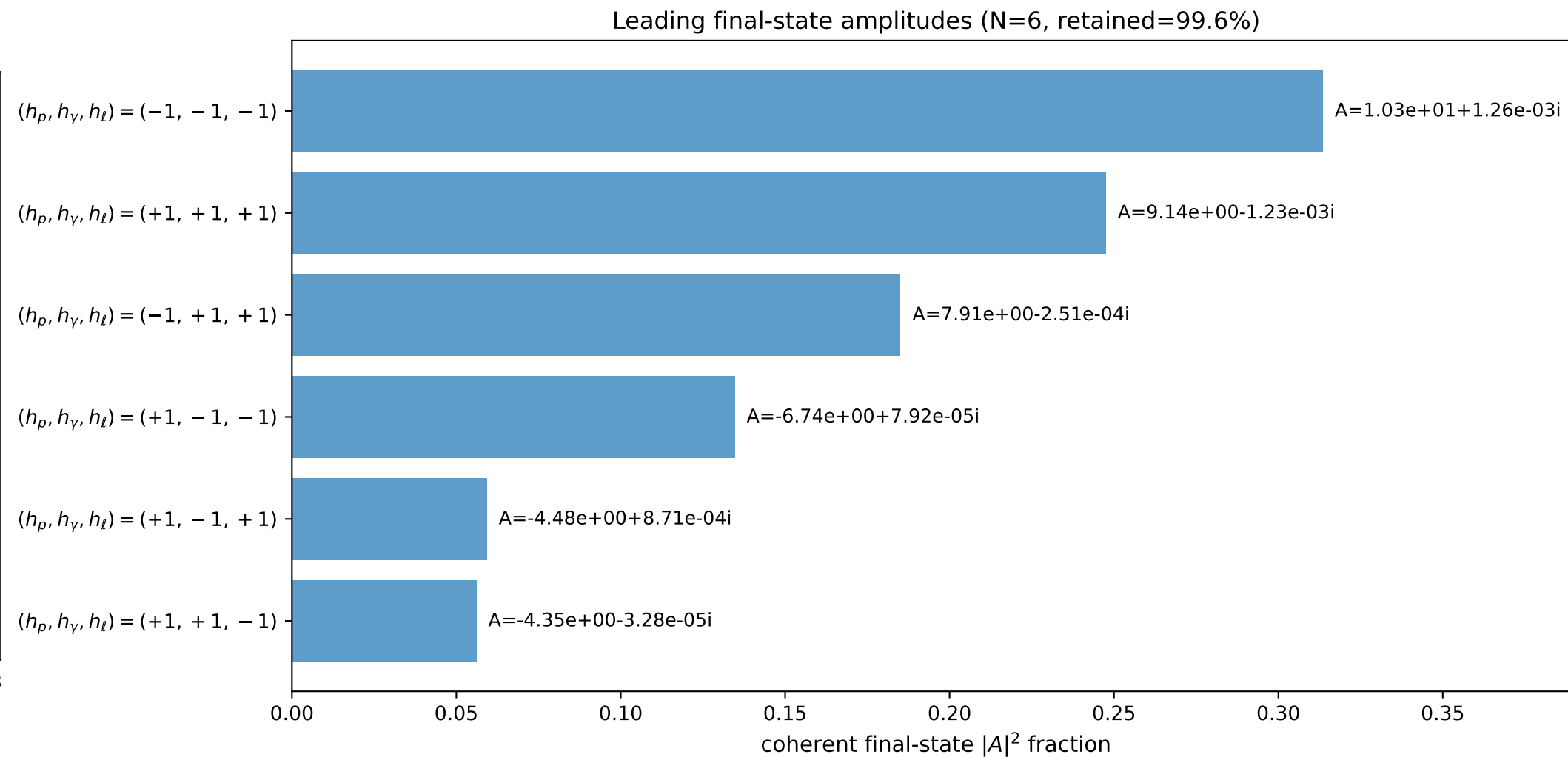
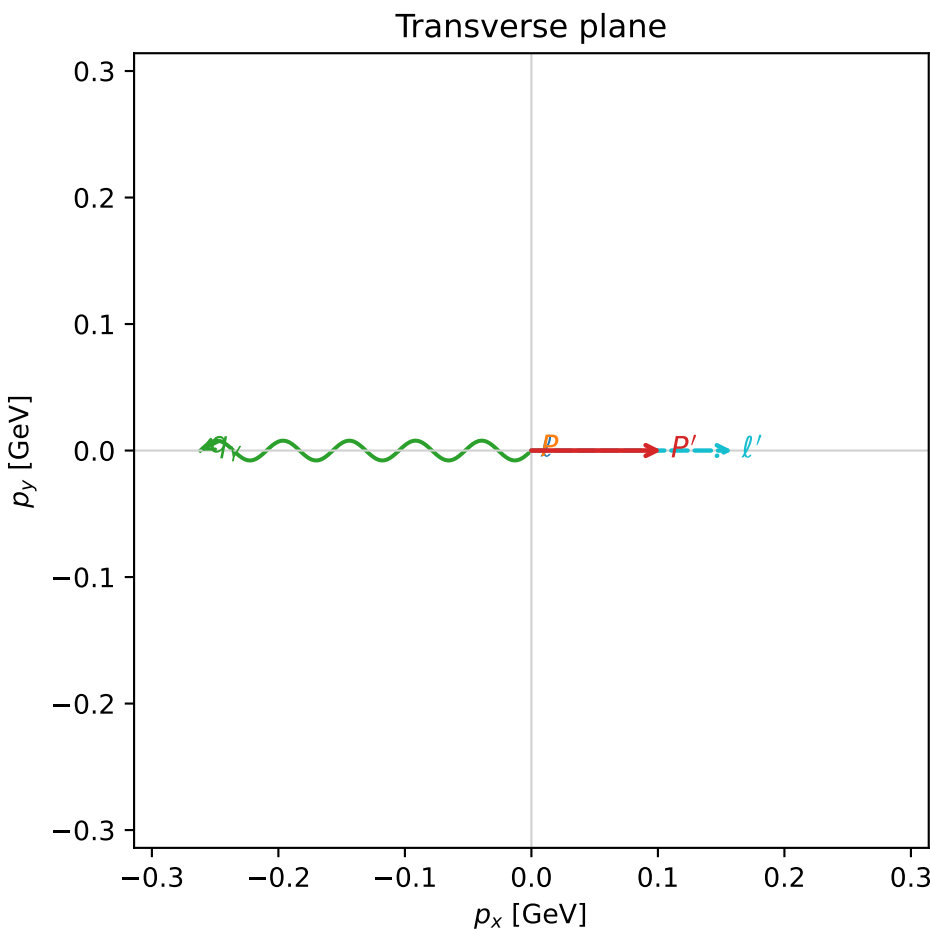
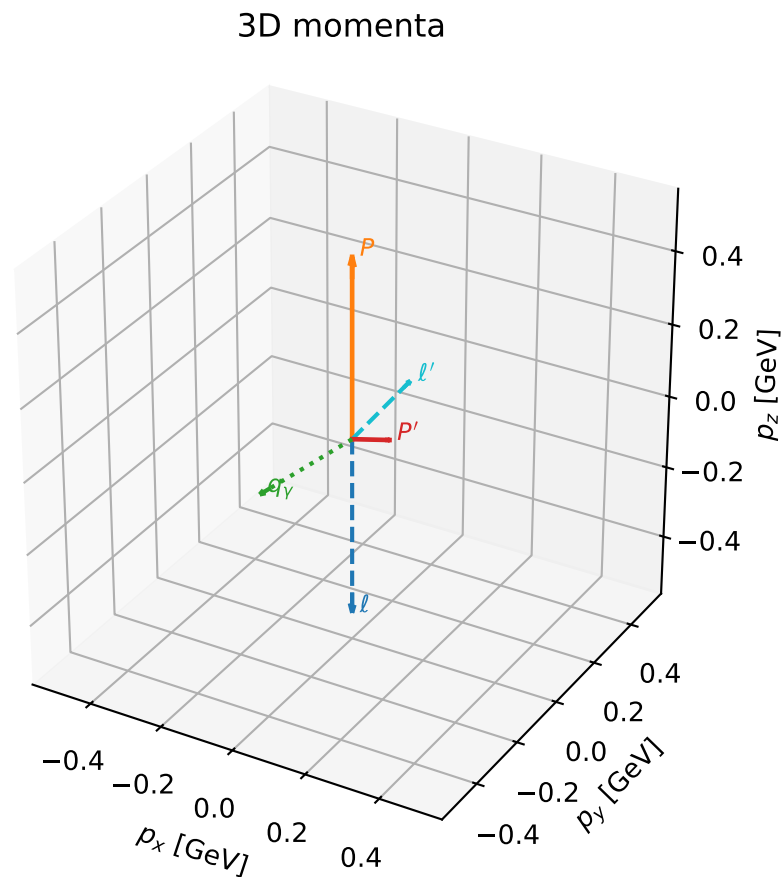
$\text{sqrt_s} = 1.617711$
 $\text{theta_p_out} = 0.6993891$
 $\text{theta_gamma_out} = 2.542999$
 $\text{q0ut} = 0.3960777$
 $\text{phi_p_out} = 6.283071$
 $\text{phi_gamma_out} = 3.14151$
 $\text{alpha_e} = 0.4302008$
 $\text{alpha_p} = 1.210173$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

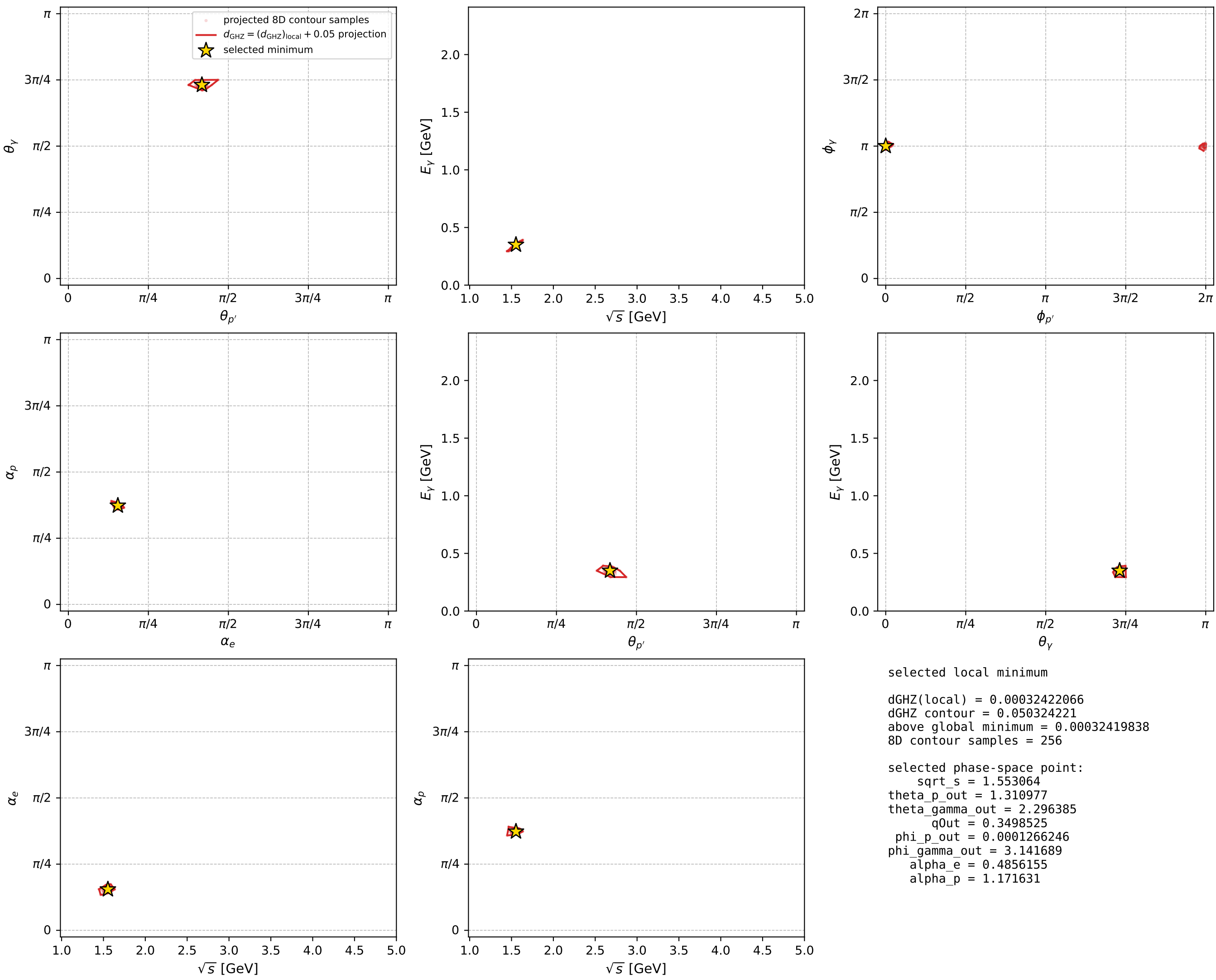
dghz_mixing_angles_polarization_01_local_minimum_301 d_{\rm GHZ}=0.000324221
 region: polarization_cluster_01_local_minimum_301
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.4856155$ rad
 incoming lepton: $0.884388|+\rangle +0.466753|-\rangle$
 proton mixing angle: $\alpha_p=1.1716313$ rad
 incoming proton: $0.388649|+\rangle +0.921386|-\rangle$

$s=2.41201$, $\sqrt{s}=1.55306$, $|\vec{P}|=0.493271$, $|\vec{P}'|=0.106449$
 $\theta_{p'}=1.31098$, $\theta_Y=2.29639$, $\phi_{p'}=0.000126625$, $\phi_Y=3.14169$
 $E_Y=0.349853$, $\theta_{\ell'}=0.659695$, $\phi_{\ell'}=7.69489\text{e-}05$
 $Q^2=0.457751$, $x_B=0.386341$, $t=-0.214265$, $W^2=1.60693$, $y=0.77331$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4933$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4933$
 P $E= 1.0598$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4933$
 ℓ' $E= 0.2592$ $p_x= 0.1589$ $p_y= 0.0000$ $p_z= 0.2048$
 P' $E= 0.9440$ $p_x= 0.1029$ $p_y= 0.0000$ $p_z= 0.0273$
 q_Y $E= 0.3499$ $p_x= -0.2617$ $p_y= -0.0000$ $p_z= -0.2322$



electron: polarization cluster P1, local minimum 301; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



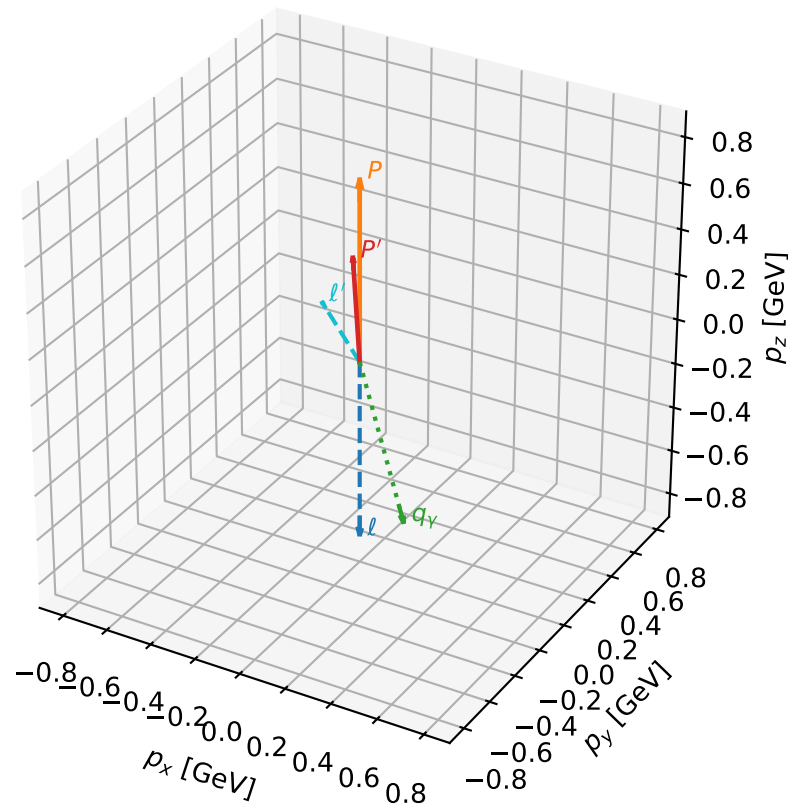
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_302 $d_{\{\text{rm GHZ}\}}=0.000327593$
 region: polarization_cluster_01_local_minimum_302
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.93037848$ rad
 incoming lepton: $0.597531|+\rangle +0.801846|-\rangle$
 proton mixing angle: $\alpha_p=0.54807107$ rad
 incoming proton: $0.853531|+\rangle +0.521042|-\rangle$

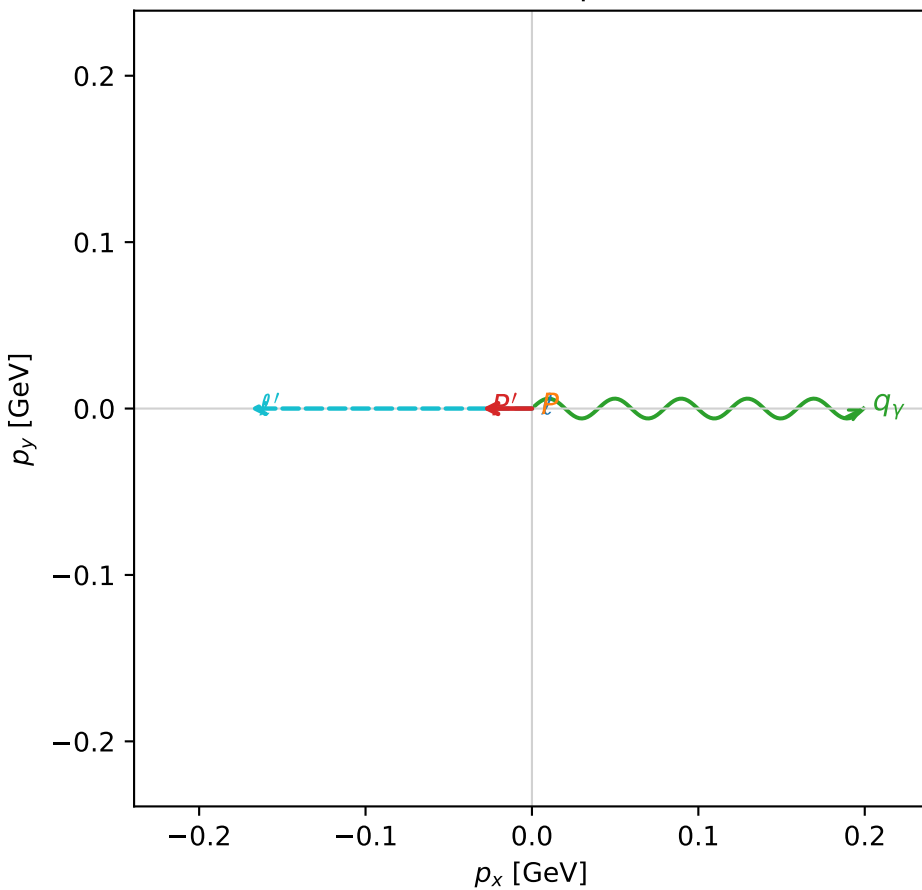
$s=4.07244$, $\sqrt{s}=2.01803$, $|\vec{P}|=0.791018$, $|\vec{P}'|=0.449885$
 $\theta_{P'}=0.0665127$, $\theta_Y=2.85259$, $\phi_{P'}=3.14156$, $\phi_Y=6.28318$
 $E_Y=0.699114$, $\theta_{\ell'}=0.65333$, $\phi_{\ell'}=3.1416$
 $Q^2=0.790759$, $x_B=0.276597$, $t=-0.0830879$, $W^2=2.94797$, $y=0.895474$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.7910$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7910$
 P $E= 1.2270$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7910$
 ℓ' $E= 0.2786$ $p_x= -0.1693$ $p_y= -0.0000$ $p_z= 0.2212$
 P' $E= 1.0403$ $p_x= -0.0299$ $p_y= 0.0000$ $p_z= 0.4489$
 q_Y $E= 0.6991$ $p_x= 0.1992$ $p_y= -0.0000$ $p_z= -0.6701$

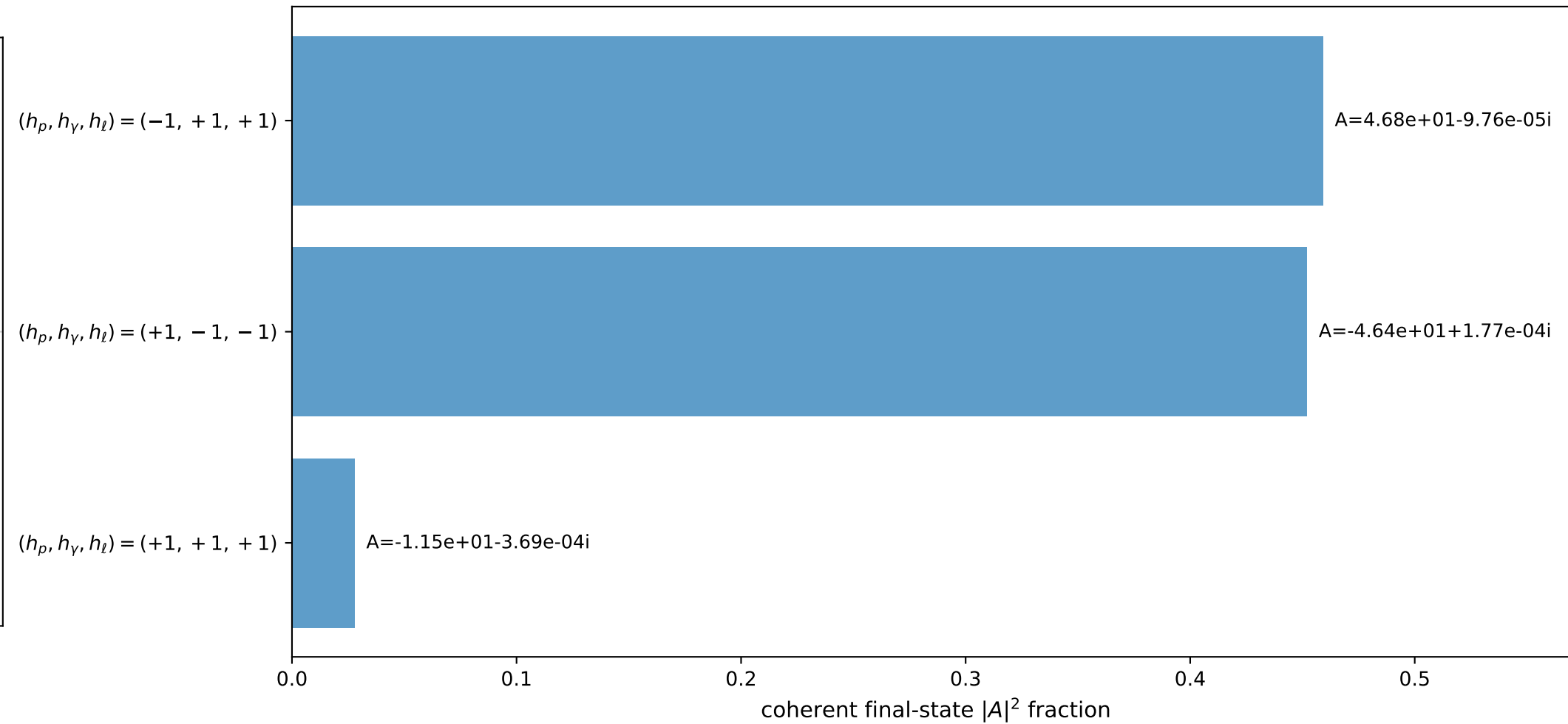
3D momenta



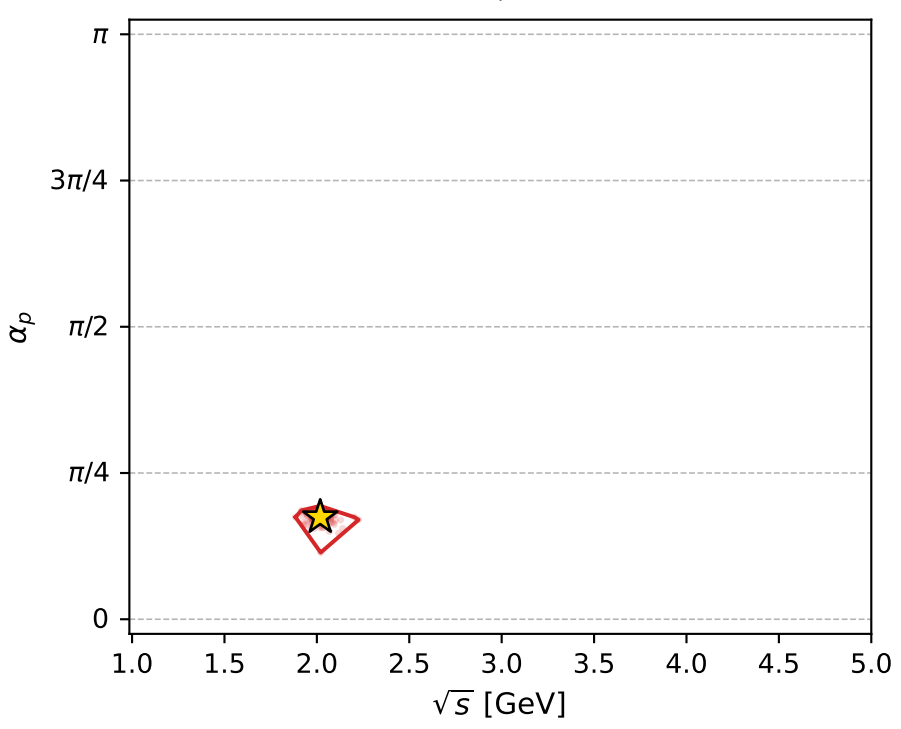
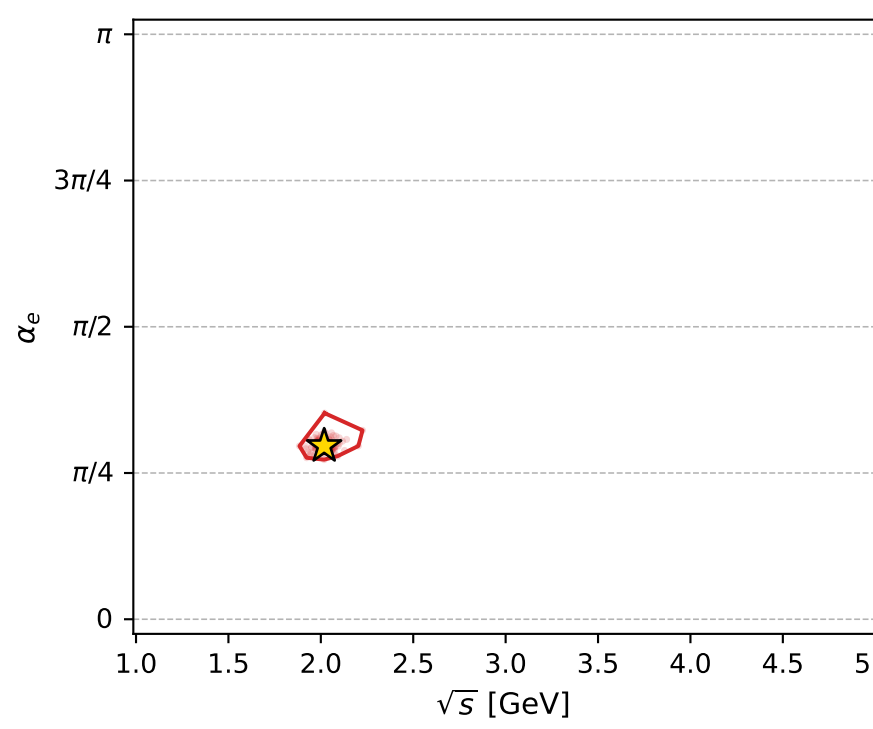
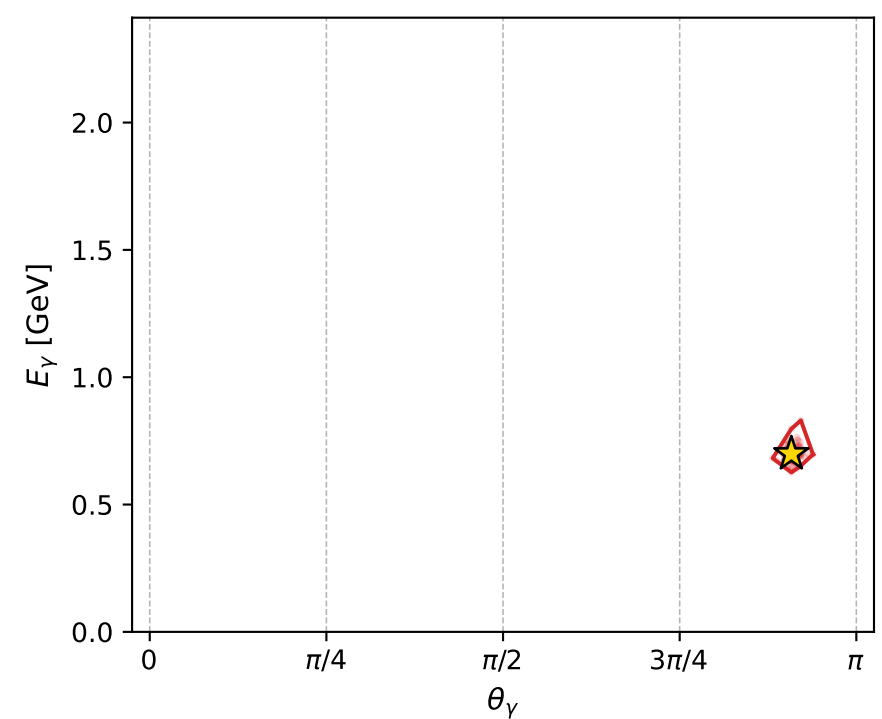
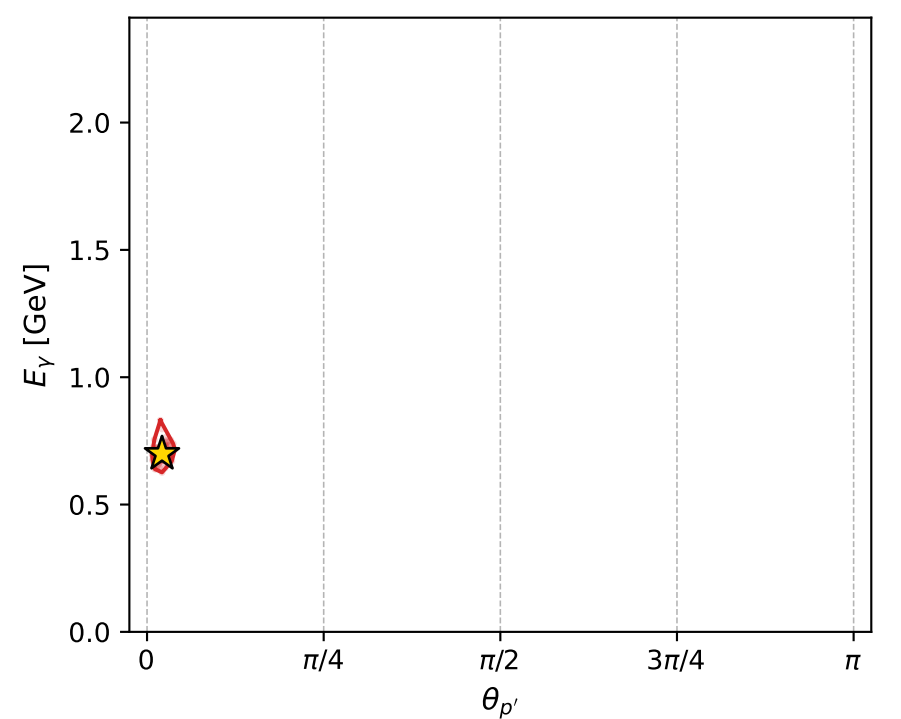
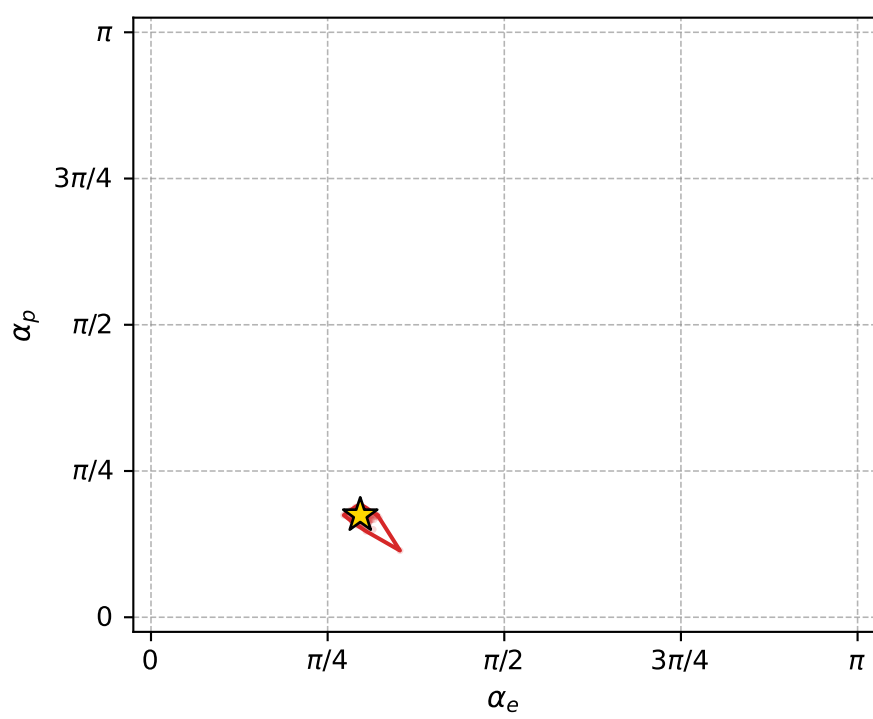
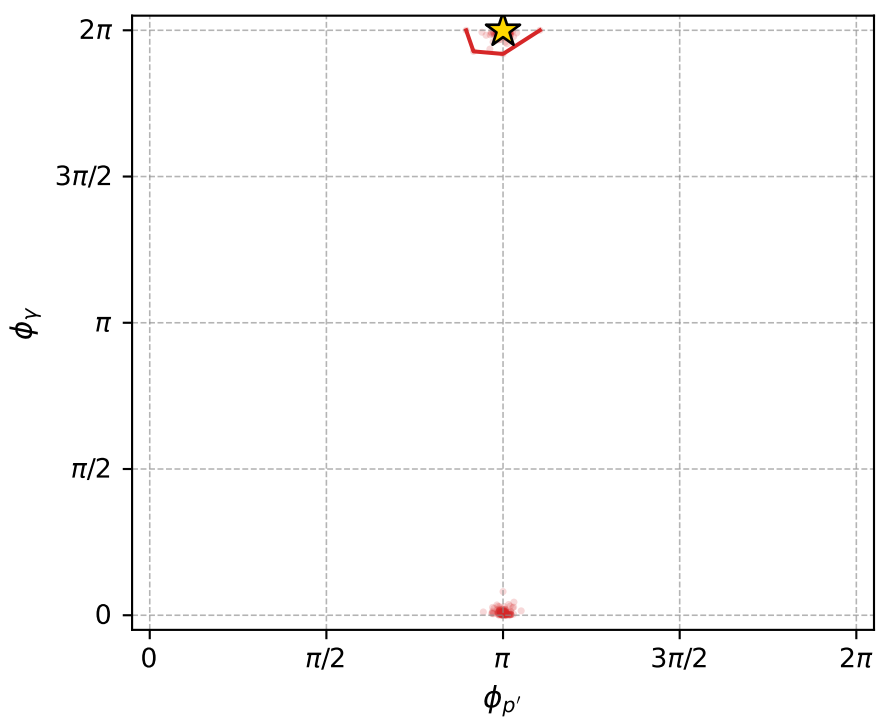
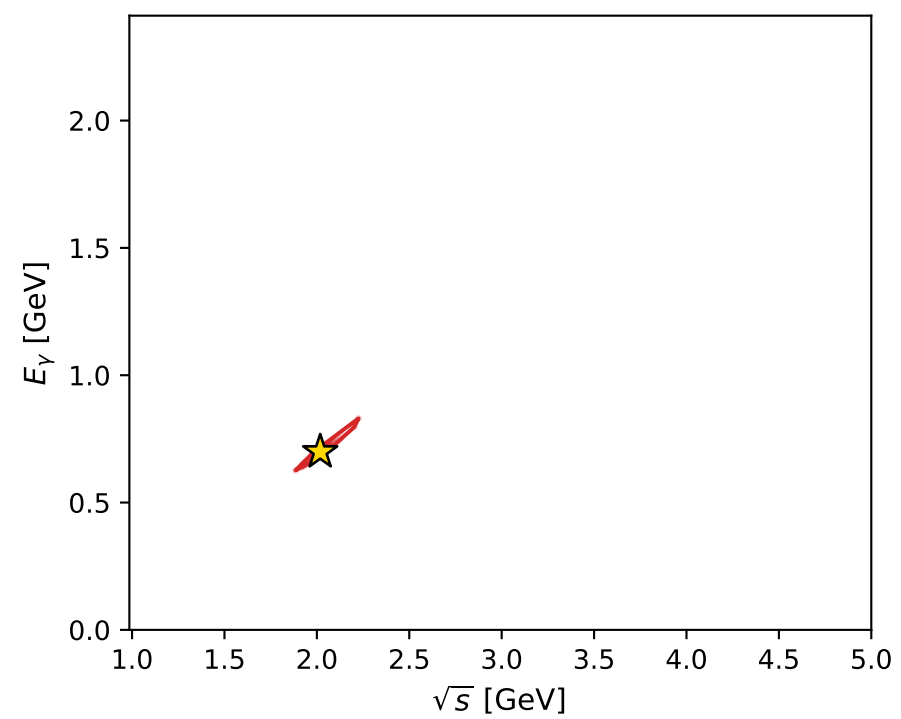
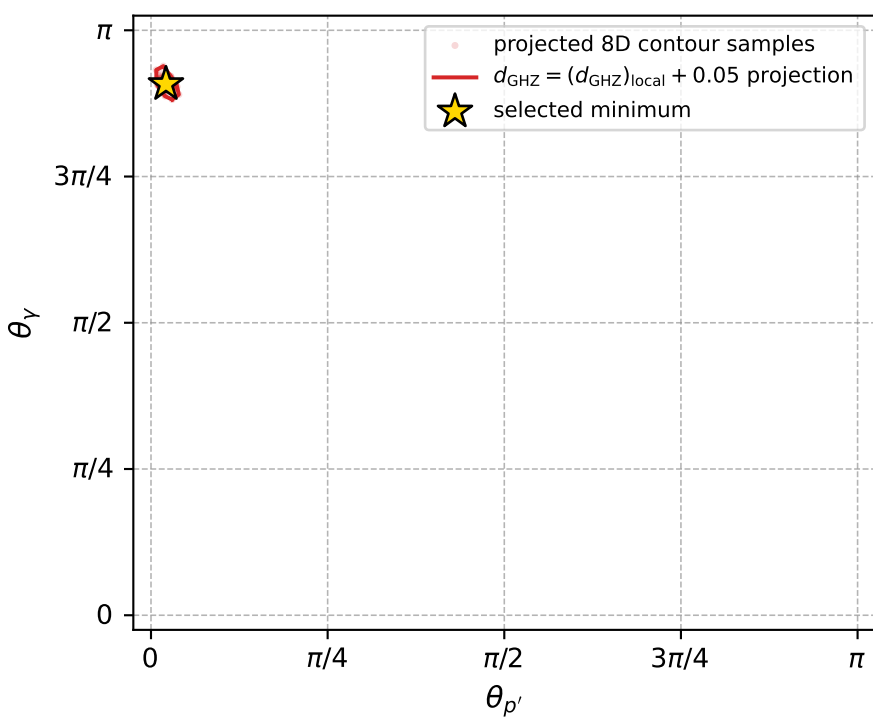
Transverse plane



Leading final-state amplitudes (N=3, retained=93.9%)



electron: polarization cluster P1, local minimum 302; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00032759311$
 $d_{\text{GHZ}} \text{ contour} = 0.050327593$
 above global minimum = 0.00032757083
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 2.018028$
 $\text{theta_p_out} = 0.06651271$
 $\text{theta_gamma_out} = 2.852587$
 $\text{q0out} = 0.6991144$
 $\text{phi_p_out} = 3.141561$
 $\text{phi_gamma_out} = 6.283183$
 $\text{alpha_e} = 0.9303785$
 $\text{alpha_p} = 0.5480711$

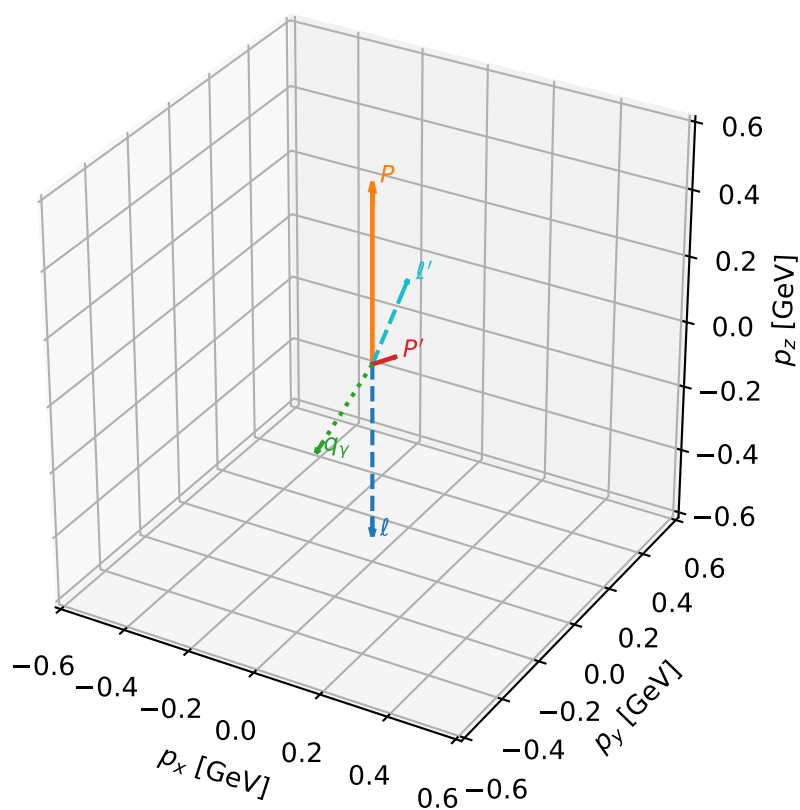
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_304 $d_{\{\text{rm GHZ}\}}=0.000330781$
 region: polarization_cluster_01_local_minimum_304
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.26405011$ rad
 incoming lepton: $0.965341|+\rangle + 0.260992|-\rangle$
 proton mixing angle: $\alpha_p=1.3534581$ rad
 incoming proton: $0.215631|+\rangle + 0.976475|-\rangle$

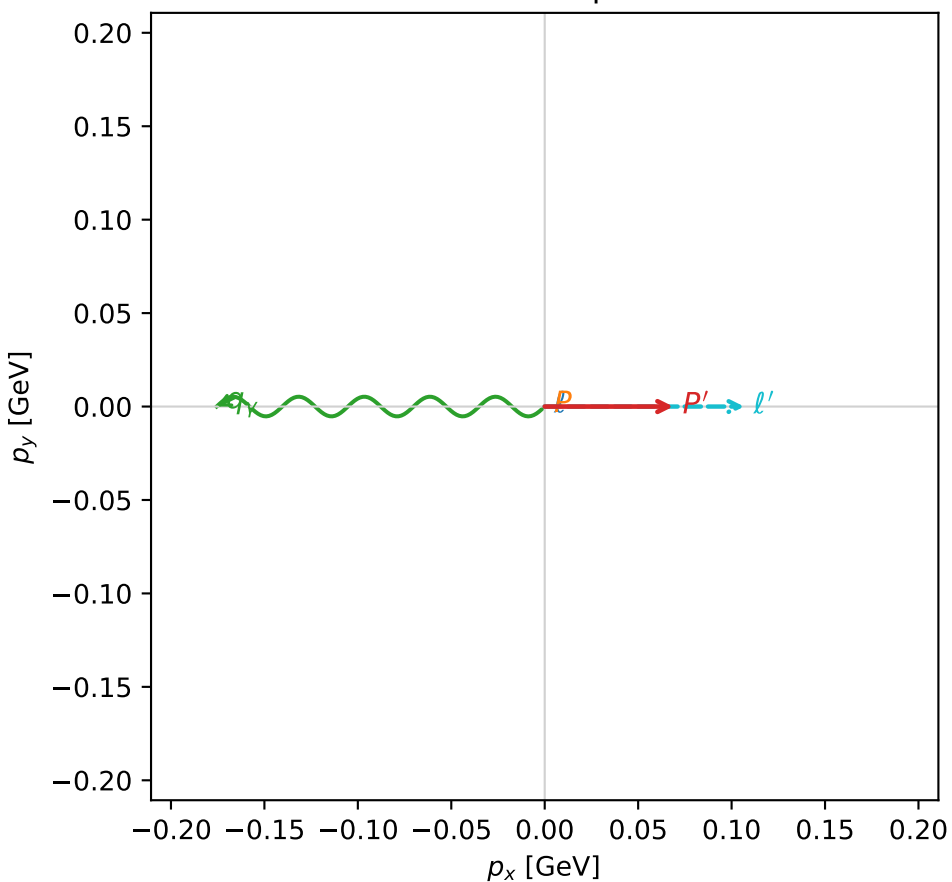
$s=2.61302$, $\sqrt{s}=1.61649$, $|\vec{P}|=0.536095$, $|\vec{P}'|=0.0805045$
 $\theta_{P'}=1.02486$, $\theta_V=2.64845$, $\phi_{P'}=6.28314$, $\phi_V=3.14155$
 $E_V=0.370838$, $\theta_{l'}=0.358555$, $\phi_{l'}=6.28314$
 $Q^2=0.631576$, $x_B=0.457236$, $t=-0.229757$, $W^2=1.62956$, $y=0.796968$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.5361$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5361$
 P $E= 1.0804$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5361$
 l' $E= 0.3042$ $p_x= 0.1067$ $p_y= -0.0000$ $p_z= 0.2849$
 P' $E= 0.9414$ $p_x= 0.0688$ $p_y= -0.0000$ $p_z= 0.0418$
 q_V $E= 0.3708$ $p_x= -0.1756$ $p_y= 0.0000$ $p_z= -0.3267$

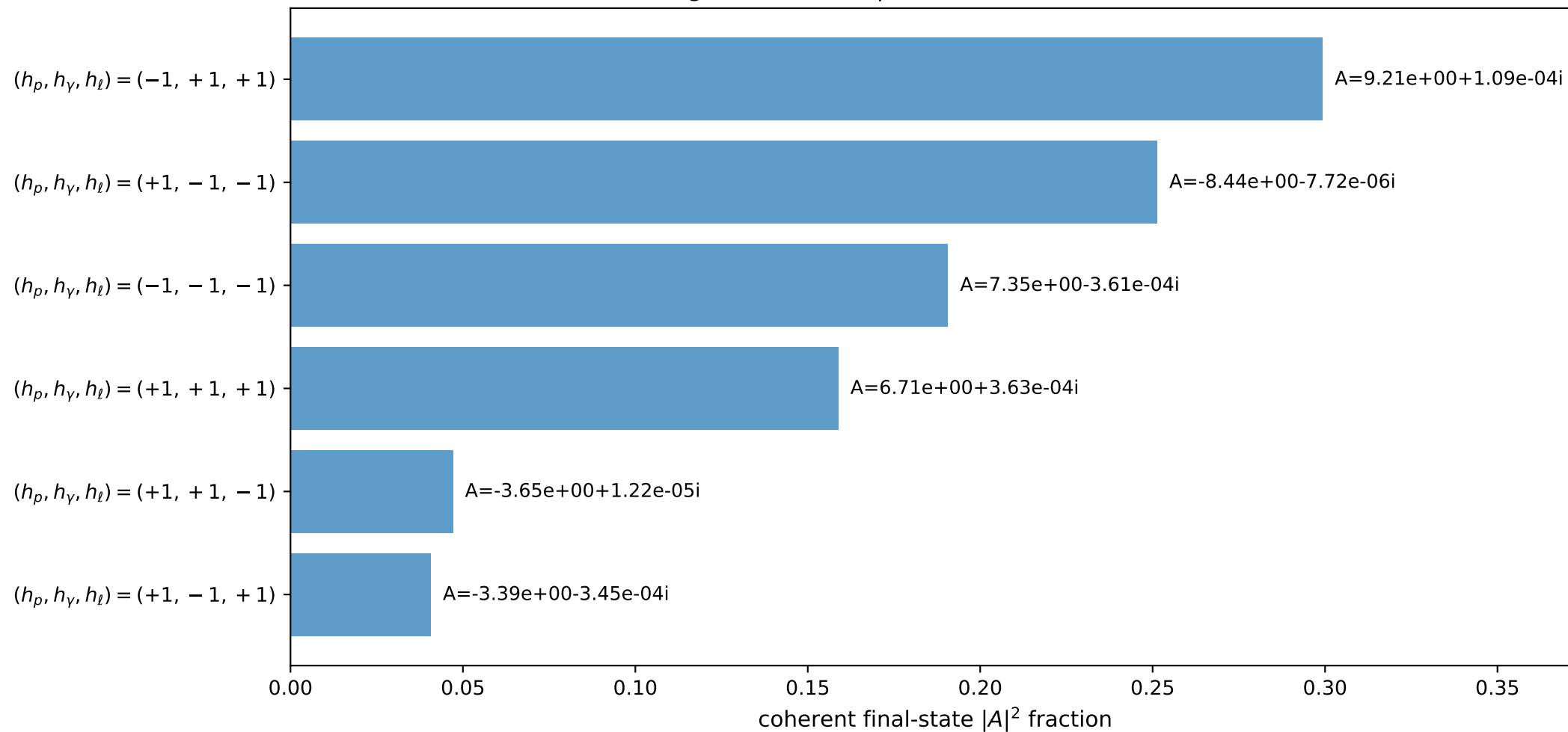
3D momenta



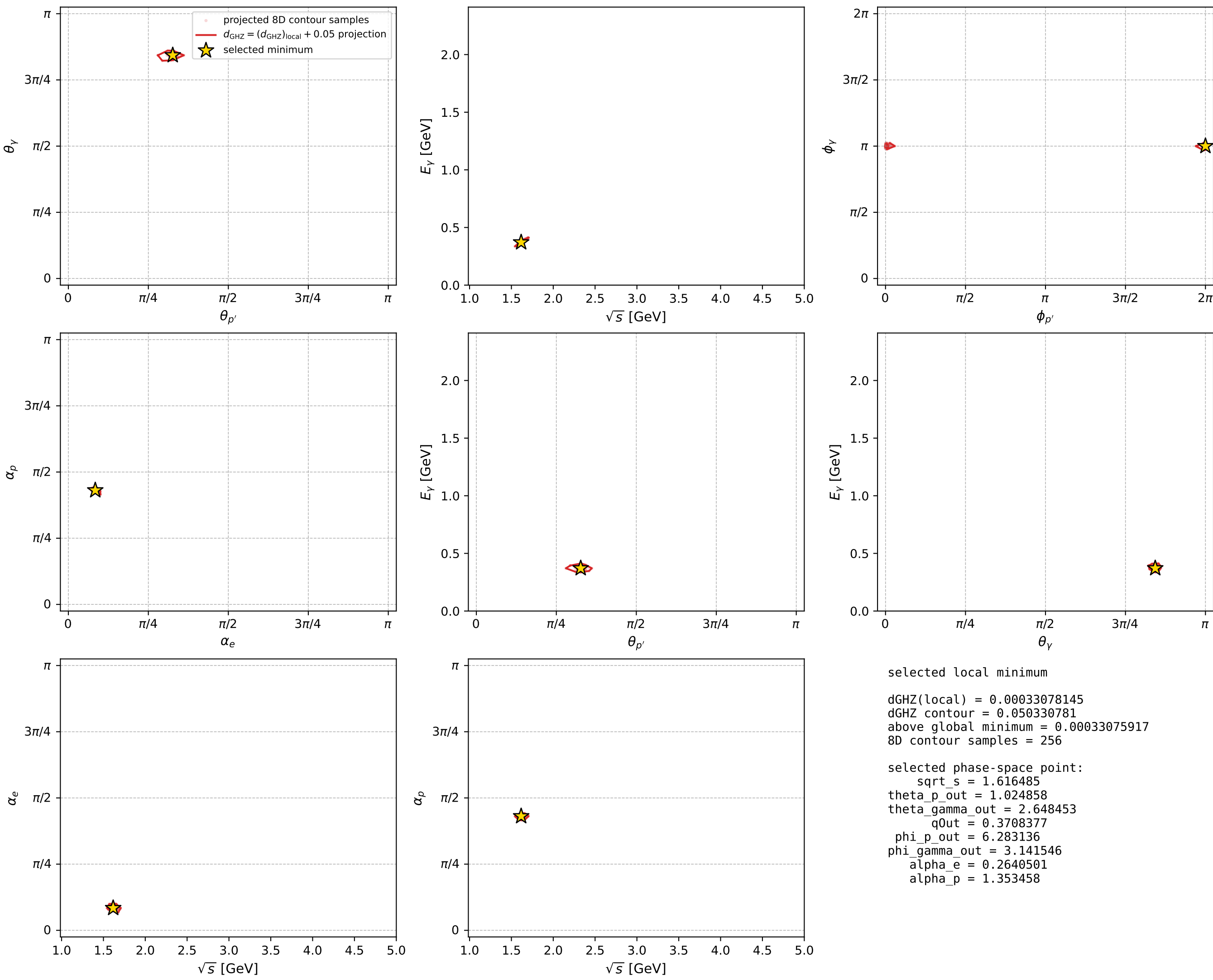
Transverse plane



Leading final-state amplitudes (N=6, retained=98.8%)



electron: polarization cluster P1, local minimum 304; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



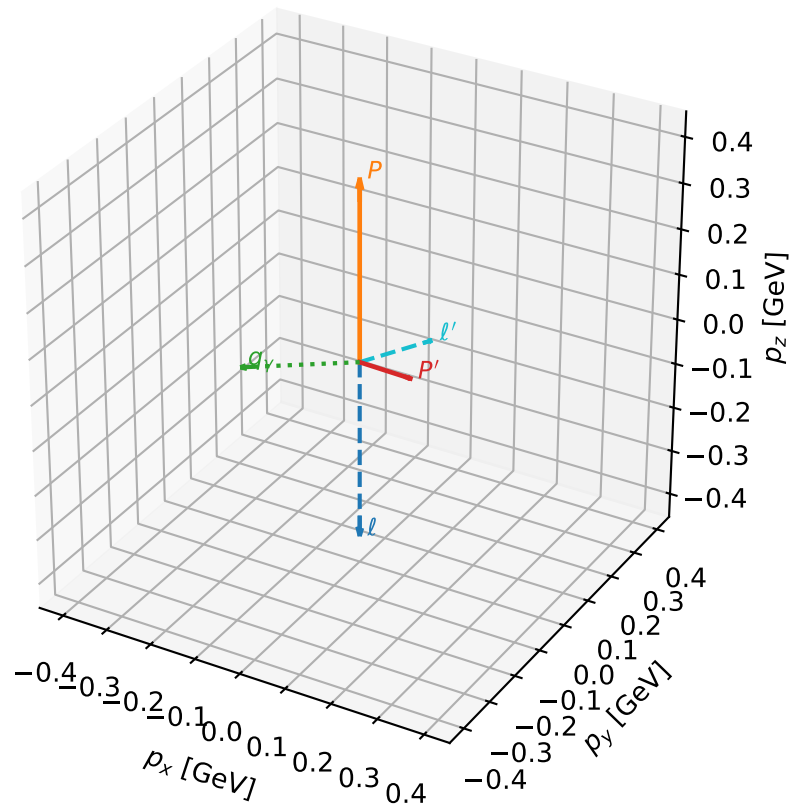
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_307 $d_{\{\text{rm GHZ}\}}=0.000345088$
 region: polarization_cluster_01_local_minimum_307
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.73376214$ rad
 incoming lepton: $0.74266|+\rangle +0.669668|-\rangle$
 proton mixing angle: $\alpha_p=0.95740588$ rad
 incoming proton: $0.575643|+\rangle +0.817701|-\rangle$

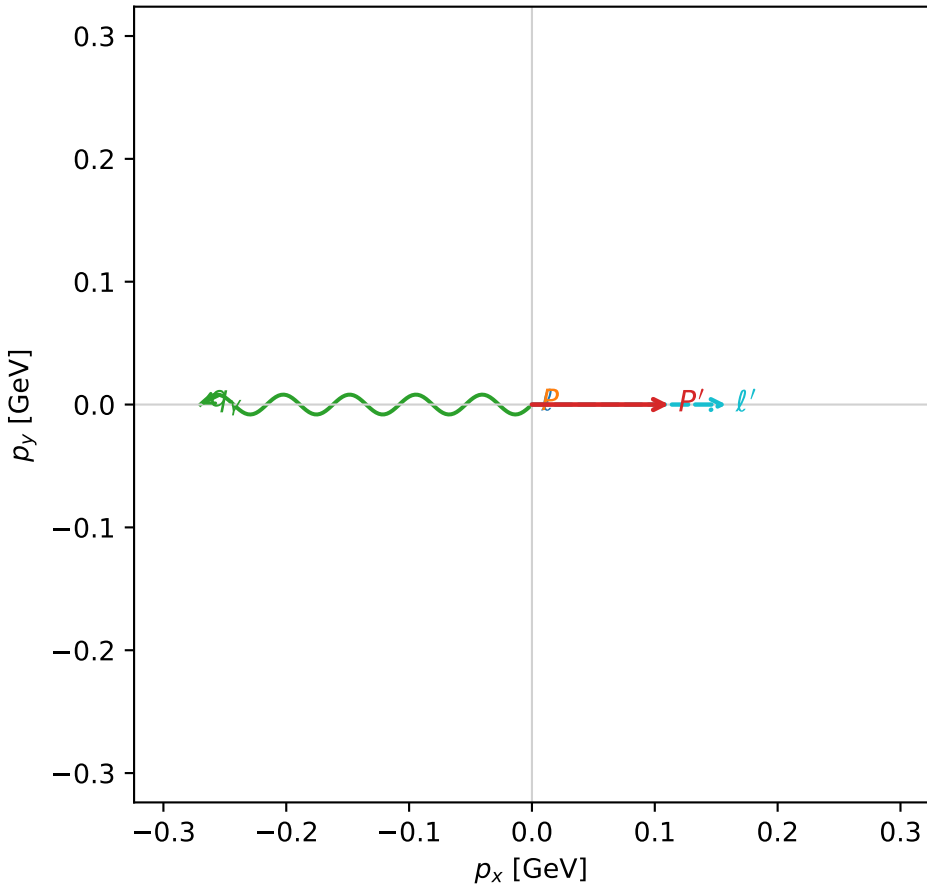
$s=1.9961$, $\sqrt{s}=1.41283$, $|\vec{P}|=0.395041$, $|\vec{P}'|=0.111693$
 $\theta_{P'}=1.60481$, $\theta_Y=1.89153$, $\phi_{P'}=4.84804\text{e-}05$, $\phi_Y=3.14162$
 $E_Y=0.284402$, $\theta_{\ell'}=1.03738$, $\phi_{\ell'}=8.25881\text{e-}06$
 $Q^2=0.219063$, $x_B=0.268477$, $t=-0.166181$, $W^2=1.47673$, $y=0.730966$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3950$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3950$
 P $E= 1.0178$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3950$
 ℓ' $E= 0.1838$ $p_x= 0.1583$ $p_y= 0.0000$ $p_z= 0.0935$
 P' $E= 0.9446$ $p_x= 0.1116$ $p_y= 0.0000$ $p_z= -0.0038$
 q_Y $E= 0.2844$ $p_x= -0.2699$ $p_y= -0.0000$ $p_z= -0.0897$

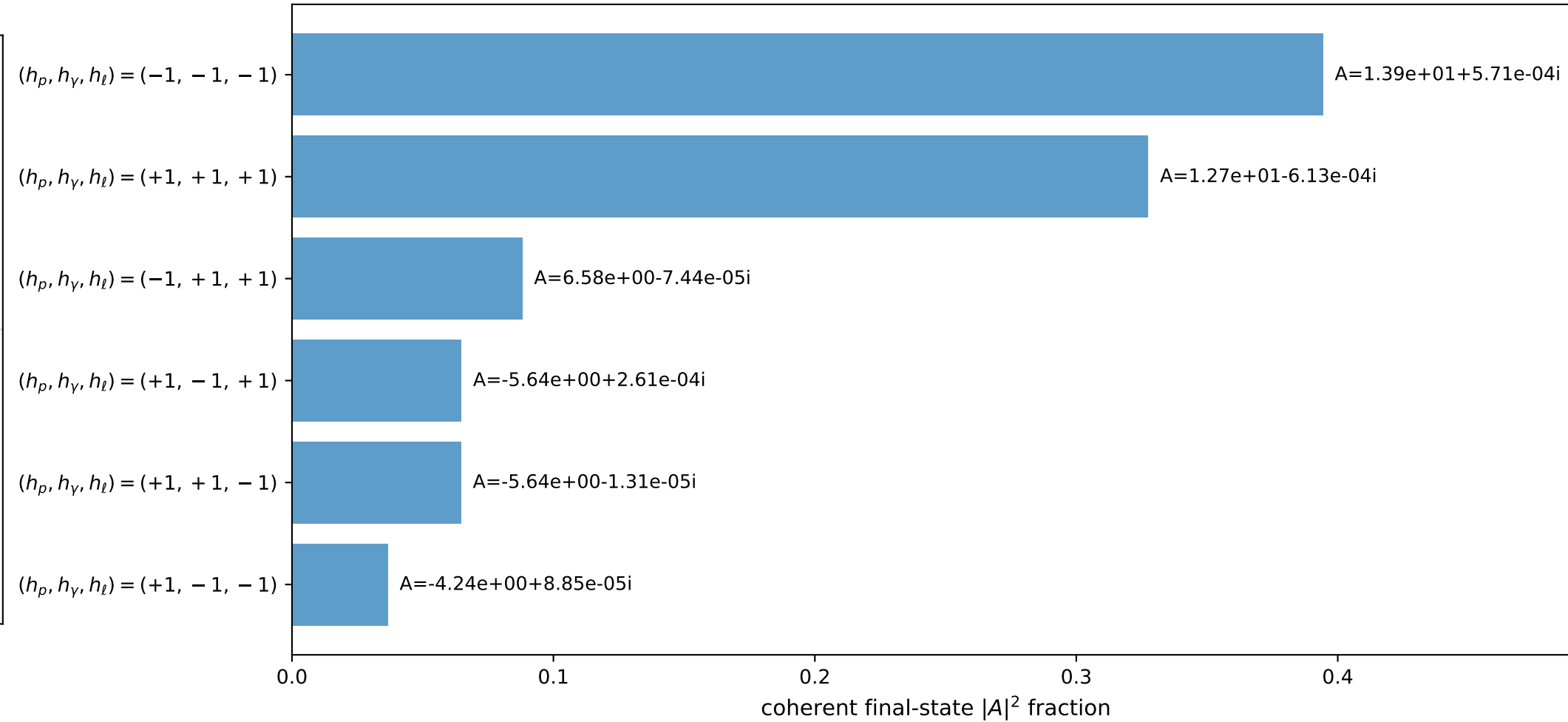
3D momenta



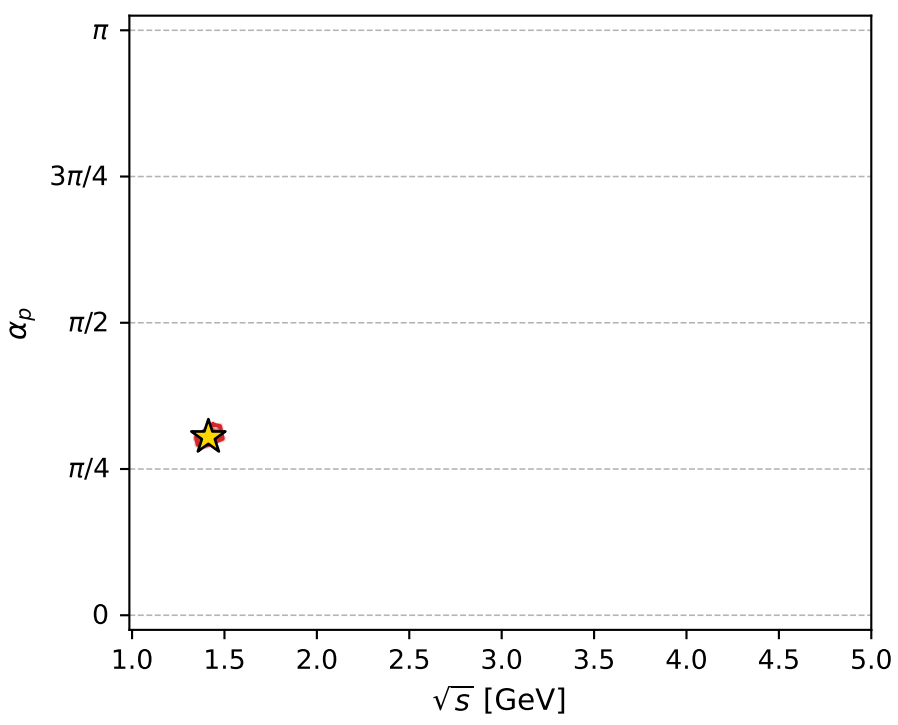
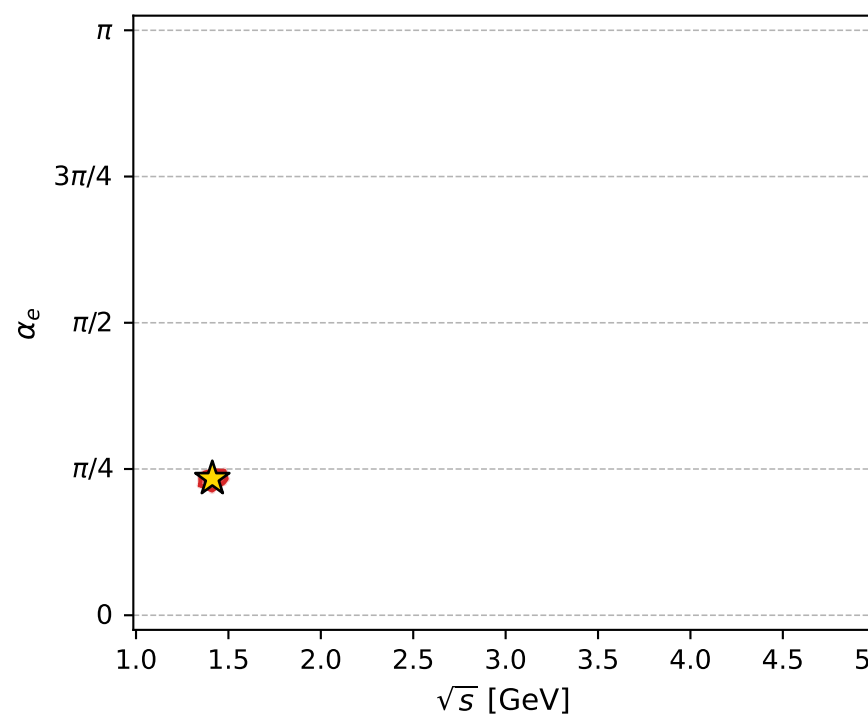
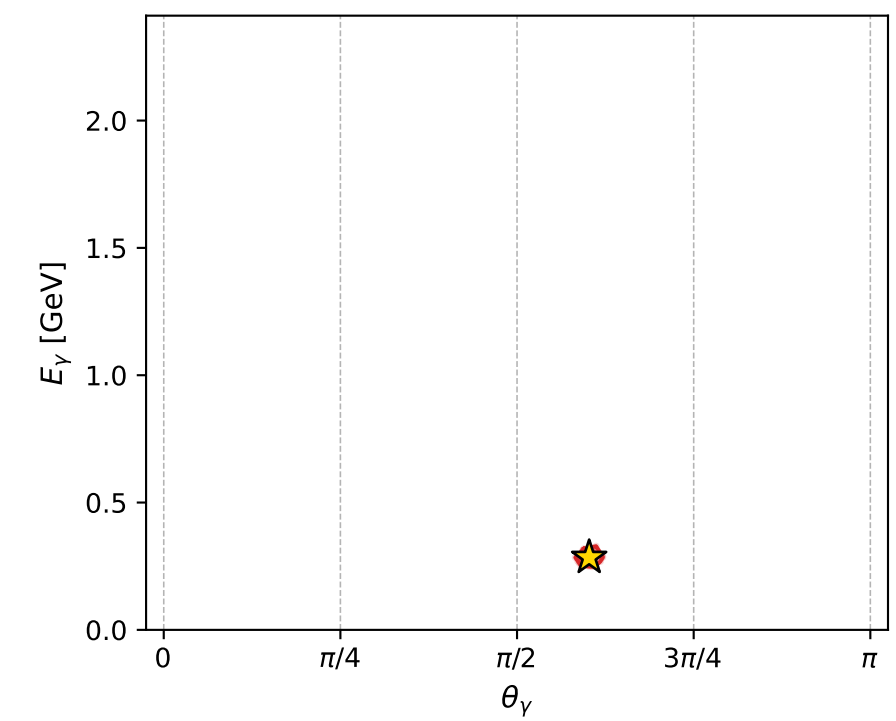
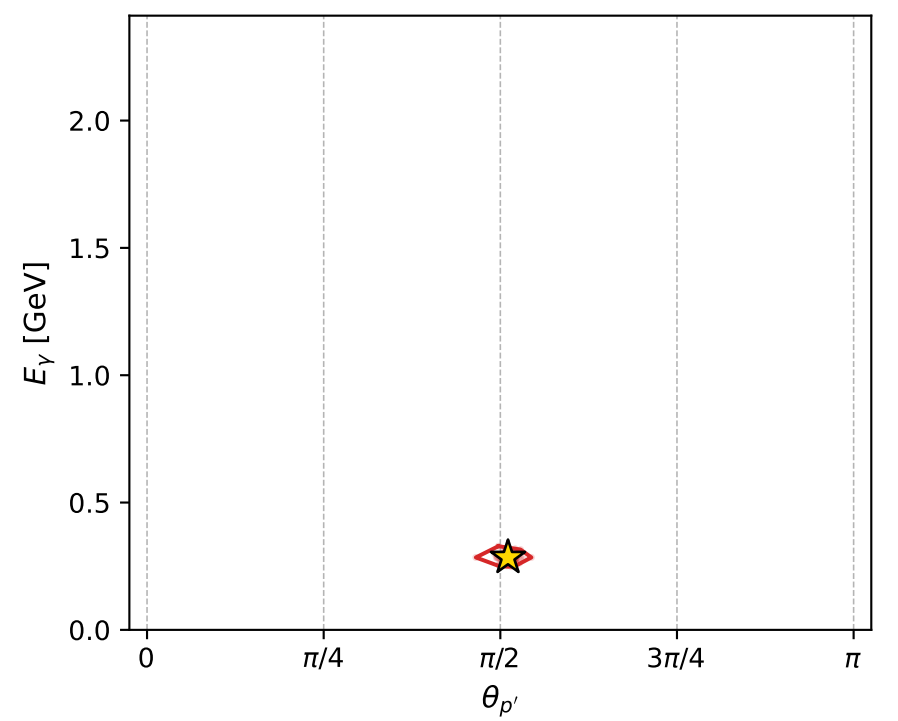
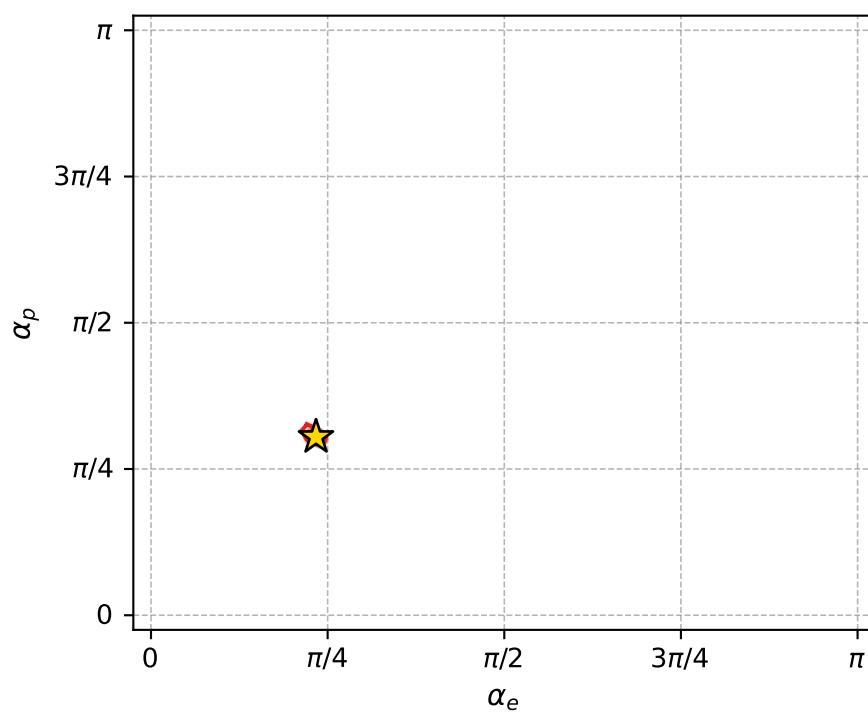
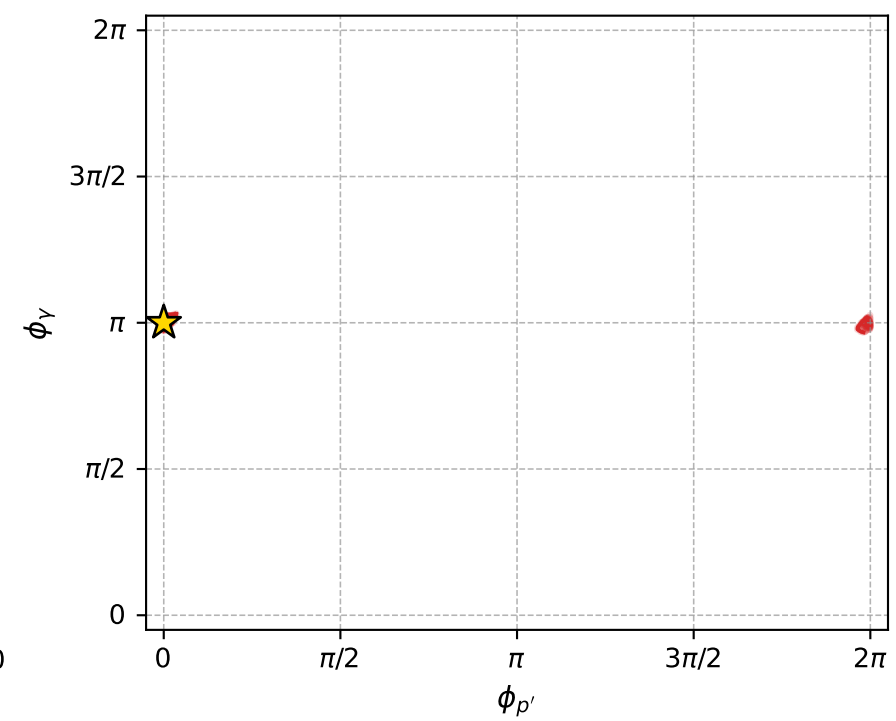
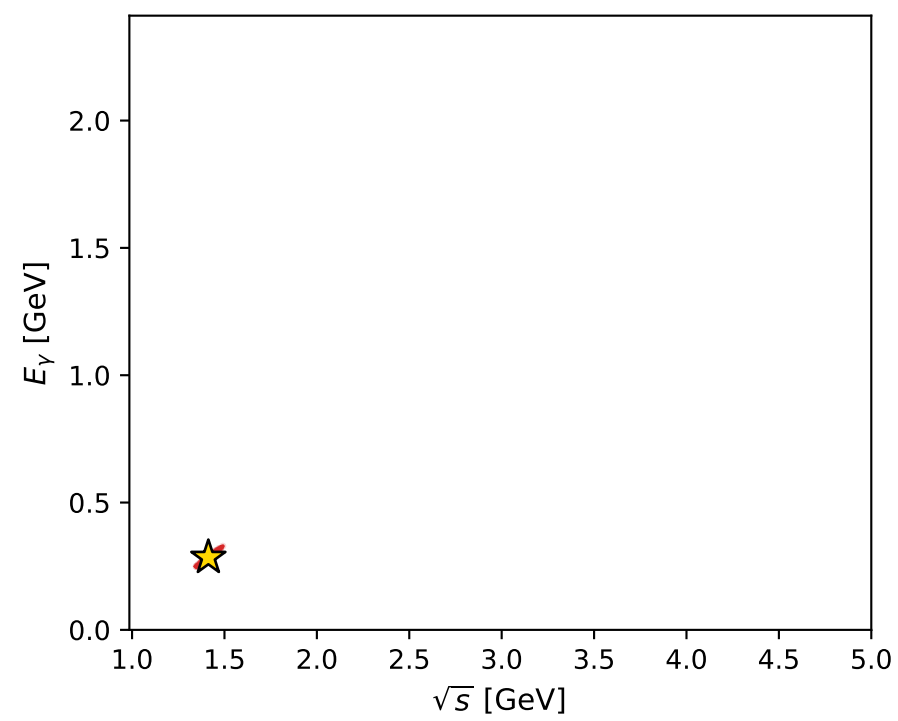
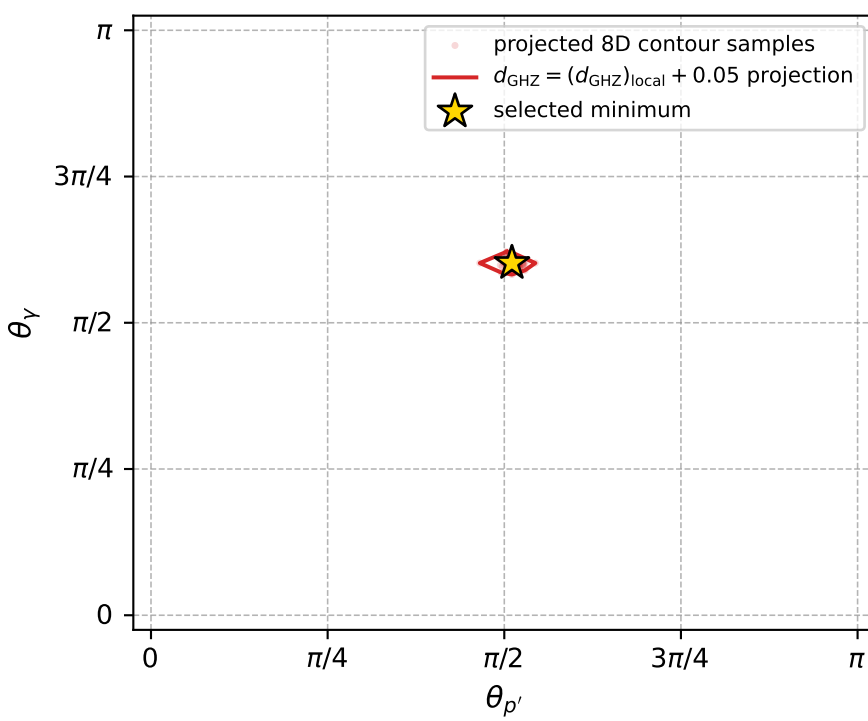
Transverse plane



Leading final-state amplitudes (N=6, retained=97.6%)



electron: polarization cluster P1, local minimum 307; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00034508818$
 $d_{\text{GHZ}} \text{ contour} = 0.050345088$
 above global minimum = 0.00034506589
 8D contour samples = 256

selected phase-space point:

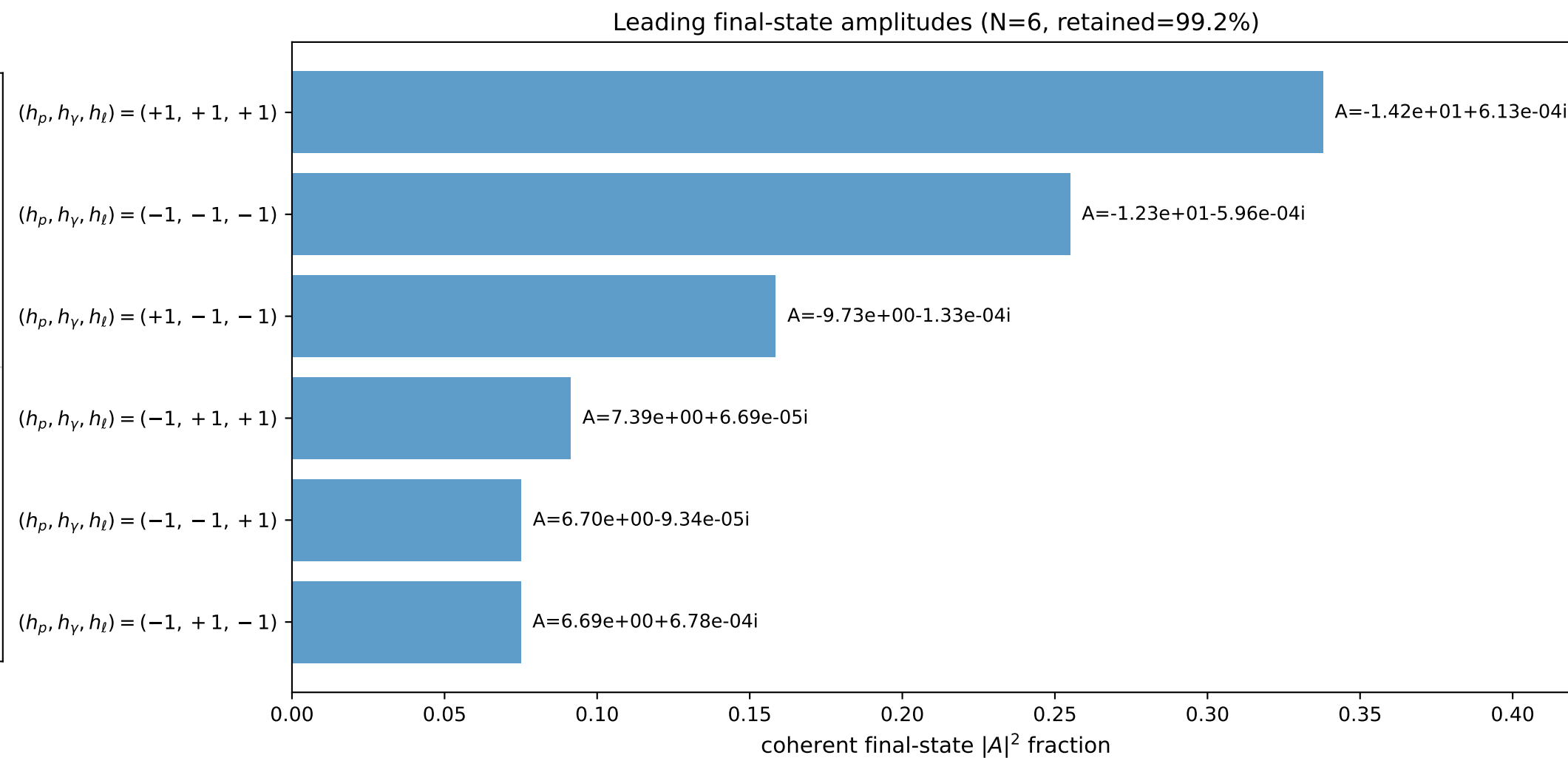
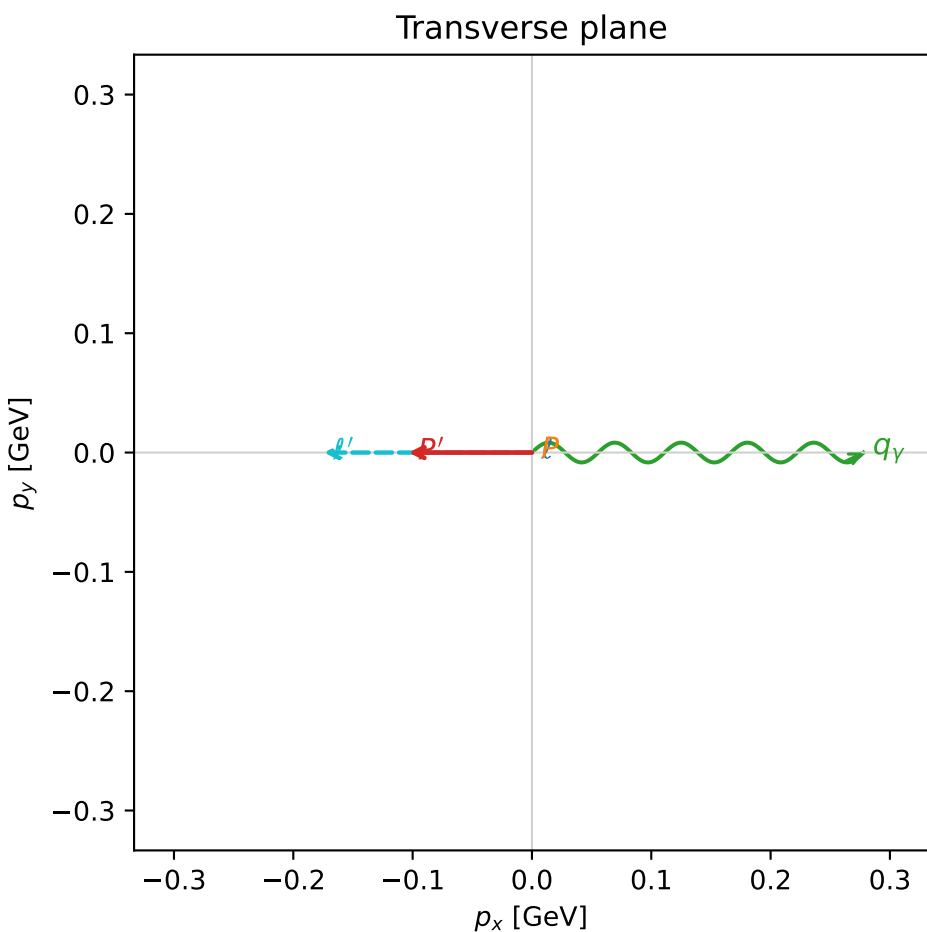
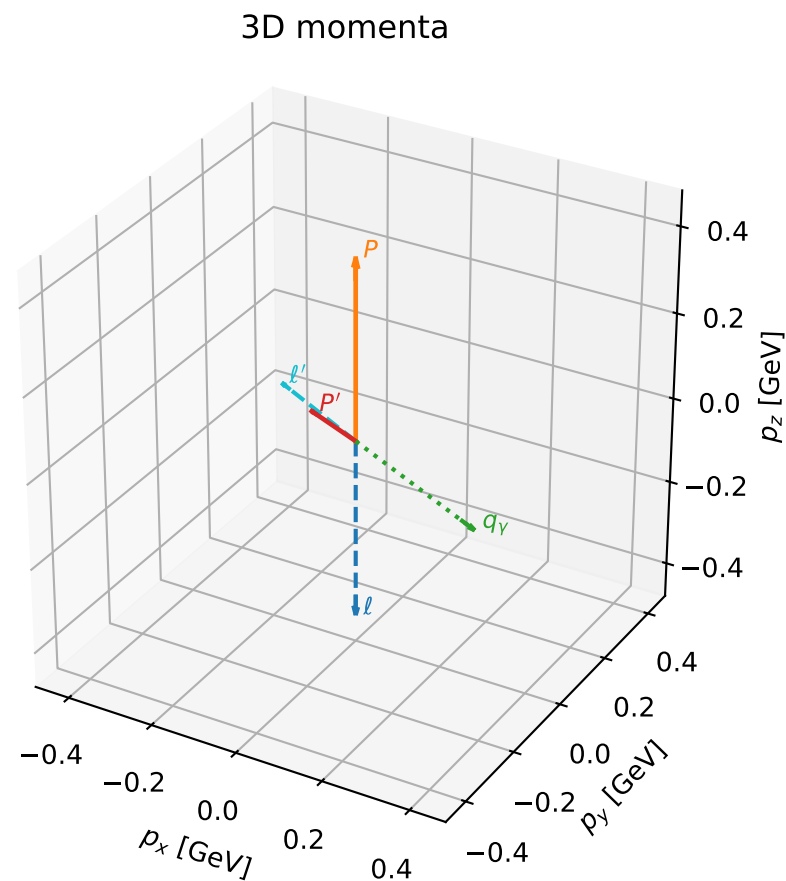
$\text{sqrt_s} = 1.412834$
 $\text{theta_p_out} = 1.604814$
 $\text{theta_gamma_out} = 1.891532$
 $\text{q0ut} = 0.2844017$
 $\text{phi_p_out} = 4.848041\text{e-}05$
 $\text{phi_gamma_out} = 3.141618$
 $\text{alpha_e} = 0.7337621$
 $\text{alpha_p} = 0.9574059$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

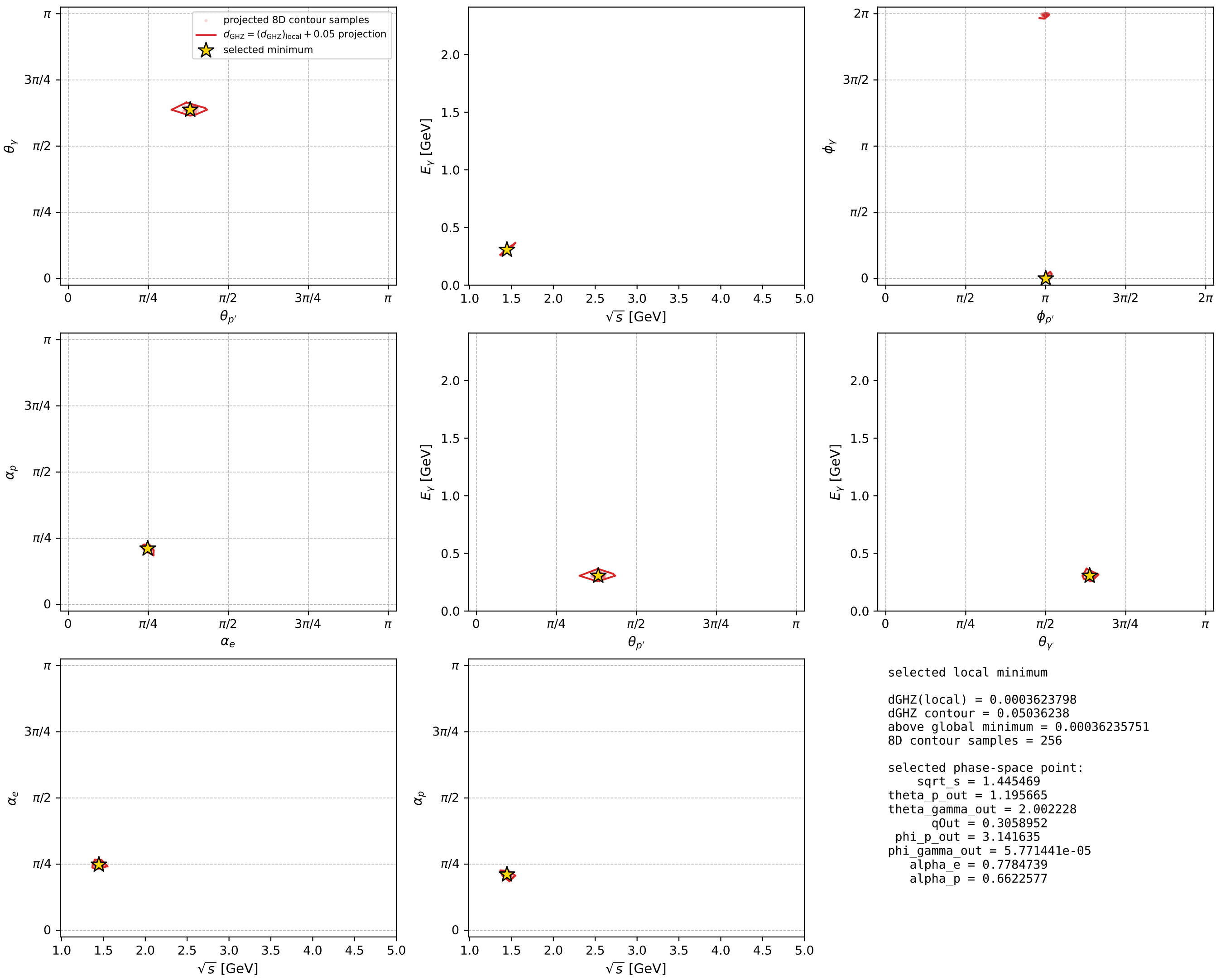
dghz_mixing_angles_polarization_01_local_minimum_310 $d_{\{\text{rm GHZ}\}}=0.00036238$
 region: polarization_cluster_01_local_minimum_310
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77847391$ rad
 incoming lepton: $0.711986|+\rangle +0.702194|-\rangle$
 proton mixing angle: $\alpha_p=0.66225766$ rad
 incoming proton: $0.788606|+\rangle +0.614899|-\rangle$

$s=2.08938$, $\sqrt{s}=1.44547$, $|\vec{P}|=0.418389$, $|\vec{P}'|=0.111163$
 $\theta_{p'}=1.19566$, $\theta_Y=2.00223$, $\phi_{p'}=3.14164$, $\phi_Y=5.77144\text{e-}05$
 $E_Y=0.305895$, $\theta_{\ell'}=1.10728$, $\phi_{\ell'}=3.14166$
 $Q^2=0.236135$, $x_B=0.267754$, $t=-0.146516$, $W^2=1.52562$, $y=0.729132$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4184$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4184$
 P $E= 1.0271$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4184$
 ℓ' $E= 0.1950$ $p_x= -0.1744$ $p_y= -0.0000$ $p_z= 0.0872$
 P' $E= 0.9446$ $p_x= -0.1034$ $p_y= -0.0000$ $p_z= 0.0407$
 q_Y $E= 0.3059$ $p_x= 0.2779$ $p_y= 0.0000$ $p_z= -0.1279$



electron: polarization cluster P1, local minimum 310; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

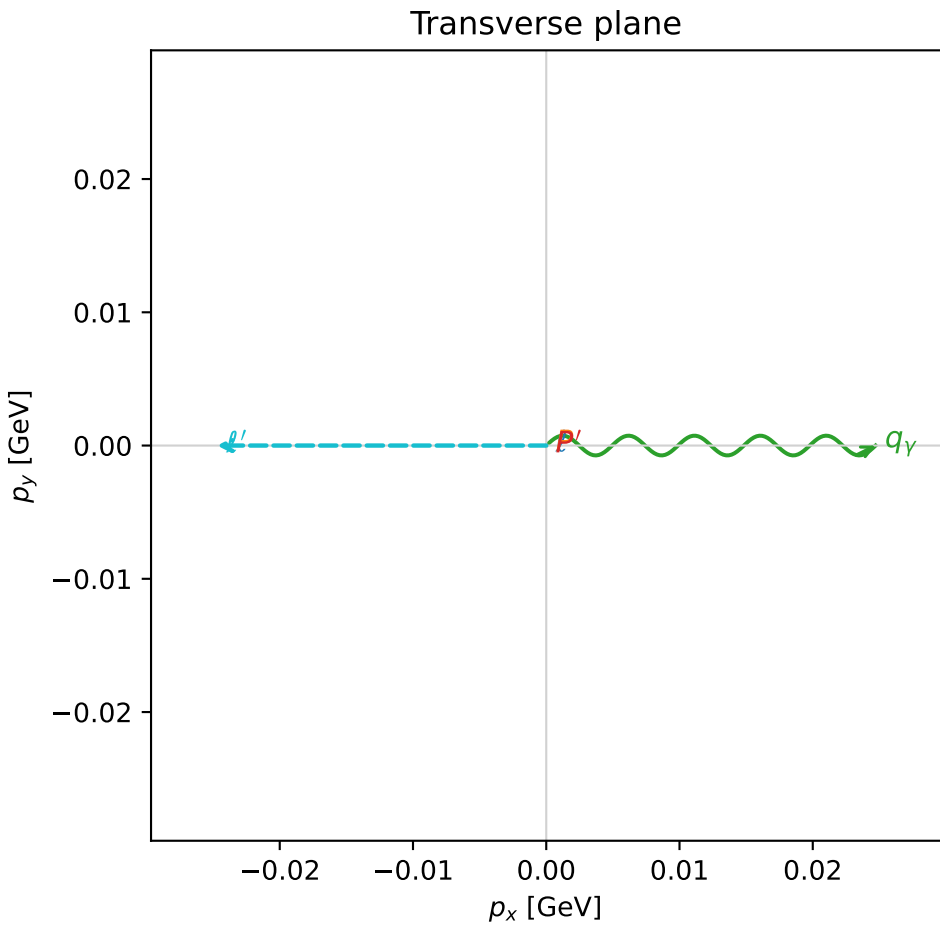
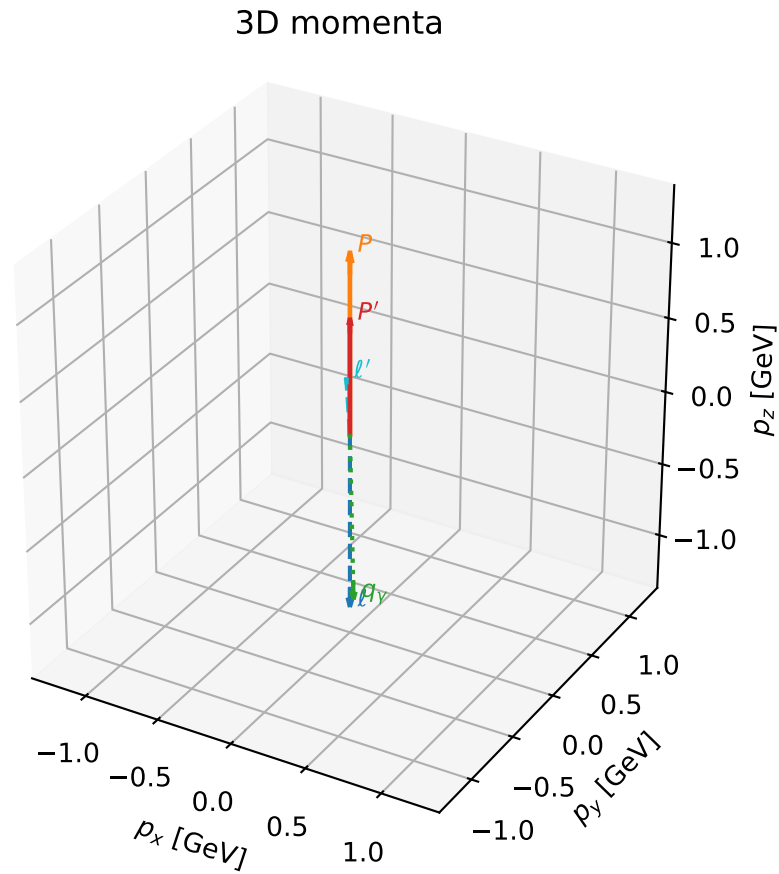


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_314 $d_{\{\text{rm GHZ}\}}=0.000387584$
 region: polarization_cluster_01_local_minimum_314
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.3236924$ rad
 incoming lepton: $0.244597|+\rangle + 0.969625|-\rangle$
 proton mixing angle: $\alpha_p=0.24858017$ rad
 incoming proton: $0.969263|+\rangle + 0.246028|-\rangle$

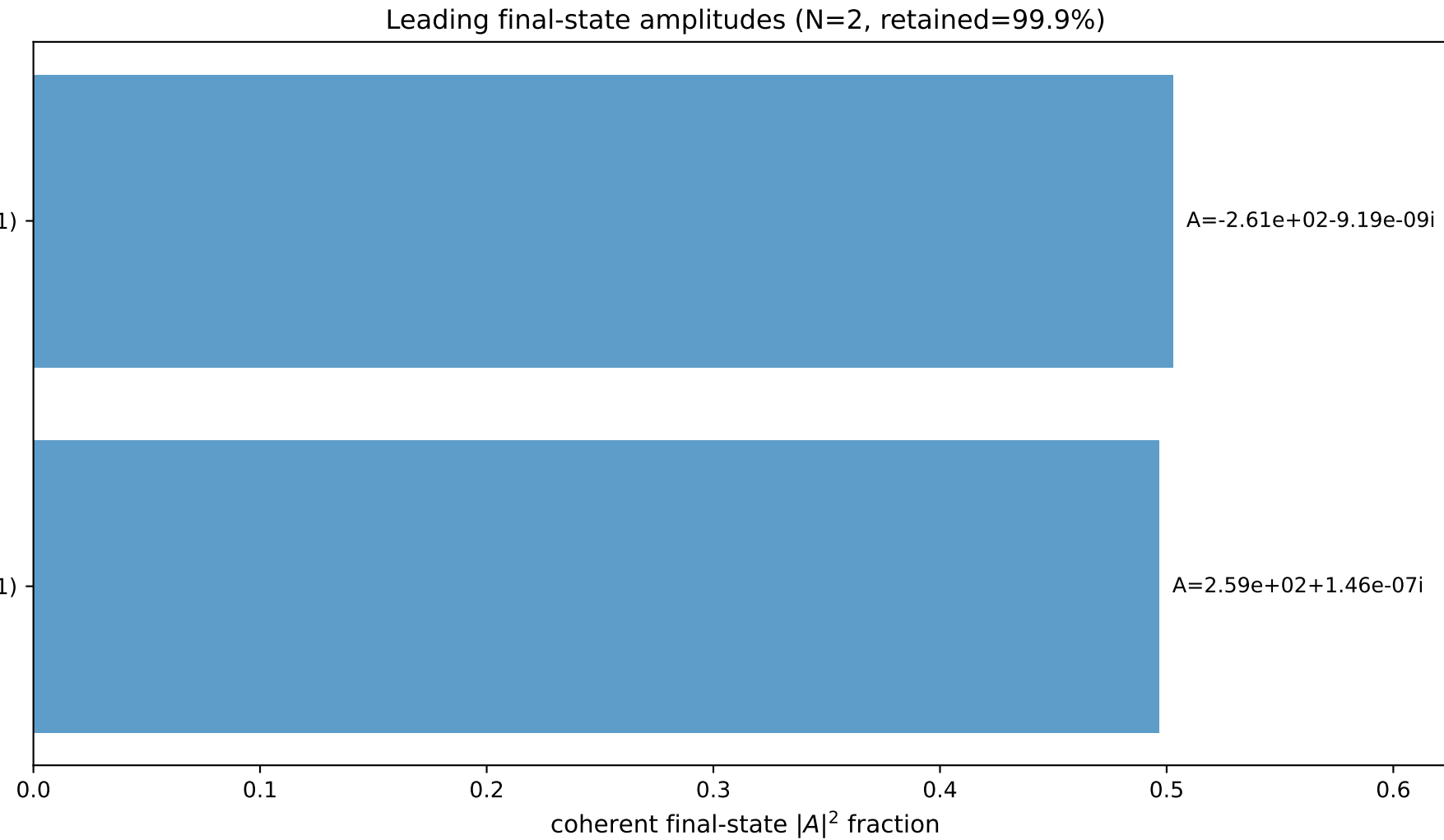
$s=7.3952$, $\sqrt{s}=2.71941$, $|\vec{P}|=1.19794$, $|\vec{P}'|=0.763049$
 $\theta_{P'}=0$, $\theta_V=3.11984$, $\phi_{P'}=3.82165$, $\phi_V=2.0169\text{e-}05$
 $E_V=1.13637$, $\theta_{\ell'}=0.0661538$, $\phi_{\ell'}=3.14161$
 $Q^2=1.78954$, $x_B=0.285346$, $t=-0.091589$, $W^2=5.36178$, $y=0.962567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1979$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1979$
 P $E= 1.5215$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1979$
 ℓ' $E= 0.3739$ $p_x= -0.0247$ $p_y= -0.0000$ $p_z= 0.3731$
 P' $E= 1.2092$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7630$
 q_Y $E= 1.1364$ $p_x= 0.0247$ $p_y= 0.0000$ $p_z= -1.1361$

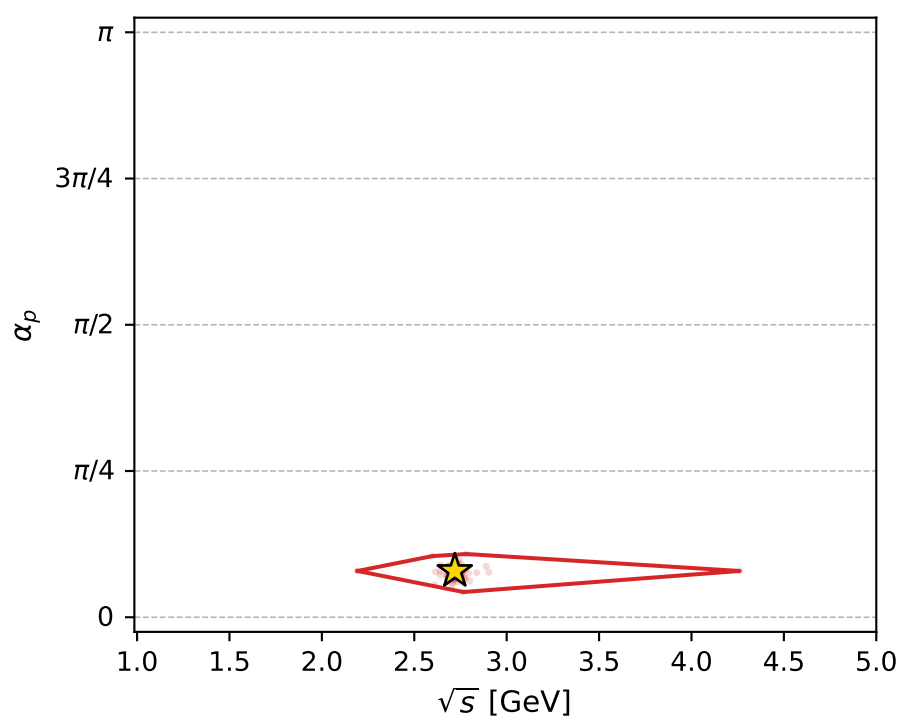
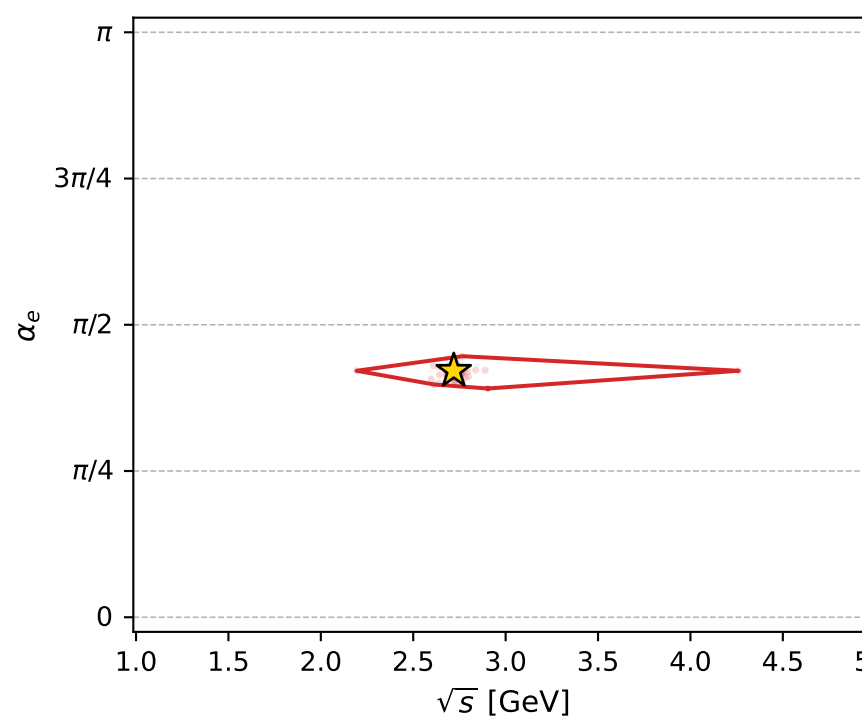
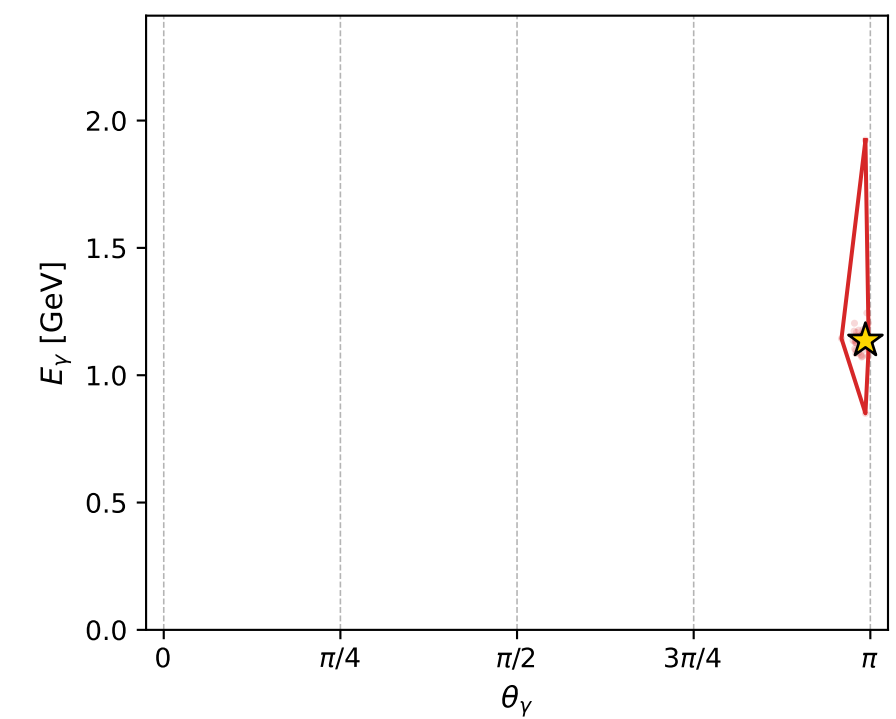
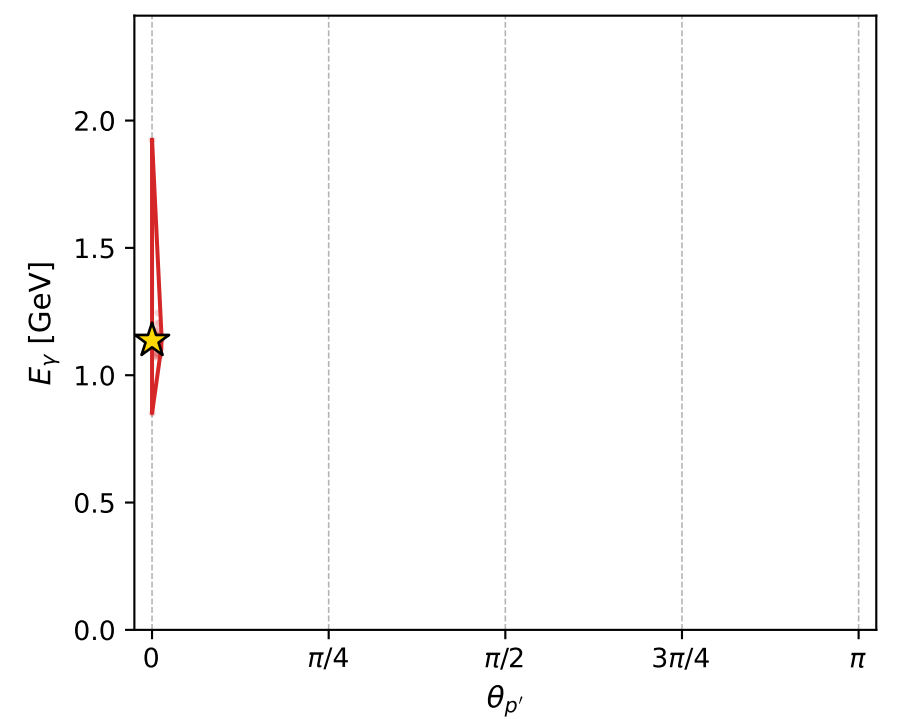
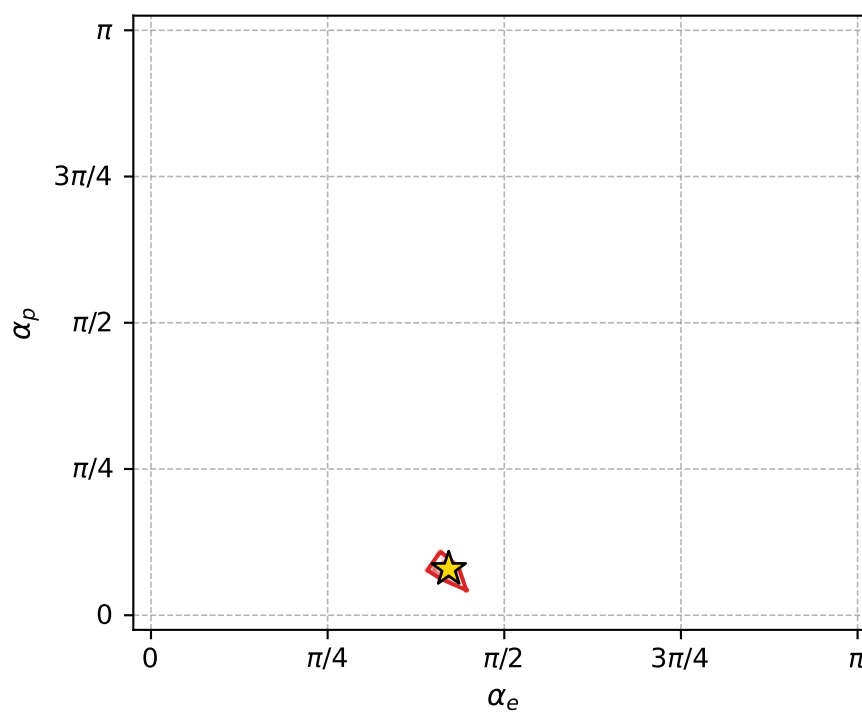
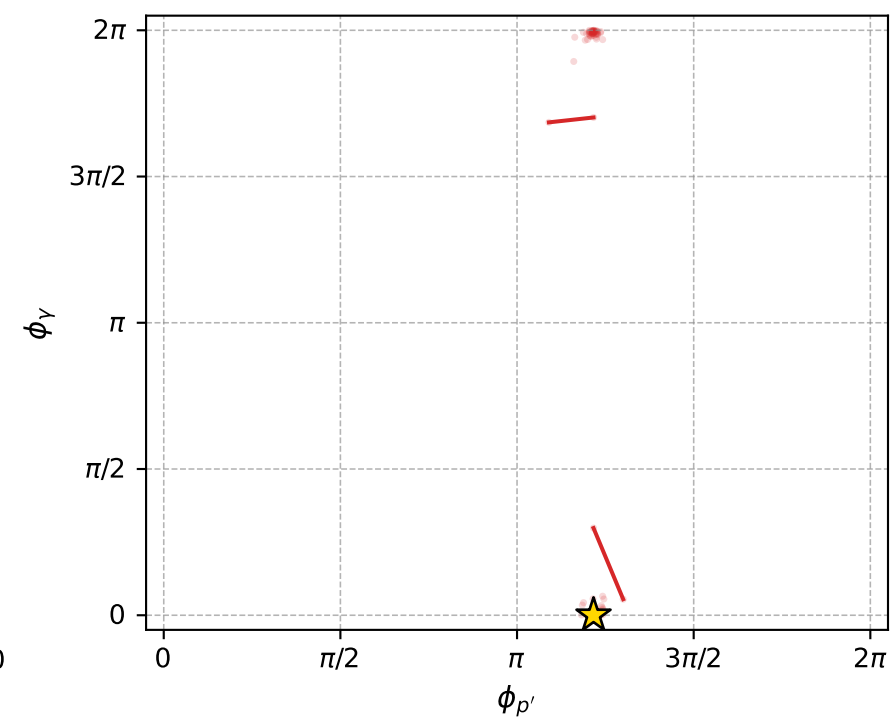
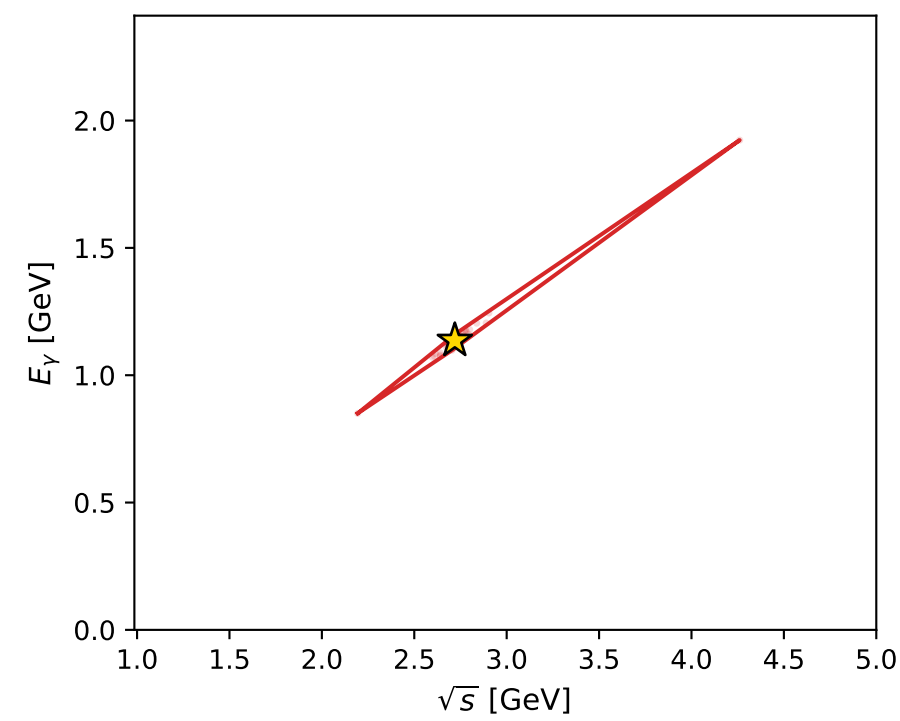
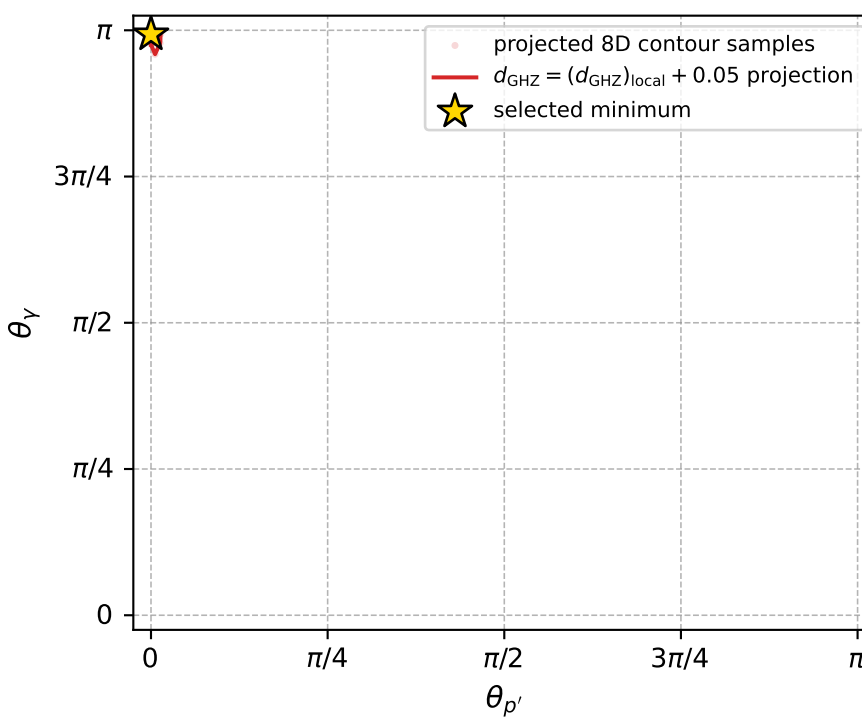


$(h_p, h_V, h_\ell) = (+1, -1, -1)$

$(h_p, h_V, h_\ell) = (-1, +1, +1)$



electron: polarization cluster P1, local minimum 314; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00038758423$
 $d_{\text{GHZ}} \text{ contour} = 0.050387584$
 above global minimum = 0.00038756195
 8D contour samples = 128

selected phase-space point:

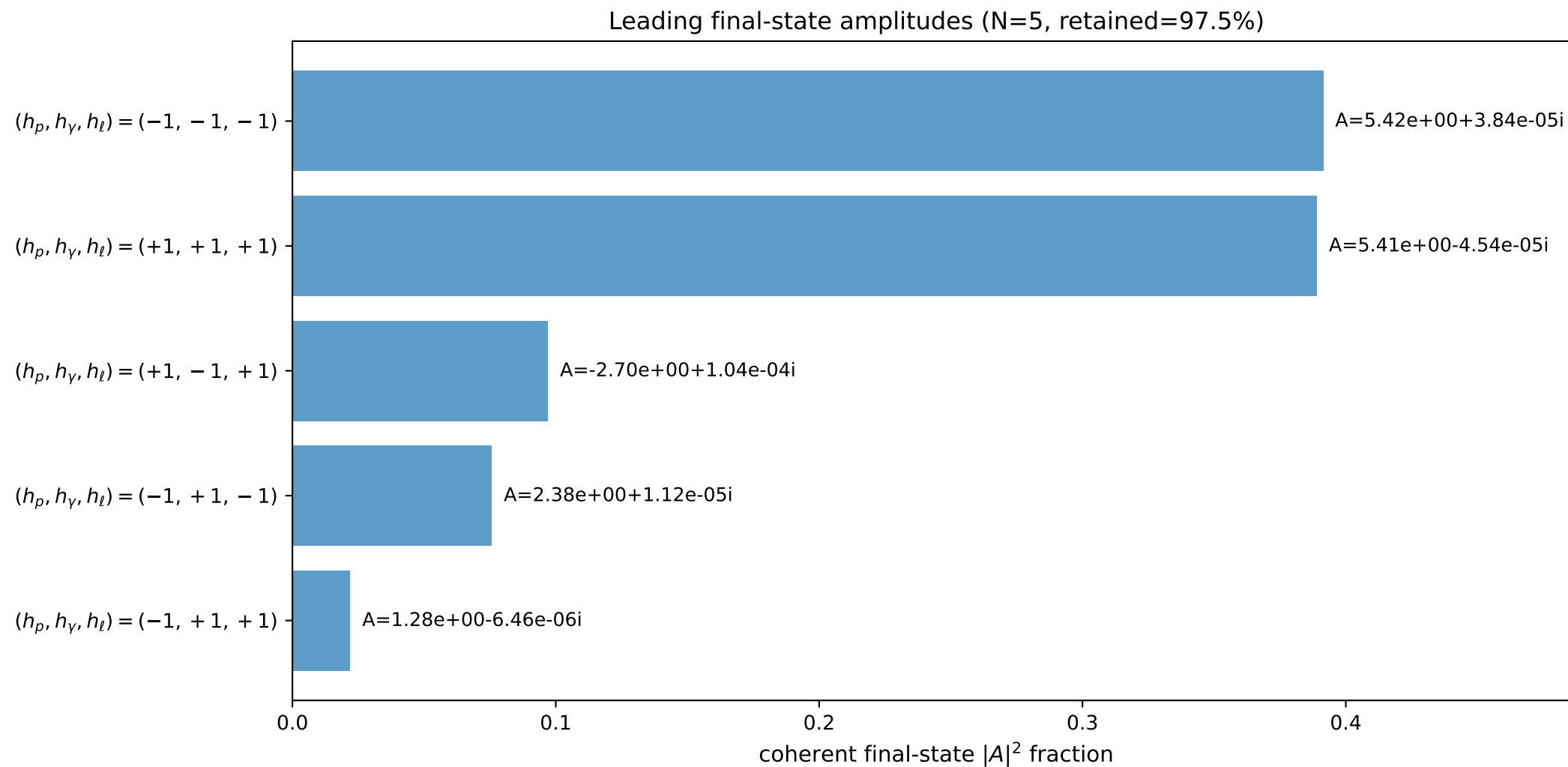
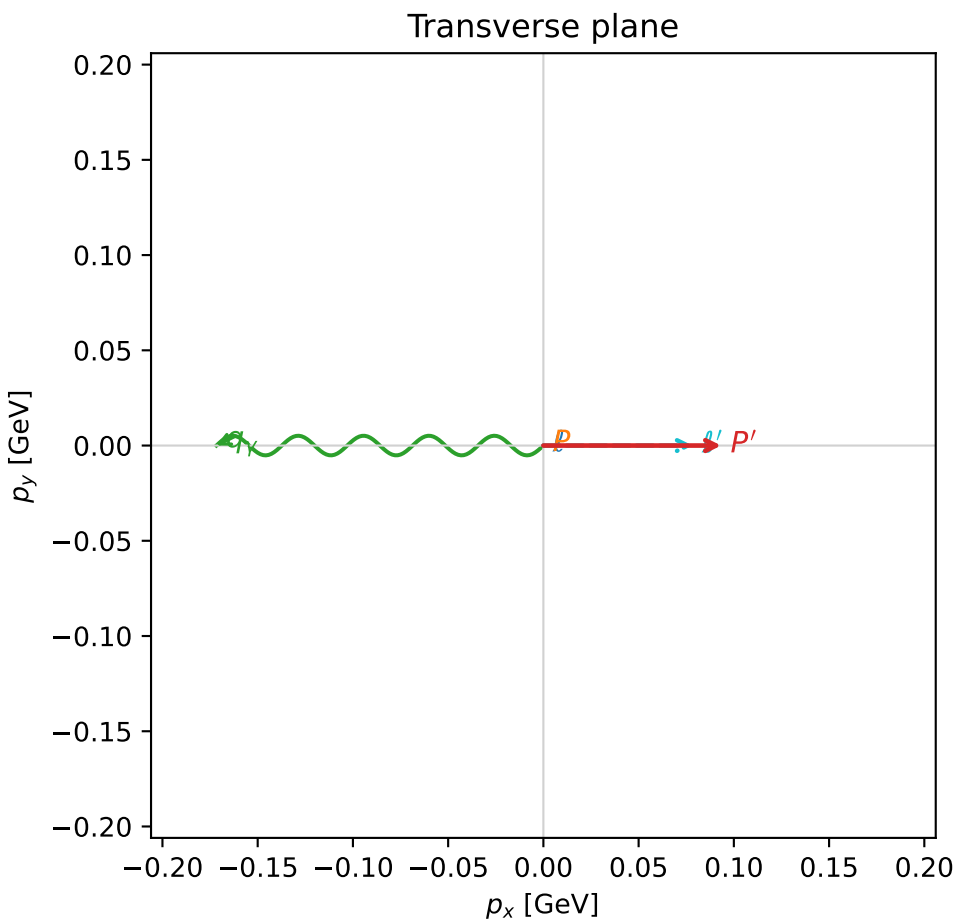
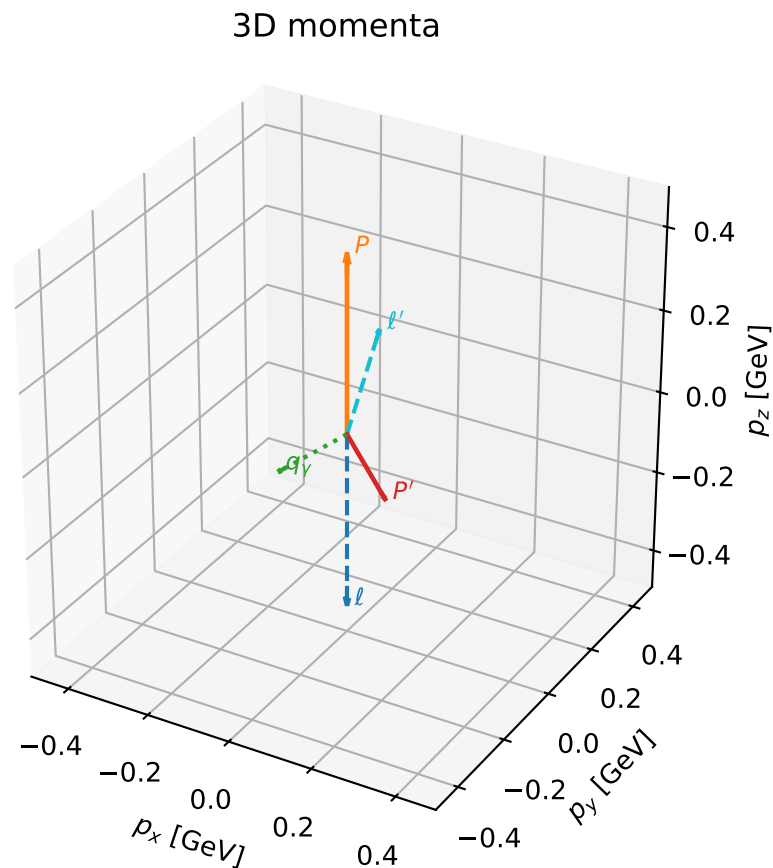
$\text{sqrt_s} = 2.719412$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 3.119842$
 $\text{q0ut} = 1.136372$
 $\text{phi_p_out} = 3.821648$
 $\text{phi_gamma_out} = 2.016898\text{e-}05$
 $\text{alpha_e} = 1.323692$
 $\text{alpha_p} = 0.2485802$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

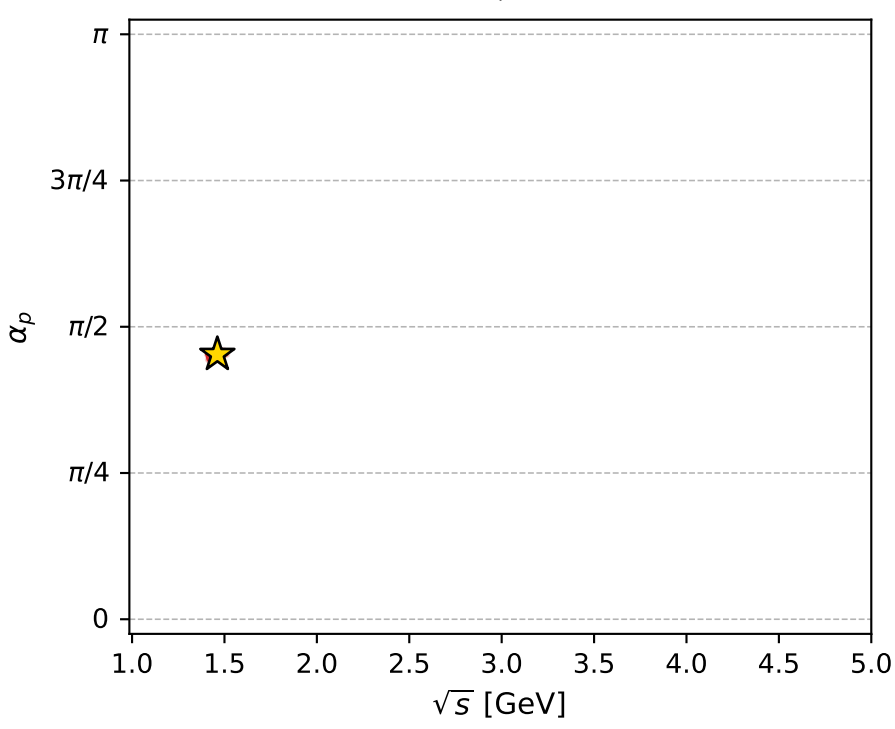
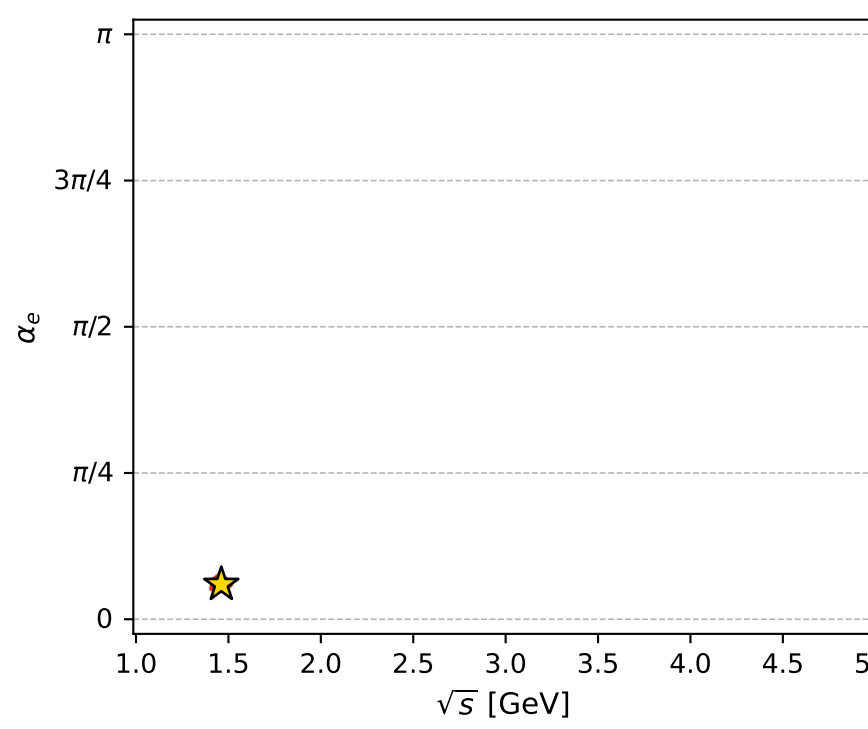
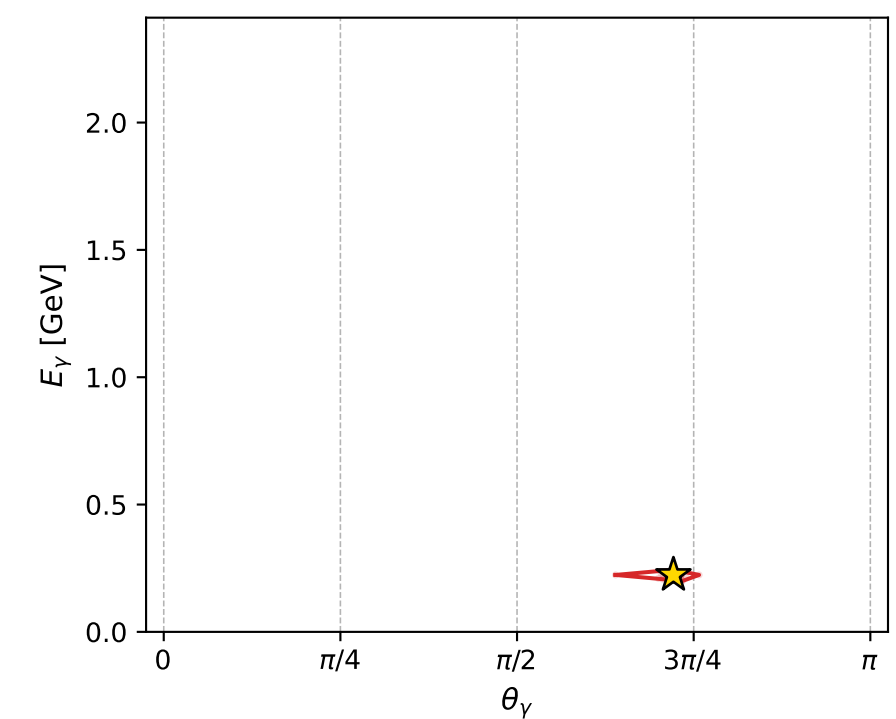
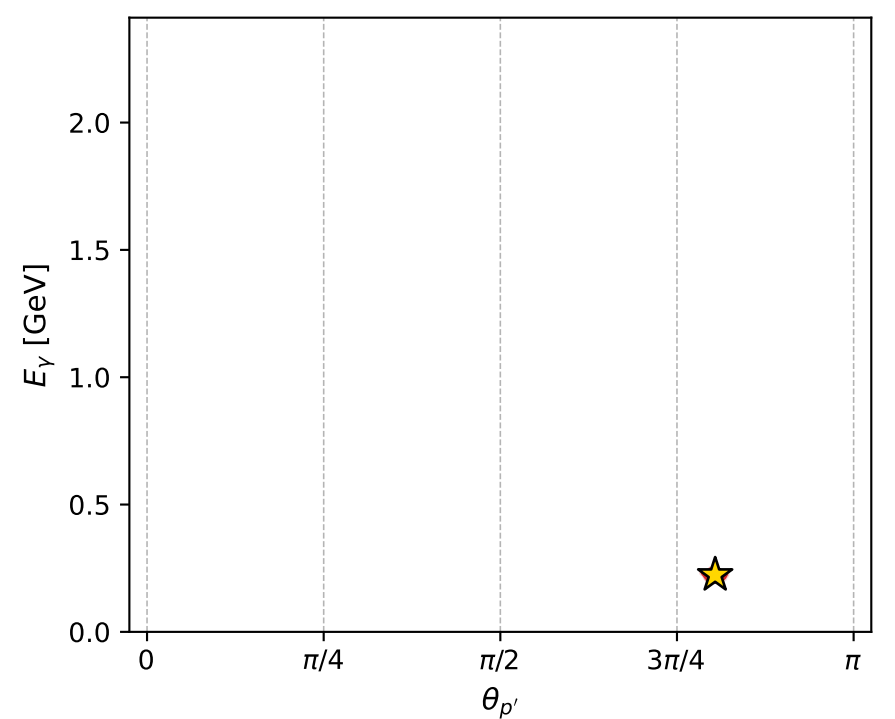
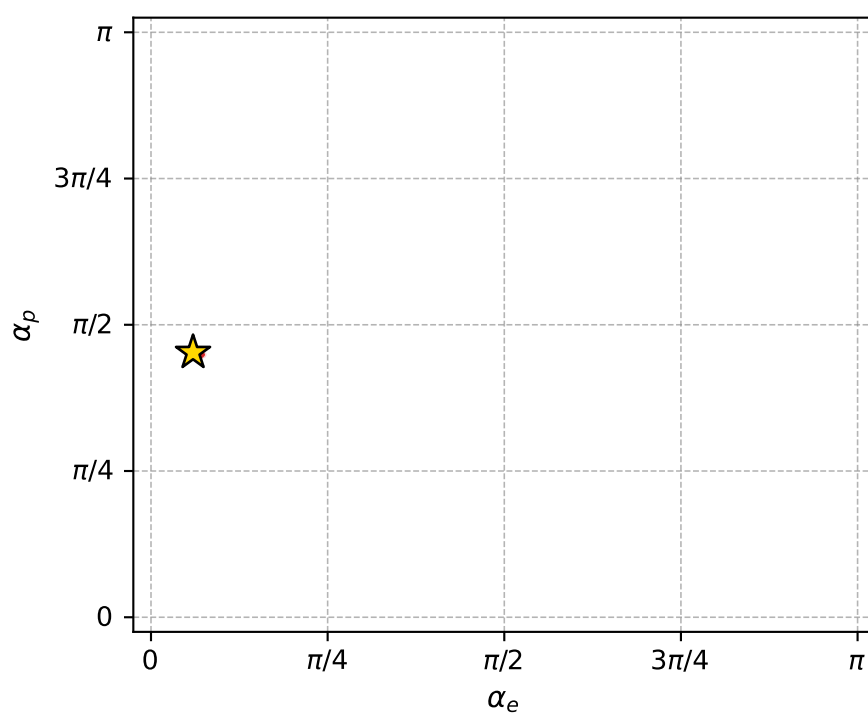
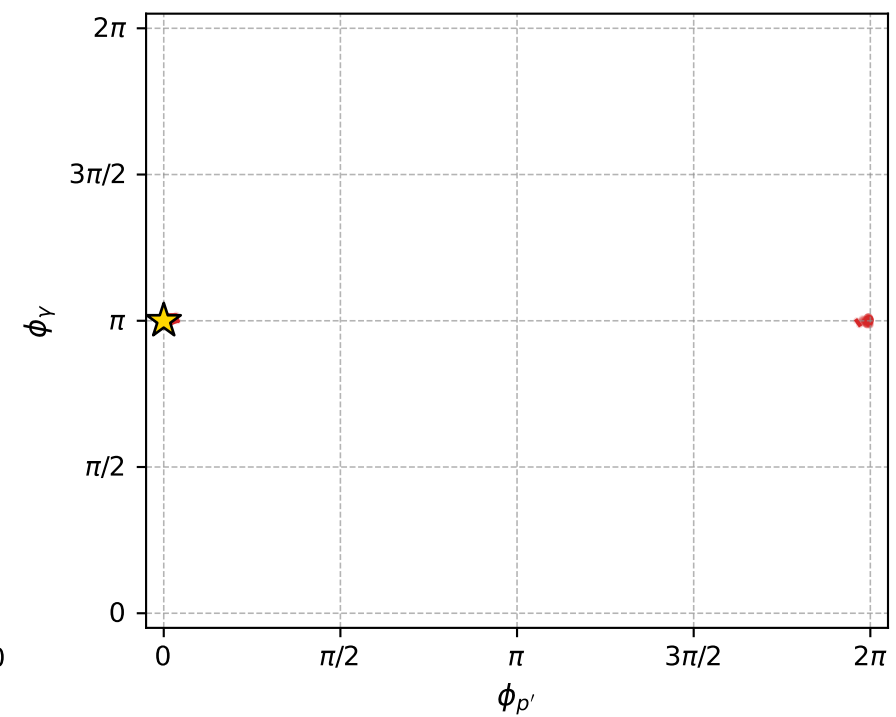
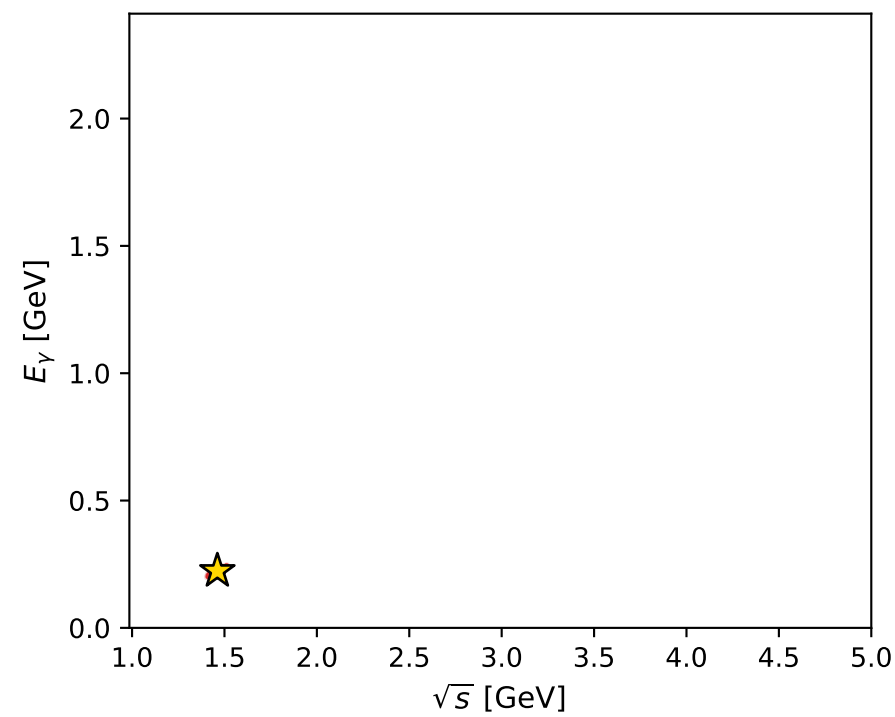
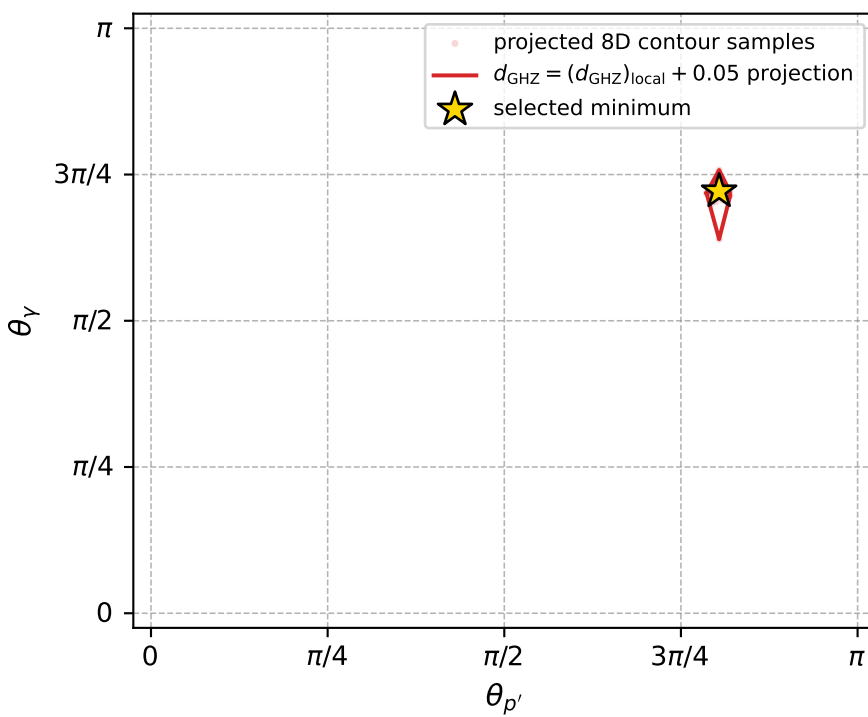
dghz_mixing_angles_polarization_01_local_minimum_315 d_{\rm GHZ}=0.000397773
 region: polarization_cluster_01_local_minimum_315
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.18693006$ rad
 incoming lepton: $0.982579|+\rangle +0.185843|-\rangle$
 proton mixing angle: $\alpha_p=1.4220063$ rad
 incoming proton: $0.148242|+\rangle +0.988951|-\rangle$

$s=2.13598$, $\sqrt{s}=1.4615$, $|\vec{P}|=0.429742$, $|\vec{P}'|=0.161637$
 $\theta_{P'}=2.52599$, $\theta_Y=2.26612$, $\phi_{P'}=6.38908\text{e-}06$, $\phi_Y=3.14161$
 $E_Y=0.22356$, $\theta_{\ell'}=0.277287$, $\phi_{\ell'}=2.83218\text{e-}05$
 $Q^2=0.482426$, $x_B=0.534692$, $t=-0.317837$, $W^2=1.29967$, $y=0.718276$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4297$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4297$
 P $E= 1.0318$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4297$
 ℓ' $E= 0.2861$ $p_x= 0.0783$ $p_y= 0.0000$ $p_z= 0.2752$
 P' $E= 0.9518$ $p_x= 0.0933$ $p_y= 0.0000$ $p_z= -0.1320$
 q_Y $E= 0.2236$ $p_x= -0.1717$ $p_y= -0.0000$ $p_z= -0.1432$



electron: polarization cluster P1, local minimum 315; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00039777303$
 $d_{\text{GHZ}} \text{ contour} = 0.050397773$
 above global minimum = 0.00039775074
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.461499$
 $\text{theta_p_out} = 2.525991$
 $\text{theta_gamma_out} = 2.266115$
 $\text{q0out} = 0.22356$
 $\text{phi_p_out} = 6.389082\text{e-}06$
 $\text{phi_gamma_out} = 3.141609$
 $\text{alpha_e} = 0.1869301$
 $\text{alpha_p} = 1.422006$

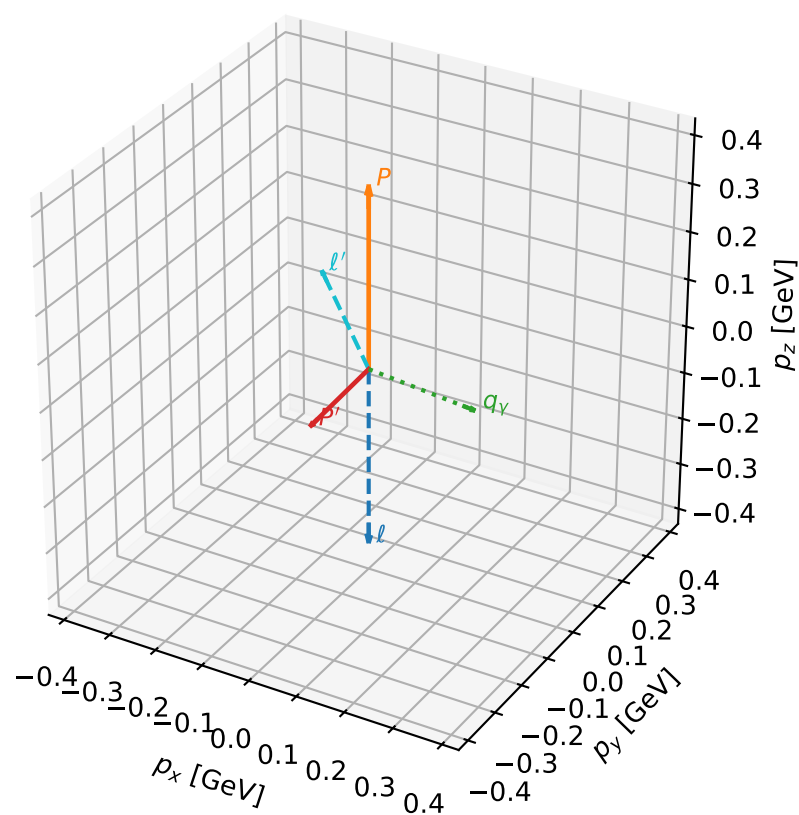
coherent mixing angles: minimum d_{GHz} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_322 $d_{\{\text{rm GHz}\}}=0.000424968$
 region: polarization_cluster_01_local_minimum_322
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2500583$ rad
 incoming lepton: $0.315267|+\rangle +0.949003|-\rangle$
 proton mixing angle: $\alpha_p=0.25599852$ rad
 incoming proton: $0.967411|+\rangle +0.253212|-\rangle$

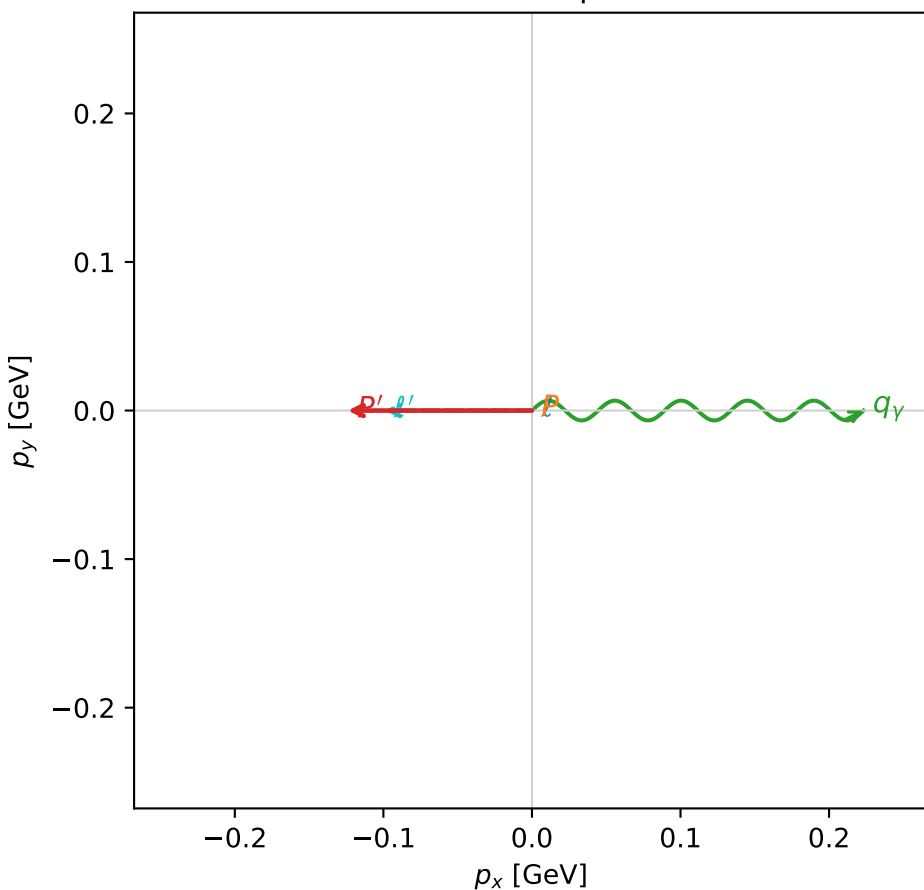
$s=1.92782$, $\sqrt{s}=1.38846$, $|\vec{P}|=0.377389$, $|\vec{P}'|=0.200003$
 $\theta_{P'}=2.47294$, $\theta_Y=1.67141$, $\phi_{P'}=3.14166$, $\phi_Y=5.10413\text{e-}05$
 $E_Y=0.224319$, $\theta_{\ell'}=0.504938$, $\phi_{\ell'}=3.14163$
 $Q^2=0.290229$, $x_B=0.377517$, $t=-0.29817$, $W^2=1.3584$, $y=0.733585$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3774$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3774$
 P $E= 1.0111$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3774$
 ℓ' $E= 0.2051$ $p_x= -0.0992$ $p_y= -0.0000$ $p_z= 0.1795$
 P' $E= 0.9591$ $p_x= -0.1240$ $p_y= -0.0000$ $p_z= -0.1569$
 q_Y $E= 0.2243$ $p_x= 0.2232$ $p_y= 0.0000$ $p_z= -0.0225$

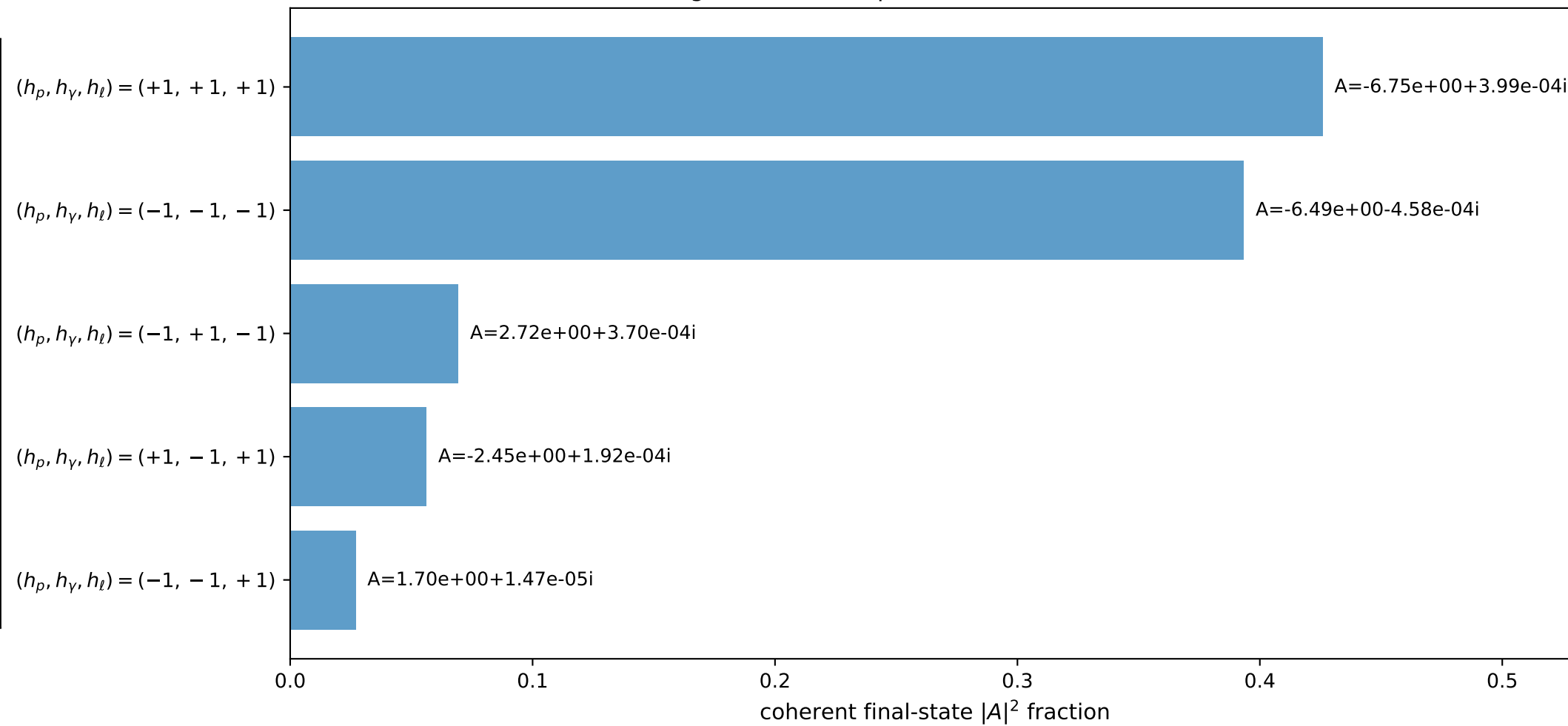
3D momenta



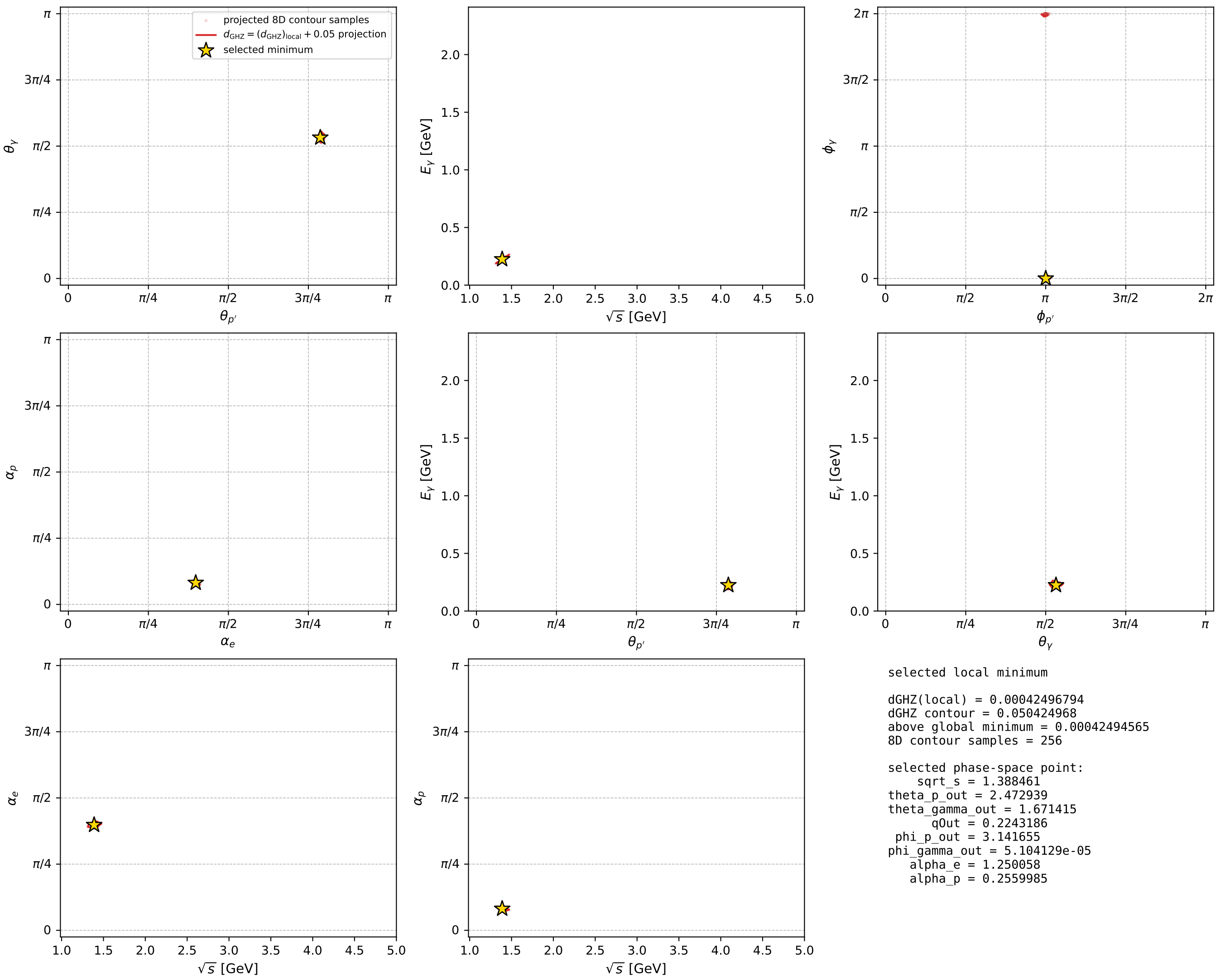
Transverse plane



Leading final-state amplitudes (N=5, retained=97.2%)



electron: polarization cluster P1, local minimum 322; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

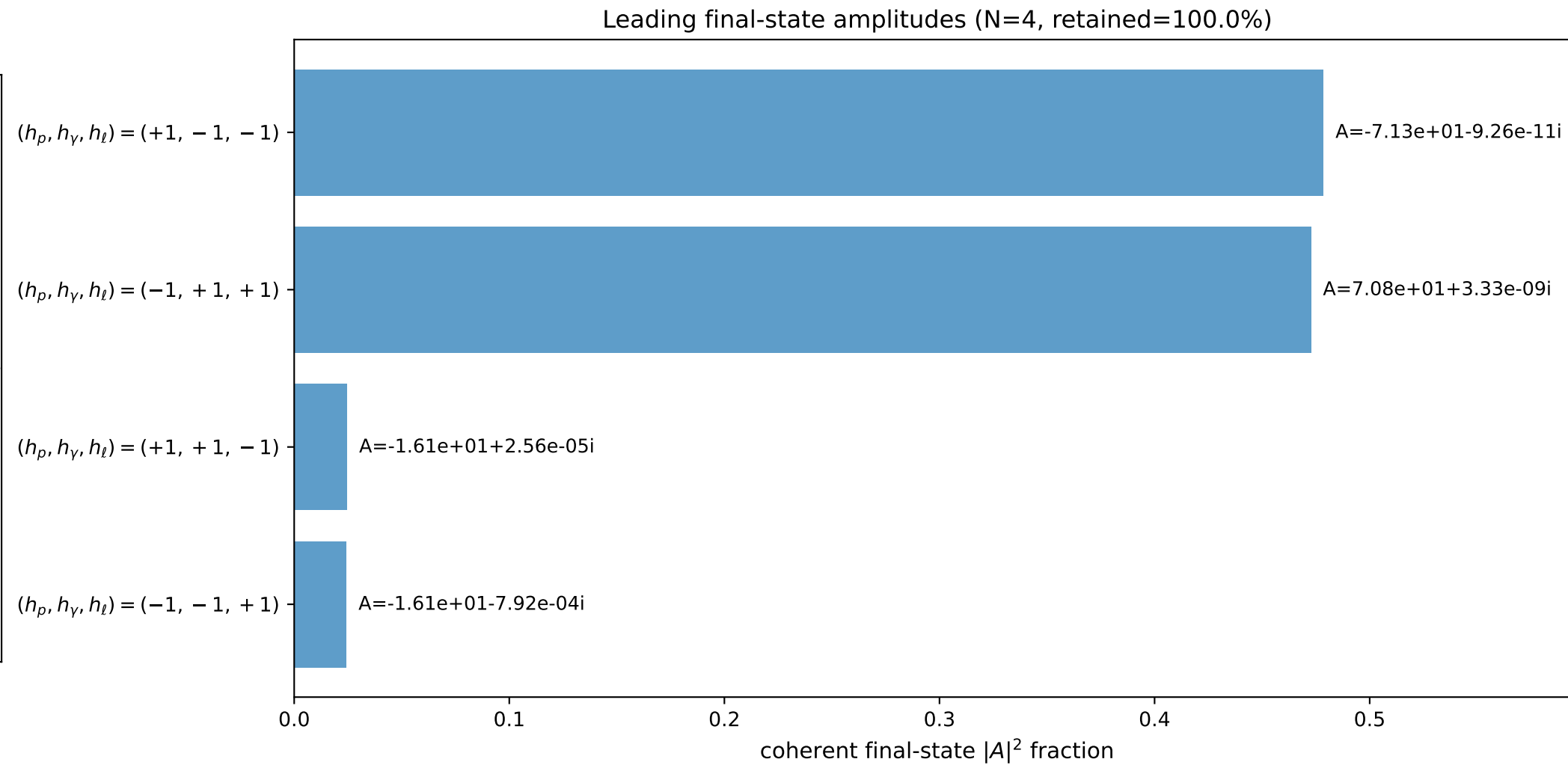
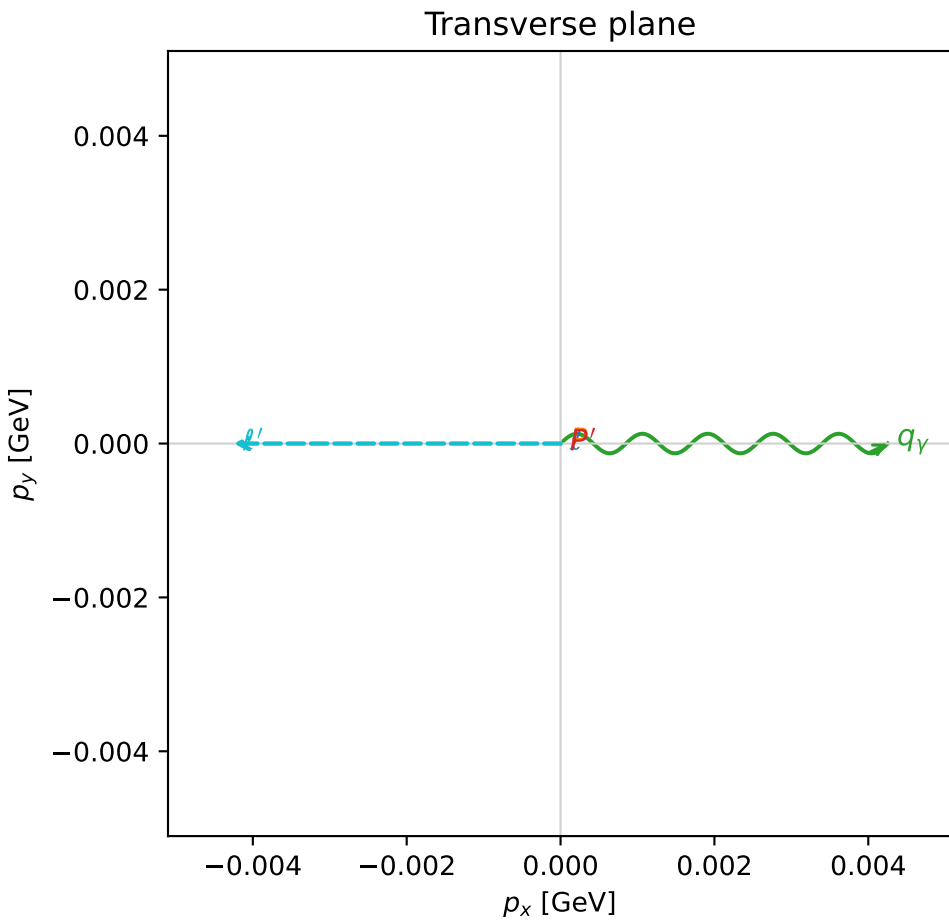
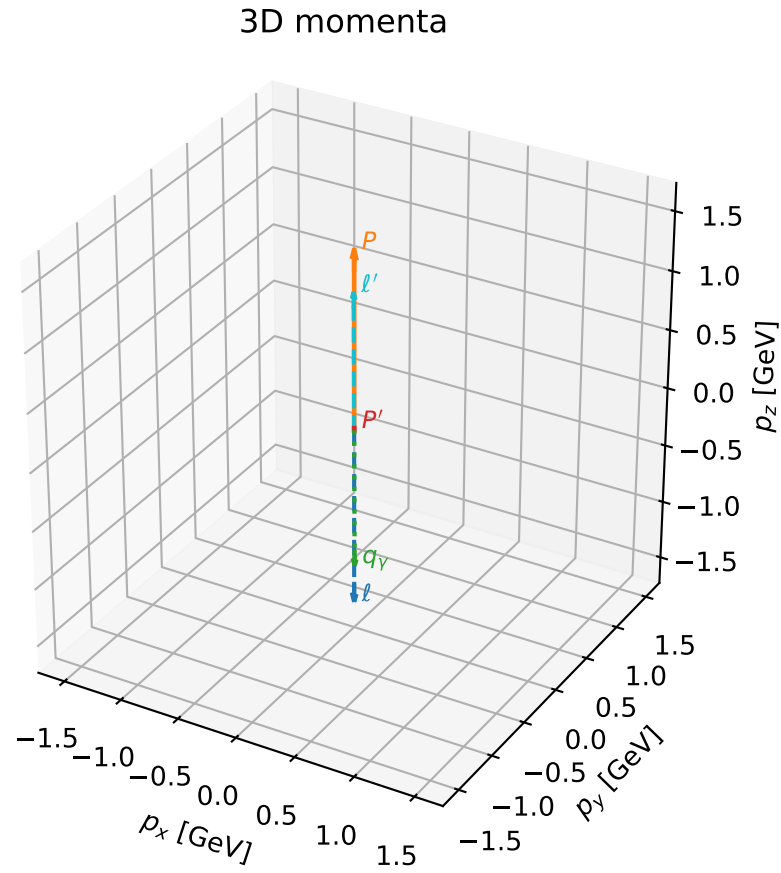


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

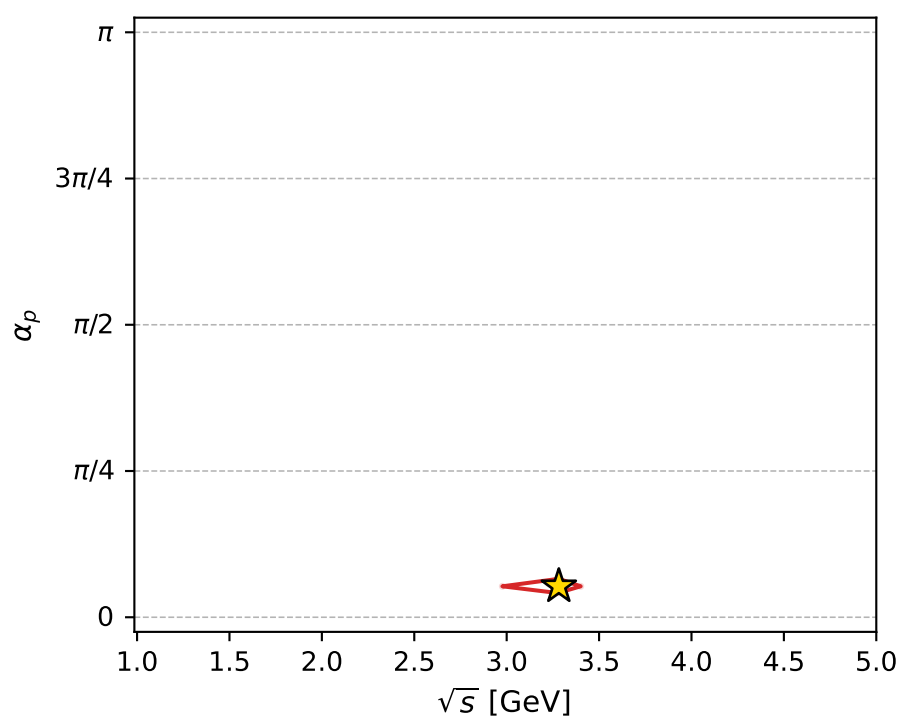
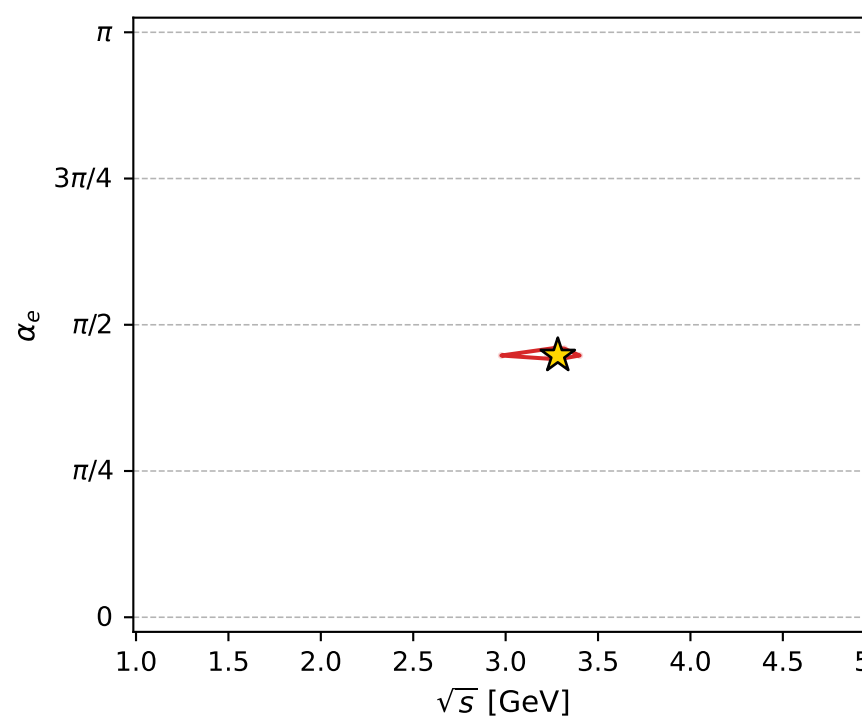
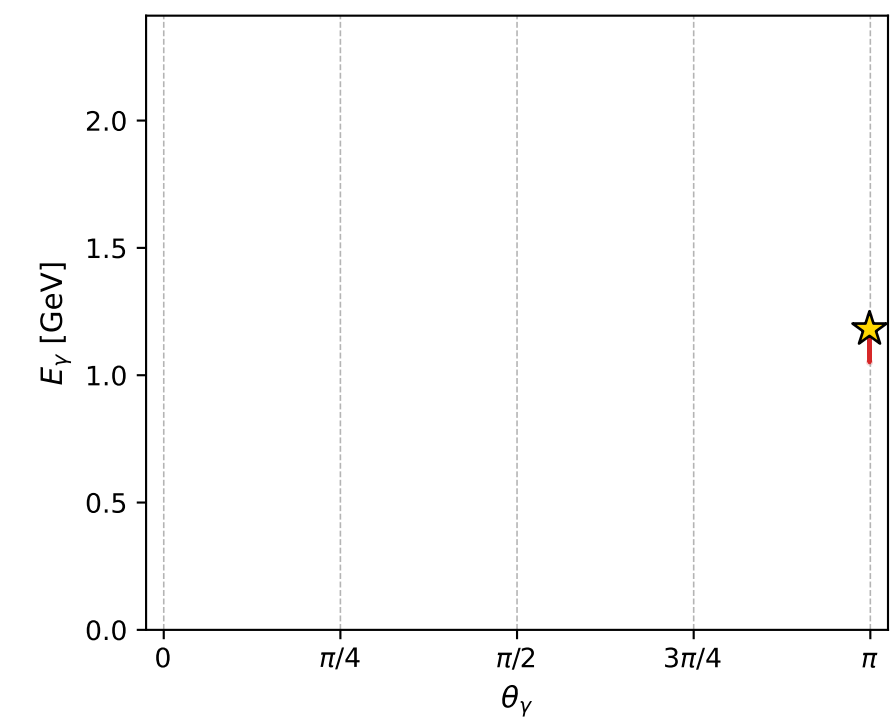
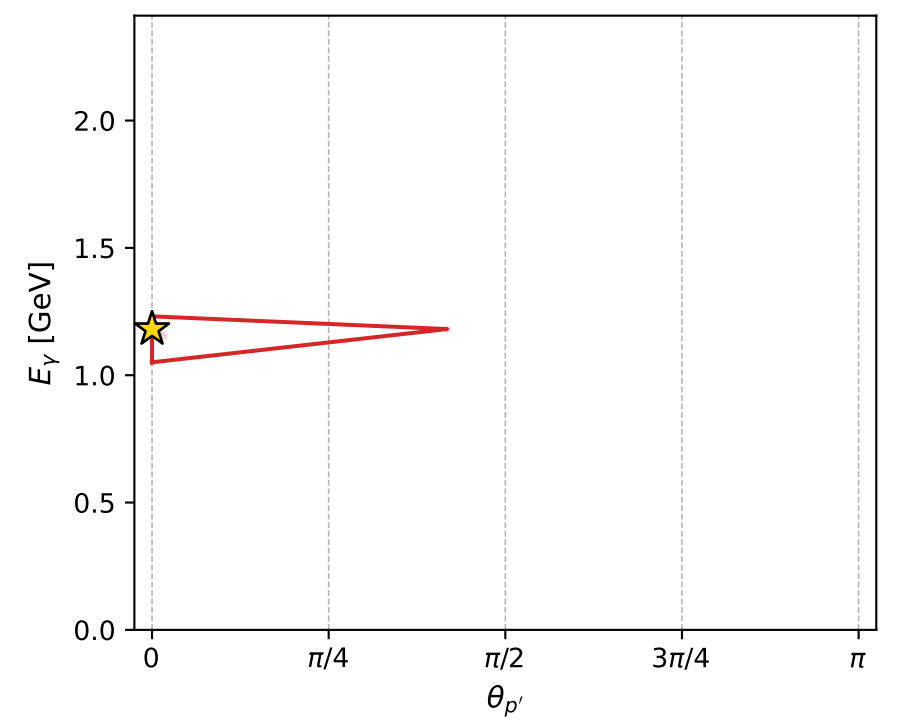
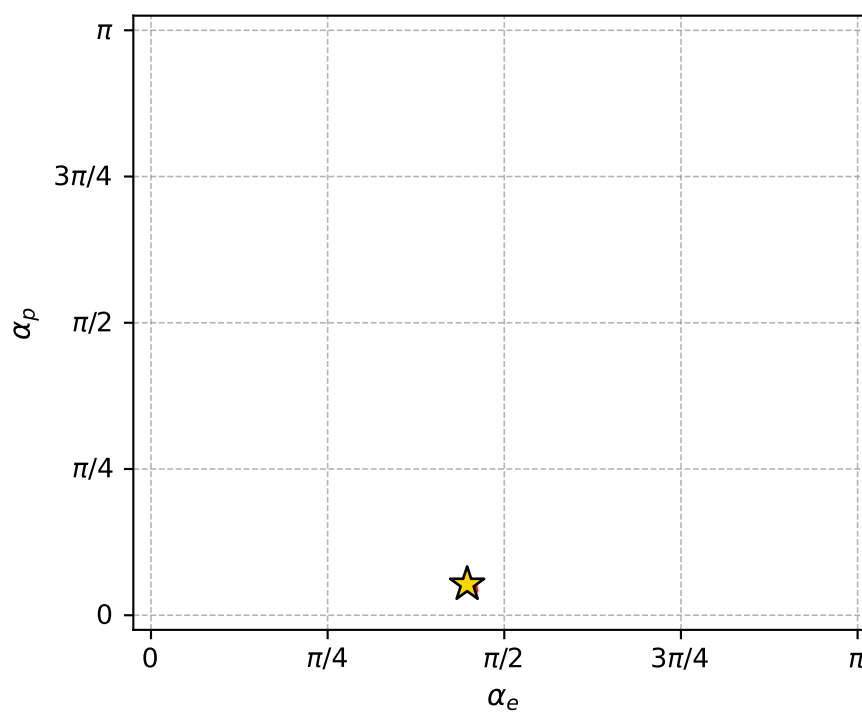
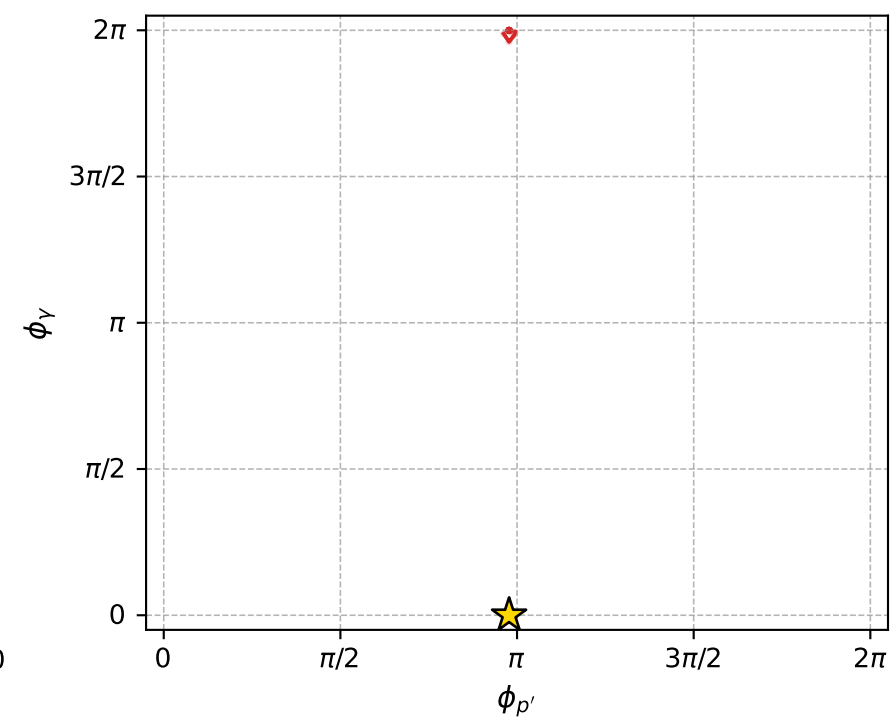
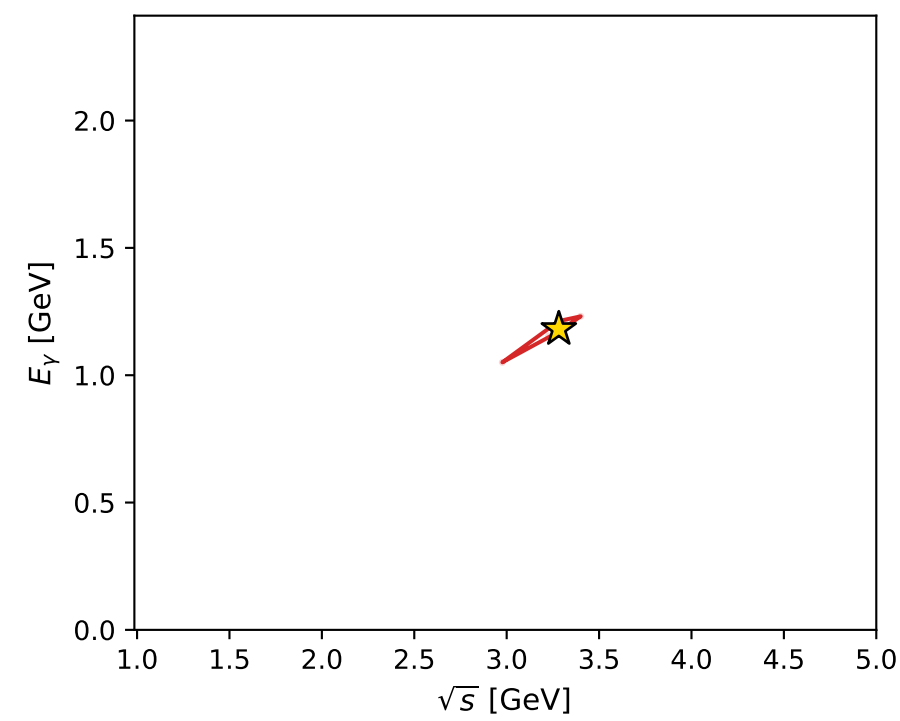
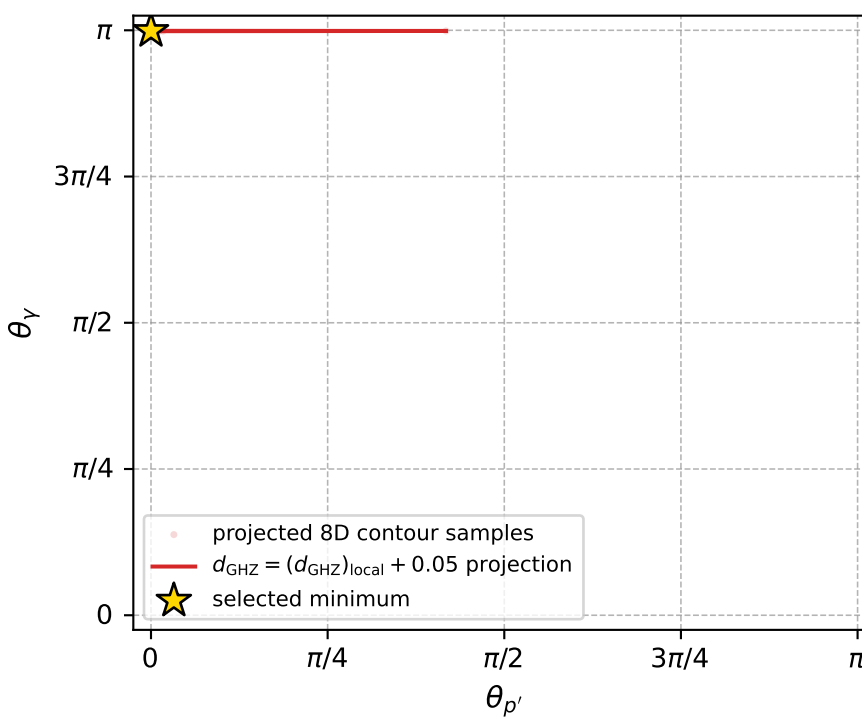
dghz_mixing_angles_polarization_01_local_minimum_324 $d_{\{\text{rm GHZ}\}}=0.000439836$
 region: polarization_cluster_01_local_minimum_324
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.4056673$ rad
 incoming lepton: $0.16438|+\rangle +0.986397|-\rangle$
 proton mixing angle: $\alpha_p=0.16612685$ rad
 incoming proton: $0.986233|+\rangle +0.165364|-\rangle$

$s=10.7733$, $\sqrt{s}=3.28227$, $|\vec{P}|=1.50711$, $|\vec{P}'|=0.0183795$
 $\theta_{p'}=0$, $\theta_\gamma=3.13799$, $\phi_{p'}=3.07095$, $\phi_\gamma=2.52488\text{e-}05$
 $E_\gamma=1.18123$, $\theta_{\ell'}=0.00365664$, $\phi_{\ell'}=3.14162$
 $Q^2=7.01016$, $x_B=0.75622$, $t=-1.51576$, $W^2=3.13969$, $y=0.936983$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5071$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5071$
 P $E= 1.7752$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5071$
 ℓ' $E= 1.1629$ $p_x= -0.0043$ $p_y= -0.0000$ $p_z= 1.1628$
 P' $E= 0.9382$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0184$
 q_γ $E= 1.1812$ $p_x= 0.0043$ $p_y= 0.0000$ $p_z= -1.1812$



electron: polarization cluster P1, local minimum 324; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00043983558$
 $d_{\text{GHZ}} \text{ contour} = 0.050439836$
 above global minimum = 0.0004398133
 8D contour samples = 130

selected phase-space point:

$\text{sqrt_s} = 3.28227$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 3.137993$
 $\text{q0ut} = 1.181235$
 $\text{phi_p_out} = 3.070947$
 $\text{phi_gamma_out} = 2.524882\text{e-}05$
 $\text{alpha_e} = 1.405667$
 $\text{alpha_p} = 0.1661269$

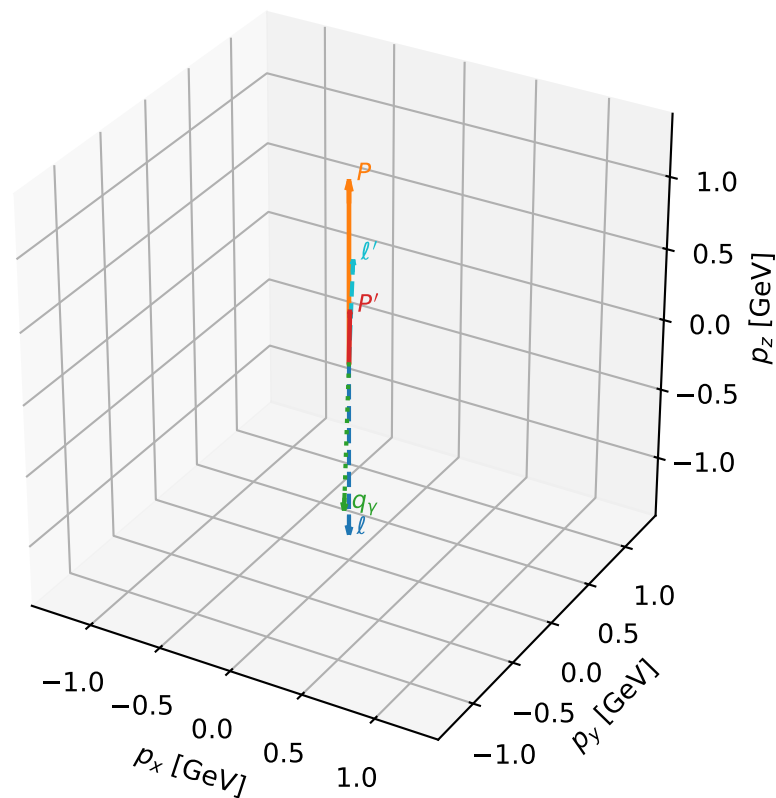
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_331 $d_{\{\text{rm GHZ}\}}=0.000498586$
 region: polarization_cluster_01_local_minimum_331
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.063309233$ rad
 incoming lepton: $0.997997|+\rangle +0.063267|-\rangle$
 proton mixing angle: $\alpha_p=1.5095946$ rad
 incoming proton: $0.0611635|+\rangle +0.998128|-\rangle$

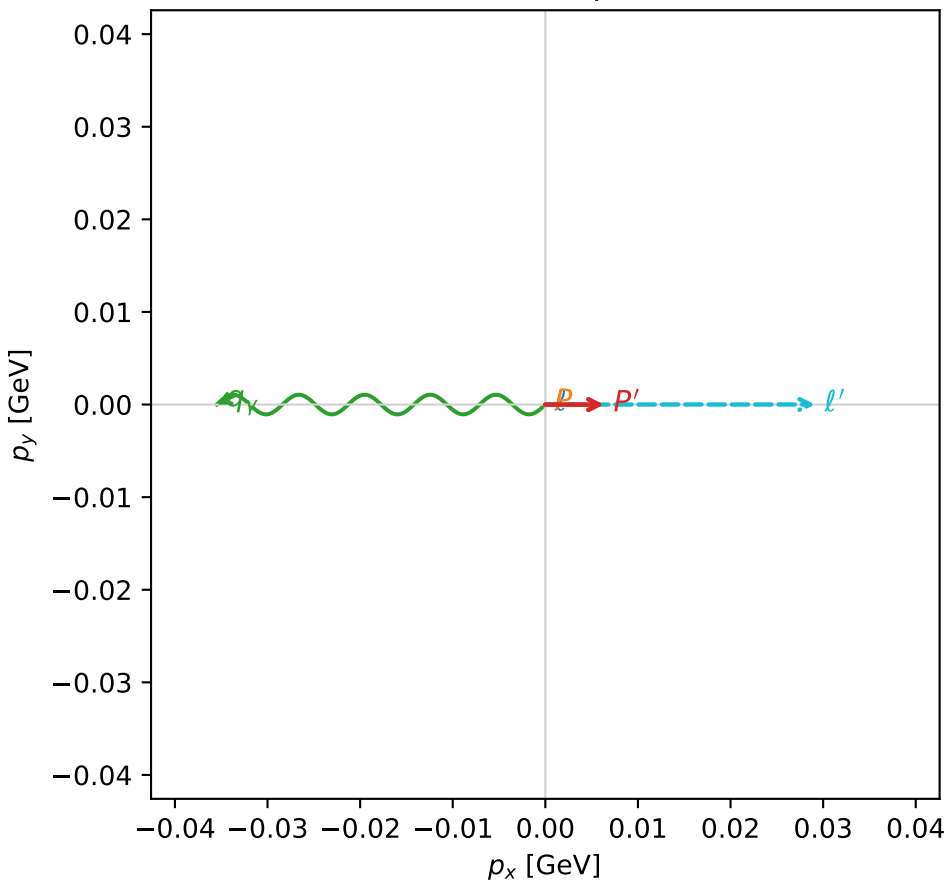
$s=7.83845$, $\sqrt{s}=2.79972$, $|\vec{P}|=1.24273$, $|\vec{P}'|=0.354558$
 $\theta_{P'}=0.0179191$, $\theta_Y=3.1086$, $\phi_{P'}=6.28227$, $\phi_Y=3.14163$
 $E_Y=1.07572$, $\theta_{\ell'}=0.0404006$, $\phi_{\ell'}=0.000247116$
 $Q^2=3.5837$, $x_B=0.551013$, $t=-0.481835$, $W^2=3.79999$, $y=0.934646$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2427$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2427$
 P $E= 1.5570$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2427$
 ℓ' $E= 0.7212$ $p_x= 0.0291$ $p_y= 0.0000$ $p_z= 0.7206$
 P' $E= 1.0028$ $p_x= 0.0064$ $p_y= -0.0000$ $p_z= 0.3545$
 q_Y $E= 1.0757$ $p_x= -0.0355$ $p_y= -0.0000$ $p_z= -1.0751$

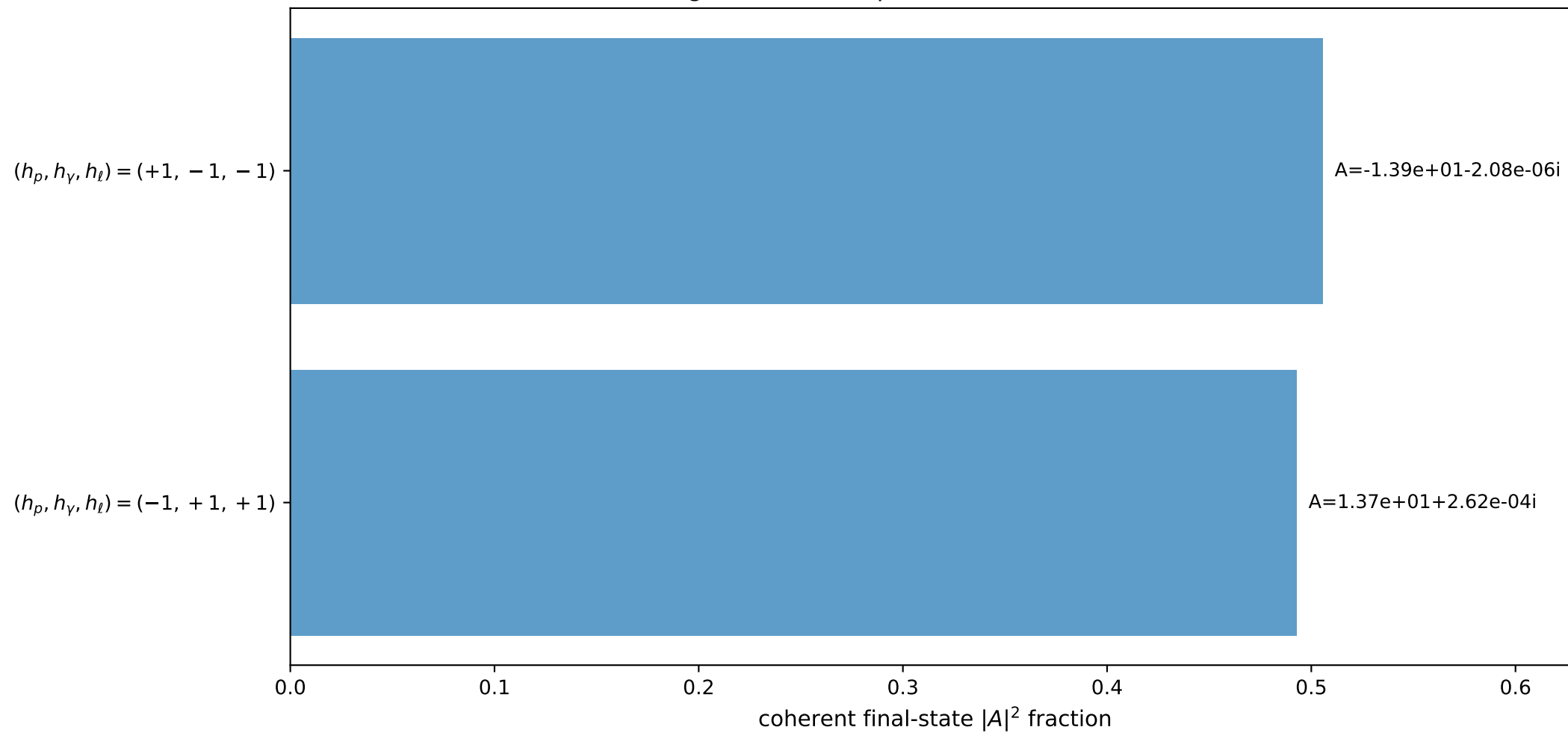
3D momenta



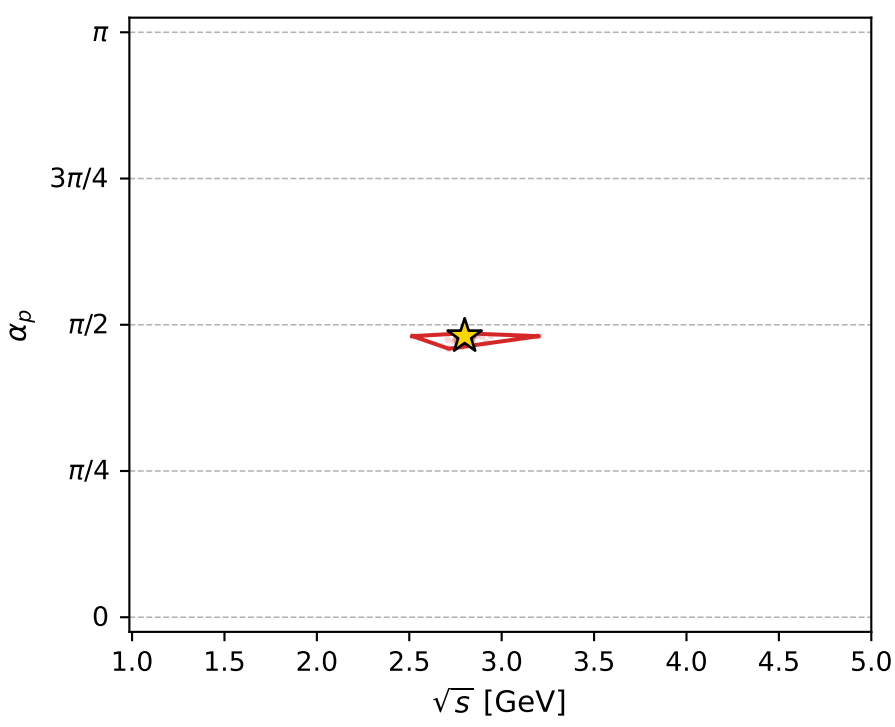
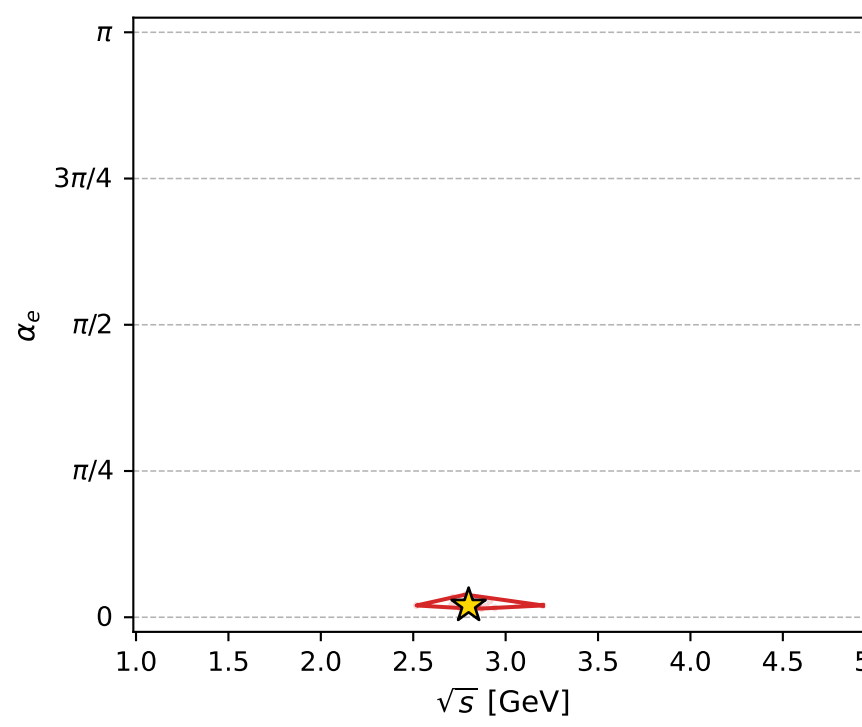
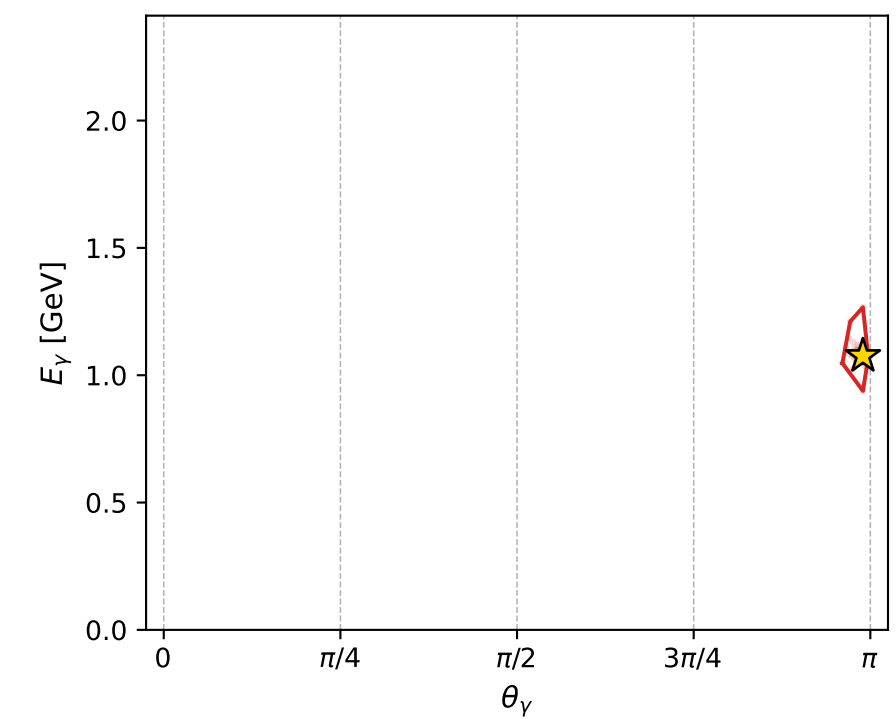
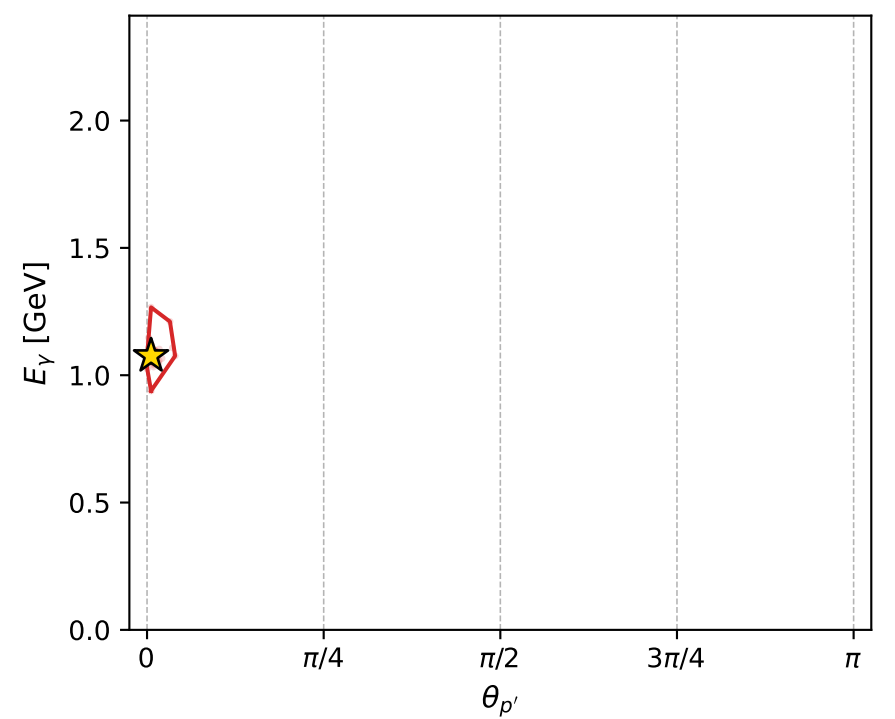
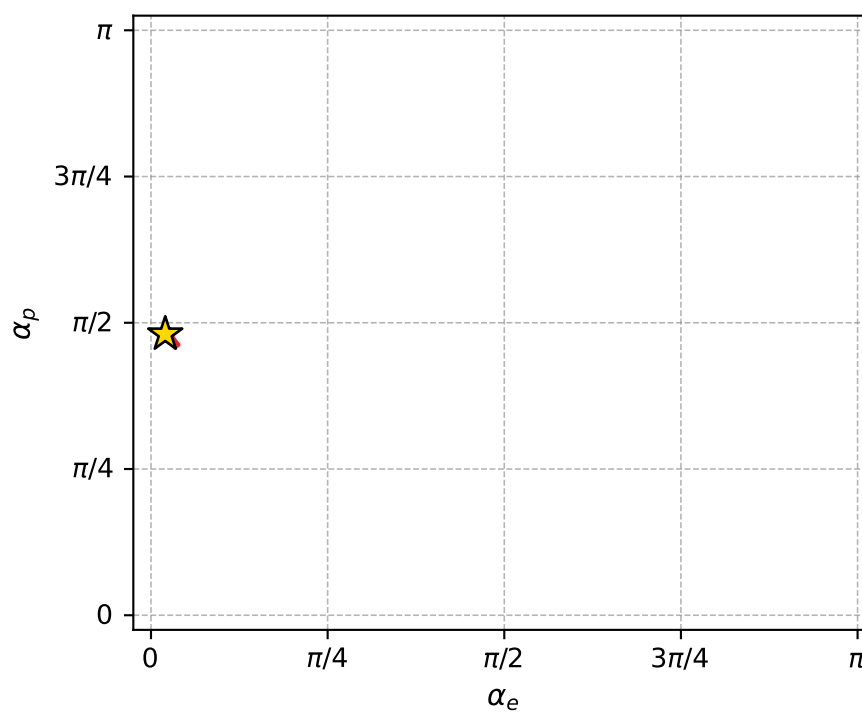
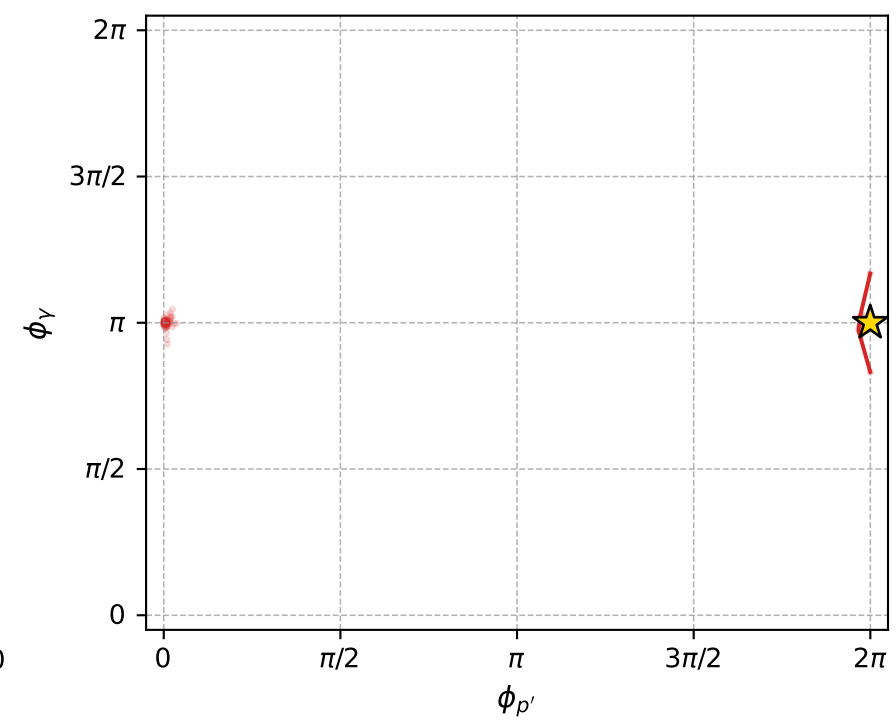
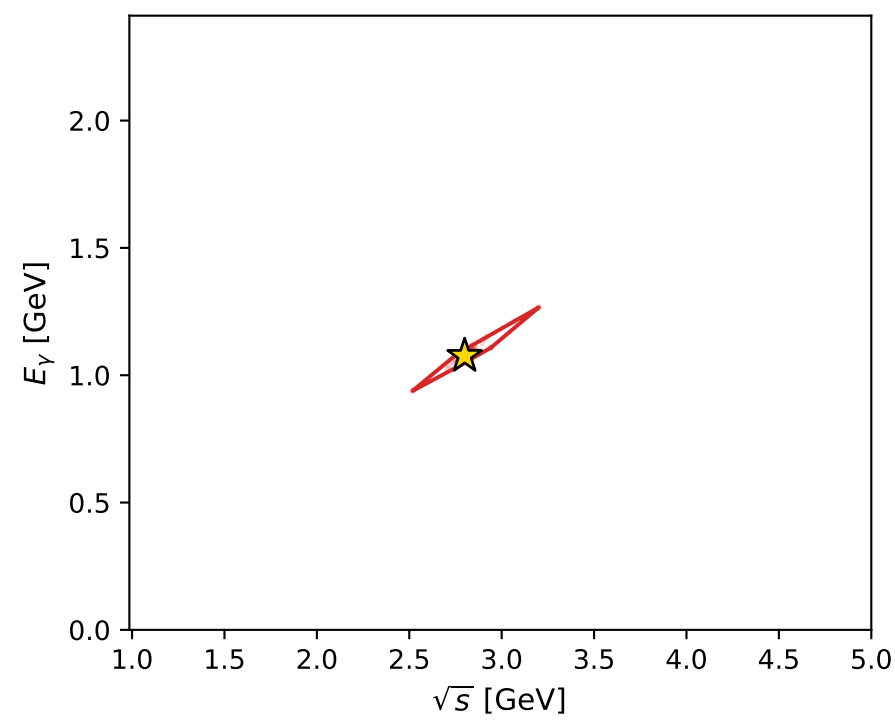
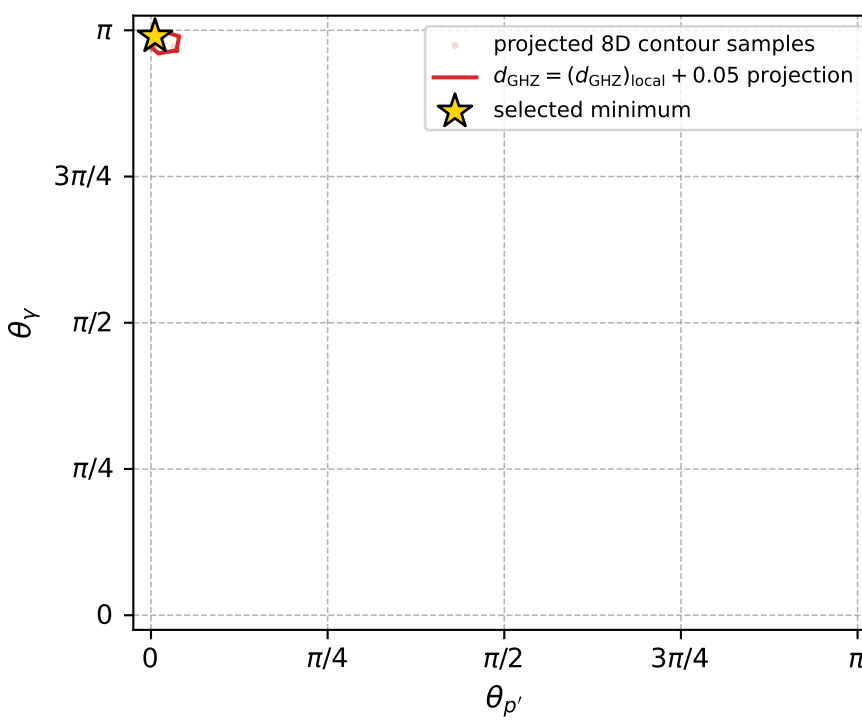
Transverse plane



Leading final-state amplitudes (N=2, retained=99.8%)



electron: polarization cluster P1, local minimum 331; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00049858574$
 $d_{\text{GHZ}} \text{ contour} = 0.050498586$
 above global minimum = 0.00049856345
 8D contour samples = 237

selected phase-space point:

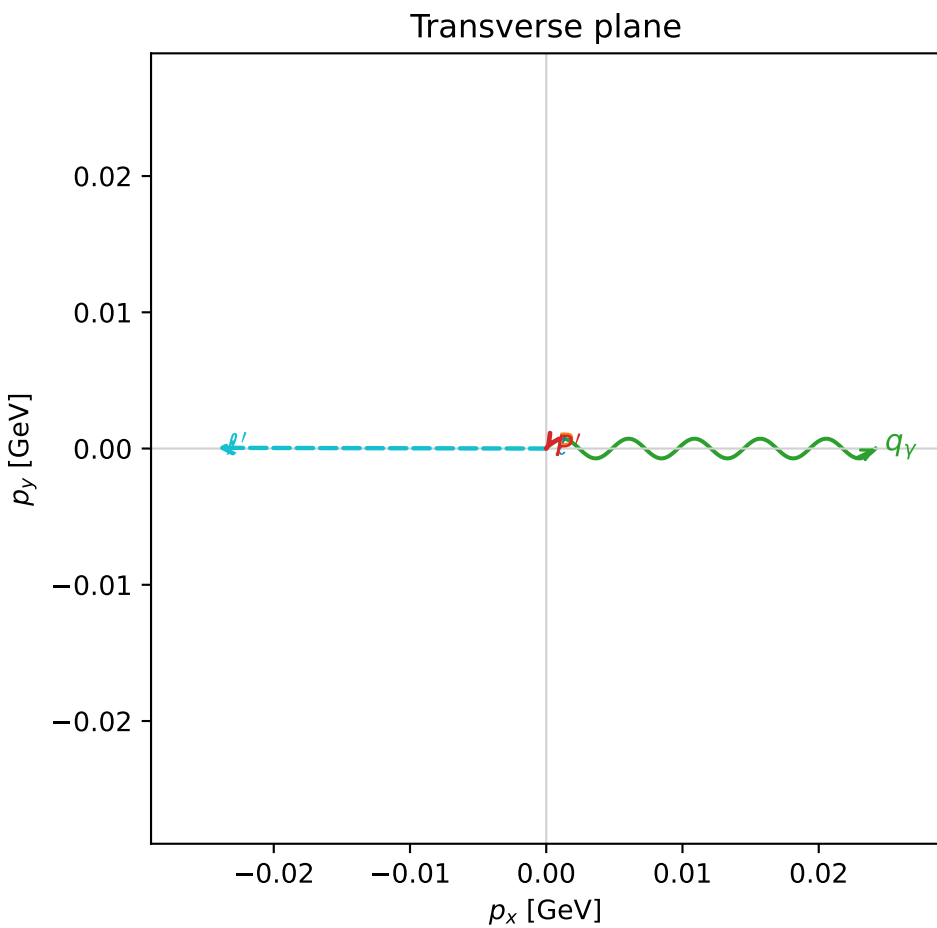
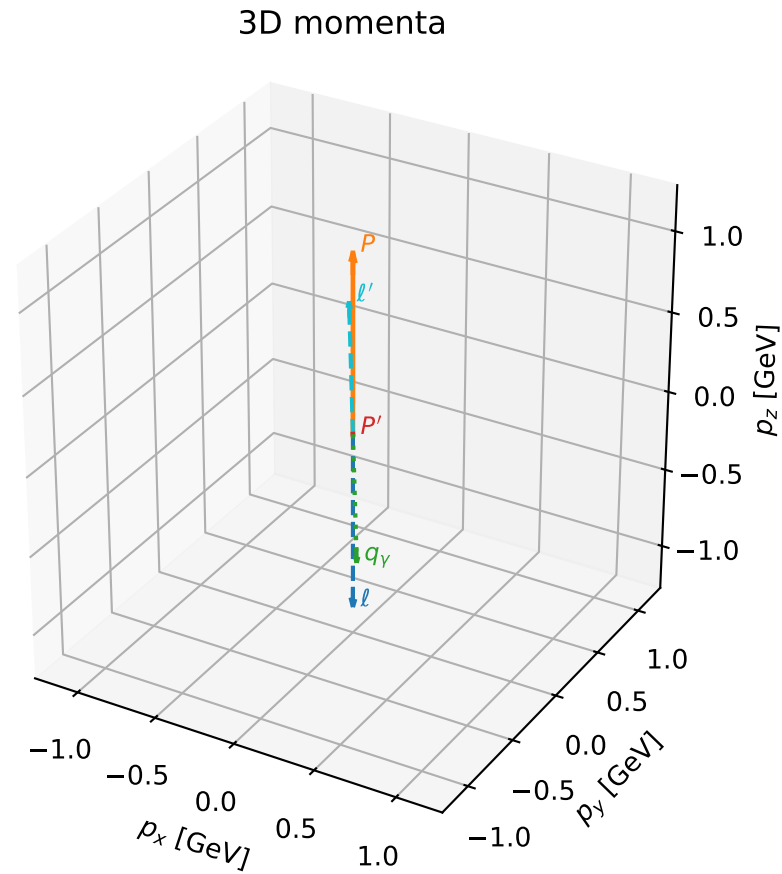
$\text{sqrt_s} = 2.799724$
 $\text{theta_p_out} = 0.01791909$
 $\text{theta_gamma_out} = 3.108601$
 $\text{q0out} = 1.075724$
 $\text{phi_p_out} = 6.282266$
 $\text{phi_gamma_out} = 3.141631$
 $\text{alpha_e} = 0.06330923$
 $\text{alpha_p} = 1.509595$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_333 d_{\rm GHZ}=0.000507492
region: polarization_cluster_01_local_minimum_333
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.5422293$ rad
incoming lepton: $0.0285631|+\rangle +0.999592|-\rangle$
proton mixing angle: $\alpha_p=0.028304256$ rad
incoming proton: $0.999599|+\rangle +0.0283005|-\rangle$

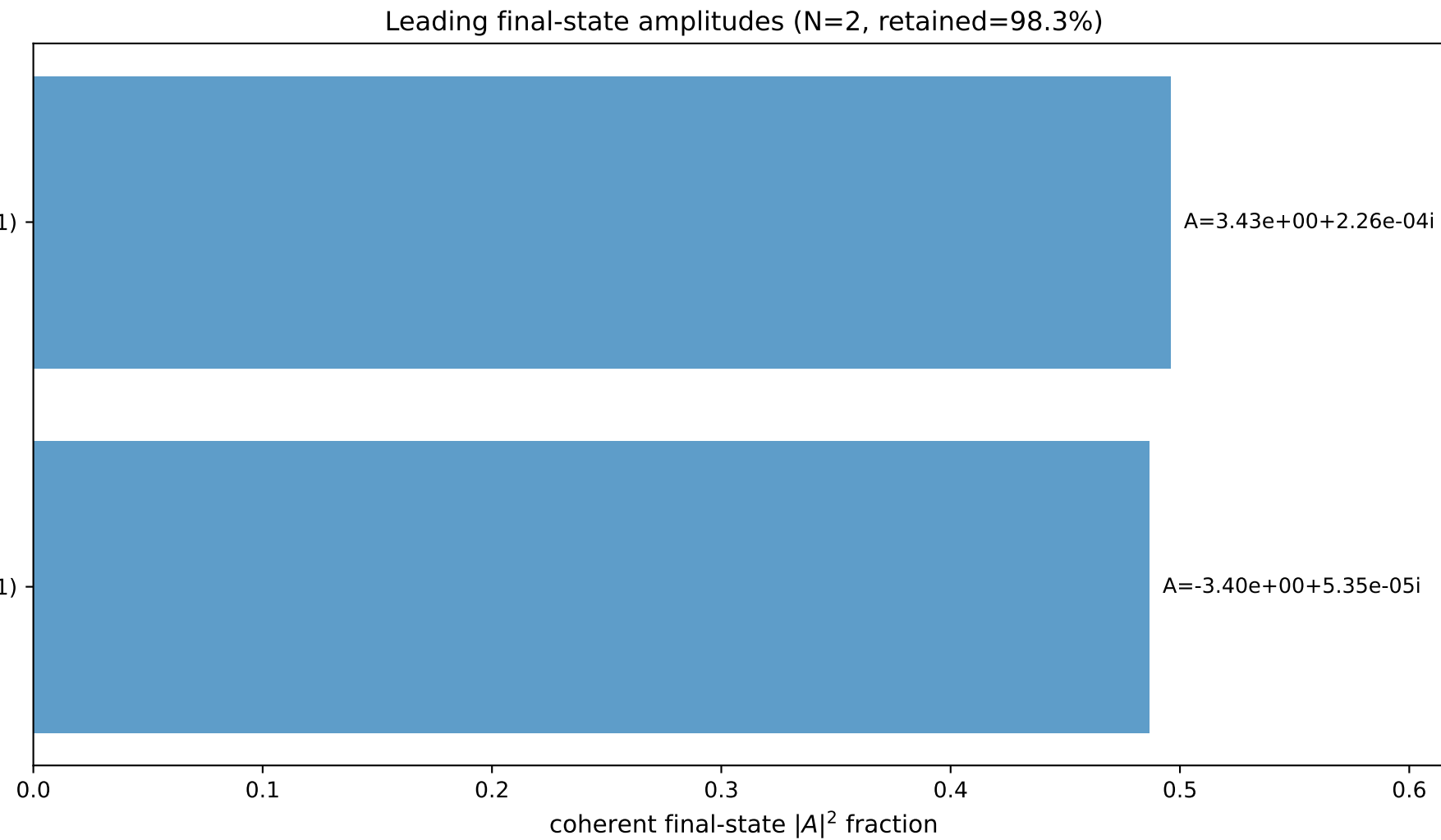
$s=6.59164$, $\sqrt{s}=2.56742$, $|\vec{P}|=1.11236$, $|\vec{P}'|=0.00112869$
 $\theta_{P'}=0.054477$, $\theta_Y=3.11194$, $\phi_{P'}=4.13085$, $\phi_Y=7.31348e-07$
 $E_Y=0.815273$, $\theta_{\ell'}=0.0296545$, $\phi_{\ell'}=3.13947$
 $Q^2=3.6217$, $x_B=0.702834$, $t=-0.967495$, $W^2=2.41114$, $y=0.902166$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1124$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1124$
 P $E= 1.4551$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1124$
 ℓ' $E= 0.8141$ $p_x= -0.0241$ $p_y= 0.0001$ $p_z= 0.8138$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 0.0011$
 q_Y $E= 0.8153$ $p_x= 0.0242$ $p_y= 0.0000$ $p_z= -0.8149$

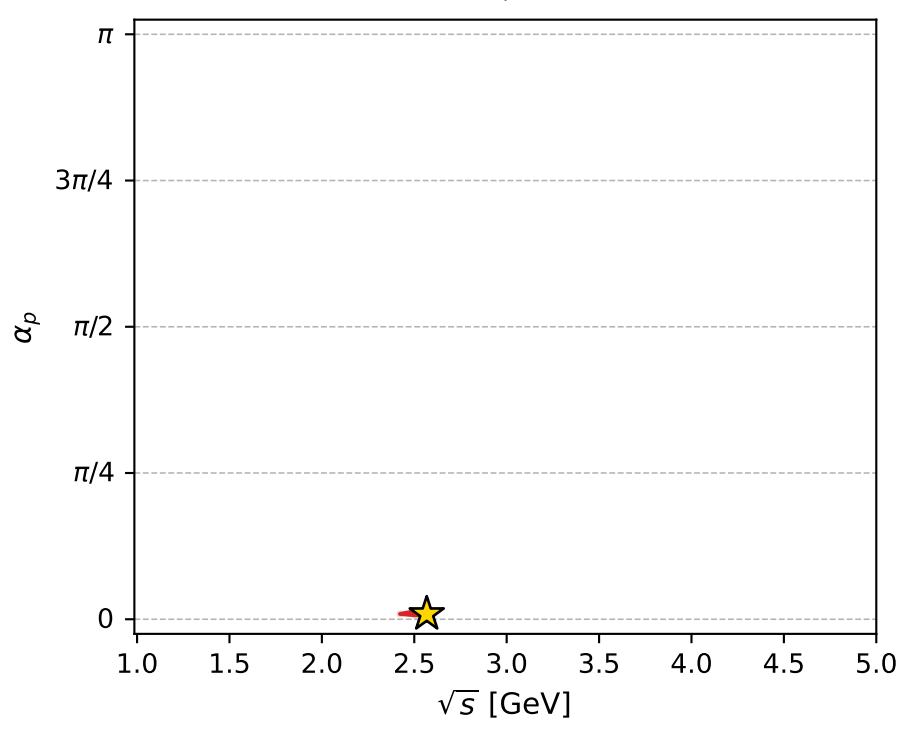
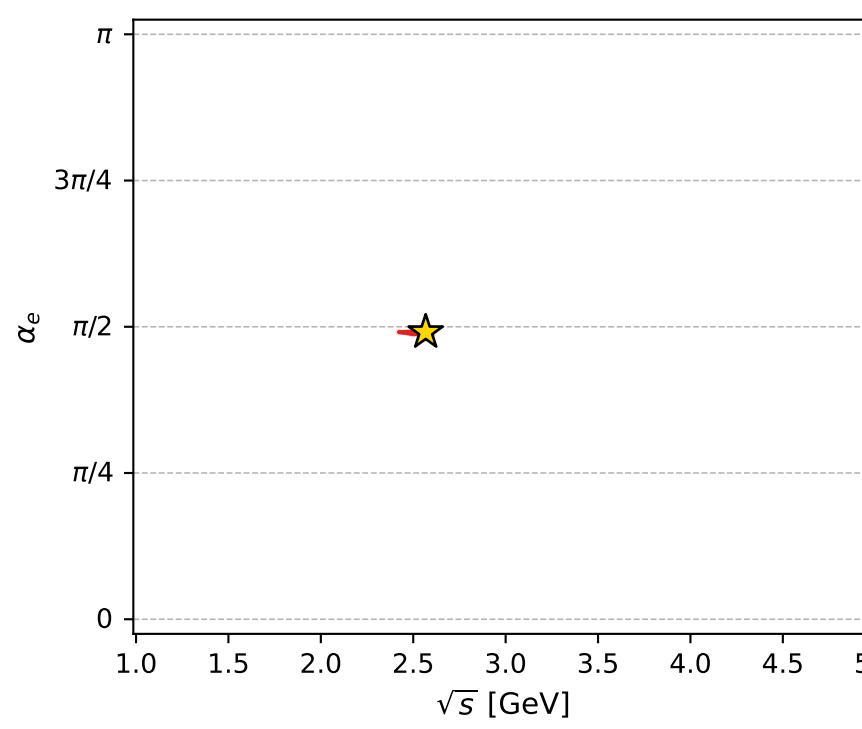
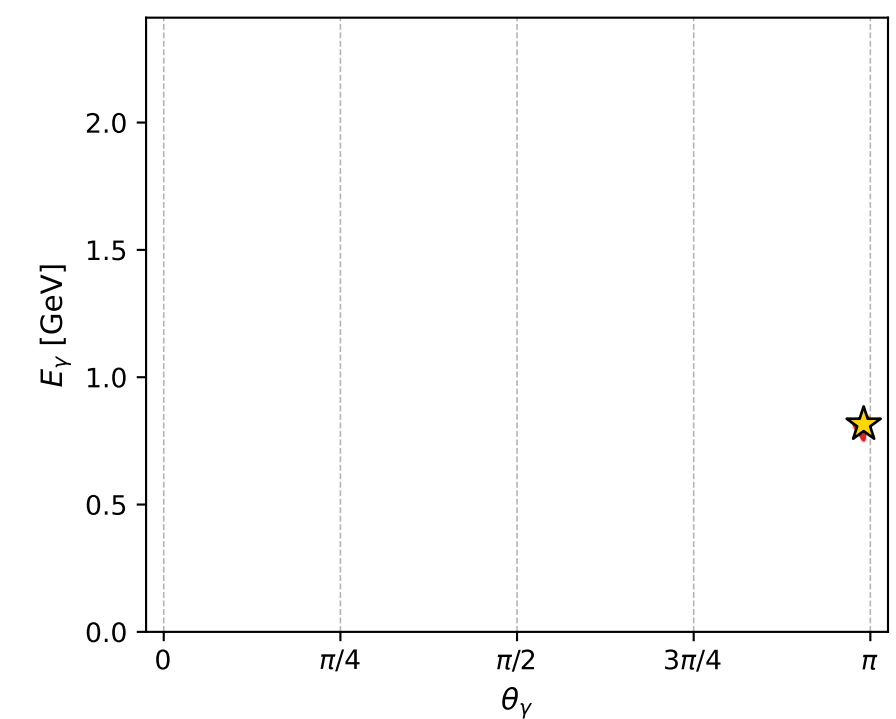
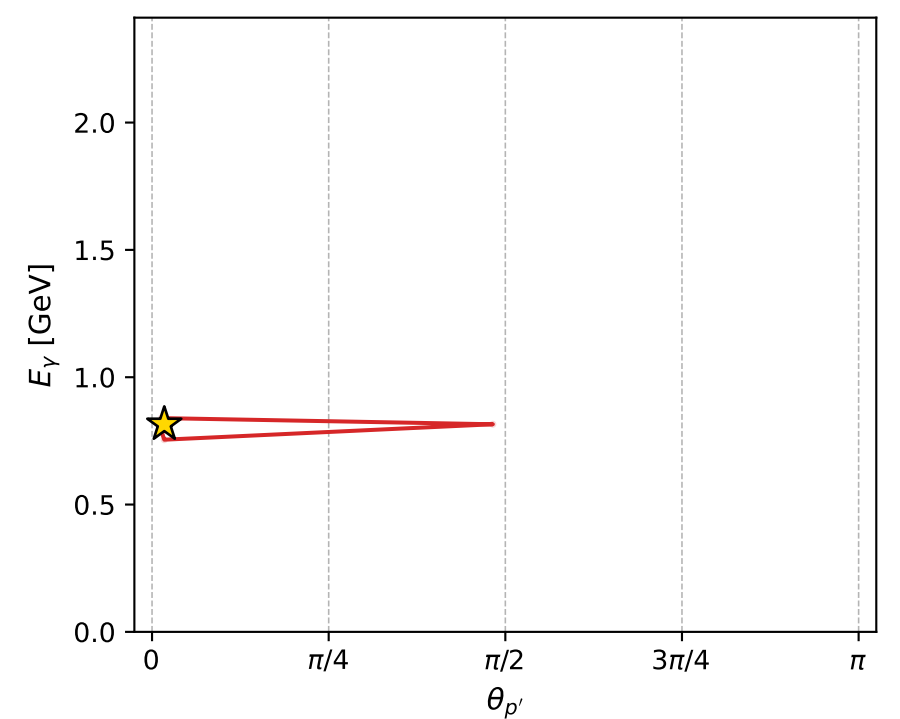
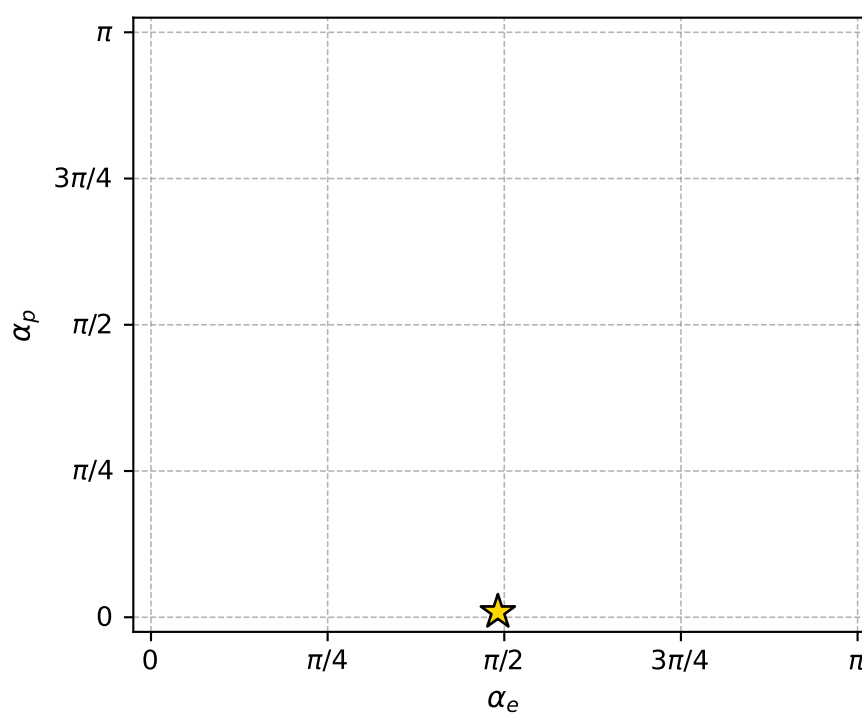
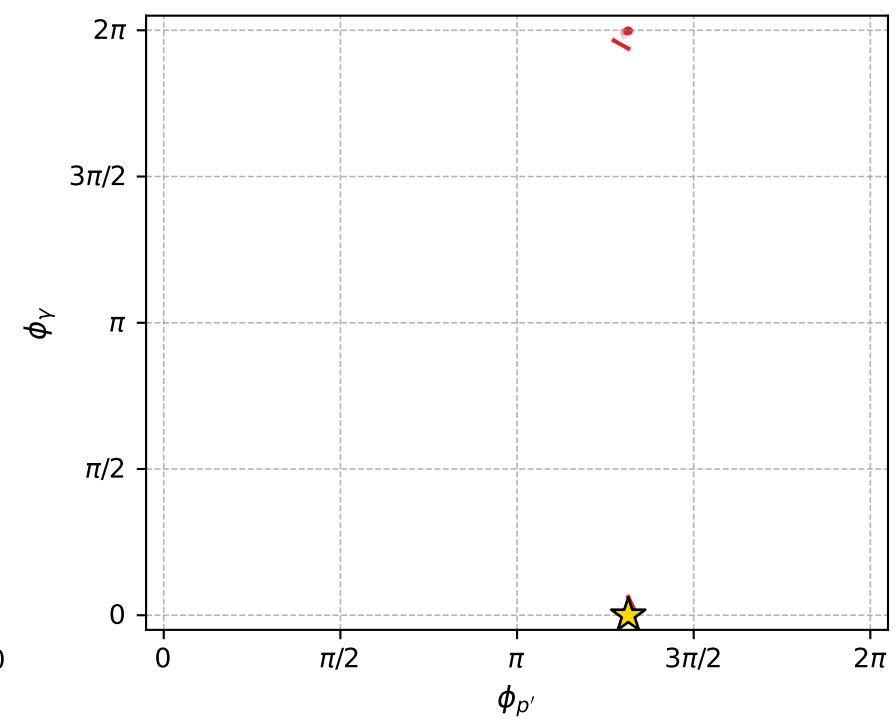
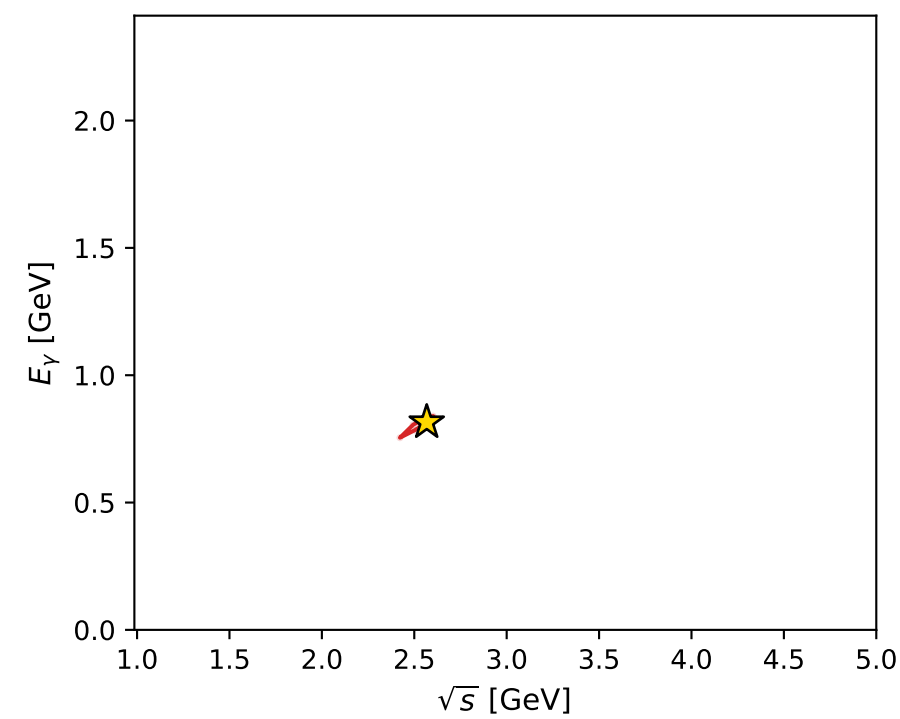
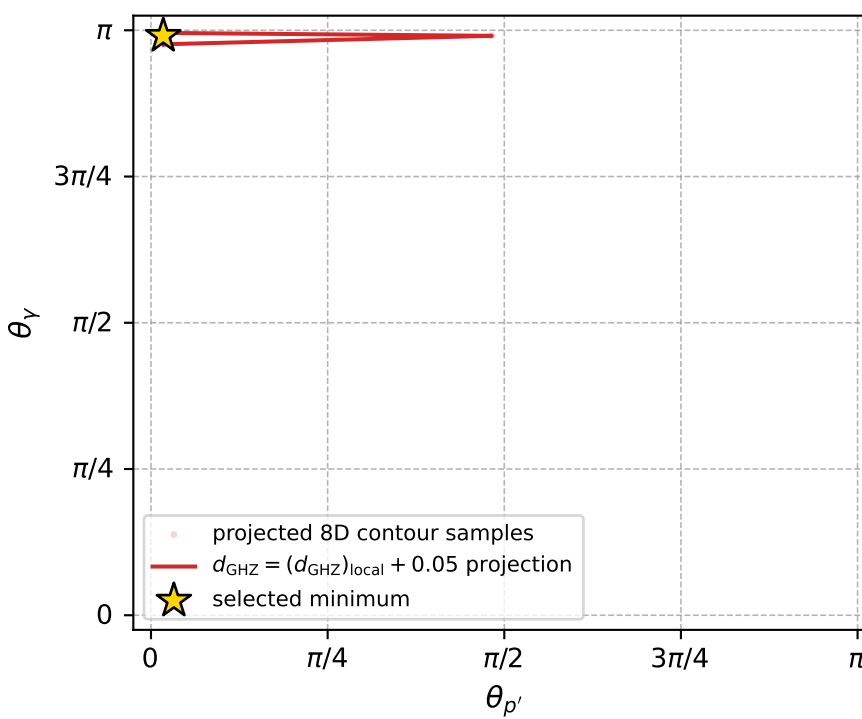


$(h_p, h_Y, h_{\ell}) = (-1, +1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, -1)$



electron: polarization cluster P1, local minimum 333; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00050749172$
 $d_{\text{GHZ}} \text{ contour} = 0.050507492$
 above global minimum = 0.00050746944
 8D contour samples = 253

selected phase-space point:

$\text{sqrt_s} = 2.567419$
 $\text{theta_p_out} = 0.054477$
 $\text{theta_gamma_out} = 3.111938$
 $\text{q0out} = 0.815273$
 $\text{phi_p_out} = 4.130846$
 $\text{phi_gamma_out} = 7.313477\text{e-}07$
 $\text{alpha_e} = 1.542229$
 $\text{alpha_p} = 0.02830426$

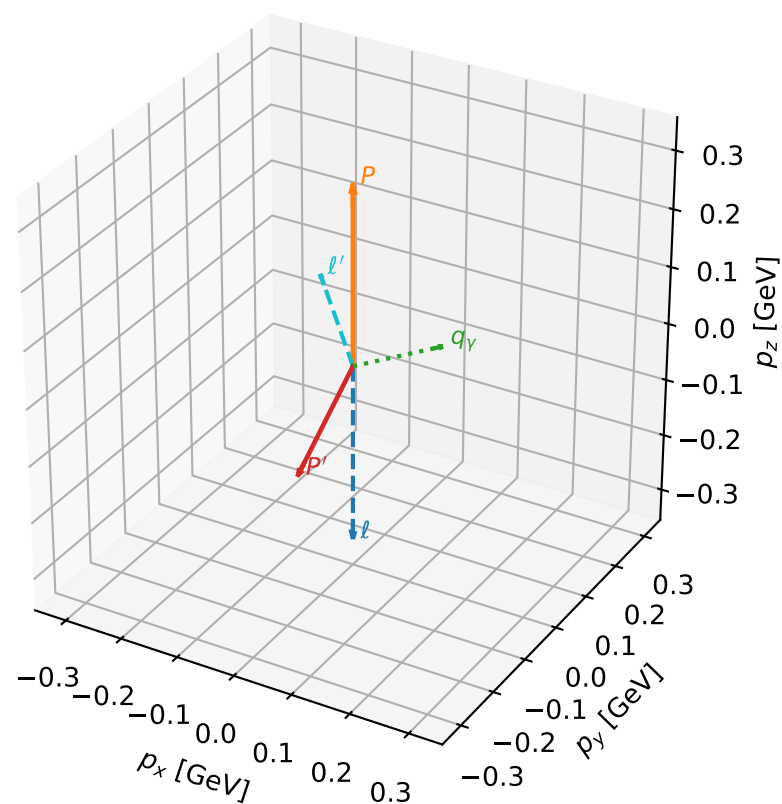
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_338 $d_{\{\text{rm GHZ}\}}=0.000560438$
 region: polarization_cluster_01_local_minimum_338
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.3443824$ rad
 incoming lepton: $0.224484|+\rangle + 0.974478|-\rangle$
 proton mixing angle: $\alpha_p=0.18189684$ rad
 incoming proton: $0.983502|+\rangle + 0.180895|-\rangle$

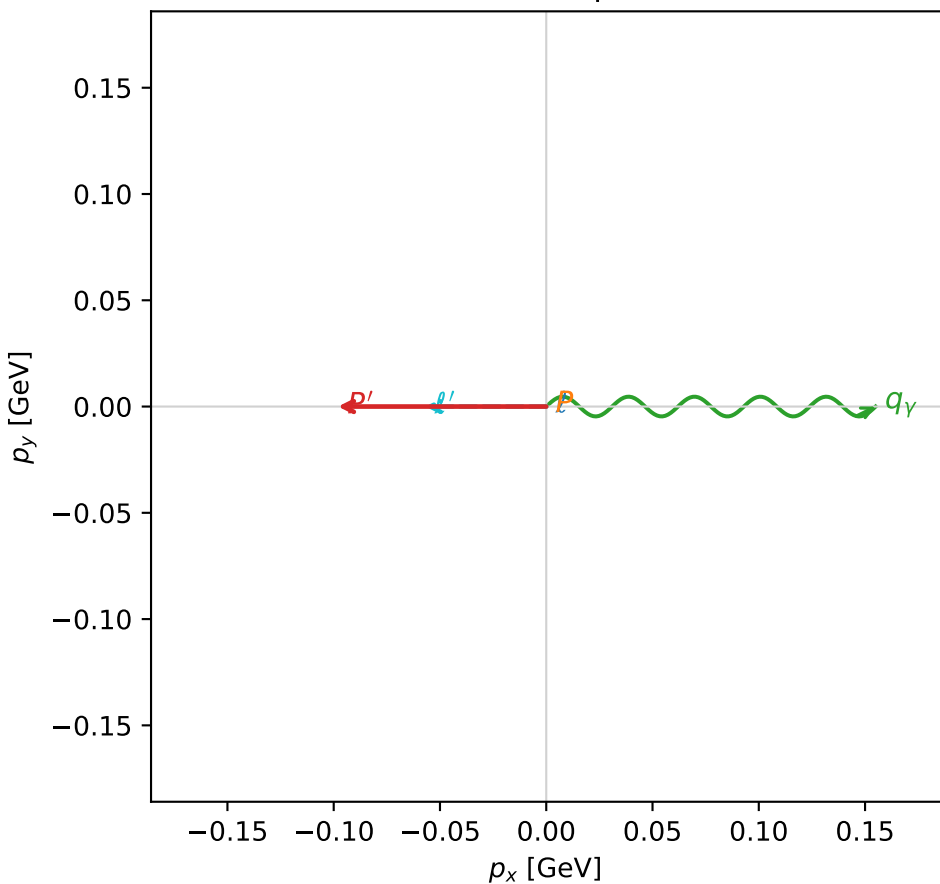
$s=1.68384$, $\sqrt{s}=1.29763$, $|\vec{P}|=0.309793$, $|\vec{P}'|=0.243325$
 $\theta_{P'}=2.72706$, $\theta_V=1.09996$, $\phi_{P'}=3.14153$, $\phi_V=6.28314$
 $E_V=0.173874$, $\theta_{\ell'}=0.377004$, $\phi_{\ell'}=3.14158$
 $Q^2=0.184976$, $x_B=0.314871$, $t=-0.292819$, $W^2=1.28233$, $y=0.730685$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3098$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3098$
 P $E= 0.9878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3098$
 ℓ' $E= 0.1547$ $p_x= -0.0570$ $p_y= 0.0000$ $p_z= 0.1438$
 P' $E= 0.9690$ $p_x= -0.0980$ $p_y= 0.0000$ $p_z= -0.2227$
 q_V $E= 0.1739$ $p_x= 0.1550$ $p_y= -0.0000$ $p_z= 0.0789$

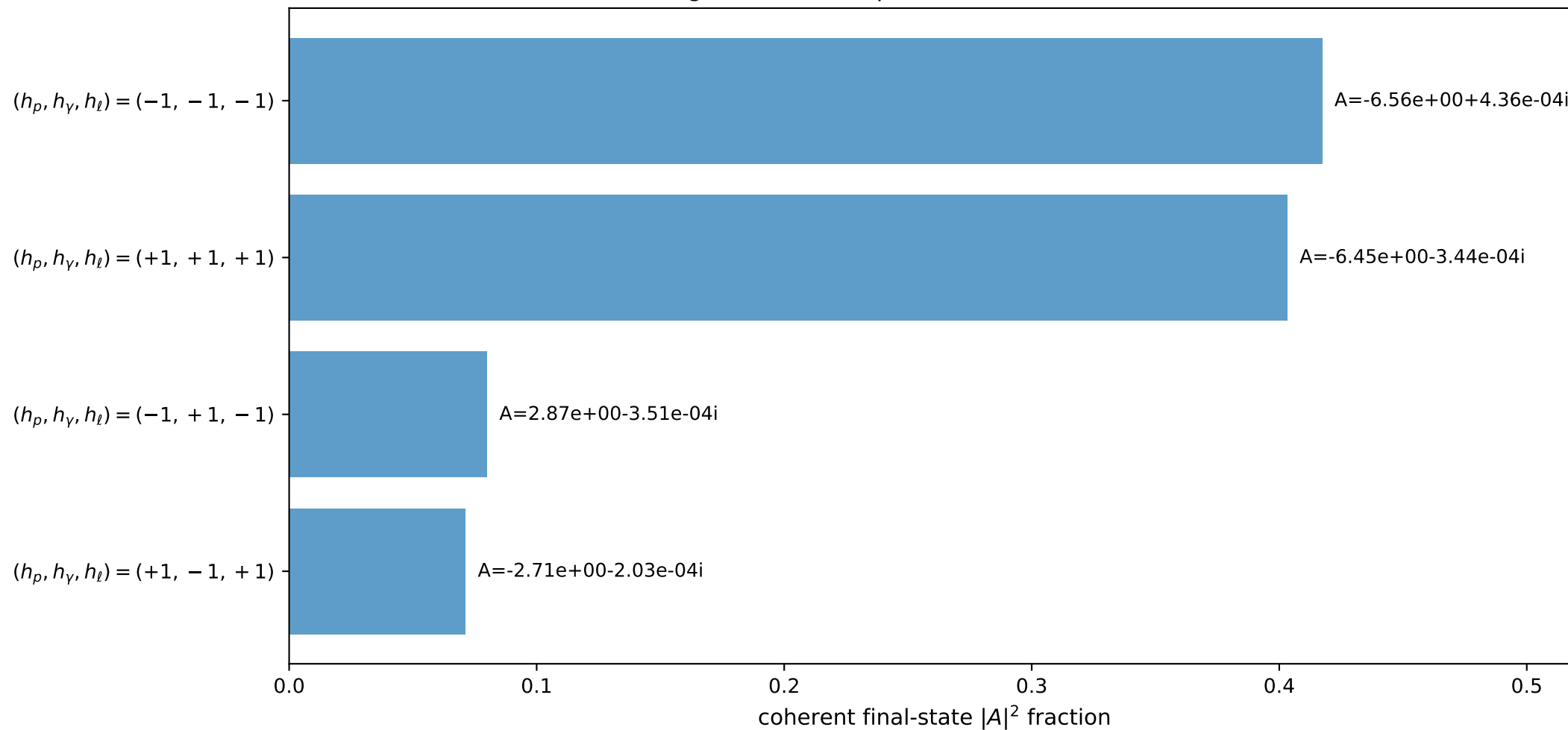
3D momenta



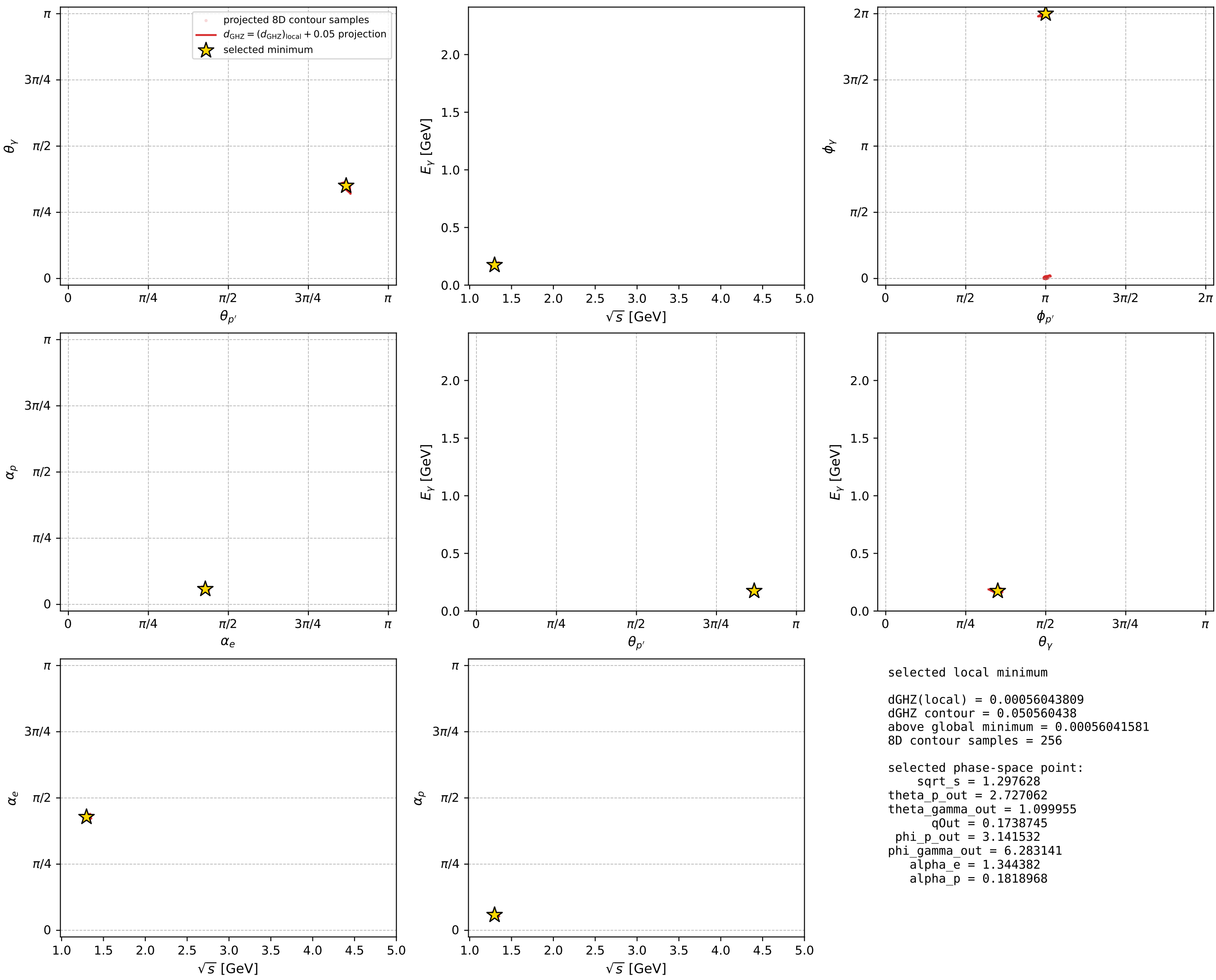
Transverse plane



Leading final-state amplitudes (N=4, retained=97.2%)



electron: polarization cluster P1, local minimum 338; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



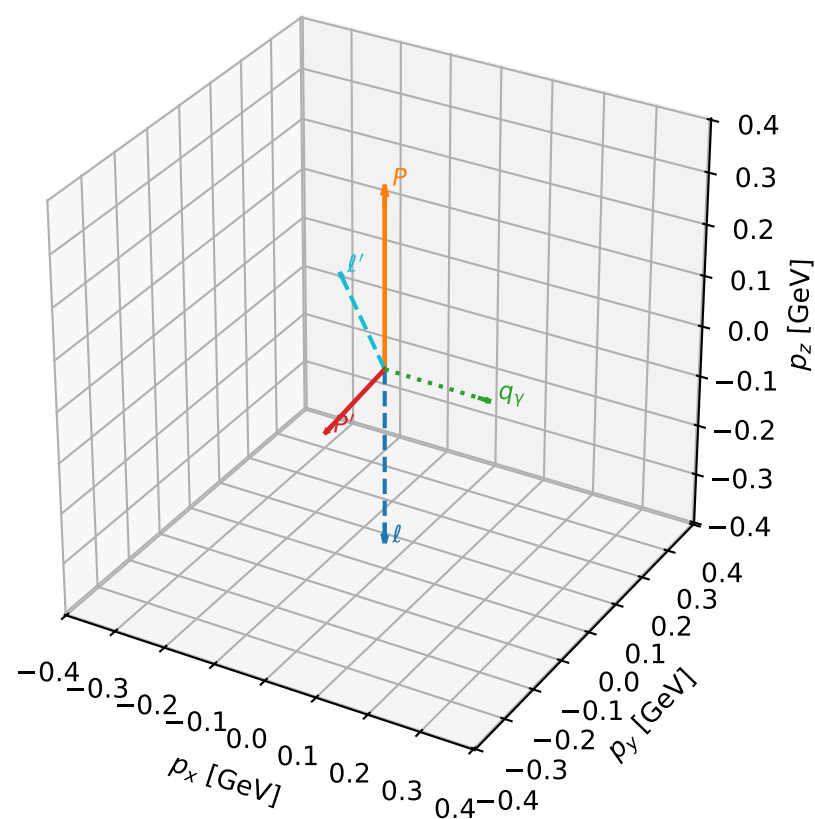
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_340 d_{\rm GHZ}=0.000563341
 region: polarization_cluster_01_local_minimum_340
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.257475$ rad
 incoming lepton: $0.30822|+\rangle +0.951315|-\rangle$
 proton mixing angle: $\alpha_p=0.25590123$ rad
 incoming proton: $0.967436|+\rangle +0.253117|-\rangle$

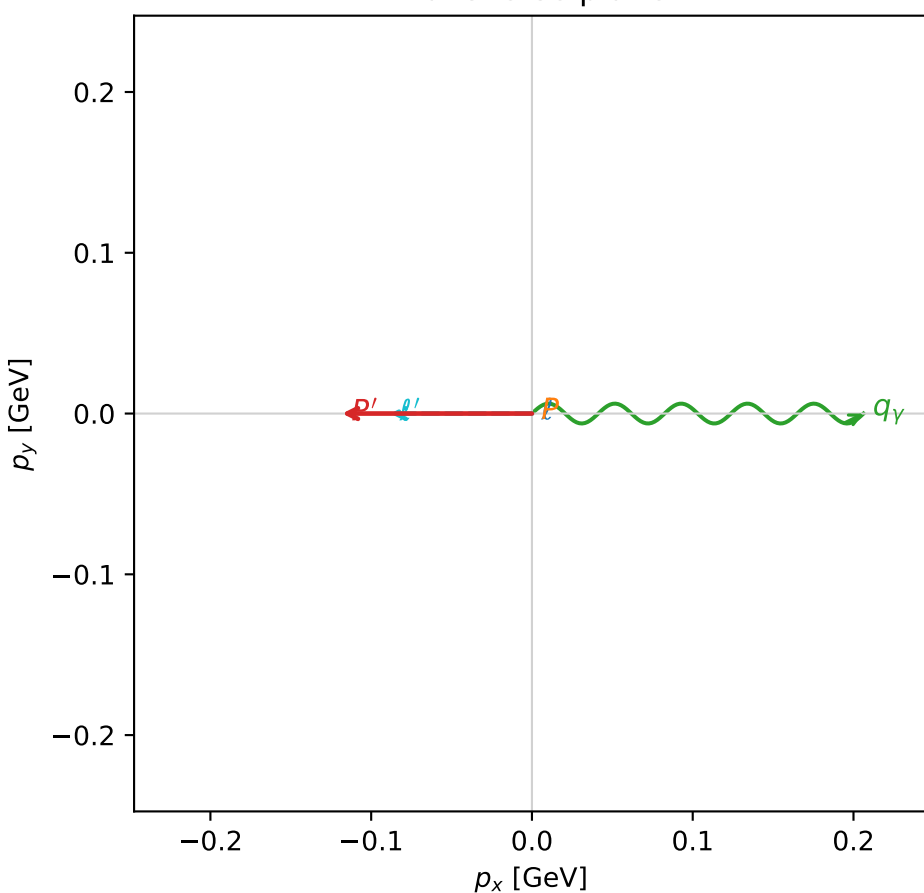
$s=1.82698$, $\sqrt{s}=1.35166$, $|\vec{P}|=0.350361$, $|\vec{P}'|=0.199548$
 $\theta_{P'}=2.5082$, $\theta_Y=1.58743$, $\phi_{P'}=3.14154$, $\phi_Y=6.28314$
 $E_Y=0.206256$, $\theta_{\ell'}=0.492354$, $\phi_{\ell'}=3.14157$
 $Q^2=0.24573$, $x_B=0.356681$, $t=-0.273487$, $W^2=1.32305$, $y=0.727388$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3504$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3504$
 P $E= 1.0013$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3504$
 ℓ' $E= 0.1864$ $p_x= -0.0881$ $p_y= 0.0000$ $p_z= 0.1643$
 P' $E= 0.9590$ $p_x= -0.1181$ $p_y= 0.0000$ $p_z= -0.1608$
 q_Y $E= 0.2063$ $p_x= 0.2062$ $p_y= -0.0000$ $p_z= -0.0034$

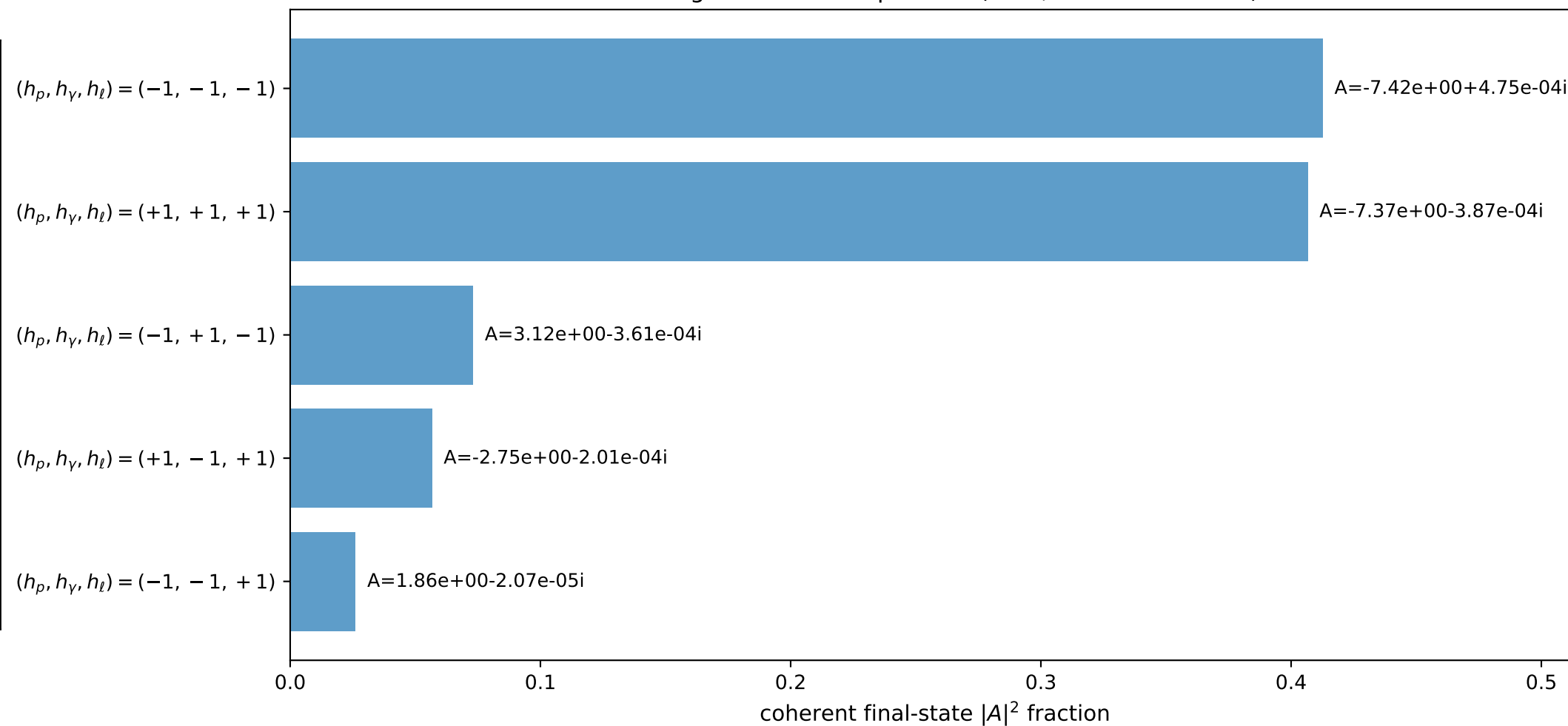
3D momenta



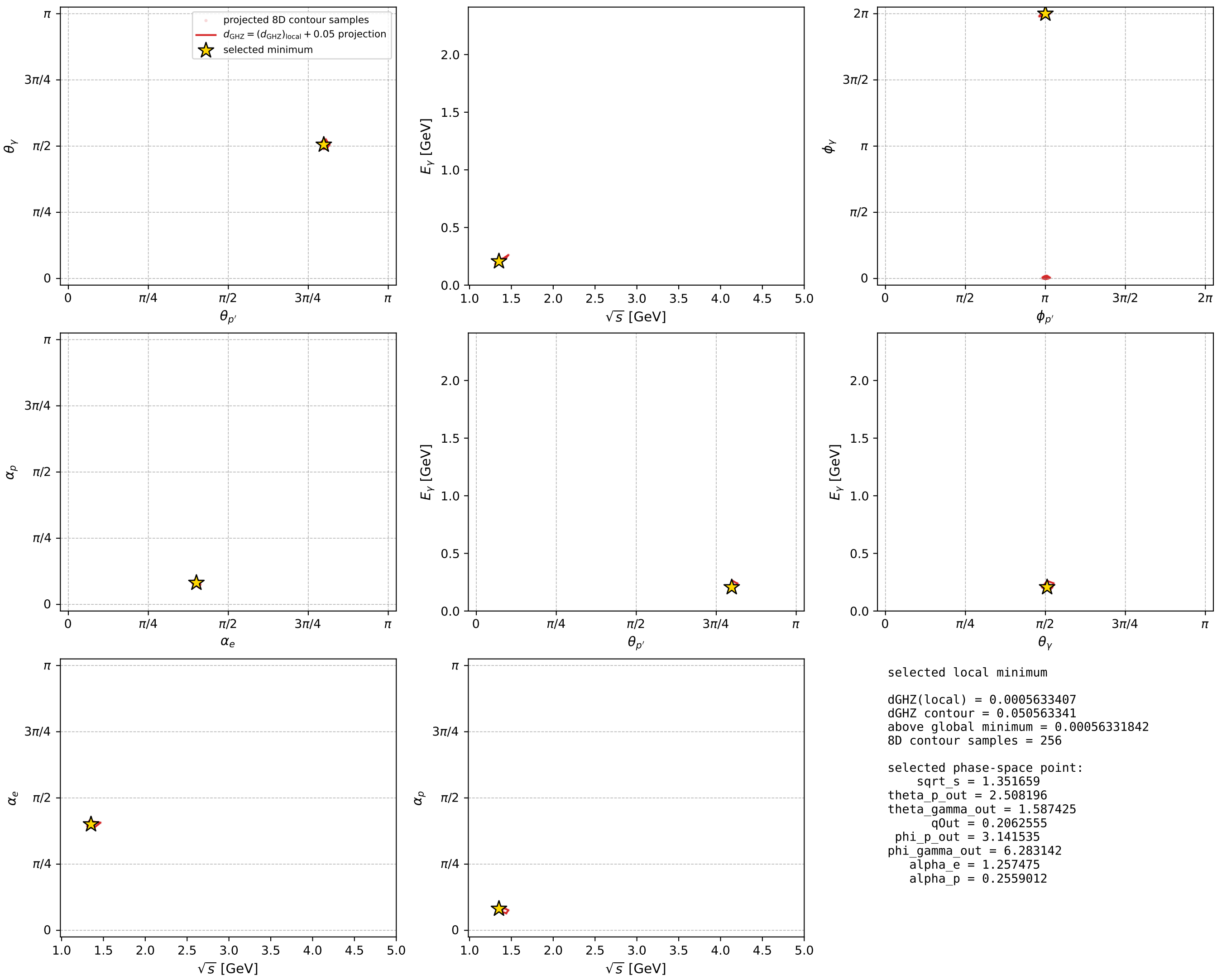
Transverse plane



Leading final-state amplitudes (N=5, retained=97.5%)



electron: polarization cluster P1, local minimum 340; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



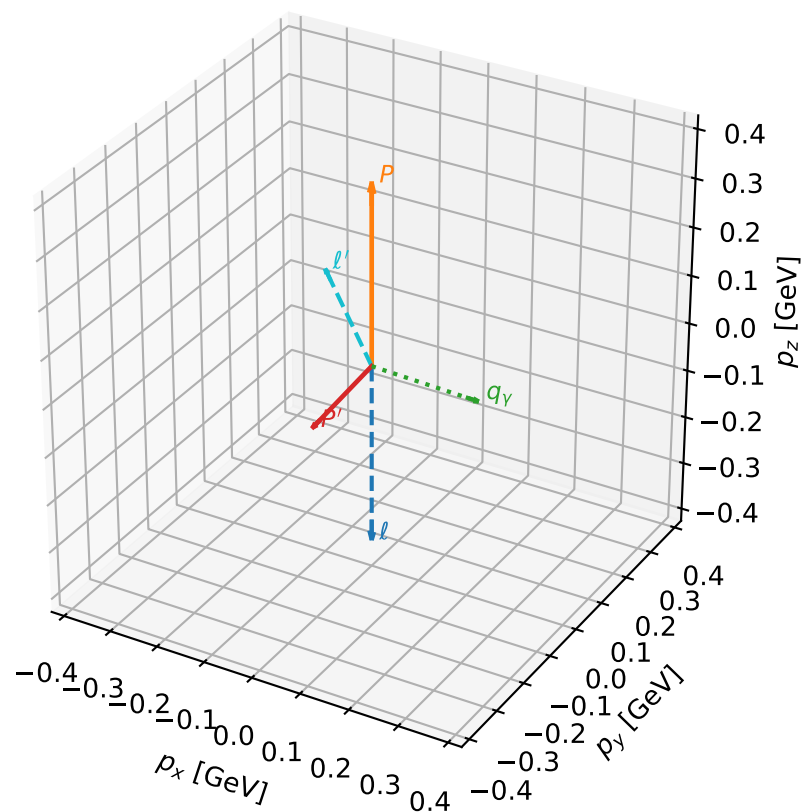
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_341 $d_{\{\text{rm GHZ}\}}=0.000579013$
region: polarization_cluster_01_local_minimum_341
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.2518472$ rad
incoming lepton: $0.313569|+\rangle +0.949565|-\rangle$
proton mixing angle: $\alpha_p=0.25399627$ rad
incoming proton: $0.967916|+\rangle +0.251274|-\rangle$

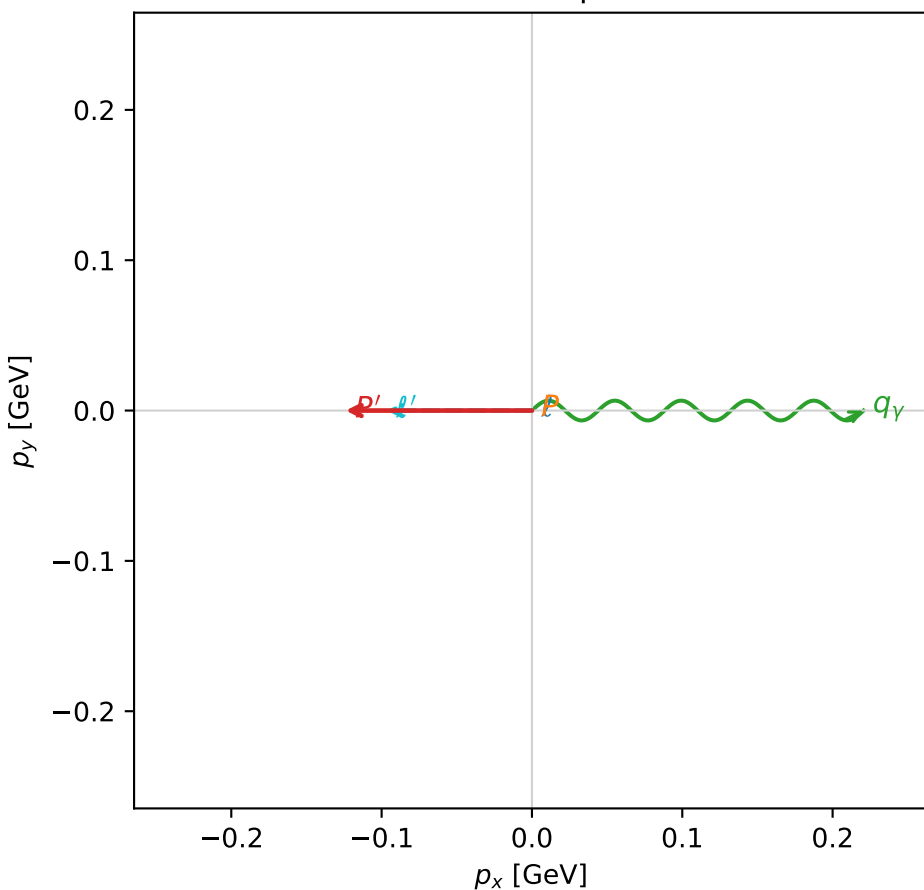
$s=1.90528$, $\sqrt{s}=1.38032$, $|\vec{P}|=0.37145$, $|\vec{P}'|=0.206249$
 $\theta_{P'}=2.49686$, $\theta_Y=1.61307$, $\phi_{P'}=3.14165$, $\phi_Y=3.58801e-05$
 $E_Y=0.220745$, $\theta_{\ell'}=0.506352$, $\phi_{\ell'}=3.1416$
 $Q^2=0.277357$, $x_B=0.368352$, $t=-0.300629$, $W^2=1.35545$, $y=0.734285$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3715$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3714$
 P $E= 1.0089$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3714$
 ℓ' $E= 0.1992$ $p_x= -0.0966$ $p_y= -0.0000$ $p_z= 0.1742$
 P' $E= 0.9604$ $p_x= -0.1240$ $p_y= -0.0000$ $p_z= -0.1648$
 q_Y $E= 0.2207$ $p_x= 0.2205$ $p_y= 0.0000$ $p_z= -0.0093$

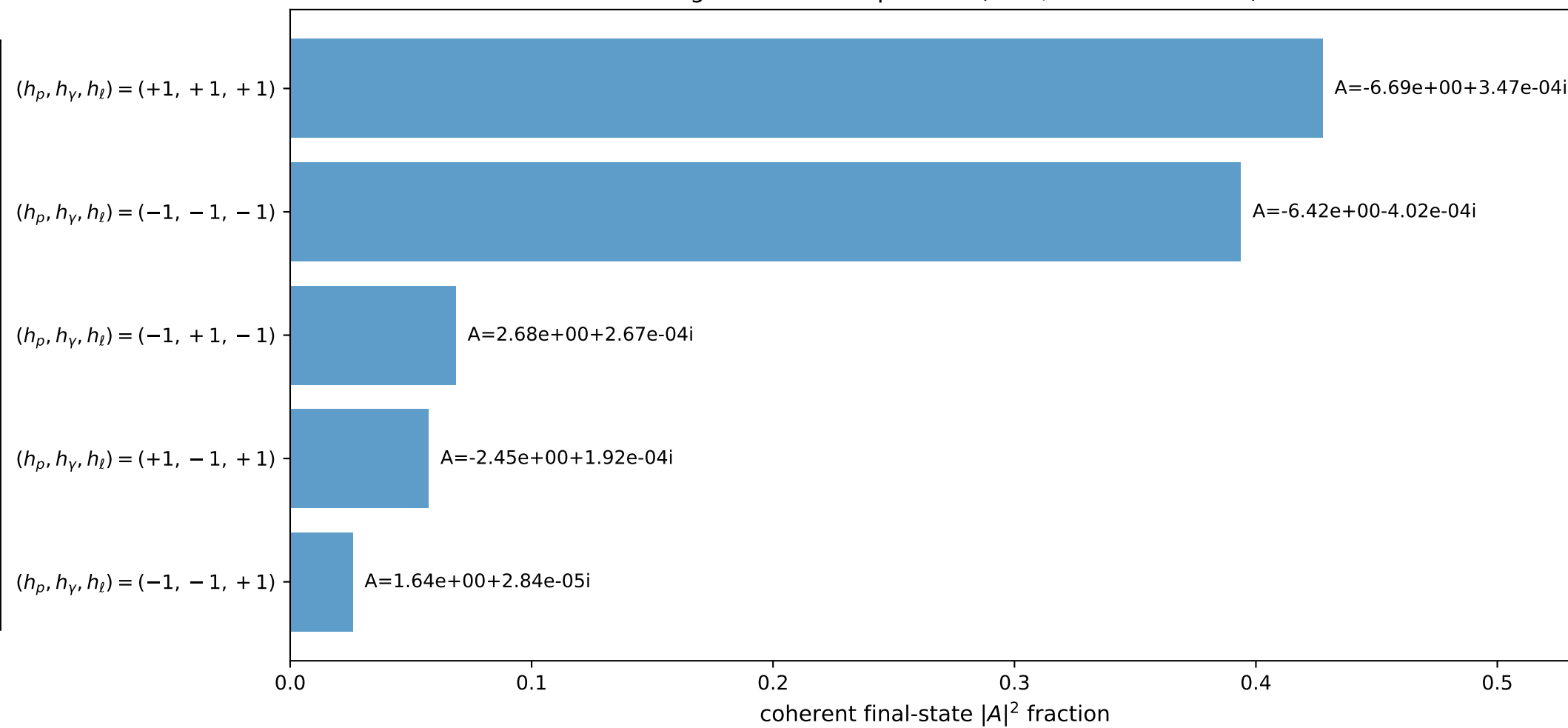
3D momenta



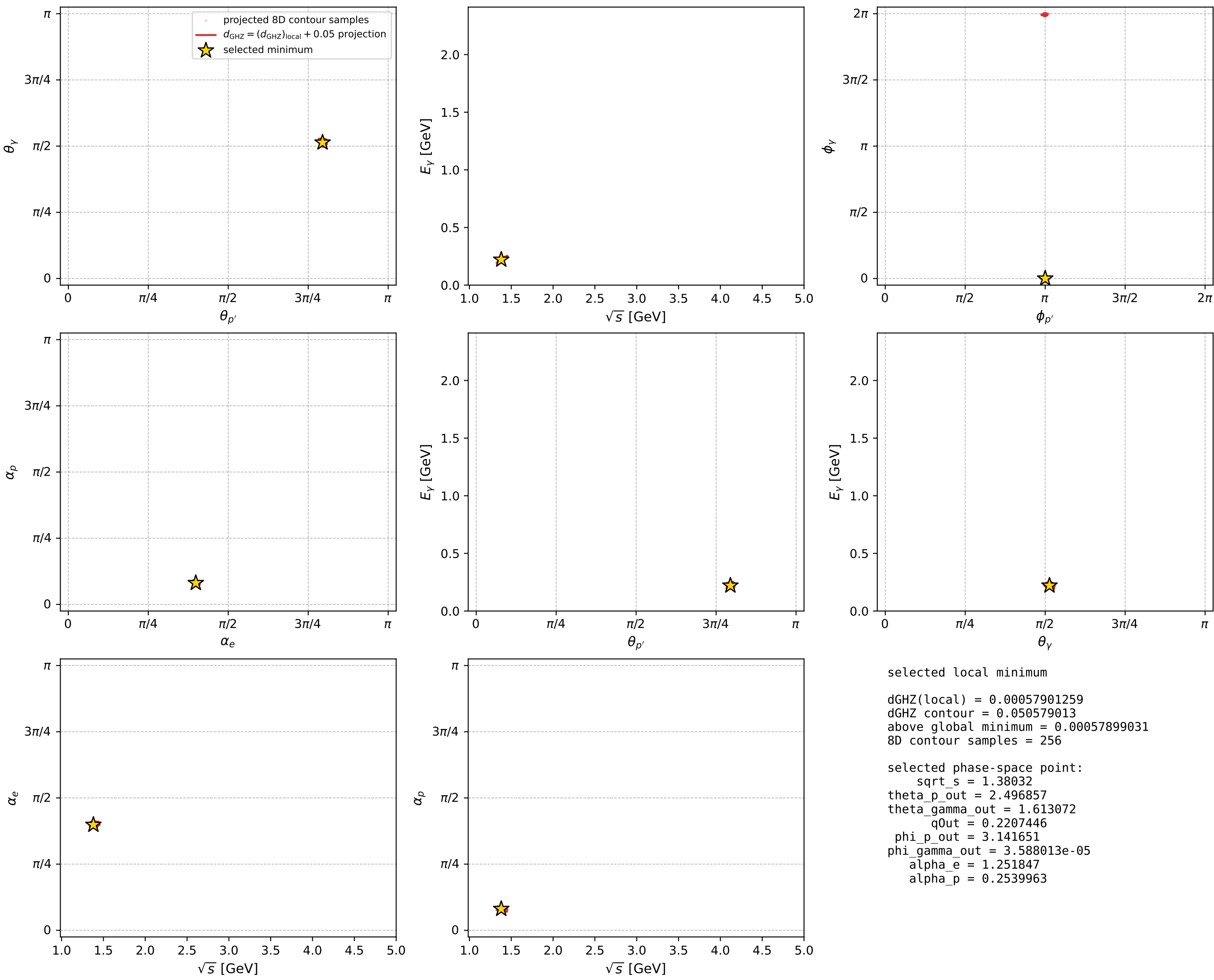
Transverse plane



Leading final-state amplitudes (N=5, retained=97.3%)



electron: polarization cluster P1, local minimum 341; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



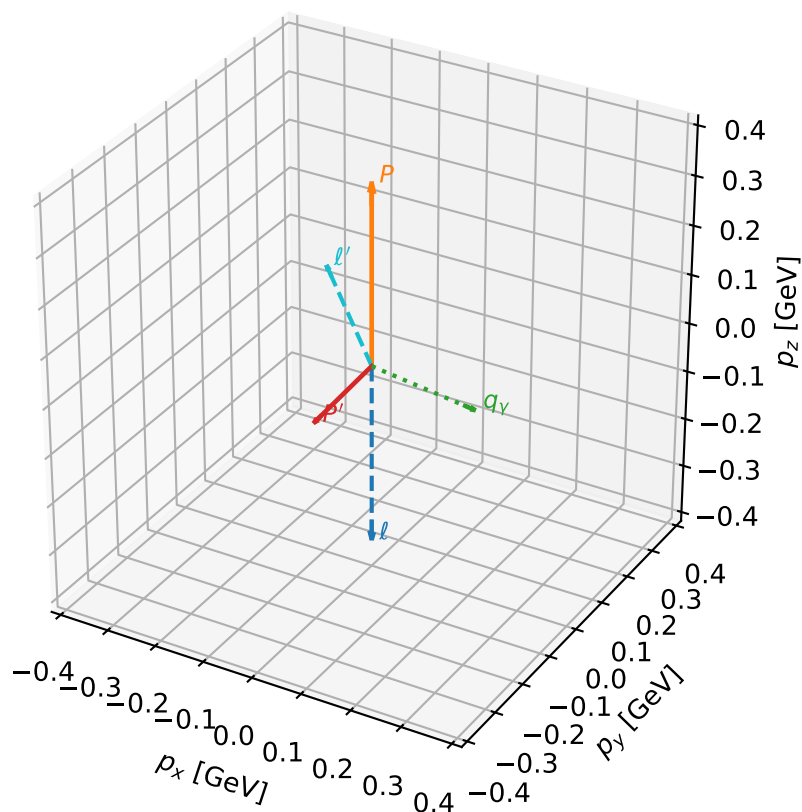
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_342 $d_{\{\text{rm GHZ}\}}=0.000604991$
 region: polarization_cluster_01_local_minimum_342
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.2641047$ rad
 incoming lepton: $0.301906|+\rangle +0.953338|-\rangle$
 proton mixing angle: $\alpha_p=0.2502507$ rad
 incoming proton: $0.96885|+\rangle +0.247647|-\rangle$

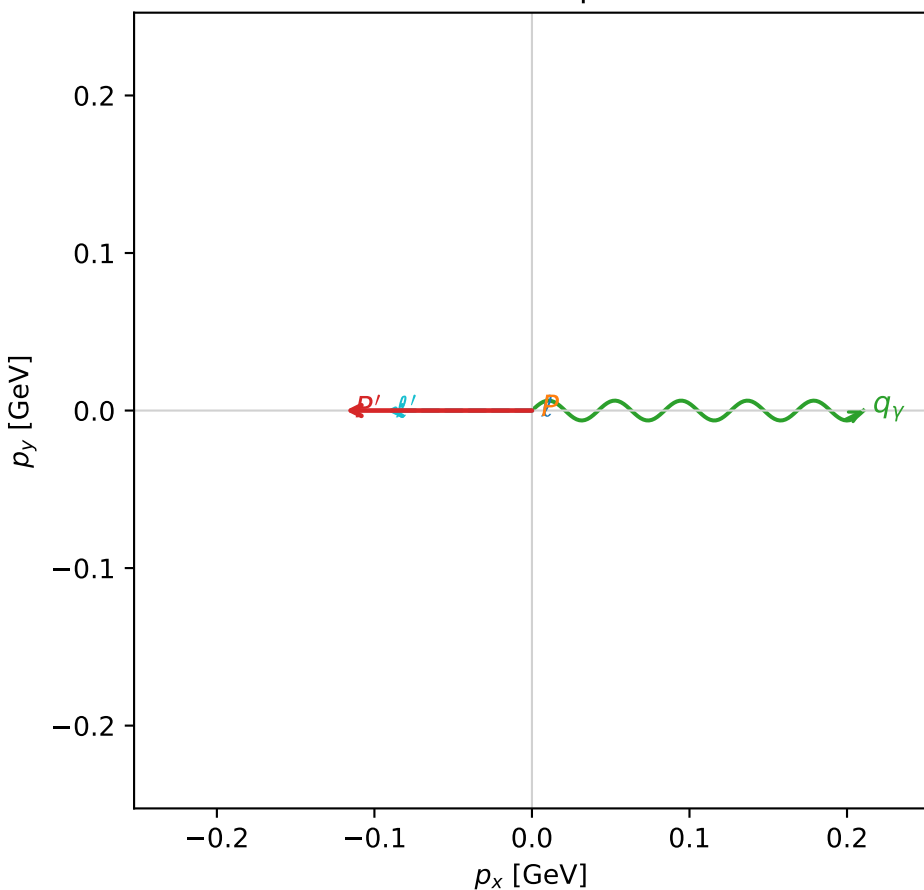
$s=1.87945$, $\sqrt{s}=1.37093$, $|\vec{P}|=0.364572$, $|\vec{P}'|=0.191131$
 $\theta_{P'}=2.47403$, $\theta_Y=1.70676$, $\phi_{P'}=3.14149$, $\phi_Y=6.28312$
 $E_Y=0.212425$, $\theta_{\ell'}=0.475616$, $\phi_{\ell'}=3.14158$
 $Q^2=0.277166$, $x_B=0.382284$, $t=-0.276481$, $W^2=1.3277$, $y=0.725312$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3646$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3646$
 P $E= 1.0064$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3646$
 ℓ' $E= 0.2012$ $p_x= -0.0921$ $p_y= 0.0000$ $p_z= 0.1789$
 P' $E= 0.9573$ $p_x= -0.1183$ $p_y= 0.0000$ $p_z= -0.1501$
 q_Y $E= 0.2124$ $p_x= 0.2105$ $p_y= -0.0000$ $p_z= -0.0288$

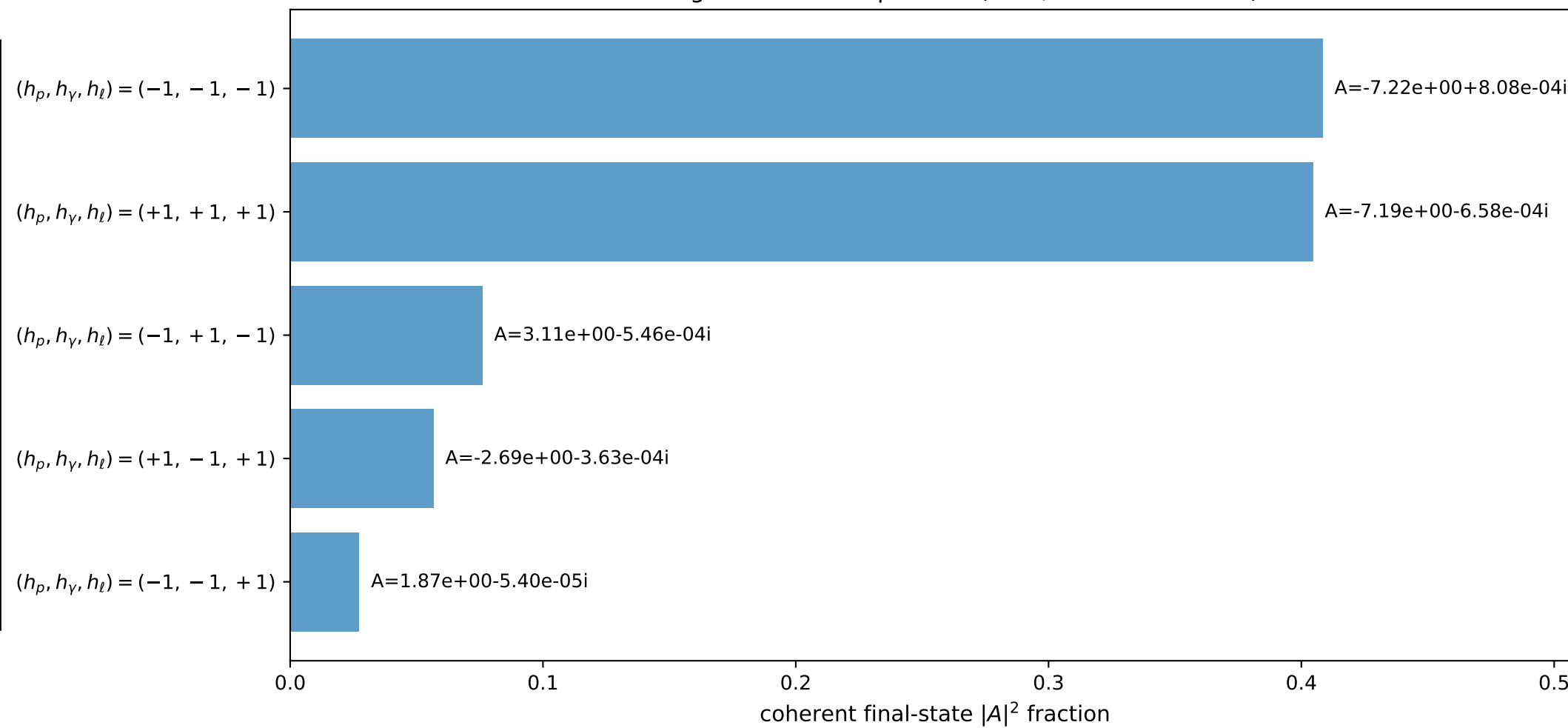
3D momenta



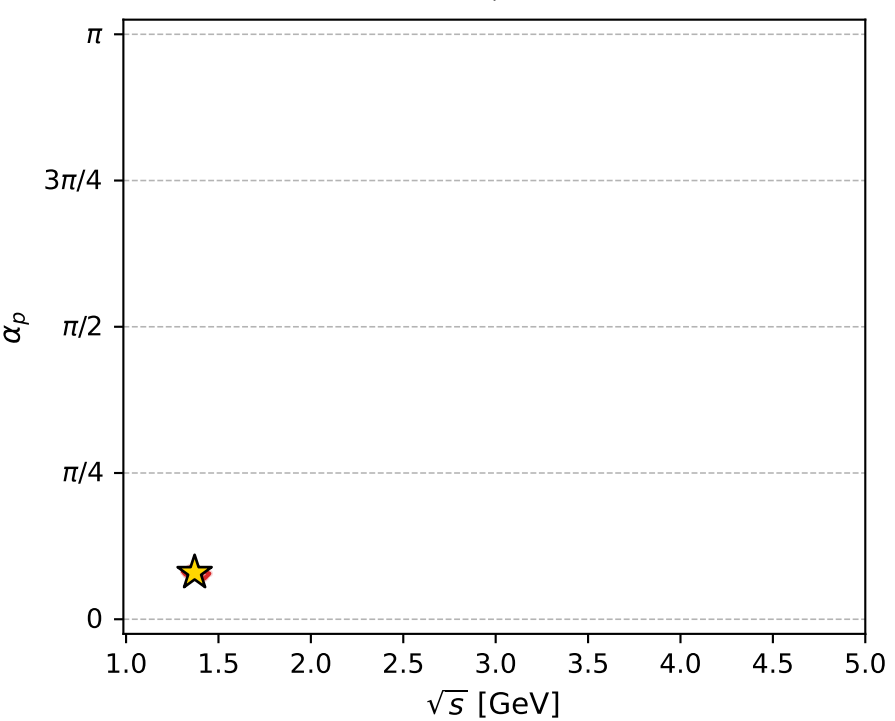
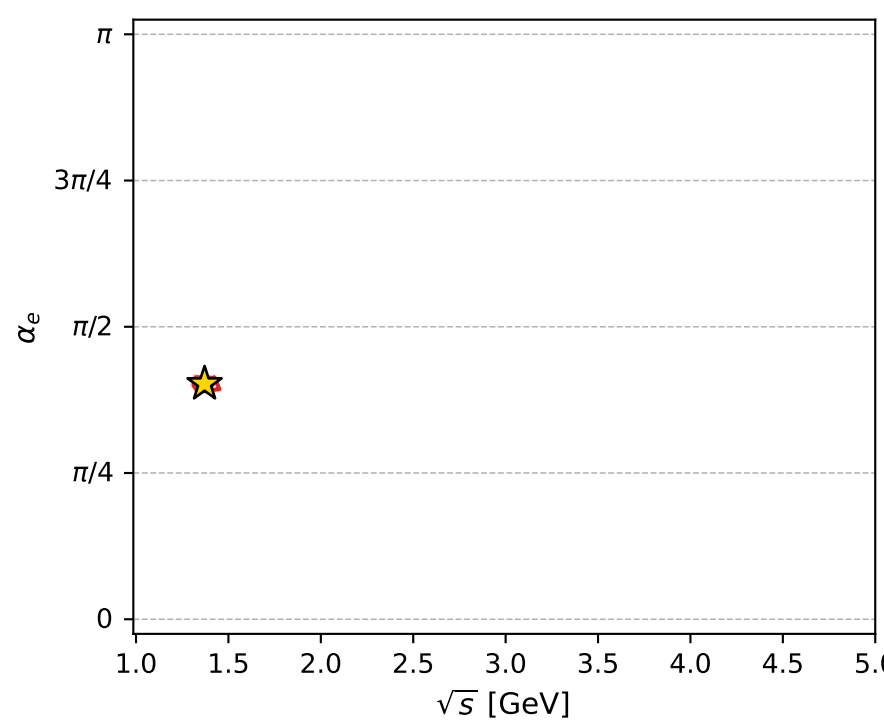
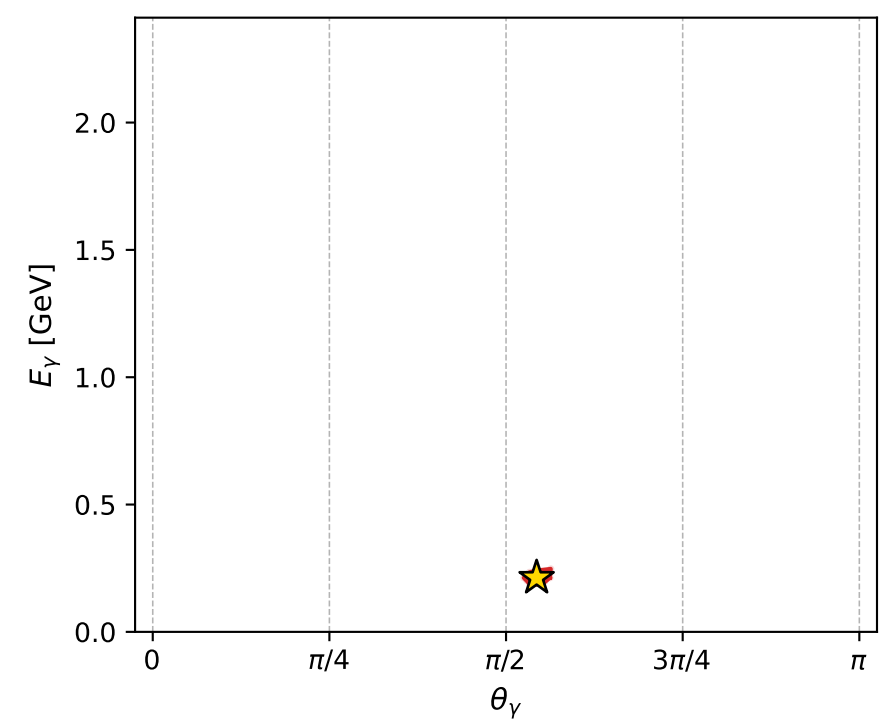
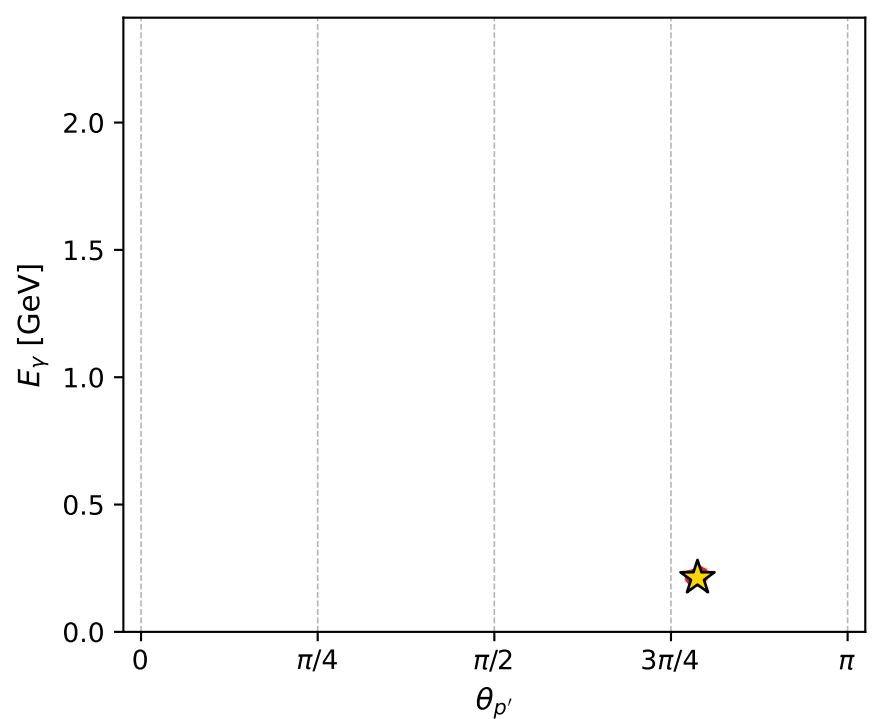
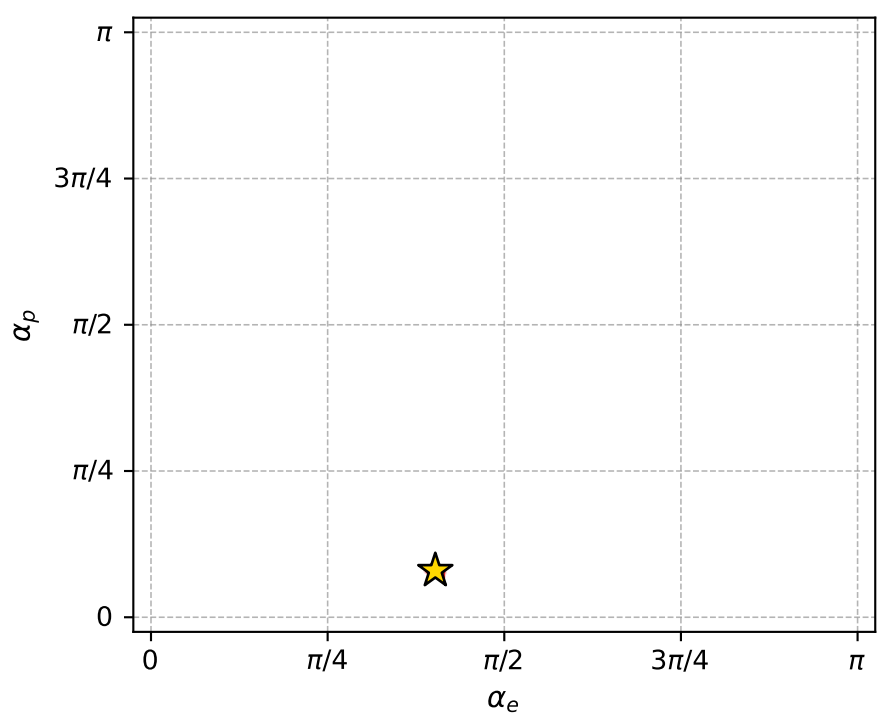
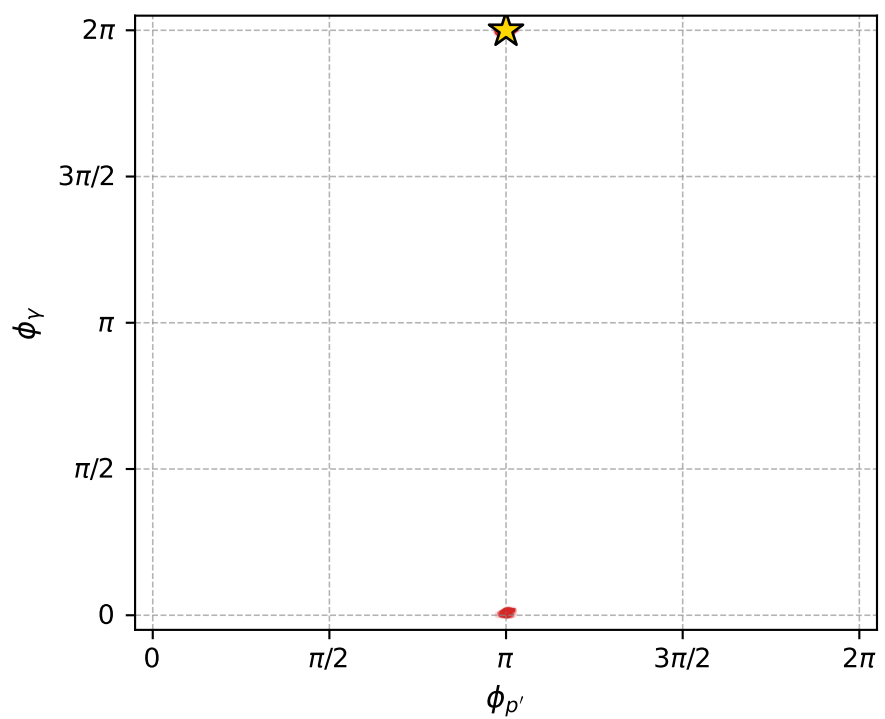
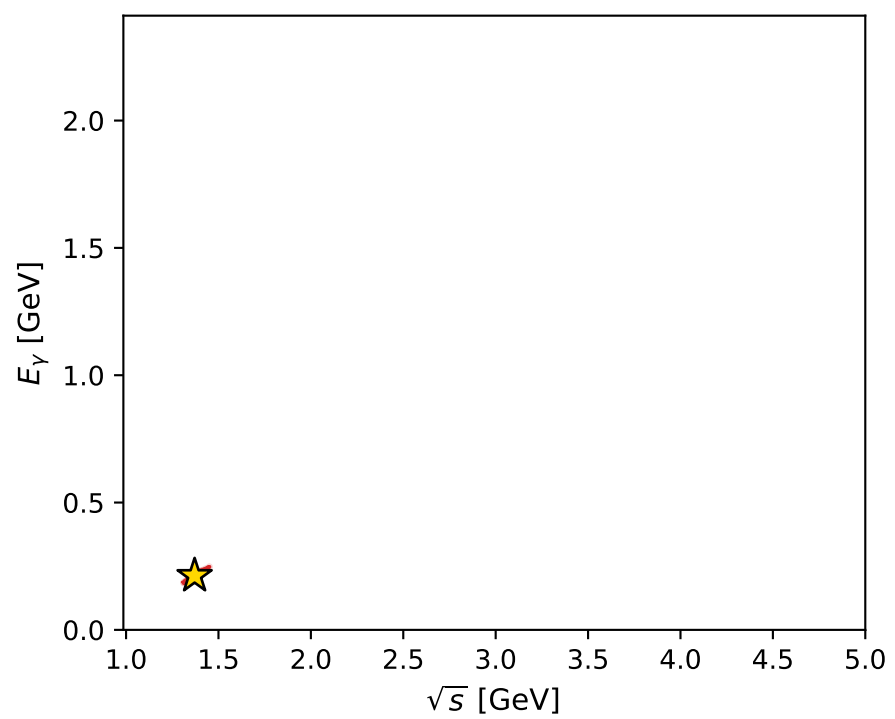
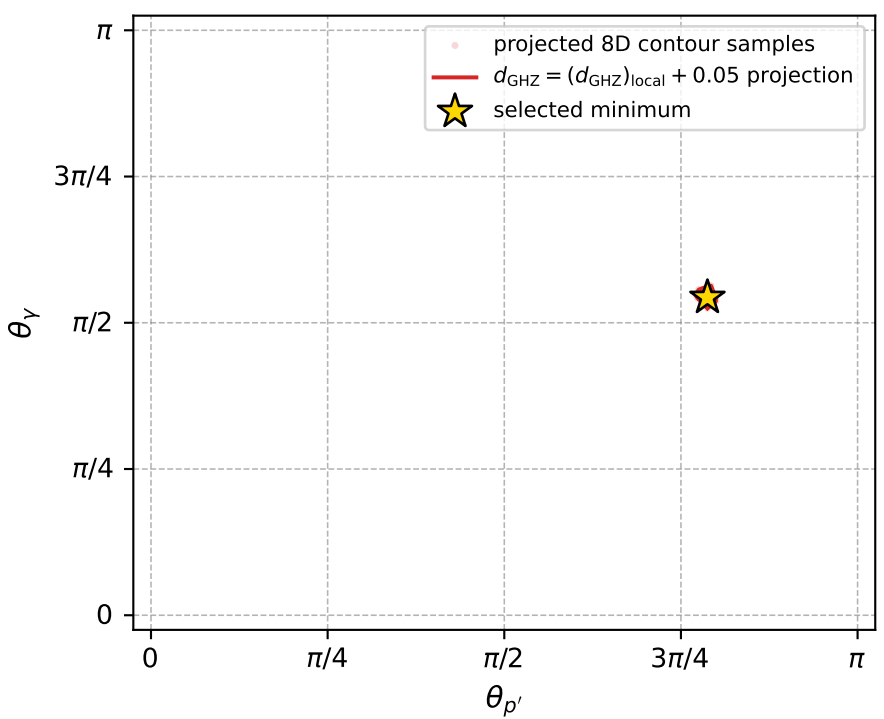
Transverse plane



Leading final-state amplitudes (N=5, retained=97.3%)



electron: polarization cluster P1, local minimum 342; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00060499071$
 $d_{\text{GHZ}} \text{ contour} = 0.050604991$
 above global minimum = 0.00060496843
 8D contour samples = 256

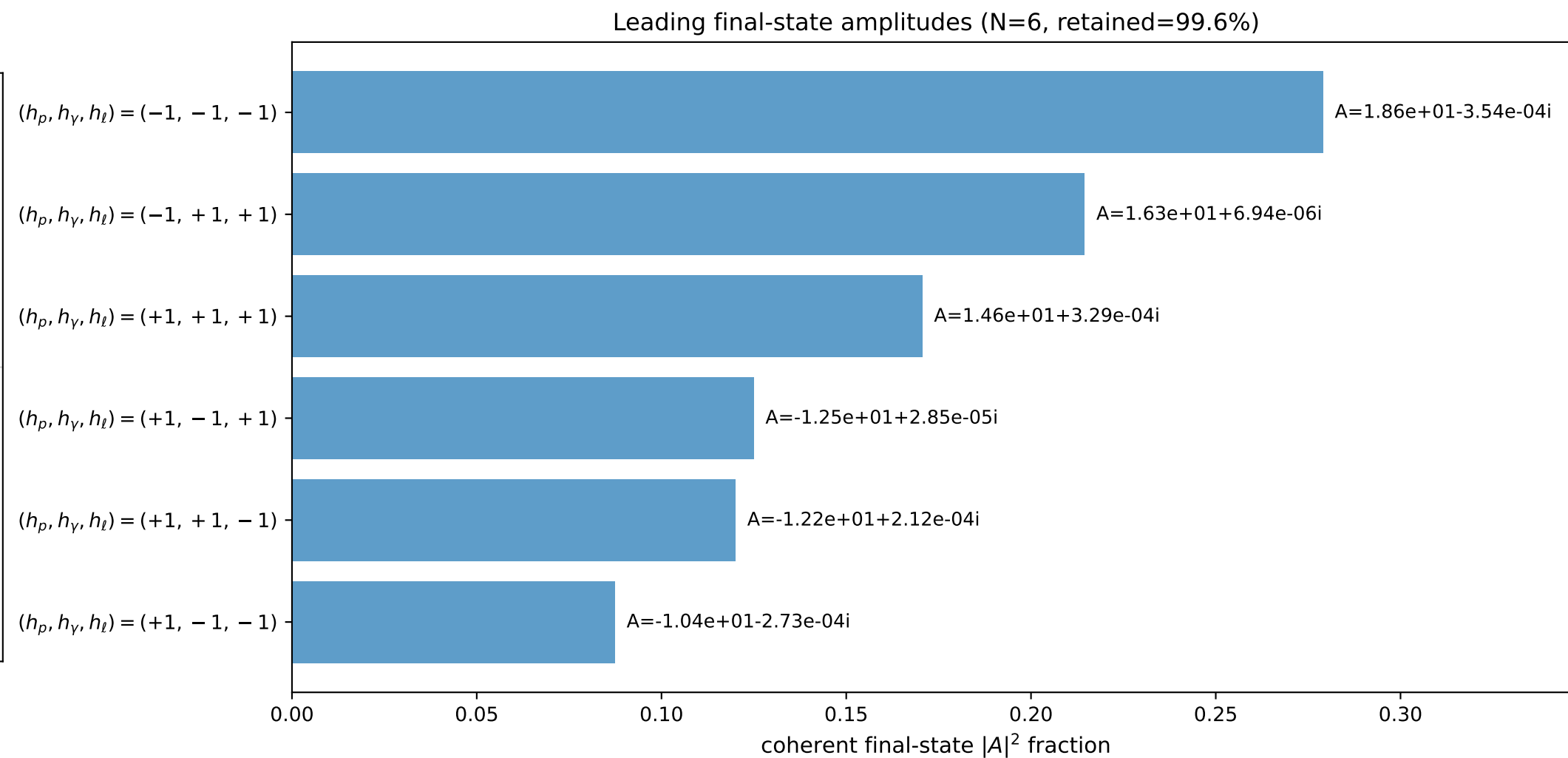
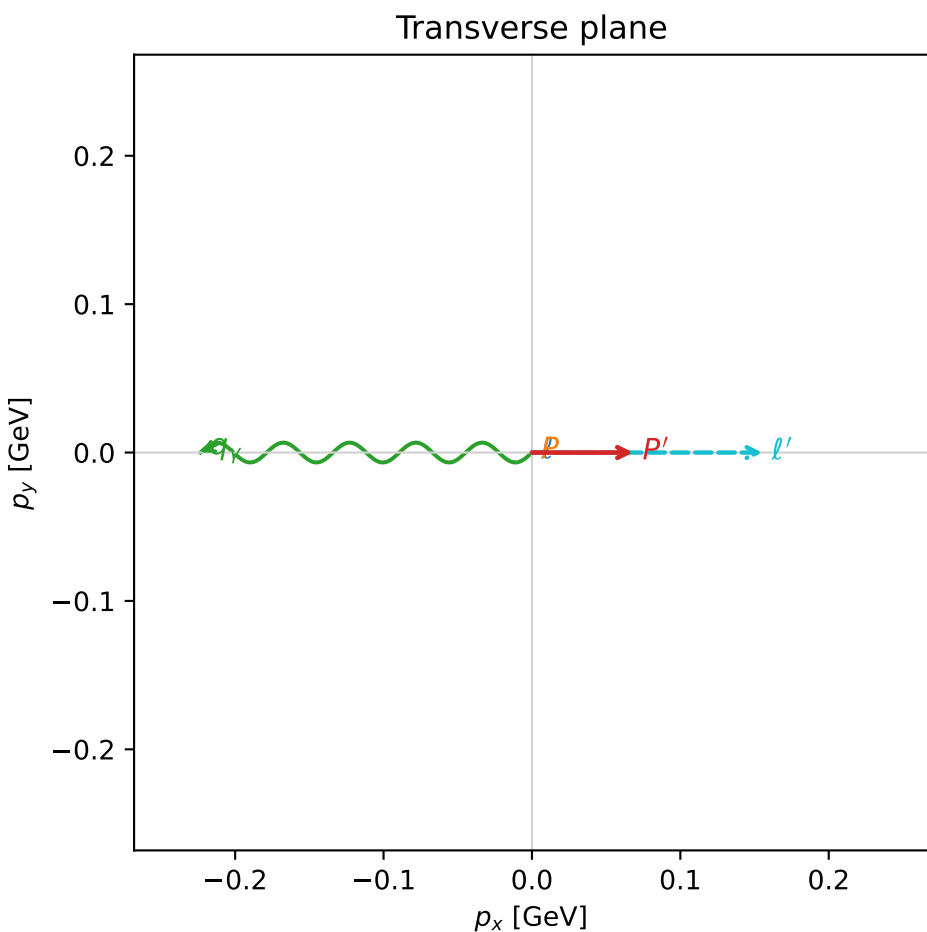
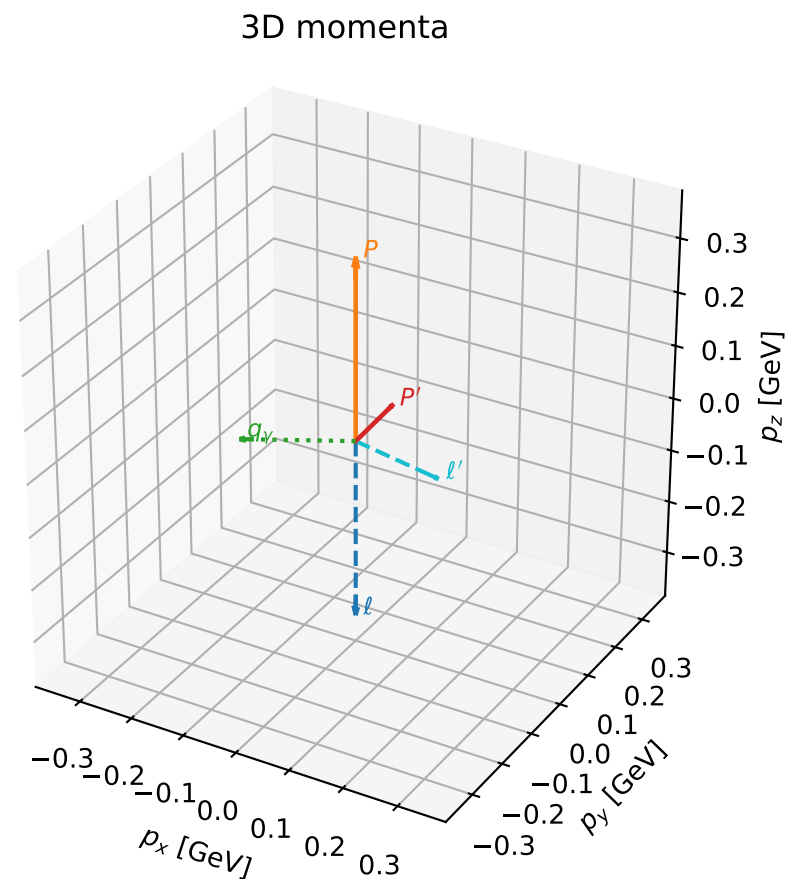
selected phase-space point:
 $\text{sqrt_s} = 1.37093$
 $\text{theta_p_out} = 2.474032$
 $\text{theta_gamma_out} = 1.706762$
 $\text{q0out} = 0.212425$
 $\text{phi_p_out} = 3.141491$
 $\text{phi_gamma_out} = 6.283122$
 $\text{alpha_e} = 1.264105$
 $\text{alpha_p} = 0.2502507$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

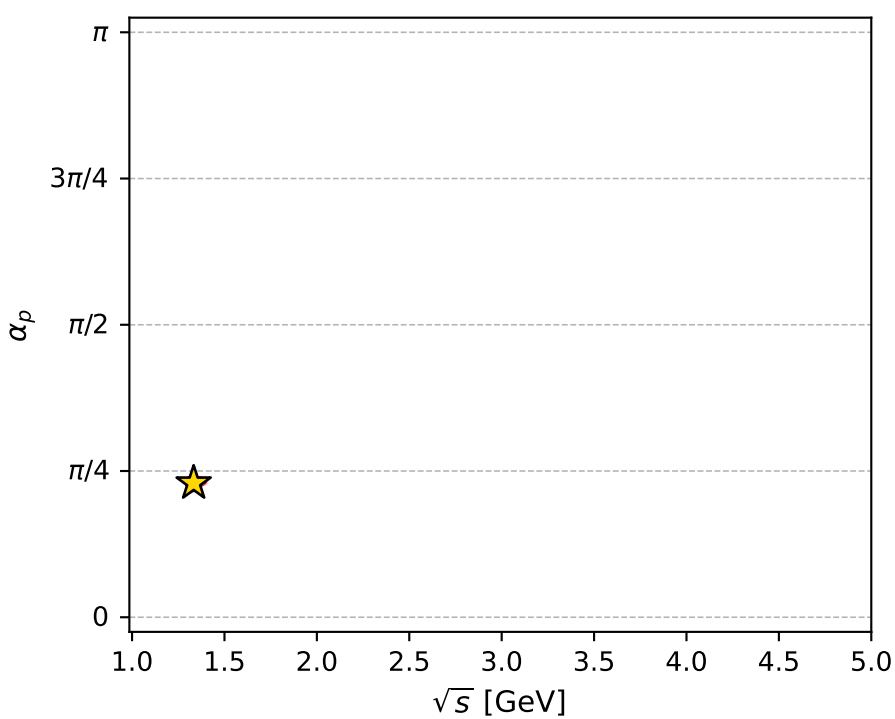
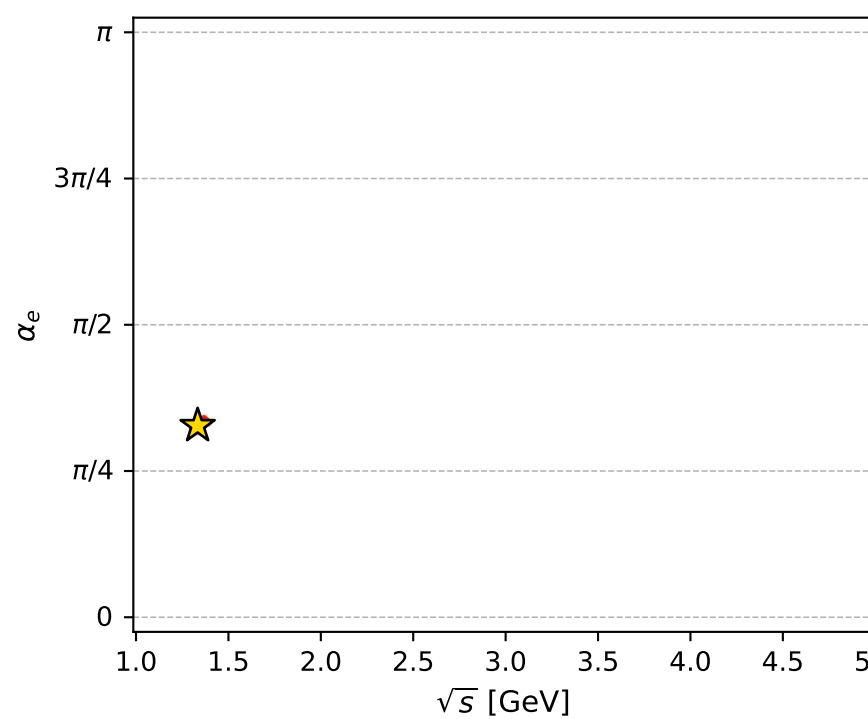
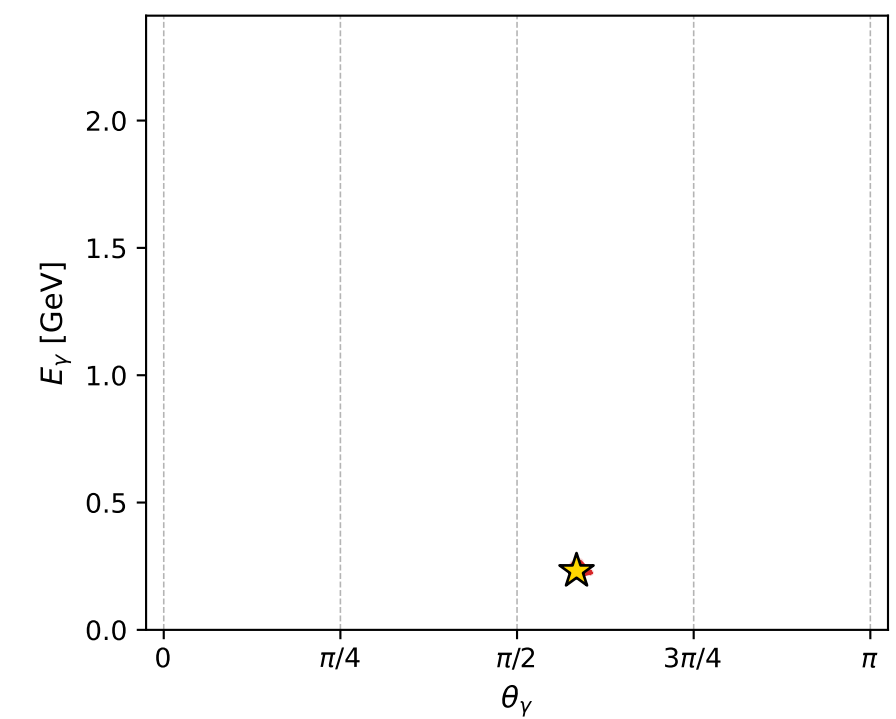
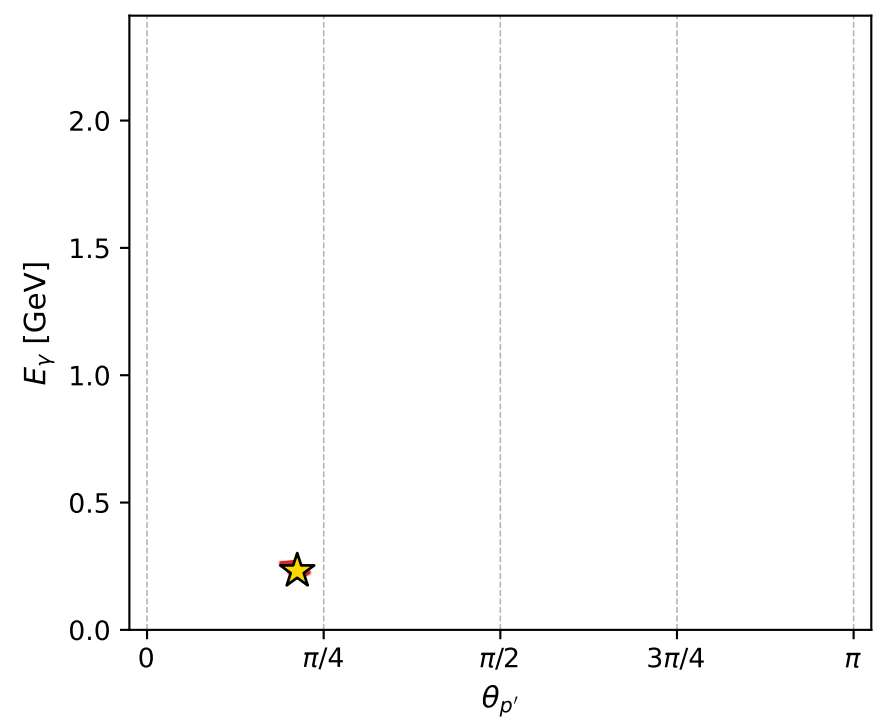
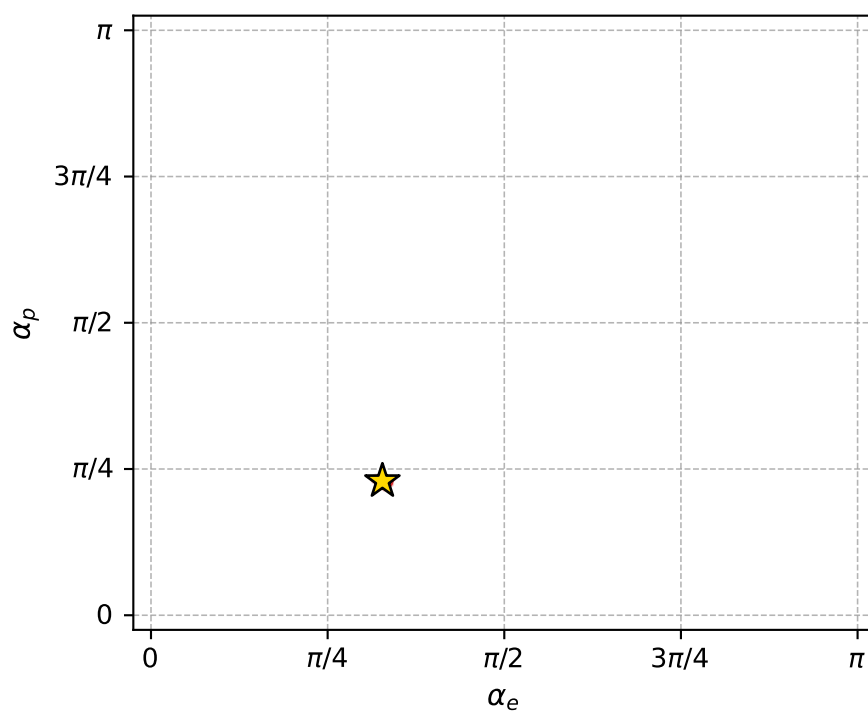
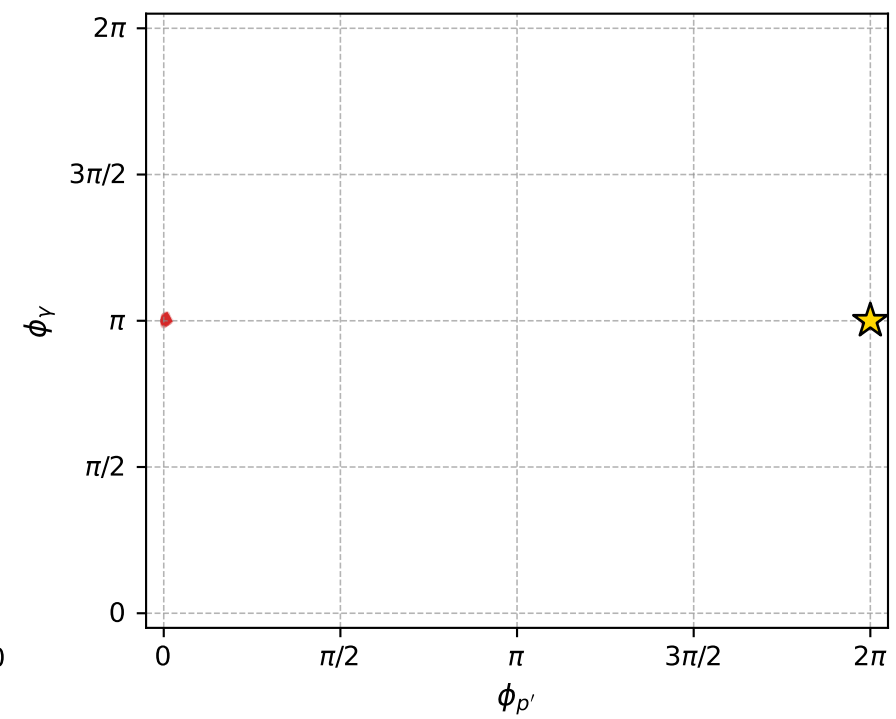
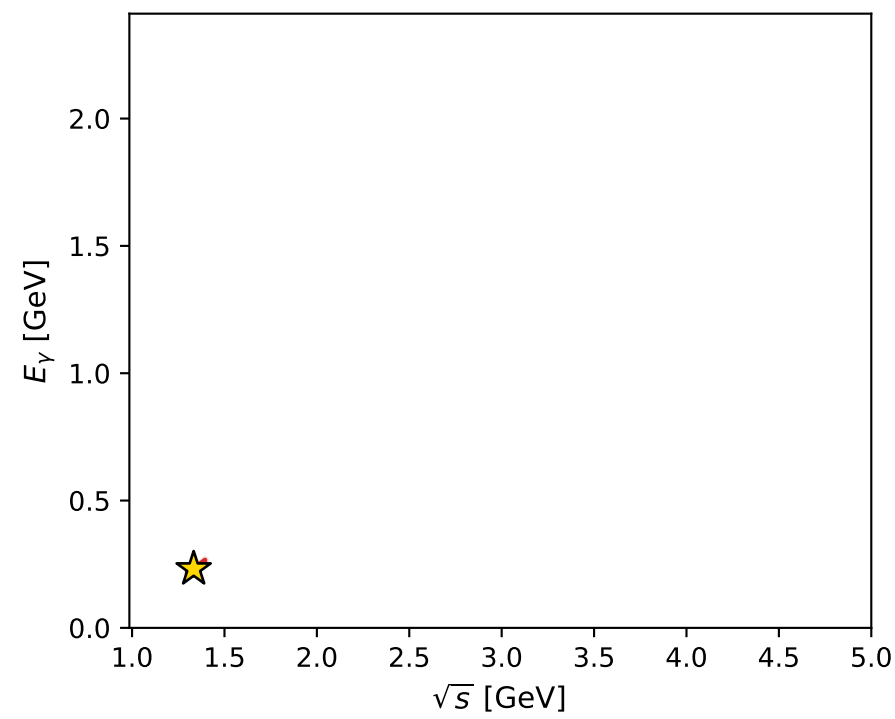
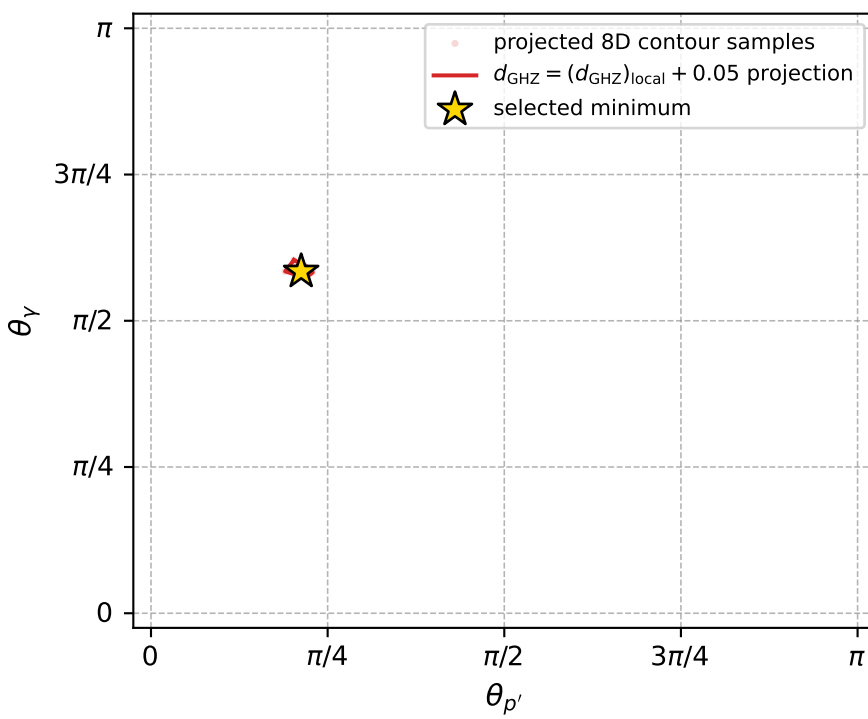
dghz_mixing_angles_polarization_01_local_minimum_346 $d_{\{\text{rm GHZ}\}}=0.000693986$
 region: polarization_cluster_01_local_minimum_346
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.0289044$ rad
 incoming lepton: $0.515758|+\rangle +0.856734|-\rangle$
 proton mixing angle: $\alpha_p=0.72047289$ rad
 incoming proton: $0.751494|+\rangle +0.65974|-\rangle$

$s=1.77775$, $\sqrt{s}=1.33332$, $|\vec{P}|=0.336716$, $|\vec{P}'|=0.110152$
 $\theta_{P'}=0.667691$, $\theta_Y=1.8353$, $\phi_{P'}=6.28316$, $\phi_Y=3.1416$
 $E_Y=0.231485$, $\theta_{\ell'}=1.73663$, $\phi_{\ell'}=1.59494\text{e-}05$
 $Q^2=0.0884955$, $x_B=0.156161$, $t=-0.0645404$, $W^2=1.35804$, $y=0.63113$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3367$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3367$
 P $E= 0.9966$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3367$
 ℓ' $E= 0.1574$ $p_x= 0.1552$ $p_y= 0.0000$ $p_z= -0.0260$
 P' $E= 0.9444$ $p_x= 0.0682$ $p_y= -0.0000$ $p_z= 0.0865$
 q_Y $E= 0.2315$ $p_x= -0.2234$ $p_y= -0.0000$ $p_z= -0.0605$



electron: polarization cluster P1, local minimum 346; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00069398647$
 $d_{\text{GHZ}} \text{ contour} = 0.050693986$
 above global minimum = 0.00069396418
 8D contour samples = 256

selected phase-space point:

$\text{sqrt_s} = 1.333322$
 $\text{theta_p_out} = 0.6676909$
 $\text{theta_gamma_out} = 1.835298$
 $\text{q0out} = 0.2314849$
 $\text{phi_p_out} = 6.283158$
 $\text{phi_gamma_out} = 3.141595$
 $\text{alpha_e} = 1.028904$
 $\text{alpha_p} = 0.7204729$

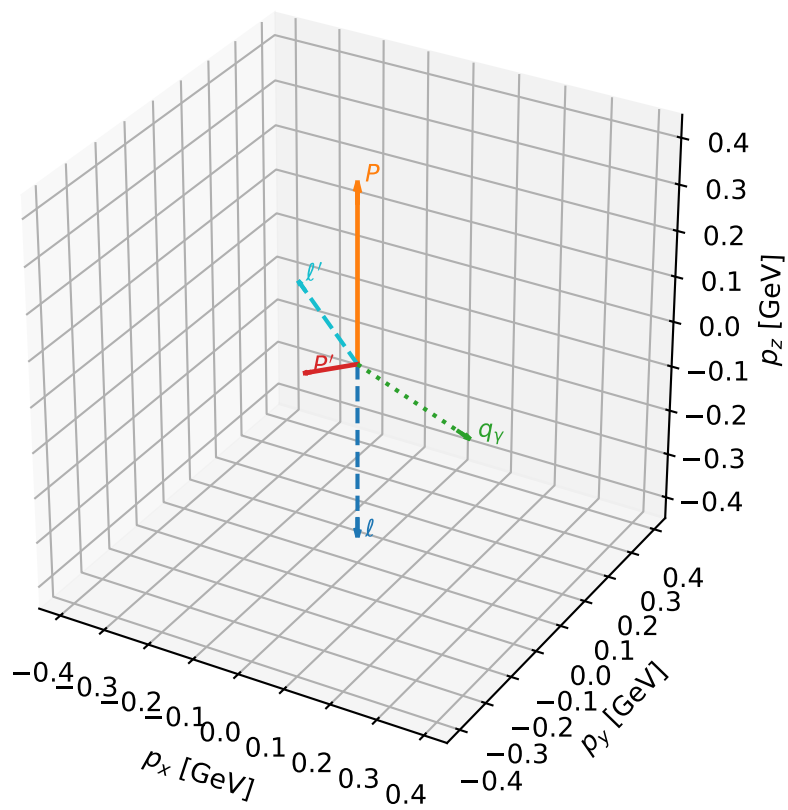
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_350 d_{\rm GHZ}=0.000879306
 region: polarization_cluster_01_local_minimum_350
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.0618757$ rad
 incoming lepton: $0.487235|+\rangle +0.873271|-\rangle$
 proton mixing angle: $\alpha_p=0.42969074$ rad
 incoming proton: $0.909095|+\rangle +0.41659|-\rangle$

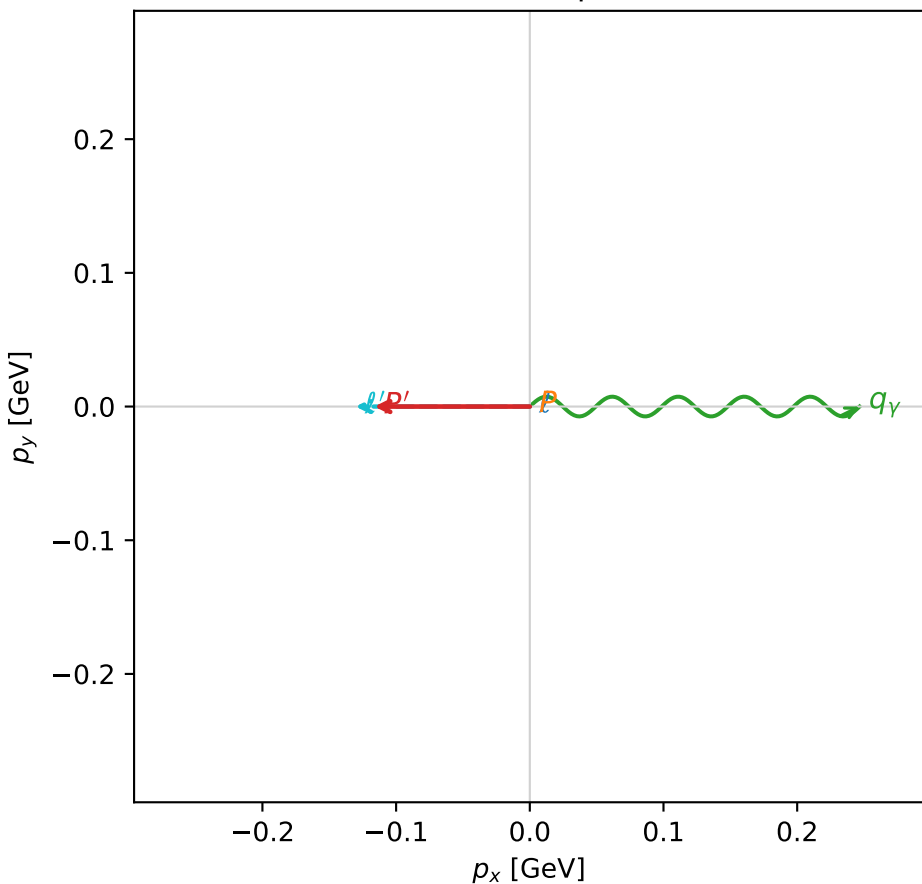
$s=1.97653$, $\sqrt{s}=1.40589$, $|\vec{P}|=0.390033$, $|\vec{P}'|=0.127839$
 $\theta_{P'}=1.99929$, $\theta_V=1.93061$, $\phi_{P'}=3.14161$, $\phi_V=6.28318$
 $E_V=0.263543$, $\theta_{l'}=0.729268$, $\phi_{l'}=3.14157$
 $Q^2=0.266461$, $x_B=0.327772$, $t=-0.205116$, $W^2=1.42633$, $y=0.741272$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.3900$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3900$
 P $E= 1.0159$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3900$
 l' $E= 0.1957$ $p_x= -0.1304$ $p_y= 0.0000$ $p_z= 0.1459$
 P' $E= 0.9467$ $p_x= -0.1163$ $p_y= -0.0000$ $p_z= -0.0531$
 q_V $E= 0.2635$ $p_x= 0.2467$ $p_y= -0.0000$ $p_z= -0.0928$

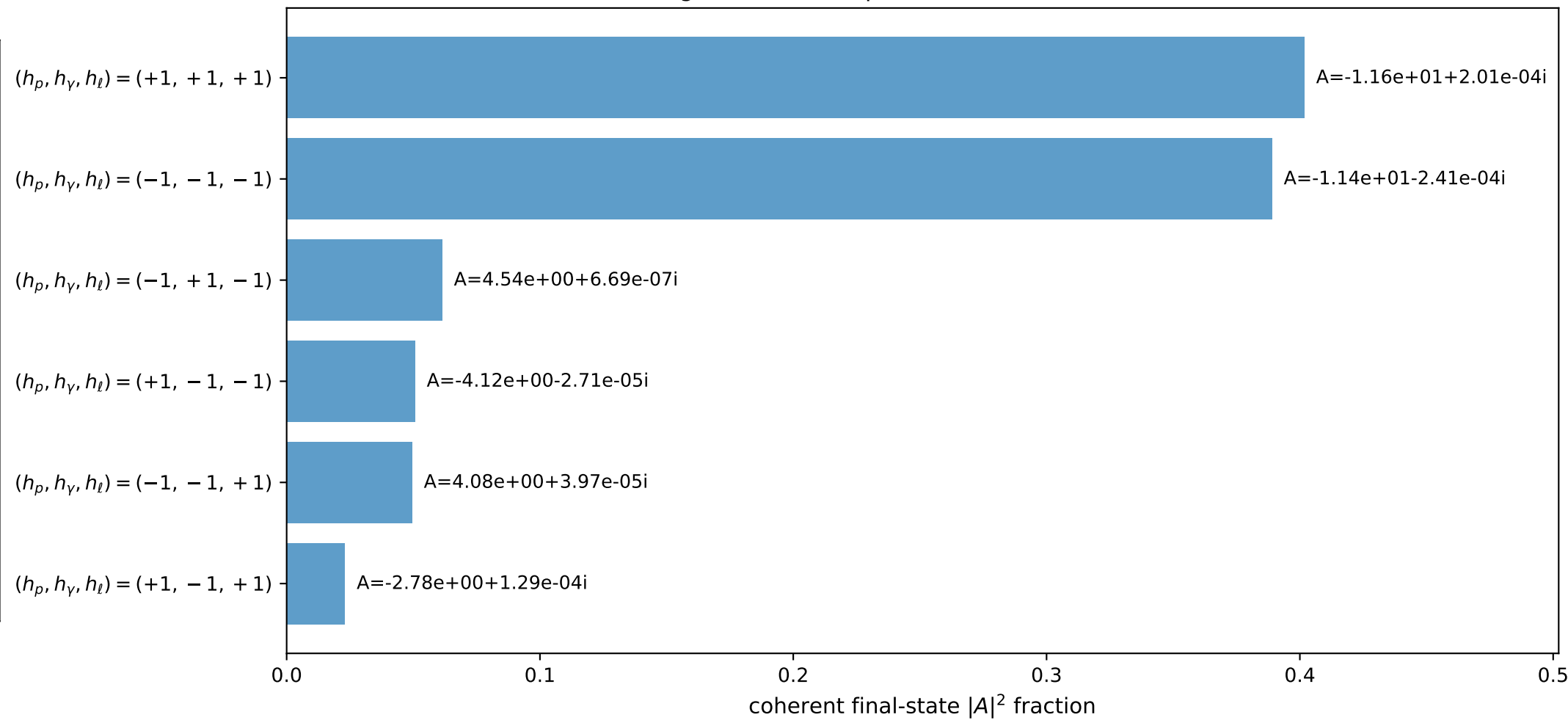
3D momenta



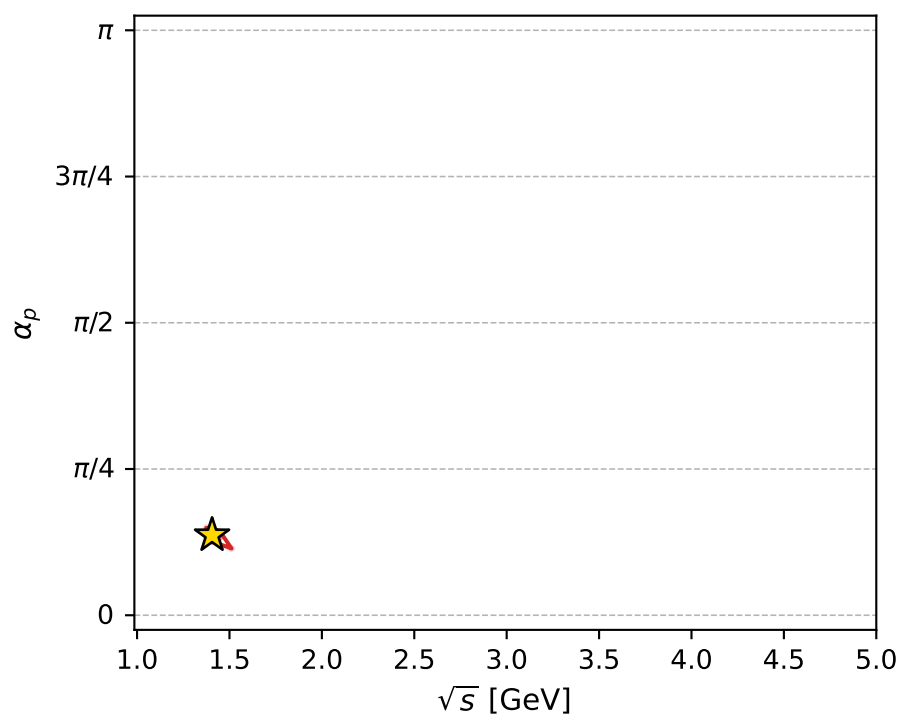
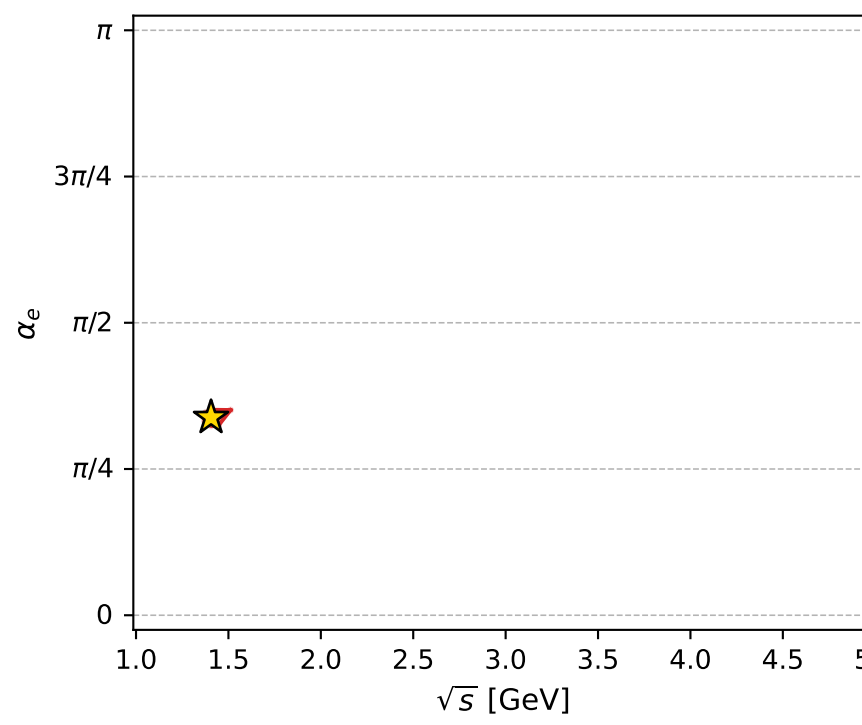
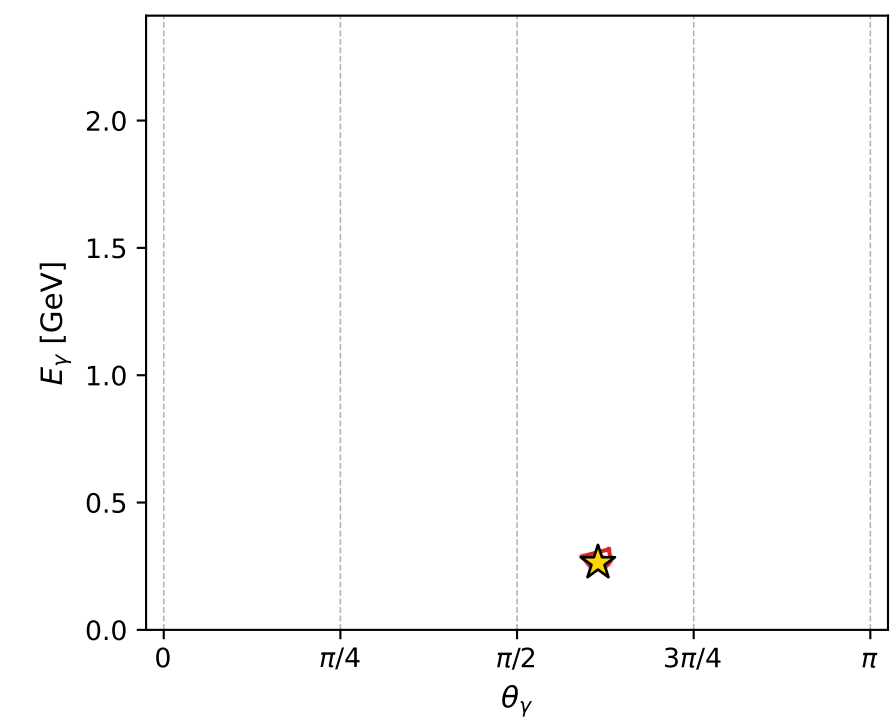
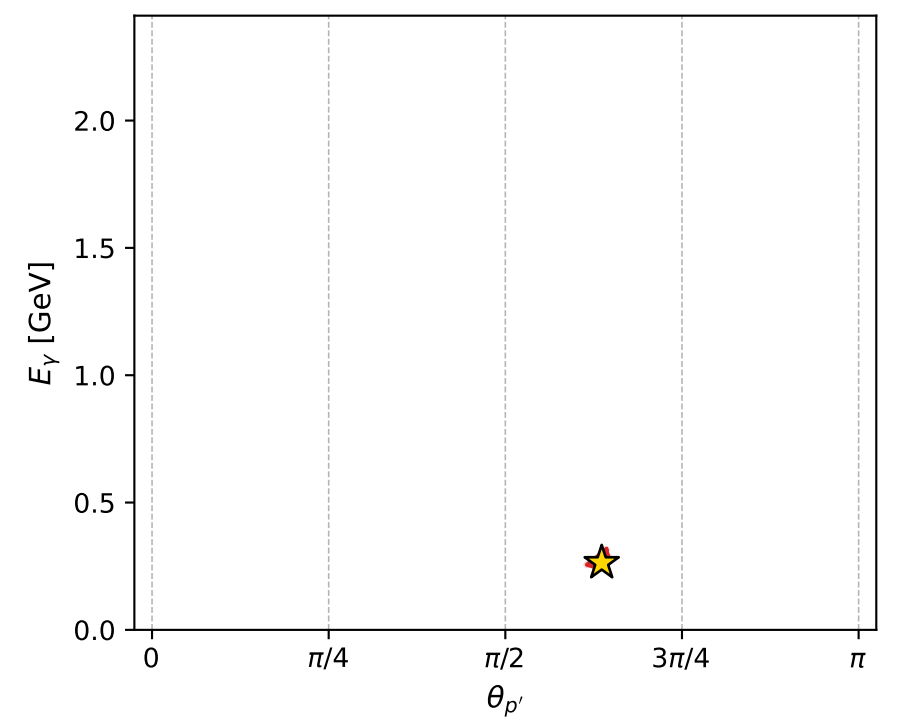
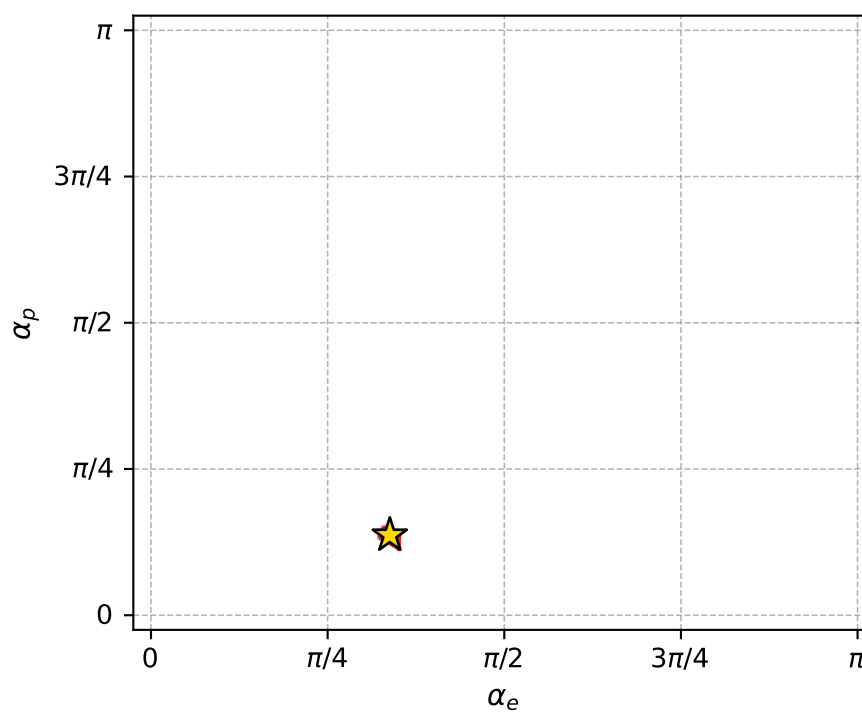
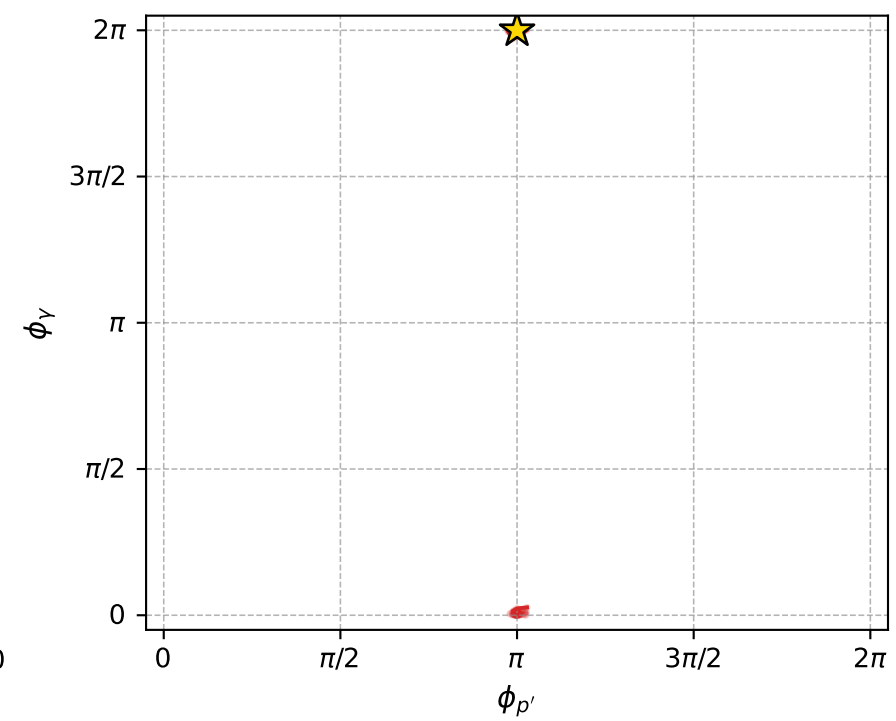
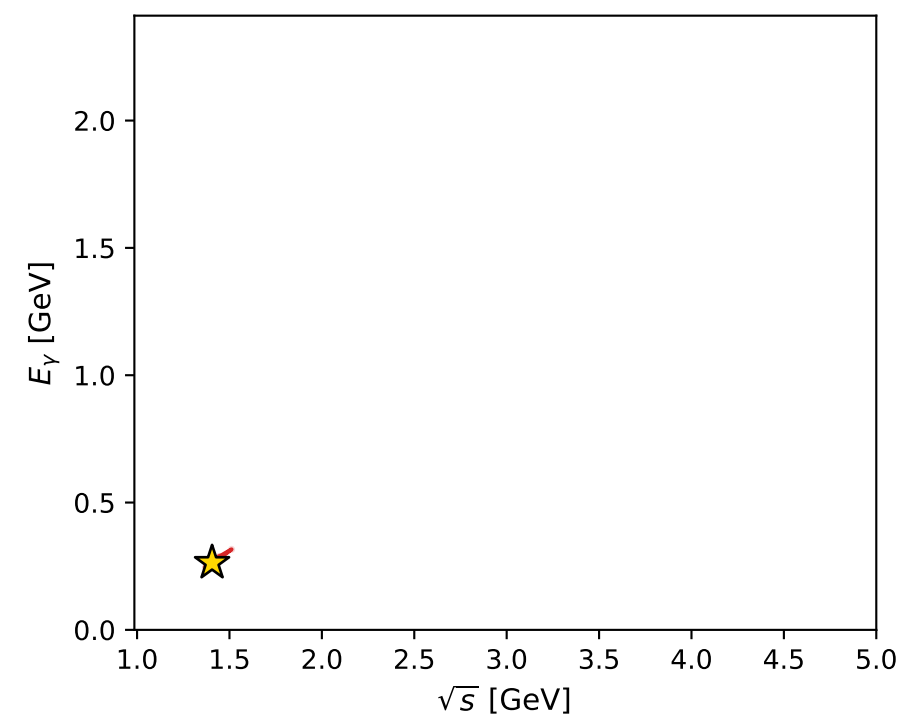
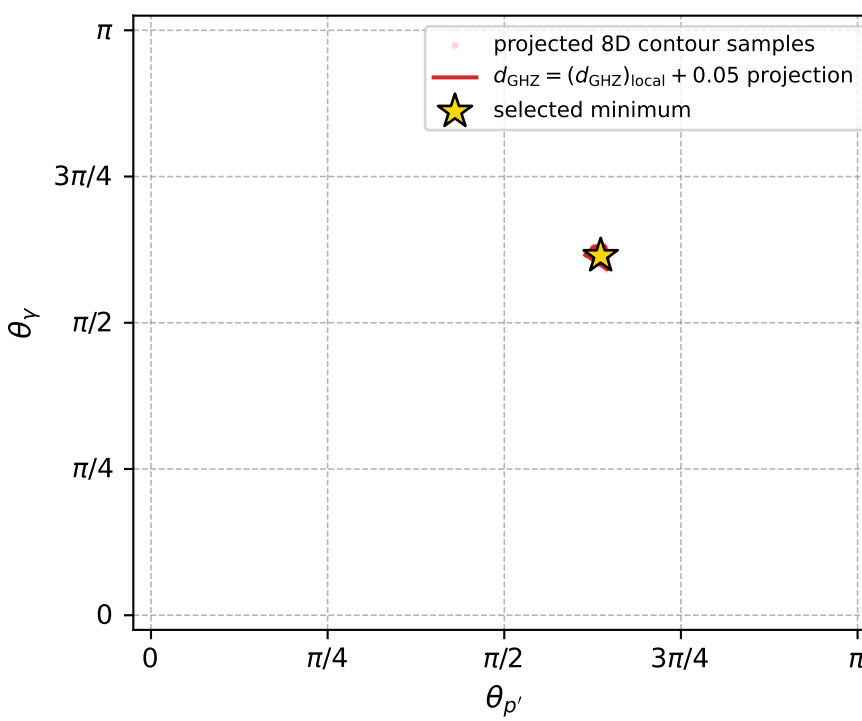
Transverse plane



Leading final-state amplitudes (N=6, retained=97.5%)



electron: polarization cluster P1, local minimum 350; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.000879306$
 $d_{\text{GHZ}} \text{ contour} = 0.050879306$
 above global minimum = 0.00087928371
 8D contour samples = 256

selected phase-space point:

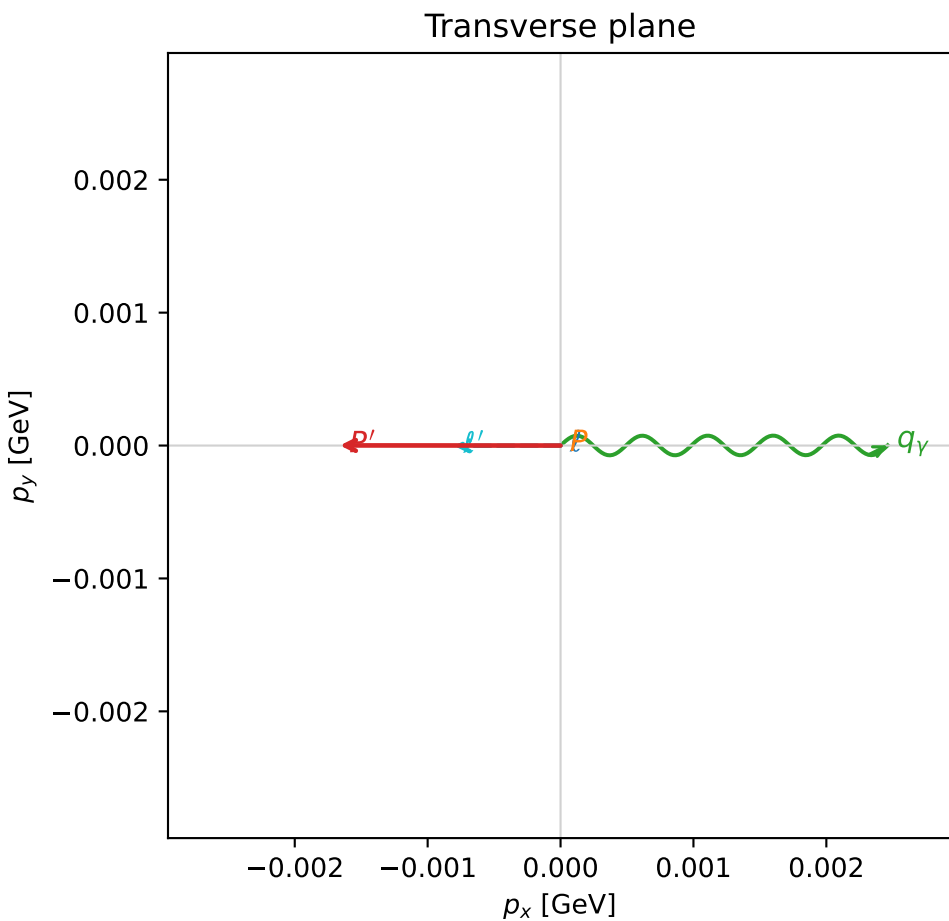
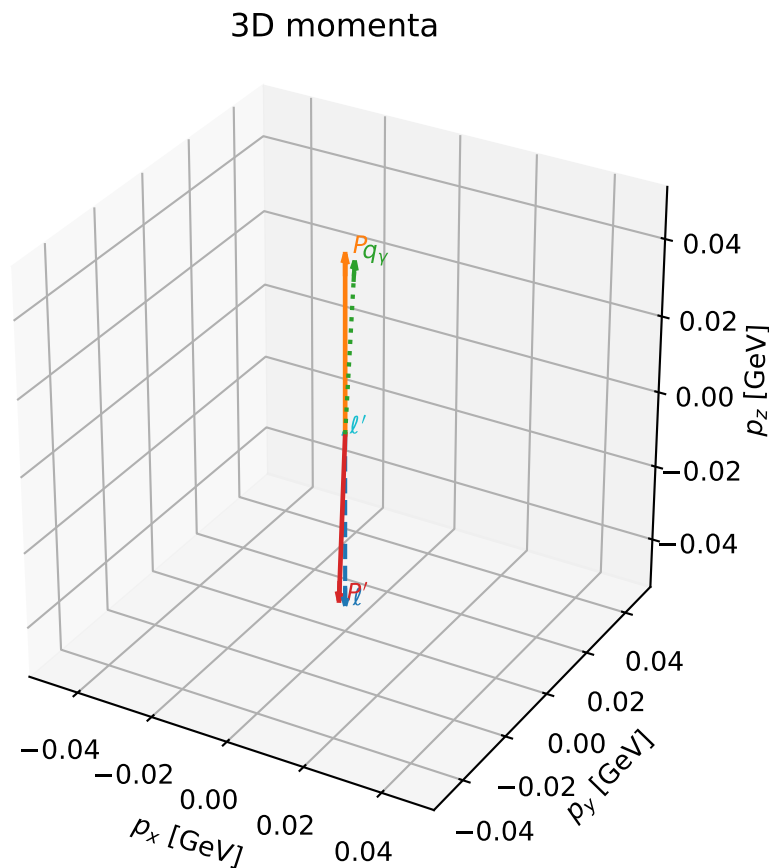
$\text{sqrt}_s = 1.405893$
 $\theta_{p_{\text{out}}} = 1.999287$
 $\theta_{\gamma_{\text{out}}} = 1.930608$
 $q_{\text{out}} = 0.2635426$
 $\phi_{p_{\text{out}}} = 3.141615$
 $\phi_{\gamma_{\text{out}}} = 6.283185$
 $\alpha_e = 1.061876$
 $\alpha_p = 0.4296907$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_351 d_{\rm GHZ}=0.000879976
region: polarization_cluster_01_local_minimum_351
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.76448436$ rad
incoming lepton: $0.721739|+\rangle +0.692165|-\rangle$
proton mixing angle: $\alpha_p=0.73414056$ rad
incoming proton: $0.742407|+\rangle +0.669949|-\rangle$

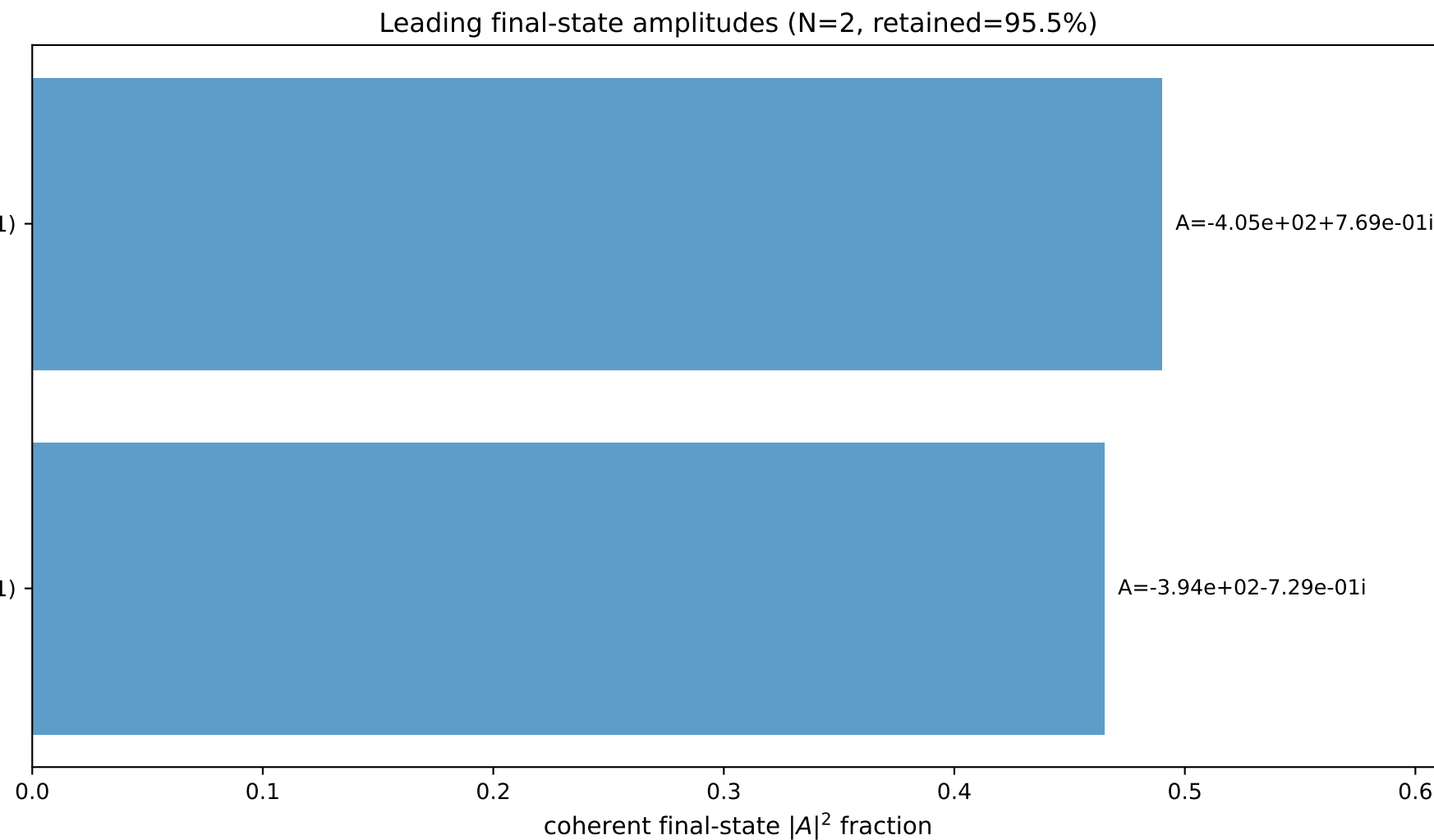
$s=0.971085$, $\sqrt{s}=0.985437$, $|\vec{P}|=0.0462921$, $|\vec{P}'|=0.0453374$
 $\theta_{p'}=3.10496$, $\theta_Y=0.0542554$, $\phi_{p'}=3.141$, $\phi_Y=0.000630486$
 $E_Y=0.045391$, $\theta_{\ell'}=1.59229$, $\phi_{\ell'}=3.14476$
 $Q^2=8.58888\text{e-}05$, $x_B=0.000960145$, $t=-0.00839316$, $W^2=0.969212$, $y=0.980416$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0463$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0463$
 P $E= 0.9391$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0463$
 ℓ' $E= 0.0010$ $p_x= -0.0008$ $p_y= -0.0000$ $p_z= -0.0000$
 P' $E= 0.9391$ $p_x= -0.0017$ $p_y= 0.0000$ $p_z= -0.0453$
 q_Y $E= 0.0454$ $p_x= 0.0025$ $p_y= 0.0000$ $p_z= 0.0453$

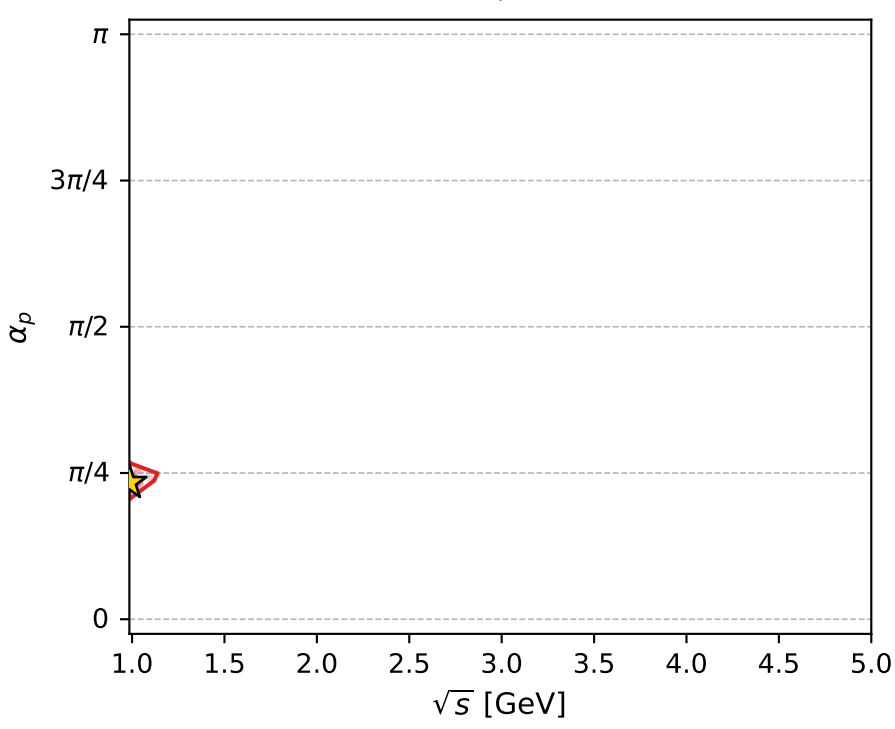
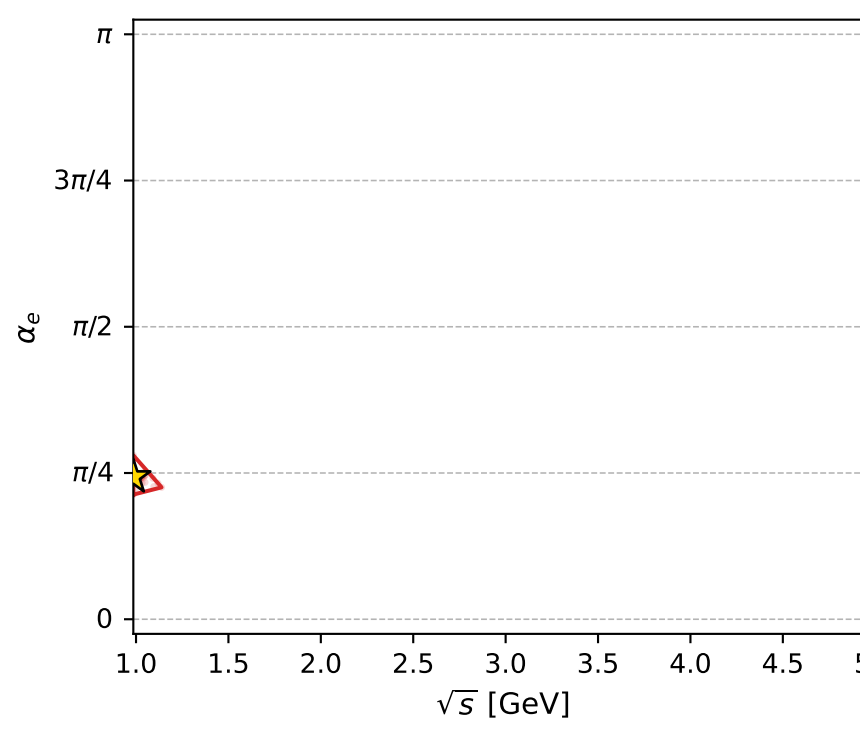
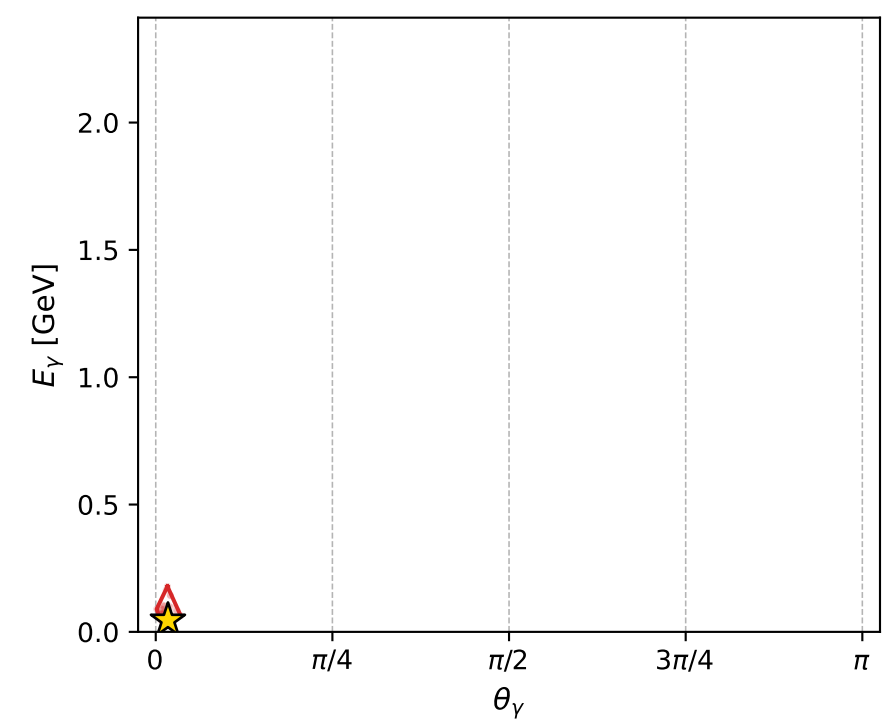
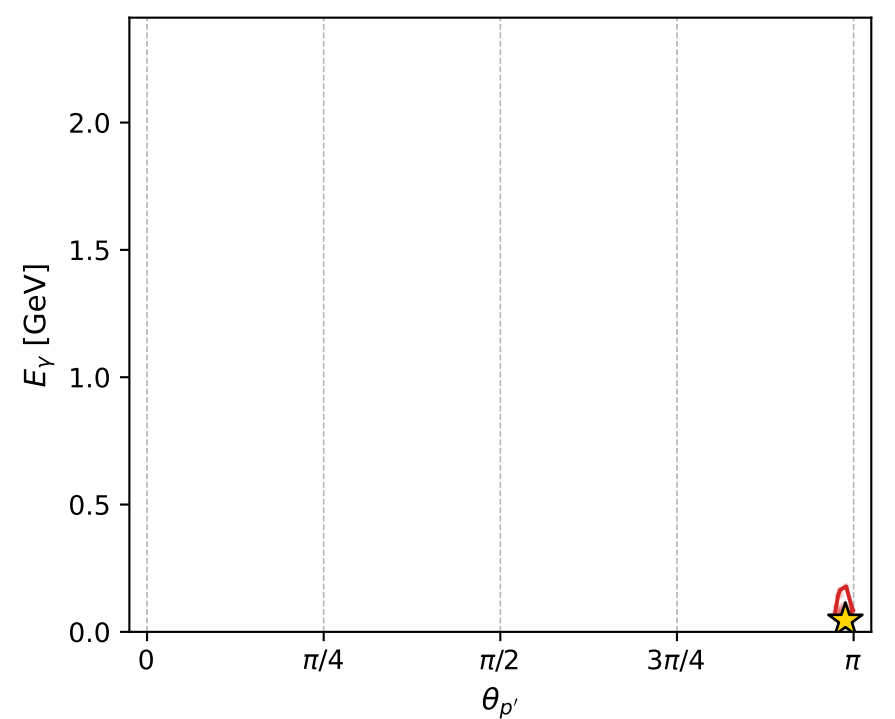
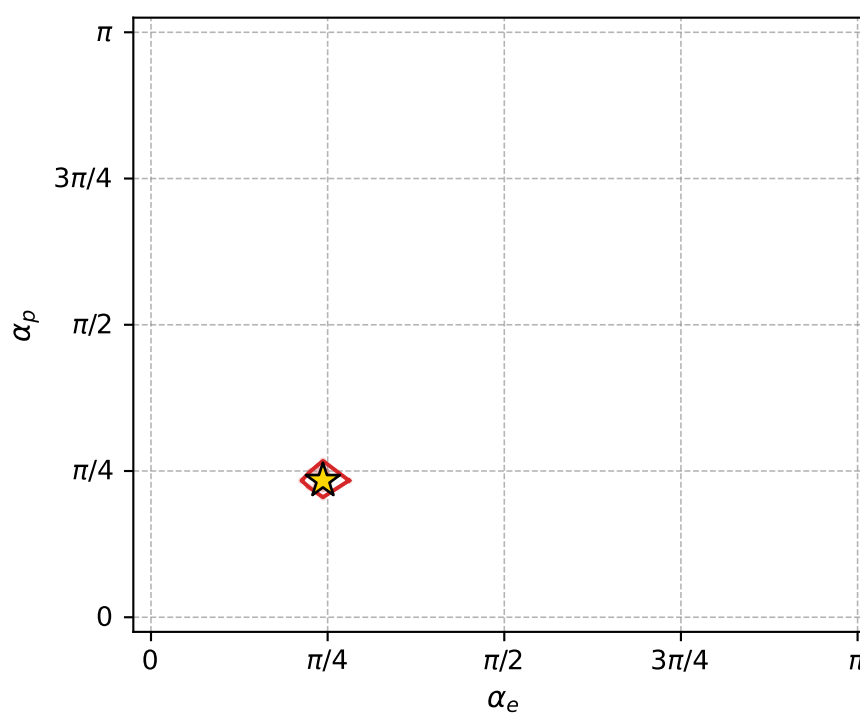
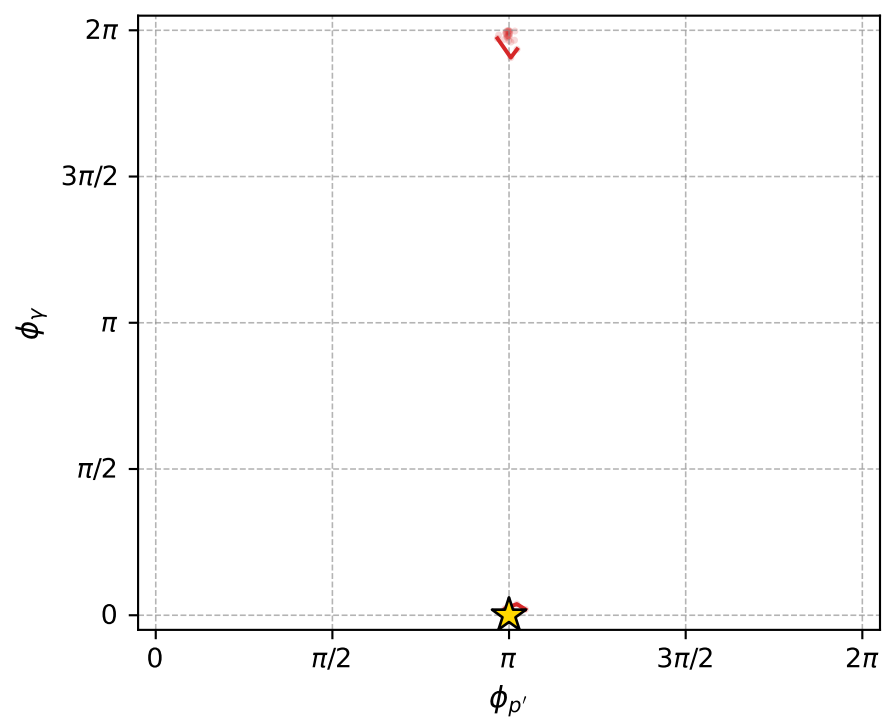
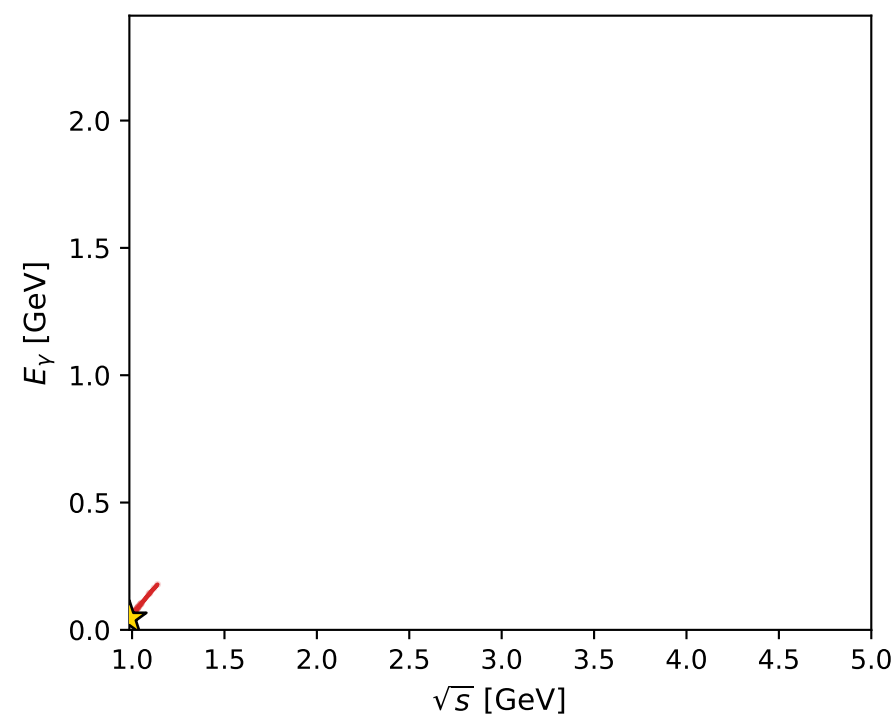
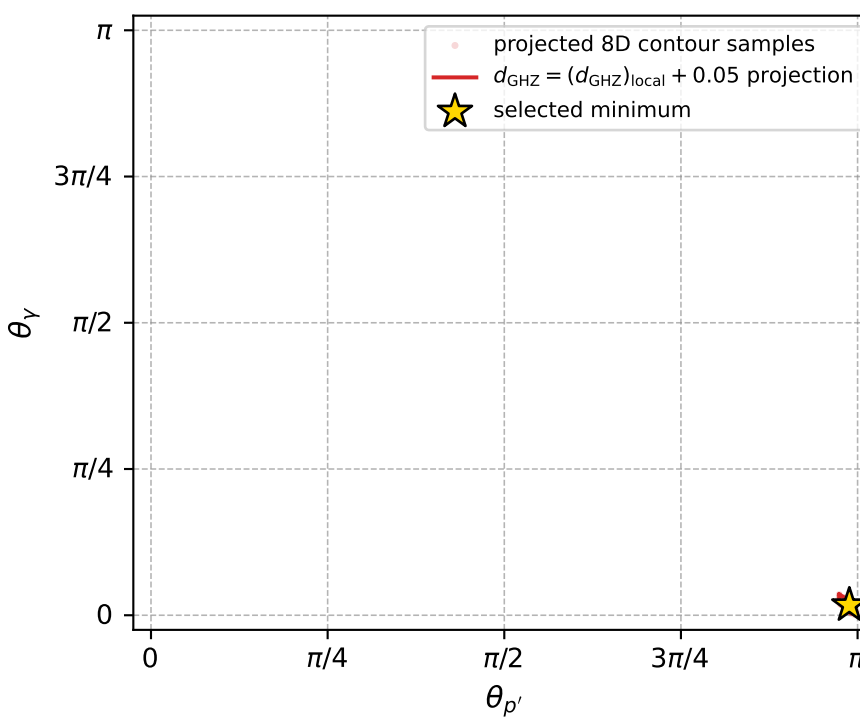


$(h_p, h_Y, h_\ell) = (+1, +1, +1)$

$(h_p, h_Y, h_\ell) = (-1, -1, -1)$



electron: polarization cluster P1, local minimum 351; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.00087997604$
 $d_{\text{GHZ}} \text{ contour} = 0.050879976$
 above global minimum = 0.00087995376
 8D contour samples = 118

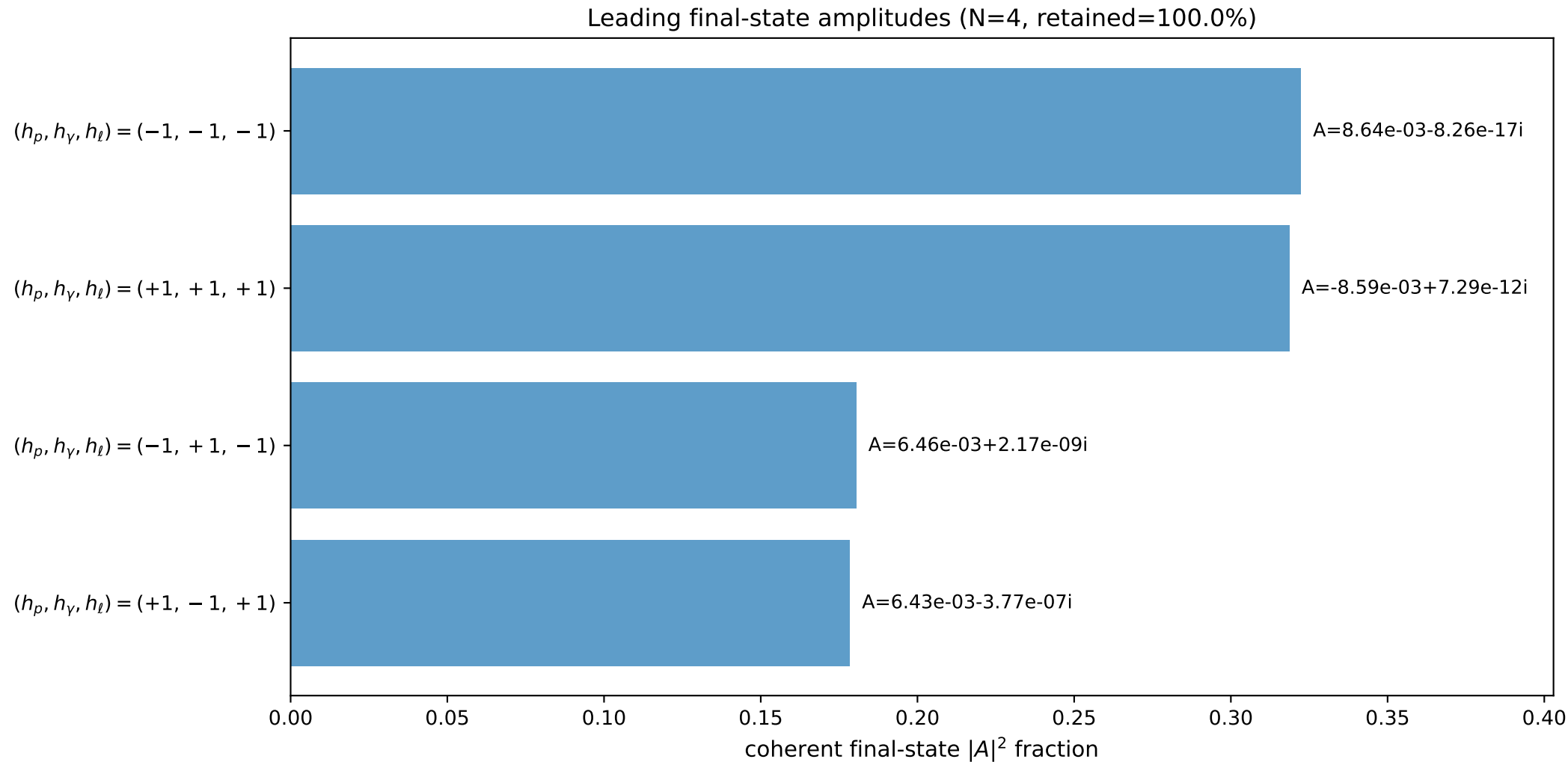
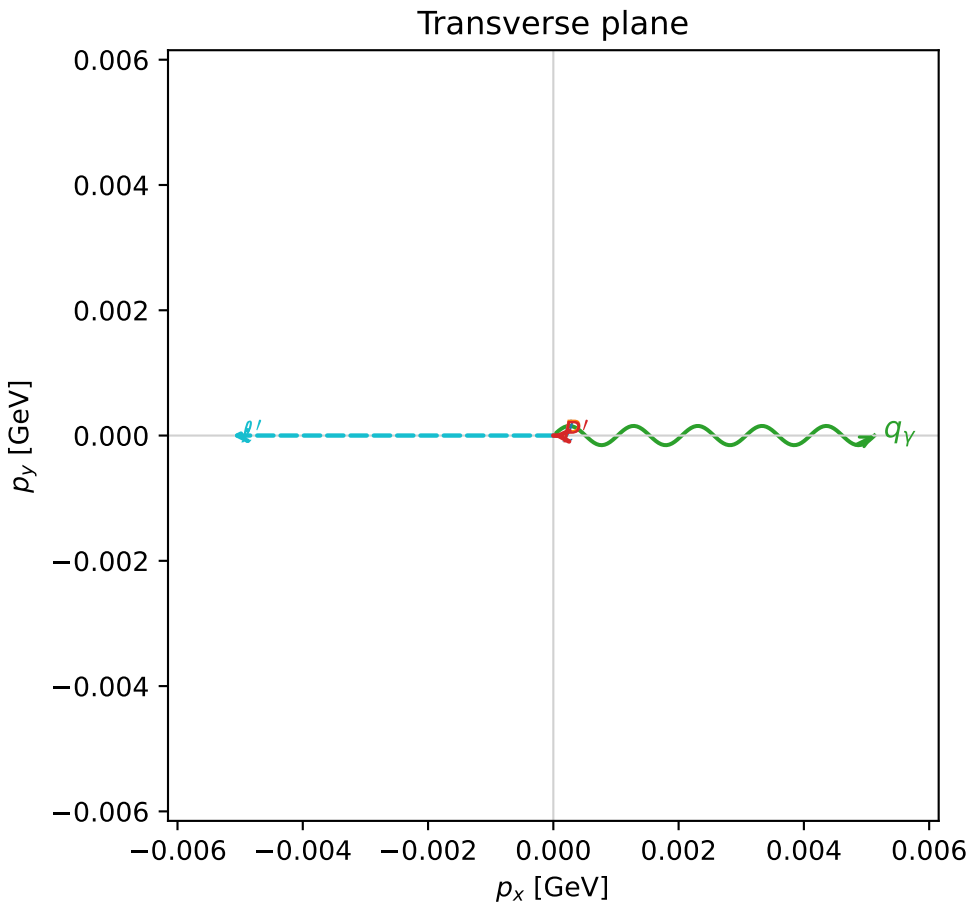
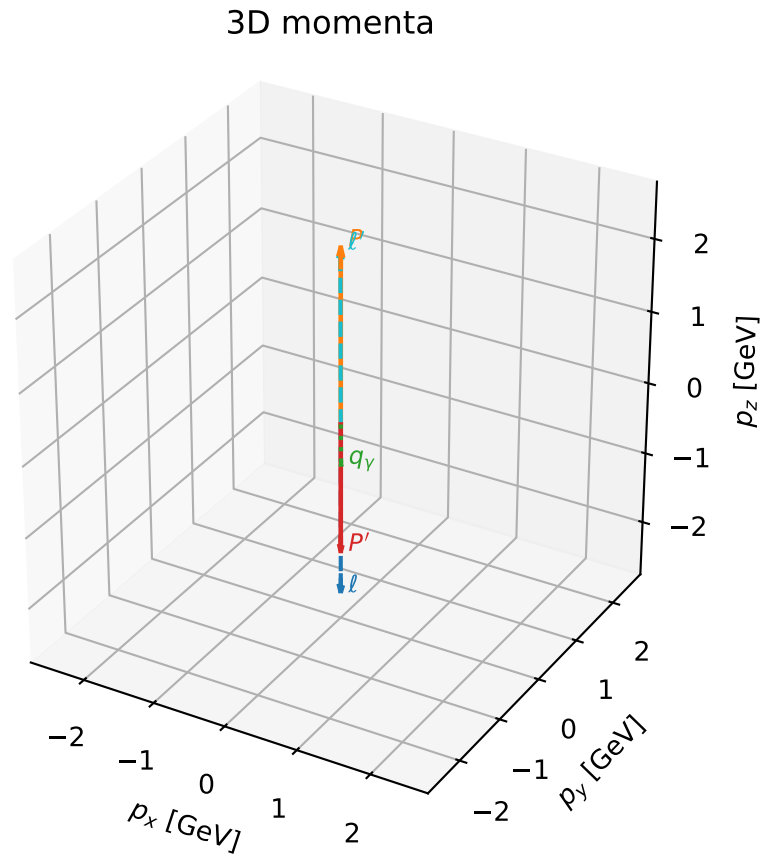
selected phase-space point:
 $\text{sqrt_s} = 0.9854365$
 $\text{theta_p_out} = 3.104965$
 $\text{theta_gamma_out} = 0.05425539$
 $\text{q0out} = 0.04539103$
 $\text{phi_p_out} = 3.141$
 $\text{phi_gamma_out} = 0.0006304865$
 $\text{alpha_e} = 0.7644844$
 $\text{alpha_p} = 0.7341406$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

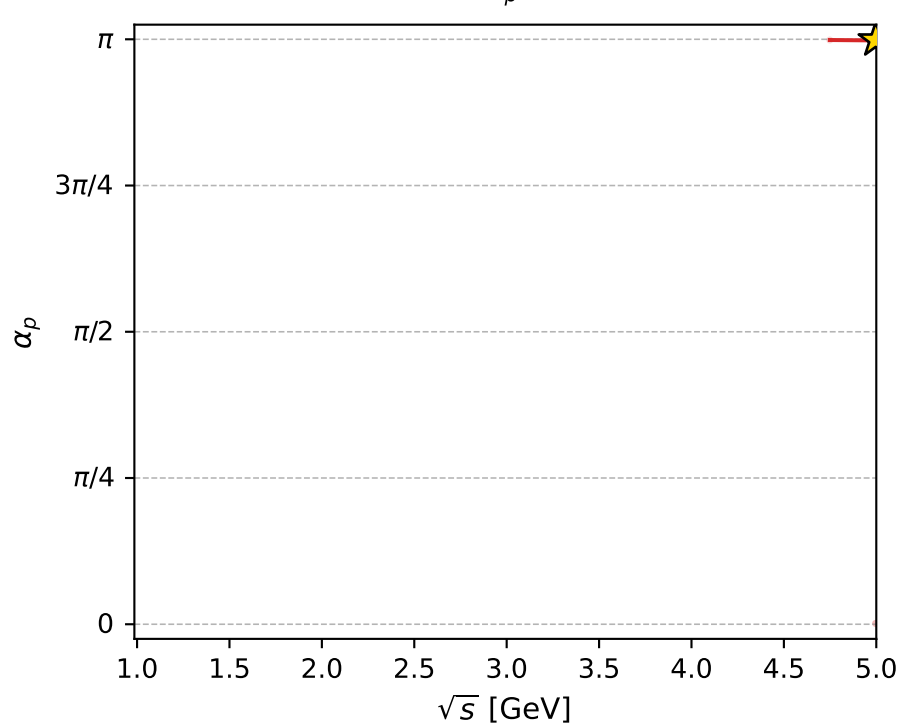
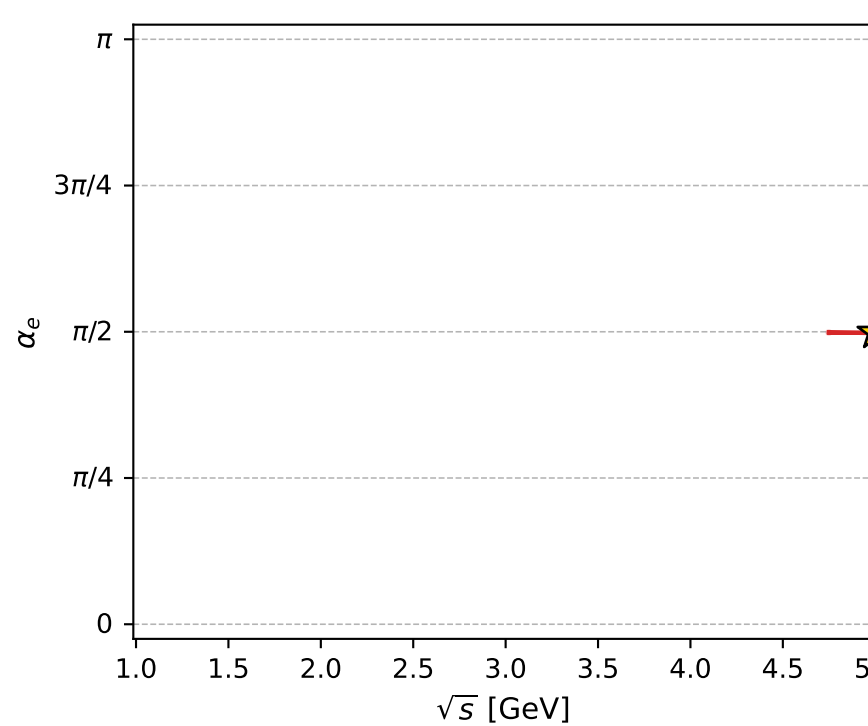
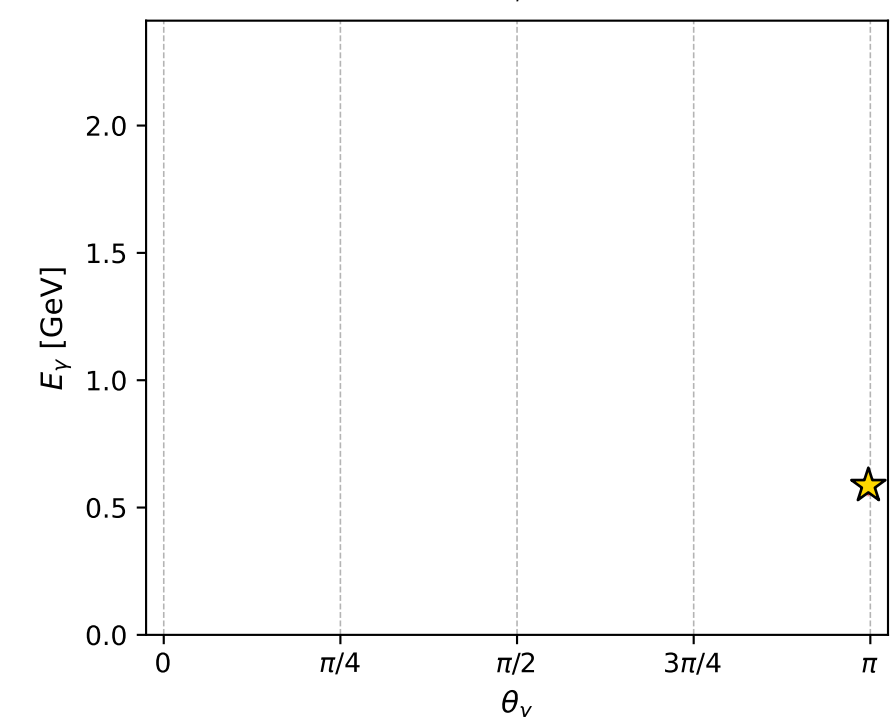
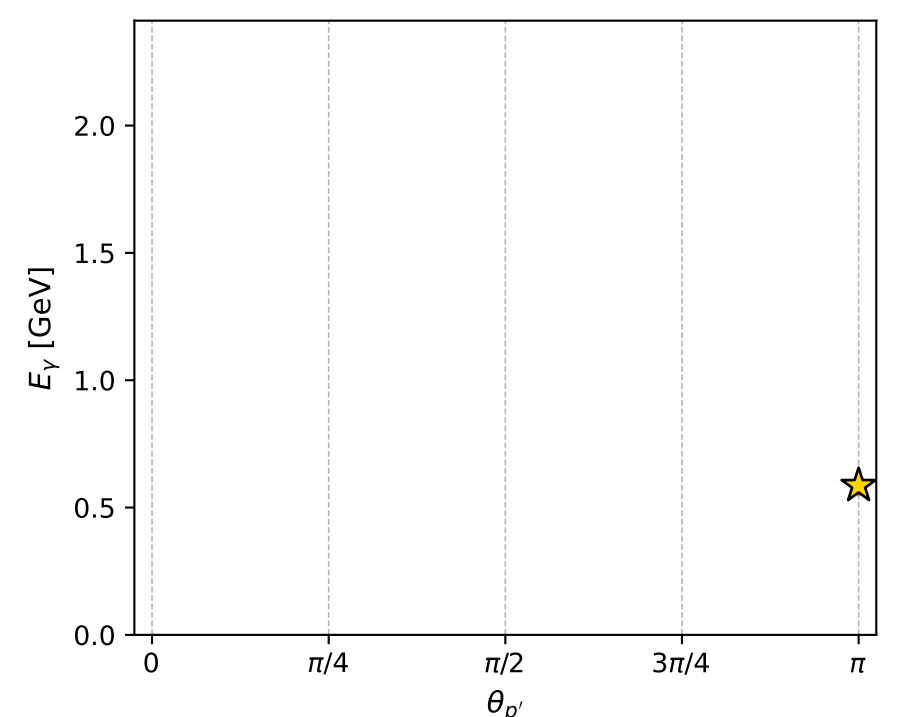
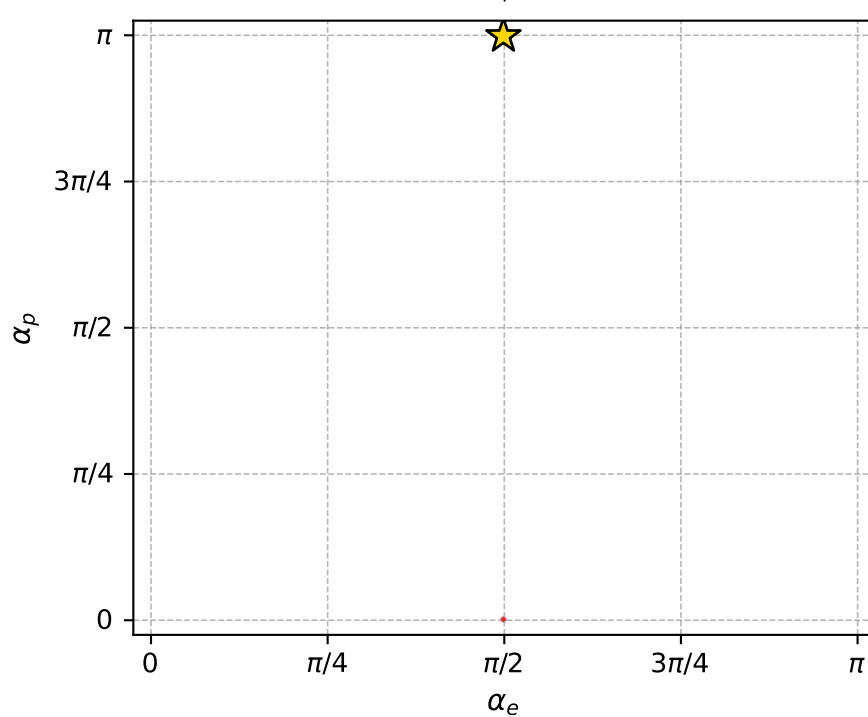
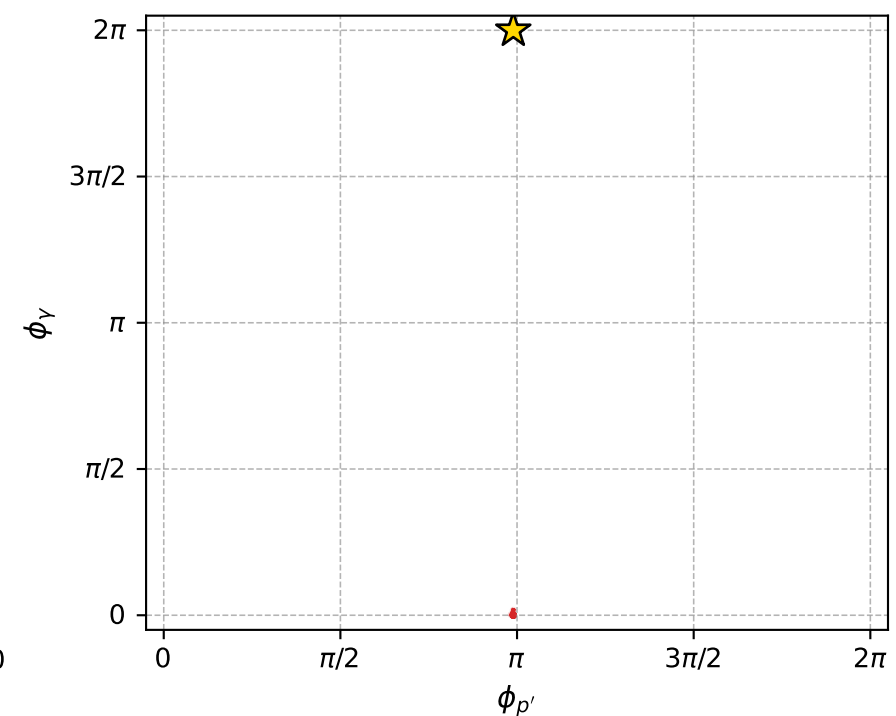
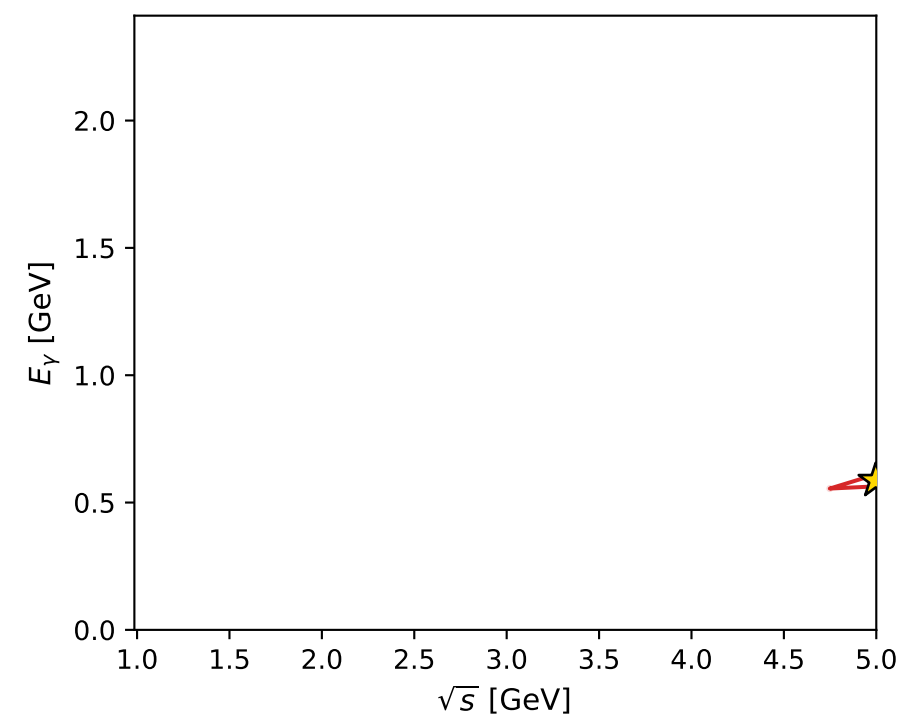
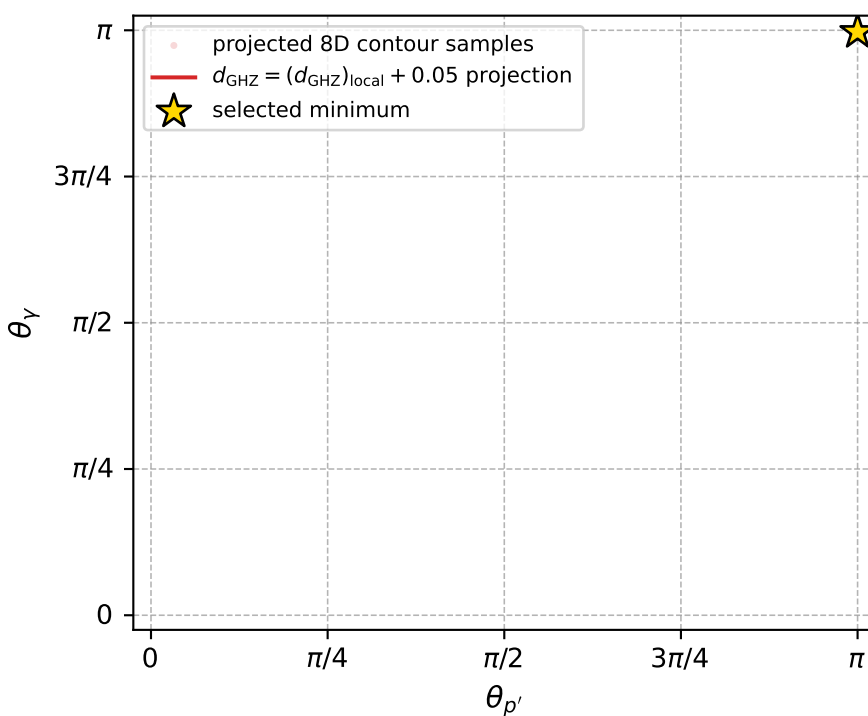
dghz_mixing_angles_polarization_01_local_minimum_353 $d_{\{\text{rm GHZ}\}}=0.000914402$
region: polarization_cluster_01_local_minimum_353
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.5668793$ rad
incoming lepton: $0.00391706|+\rangle +0.999992|-\rangle$
proton mixing angle: $\alpha_p=3.1376532$ rad
incoming proton: $-0.999992|+\rangle +0.00393944|-\rangle$

$s=24.9695$, $\sqrt{s}=4.99694$, $|\vec{P}|=2.41043$, $|\vec{P}'|=1.79723$
 $\theta_{p'}=3.14159$, $\theta_V=3.13285$, $\phi_{p'}=3.10822$, $\phi_V=6.28316$
 $E_V=0.586223$, $\theta_{l'}=0.00215086$, $\phi_{l'}=3.14156$
 $Q^2=22.9804$, $x_B=0.988396$, $t=-17.3917$, $W^2=1.14965$, $y=0.965157$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4104$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4104$
 P $E= 2.5865$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4104$
 l' $E= 2.3834$ $p_x= -0.0051$ $p_y= 0.0000$ $p_z= 2.3834$
 P' $E= 2.0273$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.7972$
 q_V $E= 0.5862$ $p_x= 0.0051$ $p_y= -0.0000$ $p_z= -0.5862$



electron: polarization cluster P1, local minimum 353; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0009144024$
 $d_{\text{GHZ}} \text{ contour} = 0.050914402$
 above global minimum = 0.00091438012
 8D contour samples = 142

selected phase-space point:

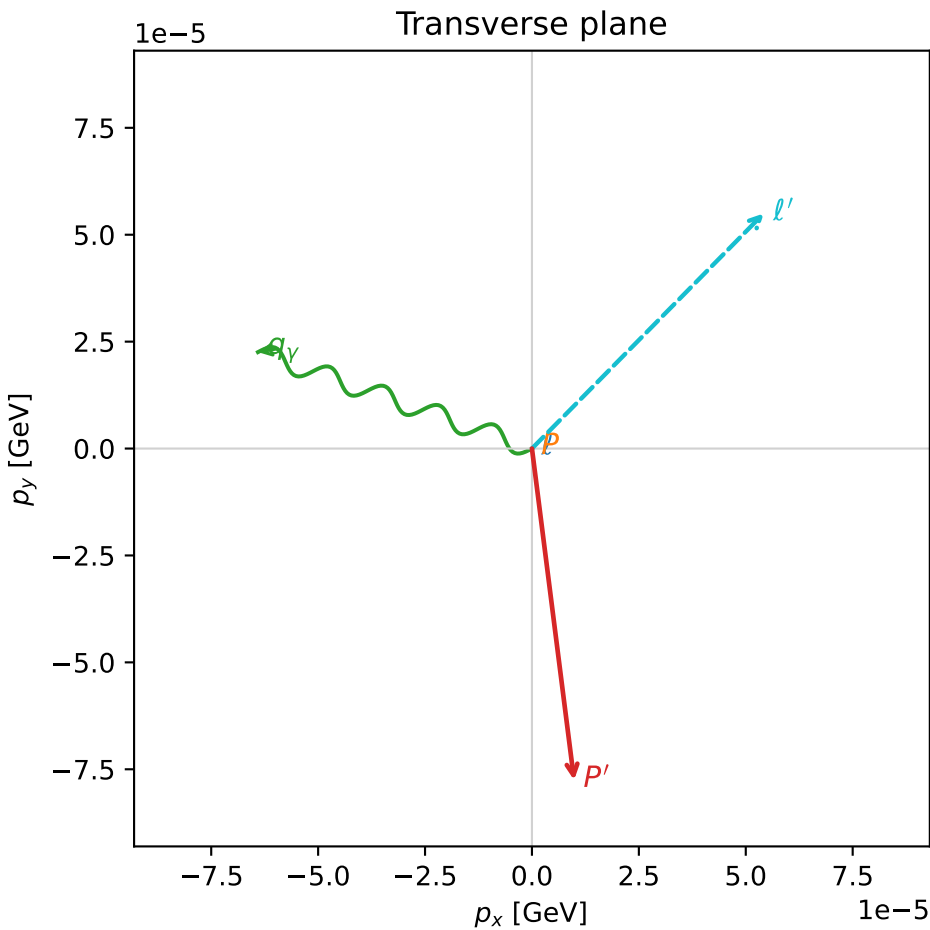
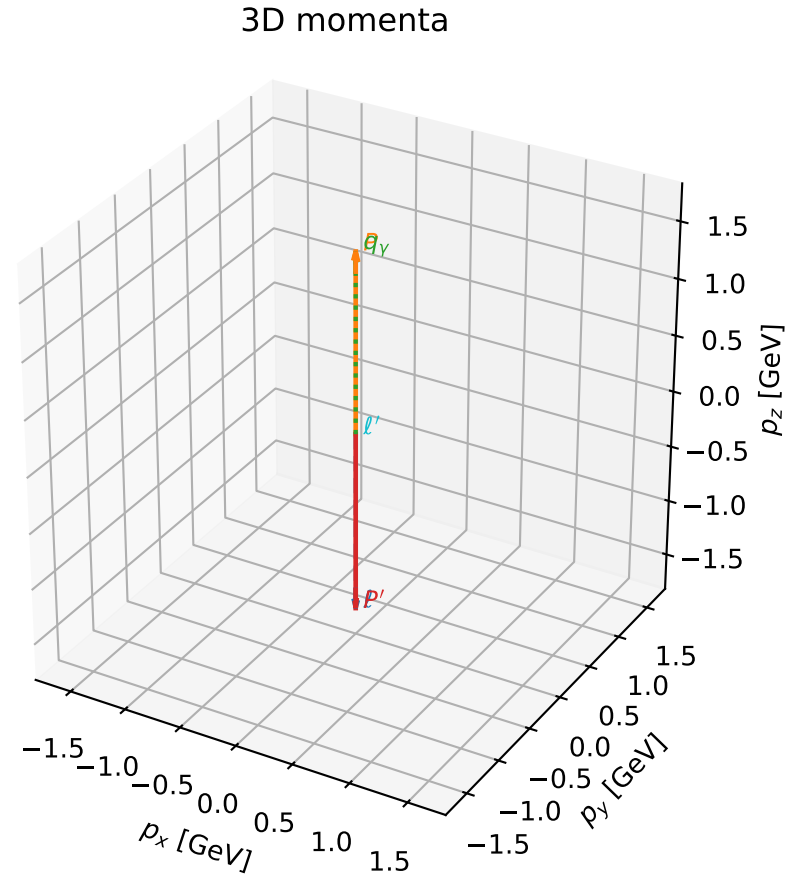
$\text{sqrt_s} = 4.996945$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.132848$
 $\text{q0out} = 0.586223$
 $\text{phi_p_out} = 3.108216$
 $\text{phi_gamma_out} = 6.283156$
 $\text{alpha_e} = 1.566879$
 $\text{alpha_p} = 3.137653$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_356 $d_{\{\text{rm GHZ}\}}=0.0011903$
 region: polarization_cluster_01_local_minimum_356
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78547747$ rad
 incoming lepton: $0.707051|+\rangle +0.707163|-\rangle$
 proton mixing angle: $\alpha_p=0.78569818$ rad
 incoming proton: $0.706895|+\rangle +0.707319|-\rangle$

$s=11.7704$, $\sqrt{s}=3.4308$, $|\vec{P}|=1.58717$, $|\vec{P}'|=1.58616$
 $\theta_{P'}=3.14154$, $\theta_Y=4.27948\text{e-}05$, $\phi_{P'}=4.8388$, $\phi_Y=2.80317$
 $E_Y=1.58703$, $\theta_{l'}=3.05347$, $\phi_{l'}=0.792465$
 $Q^2=0.000448243$, $x_B=4.11835\text{e-}05$, $t=-10.07$, $W^2=11.7634$, $y=0.999402$

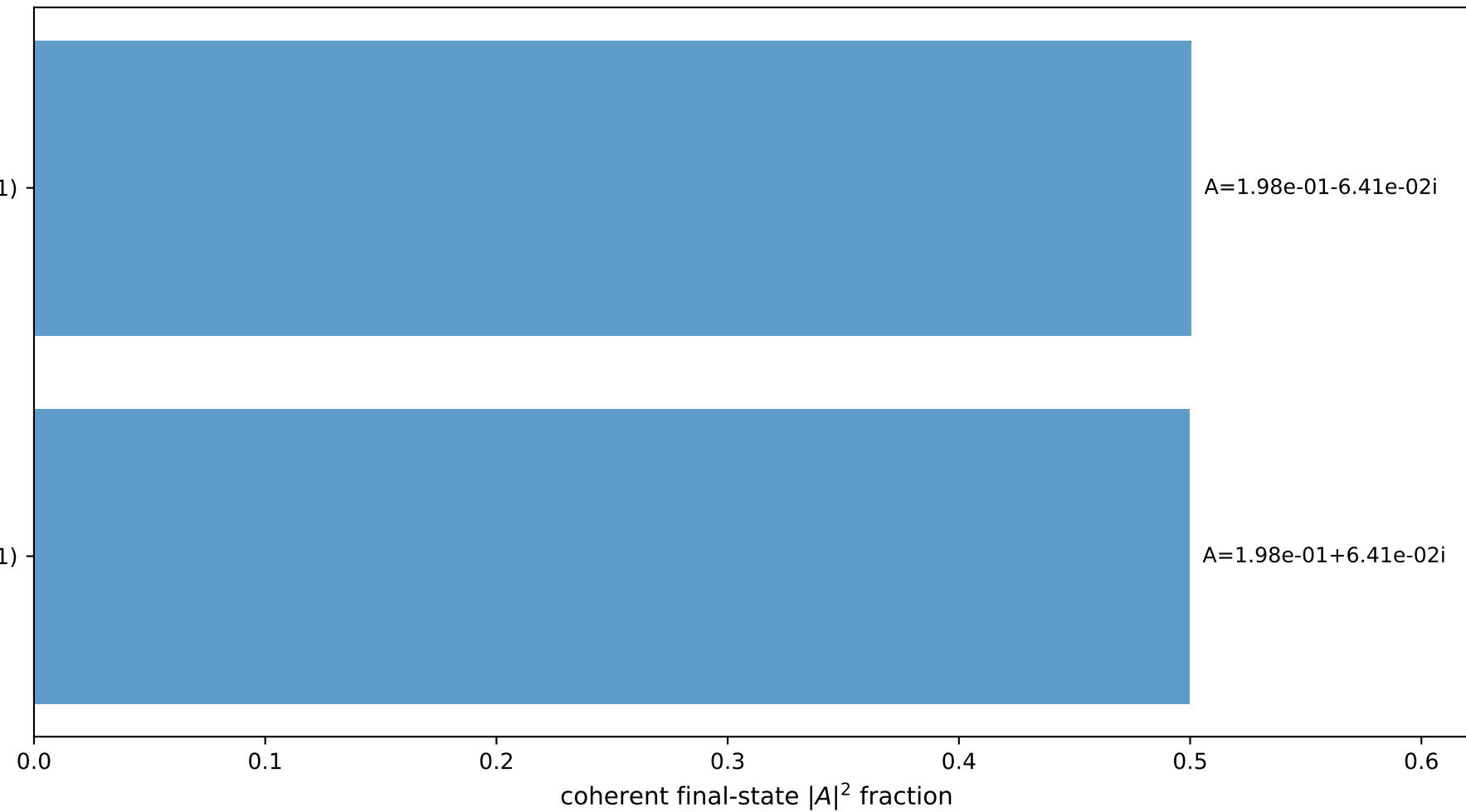
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.5872$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5872$
 P $E= 1.8436$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5872$
 l' $E= 0.0010$ $p_x= 0.0001$ $p_y= 0.0001$ $p_z= -0.0009$
 P' $E= 1.8428$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= -1.5862$
 q_Y $E= 1.5870$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 1.5870$



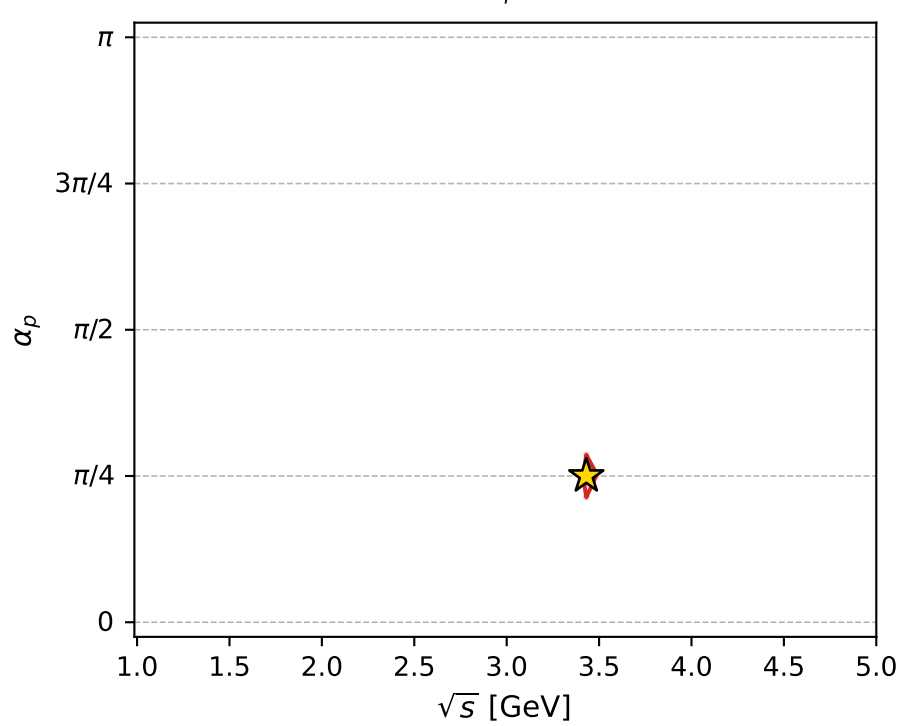
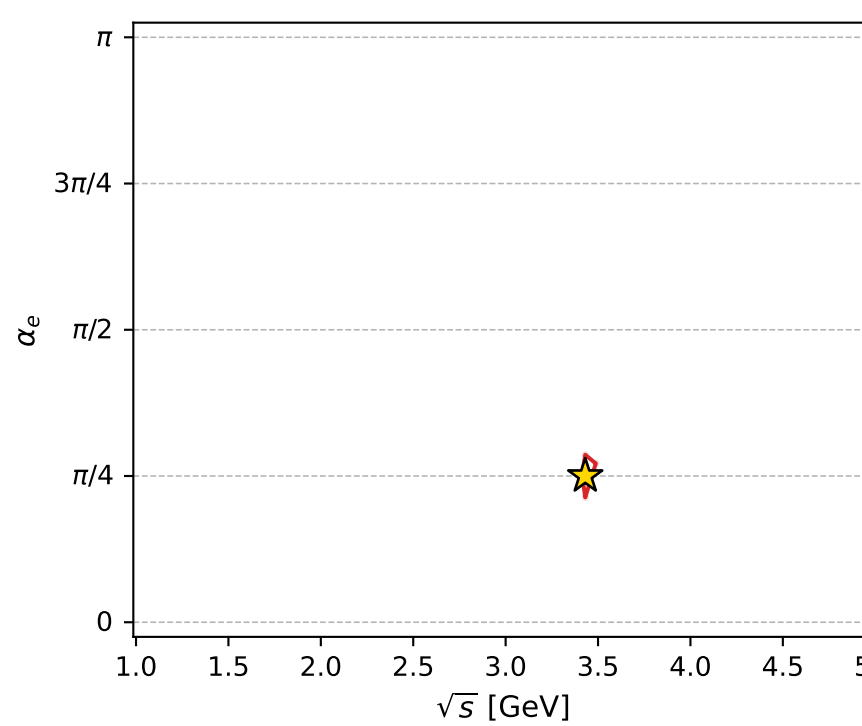
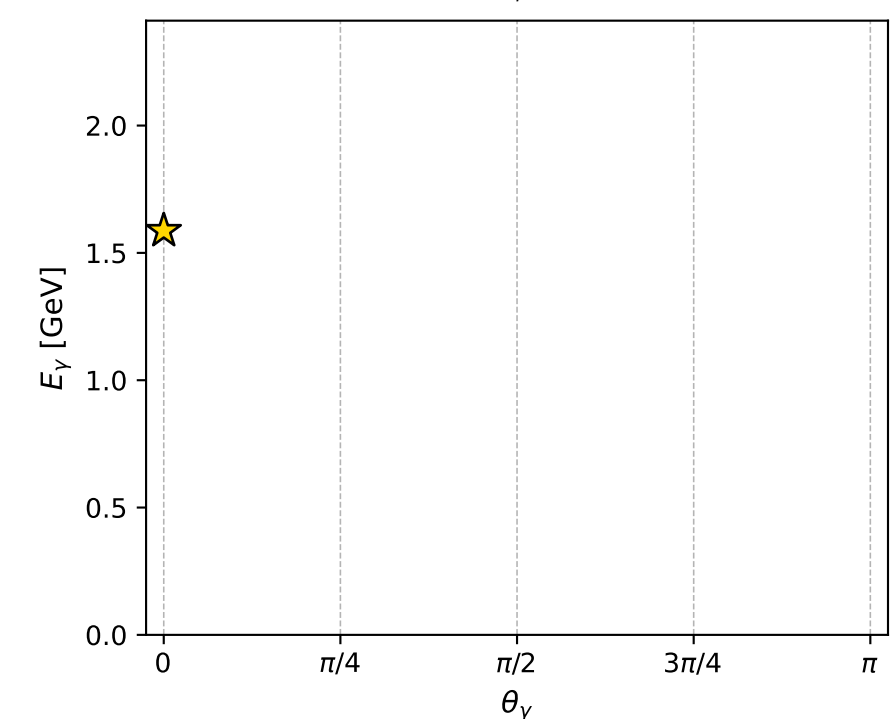
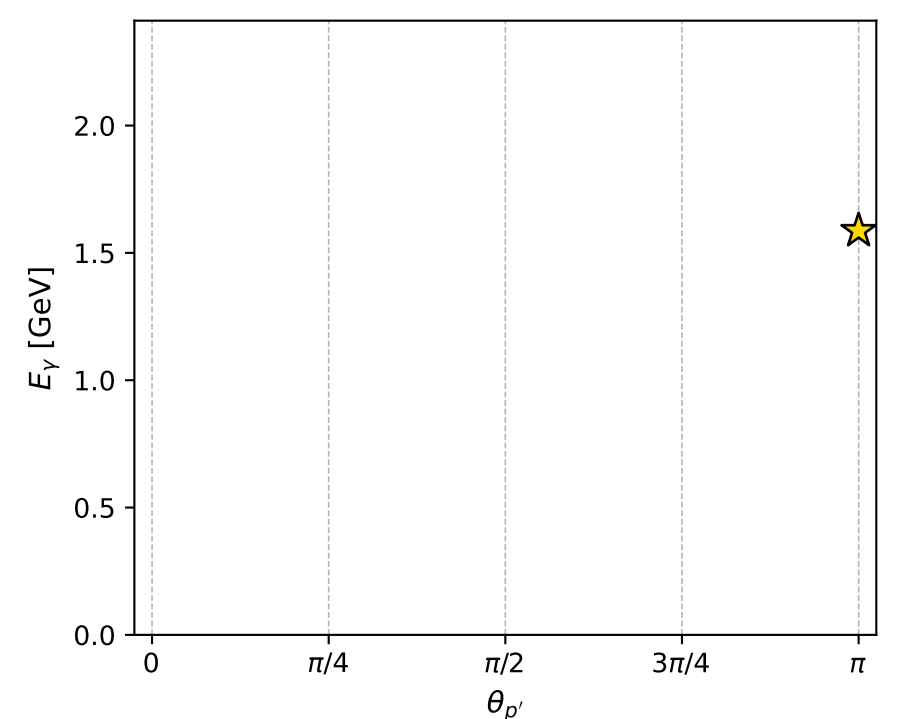
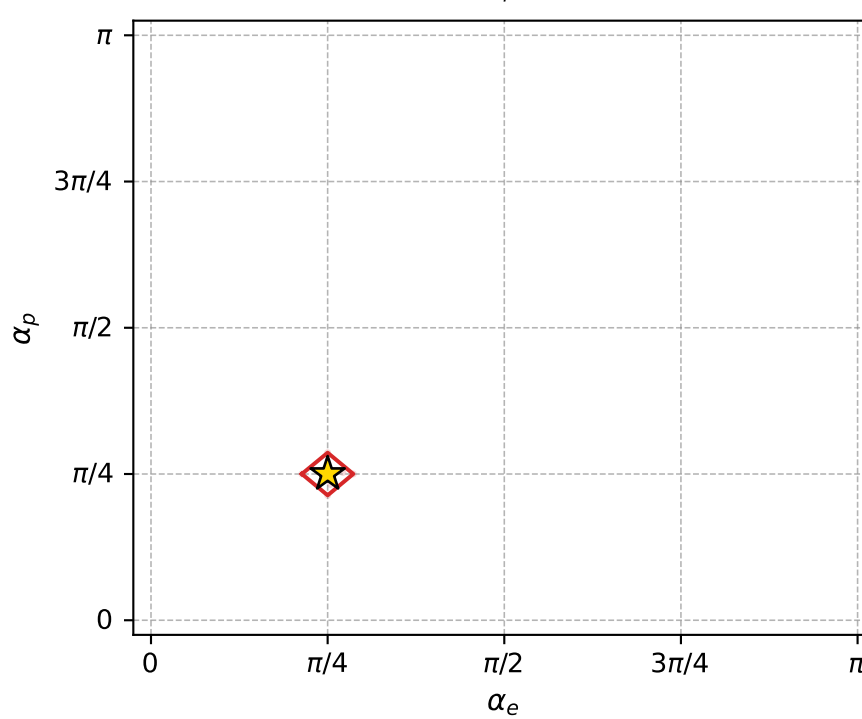
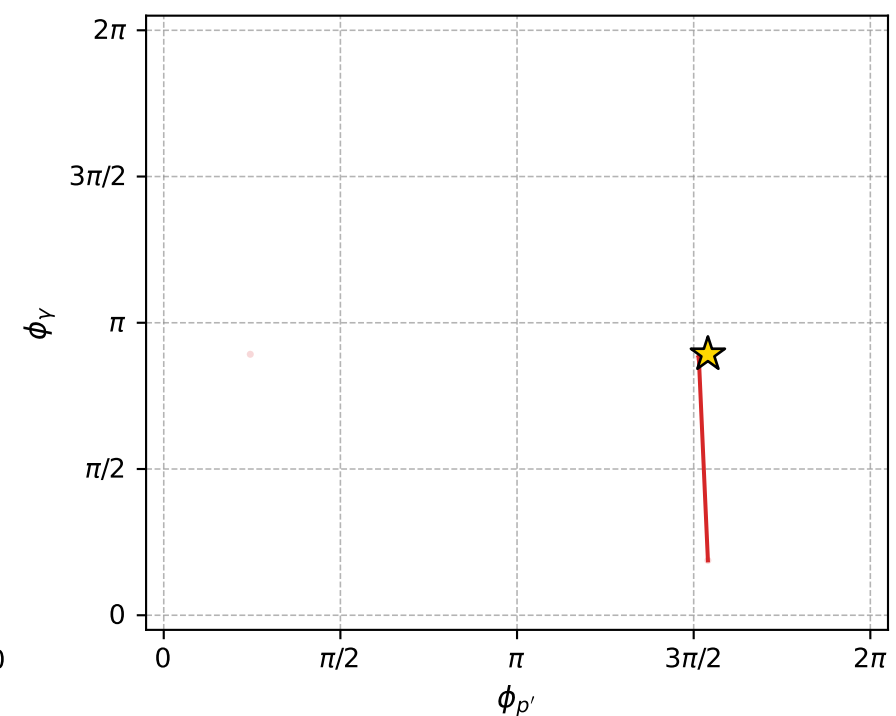
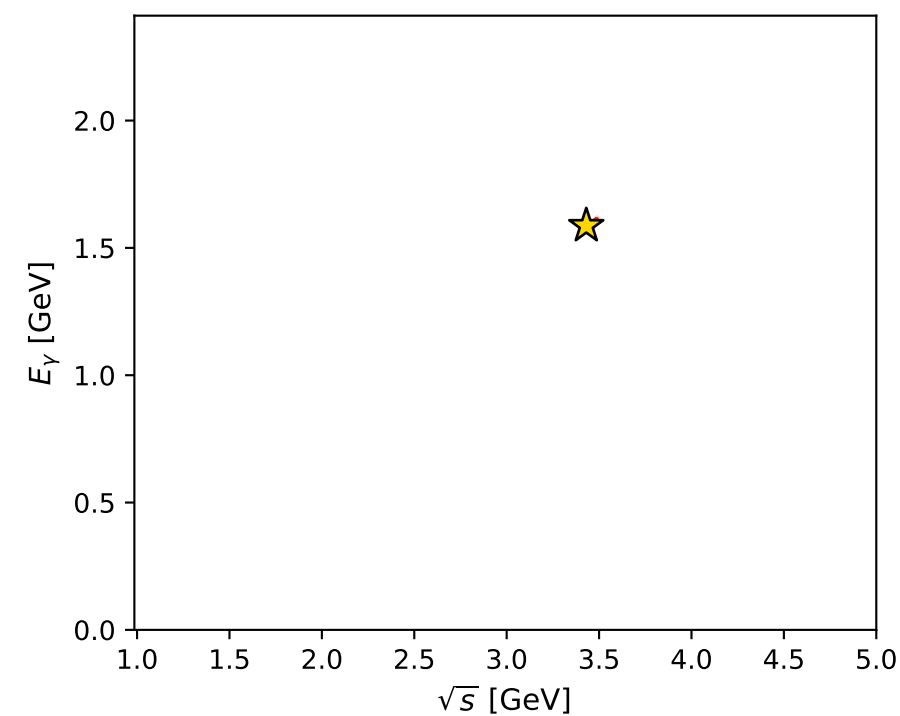
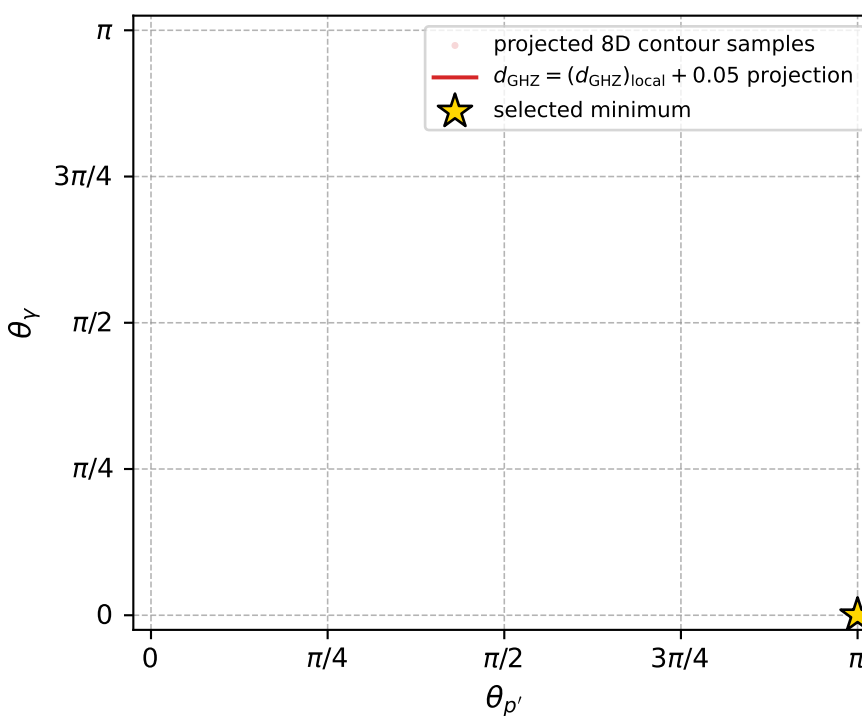
$(h_p, h_Y, h_l) = (-1, -1, -1)$

$(h_p, h_Y, h_l) = (+1, +1, +1)$

Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 356; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0011902974$
 $d_{\text{GHZ}} \text{ contour} = 0.051190297$
 above global minimum = 0.0011902751
 8D contour samples = 46

selected phase-space point:

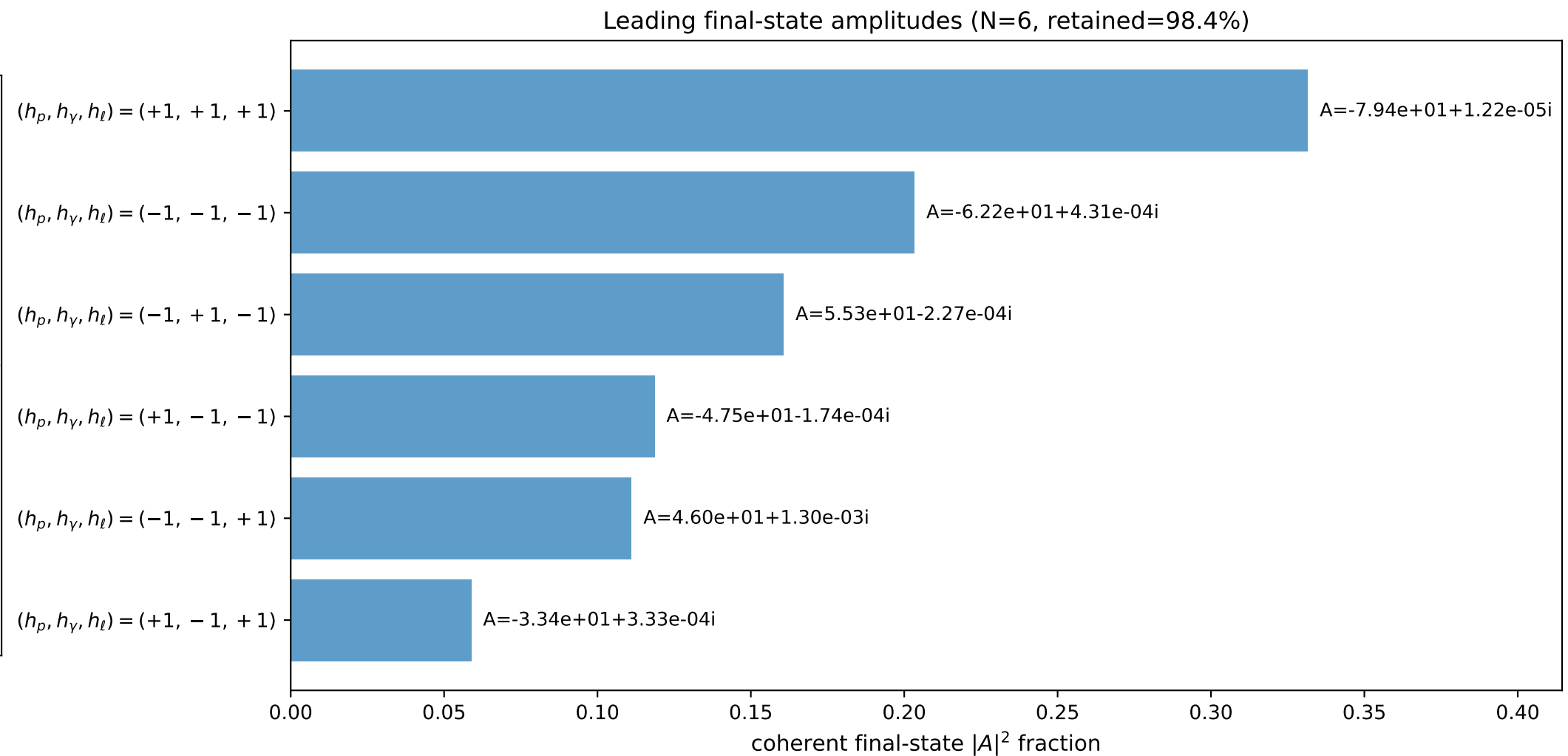
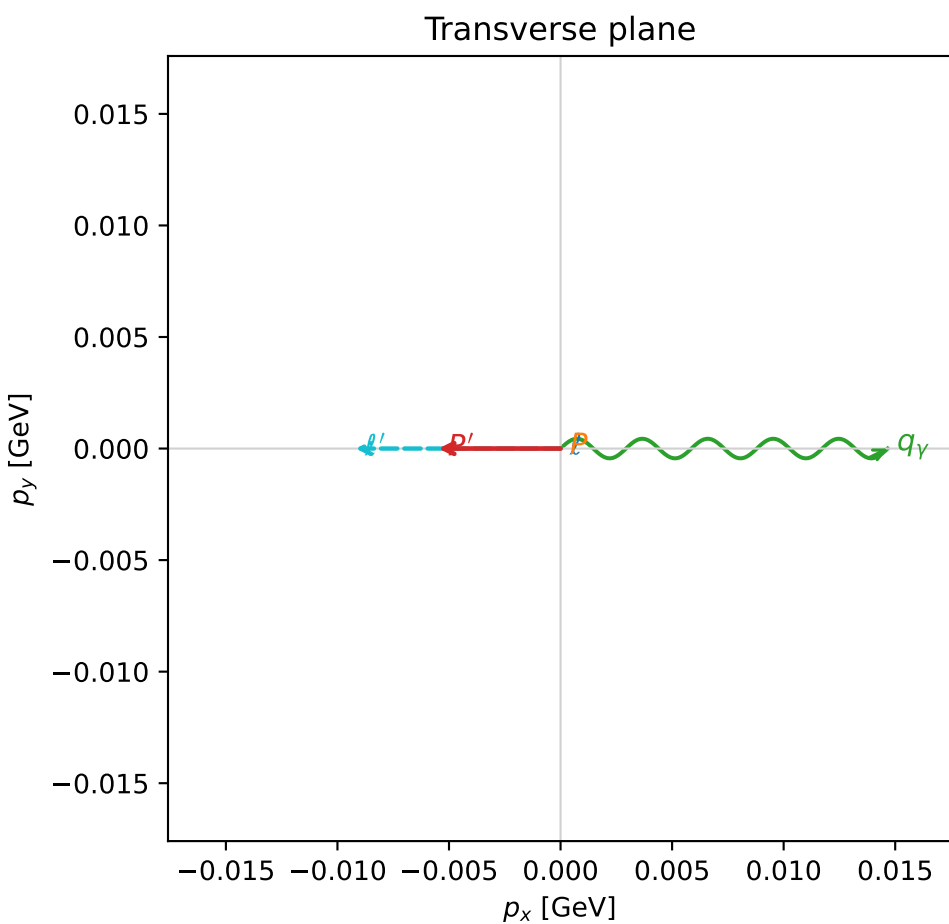
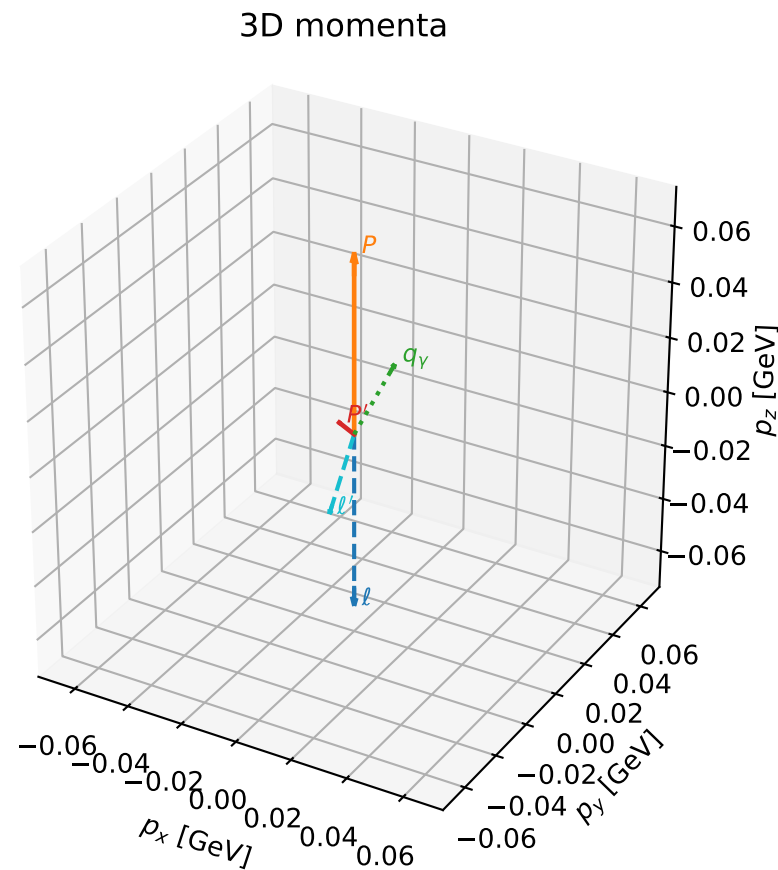
$\text{sqrt_s} = 3.430802$
 $\text{theta_p_out} = 3.141543$
 $\text{theta_gamma_out} = 4.27948\text{e-}05$
 $\text{q0ut} = 1.587032$
 $\text{phi_p_out} = 4.838798$
 $\text{phi_gamma_out} = 2.803167$
 $\text{alpha_e} = 0.7854775$
 $\text{alpha_p} = 0.7856982$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

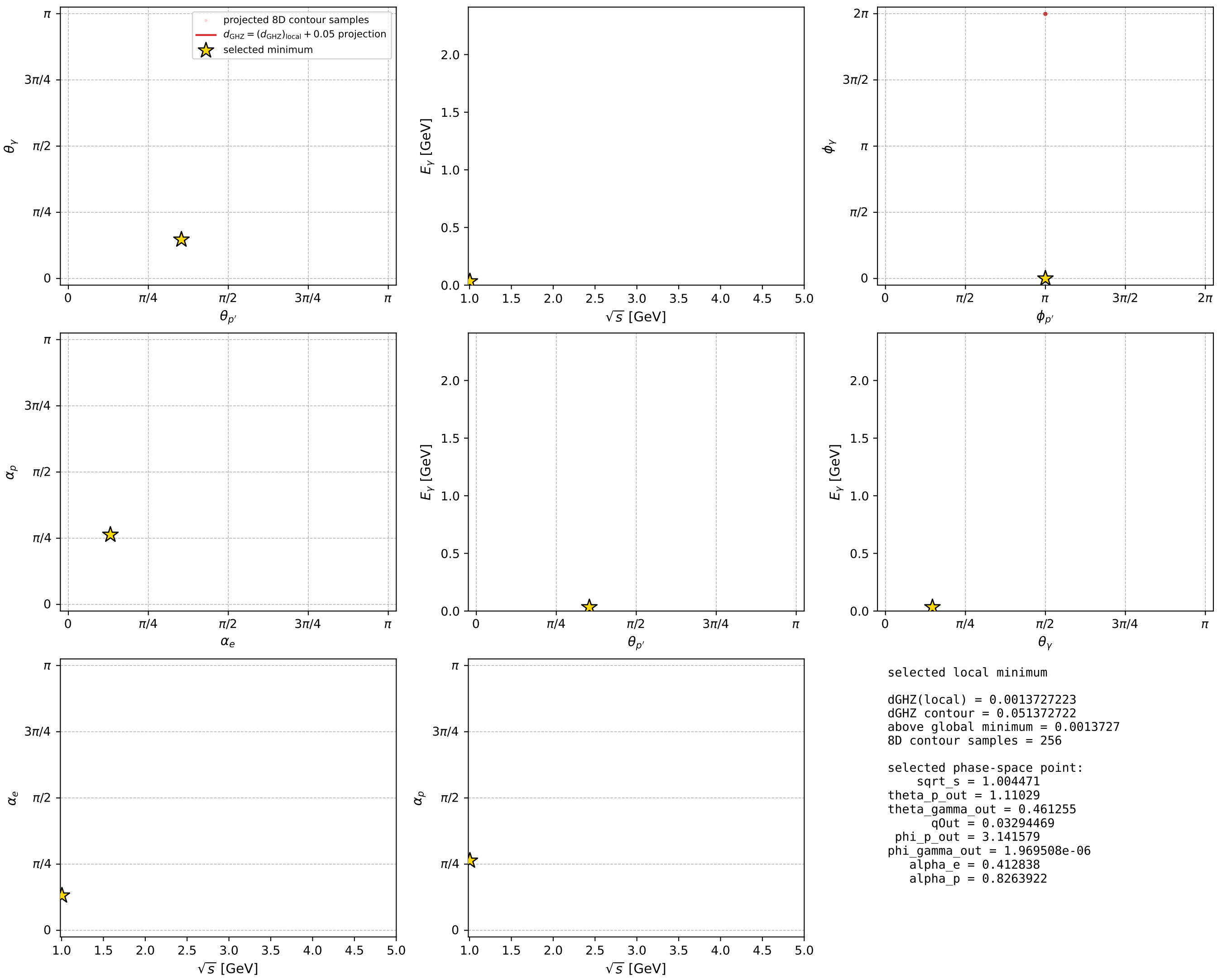
dghz_mixing_angles_polarization_01_local_minimum_358 $d_{\text{GHZ}}=0.00137272$
 region: polarization_cluster_01_local_minimum_358
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.41283802$ rad
 incoming lepton: $0.915986|+\rangle +0.401211|-\rangle$
 proton mixing angle: $\alpha_p=0.82639218$ rad
 incoming proton: $0.677534|+\rangle +0.735492|-\rangle$

$s=1.00896$, $\sqrt{s}=1.00447$, $|\vec{P}|=0.0642693$, $|\vec{P}'|=0.00611045$
 $\theta_{p'}=1.11029$, $\theta_Y=0.461255$, $\phi_{p'}=3.14158$, $\phi_Y=1.96951e-06$
 $E_Y=0.0329447$, $\theta_{\ell'}=2.86375$, $\phi_{\ell'}=3.1416$
 $Q^2=0.000165259$, $x_B=0.00266671$, $t=-0.00381409$, $W^2=0.94165$, $y=0.479961$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.0643$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.0643$
 P $E=0.9402$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.0643$
 ℓ' $E=0.0335$ $p_x=-0.0092$ $p_y=-0.0000$ $p_z=-0.0322$
 P' $E=0.9380$ $p_x=-0.0055$ $p_y=0.0000$ $p_z=0.0027$
 q_Y $E=0.0329$ $p_x=0.0147$ $p_y=0.0000$ $p_z=0.0295$



electron: polarization cluster P1, local minimum 358; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

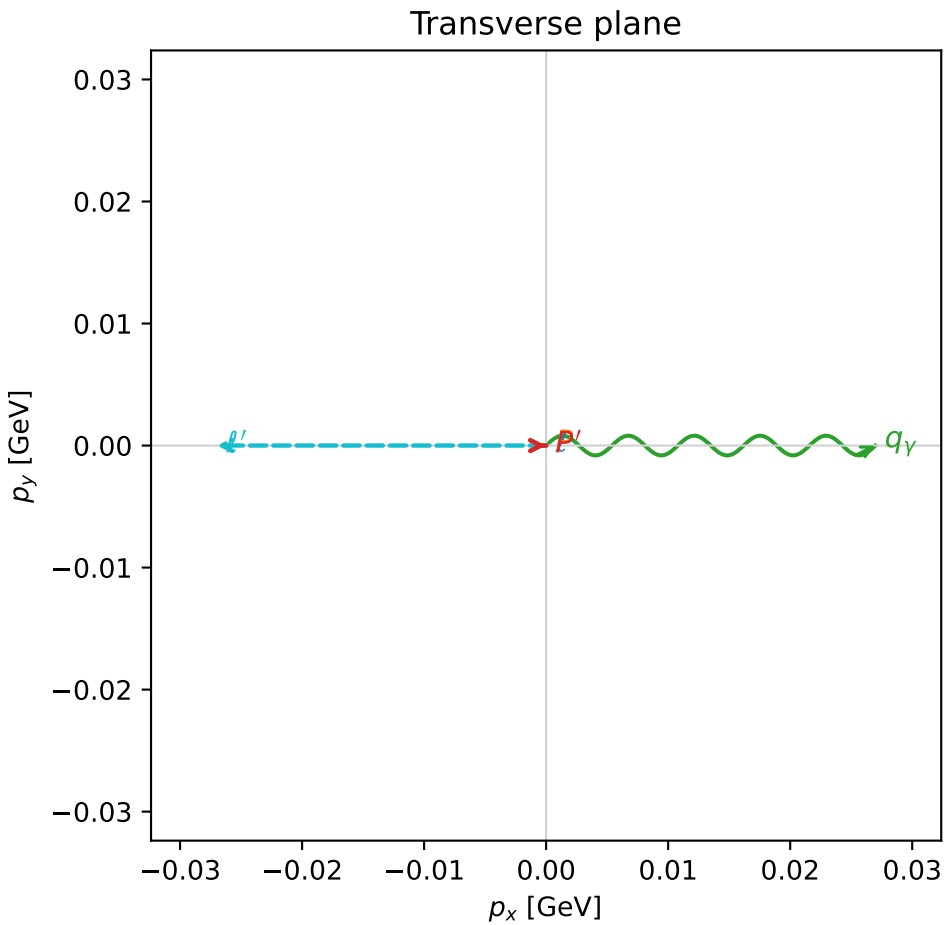
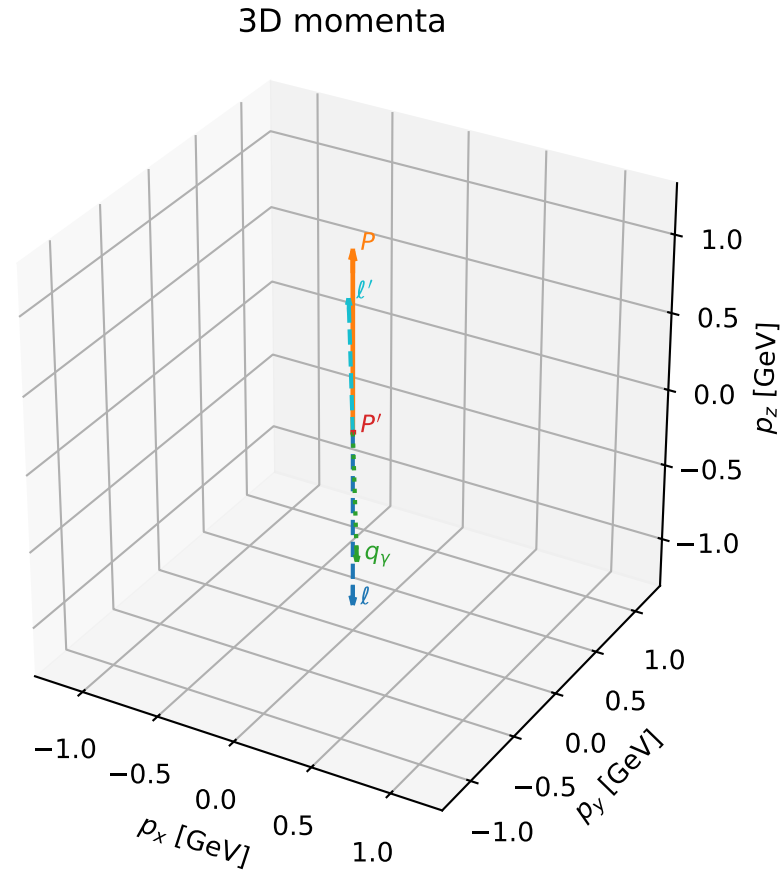


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_359 d_{\rm GHZ}=0.00157373
region: polarization_cluster_01_local_minimum_359
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.5434269$ rad
incoming lepton: $0.027366|+\rangle +0.999625|-\rangle$
proton mixing angle: $\alpha_p=0.026716948$ rad
incoming proton: $0.999643|+\rangle +0.0267138|-\rangle$

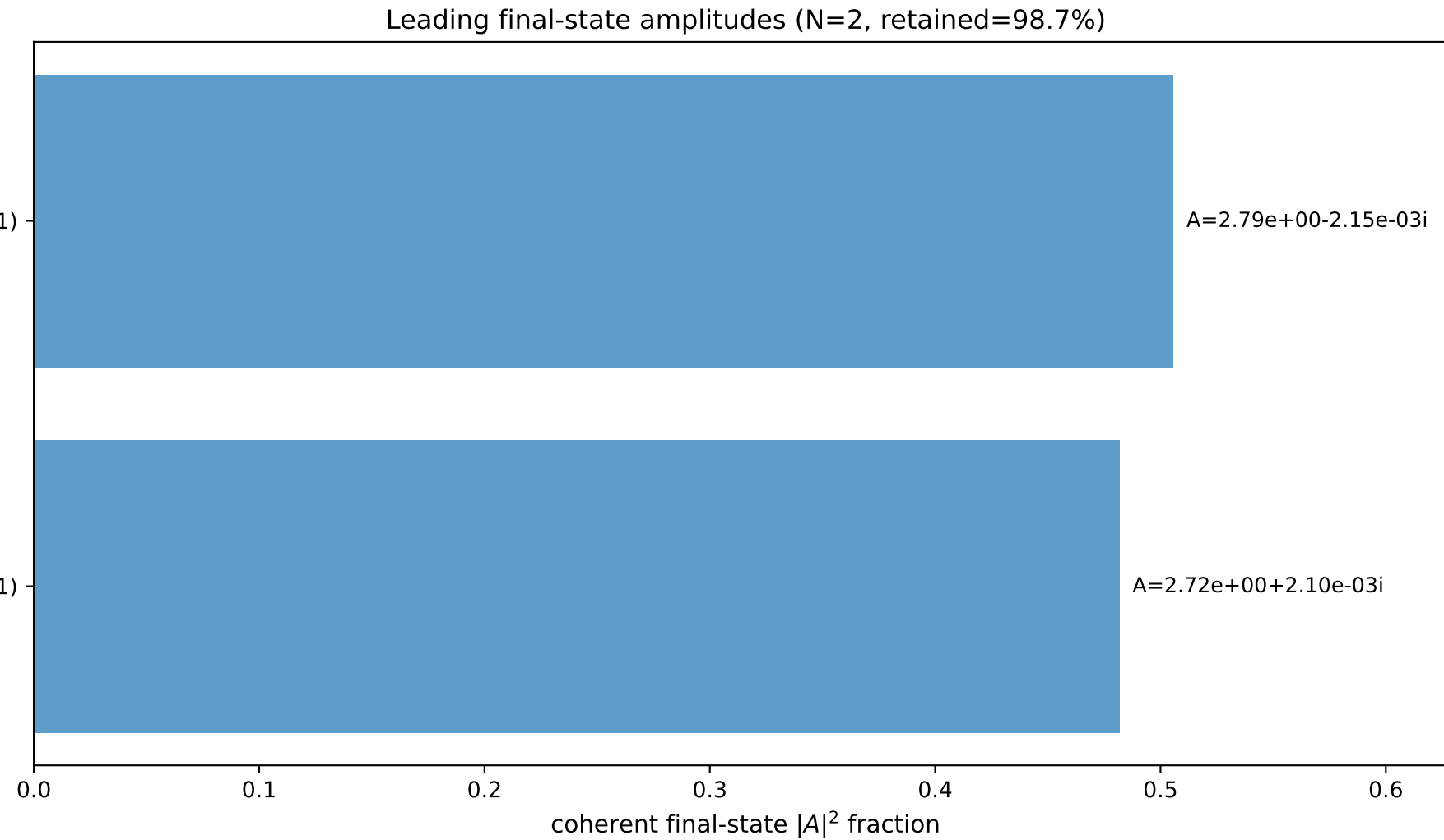
$s=6.9343$, $\sqrt{s}=2.63331$, $|\vec{P}|=1.14959$, $|\vec{P}'|=3.60035\text{e-}07$
 $\theta_{P'}=3.14082$, $\theta_V=3.10976$, $\phi_{P'}=0.000770639$, $\phi_V=6.28315$
 $E_V=0.847653$, $\theta_{\ell'}=0.0318365$, $\phi_{\ell'}=3.14156$
 $Q^2=3.89684$, $x_B=0.71019$, $t=-1.02376$, $W^2=2.47004$, $y=0.90628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1496$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1496$
 P $E= 1.4837$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1496$
 ℓ' $E= 0.8477$ $p_x= -0.0270$ $p_y= 0.0000$ $p_z= 0.8472$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_Y $E= 0.8477$ $p_x= 0.0270$ $p_y= -0.0000$ $p_z= -0.8472$

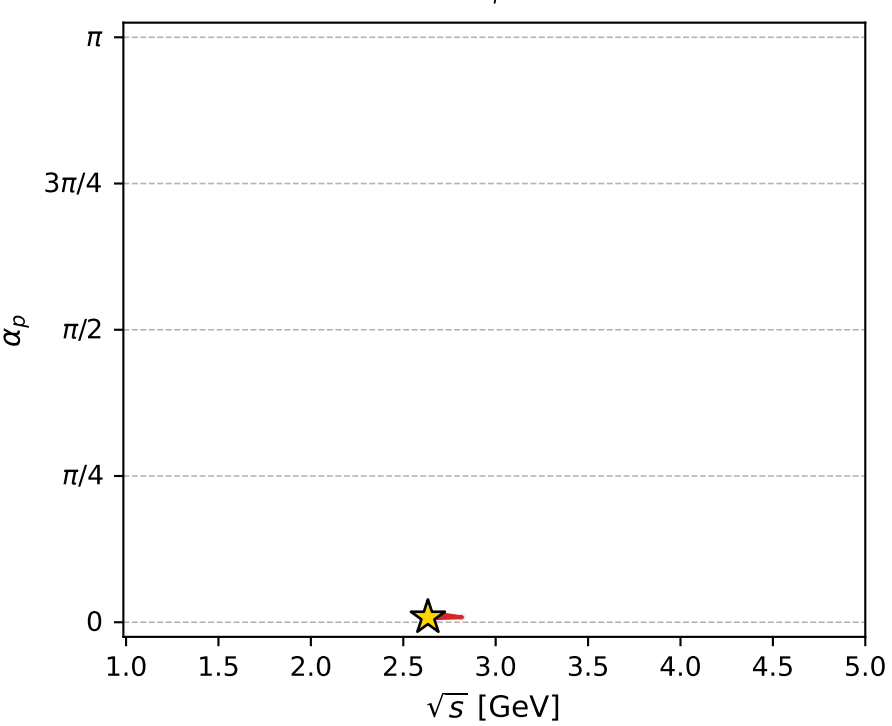
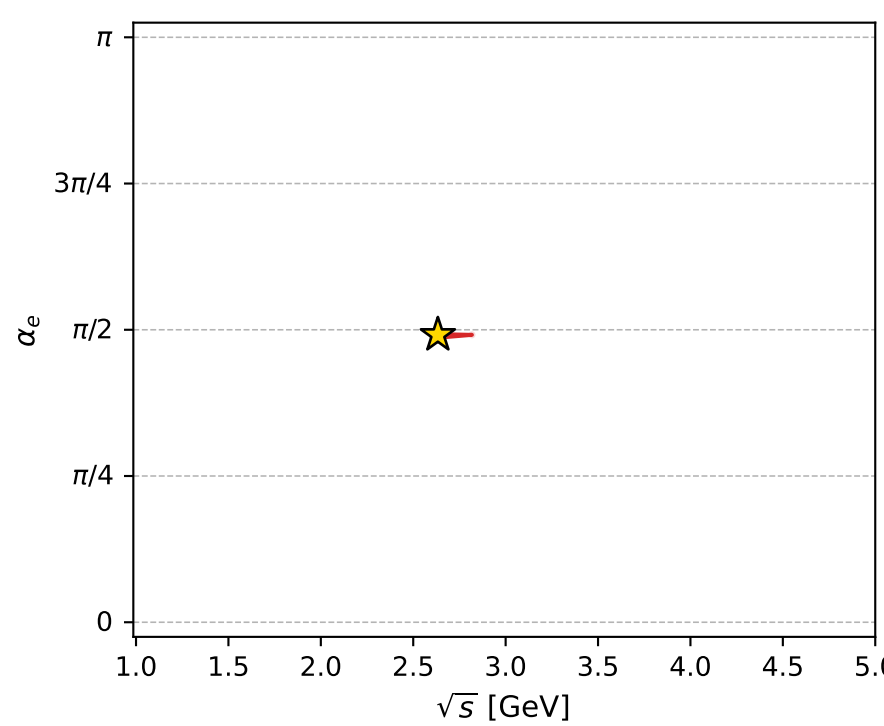
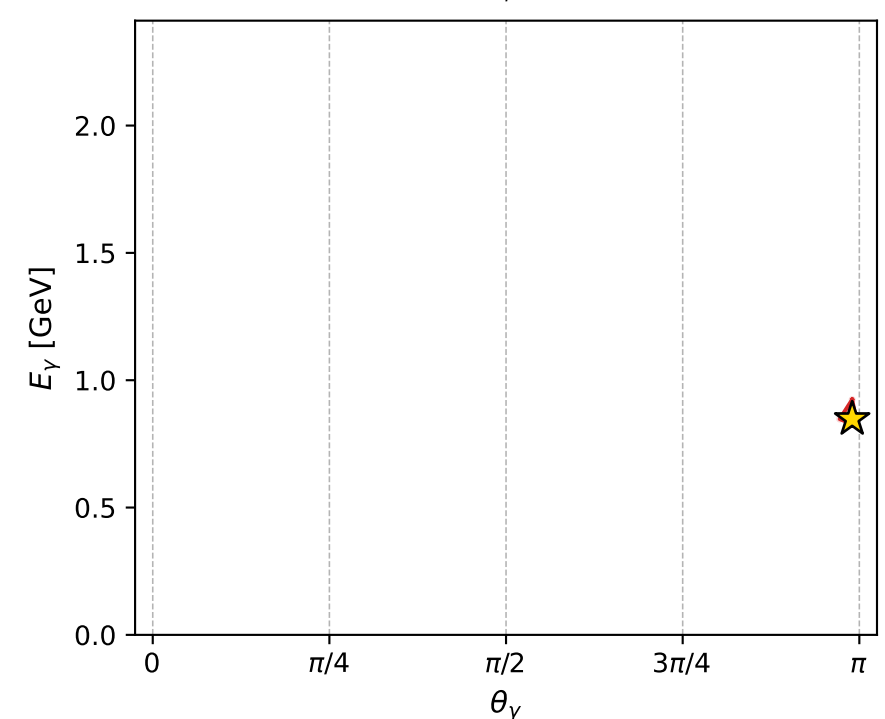
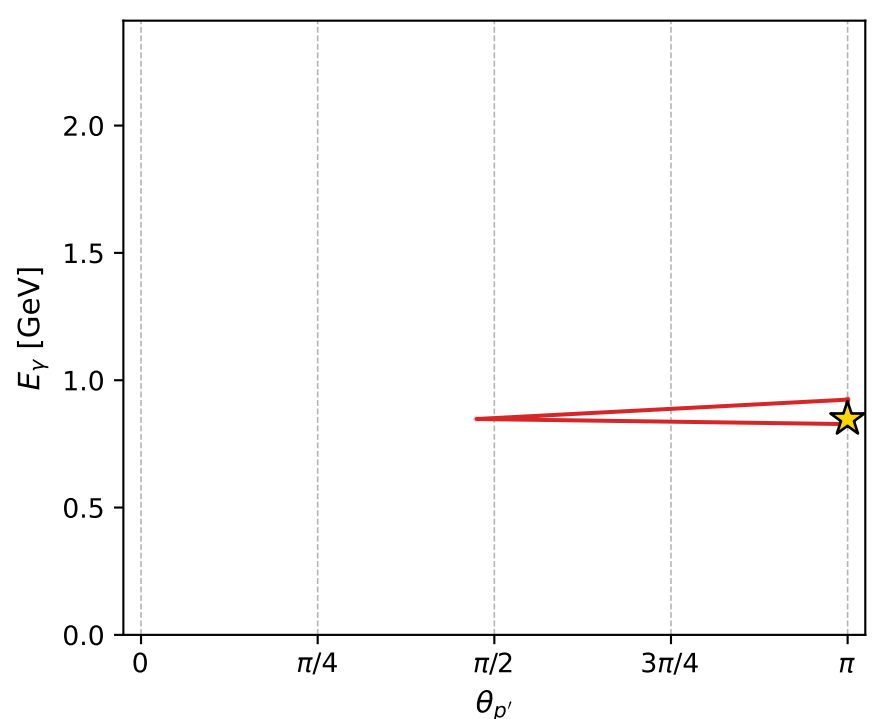
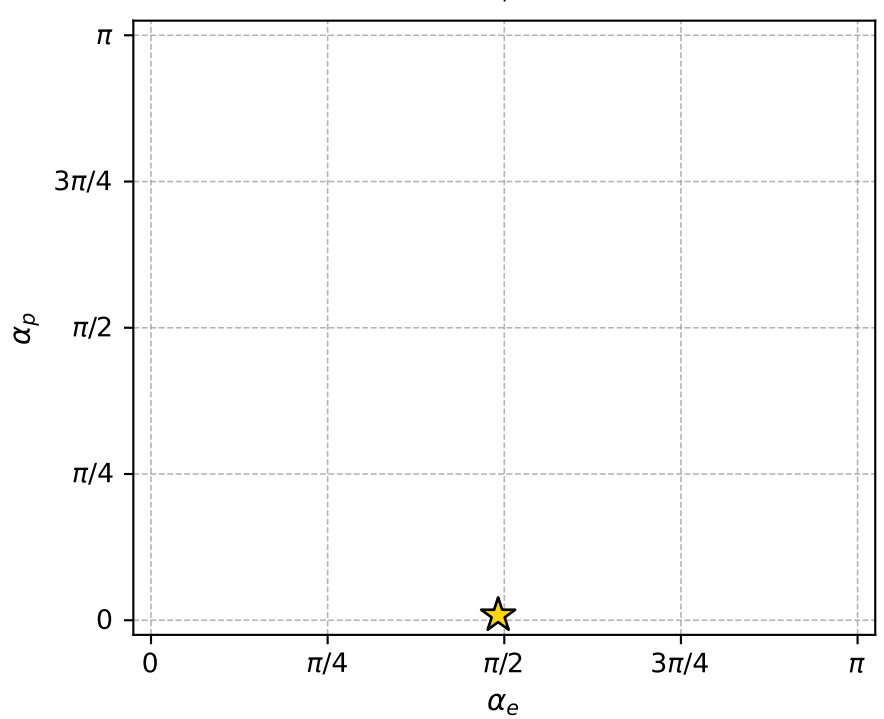
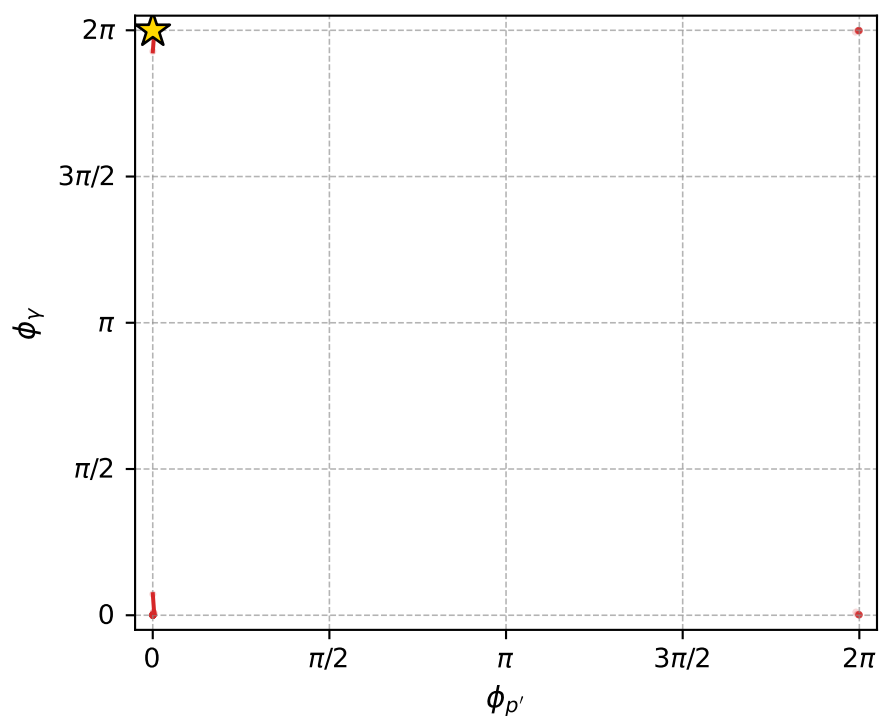
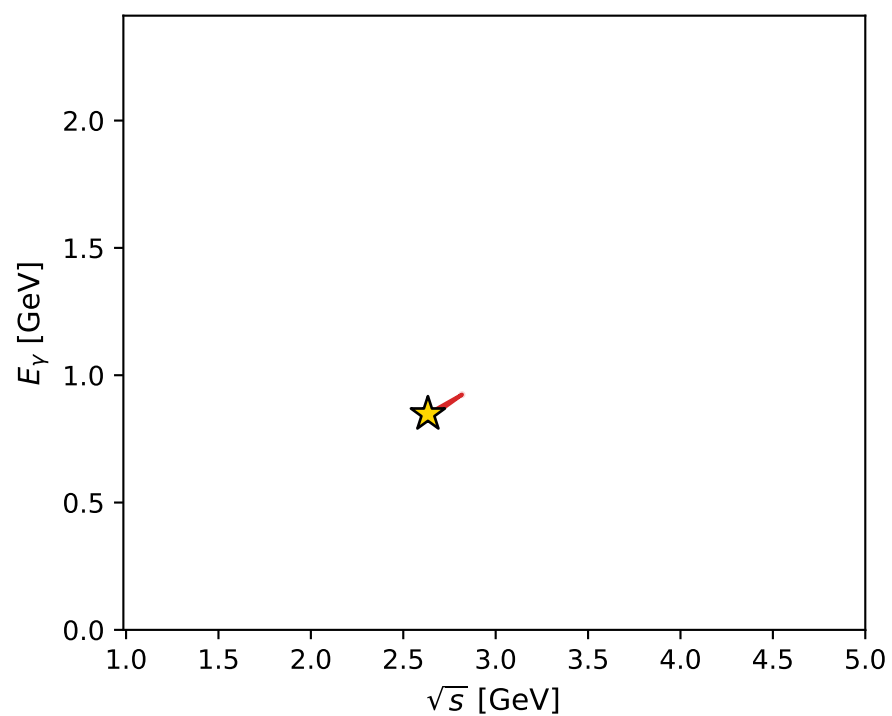
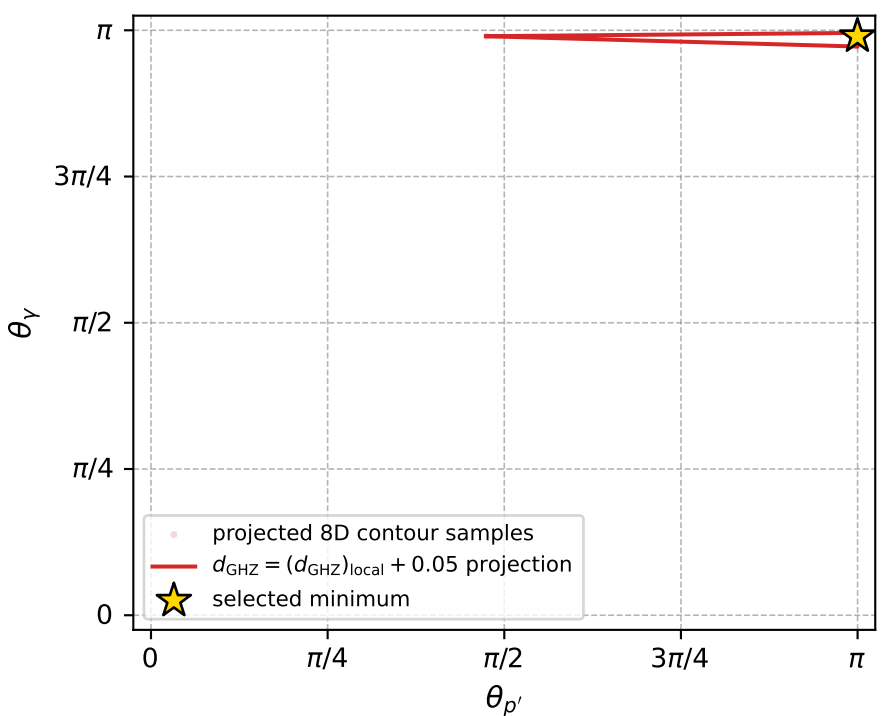


$(h_p, h_Y, h_\ell) = (+1, +1, +1)$

$(h_p, h_Y, h_\ell) = (-1, -1, -1)$



electron: polarization cluster P1, local minimum 359; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0015737309$
 $d_{\text{GHZ}} \text{ contour} = 0.051573731$
 above global minimum = 0.0015737086
 8D contour samples = 204

selected phase-space point:
 $\text{sqrt_s} = 2.633307$
 $\text{theta_p_out} = 3.140817$
 $\text{theta_gamma_out} = 3.109756$
 $\text{q0out} = 0.8476531$
 $\text{phi_p_out} = 0.0007706389$
 $\text{phi_gamma_out} = 6.283155$
 $\text{alpha_e} = 1.543427$
 $\text{alpha_p} = 0.02671695$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_360 d_{\rm GHZ}=0.00187001
region: polarization_cluster_01_local_minimum_360
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.5430613$ rad
incoming lepton: $0.0277314|+\rangle +0.999615|-\rangle$
proton mixing angle: $\alpha_p=0.027451861$ rad
incoming proton: $0.999623|+\rangle +0.0274484|-\rangle$

$s=5.25221$, $\sqrt{s}=2.29177$, $|\vec{P}|=0.953927$, $|\vec{P}'|=0.00502558$

$\theta_{P'}=3.14128$, $\theta_V=3.10607$, $\phi_{P'}=5.2495$, $\phi_V=6.28316$

$E_V=0.674367$, $\theta_{\ell'}=0.0352562$, $\phi_{\ell'}=3.14151$

$Q^2=2.59155$, $x_B=0.673146$, $t=-0.759728$, $W^2=2.1382$, $y=0.880508$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.9539$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9539$

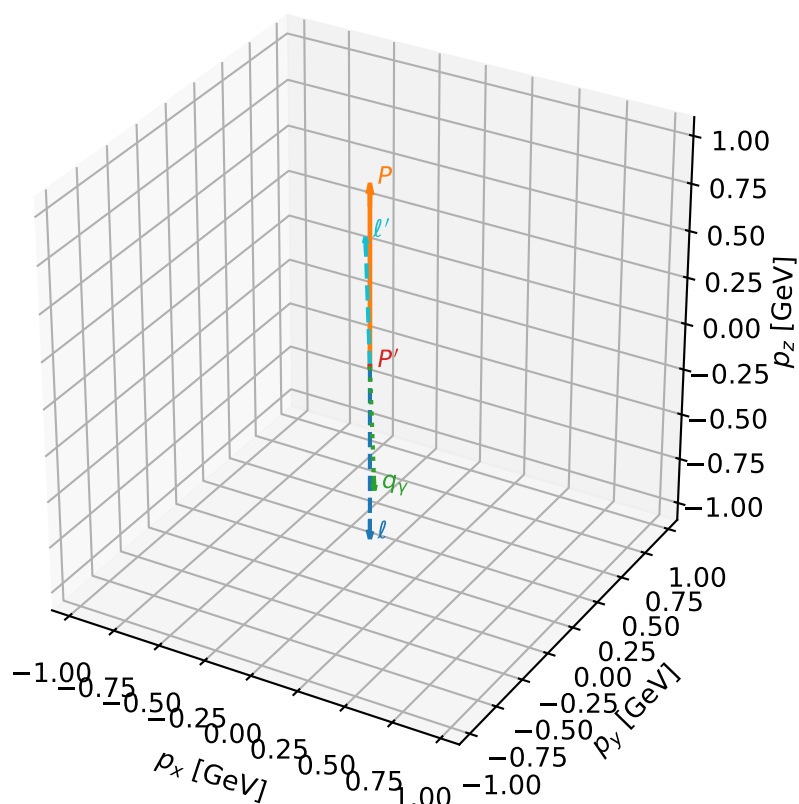
P $E= 1.3378$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9539$

ℓ' $E= 0.6794$ $p_x= -0.0239$ $p_y= 0.0000$ $p_z= 0.6790$

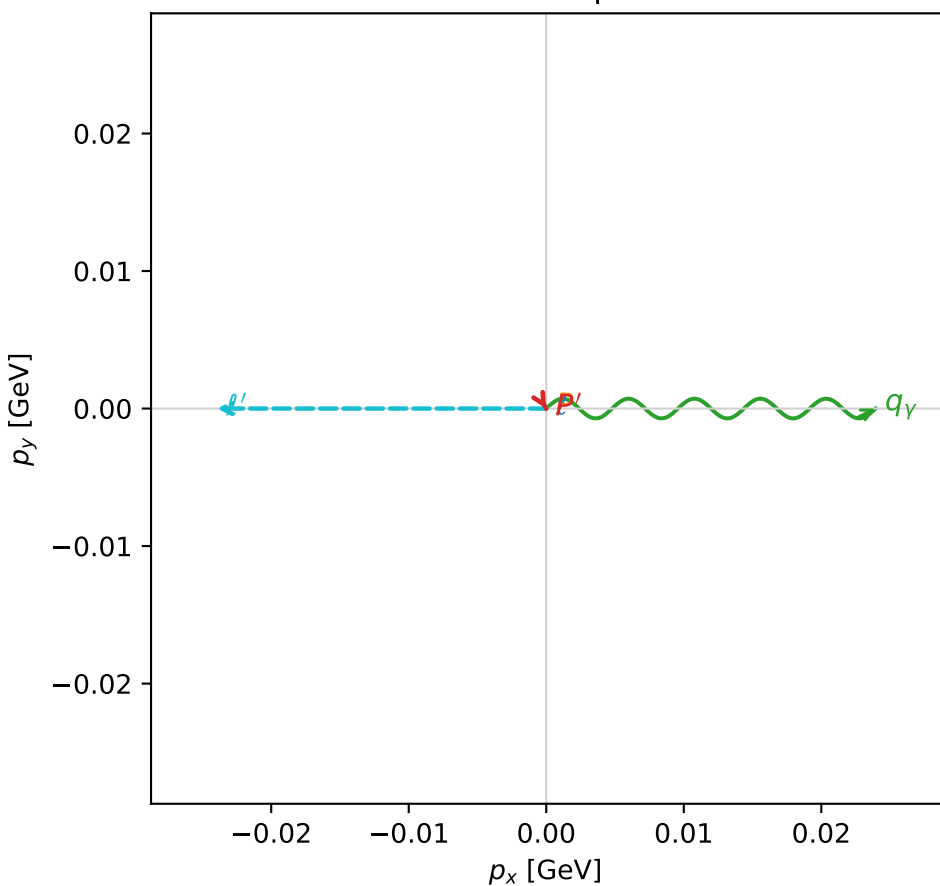
P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0050$

q_V $E= 0.6744$ $p_x= 0.0239$ $p_y= -0.0000$ $p_z= -0.6739$

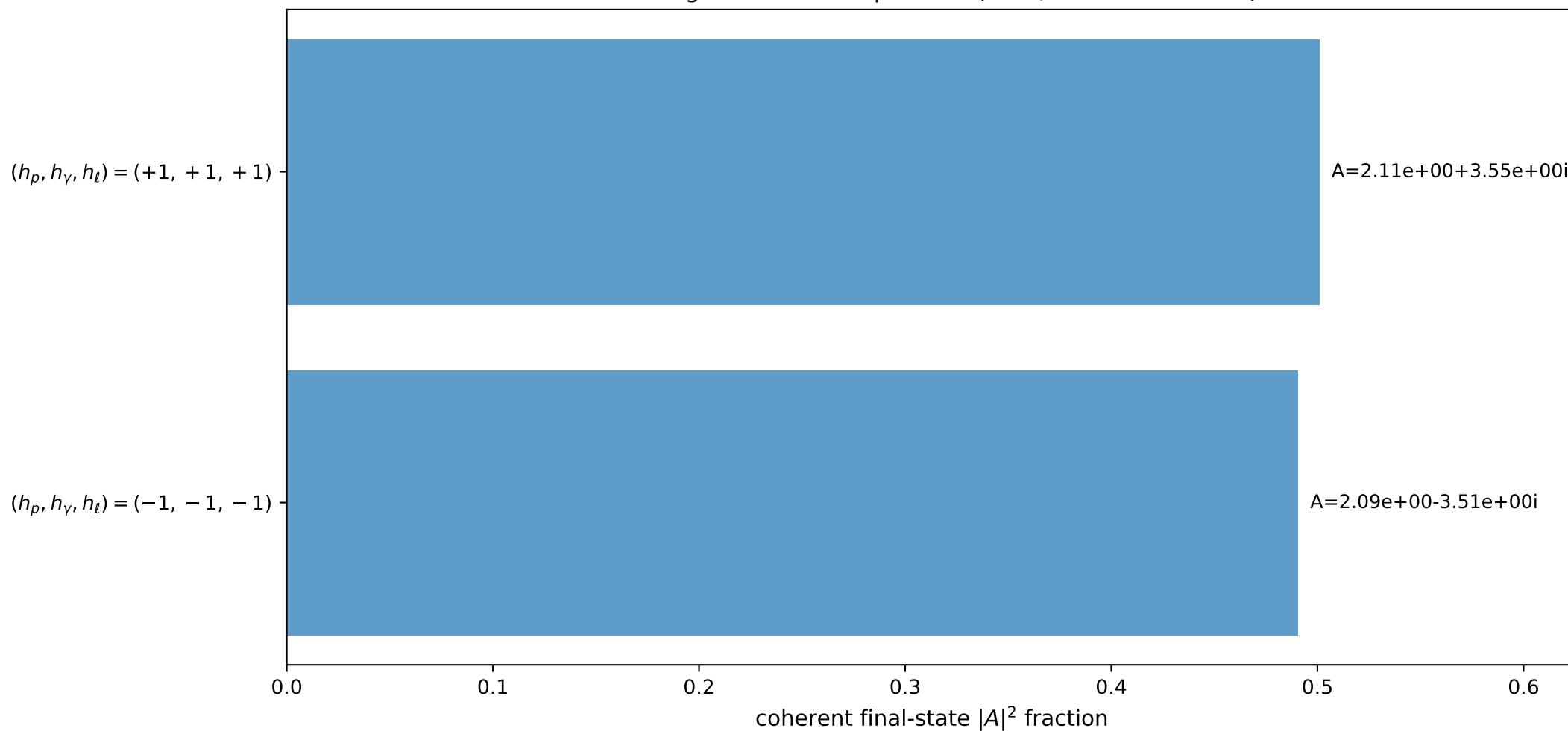
3D momenta



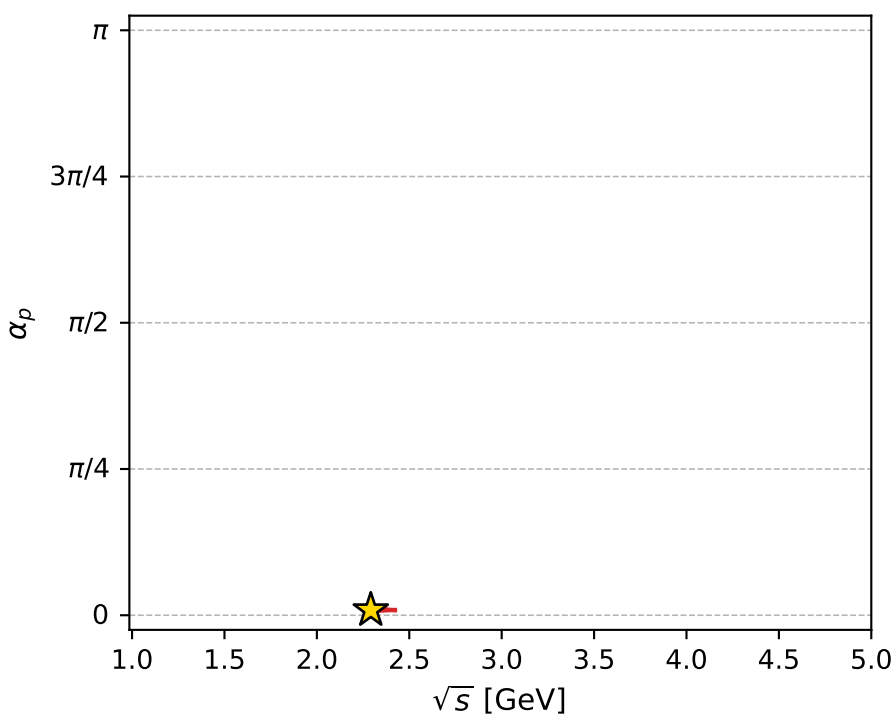
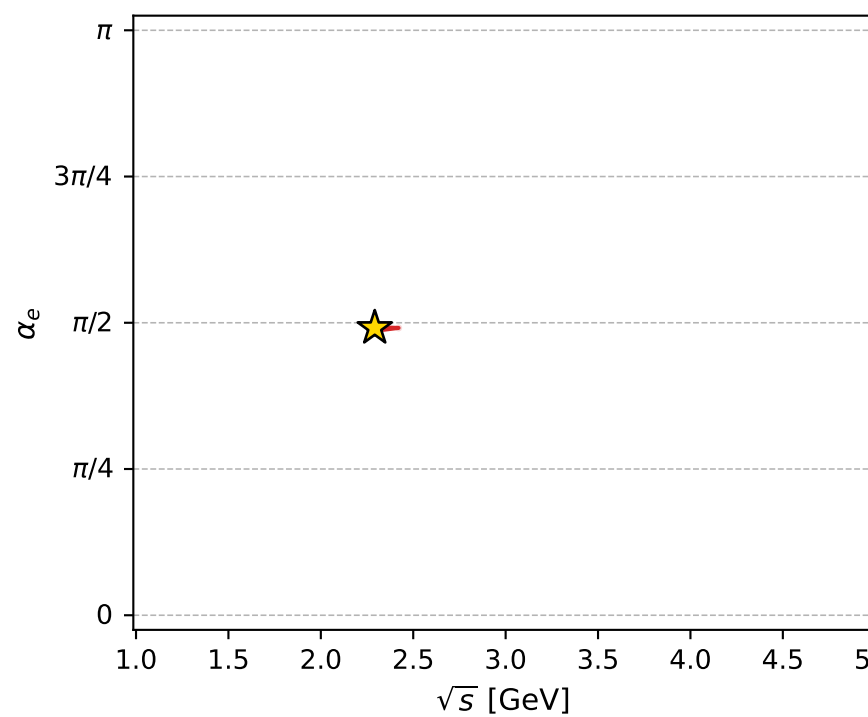
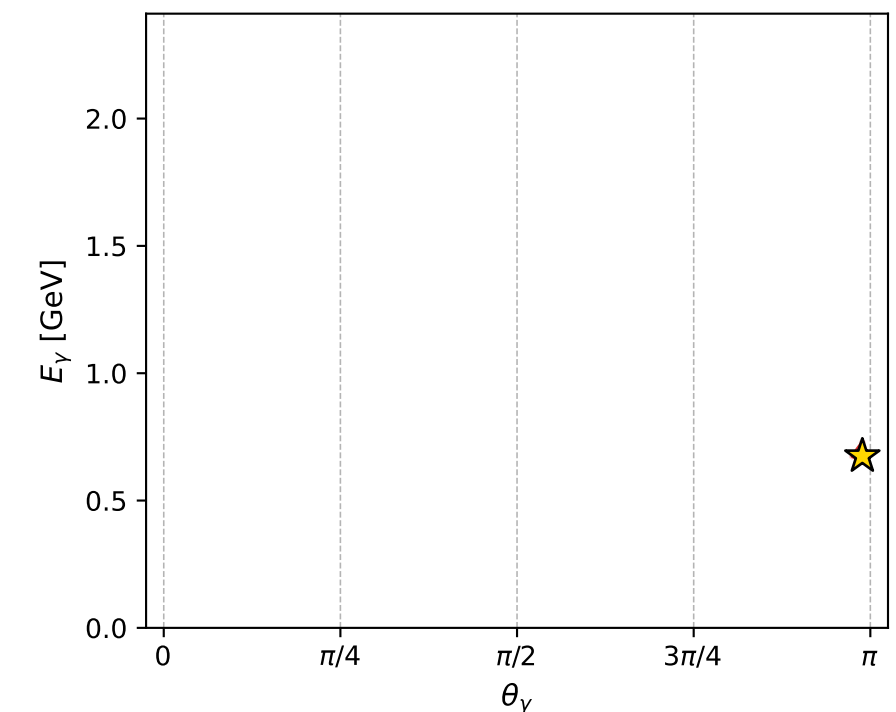
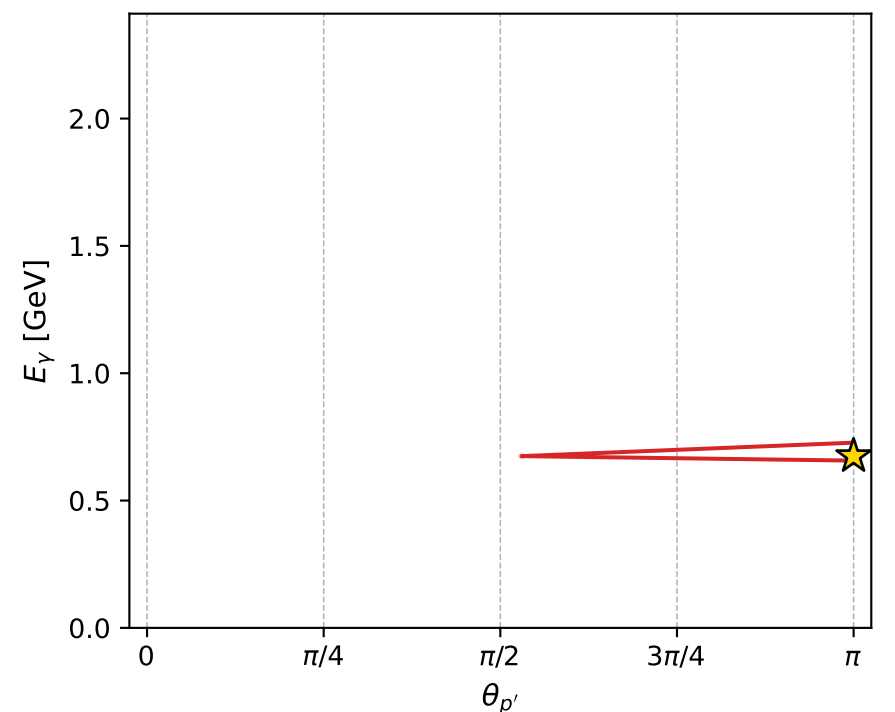
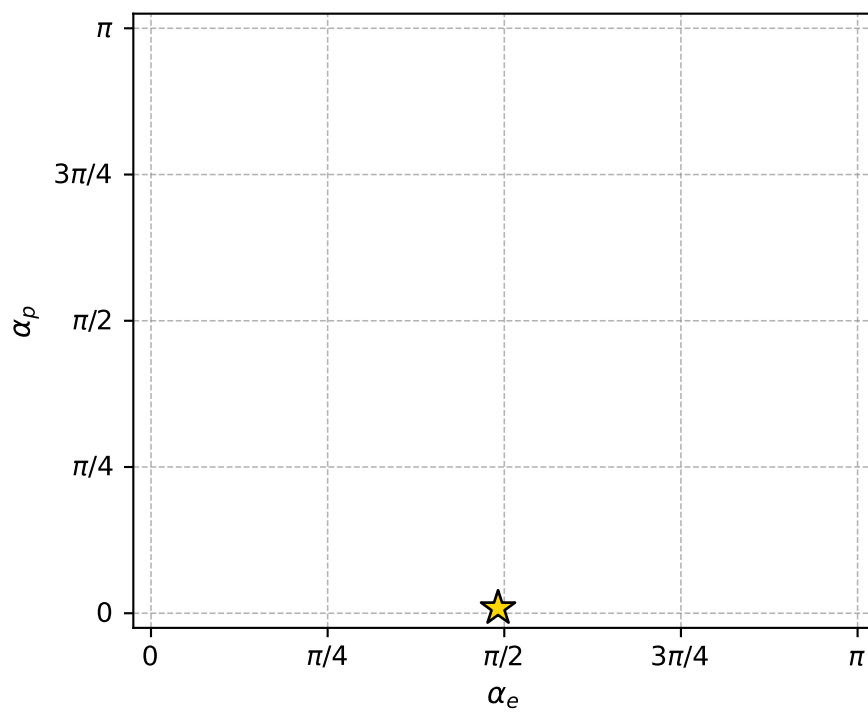
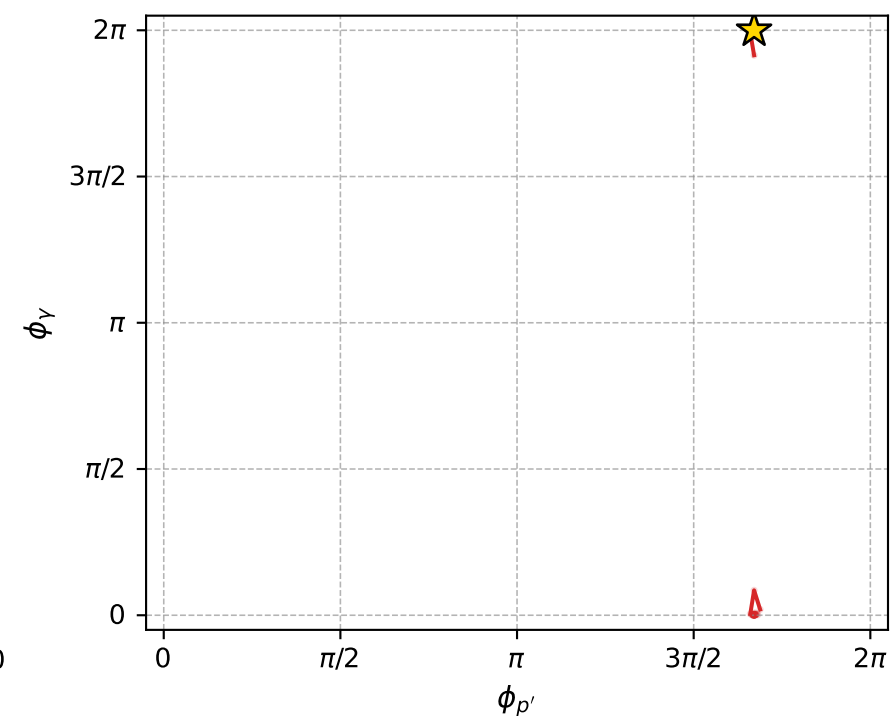
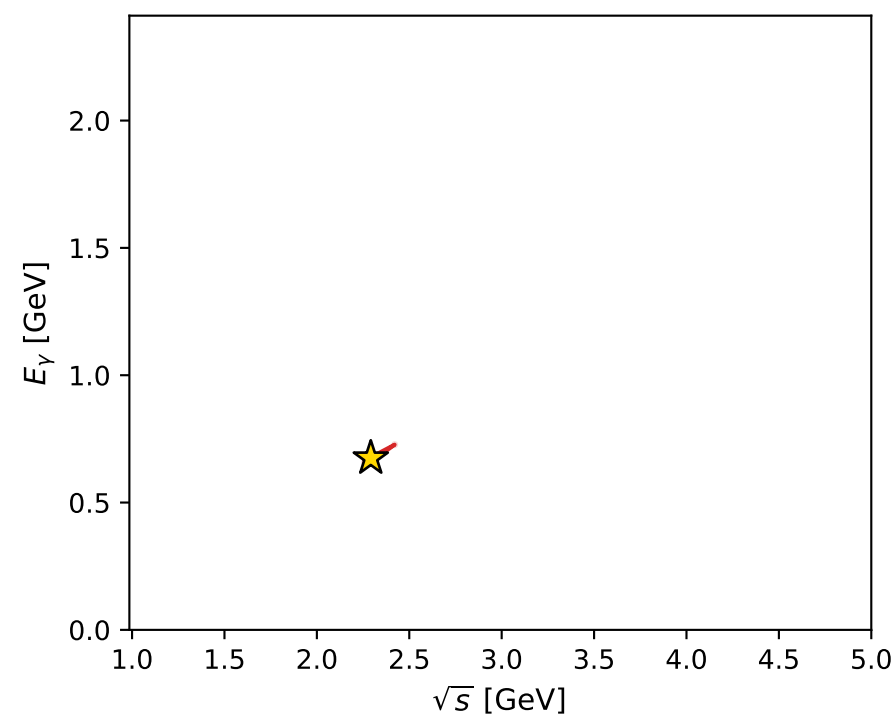
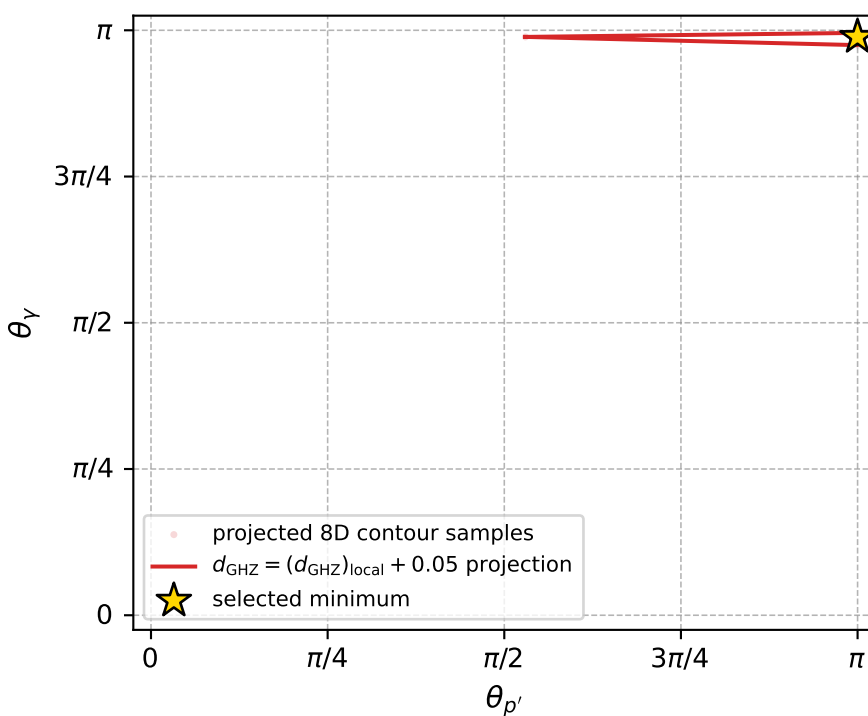
Transverse plane



Leading final-state amplitudes (N=2, retained=99.2%)



electron: polarization cluster P1, local minimum 360; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0018700053$
 $d_{\text{GHZ}} \text{ contour} = 0.051870005$
 above global minimum = 0.001869983
 8D contour samples = 144

selected phase-space point:

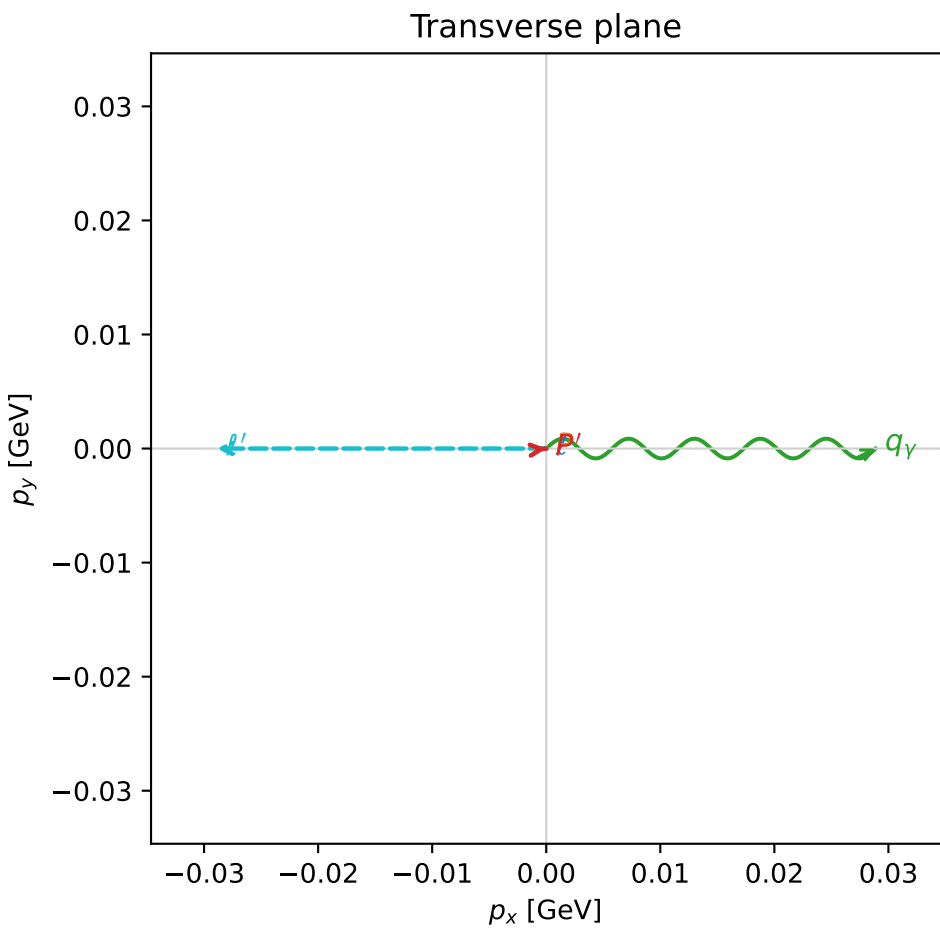
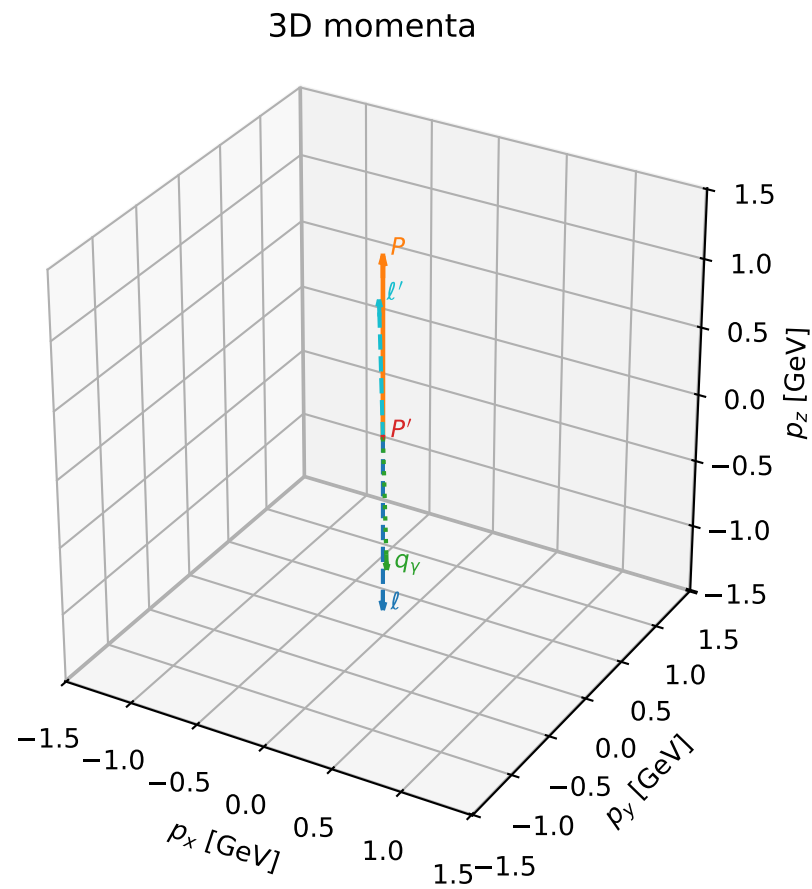
$\text{sqrt_s} = 2.29177$
 $\text{theta_p_out} = 3.141278$
 $\text{theta_gamma_out} = 3.106075$
 $\text{q0out} = 0.6743668$
 $\text{phi_p_out} = 5.2495$
 $\text{phi_gamma_out} = 6.283159$
 $\text{alpha_e} = 1.543061$
 $\text{alpha_p} = 0.02745186$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_361 d_{\rm GHZ}=0.0018947
region: polarization_cluster_01_local_minimum_361
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.5434414$ rad
incoming lepton: $0.0273516|+\rangle +0.999626|-\rangle$
proton mixing angle: $\alpha_p=0.026758462$ rad
incoming proton: $0.999642|+\rangle +0.0267553|-\rangle$

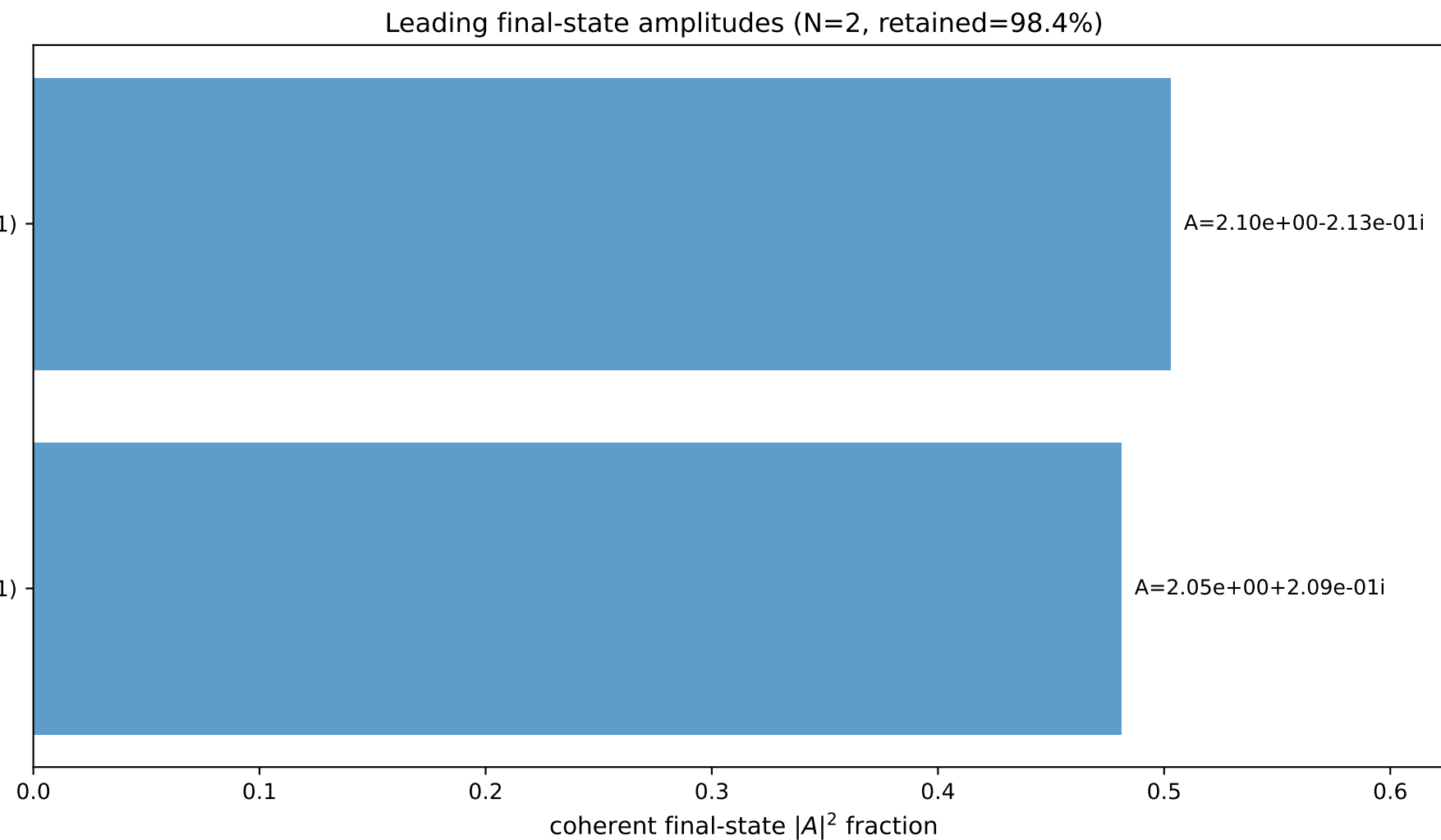
$s=8.54507$, $\sqrt{s}=2.9232$, $|\vec{P}|=1.3111$, $|\vec{P}'|=2.76355\text{e-}06$
 $\theta_{P'}=3.13818$, $\theta_V=3.1125$, $\phi_{P'}=0.101143$, $\phi_V=6.28318$
 $E_V=0.992596$, $\theta_{\ell'}=0.0290944$, $\phi_{\ell'}=3.14159$
 $Q^2=5.2045$, $x_B=0.736492$, $t=-1.2646$, $W^2=2.74195$, $y=0.921904$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3111$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3111$
 P $E= 1.6121$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3111$
 ℓ' $E= 0.9926$ $p_x= -0.0289$ $p_y= 0.0000$ $p_z= 0.9922$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_Y $E= 0.9926$ $p_x= 0.0289$ $p_y= -0.0000$ $p_z= -0.9922$

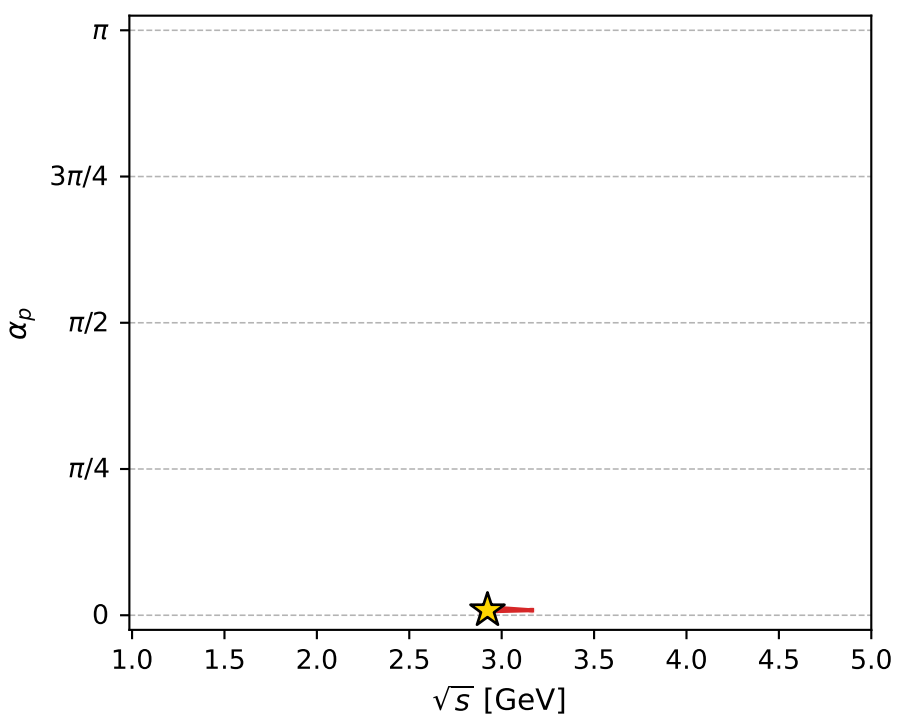
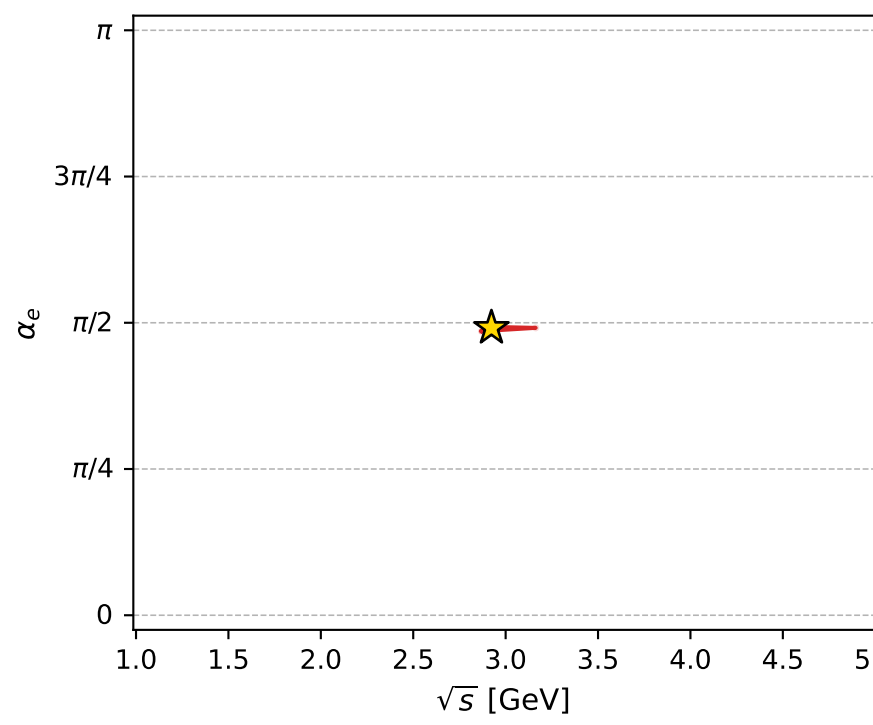
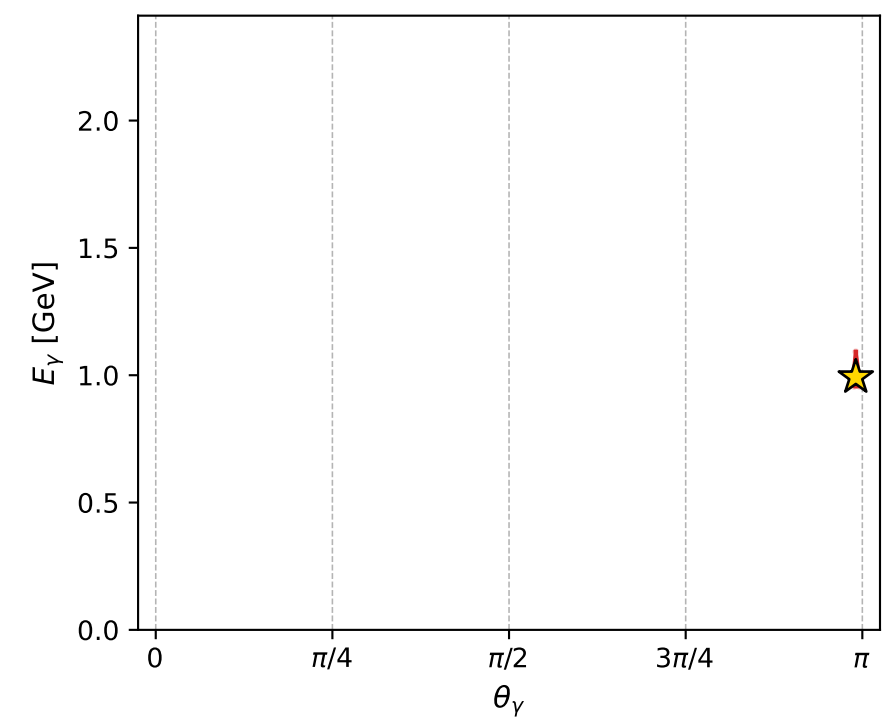
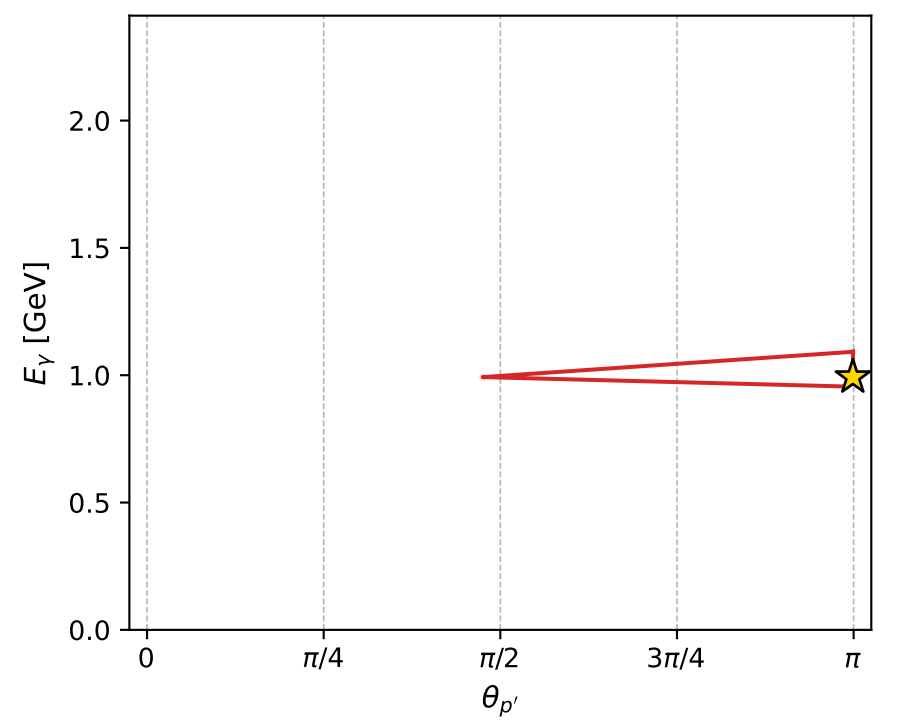
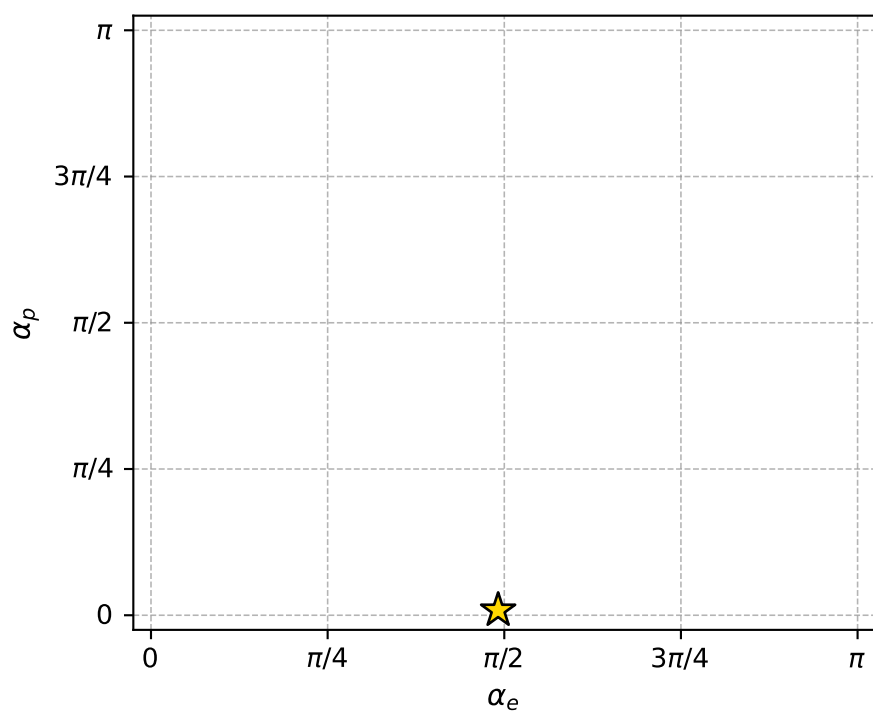
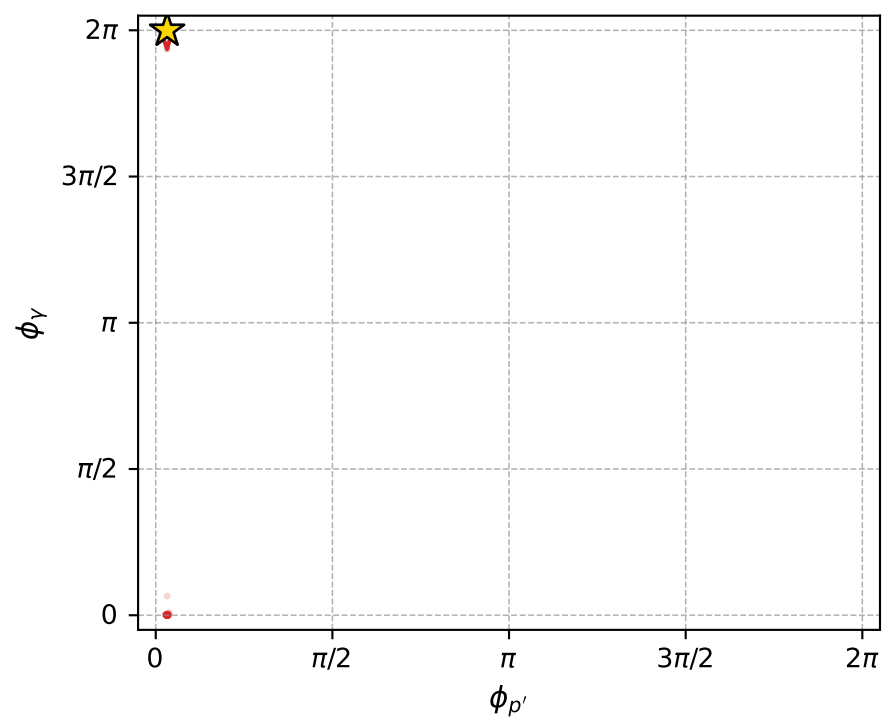
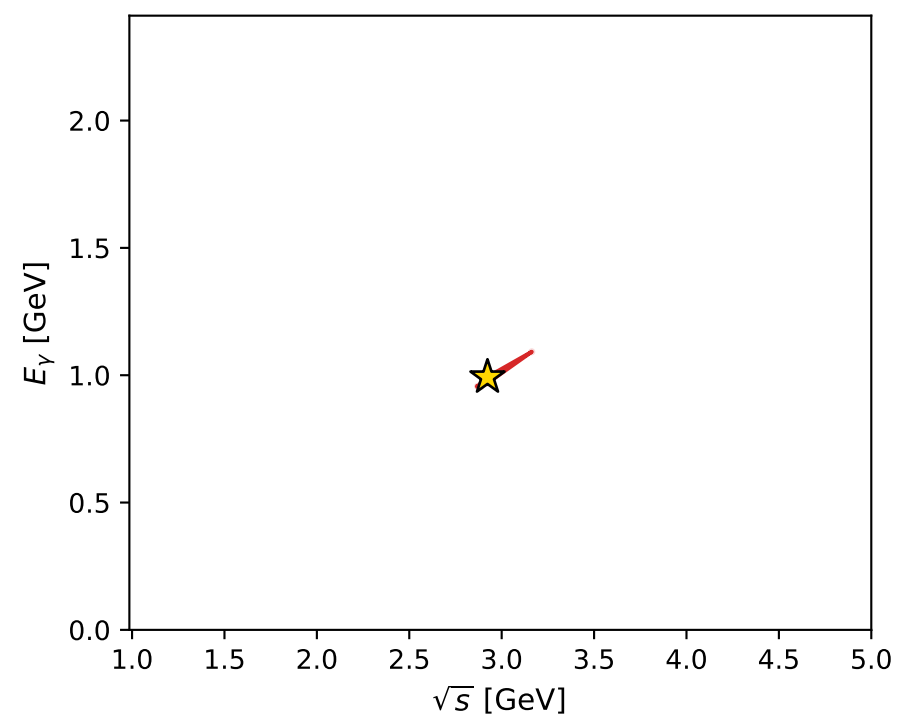
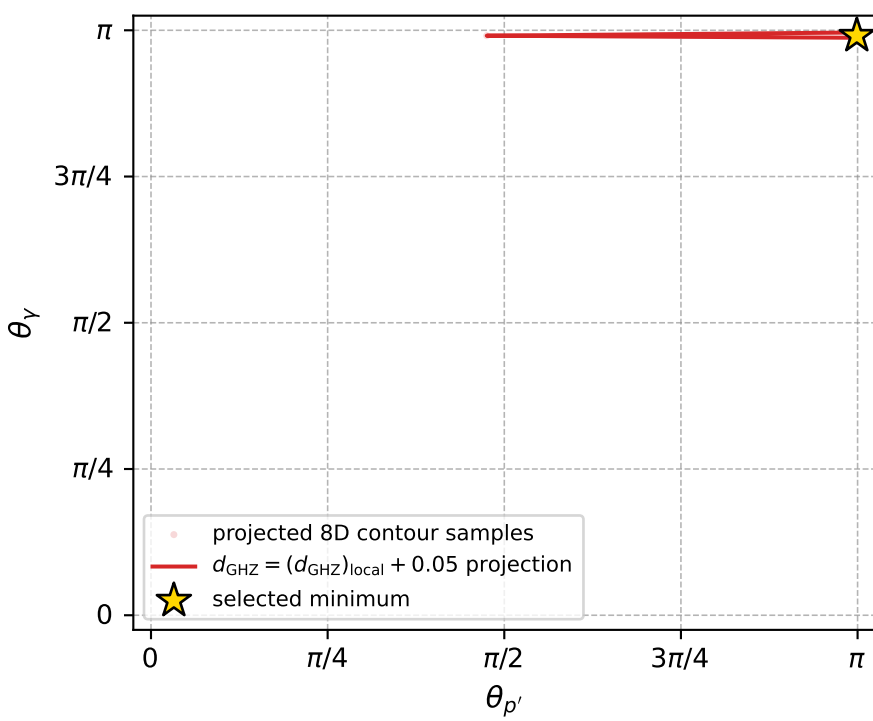


$(h_p, h_Y, h_\ell) = (+1, +1, +1)$

$(h_p, h_Y, h_\ell) = (-1, -1, -1)$



electron: polarization cluster P1, local minimum 361; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.0018947042$
 $d_{\text{GHZ}} \text{ contour} = 0.051894704$
 above global minimum = 0.0018946819
 8D contour samples = 233

selected phase-space point:

$\text{sqrt_s} = 2.923195$
 $\text{theta_p_out} = 3.13818$
 $\text{theta_gamma_out} = 3.112498$
 $\text{q0ut} = 0.9925963$
 $\text{phi_p_out} = 0.1011434$
 $\text{phi_gamma_out} = 6.283183$
 $\text{alpha_e} = 1.543441$
 $\text{alpha_p} = 0.02675846$

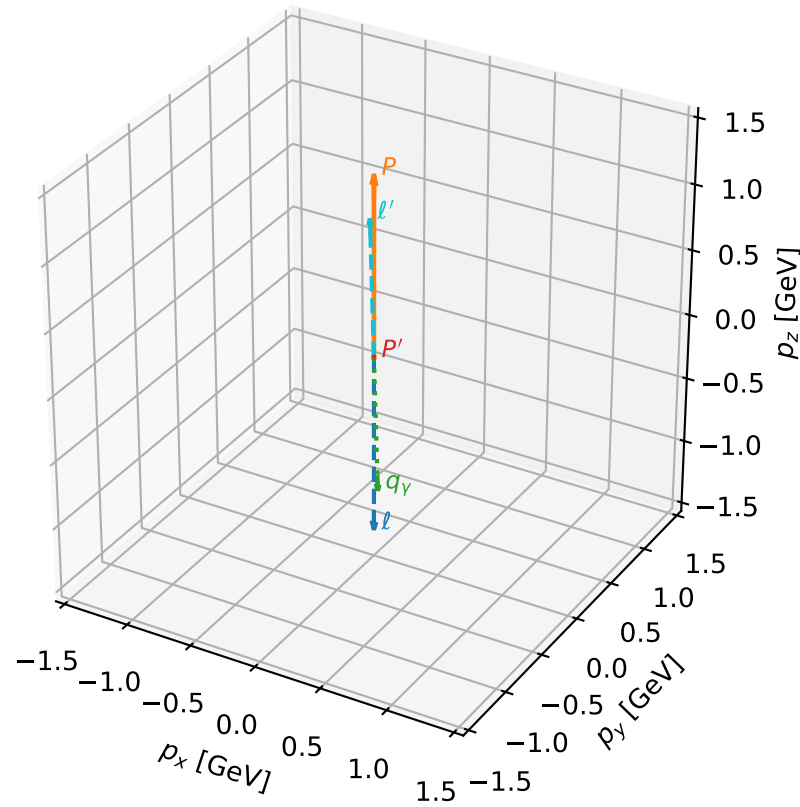
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_364 $d_{\{\text{rm GHZ}\}}=0.00240137$
region: polarization_cluster_01_local_minimum_364
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=1.5438592$ rad
incoming lepton: $0.0269338|+\rangle +0.999637|-\rangle$
proton mixing angle: $\alpha_p=0.026354725$ rad
incoming proton: $0.999653|+\rangle +0.0263517|-\rangle$

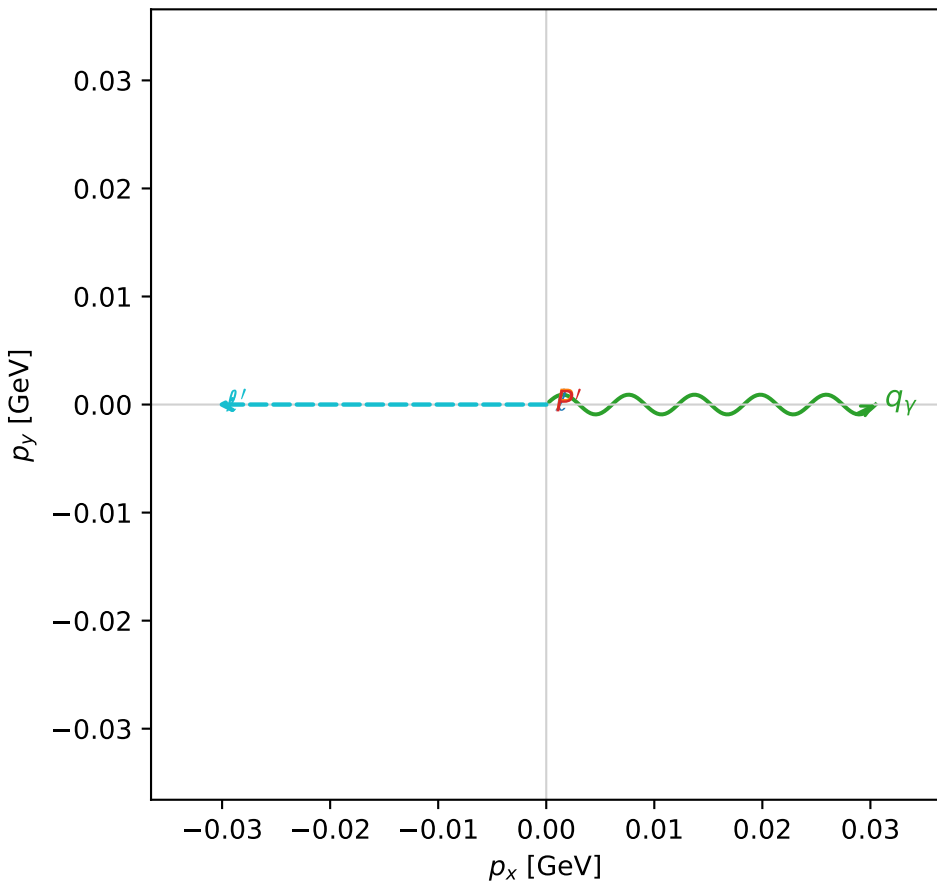
$s=9.12098$, $\sqrt{s}=3.0201$, $|\vec{P}|=1.36438$, $|\vec{P}'|=4.15695\text{e-}06$
 $\theta_{P'}=3.14159$, $\theta_V=3.11231$, $\phi_{P'}=0.518887$, $\phi_V=6.28317$
 $E_V=1.04105$, $\theta_{\ell'}=0.0292827$, $\phi_{\ell'}=3.14158$
 $Q^2=5.68034$, $x_B=0.74415$, $t=-1.34644$, $W^2=2.83284$, $y=0.926248$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3644$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3644$
 P $E= 1.6557$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3644$
 ℓ' $E= 1.0410$ $p_x= -0.0305$ $p_y= 0.0000$ $p_z= 1.0406$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$
 q_Y $E= 1.0410$ $p_x= 0.0305$ $p_y= -0.0000$ $p_z= -1.0406$

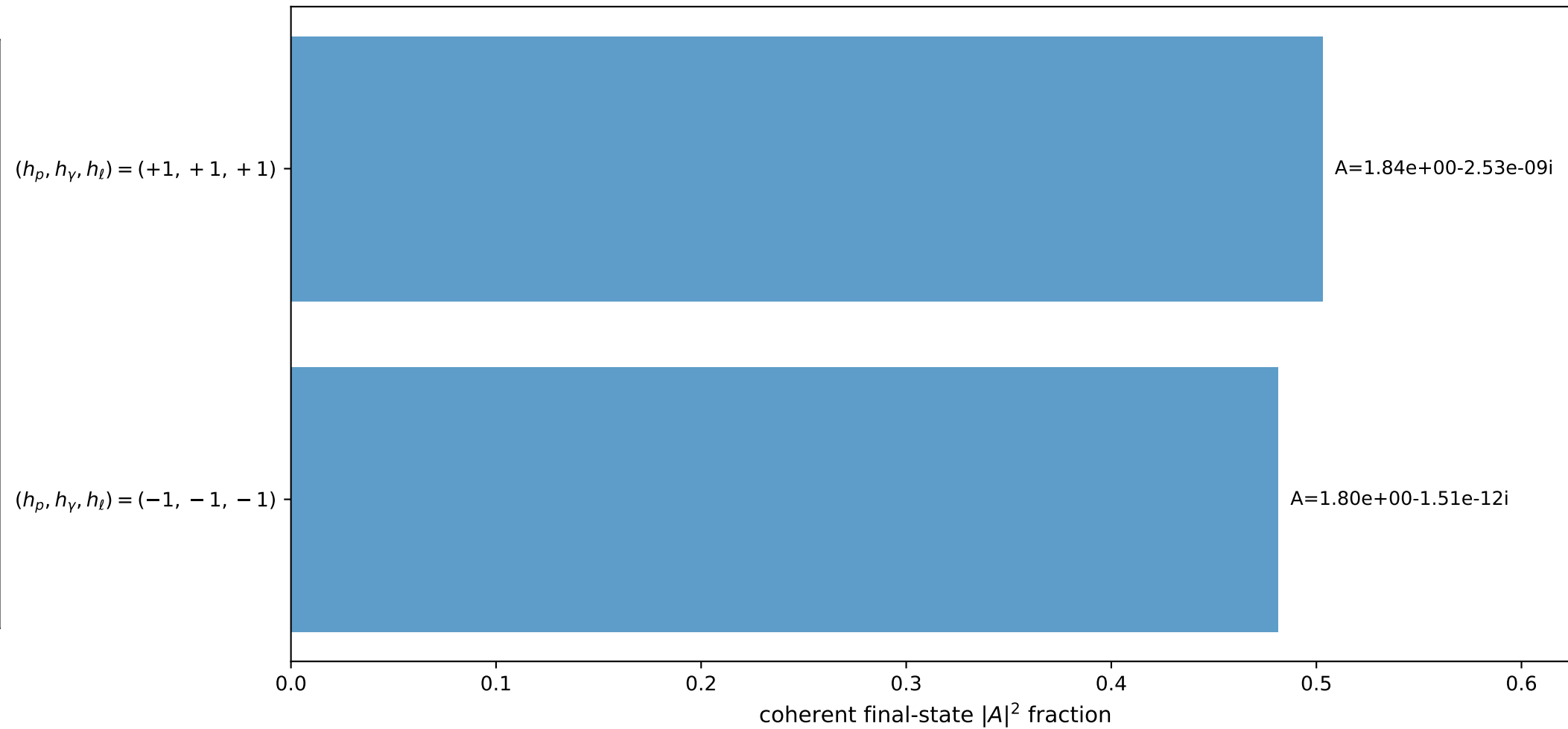
3D momenta



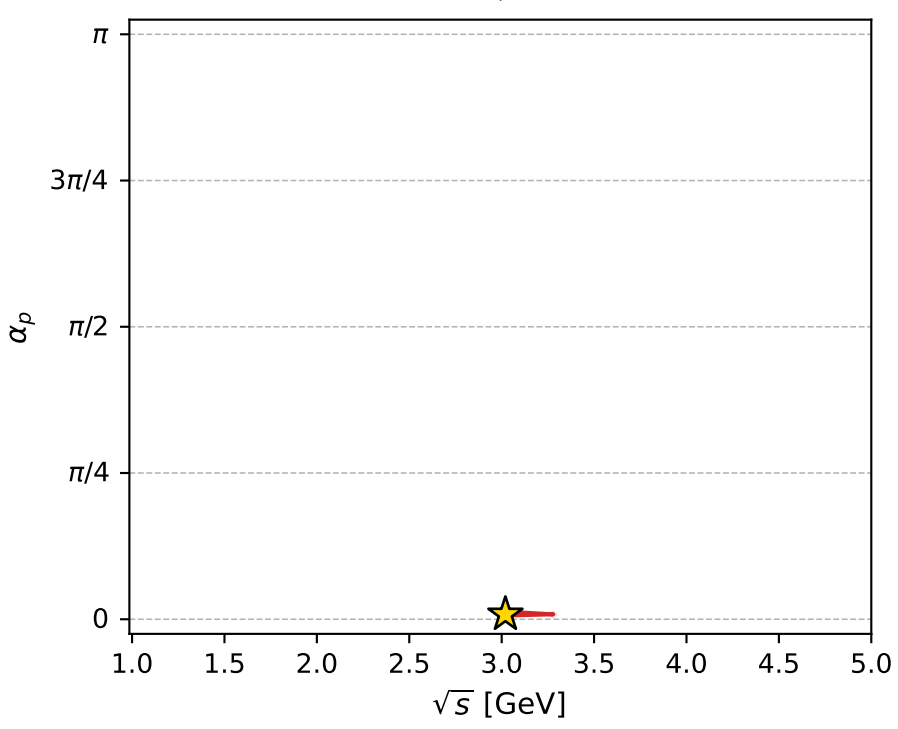
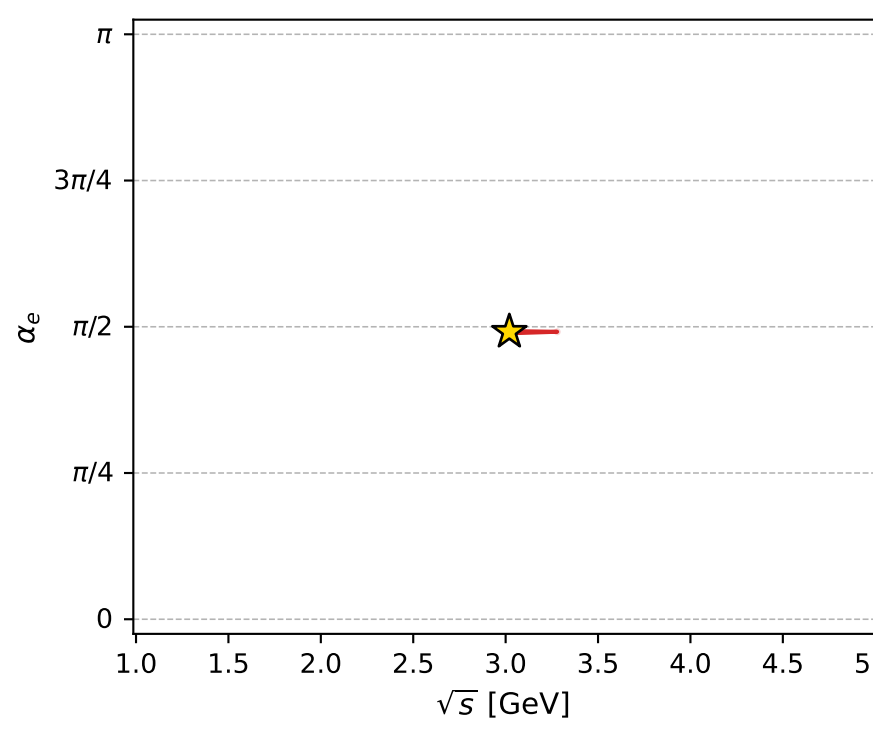
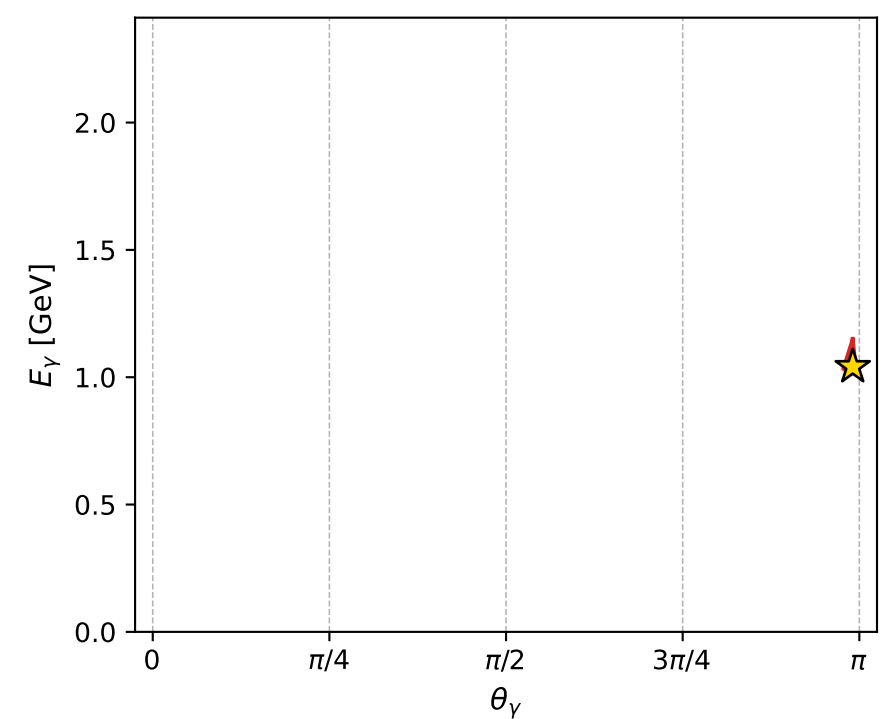
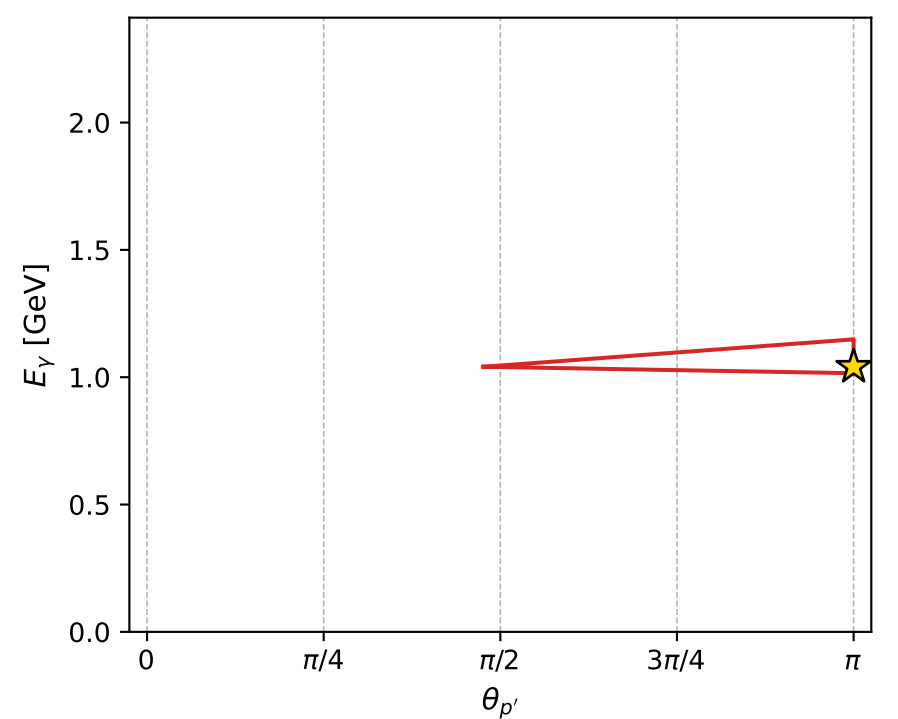
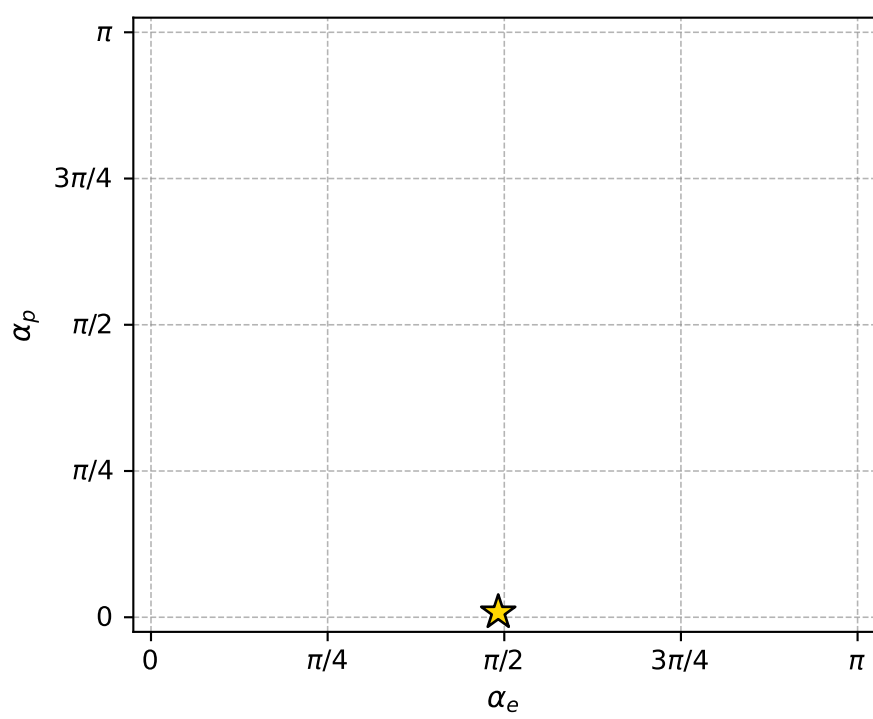
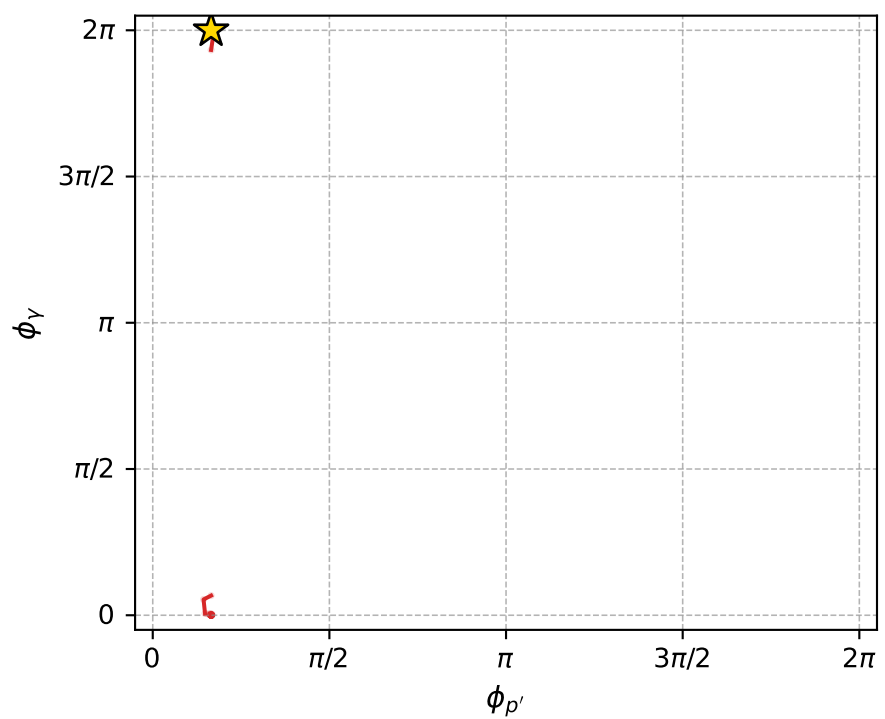
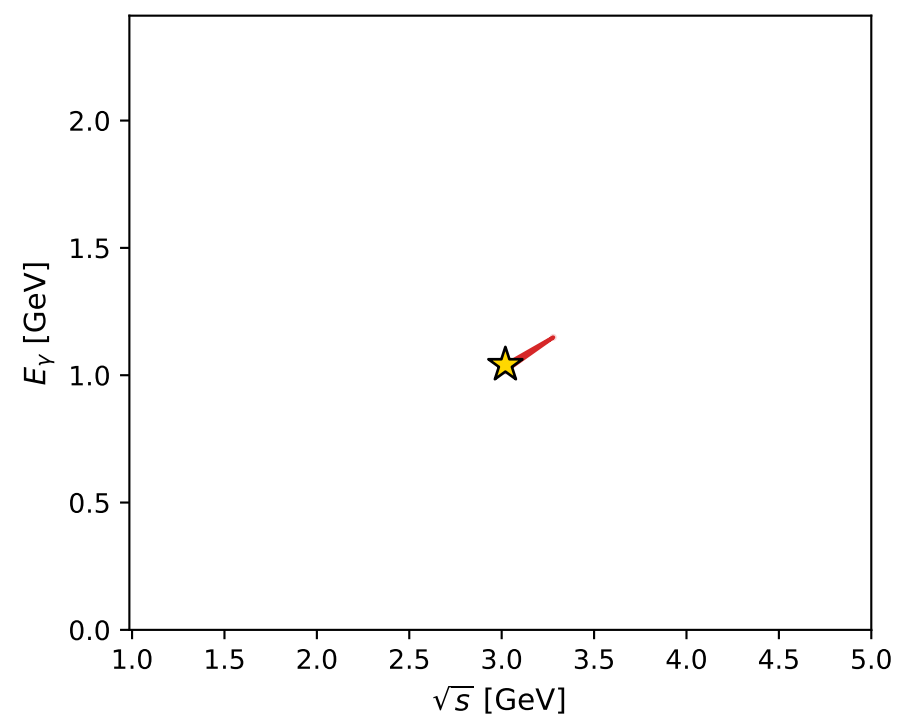
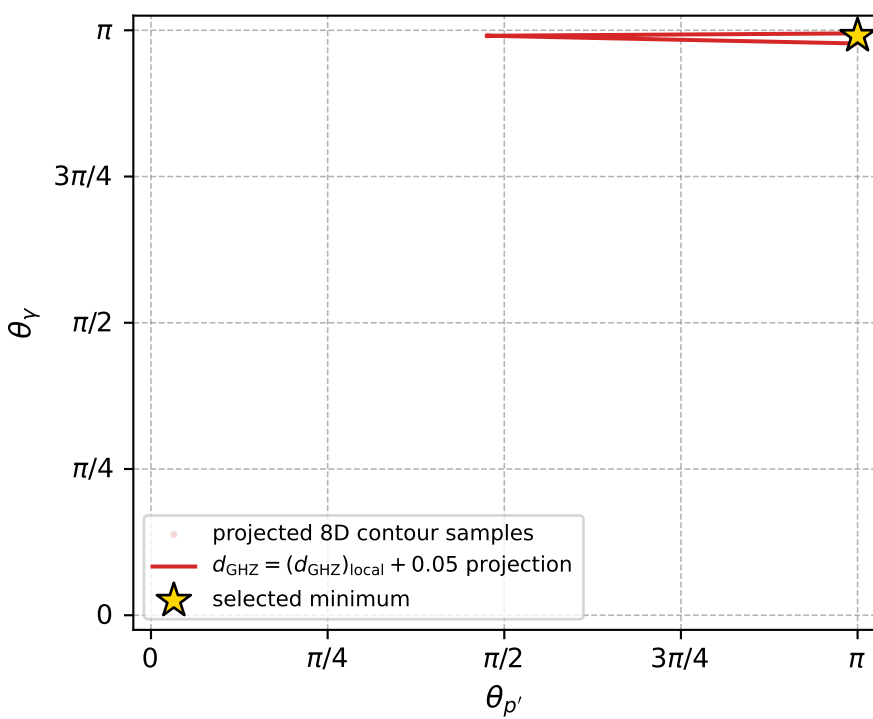
Transverse plane



Leading final-state amplitudes (N=2, retained=98.5%)



electron: polarization cluster P1, local minimum 364; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.002401373$
 $d_{\text{GHZ}} \text{ contour} = 0.052401373$
 above global minimum = 0.0024013507
 8D contour samples = 126

selected phase-space point:

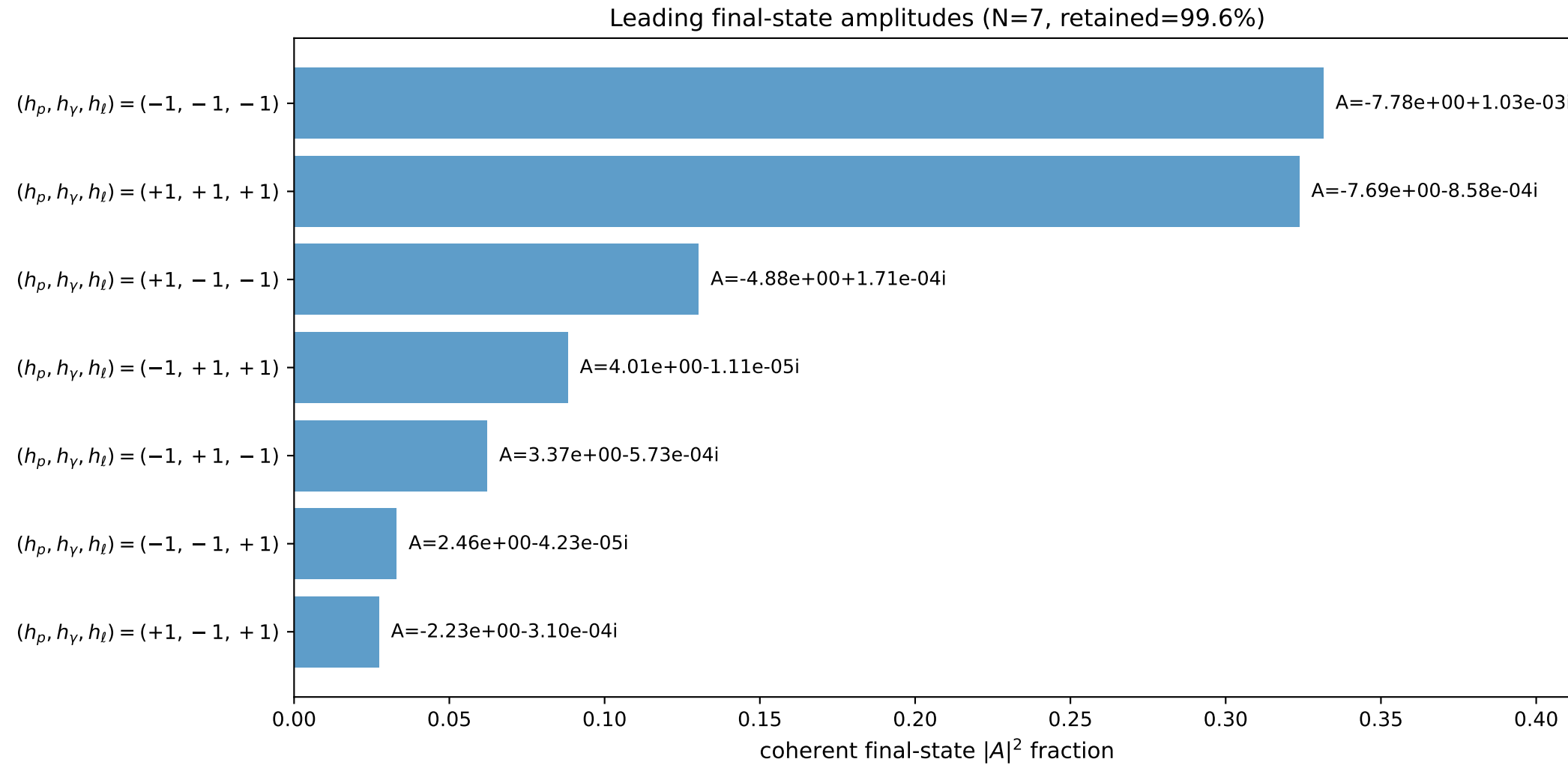
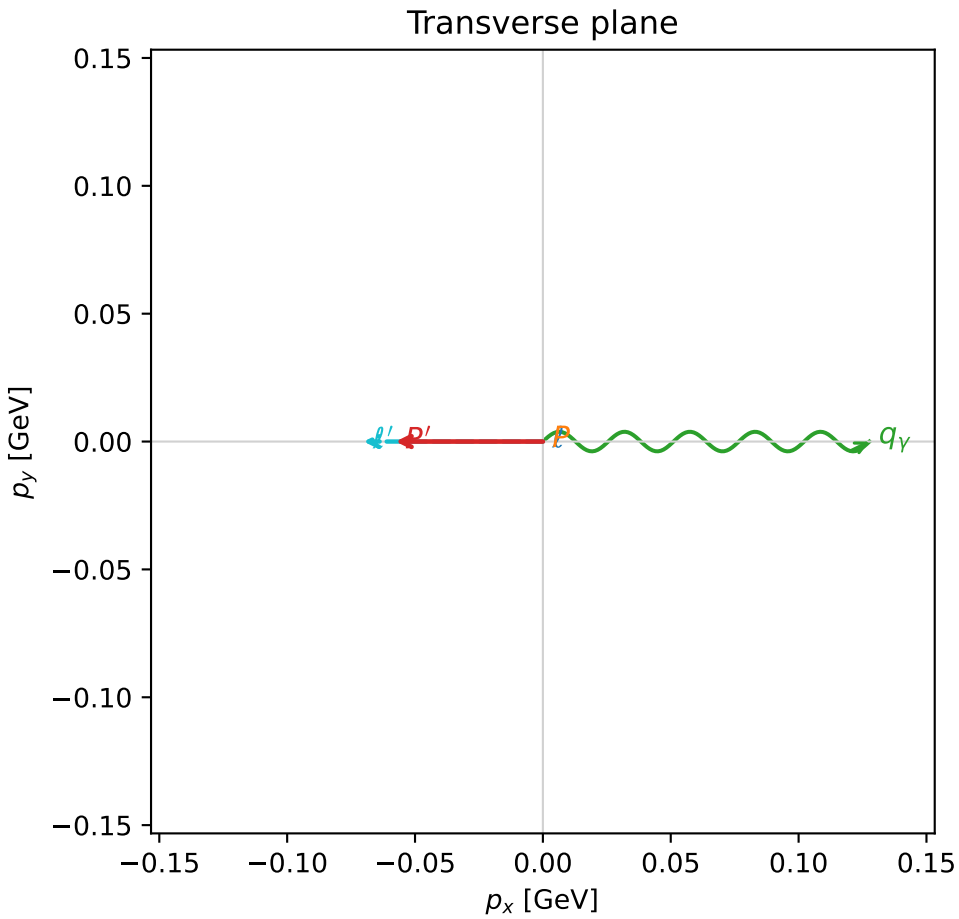
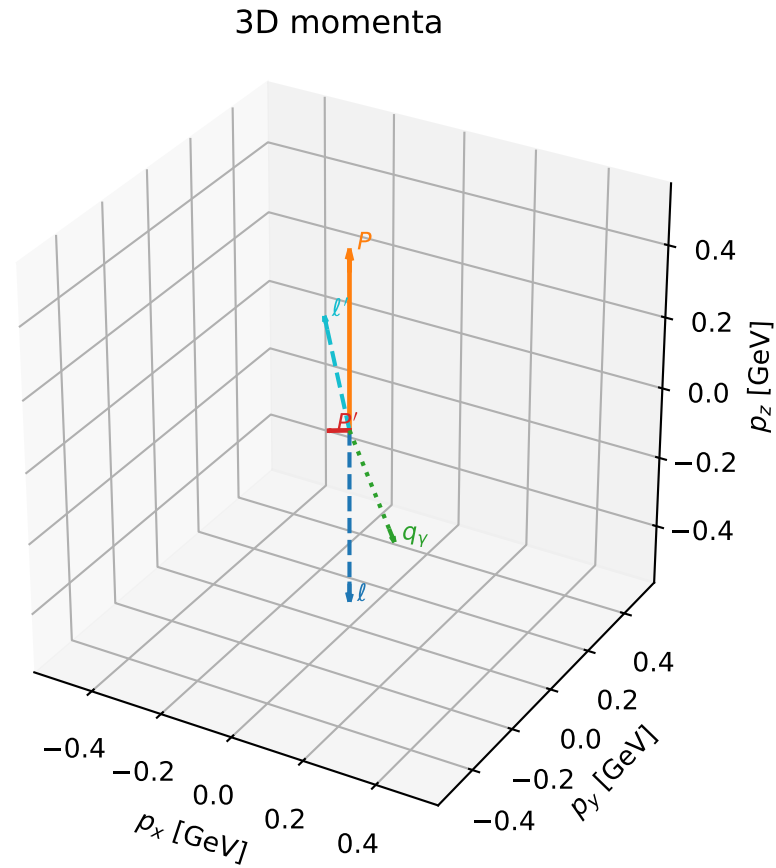
$\text{sqrt_s} = 3.020096$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 3.11231$
 $\text{q0out} = 1.041046$
 $\text{phi_p_out} = 0.5188867$
 $\text{phi_gamma_out} = 6.283172$
 $\text{alpha_e} = 1.543859$
 $\text{alpha_p} = 0.02635473$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

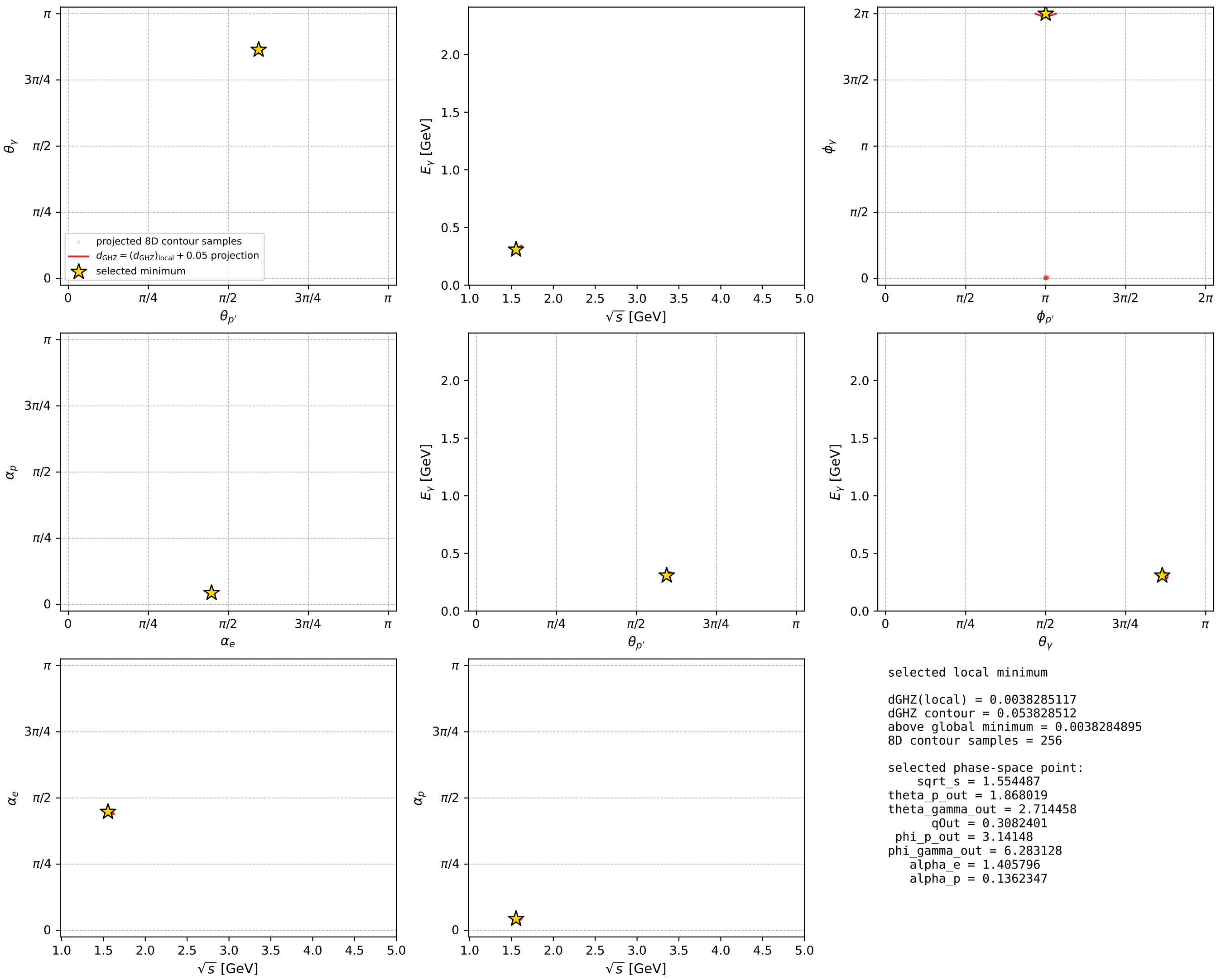
dghz_mixing_angles_polarization_01_local_minimum_365 $d_{\{\text{rm GHZ}\}}=0.00382851$
 region: polarization_cluster_01_local_minimum_365
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.4057965$ rad
 incoming lepton: $0.164252|+\rangle +0.986418|-\rangle$
 proton mixing angle: $\alpha_p=0.13623467$ rad
 incoming proton: $0.990734|+\rangle +0.135814|-\rangle$

$s=2.41643$, $\sqrt{s}=1.55449$, $|\vec{P}|=0.494242$, $|\vec{P}'|=0.0602196$
 $\theta_{P'}=1.86802$, $\theta_Y=2.71446$, $\phi_{P'}=3.14148$, $\phi_Y=6.28313$
 $E_Y=0.30824$, $\theta_{\ell'}=0.230941$, $\phi_{\ell'}=3.14158$
 $Q^2=0.597537$, $x_B=0.505618$, $t=-0.250859$, $W^2=1.4641$, $y=0.769105$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4942$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4942$
 P $E= 1.0602$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4942$
 ℓ' $E= 0.3063$ $p_x= -0.0701$ $p_y= 0.0000$ $p_z= 0.2982$
 P' $E= 0.9399$ $p_x= -0.0576$ $p_y= 0.0000$ $p_z= -0.0176$
 q_Y $E= 0.3082$ $p_x= 0.1277$ $p_y= -0.0000$ $p_z= -0.2805$



electron: polarization cluster P1, local minimum 365; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour

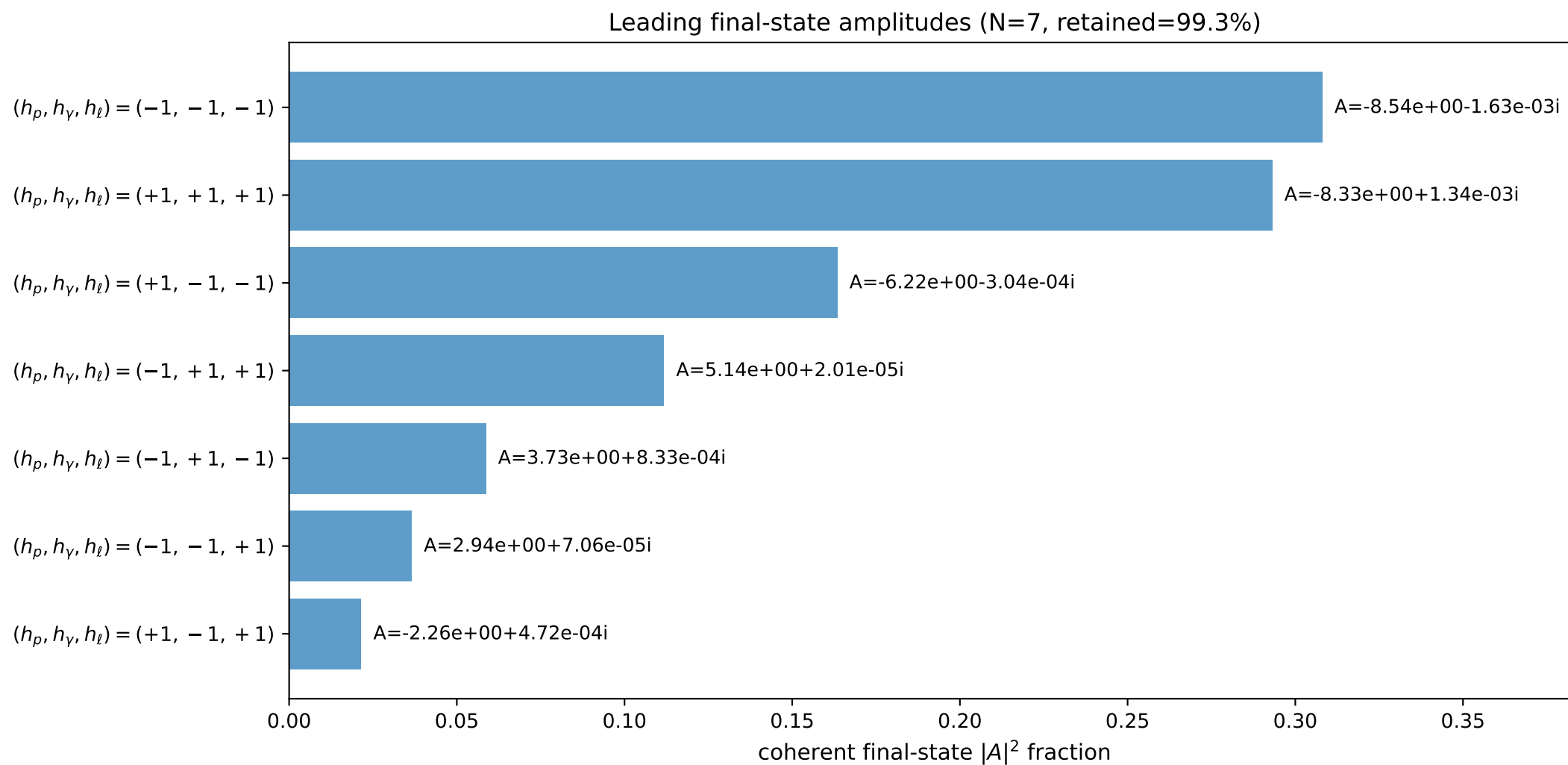
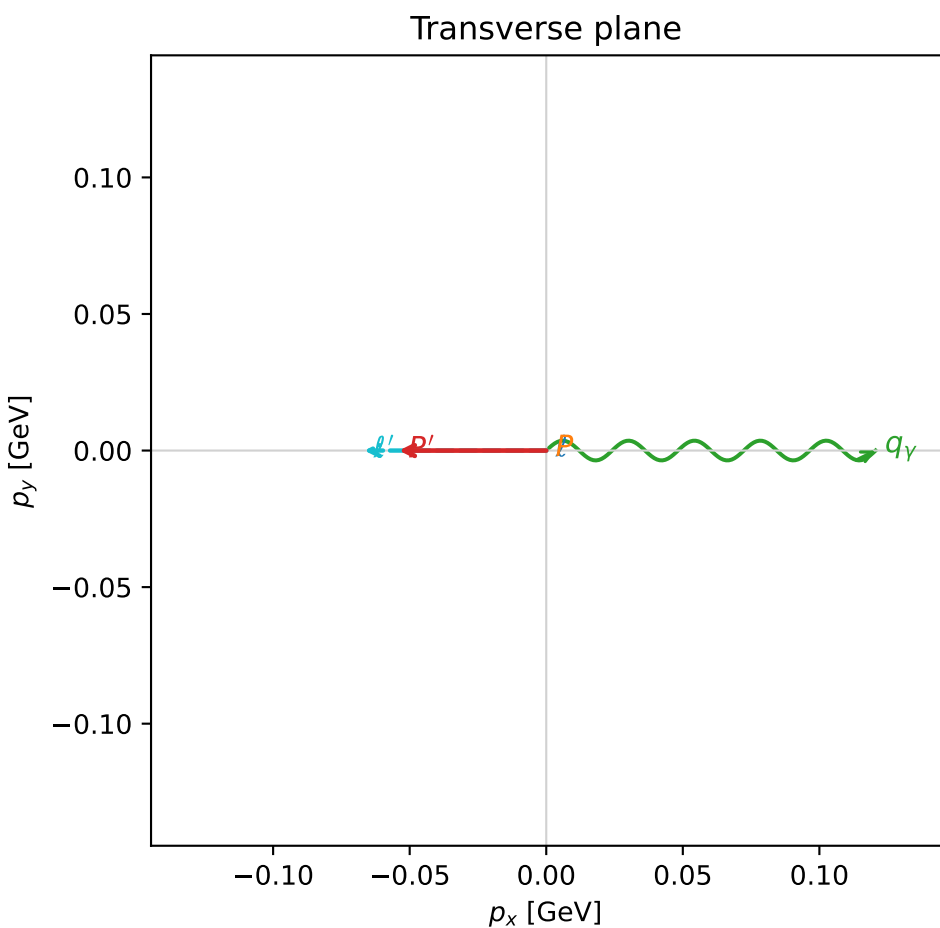
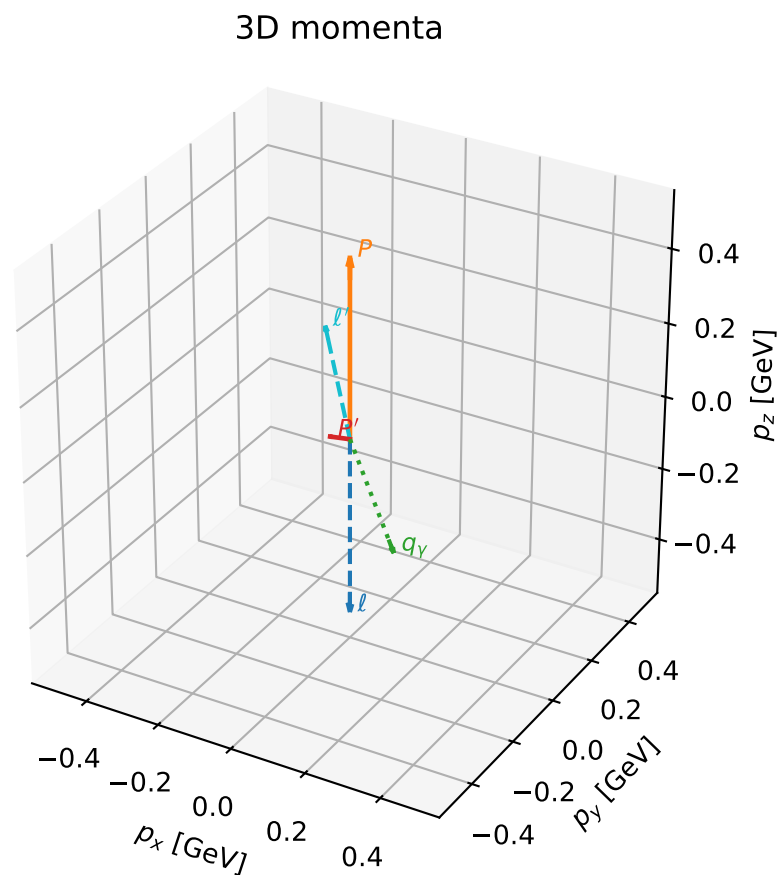


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

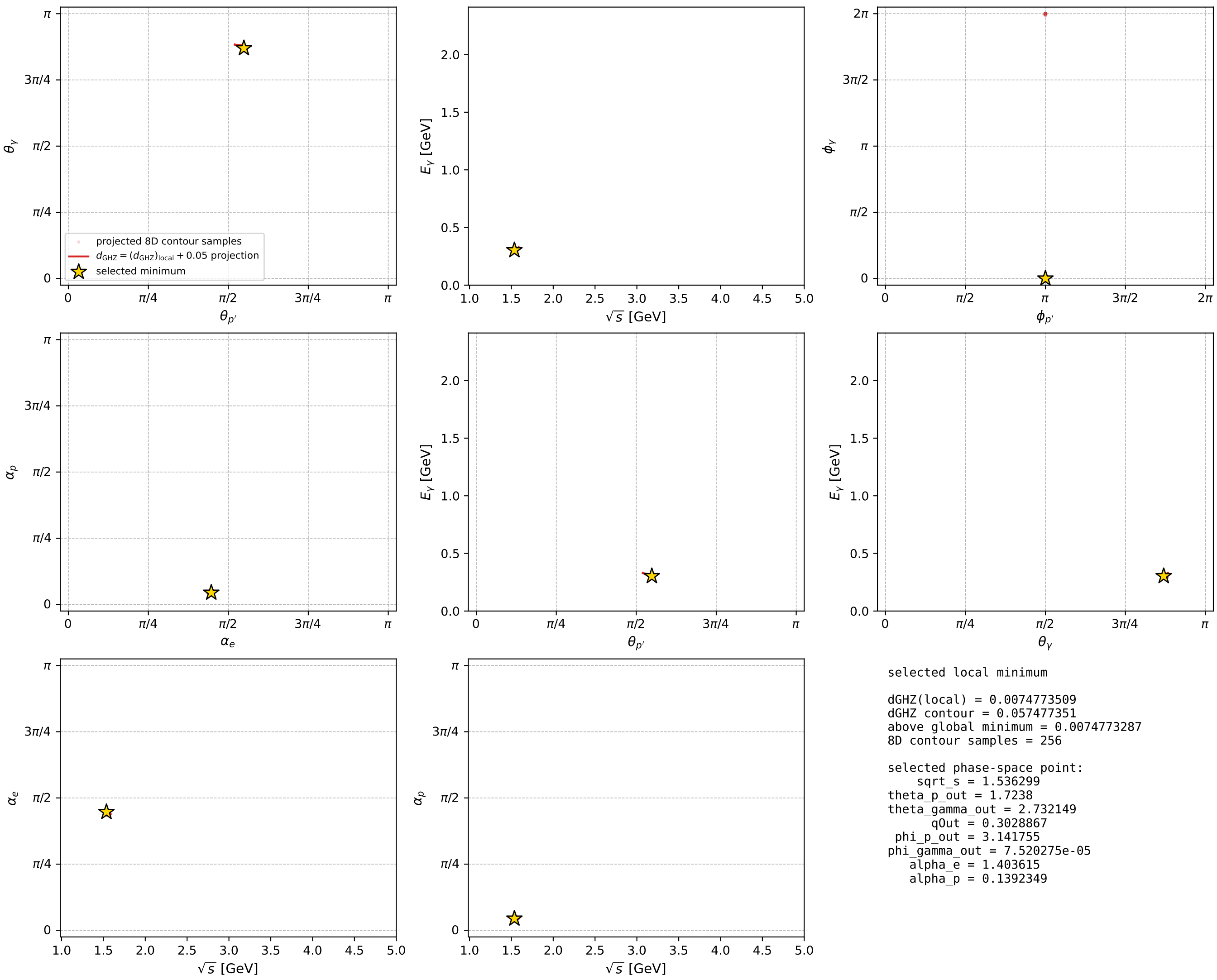
dghz_mixing_angles_polarization_01_local_minimum_366 $d_{\{\text{rm GHZ}\}}=0.00747735$
 region: polarization_cluster_01_local_minimum_366
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.4036152$ rad
 incoming lepton: $0.166403|+\rangle + 0.986058|-\rangle$
 proton mixing angle: $\alpha_p=0.13923488$ rad
 incoming proton: $0.990322|+\rangle + 0.138785|-\rangle$

$s=2.36021$, $\sqrt{s}=1.5363$, $|\vec{P}|=0.481797$, $|\vec{P}'|=0.0545814$
 $\theta_P=1.7238$, $\theta_Y=2.73215$, $\phi_{P'}=3.14176$, $\phi_Y=7.52028\text{e-}05$
 $E_Y=0.302887$, $\theta_{\ell'}=0.228776$, $\phi_{\ell'}=3.1416$
 $Q^2=0.55888$, $x_B=0.49178$, $t=-0.229918$, $W^2=1.45741$, $y=0.767675$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4818$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4818$
 P $E= 1.0545$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4818$
 ℓ' $E= 0.2938$ $p_x= -0.0666$ $p_y= -0.0000$ $p_z= 0.2862$
 P' $E= 0.9396$ $p_x= -0.0539$ $p_y= -0.0000$ $p_z= -0.0083$
 q_Y $E= 0.3029$ $p_x= 0.1206$ $p_y= 0.0000$ $p_z= -0.2779$



electron: polarization cluster P1, local minimum 366; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



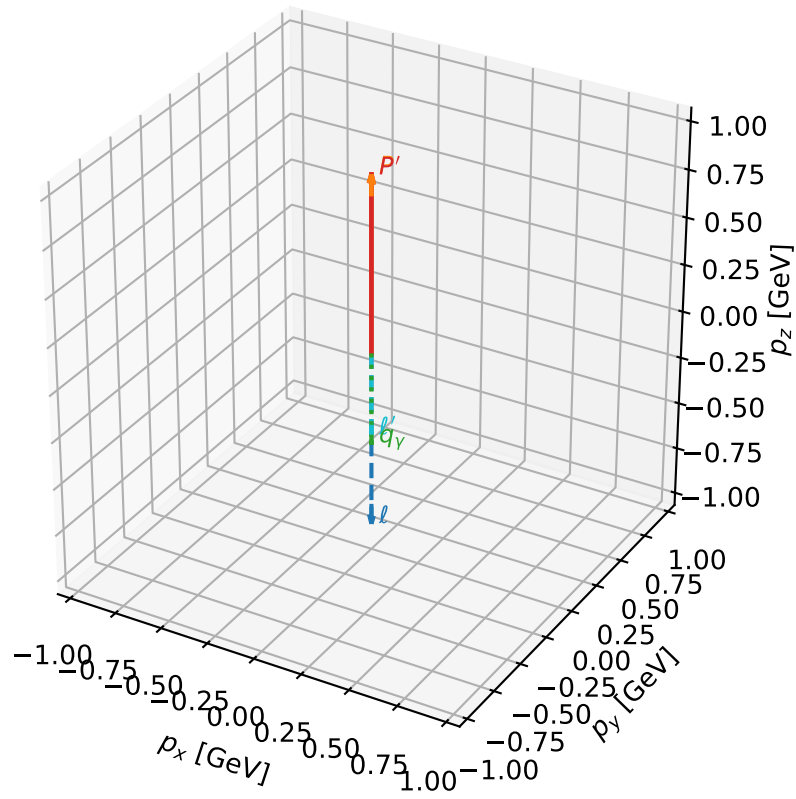
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_367 $d_{\{\text{rm GHZ}\}}=0.0171798$
 region: polarization_cluster_01_local_minimum_367
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=6.1169479\text{e-}06$ rad
 incoming lepton: $1|+\rangle +6.11695\text{e-}06|-\rangle$
 proton mixing angle: $\alpha_p=0$ rad
 incoming proton: $1|+\rangle +0|-\rangle$

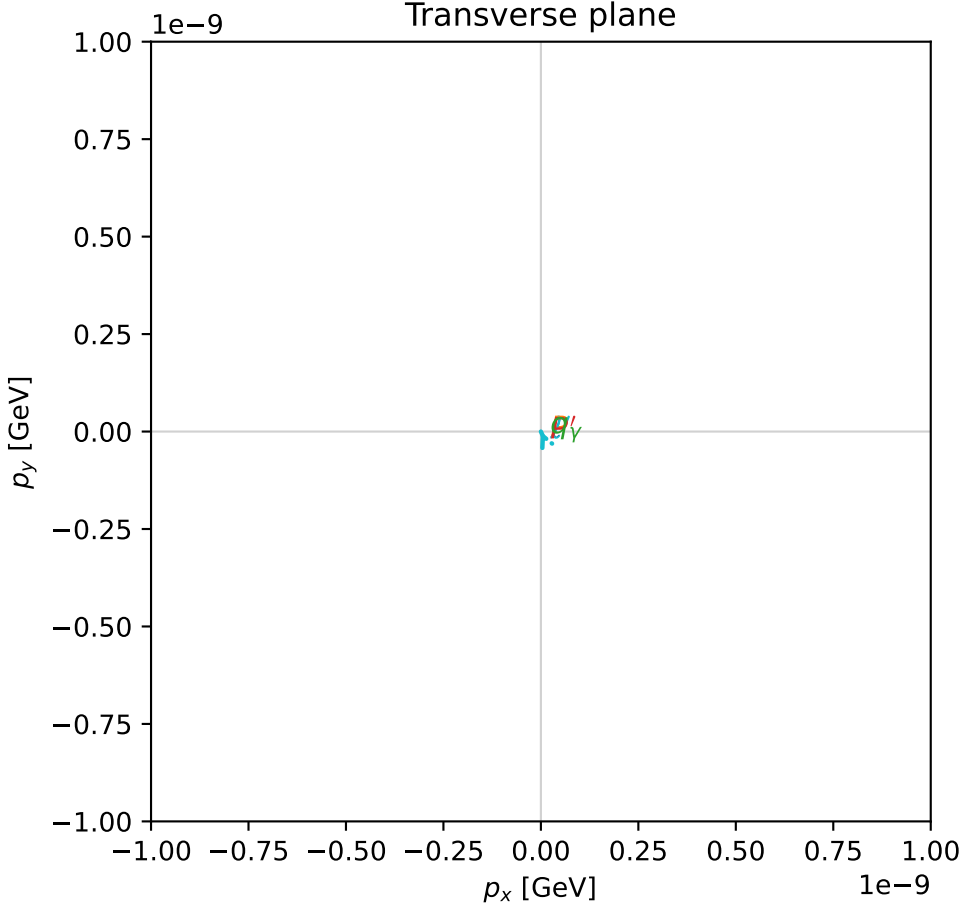
$s=5.06282$, $\sqrt{s}=2.25007$, $|\vec{P}|=0.929521$, $|\vec{P}'|=0.929521$
 $\theta_{P'}=0$, $\theta_V=3.14159$, $\phi_{P'}=2.9999$, $\phi_V=5.13056$
 $E_V=0.493368$, $\theta_{l'}=3.14159$, $\phi_{l'}=1.98897$
 $Q^2=1.56777\text{e-}07$, $x_B=7.06132\text{e-}08$, $t=-4.3871\text{e-}15$, $W^2=3.10007$, $y=0.530776$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.9295$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9295$
 P $E= 1.3206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9295$
 l' $E= 0.4362$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.4362$
 P' $E= 1.3206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.9295$
 q_V $E= 0.4934$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.4934$

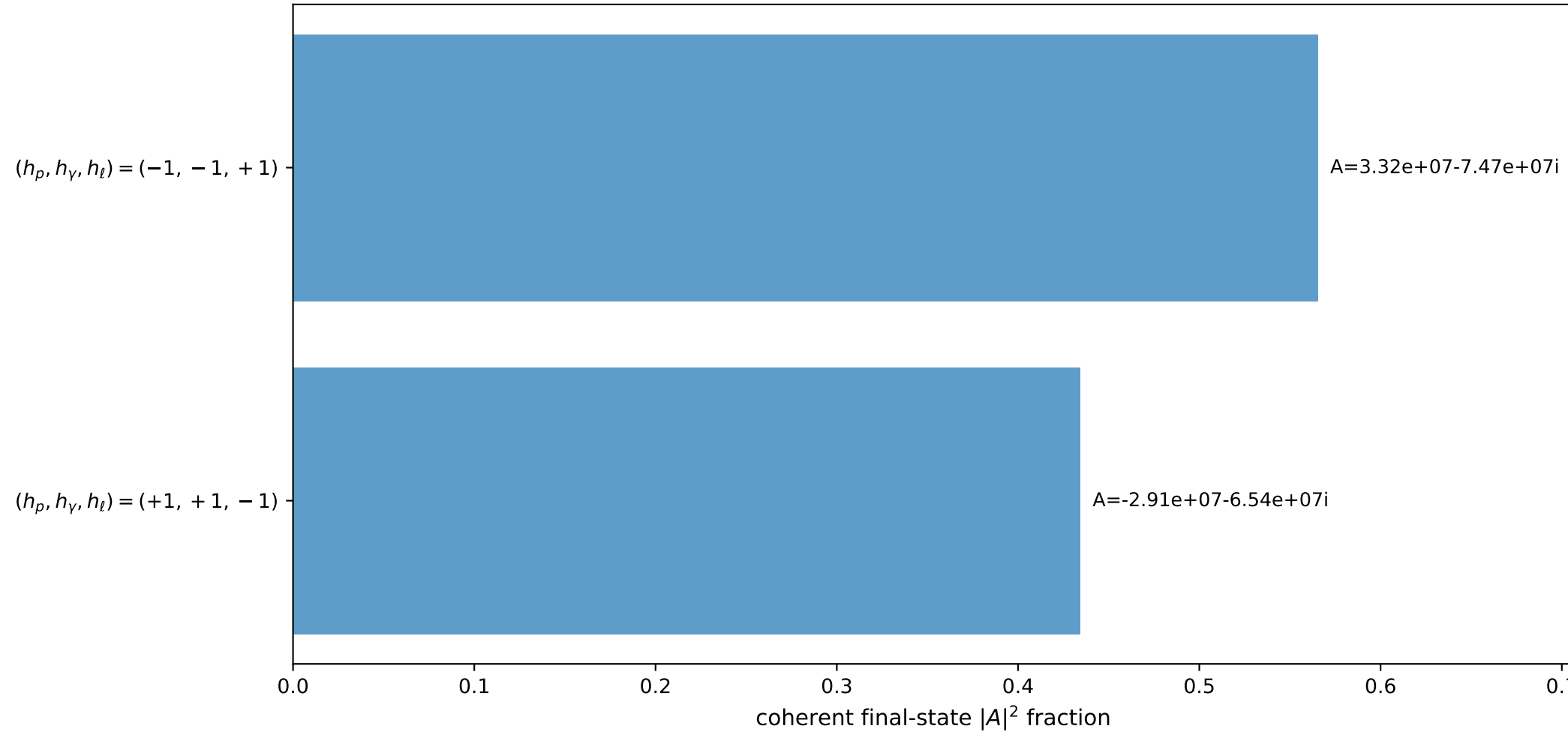
3D momenta



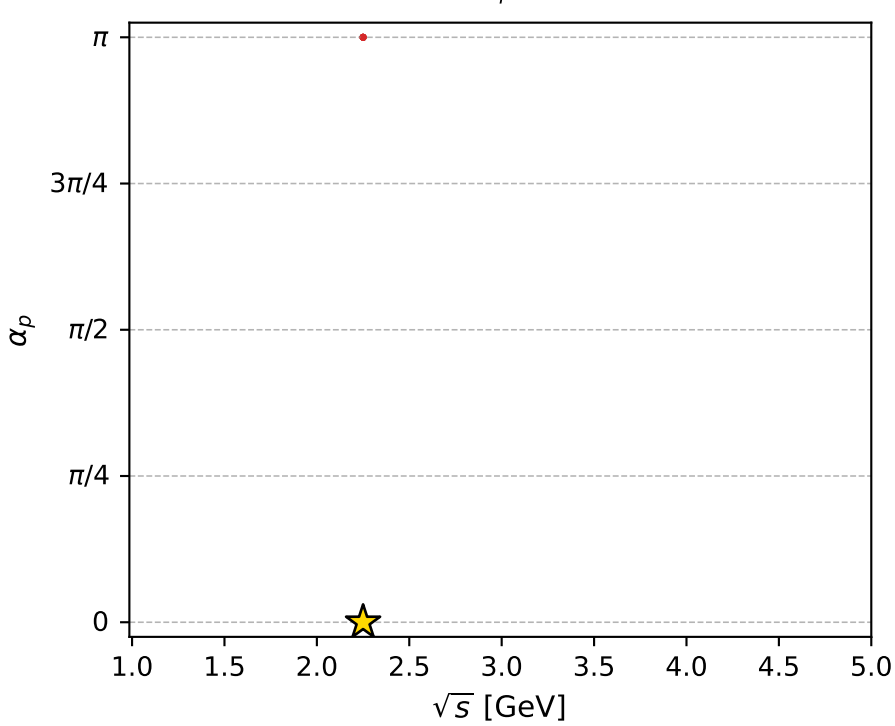
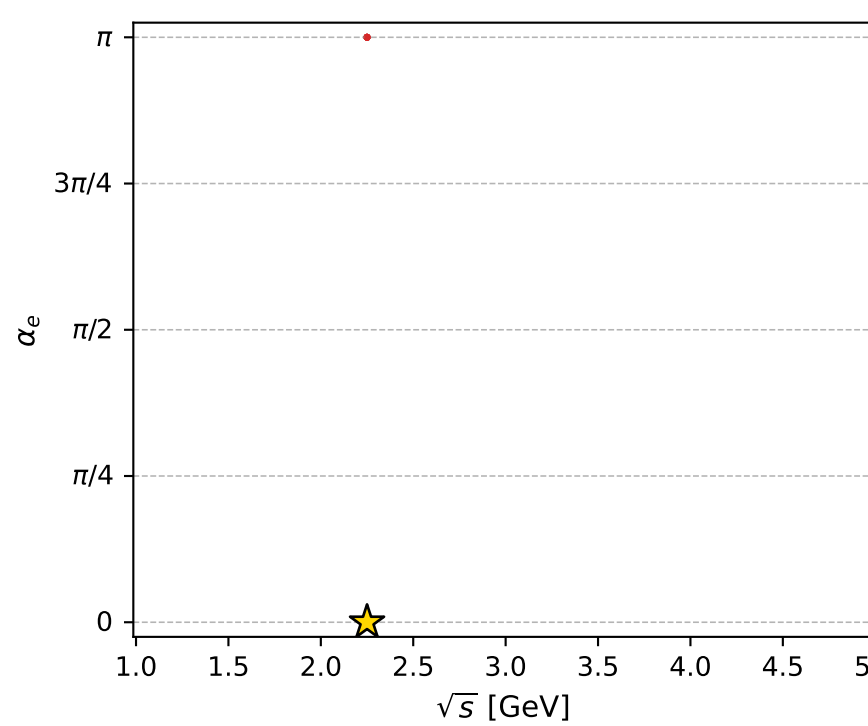
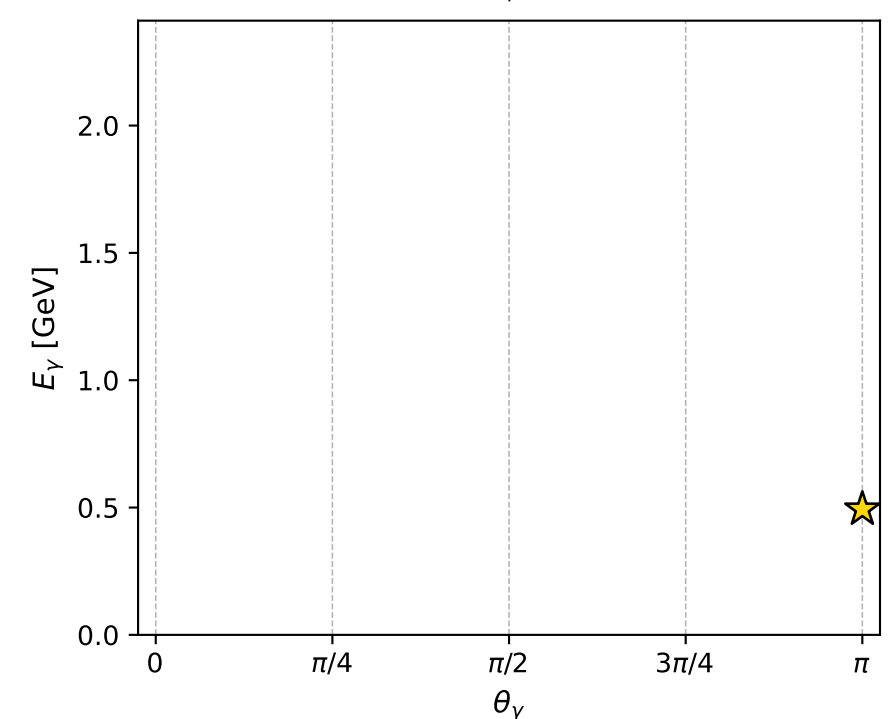
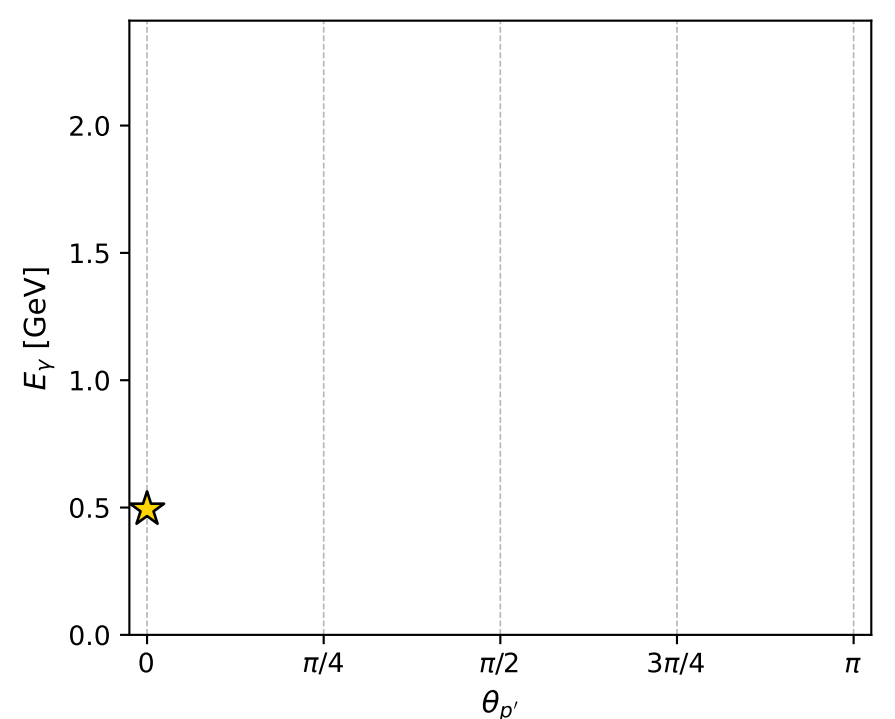
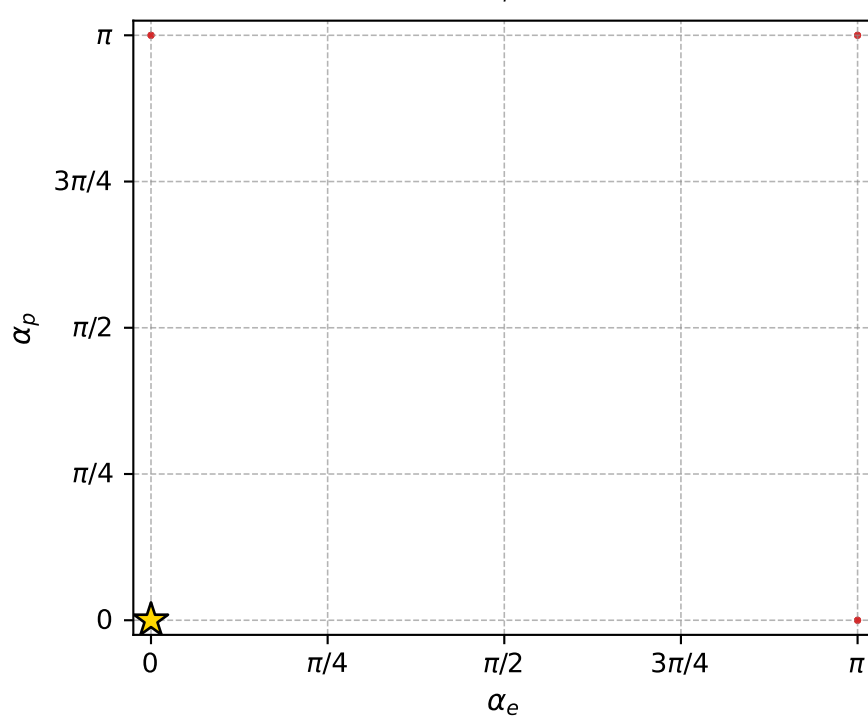
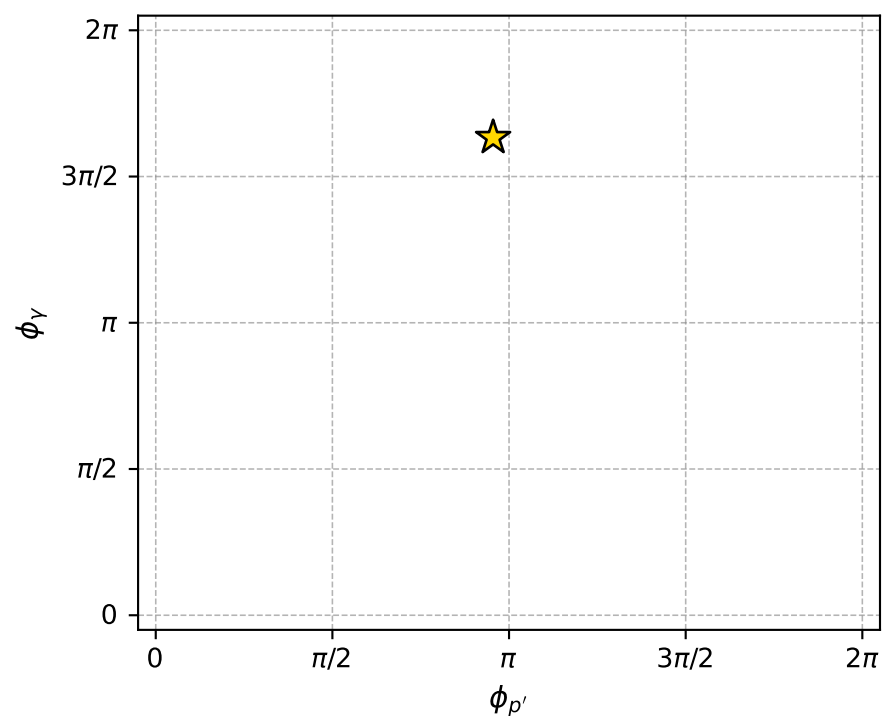
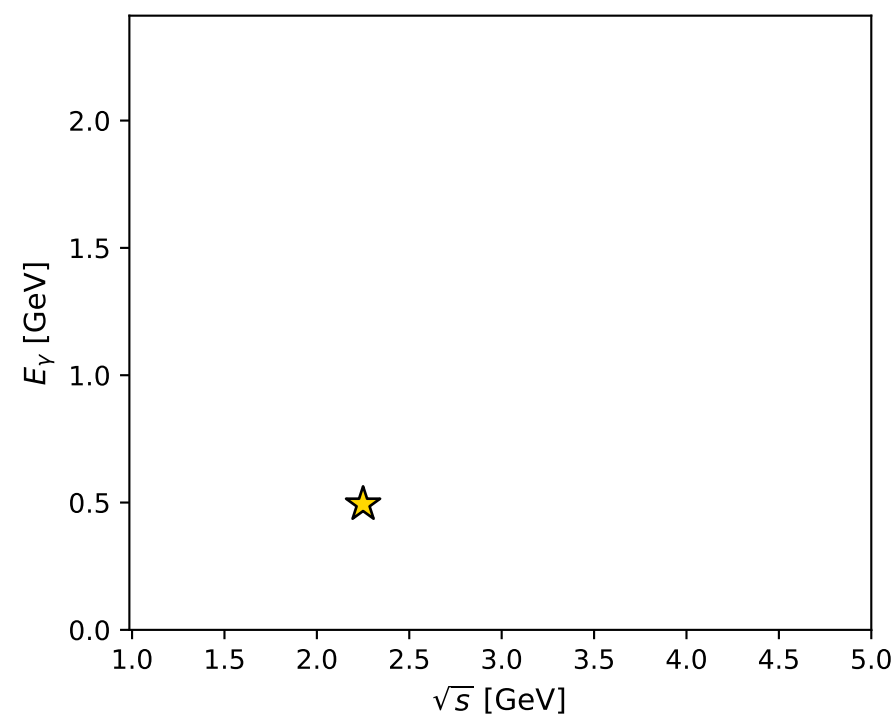
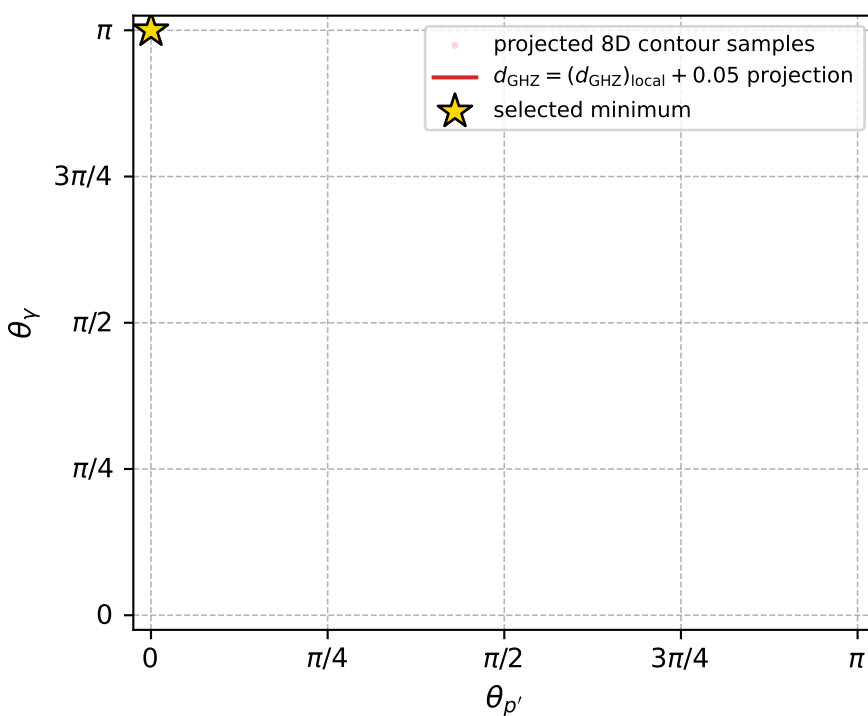
Transverse plane



Leading final-state amplitudes (N=2, retained=100.0%)



electron: polarization cluster P1, local minimum 367; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.017179829$
 $d_{\text{GHZ}} \text{ contour} = 0.067179829$
 above global minimum = 0.017179807
 8D contour samples = 80

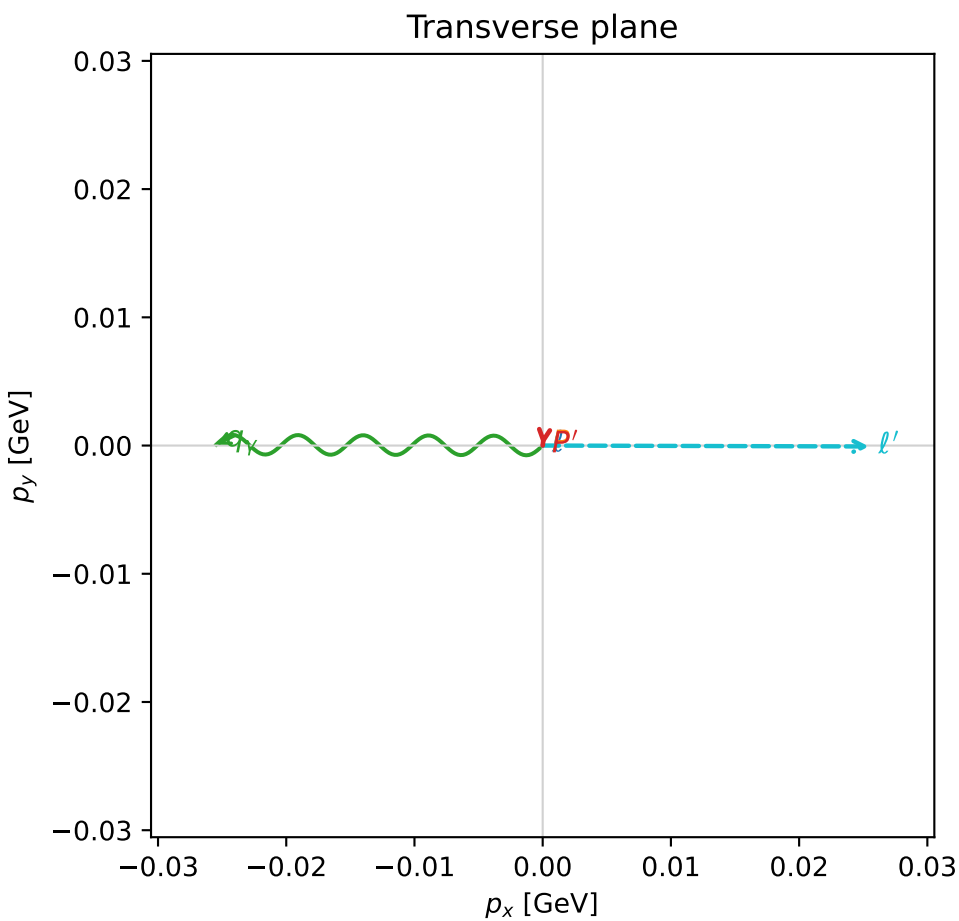
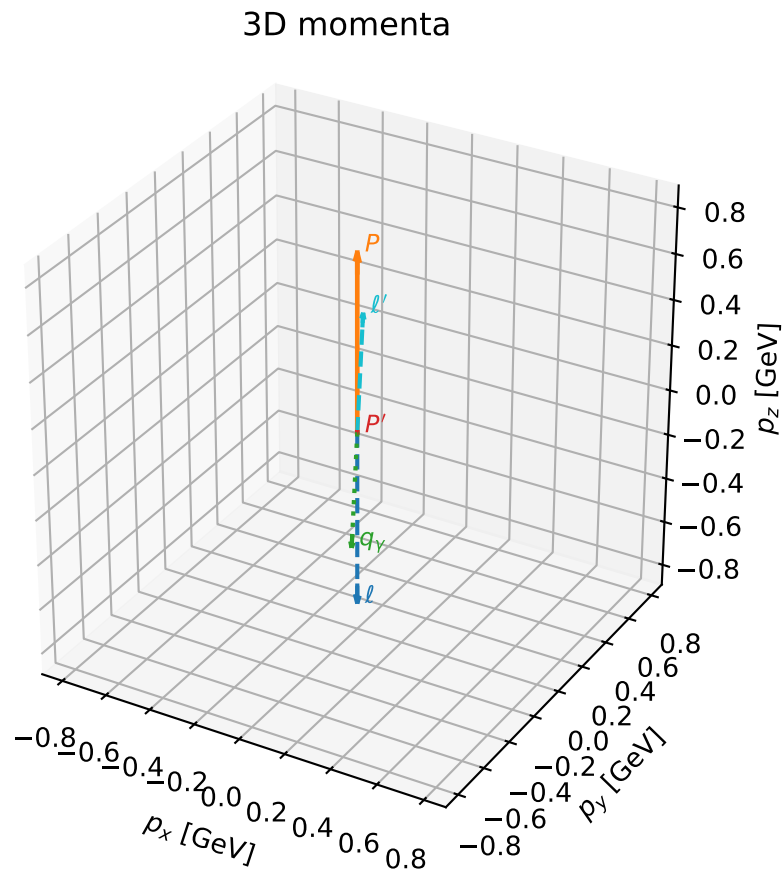
selected phase-space point:
 $\text{sqrt_s} = 2.250072$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 3.141593$
 $\text{q0out} = 0.4933678$
 $\text{phi_p_out} = 2.999902$
 $\text{phi_gamma_out} = 5.130563$
 $\text{alpha_e} = 6.116948\text{e-}06$
 $\text{alpha_p} = 0$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_369 d_{\rm GHZ}=0.0208267
 region: polarization_cluster_01_local_minimum_369
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.5707901$ rad
 incoming lepton: $6.21409\text{e-}06|+\rangle +1|-\rangle$
 proton mixing angle: $\alpha_p=3.1415775$ rad
 incoming proton: $-1|+\rangle +1.51327\text{e-}05|-\rangle$

$s=3.97747$, $\sqrt{s}=1.99436$, $|\vec{P}|=0.776596$, $|\vec{P}'|=4.18979\text{e-}06$
 $\theta_{P'}=3.14157$, $\theta_Y=3.09338$, $\phi_{P'}=4.59353$, $\phi_Y=3.13875$
 $E_Y=0.528177$, $\theta_{\ell'}=0.0482106$, $\phi_{\ell'}=6.28034$
 $Q^2=1.63978$, $x_B=0.62334$, $t=-0.524841$, $W^2=1.8707$, $y=0.849244$

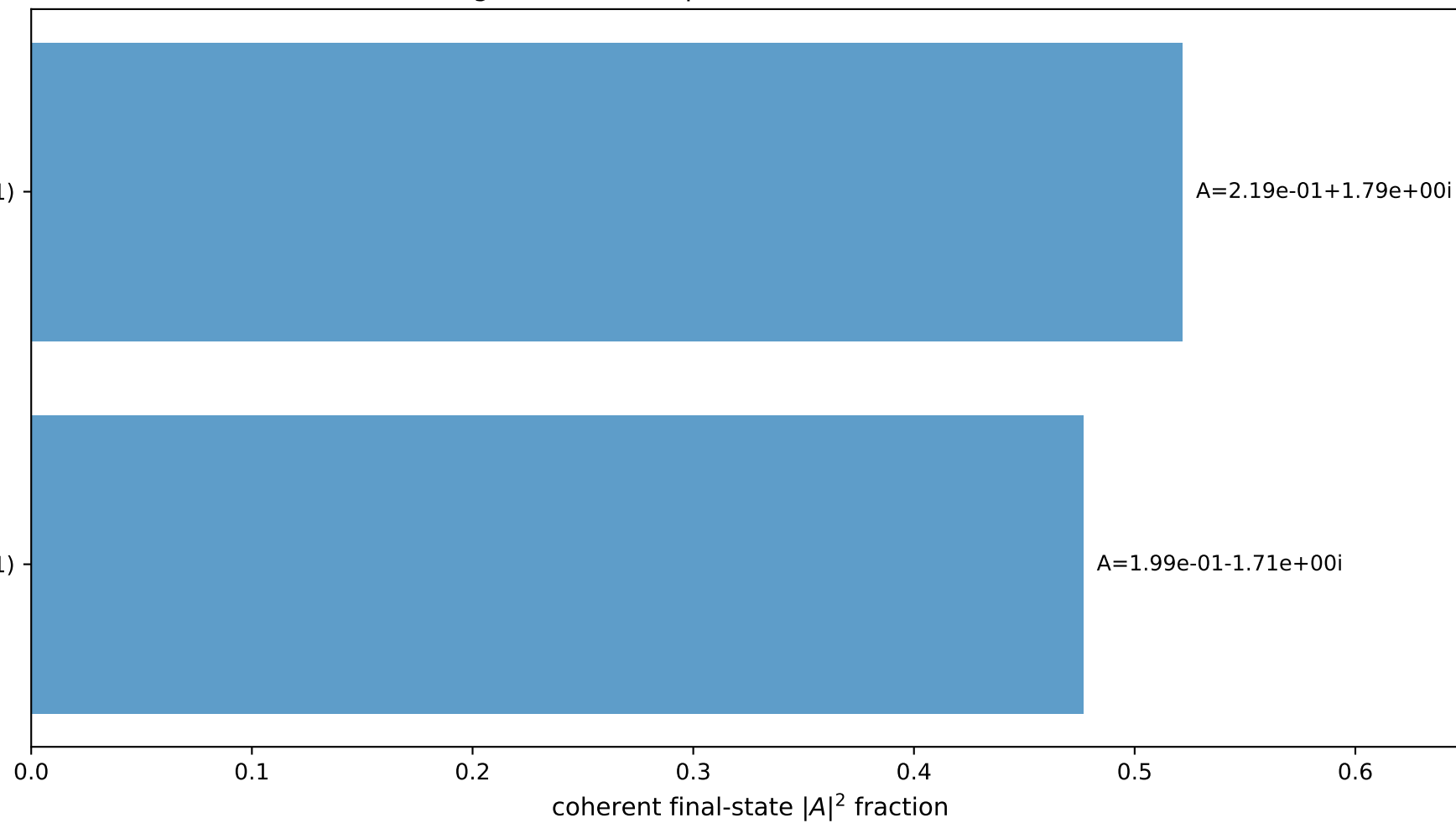
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.7766$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7766$
 P $E= 1.2178$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7766$
 ℓ' $E= 0.5282$ $p_x= 0.0255$ $p_y= -0.0001$ $p_z= 0.5276$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_Y $E= 0.5282$ $p_x= -0.0255$ $p_y= 0.0001$ $p_z= -0.5276$



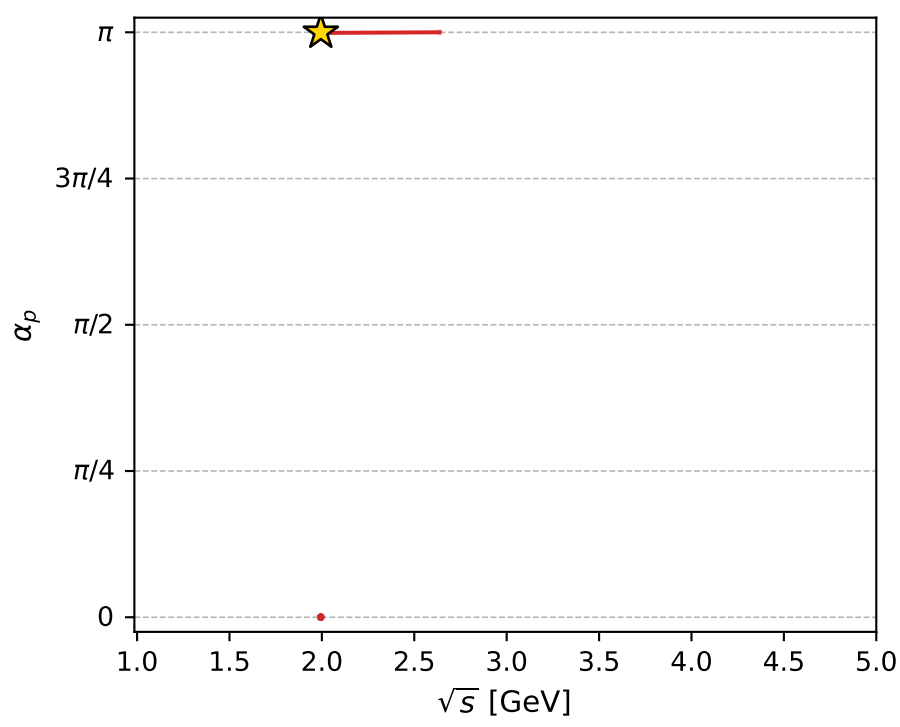
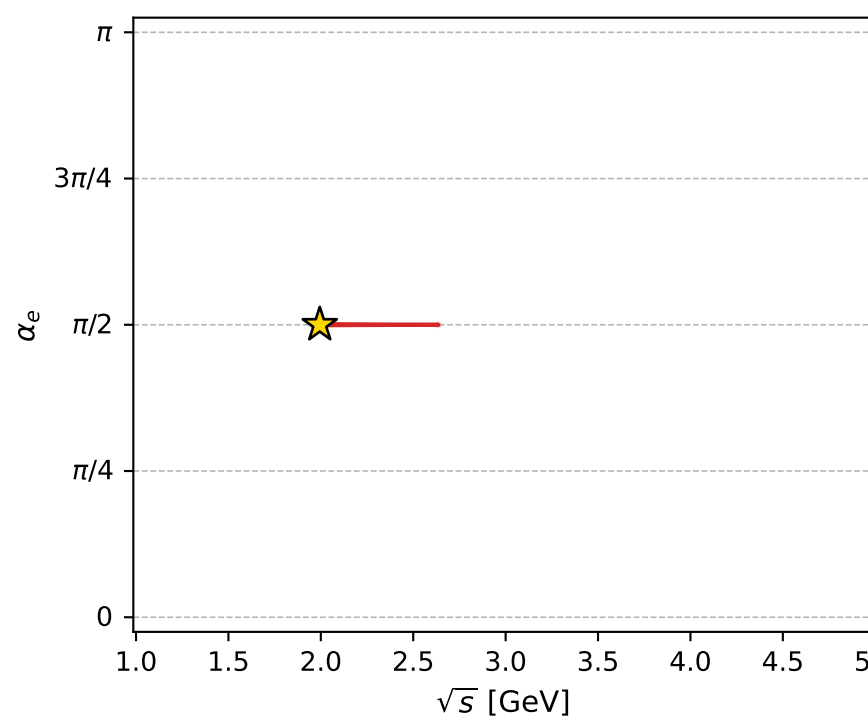
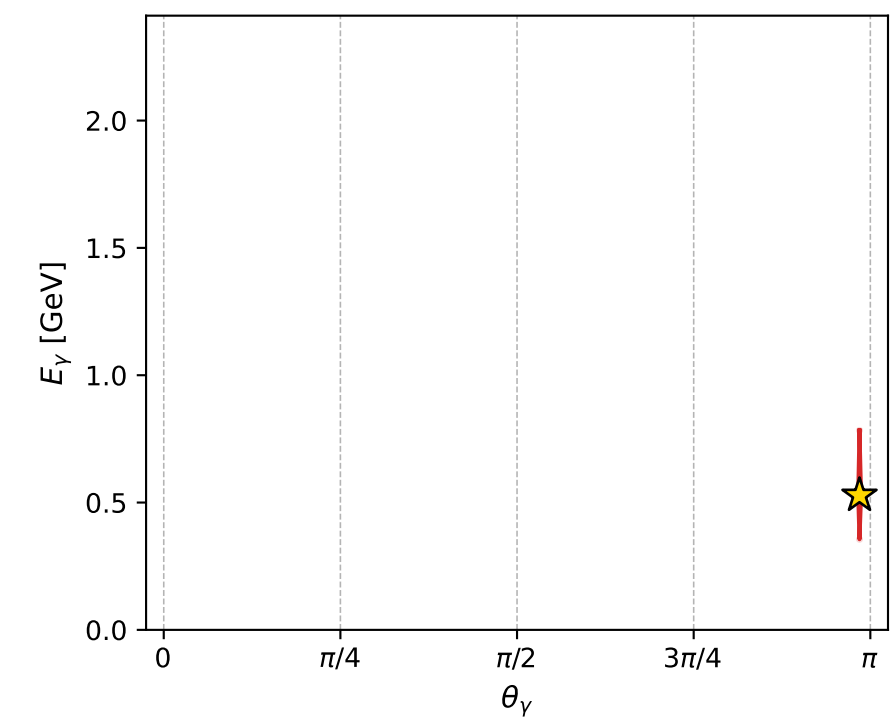
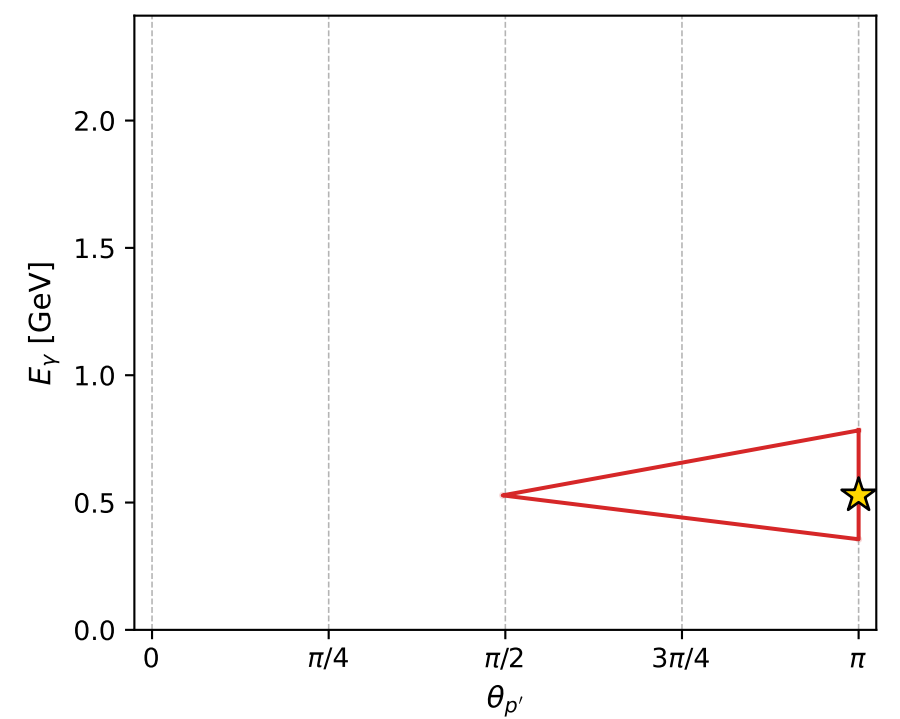
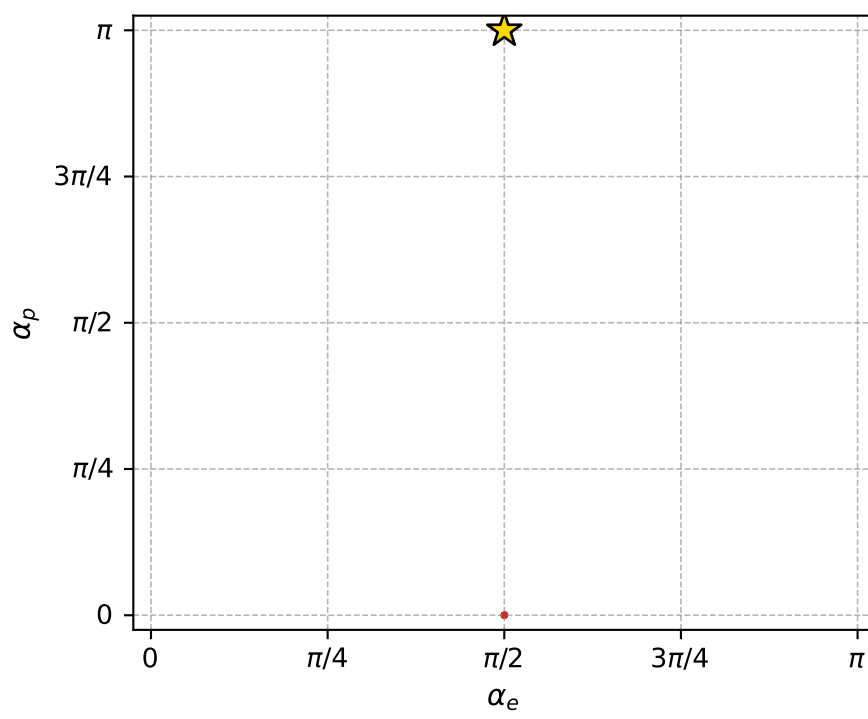
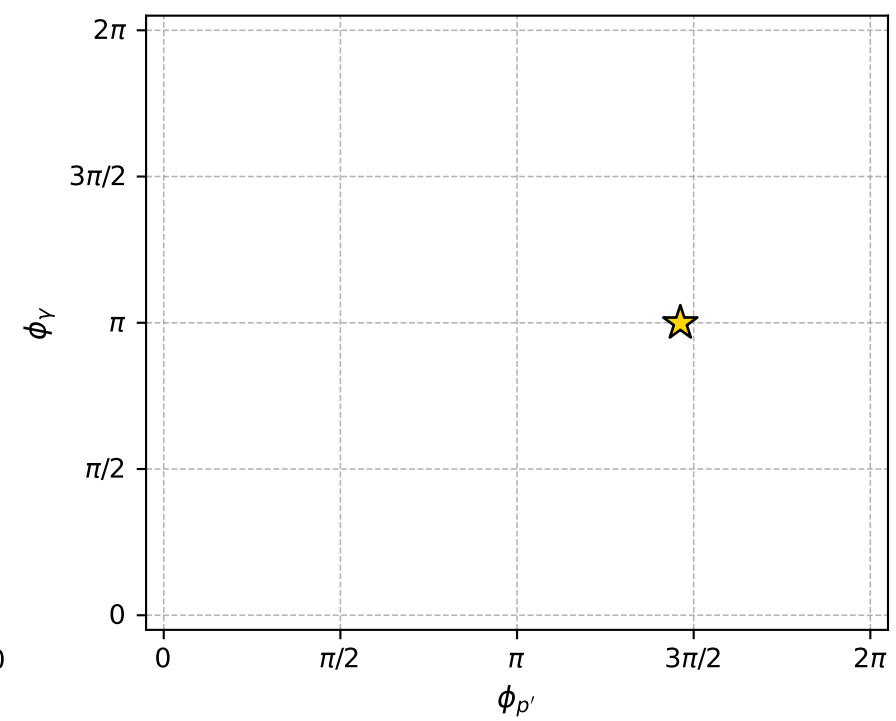
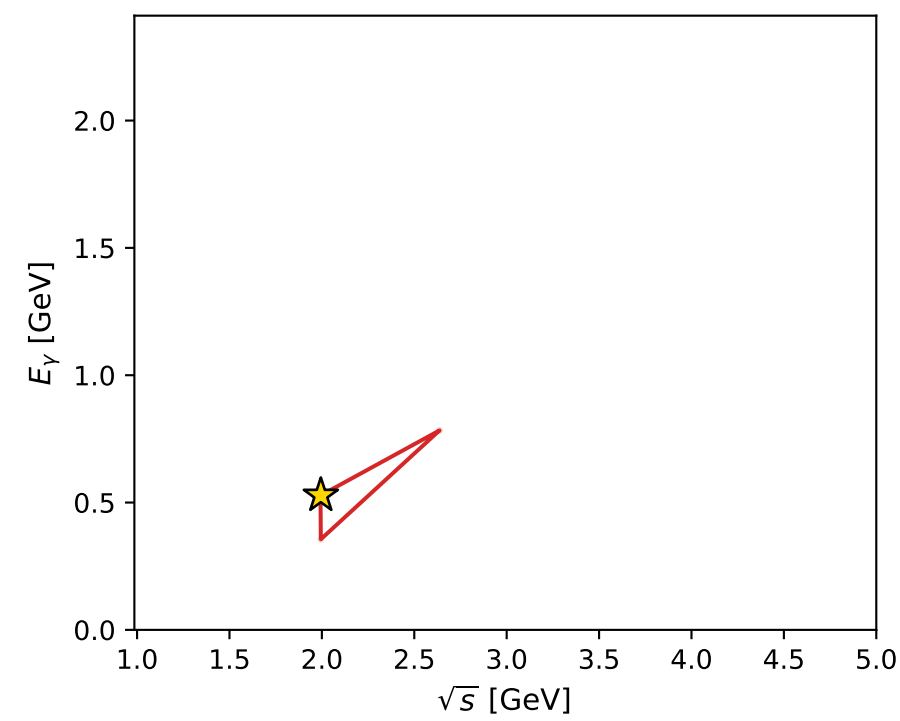
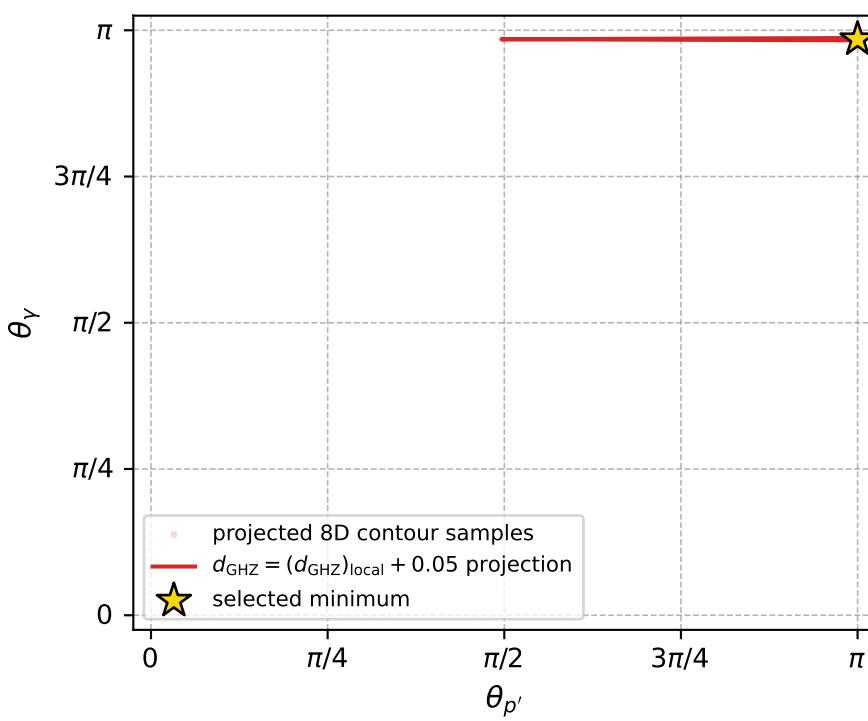
$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

Leading final-state amplitudes (N=2, retained=99.8%)



electron: polarization cluster P1, local minimum 369; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.020826668$
 $d_{\text{GHZ}} \text{ contour} = 0.070826668$
 above global minimum = 0.020826646
 8D contour samples = 176

selected phase-space point:

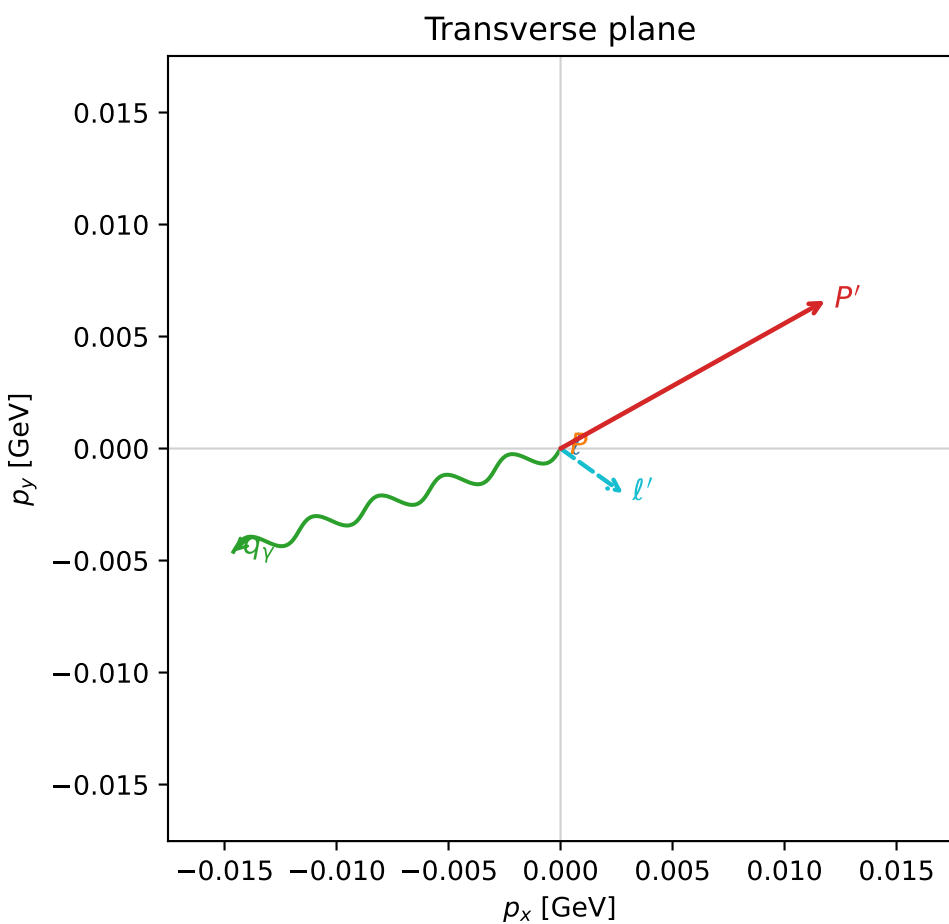
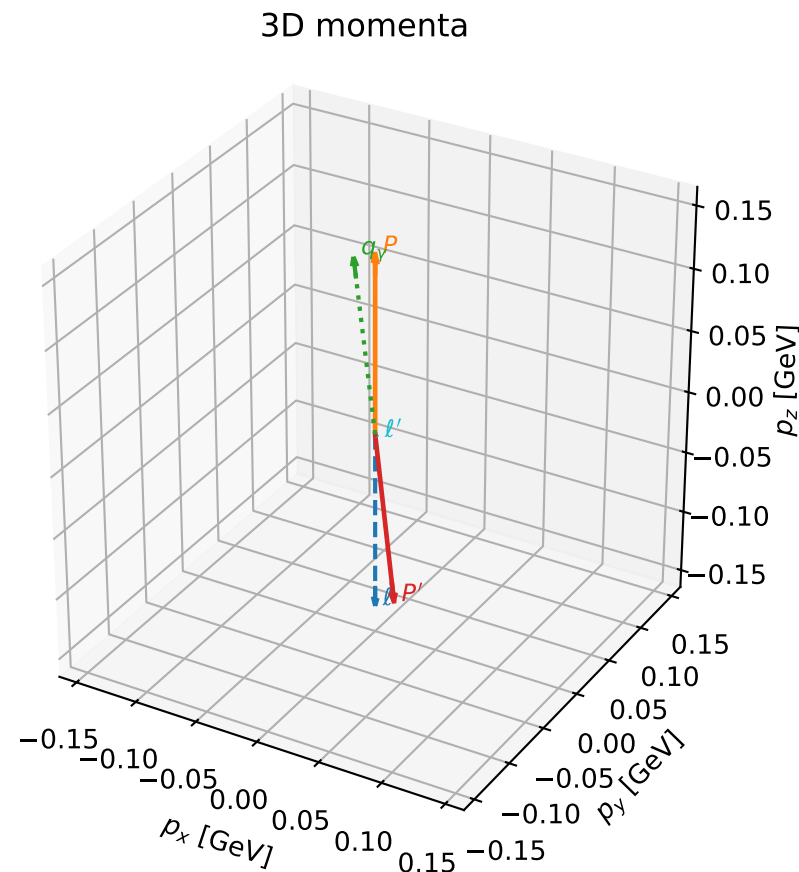
$\text{sqrt_s} = 1.994359$
 $\text{theta_p_out} = 3.141568$
 $\text{theta_gamma_out} = 3.093382$
 $\text{q0out} = 0.5281771$
 $\text{phi_p_out} = 4.593525$
 $\text{phi_gamma_out} = 3.138751$
 $\text{alpha_e} = 1.57079$
 $\text{alpha_p} = 3.141578$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

dghz_mixing_angles_polarization_01_local_minimum_373 $d_{\{\text{rm GHZ}\}}=0.0300609$
 region: polarization_cluster_01_local_minimum_373
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.74214409$ rad
 incoming lepton: $0.737021|+\rangle +0.67587|-\rangle$
 proton mixing angle: $\alpha_p=0.90015049$ rad
 incoming proton: $0.621492|+\rangle +0.78342|-\rangle$

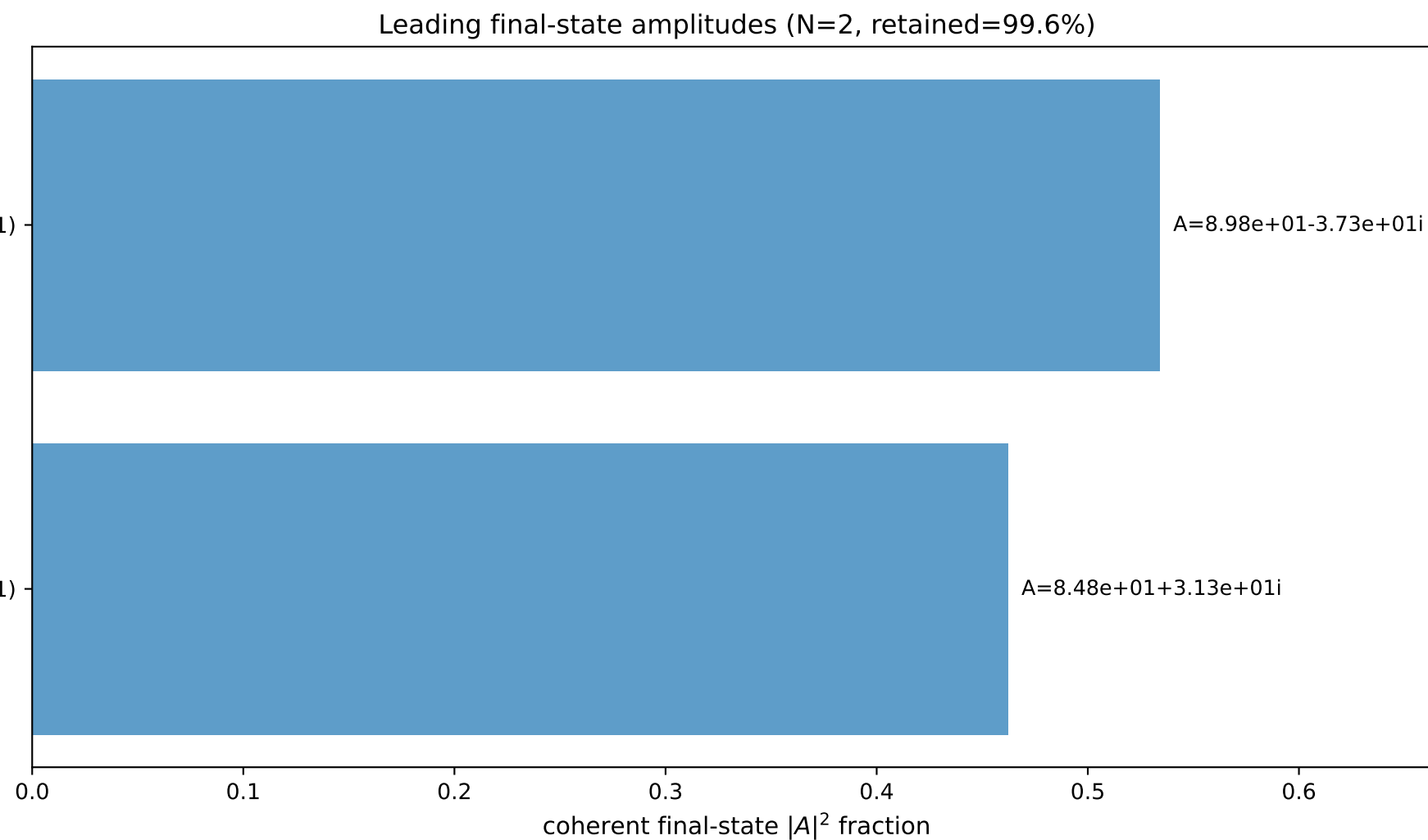
$s=1.19232$, $\sqrt{s}=1.09193$, $|\vec{P}|=0.143083$, $|\vec{P}'|=0.139544$
 $\theta_{P'}=3.04448$, $\theta_Y=0.109503$, $\phi_{P'}=0.509518$, $\phi_Y=3.44722$
 $E_Y=0.140121$, $\theta_{\ell'}=1.68565$, $\phi_{\ell'}=5.66363$
 $Q^2=0.000885013$, $x_B=0.00289466$, $t=-0.0796895$, $W^2=1.1847$, $y=0.978442$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1431$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1431$
 P $E= 0.9489$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1431$
 ℓ' $E= 0.0035$ $p_x= 0.0028$ $p_y= -0.0020$ $p_z= -0.0004$
 P' $E= 0.9483$ $p_x= 0.0118$ $p_y= 0.0066$ $p_z= -0.1389$
 q_Y $E= 0.1401$ $p_x= -0.0146$ $p_y= -0.0046$ $p_z= 0.1393$

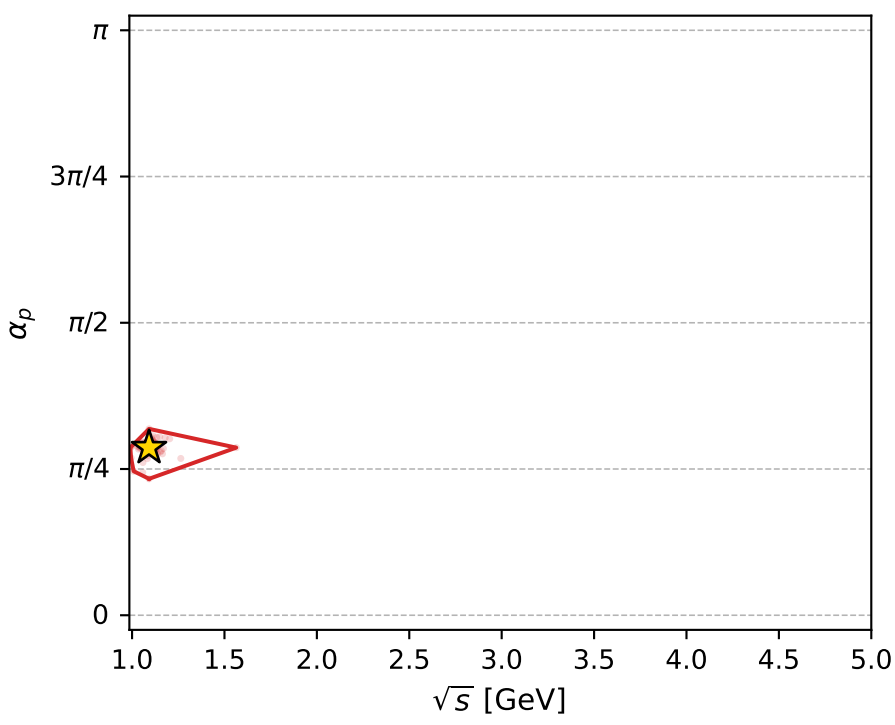
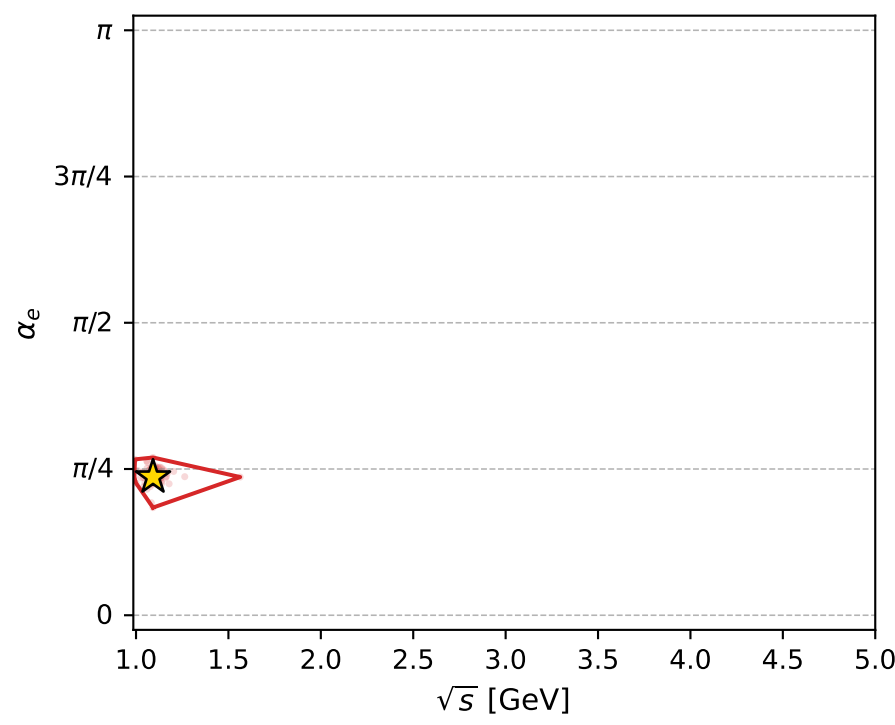
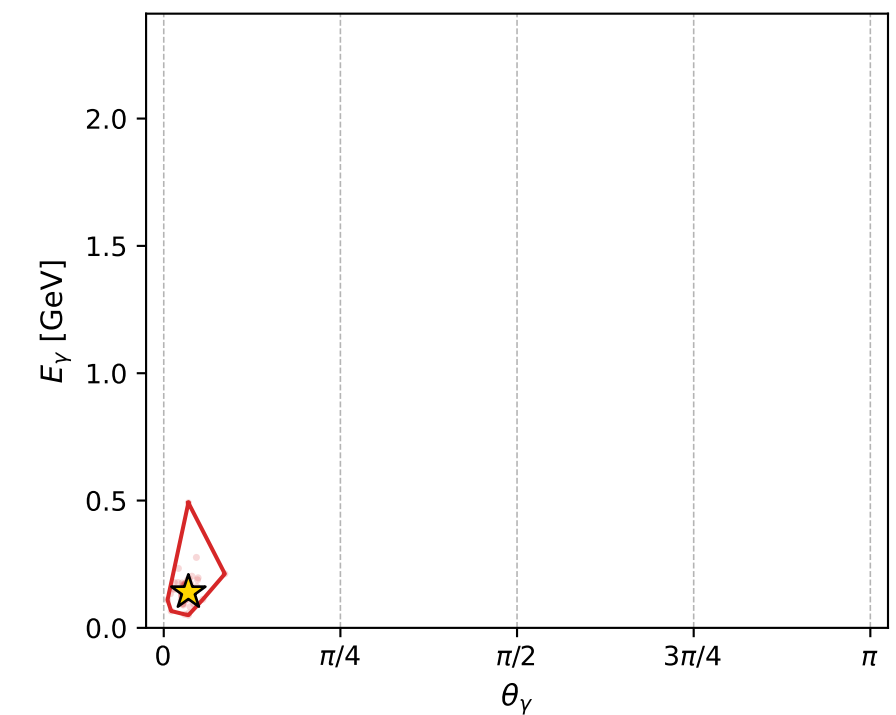
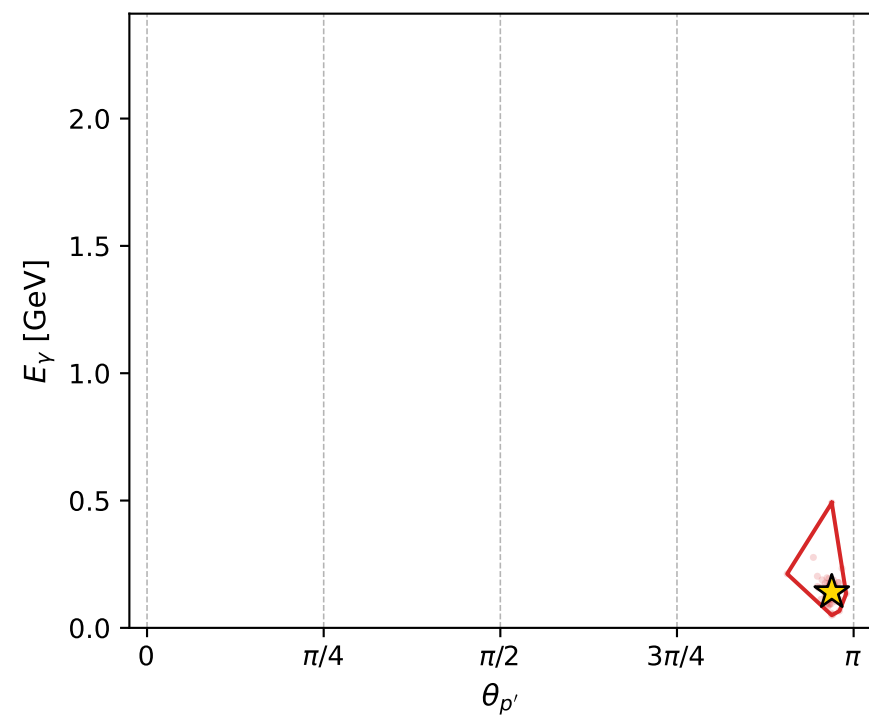
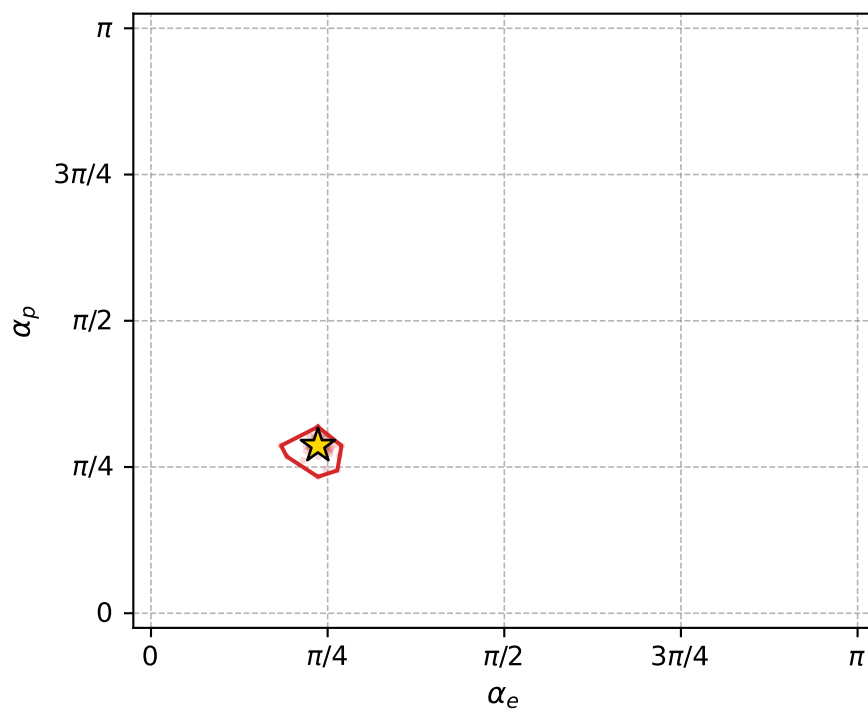
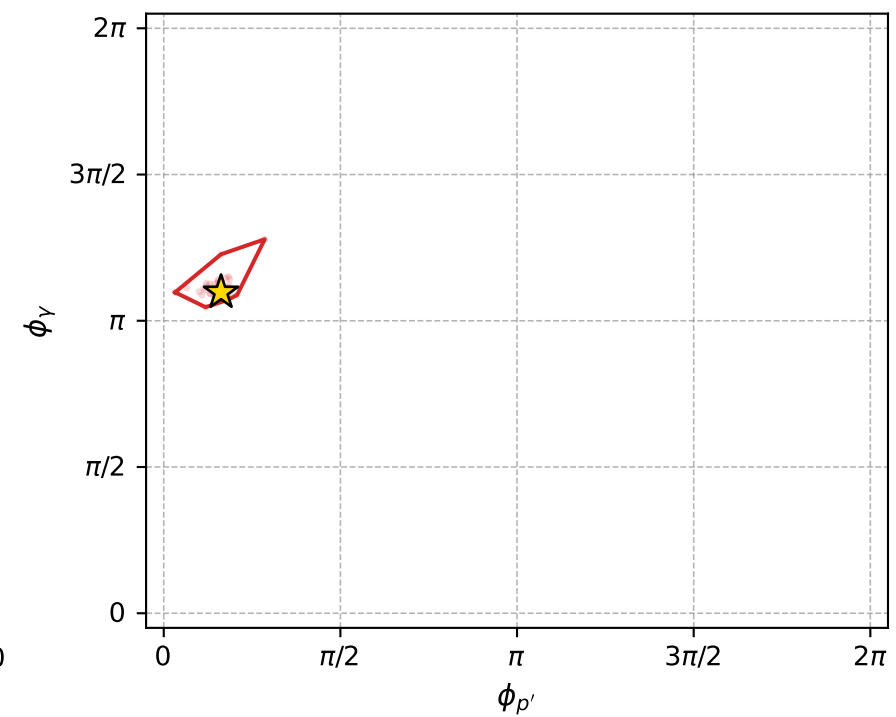
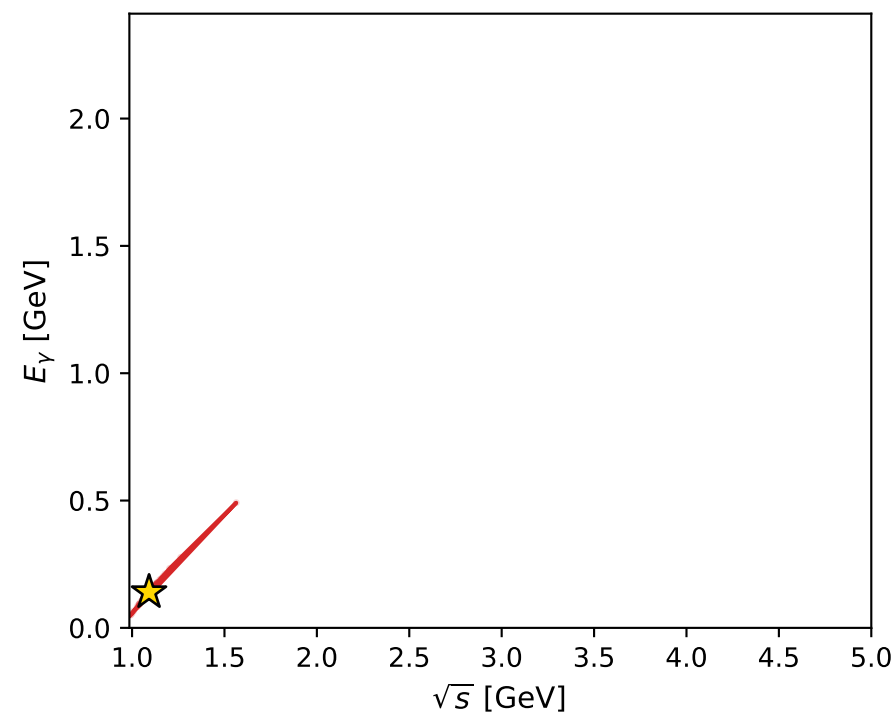
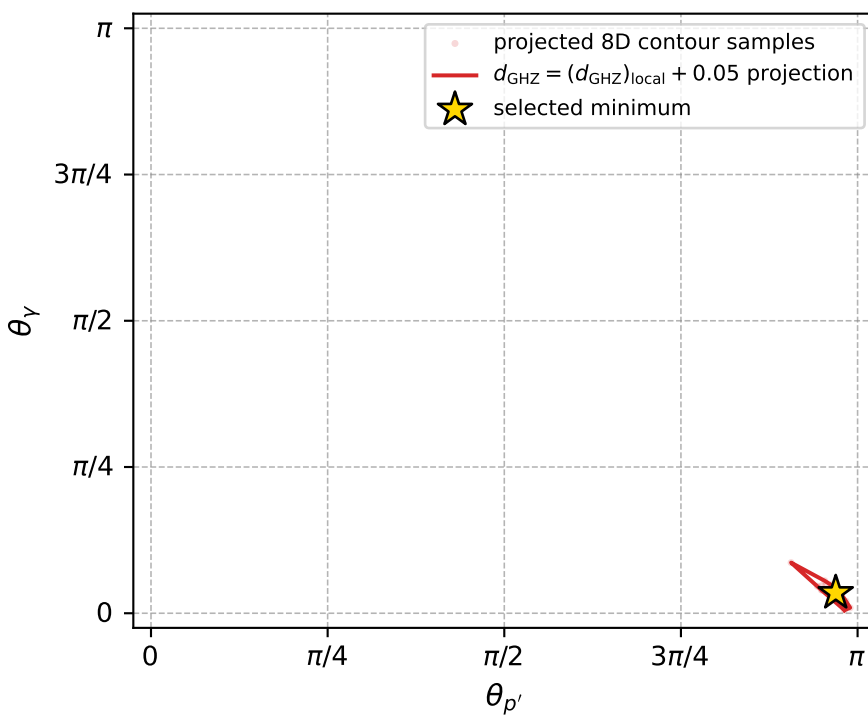


$(h_p, h_Y, h_\ell) = (-1, -1, -1)$

$(h_p, h_Y, h_\ell) = (+1, +1, +1)$



electron: polarization cluster P1, local minimum 373; pairwise projections of the 8D $d_{\text{GHZ}} = (d_{\text{GHZ}})_{\text{local}} + 0.05$ contour



selected local minimum

$d_{\text{GHZ}}(\text{local}) = 0.03006092$
 $d_{\text{GHZ}} \text{ contour} = 0.08006092$
 above global minimum = 0.030060897
 8D contour samples = 253

selected phase-space point:

$\text{sqrt_s} = 1.091934$
 $\text{theta_p_out} = 3.044484$
 $\text{theta_gamma_out} = 0.1095033$
 $\text{q0ut} = 0.1401212$
 $\text{phi_p_out} = 0.509518$
 $\text{phi_gamma_out} = 3.447224$
 $\text{alpha_e} = 0.7421441$
 $\text{alpha_p} = 0.9001505$